

Chapter 11

Connectivity

11.1 Universal Serial Bus Controller (USB)

11.1.1 Overview

The USB module is a USB 3.0-compliant serial interface engine for implementing a USB interface. This module may be connected to an external port. Collectively the module and external port are called the USB 3.0 interface. USB 3.0 supports super-speed (SS), high-speed (HS), full-speed (FS), and low-speed (LS) operations.

NOTE

Synopsys Proprietary. Used with permission.

- The upper layer is common for USB 2.0 and USB 3.0 operation. This has the bus interface, buffer management block, list processor for scheduling, and control and status register (CSR) functions
- USB 2.0 PHY and MAC layers
- USB 3.0 PHY, LINK, and MAC layers

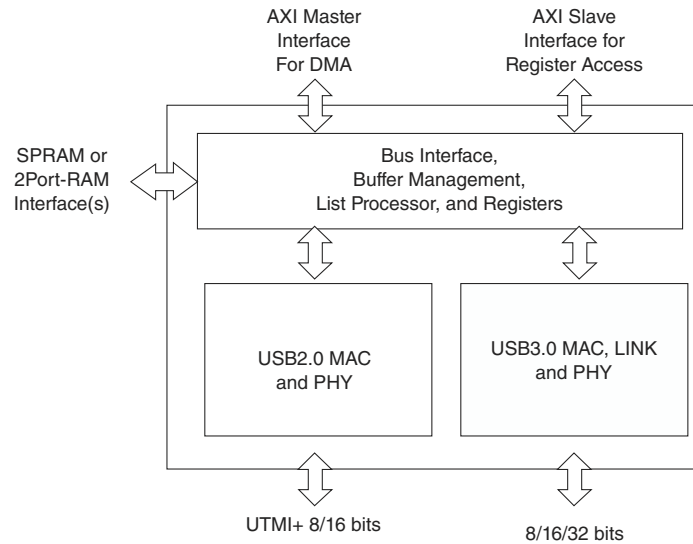


Figure 11-1. USB interface block diagram

11.1.1.1 Features

The USB 3.0 module includes the following features:

- Complies with USB specification rev 3.0 (xHCI compatible)
- Supports operation as a standalone USB host controller
- USB dual-role operation and can be configured as host or device
- Super-speed (5 Gbit/s), high-speed (480 Mbit/s), full-speed (12 Mbit/s), and low-speed (1.5 Mbit/s) operations.
- Supports operation as a standalone single port USB
- Supports four programmable, bidirectional USB endpoints
- OTG (on-the-go) 2.0 compliant, which includes both device and host capability. Super-speed operation is not supported when OTG is enabled.
- Supports system memory interface with -bit addressing capability

11.1.1.2 Modes of Operation

The USB 3.0 module operates in following modes.

- Host Mode: SS/HS/FS/LS
- Device Mode: SS/HS/FS
- OTG: HS/FS/LS

11.1.1.3 External Signals

This section contains detailed descriptions of all the USB 3.0 controller signals. Many of the signals for the PHY interfaces are muxed onto the same pins in order to reduce pin count. The following table shows the USB signals, indicating which interface supports each signal.

Table 11-1. USB 3.0 External Signals

Signal	I/O	Description
USB1_D_P	IO	USB PHY Data Plus
USB1_D_M	IO	USB PHY Data Minus
USB1_VBUS	IO	USB PHY VBUS
USB1_ID	IO	USB PHY ID Detect
USB1_TX_P	O	USB PHY 3.0 Transmit Data (positive)
USB1_TX_M	O	USB PHY 3.0 Transmit Data (negative)
USB1_RX_P	I	USB PHY 3.0 Receive Data (positive)
USB1_RX_M	I	USB PHY 3.0 Receive Data (negative)
USB1_RESREF	IO	USB PHY Impedance Calibration
USB1_DRVVBUS	O	VBus power enable.
USB1_PWRFAULT	I	Indicates that a Vbus fault has occurred.

11.1.2 Functional Description

11.1.2.1 System memory descriptor and data buffers

The software creates transfer request blocks (TRBs), and four DWORDs each, that point to the data buffers. Normally, the TRBs are allocated consecutively in system memory; only the data buffers can be scattered. In the case of a circular buffer, the link-TRB points to the next TRB. Once the TRBs and data buffers are set up in the system memory, the software driver issues a start transfer command that points to the location of the first TRB in the system memory to start the DMA operation. TRBs, though small (only four DWORDs), provide a rich set of features for the software to schedule transfers, isochronous, control, interrupt moderation, and so on.

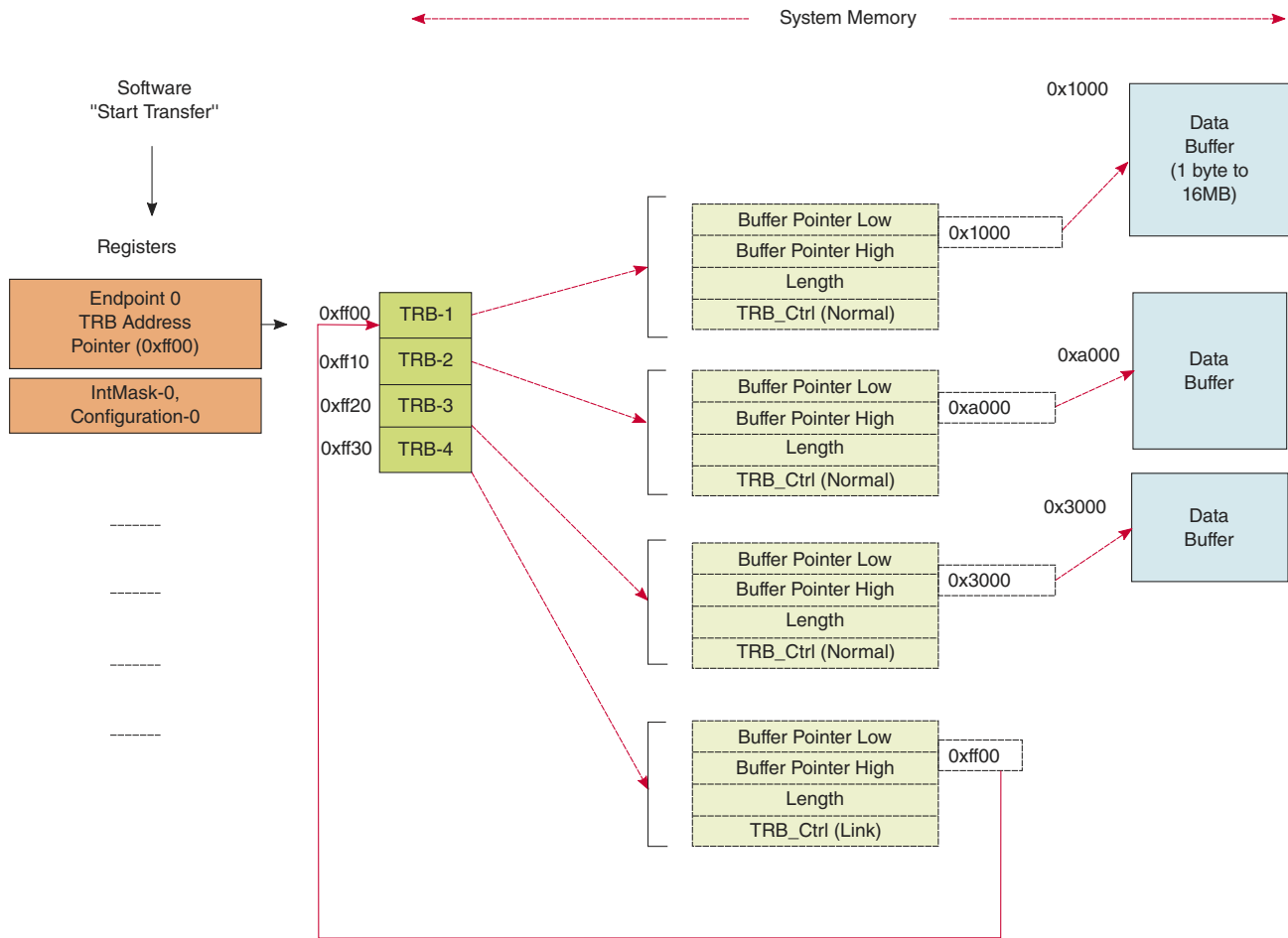


Figure 11-2. System Memory Descriptor and Data Buffers

11.1.2.2 Device descriptor structures

Device mode transfer request blocks (TRBs) are small 4 DWORDs and at the same time provide a rich set of features so that the software can efficiently manage the USB core, memory buffers, and MIPS requirements.

The following is a list of device mode TRB characteristics:

- TRBs provide support for scattered data structures. The scattered data buffers can be zero bytes to 16 MB in length.
- TRBs must be placed in system memory aligned to a 16-byte boundary.
- TRBs are kept in linear memory to enhance descriptor caching performance. If the data buffers are small (as in an Ethernet application) the core can efficiently collect the scattered data to build USB packets without wasting bus efficiency that collecting scattered descriptors requires.

- Supports TRB linking. This allows software to set up a circular ring of TRBs with the last TRB linking to the first one.
- Supports system memory interface with -bit addressing capability
- Supports byte-aligned buffers for each TRB of a transfer. This feature prevents the need for buffer copying in cases where the USB device driver receives unaligned data buffers from other applications (for example, Ethernet). Whenever the application controls buffer allocation, it must allocate SoC bus width and burst-aligned buffers to facilitate efficient bus and memory utilization. For transmitting, the core supports byte-aligned buffers on all TRBs.

For example: To use 16 bursts in a 64-bit system, you must try to allocate buffers that are $16 * 8 = 128$ bytes aligned. This is the normal buffer structure because Linux-like OSs allocate 4 KB buffers. SDR/DDR memory controllers also provide better performance when requests are burst aligned.

- Supports software queueing of multiple USB transfers (LST bit for IN/OUT transfers and CSP bit for OUT transfers control this function).
- Provides interrupt moderation capability, allowing software to selectively enable events on TRB completion, as controlled by the IOC (Interrupt on completion) bit. The IOC event is also used by software to reallocate the released buffers, allowing software to re-use just a few buffers in a circular fashion. This helps when the memory is limited but have enough MIPS to process interrupts. Larger buffers can be allocated to reduce the number of interrupts.
 - In USB 3.0 larger transfers are recommended because the raw transfer rate is almost 10 times faster, unlike USB 2.0, where drivers set up only 64 KB or 128 KB transfers. For example, a 64 KB transfer that takes 1.3 ms in USB 2.0 requires only 164 s in USB 3.0. If you set up buffers of 64 KB and enable the transfer completion event, then you receive an interrupt every 164 sec.
- Supports streaming (Stream ID field used for this purpose).

11.1.2.2.1 Structures

This figure shows the control and status field of a transfer request block.

Figure 11-3. TRB control and status fields

Table 11-2. Device descriptor structure field definitions

Field	Description	Hardware access
DW 03-00		
31:0	Buffer pointer low (BPTL)	R/W

Table continues on the next page...

Table 11-2. Device descriptor structure field definitions (continued)

Field	Description	Hardware access
	Data buffer pointer to low 32 bit address (BPTR[31:0]). Hardware may also update this field (implementation specific).	
DW 07-04		
31:0	Buffer pointer high (BPTRH) Data buffer pointer to high 32-bit address (BPTR[63:32]).	R/W
DW 0B-08		
31:28	TRB status (TRBSTS) Hardware updates this field with transfer status information before releasing the TRB. 4'h0: OK 4'h1: MissedIsoc: Isochronous interval missed or incomplete 4'h2: SetupPending - During the current control transfer data/status phase, another SETUP was received. 4'h4: TransferInProgress - During the current transfer, an end transfer command was received.	R/W
25:24	Packet count M1 (PCM1) For high-speed, High bandwidth isochronous IN endpoints, this field in an Isoc-First TRB represents the total number of packets in the Buffer Descriptor minus 1.	R/W
23:0	Buffer size (BUFSIZ) If CHN=0 and HWO=0 for the TRB, this field represents the total remaining Buffer Descriptor buffer size in bytes. Valid Range: 0 bytes to (16 MB - 1 byte). The hardware decrements this field to represent the remaining buffer size after data is transferred. For a Link TRB, the buffer size should be "0".	R/W
DW 0F-0C		
31:30	Reserved.	R/W
29:14	Stream ID / SOF Number For stream-based bulk endpoints: The Stream ID of the transfer this TRB is associated with. Stream ID must be the same in all TRBs (R/W). For isochronous endpoints: The (micro)frame number in which the last packet of this TRB's buffer was transmitted or received (debug purposes only) (RO).	R/W
13:12	Reserved.	R/W
11	Interrupt on Complete (IOC) When IOC is set in a TRB, and once the transfer for this buffer is completed, the core will issue XferInProgress event with IOC bit set in the event's status. This indicates the buffer is available for software to reuse or release.	R
10	Interrupt on Short Packet / Interrupt on Missed ISOC (ISP/IMI) Applicable to OUT endpoints when a short packet is received, and CSP=1 and LST=0. If this bit is 1, the core generates an XferInProgress event. For Isochronous endpoints: If this bit is 1, the core generates an XferInProgress event when the interval represented by the Buffer Descriptor completes with a "Missed Isoc" status.	R
9:4	TRB Control (TRBCTL) Indicates the type of TRB:	R

Table continues on the next page...

Table 11-2. Device descriptor structure field definitions (continued)

Field	Description	Hardware access
	1: Normal (Control-Data-2+ / Bulk / Interrupt) - Set TRBCTL to 1 for all TRBs used in data stage except the first TRB 2: Control-Setup 3: Control-Status-2 - Set TRBCTL to 3 for a SETUP request without data stage 4: Control-Status-3 - Set TRBCTL to 4 for a SETUP request with data stage 5: Control-Data - Set TRBCTL to 5 for the first TRB of a data stage 6: Isochronous-First - Set TRBCTL to 6 for the first TRB of a Service Interval 7: Isochronous 8: Link TRB Others: Reserved	
3	Continue on Short Packet (CSP) Applicable to OUT endpoints only when a short packet is received. If this bit is 1, the core will continue to the next Buffer Descriptor. This setting is required for isochronous endpoints. If this bit is 0, the core will generate an XferComplete event and remove the stream.	R
2	Chain Buffers (CHN) Applicable to IN and OUT endpoints. Set to 1 by software to associate this TRB with the next TRB. A Buffer Descriptor is defined as one or more TRBs. The CHN bit is used to identify the TRBs that comprise a Buffer Descriptor. The CHN bit is always 0 in the last TRB of a Buffer Descriptor and when the LST field is set to 1.	R
1	Last TRB (LST) Indicates this is the last TRB in a list. After completing the transfer for the associated buffer the core will stop the transfer for the endpoint / bulk-stream and issues an XferComplete event. The stream is automatically removed by the hardware.	R
0	Hardware Owner of Descriptor (HWO) Indicates that hardware owns the TRB. Software sets this bit to 1 when it creates the TRB, and cannot modify it until hardware resets this bit to 0. However, there are exceptions for short packets on OUT endpoints and Link TRBs. Because the hardware autonomously checks this bit to determine if the entire TRB is valid, software must set this field to '1' after preparing the other three DWORDs of the TRB with valid information.	R/W

11.1.2.2.1.1 Normal (Control-Data/Bulk/Interrupt), Isochronous, and Status Transfer Request Block Structure

The Normal TRB is used for Bulk/Control-Data/Interrupt endpoint transfers. The data buffers can be scattered anywhere and each may have different sizes. The TRB Buffer Pointer and the Buffer Size fields point to buffer address and size respectively. The Stream ID for bulk endpoints will be programmed by the application.

For isochronous endpoints, the first TRB in a service interval must have the Isoc-First type, the last TRB in a service interval must have CHN=0, and any other TRBs have CHN=1. The starting (micro)frame time is communicated via the Start Transfer command, and the core tracks the (micro)frame times of the subsequent Buffer Descriptors.

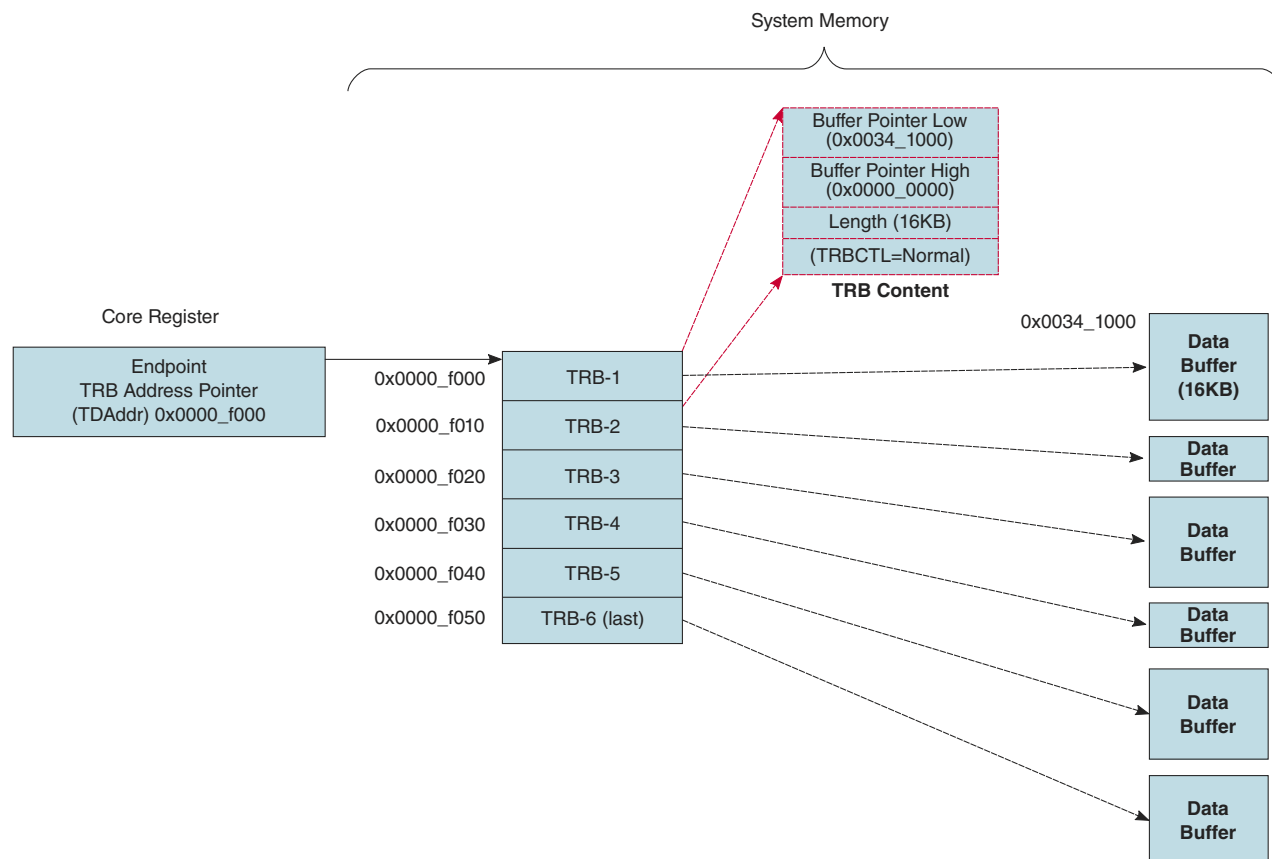


Figure 11-4. Normal (Control-Data/Bulk/Interrupt) Descriptor Structure

11.1.2.2.1.2 Setup and Status TRB Structure

To receive a SETUP packet, the driver queues up a single Setup TRB, whose buffer pointer value may be set to any address, including the address of the TRB. The buffer size must be set to 8. The core will write the 8 bytes of the received SETUP to the address requested. If the address of the TRB is used, there is no need for a separate data buffer to receive a SETUP packet. After completing Setup stage, driver will schedule Data stage and Status stage transfers. For more information, see [Control transfer programming model](#)

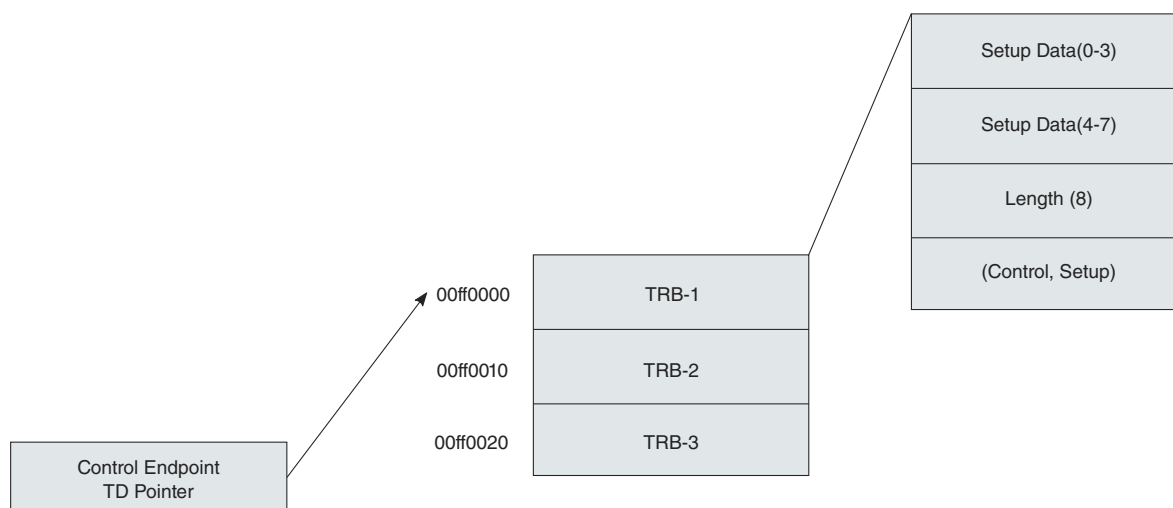


Figure 11-5. SETUP Descriptor Structure with Buffer Pointing to Setup TRB

After interpreting the SETUP bytes, software will determine if the next stage of the control transfer is a data stage or status stage.

If the SETUP bytes require a 3-stage control transfer, the TRB type used in the Data stage must be Control- Data for the first TRB of the Buffer Descriptor. When the host moves on to the Status stage, the TRB Type must be Control-Status-3.

If the SETUP bytes require a 2-stage control transfer, the TRB Type must be Control-Status-2.

The Status TRB is a zero-length TRB, with an unspecified Buffer Pointer value and a Buffer Size of zero. There is no data buffer associated with a Status TRB.

11.1.2.2.1.3 Link TRB Structure

The Link TRB is used to link back to the starting TRB for reusing TRBs in a circular fashion.

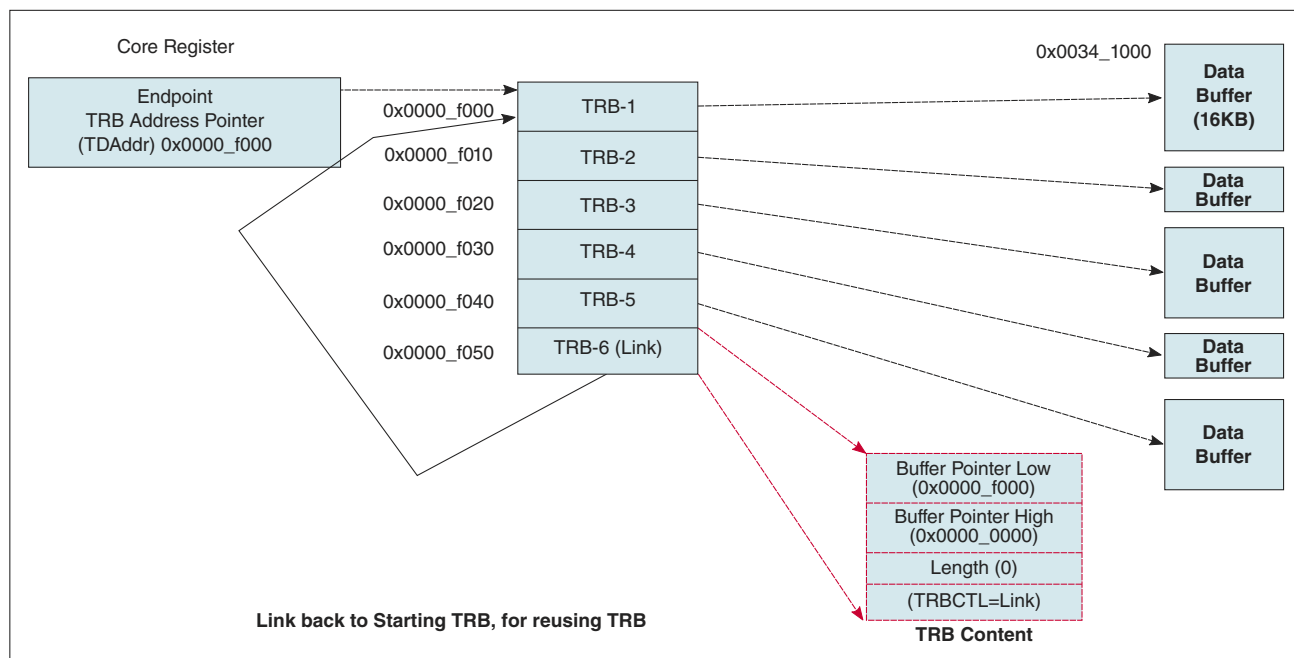


Figure 11-6. Link TRB Structure

If software prepares a circular TRB list which is shorter than the number of cached TRBs (four), the core automatically detects the loop and will not re-fetch a TRB that has already been fetched. For example, for four cached TRBs the software setup three TRBs in the following way:

- TRB-1: Normal, HWO=1
- TRB-2: Normal, HWO=1
- TRB-3: Link to TRB 1

The core will fetch TRBs 1, 2, and 3. Then the core will follow the link to TRB-1, note that its address is the same as the TRB-1 that has already been fetched into the cache, and will temporarily stop fetching TRBs. When TRB-1 completes due to traffic on the USB, the core will write TRB-1 back to memory with the HWO field set to '0', generate a **XferInProgress** event if necessary, and will automatically attempt to fetch TRB-1 again. In most cases, the core will see the HWO field set to '0' and will stop fetching until software updates the TRB, sets the HWO field back to '1', and issues an **Update Transfer** command.

11.1.2.2.1.4 Chaining buffers (CHN) and interrupt on completion (IOC) usage

This figure shows a chaining buffer example for an isochronous IN and a bulk IN transaction.

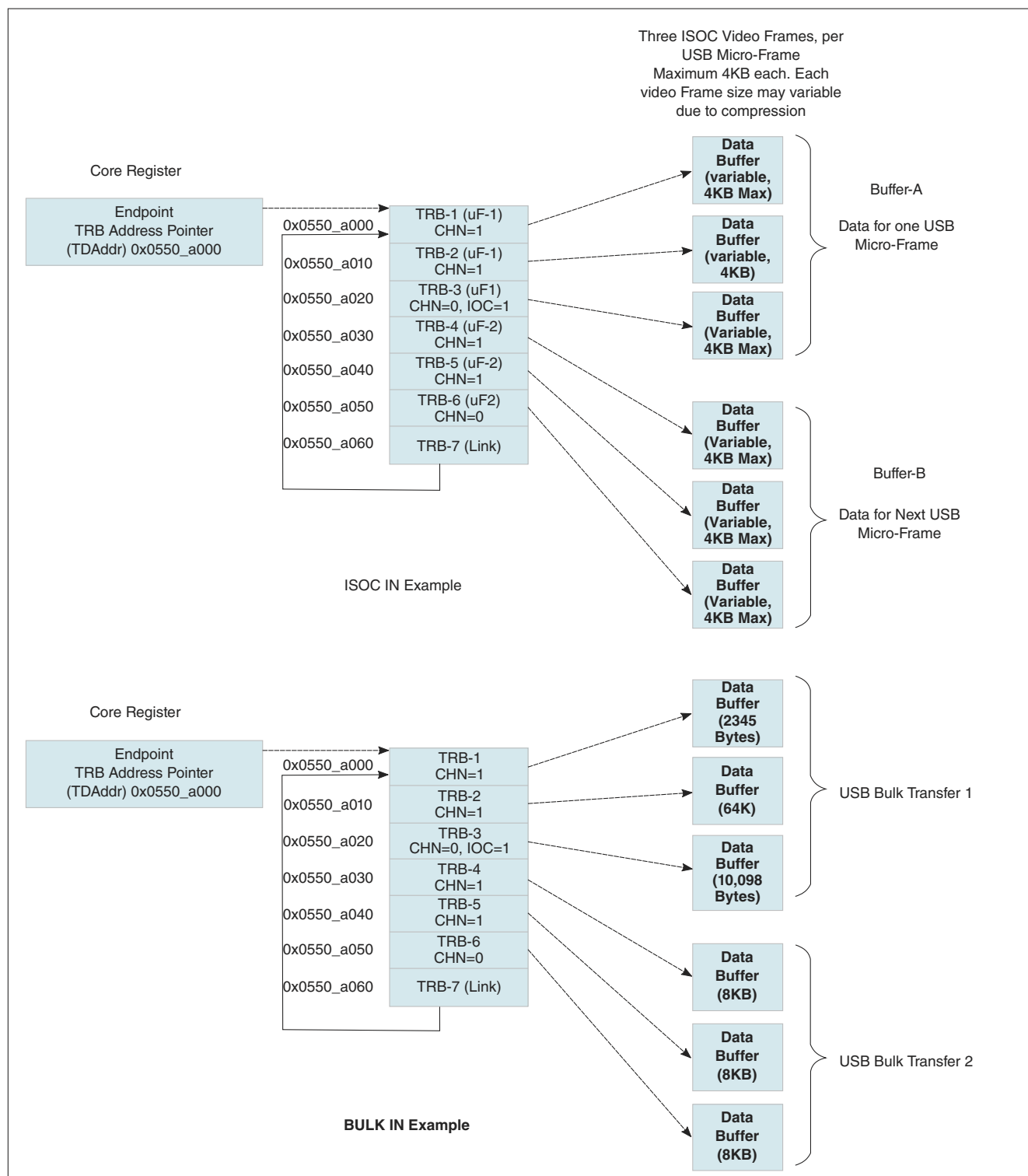


Figure 11-7. Chaining buffers for isochronous and bulk IN

In the isochronous IN application shown in the above figure, there are three video frames to be sent to the host in each microframe. Each video buffer size is maximum of 4 KB and the size may vary for each microframe depending on the compression. The CHN bit

in the TRB-3 is 0, which indicates this is the last buffer of a transfer for the given microframe. The device schedules TRB-1, 2, and 3 in the first bInterval. Similarly, TRB-6 indicates (CHN=0) microframe boundary, and TRB-4, 5, and 6 will be sent during the next bInterval.

For the last packet of last transfer in a microframe, the device internally sets last packet flag when responding at SuperSpeed for isochronous IN transfers.

The application can also use the IOC bit to receive an interrupt when the buffer-A transfer is completed, so it can use buffer-A to send data on the bInterval following the next bInterval. As long as the driver can service the interrupt and set up data before another bInterval, the USB transfers can continue without interruption. Depending upon the system interrupt latency, the buffer size can be adjusted.

If interrupt latency is high occasionally, the software may not have time to set up the linked TRB before the next bInterval. In this case, the core, on seeing HWO bit set to 0 in the TRB, stops processing further TRBs of this endpoint. When hardware receives the Update Transfer command from software, it will re-fetch the TRB. In the case of bulk application, the core will issue the NRDY signal. In isochronous transfers, zero-length packets are sent to the host until software enables the transfer again.

In the bulk IN example, the CHN bit indicates USB Transfer boundary. During a TRB transfer, CHN indicates whether to send the bytes from next TRB buffer as part of the current transfer. For example, even though TRB-1 has 2,345 bytes, the last 297 ($2345 - 2 * 1024 = 297$) bytes will be combined with the data in TRB-2 and sent as a 1,024 byte packet on the USB since CHN=1. On the other hand, when there is a short packet left in TRB-3, the short packet will be sent separately, since CHN=0.

The device writes data into TRB-1 and TRB-2 for the first transfer. The CHN bit (CHN=0) in TRB-2 indicates that this is the last buffer of the transfer, then TRB-3 and TRB-4 are written for the second transfer. Similarly, the CHN bit (CHN=0) in TRB-4 indicates the second transfer boundary.

If an interrupt latency is high, software may not have time to set up the TRB. In this case, the core, on seeing the HWO bit set to 0 in the TRB, will stop processing further TRBs for the endpoint. Hardware will re-fetch the TRB when software issues the update transfer command. In the case of bulk, control, or interrupt endpoints, the core will issue NRDY. In the case of isochronous endpoints, packets will be dropped until software enables the transfer again.

In the bulk OUT example below, when a short packet is received for TRB-1, TRB-3 will be used for the next OUT transfer and TRB-2 will be closed.

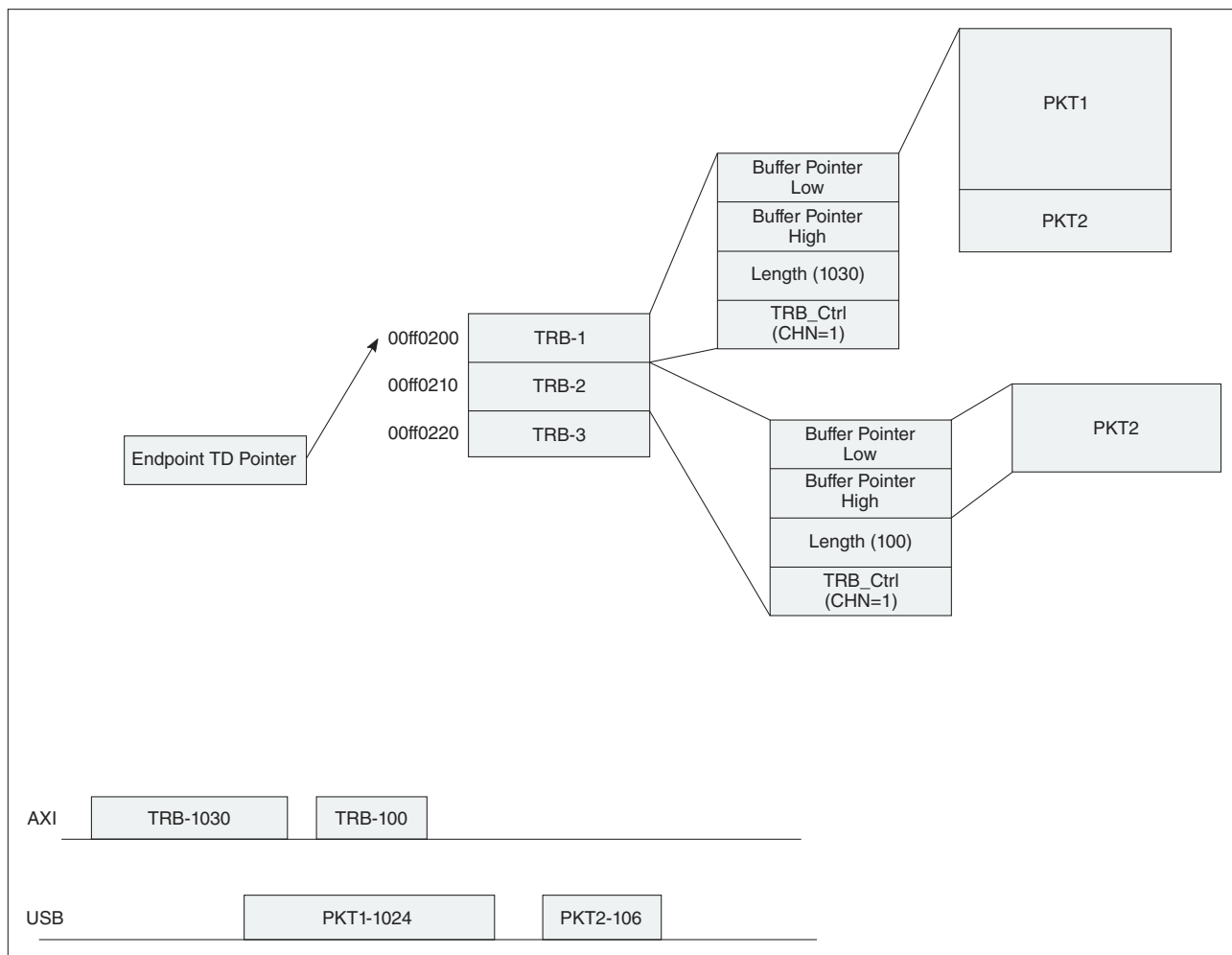


Figure 11-8. Chaining buffers, example 1

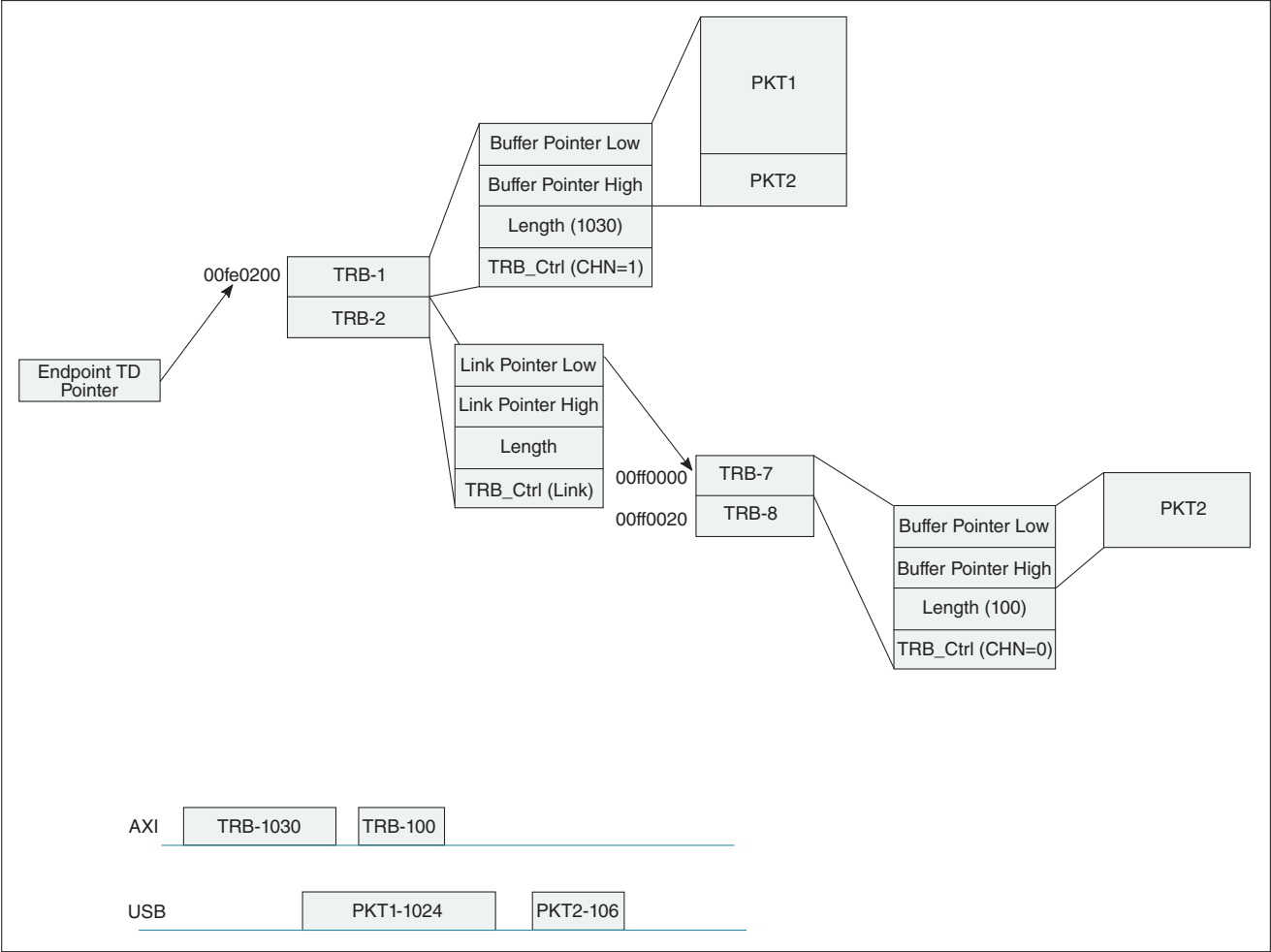


Figure 11-9. Chaining buffers, example 2

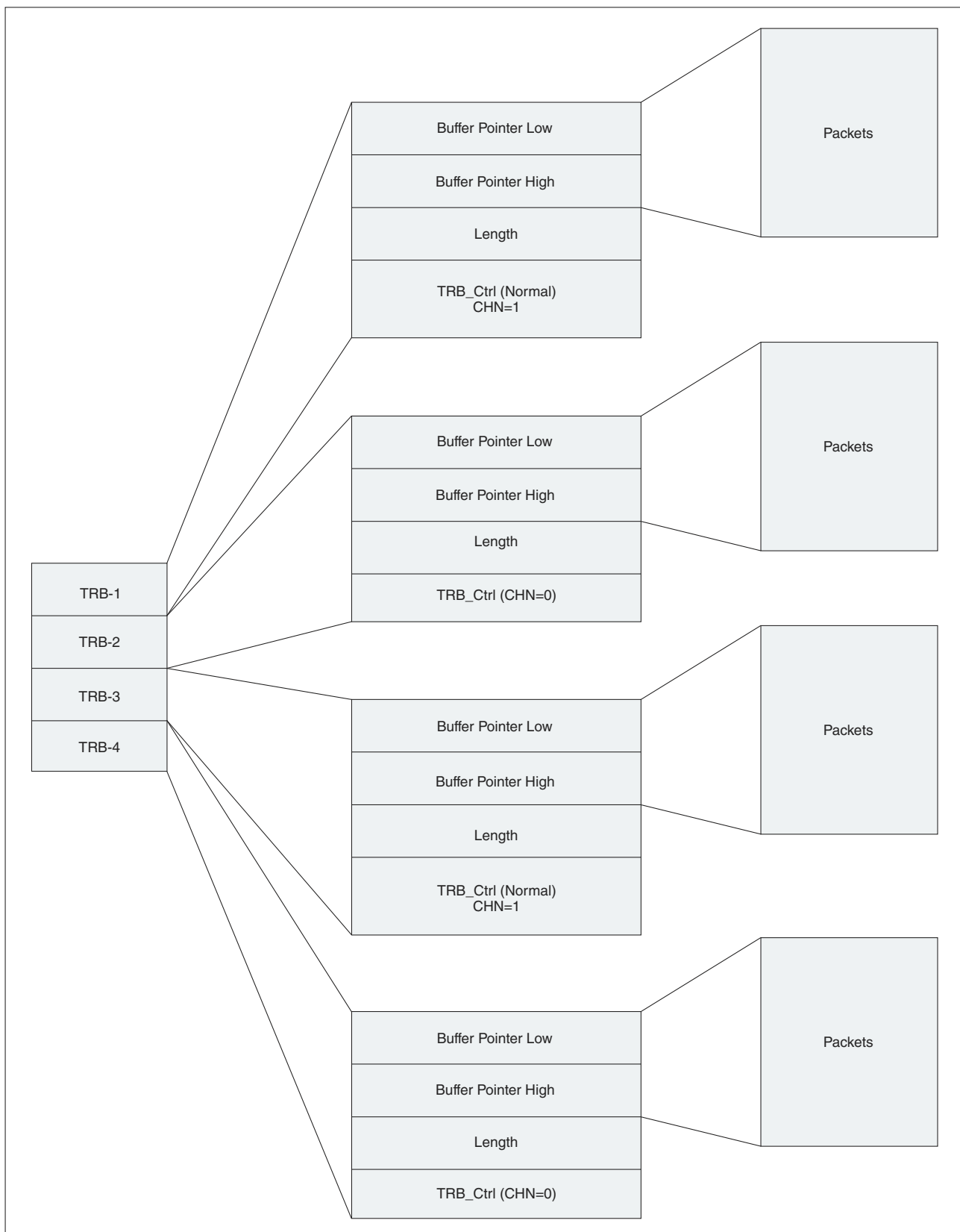


Figure 11-10. Bulk OUT with two transfers

11.1.2.2.1.5 Interrupt on Short Packet (ISP) and Continue on Short Packet (CSP) Usage

These two bits are used to schedule single or multiple OUT transfer. In most applications where only one OUT transfer per endpoint is scheduled, the software always sets ISP=1 and CSP=0. If the device receives a short packet, then it indicates a USB transfer completion through XferComplete events.

In an Ethernet-over-USB application, where software knows it is going to receive short packets, it can set up multiple transfer size buffers in one step by setting CSP=1 in all the TRBs. On a short packet, the device will update the byte count and move to next TRB. Software can also set ISP once in n number of TRBs to receive an XferInProgress event so it can process the previous short packets.

The following example shows software setting multiple 1500-byte Ethernet transfers and enabling the XferInProgress event once for four Ethernet packets to reduce the number of interrupts.

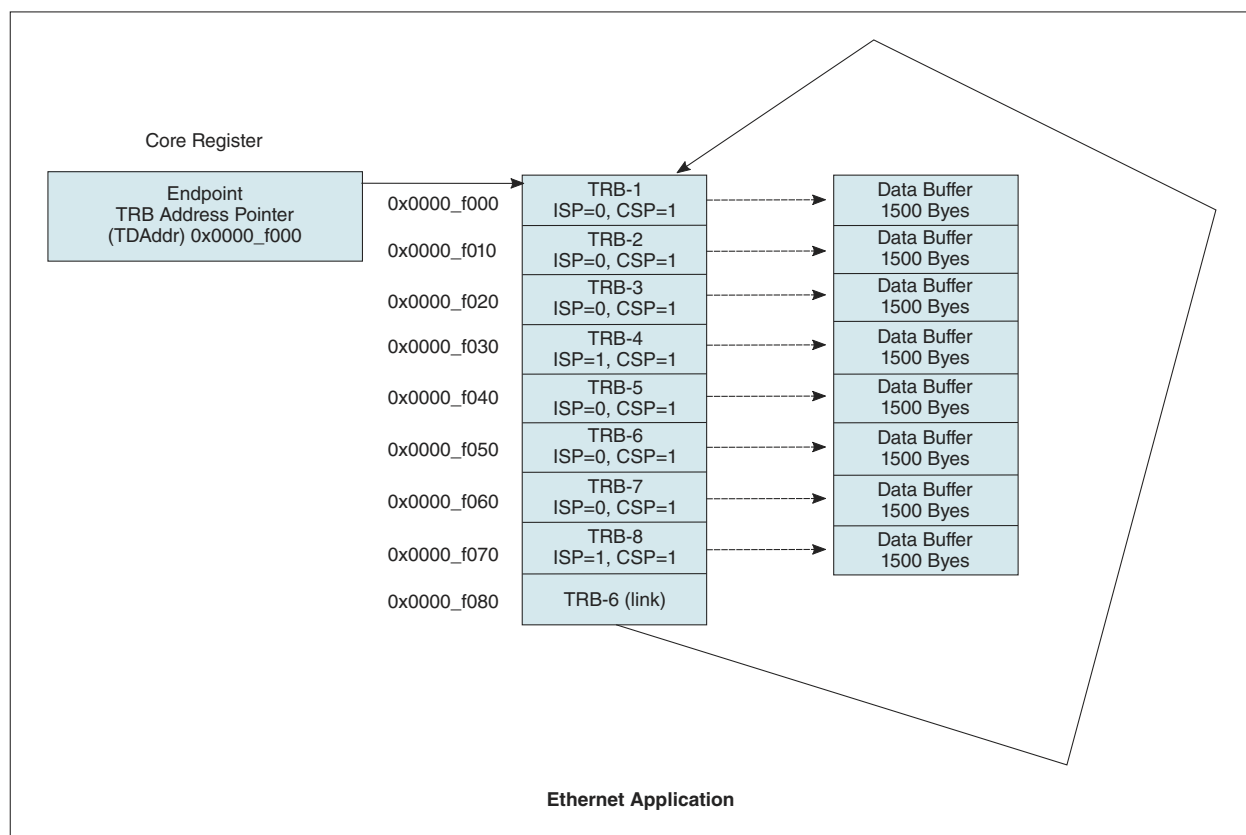
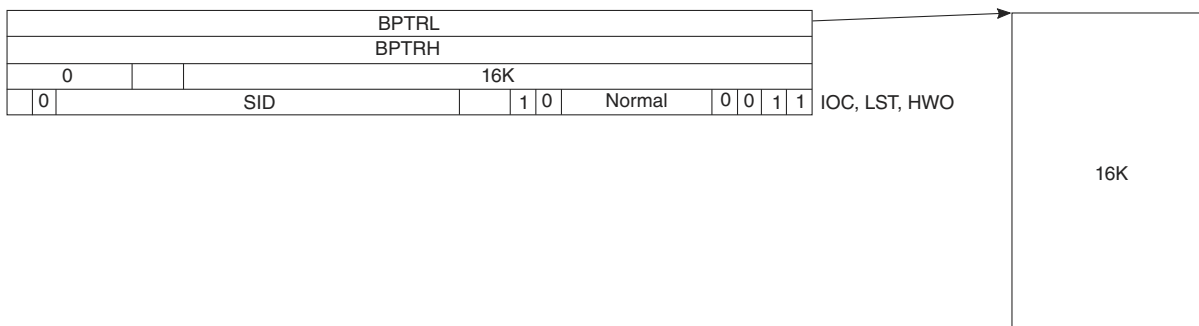


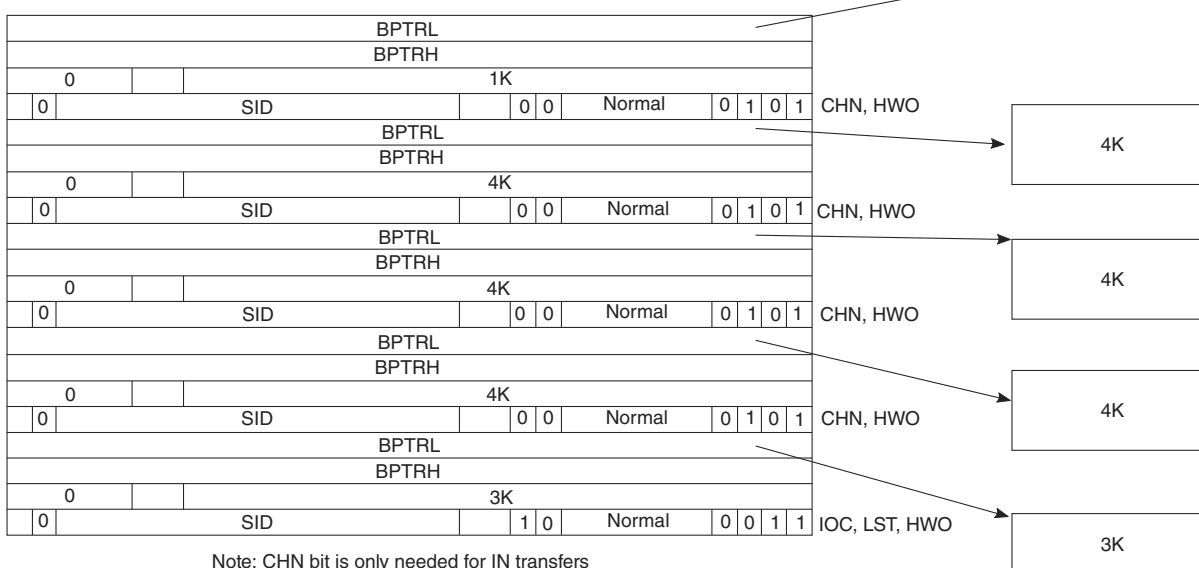
Figure 11-11. ISP and CSP Usage

11.1.2.2.1.6 Example of Setting Up TRBs

Physically Contiguous 16KB Bulk Transfer



Physically Discontiguous 16KB Bulk Transfer (e.g. Mass Storage)



Physically Discontiguous 1518 Byte Bulk Transfer (e.g. Ethernet)

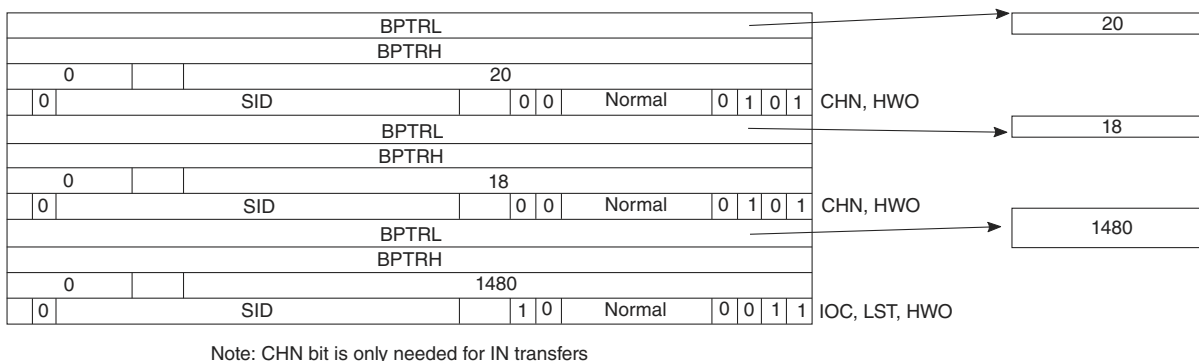


Figure 11-12. Bulk IN TRB Examples

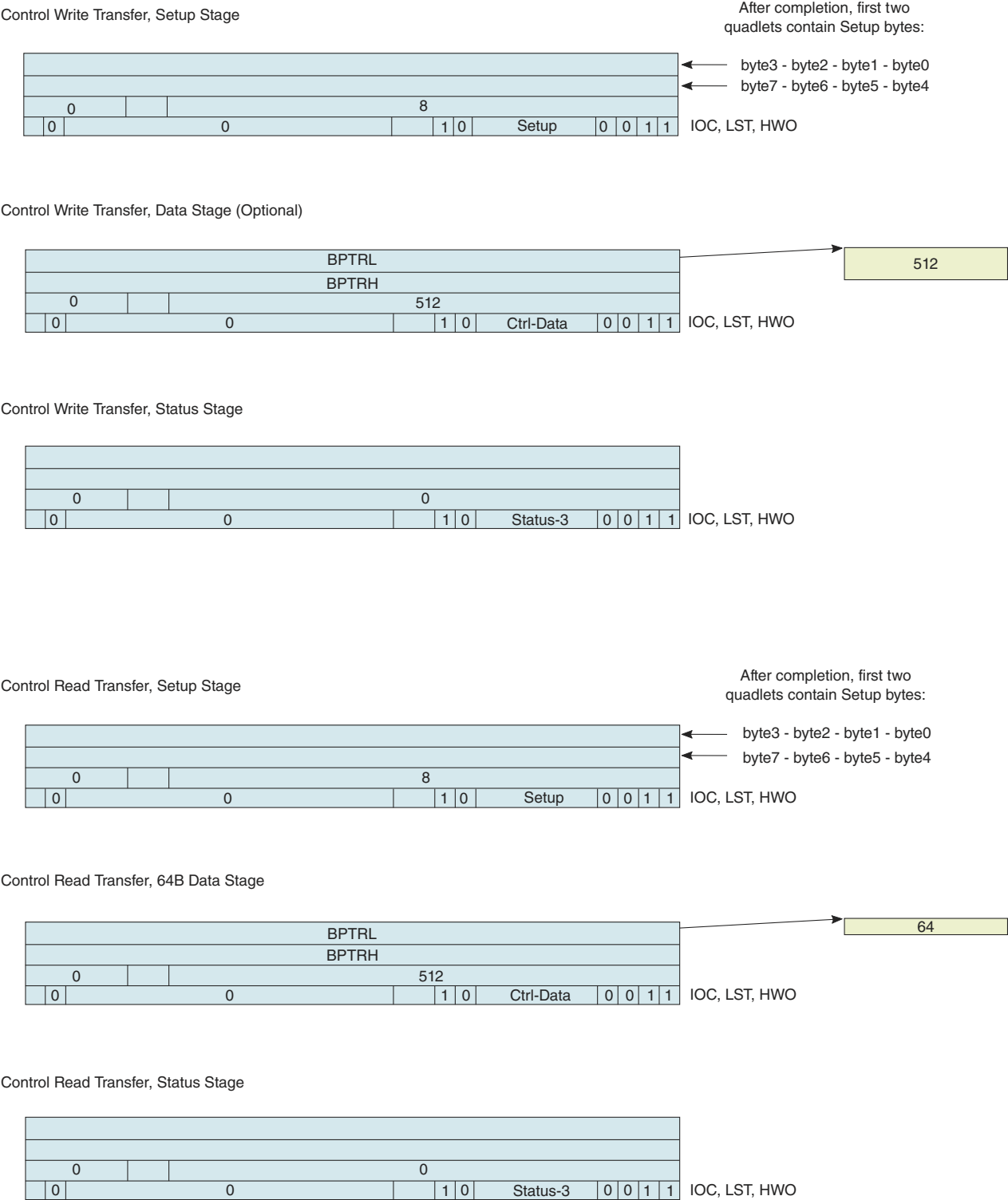


Figure 11-13. Setup/Control/Status TRB Examples

Six Isochronous transfers, one transfer every four micro frames, some physically discontinuous buffers, IOC on the 4th transfer.

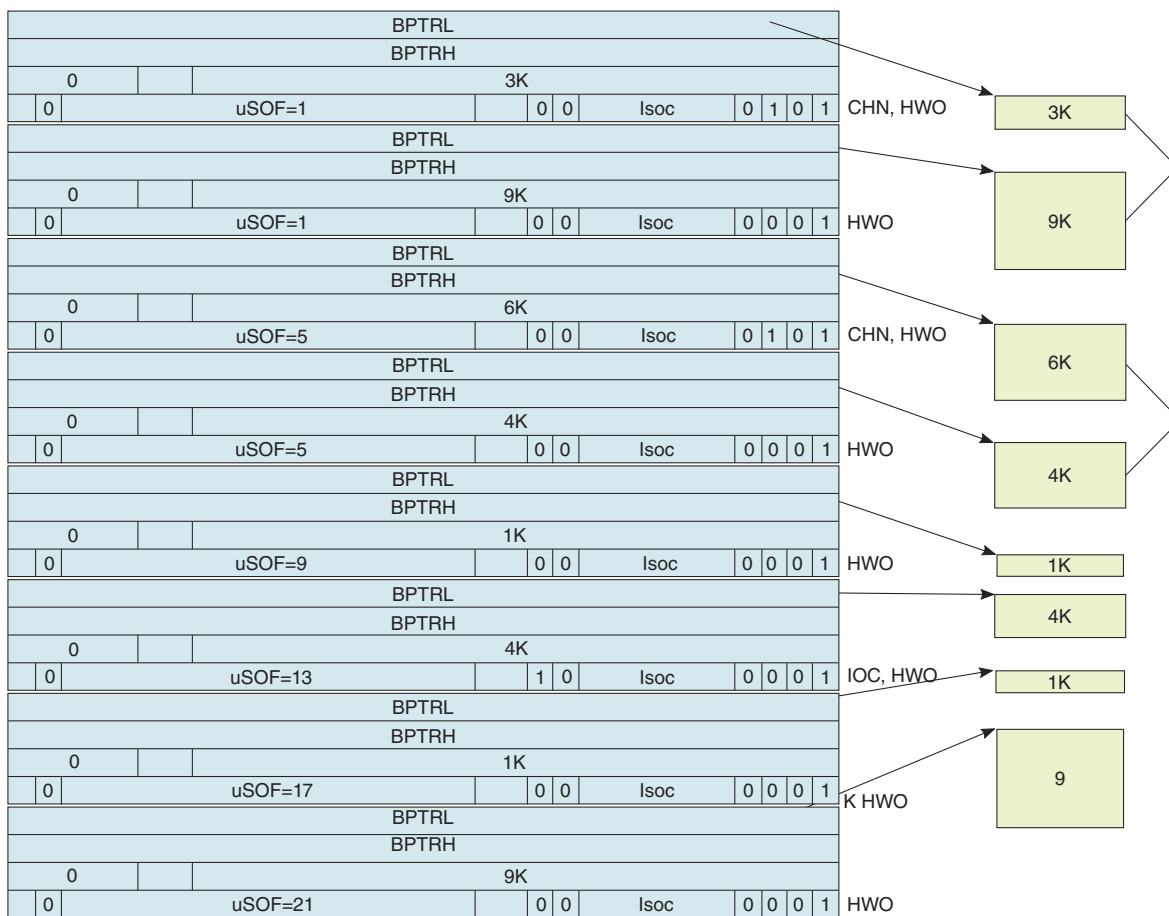


Figure 11-14. Isochronous IN TRB, Example 2

Eight Isochronous Transfers, Physically Contiguous Buffers, IOC on the 4th and 8th Transfers, Circular Linked

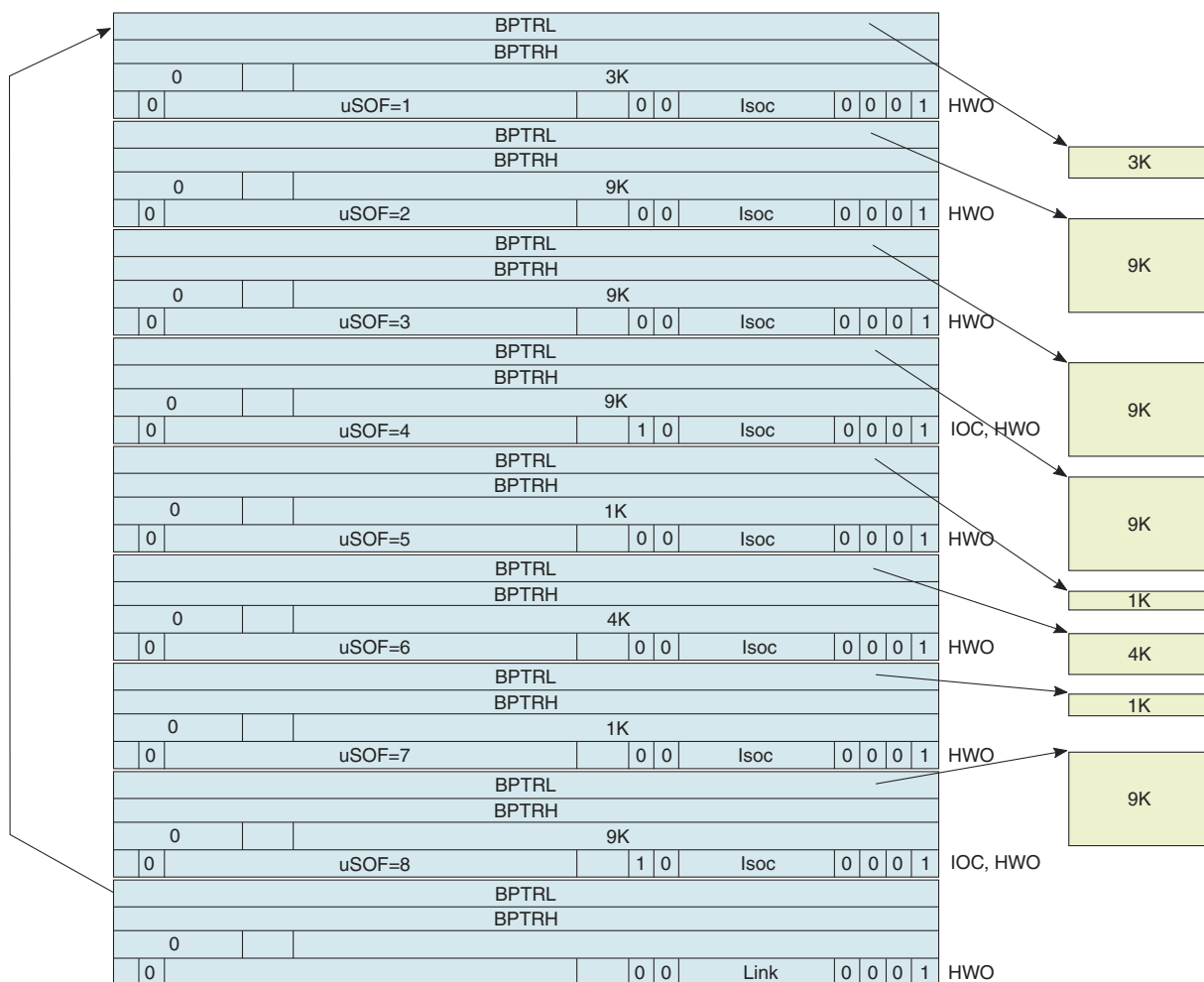


Figure 11-15. Isochronous IN TRB, Example 1

11.1.2.3 Device Programming Model

11.1.2.3.1 Register Initialization

The USB3 core contains global registers (prefixed by G*) and device registers (prefixed by D*) that are programmed to start operation and handle certain events. This section describes which registers must be accessed depending on the event that software is attempting to handle:

- Power-On or Soft Reset
- USB Reset Event
- Connect Done Event

- SetAddress Device Request
- SetConfiguration Device Request
- Disconnect Event
- Device-Initiated Disconnect and Reconnect

11.1.2.3.1.1 Device Power-On or Soft Reset

This section explains device core initialization after power-on or soft reset. The application must follow this initialization sequence for device mode operation.

When the core is first powered on, software initializes the following registers. The order of operations is not important, except for the first and last steps (DCTL[CSFTRST]=1 and DCTL[RUN_STOP]=1).

Table 11-3. Power-On or Soft Reset Register Initialization

Register	Description
DCTL	Set the CSftrSt field to '1' and wait for a read to return '0'. This resets the device core.
GSBUSCFG0/1	Leave the default values, refer Global SoC Bus Configuration Register 0 (GSBUSCFG0) and Global SoC Bus Configuration Register 1 (GSBUSCFG1)
GTXTHRCFG/ GRXTHRCFG	This is required only if threshold is enabled.
GUSB2PHYCFG	Program PHY as per Global USB2 PHY Configuration Register (GUSB2PHYCFG) or leave the default values if the correct power-on values were selected. Note: The PHY must not be enabled for auto-resume in device mode.
GUSB3PIPECTL	Program the following PHY [DatWidth] or leave the default values if the correct power-on values were selected.
GTXFIFOSIZn	Write these registers to allocate prefetch buffers for each Tx endpoint. Unless the packet sizes of the endpoints are application-specific, it is recommended to use the default value.
GRXFIFOSIZ0	Write this register to allocate the receive buffer for all endpoints. Unless the packet sizes of the endpoints are application-specific, it is recommended to use the default value.
GEVNTADR/ GEVNTSIZ/ GEVNTCOUNT	Depending on the number of interrupts allocated, program the Event Buffer Address and Size registers to point to the Event Buffer locations in system memory, the sizes of the buffers, and unmask the interrupt. Note: USB operation stops if the Event Buffer memory is insufficient, because the core stops receiving/transmitting packets.
GCTL	Program this register to override scaledown, RAM clock select, and clock gating parameters.
DCFG	Program device speed and periodic frame interval
DEV TEN	At a minimum, enable USB Reset, Connection Done, and USB/Link State Change events
DEPCMD0	Issue a DEPSTARTCFG command with DEPEVT.XferRscldx set to 0 and CmdIOC set to 0 to initialize the transfer resource allocation. Poll CmdAct for completion.
DEPCMD0/ DEPCMD1	Issue a DEPCFG command for physical endpoints 0 & 1 with the following characteristics, and poll CmdAct for completions: USB Endpoint Number = 0 or 1 (for physical endpoint 0 or 1) FIFONum= 0 XferNRdyEn and XferCmplEn = 1

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Table 11-3. Power-On or Soft Reset Register Initialization (continued)

Register	Description
	Maximum Packet Size = 512 Burst Size = 0 EPTYPE = 2'b00 (Control)
DEPCMD0/ DEPCMD1	Issue a DEPXFERCFG command for physical endpoints 0 & 1 with DEPCMDPAR0_0/1 set to 1, and poll CmdAct for completions
DEPCMD0	Prepare a buffer for a setup packet, initialize a setup TRB, and issue a DEPSTRXFER command for physical endpoint 0, pointing to the setup TRB. Poll CmdAct for completion. Note: The core attempts to fetch the setup TRB via the master interface after this command completes.
DALEPENA	Enable physical endpoints 0 & 1 by writing 0x3 to this register
DCTL	Set DCTL[RunStop] to '1' to allow the device to attach to the host. At this point, the device is ready to receive SOF packets, respond to control transfers on control endpoint 0, and generate events.

After the controller has been started, wait for the following events:

- Wait for a DEVT.USBReset event. This indicates that a reset is detected on the USB.
- Wait for a DEVT.ConnectionDone event. This event indicates the end of the reset on the USB. On this event, read the DSTS register to get the connection speed.

11.1.2.3.1.2 Initialization on USB reset

To initialize the core as a device, during USB Reset, the application must perform the following steps:

Table 11-4. Initialization on USB reset

Register	Description
DEPCMD0	If a control transfer is still in progress, complete it and get the core into the "Setup a Control-Setup TRB / Start Transfer" state
DEPCMDn	Issue a DEPENDXFER command for any active transfers (except for the default control endpoint 0)
DEPCMDn	Issue a DEPCSTALL (ClearStall) command for any endpoint in STALL mode prior to the USB Reset (excluding control endpoints)
DCFG	Set DevAddr to '0'

NOTE

- Special Reset Considerations for Default Control Endpoint 0: The default control endpoint is not affected by a USB Reset, so software must continue following the software flow for control transfers explained in "Control Transfer Programming Model" even across a USB Reset event. Resources can only be assigned to the default control endpoint once after a power on or soft reset.

11.1.2.3.1.3 Initialization on connect done

When this event is received, software must perform the following steps:

Table 11-5. Initialization on connect done

Register	Description
DSTS	Read this register to obtain the connection speed
GCTL	Program the RAMClkSel field to select the correct clock for the RAM clock domain. This field is reset to 0 after USB reset, so it must be reprogrammed each time on Connect Done.
DEPCMD0/ DEPCMD1	Issue a DEPCFG command (with Config Action set to "Modify") for physical endpoints 0 & 1 using the same endpoint characteristics from Power-On Reset, but set MaxPacketSize to 512 (SuperSpeed), 64 (High-Speed), 8/16/32/64 (Full-Speed), or 8 (Low-Speed).
GUSB2CFG/ GUSB3PIPECTL	Depending on the connected speed, write to the other PHY's control register to suspend it
GTXFIFOSIZn	(optional) Based on the new MaxPacketSize of IN endpoint 0, software may choose to re-allocate the TXFIFO sizes by writing to these registers.

11.1.2.3.1.4 Initialization on SetAddress request

When the application receives a SetAddress request in a SETUP packet, it performs the following steps:

Table 11-6. Initialization on SetAddress request

Register	Description
DCFG	Program the DCFG register with the device address received as part of the SetAddress request when SETUP packet is decoded.
DEPCMD1	After receiving the XferNotReady(Status) event, acknowledge the status stage by issuing a DEPSTRXFER command pointing to a Status TRB. This step must be done after the DCFG register is programmed with the new device address.

At this point, the device is ready to receive micro-SOF/ITP and is configured to receive control transfers on control endpoint 0 with a new address assigned.

11.1.2.3.1.5 Initialization on SetConfiguration or SetInterface Request

When the application receives a SetConfiguration or SetInterface request in a SETUP packet, it performs the following steps:

Table 11-7. Initialization on SetConfiguration or SetInterface Request

Register	Description
DALEPENA	Set this register to 0x3 to disable all endpoints other than the default control endpoint 0.
DEPCMDn	Issue a DEPENDXFER command for any active transfers (except for the default control endpoint 0).
DEPCMD1	Issue a DEPCFG command (with Config Action field set to "Modify") for physical endpoint 1 using the current endpoint characteristics to re-initialize the TXFIFO allocation.
DEPCMD0	Issue a DEPSTARTCFG command with DEPCMD0.XferRscldx set to 2 to re-initialize the transfer resource allocation.
DEPCMDn	Issue a DEPCFG command (with the Config Action field set to "Initialize") for each endpoint that is present in the new configuration (except for the default control endpoint). Note: Control endpoints are bi-directional and must use consecutive even/odd physical endpoint numbers for the OUT/IN direction (such as 2/3, or 4/5), and the FIFONum must be configured to the same value in both directions.
DEPCMDn	Issue a DEPXFERCFG command for each endpoint that is present in the new configuration (except for the default control endpoint 0).
GTXFIFOSIZn	(optional) Based on the new configuration of IN endpoints, software may choose to re-allocate the TXFIFO sizes by writing to these registers.
DALEPENA	Enable the logical endpoints that are active in the new configuration.
DEPCMD1	After receiving the XferNotReady(Status) event, acknowledge the status stage by issuing a DEPSTRTXFER command pointing to a Status TRB. This step must be done after the previous steps to ensure the host does not access any other endpoints that are being set up.

At this point, the core is prepared to accept Start Transfer commands for the newly-configured endpoints. For information on how to set up transfers, see the [Operational Model](#).

11.1.2.3.1.6 Alternate initialization on SetInterface request

When handling a SetInterface device request, another possibility is that software may reconfigure existing endpoints instead of starting the configuration from the beginning. This is only possible if no TXFIFOs need to be reassigned. This flow also assumes that the endpoints that are being removed or reconfigured in the new interface have halted their traffic prior to the host issuing the SetInterface request.

Table 11-8. Alternate initialization on SetInterface request

Register	Description
DEPCMDn	Make sure there are no transfers still active on the endpoints that are changing in the new interface. This is normally ensured by the host prior to issuing the SetInterface, but if not, issue an End Transfer command for the transfer on each endpoint that is changing.
DEPCMDn	Issue a DEPCFG command (with the Config Action field set to "Initialize") for each endpoint that is changing in the new configuration. The side effect of the DEPCFG command is that the endpoint's sequence number is automatically set to 0.
DALEPENA	Write this register with all the endpoints that are enabled (including the ones that are not changing).
DEPCMDn	Using the StartXfer command, acknowledge the status stage response for the SetInterface request.

11.1.2.3.1.7 Initialization on Disconnect Event

When the application receives a Disconnect event, it must set DCTL[ULSTCHNGREQ] to 5. Other than this, the core does not require any initialization. Because the DCTL[RUN_STOP] bit is still '1', the device attempts to reconnect to the host, at which time a USB Reset and Connect Done event occurs. However, if the application does not want to attempt to reconnect to the host, it should perform the steps in the next section.

NOTE

When DCFG[DEVSPD] is programmed for 2.0 only mode (such as, High-Speed or Full-Speed), and if the application wants to issue any commands to clear any pending transfers during a Disconnect interrupt, then it has to disable USB_GUSB2PHYCFG[SUSPENDUSB20] before issuing any commands and re-enable it after the commands have completed.

11.1.2.3.1.8 Device-initiated disconnect

If the application wants to disconnect from the host, it should perform the following actions:

Table 11-9. Initialization on device-initiated disconnect

Register	Description
DEPCMD0	If a control transfer is still in progress, complete it and get the core into the "Setup a Control-Setup TRB / Start Transfer" state
DEPCMDn	Issue a DEPENDXFER command for any active transfers (except for the default control endpoint 0)
DCTL	Set DCTL[RUN_STOP] to '0' to disconnect from the host
DSTS	Poll [DEVCTRLHLT] until it is '1'

At this point, the device is disconnected from the host and will not attempt to reconnect.

11.1.2.3.1.9 Reconnect after Device-Initiated Disconnect

If the application decides to reconnect to the host, it must follow the steps in [Device Power-On or Soft Reset](#).

11.1.2.3.2 Operational Model

The following sections describes the processes and data structures that the core uses to implement the USB 3.0 specification.

11.1.2.3.2.1 USB and Physical Endpoints

Endpoints are referred to in two ways:

- As a USB endpoint number: USB endpoints are defined in the USB specification.
- As a physical endpoint resource number: The hardware has a fixed number of physical endpoint resources. Each resource is unidirectional and can be configured to refer to either direction of any USB endpoint.

The software always works on physical endpoints and it knows how physical endpoint corresponds to USB endpoints. DALEPENA is the only register that has one enable bit per USB endpoint.

During SetConfiguration, software maps physical endpoint resources to the required USB endpoints. When a USB request comes in, the USB endpoint number gets converted to a physical endpoint number in the core. Similarly, when a USB packet is sent out, the physical endpoint number is converted to the USB endpoint and sent out.

A USB control endpoint requires two physical endpoints. One physical endpoint is mapped to the OUT direction of the Control endpoint, and the other one is mapped to the IN direction of the Control endpoint. Specifically the two physical endpoints are used as the following:

- The Setup stage of any control transfer uses the OUT direction physical endpoint.
- For a control write or 2-stage transfer, the Data stage (if present) uses the OUT direction physical endpoint and the Status stage uses the IN direction physical endpoint.
- For a control read transfer, the Data stage uses the IN direction physical endpoint and the Status stage uses the OUT direction physical endpoint.

11.1.2.3.2.2 Event Buffers

Hardware passes command completion, transfer progress, and asynchronous events to software through one or more Event Buffers.

To configure an Event Buffer, the software performs the following steps:

1. Sets up an empty buffer in system memory.

2. Writes the address of the beginning of the buffer into GEVNTADR. This address must be aligned to the Event Buffer size.
3. Writes the size of the buffer and interrupt mask into GEVNTSIZ. Depending on your system interrupt latency, enough Event Buffer space must be allocated to avoid lost interrupts or reduced performance.
4. Write a 0 into the GEVNTCOUNT register. This must be the last step, as it enables the Event Buffer. After the Event Buffer has been configured, software must not change the size or address.

There is one interrupt line per Event Buffer that indicates there are one or more events present. Software reads one or more events out of the buffer and indicates to hardware how many events it processed by writing the byte count to the GEVNTCOUNT register.

Clock crossing delays may result in the interrupt's continual assertion after software acknowledges the last event. Therefore, when the interrupt line is asserted, software must read the GEVNTCOUNT register and only process events if the GEVNTCOUNT is greater than 0.

The first event produced by the core after the Event Buffer is configured will be written to the address specified in GEVNTADR. Most events are 32 bits, and subsequent events will be written to the address (PreviousEventAddress + 4). When that address exceeds the sum of GEVNTADR and GEVNTSIZ, the core wraps around to the first GEVNTADR value. In this way, the Event Buffer operates like a circular buffer with hardware writing to the "tail" of the buffer and software reading from the "head."

Most events are exactly 4 bytes in size, but there is one exception: The Vendor Device Test LMP Received Event (VndrDevTstRcvd) is a 12 byte event that includes a header in the first 4 bytes and the contents of the LMP in the following 8 bytes.

When an event occurs within the core, hardware checks the enable bit that corresponds to the event to decide whether the event will be written to the Event Buffer or not. The Event Buffer contains one of the following types of information, depending on the value of the lower bits of the event:

- Device Endpoint-Specific Event (DEPEVT) (Event[0] = 0x0)
 - The DEPCFG endpoint-specific command specifies the bits that are enabled and which Event Buffer to use for these events.
- Device-Specific Event (DEVT) (Event[0] = 0x1, Event[7:1] = 0x00)
 - The Generic Command Complete event is enabled through the DGCMD[CmdIOC] field when the command is issued.
 - The Event Buffer Overflow Event cannot be disabled and is written to the Event Buffer that encounters the overflow.

- The rest of the Device-Specific events are enabled through the DEVTEN register.
- Except for the Event Buffer Overflow Event, these events are written to the Event Buffer specified in the DCFG[IntrNum] field.

The core always leaves one entry free in each Event Buffer. When the Event Buffer is almost full, hardware writes the Event Buffer Overflow event and the USB will eventually get stalled when endpoints start responding NRDY or the link layer will stop returning credits (in SuperSpeed). This event is an indication to software that it is not processing events quickly enough. During this time, events will be queued up internally. When software frees up Event Buffer space, the queued up events will be written out and the USB will return to normal operation.

Event Buffer Content for Device Endpoint-Specific Events (DEPEVT)

Table 11-10. Device Endpoint-n Events: DEPEVT

Field	Description
31–16	Event Parameters (EventParam) For XferNotReady, XferComplete, and Stream events on Bulk Endpoints: [31-16]StreamID. Applies only to bulk endpoints that support streams. This indicates the Stream ID ¹ of the transfer for which the event is generated For XferInProgress: [31-16] Isochronous Microframe Number (IsocMicroFrameNum). Indicates the microframe number of the beginning of the interval that generated the XferInProgress event (debug purposes only) For XferNotReady events on Isochronous Endpoints: [31-16] Isochronous Microframe Number (IsocMicroFrameNum). Indicates the microframe number during which the endpoint was not ready NOTE: USB 3.0 core represents USB bus time as a 14-bit value on the bus and also in the DSTS register (DSTS[SOFFN]), but as a 16-bit value in the XferNotReady event. Use the 16-bit value to interact with Isochronous endpoints via the StartXfer command. The extra two bits that the USB 3.0 core produces will be necessary for handling wrap-around conditions in the interaction between software and hardware. EPCmdCmpl events For all EPCmdCmpl events [27-24]- Command Type. The command type that completed (Valid only in a DEPEVT event. Undefined when read from the DEPCMD.EventParam field). For EPCmdCmpl event in response to Start Transfer command. [22-16] Transfer Resource Index (XferRsclIdx). The internal hardware transfer resource index assigned to this transfer. This index must be used in all Update Transfer and End Transfer commands.
15–12	Event Status (EventStatus) Within an XferNotReady event- [15]: Indicates the reason why the XferNotReady event is generated: 0- XferNotActive- Host initiated a transfer, but the requested transfer is not present in the hardware 1- XferActive- Host initiated a transfer, the transfer is present, but no valid TRBs are available

Table continues on the next page...

Table 11-10. Device Endpoint-n Events: DEPEVT (continued)

Field	Description
	[14]: Not Used [13:12]: For control endpoints, indicates what stage was requested when the transfer was not ready: 2'b01- Control Data Request 2'b10- Control Status Request Within an XferComplete or XferInProgress event- [15]: LST bit of the completed TRB (XferComplete only) [15]: MissedIsoc: Indicates the interval did not complete successfully (XferInProgress only) [14]: IOC bit of the TRB that completed [13]: Indicates the TRB completed with a short packet reception or the last packet of an isochronous interval [12]: Reserved If the host aborts the data stage of a control transfer, software may receive a XferComplete event with the EventStatus field equal to '0'. This is a valid event that must be processed as a part of the Control transfer programming model. Within a Stream Event- [15:12]: - 4'h2- StreamNotFound- This stream event is issued when the stream-capable endpoint performed a search in its transfer resource cache, but could not find an active and ready stream. - 4'h1- StreamFound- This stream event is issued when the stream-capable endpoint found an active and ready stream in its transfer resource cache, and initiated traffic for that stream to the host. The ID of the selected Stream is in the EventParam field. In response to a Start Transfer command- 4'h2- Indicates expiry of the bus time reflected in the Start Transfer command. 4'h1- Indicates there is no transfer resource available on the endpoint. In response to a Set Transfer Resource (DEPXFRCFG) command- 4'h1- Indicates an error has occurred because software is requesting more transfer resources to be assigned than have been configured in the hardware. In response to a End Transfer command- 4'h1- Indicates an invalid transfer resource was specified.
11–10	This field is reserved.
9–6	4'h7- Endpoint Command Complete (EPCmdCmplt) Indicates software may issue another Device Endpoint command to the endpoint. When issued in response to an End Transfer command, indicates that DMA stopped for the endpoint. For all other commands, this event does not imply that all effects of the command took place. The DEPCMD register contains the same status information present in the Event Status Bits field. 4'h6- Stream Event (StreamEvt) Indicates that a stream-capable endpoint initiated a search within its transfer resource cache. The result of the search is in the EventStatus field (Found or NotFound). 4'h5- Reserved 4'h4 - Reserved

Table continues on the next page...

Table 11-10. Device Endpoint-n Events: DEPEVT (continued)

Field	Description
	4'h3- XferNotReady Event (XferNotReady) Indicates receipt of a transaction when no TRBs are available for the endpoint. For isochronous IN endpoints, a zero-length packet is automatically sent by hardware and NRDY for non-isochronous endpoints. The application must enable this event if it plans to issue Start Transfer on demand. This event can happen when software issues a Start Transfer or Update Transfer. In this case, software must ignore this event because it has already issued the Start Transfer or Update Transfer. XferNotReady is generated when the core responds NRDY to the host on the USB. It is useful in the beginning of a transfer when software wants to wait for the host to start polling the endpoint before setting up TRBs, but it is not efficient to use this event to determine when the core has run out of TRBs. For determining when the core has processed TRBs and needs software to setup more, use the IOC field in a TRB along with the XferInProgress event that is generated when the core completes a TRB with IOC=1. For non-stream-capable endpoints, the hardware filters multiple events if the host continues transactions, even after the core responds with NRDY. This internal filter is reset after software issues any endpoint command to this endpoint. For stream-capable endpoints, this event is generated each time the host attempts a transaction, even if the core responds with NRDY. For isochronous endpoints, this event is generated only once prior to the Start Transfer command to communicate the current bus time. For additional information, see the Event Status field. 4'h2- XferInProgress Event (XferInProgress) Applies to IN and OUT endpoints. Indicates an EP/Stream specific event happened and it is continuing the transfer. For additional information, see the Device Programming Model. 4'h1- XferComplete Event (XferComplete) Applies to IN and OUT endpoints. Indicates that a EP/Stream transfer completed and the core stopped the transfer. For additional information, see the Device Programming Model. 4'h0- Reserved
5–1	Physical Endpoint Number (0-31)
0	1'b0- Indicates that this is an endpoint-specific event.

1. Note that term Stream ID used in USB chapter is different from Arm SMMU terminology.

Event Buffer Content for Device-Specific Events (DEVT)

Table 11-11. Device-specific events: DEVT

Field	Description
31-25	Reserved
24-16	Event Information Bits (EvtInfo) For a USB/Link State change Event, this field indicates the state of the link- EvtInfo[8:5] - HIRD value received from the LPM token (valid for a Hibernation Request Event) EvtInfo[4] - SuperSpeed event. Set to 1 for SS; Set to 0 for non-SS.

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Table 11-11. Device-specific events: DEVT (continued)

Field	Description
	EvtInfo[3:0] - Link State. Indicates link state at the time of the event. Follows same encoding as DSTS link state bits.
15-12	Reserved
11-8	11- Event Buffer Overflow Event (EvtOverflow) The core writes this event to indicate there is no space in the event buffer, and one or more Device-specific events may have been dropped after this event. Endpoint-Specific events will not be dropped, but they will be delayed which can cause a drop in performance on the USB. Software uses this event as a warning that the Event Buffer is too small and the core should be reconfigured with a larger Event Buffer or software needs to process events more quickly. In order to avoid repeated Event Buffer Overflow Events, software must free up space in the Event Buffer by acknowledging more than 1 event (writing a value greater than 4 to the GEVNTCOUNT register). This event cannot be disabled, and is always written to the Event Buffer that encounters the overflow. 10- Generic Command Complete Event (CmdCmpl) The core writes this event to indicate the generic command is complete. 9- Erratic Error Event (ErrticErr) The core writes this event to report erratic errors (phy_rxvalid_i/phy_rxvldh_i or phy_rxactive_i is asserted for at least 2 ms, due to PHY error) seen on the UTMI+. Due to erratic errors, the USB 3.0 core goes into Suspended state and a USB/Link state change event (ULStChng) is generated to the application. If the early suspend is asserted due to an erratic error, the application can only perform a soft disconnect recover. For SuperSpeed operation, the PHY erratic error event indicates that the PIPE is not responding to the PHY command (for example, pipe3_PowerDown change or receiver detection). After the core requested any PHY command, if it did not receive pipe3_PhyStatus within 100ms, the core will timeout and generate the erratic error event. After generating the erratic error, the core assumes that the PHY command is completed successfully and proceeds with the next pending PHY command. Software must reset the controller on receiving the erratic error. 8- Reserved 7- Start of (micro)Frame (Sof) 6- USB Suspend Entry Event This event provides a notification that the link has gone to a suspend state (L2, U3, or L1). This event is generated when hibernation mode is disabled. The existing Link State Change event (3) provides the same information, but is generated for every link state change. 5- Hibernation Request Event This event provides a notification that the link has gone to a suspend state where hibernation is supported (L2, U3, or L1) and requires software intervention - When DCTL.KeepConnect=1 and L1HibernationEn=0, this event is generated only when the link goes to L2 or U3 - When DCTL.KeepConnect=1, L1HibernationEn=1, and the core is configured to support LPM (DCTL.LPMCap=1 and DCTL.HIRD_Thres[4]=1), this event is generated when the link goes to L2, U3, or L1 with an HIRD value greater than the DCTL.HIRD_Thres[3-0] value The existing Link State Change event (3) provides the same information, but is generated for every link state change. If both events are enabled, only the Hibernation Request event will be generated if the conditions are met. When this event occurs, software is required to start the hibernation process

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Table 11-11. Device-specific events: DEVT (continued)

Field	Description
	EvtInfo[4:0] indicates the Link State and Speed which caused the Hibernation Event (using the same encoding as the Link State Change event. Suspend is 3, and L1 is 2) EvtInfo[8:5] provides the HIRD value from the host's LPM tokenFor L2 and U3, this event is only generated when DCTL.KeepConnect=1 For L1, this event is only generated when DCTL.L1HibernationEn=1 and DCTL.KeepConnect=1 In addition to the suspend states that trigger this event, hibernation is also supported when the link is disconnected. See 'Device-Initiated Disconnect and Low Power While Disconnected' for details. 4- Resume/Remote Wakeup Detected Event (WkUpEvt) This event is generated when the host initiates a resume condition on the USB bus. This event is not generated by the device initiating a remote wakeup to the host. 3- USB/Link State Change (ULStChng) The core writes this event to indicate a change of USB or Link state. Bits 23-16 (EvtInfo) of this event indicate the Link Status. The event is generated in SuperSpeed when the link LTSSM state changes, except when LTSSM exits from HOT_RESET, POLL or LTSSM enters into RECOVERY. 2- Connection Done (ConnectDone) 1- USB Reset (USBRst) 0- Disconnect Detected Event (DisconnEvt) Note: This event is generated only if VBUS is off.
7-1	7'h00 - Device Specific Event
0	1'b1 - Non-Endpoint-Specific Event

11.1.2.3.2.3 Transfer and Buffer Rules

Buffers that are used to transfer data to and from an endpoint are defined using a Buffer Descriptor, which consists of one or more Transfer Request Blocks (TRBs). A CHN flag in the TRB is used to identify the TRBs that comprise a Buffer Descriptor. Therefore, a Buffer Descriptor refers to a consecutive set of TRB data structures where the CHN flag is set in all TRBs, except the last. Note that a Buffer Descriptor may consist of a single TRB, whose CHN flag will not be set.

Software communicates the location of the first TRB by using the Start Transfer command on an endpoint. Even endpoints that are not stream-capable use this command. The core fetches TRBs from external memory starting at the address provided in the Start Transfer command parameters, continuing in a linear fashion and following the Link TRB until a TRB is encountered that has its LST bit set to '1' or HWO bit set to '0'. Therefore, software must ensure that it has valid TRBs prepared before issuing the Start Transfer command to prevent the core from reading uninitialized memory.

Descriptor fetch requests are buffered within the hardware and handled separately from the progress of the current transfer. Therefore, it is possible that the core completes a transfer with XferComplete but still continues reading TRBs that it has not cached yet. If

software is immediately de-allocating the memory for TRBs based on the XferComplete event, it is recommended that software issue an End Transfer command for the endpoint/transfer resource prior to de-allocating the memory. The completion of the End Transfer command flushes out any pipelined descriptor fetches and avoids a potential bus error.

While processing TRBs, two conditions may cause the core to write out an event and raise an interrupt line:

- **TRB Complete:**
 - For OUT endpoints, a packet is received which reduces the remaining byte count in the TRB buffer to zero.
 - For IN endpoints, an acknowledgement is received for a transmitted packet which reduces the remaining byte count in the TRB buffer to zero.
- **Short Packet Received:**
 - For OUT endpoints only. While writing to a TRB buffer, the endpoint receives a packet that is smaller than the endpoint's MaxPacketSize.

This table shows the USB 3.0 Core Actions Based on TRB Control Bits.

Table 11-12. USB 3.0 Core Actions Based on TRB Control Bits

Direction	TRB Complete Packet	ISP	IOC	CHN	LST	CSP	Action
IN	Yes	-	X	0	1	-	XferComplete event
IN	Yes	-	0	X	0	-	No event
IN	Yes	-	1	X	0	-	XferInProgress event
OUT	Yes No	-	X	0	1	-	XferComplete event
OUT	Yes No	-	0	X	0	-	No event
OUT	Yes No	-	1	X	0	-	XferInProgress event
OUT	X Yes	X	X	X	X	0	XferComplete event ¹
OUT	X Yes	X	X	0	1	1	XferComplete event
OUT	X Yes	X	X	1	0	1	Search for CHN=0, accumulate IOC and ISP, then follow CHN=0 rules ²
OUT	X Yes	X	1	0	0	1	XferInProgress event
OUT	X Yes	0	0	0	0	1	No event
OUT	X Yes	1	0	0	0	1	XferInProgress event

1. When a TRB receives whose CSP bit is 0 and CHN bit is 1 receives a short packet, the chained TRBs that follow it are not written back (for example, the BUFSIZ and HWO fields remain the same as the software-prepared value)
2. In the case of an OUT endpoint, if the CHN bit is set (and CSP is also set), and a short packet is received, the core retires the TRB in progress and skip past the TRB where CHN=0, accumulating the ISP and IOC bits from each TRB. If ISP or IOC is set in any TRB, the core generates an XferInProgress event. Hardware does not set the HWO bit to 0 in skipped TRBs. If the endpoint type is isochronous, the CHN=0 TRB will also be retired and its buffer size field updated with the total number of bytes remaining in the BD.

On XferComplete and XferInProgress events, status bits in the event indicate LST, IOC, Short Packet Received, or Bus Error status. In the "fast-forward" case, the IOC status is accumulated from the skipped TRBs, not just the TRB that received the short packet.

When the hardware writes back the TRBs, it updates the BUFSIZ field to represent the remaining unused buffer.

11.1.2.3.2.3.1 *Number of TRBs Rule*

- Software must set up only one TRB for a control setup or status stage.
- If software is preparing multiple transfers for an IN endpoint, it may be necessary to place a 0-length TRB between transfers that are MaxPacketSize aligned to indicate transfer boundaries to the host. This is not necessary, if the class driver and the host can handle bursting between transfers.
- If software is preparing multiple transfers for an OUT endpoint, it needs to place a MaxPacketSize TRB between transfers, if it expects the host to transmit a 0-length packet between transfers.

11.1.2.3.2.3.2 *TRB Control Bit Rules*

Transfer control bits must conform to the following restrictions:

For OUT endpoints, the CSP bit must be the same in every TRB within a Buffer Descriptor (either set or clear).

- The core autonomously checks the HWO field of a TRB to determine if the entire TRB is valid.
- Therefore, software must ensure that the rest of the TRB is valid before setting the HWO field to '1'. In most systems, this means that software must update the fourth DWORD of a TRB last. Every time software validates a TRB by setting HWO=1, it must also issue an Update Transfer command to the core.
- Software sets the HWO bit to 1 when it creates the TRB, and cannot modify it until hardware resets it to 0. However,
 - Software must detect when a "fast-forward" occurs on an OUT endpoint that receives a short packet, since some TRBs in a chain may still have their HWO bit set to 1 while belonging to software.
 - Hardware will not clear the HWO bit of a Link TRB. Therefore, software can only modify a Link TRB if the TRB prior to the Link TRB has its HWO bit set to 0.
- The LST bit must not be set to 1 for isochronous endpoints.
- For a Setup or Status TRB, set CHN=0, LST=1, and CSP=0.
- For the data stage of a control transfer, set CSP=0 in all TRBs and LST=1 in the last TRB of the data stage.

- For Link TRBs, the LST, CHN, IOC, ISP, CSP, and Stream ID fields are ignored and must be set to 0.
- The Link TRB's chain bit is implicitly equal to the chain bit of the TRB before it. If the TRB before it has CHN=1, then the Link acts as if it's CHN=1. If the TRB before it has CHN=0, then the Link acts as if it's CHN=0. A Link TRB cannot be the last TRB in a Buffer Descriptor.
- When CSP=1 and the core receives a short packet, it searches for the end of the Buffer Descriptor by finding the Normal TRB that has CHN=0. Therefore, it is illegal to setup a circular buffer with all Normal TRBs with CHN=1.

11.1.2.3.2.3.3 *Buffer size rules and zero-length packets*

The hardware contains a cache that holds four TRBs per transfer.

For IN endpoints, the following rules apply:

- The number of chained TRBs necessary to construct a single packet must never exceed 3 . A maximum of one link TRB can be present in the chain.
- If software wants to indicate a transfer completion to the host by sending a zero-length packet after a multiple of MAXPACKETSIZE, it must set up a zero-length TRB following the last TRB in the transfer.

For OUT endpoints, the following rules apply:

- If the first TRB has CHN=1, its buffer size must be ≥ 1 byte
- The total size of a buffer descriptor must be a multiple of MAXPACKETSIZE
- A received zero-length packet still requires a MAXPACKETSIZE buffer. Therefore, if the expected amount of data to be received is a multiple of MAXPACKETSIZE, software should add MAXPACKETSIZE bytes to the buffer to sink a possible zero-length packet at the end of the transfer.

For IN and OUT endpoints, the following rule applies:

- The BUFSIZ field in a link TRB must be set to 0.

11.1.2.3.2.3.4 *Transfer setup recommendations*

Software can either set up transfers before the host attempts to move data on an endpoint ("preset" transfers) or can set up transfers on demand ("on-demand" transfers). When using preset transfers, software can safely disable the XferNotReady event in the endpoint configuration. However, when using on-demand transfers, the XferNotReady event must be enabled and software may not use the "No Response" variant of the Update Transfer command. The XferNotReady event is issued when the host attempts to move data on an endpoint when one of the following conditions is present:

- No previous transfer was started with Start Transfer.
- Not enough hardware-owned (HWO=1) TRBs are available to handle the requested data movement. The XferNotReady event must not be disabled for control endpoints because the event is an integral part of control transfer handling.

Although there are many valid ways to set up transfers, it is recommended that you choose one of three general mechanisms:

- When software wants to set up one transfer at a time and has the entire buffer available for transfer, it must set up TRBs that point to the data buffers and in the last TRB it must set the LST bit and issue Start Transfer, which points to the first TRB location.
 - When the USB transfer completes, the core will notify the software through the XferComplete event. The LST bit will be set in the status field of the event. The XferComplete will also release the Transfer Resource.
 - A premature XferComplete event can happen before all the data buffers are exhausted, if there is a Bus Error or, in an OUT transfer a short packet has been received and CSP=0. During these conditions the TRB will have an updated BUFSIZ field which represents the amount of buffer remaining after the successful part of data transfer.
 - Software can also set up an IOC bit TRB, so that it gets notified when data buffers are used up and can free them up sooner than waiting for the entire transfer to complete. On completing a TRB with IOC set, and not the last one, the core will issue an XferInProgress event with IOC set in the status field of the event.
- When software wants to set up one transfer at a time, but it has fewer data buffers available than the full transfer size, it must set up circular TRBs (using a Link TRB), and also set up IOC bits in the TRBs. Depending on buffer allocation and interrupt frequency it can set IOC for once in "x" number of TRBs.
 - As soon as the USB transfer for a TRB is completed, and if IOC is set, the core will generate an XferInProgress event with IOC set in the status field.
 - On seeing the event, software reuses the TRB and updates it with the next data buffer. It also issues an Update Transfer command, indicating it has updated a TRB.
 - If software is slow and hardware finds a TRB with HWO reset to 0, it waits for an Update Transfer command. Upon seeing the Update Transfer, it prefetches the TRB and continues the transfer.
 - When software reaches the end of the transfer, it sets the TRB LST bit. When hardware completes the TRB, it issues an XferComplete event and releases the Transfer Resource.
- When the Device software has multiple transfers to set up, it must set up circular TRBs and also set up CHN bits in all TRBs, except the last of each transfer. For

OUT endpoints, multiple transfers can be supported when the CSP field is set to '0' if each transfer has a multiple of MaxPacketSize bytes, since a short packet will end the transfer. If multiple transfers may contain short packets, the CSP field must be set to '1' to enable the next transfer to continue even if a short packet is received.

Depending on buffer allocation and interrupt frequency, it can either set the IOC for once in "x" number of TRBs or in the last TRB of each transfer.

- For OUT endpoints, the transfer can finish prematurely due to a short packet from the host. In this case, the core processes the remaining TRBs, skipping the updates to these TRBs until it reaches TRB of the next transfer. While skipping these TRBs, if any of the TRB it encounters the interrupt setting of ISP or IOC, it will generate the corresponding event once and ignore the interrupt settings of the remaining TRBs until a TRB of next transfer is reached.
- Device software can reclaim these skipped TRBs even though the HWO still indicates 1 (hardware-owned).
- For IN endpoints, software cannot stop providing transfers while it is ending transfers with CHN=0 and LST=0, otherwise it is possible that the endpoint is left in a flow-controlled state on the USB.

11.1.2.3.2.4 Transfer Resource Usage and Transfer State

Software allocates one Transfer Resource for an endpoint during initialization. When software issues a Start Transfer command, this Transfer Resource is used. When an XferComplete event happens, the Transfer Resource is released back to software. Similarly, when End Transfer completes, the Transfer Resource is released.

Following is a typical transfer usage summary (not complete usage model):

- Software sets up data buffer(s) and TRB(s) in external memory with the TRB(s) pointing to the buffer(s).
- Software issues Start Transfer with the pointer to the first TRB in the command parameters.
- If software sets up all the buffers and TRBs at the same time (Host and Device Software negotiate the transfer size), then just one Start Transfer is enough, and software waits for a XferComplete event.
- If software did not set up all the TRBs, then every time software adds a new TRB (by setting HWO=1) it must issue an Update Transfer command. The hardware will fetch the TRB again and continue the transfer.
- The End Transfer command is used only during error conditions and not used during normal transfers. For example, if software has set up multiple transfers and if a USB Reset event happens, it must remove all the transfers from the queue using End Transfer with the ForceRM bit set 1.

- A transfer is completed when all the data has been transferred or a short packet has been received. If software is queuing in only single transfers at a time (normal method for mass storage - for ethernet over USB, software can set up multiple transfers using the Chain bit and Continue on Short Packet bit), once XferComplete interrupt has happened, the transfer is completed. When starting a new transfer, software now needs to issue the Start Transfer command.
- An OUT transfer's transfer size (Total TRB buffer allocation) must be a multiple of MaxPacketSize even if software is expecting a fixed non-multiple of MaxPacketSize transfer from the Host.

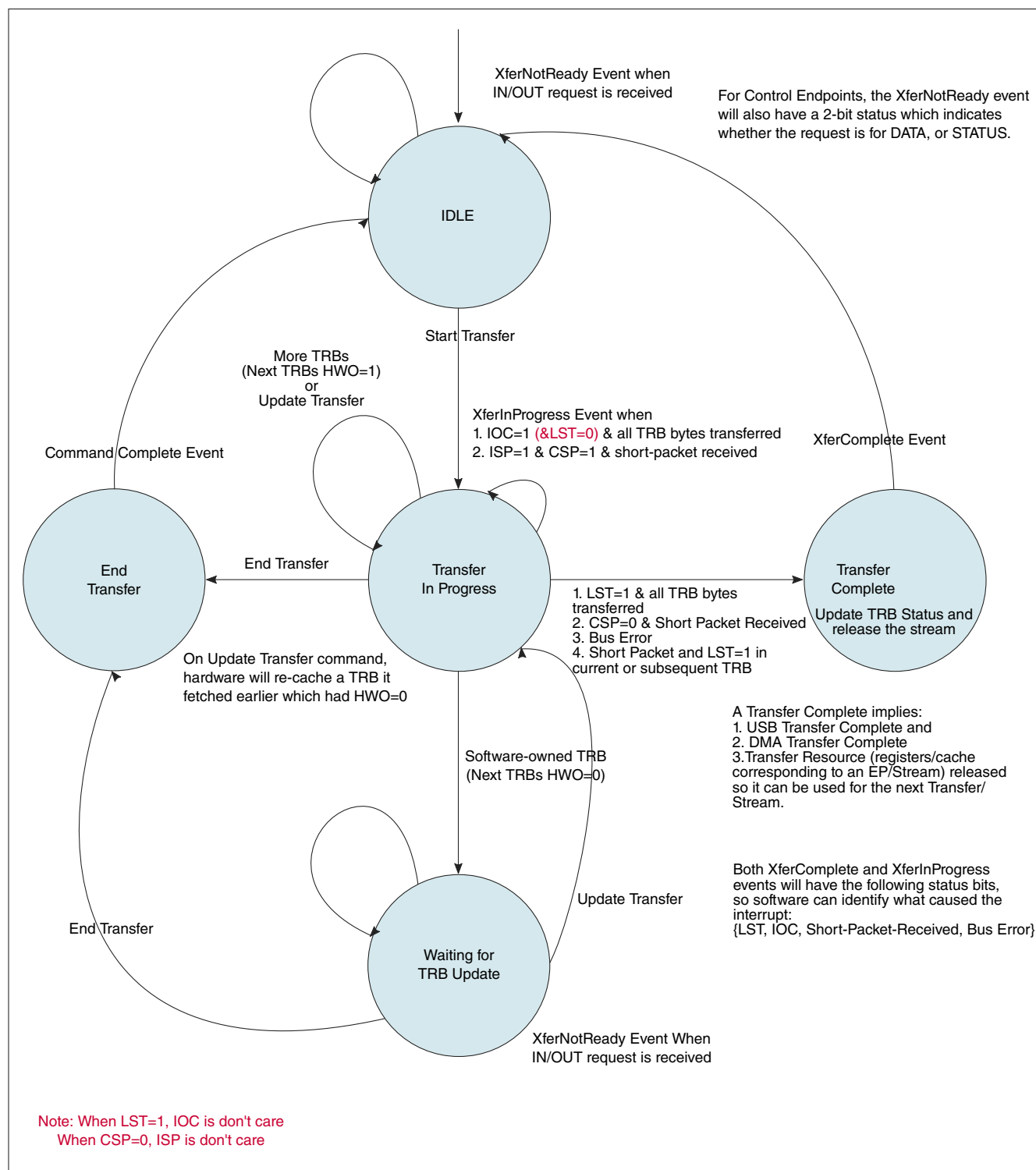


Figure 11-16. Transfer Usage State Machine.

11.1.2.3.2.5 Transfer Descriptions

The following sections describe typical transfer processes.

11.1.2.3.2.5.1 Non-Isochronous OUT Transfers

The sequence below describes a regular, non-isochronous OUT transfer (control-data/bulk/interrupt).

Table 11-13. Non-Isochronous OUT Transfer Sequence

Steps	Host	Device Core	Device Software
Software sets up data buffers before a host request			
Step 1	-	-	Software sets up Normal TRBs and enables DMA by issuing Start Transfer.
Step 2	-	Device fetches the TRB and caches it.	-
Step 3	Host sends data packets	-	-
Step 4	-	Core receives the packets in RXFIFO and stores them in to TRB data buffers through DMAs. While the RXFIFO has space and the BUS DMA keeps up with the host data transfers, the device receives all the packets. Once all the data is received, the core updates the TRB status and then generates a XferComplete event. In SS mode, if GRXTHRCFG.USBRxPktCntSel is set to, • 0: the NumP sent depends on the DCFG.NumP and bMaxBurstSize • 1: the NumP sent depends on the RXFIFO space available when the packet is received When DMA is slow or CRC fails: • If the RXFIFO overruns or CRC error is detected, device times out in Non-SS mode or asks for the same data packet (through ACK/SeqNumber/Retry Bit) in SS mode. • In HS mode, on NAK the host will issue PING and device will ACK it (if there is enough space in the RXFIFO) and the host resumes with data again. In SS mode, the host will send the requested data. If the device receives any packet with other than the requested data sequence, it will ignore it.	-
Step 5	-	-	Software processes the XferComplete event.
Software sets up data buffers after the host request.			
Step 6	Host sends data packets.	-	-
Step 7	-	Since DMA is not set up, the device will send NAK in Non-SS mode and will send NRDY in SS mode to the host. It also generates an XferNotReady event.	-
Step 8	-	-	On seeing the XferNotReady event, Software sets up TRBs and enables DMA by issuing Start Transfer.
Step 9	-	Steps 2-5 are followed.	-

Table continues on the next page...

Table 11-13. Non-Isochronous OUT Transfer Sequence (continued)

Steps	Host	Device Core	Device Software
Step 10	-	If the core encounters a short packet before the transfer size data programmed in the TRB is received, it skips the remaining TRBs of the transfer. If the core encounters control bits LST, IOC, or ISP it will generate events and, in the case of an LST TRB, the stream will be completed. Software has to reclaim any TRBs skipped with HWO=1.	On detecting a skipped TRB condition, software reclaims the TRBs with HWO=1.

11.1.2.3.2.5.2 Isochronous OUT Transfers

This section describes the difference between isochronous and non-isochronous OUT transfers referring to the sequence in [Non-Isochronous OUT Transfers](#)

- In Step 2, if the TRB is not fetched in time, the core discards the packet. In addition subsequent packets in the service interval also will be dropped.
- In Step 4, if DMA is enabled and if RXFIFO overrun happens or CRC fails, the core discards the packet. In addition subsequent packets in the service interval also will be dropped.
- In Step 7, if DMA is not enabled, the core drops the data and generates XferNotReady event.

In general, the TRB chain of a service interval will be released if the core receives DATA0 PID in HS USB 2.0 mode or receives a packet with LPF set in SuperSpeed mode.

Special scenarios are:

- If the core misses this last packet indication, it releases TRB at the service interval boundary.
- If the core encounters a short packet before the transfer size data programmed in TRB is received, it skips the remaining TRBs of the transfer.
- If the core encounters control bits 'LST' or 'IOC' or 'ISP' it will generate event and, in the case of LST TRB, the stream will be completed, but software has to reclaim TRBs if any TRBs skipped with HWO=1.

11.1.2.3.2.5.3 Non-isochronous IN transfers

The sequence below describes a regular, non-isochronous IN data transfer (control-data/bulk/interrupt).

Table 11-14. Non-isochronous IN transfers

Steps	Host	Device Core	Device Software
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Table continues on the next page...

Table 11-14. Non-isochronous IN transfers (continued)

	Software sets up data buffers before a host request		
Step 1	-	-	Software sets up TRBs and enables DMA by issuing Start Transfer.
Step 2	-	Device fetches the TRB, caches it, and prefetches the data into TXFIFO. Only in SS mode, the core sends ERDY (Async ERDY, even before host requesting data provided endpoint is in flow controlled state. LSP flow is same regardless of if the host asked for it.)	-
Step 3	Non-SS host issues a token request, SS host issues ACK TP to request data.	-	-
Step 4	-	If DMA fetches keep up with host requests, the device sends all the data, updates the TRB status, and then generates an XferComplete event when ACKs for all the packets are received. When DMA is slow: • If the TXFIFO becomes empty on a host data request, the device sends a NAK in Non-SS mode. In SS mode, the device sends NRDY, and when data is available in TXFIFO it sends ERDY and waits for the host request again. • If the TXFIFO becomes empty in the middle of a packet transfer (happens only in Threshold mode), in Non-SS mode the device will invert the CRC. In SS mode, it terminates with EDB after appending the corrupted CRC. The core waits for the host to retry the packet before fetching the data again. If the host retries a packet: • In Non-SS mode, a transmitted packet is kept in the TXFIFO until the ACK arrives (this is immediate in Non-SS mode). If the host retries a packet, it is immediately sent out from the TXFIFO. • In SS mode, before ACK can be received, the core could have sent additional packets in a burst. When a retry happens, NRDY is sent to the host and the TXFIFO is flushed. A DMA command is issued on the master bus to prefetch data from where the host is asking. Once data is fetched into the TXFIFO, the device sends ERDY and waits for host request again. • In Threshold mode, when the TX threshold amount of data is fetched, the device sends ERDY to the host.	-
Step 5	-	-	Software processes XferComplete event.
	Software sets up data buffers after a host request.		

Table continues on the next page...

Table 11-14. Non-isochronous IN transfers (continued)

Step 6	Non-SS host issues a token request. SS host issues ACK TP to request data.	-	-
Step 7	-	Since DMA is not set up, the device sends a NAK in Non- SS mode, or NRDY in SS mode, to the host. The core also generates an XferNotReady event	-
Step 8	-	-	On seeing the XferNotReady event, software sets up Normal TRBs and enables DMA by issuing Start Transfer.
Step 9	-	Steps 2-5 are followed.	-
Step 10	-	When software knows it is talking to a host that always ends transfers with short packets, and therefore always requires a zero-byte packet to end a transfer, it includes a zero-byte TRB (CHN=0) after the last TRB which is an even multiple of MaxPacketSize. When software does not know if the host is going to do another IN token after receiving the exact number of bytes it expected (and it was a multiple of MPS), it does not include a zero-byte TRB, and sets LST in the last TRB. If the host returns with another IN for a zero-byte packet, it uses the XferNotReady/Start Transfer mechanism to add one zero-byte TRB. This is called "On- Demand."	-

11.1.2.3.2.5.4 Isochronous IN Transfers

This section describes the difference between isochronous and non-isochronous IN transfers referring to the sequence in [Non-isochronous IN transfers](#).

- In Step 4, if the TXFIFO is empty when host IN request arrives, the core returns a zero length packet in both Non-SS and SS modes.
- In Step 7, the device returns zero length packet and generates an XferNotReady event.

Special scenarios:

- When data is fetched and there is no request from the host (possibly due to a missing request or corrupted request) the data will be flushed at the end of the service interval (corresponding Micro- Frame) boundary and the core moves onto fetching next service interval data. If there are some TRBs that are not serviced in the service interval, the core will skip these TRBs.
- If the core encounters control bits LST or IOC, it will generate an event, and in the case of LST TRB, the stream will be completed, but has to reclaim TRBs if any TRBs skipped with HWO=1.

11.1.2.3.2.6 Handling ENDPOINT_HALT

On receiving a SetFeature (ENDPOINT_HALT) control transfer, software issues a Set Stall command on the endpoint. Because of the delay from when software issues the command and when it is recognized by the core, other transfer events (XferInProgress, XferComplete) may be received prior to the endpoint being halted. For a description of the Set Stall endpoint command and its effects, see [Commands 4 and 5: Set Stall and Clear Stall \(DEPSSTALL, DEPCSTALL\)](#).

On ClearFeature (ENDPOINT_HALT), software must first remove all pending transfers for the endpoint through the End Transfer command. It may then issue a Clear Stall command on the endpoint followed by Start Transfer to start transfers again. For a description of the Clear Stall endpoint command and its effects, see [Commands 4 and 5: Set Stall and Clear Stall \(DEPSSTALL, DEPCSTALL\)](#).

11.1.2.3.2.7 Handling L1 Event During a Transfer

This section discusses the scenario in which the software already issued the Start Transfer command but the LPM occurred before or during the data transfer.

When an IN transfer is in progress, (that is, started and not completed yet), and the software sees a link state change event to L1, then the software can use the GDBGFIFOSPACE register (write followed by a read) to determine if the TXFIFO is empty or not. If the TXFIFO is not empty, it can initiate a remote wakeup.

NOTE

- The GDBGFIFOSPACE register is a debug register which gives the 'Space Available' for the endpoint selected with a write prior to the read. If this space available is less than the particular TXFIFO's maximum size, then the TXFIFO is not empty.
- Software can enable the Link State change event for USB 2.0 as there are not a lot of events compared to SS.

11.1.2.3.3 Isochronous Transfer Programming Model

Isochronous endpoints get guaranteed service by the host during a "Service Interval". The service interval for an endpoint is communicated to the host via the endpoint descriptor, and it is configured in the hardware through the DEPCFG command. However, unexpected behavior on the bus may prevent the host from servicing the endpoint as frequently as expected.

Although the endpoint receives "guaranteed" service during the interval, there are no bus-level acknowledgments, which means that there is no guarantee that the host (IN endpoints) or the device (OUT endpoints) receives correct data.

Therefore, the isochronous programming model differs from the bulk programming model to accommodate these two factors:

- No bus-level acknowledgement of packet transmission or reception
- No guarantee that all the data setup for an interval has been transmitted or received

For isochronous endpoints, software has the following responsibilities:

- Set up enough TRBs to keep up with the rate of data transmission or reception
- For FIFO-based IN endpoints, guarantee the validity of the TX data at least 1 micro Frame before the beginning of the interval that the data will be transmitted

Hardware has the following responsibilities:

- Transmit or receive data at the appropriate bus time
- For FIFO-based IN endpoints, delay fetching until 1 micro Frame before the beginning of the interval that the data will be transmitted
- Maintain a 1-to-1 correspondence between Buffer Descriptors and the intervals on the bus
- Report missed isoc intervals and update the buffer sizes in TRBs to reflect the amount of data moved

11.1.2.3.3.1 Definitions

- **bInterval:** The field in an endpoint descriptor used to communicate the service interval of the device. Its value ranges from 1 to 16, however, the USB 3.0 core supports a range of 1 to 14.
- **Service interval:** An integral number of microframes ($2^{[bInterval-1]}$) during which the host will poll the device to move data.
- **Beginning of interval:** The interval begins when the least significant (bInterval-1) bits of the bus time are all 0's.
- **End of interval:** The interval ends when the least significant (bInterval-1) bits of the bus time are all 1's.
- **Data payload:** The number of bytes to be moved during the service interval. It will be transmitted using MaxPacketSize packets.
- **Buffer descriptor:** A set of one or more TRBs with CHN=1 and the last one having CHN=0. The group of TRBs represents the data buffers for one service interval.
- **Last TRB of a Buffer Descriptor:** The TRB within a Buffer Descriptor that has CHN=0. When an interval has completed, this TRB will contain the total remaining buffer size of the Buffer Descriptor and thus also the completion status of the interval.

- Microframe (μF): A unit of time on the USB that lasts 125 μs .
- Start of Frame (SOF): The beginning of a microframe.
- FIFO-Based Isoc IN: A sub-type of Isoc IN endpoints that has the characteristic of always assuming the HWO bit in a TRB is '1' and delays fetching of transmit data.
- Prefetch delta: For FIFO-based IN endpoints, TX data may be prefetched as early as 1 μF before the beginning of a Service Interval, but never earlier.
- Retire a TRB: When HW sets the HWO bit to 0 in a TRB and writes it back.
- Retire a Buffer Descriptor: When HW retires at least the first and last TRB of a Buffer Descriptor.
 - Other TRBs of the Buffer Descriptor may not be retired if an interval was missed (IN or OUT) or if a short packet/last packet is received (OUT). In this case, software may reclaim those TRBs even though HWO=1.
- Missed interval: When the device does not move all the data in an interval. This may occur when the host does not poll for all the data, or because of application-side delays that prevent all the data from being moved.
 - For IN endpoints, this occurs when the host does not request all the data prepared in the Buffer Descriptor, or not all the data is fetched from the system bus in time.
 - For OUT endpoints, this occurs when the host does not send a packet, a packet is dropped due to CRC error, or the system bus is too busy to accept the data.

11.1.2.3.3.2 Endpoint configuration

An isochronous endpoint is setup in much the same way as a bulk endpoint, with the following exceptions:

- The bInterval_m1 value in DEPCFG Parameter 1 must be set to the value reported in the endpoint descriptor minus 1. When the core is operating in Full Speed, bInterval_m1 must be set to 0.
- The MaxPacketSize value in DEPCFG Parameter 0 must be set to the same value reported in the endpoint descriptor.
- The "FIFO-based" bit in DEPCFG Parameter 1, bit 31, must be set for FIFO-based isochronous endpoints.
- For the best performance, set the TXFIFOs corresponding to Isochronous IN endpoints to a high priority.

After the endpoint is configured, it can be enabled by setting the appropriate bit in the DALEPENA register.

11.1.2.3.3.3 Transfer configuration

Software describes isochronous transfers through a series of Buffer Descriptors. Each Buffer Descriptor corresponds directly to one service interval of data. The fields within an isochronous TRB are the same as bulk TRBs with the following exceptions:

- The SOF field in an IN endpoint TRB is a don't care. For an OUT endpoint, the core will write this field with the timestamp of the last packet received into the TRB's buffer. This information is not needed for normal operation, only for debug purposes.
- For High-Speed, High-Bandwidth IN endpoints, a maximum of 3 packets can be sent during an interval. The PktCntM1 field ([25:24] of the 3rd DWORD) must be set to the (number of packets in the Buffer Descriptor - 1). For example, if 3 packets are to be transmitted during the interval, this field must be set to 2. For Super-Speed and OUT endpoints, this field is a don't care.
- The first TRB in a Buffer Descriptor must have the TRBCTL field set to the "Isochronous-First" type while all others have this field set to "Isochronous".
- The ISP bit is renamed IMI (Interrupt on Missed Interval) and should be set if software wants to receive an XferInProgress event when at least 1 packet is missed within an interval.
- The IMI bit should be set to the same value in all TRBs of a Buffer Descriptor.
- The CSP bit must be set to 1 (short packets cause the hardware to move to the next Buffer Descriptor, they do not end an isochronous transfer).
- The LST bit should be set to 0 (isochronous transfers normally continue until the endpoint is removed entirely, at which time an End Transfer command is used to stop the transfer).

All other bits and fields (IOC, HWO, CHN, BPTR, BUFSIZ) retain the same behavior as they have for bulk endpoints. If software needs to receive an interrupt after every service interval, it should set the IOC bit to '1' in each TRB. However, if software wants to reduce the interrupt frequency and the application can tolerate some latency, the IOC bit can be set to '0' and the core will not generate a XferInProgress event when the TRB is completed.

For OUT endpoints, the Buffer Descriptor must describe a total buffer size of:

$$(Mult+1) * MaxBurst * MaxPacketSize$$

11.1.2.3.3.4 Starting a Transfer

The hardware will report the bus time that the host starts polling the endpoint inside the XferNotReady event. Software will use this value as a reference when issuing the Start Transfer command to serve as time synchronization with the host.

1. Set up TRBs and data buffer for the endpoint.

2. After configuring and enabling the endpoint, wait for an XferNotReady event.

The XferNotReady event will contain the time that the host started polling the endpoint in the upper 16 bits.

3. Issue the Start Transfer command to the endpoint with a future microframe time written into the upper 16 bits of the DEPCMD register. The future microframe time must be a value that is an integral multiple of intervals after the time reported in the XferNotReady event and aligned to the beginning of an interval. For example, if bInterval is 3 (4 microframes), and the XferNotReady time is 2, the value can be 4, 8, 12, and so on. The future microframe time must also be no greater than 4 seconds past the time reported in the XferNotReady event.
4. If the future microframe time has already passed when the command is received, the core will respond with an error (bit 13 in the Command Complete event).

In this case, software must issue End Transfer, then wait for another XferNotReady event and attempt the command again with a time that is further in the future. Otherwise, the first Buffer Descriptor will be used for the interval starting with the microframe time specified in the Start Transfer command.

11.1.2.3.3.5 Core Behavior During an Interval

After a transfer has been started, the hardware will perform the following functions for IN endpoints:

1. Fetch TX data as early as one interval prior to the beginning of the interval (call it A) if the HWO bit is set to one in the TRB.
2. Decrement the buffer size of each TRB as packets are transmitted.
3. Retire TRBs when their buffer size has reached 0, issuing an XferInProgress event if the IOC bit is set.
4. If the next interval (B) starts before all the packets have been transmitted for interval A:
 - a. Flush the TXFIFO.
 - b. Retire the Buffer Descriptor of interval A with a "Missed Isochronous" status.
 - c. Retire the Buffer Descriptor of interval B with a "Missed Isochronous" status.
 - d. See [Checking interval status](#) for a description of how software can determine that an interval ended unexpectedly.
 - e. Go to step 1 to prepare for interval C
5. Otherwise, if all the TRBs of interval A are completed, the hardware will prepare for interval B.
6. If the host completes an interval by polling for all the data, but then it polls the endpoint again during the same interval, the hardware will respond with a zero-length packet and no interrupt will be made to software.

7. If the host polls the endpoint prior to the time specified in the Start Transfer command, the hardware will respond with a zero-length packet and no interrupt will be made to software.
8. If the host polls the endpoint during the expected interval and the hardware has no data prepared, the core will respond with a zero-length packet, but no XferNotReady event will be generated because the transfer is active. When the next interval starts, the XferInProgress event is generated (based on ISP and IMI) with the "Missed Isochronous" status, which is the way software finds out that this occurred.

For OUT endpoints, the hardware performs the following functions:

1. If a packet is received for the correct interval represented by the Buffer Descriptor (call it A).
 - a. Write the packet into the buffer.
 - b. Decrement the buffer size of the TRB.
 - c. Write the timestamp of the received packet into the TRB.
 - d. Retire a TRB when its buffer size reaches 0, issuing an XferInProgress event if the IOC bit is set.
2. If a short packet or packet with the last packet flag (lpf) is received for the correct interval.
 - a. Write the packet into the buffer.
 - b. Retire the Buffer Descriptor of interval A with the TRBSTS set to 0.
 - c. If the IOC bit is set in the last TRB of the Buffer Descriptor, issue an XferInProgress event with bit [13] set.
 - d. Go to step 1 to prepare for interval B.
3. If a packet is received for the correct interval but it has an error (CRC error, RXFIFO overflow), the core will not increment its expected sequence number value which causes future packets within the same interval to be dropped.
4. If a packet is received at a bus time prior to the time specified in the Start Transfer command, it will be dropped.
5. If a packet is received for the correct interval (A), but not enough buffer space is available for the core to write the packet (due to the HWO bit still '0' in one or more of the TRBs of the Buffer Descriptor), no XferNotReady event will be generated. The core will behave as follows:
 - a. Wait until software sets the HWO bit to '1' and issues an Update Transfer command
 - b. Retire the Buffer Descriptor of interval A with a "Missed Isochronous" status
 - c. Go to step 1 to prepare for interval B.
6. If a packet is received for the next interval (B) before all the packets have been received for interval A:
 - a. Retire the Buffer Descriptor of interval A with a "Missed Isochronous" status.
 - b. Retire the Buffer Descriptor of interval B with a "Missed Isochronous" status.

- c. See section [Checking interval status](#) for a description of how software can determine that an interval ended unexpectedly.
- d. Go to step 1 to prepare for interval C.

Special Considerations for Isochronous OUT Endpoints

For OUT isochronous endpoints, the hardware detects a missed interval when the host sends a data packet in a future interval. It does not detect the missed interval at the exact interval boundary. Therefore, if the host abruptly stops sending isochronous OUT packets, there will be no interrupt or event indicating this to software. Software should use one of the following mechanisms to detect the interruption of isochronous OUT traffic:

1. The ultimate consumer of the isochronous OUT data will detect an underflow.
2. The device driver can use a timer detect that no XferInProgress events have been received for multiple intervals.

If this occurs, software should issue an End Transfer command to the endpoint and wait for a XferNotReady event which signals that the host is ready to resume isochronous traffic.

11.1.2.3.3.6 Checking interval status

As packets are transmitted or received during an interval, the buffer size of each TRB will be decremented. When the host is operating normally and polling for all the interval data, the buffer size of all the TRBs of an IN endpoint Buffer Descriptor will be zero after the interval has completed. For OUT endpoints, if the host sends less data than the Buffer Descriptor was setup for, the remaining buffer size may be greater than 0.

When the interval completes, the final remaining buffer size and missed isoc status is written to the last TRB of the Buffer Descriptor. If MissedIsoc is set, then it means the BUFSIZ is not accurate and may indicate that more data was transmitted or received than in reality. If the MissedIsoc is not set, it means the BUFSIZ field is correct.

When hardware detects the end of an interval (including normal and abnormal ends), it performs the following:

- If it has not already been retired, the first TRB of the Buffer Descriptor will be retired.
- Any non-first TRBs with CHN=1 that had not already been retired will not be written back. HWO will still be 1, and software can reclaim them for another transfer.
- The last TRB of the Buffer Descriptor will be retired with:
 - HWO = 0.

- **BUFSIZ** = The total remaining buffer size of the Buffer Descriptor.
- **TRBSTS** = "Missed Isoc" if any packets were missed, zero otherwise.
- If the IOC bit is set in the last TRB, or the IMI bit is set and packets were missed, hardware will issue an XferInProgress event

Software can get notification of an interval completing by setting the IOC bit in the last TRB of the Buffer Descriptor. The IOC bit may also be set in any other TRB of the Buffer Descriptor if software wants earlier notification that a TRB has completed. Software can also get only a notification of an interval completing unexpectedly by setting the IMI (Interrupt on Missed Isoc) bit in the last TRB of the Buffer Descriptor. When an interval completes unexpectedly and either the IOC or IMI bit is set in the TRB, the MissedIsoc bit (15) of the XferInProgress event will be set. The following table shows which events are generated in each scenario:

Table 11-15. TRB event generation events

Scenario	IOC	IMI	Action
TRB completed with a MaxPacketSize packet	0	0	No interrupt
	0	1	No interrupt
	1	0	XferInProgress (IOC=1)
	1	1	XrefInProgress (IOC=1)
TRB completed with a short packet	0	0	No interrupt
	0	1	No interrupt
	1	0	XferInProgress (Short=1, IOC=1)
	1	1	XrefInProgress (Short=1, IOC=1)
Interval completed due to missed packet (missed isoc)	0	0	No interrupt
	0	1	XferInProgress (MissedIsoc=1, IOC=1)
	1	0	XferInProgress (MissedIsoc=1, IOC=1)
	1	1	XrefInProgress (MissedIsoc=1, IOC=1)

It is normal to lose two intervals at a time when an error occurs during one interval. One interval is lost because of the host (which may not be polling enough) and the second interval is dropped because the device needs time to synchronize up to the next (third) interval.

11.1.2.3.3.7 Adding intervals to a transfer

Because isochronous endpoints represent a stream of data, the TRBs of an isochronous endpoint will normally be setup in a circular list, such as:

- TRB 1 (CHN=0, IOC=1)

- TRB 2 (CHN=0, IOC=1)
- Link (to TRB 1)

Example: Assume TRB 1 represents the data for the first interval and TRB 2 represents the data for the second interval. Because IOC=1, when the core completes the first interval, it will issue an XferInProgress event which indicates to software that TRB 1 can be analyzed to retrieve the results from interval 1. TRB 1 can be re-used to represent the data for the third interval by setting HWO=1 and issuing an Update Transfer command.

Each time software sets up TRBs for a new interval, it must follow the guidelines in the Transfer Configuration section. Software will set up TRBs for future intervals when it obtains those buffers from another software layer or when TRBs are retired by the core.

11.1.2.3.3.8 Moderating Events

During a transfer, software may enable or disable any endpoint-specific event by re-issuing the DEPCFG command with the "Config Action" set to "Modify" and a modified DEPEVTEN field. All other fields in all other parameters must remain the same as the initial endpoint configuration.

11.1.2.3.3.9 Other Types of Isochronous Endpoints

The above description of the isochronous endpoint programming model assumes that software is setting up buffers in external memory that apply to certain intervals of data. This is called a "buffer-based" isochronous endpoint.

However, there is another mode that the hardware supports to accommodate other mechanisms of sourcing or sinking isochronous data, called "FIFO-based" and the types are defined by bits [31:30] of DEPCFG Parameter 1. This table illustrates the differences between the models:

Table 11-16. Isochronous Endpoints Models

Type	FIFO-Based [31]	Bulk-based [30]	TRB Fetch	TRB Write back	TX DMA	Retire Buffer Descriptor at passed interval
Buffer-based	0	0	When HWO=1	Yes	When HWO=1, no earlier than one interval ahead of time.	Yes
FIFO-based	1	0	When internal cache space available	No	No earlier than 1 μ F before the interval	Yes
Invalid	1	1				

11.1.2.3.3.9.1 *FIFO-based isochronous IN Endpoints*

The software of some applications is unable to keep up with the short latency and high bandwidth of isochronous traffic and require the data to be sourced and done a sync through external FIFOs. One example is that there are two external TXFIFOs, one for each interval, and an address-to-FIFO-pop logic is implemented so that when the core attempts to read from interval 1's address, the translation logic converts this into a pop signal for interval 1's TXFIFO. When the core attempts to read from interval 2's address, the translation logic converts this into a pop signal for interval 2's TXFIFO.

To describe a FIFO-based implementation, software sets up an endpoint with the "FIFO-based" configuration bit set. This type of endpoint ignores the HWO bit in all TRBs, assuming that the TRB is always valid, and that the core should never write it back. Software chooses the buffer pointers within the TRBs to correspond to the address needed by the translation logic to specify which FIFO should be popped. By also using the CHN bit, headers and payload can be concatenated from different FIFOs. For example, if there are 4 external FIFOs: HA, PA, HB, PB, where HA/HB contain the 4 byte header for 2 intervals and PA/PB contain the 512 byte payload for 2 intervals, this can be described by using the following 5 TRBs:

1. BUFPTR=HA, IOC=0, CHN=1, BUFSIZ=4
2. BUFPTR=PA, IOC=1, CHN=0, BUFSIZ=512
3. BUFPTR=HB, IOC=0, CHN=1, BUFSIZ=4
4. BUFPTR=PB, IOC=1, CHN=0, BUFSIZ=512
5. Link to (1)

Every interval, the core will be creating a 516 byte packet that consists of 4 bytes from the header FIFO and 512 bytes from the payload FIFO. However, if the host does not poll for the packet, the core will skip 1 or 2 intervals of popping, depending on whether it has already started reading the next interval. External logic (or software) is responsible for flushing and refilling alternate FIFOs so that the core is always reading the correct data for the next interval. In normal intervals, the FIFOs will be empty, but when an interval is missed, there may still be data present in the FIFOs.

Software Requirements:

- The "FIFO-based" bit must be set in the DEPCFG when configuring the endpoint.
- No field within any TRB may be changed after the Start Transfer command is issued.
- Data must be valid in the external FIFO at least 1 uF before the beginning of the interval for which the data is intended.

Core Behavior:

- The earliest the core will read from the FIFO will be 1 μ F before the beginning of the interval for which the data is intended.
- The core will not write back the TRB after it has completed it.
- The core will re-read the TRB from external memory even though it is not allowed to change.
- The only indication of missed intervals is the XferInProgress event if IOC or IMI is set in the TRBs. No indication will be made of how much data was actually transmitted.

11.1.2.3.3.9.2 *FIFO-based isochronous OUT Endpoints*

The FIFO-based model is also supported for OUT endpoints. In this case, core writes to specific addresses will be translated into external FIFO pushes. The same example from above (4 byte headers and 512 byte payloads) can be applied to OUT endpoints where the headers are split from the payload as the core receives 516 byte packets. The same software requirements and core behavior apply (for example, no writebacks and no indication of how much data was received). External logic assumes that if less than the expected amount of data is pushed into an interval's FIFO, the interval is invalid.

11.1.2.3.3.10 **End a Transfer**

Since isochronous endpoints have no bus-level acknowledgements, it is not necessary to set LST=1 in a TRB to gracefully end an isochronous transfer. Software can issue an End Transfer command to end an isochronous transfer. The core will wait until it can complete operations for the endpoint before returning the Command Complete event in response.

11.1.2.3.4 **Control transfer programming model**

Control transfers follow a flow of setup/data/status. On the USB bus, there are many error scenarios that can disrupt this flow. The hardware has a special mechanism that automatically recovers from these scenarios as long as software follows this single control transfer programming model.

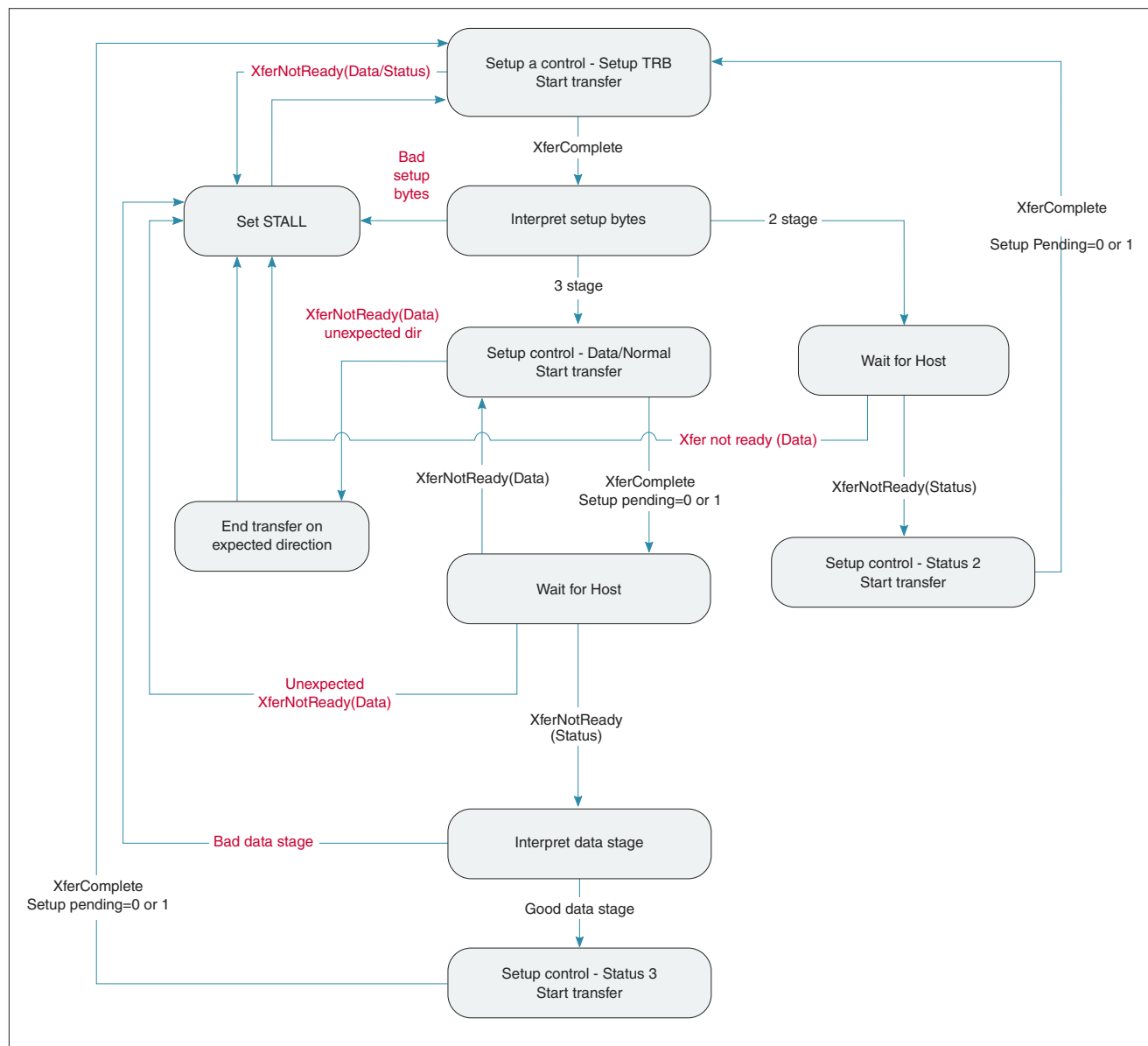


Figure 11-17. Software flow for control transfers

If hardware does not receive a setup packet, it discards the unexpected data and status stages. It does not issue XferNotReady to the application. The hardware will also not generate a XferNotReady event if it receives a setup packet before a TRB is prepared for it. As control endpoints are bi-directional, the IN and OUT directions of the endpoint are used for the different stages as follows:

Control read:

- EP0 - Setup OUT
- EP1 - Data IN
- EP0 - Status OUT/Status TP

Control write or 2-stage transfer:

- EP0 - Setup OUT
- EP0 - Data OUT (if present)
- EP1 - Status IN/Status TP

Set STALL is always issued on EP0, and the XferComplete/XferNotReady events are generated on the correct direction. For other control endpoints, substitute EP0 with the OUT direction (such as, 2, 4, 6) and EP1 with the IN direction (such as, 3, 5, 7). The same programming model is used for USB 2.0 and USB 3.0 control transfers.

11.1.2.3.4.1 Two-stage control transfer programming model

1. Software sets up a Setup TRB and issues Start Transfer on EP0 pointing to the Setup TRB.
2. After XferComplete is received, software interprets the setup bytes. If they are bad, issue Set Stall on EP0 and go back to the step 1. If a XferNotReady (Data/Status) event is received before the XferComplete event for the Setup stage, issue Set Stall. This is an error case where the host is attempting to move data or start the status stage for a previous control transfer that has already completed.
3. Wait for an XferNotReady event for the Status stage which will occur on EP1. If an XferNotReady event for the Data stage is received (either direction), issue Set Stall on EP0 and go back to the step 1. This is an error case where the host is attempting to start the data stage when the setup bytes did not indicate a data stage was present.
4. After XferNotReady (Status) is received, start the status stage by issuing Start Transfer on EP1 pointing to a valid Status-2 TRB.
5. After XferComplete is received on EP1, go back to the step 1.

Although not required, software may look at the Status TRB and examine the Setup Pending bit in the TRB status field. If it is set, it means that this control transfer was aborted on the USB bus and that the host did not receive the Status stage ACK.

11.1.2.3.4.2 Three-stage control transfer programming model

1. Software sets up a Setup TRB and issues Start Transfer on EP0 pointing to the Setup TRB.
2. After XferComplete is received, software interprets the setup bytes. If they are bad, issue Set Stall on EP0 and go back to the step 1. If a XferNotReady (Data/Status) event is received before the XferComplete event for the Setup stage, issue Set Stall. This is an error case where the host is attempting to move data or start the status stage for a previous control transfer that has already completed.
3. Start the data stage by issuing Start Transfer on EP0 (control write) or EP1 (control read) pointing to a valid Control-Data TRB.

- a. If an XferNotReady (Data) event is received for the incorrect direction, software must issue an End Transfer for the data stage it has already started, then issue Set Stall. This is an error case where the host is attempting to move data in the wrong direction.
 - b. If a XferNotReady (Data) event is received for the correct direction, ignore the event and continue to wait for a XferComplete event.
4. After the XferComplete event is received, software waits for an XferNotReady event. Although not required, software may look at the completed Data TRB(s) and examine the Setup Pending bit in the TRB status field and the BUFSIZ. If the Setup Pending bit is set or the BUFSIZ is non-zero, it means that this control transfer was aborted on the USB bus and that the host did not complete the data stage.
5. If a XferNotReady (Data) event is received after the XferComplete for the Data stage, it could mean one of two things:
 - a. This host is trying to complete the data stage by moving a 0-length packet. This can occur if the data stage was an exact multiple of max packet size. If this is the case, software sets up an extra TRB (with a BUFSIZ of max packet size for control writes, or 0 for control reads), issues Start Transfer, and goes back to the step 4.
 - b. This host is trying to move more data than specified in the wLength field of the setup bytes. In this case, software issues Set Stall on EP0 and goes back to the step 1.
6. When the XferNotReady (Status) event is received, software interprets the data stage (for example, number of bytes transferred, content of the control write buffer) and decides if the data stage was successful. If not, it issues Set Stall on EP0 and goes back to the step 1.
7. Start the status stage by issuing Start Transfer on EP1 (control write) or EP0 (control read), pointing to a valid Status-3 TRB.
8. After XferComplete is received, go back to the step 1.

Although not required, software may look at the Status TRB and examine the Setup Pending bit in the TRB status field. If it is set, it means that this control transfer was aborted on the USB bus and that the host did not actually receive the Status stage ACK. When device software sets STALL with the Set Stall command, it is possible that this command can be delayed on the chip bus due to latency or the chip bus servicing other agents. In this case, software can receive an XferNotReady event if the core detects no TRB active before it detected the Set Stall. Software must service these events knowing that it stalled the control command.

11.1.2.3.4.3 Handling fewer requests than wLength

In a control IN transfer, if the host asks for less data than wLength, the core skips the TRBs of the data stage and generates an XferComplete event after encountering the TRB that has LST=1. Hardware sets HWO to 0 in those TRBs that it skipped and in the last TRB. In a control OUT transfer, if the host sends less data than wLength, the core treats this like a short packet received into a TRB with CSP=0. It immediately generates a XferComplete event, not setting the HWO bit in the remaining TRBs to 0. Software needs to reclaim those TRBs that still have HWO=1.

11.1.2.3.4.4 Control OUT transfer examples

The sequence in the following table describes two and three stage control OUT transfers.

Table 11-17. Control OUT transfer

Steps	Host	Device core	Device software
Setup stage			
1	-	-	Software sets up the Setup TRB and enables DMA by issuing Start Transfer to EP0 (endpoint 0, OUT direction).
2	-	Device fetches the TRB and caches it.	-
3	Host sends SETUP packet.	-	-
4	-	SETUP packet is received in the RXFIFO. There are 24 bytes allocated for each control endpoint, which guarantees reception of the three back-to-back SETUP packets. An ACK handshake is returned in non-SS mode. An ACK TP with NumP=1 is returned in SS mode. After SETUP, bytes are transferred through the DMA, and the core issues an XferComplete event.	-
5	-	If the host starts a Data/Status stage, the core returns NAK in Non-SS mode or NRDY TP in SS mode. If the SETUP request is invalid, core returns STALL (when software sets STALL) in non-SS mode. In SS mode the core returns ERDY, and when the Data/Status stage is received it returns STALL.	Software processes the XferComplete event and decodes the SETUP bytes. If a Data stage is required, it will set up one Control-Data TRB and possibly subsequent Normal TRBs, then issue Start Transfer to EP0 (endpoint 0, OUT direction). If a Data stage is not needed, it will skip to the Status stage. If the SETUP request is invalid, software instead sets the STALL bit by issuing Set Stall command for EP0 (OUT direction).
Data stage			
6	-	Device fetches the TRB and caches it. In SS mode only, the core will send ERDY.	-

Table continues on the next page...

Table 11-17. Control OUT transfer (continued)

Steps	Host	Device core	Device software
7	Host sends data packets.	-	-
8	-	The NumP for control transfer will be 1. In addition, if another SETUP is received before the data transfer happens on USB, the core will skip the data transfer that software sets up without waiting for it to happen on the USB bus, and send XferComplete (when LST=1 TRB encountered) with the SetupPndg status bit set. Software has to reclaim the TRBs with HWO=1 in the skipped TRBs.	-
9	-	If the host starts a Status stage, the core returns NAK in Non-SS mode or NRDY TP in SS mode.	On seeing the event: - Software decodes the control write data. If the Data stage is OK, then it will wait for XferNotReady with "Control Status Request" set, set up a Control-Status-3 TRB (0-bytes), and enable DMA by issuing Start Transfer to EP1 (endpoint 0, IN direction). - If the Data stage is not OK, then software sets STALL by issuing Set Stall command to EP0.
Status stage			
10	-	Device fetches the TRB and caches it. In SS mode only, the core will send ERDY	Software sets up a Status stage TRB only after the Data stage of a three-stage control transfer is completed and an XferNotReady event occurred.
11	Host requests status packets.	-	-
12	-	In Non-SS mode, the device will send a zero-length packet if STALL is not set; otherwise, it will send STALL. In SS mode, if STALL is not set, then it sends ACK TP with NUMP=1 else it sends STALL. If another SETUP is received before the status transfer on USB, the core will send XferComplete with the SetupPndg status bit set. The STALL bit will be cleared by the core whenever it receives a SETUP packet. SETUPS are always accepted.	-
13	-	-	On seeing the event: If there are no other errors, then the transfer has completed successfully.

11.1.2.3.4.5 Control IN transfer examples

The sequence in the following table describes two and three stage control IN transfers.

Table 11-18. Control IN transfer

Steps	Host	Device Core	Device Software
	Setup stage		
1-5	Steps 1-5 are same as Control OUT transfer examples , except that software sets up Tx buffers and Start Transfer is issued to EP1 (endpoint 0, IN direction).		
	Data stage		
6	-	Device fetches the TRB and caches it. It prefetches the data and puts it in the TXFIFO. The core sends ERDY only in SS mode.	-
7	Host requests data packets	-	-
8	-	This step is similar to the step 4 of Non-isochronous IN transfers . The NumP for control transfer is 1. In addition, if another SETUP is received before the data transfer happens on USB, the core will skip the data transfer software has set up without waiting for it to happen on the USB bus and send XferComplete (when LST=1 TRB encountered) with the SetupPndg status bit set. Software has to reclaim the TRBs with HWO=1 in the skipped TRBs and flush the TXFIFO.	-
9	-	If the host starts the Status stage, the core returns NAK in non-SS mode or NRDY TP in SS mode.	On seeing the event: • If the Data stage is OK, then it sets up Control-Status-3 TRB (zero-bytes) and enables DMA by issuing Start Transfer to EP0 (endpoint 0, OUT direction). • If the Data stage is not OK, then software will set a STALL by issuing Set Stall command to EP0 (OUT direction).
	Status stage		
	Same as in Control OUT transfer examples .		

11.1.2.3.5 Stream handling in SuperSpeed

The SuperSpeed stream protocol consists of a negotiation between the host and device concerning the selection of a stream, followed by data movement on the selected stream. The core automatically handles the stream protocol by initiating stream selection and also taking into account streams initiated by the host.

NOTE

This section is not applicable for USB 2.0 mode.

11.1.2.3.5.1 Stream IDs and transfer resources

A stream ID is a 16-bit value which represents an individual flow of data between host and device. For stream-capable endpoints, the stream ID must be non-zero and is used in:

- The Command Parameter field in the DEPCMD register used when issuing a Start Transfer command.
- The Stream ID field in a TRB.
- The EventParam field in an endpoint-specific event (DEPEVT): XferComplete, XferInProgress, XferNotReady, and StreamEvt.

It is necessary to allocate 1 Transfer Resource per endpoint. The cost of each transfer resource is $(32 + N_TRBS_PER_XFER \times 16)$ bytes of memory in the descriptor cache. Core can cache maximum four TRBs per transfer.

After the Start Transfer command completes, the hardware returns the 7-bit Transfer Resource Index in the command complete event (and [Device Physical Endpoint-n Command Register \(DEPCMD\)](#)). This index is used only for the Update Transfer and End Transfer commands. In all other places, the Stream ID is used.

11.1.2.3.5.2 Stream selection and stream programming model

Referring to the USB 3.0 Device Stream Protocol State Machine, the Idle state is the state where a device does not have a current stream to work on. When the hardware detects that it is in the Idle state, it will attempt to initiate and move data on a stream by transmitting ERDY to the host.

In order for this to occur, the endpoint must be "primed" and the stream must be both active and ready.

- **Primed:** After a stream-capable endpoint is configured, it is in the disabled state, which prevents the device from initiating any stream selection. After the host performs a Prime transaction, the endpoint is moved into the "Primed" state where it is eligible to initiate stream selection.
- **Active:** Software makes a stream active by issuing a Start Transfer command for the stream. A stream becomes inactive after an XferComplete event or after software issues an End Transfer command.
- **Ready:** After a stream is added via the Start Transfer command, its default state is "ready." The hardware will label the stream as "not ready" if the host gives a NoStream rejection to the hardware's attempt to initiate the stream. If the host

performs a Prime, all currently active streams are automatically set to the "ready" state by hardware.

In the Idle state, the hardware selects a stream from its cache to initiate. Once the core has chosen an active stream, it attempts to initiate and move data on that stream until the endpoint enters the Idle state, which occurs when the host rejects the stream selection (NoStream) or the stream is terminated (PP=0) by the host or device. Therefore, software is required to prepare enough HWO=1 TRBs for at least one packet prior to issuing the StartXfer command.

In device-oriented stream selection class drivers (such as UASP), the host will not reject the device's stream selection. However, other class drivers may be developed in which the host initiates stream selection. In this situation, software will not start any transfers and will wait for a XferNotReady event. When that event is received, software will look at the StreamID in the event and start the transfer associated with the given Stream ID. If for some reason the class driver allows the host to reject a stream that has already been initiated by the host or device, software will receive a Stream (NotFound) event. If this occurs, software may replace the existing stream by issuing End Transfer for it and then issuing Start Transfer for another stream.

Some host implementation scenarios exist where the host and device become out of sync in the stream protocol, creating a deadlock condition where the device waits for the host to issue a Prime transaction while the host waits for the device to issue an ERDY. To resolve potential deadlock conditions, software should perform the following:

1. Implement a time out for stalled transfers on stream-capable endpoints.
2. When the timeout elapses with no data transfer for any stream after a StreamEvent (NotFound) event, restart one of the streams by issuing an End Transfer command followed by a Start Transfer command.

This places the stream in the Ready state and causes the core to transmit an ERDY to the host.

11.1.2.3.5.3 Data movement within a stream

Data movement within a stream is the same as data movement using the Bulk endpoint type (Start Transfer, Update Transfer, and updating TRBs). The XferInProgress, XferComplete, and XferNotReady events are issued as data is transferred between host and device. The Stream ID field in the events indicate which stream within the endpoint is generating the events. If software receives an XferNotReady event with the EventStatus field indicating XferNotActive, it means that the host is attempting to initiate a stream that has not been added to the hardware's cache through Start Transfer. In response to this event, software should add the requested stream if it is available.

11.1.2.3.6 Low power operation

11.1.2.3.6.1 Low power operation of USB

The USB 3.0 protocol allows a SuperSpeed host and device to negotiate with each other and decide when to bring their link into a lower power state (U1 or U2). This is accomplished by each side making individual decisions about whether they have any traffic currently pending or if they expect any traffic soon. The device requests low power entry by transmitting an LGO_U1 (or LGO_U2) Link Command and exits low power by using an LFPS handshake. The USB 3.0 Device Controller uses information from the host (bus side) and from software (application side) to decide when the best time to request low power entry is and when to exit from low power. The same calculation is used when the controller decides whether to accept a low power request from the host. The controller maintains state information about each endpoint and whether it is "active" or "paused". If all enabled endpoints are paused from either the application side or the bus side, low power entry is allowed. If at least one endpoint is not paused, low power entry is not allowed. To indicate that it does not have any pending traffic and allow the link to go into a low power state, the software may pause all enabled endpoints using one of the following mechanisms:

- Allow transfers to complete and not start any new ones
- Not setting the HWO bit in the TRBs to '1'

Exception: Software may prepare a Setup TRB for control endpoints without affecting the ability of the link going into low power. Otherwise, an endpoint is considered active on the application side. For these endpoints, the host indicates that it is paused in one of the following ways:

- Setting the Packet Pending (PP) flag to '0' in the DPH
- Setting the PP flag to '0' and NumP to '0' in the ACK TP
- For interrupt endpoints, by not polling the endpoint for two service intervals
- Note: After an endpoint is configured using the DEPCFG command and before the host sends any packets to it, the endpoint is considered to be paused on the bus side

Exception: If the data packet payload has a CRC error, the core responds with an ACK (Retry=1), and the endpoint is not paused. Otherwise, if PP=1 or Deferred=1, the endpoint is considered to be active on the bus side. Control endpoints have an additional condition as follows:

The core does not allow low power entry during a control transfer (starting from the reception of setup packet and ending at the completion of the status stage). If the core is configured for isochronous support, the following additional rules apply:

- The controller enters U0 at SystemExitLatency microseconds prior to the beginning of every microframe to receive the Isochronous Timestamp Packet (ITP).
- A PING causes every isochronous endpoint to wake up and be considered active, preventing low power entry.
- An isochronous endpoint is paused automatically when it transmits or receives the last packet of its interval (lpf=1) or if two intervals pass and the endpoint has not transmitted or received its last packet. In this way, after the PING, the core does not enter low power until all isochronous endpoints have moved their last packet or two intervals pass without another PING.

If all enabled endpoints are paused by software or the host, the core issues an LGO_U1 (or LGO_U2) link command or it responds with an LAU link command to the low power request of the host or hub. Otherwise, the core rejects the low power requests with an LXU link command.

Additionally,

- For bulk endpoints, the USB 3.0 specification says that the device may go to U1/U2 500ms (tERDYTimeout) after transmitting ERDY. It is the responsibility of the software to detect no activity on a bulk endpoint for 500ms or more and issue End Transfer if there is no activity. When the transfer is removed, the endpoint allows U1/U2 entry again.
- Because U1/U2 entry is based on the presence of packets in the TX FIFO and not based on the TRBs that are available to the core, it is possible that an IN endpoint responds with a NRDY, goes to U1, then U0, then transmits an ERDY if the timing and TXFIFO size works out such that the core cannot keep at least one packet in the TXFIFO at all times.
- In addition, before the final decision made to either initiate or accept low power state, device checks the state of DCTL[INITU2ENA][ACCEPTU2ENA][INITU1ENA][ACCEPTU1ENA]. If the device receives Set Link Function LMP with Force_LinkPM_Accept bit asserted, the device accepts low power request from its link partner independent of endpoints state and DCTL Power control settings.

11.1.2.3.6.2 Low power operation of core

When the link is in the U1, U2, U3, SS.Inactive, SS.Rxdetect, or SS.Disabled state (for USB 3.0) or the USB is in Sleep, Suspend, or V_{BUS}-off mode (for USB 2.0), the core can be put into a low power mode to save power. The core supports the following low power mode:

- **Clock-gating mode:** In this mode, the clocks connected to most of the core modules are gated off. Modules that still get clocks in the low power mode detect wakeup conditions and remove clock gating to other modules when a wakeup condition is

detected. Clock-Gating mode can be used when the USB is in any low power state. Clock-Gating mode is enabled by setting the Disable Clock Gating bit in GCTL to 0.

11.1.2.3.6.2.1 Clock-Gating Mode

Clock-Gating mode is enabled if the Disable Clock Gating bit in the GCTL is set to 0. When the USB/link is in a low power state, the clocks to most of the core modules are gated off automatically by the core. When a wakeup condition is detected by the core, or when the software schedules new work to the core (in U1 or U2 states) or changes the USB/Link state, the core turns on clocks and the entire core goes into active mode.

11.1.2.4 Host programming model

Please refer xHCI specification for detailed information.

11.1.2.4.1 Initializing host registers

In order to initialize the core as a host, the application should perform the steps described in the xHCI specification. Global registers need to be re-initialized as described in [Initializing global registers](#).

11.1.2.4.2 Host controller capability registers

For register definition, refer to [Capability registers length and Host Controller Operational Registers \(CAPLENGTH\)](#) to [Runtime Register Space Offset Register \(RTSOFF\)](#).

11.1.2.4.3 xHCI implementation details

This section discusses xHCI implementation details specific to the USB 3.0 controller.

11.1.2.4.3.1 LHCRST behavior

The xHCI Specification does not describe the programming model of light reset. It only specifies that the Operational and Runtime Registers that are not contained in the Aux Power well are at their default values.

Light reset must only be applied when USB_DCTL[Run_Stop] is equal to '0' to ensure that there is no discrepancy between hardware and software states. If light reset is applied during a transfer, the behavior on the USB is undefined, and may cause a packet to be terminated abruptly.

On light reset, the hardware resets all non-Aux registers to their default values which means that the port state (which is contained in PORTSC) does not change, but USB_DCTL[Run_Stop] is immediately set to '0' and all the context information about connected devices is lost. It is the responsibility of software to re-initialize the controller.

11.1.2.4.3.2 ENT requirements

In the xHCI Specification, the rules about when the ENT (Evaluate Next TRB) flag in a TRB must be set or cleared are not very strict. However, it states that the ENT flag must be set to '1' in the last Normal TRB when a TD ends with an Event Data TRB.

The controller requires software to follow these rules:

1. Regardless of whether the endpoint is stream-capable or not, if a TD ends with an Event Data TRB, the Normal TRB that precedes it must have ENT set to '1'. This Normal TRB may be separated from the Event Data TRB by a Link TRB
2. In all other cases, the ENT flag must be set to '0'. Failure to follow these rules results in undefined behavior.

11.1.2.4.3.3 Behavior on babble error

If a babble error is detected and the received data passes all integrity checks, the host controller may write the received data (up to the expected data length) to the data buffer, and the value of the TRB transfer length field in the babble detected error transfer event shall be consistent with the number of data bytes written to the buffer.

The controller does not write the received data to the data buffer. The entire packet is discarded and the residual byte count is written to the TRB transfer length field.

11.1.2.4.3.4 Max_exit_latency_too_large message

If periodic transfers are on every microframe (Binterval 1), the controller reports max_exit_latency_too_large if PING scheduling is enabled. The xHCI scheduler must prefetch data from the system memory ahead of next microframe, manage non-periodic transfer, and schedule PING before ISOC. Because the system memory access and USB responses are not predictable Quality of Service (QoS) is not guaranteed if the xHCI host needs to perform all of the above for every microframe when U1/U2 is enabled. Therefore, the expectation is to disable U1/U2 if periodic data needs to be scheduled every microframe. In addition, because the U1 exit latency is in the order of 10usec, only very low system-level power saving is achieved even if the scheduler allows U1 transition by reducing QoS.

U1, U2, and U3 link states and exit time observed during lab testing:

- U1: Lowest power saving
 - In this state the chip is active, USB controller is inactive, only the Tx transmitter of the PHY is inactive.
 - U1 exit time per link partner: ~10 - 19 usec
- U2: Higher power saving than U1
 - In this state the chip is active, USB controller is inactive, both the Rx and Tx transceivers of the PHY are inactive.
 - U2 exit time per link partner: ~85 - 115 usec
- U3: Highest power saving
 - In this state, the chip and USB controller/PHY are inactive.
 - U3 exit time per link partner: 200 - 450 usec

Following are the recommendations for driver to handle periodic endpoints and "max_exit_latency_too_large" message:

- If Binterval of any endpoint is '1', then disable U1/U2 and do not schedule PING ("max_exit_latency" set to 0).
- If periodic scheduling is expected on every microframe, then disable U1/U2 and do not schedule PING.
- If periodic scheduling is expected only on every other microframe, then disable U2 and use U1 exit latency for "max_exit_latency" calculation.
- If $((\text{number of hubs} + 1) * \text{U2 exit latency}) \geq (2 * (\text{Binterval} - 1) * 125 \text{ usec})$, then disable U2.
- For all other periodic scheduling use U2 exit latency for "max_exit_latency" calculation.
- If the "max_exit_latency_too_large" error is sent to software, then disable U2 and re-issue ep- config with U1 exit latency.
- If "max_exit_latency_too_large" still happens, then disable U1 and re-issue ep_config with max_exit_latency set to 0 (no PING).

11.1.2.5 Device physical endpoint-specific commands

11.1.2.5.1 Command 1: Set endpoint configuration: DEPCFG

This command sets the physical endpoint configuration information.

NOTE

If GUSB2PHYCFG[6] is set to '1', it must be set to '0' prior to issuing this command and may be set to '1' after the command completes.

Table 11-19. Command 1: Set endpoint configuration parameters: DEPCFG

Field	Description
Parameter 2	
31-0	Set to endpoint state when the Config Action field of Parameter 0 is 1. Otherwise, this is Reserved.
Parameter 1	
31	FIFO-based Set to '1' if this isochronous endpoint represents a FIFO-based data stream where TRBs have fixed values and are never written back by the core.
30	Bulk-based Set to '1' if this isochronous endpoint represents a bulk data stream that ignores the relationship of bus time to the intervals programmed in TRBs.
29-25	USB Endpoint Number bit[29:26]: Endpoint number bit[25]: Endpoint direction: 0 OUT 1 IN Physical endpoint 0 (EP0) must be allocated for control endpoint 0 OUT. Physical endpoint 1 (EP1) must be allocated for control endpoint 0 IN.
24	Stream Capable (StrmCap) Indicates this endpoint is stream-capable (MaxStreams != 0).
23-16	bInterval_m1 Set to the bInterval value minus 1. The valid values for this field are 0 through 13. The bInterval value is reported in the endpoint descriptor. When the core is operating in Full-Speed mode, this field must be set to 0.
15	Set to '1' if this bulk endpoint utilizes the External Buffer Control (EBC) mode. This limits the number of outstanding DMA transfers to one read and one write. This is a requirement for the EBC to operate properly. Using EBC without setting this bit to '1' results in undefined behavior.
14	Reserved
13-8	Bits 13-8 are used for Device Endpoint Specific Event Enable (DEPEVTEN). If an enable bit is set to 0, the corresponding event is not generated. Bit 13 Stream Event Enable (StreamEvtEn) Bit 12 Reserved Bit 11 Reserved Bit 10 XferNotReady Enable (XferNRdyEn) Bit 9 XferInProgress Enable (XferInProgEn) Bit 8 XferComplete Enable (XferCmplEn)
7-5	Reserved
4:0	Interrupt number (IntrNum) Applies to IN and OUT endpoints.

Table continues on the next page...

Table 11-19. Command 1: Set endpoint configuration parameters: DEPCFG (continued)

Field	Description
	Indicates interrupt/Event Buffer number on which endpoint related interrupts for this endpoint are generated. This must be set to 0.
Parameter 0	
31-30	Config Action 0 Initialize endpoint state: This action is used when an endpoint is configured the first time. It will cause the data sequence number and flow control state to be reset. DEPCMDPAR2 will be ignored. The encoding of this action is backward compatible with software that previously set "Ignore Sequence Number" to 0. 1 Restore endpoint state: This action is used when reconfiguring an endpoint with saved state after hibernation. It will cause the data sequence number and flow control state to be restored from DEPCMDPAR2. 2 Modify endpoint state: This action is used when modifying an existing endpoint configuration, such as changing the DEPEVTEN event enable bits, interrupt number, or MaxPacketSize. The data sequence number and flow control state will be unchanged, and DEPCMDPAR2 will be ignored. The encoding of this action is backward compatible with software that previously set "Ignore Sequence Number" to 1. When issuing a Endpoint Configuration command with Config Action=Restore, Parameter 2 must be filled in with the same value returned by the Get Endpoint State command prior to hibernation.
29-26	Reserved
25-22	Burst Size (BrstSiz) When field is set to- 0 Burst length = 1 1 Burst length = 2, and so on, up to, 15 Burst length = 16 For IN transfers, this value represents the maximum length of the burst that the device attempts when the Host initiates an IN transfer with a TP_ACK. If BrstSiz < the NumP value in the initiating TP_ACK, then the device controller limits the burst length to BrstSiz. If BrstSiz >= the NumP value in the initiating TP_ACK, then the device controller attempts a burst length of NumP. For OUT transfers, this value represents the NumP value utilized in the response TP_ACK from the device controller. The NumP value indicates (to the host) the burst length the device desires. Note: In the special case of BrstSiz=0 (burst length = 1), the TP_ACK from the device controller has NumP=0; which is a flow-control condition. If this value is utilized, then the endpoint enters flow control for each DP received and a subsequent ERDY is transmitted (according to the USB Specification). However, this may have an undesirable impact on the system throughput. To avoid this impact, it is recommended to set BrstSize=1 to prevent the flow-control condition. The trade-off is that the host could potentially burst two OUT data packets to the device, resulting in an ACK for the first packet, and an NRDY for the second packet.
21-17	FIFO Number (FIFONum) Indicates which transmit FIFO is assigned to this endpoint. For control endpoints, the FIFONum value in the OUT direction must be programmed to the same value as the IN direction. This field should be set to 0 for all other OUT endpoints. Even though there may be more than 16 TXFIFOs in DRD mode, the device mode must use lower 16 TXFIFOs.
16-14	Reserved

Table continues on the next page...

Table 11-19. Command 1: Set endpoint configuration parameters: DEPCFG (continued)

Field	Description
13-3	Maximum Packet Size (MPS) Applies to IN and OUT endpoints. The application must program this field with the maximum packet size (in bytes) for the current USB endpoint. USB 3.0 supports up to 1024 bytes.
2-1	Endpoint Type (EPTYPE) This is the transfer type supported by this USB endpoint. 00 Control 01 Isochronous 10 Bulk 11 Interrupt
0	Reserved

11.1.2.5.2 Command 2: Set endpoint transfer resource configuration (DEPXFRCFG)

There must be only one transfer resource allocated per endpoint. Start transfer causes the use of the transfer resource. End Transfer or an XferComplete event releases the transfer resource. If software attempts to allocate more transfer resources than have been configured in the hardware, this command will return an error in the CmdStatus/EventStatus field.

NOTE

If GUSB2PHYCFG[SUSPENDUSB20] is set to '1', it must be set to '0' prior to issuing this command and may be set to '1' after the command completes.

Table 11-20. Command 2: Set endpoint transfer resource configuration parameters: DEPXFRCFG

Field	Description
Parameter 2	
31-0	Reserved
Parameter 1	
31-0	Reserved
Parameter 0	
31-16	Reserved
15-0	Number of Transfer Resources (NumXferRes) Defines the number of Transfer Resources allocated to this endpoint. This field must be set to 1.

11.1.2.5.3 Command 3: Get endpoint state (DEPGETSTATE)

After Get Endpoint State (DEPGETSTATE) command completes, the 32 bits in DEPCMDPAR2 represent the endpoint state to be saved. This includes the current data sequence number, flow control state, and control transfer state (for control endpoints).

This command is only used when the controller enters hibernation. The controller requires the 32 bits of state information when exiting hibernation as part of the DEPCFG command with the Config Action field set to "Restore endpoint state".

NOTE

If GUSB2PHYCFG[6] is set to '1', it must be set to '0' prior to issuing this command and may be set to '1' after the command completes.

11.1.2.5.4 Commands 4 and 5: Set Stall and Clear Stall (DEPSSTALL, DEPCSTALL)

These commands apply to non-isochronous IN and OUT endpoints only.

The application issues a Set Stall command to stall all tokens from the USB host to this endpoint. If the endpoint is in the NRDY state, the STALL state takes priority.

If a transaction is currently in progress when the software issues Set Stall, the behavior depends on the direction of the endpoint:

- For OUT endpoints, the current packet will complete. The core will respond with STALL to the next OUT DP or token.
- For IN endpoints in SuperSpeed, the current burst transaction will complete, and the core will respond with STALL to the next ACK TP. For other speeds, the core will complete the current packet and respond STALL to the next IN token.

For non-control endpoints, the application is responsible for both setting and clearing STALL via the Set Stall/Clear Stall commands. When the application clears the STALL, the endpoint's data sequence number is reset to zero.

For control endpoints, the application issues only the Set Stall command, and only on the OUT direction of the control endpoint. The controller automatically clears the STALL when it receives a SETUP token for the endpoint.

The application must not issue the Clear Stall command on a control endpoint. If the endpoint is in flow control (NRDY) and also in the STALL state, it responds with a STALL for any packet. The only exception is that the controller always responds to SETUP data packets with an ACK handshake, independent of the STALL state.

NOTE

- If GUSB2PHYCFG[6] is '1', it must be set to '0' prior to issuing this command and may be set to '1' after the command completes.
- When issuing Clear Stall command for IN endpoints in SuperSpeed mode, the software must set the "ClearPendIN" bit to '1' to clear any pending IN transactions, so that the device does not expect any ACK TP from the host for the data sent earlier.

11.1.2.5.5 Command 6: Start transfer (DEPSTRXFER)

This command applies to IN and OUT endpoints.

A Transfer Descriptor (TD) in Device mode is a list of TRBs that comprises one or more transfers. The pointer to a TD is added to the core's cache by the Start Transfer command and removed by an XferComplete event or by the End Transfer command.

Software issues this command indicating that a descriptor is ready to be processed and DMA can start for this endpoint/stream combination. For IN endpoints, this causes the descriptor to be processed, and data is moved from the corresponding memory buffer to corresponding transmit FIFO when the transfer is started. For OUT endpoints, this causes the descriptor to be processed and data is moved from the receive FIFO to corresponding memory buffer.

In response to the Start Transfer command, the hardware assigns this transfer a resource index number (XferRscIdx) and returns the index in the DEPCMDn register and in the Command Complete event. This index must be used in subsequent Update and End Transfer commands.

It is illegal to issue a Start Transfer command for the same TD if it is already present in the core's cache.

Non-stream capable endpoints rules:

- The Stream ID field must be set to 0.
- The HighPri field is reserved.

Stream-capable endpoint rules:

- The Stream ID field must be set to non-zero and match the Stream ID passed into the Start Transfer endpoint command associated with this transfer. In all the TRBs of a transfer, the Stream ID fields must be the same.
- The HighPri field is reserved.

NOTE

- If GUSB2PHYCFG[6] is set to '1', it must be set to '0' prior to issuing this command and may be set to '1' after the command completes.
- Before issuing a start transfer command, software needs to verify whether the link is in U0. If the link is not in U0, software needs to bring the link to U0 by performing a remote wakeup. This is applicable to both SS and USB 2.0 speeds.
- When an IN transfer is in progress, (that is, started and not completed yet), and the software sees a link state change event to L1, then the software can use the GDBGFIFOSPACE register (write followed by a read) to determine if the TXFIFO is empty or not. If the TXFIFO is not empty, it can initiate a remote wakeup.

Table 11-21. Command 6: Start transfer parameters: DEPSTRTXFER

Field	Description
Parameter 2	
31-0	Reserved
Parameter 1	
31-0	Transfer Descriptor Address (TDAAddr Low) Indicates the lower 32 bits of the external memory's start address for the transfer descriptor. Because TRBs must be aligned to a 16-byte boundary, the lower 4 bits of this address must be 0.
Parameter 0	
31-0	Transfer Descriptor Address (TDAAddr High) Indicates the higher 32 bits of the external memory's start address for the transfer descriptor.

11.1.2.5.6 Command 7: Update transfer (DEPUPDXFER)

If software uses circular TRB buffers and updates a TRB, whose Hardware Owner (HWO) bit was 0, by setting HWO=1, it must execute the Update Transfer command, specifying the transfer resource index of the TRB in the DEPCMD register. The hardware uses this information to re-cache the TRB.

Software may issue a special “No Response Update Transfer” command by setting CmdAct=0 and CmdIOC=0. In this case, the hardware does not generate a Command Complete event, does not set the CmdAct bit to '0' (because it will be '0'), and software may immediately issue another command to the same endpoint following this one. This

special type of Update Transfer may not be used when software depends on the XferNotReady event to setup TRBs (see “On Demand” transfers in [Transfer setup recommendations](#)).

Software may issue an Update Transfer command for a transfer resource that has already completed (either due to XferComplete event or an End Transfer command), and the core will detect that the Update Transfer is unnecessary. However, software must not issue an Update Transfer command for a transfer resource index that has never been started.

NOTE

If GUSB2PHYCFG[6] is set to '1', it must be set to '0' prior to issuing this command and may be set to '1' after the command completes.

11.1.2.5.7 Command 8: End transfer (DEPENDXFER)

This command applies to IN and OUT endpoints. Software issues this command requesting DMA to stop for the endpoint/stream specifying the transfer resource index of the TRB and the ForceRM parameter to be set to 1 in the DEPCMD register.

When issuing an End Transfer command, software must set the CmdIOC bit (field 8) so that an Endpoint Command Complete event is generated after the transfer ends. This is necessary to synchronize the conclusion of system bus traffic before the End Transfer command is completed.

For IN endpoints, this command causes descriptor processing to stop and the core stops fetching new data for the endpoint and stops transfers on the USB. The core may truncate a packet being transmitted with the DPPABORT ordered set, does not wait for any pending ACKs from the USB, and does not update the TRB status. For OUT endpoints, this command causes descriptor processing to stop. If there is currently data, it is moved from the receive FIFO to corresponding memory buffer and completed at packet boundary. The core does not update the TRB status.

Use this command under the following conditions:

- When handling USBReset or SetConfiguration, endpoints are closed using this command.
- After receiving a ClearFeature (STALL) control transfer, software issues End Transfer followed by Clear Stall, followed by Start Transfer.
- After an XferInProgress event when the TRB after the one that caused the XferInProgress event has its HWO bit set to '0', this command can be used.
- For isochronous endpoints, if the host stops moving data for many intervals, software may force the end of the transfer and wait for the host to restart.

The hardware does not issue a XferComplete event on End Transfer, but only issues a CommandComplete event.

NOTE

If GUSB2PHYCFG[6] is set to '1', it must be set to '0' prior to issuing this command and may be set to '1' after the command completes.

Table 11-22. Command 8: End Transfer Parameters: DEPENDXFER

Field	Description
Parameter 2	
31-0	Reserved
Parameter 1	
31-0	Reserved
Parameter 0	
31-0	Reserved

End Transfer Command With Hibernation Enabled The following applies only when hibernation is enabled. Prior to hibernation, software will set ForceRM to '0' when it issues End Transfer commands for active transfers. This command may cause a TRB to be written back with intermediate state. When software issues an End Transfer command with the ForceRM bit set to 0, and the core has at least one valid current TRB, it will perform the following:

- Write back the TRB updating the Buffer Pointer and Buffer Size fields to reflect the current state of the TRB.
- Write the value '4' (TRBInProgress) into the TRBSTS field to indicate that this TRB may have been partially used and is still active. This is an indication to software that the original buffer pointer field may have been changed by hardware.
- Set the HWO bit to '0' unless it is a Link TRB. The HWO bit in a Link TRB will be unchanged.

After the state is restored, software may restart a transfer from a partially completed TRB by setting TRBSTS to '0', setting the HWO bit back to '1', and issuing the Start Transfer command. Software is not required to perform any special manipulation on partially completed TRBs. When the core completes a restarted partially complete TRB, it will set TRBSTS back to '0'.

11.1.2.5.8 Command 9: Start new configuration (DEPSTARTCFG)

Software issues this command under the following conditions:

- After power-on-reset with XferRscIdx=0 before starting to configure Endpoints 0 and 1. CmdIOC must be set to '0' and software must poll the CmdAct bit to determine when the command is complete because Endpoint 0 is not yet configured with a valid interrupt number.
- With XferRscIdx=2 when it receives SetConfiguration before starting to configure Endpoints > 1. CmdIOC may be set to '0' or '1'.

This command should always be issued to Endpoint 0 (DEPCMD0).

Hardware resets the transfer resource allocation to the value in the XferRscIdx parameter (must be 0 or 2) upon receiving this command.

NOTE

If GUSB2PHYCFG[6] is set to '1', it must be set to '0' prior to issuing this command and may be set to '1' after the command completes.

11.1.2.6 OTG

This section describes OTG for USB 3.0, and the programming requirements for the USB 3.0 core in OTG mode.

11.1.2.6.1 OTG 2.0 for USB 3.0 Functionality

The following sections describe the OTG 2.0 functions for USB 3.0.

11.1.2.6.1.1 Core OTG functions

The OTG Interface block within the U2MAC handles the core OTG functions: Session Request Protocol (SRP) and Host Negotiation Protocol (HNP). These OTG protocols are implemented through the regular UTMI+ OTG interface.

The MAC handles HNP for host and peripheral role swapping.

The MAC also handles SRP, which allows an A-Device to turn off VBUS to save power when the USB is not used, and provides a means for a B-Device to monitor VBUS and request the A-Device to activate VBUS.

It includes the following logic necessary to achieve SRP and HNP:

- Control of VBUS through Vbus drive enable (utmiotg_drvvbus) as A-Device
- Control of IDDIG line sampling enable control output (utmiotg_idpullup)

- Control of D+/D- pull-down resistor enables (utmiotg_dppulldown/utmiotg_dmpulldown)
- Generates SRP (data line pulsing) as B-Device when the session is off.

NOTE

- When the application programs the USB_GCTL[PRTCAPDIR] as 2'b11 to enable OTG mode, then in the A- Device mode, the core will only enumerate as a HS device and hence will not support any SS-capable device to be connected. This restriction is not valid when the USB_GCTL[PRTCAPDIR] is programmed to 2'b01 or 2'b10.
- The utmisrp_chrgvbus and utmisrp_dischrgvbus outputs are provided for legacy PHY connections but they are both set to 1'b0 from the core since VBUS charging is not supported in OTG 2.0 specifications.
- The application should not program GCTL[0] as 1'b0 when it wants to do a ADP or HNP. In such a case, the application should disable the Clock Gating feature by programming GCTL[0] as 1'b1. The application should re-enable the Clock Gating feature only when the core returns to its original role of operation.

11.1.2.6.1.1.1 HNP Polling and Enable

This HNP Polling activity involves an OTG device acting as the current host to periodically poll the remote device to check if the remote device wishes to take the host role. It will then grant the role swap opportunity at the earliest opportunity. Being a software activity, HNP Polling is expected to be implemented through software timers and periodic exchange of SetFeature.SetHNPEnable packets.

The core is then informed of successful exchange of these packets to Enable HNP activity within the core.

11.1.2.6.1.2 ADP functions

The ADP functions involve the following two processes:

- ADP sensing
- ADP probing

ADP probing capability is required in both A and B devices, while sensing is a must only for B-Device. The main functional unit is the ADP controller. You can choose to have the ADP controller logic supplied along with the USB 3.0 core or can choose to implement proprietary ADP controller logic.

11.1.2.6.1.2.1 Internal ADP controller

In this option, ADP controller logic is supplied alongside the USB 3.0 controller by enabling `DWC_USB3_EN_ADP` coreConsultant parameter. ADP controller logic is inferred along with outside USB 3.0 core functions within the controller.

- All ADP timers are maintained in pwrn (Power) module and generate the enable and disable signaling to PHY for ADP probing and sensing.
- PHY contains the following circuitry related to ADP functionality:
 - Comparators for PRB and SNS
 - I_ADP_SRC and I_ADP_SNK
 - Vbus circuitry
- Application software programs the ADP timing registers residing in the pwrn module.
- pwrn module provides a mechanism to application through the interrupts to log and report events pertaining to ADP probing and sensing.
- Only pwrn module needs to be always powered on in this option. USB 3.0 controller may or may not be powered on.
- As a product, both the ADP controller logic and OTG controller are packaged into USB3 core. In the USB3 core, ADP sensing and probing is loosely coupled with the rest of the logic and is directly under software control. This achieves maximum flexibility to interact with OTG core functions.

The ADP controller in the block diagram is a fully digital controller. This has mainly timers inside it to help ADP operation. All ADP related timers are part of pwrn module.

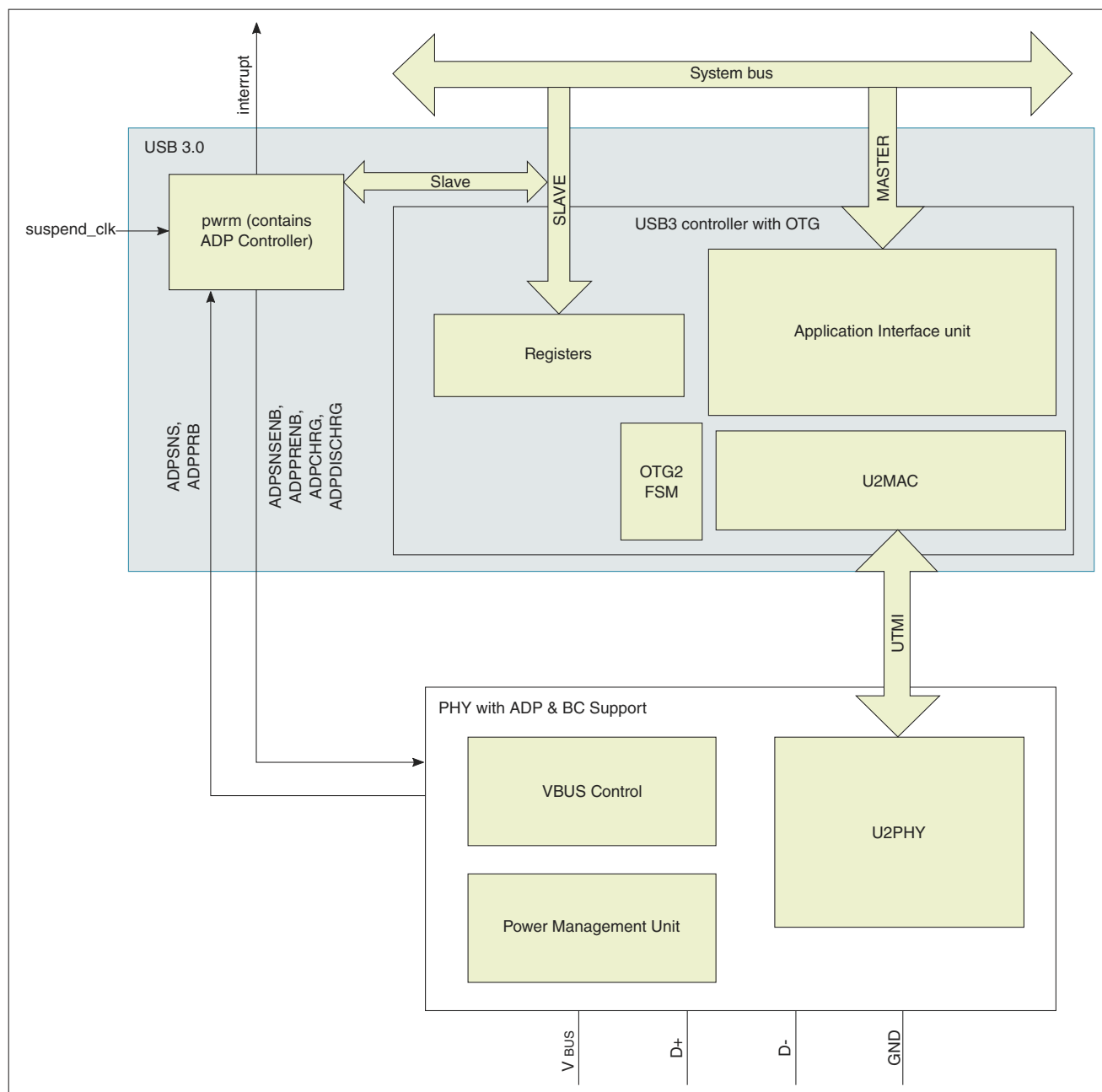


Figure 11-18. Internal ADP controller

For the PHY to generate ADPPRB and ADPSNS, the PHY needs to know the value to compare with VADP_PRB and VADP_SNS respectively. These values can be hardcoded in the PHY or the PHY can have a limited programmable option by which these reference voltage values can be changed. It is assumed that the PHY takes care of this. The pwrn module also implements SRP detection in A-Device mode. This is required if pwrdown module of the USB 3.0 controller is powered down completely while ADP is in progress and B-Device initiates SRP.

NOTE

- The ADP block works with the suspend clock.
- The ADP timers run with the suspend clock along with the programmed USB_GCTL[PWRDNSCALE] value. The USB_GCTL[PWRDNSCALE] value is used internally to generate 32 KHz pulse for the timers.

11.1.2.6.1.2.2 External ADP Controller Option

In this option, ADP controller logic supplied with USB 3.0 controller is not required. The following points apply:

- ADP logic is controlled directly by software.
- Application software controls ADP probe/sense enable and disable directly. The USB 3.0 controller can be completely powered off during this period.
- All ADP timers programmed are programmed by application through a dedicated interface to the ADP controller.
- Any ADP_CHANGE status from the external ADP controller is handled by the application directly.

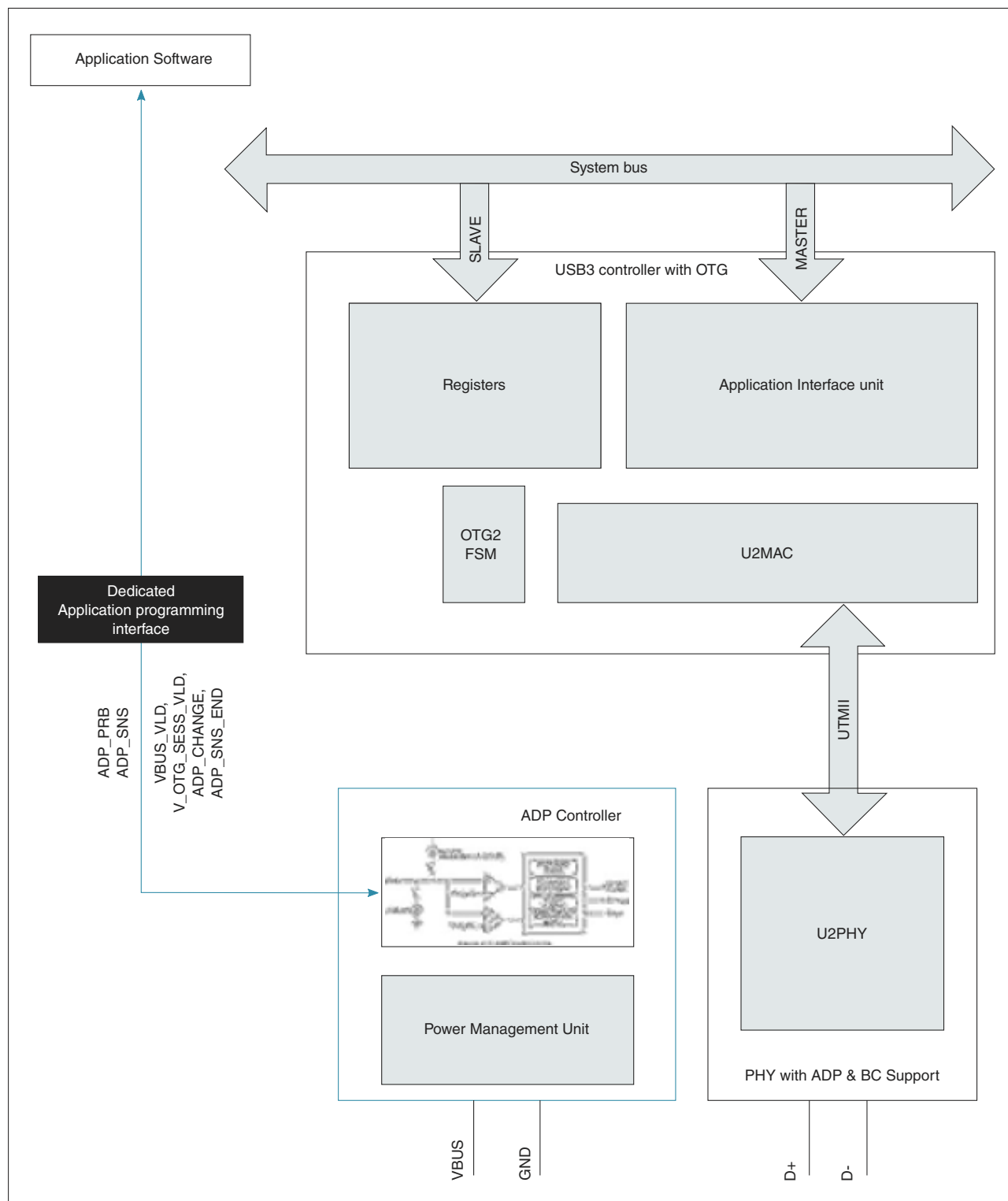


Figure 11-19. External ADP Controller Option

11.1.2.6.1.3 Software flow

The software flow diagram is illustrated in the following figure.

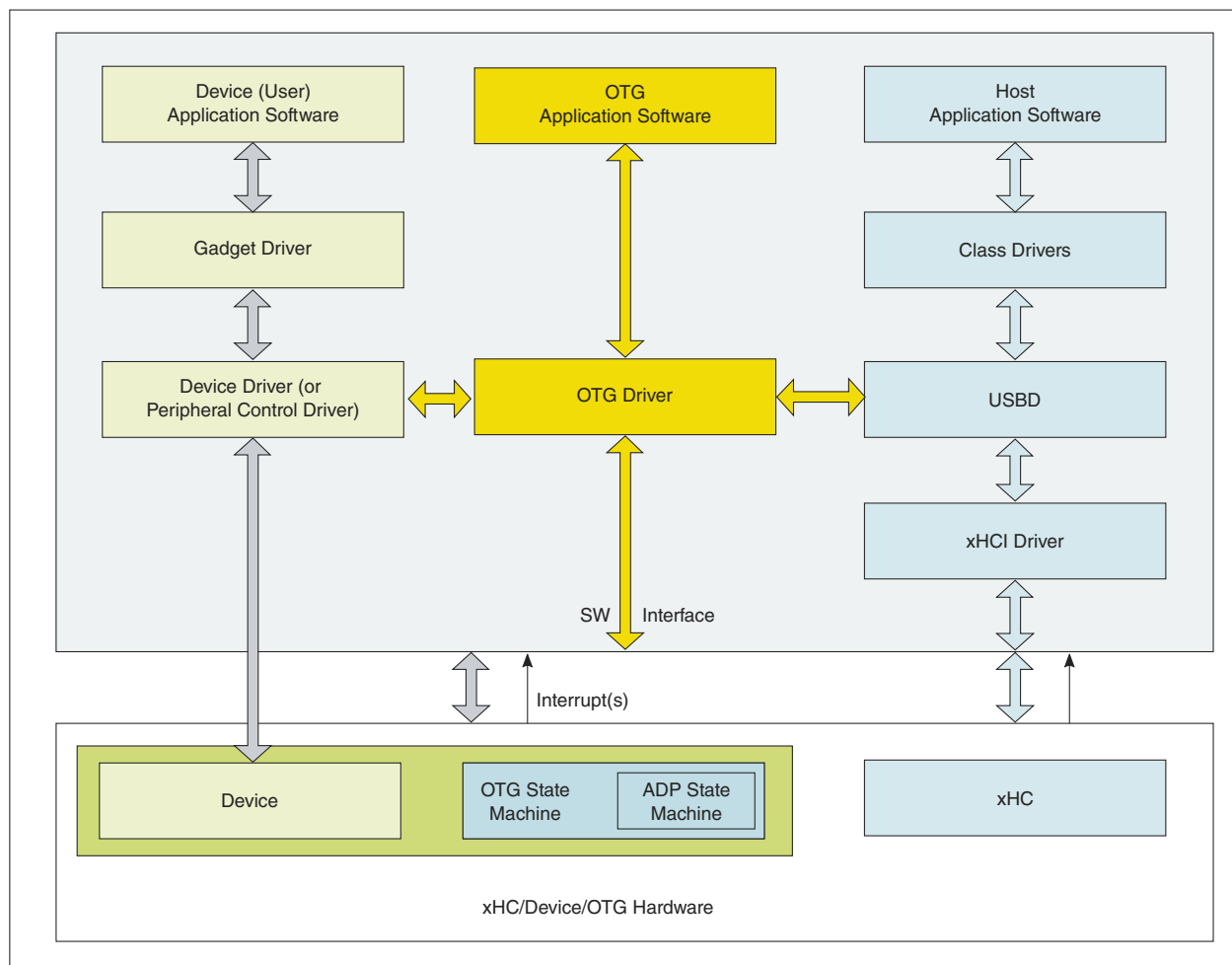


Figure 11-20. Software flow diagram

The following two sections describe the A-device and B-device flows briefly with respect to the above flow diagram.

NOTE

The following concise flows are not very detailed, and are provided here to help understand at an abstract level.

11.1.2.6.1.3.1 A-device activity concise flow

- Power-on-reset
- Only the OTG driver is active after power-on-reset. The host and device drivers are inactive after power-on-reset since at this point of time it is unsure what role the device assumes.
- The OTG driver initializes ADP and waits until a successful ADP event is received.

- The OTG driver senses ID pin = 0 and initializes the OTG registers.
- The OTG driver waits until a successful SRP event is received and turns on the VBUS.
- The OTG driver requests the USB D (USB driver) to bring up the xHC.
- After successful xHC bring up, device enumeration (and data transfers) occurs during which the device capabilities are exchanged for determining a possible role switch.
- In the event where a role switch is possible, the OTG application software or the B-device may initiate a role switch (The OTG application software may request via the Host application software to create favorable conditions for a role switch).
- The xHC is shut down and the control is passed on to the OTG driver.
- The OTG driver requests the PCD (Peripheral driver) to bring up the Peripheral.
- After successful Peripheral bring-up, device enumeration (and data transfers) occurs.
- In an event where a role switch is possible, OTG application software may initiate a role switch (The OTG application software may request via the Device application software for a role switch).
- The PCD is shut down and the control is passed on to the OTG driver.
- The OTG driver requests the USB D to bring-up the xHC, thus reverting to the original roles.

11.1.2.6.1.3.2 B-device activity concise flow

- Power-on-reset
- Only the OTG driver is active after power-on-reset. The host and device drivers are inactive after power-on-reset since at this point of time it is unsure what role the device assumes.
- The OTG driver initializes ADP and waits until a successful ADP event is received.
- The OTG driver senses ID pin = 1 and initializes the OTG registers.
- The OTG driver initiates a SRP and waits for the VBUS to be turned on by the A-device.
- The OTG driver requests the PCD to bring up the peripheral.
- After successful peripheral bring up, device enumeration (and data transfers) occurs during which the device capabilities are exchanged for determining a possible role switch.
- In an event where a role switch is possible, the OTG application software or the A-device may initiate a role switch (The OTG application software may request through the device application software for a role switch).
- The PCD is shut down and the control is passed on to the OTG driver.
- The OTG driver requests the USB D to bring up the xHC.
- After successful xHC bring-up, device enumeration (and data transfers) occurs.
- In an event where a role switch is possible, the OTG application software or A-device may initiate a role switch (The OTG application software may request the host application software for a role switch).

- The xHC is shut down and the control is passed on to the OTG driver.
- The OTG driver requests the PCD to bring-up the Peripheral, thus reverting to the original roles.

11.1.2.6.1.4 Programming model

This section describes the programming requirements for the USB 3.0 core in OTG mode. The core can be configured either as a DRD-device or an OTG, based on bit [13:12] of the Global Control Register's (GCTL) Power-On Initialization Value. If power-on configuration option is chosen as DRD-device, then `PrtCapDir` has to be explicitly programmed to 2'b11 for operating as an OTG device. The premise for the programming model is as follows.

1. The OTG driver is expected to run concurrently and independently with the PCD and xHCI drivers.
2. The OTG driver doesn't get involved in the actual data transfers but is responsible for communicating messages between the xHCI driver and PCD during role changes.
3. The OTG driver is responsible for handling the HNP, SRP and ADP (internal) events from the core. After power on, the OTG driver is responsible for loading, starting and switching between the xHCI driver and PCD. When started, the xHCI driver initializes the xHCI register set before enumerating the connected device. Similarly, when started, the PCD initializes the device register set and waits for the USB reset event to continue. Once the connection is established between the xHCI driver or the PCD by the OTG driver, the corresponding driver takes over for data transfers.

After every successful HNP switching, it is necessary that the active driver is changed from the xHCI driver to PCD or vice-versa. In such cases, the application may also unload the active driver from the memory and re-use the same memory area for loading the next active driver. The subsequent flow described in this section assumes that the core can get enumerated in any of the speeds (SS or HS/FS) and bases its discussion accordingly. For example, if the core enumerates in HS/FS, follow the OTG 2.0 flow.

11.1.2.6.1.4.1 Initializing global registers

The global registers of the USB 3.0 core are shared between the device and host modes. It is the responsibility of the individual driver to initialize the core's global register specific to its functions. This is not explicitly mentioned in the flows described in this chapter. This specifically includes initializing the `GTXTTHRCFG/GRXTTHRCFG`, `GEVNTEN`, and `GPRTBIMAP` registers, and the Global FIFO Size and Global Event Buffer Registers. For more details, refer to Global registers in the register content. However, these global registers that are programmed once at power on can be initialized in the global initialization routine in the OTG driver. Examples of one-time programmable

registers are GSBUSCFG, GCTL, GSNPSID, and GUCTL. Examples of fields that are one-time programmable are the PHY interface (UTMI), AXI parameters like burst size, and so on. The standard xHCI host driver does not get involved in the core-specific global register programming, and therefore, needs to be handled by the Board Support Package software.

11.1.2.6.1.4.2 *Initializing host registers*

In order to initialize the core as a host, the application should perform the steps described in the xHCI specification. Global registers need to be re-initialized as described in [Initializing global registers](#).

11.1.2.6.1.4.3 *Initializing device registers*

In order to initialize the core as a device, the application should perform the steps described in the section [Device Programming Model](#). Global registers need to be re-initialized as described in [Initializing global registers](#). For specific registers, refer to [Device Programming Model](#).

11.1.2.6.1.4.4 *Initializing OTG registers*

The application should initialize the OTG registers based on the initial value of the OSTS.ConIDSts after power on. The following sections depict the programming flow for the OTG application in detail.

11.1.2.6.1.4.5 *Programming flow for OTG in USB 3.0*

The following figure shows the programming flow after power-on reset.

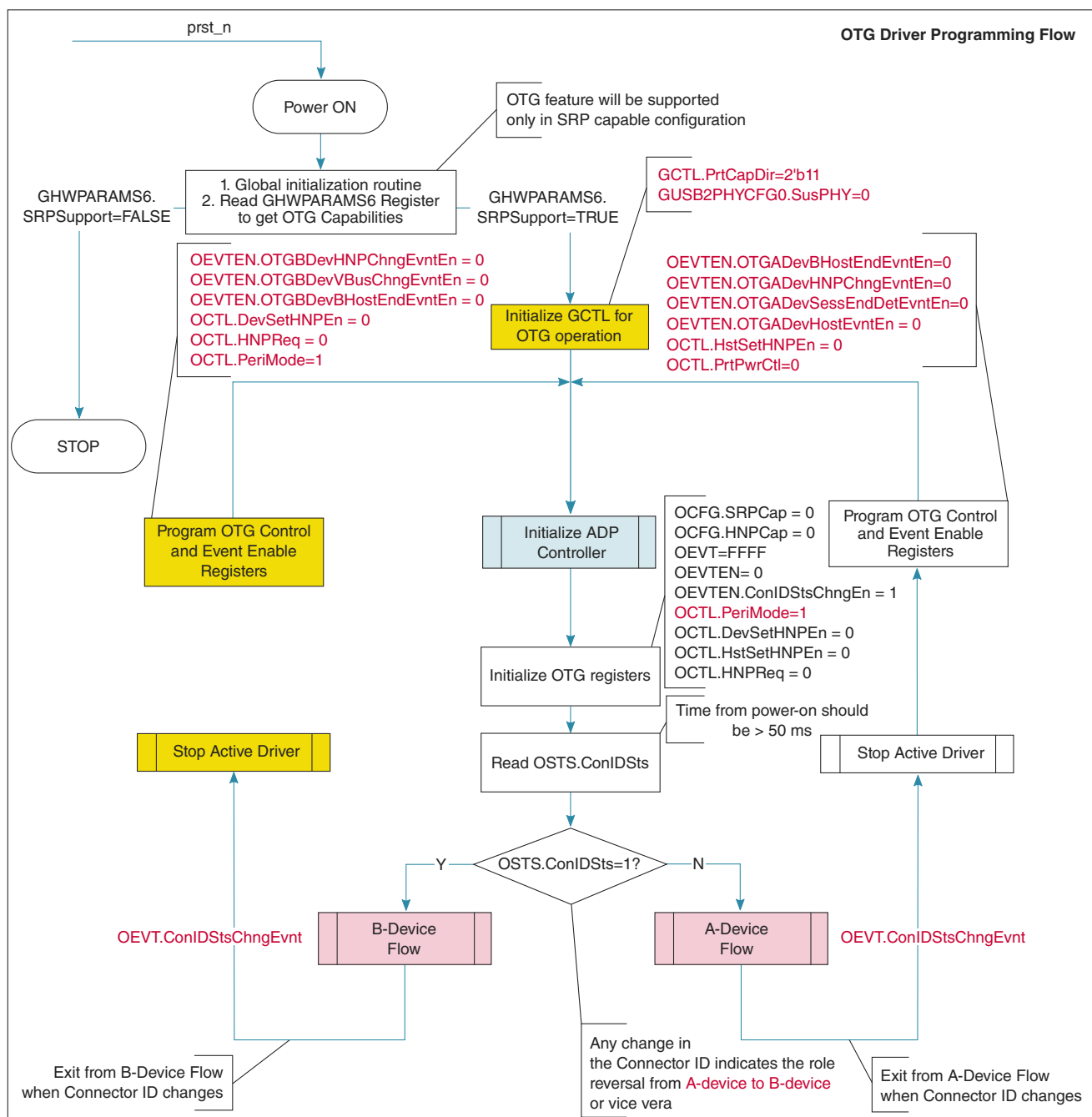


Figure 11-21. OTG driver overall programming flow

Following steps describe overall programming flow:

1. During global initialization, read the GHWPARAMS6 register to see if SRP support is enabled in the configured core. If SRP support is enabled, then proceed with the OTG programming flow by programming USB_GCTL[PRTCAPDIR] as 2'b11. If SRP support is not enabled, do not proceed further in the OTG programming flow. The configured device must at least support SRP for cable connection-based (Connector ID) role of operation.

2. Initialize the ADP controller as explained in ADP programming flow.
3. The OTG 2.0 state machine is initially in B-IDLE if IDDIG is 1, or A-IDLE if IDDIG is 0. Enable USB_OEVTEN[ConIDStsChngEvntEn] for any change in the Connector ID status.
4. Initialize the OTG control and configuration register to default values.
5. Read the OSTS register to find out the current status of the Connector ID (ConIDSts). After power-on or soft reset, the ConIDSts will be valid only after the PHY delay from IDPULL=1 to IDDIG active and any filter delay for IDDIG inside or outside the USB 3.0 core.
6. If USB_OSTS[ConIDSts] is 1, the OTG 2.0 state machine is in B-IDLE, follow the B-Device flow. Otherwise, if OTG 2.0 state machine is in A-IDLE, follow the A-Device flow.
7. When there is any connector ID (IDDIG) change resulting in USB_OEVT[ConIDStsChngEvnt], then exit the A-Device/B-Device flow. Stop the active driver and re-initialize the OTG control and status registers.

11.1.2.6.1.5 Common driver tasks

The common tasks for both A-Device and B-Device flow are as follows:

- Start xHCI driver
- Start peripheral control driver (PCD)
- Switch peripheral
- Switch host
- Stop active driver
- Enable HNP change

The Stop Active Driver is already introduced in the common flow diagram. Other tasks are useful for the A-Device/B-Device flows.

Start xHCI driver

The OTG driver, being responsible for starting the xHCI driver, executes this task. The OTG driver loads the xHCI driver after initializing the global registers, and the xHCI driver is responsible for all the functions thereafter.

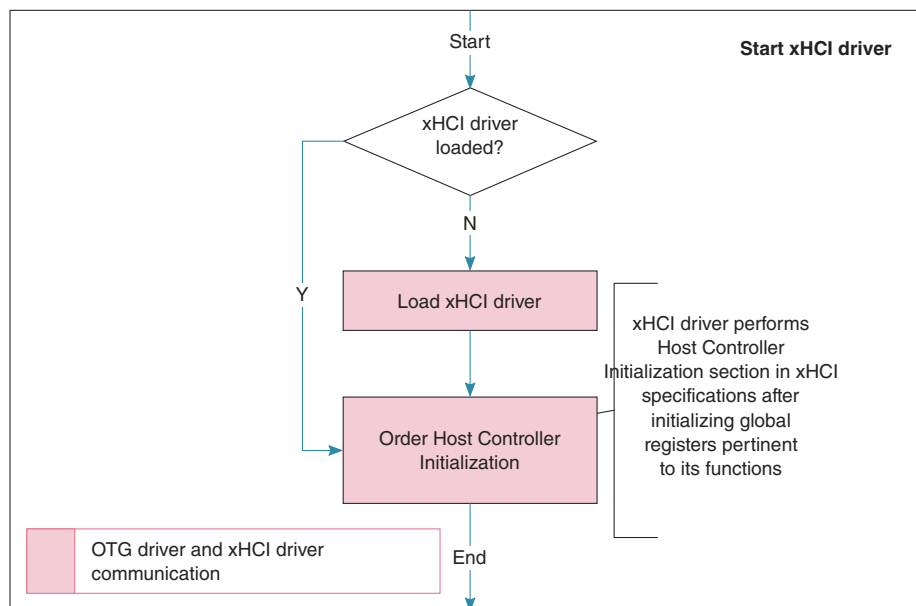


Figure 11-22. Start xHCI driver task

Start peripheral control driver

The OTG driver, being responsible for starting the Peripheral Control Driver (PCD), executes this task. The OTG driver only loads the PCD, which is responsible for all the functions thereafter, including the global registers initialization specific to the USB 3.0 core. Note that except the GTXFIFOSIZ and GEVNTADR, none of the other global registers need to be potentially re-programmed with different values.

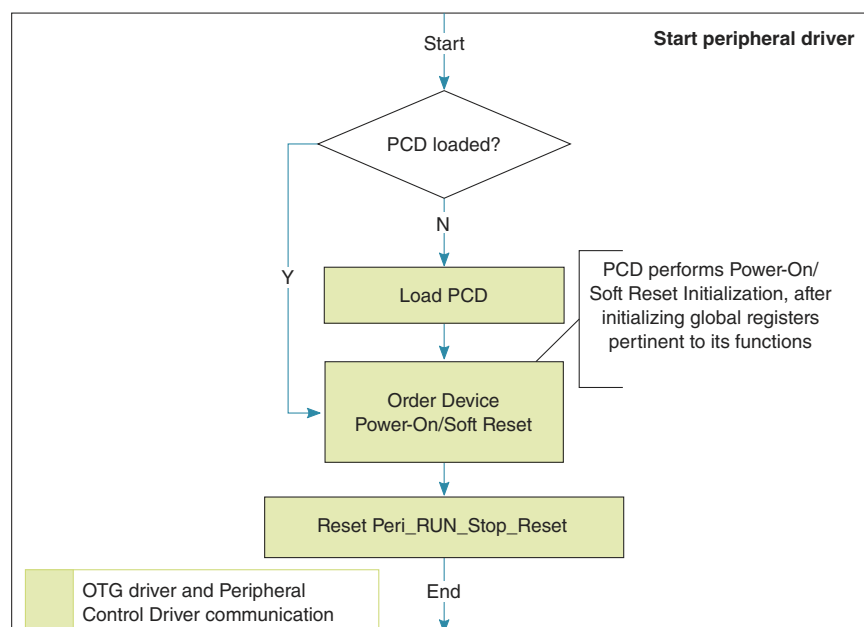


Figure 11-23. Start peripheral driver task

Stop active driver

This task checks which driver is currently active and stops the active driver. Optionally, it may unload the driver if the OS supports dynamic loading of drivers.

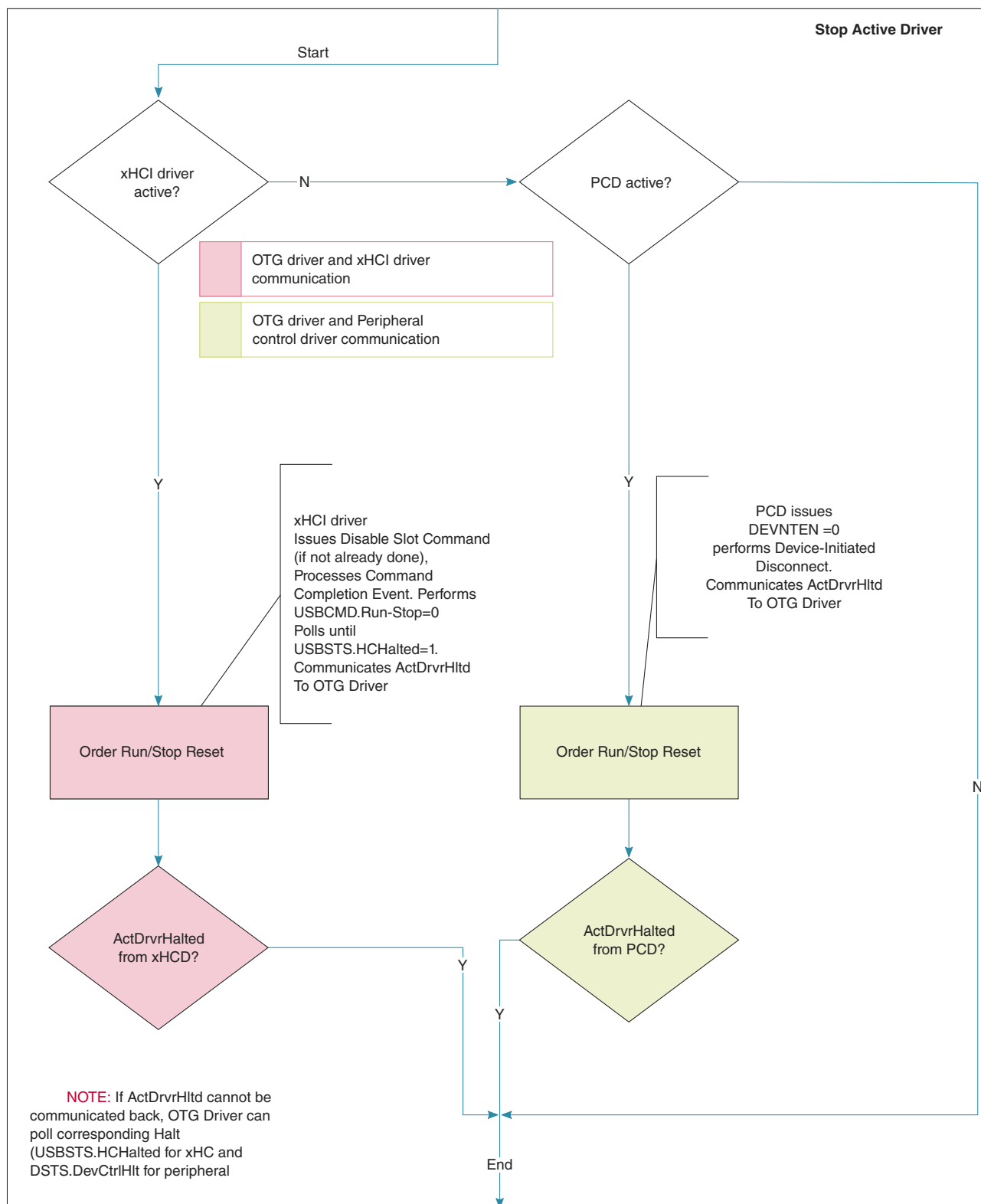


Figure 11-24. Stop active driver task

Switch peripheral and switch host

These tasks are used during HNP and each consists of two individual tasks.

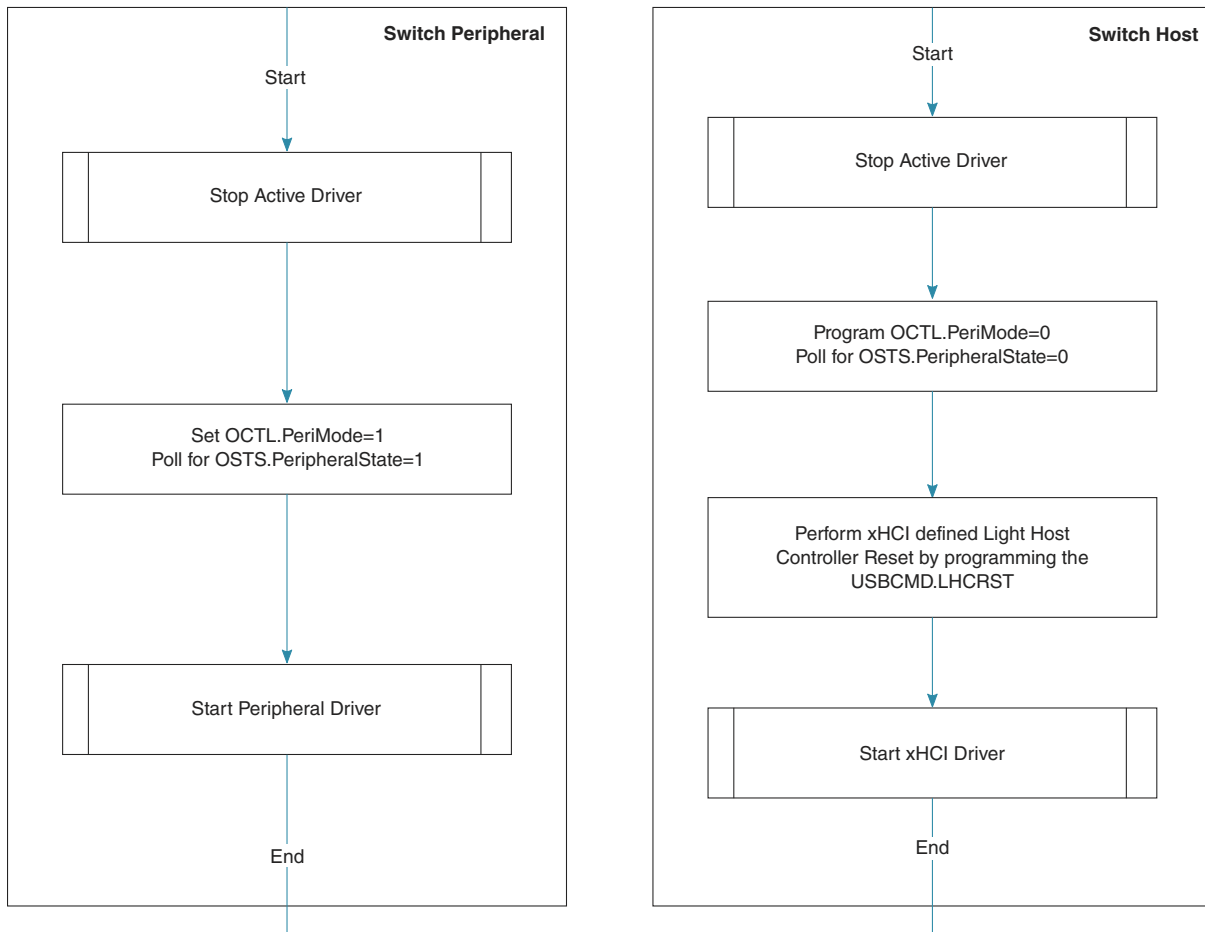


Figure 11-25. Switch peripheral task and switch host task

Programming flow for switch peripheral in SS mode:

This programming flow indicates the steps to be followed by the software when Host to Peripheral role change is intended. Note that the handover of control from the xHCI driver to PCD is accomplished using the OTG driver. The OTG driver plays the role of a mediator here. The steps indicated below is the sequence in which the xHCI driver stops its operation.

1. The xHCI driver decides to end all data transfers.
2. The xHCI driver services all pending interrupts.
3. The xHCI driver issues disable slot command.
4. The xHCI driver waits for the successful completion of disable slot command.
5. The xHCI driver clears the connect status change and port enabled status change bits of the PORTSC0 register.
6. The xHCI driver programs run/stop as 1'b0.
7. The xHCI driver waits for the HCH bit to be set in the USBSTS register.

- When the HCH bit is set, the xHCI driver hands over the control to the OTG driver.

Programming flow for switch host in SS mode:

This programming flow indicates the steps to be followed by the software when Peripheral to Host role change is intended. Note that the handover of control from the PCD to xHCI driver is accomplished using the OTG driver. The OTG driver plays the role of a mediator here. The steps indicated below is the sequence in which the PCD stops its operation.

- The PCD sends a Device Notification packet for role swap.
- The PCD services all pending interrupts.
- The PCD programs DCTL.Run bit to 1'b0.
- The PCD waits for DevCtrlHlt = 1'b1.
- When the DevCtrlHlt bit is set, the PCD hands over the control to the OTG driver.

Enable HNP change

For HS/FS, this task is called when SetFeature.SetHNP is successfully exchanged between the host and the device. The set Peri_Run_Stop_Reset indicates to the device driver to automatically disable all events and perform a peripheral-initiated disconnect when the host puts the device in Suspend. Program the OCTL.DevSetHNPEn as 1, and OCTL.DevSetHNPREq as 1 immediately after the successful completion of SetFeature.SetHNP on the USB. This ensures the HNP attempt from the OTG core in the following Suspend state.

NOTE

By default, at power-on, program GUSB2PHYCFG.SusPHY as 1 to ensure that the PHY enters low power mode during normal suspend/resume without role switch.

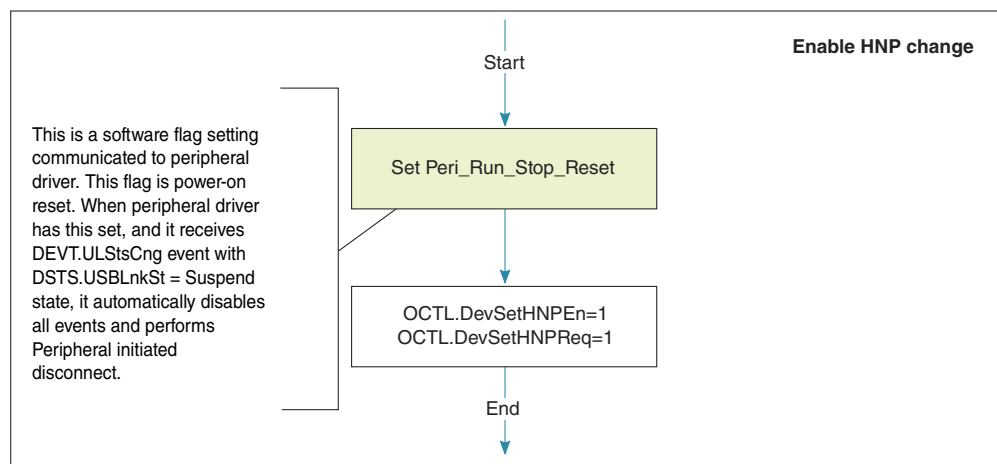


Figure 11-26. Enable HNP change

11.1.2.6.1.6 A-device flow

The following figure shows the A-device flow.

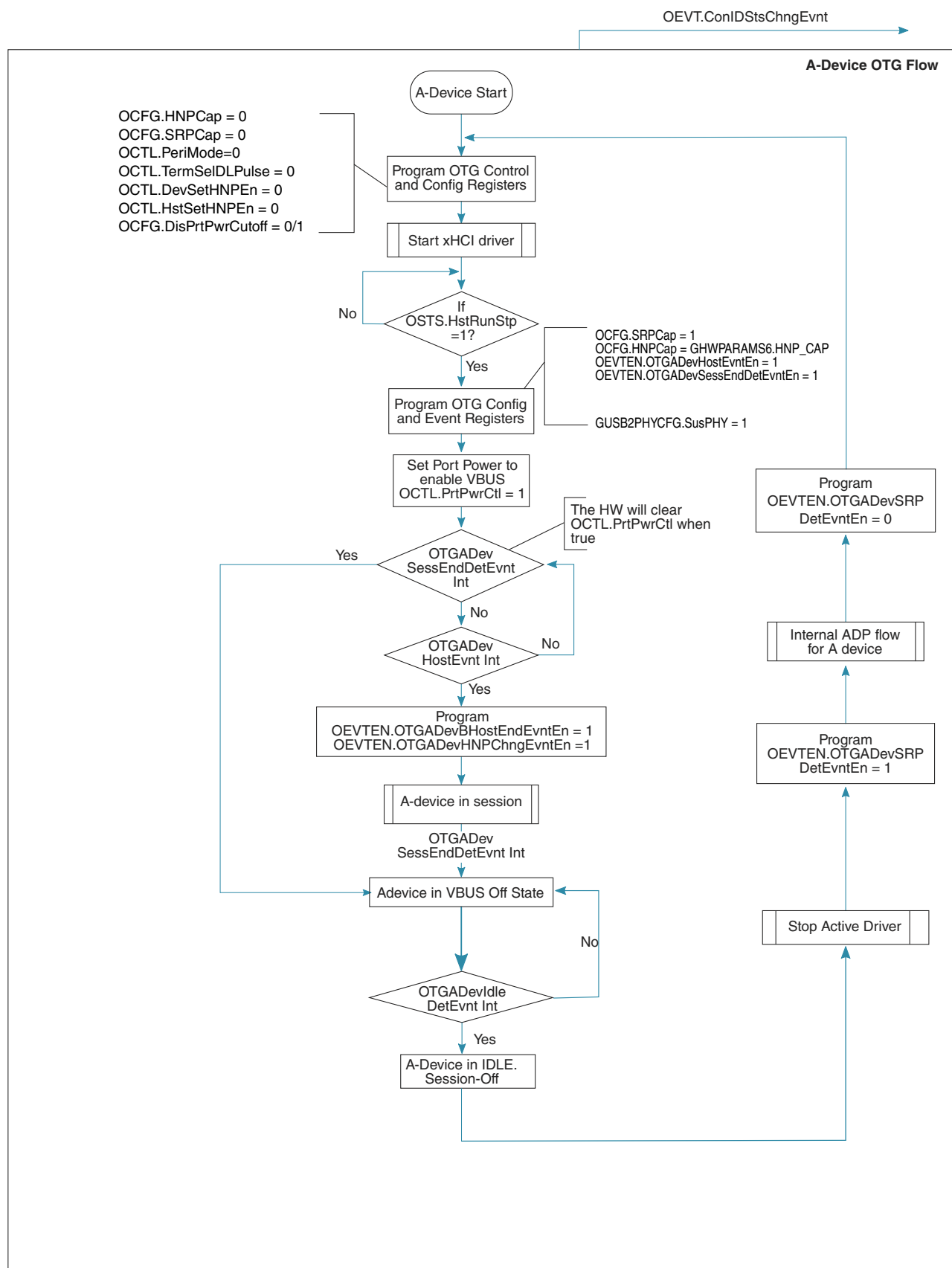


Figure 11-27. A-Device flow diagram

A-device flow:

1. At Start, the A-device core will be in A_IDLE VBUS off state.
2. The OTG software should program the OTG control and configuration registers for A-device operation:
 - OCFG[HNPCAP] = 0
 - OCFG[SRPCAP] = 0
 - OCTL[PERIMODE] = 0
 - OCTL[TERMSELDLPULSE] = 0
 - OCTL[DEVHNPEN] = 0
 - OCTL[HSTSETHNPEN] = 0
 - OCFG[DISPRTPWRCUTOFF] = 0 or 1, based on whether the application requires the core disable the feature to automatically switch off VBUS after 1 second if port is disconnected.
3. Start the xHCI driver.
4. The OTG software should wait for OSTS[HSTRUNSTP] to become 1 to continue. If OSTS[HSTRUNSTP] is 0, the OTG Software should wait for a software event that indicates the change in the xHCI RUN_STOP bit to continue. Alternatively, the OTG Software can poll for OSTS[HSTRUNSTP] bit to be set.
5. OTG Software should program the OTG configuration and event registers to enable A-device events to detect peripheral connect and session end. Refer to A-device flow diagram.
6. Program GUSB2PHYCFG0[SUSPHY] = 1 to enable the core to switch off the PHY clock in the possible non-HNP suspend scenarios.
7. Set OCTL[PRTPWRCTL] when OSTS[XHCIPRTPOWER] is 1. This enables driving of VBUS to the B-device.
8. The A-device may enumerate in HS/FS based on the device signaling.
9. The OTG 2.0 state machine enters a3_ds_host/A-host for HS (FS) if there is a successful B-device connect (ADEVHOSTEVNT).
10. The OTG software should enable the ADEVHNPCHNG and ADEVBHOSTEND events to detect the transition from A-host to A-peripheral and the end of A-peripheral events.
11. If there is an ADEVSESENDDDETEVNT interrupt indicating that the core has now stopped driving the VBUS, the software should wait for ADEVIDLEDETEVNT interrupt and then disable all the OTG events.
12. The Software can enable the ADP flow.
13. After an ADEVHOSTEVNT interrupt, the core is in session. The flow for the A-device in session is shown in earlier in this chapter.

NOTE

- At any step, if CONIDSTSCHNGEVNT occurs, A-device flow is terminated as per the figure "OTG Driver Overall Programming Flow".
- Over current is not detected by the OTG state machine. The xHCI controller detects it, and the OTG driver in turn sees an ADEVSESENDDDETEVNT (from PORTSC[PP] = 0) due to an overcurrent.

OTG state machine mapping to the software flow

- Steps 1-5 are A-IDLE.
- Step 7 is A_WAIT_VRISE in which A-device has detected SRP and therefore waits for VBUS to stabilize.
- Step 8 and step 10 are primarily A_HOST through A_WAIT_VRISE and A_WAIT_BCON. If there is an error in either A_WAIT_VRISE or A_WAIT_BCON, ADEVSESENDDDETEVNT occurs making the state-machine fall back to A-IDLE.

Handling over-current in OTG mode of operation

- When the core is operating in USB 2.0 mode, an over-current condition causes PORTSC[OCA] = 1 and PORTSC[OCC] = 1 on the corresponding USB 2.0 port number. The core also updates a port status change event TRB for the corresponding USB 2.0 port number.
- When the core is operating in USB 3.0 mode, an over-current condition causes PORTSC[OCA] = 1 and PORTSC[OCC] = 1 on the corresponding USB 2.0 and USB 3.0 port numbers. The core also updates a port status change event TRB, one each for the corresponding USB 2.0 port number and the corresponding USB 3.0 port number.
- Note that PORTSC[OCA] may transition quickly from 0 to 1 and 1 to 0 before xHCI driver can see two distinct interrupts. This may occur due to over-current condition being removed quickly. In such a case, the xHCI driver will receive a single interrupt (PORTSC[OCC]) for over-current.

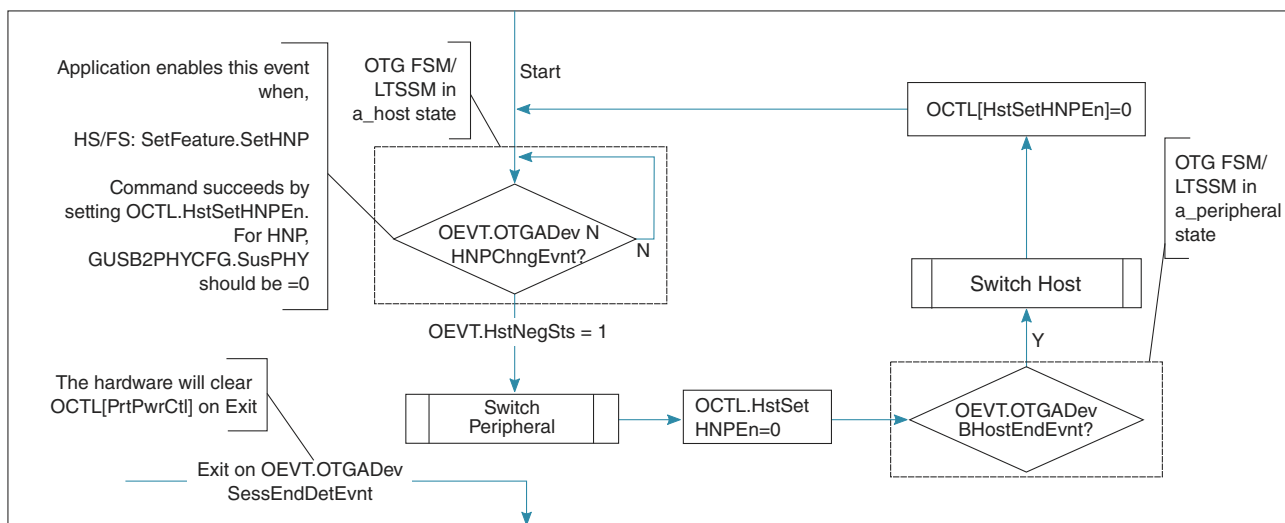


Figure 11-28. A-host -> A-peripheral -> A-host flow diagram

For HS/FS mode:

1. When the core starts the session with A-device as host, the application may periodically exchange SetFeature.SetHNPEnable packets. If B-device wants to become the host, this packet exchange results in the host application knowing about B-Device's intent to become the host, and it directs the OTG driver to set OCTL[HSTSETHNPEN]. This enables the core to cater for HNP from B-device when suspend is initiated. During suspend, ADEVHNPCHNGEVNT is set.
2. If ADEVHNPCHNGEVNT is set, the OTG driver reads OEVT[HSTNEGSTS]. If OEVT[HSTNEGSTS] is set, then the core transitions to A-Peripheral state by executing Switch Peripheral task. Otherwise, wait for ADEVSESSENDDDETEVNT and go to ADP probing.
3. When in A-peripheral state, if there is ADEVBHOSTENDEVNT, it signifies that the B-device no longer wants to be the host. Therefore, the driver goes back to Step 1 by executing the switch host task and assumes an A-host role.

Alternatively, if there is an xHCI[PRTPOWER] de-assertion, the state machine goes to start ADP probing due to the occurrence of ADEVSESSENDDDETEVNT.

4. If ADEVBHOSTENDEVNT is set, then the core would be in A-host state by executing switch host task. Otherwise, perform either of the following:
 - a. Drop VBUS and go to ADP probing.

If the role change is successful, the OTG driver hands over the control to the xHCI driver.

Internal ADP flow for A-device

The internal ADP flow of A-device is as follows:

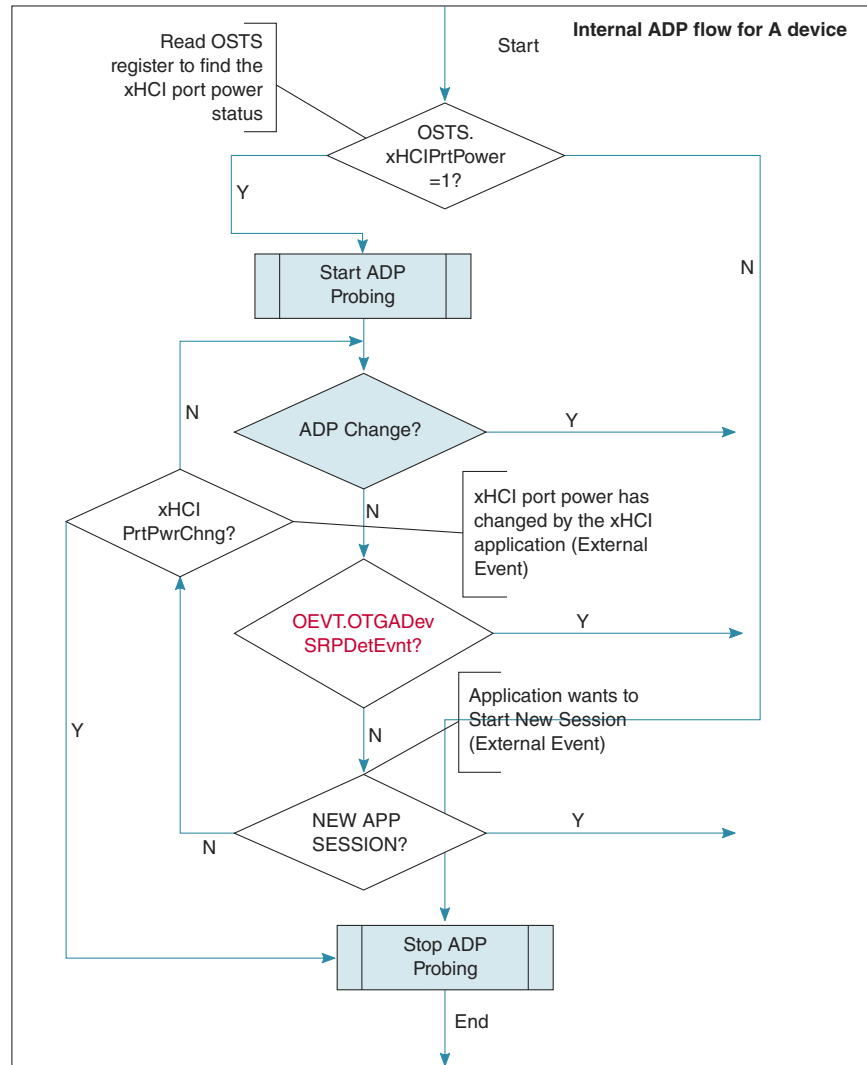


Figure 11-29. Internal ADP flow for A-device

- ADEVSESENDDDETEVNT occurs due to:
 - PORTSC[PRTPOWER] = 1'b0
 - OTG driver stopping VBUS due to:
 - BCON_TOUT when in A_WAIT_BCON
 - A_AIDL_BDIS_TOUT when in A_SUSPEND
- At any step, if ADEVSESENDDDETEVNT occurs, the software starts with ADP probing.
- Once the ADP probing (both external and internal) is started, the software should wait for one of the following events to occur (to stop ADP probing and proceed to Step 2 in A-device flow) to continue.
 - ADP change event
 - SRP detect event
 - New session event from A-device application
 - xHCI port power change event from A-device application

11.1.2.6.1.7 SRP detection by the core (Timeline for ADevSRPDetEvt)

ADevSRPDetEvt is triggered by a valid DLINE pulsing on the USB D+ line by B-Device. The A-Device looks at both the positive and negative edge transition occurrence for a valid data-line pulse time before detecting SRP.

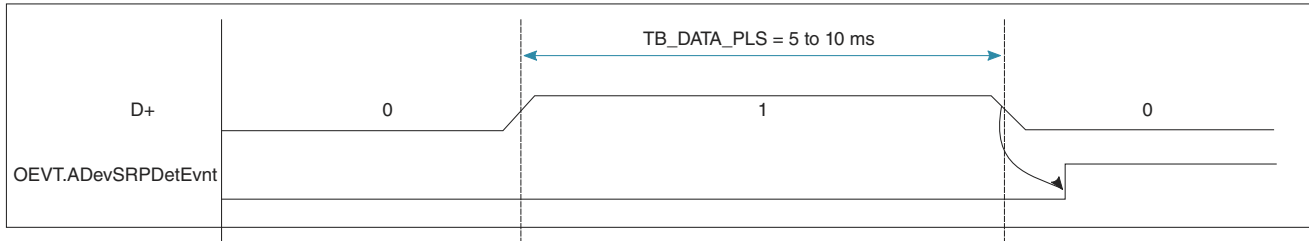


Figure 11-30. Timeline for ADevSRPDetEvt

11.1.2.6.1.8 VBUS turned ON by the core (Timeline for ADevBSessEndEvt)

When the A-device turns on the VBUS, it is unsure at this point of time at what speed the USB enumerates to. So, both the LTSSM and the OTG 2.0 FSMs start concurrently. The LTSSM proceeds with the training sequence after Rx_detect and the OTG 2.0 FSM waits for A_WAIT_BCON.

SS Mode: Before entering a3_ds_host, the training sequence is started. If the training sequence fails, the core automatically falls back to operate in HS/FS mode.

HS/FS Mode: Before entering A-Host, there is a check done in the OTG state machine to ensure B-Device is connected. If B-Device is not connected within TA_WAIT_BCON, ADevBSessEndDetEvt is set. The core resets OCTL.PrtPwrCtl, and the OTG state machine enters A_WAIT_VFALL. Then VBUS to B-Device is removed and this results in ADevSessEndEvt. Note that the A_WAIT_VFALL transition can occur due to over current, which is not depicted in the following figure.

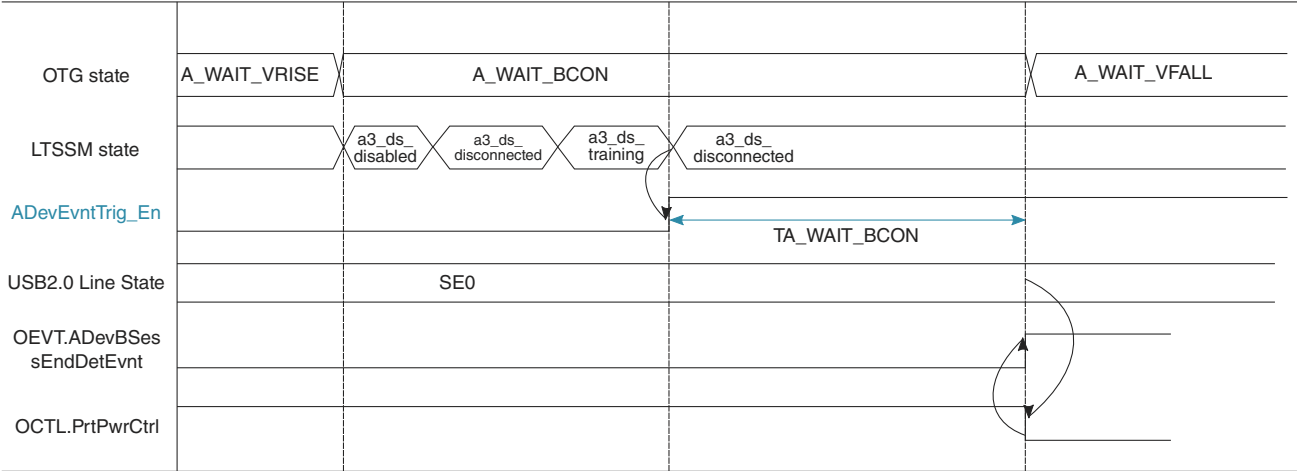


Figure 11-31. Timeline for ADevBSessEndEvt

11.1.2.6.1.9 Core entering a3_ds_host in SS mode or A-Host in HS/FS mode (Timeline for ADevBHostEvt)

The core gives the first chance to connect in SS mode. If SS connection times out (2 ms of polling timeout in LTSSM), the core enables the OTG 2.0 FSM to detect any connect on D+ line (TA_WAIT_BCON).

SS Mode: ADevHostEvt is not set while operating in SS mode.

HS/FS Mode: If the B-Device is connected within TA_WAIT_BCON, ADevBHostEvt is set.

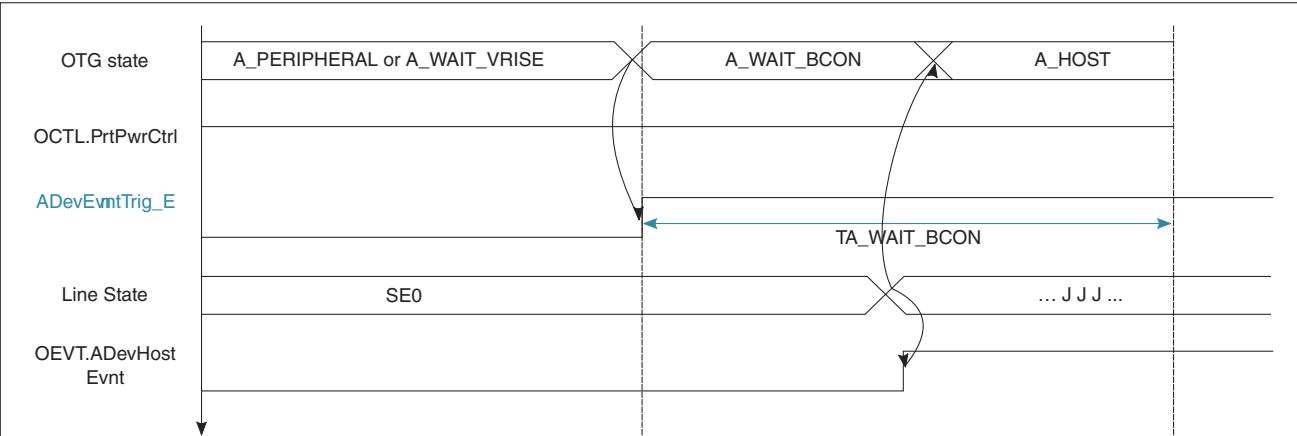


Figure 11-32. Timeline for ADevBHostEvt

11.1.2.6.1.10 B-device flow

The following figure shows the B-device flow.

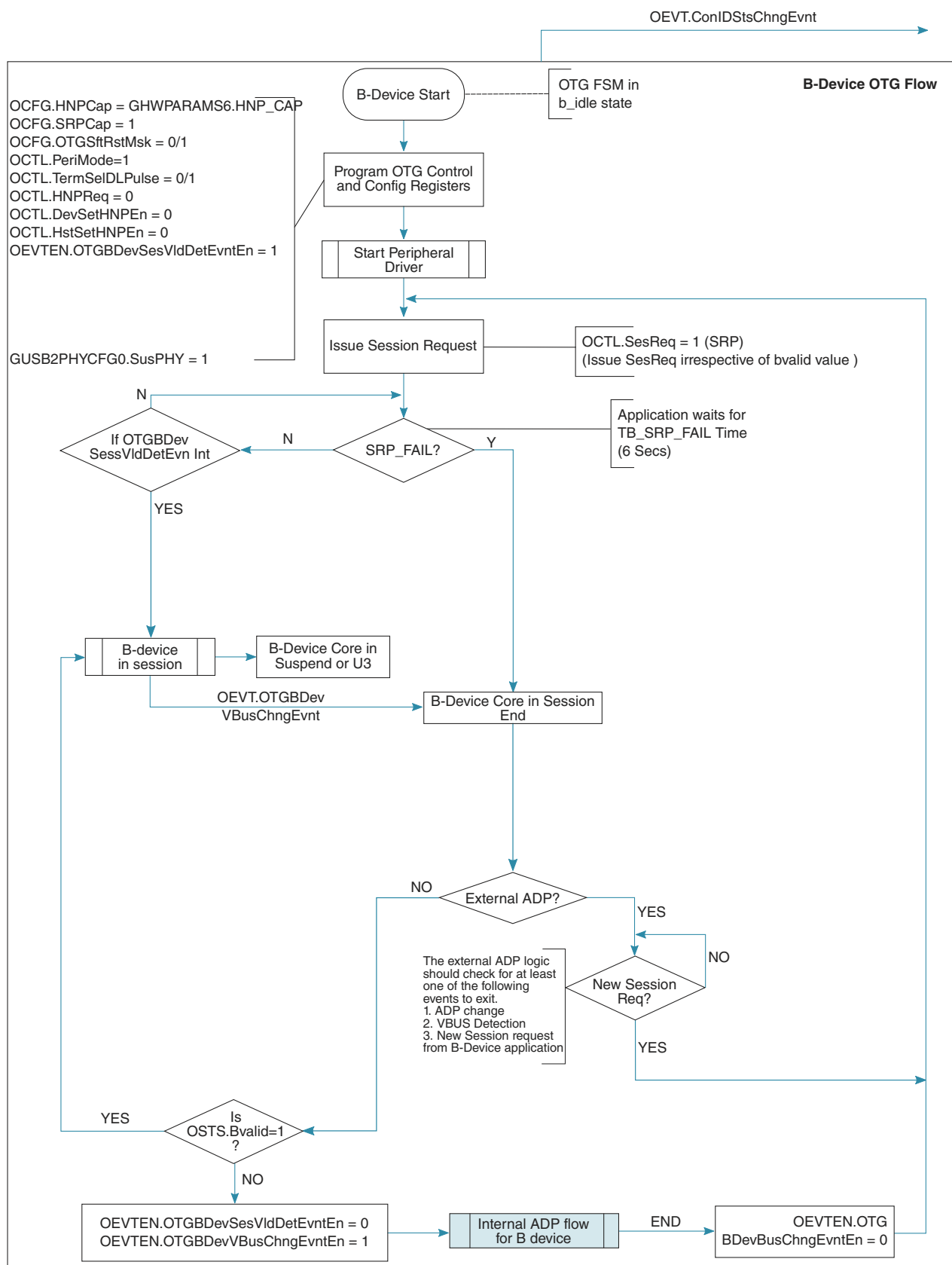


Figure 11-33. B-device flow diagram

1. At start, the core is in B-IDLE state. Note that for OTG functionality in peripheral mode, DCTL[RUN_STOP] should be set, otherwise the core as B-peripheral does not initiate SRP pulsing. Also, it is important for the device driver to program the speed of operation as B-peripheral (DCFG[SPD]).
2. The OTG software should program the following OTG control registers for B-device operation:
 - OCFG[HNPCAP] = GHWPARAMS6[HNPSUPPORT]
 - OCFG[SRPCAP] = 1
 - OCTL[PERIMODE] = 1
 - OCTL[TERMSELDLPULSE] = 0
 - OCTL[HNPREQ] = 0
 - OCTL[DEVHNPEN] = 0
 - OCTL[HSTSETHNPEN] = 0
 - OEVTEN[OTGBDEVSESVLDEVNTEN] = 1
 - GUSB2PHYCFG0[SUSPHY] = 1
 - GCTL[GBLHIBERNATIONEN] = 0 to prevent hibernation entry so that the OTG software can perform SRP.
3. Start peripheral driver.
4. Program OCTL[SESREQ] = 1 to initiate SRP by D-line pulsing. The core initiates SRP and waits for a valid VBUS (BVALID).
5. The B-device core continues to wait for BVALID if the A-device does not respond. The application should timeout if SESSVLDDDETEVNT does not come within the SRP fail time (TB_SRP_FAIL is 6 seconds).

If hibernation feature is enabled, the OTG Software can program the GCTL[GBLHIBERNATIONEN] = 1 to enable hibernation entry.

6. When SRP fails, the application should enable ADP probing.

If the core had entered Hibernation after SRP Failure, the OTG Software should direct the power management software to exit hibernation if the OTG software decides to perform internal ADP. On hibernation exit, the OTG software should program the GCTL[GBLHIBERNATIONEN] = 0 before starting the internal ADP Flow.

7. If the A-device asserts a valid VBUS in response to SRP, the core reports an BDEVSESSVLDDDETEVNT to indicate the start of a session.
8. If there is a BDEVSESSVLDDDETEVNT, the core enters a session on state. For HNP flows in B-peripheral -> B-host -> B-peripheral flow diagram.
9. If there is a BDEVVBUSCHNGEVNT anytime, indicating VBUS is no longer valid, the core enters session-end state.

If hibernation feature is enabled, the OTG Software can program the `GCTL[GBLHIBERNATIONEN] = 1` to enable hibernation entry.

NOTE

- For OTG 2.0, during HNP process where the B-device is going to assume the role of a host, the B-device application needs to ensure that a USB reset process is programmed within 150 ms (`TB_ACON_BSE0`) of getting a device connect interrupt.
- At any step, if `CONIDSTSCHNGEVNT` occurs, the A-device flow is terminated as per "A-device Flow Diagram".

The OTG state machine mapping to the software flow is as follows:

- Steps 1-3 and 6 are B-IDLE
- Step 4 is B-SRP-INIT
- Step 7 and 8 are B-PERIPHERAL

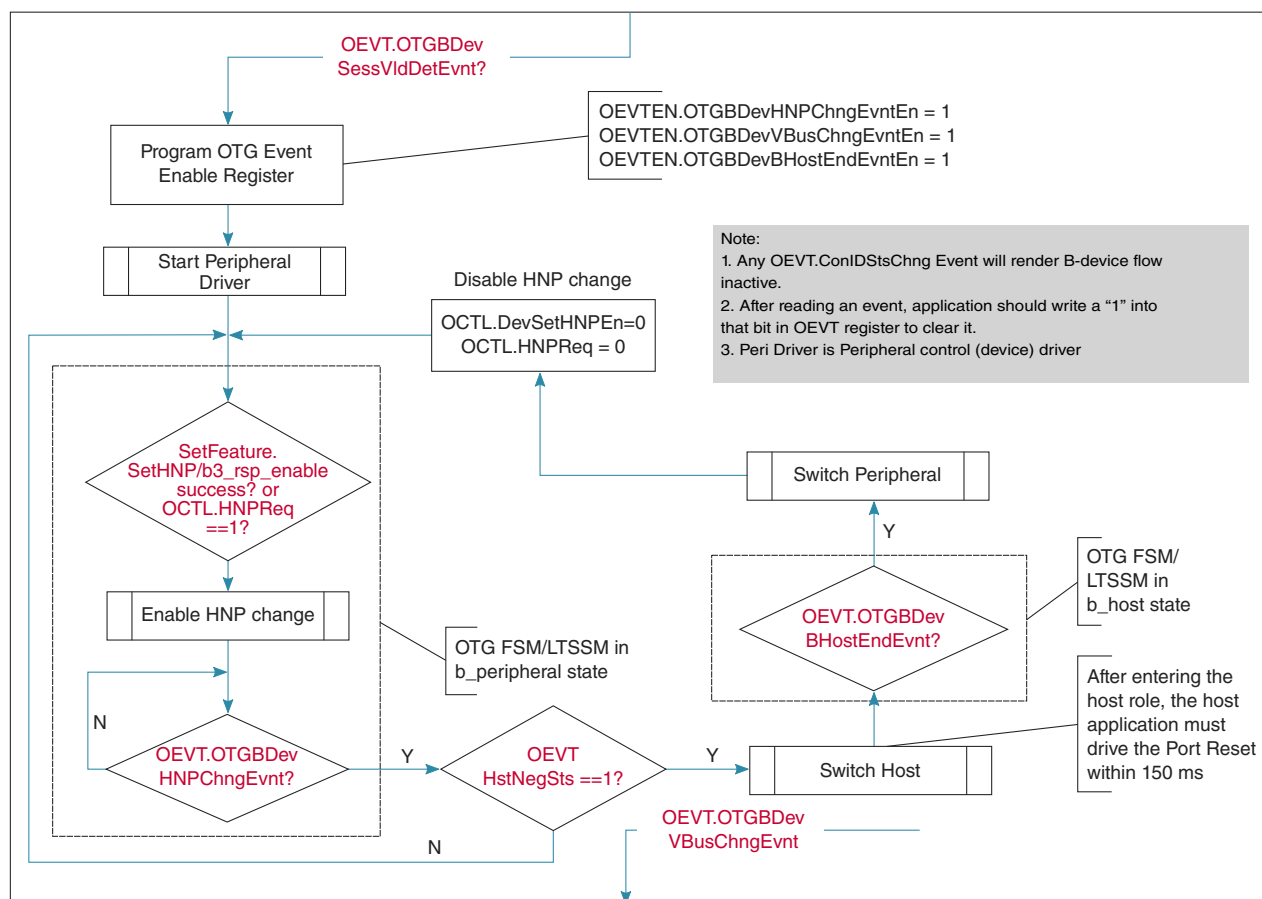


Figure 11-34. B-peripheral -> B-host -> B-peripheral flow diagram

For HS/FS mode:

1. The OTG state machine starts operation as a B-peripheral. The following events are enabled.
 - OEVTEN[OTGBDEVHNPCHNGEVNTEN] = 1
 - OEVTEN[OTGBDEVVBUSCHNGEVNTEN] = 1
 - OEVTEN[OTGBDEVBHOSTENDEVNTEN] = 1

In this state, the application periodically exchanges SetFeature.SetHNPEnable packets. If the B-device wants to become host, this packet exchange results in a device application sending an ACK to this packet, and it directs the OTG driver to execute Enable HNP change. Ensure that GUSB2PHYCFG0[SUSPHY] is 1'b0 when SetFeature.SetHNP command succeeds. This enables the core to initiate HNP from B-device when a suspend is initiated. During a suspend, BDEVHNPCHNGEVNT is set. OTGBDEVHNPCHNGEVNT starts the transition from B-peripheral ->B-WAIT-ACON to either B-host or B-peripheral.

2. If BDEVHNPCHNGEVNT is set, the OTG driver reads OEVT[HSTNEGSTS]. If OEVT[HSTNEGSTS] is set, it indicates successful connection of A-device as peripheral, the core therefore transitions to B-host state. After entering the host role, the host application must drive the port reset within 150 ms. If OEVT[HSTNEGSTS] is not set, it indicates failure of HNP and it stays as B-peripheral.
3. When in B-host state, if there is BDEVBHOSTENDEVNT, it signifies that the B-device no longer wants to be the host, therefore suspended the bus and A-device has stopped signaling connect. Therefore, the driver goes back to step 1 and assumes B-peripheral role.
4. The core generates BDEVBHOSTENDEVNT.

NOTE

- Step 3 above is B-host and B-host to B-peripheral.
- OTGBDEVVBUSCHNGEVNT indicates transition back to B-IDLE.

For SS mode:

1. The OTG state machine starts operation as b3_us_peripheral. The following events are enabled.
 - OEVTEN[OTGBDEVHNPCHNGEVNTEN] = 1
 - OEVTEN[OTGBDEVVBUSCHNGEVNTEN] = 1
 - OEVTEN[OTGBDEVBHOSTENDEVNTEN] = 1
 - a. The A-device (host) should send a SET_FEATURE with NTF_HOST_REL feature selector. This SET_FEATURE needs to be sent by the A-device within TRSP_ACK. The core receives this information and is passed onto PCD. If this SET_FEATURE is not received within TRSP_ACK_ERR the OTG driver needs to program OCTL[OTG3_GOERR] bit for instructing the core's LTSSM to go to

- b3_us_error state. The TRSP_ACK_ERR timer needs to be maintained in software.
- b. As a response, the PCD programs the device generic command register (command value = 07h). For programming the device generic command register, refer to "Device Programming Model". The PCD should program the core to send this device notification packet within TRSP_CNF.
 - c. The PCD programs OCTL[DEVSETHNPEN] = 1 as a result of the above step.
 - d. The core is responsible for internally detecting warm reset and generating OTGBDEVHNPCHNGEVNT. If the warm reset is not received within TRSP_WRST, the core doesn't generate OTGBDEVHNPCHNGEVNT and hence the PCD should program OCTL[OTG3_GOERR] which will direct the core to b3_us_error state.
 - e. As a result of above step and if OTGBDEVHNPCHNGEVNT is generated, the OTG driver informs the PCD and in turn, the PCD completes its activities, programs the core to stop the activities as a peripheral and hands over the control to OTG driver. The OTG driver programs OCTL[PERIMODE] as 0. Ensure that GCTL[DSBLCLKGTNG] is 1 before role change.
2. If BDEVHNPCHNGEVNT is set, it indicates that the core has transitioned to b3_ds_disconnected state. The OTG driver now hands over the control to xHCI driver. Note that after xHCI driver has taken control, if there is a transition to b3_ds_error state by the A-device, this error handling is performed by the xHCI driver.
 3. When in b3_ds_host_state
 4. The core generates BDEVBHOSTENDEVNT.

NOTE

- Step 3 above is B-host and B-host to B-peripheral.
- OTGBDEVVBUSCHNGEVNT indicates transition back to B-IDLE.
- If there is any unexpected error scenario like VBUS drop after the OCTL[DEVSETHNPEN] bit is set, the OTG driver must clear the OCTL[DEVSETHNPEN] bit.

Internal ADP flow for B-device

The internal ADP flow of B-device is as follows:

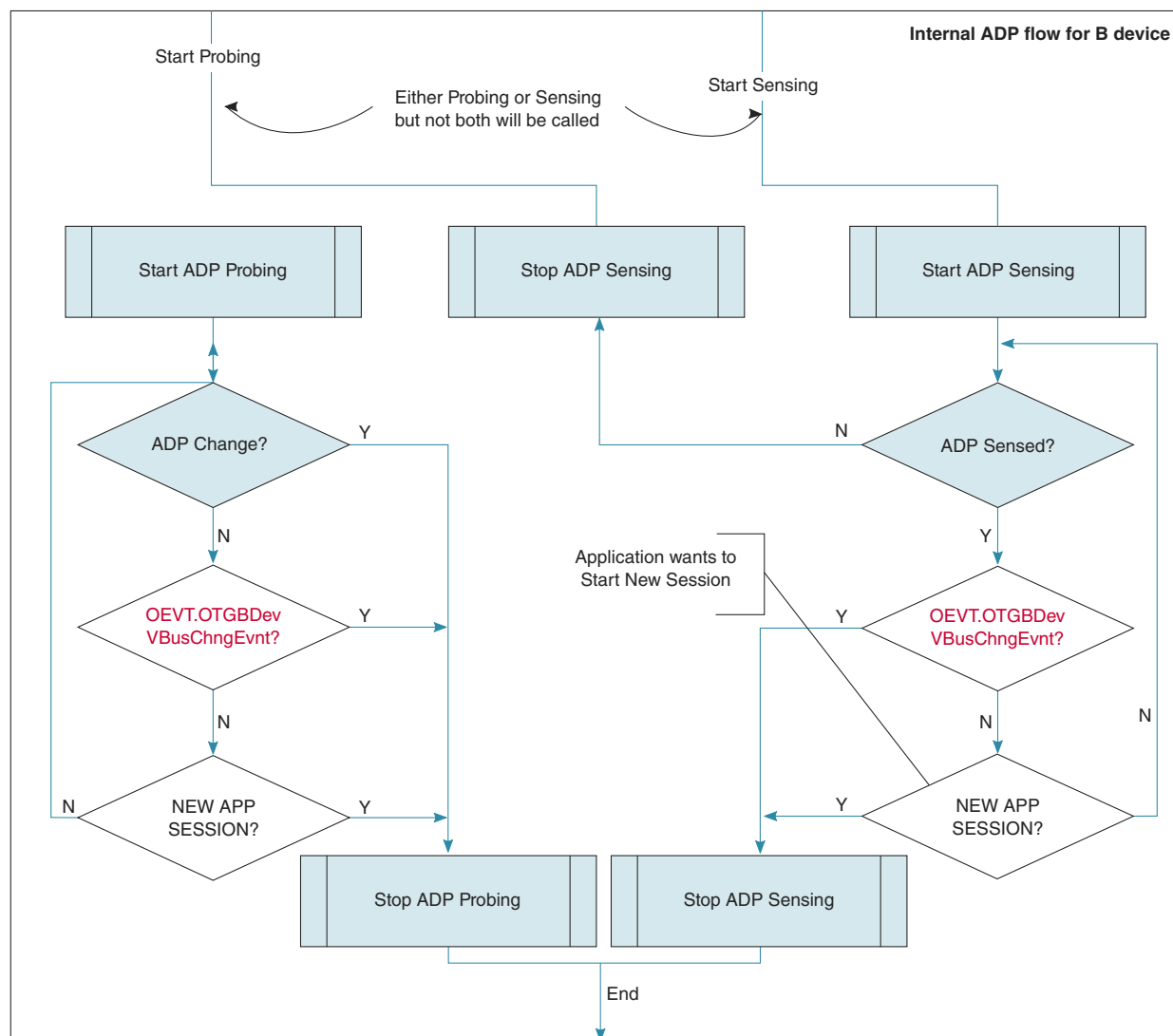


Figure 11-35. Internal ADP flow for B-device

While probing, the application should wait for one of the following events to exit probing and start SRP request again.

- ADP change
- VBus detection
- New session request from B-device application

While sensing, the application should wait for one of the following events to exit sensing.

- ADP sensed failed
- VBUS detection
- New session request from B-device application

The application should continue with ADP probing if ADP sense failed. Else, the application should exit sensing and start SRP request again

11.1.2.6.1.11 Core entering b3_us_peripheral in SS mode or B-Peripheral in HS/FS mode (Timeline for BDevSessVldDetEvt)

The PCD decides the speed in which the USB3 core needs to operate by programming DCFG.DevSpd bits. Accordingly, the core decides to operate either in SS or FS to start with.

SS Mode: OEVT.BDevSessVldDetEvt is triggered when the LTSSM enters b3_us_disconnected state after detecting VBUS turned on by the A-device

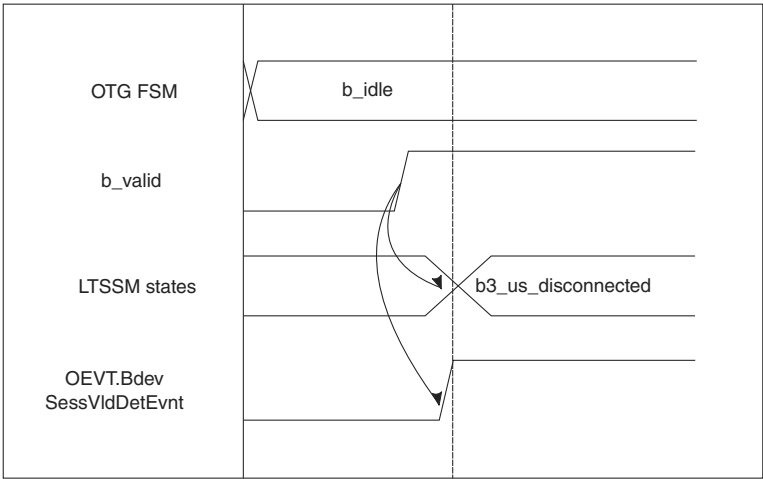


Figure 11-36. Timeline for BDevSessVldDetEvt in SS mode

HS/FS Mode: BDevSessVldDetEvt is triggered by a valid VBUS detection by B-Device in B-IDLE state. The B-Device will transition to B-PERIPHERAL state before triggering this event. Note that this is common to both SS and HS(FS).

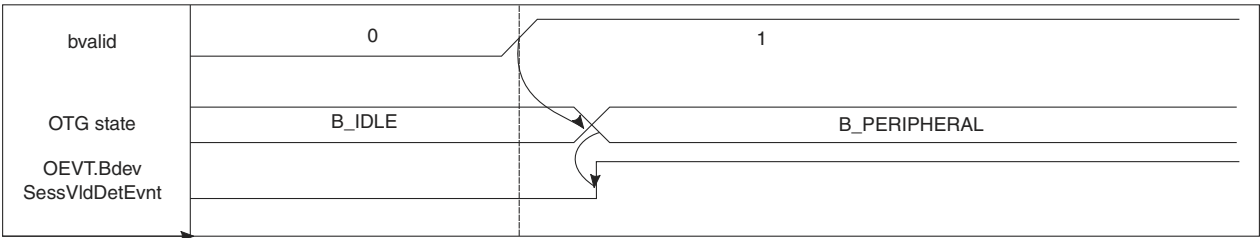


Figure 11-37. Timeline for BDevSessVldDetEvt in HS/FS mode

11.1.2.6.1.12 VBUS change detected on USB (Timeline for BDevVBusChngEvt)

BDevVBusChngEvt is triggered when a change in bvalid is detected irrespective of OTG state machine.

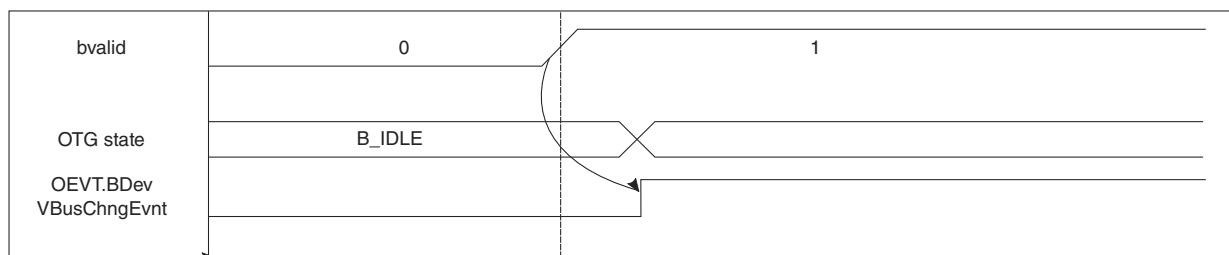


Figure 11-38. Timeline for BDevVBusChngEvt

11.1.2.6.1.13 Internal ADP controller logic

The ADP controller has timers that assist in ADP operation. The ADP-related registers are a part of the `pwrn_otgif` module. For the PHY to generate ADPPRB and ADPSNS, the PHY needs to know the value to compare with VADP_PRB and VADP_SNS, respectively. These values can be hardcoded in the PHY or the PHY can have limited programmable option by which these reference voltage values can be changed. The "Wakeup Logic" module implements SRP detection in A-Device mode. This is required if the USB 3.0 controller is powered down completely while ADP is still in progress and B-Device initiates SRP.

NOTE

ADP controller logic is inferred along with, but outside USB 3.0 controller.

- As a product, both ADP controller logic and OTG controller are packaged into USB 3.0 core.
- All ADP timers are maintained in the ADP controller module, which generates the enable and disable signaling to the PHY for ADP probing and sensing.
- PHY contains the following circuitry related to ADP functionality:
 - Comparators for PRB and SNS
 - I_ADP_SRC and I_ADP_SNK
 - V_{BUS} circuitry
- Application software programs the ADP timing registers that resides in the `pwrn_otgif` module.
- The `pwrn_otgif` module provides a mechanism to the application via interrupts to log and report events pertaining to ADP probing and sensing.
- In this operation, only the `pwrn` module needs to be always powered on. The `pwrdsn` module can be either powered on or off.

11.1.2.6.1.14 Initializing ADP Controller

Initializing the ADP controller comprises of the following steps:

1. Set ADPEVTEN.ADPRstCmpltEvntEn to 1'b1.
2. Set ADPCTL.ADPRes to 1'b1. This initiates reset of the ADP controller.
3. Wait for ADPEVT.ADPRstCmpltEvnt to ensure that the reset process is complete.
4. Program the appropriate ADPCFG.PrbDschg (Discharge time), ADPCFG.PrbDelta (Resolution unit), and ADPCFG.PrbPer (Probe period) values.
5. Read the ADPCTL register to see if the ADPCFG values have been written into the ADP controller (Suspend clock domain) by looking at the ADPCTL.WB bit. Wait for the ADPCTL.WB bit to be cleared by the hardware to confirm that the write operation is complete.
6. Clear ADPEVTEN.ADPRstCmpltEvntEn to 1'b0 in order to complete the ADP initialization sequence.

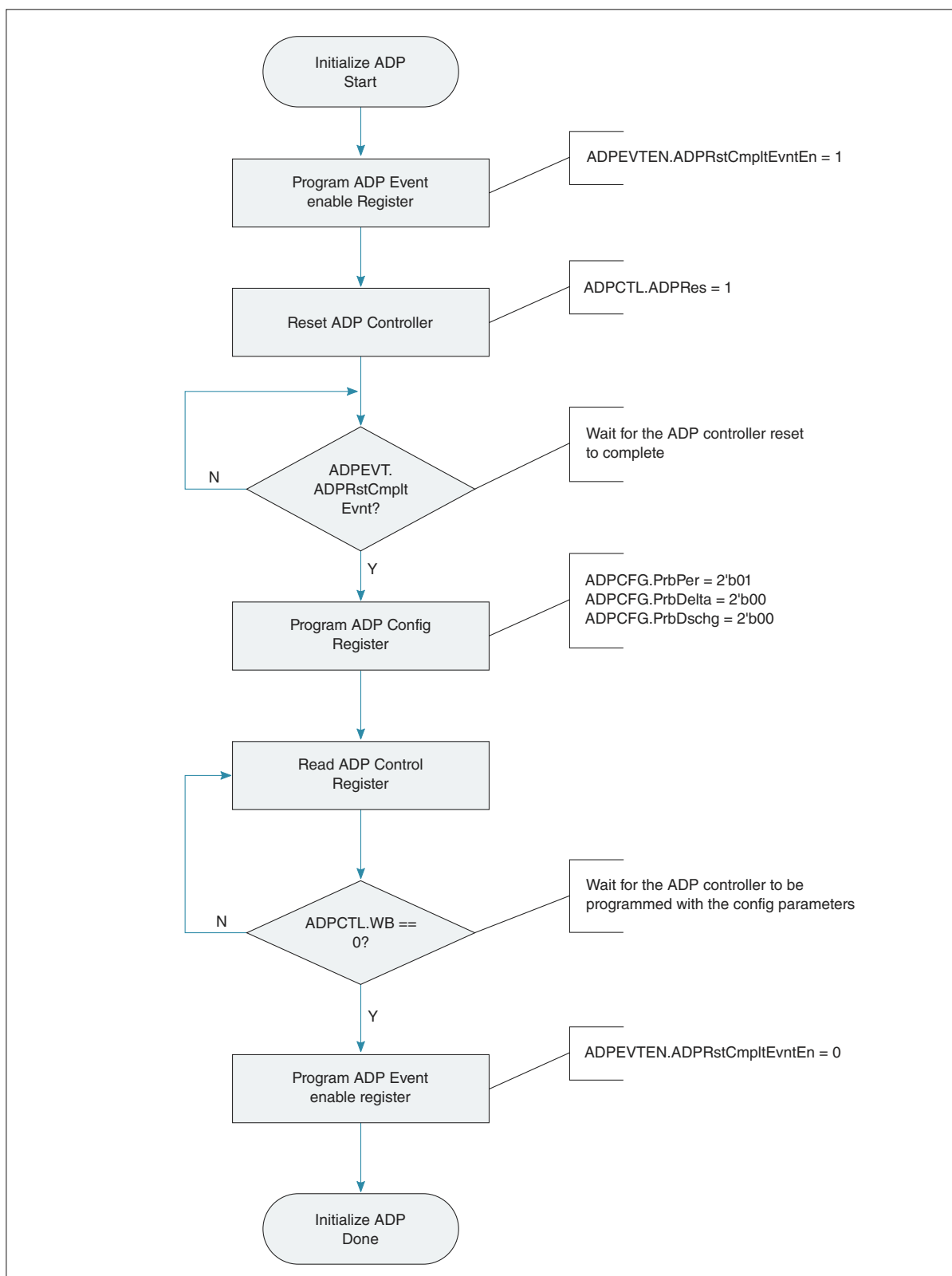


Figure 11-39. Flow Diagram for Initializing ADP

11.1.2.6.1.15 ADP sensing

ADP sensing is allowed only when operating as a B-device and not allowed when operating as an A-device. ADP sensing is done with the following three steps:

1. Start ADP sense
2. Detect ADP sense
3. Stop ADP sense

Start ADP sense

To start ADP sensing, perform the following steps:

1. Set ADPEVTEN[ADPSNSEVNTEN] to 1'b1.
2. Set ADPCTL[ADPEN] and ADPCTL[ENASNS] to 1'b1.
3. Read the ADPCTL register to see if the ADPCTL values have been written into the ADP controller (Suspend clock domain) by looking at the ADPCTL[WB] bit. Wait for the ADPCTL[WB] bit to be cleared by the hardware to confirm that the write operation is complete. Now the core starts an ADP sense to the PHY.

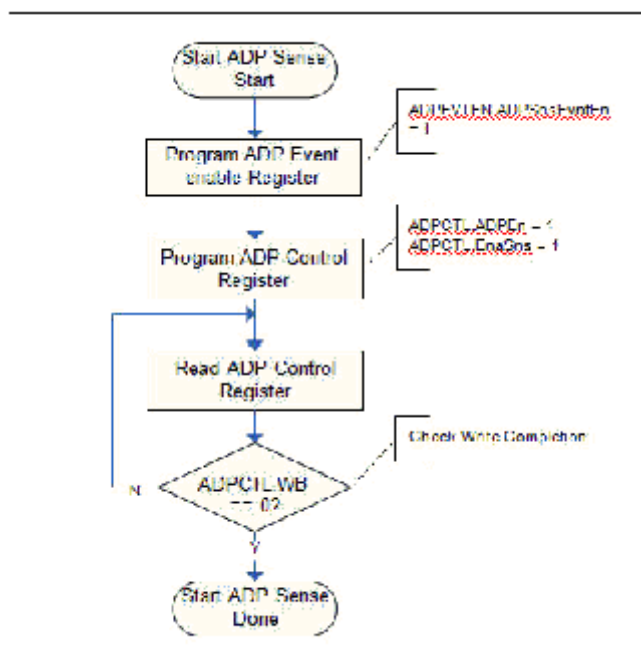


Figure 11-40. Start ADP sense

Detect ADP sense

To detect ADP sensing, perform the following steps:

1. Wait for ADPEVTEN[ADPSNSEVNT] to conclude that A-host is performing ADP probing. After starting the ADP sense, ADPEVTEN[ADPSNSEVNT] should occur at every probe period of A-host.

2. If ADPEVTEN[ADPSNSEVNT] is not detected within TB_ADP_DETACH time (3 seconds), conclude that the ADP sense has failed.

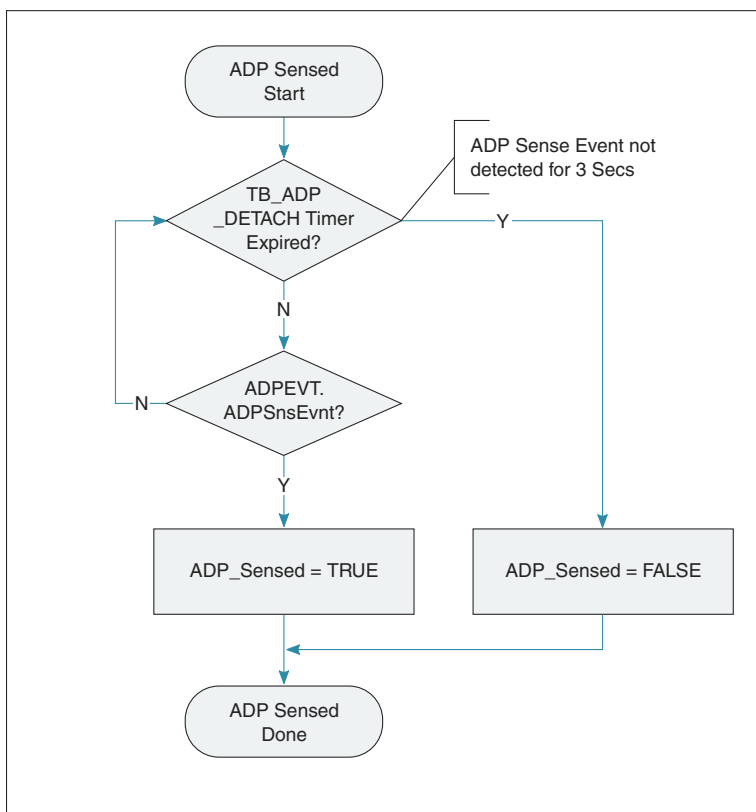


Figure 11-41. Flow diagram to detect ADP sense

Stop ADP sense

To stop ADP sensing, perform the following steps:

1. Clear ADPEVTEN[ADPSNSEVNTEN] to 1'b0.
2. CLEARADPCTL[ADPEN] and ADPCTL[ENASNS] to 1'b0.
3. Read the ADPCTL register to see if the ADPCTL values have been written into the ADP controller (Suspend clock domain) by looking at the ADPCTL[WB] bit. Wait for the hardware to clear the ADPCTL[WB] bit to confirm that the write operation is complete. Now the core will stop ADP sense to PHY.

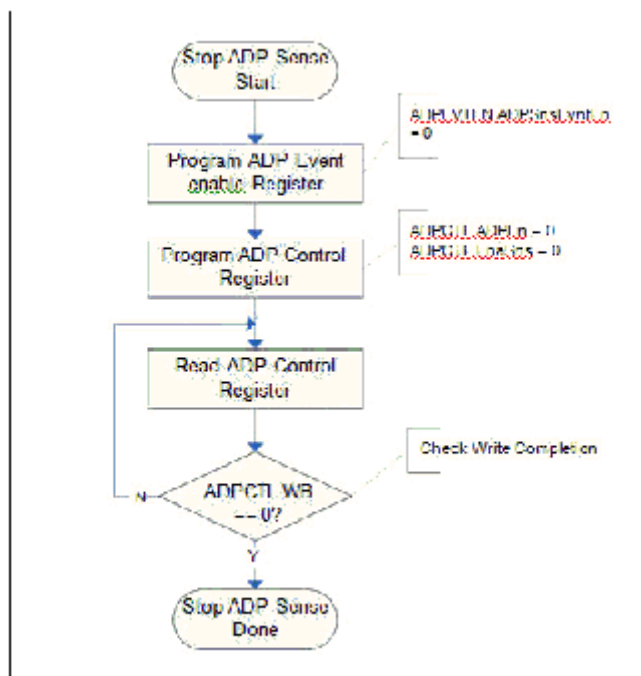


Figure 11-42. Stop ADP sense

11.1.2.6.1.16 ADP probing

ADP probing is allowed in both A-device and B-device. ADP probing is handled with the following three steps:

1. Start ADP probe
2. Detect ADP change
3. Stop ADP probe

Start ADP probe

To start ADP probing, perform the following steps:

1. Set ADPEVTEN[ADPPRBEVTEN] and ADPEVTEN[ADPTMOUTEVNTEN] to 1'b1.
2. Set ADPCTL[ADPEN] and ADPCTL[ENAPRB] to 1'b1.
3. Read the ADPCTL register to see if the ADPCTL values have been written into the ADP controller (suspend clock domain) by looking at the ADPCTL[WB] bit. Wait for the hardware to clear the ADPCTL[WB] bit to confirm that the write operation is complete. Now the core will start ADP probing to PHY.

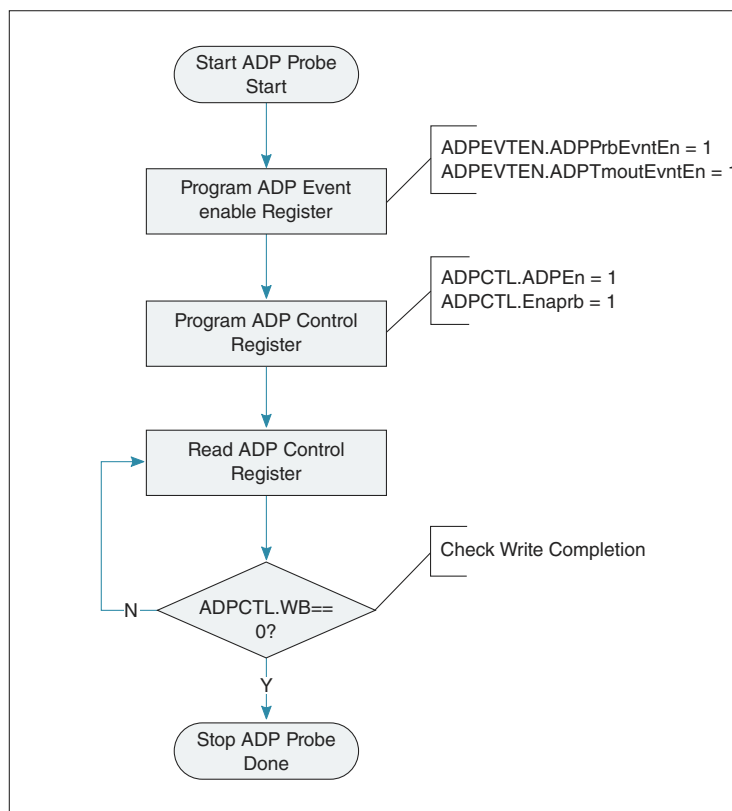


Figure 11-43. Flow diagram for start ADP probe

Detect ADP change

To detect ADP change, perform the following steps:

1. Wait for ADPEVT[ADPPRBEVNT].
2. Read the Ramp Time (RTIM) value and compare it with the previous read value. If the difference is greater than the minimum ramp value of attach/detach (including the noise delta), then report ADP change = TRUE, else report ADP change = FALSE.

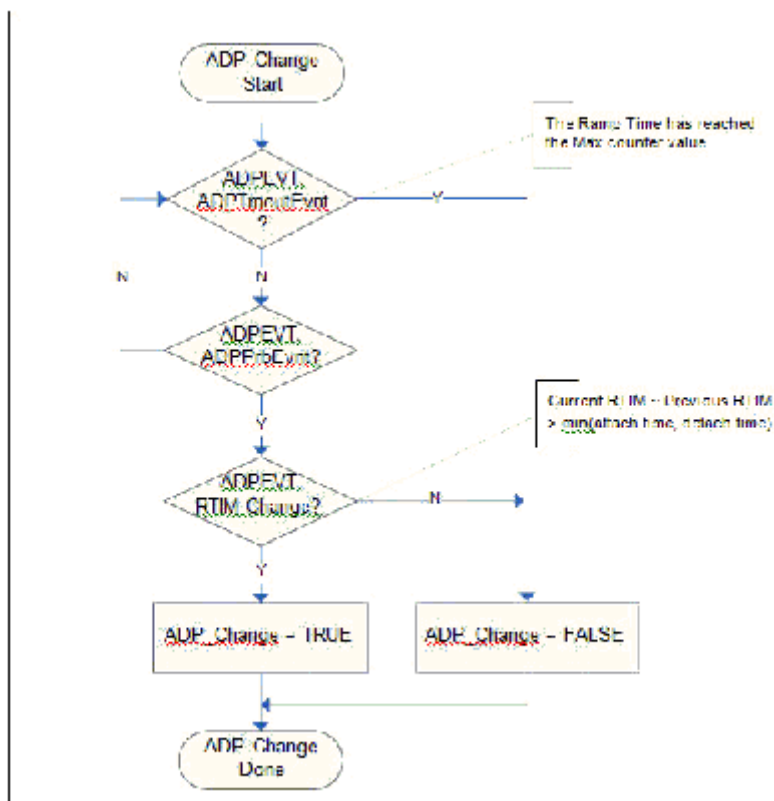


Figure 11-44. Detect ADP change

Table 11-23. Cause of interrupt

Status bit values	Explanation
ADP_PRB_INT=0, ADP_TMOUT_INT=0	Not applicable
ADP_PRB_INT=0, ADP_TMOUT_INT=1	During an ADP probe, the ADPEVT[RTIM] has reached the terminal value (while trying to charge VBUS) and the VADP PRB value has not been reached. The application software must note down the ADPEVT[RTIM] value for this charging cycle.
ADP_PRB_INT=1, ADP_TMOUT_INT=0	VADP PRB voltage has been reached on VBUS. The application software can note down the intermediate ADPEVT[RTIM] value. The timer value indicates the time taken for the ramp to reach VADP PRB voltage.
ADP_PRB_INT=1, ADP_TMOUT_INT=1	The ramp time has reached the terminal value (0x7FF) and VADP PRB voltage is also reached. The application must note down the RTIM value.

Stop ADP probe

To stop ADP probing, perform the following steps:

1. Clear ADPEVTEN[ADPPRB] and ADPEVTEN[ADPTMOUT] to 1'b0.
2. Clear ADPCTL[ADPEN] and ADPCTL[ENAPRB] to 1'b0.

3. Read the ADPCTL register to see if the ADPCTL values have been written into the ADP controller (Suspend clock domain) by looking at the ADPCTL[WB] bit. Wait for the hardware to clear the ADPCTL[WB] bit to confirm that the write operation is complete. Now the core stops ADP probing to PHY.

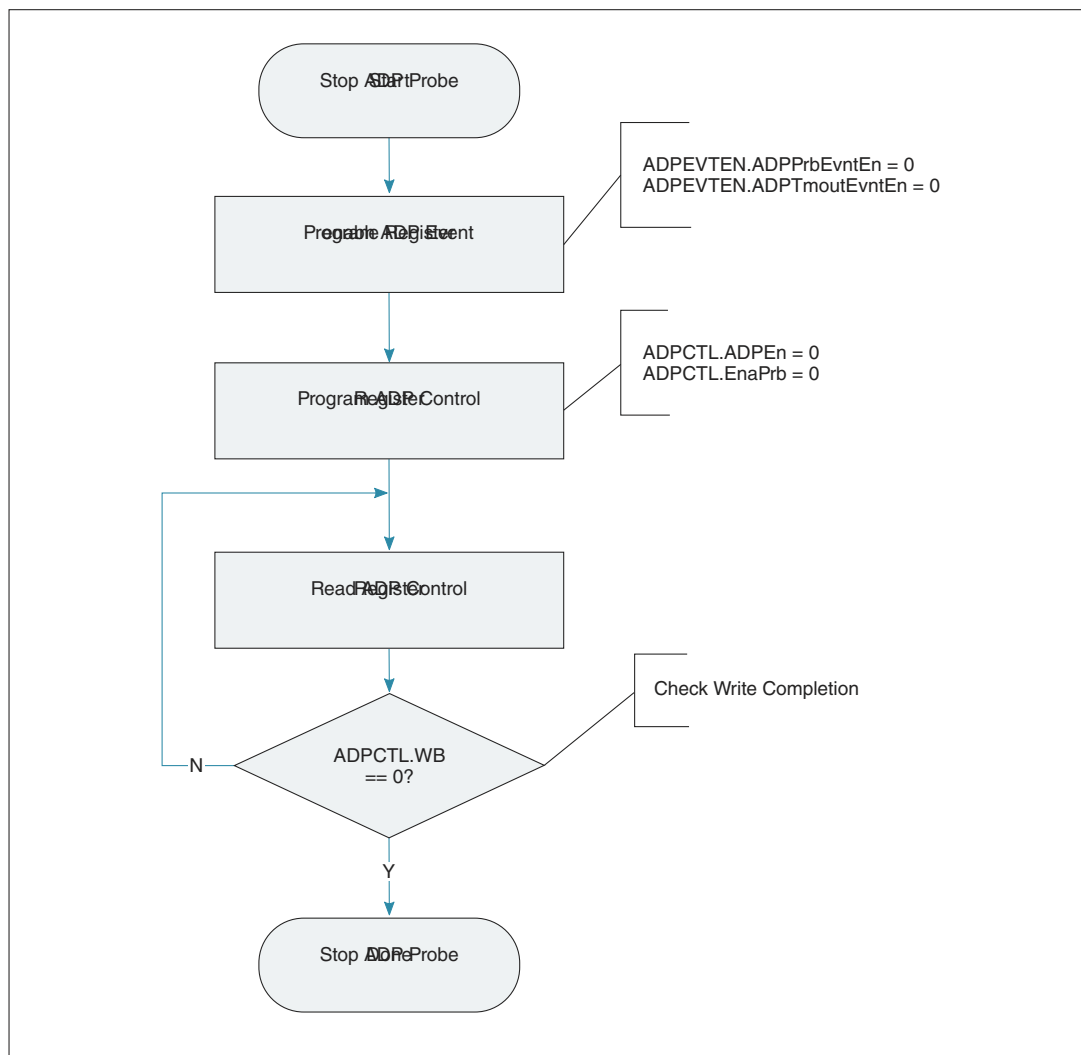


Figure 11-45. Flow diagram to stop ADP probe

11.1.2.7 Initialization/application information

The PHY can be configured for fixed equalization by programming relevant control registers in the USB 3.0 PHY.

In order to initialize the USB 3.0 PHY, software should perform the following steps:

1. Write 1'b0 to RX_OVRD_IN_HI.RX_EQ_EN [address 16'h1006: bit 6].

2. Write 1'b1 to RX_OVRD_IN_HI.RX_EQ_EN_OVRD [address 16'h1006: bit 7].
3. Write a fixed value to RX_OVRD_IN_HI.RX_EQ [address 16'h1006: bits 10–8] [equalization setting] (generally 2–4, based on testing in customer environment).
4. Write 1'b1 to RX_OVRD_IN_HI.RX_EQ_OVRD [address 16'h1006: bit 11].

11.1.2.8 Power management overview

The following power management features are available in the USB 3.0 core:

- Hibernation (save/restore) in device (L1, L2, U3, Disconnected) and host (U3, Disconnected)
- Hardware-controlled LPM in USB 2.0 host
- Clock gating in device (U1/U2/U3) and host (U1/U2/U3)

When hibernation is supported, the core saves some of its internal state to an external location and allows the USB3 core to be powered off. Small modules, namely the PMUs, detect wakeup conditions and request power be reapplied to the core, at which time the external state is brought back into the core.

Since the PMU modules are not inside the USB 3.0 core, the PMUs can easily be placed anywhere in the chip during floor-planning, adding isolation cell/voltage shifter cells. The PMU modules can be placed in the always ON power domain which could be operating at a different voltage than the USB 3.0 power domain.

Hibernation is described in the xHCI Specification as an optional feature that can be supported.

In addition, this feature is also added to the device controller.

NOTE

- The PMU does not support the following features in Hibernation mode:
 - ADP
 - BC
 - SRP signalling
- It is required that the SuperSpeed PHY must support receiver detection in P3 PowerDown mode.
- Hibernation is not supported in OTG mode when the roles are reversed.

- In the host mode, the USB 3.0 core does not support clock gating and hibernation when the Debug Capability is enabled during runtime (DCCTRL.DCE=1).
- During the normal host mode of operation, the USB 3.0 core supports clock gating and hibernation features when the Debug Capability is not enabled by the software (DCCTRL.DCE=0).

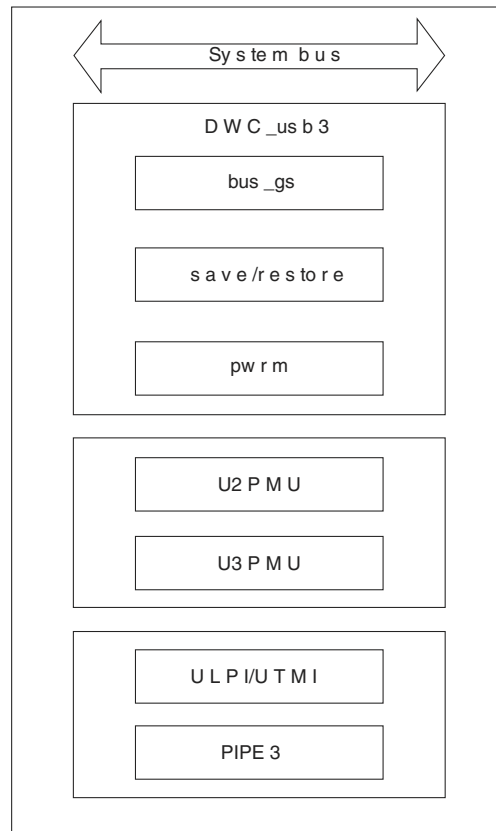


Figure 11-46. Power management hibernation architecture

The second power saving mechanism is hardware-controlled LPM. This is a USB 2.0 host-only feature where the host controller automatically detects the downstream port is idle for a certain amount of time and autonomously initiates an LPM token to the connected device. If the device accepts, the link is put into the standard USB 2.0 Suspend state.

Finally, clock gating is also supported in the host controller as well as in the device controller. The following table gives the list of power saving mechanisms available in USB 3.0 core.

Table 11-24. Power savings support

USB 3.0 State	USB 2.0 State	Device Power Savings	Host Power Savings
U1 (Hardware Initiated)	None	Core: Clock Gating PHY: P1	Core: Clock Gating PHY: P1
U2 (Hardware Initiated)	LPM-L1 (Hardware Initiated)	Core: Clock Gating, LPM-L1 Hibernation PHY: P2/Sleep	Core: Clock Gating, PHY: P2/ Sleep
U3 (Software Initiated)	Suspend (Software Initiated)	Core: Clock Gating, Hibernation PHY: P3/Suspend	Core: Clock Gating, Hibernation PHY: P3/Suspend
Rx.Detect (Software Initiated)	None	None	Core: Clock Gating, Hibernation PHY: P3/Suspend
SS.Disabled (Software Initiated)	None	Core: Clock Gating, Hibernation PHY: P3/Suspend	None

11.1.2.8.1 Clock gating

The core implements clock gating in the following situations:

- In USB 2.0 device mode
 - When the UTMI suspend or ll_suspend signal is asserted and the core is idle
- In USB 3.0 device mode, when the link is in U1, U2 or U3 state and the core is idle.

In host mode, clock gating is enabled in the following situations:

- When all USB 2.0 ports are in the suspend or L1 Suspend state, and all USB 3.0 ports are in the U1/U2/U3 state, and the core is idle.

Internal clock gating will apply to:

- Modules that use the ram_clk
- Some modules that use bus_clk

Internal clock gating will not apply to:

- Modules that use mac3_clk: When the USB 3.0 PHY is suspended, these modules switch to using suspend_clk, so their power consumption is reduced but the clock is not completely stopped.
- Modules that use mac2_clk: When the USB 2.0 PHY is suspended, these modules switch to using suspend_clk, so their power consumption is reduced but the clock is not completely stopped.
- The bus_gs module, which uses bus_clk. This module detects wakeup on the slave interface and thus needs a non-gated clock.

Clock gating can be enabled or disabled using the GCTL register. The GCTL register is "sticky," it retains its value across soft resets and hibernation.

11.1.2.8.2 Hardware-Controlled LPM

In USB 2.0, the main link power management mechanism was for the host to suspend the link. However, suspending the link requires a long period of idle and was wasteful of power, especially because the host knows ahead of time whether it is going to schedule any traffic. It also required a long period of resume signaling.

The USB 2.0 Link Power Management Addendum [LPM-ECN] added a new token to the USB 2.0 specification, the LPM token. The host has the option of suspending the link quickly instead of waiting for a long period of idle, and communicates its desire to the device by sending an LPM token. The ECN also introduced a terminology for different link states. This new link state was called L1. The traditional USB 2.0 suspend was called L2. The L1 state requires a much shorter resume signaling period too.

An xHCI driver can initiate LPM by writing '2' to the PORTSC.PLS field. However, there is a lot of overhead for an xHCI driver to be constantly trying to keep the link in a low power state. Because of this, the xHCI specification also allows a compatible xHC to be enabled with "Hardware-Controlled LPM" by setting PORTPMSC.HLE as 1. When this happens, the xHC is expected to automatically send LPM tokens when it has no pending transfers and thinks that the link is idle enough to enter L1. This feature is supported by the host core.

11.1.2.8.2.1 Special consideration for OTG

When the core is configured as an OTG device, it will not initiate LPM when acting in a reverse role as a B- Host. This is because suspend on the link is treated as a request to switch back to its original role.

11.1.2.8.3 USB PHY power gating

The USB PHY power gating is used to save the leakage power of PHY if it is not required. Follow the steps given below to take PHY into power gating state:

1. Assert PHY_RESET by writing 1'b0 to the USB_PHY_CTRL[RST] bit.
2. Set the USB_PHY_CTRL[PDN_SSP] and USB_PHY_CTRL[PDN_HSP] to 2'b11.

NOTE

Both the above steps must have separate writes.

11.1.3 Memory Map and Register Definition

This section includes the memory map and detailed descriptions of all registers.

NOTE

Synopsys Proprietary. Used with permission.

11.1.3.1 USB3 register descriptions

11.1.3.1.1 DWC_usb3 Memory map

Base Address USB1-38100000 USB2-38200000

USB3 base address: 3810_0000h

Offset	Register	Width (In bits)	Access	Reset value
0h	Capability registers length and Host Controller Operational Registers (CAPLENGTH)	32	RO	Table 11-24
4h	Structural Parameters 1 Register (HCSPARAMS1)	32	RO	Table 11-24
8h	Structural Parameters 2 Register (HCSPARAMS2)	32	RO	Table 11-24
Ch	Structural Parameters 3 Register (HCSPARAMS3)	32	RO	Table 11-24
10h	Capability Parameters 1 Register (HCCPARAMS1)	32	RO	0220_FE6Ch
14h	Doorbell Offset Register (DBOFF)	32	RO	Table 11-24
18h	Runtime Register Space Offset Register (RTSOFF)	32	RO	Table 11-24
1Ch	Host Controller Capability Parameters 2 (HCCPARAMS2)	32	RO	0000_002Fh
20h	USB Command Register (USBCMD)	32	RW	Table 11-24
24h	USB Status Register (USBSTS)	32	RW	Table 11-24
28h	Page Size Register (PAGESIZE)	32	RO	Table 11-24
34h	Device Notification Register (DNCTRL)	32	RW	0000_0000h
38h	CRCR_LO (CRCR_LO)	32	RW	Table 11-24
3Ch	(CRCR_HI)	32	RW	0000_0000h
50h	DCBAAP_LO (DCBAAP_LO)	32	RW	0000_0000h
54h	DCBAAP_HI (DCBAAP_HI)	32	RW	0000_0000h
58h	Configuration Register (CONFIG)	32	RW	Table 11-24
420h	Port Status and Control Register (PORTSC_20)	32	RW	Table 11-24
424h	USB3 Port Power Management Status and Control Register (PORT PMSC_20)	32	RW	Table 11-24
428h	(PORTLI_20)	32	RO	0000_0000h
42Ch	(PORTHLPMP_20)	32	RW	Table 11-24

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
430h	(PORTSC_30)	32	RW	Table 11-24
434h	USB3 Port Power Management Status and Control Register (PORT PMSC_30)	32	RW	Table 11-24
438h	Port Link Info Register (PORTLI_30)	32	RO	Table 11-24
43Ch	USB2 Port Hardware LPM Control Register (PORTHLPSC_30)	32	RO	0000_0000h
440h	Microframe Index Register (MFINDEX)	32	RO	Table 11-24
460h	Interrupter Management Register (IMAN)	32	RW	Table 11-24
464h	Interrupter Moderation Register (IMOD)	32	RW	0000_0FA0h
468h	ERSTSZ (ERSTSZ)	32	RW	Table 11-24
470h	ERSTBA_LO (ERSTBA_LO)	32	RW	0000_0000h
474h	ERSTBA_HI (ERSTBA_HI)	32	RW	0000_0000h
478h	ERDP_LO (ERDP_LO)	32	RW	0000_0000h
47Ch	ERDP_HI (ERDP_HI)	32	RW	0000_0000h
480h	Doorbell Register (DB)	32	RW	Table 11-24
880h	USBLEGSUP (USBLEGSUP)	32	RW	Table 11-24
884h	USBLEGCTLSTS (USBLEGCTLSTS)	32	RW	Table 11-24
890h	SUPTPRT2_DW0 (SUPTPRT2_DW0)	32	RO	0200_0402h
894h	SUPTPRT2_DW1 Register (SUPTPRT2_DW1)	32	RO	2042_5355h
898h	xHCI Supported Protocol Capability_ Data Word 2 (SUPTPRT2_DW2)	32	RO	Table 11-24
89Ch	SUPTPRT2_DW3 Register (SUPTPRT2_DW3)	32	RO	Table 11-24
8A0h	(SUPTPRT3_DW0)	32	RO	0300_0402h
8A4h	SUPTPRT3_DW1 Register (SUPTPRT3_DW1)	32	RO	2042_5355h
8A8h	SUPTPRT3_DW2 (SUPTPRT3_DW2)	32	RO	0000_0102h
8ACh	SUPTPRT3_DW3 (SUPTPRT3_DW3)	32	RO	Table 11-24
C100h	Global SoC Bus Configuration Register 0 (GSBUSCFG0)	32	RW	Table 11-24
C104h	Global SoC Bus Configuration Register 1 (GSBUSCFG1)	32	RW	Table 11-24
C108h	Global Tx Threshold Control Register (GTXTHRCFG)	32	RW	Table 11-24
C10Ch	Global Rx Threshold Control Register (GRXTHRCFG)	32	RW	Table 11-24
C110h	Global Core Control Register (GCTL)	32	RW	30C1_3004h
C118h	Global Status Register (GSTS)	32	RW	Table 11-24
C11Ch	(GUCTL1)	32	RW	Table 11-24
C128h	Global User ID Register (GUID)	32	RW	A000_0000h
C12Ch	Global User Control Register (GUCTL)	32	RW	Table 11-24
C130h	Gobal SoC Bus Error Address Register - Low (GBUSERRADDRLO)	32	RO	0000_0000h
C134h	Gobal SoC Bus Error Address Register - High (GBUSERRADDRHI)	32	RO	0000_0000h
C138h	Global SS Port to Bus Instance Mapping Register - Low (GPRTBIMA PLO)	32	RW	0000_0000h
C13Ch	Global SS Port to Bus Instance Mapping Register - High (GPRTBIMA PHI)	32	RW	Table 11-24
C140h	Global Hardware Parameters Register 0 (GHWPARAMS0)	32	RO	2020_400Ah

Table continues on the next page...

Universal Serial Bus Controller (USB)

Offset	Register	Width (In bits)	Access	Reset value
C144h	Global Hardware Parameters Register 1 (GHWPARAMS1)	32	RO	8160_C93Bh
C148h	Global Hardware Parameters Register 2 (GHWPARAMS2)	32	RO	A000_0000h
C14Ch	Global Hardware Parameters Register 3 (GHWPARAMS3)	32	RO	0821_0085h
C150h	Global Hardware Parameters Register 4 (GHWPARAMS4)	32	RO	4782_2004h
C154h	Global Hardware Parameters Register 5 (GHWPARAMS5)	32	RO	0420_4108h
C158h	Global Hardware Parameters Register 6 (GHWPARAMS6)	32	RO	0A58_DC20h
C15Ch	Global Hardware Parameters Register 7 (GHWPARAMS7)	32	RO	0608_11ECh
C180h	Global High-Speed Port to Bus Instance Mapping Register - Low (GPRTBIMAP_HSLO)	32	RW	0000_0000h
C184h	Global High-Speed Port to Bus Instance Mapping Register - High (GPRTBIMAP_HSHI)	32	RW	Table 11-24
C188h	Global Full-Speed Port to Bus Instance Mapping Register - Low (GPRTBIMAP_FSLO)	32	RW	0000_0000h
C18Ch	Global Full-Speed Port to Bus Instance Mapping Register - High (GPRTBIMAP_FSHI)	32	RW	Table 11-24
C19Ch	Global User Control Register 2 (GUCTL2)	32	RW	Table 11-24
C200h	Global USB2 PHY Configuration Register (GUSB2PHYCFG)	32	RW	Table 11-24
C280h	Global USB 2.0 UTMI PHY vendor control register (GUSB2PHYACC_ULPI)	32	RO	Table 11-24
C2C0h	Global USB 3.0 PIPE control register (GUSB3PIPECTL)	32	RW	010C_0002h
C300h - C31Ch	Global transmit FIFO size register (GTXFIFOSIZ0 - GTXFIFOSIZ7)	32	RW	0000_0042h
C380h - C388h	Global receive FIFO size register (GRXFIFOSIZ0 - GRXFIFOSIZ2)	32	RW	0000_0285h
C400h	Global Event Buffer Address (Low) Register (GEVNTADRLO)	32	RW	0000_0000h
C404h	Global Event Buffer Address (High) Register (GEVNTADRHI)	32	RW	0000_0000h
C408h	Global event buffer size register (GEVNTSIZ)	32	RW	Table 11-24
C40Ch	Global event buffer count register (GEVNTCOUNT)	32	RW	0000_0000h
C600h	Global Hardware Parameters Register 8 (GHWPARAMS8)	32	RO	0000_0A58h
C610h	Global Device TX FIFO DMA Priority Register (GTXFIFOPRIDEV)	32	RW	Table 11-24
C618h	Global Host TX FIFO DMA Priority Register (GTXFIFOPRIHST)	32	RW	Table 11-24
C61Ch	Global Host RX FIFO DMA Priority Register (GRXFIFOPRIHST)	32	RW	Table 11-24
C620h	Global Host Debug Capability DMA Priority Register (GFIFOPRIDBC)	32	RW	Table 11-24
C624h	(GDMAHLRATIO)	32	RW	Table 11-24
C630h	Global Frame Length Adjustment Register (GFLADJ)	32	RW	Table 11-24
C700h	Device Configuration Register (DCFG)	32	RW	Table 11-24
C704h	Device control register (DCTL)	32	RW	Table 11-24
C708h	Device Event Enable Register (DEV TEN)	32	RW	Table 11-24
C70Ch	Device Status Register (DSTS)	32	RW	Table 11-24
C710h	Device Generic Command Parameter Register (DGCMDPAR)	32	RW	0000_0000h
C714h	(DGCMD)	32	RW	Table 11-24
C720h	Device Active USB Endpoint Enable Register (DALEPENA)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
C800h	Device physical endpoint-n command parameter 2 register (DEPCMDPAR2)	32	RW	0000_0000h
C804h	Device Physical Endpoint-n Command Parameter 1 Register (DEPCMDPAR1)	32	RW	0000_0000h
C808h	Device Physical Endpoint-n Command Parameter 0 Register (DEPCMDPAR0)	32	RW	0000_0000h
C80Ch	Device Physical Endpoint-n Command Register (DEPCMD)	32	RW	0000_0000h
CA00h	Device Interrupt Moderation Register (DEV_IMOD)	32	RW	0000_0000h
CC00h	OTG Configuration Register (OCFG)	32	RW	Table 11-24
CC04h	OTG Control Register (OCTL)	32	RW	Table 11-24
CC08h	OTG Events Register (OEVT)	32	RW	Table 11-24
CC0Ch	OTG Events Enable Register (OEVTEN)	32	RW	Table 11-24
CC10h	OTG Status Register (OSTS)	32	RO	Table 11-24
CC20h	ADP Configuration Register (ADPCFG)	32	RW	Table 11-24
CC24h	ADP Control Register (ADPCTL)	32	RW	Table 11-24
CC28h	ADP Event Register (ADPEVT)	32	RW	Table 11-24
CC2Ch	(ADPEVTEN)	32	RW	Table 11-24
CC30h	BCFG (BCFG)	32	RW	Table 11-24
CC38h	BCEVT (BCEVT)	32	RW	0000_0000h
CC3Ch	BCEVTEN (BCEVTEN)	32	RW	Table 11-24

11.1.3.1.2 Capability registers length and Host Controller Operational Registers (CAPLENGTH)

11.1.3.1.2.1 Offset

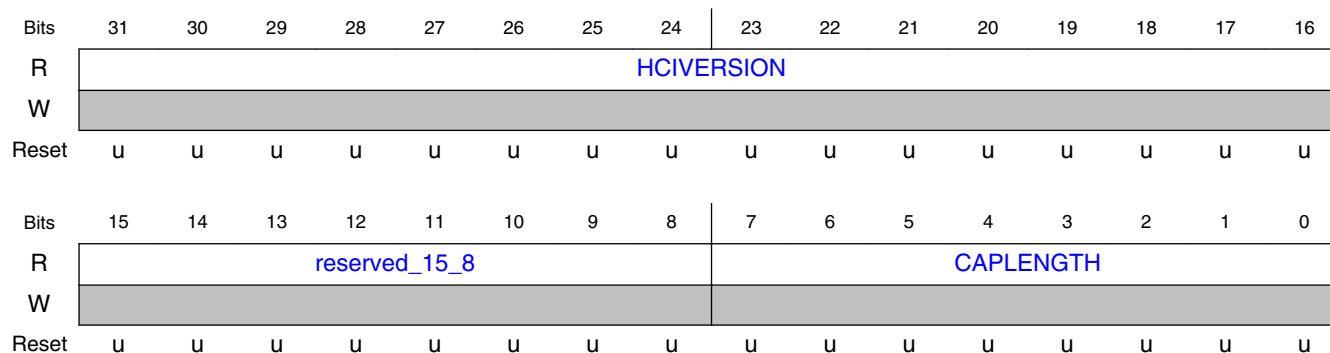
Register	Offset
CAPLENGTH	0h

11.1.3.1.2.2 Function

Capability Registers Length = Base address + CAPLENGTH where CAPLENGTH is `DWC_USB3_HOST_CAP_REG_LEN whose default value is 20h.

Reset Mask:0xFF00

11.1.3.1.2.3 Diagram



11.1.3.1.2.4 Fields

Field	Function
31-16 HCIVERSION	HC Interface Version Number (HCIVERSION)
15-8 reserved_15_8	Reserved
7-0 CAPLENGTH	Capability Registers Length (CAPLENGTH)

11.1.3.1.3 Structural Parameters 1 Register (HCSPARAMS1)

11.1.3.1.3.1 Offset

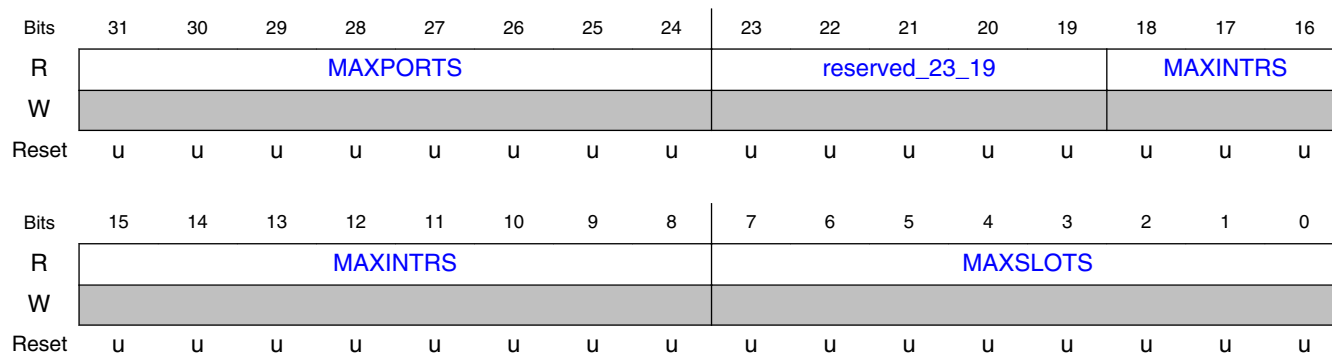
Register	Offset
HCSPARAMS1	4h

11.1.3.1.3.2 Function

For register definitions, refer to the xHCI specification.

Reset Mask:0xF80000

11.1.3.1.3.3 Diagram



11.1.3.1.3.4 Fields

Field	Function
31-24 MAXPORTS	Number of Ports (MaxPorts) - Number of ports implemented is defined by the parameter (<code>`DWC_USB3_HOST_NUM_U2_ROOT_PORTS + `DWC_USB3_HOST_NUM_U3_ROOT_PORTS</code>) - Number of ports enabled is controlled by the core input signals <code>host_num_u2_port[3:0]+host_num_u3_port[3:0]</code> Note: In USB 2.0-only mode, the <code>host_num_u3_port</code> signal is zero.
23-19 reserved_23_19	Reserved
18-8 MAXINTRS	Number of Interrupters (MaxIntrs) Defined by the configurable parameter <code>`DWC_USB3_HOST_NUM_INTERRUPTER_SUPT</code>
7-0 MAXSLOTS	Number of device slots (MaxSlots) Defined by configurable parameter <code>`DWC_USB3_NUM_DEVICE_SUPT</code>

11.1.3.1.4 Structural Parameters 2 Register (HCSPARAMS2)

11.1.3.1.4.1 Offset

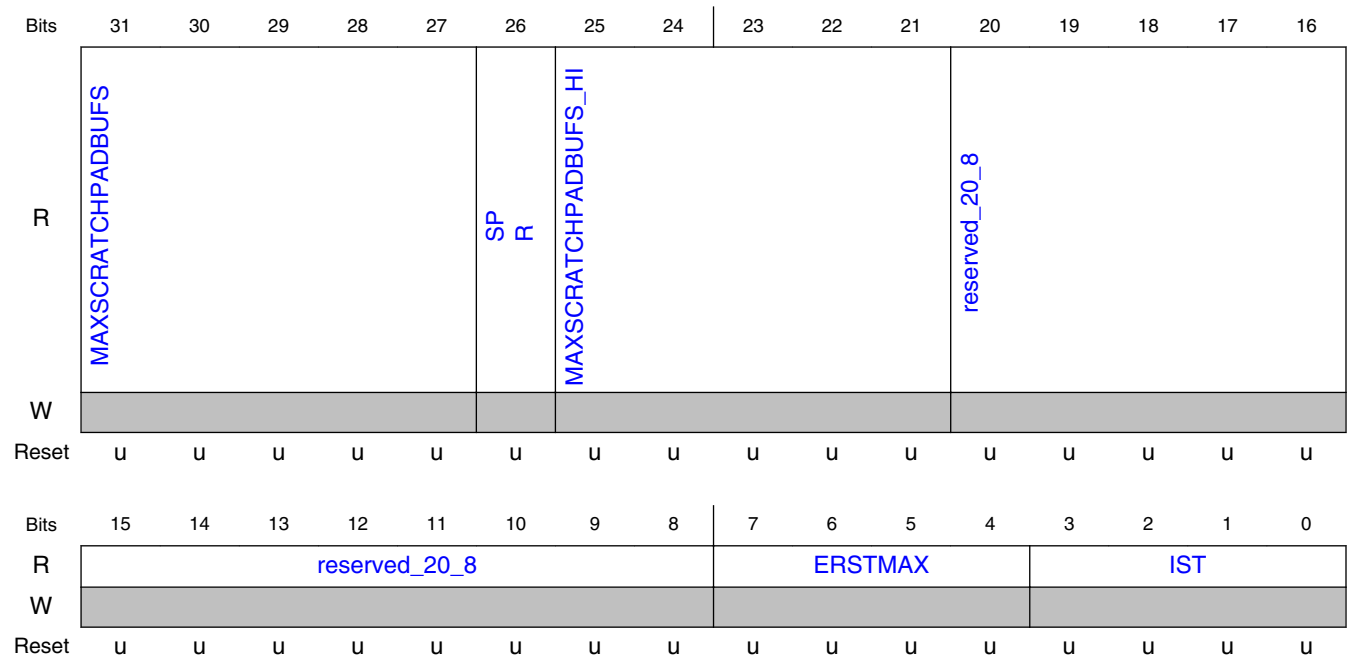
Register	Offset
HCSPARAMS2	8h

11.1.3.1.4.2 Function

For register definitions, refer to the xHCI specification.

Reset Mask:0x1FFF00

11.1.3.1.4.3 Diagram



11.1.3.1.4.4 Fields

Field	Function
31-27 MAXSCRATCH PADBUFS	Max Scratchpad Bufs Lo The value is calculated based on chosen configuration parameter values. Possible values are 1-4.
26 SPR	Scratchpad Restore (SPR)
25-21 MAXSCRATCH PADBUFS_HI	Max Scratchpad Bufs HI The core automatically updates this field.
20-8 reserved_20_8	Reserved
7-4 ERSTMAX	Event Ring Segment Table Max (ERST Max)
3-0 IST	Isochronous Scheduling Threshold (IST)

11.1.3.1.5 Structural Parameters 3 Register (HCSPARAMS3)

11.1.3.1.5.1 Offset

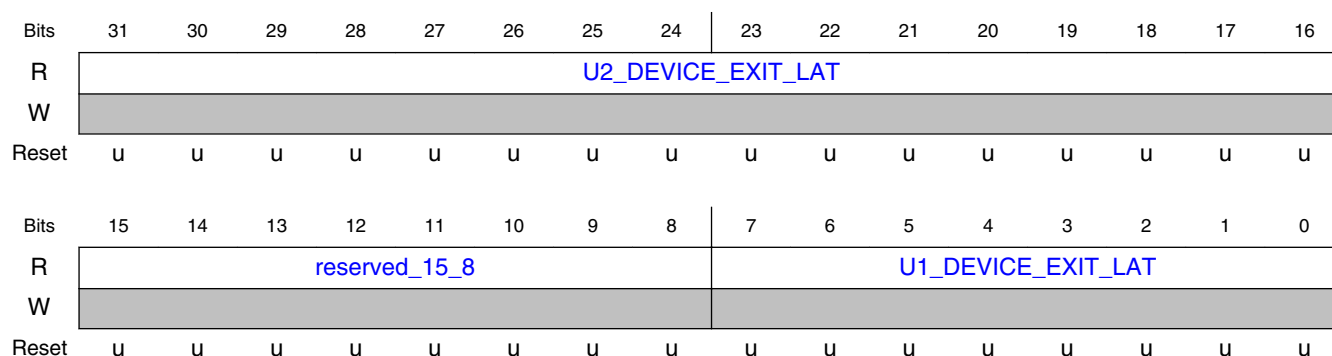
Register	Offset
HCSPARAMS3	Ch

11.1.3.1.5.2 Function

For register definitions, refer to the xHCI specification.

Reset Mask:0xFF00

11.1.3.1.5.3 Diagram



11.1.3.1.5.4 Fields

Field	Function
31-16 U2_DEVICE_EXIT_LAT	U2 Device Exit Latency
15-8 reserved_15_8	Reserved
7-0 U1_DEVICE_EXIT_LAT	U1 Device Exit Latency

11.1.3.1.6 Capability Parameters 1 Register (HCCPARAMS1)

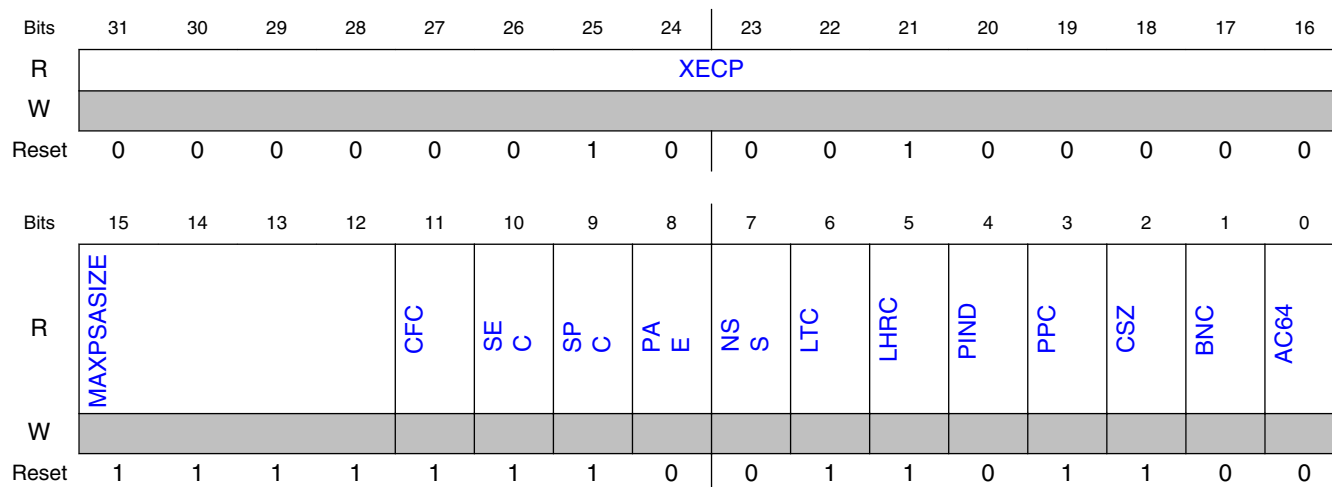
11.1.3.1.6.1 Offset

Register	Offset
HCCPARAMS1	10h

11.1.3.1.6.2 Function

For register definitions, refer to the xHCI specification.

11.1.3.1.6.3 Diagram



11.1.3.1.6.4 Fields

Field	Function
31-16 XECP	xHCI Extended Capabilities Pointer (xECP) Based on configuration, core automatically updates it. Refer to <workspace>/src/DWC_usb3_params.v for details on DWC_USB3_HC_XECP.
15-12 MAXPSASIZE	Maximum Primary Stream Array Size (MaxPSASize) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
11 CFC	Contiguous Frame ID Capability (CFC)
10 SEC	Stopped EDLTA Capability (SEC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
9 SPC	Short Packet Capability (SPC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
8 PAE	Parse All Event Data (PAE) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

Table continues on the next page...

Field	Function
7 NSS	No Secondary SID Support (NSS) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
6 LTC	Latency Tolerance Messaging Capability (LTC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
5 LHRC	Light HC Reset Capability For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
4 PIND	Port Indicators (PIND) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
3 PPC	Port Power Control For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
2 CSZ	Context Size (CSZ) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
1 BNC	BW Negotiation Capability (BNC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
0 AC64	64-bit Addressing Capability (AC64) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.7 Doorbell Offset Register (DBOFF)

11.1.3.1.7.1 Offset

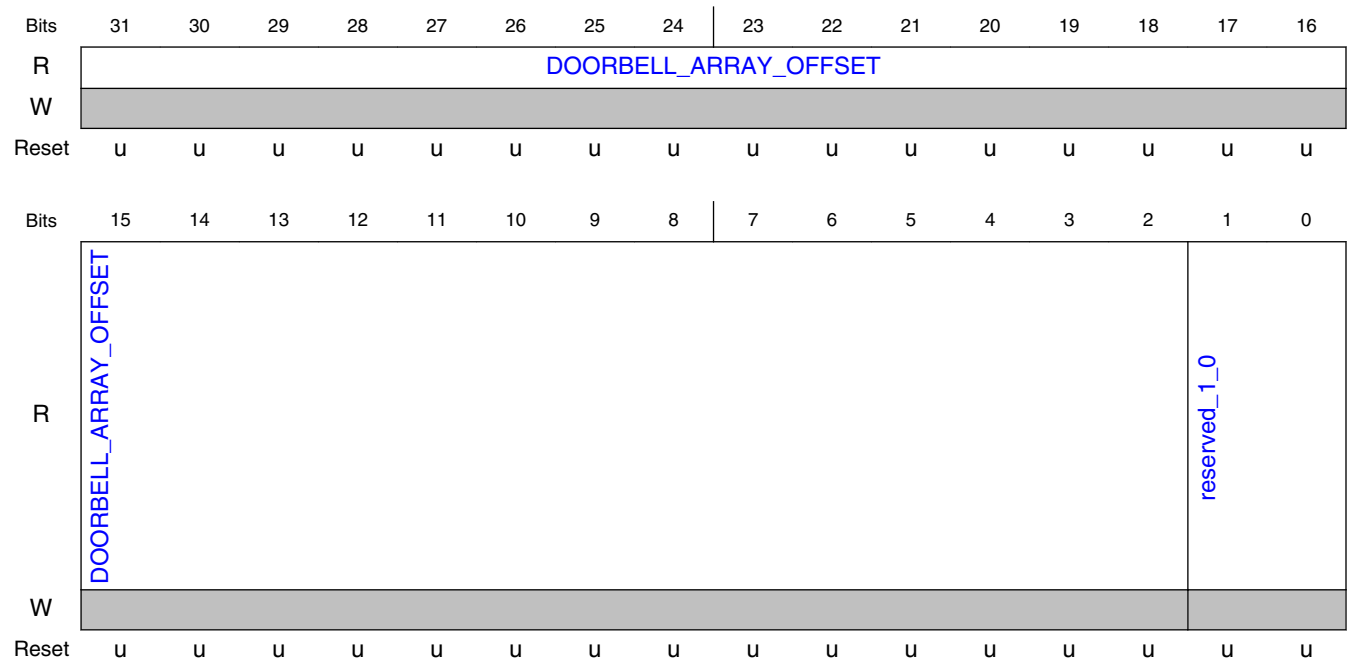
Register	Offset
DBOFF	14h

11.1.3.1.7.2 Function

For register definitions, refer to the xHCI specification.

Reset Mask:0x3

11.1.3.1.7.3 Diagram



11.1.3.1.7.4 Fields

Field	Function
31-2 DOORBELL_ARRAY_OFFSET	Doorbell Array Offset - RO Based on configuration, core automatically updates it. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
1-0 reserved_1_0	Reserved

11.1.3.1.8 Runtime Register Space Offset Register (RTSOFF)

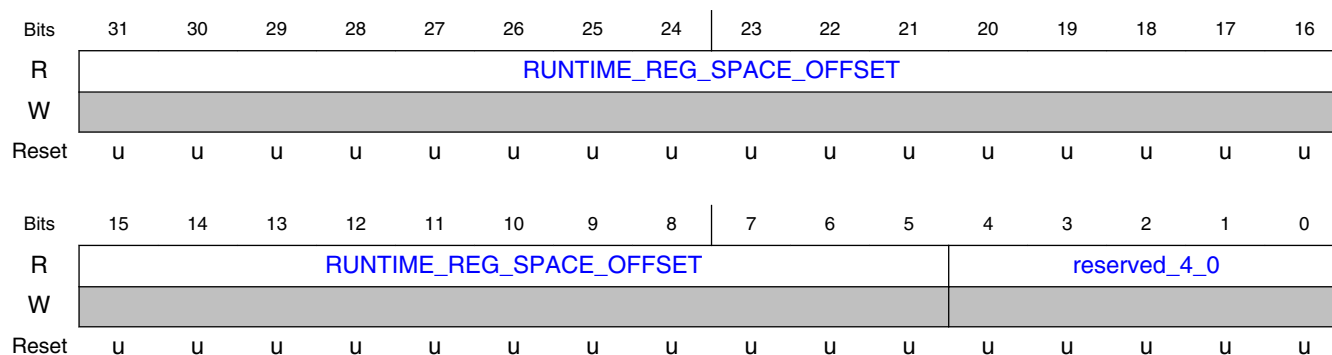
11.1.3.1.8.1 Offset

Register	Offset
RTSOFF	18h

11.1.3.1.8.2 Function

Reset Mask:0x1F

11.1.3.1.8.3 Diagram



11.1.3.1.8.4 Fields

Field	Function
31-5 RUNTIME_REG_SPACE_OFFSET	Runtime Register Space Offset Based on configuration, core automatically updates it. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
4-0 reserved_4_0	Reserved

11.1.3.1.9 Host Controller Capability Parameters 2 (HCCPARAMS2)

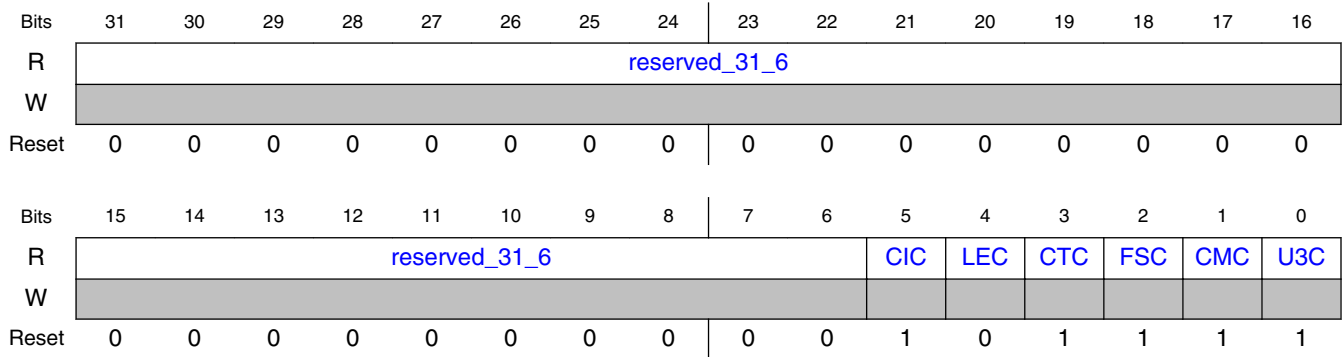
11.1.3.1.9.1 Offset

Register	Offset
HCCPARAMS2	1Ch

11.1.3.1.9.2 Function

For register definitions, refer to the xHCI specification.

11.1.3.1.9.3 Diagram



11.1.3.1.9.4 Fields

Field	Function
31-6 reserved_31_6	Reserved
5 CIC	Configuration Information Capability (CIC) For a description of this standard USB register field, see the eXtensible Host Controller I nterface for Universal Serial Bus (USB) Specification 3.0.
4 LEC	Large ESIT Payload Capability (LEC) For a description of this standard USB register field, see the eXtensible Host Controller I nterface for Universal Serial Bus (USB) Specification 3.0.
3 CTC	Compliance Transition Capability (CTC) For a description of this standard USB register field, see the eXtensible Host Controller I nterface for Universal Serial Bus (USB) Specification 3.0.
2 FSC	Force Save Context Capability (FSC) For a description of this standard USB register field, see the eXtensible Host Controller I nterface for Universal Serial Bus (USB) Specification 3.0.
1 CMC	Configure Endpoint Command Max Exit Latency Too Large Capability (CMC) For a description of this standard USB register field, see the eXtensible Host Controller I nterface for Universal Serial Bus (USB) Specification 3.0.
0 U3C	U3 Entry Capability (U3C) For a description of this standard USB register field, see the eXtensible Host Controller I nterface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.10 USB Command Register (USBCMD)

11.1.3.1.10.1 Offset

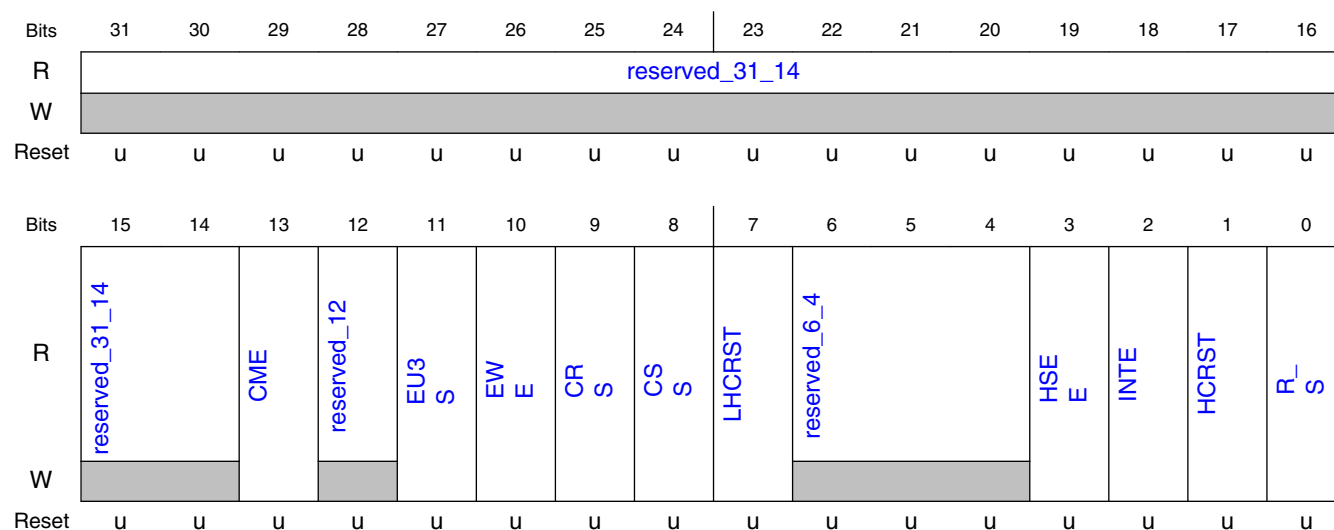
Register	Offset
USBCMD	20h

11.1.3.1.10.2 Function

For a description of this standard USB register field see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

Reset Mask:0xFFFFC070

11.1.3.1.10.3 Diagram



11.1.3.1.10.4 Fields

Field	Function
31-14 reserved_31_14	Reserved
13 CME	CEM Enable For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
12 reserved_12	Reserved
11 EU3S	EU3S For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
10 EWE	EWE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
9 CRS	Controller Restore State This command is similar to the USBCMD.CRS bit in host mode and initiates the restore process. When software sets this bit to '1', the controller immediately sets DSTS.RSS to '1'. When the controller has finished the restore process, it sets DSTS.RSS to '0'. Note: When read, this field always returns '0'.
8 CSS	Controller Save State This command is similar to the USBCMD.CSS bit in host mode and initiates the save process. When software sets this bit to '1', the controller immediately sets DSTS.SSS to '1'. When

Table continues on the next page...

Field	Function
	the controller has finished the save process, it sets DSTS.SSS to '0'. Note: When read, this field always returns '0'.
7 LHCRST	Light Host Controller Reset For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0. The following bits reset the internal logic of the host controller. Under soft reset, some CSR accesses may fail (Timeout). - HCRST - LHCRST Bit Bash register testing is not recommended.
6-4 reserved_6_4	Reserved
3 HSEE	HSEE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
2 INTE	INTE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
1 HCRST	HCRST The following bits reset the internal logic of the host controller. Under soft reset, some CSR accesses may fail (Timeout). - HCRST - LHCRST Bit Bash register testing is not recommended.
0 R_S	R_S For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0. Due to side-effects this register field is not recommended for Bit-Bash testing.

11.1.3.1.11 USB Status Register (USBSTS)

11.1.3.1.11.1 Offset

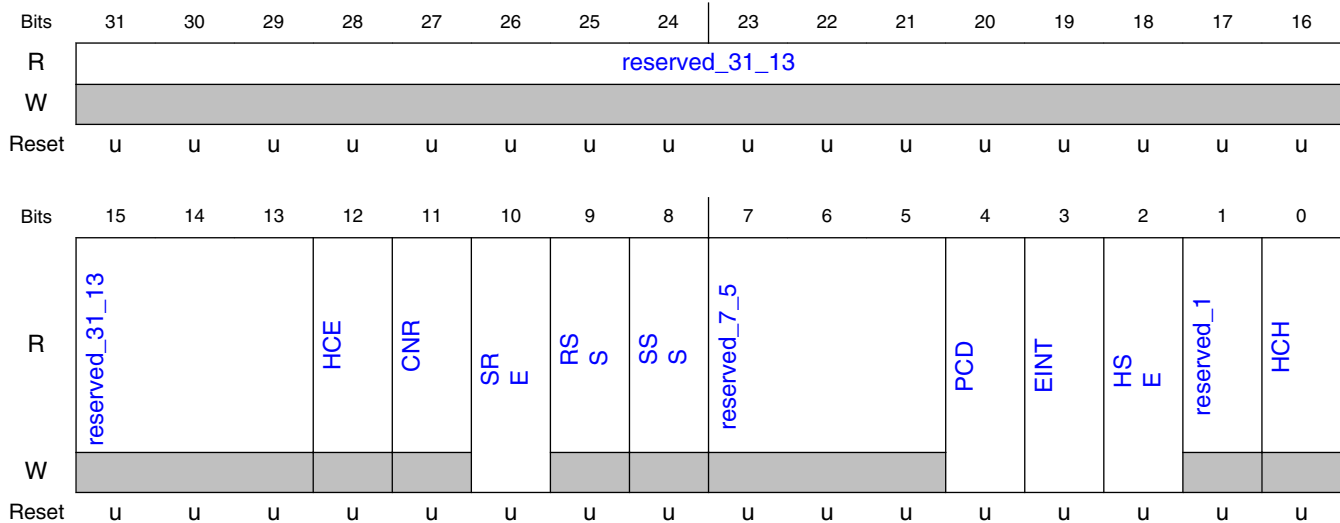
Register	Offset
USBSTS	24h

11.1.3.1.11.2 Function

Bit Definitions For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

Reset Mask:0xFFFFE0E2

11.1.3.1.11.3 Diagram



11.1.3.1.11.4 Fields

Field	Function
31-13 reserved_31_13	Reserved
12 HCE	Host Controller Error (HCE) - RO Default = 0. '0' = No internal xHC error conditions exist and '1' = Internal xHC error condition. This flag must be set to indicate that an internal error condition has been detected which requires software to reset and reinitialize the xHC. Refer to section 4.24.1 of the DesignWare Cores SuperSpeed USB 3.0 Controller Databook for more information.
11 CNR	Controller Not Ready (CNR) - RO Default = '1'. '0' = Ready and '1' = Not Ready. Software must not write to the Doorbell or Operational register of the xHC, other than the USBSTS register, until CNR = '0'. This flag is set by the xHC after a Chip Hardware Reset and cleared when the xHC is ready to begin accepting register writes. This flag remains cleared ('0') until the next Chip Hardware Reset.
10 SRE	Save/Restore Error This bit is currently not supported.
9 RSS	Restore State Status This bit is similar to the USBSTS.RSS in host mode. When the controller has finished the restore process, it completes the command by setting DSTS.RSS to '0'.
8 SSS	Save State Status This bit is similar to the USBSTS.SSS in host mode. When the controller has finished the save process, it completes the command by setting DSTS.SSS to '0'.
7-5 reserved_7_5	Reserved
4 PCD	Reset Value for PCD
3 EINT	Reset Value for EINT
2	HSE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0..

Table continues on the next page...

Universal Serial Bus Controller (USB)

Field	Function
HSE	
1 reserved_1	Reserved
0 HCH	HCH For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.12 Page Size Register (PAGESIZE)

11.1.3.1.12.1 Offset

Register	Offset
PAGESIZE	28h

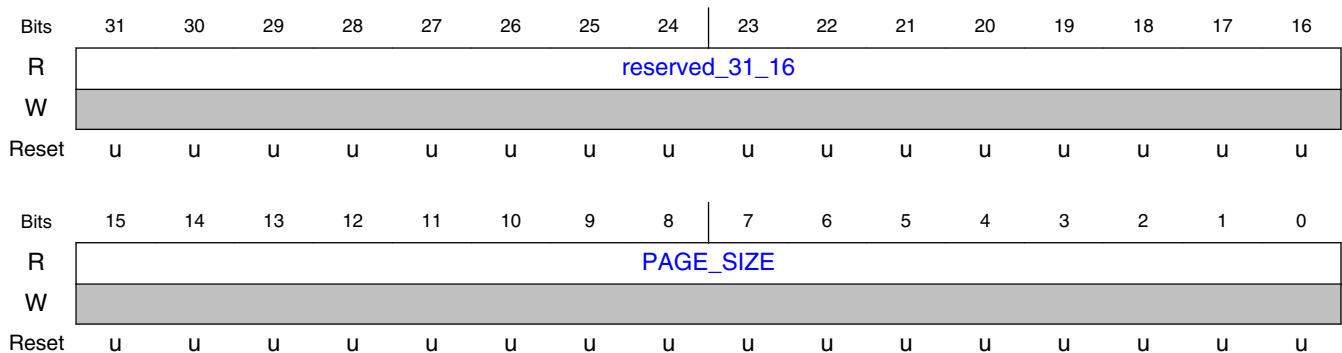
11.1.3.1.12.2 Function

Bit Definitions Use this register to enable or disable the reporting of specific USB Device Notification Transaction Packets being received.

Reset Mask:0xFFFF0000

A Notification Enable (Nx, where x = 0 to 15) flag is defined for each of the 16 possible device notification types. If a flag is set for a specific notification type, a Device Notification Event is generated when the respective notification packet is received. After reset, all notifications are disabled. Refer to section 6.4.2.7 of the DesignWare Cores SuperSpeed USB 3.0 Controller Databook for more information. This register is written as a Dword. Byte writes produce undefined results.

11.1.3.1.12.3 Diagram



11.1.3.1.12.4 Fields

Field	Function
31-16 reserved_31_16	Reserved
15-0 PAGE_SIZE	PAGE_SIZE

11.1.3.1.13 Device Notification Register (DNCTRL)

11.1.3.1.13.1 Offset

Register	Offset
DNCTRL	34h

11.1.3.1.13.2 Function

Bit Definitions For a description of this standard USB register field see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

11.1.3.1.13.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_31_16															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	N0_N15															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.1.3.1.13.4 Fields

Field	Function
31-16 reserved_31_16	Reserved

Table continues on the next page...

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Field	Function
15-0 N0_N15	N0_N15 For a description of this standard USB register field see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.14 CRCR_LO (CRCR_LO)

11.1.3.1.14.1 Offset

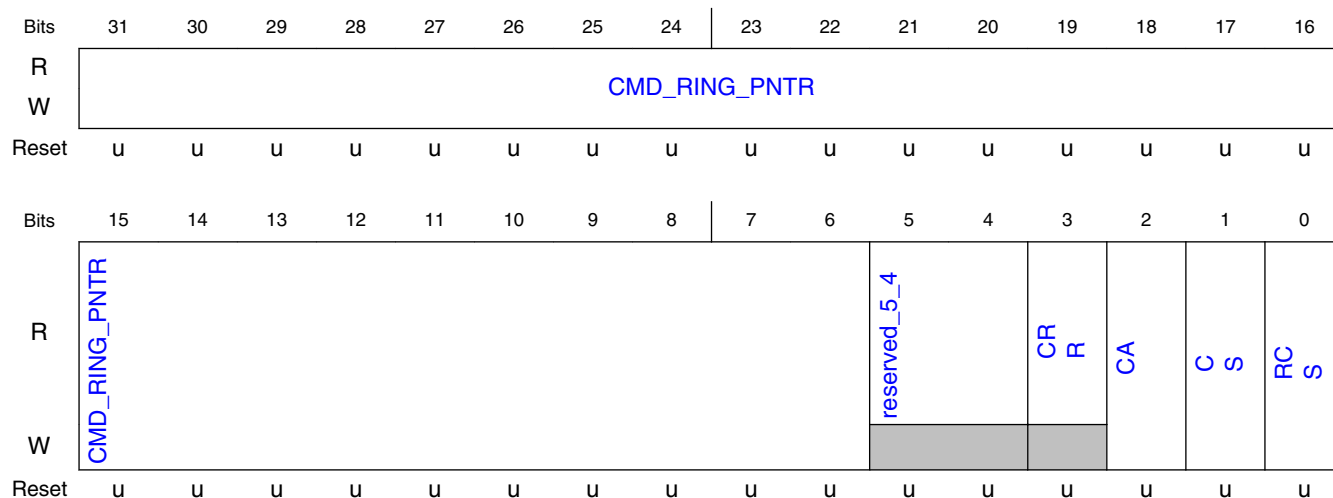
Register	Offset
CRCR_LO	38h

11.1.3.1.14.2 Function

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

Reset Mask:0x30

11.1.3.1.14.3 Diagram



11.1.3.1.14.4 Fields

Field	Function
31-6	CMD_RING_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

Table continues on the next page...

Field	Function
CMD_RING_PNTR	
5-4 reserved_5_4	Reserved
3 CRR	CRR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
2 CA	CA For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
1 CS	CS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
0 RCS	RCS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.15 (CRCR_HI)

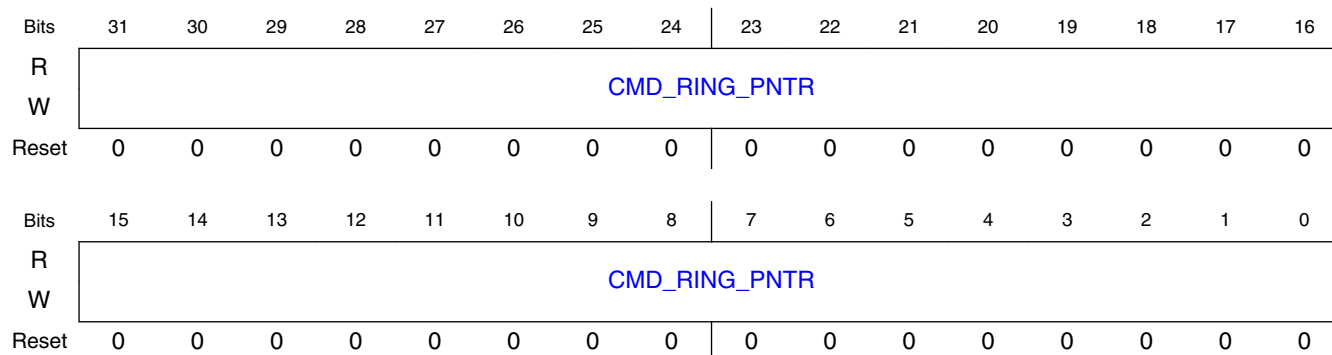
11.1.3.1.15.1 Offset

Register	Offset
CRCR_HI	3Ch

11.1.3.1.15.2 Function

CRCR_HI

11.1.3.1.15.3 Diagram



11.1.3.1.15.4 Fields

Field	Function
31-0 CMD_RING_PN TR	COMMAND_RING_POINTER Reading this field always returns '0'. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.16 DCBAAP_LO (DCBAAP_LO)

11.1.3.1.16.1 Offset

Register	Offset
DCBAAP_LO	50h

11.1.3.1.16.2 Function

DCBAAP_LO

11.1.3.1.16.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DEVICE_CONTEXT_BAAP															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEVICE_CONTEXT_BAAP								reserved_5_0							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.1.3.1.16.4 Fields

Field	Function
31-6 DEVICE_CONTEXT_BAAP	DEVICE_CONTEXT_BAAP For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
5-0 reserved_5_0	Reserved

11.1.3.1.17 DCBAAP_HI (DCBAAP_HI)

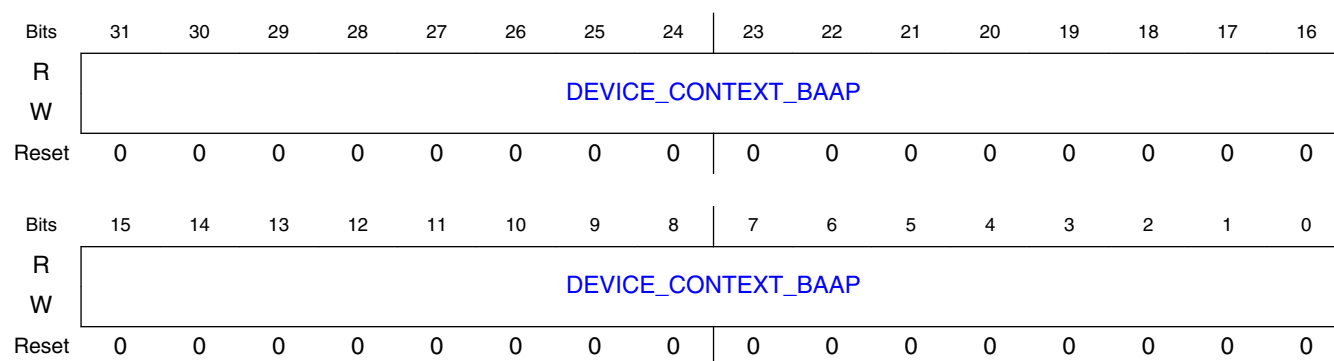
11.1.3.1.17.1 Offset

Register	Offset
DCBAAP_HI	54h

11.1.3.1.17.2 Function

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

11.1.3.1.17.3 Diagram



11.1.3.1.17.4 Fields

Field	Function
31-0 DEVICE_CONTEXT_BAAP	DEVICE_CONTEXT_BAAP

11.1.3.1.18 Configuration Register (CONFIG)

11.1.3.1.18.1 Offset

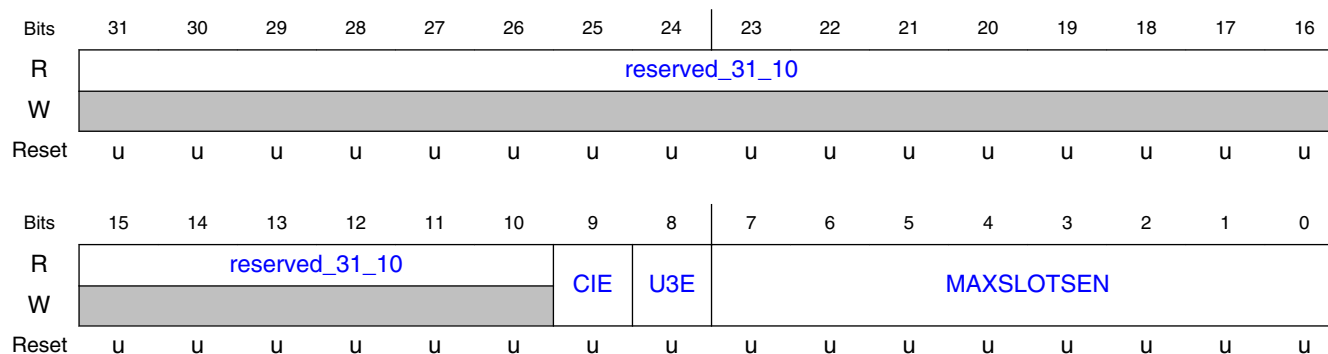
Register	Offset
CONFIG	58h

11.1.3.1.18.2 Function

This register is in the Aux Power well.

Reset Mask:0xFFFFFC00

11.1.3.1.18.3 Diagram



11.1.3.1.18.4 Fields

Field	Function
31-10 reserved_31_10	Reserved
9 CIE	Configuration Information Enable For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
8 U3E	U3 Entry Enable For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
7-0 MAXSLOTSEN	MAXSLOTSEN For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.19 Port Status and Control Register (PORTSC_20)

11.1.3.1.19.1 Offset

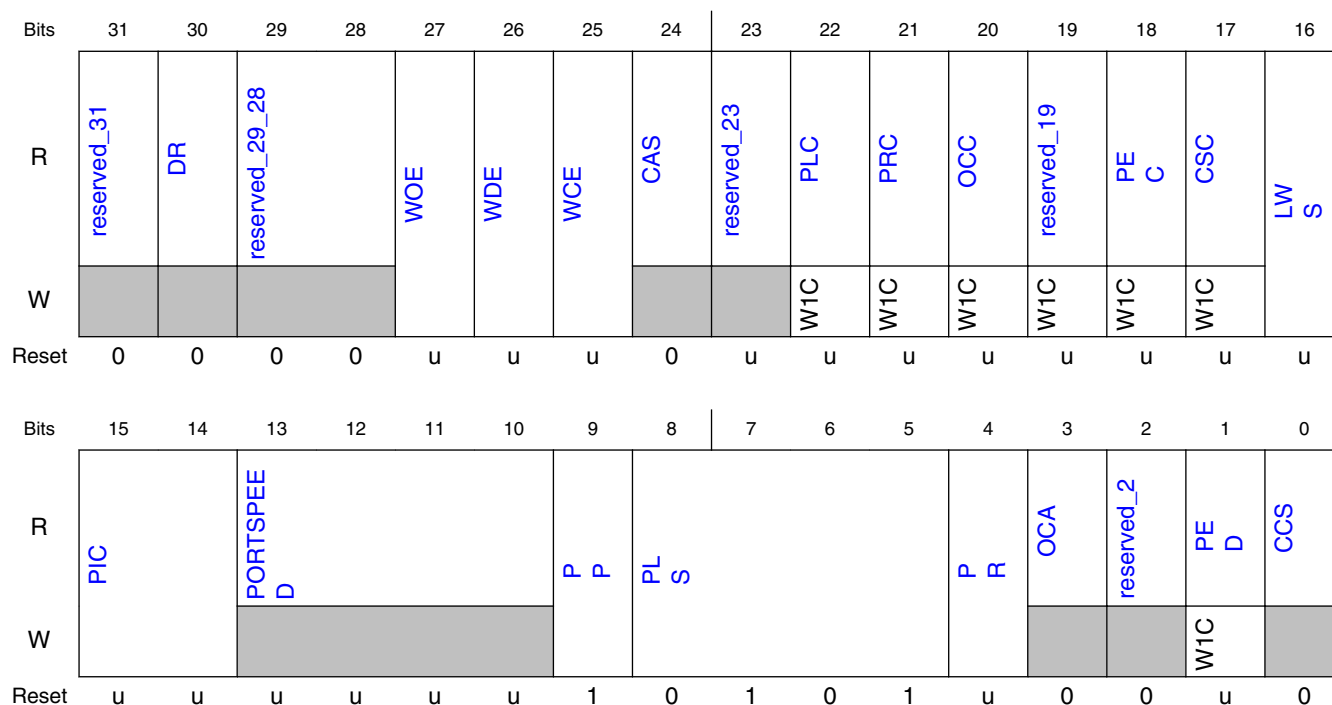
Register	Offset
PORTSC_20	420h

11.1.3.1.19.2 Function

Access fails (Timeout) if the UTMI/ULPI clock is not running or one of the following bits is asserted.

Reset Mask:0xF10003ED

11.1.3.1.19.3 Diagram



11.1.3.1.19.4 Fields

Field	Function
31 reserved_31	
30 DR	Reset value For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
29-28 reserved_29_28	Reserved
27 WOE	WOE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
26 WDE	

Table continues on the next page...

Universal Serial Bus Controller (USB)

Field	Function
WDE	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
25 WCE	WCE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
24 CAS	CAS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
23 reserved_23	
22 PLC	PLC PLC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
21 PRC	PRC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
20 OCC	OCC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
19 reserved_19	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
18 PEC	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
17 CSC	CSC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
16 LWS	LWS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
15-14 PIC	PIC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
13-10 PORTSPEED	PORTSPEED For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
9 PP	PP For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
8-5 PLS	PLS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
4 PR	PR

Table continues on the next page...

Field	Function
PR	DWC_usb3_map/DWC_usb3_block_Host_Cntrl_Port_Reg_Set/PORTSC_20_REGS/PORTSC_20/PR VolatileMemory 1 Programming this field with random data will cause side effect.
3 OCA	OCA For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
2 reserved_2	Reserved
1 PED	PED For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
0 CCS	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3

11.1.3.1.20 USB3 Port Power Management Status and Control Register (PORTPMSC_20)

11.1.3.1.20.1 Offset

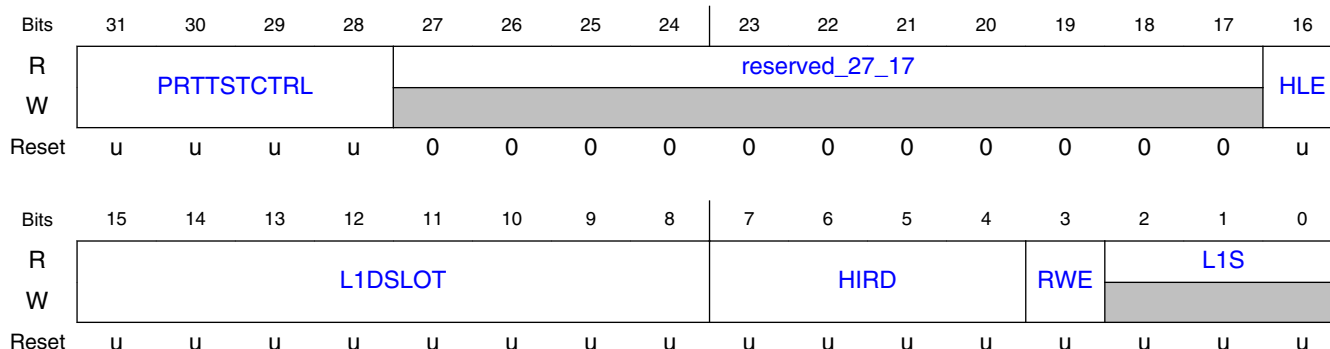
Register	Offset
PORTPMSC_20	424h

11.1.3.1.20.2 Function

This register is in the Aux Power well.

Reset Mask:0xFFE0000

11.1.3.1.20.3 Diagram



11.1.3.1.20.4 Fields

Field	Function
31-28 PRTTSTCTRL	Port Test Control . When this field is '0', the port is NOT operating in a test mode. A non-zero value indicates that it is operating in test mode and the specific test mode is indicated by the specific value. A non-zero Port Test Control value is only valid to a port that is in the Powered-Off state (PLS = Disabled). If the port is not in this state, the xHC shall respond with the Port Test Control field set to Port Test Control Error. Refer to 'Operational Model' topic for using these test modes. The encoding of the Test Mode bits for a USB2 protocol port are: 0: Test mode not enabled 1: Test J_STATE 2: Test K_STATE 3: Test SE0_NAK 4: Test Packet 5: Test FORCE_ENABLE 6-14: Reserved. 15: Port Test Control Error.
27-17 reserved_27_17	Reserved
16 HLE	Port Test Control For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
15-8 L1DSLOT	L1DSLOT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
7-4 HIRD	Host Initiated Resume Duration (HIRD)
3 RWE	RWE Port Test Control For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
2-0 L1S	L1 Status (L1S)

11.1.3.1.21 (PORTLI_20)

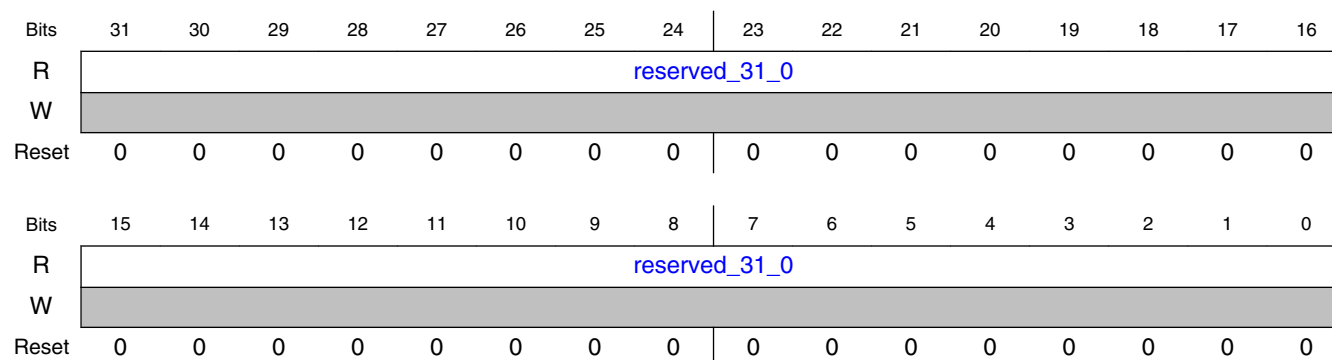
11.1.3.1.21.1 Offset

Register	Offset
PORTLI_20	428h

11.1.3.1.21.2 Function

Port Link Info Register Programming this field with random data will cause side effect i.

11.1.3.1.21.3 Diagram



11.1.3.1.21.4 Fields

Field	Function
31-0 reserved_31_0	Reserved

11.1.3.1.22 (PORTHLPMC_20)

11.1.3.1.22.1 Offset

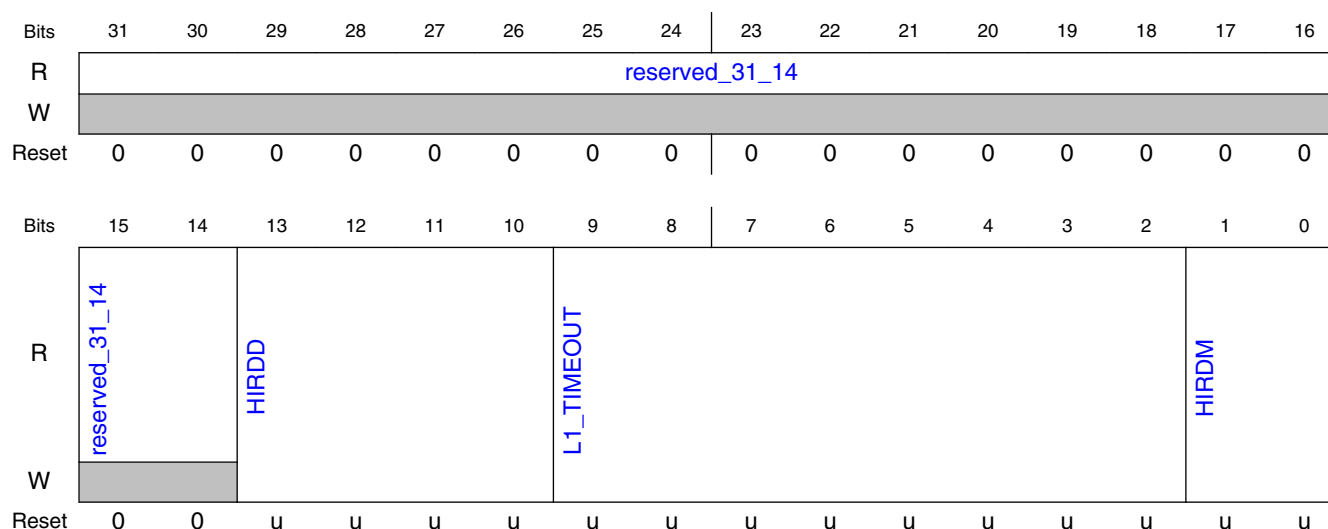
Register	Offset
PORTHLPMC_20	42Ch

11.1.3.1.22.2 Function

USB2 Port Hardware LPM Control Register Bit Definitions For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

Reset Mask:0xFFFFC000

11.1.3.1.22.3 Diagram



11.1.3.1.22.4 Fields

Field	Function
31-14 reserved_31_14	Reserved For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
13-10 HIRDD	PORTHLPMC_20 HIRDD
9-2 L1_TIMEOUT	PORTHLPMC_20 L1_TIMEOUT.
1-0 HIRDM	Host Initiated Resume Duration Mode (HIRDM) - RWS.

11.1.3.1.23 (PORTSC_30)

11.1.3.1.23.1 Offset

Register	Offset
PORTSC_30	430h

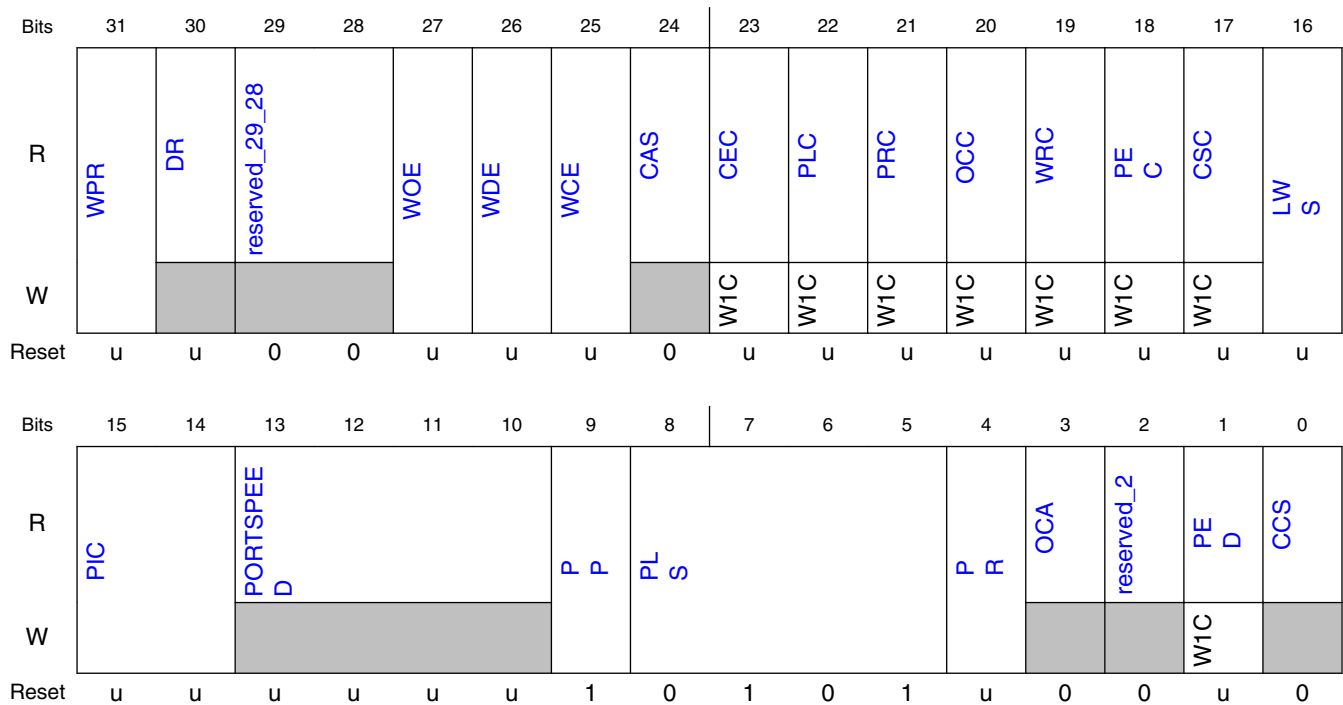
11.1.3.1.23.2 Function

Port Status and Control Register Bit Definitions The PORTSC Register

Access fails (Timeout) if the UTMI/ULPI clock is not running or one of the following bits is asserted.

Reset Mask:0x310003ED

11.1.3.1.23.3 Diagram



11.1.3.1.23.4 Fields

Field	Function
31 WPR	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
30 DR	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
29-28 reserved_29_28	Reserved
27 WOE	WOE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

Table continues on the next page...

Universal Serial Bus Controller (USB)

Field	Function
26 WDE	WDE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
25 WCE	WCE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
24 CAS	Cold Attach Status For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
23 CEC	CEC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
22 PLC	PLC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
21 PRC	PRC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
20 OCC	OCC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
19 WRC	WRC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
18 PEC	PEC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
17 CSC	CSC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
16 LWS	LWS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
15-14 PIC	PIC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
13-10 PORTSPEED	PORTSPEED For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
9 PP	PP For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification

Table continues on the next page...

Field	Function
	3.
8-5 PLS	PLS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
4 PR	PR set_register_field_attribute DWC_usb3_map/DWC_usb3_block_Host_Cntrl_Port_Reg_Set/PORTSC_30_REGS/PORTSC_30/PR VolatileMemory 1 Programming this field with random data will cause side effect.
3 OCA	OCA For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
2 reserved_2	Reserved
1 PED	PED For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
0 CCS	CCS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

11.1.3.1.24 USB3 Port Power Management Status and Control Register (PORTPMSC_30)

11.1.3.1.24.1 Offset

Register	Offset
PORTPMSC_30	434h

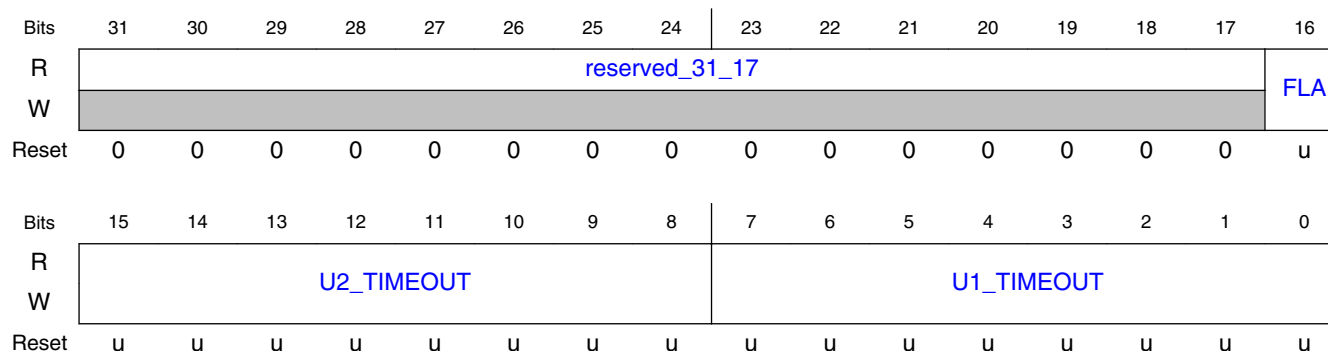
11.1.3.1.24.2 Function

USB3 Port Power Management Status and Control Register Bit Definitions

This register is in the Aux Power well.

Reset Mask:0xFFFE0000

11.1.3.1.24.3 Diagram



11.1.3.1.24.4 Fields

Field	Function
31-17 reserved_31_17	Reserved
16 FLA	FLA For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
15-8 U2_TIMEOUT	U2_TIMEOUT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
7-0 U1_TIMEOUT	U1_TIMEOUT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

11.1.3.1.25 Port Link Info Register (PORTLI_30)

11.1.3.1.25.1 Offset

Register	Offset
PORTLI_30	438h

11.1.3.1.25.2 Function

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

Reset Mask:0xFFFF0000

11.1.3.1.25.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	reserved_31_16															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	LINK_ERROR_COUNT															
W																
Reset	u	u	u	u	u	u	u	u	u	u	u	u	u	u	u	u

11.1.3.1.25.4 Fields

Field	Function
31-16 reserved_31_16	Reserved
15-0 LINK_ERROR_COUNT	LINK_ERROR_COUNT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

11.1.3.1.26 USB2 Port Hardware LPM Control Register (PORTHLPMC_30)

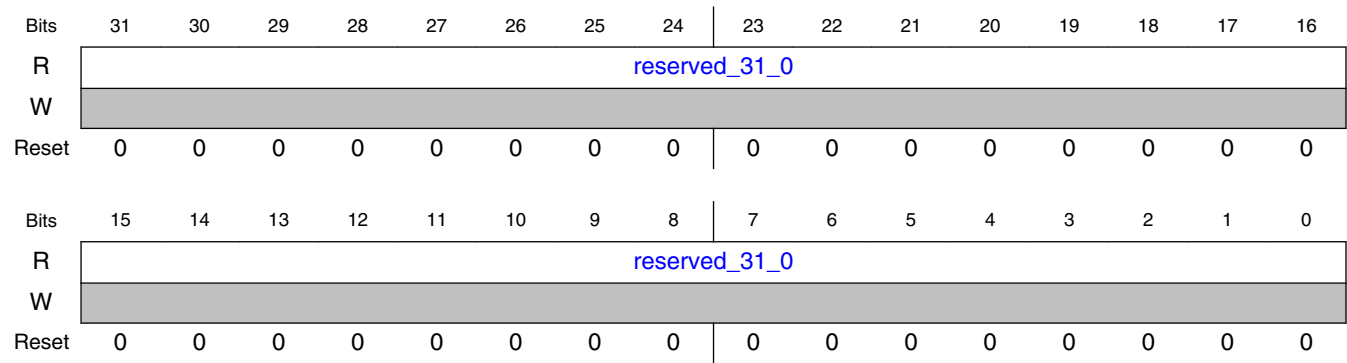
11.1.3.1.26.1 Offset

Register	Offset
PORTHLPMC_30	43Ch

11.1.3.1.26.2 Function

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

11.1.3.1.26.3 Diagram



11.1.3.1.26.4 Fields

Field	Function
31-0 reserved_31_0	Reserved For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

11.1.3.1.27 Microframe Index Register (MFINDEX)

11.1.3.1.27.1 Offset

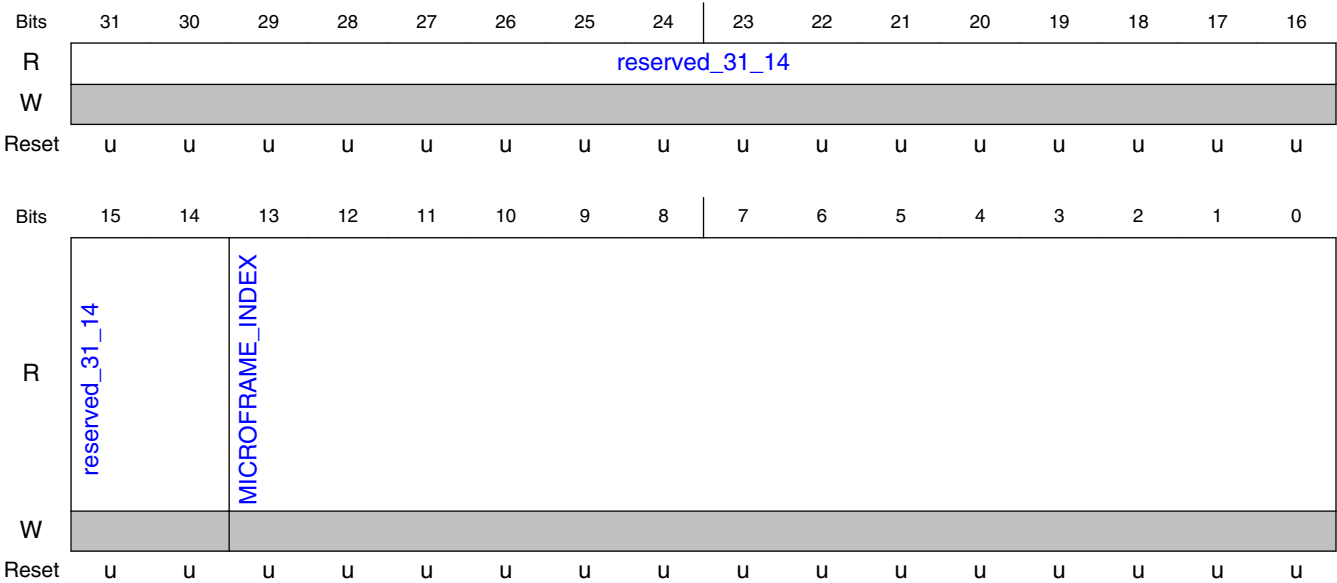
Register	Offset
MFINDEX	440h

11.1.3.1.27.2 Function

See the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

Reset Mask:0xFFFFC000

11.1.3.1.27.3 Diagram



11.1.3.1.27.4 Fields

Field	Function
31-14 reserved_31_14	Reserved
13-0 MICROFRAME_INDEX	MICROFRAME_INDEX For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.28 Interrupter Management Register (IMAN)

11.1.3.1.28.1 Offset

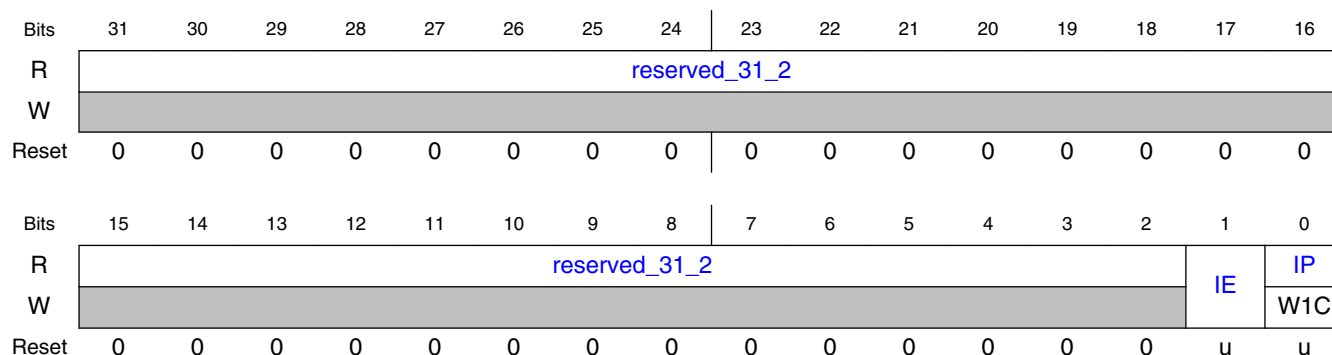
Register	Offset
IMAN	460h

11.1.3.1.28.2 Function

Reset Mask:0xFFFFF0FC

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

11.1.3.1.28.3 Diagram



11.1.3.1.28.4 Fields

Field	Function
31-2 reserved_31_2	Reserved
1 IE	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
0 IP	IP Interrupt Pending For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

11.1.3.1.29 Interrupter Moderation Register (IMOD)

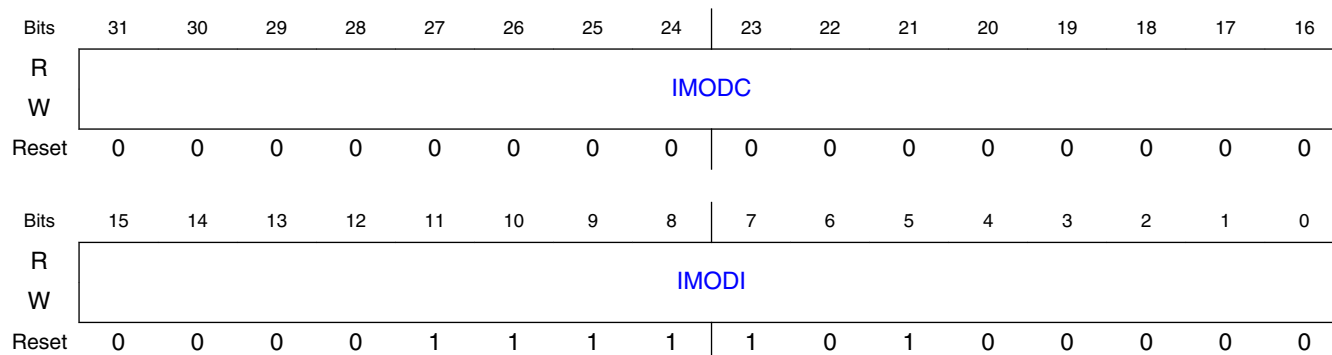
11.1.3.1.29.1 Offset

Register	Offset
IMOD	464h

11.1.3.1.29.2 Function

The software may use this register to
pace (or even out) the delivery of interrupts to the host CPU.

11.1.3.1.29.3 Diagram



11.1.3.1.29.4 Fields

Field	Function
31-16 IMODC	Interrupt Moderation Counter (IMODC) - RW.
15-0 IMODI	Interrupt Moderation Interval (IMODI) - RW.

11.1.3.1.30 ERSTSZ (ERSTSZ)

11.1.3.1.30.1 Offset

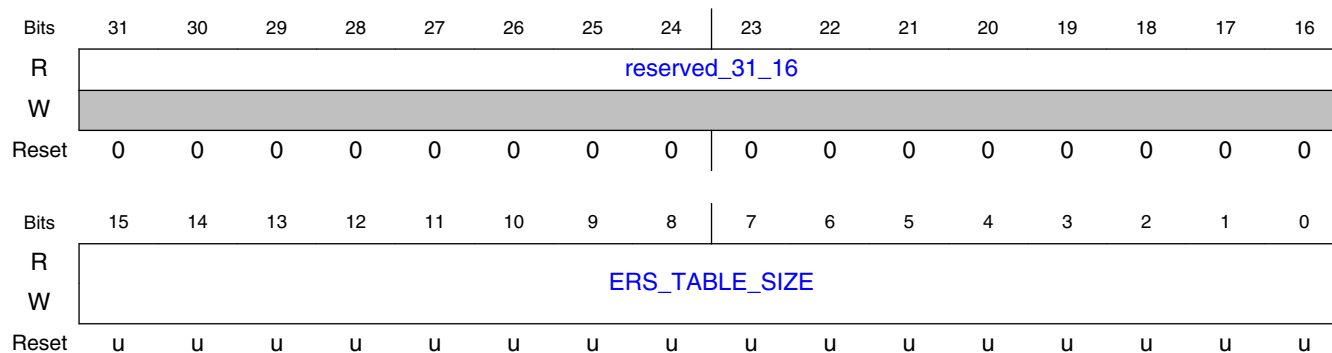
Register	Offset
ERSTSZ	468h

11.1.3.1.30.2 Function

Event Ring Segment Table Size Register Bit Definitions For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

Reset Mask:0xFFFF0000

11.1.3.1.30.3 Diagram



11.1.3.1.30.4 Fields

Field	Function
31-16 reserved_31_16	Reserved
15-0 ERS_TABLE_SIZE	ERS_TABLE_SIZE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

11.1.3.1.31 ERSTBA_LO (ERSTBA_LO)

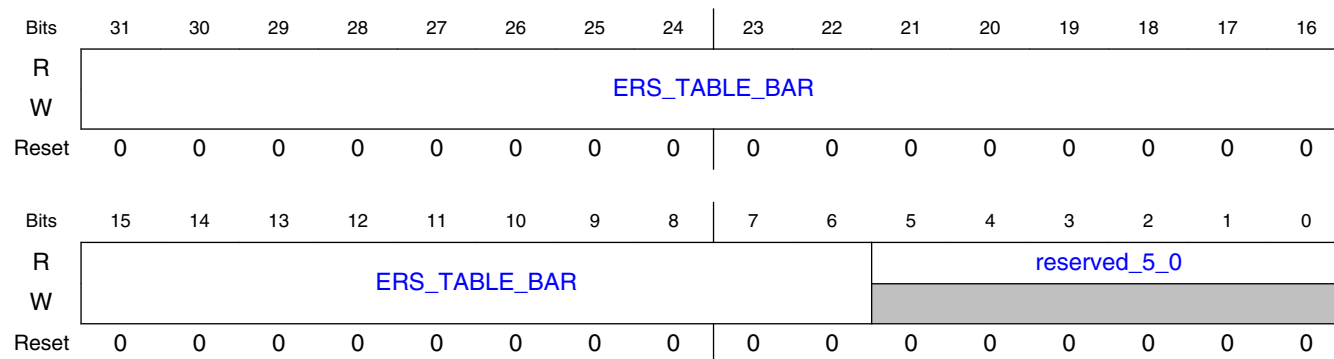
11.1.3.1.31.1 Offset

Register	Offset
ERSTBA_LO	470h

11.1.3.1.31.2 Function

ERSTBA_LO

11.1.3.1.31.3 Diagram



11.1.3.1.31.4 Fields

Field	Function
31-6 ERS_TABLE_B AR	ERS_TABLE_BAR
5-0 reserved_5_0	Reserved

11.1.3.1.32 ERSTBA_HI (ERSTBA_HI)

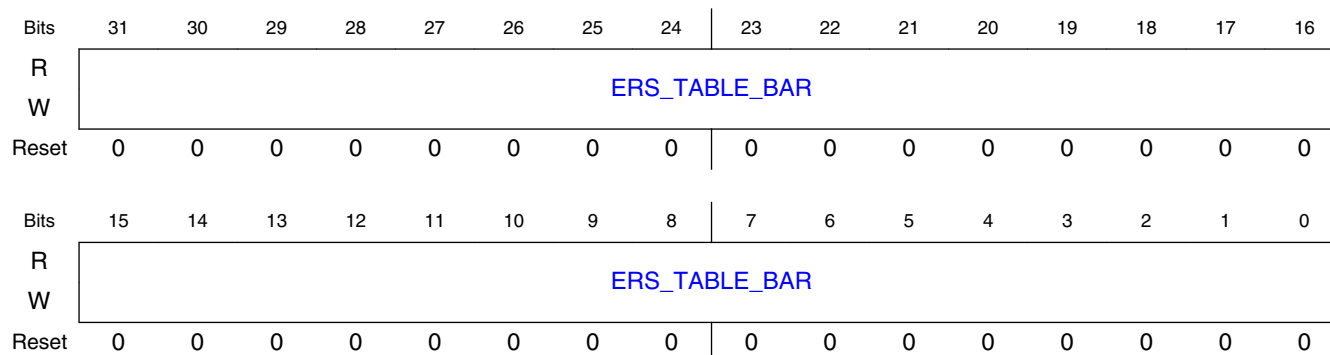
11.1.3.1.32.1 Offset

Register	Offset
ERSTBA_HI	474h

11.1.3.1.32.2 Function

ERSTBA_HI

11.1.3.1.32.3 Diagram



11.1.3.1.32.4 Fields

Field	Function
31-0 ERS_TABLE_B AR	ERS_TABLE_BAR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0

11.1.3.1.33 ERDP_LO (ERDP_LO)

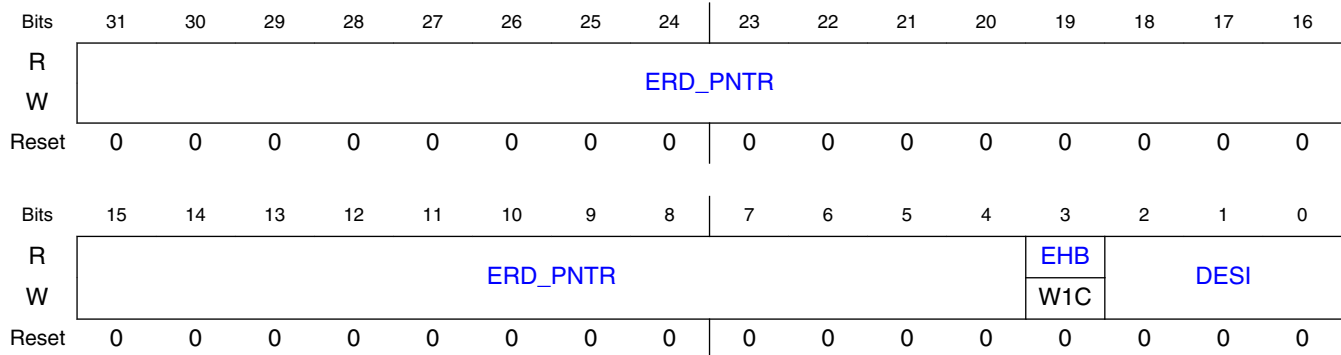
11.1.3.1.33.1 Offset

Register	Offset
ERDP_LO	478h

11.1.3.1.33.2 Function

ERDP_LO For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

11.1.3.1.33.3 Diagram



11.1.3.1.33.4 Fields

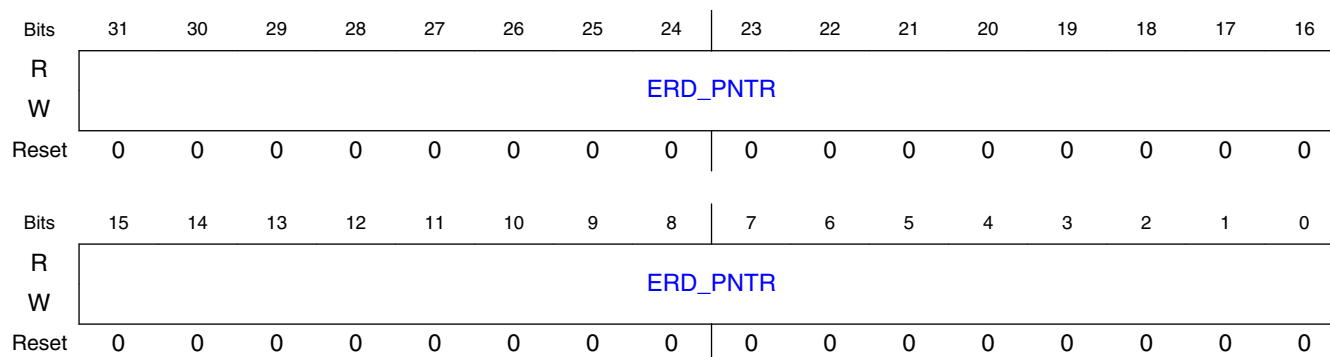
Field	Function
31-4 ERD_PNTR	ERD_PNTR ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.
3 EHB	EHB HC OS Owned SemaphoreERS_TABLE_SIZEHC BIOS Owned Semaphore USB SMI Enable SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count - HC OS Owned Semaphore - HC BIOS Owned Semaphore - USB SMI Enable - SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count
2-0 DESI	DESI - For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

11.1.3.1.34 ERDP_HI (ERDP_HI)

11.1.3.1.34.1 Offset

Register	Offset
ERDP_HI	47Ch

11.1.3.1.34.2 Diagram



11.1.3.1.34.3 Fields

Field	Function
31-0 ERD_PNTR	ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

11.1.3.1.35 Doorbell Register (DB)

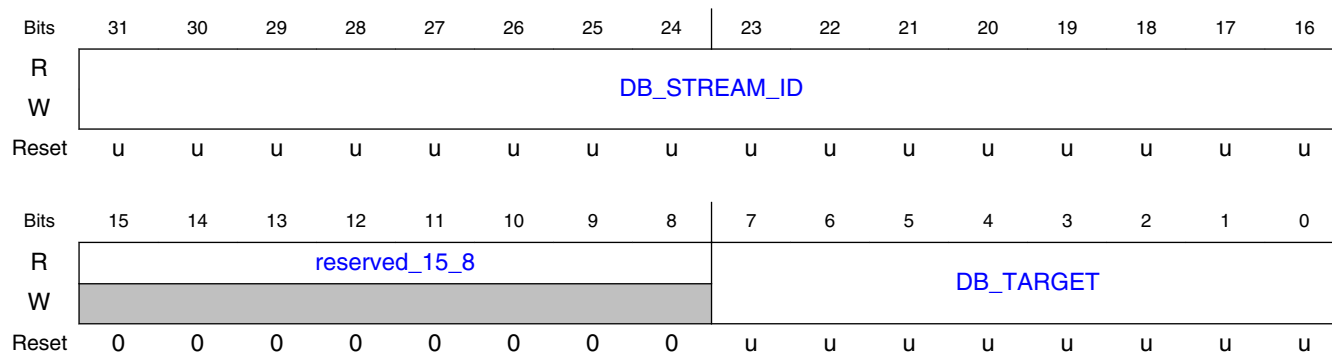
11.1.3.1.35.1 Offset

Register	Offset
DB	480h

11.1.3.1.35.2 Function

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3. Reset Mask:0xFF00

11.1.3.1.35.3 Diagram



11.1.3.1.35.4 Fields

Field	Function
31-16 DB_STREAM_ID	DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0
15-8 reserved_15_8	Reserved
7-0 DB_TARGET	DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

11.1.3.1.36 USBLEGSUP (USBLEGSUP)

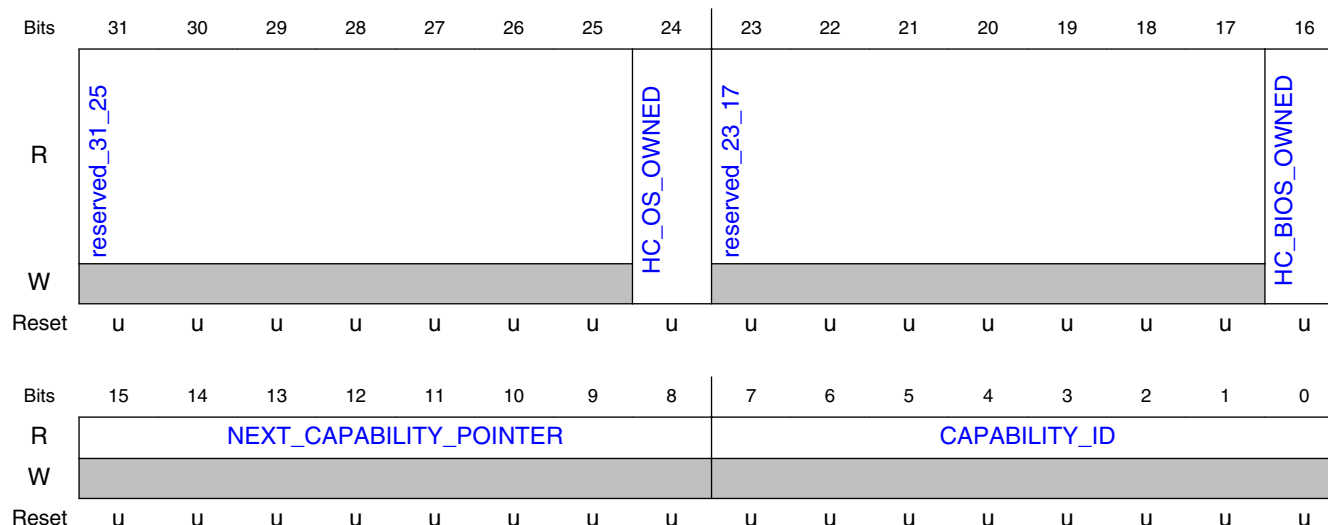
11.1.3.1.36.1 Offset

Register	Offset
USBLEGSUP	880h

11.1.3.1.36.2 Function

Reset Mask:0x7E7E0000

11.1.3.1.36.3 Diagram



11.1.3.1.36.4 Fields

Field	Function
31-25 reserved_31_25	Reserved
24 HC_OS_OWNE D	HC_OS_OWNED SEMAPHORE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
23-17 reserved_23_17	Reserved
16 HC_BIOS_OWNE D	HC_BIOS_OWNED SEMAPHORE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
15-8 NEXT_CAPABIL ITY_POINTER	NEXT_CAPABILITY_POINTER For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
7-0 CAPABILITY_ID	CAPABILITY_ID set_register_field_attribute DWC_usb3_map/ DWC_usb3_block_HC_Extended_Capability_Register/USBLEGSUP/CAPABILITY_ID RegisterResetValue 0x1 For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.37 USBLEGCTLSTS (USBLEGCTLSTS)

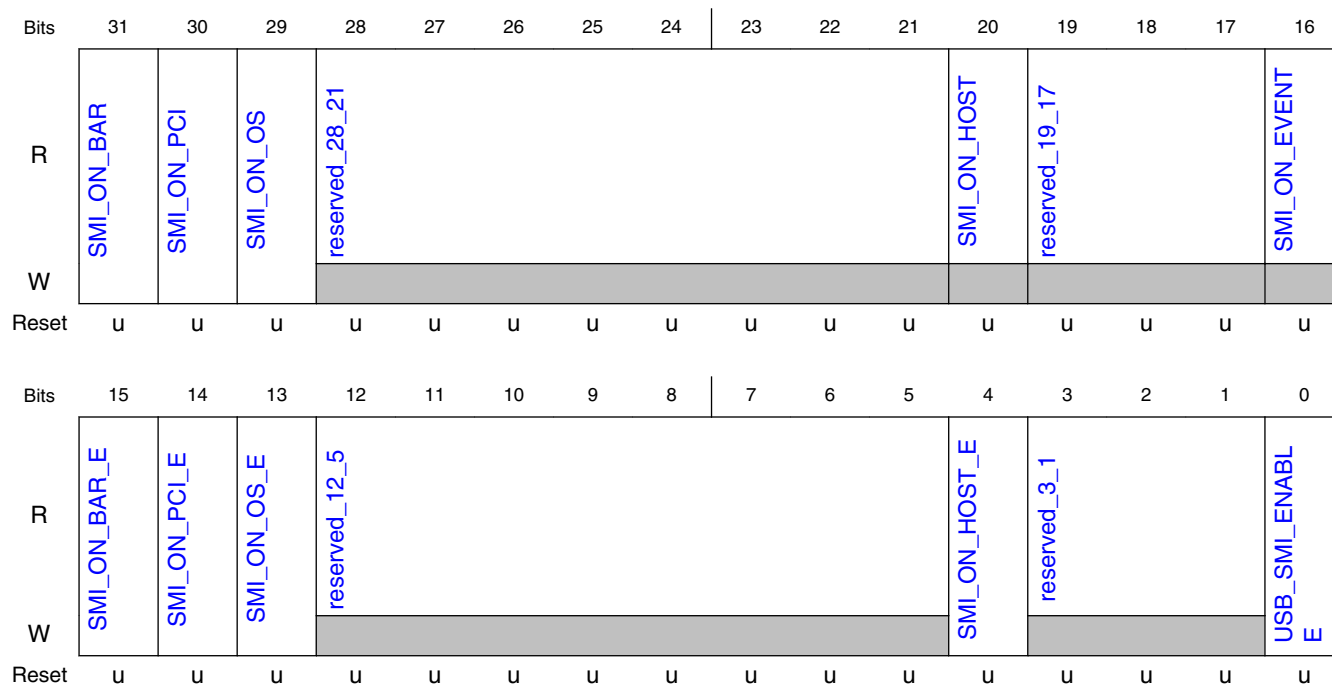
11.1.3.1.37.1 Offset

Register	Offset
USBLEGCTLSTS	884h

11.1.3.1.37.2 Function

Reset Mask:0x1FEE1FEE

11.1.3.1.37.3 Diagram



11.1.3.1.37.4 Fields

Field	Function
31 SMI_ON_BAR	SMI_ON_BAR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
30 SMI_ON_PCI	SMI_ON_PCI COMMAND For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
29 SMI_ON_OS	SMI_ON_OS OWNERSHIP CHANGE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
28-21	Reserved

Table continues on the next page...

Universal Serial Bus Controller (USB)

Field	Function
reserved_28_21	
20 SMI_ON_HOST	SMI_ON_HOST SYSTEM ERROR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
19-17 reserved_19_17	Reserved
16 SMI_ON_EVENT T	SMI_ON_EVENT INTERRUPT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
15 SMI_ON_BAR_ E	SMI_ON_BAR ENABLE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
14 SMI_ON_PCI_E	SMI_ON_PCI COMMAND ENABLE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
13 SMI_ON_OS_E	SMI_ON_OS OWNERSHIP ENABLE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
12-5 reserved_12_5	Reserved
4 SMI_ON_HOST_ E	SMI_ON_HOST SYSTEM ERROR ENABLE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
3-1 reserved_3_1	Reserved
0 USB_SMI_ENA BLE	USB_SMI_ENABLE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.38 SUPTPRT2_DW0 (SUPTPRT2_DW0)

11.1.3.1.38.1 Offset

Register	Offset
SUPTPRT2_DW0	890h

11.1.3.1.38.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MAJOR_REVISION								MINOR_REVISION							
W																
Reset	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	NEXT_CAPABILITY_POINTER								CAPABILITY_ID							
W																
Reset	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0

11.1.3.1.38.3 Fields

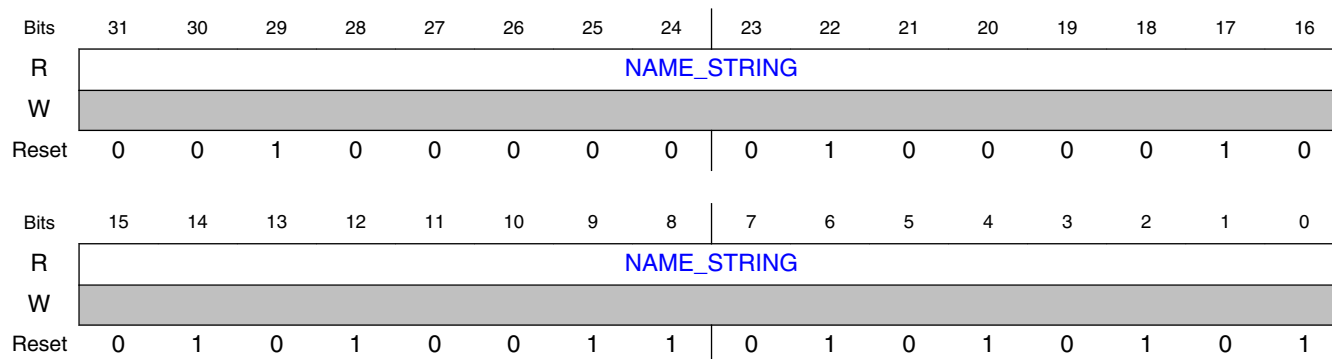
Field	Function
31-24 MAJOR_REVISION	MAJOR_REVISION
23-16 MINOR_REVISION	MINOR_REVISION
15-8 NEXT_CAPABILITY_POINTER	NEXT_CAPABILITY_POINTER
7-0 CAPABILITY_ID	CAPABILITY_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.39 SUPTPRT2_DW1 Register (SUPTPRT2_DW1)

11.1.3.1.39.1 Offset

Register	Offset
SUPTPRT2_DW1	894h

11.1.3.1.39.2 Diagram



11.1.3.1.39.3 Fields

Field	Function
31-0 NAME_STRING	NAME_STRING For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.40 xHCI Supported Protocol Capability_ Data Word 2 (SUPTPRT2_DW2)

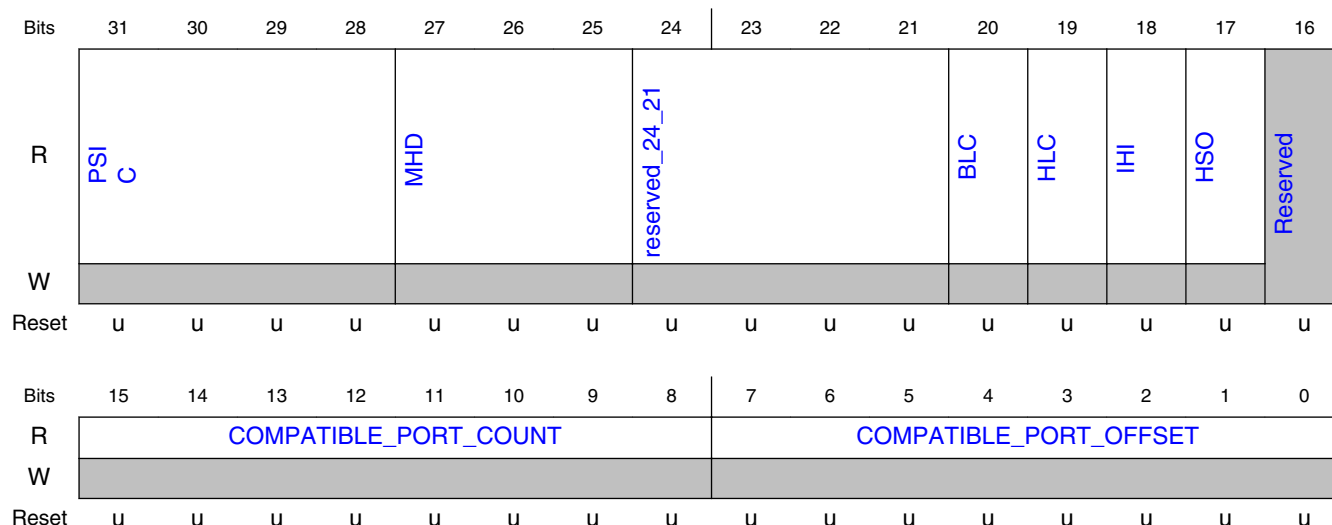
11.1.3.1.40.1 Offset

Register	Offset
SUPTPRT2_DW2	898h

11.1.3.1.40.2 Function

Reset Mask:0x1E00000

11.1.3.1.40.3 Diagram



11.1.3.1.40.4 Fields

Field	Function
31-28 PSIC	PSIC
27-25 MHD	Hub Depth For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
24-21 reserved_24_21	Reserved
20 BLC	BESL LPM Capability When this bit is set to: - 1: The ports described by this xHCI Supported Protocol Capability applies BESL timing to the BESL and BESLD fields of the PORTPMSC and PORTHLPMC registers. - 0: The ports described by this xHCI Supported Protocol Capability applies HIRD timing to the BESL and BESLD fields of the PORTPMSC and PORTHLPMC registers.
19 HLC	Compatible Port Offset. Compatible Port Count Refer to Table 7-3 in the DWC Cores SuperSpeed USB 3.0 Controller Databook.
18 IHI	IHI
17 HSO	HSO
16 —	Reserved
15-8 COMPATIBLE_PORT_COUNT	COMPATIBLE_PORT_COUNT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

Table continues on the next page...

Field	Function
7-0 COMPATIBLE_PORT_OFFSET	COMPATIBLE_PORT_OFFSET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.41 SUPTPRT2_DW3 Register (SUPTPRT2_DW3)

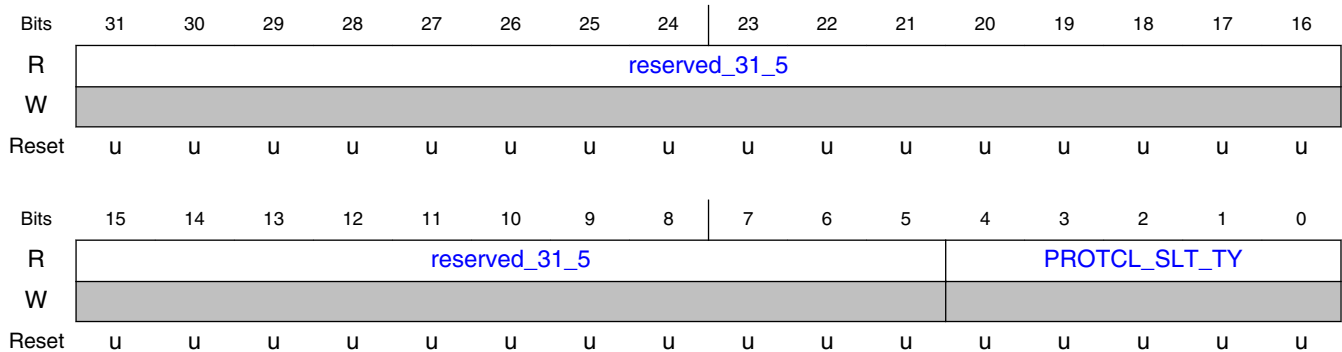
11.1.3.1.41.1 Offset

Register	Offset
SUPTPRT2_DW3	89Ch

11.1.3.1.41.2 Function

Reset Mask:0xFFFFFEE0

11.1.3.1.41.3 Diagram



11.1.3.1.41.4 Fields

Field	Function
31-5 reserved_31_5	Reserved
4-0 PROTCL_SLT_TY	Protocol Slot Type

11.1.3.1.42 (SUPTPRT3_DW0)

11.1.3.1.42.1 Offset

Register	Offset
SUPTPRT3_DW0	8A0h

11.1.3.1.42.2 Function

Register SUPTPRT3_DW0

11.1.3.1.42.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MAJOR_REVISION								MINOR_REVISION							
W																
Reset	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	NEXT_CAPABILITY_POINTER								CAPABILITY_ID							
W																
Reset	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0

11.1.3.1.42.4 Fields

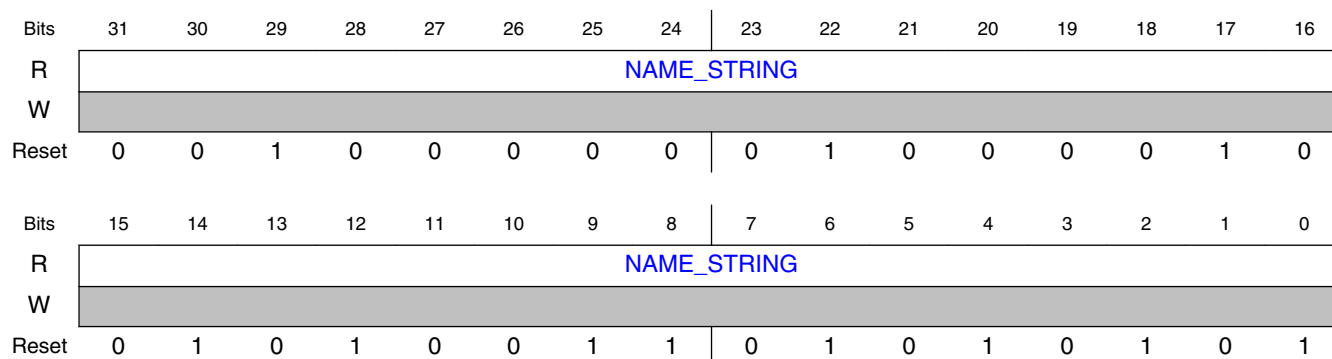
Field	Function
31-24 MAJOR_REVISION	MAJOR_REVISION For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
23-16 MINOR_REVISION	MINOR_REVISION For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
15-8 NEXT_CAPABILITY_POINTER	NEXT_CAPABILITY_POINTER For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
7-0 CAPABILITY_ID	CAPABILITY_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.43 SUPTPRT3_DW1 Register (SUPTPRT3_DW1)

11.1.3.1.43.1 Offset

Register	Offset
SUPTPRT3_DW1	8A4h

11.1.3.1.43.2 Diagram



11.1.3.1.43.3 Fields

Field	Function
31-0	NAME_STRING
NAME_STRING	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

11.1.3.1.44 SUPTPRT3_DW2 (SUPTPRT3_DW2)

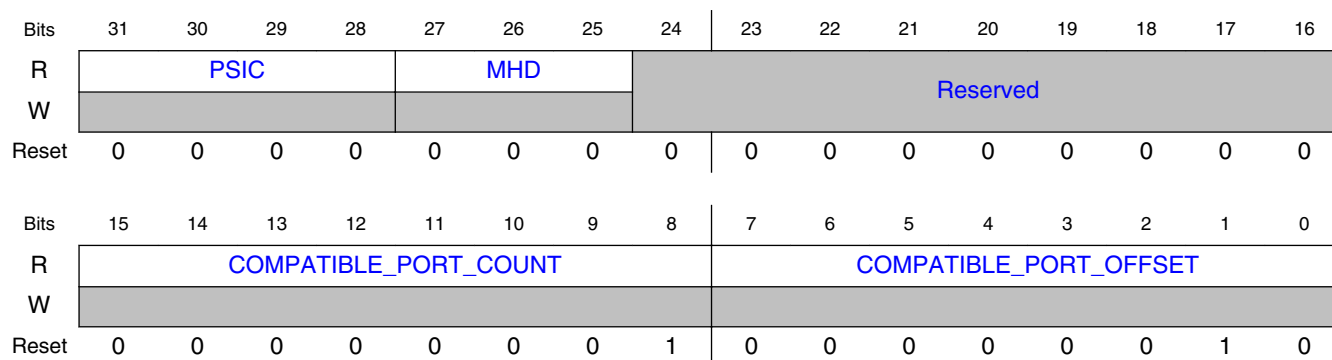
11.1.3.1.44.1 Offset

Register	Offset
SUPTPRT3_DW2	8A8h

11.1.3.1.44.2 Function

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.

11.1.3.1.44.3 Diagram



11.1.3.1.44.4 Fields

Field	Function
31-28 PSIC	PSIC
27-25 MHD	Hub Depth For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
24-16 —	Reserved
15-8 COMPATIBLE_PORT_COUNT	COMPATIBLE_PORT_COUNT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.
7-0 COMPATIBLE_PORT_OFFSET	COMPATIBLE_PORT_OFFSET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.0.

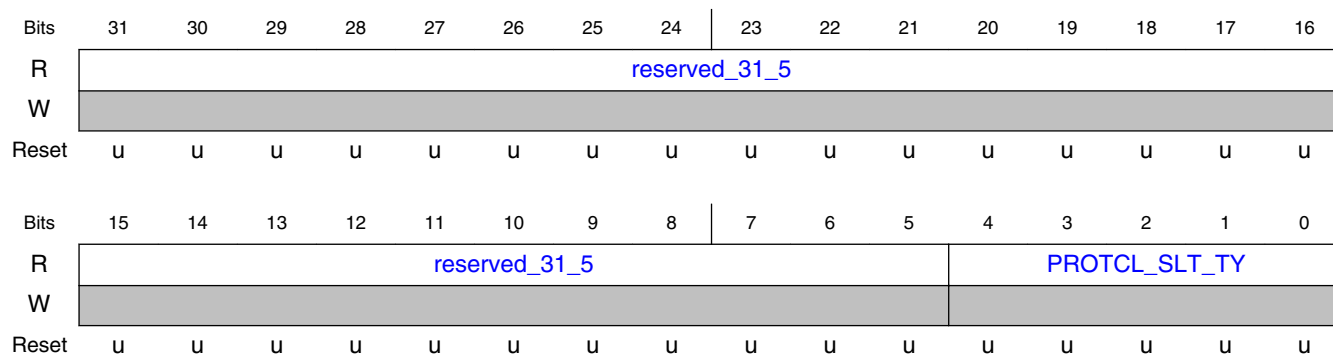
11.1.3.1.45 SUPTPRT3_DW3 (SUPTPRT3_DW3)

11.1.3.1.45.1 Offset

Register	Offset
SUPTPRT3_DW3	8ACh

11.1.3.1.45.2 Function

Reset Mask:0xFFFFFE0

11.1.3.1.45.3 Diagram**11.1.3.1.45.4 Fields**

Field	Function
31-5 reserved_31_5	Reserved
4-0 PROTCL_SLT_TY	Protocol Slot Type Protocol Slot Type

11.1.3.1.46 Global SoC Bus Configuration Register 0 (GSBUSCFG0)**11.1.3.1.46.1 Offset**

Register	Offset
GSBUSCFG0	C100h

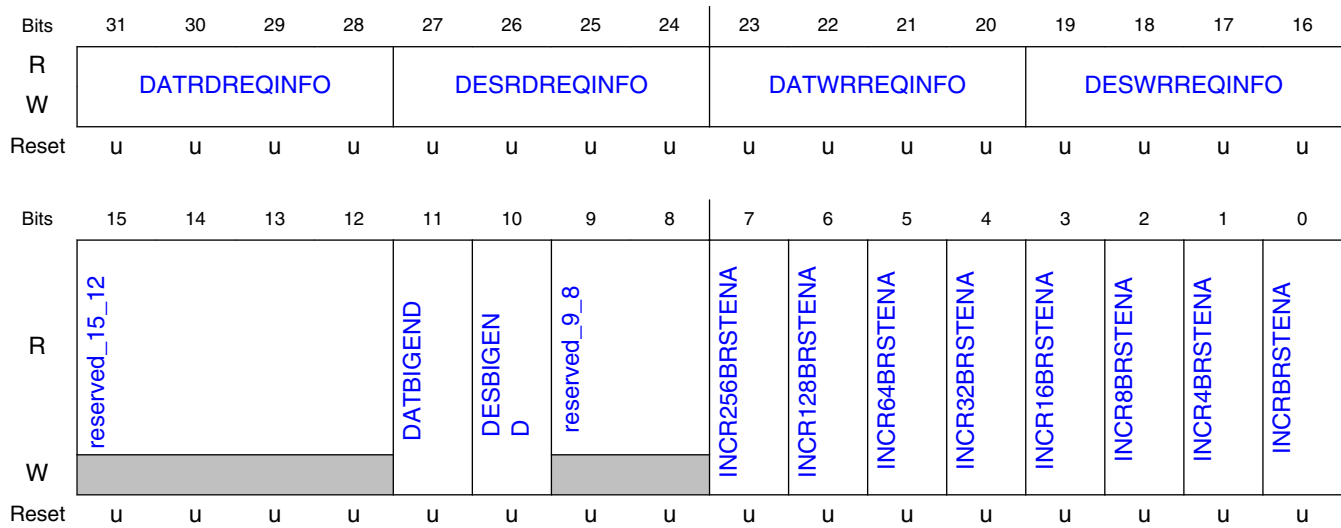
11.1.3.1.46.2 Function

This register configures system bus DMA options for the master bus, which may be configured as AHB, AXI, or Native.

Reset Mask:0xF300

Reset Default Value:0x1

11.1.3.1.46.3 Diagram



11.1.3.1.46.4 Fields

Field	Function
31-28 DATRDREQINFO	DATRDREQINFO AHB-prot/AXI-cache/OCP-ReqInfo for Data Read (DatRdReqInfo) Input to BUS-GM.
27-24 DESRDREQINFO	DESRDREQINFO AHB-prot/AXI-cache/OCP-ReqInfo for Descriptor Read (DesRdReqInfo). Input to BUS-GM.
23-20 DATWRREQINFO	DATWRREQINFO AHB-prot/AXI-cache/OCP-ReqInfo for Data Write (DatWrReqInfo). Input to BUS-GM.
19-16 DESWRREQINFO	DESWRREQINFO AHB-prot/AXI-cache/OCP-ReqInfo for Descriptor Write (DesWrReqInfo) Input to BUS-GM.
15-12 reserved_15_12	Reserved for future use
11 DATBIGEND	Data Access is Big Endian This bit controls the endian mode for data accesses. - Little-endian (default); - Big-endian; In big-endian mode, DMA access (both read and write) for packet data a Byte Invariant Big-Endian mode (see "Little-Endian and Big-Endian" section in the DWC SuperSpeed USB 3.0 Controller User Guide). Note: Since AXI requires byte invariant endianness, setting DescBigend and DatBigEnd to one causes an address invariant transform to be applied, which is not appropriate. See section 9.3 and 9.4 of the AMBA AXI Specification. Hence for an AXI master (DWC_USB3_MBUS_TYPE=1), this bit must be set to zero.
10 DESBIGEND	Descriptor Access is Big Endian This bit controls the endian mode for descriptor accesses. - Little-endian (default) - Big-endian In big-endian mode, DMA access (both read and write) for descriptors uses a Byte Invariant Big-Endian mode (see "Little-Endian and Big-Endian" section in the DWC SuperSpeed USB 3.0

Table continues on the next page...

Field	Function
	Controller User Guide. Data is considered as 'embedded data' in the descriptors in the following cases: - Device mode: The buffer pointer of a Setup TRB points to the Setup TRB itself. - Host mode: The Immediate Data (IDT) bit in a Transfer TRB is set to 1. In device mode, if the system uses different endian modes for descriptor and data, software must not use 'embedded' data. In host mode, if the system uses different endian modes for data and descriptors, the controller treats 'embedded data' as descriptor (not as data) in terms of endian mode handling. If this is not the expectation of the system, the software must manipulate the 'embedded data' accordingly. Note: Since AXI requires byte invariant endianness, setting DescBigend and DatBigEnd to one causes an address invariant transform to be applied, which is not appropriate. See section 9.3 and 9.4 of the AMBA AXI Specification. Hence for an AXI master (DWC_USB3_MBUS_TYPE=1), this bit must be set to zero.
9-8 reserved_9_8	Reserved
7 INCR256BRSTENA	INCR256 Burst Type Enable Input to BUS-GM. For the AHB/AXI configuration, if software set this bit to 1, the AHB/AXI master uses INCR to do the 256-beat burst.
6 INCR128BRSTENA	INCR128 Burst Type Enable Input to BUS-GM; For the AHB/AXI configuration, if software set this bit to 1, the AHB/AXI master uses INCR to do the 128-beat burst.
5 INCR64BRSTENA	INCR64 Burst Type Enable - Input to BUS-GM; For the AHB/AXI configuration, if software set this bit to 1, the AHB/AXI master uses INCR to do the 64-beat burst.
4 INCR32BRSTENA	INCR32 Burst Type Enable Input to BUS-GM; For the AHB/AXI configuration, if software set this bit to 1, the AHB/AXI master uses INCR to do the 32-beat burst.
3 INCR16BRSTENA	INCR16 Burst Type Enable Input to BUS-GM. For the AHB/AXI configuration, if software set this bit to '1', the AHB/AXI master uses INCR to do the 16-beat burst.
2 INCR8BRSTENA	INCR8 Burst Type Enable Input to BUS-GM; For the AHB/AXI configuration, if software set this bit to "1", the AHB/AXI master uses INCR to do the 8-beat burst.
1 INCR4BRSTENA	INCR4 Burst Type Enable Input to BUS-GM; For the AXI configuration, when this bit is enabled the controller is allowed to do bursts of beat length 1, 2, 3, and 4. It is highly recommended that this bit is enabled to prevent descriptor reads and writes from being broken up into separate transfers.
0 INCRBRSTENA	INCRBRSTENA Undefined Length INCR Burst Type Enable (INCRBrstEna) Input to BUS-GM; This bit determines the set of burst lengths the master interface uses. It works in conjunction with the GSBUSCFG0[7:1] enables (INCR256/128/64/32/16/8/4). 0: INCRX burst mode HBURST (for AHB configurations) and ARLEN/AWLEN (for AXI configurations) do not use INCR except in case of non-aligned burst transfers. In the case of address-aligned transfers, they use only the following burst lengths: - 1 - 4 (if GSBUSCFG0.INCR4BrstEna = 1) - 8 (if GSBUSCFG0.INCR8BrstEna = 1) - 16 (if GSBUSCFG0.INCR16BrstEna = 1) - 32 (if GSBUSCFG0.INCR32BrstEna = 1) - 64 (if GSBUSCFG0.INCR64BrstEna = 1) - 128 (if GSBUSCFG0.INCR128BrstEna = 1) - 256 (if GSBUSCFG0.INCR256BrstEna = 1) Note: - In case of non-address-aligned transfers, INCR may get generated at the beginning and end of the transfers to align the address boundaries, even though INCR is disabled. - In AHB mode, if INCRX burst mode is enabled, but none of the supported INCRx bursts bits are enabled, then the controller will perform (undefined length) INCR bursts. 1: INCR (undefined length) burst mode - AHB configurations: HBURST uses SINGLE or INCR of any length with handling 1KB boundary breakup. - AXI configurations: ARLEN/AWLEN uses any length less than or equal to the

Field	Function
	largest-enabled burst length of INCR32/64/128/256. For cache line-aligned applications, this bit is typically set to 0 to ensure that the master interface uses only power-of-2 burst lengths (as enabled via GSBUSCFG0[7:0]).

11.1.3.1.47 Global SoC Bus Configuration Register 1 (GSBUSCFG1)

11.1.3.1.47.1 Offset

Register	Offset
GSBUSCFG1	C104h

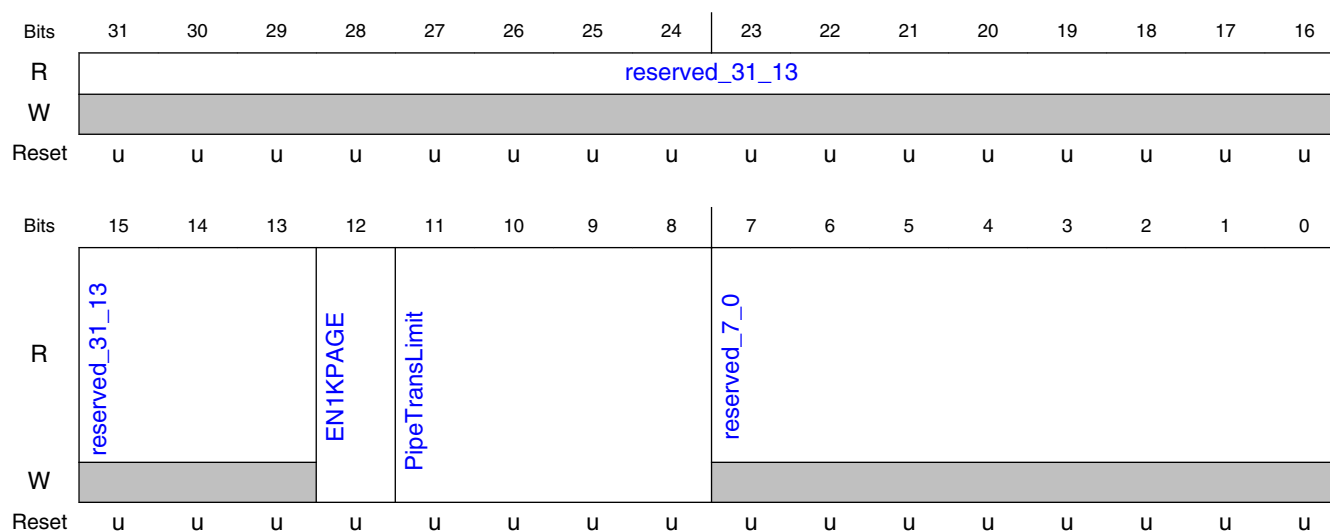
11.1.3.1.47.2 Function

xHCI Register Power-On Value: If you are using a standard xHCI host driver, make sure to set the register's power-on value during coreConsultant configuration (DWC_USB3_GSBUSCFG1_INIT parameter) because the standard xHCI driver does not access this register.

Reset Mask:0xFFFFE0FF

Reset Default Value:0x300

11.1.3.1.47.3 Diagram



11.1.3.1.47.4 Fields

Field	Function
31-13 reserved_31_13	Reserved
12 EN1KPAGE	1k Page Boundary Enable By default (this bit is disabled) the AXI breaks transfers at the 4k page boundary. When this bit is enabled, the AXI master (DMA data) breaks transfers at the 1k page boundary.
11-8 PipeTransLimit	AXI Pipelined Transfers Burst Request Limit The field controls the number of outstanding pipelined transfer requests the AXI master pushes to the AXI slave. When the AXI master reaches this limit, it does not make any more requests on the AXI ARADDR and AWADDR buses until the associated data phases complete. This field is encoded as follows: - 'h0: 1 request - 'h1: 2 requests - 'h2: 3 requests - 'h3: 4 requests - ... - 'hF: 16 requests
7-0 reserved_7_0	Reserved

11.1.3.1.48 Global Tx Threshold Control Register (GTXTHRCFG)**11.1.3.1.48.1 Offset**

Register	Offset
GTXTHRCFG	C108h

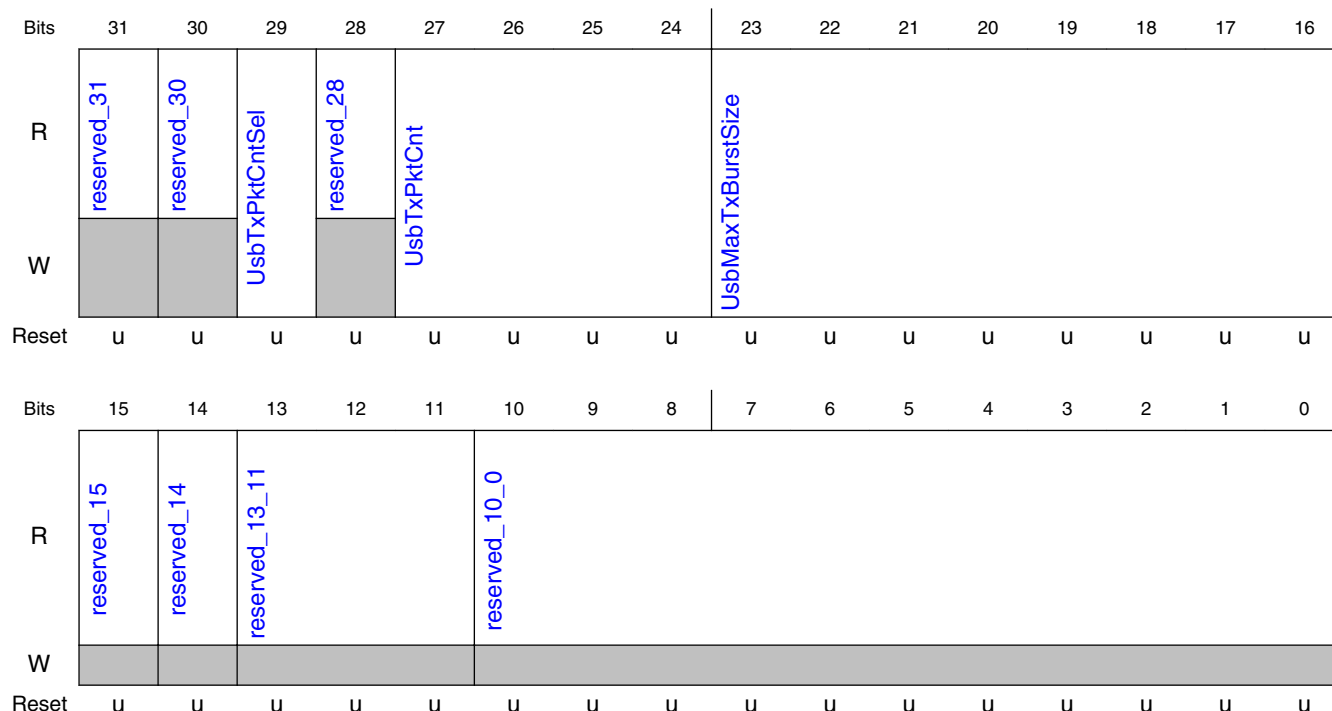
11.1.3.1.48.2 Function

All bits in the GTXTHRCFG register are valid only in host mode. This register is not applicable for debug target and USB 2.0-only mode.

Reset Mask:0xD000FFFF

Reset Default Value:0x0

11.1.3.1.48.3 Diagram



11.1.3.1.48.4 Fields

Field	Function
31 reserved_31	Reserved
30 reserved_30	Reserved
29 UsbTxPktCntSel	USB Transmit Packet Count Enable This field enables/disables the USB transmission multi-packet thresholding: - 0: USB transmission multi-packet thresholding is disabled; the core can start transmission on the USB after the entire (one full) packet has been fetched into the corresponding TXFIFO. - 1: USB transmission multi-packet thresholding is enabled. The core can only start transmission on the USB after USB Transmit Packet Count amount of packets for the USB transaction (burst) are already in the corresponding TXFIFO. This mode is valid in both host and device modes. It is only used for SuperSpeed operation.
28 reserved_28	Reserved
27-24 UsbTxPktCnt	USB Transmit Packet Count This field specifies the number of packets that must be in the TXFIFO before the core can start transmission for the corresponding USB transaction (burst). This field is only valid when the USB Transmit Packet Count Enable field is set to one. Valid values are from 1 to 15. Note: - In device mode, if device controller does not have the TRBs for the number of packets or if it cannot fetch the TRBs because of high latency or switching between other endpoints, then it does not wait for the threshold number of packets. The threshold number of packets will be honored only when the TRBs are available in the controller for the number of packets before it starts the data fetch. - This field must be less than or equal to the USB Maximum TX Burst Size field.

Table continues on the next page...

Field	Function
23-16 UsbMaxTxBurst Size	USB Maximum TX Burst Size When UsbTxPktCntSel is one, this field specifies the Maximum Bulk OUT burst the core can do. When the system bus is slower than the USB, TX FIFO can underrun during a long burst. User can program a smaller value to this field to limit the TX burst size that the core can do. Host mode: It only applies to SS Bulk, Isochronous, and Interrupt OUT endpoints. Device mode: This value is not used in device mode, but users need to program a value when using the TX threshold feature to make sure that the value programmed in UsbTxPktCnt is less than this value. Valid values are from 1 to 16.
15 reserved_15	Reserved_15
14 reserved_14	Reserved1(Rsvd/Rs) Register field must write only 0 by the application. The read value must be treated as X (unknown).
13-11 reserved_13_11	Reserved (Rsvd/Rs) The register field must write only 0 by the application. The read value must be treated as X (unknown).
10-0 reserved_10_0	Reserved for future use

11.1.3.1.49 Global Rx Threshold Control Register (GRXTHRCFG)

11.1.3.1.49.1 Offset

Register	Offset
GRXTHRCFG	C10Ch

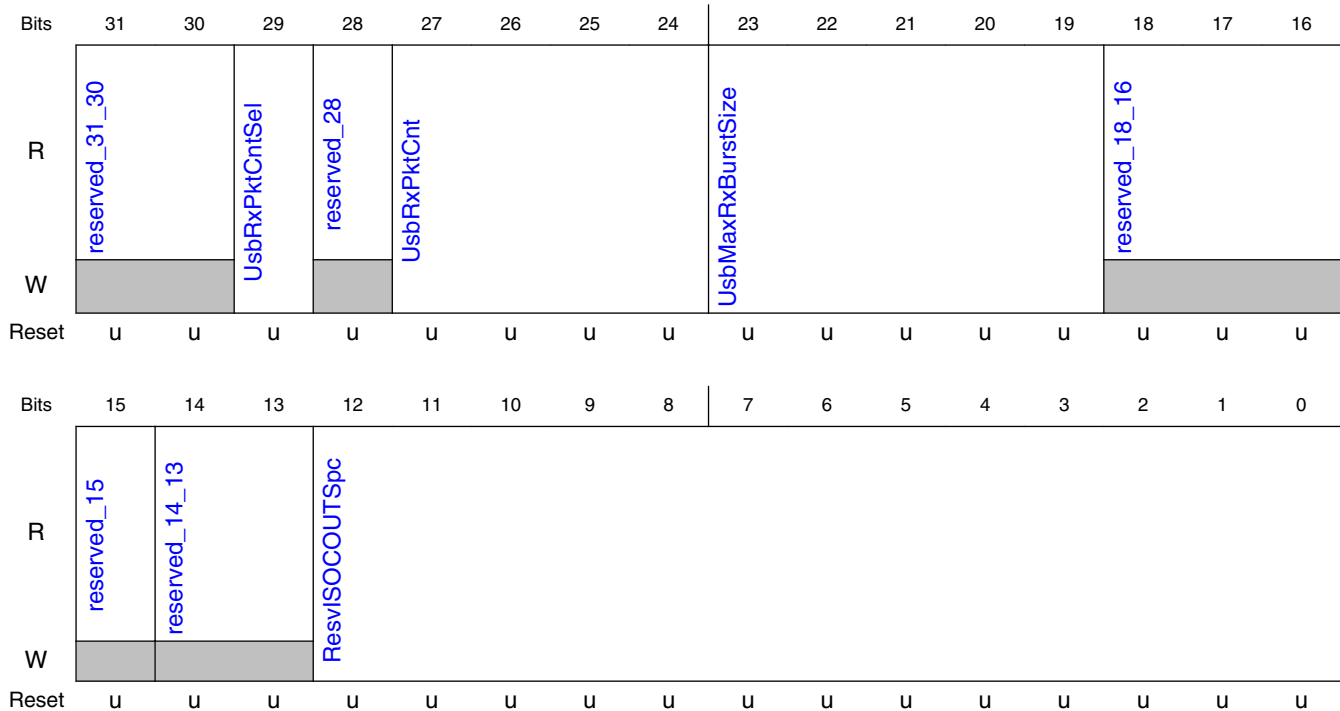
11.1.3.1.49.2 Function

This register is not applicable for debug target and USB 2.0-only mode. In a normal case, a Tx burst starts as soon as one packet is prefetched; an Rx burst starts as soon as 1-packet space is available.

Reset Mask:0xD007E000

Reset Default Value:0x0

11.1.3.1.49.3 Diagram



11.1.3.1.49.4 Fields

Field	Function
31-30 reserved_31_30	Reserved
29 UsbRxPktCntSel	USB Receive Packet Count Enable This field enables/disables the USB reception multi-packet thresholding: - 0: The core can only start reception on the USB when the RX FIFO has space for at least one packet. - 1: The core can only start reception on the USB when the RX FIFO has space for at least UsbRxPktCnt amount of packets. This mode is valid in both host and device mode. It is only used for SuperSpeed. In device mode, - Setting this bit to 1 also enables the functionality of reporting NUMP in the ACK TP based on the RX FIFO space instead of reporting a fixed NUMP derived from DCFG.NUMP for non-control endpoints. - If you are using external buffer control (EBC) feature, disable this mode by setting UsbRxPktCntSel to 0.
28 reserved_28	Reserved
27-24 UsbRxPktCnt	USB Receive Packet Count In host mode, this field specifies the space (in terms of the number of packets) that must be available in the RX FIFO before the core can start the corresponding USB RX transaction (burst). In device mode, this field specifies the space (in terms of the number of packets) that must be available in the RX FIFO before the core can send ERDY for a flow-controlled endpoint. This field is valid only when the USB Receive Packet Count Enable field is set to 1. The valid values for this field are from 1 to 15. Note: This field must be less than or equal to the USB Maximum Receive Burst Size field.
23-19	USB Maximum Receive Burst Size In host mode, this field specifies the Maximum Bulk IN burst the DWC_usb3 core can perform. When the system bus is slower than the USB, RX FIFO can overrun during a long burst. You can program a smaller value to this field to limit the RX burst size that the core can

Table continues on the next page...

Universal Serial Bus Controller (USB)

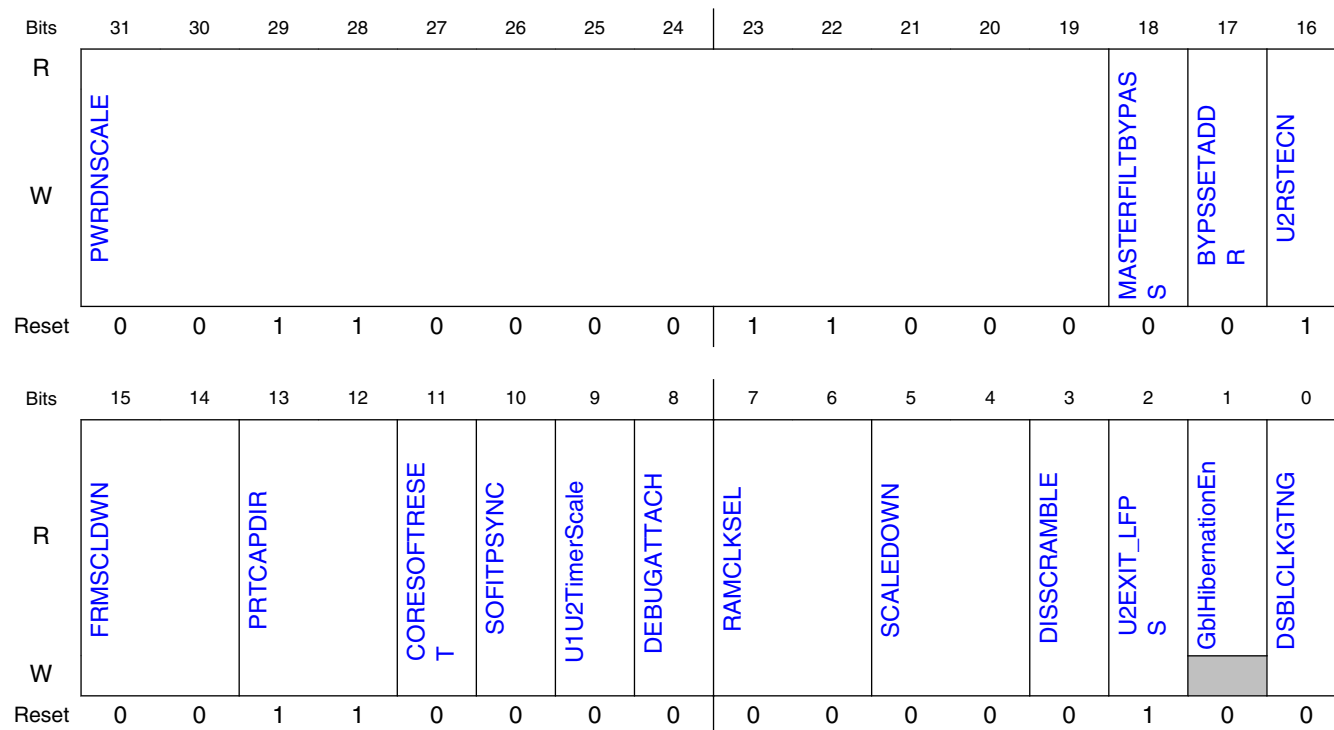
Field	Function
UsbMaxRxBurst Size	perform. It only applies to SS Bulk, Isochronous, and Interrupt IN endpoints in the host mode. In device mode, this field specifies the NUMP value that is sent in ERDY for an OUT endpoint. The programmed value should not exceed the RXFIFO size. This field is valid only when UsbRxPktCntSel is one. The valid values for this field are from 1 to 16.
18-16 reserved_18_16	Reserved
15 reserved_15	Reserved
14-13 reserved_14_13	Reserved
12-0 ResvISOCOUT Spc	Space reserved in Rx FIFO for ISOC OUT In host mode, this field is not applicable and must be programmed to 0. In device mode, this value represents the amount of space to be reserved for ISOC OUT packets. The value to be programmed should be chosen so as to ensure that non ISOC packets are not completely dropped. If no space needs to be reserved for ISOC OUT packets, program this field to 0. This field is valid only in device mode. The maximum configurable depth of RX FIFO is 8192. Therefore, this field is 13 bits wide. The value of space reserved is in terms of DWC_USB3_MDWIDTH. For SS, the space reservation is always rounded off to the nearest packet boundary. Therefore, it is always recommended to program a value corresponding to MPS or its multiples. For HS/FS, the space reservation is the actual value. Note: For SS, reserve space for ISOC when the Rx FIFO space can accommodate two MPS or more. Otherwise, this may result in degraded performance for non-ISOC packets. If the space is entirely allocated for ISOC, the non-ISOC packets will be completely dropped. To help you decide during the time of configuring the core, refer to the section "Device-Mode Receive Path" in Chapter 3.

11.1.3.1.50 Global Core Control Register (GCTL)

11.1.3.1.50.1 Offset

Register	Offset
GCTL	C110h

11.1.3.1.50.2 Diagram



11.1.3.1.50.3 Fields

Field	Function
31-19 PWRDNSCALE	Power Down Scale (PwrDnScale) The USB3 suspend_clk input replaces pipe3_rx_pclk as a clock source to a small part of the USB3 core that operates when the SS PHY is in its lowest power (P3) state, and therefore does not provide a clock. The Power Down Scale field specifies how many suspend_clk periods fit into a 16 kHz clock period. When performing the division, round up the remainder. For example, when using an 8-bit/16-bit/32-bit PHY and 25-MHz Suspend clock, Power Down Scale = 25000 kHz/16 kHz = 13'd1563 (rounder up) Note: - Minimum Suspend clock frequency is 32 kHz - Maximum Suspend clock frequency is 125 MHz The LTSSM uses Suspend clock for 12-ms and 100-ms timers during suspend mode. According to the USB 3.0 specification, the accuracy on these timers is 0% to +50%. - 12 ms + 0~+50% accuracy = 18 ms (Range is 12 ms - 18 ms) - 100 ms + 0~+50% accuracy = 150 ms (Range is 100 ms - 150 ms). The suspend clock accuracy requirement is: - (12,000/62.5) * (GCTL[31:19]) * actual suspend_clk_period must be between 12,000 and 18,000 - (100,000/62.5) * (GCTL[31:19]) * actual suspend_clk_period must be between 100,000 and 150,000 For example, if your suspend_clk frequency varies from 7.5 MHz to 10.5MHz, then the value needs to programmed is: Power Down Scale = 10500/16 = 657 (rounded up; and fastest frequency used).
18 MASTERFILTBPASS	Master Filter Bypass When this bit is set to 1'b1, irrespective of the parameter `DWC_USB3_EN_BUS_FILTERS chosen, all the filters in the DWC_usb3_filter module are bypassed. The double synchronizers to mac_clk preceding the filters are also bypassed. For enabling the filters, this bit must be 1'b0.
17 BYPSSSETADDR	Bypass SetAddress in Device Mode. When BYPSSSETADDR bit is set, the device core uses the value in the DCFG[DevAddr] bits directly for comparing the device address in the tokens. For simulation, you can use this feature to avoid sending an actual SET ADDRESS control transfer on the USB, and make the device core respond to a new address. When the xHCI Debug capability is enabled and this bit is set, the Debug Target immediately enters the configured state without requiring the Debug Host to send a

Table continues on the next page...

Field	Function
	SetAddress or SetConfig request. Note: You can set this bit for simulation purposes only. In the actual hardware, this bit must be set to 1'b0.
16 U2RSTECN	U2RSTECN If the SuperSpeed connection fails during POLL or LMP exchange, the device connects at non-SS mode. If this bit is set, then device attempts three more times to connect at SS, even if it previously failed to operate in SS mode. For each attempt, the device checks receiver termination eight times. From 2.60a release, this bit controls whether to check for Rx.Detect eight times or one time for every attempt. Device controller on USB 2.0 reset checks for receiver termination eight times per attempt if this bit is set to zero, or only once per attempt if the bit is set to one. Note: This bit is applicable only in device mode.
15-14 FRMSCLDWN	FRMSCLDWN This field scales down device view of a SOF/USOF/ITP duration. For SS/HS mode: - Value of 2'h3 implements interval to be 15.625 us - Value of 2'h2 implements interval to be 31.25 us - Value of 2'h1 implements interval to be 62.5 us - Value of 2'h0 implements interval to be 125us For FS mode, the scale-down value is multiplied by 8. When xHCI Debug Capability is enabled, this field also scales down the MaxPacketSize of the IN and OUT bulk endpoint to allow more traffic during simulation. It can only be changed from a non-zero value during simulation. - 2'h0: 1024 bytes - 2'h1: 512 bytes - 2'h2: 256 bytes - 2'h3: 128 bytes
13-12 PRTCAPDIR	PRTCAPDIR: Port Capability Direction (PrtCapDir) - 2'b01: for Host configurations - 2'b10: for Device configurations - 2'b11: for OTG configurations When DWC_USB3_EN_OTG is 0, then the core acts as a DRD. When DWC_USB3_EN_OTG is 1, - If PrtCapDir is 2'b11, it acts as an OTG 2.0 device with A-device or B-device determined by the IDDIG input, and Host or Peripheral role based on HNP. - If PrtCapDir is 2'b01, it acts as a DRD in host mode. - If PrtCapDir is 2'b10, it acts as a DRD in device mode. The OTG device can be programmed to enable/disable SRP and HNP by using the fields present in OCFG register. When DWC_USB3_EN_OTG is 2, - If PrtCapDir is 2'b11, it acts as an OTG 3.0 device with A-device or B-device determined by the IDDIG input, and Host or Peripheral role based on HNP/RSP. - If PrtCapDir is 2'b01, it acts as a DRD in host mode. In this mode, the VBUS is driven immediately after power-on reset. When the DWC_usb3 controller is plugged into a PC and the application configures the device later on, VBUS is driven on the USB from both sides of the cable. This will potentially damage the PC host. - If PrtCapDir is 2'b10, it acts as a DRD in device mode. The OTG device can be programmed to enable/disable SRP and HNP/RSP using the fields present in the OCFG register. Note: For static Host-only/Device-only applications, use DRD Host or DRD Device mode. The combination of GCTL.PrtCapDir=2'b11 with SRP and HNP/RSP disabled is not recommended for these applications. The sequence for switching modes in DRD configuration is as follows: Switching from Device to Host: 1. Reset the controller using GCTL[11] (CoreSoftReset). 2. Set GCTL[13:12] (PrtCapDir) to 2'b01 (Host mode). 3. Reset the host using USBCMD.HCRESET. 4. Follow the steps in "Initializing Host Registers" section of the databook. Switching from Host to Device: 1. Reset the controller using GCTL[11] (CoreSoftReset). 2. Set GCTL[13:12] (PrtCapDir) to 2'b10 (Device mode). 3. Reset the device by setting DCTL[30] (CSftRst). 4. Follow the steps in "Register Initialization" section of the databook. Programming this field with random data causes the core to keep toggling between the host mode and the device mode. Bit Bash register testing is not recommended.
11 CORESOFTRE SET	Core Soft Reset (CoreSoftReset) - 1'b0 - No soft reset - 1'b1 - Soft reset to core Clears the interrupts and all the CSRs except the following registers: - GCTL - GUCTL - GSTS - GSNPSID - GGPIIO - GUID - GUSB2PHYCFGn registers - GUSB3PIPECTLn registers - DCFG - DCTL - DEVTEN - DSTS When you reset PHYs (using GUSB3PHYCFG or GUSB3PIPECTL registers), you must keep the core in reset state until PHY clocks are stable. This controls the bus, ram, and mac domain resets. Refer to the "Reset Generation" section in the Architecture Details chapter in the DesignWare Cores SuperSpeed USB 3.0 Controller Databook. Note: This bit is for debug purposes only. Use USBCMD.HCRESET in xHCI Mode and DCTL.SoftReset in device mode for soft reset. Programming this field with random data will reset the internal logic of the host controller. Due to this side effect Bit Bash register testing is not recommended.
10 SOFITPSYNC	SOFITPSYNC If this bit is set to '0' operating in host mode, the core keeps the UTMI/ULPI PHY on the first port in a non-suspended state whenever there is a SuperSpeed port that is not in Rx.Detect, SS.Disable and U3. If this bit is set to '1' operating in host mode, the core keeps the UTMI/ULPI PHY on the first port in a non-suspended state whenever the other non-SuperSpeed ports are not in a suspended state. This feature is useful because it saves power by suspending UTMI/ULPI when SuperSpeed only is active, and it helps resolve when the PHY does not transmit a host resume unless it is placed in suspend state. This bit must be programmed as a part of initialization at power-on reset, and must not be

Table continues on the next page...

Field	Function
	dynamically changed afterwards. Note: - USB2PHYCFGn[6].PhySusp eventually decides to put the UTMI/ULPI PHY in to suspend state. In addition, when this bit is set to '1', the core generates ITP from the ref_clk based counter. Otherwise, ITP and SOF are generated from utmi/ulpi_clk[0] based counter. To program the reference clock period inside the core, refer to GUCTL[31:22].REFCLKPER. - This feature is valid in Host and DRD/OTG configurations and used only in Host mode operation. - If you never use this feature or the GFLADJ.GFLADJ_REFCLK_LPM_SEL, the minimum frequency for the ref_clk can be as low as 32KHz. You can connect the suspend_clk (as low as 32 KHz) to the ref_clk. - If you plan to enable hardware-based LPM or software-based LPM (PORTPMSC. HLE=1), then you cannot use this feature. Turn off this feature by setting this bit to '0' and use the GFLADJ.GFLADJ_REFCLK_LPM_SEL feature. - If you set this bit to '1', the GUSB2PHYCFG.U2_FREECLK_EXISTS bit and the DWC_USB3_FREECLK_USB2_EXIST parameter must be set to '0'. Program this bit to 0 if the core is intended to be operated in USB 3.0 mode.
9 U1U2TimerScale	Disable U1/U2 timer Scaledown (U1U2TimerScale). If set to '1' along with GCTL[5:4] (ScaleDown) = 2'bX1, disables the scale down of U1/U2 inactive timer values. This is for simulation mode only.
8 DEBUGATTACH	Debug Attach When this bit is set, - SS Link proceeds directly to the Polling link state (after RUN/STOP in the DCTL register is asserted) without checking remote termination; - Link LFPS polling timeout is infinite; - Polling timeout during TS1 is infinite (in case link is waiting for TXEQ to finish).
7-6 RAMCLKSEL	RAM Clock Select (RAMClkSel) - 2'b00: bus clock - 2'b01: pipe clock (Only used in device mode) - 2'b10: In device mode, pipe/2 clock. In Host mode, controller switches ram_clk between pipe/2 clock, mac2_clk and bus_clk based on the status of the U2/U3 ports - 2'b11: In device mode, selects mac2_clk as ram_clk (when 8-bit UTMI or ULPI used. Not supported in 16-bit UTMI mode) In Host mode, controller switches ram_clk between pipe_clk, mac2_clk and bus_clk based on the status of the U2/U3 ports. In device mode, upon a USB reset and USB disconnect, the hardware clears these bits to 2'b00. For more information on how to select the RAM clock, see "Clock Generation and Clock Tree Synthesis (CTS) Requirements" in the Architectural Details chapter in the DesignWare Cores SuperSpeed USB 3.0 Controller Databook. Note: - In device mode, if you set RAMClkSel to 2'b11 (mac2_clk), the controller internally switches the ram_clk to bus_clk when the link state changes to Suspend (L2 or L3), and switches the ram_clk back to mac2_clk when the link state changes to resume or U2. - In host mode, if a value of 2/3 is chosen, then controller switches ram_clk between bus_clk, mac2_clk and pipe_clk, pipe_clk/2, based on the state of the U2/U3 ports. For example if only the U2 port is active and the U3 ports are suspended, then the ram_clk is switched to mac2_clk. When only the U3 ports are active and the U2 ports are suspended, then the core internally switch the ram_clk to pipe3 clock and when all U2 and U3 ports are suspended, it switch the ram_clk to bus_clk. This allows de coupling the ram_clk from the bus_clk and depending on the bandwidth requiredmnet allows the bus_clk to be run at a lower frequency than the ram_clk requirements. bus_clk frequency still cannot go below 60Mhz in host mode, and this is not verified. A value of 2 can be chosen only if the pipe data width is 8 or 16 bits. In this case the when the ram_clk is switched to pipe_clk, it uses pipe_clk/2 instead of pipe_clk. If a value of 3 is chosen for RAMClkSel, then when ram_clk is switched to pipe_clk, then pipe_clk is used without any divider. - In device mode, when RAMClkSel != 2'b00, the bus_clk_early frequency can be a minimum of 1 MHz. This is tested in simulation and also in hardware with Linux, Microsoft Windows 8, and MCCI Windows7 host drivers. Only control and non periodic transfers are supported when bus_clk is 1 MHz. For periodic applications, the bus_clk_early minimum frequency is higher depending on your application and SoC bus. Even though 1 MHz has been tested with standard host drivers, Synopsys recommends 5 MHz minimum for ASIC designs to provide a margin or at least have a backup option to increase the bus_clk frequency to 5 MHz if needed. Programming this field with random data will cause side effect. Bit Bash register testing is not recommended.
5-4 SCALEDOWN	Scale-Down Mode (ScaleDown) When Scale-Down mode is enabled for simulation, the core uses scaled-down timing values, resulting in faster simulations. When Scale-Down mode is disabled, actual timing values are used. This is required for hardware operation. HS/FS/LS Modes - 2'b00: Disables all scale-downs. Actual timing values are used. - 2'b01: Enables scale-down of all timing values except Device mode suspend and resume. These include Speed enumeration, HNP/SRP, and Host mode suspend and resume - 2'b10: Enables scale-down of Device mode suspend and resume timing values only. - 2'b11: Enables bit 0 and bit 1 scale-down timing values. SS Mode - 2'b00: Disables all scale-downs. Actual timing values are used. - 2'b01: Enables scaled down SS timing and repeat values including: (1) Number

Table continues on the next page...

Universal Serial Bus Controller (USB)

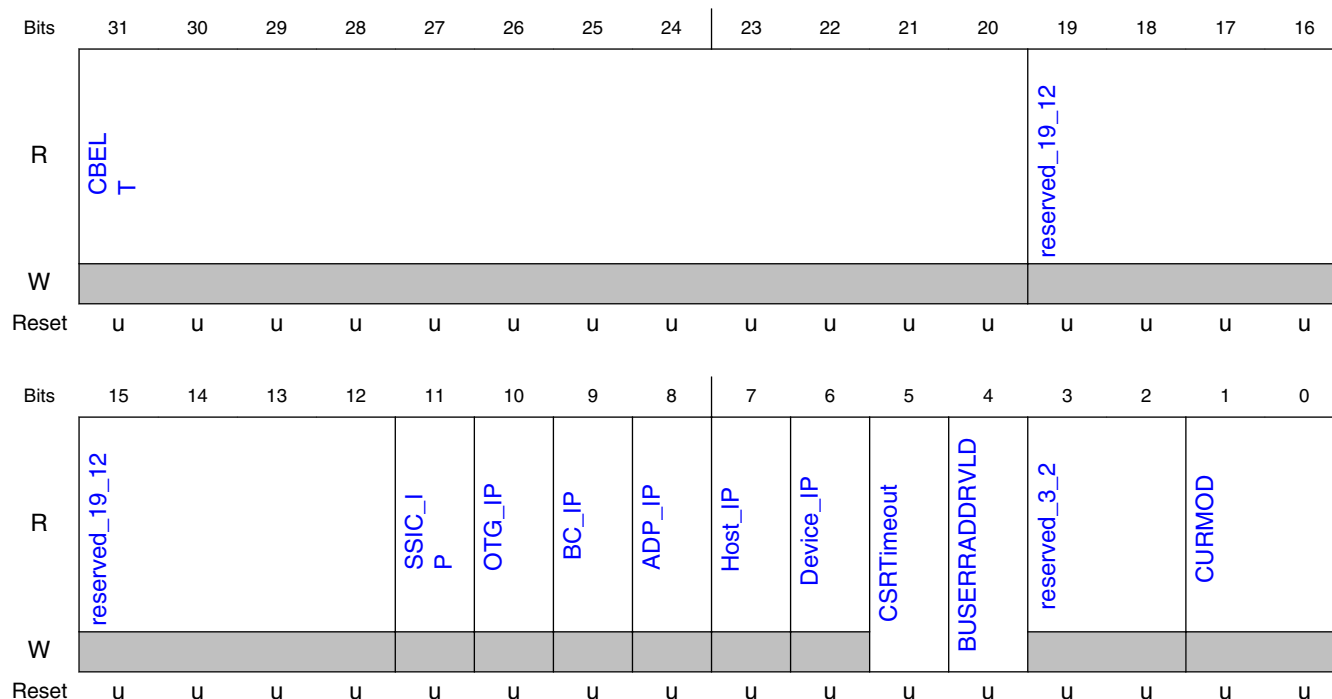
Field	Function
	of TxEq training sequences reduce to 8; (2) LFPS polling burst time reduce to 256 nS; (3) LFPS warm reset receive reduce to 30 uS. Refer to the rtl_vip_scaledown_mapping.xls file under <workspace>/sim/ SoC_sim directory for the complete list. - 2'b10: No TxEq training sequences are sent. Overrides Bit 4. - 2'b11: Enables bit 0 and bit 1 scale-down timing values.
3 DISSCRAMBLE	Disable Scrambling (DisScramble) Transmit request to Link Partner on next transition to Recovery or Polling.
2 U2EXIT_LFPS	U2EXIT_LFPS If this bit is, - 0: the link treats 248ns LFPS as a valid U2 exit. - 1: the link waits for 8us of LFPS before it detects a valid U2 exit. This bit is added to improve interoperability with a third-party host/device controller. This host/device controller in U2 state while performing receiver detection generates an LFPS glitch of about 4ms duration. This causes the host/device to exit from U2 state because the LFPS filter value is 248ns. With the new functionality enabled, the host/device can stay in U2 while ignoring this glitch from the host/device controller. This bit is applicable for both host and device controller. This bit is added to improve interoperability with a third party host controller. This host controller in U2 state while performing receiver detection generates an LFPS glitch of about 4ms duration. This causes the device to exit from U2 state because the LFPS filter value is 248ns. With the new functionality enabled, the device can stay in U2 while ignoring this glitch from the host controller.
1 GblHibernationEn	GblHibernationEn This bit enables hibernation at the global level. If hibernation is not enabled through this bit, the PMU immediately accepts the D0->D3 and D3->D0 power state change requests, but does not save or restore any core state. In addition, the PMUs never drive the PHY interfaces and let the core continue to drive the PHY interfaces.
0 DSBLCLKGTNG	Disable Clock Gating (DsbClkGtng) This bit is set to 1 and the core is in Low Power mode, internal clock gating is disabled. You can set this bit to 1'b1 after Power On Reset.

11.1.3.1.51 Global Status Register (GSTS)

11.1.3.1.51.1 Offset

Register	Offset
GSTS	C118h

11.1.3.1.51.2 Diagram



11.1.3.1.51.3 Fields

Field	Function
31-20 CBELT	Current BELT Value In Host mode, this field indicates the minimum value of all received device BELT values and the BELT value that is set by the Set Latency Tolerance Value command.
19-12 reserved_19_12	Reserved
11 SSIC_IP	SSIC interrupt pending (SSIC_IP) This field indicates that there is a pending interrupt related to SSIC in the SEVT register. Note: When the DWC_USB3_NUM_SSIC_PORTS parameter is set to zero, this bit is reserved.
10 OTG_IP	OTG Interrupt Pending This field indicates that there is a pending interrupt pertaining to OTG in OEVT register.
9 BC_IP	Battery Charger Interrupt Pending This field indicates that there is a pending interrupt pertaining to BC in BCEVT register.
8 ADP_IP	ADP Interrupt Pending This field indicates that there is a pending interrupt pertaining to ADP in ADPEVT register.
7 Host_IP	Host Interrupt Pending: This field indicates that there is a pending interrupt pertaining to xHC in the Host event queue.
6 Device_IP	Device Interrupt Pending This field indicates that there is a pending interrupt pertaining to peripheral (device) operation in the Device event queue.

Table continues on the next page...

Universal Serial Bus Controller (USB)

Field	Function
5 CSRTIMEOUT	CSR Timeout When this bit is 1'b1, it indicates that the software performed a write or read to a core register that could not be completed within `DWC_USB3_CSR_ACCESS_TIMEOUT bus clock cycles (default: h1FFFF).
4 BUSERRADDR VLD	Bus Error Address Valid (BusErrAddrVld) Indicates that the GBUSERRADDR register is valid and reports the first bus address that encounters a bus error. Note: Only supported in AHB and AXI configurations.
3-2 reserved_3_2	Reserved
1-0 CURMOD	Current Mode of Operation (CurMod) Indicates the current mode of operation: - 2'b00: Device mode - 2'b01: Host mode

11.1.3.1.52 (GUCTL1)

11.1.3.1.52.1 Offset

Register	Offset
GUCTL1	C11Ch

11.1.3.1.52.2 Function

Global User Control Register 1

11.1.3.1.52.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DEV_DECOUPLE_L1L2_EVT	DS_RXDET_MAX_TOUT_CTRL	FILTER_SE0_FSLs_EOP	TX_IPGAP_LINECHECK_DIS	DEV_TRB_OUT_SPR_IND	DEV_FORCE_20_CLK_FOR_30_CLK	P3_IN_U2	DEV_L1_EXIT_BY_HW	IP_GAP_ADD_ON			DEV_LSP_TAIL_LOCK_DIS	NAK_PER_ENH_FS	NAK_PER_ENH_HS	PARKMODE_DISABLE_SS	PARKMODE_DISABLE_HS
W																
Reset	u	u	u	u	u	u	u	u	u	u	u	u	u	u	u	u

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PARKMODE_DISABLE_FSLs	reserved_14_11				RESUME_OPMODE_HS_HOST	DEV_HS_NYET_BULK_SP	L1_SUSP_THRLD_EN_FOR_HOST	L1_SUSP_THRLD_FOR_HOST				HC_ERRATA_ENABLE	HC_PARCHK_DISABLE	OVRD_L1_SUSP_COM	LOA_FILTER_EN
W																
Reset	u	u	u	u	u	u	u	u	u	u	u	u	u	u	u	u

11.1.3.1.52.4 Fields

Field	Function
31 DEV_DECOUPLE_L1L2_EVT	DEV_DECOUPLE_L1L2_EVT - 0: Default behavior, no change in device events L1/L2U3 events are not decoupled (old behavior of v2.90a and before) - 1: Feature enabled, L1 and L2 events are separated when operating in 2.0 mode. Separate event enable bits for L1 suspend and wake events. This bit is applicable for device mode only. If this feature is enabled, L1 suspend and wake events have individual controls to enable/mask them. Enable this feature if you want to get L1 (LPM) events separately and not combined with L2 events when operating in 2.0 speeds.
30 DS_RXDET_MAX_TOUT_CTRL	DS_RXDET_MAX_TOUT_CTRL This bit is used to control the tRxDetectTimeoutDFP timer for the SuperSpeed link. - 0: Default behavior; 12ms is used as tRxDetectTimeoutDFP. - 1: 120ms is used as the tRxDetectTimeoutDFP. This bit is used only in host mode. For more details, refer to ECN020 for USB 3.0 Specification.
29 FILTER_SE0_FSLs_EOP	FILTER_SE0_FSLs_EOP - 0: Default behavior, no change in Linestate check for SE0 detection in FS/LS - 1: Feature enabled, FS/LS SE0 is filtered for 2 clocks for detecting EOP This bit is applicable for FS/LS operation. If this feature is enabled, then SE0 on the linestate is validated for 2 consecutive utmi/ulpi

Table continues on the next page...

Field	Function
FILTER_SE0_F SLS_EOP	clock edges for EOP detection. This feature is applicable only in FS in device mode and FS/LS mode of operation in host mode. Device mode: FS - If GUCTL1.FILTER_SE0_FSL_SLS_EOP is set, then for device LPM handshake, the core will ignore single SE0 glitch on the linestate during transmit. Only 2 or more SE0 is considered as a valid EOP on FS. Host mode: FS/LS - If GUCTL1.FILTER_SE0_FSL_SLS_EOP is set, then the core will ignore single SE0 glitch on the linestate during transmit. Only 2 or more SE0 is considered as a valid EOP on FS/LS port. Enable this feature if the LineState has SE0 glitches during transmission. This bit is quasi-static, that is, it must not be changed during device operation.
28 TX_IPGAP_LIN ECHECK_DIS	TX_IPGAP_LINECHECK_DIS - 0: Default behavior, no change in Linestate check - 1: Feature enabled, 2.0 MAC disables Linestate check during HS transmit This bit is applicable for HS operation of u2mac. If this feature is enabled, then the 2.0 mac operating in HS ignores the UTMI/ULPI Linestate during the transmit of a token (during token-to-token and token-to-data IPGAP). When enabled, the controller implements a fixed 40-bit TxEndDelay after the packet is given on UTMI and ignores the Linestate during this time. This feature is applicable only in HS mode of operation. Device mode: If GUCTL1.TX_IPGAP_LINECHECK_DIS is set, then for device LPM handshake, the core will ignore the linestate after TX and wait for fixed clocks (40 bit times equivalent) after transmitting ACK on utmi. Host mode: If GUCTL1.TX_IPGAP_LINECHECK_DIS is set, then the ipgap between (tkn to tkn/data) is added by 40 bit times of TXENDDELAY, and linestate is ignored during this 40 bit times delay. Enable this bit if the LineState will not reflect the expected line state (J) during transmission. This bit is quasi-static, that is, it must not be changed during device operation.
27 DEV_TRB_OUT _SPR_IND	DEV_TRB_OUT_SPR_IND - 0: Default behavior, no change in TRB status dword - 1: Feature enabled, OUT TRB status indicates Short Packet This bit is applicable for device mode only (and ignored in host mode). If the device application (software/hardware) wants to know if a short packet was received for an OUT in the TRB status itself, then this feature can be enabled, so that a bit is set in the TRB writeback in the buf_size dword. Bit[26] - SPR of the {trbstatus, RSVD, SPR, PCM1, bufsize} dword will be set during an OUT transfer TRB write back if this is the last TRB used for that transfer descriptor. This bit is quasi-static, that is, it must not be changed during device operation.
26 DEV_FORCE_2 0_CLK_FOR_30 _CLK	DEV_FORCE_20_CLK_FOR_30_CLK - 0: Default behavior, Uses 3.0 clock when operating in 2.0 mode - 1: Feature enabled This bit is applicable (and to be set) for device mode (DCFG.Speed != SS) only. In the 3.0 device core, if the core is programmed to operate in 2.0 only (that is, Device Speed is programmed to 2.0 speeds in DCFG[Speed]), then setting this bit makes the internal 2.0 (utmi/ulpi) clock to be routed as the 3.0 (pipe) clock. Enabling this feature allows the pipe3 clock to be not-running when forcibly operating in 2.0 device mode. Note: - When using this feature, all pipe3 inputs must be in inactive mode. In particular, the pipe3 clocks must not be running and the pipe3_phystatus_async must be tied to 0. This bit should not be set if the core is programmed to operate in SuperSpeed mode (even when it falls back to 2.0). - This bit is quasi-static, that is, it must not be changed during operation. - If the parameter "DWC_USB3_REMOVE_PIPE_CLK_MUX_FOR_20_MODE" is enabled, then muxing 2.0 clock/signals to pipe_clk/signals have to be done outside the controller for this feature to work.
25 P3_IN_U2	P3_IN_U2 - 0: Default behavior, When SuperSpeed link is in U2, PowerState P2 is attempted on the PIPE Interface. - 1: When SuperSpeed link is in U2, PowerState P3 is attempted if GUSB3PIPECTL[17] is set. Setting this bit enables P3 Power State when the SuperSpeed link is in U2. Another Power Saving option. Check with your PHY vendor before enabling this option. When setting this bit to 1 to enable P3 in P2, GUSB3PIPECTL[27] should be set to 0 to make sure that the U2 exit is attempted in P0. This bit should be set only when GCTL.SOFITPSYNC=1 or GFLADJ.GFLADJ_REFCLK_LPM_SEL=1.
24 DEV_L1_EXIT_ BY_HW	DEV_L1_EXIT_BY_HW - 0: Default behavior, disables device L1 hardware exit logic - 1: feature enabled This bit is applicable for device mode (2.0) only. This field enables device controller sending remote wakeup for L1 if the device becomes ready for sending/accepting data when in L1 state. If the host expects the device to send remote wakeup signaling to resume after going into L1 in flow controlled state, then this bit can be set to send the remote wake signal automatically when the device controller becomes ready. This hardware remote wake feature is applicable only to bulk and interrupt transfers, and not for Isoch/Control - When control transfers are in progress, the LPM will be rejected (NYET response). Only after control transfers are completed (either with ACK/STALL), LPM will be accepted - For Isoch transfers, the host needs to do the wake-up and start the transfer. Device controller will not do remote-wakeup when Isoch endpoints get ready. The device SW needs to keep the GUSB2PHYCFG[EnbSlpM] reset in order to keep the PHY clock to be running for keeping track of SOF intervals. - When L1

Table continues on the next page...

Field	Function
	hibernation is enabled, the controller will not do automatic exit for hibernation requests thru L1. This bit is quasi-static, that is, it must not be changed during device operation.
23-21 IP_GAP_ADD_ON	This register field is used to add on to the default inter packet gap setting in the USB 2.0 MAC. This should be programmed to a non zero value only in case where you need to increase the default inter packet delay calculations in the USB 2.0 MAC module DWC_usb3_u2mac.v
20 DEV_LSP_TAIL_LOCK_DIS	DEV_LSP_TAIL_LOCK_DIS - 0: Default behavior, enables device lsp lock logic for tail TRB update - 1: Fix disabled This is a bug fix for STAR 9000716195 that affects the CSP mode for OUT endpoints in device mode. The issue is that tail TRB index is not synchronized with the cache Scratchpad bytecount update. If the fast-forward request comes in-between the bytecount update on a newly fetched TRB and the tail-index write update in TPF, the RDP works on an incorrect tail index and misses the byte count decrement for the newly fetched TRB in the fast-forwarding process. This fix needs to be present all the times.
19 NAK_PER_ENH_FS	NAK_PER_ENH_FS - 1: Enables performance enhancement for FS async endpoints in the presence of NAKs - 0: Enhancement not applied If a periodic endpoint is present , and if a bulk endpoint which is also active is being NAKed by the device, then this could result in a decrease in performance of other Full Speed bulk endpoint which is ACKed by the device. Setting this bit to 1, will enable the host controller to schedule more transactions to the async endpoints (bulk/ control) and hence will improve the performance of the bulk endpoint. This control bit should be enabled only if the existing performance with the default setting is not sufficient for your FullSpeed application. Setting this bit will only control, and is only required for Full Speed transfers.
18 NAK_PER_ENH_HS	NAK_PER_ENH_HS - 1: Enables performance enhancement for HS async endpoints in the presence of NAKs - 0: Enhancement not applied If a periodic endpoint is present, and if a bulk endpoint which is also active is being NAKed by the device, then this could result in decrease in performance of other High Speed bulk endpoint which is ACKed by the device. Setting this bit to 1, will enable the host controller to schedule more transactions to the async endpoints (bulk/ control) and hence will improve the performance of the bulk endpoint. This control bit should be enabled only if the existing performance with the default setting is not sufficient for your HighSpeed application. Setting this bit will only control, and is only required for High Speed transfers.
17 PARKMODE_DISABLE_SS	PARKMODE_DISABLE_SS This bit is used only in host mode, and is for debug purpose only. When this bit is set to '1' all SS bus instances in park mode are disabled.
16 PARKMODE_DISABLE_HS	PARKMODE_DISABLE_HS This bit is used only in host mode. When this bit is set to '1' all HS bus instances park mode are disabled. To improve performance in park mode, the xHCI scheduler queues in three requests of 4 packets each for High Speed asynchronous endpoints in a micro-frame. But if a device is slow and if it NAKs more than 3 times, then it is rescheduled only in the next micro-frame. This could decrease the performance of a slow device even further. In a few high speed devices (such as Sandisk Cruzer Blade 4GB VID:1921, PID:21863 and Flex Drive VID:3744, PID:8552) when an IN request is sent within 900ns of the ACK of the previous packet, these devices send a NAK. When connected to these devices, if required, the software can disable the park mode if you see performance drop in your system. When park mode is disabled, pipelining of multiple packet is disabled and instead one packet at a time is requested by the scheduler. This allows up to 12 NAKs in a micro-frame and improves performance of these slow devices.
15 PARKMODE_DISABLE_FSLs	PARKMODE_DISABLE_FSLs This bit is used only in host mode, and is for debug purpose only. When this bit is set to '1' all FS/LS bus instances in park mode disabled.
14-11 reserved_14_11	Reserved
10 RESUME_OPMODE_HS_HOST	RESUME_OPMODE_HS_HOST This bit is used only in host mode, and is for USB 2.0 opmode behavior in HS Resume. - When this bit is set to '1', the UTMI/ULPI opmode will be changed to "normal" along with HS terminations after EOR. This option is to support certain legacy UTMI/ULPI PHYs. - When this bit is set to '0', the UTMI/ULPI opmode will be changed to "normal" 2us after HS terminations change after EOR. This is the default behavior.

Table continues on the next page...

Field	Function
9 DEV_HS_NYET_BULK_SPR	DEV_HS_NYET_BULK_SPR - 0: Default behavior, no change in device response - 1: Feature enabled, HS bulk OUT short packet gets NYET response This bit is applicable for device mode only (and ignored in host mode) to be used in 2.0 operation. If this bit is set, the device core sends NYET response instead of ACK response for a successfully received bulk OUT short packet. If NYET is sent after receiving short packet, then the host would PING before sending the next OUT; this improves the performance as well as clears up the buffer/cache on the host side. Internal to the device core, short packet (SPR=1) processing takes some time, and during this time, the USB is flow controlled. With NYET response instead of ACK on short packet, the host does not send another OUT-DATA without pinging in HS mode. This bit is quasi-static, that is, it must not be changed during device operation.
8 L1_SUSP_THRLD_EN_FOR_HOST	L1_SUSP_THRLD_EN_FOR_HOST This bit is used only in host mode. The host controller asserts the utmi_l1_suspend_n and utmi_sleep_n output signals (see "LPM Interface Signals" table in the Databook) as follows: The controller asserts the utmi_l1_suspend_n signal to put the PHY into deep low-power mode in L1 when both of the following are true: - The HIRD/BESL value used is greater than or equal to the value in L1_SUSP_THRLD_FOR_HOST field. - The L1_SUSP_THRLD_EN_FOR_HOST bit is set to 1'b1. The controller asserts utmi_sleep_n on L1 when one of the following is true: - The HIRD/BESL value used is less than the value in L1_SUSP_THRLD_FOR_HOST field. - The L1_SUSP_THRLD_EN_FOR_HOST bit is set to 1'b0.
7-4 L1_SUSP_THRLD_FOR_HOST	L1_SUSP_THRLD_FOR_HOST This field is effective only when the L1_SUSP_THRLD_EN_FOR_HOST bit is set to 1. For more details, refer to the description of the L1_SUSP_THRLD_EN_FOR_HOST bit.
3 HC_ERRATA_ENABLE	Host ELD Enable (HELDn) When this bit is set to 1, it enables the Exit Latency Delta (ELD) support defined in the xHCI 1.0 Errata. This bit is used only in the host mode. This bit has to be set to 1 in Host mode.
2 HC_PARCHK_DISABLE	Host Parameter Check Disable (HParChkDisable) When this bit is set to '0' (by default), the xHC checks that the input slot/EP context fields comply to the xHCI Specification. Upon detection of a parameter error during command execution, the xHC generates an event TRB with completion code indicating 'PARAMETER ERROR'. When the bit is set to '1', the xHC does not perform parameter checks and does not generate 'PARAMETER ERROR' completion code.
1 OVRD_L1_SUSP_COM	OVRD_L1_SUSP_COM If this bit is set, the utmi_l1_suspend_com_n is overloaded with the utmi_sleep_n signal. This bit is usually set if the PHY stops the port clock during L1 sleep condition. Note: The recommended connection for the SUSPENDM/SLEEPN signals to the PHY with respect to this bit is as follows. For non-zero ports: Connect: - utmi_sleep_n[n] to SLEEPN[n] - (utmi_suspend_n[n] & utmi_l1_suspend_n[n]) to SUSPENDM[n] - USB2 PHYCLK[n] to utmi_clk[n] GUCTL1.OVRD_L1_SUSP_COM impacts only Port0. For Port0: For Synopsys PHY, GUSB2PHYCFGn.U2_FREECLK_EXISTS=1; With this connection, the PHY keeps PLL active so that FREECLK is always available irrespective of suspend/sleep. - Connect USB2 PHY COMMONONN to 0. - Connect utmi_sleep_n[0] to SLEEPN[0]. - Connect (utmi_suspend_n[0] & utmi_l1_suspend_n[0]) to SUSPENDM[0]. - Connect USB2 PHY FREECLK to utmi_clk[0]. - Leave utmi_suspend_com_n, utmi_l1_suspend_com_n unconnected. - GUCTL1.OVRD_L1_SUSP_COM can be set to any value. For Third Party PHY, GUSB2PHYCFGn.U2_FREECLK_EXISTS=0; With this connection the PHY can shut off all the clocks when the required conditions are met (like, GUSB2PHYCFGn[8,6], GUCTL1[1], GFLADJ[23], GCTL[10], Suspend condition, HW LPM enable etc). - Connect -utmi_suspend_com_n to SUSPENDM[0] (or equivalent). - Connect -utmi_l1_suspend_com_n to SLEEPN[0] (or equivalent). - Connect PHYCLK0 (first port clock) to utmi_clk[0]. - Leave utmi_suspend_n[0], utmi_l1_suspend_n[0], utmi_sleep_n[0] unconnected. - Set GUCTL1.OVRD_L1_SUSP_COM to 1'b1.
0 LOA_FILTER_EN	LOA_FILTER_EN If this bit is set, the USB 2.0 port babble is checked at least three consecutive times before the port is disabled. This prevents false triggering of the babble condition when using low quality cables. Note: This bit is valid only in host mode.

11.1.3.1.53 Global User ID Register (GUID)

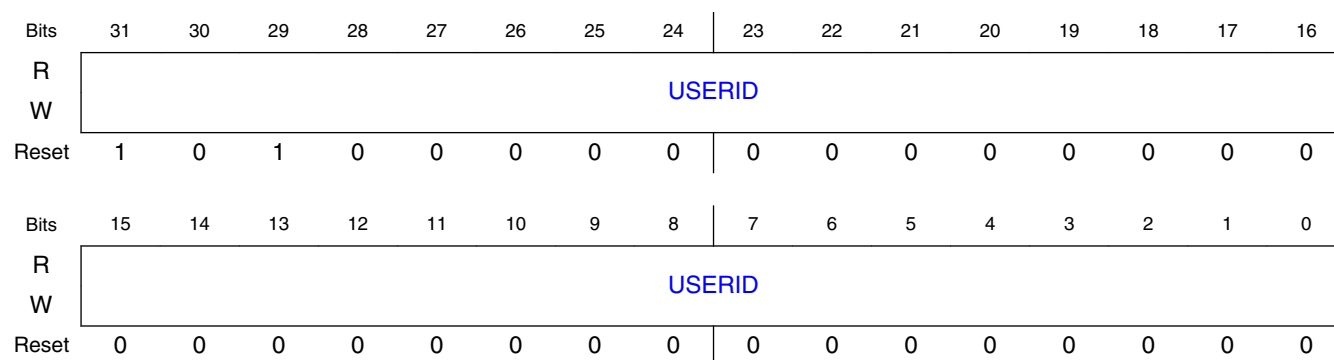
11.1.3.1.53.1 Offset

Register	Offset
GUID	C128h

11.1.3.1.53.2 Function

This is a read/write register containing the User ID.

11.1.3.1.53.3 Diagram



11.1.3.1.53.4 Fields

Field	Function
31-0 USERID	USERID Application-programmable ID field.

11.1.3.1.54 Global User Control Register (GUCTL)

11.1.3.1.54.1 Offset

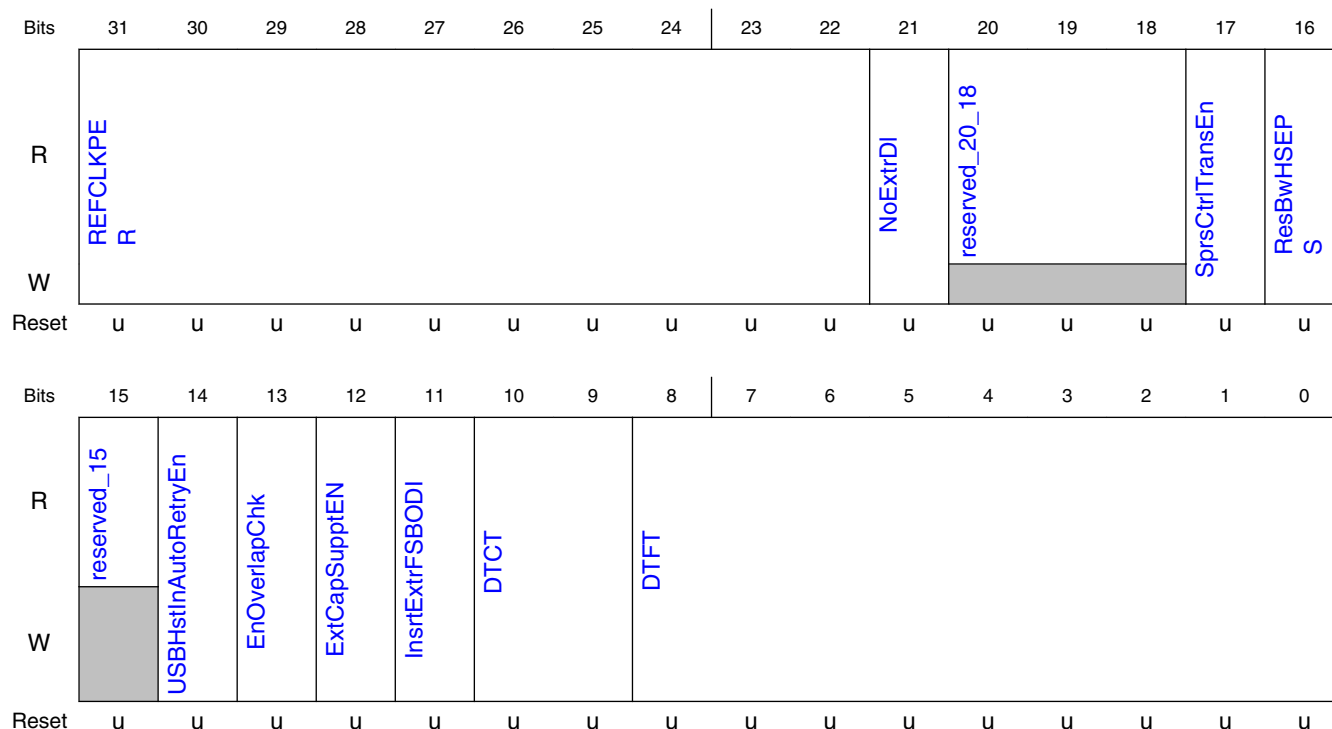
Register	Offset
GUCTL	C12Ch

11.1.3.1.54.2 Function

This register provides a few options for the software to control the core behavior in the Host mode.

Reset Mask: 0x1C8000

11.1.3.1.54.3 Diagram



11.1.3.1.54.4 Fields

Field	Function
31-22 REFCLKPER	REFCLKPER This field indicates in terms of nano seconds the period of ref_clk. The default value of this register is set to 'h8 (8ns/125 MHz). This field needs to be updated during power-on initialization, if GCTL.SOFITPSYNC or GFLADJ.GFLADJ_REFCLK_LPM_SEL is set to '1'. The programmable maximum value is 62ns, and the minimum value is 8ns. You must use a reference clock with a period that is an integer multiple, so that ITP can meet the jitter margin of 32ns. The allowable ref_clk frequencies whose period is not integer multiples are 16/17/19.2/24/39.7MHz. This field must not be set to '0' at any time. If you never plan to use this feature, then set this field to 'h8, the default value.
21 NoExtrDI	No Extra Delay Between SOF and the First Packet(NoExtrDI) Some HS devices misbehave when the host sends a packet immediately after a SOF. However, adding an extra delay between a SOF and the first packet can reduce the USB data rate and performance. This bit is used to control whether the host must wait for 2 microseconds before it sends the first packet after a SOF, or not. User can set this bit to one to improve the performance if those problematic devices are not a concern in the user's host environment. - 1'b0: Host waits for 2 microseconds after a SOF before it sends the first USB packet. - 1'b1: Host doesn't wait after a SOF before it sends the first USB packet.

Table continues on the next page...

Field	Function
20-18 reserved_20_18	Reserved
17 SprsCtrlTransEn	Sparse Control Transaction Enable Some devices are slow in responding to Control transfers. Scheduling multiple transactions in one microframe/frame can cause these devices to misbehave. If this bit is set to 1'b1, the host controller schedules transactions for a Control transfer in different microframes/frames.
16 ResBwHSEPS	Reserving 85% Bandwidth for HS Periodic EPs (ResBwHSEPS) By default, HC reserves 80% of the bandwidth for periodic EPs. If this bit is set, the bandwidth is relaxed to 85% to accommodate two high speed, high bandwidth ISOC EPs. USB 2.0 required 80% bandwidth allocated for ISOC traffic. If two High-bandwidth ISOC devices (HD Webcams) are connected, and if each requires 1024-bytes X 3 packets per Micro-Frame, then the bandwidth required is around 82%. If this bit is set, then it is possible to connect two Webcams of 1024bytes X 3 payload per Micro-Frame each. Otherwise, you may have to reduce the resolution of the Webcams. This bit is valid in Host and DRD configuration and is used in host mode operation only. Ignore this bit in device mode.
15 reserved_15	Reserved
14 USBHstInAutoRetryEn	Host IN Auto Retry (USBHstInAutoRetryEn) When set, this field enables the Auto Retry feature. For IN transfers (non-isochronous) that encounter data packets with CRC errors or internal overrun scenarios, the auto retry feature causes the Host core to reply to the device with a non-terminating retry ACK (that is, an ACK transaction packet with Retry = 1 and NumP != 0). If the Auto Retry feature is disabled (default), the core will respond with a terminating retry ACK (that is, an ACK transaction packet with Retry = 1 and NumP = 0). - 1'b0: Auto Retry Disabled - 1'b1: Auto Retry Enabled Note: When enabling Auto Retry feature, if the system latency is large enough to cause the internal PSQ full (PSQ can be full as the result of messages not being processed because of pending fetches before flushing the TxQ due to NRDY/ERDY conditions), then the host controller can generate a transaction error.
13 EnOverlapChk	Enable Check for LFPS Overlap During Remote Ux Exit: If this bit is set to, - 1'b1: The SuperSpeed link when exiting U1/U2/U3 waits for either the remote link LFPS or TS1/TS2 training symbols before it confirms that the LFPS handshake is complete. This is done to handle the case where the LFPS glitch causes the link to start exiting from the low power state. Looking for the LFPS overlap makes sure that the link partner also sees the LFPS. - 1'b0: When the link exists U1/U2/U3 because of a remote exit, it does not look for an LFPS overlap.
12 ExtCapSuptEN	External Extended Capability Support Enable (ExtCapSuptEN) When set, this field enables extended capabilities to be implemented outside the core. When the ExtCapSuptEN is set and the Debug Capability is enabled, the Next Capability pointer in "Debug Capability" returns 16. A read to the first DWORD of the last internal extended capability (the "xHCI Supported Protocol Capability for USB 3.0" when the Debug Capability is not enabled) returns a value of 4 in the Next Capability Pointer field. This indicates to software that there is another capability four DWORDs after this capability (for example, at address N+16 where N is the address of this DWORD). If enabled, an external address decoder that snoops the xHC slave interface must be implemented. If it sees an access to N+16 or greater, the slave access is re-routed to a piece of hardware which returns the external capability pointer register of the new capability and also handles reads/writes to this new capability and the side effects. If disabled, a read to the first DWORD of the last internal extended capability returns 0 in the 'Next Capability Pointer field. This indicates there are no more capabilities.
11 InsrtExtrFSBODI	Insert Extra Delay Between FS Bulk OUT Transactions (InsrtExtrFSBODI). Some FS devices are slow to receive Bulk OUT data and can get stuck when there are consecutive Bulk OUT transactions with short inter-transaction delays. This bit is used to control whether the host inserts extra delay between consecutive Bulk OUT transactions to a FS Endpoint. - 1'b0: Host doesn't insert extra delay between consecutive Bulk OUT transactions to a FS Endpoint. - 1'b1: Host inserts about 12us extra delay between consecutive Bulk OUT transactions to a FS Endpoint to work around the device issue. Note: Setting this bit to one will reduce the Bulk OUT transfer performance for most of the FS devices.
10-9 DTCT	Device Timeout Coarse Tuning (DTCT) This field is a Host mode parameter which determines how long the host waits for a response from device before considering a timeout. The core first checks the DTCT value. If it is 0, then the timeout value is defined by the DTFT. If it is non-zero, then it uses the following timeout values: - 2'b00: 0 usec -> use DTFT value instead - 2'b01: 500 usec - 2'b10: 1.5 msec - 2'b11:

Table continues on the next page...

Field	Function
	6.5 msec Note: When the system latency is larger than the programmed DTCT/DTFT value, if the host controller is not able to accept certain transactions on the bus (because of system bus delays), the controller may not release header credits which in turn can cause the host to report a transaction error. Therefore, program this value to be larger than your system delay.
8-0 DTFT	Device Timeout Fine Tuning (DTFT) This field is a Host mode parameter which determines how long the host waits for a response from device before considering a timeout. For the DTFT field to take effect, DTCT must be set to 2'b00. The DTFT value is the number of 125 MHz clocks * 256 to count before considering a device timeout. The minimum value of DTFT is 2. For example, if the mac3_clk is 125 MHz clk (8 ns period), this is calculated as follows: (DTFT value) * 256 * (8 ns) Quick Reference: - if DTFT = 0x2, 2*256*8 = 4usec timeout - if DTFT = 0x5, 5*256*8 = 10usec timeout - if DTFT = 0xA, 10*256*8 = 20usec timeout - if DTFT = 0x10, 16*256*8 = 32usec timeout - if DTFT = 0x19, 25*256*8 = 51usec timeout - if DTFT = 0x31, 49*256*8 = 100usec timeout - if DTFT = 0x62, 98*256*8 = 200usec timeout Note: - When SSIC is enabled, in HS_G1_G2_G3 mode when ssic_soc_pa_clk_freq = 2'b01, then DTFT value must be calculated for 156.25 MHz clock, for example, for equivalent delay of 32usec DTFT value must be 9'h14. - When the system latency is larger than the programmed DTCT/DTFT value, if the host controller is not able to accept certain transactions on the bus (because of system bus delays), the controller may not release header credits which in turn can cause the host to report a transaction error. Therefore, program this value to be larger than your system delay.

11.1.3.1.55 GobaI SoC Bus Error Address Register - Low (GBUSERRA DDRLO)

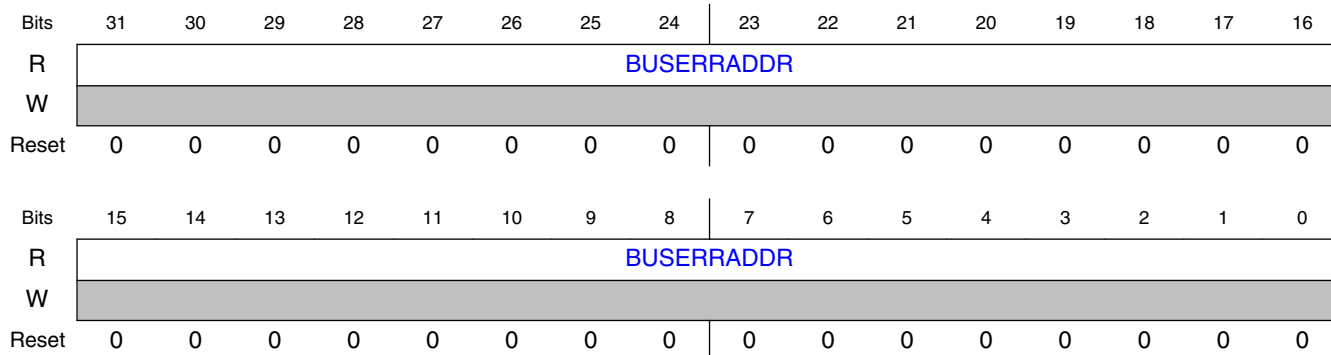
11.1.3.1.55.1 Offset

Register	Offset
GBUSERRADDRLO	C130h

11.1.3.1.55.2 Function

This is an alternate register for the GBUSERRADDR register.

11.1.3.1.55.3 Diagram



11.1.3.1.55.4 Fields

Field	Function
31-0 BUSERRADDR	Bus Address - Low (BusAddrLo) This register contains the lower 32 bits of the first bus address that encountered a SoC bus error. It is valid when the GSTS.BusErrAddrVld field is 1. It can only be cleared by resetting the core. Note: Only supported in AHB and AXI configurations.

11.1.3.1.56 Global SoC Bus Error Address Register - High (GBUSERRADDRHI)

11.1.3.1.56.1 Offset

Register	Offset
GBUSERRADDRHI	C134h

11.1.3.1.56.2 Function

This is an alternate register for the GBUSERRADDR register.

11.1.3.1.56.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	BUSERRADDR															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	BUSERRADDR															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.1.3.1.56.4 Fields

Field	Function
31-0 BUSERRADDR	Bus Address - High (BusAddrHi) This register contains the higher 32 bits of the first bus address that encountered a SoC bus error. It is valid when the GSTS.BusErrAddrVld field is 1. It can only be cleared by resetting the core. Note: Only supported in AHB and AXI configurations.

11.1.3.1.57 Global SS Port to Bus Instance Mapping Register - Low (GPRTBIMAPLO)

11.1.3.1.57.1 Offset

Register	Offset
GPRTBIMAPLO	C138h

11.1.3.1.57.2 Function

This is an alternate register for the GPRTBIMAP register.

11.1.3.1.57.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	BINUM8				BINUM7				BINUM6				BINUM5			
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	BINUM4				BINUM3				BINUM2				BINUM1			
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.1.3.1.57.4 Fields

Field	Function
31-28 BINUM8	BINUM8: SS USB Instance Number for Port 8. Application-programmable ID field.
27-24 BINUM7	BINUM7: SS USB Instance Number for Port 7. Application-programmable ID field.
23-20 BINUM6	BINUM6: SS USB Instance Number for Port 6. Application-programmable ID field.
19-16 BINUM5	BINUM5: SS USB Instance Number for Port 5. Application-programmable ID field.
15-12 BINUM4	BINUM4: SS USB Instance Number for Port 4. Application-programmable ID field.
11-8 BINUM3	BINUM3: SS USB Instance Number for Port 3. Application-programmable ID field.

Table continues on the next page...

Field	Function
7-4 BINUM2	BINUM2: SS USB Instance Number for Port 2. Application-programmable ID field.
3-0 BINUM1	BINUM1: SS USB Instance Number for Port 1. Application-programmable ID field.

11.1.3.1.58 Global SS Port to Bus Instance Mapping Register - High (GPRTBIMAPHI)

11.1.3.1.58.1 Offset

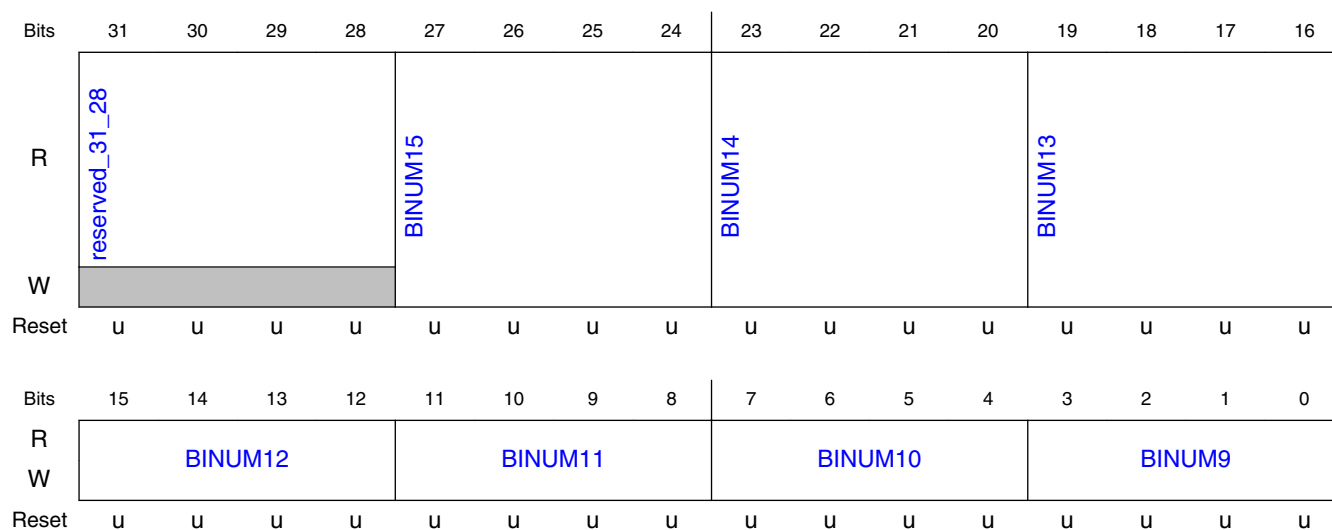
Register	Offset
GPRTBIMAPHI	C13Ch

11.1.3.1.58.2 Function

This is an alternate register for the GPRTBIMAP register.

Reset Mask: 0xF0000000

11.1.3.1.58.3 Diagram



11.1.3.1.58.4 Fields

Field	Function
31-28 reserved_31_28	Reserved
27-24 BINUM15	BINUM15: SS USB Instance Number for Port 15. Application-programmable ID field.
23-20 BINUM14	BINUM14: SS USB Instance Number for Port 14. Application-programmable ID field.
19-16 BINUM13	BINUM13: SS USB Instance Number for Port 13. Application-programmable ID field.
15-12 BINUM12	BINUM12: SS USB Instance Number for Port 12. Application-programmable ID field.
11-8 BINUM11	BINUM11: SS USB Instance Number for Port 11. Application-programmable ID field.
7-4 BINUM10	BINUM10: SS USB Instance Number for Port 10. Application-programmable ID field.
3-0 BINUM9	BINUM9: SS USB Instance Number for Port 9. Application-programmable ID field.

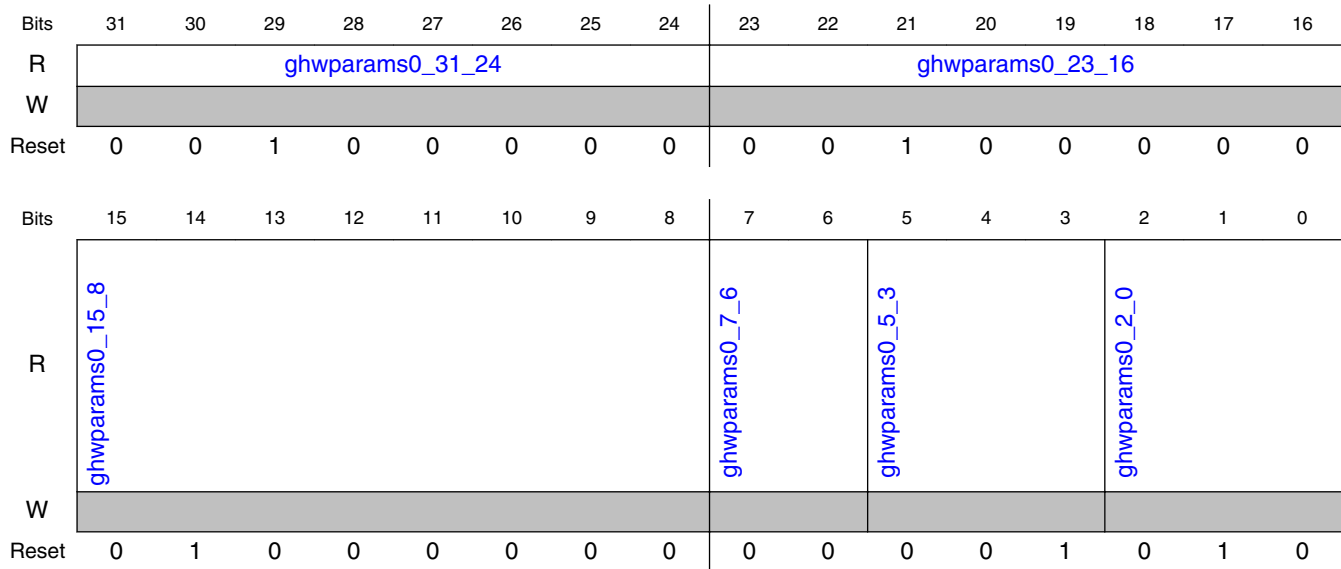
11.1.3.1.59 Global Hardware Parameters Register 0 (GHWPARAMS0)**11.1.3.1.59.1 Offset**

Register	Offset
GHWPARAMS0	C140h

11.1.3.1.59.2 Function

This register contains the hardware configuration options that you can select in the coreConsultant GUI.

11.1.3.1.59.3 Diagram



11.1.3.1.59.4 Fields

Field	Function
31-24 ghwparams0_31_24	`DWC_USB3_AWIDTH
23-16 ghwparams0_23_16	`DWC_USB3_SDWIDTH
15-8 ghwparams0_15_8	`DWC_USB3_MDWIDTH
7-6 ghwparams0_7_6	`DWC_USB3_SBUS_TYPE
5-3 ghwparams0_5_3	`DWC_USB3_MBUS_TYPE
2-0 ghwparams0_2_0	`DWC_USB3_MODE

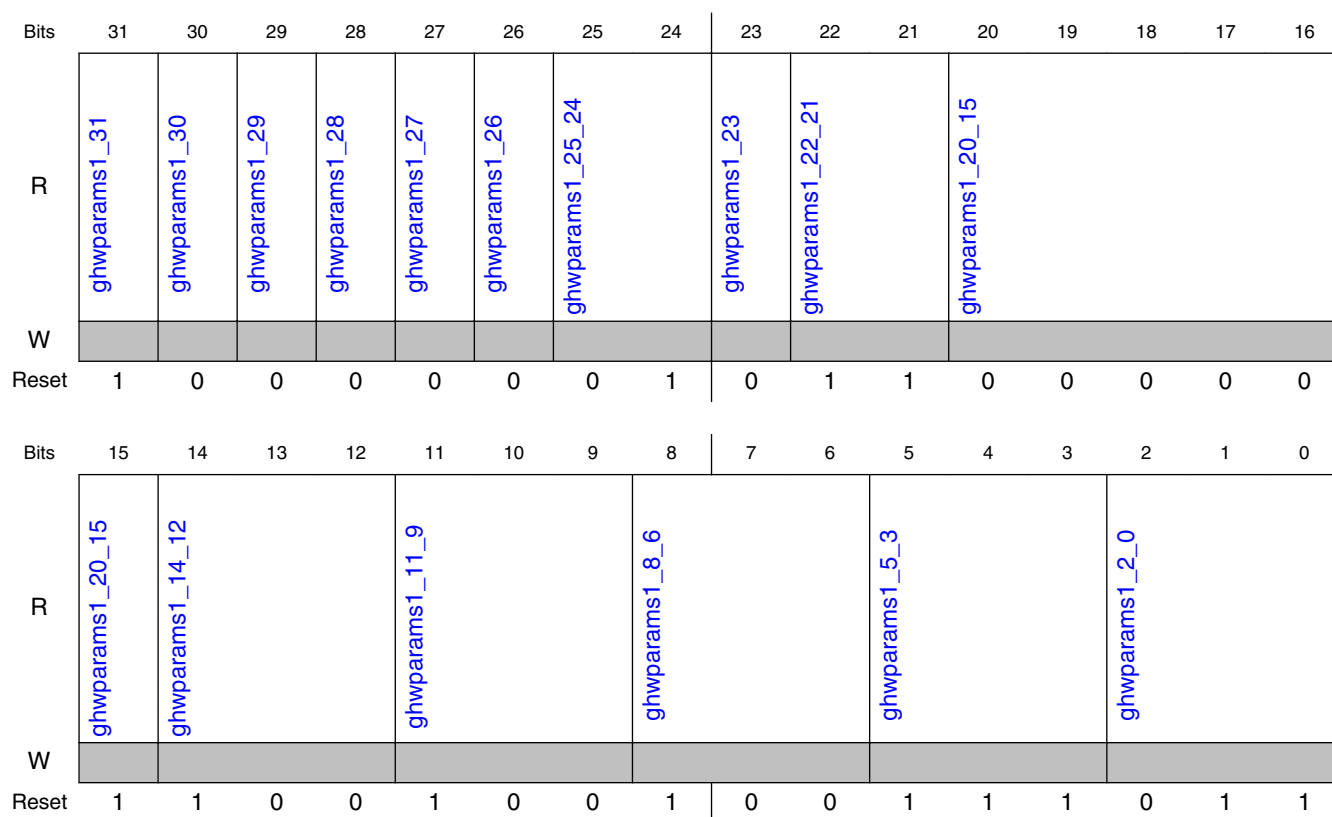
11.1.3.1.60 Global Hardware Parameters Register 1 (GHWPARAMS1)

11.1.3.1.60.1 Offset

Register	Offset
GHWPARAMS1	C144h

11.1.3.1.60.2 Function

This register contains the hardware configuration options that you can select in the coreConsultant GUI.

11.1.3.1.60.3 Diagram**11.1.3.1.60.4 Fields**

Field	Function
31 ghwparams1_31	`DWC_USB3_EN_DBC
30 ghwparams1_30	`DWC_USB3_RM_OPT_FEATURES

Table continues on the next page...

Field	Function
29 ghwparams1_29	Reserved1
28 ghwparams1_28	`DWC_USB3_RAM_BUS_CLKS_SYNC
27 ghwparams1_27	`DWC_USB3_MAC_RAM_CLKS_SYNC
26 ghwparams1_26	`DWC_USB3_MAC_PHY_CLKS_SYNC
25-24 ghwparams1_25_24	`DWC_USB3_EN_PWROPT
23 ghwparams1_23	`DWC_USB3_SPRAM_TYP
22-21 ghwparams1_22_21	`DWC_USB3_NUM_RAMs
20-15 ghwparams1_20_15	`DWC_USB3_DEVICE_NUM_INT For details on `DWC_USB3_DEVICE_NUM_INT, refer to <workspace>/src/DWC_usb3_params.v file.
14-12 ghwparams1_14_12	`DWC_USB3_ASPACEWIDTH
11-9 ghwparams1_11_9	`DWC_USB3_REQINFOWIDTH
8-6 ghwparams1_8_6	`DWC_USB3_DATAINFOWIDTH
5-3 ghwparams1_5_3	`DWC_USB3_BURSTWIDTH-1
2-0 ghwparams1_2_0	`DWC_USB3_IDWIDTH-1

11.1.3.1.61 Global Hardware Parameters Register 2 (GHWPARAMS2)

11.1.3.1.61.1 Offset

Register	Offset
GHWPARAMS2	C148h

11.1.3.1.61.2 Function

This register contains the hardware configuration options that you can select in the coreConsultant GUI.

11.1.3.1.61.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	ghwparams2_31_0															
W																
Reset	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ghwparams2_31_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.1.3.1.61.4 Fields

Field	Function
31-0 ghwparams2_31_0	`DWC_USB3_USERID

11.1.3.1.62 Global Hardware Parameters Register 3 (GHWPARAMS3)

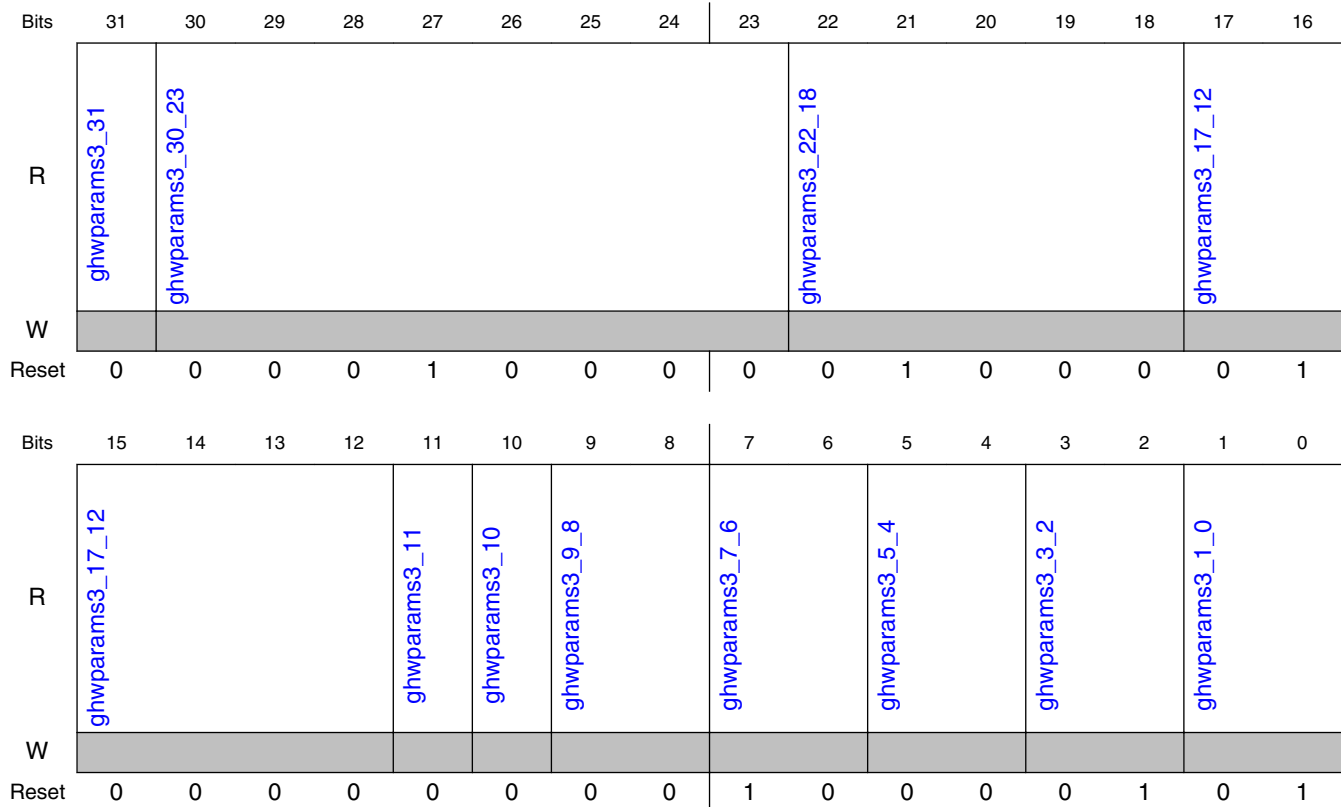
11.1.3.1.62.1 Offset

Register	Offset
GHWPARAMS3	C14Ch

11.1.3.1.62.2 Function

This register contains the hardware configuration options that you can select in the coreConsultant GUI.

11.1.3.1.62.3 Diagram



11.1.3.1.62.4 Fields

Field	Function
31 ghwparams3_31	Reserved
30-23 ghwparams3_30_23	`DWC_USB3_CACHE_TOTAL_XFER_RESOURCES
22-18 ghwparams3_22_18	`DWC_USB3_NUM_IN_EPS
17-12 ghwparams3_17_12	`DWC_USB3_NUM_EPS
11 ghwparams3_11	`DWC_USB3_ULPI_CARKIT
10 ghwparams3_10	`DWC_USB3_VENDOR_CTL_INTERFACE
9-8	Reserved

Table continues on the next page...

Field	Function
ghwparams3_9_8	
7-6 ghwparams3_7_6	`DWC_USB3_HSPHY_DWIDTH
5-4 ghwparams3_5_4	`DWC_USB3_FSPHY_INTERFACE
3-2 ghwparams3_3_2	`DWC_USB3_HSPHY_INTERFACE
1-0 ghwparams3_1_0	`DWC_USB3_SSPHY_INTERFACE

11.1.3.1.63 Global Hardware Parameters Register 4 (GHWPARAMS4)

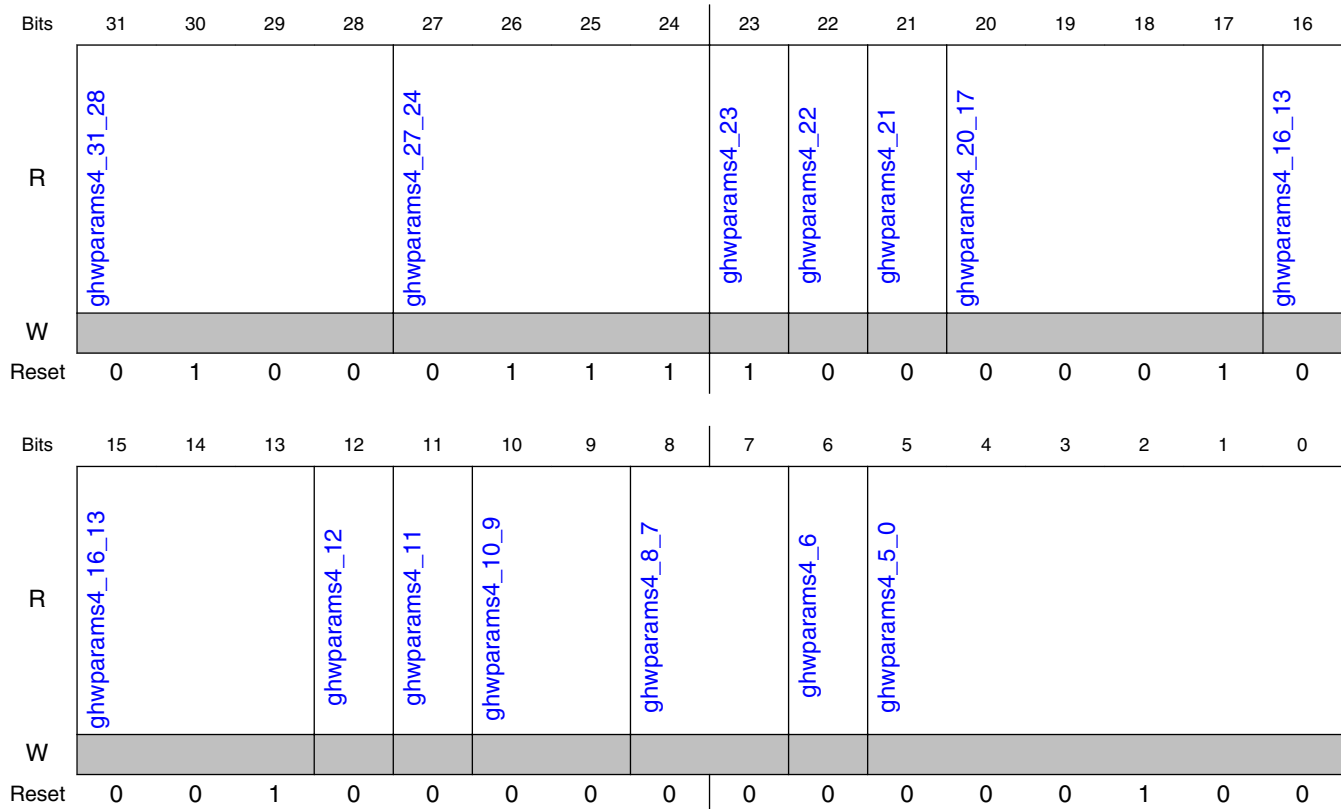
11.1.3.1.63.1 Offset

Register	Offset
GHWPARAMS4	C150h

11.1.3.1.63.2 Function

This register contains the hardware configuration options that you can select in the coreConsultant GUI.

11.1.3.1.63.3 Diagram



11.1.3.1.63.4 Fields

Field	Function
31-28 ghwparams4_31_28	`DWC_USB3_BMU_LSP_DEPTH
27-24 ghwparams4_27_24	`DWC_USB3_BMU_PTL_DEPTH-1
23 ghwparams4_23	`DWC_USB3_EN_ISOC_SUPT
22 ghwparams4_22	Reserved
21 ghwparams4_21	`DWC_USB3_EXT_BUFF_CONTROL
20-17 ghwparams4_20_17	`DWC_USB3_NUM_SS_USB_INSTANCES

Table continues on the next page...

Field	Function
16-13 ghwparams4_16_13	`DWC_USB3_HIBER_SCRATCHBUFS Number of external scratchpad buffers the core requires to save its internal state in the device mode. Each buffer is assumed to be 4KB. The scratchpad buffer array must have this many buffer pointers.
12 ghwparams4_12	`DWC_USB3_EN_SSIC - 1'b0: if DWC_USB3_EN_SSIC == 0 - 1'b1: if DWC_USB3_EN_SSIC != 0 Note: When the DWC_USB3_NUM_SSIC_PORTS parameter is set to zero, this bit is Reserved.
11 ghwparams4_11	`DWC_USB3_SSIC_NON_SNPS_MPHY This field indicates whether Synopsys M-PHY or a third-party M-PHY is used with SSIC ports. - 1'b0: Synopsys M-PHY - 1'b1: Third-party M-PHY Note: When the DWC_USB3_NUM_SSIC_PORTS parameter is set to zero, this bit is Reserved.
10-9 ghwparams4_10_9	`DWC_USB3_SSIC_GEAR This field indicates DWC_USB3_SSIC_GEAR parameter value chosen by the user - 2'b00: Reserved - 2'b01: HS-G1 - 2'b10: HS-G2 - 2'b11: HS-G3 Note: When the DWC_USB3_NUM_SSIC_PORTS parameter is set to zero, this field is Reserved.
8-7 ghwparams4_8_7	`DWC_USB3_NUM_SSIC_NUM_LANE This bit indicates `DWC_USB3_SSIC_NUM_LANE parameter value chosen by the user - 2'b00: 4 lane - 2'b01: 1 lane - 2'b10: 2 lane - 2'b11: Reserved Note: When the DWC_USB3_NUM_SSIC_PORTS parameter is set to zero, this field is Reserved.
6 ghwparams4_6	Reserved
5-0 ghwparams4_5_0	`DWC_USB3_CACHE_TRBS_PER_TRANSFER

11.1.3.1.64 Global Hardware Parameters Register 5 (GHWPARAMS5)

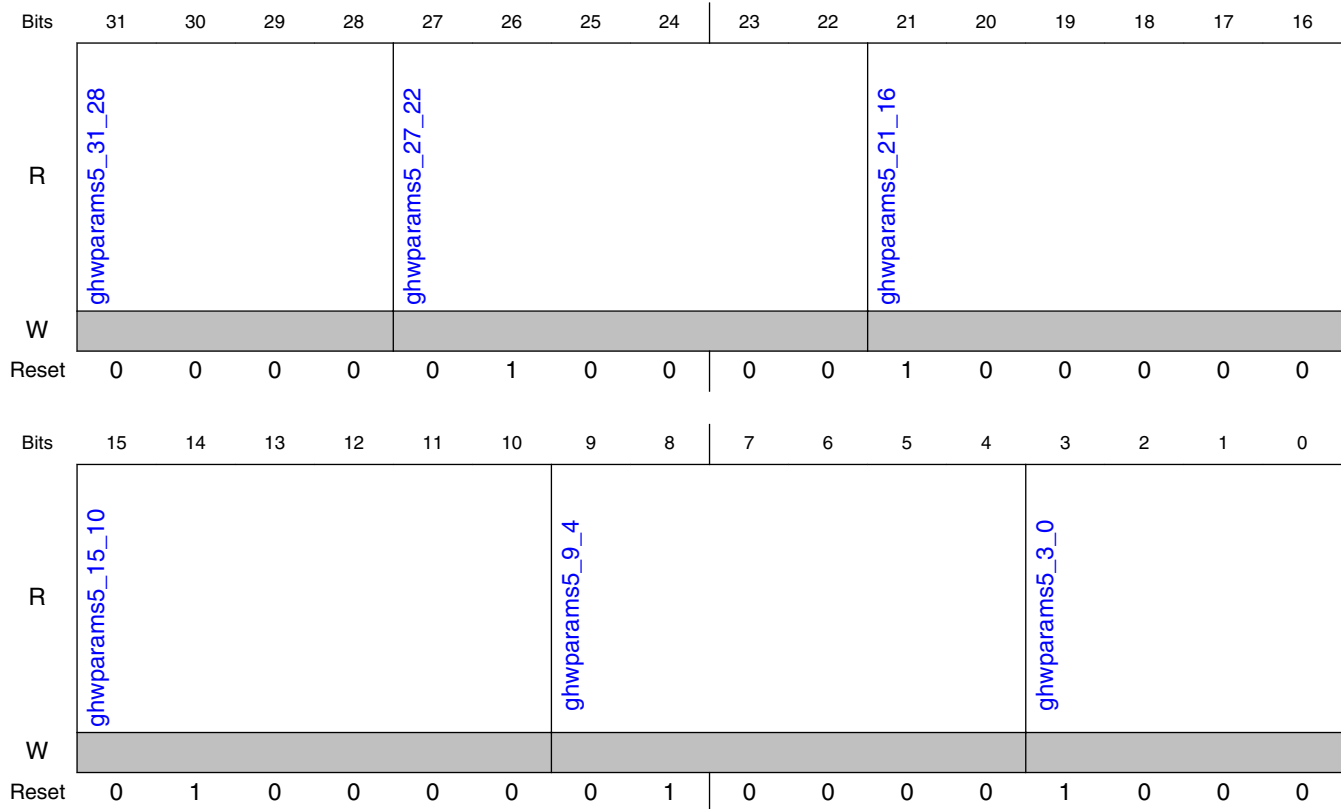
11.1.3.1.64.1 Offset

Register	Offset
GHWPARAMS5	C154h

11.1.3.1.64.2 Function

This register contains the hardware configuration options that you can select in the coreConsultant GUI.

11.1.3.1.64.3 Diagram



11.1.3.1.64.4 Fields

Field	Function
31-28 ghwparams5_31_28	Reserved
27-22 ghwparams5_27_22	`DWC_USB3_DFQ_FIFO_DEPTH
21-16 ghwparams5_21_16	`DWC_USB3_DWQ_FIFO_DEPTH
15-10 ghwparams5_15_10	`DWC_USB3_TXQ_FIFO_DEPTH
9-4 ghwparams5_9_4	`DWC_USB3_RXQ_FIFO_DEPTH
3-0	`DWC_USB3_BMU_BUSGM_DEPTH

Field	Function
ghwparams5_3_0	

11.1.3.1.65 Global Hardware Parameters Register 6 (GHWPARAMS6)

11.1.3.1.65.1 Offset

Register	Offset
GHWPARAMS6	C158h

11.1.3.1.65.2 Function

This register contains the hardware configuration options that you can select in the coreConsultant GUI.

11.1.3.1.65.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	ghwparams6_31_16															
W																
Reset	0	0	0	0	1	0	1	0	0	1	0	1	1	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	BusFtrSupport	BCSupport	OTG_SS_Support	ADPSupport	HNPSupport	SRPSupport	ghwparams6_9_8		ghwparams6_7	ghwparams6_6	ghwparams6_5_0					
W																
Reset	1	1	0	1	1	1	0	0	0	0	1	0	0	0	0	0

11.1.3.1.65.4 Fields

Field	Function
31-16 ghwparams6_31_16	DWC_USB3_RAM0_DEPTH

Table continues on the next page...

Field	Function
15 BusFltrsSupport	`DWC_USB3_EN_BUS_FILTERS
14 BCSupport	`DWC_USB3_EN_BC
13 OTG_SS_Support	OTG 3.0 Support Enabled This bit indicates whether the parameter `DWC_USB3_EN_OTG is set to 2. In other words, it indicates that whether the DWC_usb3 core supports OTG 3.0. - 1'b0: No OTG 3.0 support - 1'b1: Supports OTG 3.0
12 ADPSupport	`DWC_USB3_EN_ADP
11 HNPSupport	RSP/HNP Support Enabled The application uses this bit to determine the DWC_usb3 core's RSP/HNP support. If DWC_USB3_EN_OTG=2, - 1'b0: RSP and HNP support is not enabled. The only exception for this rule is for SSPC-OTG devices where RSP support is not enabled, but HNP support is enabled. (Refer to the OCFG.SSPC-OTG bit.) - 1'b1: RSP and HNP support is enabled If DWC_USB3_EN_OTG=1, - 1'b0: HNP support is not enabled; - 1'b1: HNP support is enabled; This bit is enabled only if HNP mode was specified for HNP Mode of Operation in coreConsultant (parameter DWC_USB3_EN_OTG is not 0, and DWC_USB3_MODE is DRD). Otherwise, it reads 0.
10 SRPSupport	SRP Support Enabled The application uses this bit to determine the DWC_usb3 core's SRP support. - 1'b0: SRP support is not enabled; - 1'b1: SRP support is enabled; This bit is 1'b1 when the parameter DWC_USB3_EN_OTG is not 0.
9-8 ghwparams6_9_8	Reserved
7 ghwparams6_7	`DWC_USB3_EN_FPGA
6 ghwparams6_6	`DWC_USB3_EN_DBG_PORTS
5-0 ghwparams6_5_0	`DWC_USB3_PSQ_FIFO_DEPTH

11.1.3.1.66 Global Hardware Parameters Register 7 (GHWPARAMS7)

11.1.3.1.66.1 Offset

Register	Offset
GHWPARAMS7	C15Ch

11.1.3.1.66.2 Function

This register contains the hardware configuration options that you can select in the coreConsultant GUI.

11.1.3.1.66.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	ghwparams7_31_16															
W																
Reset	0	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ghwparams7_15_0															
W																
Reset	0	0	0	1	0	0	0	1	1	1	1	0	1	1	0	0

11.1.3.1.66.4 Fields

Field	Function
31-16 ghwparams7_31_16	`DWC_USB3_RAM2_DEPTH
15-0 ghwparams7_15_0	`DWC_USB3_RAM1_DEPTH

11.1.3.1.67 Global High-Speed Port to Bus Instance Mapping Register - Low (GPRTBIMAP_HSLO)

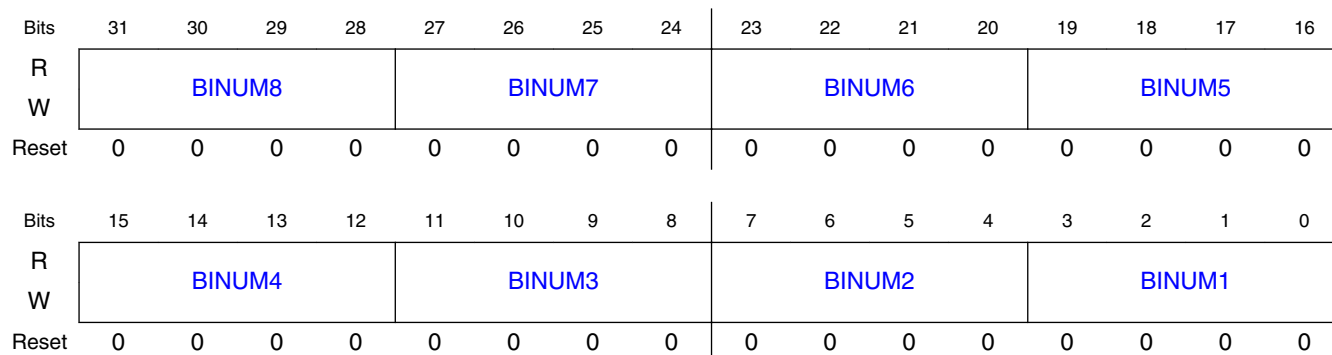
11.1.3.1.67.1 Offset

Register	Offset
GPRTBIMAP_HSLO	C180h

11.1.3.1.67.2 Function

This is an alternate register for the GPRTBIMAP_HS register.

11.1.3.1.67.3 Diagram



11.1.3.1.67.4 Fields

Field	Function
31-28 BINUM8	BINUM8: HS USB Instance Number for Port 8. Application-programmable ID field.
27-24 BINUM7	BINUM7: HS USB Instance Number for Port 7. Application-programmable ID field.
23-20 BINUM6	BINUM6: HS USB Instance Number for Port 6. Application-programmable ID field.
19-16 BINUM5	BINUM5: HS USB Instance Number for Port 5. Application-programmable ID field.
15-12 BINUM4	BINUM4: HS USB Instance Number for Port 4. Application-programmable ID field.
11-8 BINUM3	BINUM3: HS USB Instance Number for Port 3. Application-programmable ID field.
7-4 BINUM2	BINUM2: HS USB Instance Number for Port 2. Application-programmable ID field.
3-0 BINUM1	BINUM1: HS USB Instance Number for Port 1. Application-programmable ID field.

11.1.3.1.68 Global High-Speed Port to Bus Instance Mapping Register - High (GPRTBIMAP_HSHI)

11.1.3.1.68.1 Offset

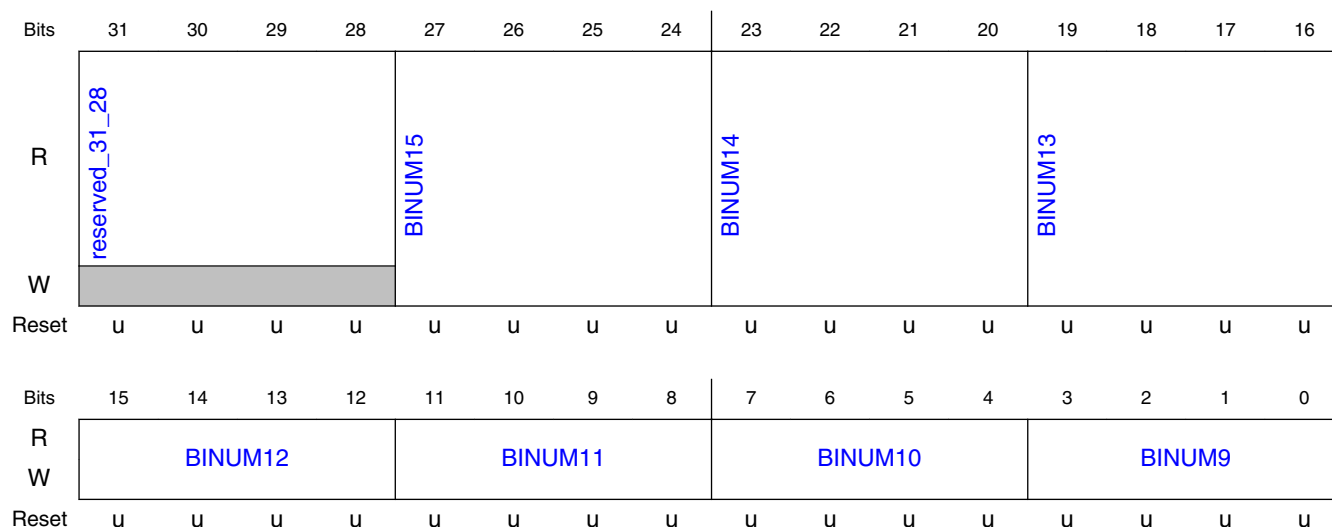
Register	Offset
GPRTBIMAP_HSHI	C184h

11.1.3.1.68.2 Function

This is an alternate register for the GPRTBIMAP_HS register.

Reset Mask: 0xF0000000

11.1.3.1.68.3 Diagram



11.1.3.1.68.4 Fields

Field	Function
31-28 reserved_31_28	Reserved
27-24 BINUM15	BINUM15: HS USB Instance Number for Port 15. Application-programmable ID field.
23-20 BINUM14	BINUM14: HS USB Instance Number for Port 14. Application-programmable ID field.
19-16 BINUM13	BINUM13: HS USB Instance Number for Port 13. Application-programmable ID field.
15-12 BINUM12	BINUM12: HS USB Instance Number for Port 12. SApplication-programmable ID field.
11-8 BINUM11	BINUM11: HS USB Instance Number for 11. Application-programmable ID field.
7-4 BINUM10	BINUM10: HS USB Instance Number for Port 10. Application-programmable ID field.
3-0 BINUM9	BINUM9: HS USB Instance Number for Port 9. Application-programmable ID field.

11.1.3.1.69 Global Full-Speed Port to Bus Instance Mapping Register - Low (GPRTBIMAP_FSLO)

11.1.3.1.69.1 Offset

Register	Offset
GPRTBIMAP_FSLO	C188h

11.1.3.1.69.2 Function

This is an alternate register for the GPRTBIMAP_FS register.

11.1.3.1.69.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	BINUM8				BINUM7				BINUM6				BINUM5			
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	BINUM4				BINUM3				BINUM2				BINUM1			
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.1.3.1.69.4 Fields

Field	Function
31-28 BINUM8	BINUM8: FS USB Instance Number for Port 8. Application-programmable ID field.
27-24 BINUM7	BINUM7: FS USB Instance Number for Port 7. Application-programmable ID field.
23-20 BINUM6	BINUM6: FS USB Instance Number for Port 6. Application-programmable ID field.
19-16 BINUM5	BINUM5: FS USB Instance Number for Port 5. Application-programmable ID field.
15-12 BINUM4	BINUM4: FS USB Instance Number for Port 4. Application-programmable ID field.
11-8	BINUM3: FS USB Instance Number for Port 3. Application-programmable ID field.

Table continues on the next page...

Field	Function
BINUM3	
7-4 BINUM2	BINUM2: FS USB Instance Number for Port 2. Application-programmable ID field.
3-0 BINUM1	BINUM1: FS USB Instance Number for Port 1. Application-programmable ID field.

11.1.3.1.70 Global Full-Speed Port to Bus Instance Mapping Register - High (GPRTBIMAP_FSHI)

11.1.3.1.70.1 Offset

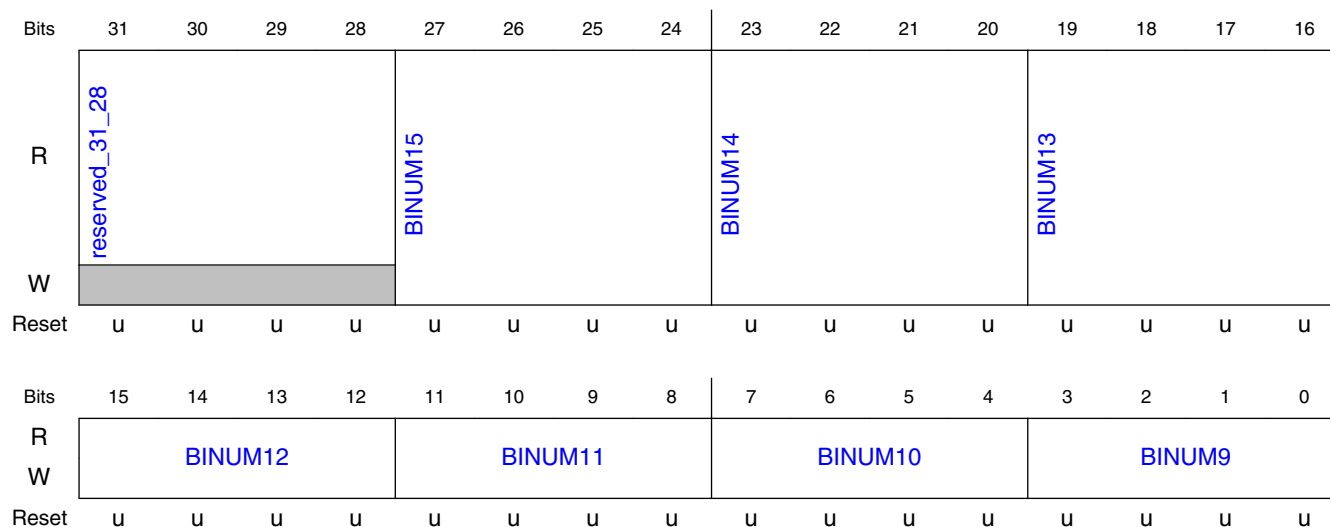
Register	Offset
GPRTBIMAP_FSHI	C18Ch

11.1.3.1.70.2 Function

This is an alternate register for the GPRTBIMAP_FS register.

Reset Mask: 0xF0000000

11.1.3.1.70.3 Diagram



11.1.3.1.70.4 Fields

Field	Function
31-28 reserved_31_28	Reserved
27-24 BINUM15	BINUM15: FS USB Instance Number for Port 15. Application-programmable ID field
23-20 BINUM14	BINUM14: FS USB Instance Number for Port 14. Application-programmable ID field
19-16 BINUM13	BINUM13: FS USB Instance Number for Port 13. Application-programmable ID field
15-12 BINUM12	BINUM12: FS USB Instance Number for Port 12. Application-programmable ID field
11-8 BINUM11	BINUM11: FS USB Instance Number for Port 11. Application-programmable ID field
7-4 BINUM10	BINUM10: FS USB Instance Number for Port 10. Application-programmable ID field
3-0 BINUM9	BINUM9: FS USB Instance Number for Port 9. Application-programmable ID field.

11.1.3.1.71 Global User Control Register 2 (GUCTL2)

11.1.3.1.71.1 Offset

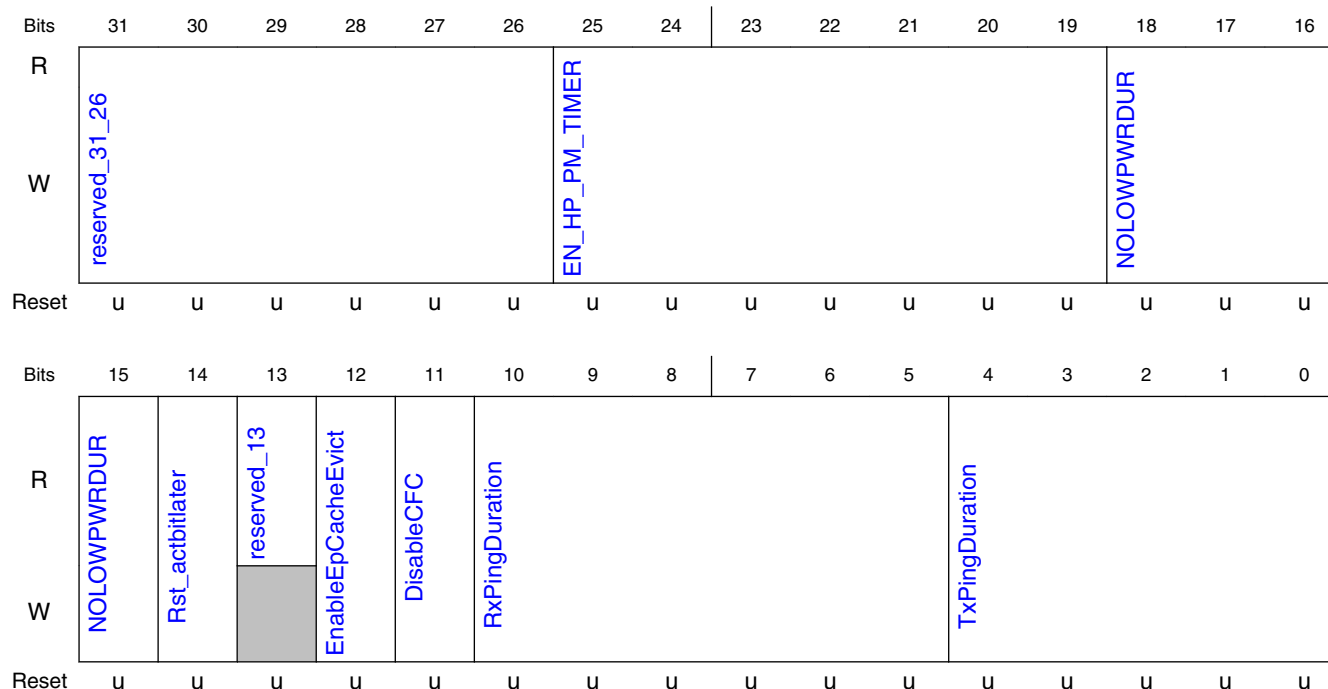
Register	Offset
GUCTL2	C19Ch

11.1.3.1.71.2 Function

This register provides a few options for the software to control the core behavior in the Host and device mode.

Reset Mask: 0x3FC008000

11.1.3.1.71.3 Diagram



11.1.3.1.71.4 Fields

Field	Function
31-26 reserved_31_26	Reserved
25-19 EN_HP_PM_TIMER	This register field is used to set new HP and PM timers. - To enable PM timer, set GUCTL2[19] bit as 1. - To enable HP timer, set GUCTL2[20] bit as 1. Default value is 4us, when new HP timer is enabled default value is 11us. Use GUCTL2[25:21] to specify HP timer value in microseconds.
18-15 NOLOWPWRDUR	No Low Power Duration (NOLOWPWRDUR) This bit is applicable for device mode only and is ignored in host mode. After starting a transfer on an ESS ISOC endpoint, the application must program these bits. Each count represents the duration in terms of milli seconds. For example, a value of 3 represents 3 ms.
14 Rst_actbitlater	Enable clearing of the command active bit for the ENDXFER command after the command execution is completed. This bit is valid in device mode only.
13 reserved_13	Reserved for future use
12 EnableEpCacheEvict	Enable Evicting Endpoint cache after Flow Control for bulk endpoints. In 3.00a release, a performance enhancement was done to keep the non-stream capable bulk IN endpoint in cache after flow control. Setting this bit will disable this enhancement. This should be set only for debug purpose.
11 DisableCFC	Disable xHCI Errata Feature Contiguous Frame ID Capability This field controls the xHCI Errata feature Contiguous FrameID capability. When set, the xHCI HCCPARAMS1 bit 11 will be set to 0 indicating that CFC is not supported. Disable this feature only if your application cannot tolerate Missed Service Error events for Isoc transfers, and your system latencies are large to cause Missed Service errors even if the software is following the Isochronous Thresholding rules.

Table continues on the next page...

Field	Function
10-5 RxPingDuration	Receive Ping Maximum Duration This field is relevant to Host mode and controls the maximum duration of received LFPS to be treated as a Ping LFPS. The Max duration of the Ping LFPS is controlled by programming this value and is in terms of 8 ns granularity. Eg: A value of 32 indicates 256 ns.
4-0 TxPingDuration	Transmit Ping Maximum Duration This field is relevant to Device mode and controls the maximum duration for which the controller should instruct the PHY to transmit a Ping LFPS. The duration of the Ping LFPS is controlled by programming this value and is in terms of 8 ns granularity. Eg: A value of 13 indicates 104 ns.

11.1.3.1.72 Global USB2 PHY Configuration Register (GUSB2PHYCFG)

11.1.3.1.72.1 Offset

Register	Offset
GUSB2PHYCFG	C200h

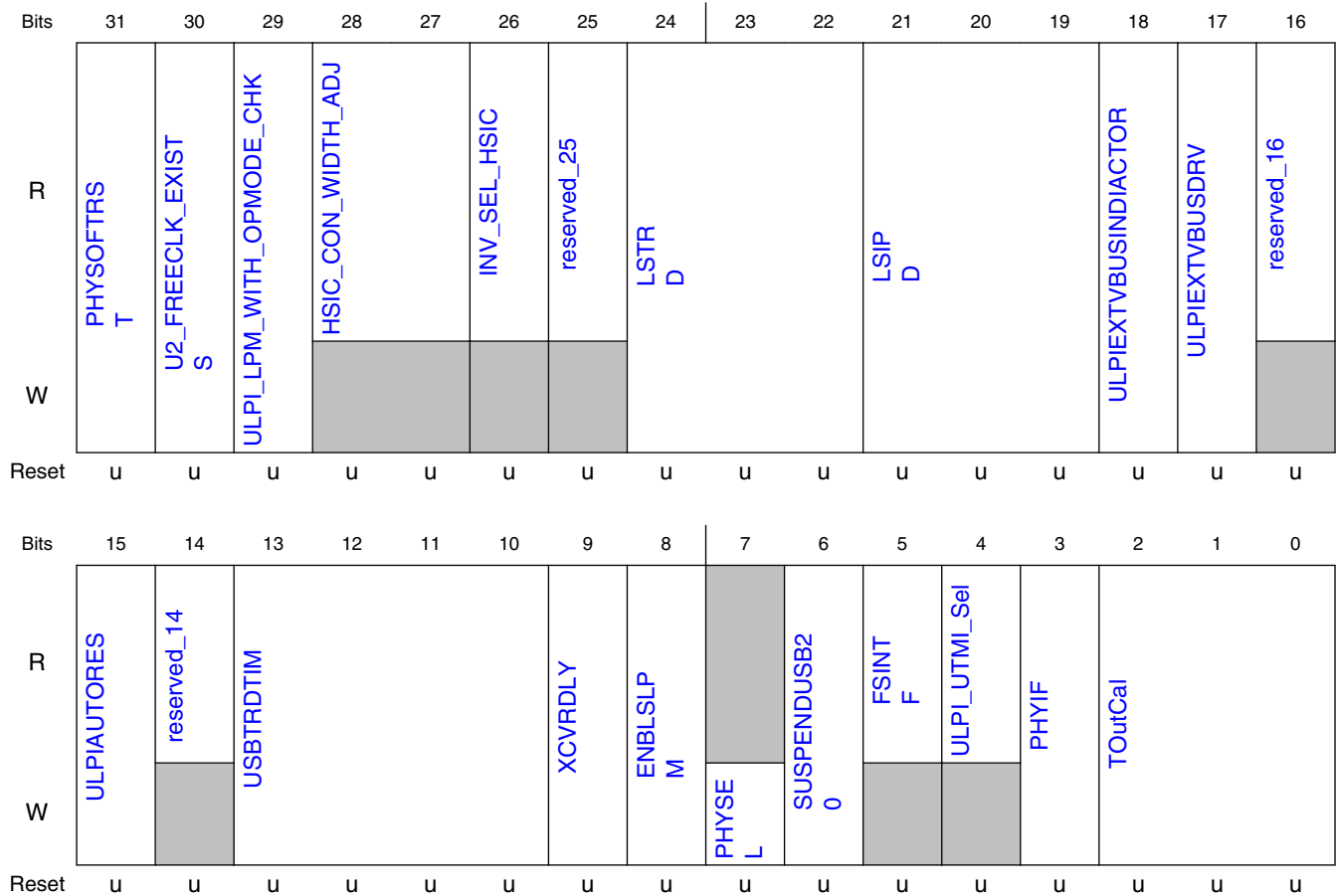
11.1.3.1.72.2 Function

The application must program this register before starting any transactions on either the SoC bus or the USB.

Reset Mask:0x2014000

this register before starting any transactions on either the SoC bus or the USB. In registers are implemented.

11.1.3.1.72.3 Diagram



11.1.3.1.72.4 Fields

Field	Function
31 PHYSOFTRST	UTMI PHY Soft Reset (PHYSoftRst) Causes the usb2phy_reset signal to be asserted to reset a UTMI PHY.
30 U2_FREECLK_EXISTS	U2_FREECLK_EXISTS Specifies whether your USB 2. free-running PHY clock, which is active when the clock control input is active. If your USB 2.0 PHY provides a free-running PHY clock, it must be connected to the utmi_clk[0] input. The remaining utmi_clk[n] must be connected to the respective port clocks. The core uses the Port-0 clock for generating the internal mac2 clock. - 1'b0: USB 2.0 free clock does not exist - 1'b1: USB 2.0 free clock exists Note: When the core is configured as device-only (DWC_USB3_MODE = 0), do not set this bit to 1.
29 ULPI_LPM_WITH_OPMODE_CHK	ULPI_LPM_WITH_OPMODE_CHK Support the LPM over ULPI without NOPID token to the ULPI PHY. token to the ULPI PHY. If this bit is set, the ULPI PHY is expected to qualify the EXT PID with OPMODE=2'b00 for LPM and not treat it as a NOPID. Check with your PHY vendor about your PHY behavior. This bit is valid only when the DWC_USB3_HSPHY_INTERFACE parameter is 2 or 3. - 1'b0: A

Table continues on the next page...

Field	Function
	NOPID is sent before sending an EXTPID for LPM; - 1'b1: An EXTPID is sent without previously sending a NOPID; Note: This bit is valid only in host mode. This bit should be '0' for Synopsys PHY.
28-27 HSIC_CON_WIDTH_ADJ	HSIC_CON_WIDTH_ADJ This bit is used in the HSIC device mode of operation.
26 INV_SEL_HSIC	INV_SEL_HSIC The application driver uses this bit to control the HSIC enable/disable function.
25 reserved_25	Reserved
24-22 LSTRD	LS Turnaround Time (LSTRDTIM) This field indicates the value of the Rx-to-Tx packet gap for LS devices.
21-19 LSIPD	LS Inter-Packet Time (LSIPD) This field indicates the value of Tx-to-Tx packet gap for LS devices.
18 ULPIEXTVBUSINDIATOR	ULPI External VBUS Indicator (ULPIExtVbusIndicator) Indicates the ULPI PHY VBUS over-current indicator.
17 ULPIEXTVBUSDRV	ULPI External VBUS Drive (ULPIExtVbusDrv) Selects supply source to drive 5V on VBUS, in the ULPI PHY.
16 reserved_16	Reserved
15 ULPIAUTORES	ULPI Auto Resume (ULPIAutoRes) Sets the AutoResume bit in Interface Control register on the ULPI PHY.
14 reserved_14	Reserved
13-10 USBTRDTIM	USB 2.
9 XCVRDLY	Transceiver Delay: Enables a delay between the assertion of the UTMI/ULPI Transceiver Select signal (for HS) and the assertion of the TxValid signal during a HS Chirp.
8 ENBLSLPM	Enable utmi_sleep_n and utmi_l1_suspend_n (EnbLSlpM) The application uses this bit to control utmi_sleep_n and utmi_l1_suspend_n assertion to the PHY in the L1 state.
7 PHYSEL	USB 2.
6 SUSPENDUSB20	Suspend USB2.
5 FSINTF	Full-Speed Serial Interface Select (FSIntf) The application uses this bit to select a unidirectional or bidirectional USB 1.
4	ULPI or UTMI+ Select (ULPI_UTMI_Sel) The application uses this bit

Table continues on the next page...

Universal Serial Bus Controller (USB)

Field	Function
ULPI_UTMI_Sel	to select a UTMI+ or ULPI Interface.
3 PHYIF	PHY Interface (PHYIf) If UTMI+ is selected, the application uses this bit to configure the core to support a UTMI+ PHY with an 8- or 16-bit interface.
2-0 TOutCal	HS/FS Timeout Calibration (TOutCal) The number of PHY clocks, as indicated by the application in this field, is multiplied by a bit-time factor; this factor is added to the high-speed/full-speed interpacket timeout duration in the core to account for additional delays introduced by the PHY.

11.1.3.1.73 Global USB 2.0 UTMI PHY vendor control register (GUSB2PHYACC_ULPI)

11.1.3.1.73.1 Offset

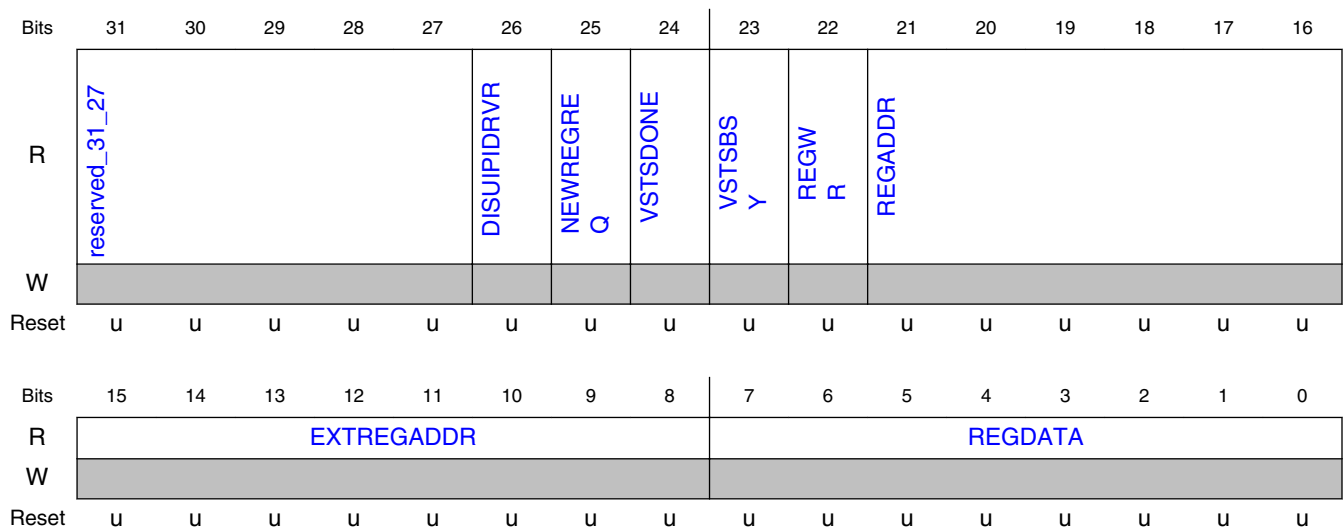
Register	Offset
GUSB2PHYACC_ULPI	C280h

11.1.3.1.73.2 Function

Global USB 2.

Reset Mask:0xF8000000

11.1.3.1.73.3 Diagram



11.1.3.1.73.4 Fields

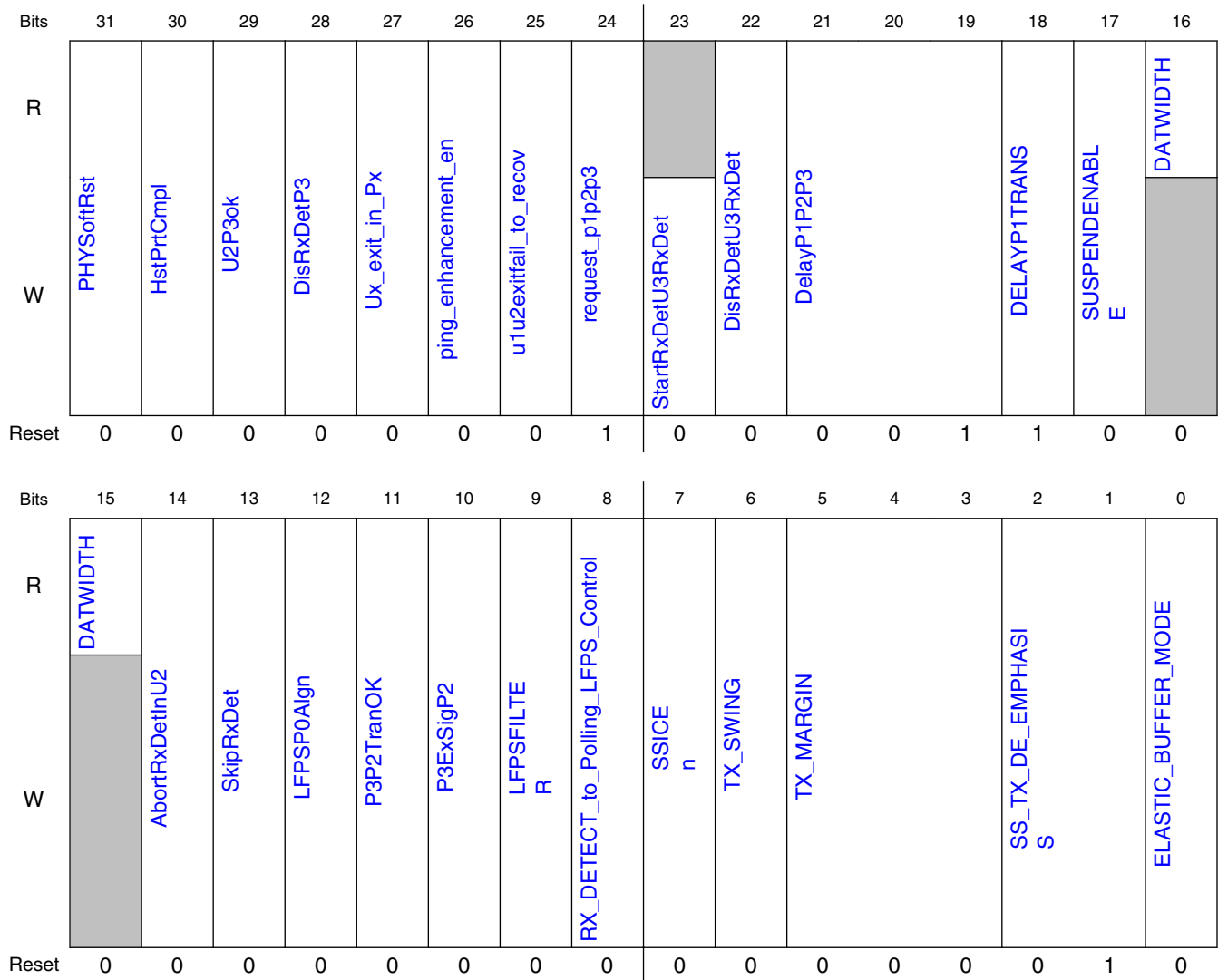
Field	Function
31-27 reserved_31_27	Reserved
26 DISUIPIDRVR	DISUIPIDRVR
25 NEWREGREQ	New Register Request The application sets this bit for a new vendor control access.
24 VSTSDONE	VSTSDONE
23 VSTSBSY	VSTSBSY
22 REGWR	Register Write The application sets this bit for register writes and clears it for register reads. clears it for register reads. Note: This bit is applicable for ULPI register read/write access only.
21-16 REGADDR	Register Address The 6-bit PHY register address for immediate PHY Register Set access. Register Set access. Set to 6'h2F for Extended PHY Register Set access. Note: These bits are applicable for ULPI only.
15-8 EXTREGADDR	EXTREGADDR
7-0 REGDATA	REGDATA

11.1.3.1.74 Global USB 3.0 PIPE control register (GUSB3PIPECTL)

11.1.3.1.74.1 Offset

Register	Offset
GUSB3PIPECTL	C2C0h

11.1.3.1.74.2 Diagram



11.1.3.1.74.3 Fields

Field	Function
31 PHYSoftRst	USB3 PHY Soft Reset After setting this bit to '1', the software needs to clear this bit.
30 HstPrtCmpl	HstPrtCmpl This feature tests the PIPE PHY compliance patterns without having to have a test fixture on the USB 3.
29 U2P3ok	P3 OK for U2 (u2P3ok) - 0: During link state U2/SS.
28 DisRxDetP3	Disabled receiver detection in P3 (DisRxDetP3) - 0: If PHY is in P3 and Core needs to perform receiver detection, The core performs receiver detection in

Table continues on the next page...

Field	Function
	P3.
27 Ux_exit_in_Px	Ux Exit in Px (Ux_exit_in_Px) - 0: The core does U1/U2/U3 exit in PHY power state P0 (default behavior).
26 ping_enhancement_en	Ping Enhancement Enable (ping_enhancement_en) When set, the Downstream port U1 ping receive timeout becomes 500 ms instead of 300 ms.
25 u1u2exitfail_to_recovery	U1U2exitfail to Recovery (u1u2exitfail_to_recov) When set, and U1/U2 LFPS handshake fails, the LTSSM transitions from U1/U2 to Recovery instead of SS Inactive.
24 request_p1p2p3	Always Request P1/P2/P3 for U1/U2/U3 (request_p1p2p3) When set, the core always requests PHY power change from P0 to P1/P2/P3 during U0 to U1/U2/U3 transition.
23 StartRxDetU3RxDet	Start Receiver Detection in U3/Rx.
22 DisRxDetU3RxDet	Disable Receiver Detection in U3/Rx.
21-19 DelayP1P2P3	Delay P1P2P3 Delay P0 to P1/P2/P3 request when entering U1/U2/U3 until (DWC_USB3_GUSB3PIPECTL_INIT[21:19]*8) 8B10B error occurs, or Pipe3_RxValid drops to 0.
18 DELAYP1TRANS	Delay PHY power change from P0 to P1/P2/P3 when link state changing from U0 to U1/U2/U3 respectively.
17 SUSPENDENABLE	Suspend USB3.
16-15 DATWIDTH	PIPE Data Width (DatWidth) - 2'b00: 32 bits - 2'b01: 16 bits - 2'b10: 8 bits One clock after reset, these bits receive the value seen on the pipe3_DataBusWidth.
14 AbortRxDetInU2	Abort Rx Detect in U2 (AbortRxDetInU2) When set, and the link state is U2, then the core will abort receiver detection if it receives U2 exit LFPS from the remote link partner.
13 SkipRxDet	Skip Rx Detect: When set, the core skips Rx Detection if pipe3_RxElecIdle is low.
12 LFPSP0Algn	LFPS P0 Align: When set, - The core deasserts LFPS transmission on the clock edge that it requests Phy power state 0 when exiting U1, U2, or U3 low power states.
11 P3P2TranOK	P3 P2 Transitions OK (P3P2TranOK) When set, the core transitions directly from Phy power state P2 to P3 or from state P3 to P2.
10	P3 Exit Signal in P2 (P3ExSigP2) When this bit is set, the core

Table continues on the next page...

Field	Function
P3ExSigP2	always changes the PHY power state to P2, before attempting a U3 exit handshake.
9 LFPSFILTER	LFPS Filter (LFPSFilt) When set, filter LFPS reception with pipe3_RxValid in PHY power state P0, that is, ignore LFPS reception from the PHY unless both pipe3_Rxelecidle and pipe3_RxValid are deasserted.
8 RX_DETECT_to_Polling_LFPS_Control	RX_DETECT to Polling.
7 SSICEn	USB3 SSIC Enable (SSICEn) This bit is valid only when coreConsultant parameter DWC_USB3_EN_SSIC=1 else this bit needs to be set to 1'b0.
6 TX_SWING	Tx Swing (TxSwing) Refer to the PIPE3 specification.
5-3 TX_MARGIN	Tx Margin[2:0] (TxMargin) Refer to Table 5-3 of the PIPE3 Specification.
2-1 SS_TX_DE_EM PHASIS	Tx Deemphasis (TxDeemphasis) The value driven to the PHY is controlled by the LTSSM during USB3 Compliance mode.
0 ELASTIC_BUFFER_MODE	Elastic Buffer Mode (ElasticBufferMode) (Refer to Table 5-3 of the PIPE3 specification.

11.1.3.1.75 Global transmit FIFO size register (GTXFIFOSIZ0 - GTXFIFOSIZ7)

11.1.3.1.75.1 Offset

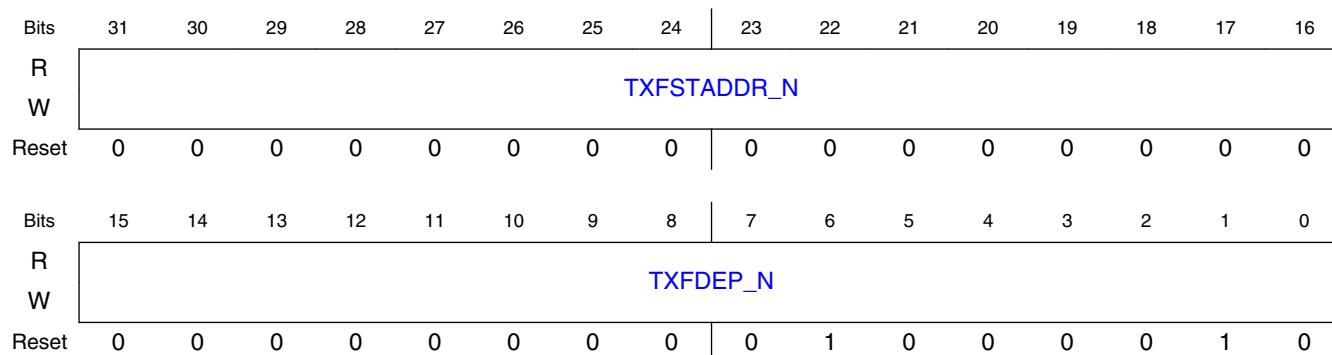
For a = 0 to 7:

Register	Offset
GTXFIFOSIZa	C300h + (a × 4h)

11.1.3.1.75.2 Function

This register specifies the RAM start address and depth (both in MDWIDTH-bit words) for each implemented TxFIFO.

11.1.3.1.75.3 Diagram



11.1.3.1.75.4 Fields

Field	Function
31-16 TXFSTADDR_N	Transmit FIFO RAM Start Address This field contains the memory start address for TxFIFO in MDWIDTH-bit words.
15-0 TXFDEP_N	TXFIFO depth This field contains the depth of TxFIFO in MDWIDTH-bit words.

11.1.3.1.76 Global receive FIFO size register (GRXFIFOSIZ0 - GRXFIFOSIZ2)

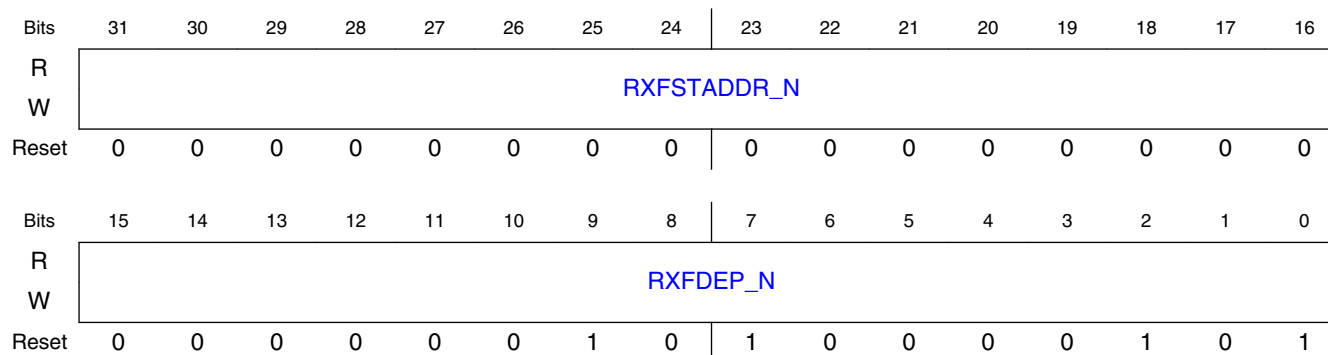
11.1.3.1.76.1 Offset

Register	Offset
GRXFIFOSIZ0	C380h
GRXFIFOSIZ1	C384h
GRXFIFOSIZ2	C388h

11.1.3.1.76.2 Function

This register specifies the RAM start address and depth (both in MDWIDTH-bit words) for each implemented RxFIFO.

11.1.3.1.76.3 Diagram



11.1.3.1.76.4 Fields

Field	Function
31-16 RXFSTADDR_N	RxFIFO RAM Start Address (RxFStAddr_n) This field contains the memory start address for RxFIFO in MDWIDTH-bit words.
15-0 RXFDEP_N	RxFIFO Depth (RxFDep_n) This field contains the depth of RxFIFO in MDWIDTH-bit words.

11.1.3.1.77 Global Event Buffer Address (Low) Register (GEVNTADRLO)

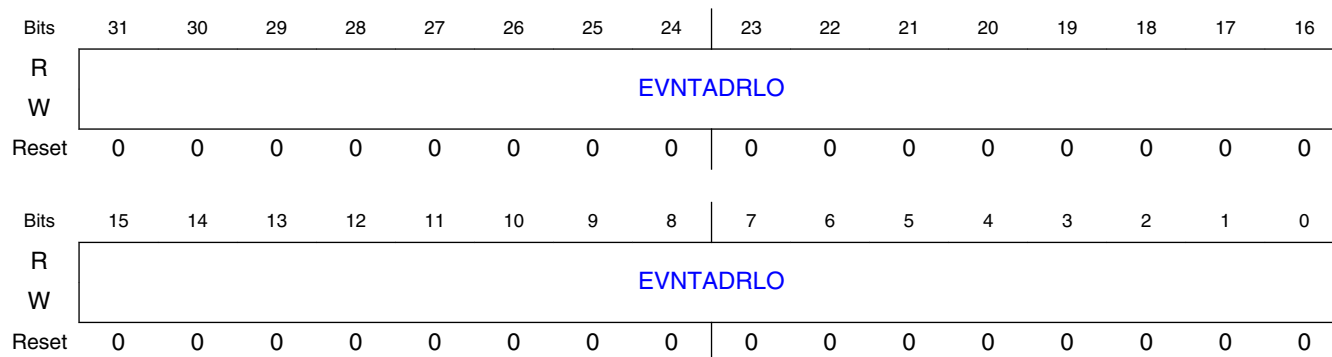
11.1.3.1.77.1 Offset

Register	Offset
GEVNTADRLO	C400h

11.1.3.1.77.2 Function

This is an alternate register for the GEVNTADRn register.

11.1.3.1.77.3 Diagram



11.1.3.1.77.4 Fields

Field	Function
31-0 EVNTADRLO	Event Buffer Address (EvtAdrLo) Holds the lower 32 bits of start address of the external memory for the Event Buffer.

11.1.3.1.78 Global Event Buffer Address (High) Register (GEVNTADRHI)

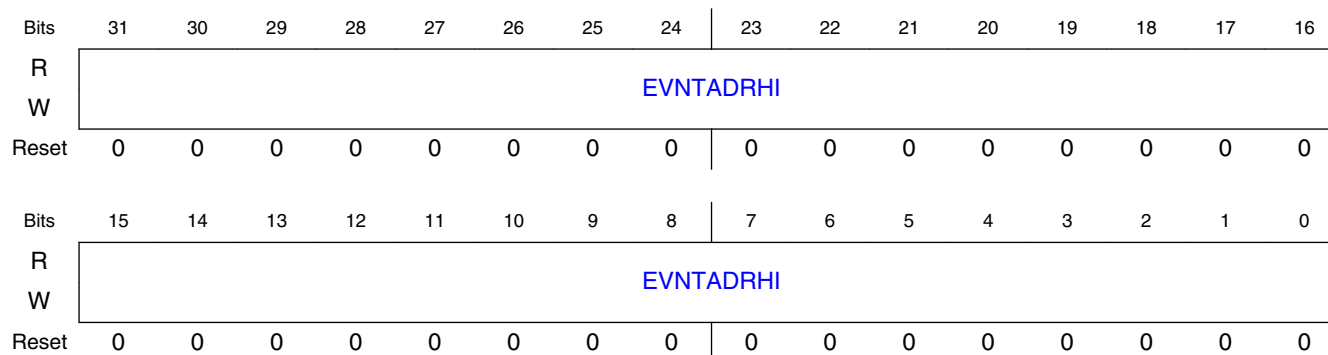
11.1.3.1.78.1 Offset

Register	Offset
GEVNTADRHI	C404h

11.1.3.1.78.2 Function

This is an alternate register for the GEVNTADRn register.

11.1.3.1.78.3 Diagram



11.1.3.1.78.4 Fields

Field	Function
31-0 EVNTADRHI	Event Buffer Address (EvtAdrHi) Holds the higher 32 bits of start address of the external memory for the Event Buffer.

11.1.3.1.79 Global event buffer size register (GEVNTSIZ)

11.1.3.1.79.1 Offset

Register	Offset
GEVNTSIZ	C408h

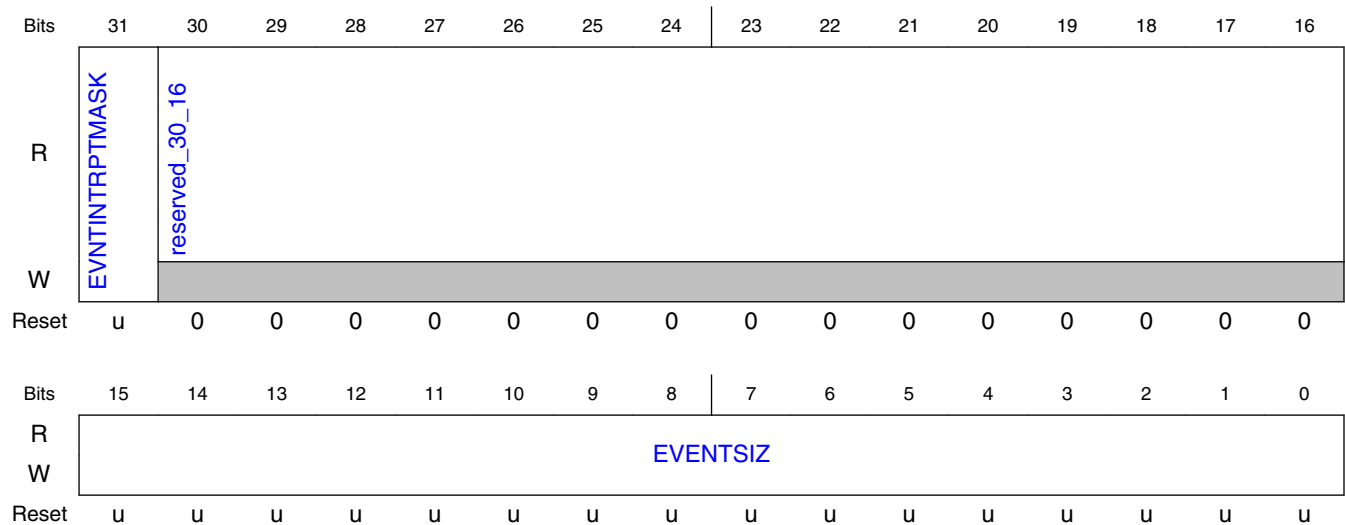
11.1.3.1.79.2 Function

This register holds the Event Buffer Size and the Event Interrupt Mask bit.

Reset Mask: 0x7FFF0000

Global Event Buffer Size Register This register holds the Event Buffer Size and the Event Interrupt Mask bit. During power-on initialization, software must initialize the size with the number of bytes allocated for the Event Buffer. The Event Interrupt Mask will mask the interrupt, but events are still queued. After configuration, software must preserve the Event Buffer Size value when changing the Event Interrupt Mask.

11.1.3.1.79.3 Diagram



11.1.3.1.79.4 Fields

Field	Function
31 EVNTINTRPTMASK	Event Interrupt Mask (EvtIntMask).
30-16 reserved_30_16	Reserved
15-0 EVENTSIZ	Event Buffer Size in bytes (EVNTSiz) Holds the size of the Event Buffer in bytes; must be a multiple of four.

11.1.3.1.80 Global event buffer count register (GEVNTCOUNT)

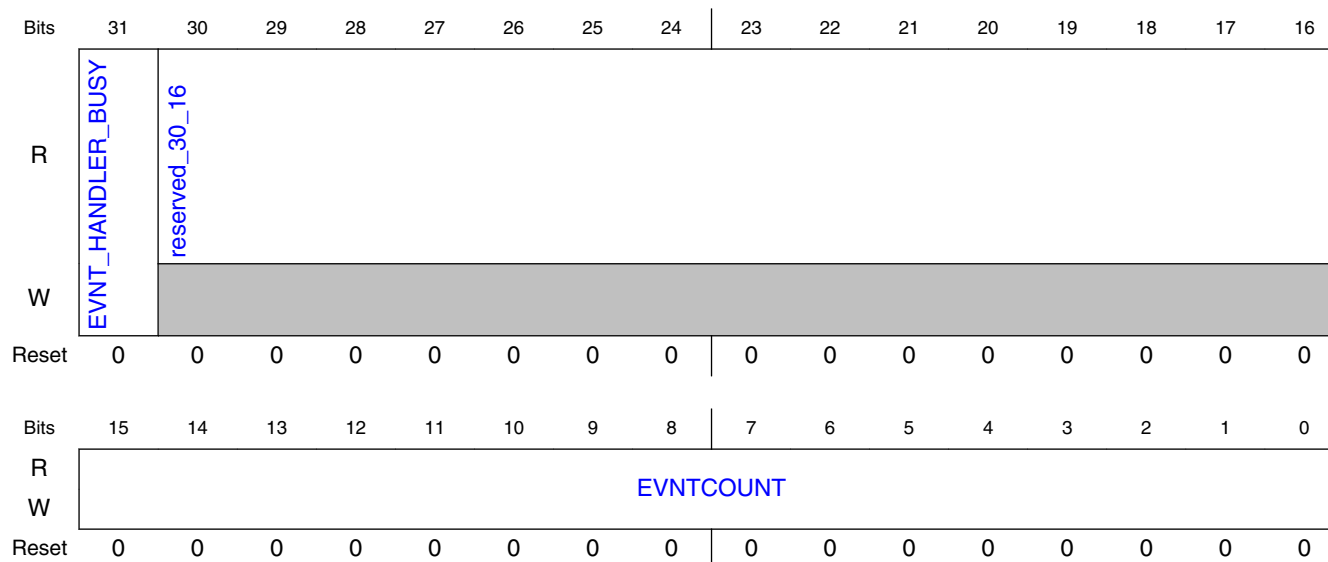
11.1.3.1.80.1 Offset

Register	Offset
GEVNTCOUNT	C40Ch

11.1.3.1.80.2 Function

This register holds the number of valid bytes in the Event Buffer.

11.1.3.1.80.3 Diagram



11.1.3.1.80.4 Fields

Field	Function
31 EVNT_HANDLE R_BUSY	Event Handler Busy Device software event handler busy indication.
30-16 reserved_30_16	Reserved
15-0 EVNTCOUNT	Event Count (EVNTCount) When read, returns the number of valid events in the Event Buffer (in bytes).

11.1.3.1.81 Global Hardware Parameters Register 8 (GHWPARAMS8)

11.1.3.1.81.1 Offset

Register	Offset
GHWPARAMS8	C600h

11.1.3.1.81.2 Function

This register contains the hardware configuration options that you can select in the coreConsultant GUI.

11.1.3.1.81.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	ghwparams8_32_0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ghwparams8_32_0															
W																
Reset	0	0	0	0	1	0	1	0	0	1	0	1	1	0	0	0

11.1.3.1.81.4 Fields

Field	Function
31-0 ghwparams8_32_0	DWC_USB3_DCACHE_DEPTH_INFO

11.1.3.1.82 Global Device TX FIFO DMA Priority Register (GTXFIFOPRIDEV)

11.1.3.1.82.1 Offset

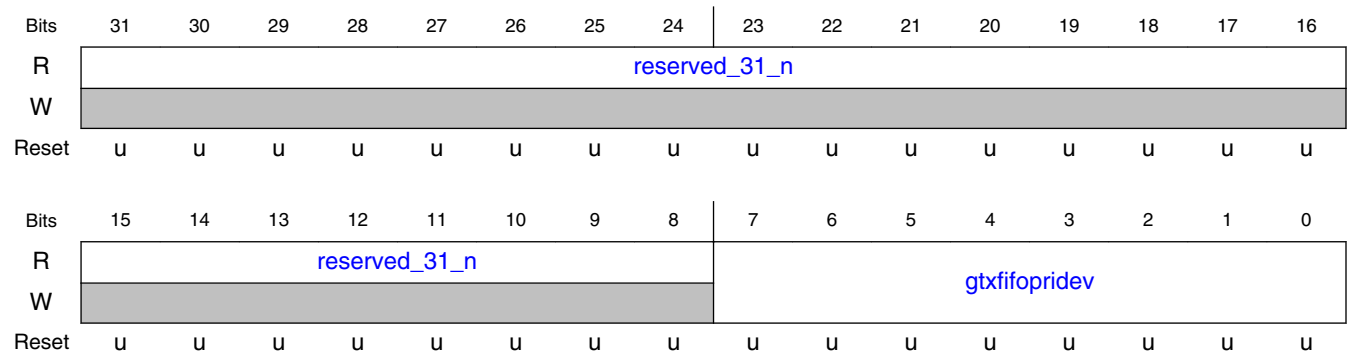
Register	Offset
GTXFIFOPRIDEV	C610h

11.1.3.1.82.2 Function

This register specifies the relative DMA priority level among the Device TXFIFOs (one per IN endpoint).

Reset Mask: 0xFFFF00

11.1.3.1.82.3 Diagram



11.1.3.1.82.4 Fields

Field	Function
31-8 reserved_31_n	Reserved
7-0 gtxfifoprdev	Device TxFIFO priority

11.1.3.1.83 Global Host TX FIFO DMA Priority Register (GTXFIFOPRIHST)

11.1.3.1.83.1 Offset

Register	Offset
GTXFIFOPRIHST	C618h

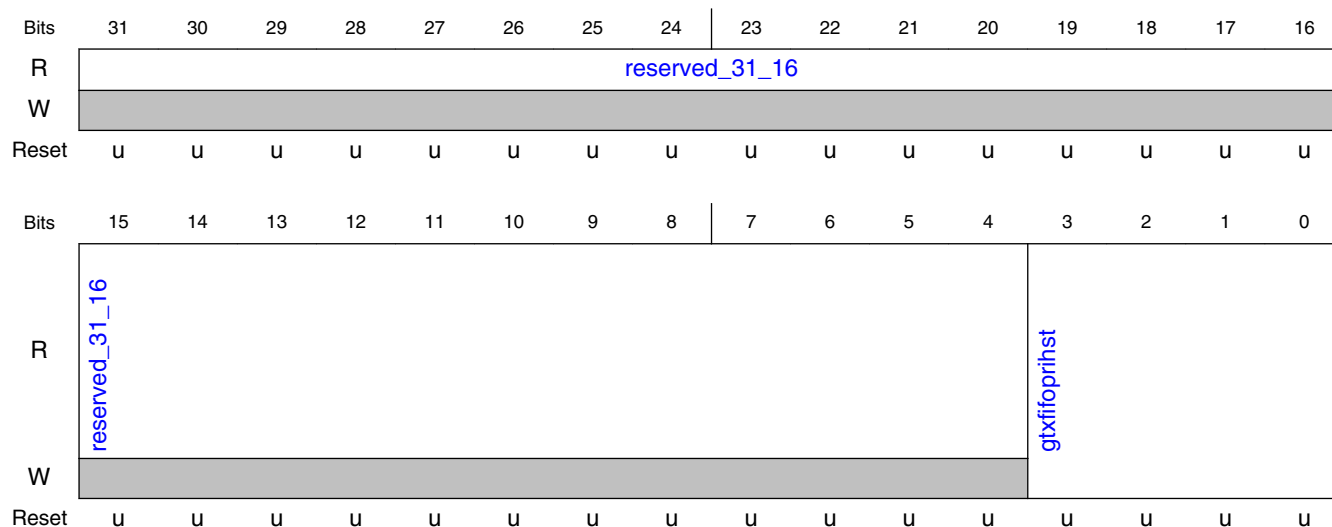
11.1.3.1.83.2 Function

This register specifies the relative DMA priority level among the Host TXFIFOs (one per USB bus instance) within the associated speed group (SS or HS/FSLs).

Reset Mask: 0xFFFF0

Reset Default Value: 0x0

11.1.3.1.83.3 Diagram



11.1.3.1.83.4 Fields

Field	Function
31-4 reserved_31_16	Reserved
3-0 gtxfifoprihst	Host TxFIFO priority

11.1.3.1.84 Global Host RX FIFO DMA Priority Register (GRXFIFOPRIHST)

11.1.3.1.84.1 Offset

Register	Offset
GRXFIFOPRIHST	C61Ch

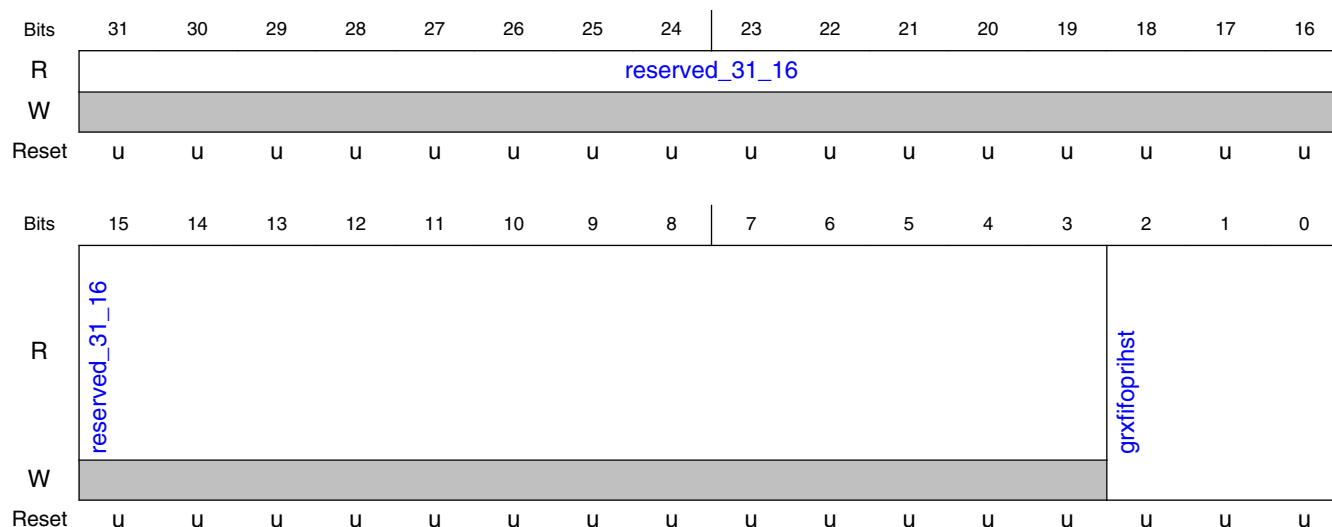
11.1.3.1.84.2 Function

This register specifies the relative DMA priority level among the Host RXFIFOs (one per USB bus instance) within the associated speed group (SS or HS/FSLs).

Reset Mask: 0x7FFF8

Reset Default Value: 0x0

11.1.3.1.84.3 Diagram



11.1.3.1.84.4 Fields

Field	Function
31-3 reserved_31_16	Reserved
2-0 grxfifoprihst	Host RxFIFO priority

11.1.3.1.85 Global Host Debug Capability DMA Priority Register (GFIFOPRIDBC)

11.1.3.1.85.1 Offset

Register	Offset
GFIFOPRIDBC	C620h

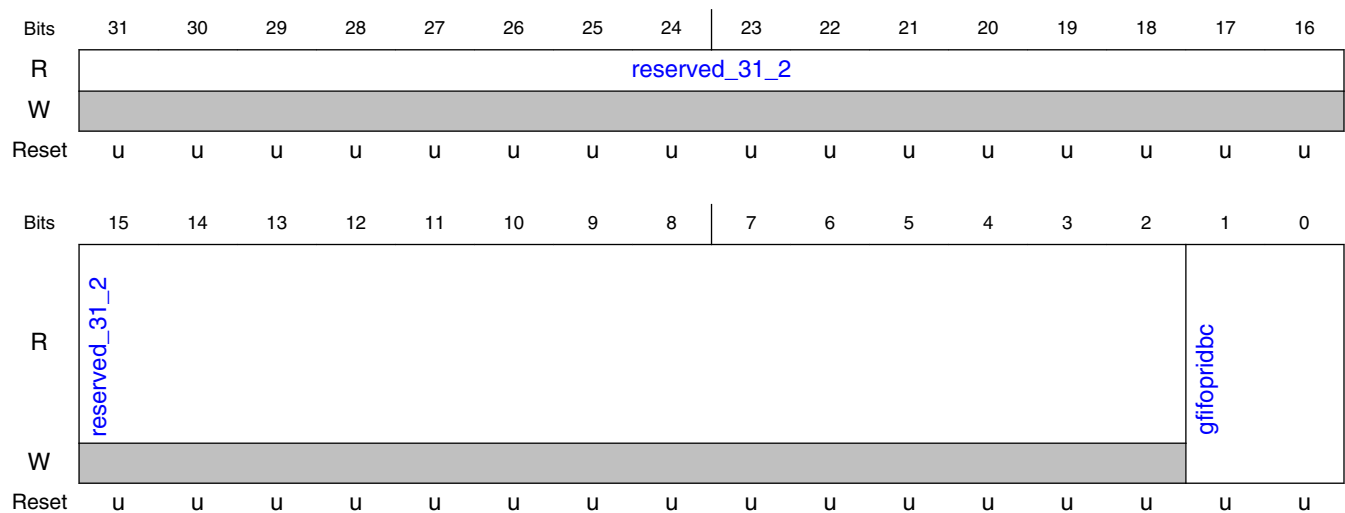
11.1.3.1.85.2 Function

This register specifies the relative priority of the RXFIFOs and TXFIFOs associated with the DbC mode.

Reset Mask: 0xFFFFFFFFFC

Reset Default Value: 0x0

11.1.3.1.85.3 Diagram



11.1.3.1.85.4 Fields

Field	Function
31-2 reserved_31_2	Reserved
1-0 gfifopridbc	Host DbC DMA priority

11.1.3.1.86 (GDMAHLRATIO)

11.1.3.1.86.1 Offset

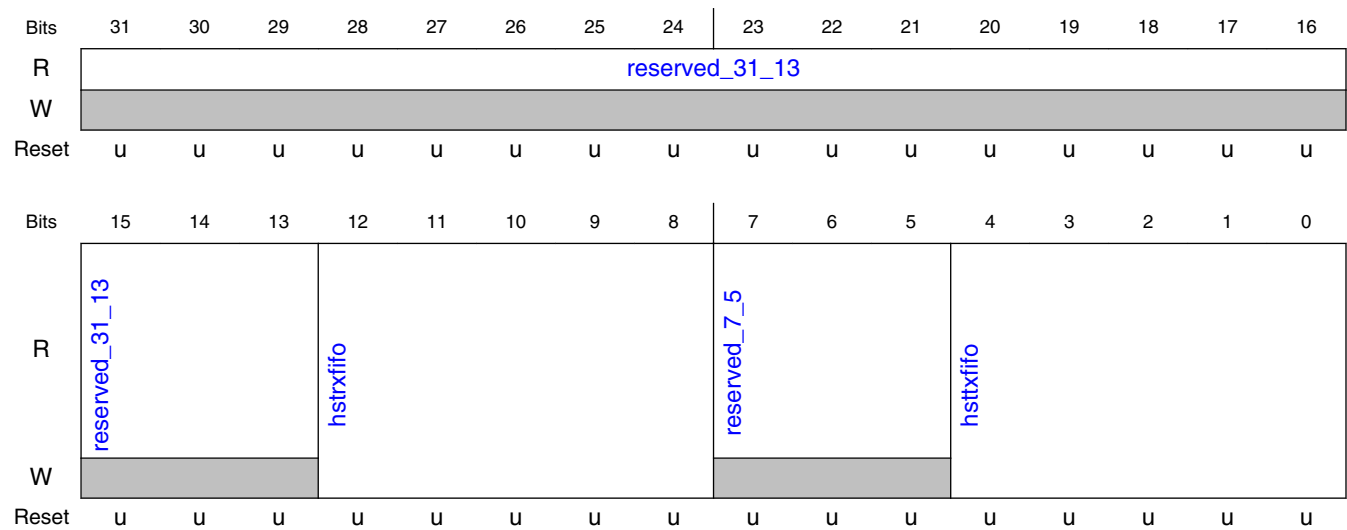
Register	Offset
GDMAHLRATIO	C624h

11.1.3.1.86.2 Function

Global Host FIFO DMA High-Low Priority Ratio Register This register specifies the relative priority of the SS FIFOs with respect to the HS/FSLs FIFOs.

Reset Mask:0xFFFFFE0E0

11.1.3.1.86.3 **Diagram**



11.1.3.1.86.4 **Fields**

Field	Function
31-13 reserved_31_13	Reserved
12-8 hstrxfifo	Host RXFIFO DMA High-Low Priority
7-5 reserved_7_5	Reserved
4-0 hsttxfifo	Host TXFIFO DMA High-Low Priority

11.1.3.1.87 **Global Frame Length Adjustment Register (GFLADJ)**

11.1.3.1.87.1 **Offset**

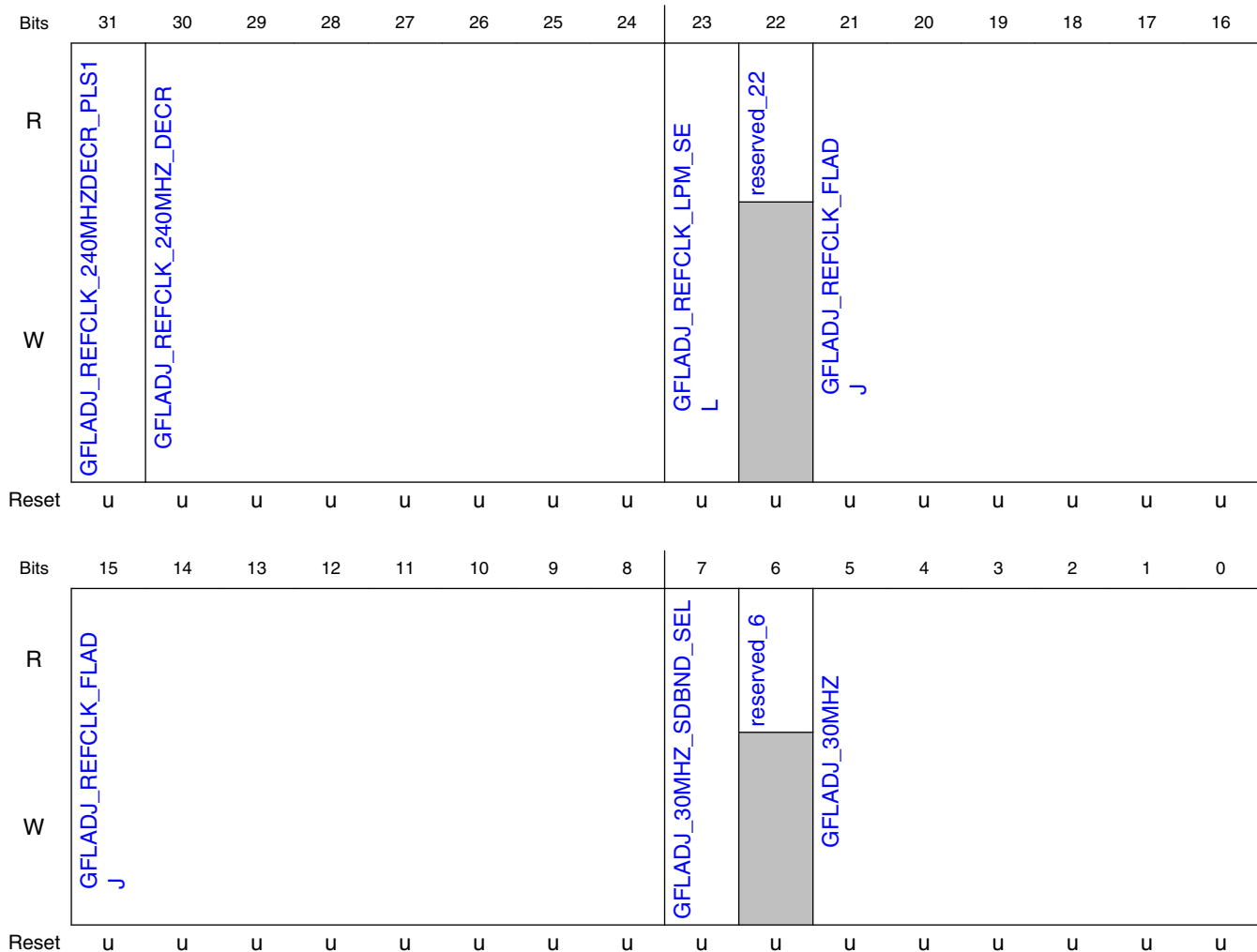
Register	Offset
GFLADJ	C630h

11.1.3.1.87.2 Function

This register provides options for the software to control the core behavior with respect to SOF (Start of Frame) and ITP (Isochronous Timestamp Packet) timers and frame timer functionality.

Reset Mask:0x400040

11.1.3.1.87.3 Diagram



11.1.3.1.87.4 Fields

Field	Function
31 GFLADJ_REFCLK_240MHZDECR_PLS1	GFLADJ_REFCLK_240MHZDECR_PLS1 This field indicates that the decrement value that the controller applies for each ref_clk must be GFLADJ_REFCLK_240MHZ_DECR and GFLADJ_REFCLK_240MHZ_DECR +1 alternatively on each ref_clk. Set this bit to a '1' only if GFLADJ_REFCLK_LPM_SEL is set to '1' and the fractional component of 240/ref_frequency is greater

Table continues on the next page...

Field	Function
	than or equal to 0.5. Examples: If the ref_clk is 19.2 MHz then - GUCTL.REF_CLK_PERIOD = 52 - GFLADJ.GFLADJ_REFCLK_240MHZ_DECR = $(240/19.2) = 12.5$ - GFLADJ.GFLADJ_REFCLK_240MHZDECR_PLS1 = 1 If the ref_clk is 24 MHz then - GUCTL.REF_CLK_PERIOD = 41 - GFLADJ.GFLADJ_REFCLK_240MHZ_DECR = $(240/24) = 10$ - GFLADJ.GFLADJ_REFCLK_240MHZDECR_PLS1 = 0
30-24 GFLADJ_REFCLK_240MHZ_DECR	This field indicates the decrement value that the controller applies for each ref_clk in order to derive a frame timer in terms of a 240-MHz clock. This field must be programmed to a non-zero value only if GFLADJ_REFCLK_LPM_SEL is set to '1'. The value is derived as follows: GFLADJ_REFCLK_240MHZ_DECR = $240/\text{ref_clk_frequency}$ Examples: If the ref_clk is 24 MHz then - GUCTL.REF_CLK_PERIOD = 41 - GFLADJ.GFLADJ_REFCLK_240MHZ_DECR = $240/24 = 10$ If the ref_clk is 48 MHz then - GUCTL.REF_CLK_PERIOD = 20 - GFLADJ.GFLADJ_REFCLK_240MHZ_DECR = $240/48 = 5$ If the ref_clk is 17 MHz then - GUCTL.REF_CLK_PERIOD = 58 - GFLADJ.GFLADJ_REFCLK_240MHZ_DECR = $240/17 = 14$
23 GFLADJ_REFCLK_LPM_SEL	This bit enables the functionality of running SOF/ITP counters on the ref_clk. This bit must not be set to '1' if GCTL.SOFITPSYNC bit is set to '1'. Similarly, if GFLADJ_REFCLK_LPM_SEL is set to '1', GCTL.SOFITPSYNC must not be set to '1'. When GFLADJ_REFCLK_LPM_SEL is set to '1' the overloading of the suspend control of the USB 2.0 first port PHY (UTMI/ULPI) with USB 3.0 port states is removed. For example, for Synopsys PHY, the COMMONONN signal can be tied to '1'. Note that the ref_clk frequencies supported in this mode are 16/17/19.2/20/24/39.7/40 MHz. The utmi_clk[0] signal of the core must be connected to the FREECLK of the PHY. Note: If you set this bit to '1', the GUSB2PHYCFG.U2_FREECLK_EXISTS bit must be set to '0'.
22 reserved_22	Reserved for future use
21-8 GFLADJ_REFCLK_FLADJ	This field indicates the frame length adjustment to be applied when SOF/ITP counter is running on the ref_clk. This register value is used to adjust the ITP interval when GCTL[SOFITPSYNC] is set to '1'; SOF and ITP interval when GLADJ.GFLADJ_REFCLK_LPM_SEL is set to '1'. This field must be programmed to a non-zero value only if GFLADJ_REFCLK_LPM_SEL is set to '1' or GCTL.SOFITPSYNC is set to '1'. The value is derived as follows: FLADJ_REF_CLK_FLADJ = $((125000/\text{ref_clk_period_integer}) - (125000/\text{ref_clk_period})) * \text{ref_clk_period}$ where - the ref_clk_period_integer is the integer value of the ref_clk period got by truncating the decimal (fractional) value that is programmed in the GUCTL.REF_CLK_PERIOD field. - the ref_clk_period is the ref_clk period including the fractional value. Examples: If the ref_clk is 24 MHz then - GUCTL.REF_CLK_PERIOD = 41 - GFLADJ.GFLADJ_REFCLK_FLADJ = $((125000/41) - (125000/41.6666)) * 41.6666 = 2032$ (ignoring the fractional value) If the ref_clk is 48 MHz then - GUCTL.REF_CLK_PERIOD = 20 - GFLADJ.GFLADJ_REFCLK_FLADJ = $((125000/20) - (125000/20.8333)) * 20.8333 = 5208$ (ignoring the fractional value)
7 GFLADJ_30MHZ_SDBND_SEL	GFLADJ_30MHZ_SDBND_SEL This field selects whether to use the input signal fladj_30mhz_reg or the GFLADJ.GFLADJ_30MHZ to adjust the frame length for the SOF/ITP. When this bit is set to, - 1, the controller uses the register field GFLADJ.GFLADJ_30MHZ value - 0, the controller uses the input signal fladj_30mhz_reg value
6 reserved_6	Reserved for future use
5-0 GFLADJ_30MHZ	GFLADJ_30MHZ This field indicates the value that is used for frame length adjustment instead of considering from the sideband input signal fladj_30mhz_reg. This enables post-silicon frame length adjustment in case the input signal fladj_30mhz_reg is connected to a wrong value or is not valid. For details on how to set this value, refer to section 5.2.4, "Frame Length Adjustment Register (FLADJ)," of the xHCI Specification.

11.1.3.1.88 Device Configuration Register (DCFG)

11.1.3.1.88.1 Offset

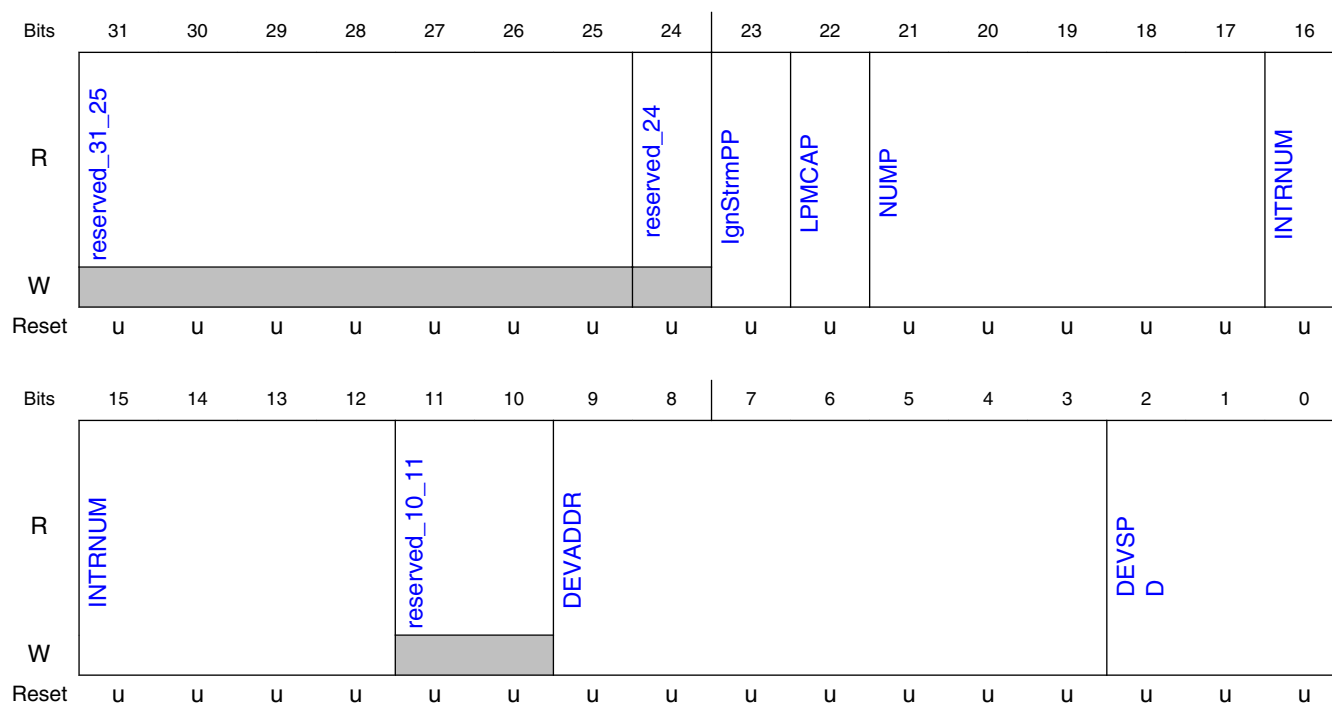
Register	Offset
DCFG	C700h

11.1.3.1.88.2 Function

Reset Mask:0x7F000C00

This register configures the core in device mode after power-on or after certain control commands or enumeration. Do not make changes to this register after initial programming.

11.1.3.1.88.3 Diagram



11.1.3.1.88.4 Fields

Field	Function
31-25 reserved_31_25	Reserved
24	reserved

Table continues on the next page...

Field	Function
reserved_24	
23 IgnStrmPP	IgnoreStreamPP This bit only affects stream-capable bulk endpoints. When this bit is set to '0' and the controller receives a Data Packet with the Packet Pending (PP) bit set to 0 for OUT endpoints, or it receives an ACK with the NumP field set to 0 and PP set to 0 for IN endpoints, the core attempts to search for another stream (CStream) to initiate to the host. However, there are two situations where this behavior is not optimal: - When the host is setting PP=0 even though it has not finished the stream, or - When the endpoint on the device is configured with one transfer resource and therefore does not have any other streams to initiate to the host. When this bit is set to '1', the core ignores the Packet Pending bit for the purposes of stream selection and does not search for another stream when it receives DP(PP=0) or ACK(NumP=0, PP=0). This can enhance the performance when the device system bus bandwidth is low or the host responds to the core's ERDY transmission very quickly.
22 LPMCAP	LPM Capable The application uses this bit to control the DWC_usb3 core LPM capabilities. If the core operates as a non-LPM-capable device, it cannot respond to LPM transactions. - 1'b0: LPM capability is not enabled. - 1'b1: LPM capability is enabled.
21-17 NUMP	Number of Receive Buffers. This bit indicates the number of receive buffers to be reported in the ACK TP. The DWC_usb3 controller uses this field for non-control endpoints if GRXTHRCFG.UsbRxPktCntSel is set to '0'. The application can program this value based on RxFIFO size, buffer sizes programmed in descriptors, and system latency. For an OUT endpoint, this field controls the number of receive buffers reported in the NumP field of the ACK TP transmitted by the core. Note: This bit is used in host mode when Debug Capability is enabled.
16-12 INTRNUM	Interrupt number Indicates interrupt/EventQ number on which non-endpoint-specific device-related interrupts (see DEVT) are generated.
11-10 reserved_10_11	Reserved
9-3 DEVADDR	Device Address. The application must perform the following: - Program this field after every SetAddress request. - Reset this field to zero after USB reset.
2-0 DEVSPD	Device Speed. Indicates the speed at which the application requires the core to connect, or the maximum speed the application can support. However, the actual bus speed is determined only after the chirp sequence is completed, and is based on the speed of the USB host to which the core is connected. - 3'b100: SuperSpeed (USB 3.0 PHY clock is 125 MHz or 250 MHz) - 3'b000: High-speed (USB 2.0 PHY clock is 30 MHz or 60 MHz) - 3'b001: Full-speed (USB 2.0 PHY clock is 30 MHz or 60 MHz) 000b - 001b - 100b -

11.1.3.1.89 Device control register (DCTL)

11.1.3.1.89.1 Offset

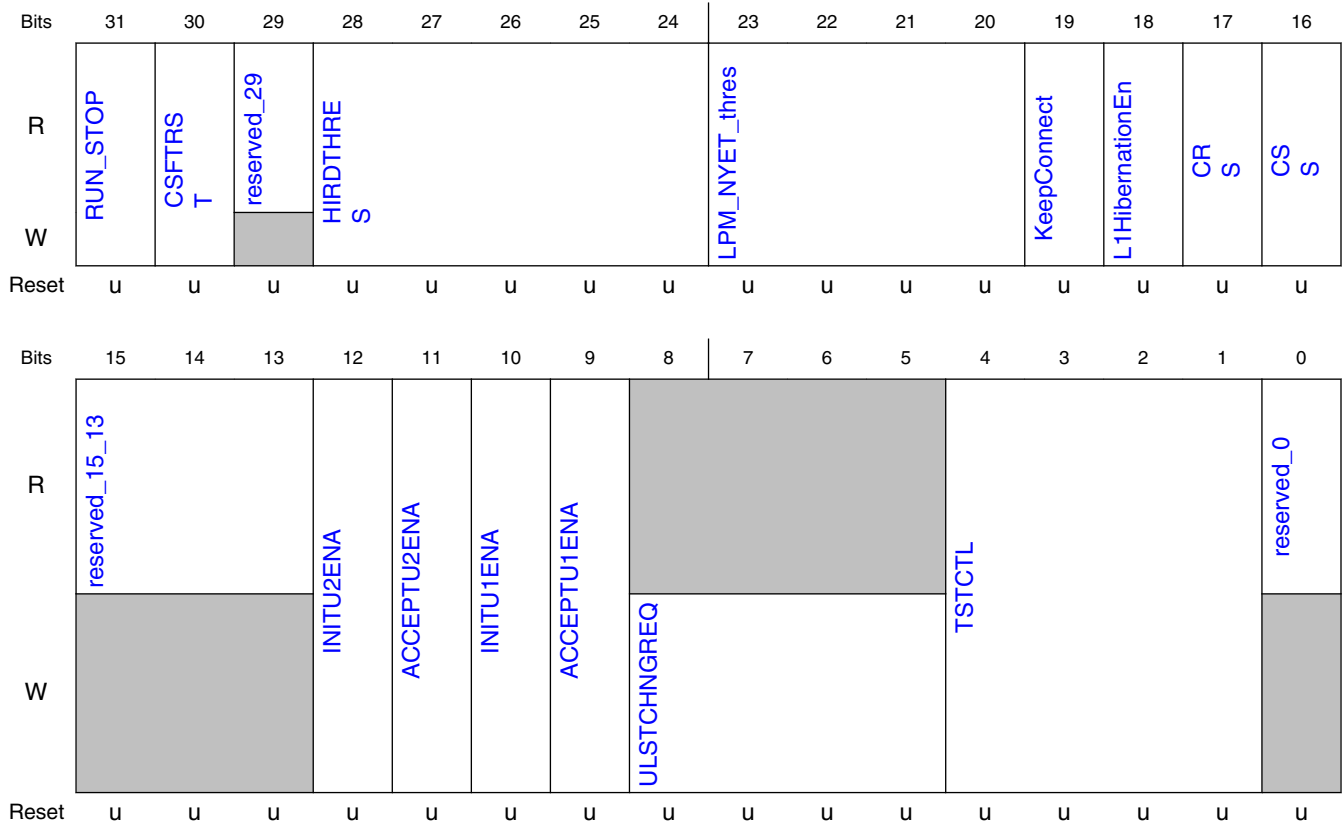
Register	Offset
DCTL	C704h

11.1.3.1.89.2 Function

When hibernation is not enabled using GCTL[GBLHIBERNATIONEN] bit,

- Any value can be written to CSS, CRS, L1HIBERNATIONEN, and KEEPCONNECT bits
- L1HIBERNATIONEN and KEEPCONNECT bits always return 0 when read in this hibernation-disabled state

11.1.3.1.89.3 Diagram



11.1.3.1.89.4 Fields

Field	Function
31 RUN_STOP	Run/Stop The software writes 1 to this bit to start the device controller operation. To stop the device controller operation, the software must remove any active transfers and write 0 to this bit. When the controller is stopped, it sets the DSTS.DevCtrlHlt bit when the core is idle and the lower layer finishes the disconnect process. The Run/Stop bit must be used in following cases as specified: - After power-on reset and CSR initialization, the software must write 1 to this bit to start the device controller. The controller does not signal connect to the host until this bit is set. - The software uses this bit to control the device controller to perform a soft disconnect. When the software writes 0 to this bit, the host does not see that the device is connected. The device controller stays in the disconnected state until the software writes 1 to this bit. The minimum duration of keeping this bit cleared is specified in the Note below. If the software attempts a connect after the soft disconnect or detects a disconnect event, it must set DCTL[8:5] to 5 before reasserting the Run/Stop bit. - When the USB or Link is in a lower power state and the Two Power Rails configuration is selected, software writes 0 to this bit to indicate that it is going to turn off the Core Power Rail. After the software turns on the Core Power Rail again and re-initializes the device

Table continues on the next page...

Field	Function
	controller, it must set this bit to start the device controller. For more details, see "Low Power Operation" section of the Databook. Note: The following is the minimum duration under various conditions for which the soft disconnect (SftDiscon) bit must be set for the USB host to detect a device disconnect: 30ms: - For SuperSpeed, when the device state is Suspended, Idle, Transmit, or Receive. 10ms: - For high-speed, when the device state is Suspended, Idle, or not Idle/Suspended (performing transactions) - For full-speed/low-speed, when the device state is Suspended, Idle, or not Idle/Suspended (performing transactions) To accommodate clock jitter, it is recommended that the application add extra delay to the specified minimum duration.
30 CSFTRST	Core Soft Reset Resets the all clock domains as follows: - This bit clears the interrupts and all the CSRs except GSTS, GSNPSID, GGPIO, GUID, GUSB2PHYCFGn registers, GUSB3PIPECTLn registers, DCFG, DCTL, DEVTEN, and DSTS registers. - All module state machines (except the SoC Bus Slave Unit) are reset to the IDLE state, and all the TxFIFOs and the Rx FIFO are flushed. - Any transactions on the SoC bus Master are terminated as soon as possible, after gracefully completing the last data phase of a SoC bus transfer. Any transactions on the USB are terminated immediately. The application can write this bit at any time to reset the core. This is a self-clearing bit; the core clears this bit after all necessary logic is reset in the core, which may take several clocks depending on the core's current state. Once this bit is cleared, the software must wait at least 3 PHY clocks before accessing the PHY domain (synchronization delay). Typically, software reset is used during software development and also when you dynamically change the PHY selection bits in the USB configuration registers listed above. When you change the PHY, the corresponding clock for the PHY is selected and used in the PHY domain. Once a new clock is selected, the PHY domain must be reset for proper operation. Note: Programming this field with random data causes side effect . Bit Bash register testing is not recommended.
29 reserved_29	Reserved1
28-24 HIRDTHRES	HIRD Threshold (HIRD_Thres) The core asserts output signals utmi_l1_suspend_n and utmi_sleep_n (see "LPM Interface Signals" table in the Databook) on the basis of this signal: The core asserts utmi_l1_suspend_n to put the PHY into Deep Low-Power mode in L1 when both of the following are true: - HIRD value is greater than or equal to the value in DCTL.HIRD_Thres[3:0] - HIRD_Thres[4] is set to 1'b1. The core asserts utmi_sleep_n on L1 when one of the following is true: - If the HIRD value is less than HIRD_Thres[3:0] or - HIRD_Thres[4] is set to 1'b0. Note: This field must be set to '0' during SuperSpeed mode of operation.
23-20 LPM_NYET_thres	LPM NYET Threshold When LPM Errata is enabled: Bits [23:20]: LPM NYET Response Threshold (LPM_NYET_thres) Handshake response to LPM token specified by device application. Response depends on DCFG.LPMCap. - DCFG.LPMCap is 1'b0 - The core always responds with Timeout (that is, no response). - DCFG.LPMCap is 1'b1 - The core responds with an ACK on successful LPM transaction, which requires that all of the following are satisfied: - There are no PID or CRC5 errors in both the EXT token and the LPM token (if not true, inactivity results in a timeout ERROR). - No data is pending in the TxFIFO and Rx FIFO is empty (else NYET). - The BESL value in the LPM token is less than or equal to LPM_NYET_thres[3:0]
19 KeepConnect	Keep Connect When '1', this bit enables the save and restore programming model by preventing the core from disconnecting from the host when DCTL.RunStop is set to '0'. It also enables the Hibernation Request Event to be generated when the link goes to U3 or L2. The device core disconnects from the host when DCTL.RunStop is set to '0'. This bit indicates whether to preserve this behavior ('0'), or if the core must not disconnect when RunStop is set to 0 ('1'). This bit also prevents the LTSSM from automatically going to U0/L0 when the host requests resume from U3/L2. Note: If Hibernation is disabled, that is, GCTL[1].GblHibernationEn = 0, this bit is tied to zero.
18 L1HibernationEn	L1HibernationEn When this bit is set along with KeepConnect, the device core generates a Hibernation Request Event if L1 is enabled and the HIRD value in the LPM token is larger than the threshold programmed in DCTL.HIRD_Thres. The core does not exit the LPM L1 state until software writes Recovery into the DCTL.ULStChngReq field. This prevents corner cases where the device is entering hibernation at the same time the host is attempting to exit L1. Note: If Hibernation is disabled, that is, GCTL[1].GblHibernationEn = 0, this bit is tied to zero.
17	Controller Restore State (CRS) This command is similar to the USBCMD.CRS bit in host mode and initiates the restore process. When software sets this bit to '1', the controller immediately sets DSTS.RSS

Table continues on the next page...

Field	Function
CRS	to '1'. When the controller has finished the restore process, it sets DSTS.RSS to '0'. Note: When read, this field always returns '0'.
16 CSS	Controller Save State (CSS) This command is similar to the USBCMD.CSS bit in host mode and initiates the save process. When software sets this bit to '1', the controller immediately sets DSTS.SSS to '1'. When the controller has finished the save process, it sets DSTS.SSS to '0'. Note: When read, this field always returns '0'.
15-13 reserved_15_13	Reserved
12 INITU2ENA	Initiate U2 Enable - 1'b0: May not initiate U2 (default) - 1'b1: May initiate U2 On USB reset, hardware clears this bit to 0. Software sets this bit after receiving SetFeature(U2_ENABLE), and clears this bit when ClearFeature(U2_ENABLE) is received. If DCTL[11] (AcceptU2Ena) is 0, the link immediately exits U2 state.
11 ACCEPTU2ENA	Accept U2 Enable - 1'b0: Reject U2 except when Force_LinkPM_Accept bit is set (default) - 1'b1: Core accepts transition to U2 state if nothing is pending on the application side. On USB reset, hardware clears this bit to 0. Software sets this bit after receiving a SetConfiguration command.
10 INITU1ENA	Initiate U1 Enable - 1'b0: May not initiate U1 (default); - 1'b1: May initiate U1. On USB reset, hardware clears this bit to 0. Software sets this bit after receiving SetFeature(U1_ENABLE), and clears this bit when ClearFeature(U1_ENABLE) is received. If DCTL[9] (AcceptU1Ena) is 0, the link immediately exits U1 state.
9 ACCEPTU1ENA	Accept U1 Enable - 1'b0: Core rejects U1 except when Force_LinkPM_Accept bit is set (default) - 1'b1: Core accepts transition to U1 state if nothing is pending on the application side. On USB reset, hardware clears this bit to 0. Software sets this bit after receiving a SetConfiguration command.
8-5 ULSTCHNGREQ	ULSTCHNGREQ Software writes this field to issue a USB/Link state change request. A change in this field indicates a new request to the core. If software wants to issue the same request back-to-back, it must write a 0 to this field between the two requests. The result of the state change request is reflected in the USB/Link State in DSTS. These bits are self-cleared on the MAC Layer exiting suspended state. If software is updating other fields of the DCTL register and not intending to force any link state change, then it must write a 0 to this field. SS Compliance mode is normally entered and controlled by the remote link partner. Refer to the USB 3.0 specification. Alternatively, you can force the local link directly into compliance mode, by resetting the SS link with the RUN/STOP bit set to zero. If you then write '10' to the USB/Link State Change field and '1' to RUN/STOP, the link goes to compliance mode. Once you are in compliance, you may alternately write zero and '10' to this field to advance the compliance pattern. In SS mode: - Value Requested Link State Transition/Action - 0 No Action - 4 SS.Disabled - 5 Rx.Detect - 6 SS.Inactive - 8 Recovery - 10 Compliance - Others: Reserved In HS/FS/LS mode: - ValueRequested USB state transition - 8 Remote wakeup request - Others: Reserved The Remote wakeup request must be issued 2us after the device goes into suspend state (DSTS[21:18] is 3 - refer to Table "Fields for Register: DSTS" in the Databook). Note: After coming out of hibernation, software must write 8 (Recovery) into this field to confirm exit from the suspended state.
4-1 TSTCTL	Test Control - 4'b000: Test mode disabled - 4'b001: Test_J mode - 4'b010: Test_K mode - 4'b011: Test_SE0_NAK mode - 4'b100: Test_Packet mode - 4'b101: Test_Force_Enable - Others: Reserved
0 reserved_0	Reserved

11.1.3.1.90 Device Event Enable Register (DEV TEN)

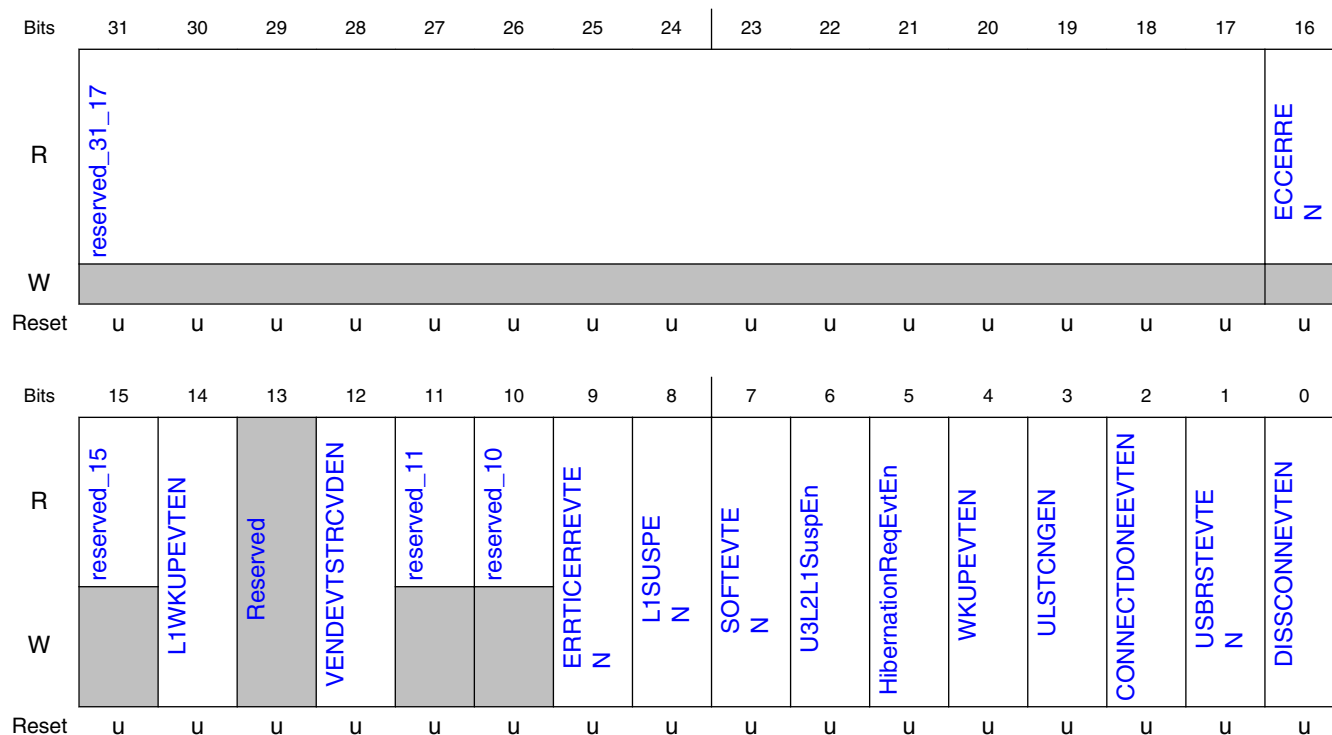
11.1.3.1.90.1 Offset

Register	Offset
DEV TEN	C708h

11.1.3.1.90.2 Function

This register controls the generation of device-specific events (see "Event Buffer Content for Device-Specific Events (DEVT)")

Reset Mask:0xFFFE0C00

11.1.3.1.90.3 Diagram**11.1.3.1.90.4 Fields**

Field	Function
31-17 reserved_31_17	Reserved
16 ECCERREN	ECC Error Enable. If this bit is set to 1, the controller reports an ECC error to the software when an uncorrectable ECC occurs internally.

Table continues on the next page...

Field	Function
15 reserved_15	Reserved
14 L1WKUPEVTEN	L1 Resume Detected Event Enable. Note: If GUCTL1[DEV_DECOUPLE_L1L2_EVT] is enabled, then this bit is for L1 Resume Detected Event Enable.
13 —	Reserved
12 VENDEVTSTRC VDEN	Vendor Device Test LMP Received Event (VndrDevTstRcvdEn)
11 reserved_11	Reserved
10 reserved_10	Reserved
9 ERRTICERREV TEN	Erratic Error Event Enable
8 L1SUSPEN	L1 Suspend Event Enable Note: Only if GUCTL1[DEV_DECOUPLE_L1L2_EVT] is enabled, this bit is for L1 Suspend Event Enable.
7 SOFTEVTEN	Start of (u)frame
6 U3L2L1SuspEn	U3/L2 or U3/L2L1 Suspend Event Enable. Note: - If GUCTL1[DEV_DECOUPLE_L1L2_EVT] is enabled, then this bit is for U3/L2 Suspend Event Enable. - If GUCTL1[DEV_DECOUPLE_L1L2_EVT] is not enabled, then this bit is for U3/L2L1 Suspend Event Enable.
5 HibernationReq EvtEn	This bit enables/disables the generation of the Hibernation Request Event.
4 WKUPEVTEN	U3/L2 or U3/L2L1 Resume Detected Event Enable. Note: - If GUCTL1[DEV_DECOUPLE_L1L2_EVT] is enabled, then this bit is for U3/L2 Resume Detected Event Enable. - If GUCTL1[DEV_DECOUPLE_L1L2_EVT] is not enabled, then this bit is for U3/L2L1 Resume Detected Event Enable.
3 ULSTCNGEN	USB/Link State Change Event Enable
2 CONNECTDON EEVTEN	Connection Done Enable
1 USBRSTEV TEN	USB Reset Enable
0 DISSCONNEVT EN	Disconnect Detected Event Enable

11.1.3.1.91 Device Status Register (DSTS)

11.1.3.1.91.1 Offset

Register	Offset
DSTS	C70Ch

11.1.3.1.91.2 Function

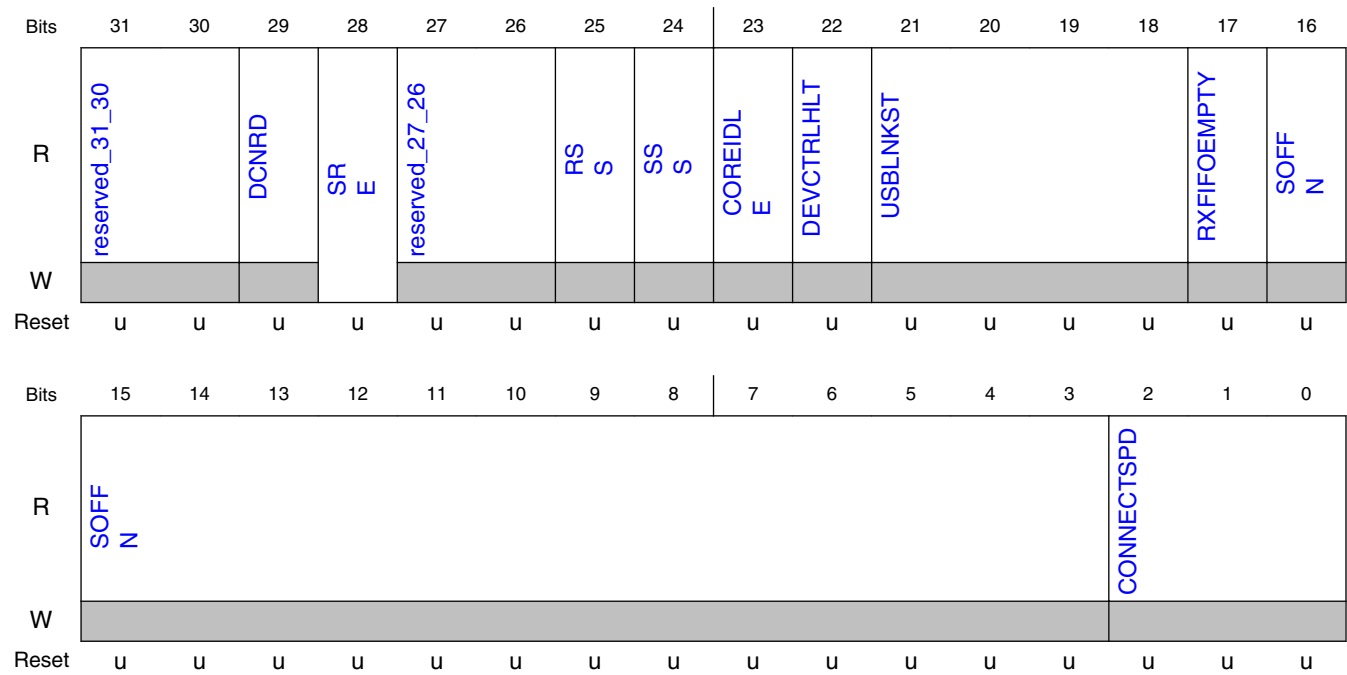
Device Status Register This register indicates the status of the device controller with respect to USB-related events.

Reset Mask:0xDC000000

NOTE

When hibernation is not enabled, RSS and SSS bits always return 0 when read.

11.1.3.1.91.3 Diagram



11.1.3.1.91.4 Fields

Field	Function
31-30 reserved_31_30	Reserved
29 DCNRD	Device Controller Not Ready The bit indicates that the core is in the process of completing the state transitions after exiting from hibernation. To complete the state transitions, it takes 256 bus clock cycles from the time DCTL[31].Run/Stop is set. During hibernation, if the UTMI/ULPI PHY is in suspended state, then the 256-bus clock cycle delay starts after the PHY exited suspended state. Software must set DCTL[31].Run/Stop to '1' and wait for this bit to be de-asserted to zero before processing DSTS.USBLnkSt. This bit is valid only when DWC_USB3_EN_PWROPT is set to 2 and GCTL[1].GblHibernationEn = 1.
28 SRE	Save Restore Error. Currently not supported.
27-26 reserved_27_26	Reserved
25 RSS	RSS Restore State Status This bit is similar to the USBSTS.RSS in host mode. When the controller finishes the restore process, it completes the command by setting DSTS.RSS to '0'.
24 SSS	SSS Save State Status This bit is similar to the USBSTS.SSS in host mode. When the controller has finished the save process, it completes the command by setting DSTS.SSS to '0'.
23 COREIDLE	Core Idle The bit indicates that the core finished transferring all RxFIFO data to system memory, writing out all completed descriptors, and all Event Counts are zero. Note: While testing for Reset values, mask out the read value. This bit represents the changing state of the core and does not hold a static value.
22 DEVCTRLHLT	Device Controller Halted This bit is set to 0 when the Run/Stop bit in the DCTL register is set to 1. The core sets this bit to 1 when, after SW sets Run/Stop to 0, the core is idle and the lower layer finishes the disconnect process. When Halted=1, the core does not generate Device events. Note: - The core does not set this bit to 1 if GEVNTCOUNTn has some valid value. Software needs to acknowledge the events that are generated (by writing to GEVNTCOUNTn) while it is waiting for this bit to be set to 1. - When Interrupt Moderation is enabled, there could be delay in raising the interrupt line when the event count is non-zero. Software should read the GEVNTCOUNT register directly and acknowledge them.
21-18 USBLNKST	USBLNKST. USB/Link State In SS mode: LTSSM State - 4'h0: U0 - 4'h1: U1 - 4'h2: U2 - 4'h3: U3 - 4'h4: SS_DIS - 4'h5: RX_DET - 4'h6: SS_INACT - 4'h7: POLL - 4'h8: RECOV - 4'h9: HRESET - 4'ha: CPLY - 4'hb: LPBK - 4'hf: Resume/Reset In HS/FS/LS mode: - 4'h0: On state - 4'h2: Sleep (L1) state - 4'h3: Suspend (L2) state - 4'h4: Disconnected state (Default state) - 4'h5: Early Suspend state (valid only when Hibernation is disabled, GCTL[1].GblHibernationEn = 0) - 4'he: Reset (valid only when Hibernation is enabled, GCTL[1].GblHibernationEn = 1) - 4'hf: Resume (valid only when Hibernation is enabled, GCTL[1].GblHibernationEn = 1) The link state Resume/Reset indicates that the core received a resume or USB reset request from the host while the link was in hibernation. Software must write '8' (Recovery) to the DCTL.ULStChngReq field to acknowledge the resume/reset request. When Hibernation is enabled, GCTL[1].GblHibernationEn = 1, this field USBLnkSt is valid only when DCTL[31].Run/Stop set to '1' and DSTS[29].DCNRD = 0. Note: If SSIC is enabled, while exiting a low power state, the USBLnkSt field indicates Resume/Reset even for a disconnect condition because a resume precedes the disconnect.
17 RXFIFOEMPTY	RxFIFO Empty.
16-3 SOFFN	Frame/Microframe Number of the Received SOF. When the core is operating at SuperSpeed, - [16:3] indicates the uframe/ITP number When the core is operating at high-speed, - [16:6] indicates the frame number - [5:3] indicates the microframe number When the core is operating at full-speed, - [16:14] is not used. Software can ignore these 3 bits - [13:3] indicates the frame number
2-0 CONNECTSPD	Connected Speed (ConnectSpd) Indicates the speed at which the DWC_usb3 core has come up after speed detection through a chirp sequence. - 3'b100: SuperSpeed (PHY clock is running at 125 or 250

Field	Function
	MHz) - 3'b000: High-speed (PHY clock is running at 30 or 60 MHz) - 3'b001: Full-speed (PHY clock is running at 30 or 60 MHz) Low-speed is not supported for devices using a UTMI+ PHY. 000b - 001b - 100b -

11.1.3.1.92 Device Generic Command Parameter Register (DGCMDPAR)

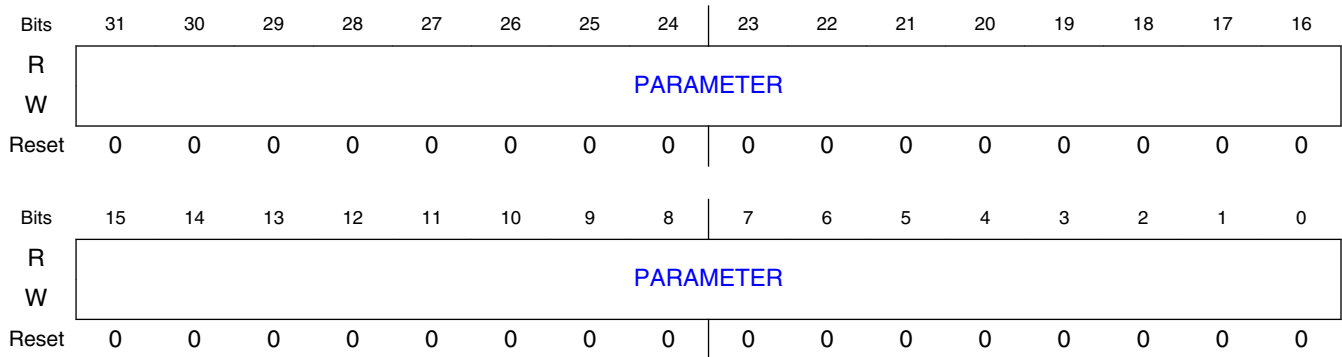
11.1.3.1.92.1 Offset

Register	Offset
DGCMDPAR	C710h

11.1.3.1.92.2 Function

This register indicates the device command parameter.

11.1.3.1.92.3 Diagram



11.1.3.1.92.4 Fields

Field	Function
31-0 PARAMETER	PARAMETER

11.1.3.1.93 (DGCMD)

11.1.3.1.93.1 Offset

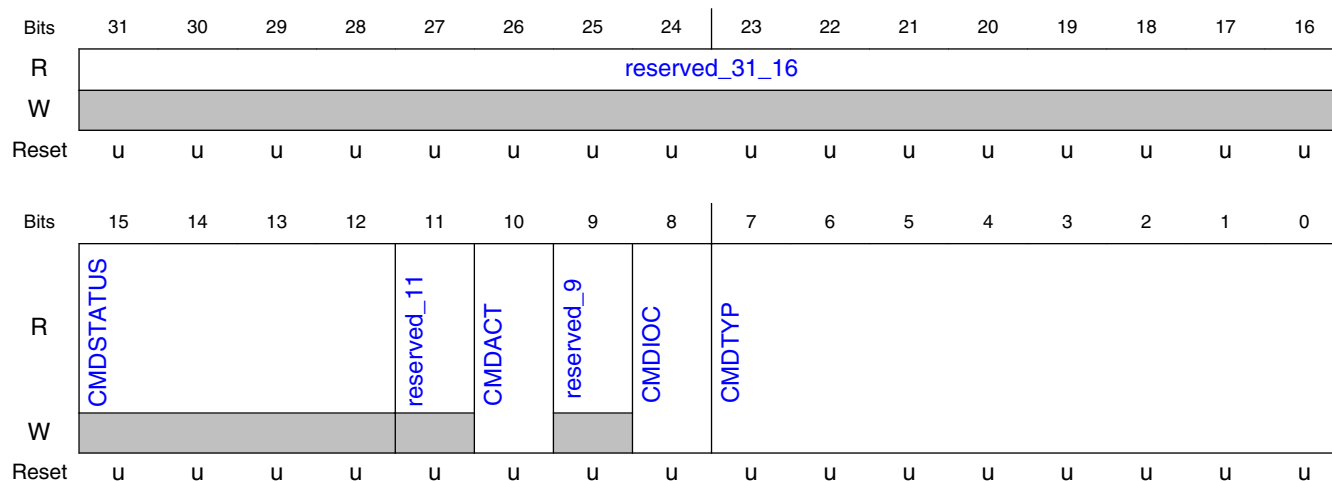
Register	Offset
DGCMD	C714h

11.1.3.1.93.2 Function

Device Generic Command Register This register enables software to program the core using a single generic command interface to send link management packets and notifications.

Reset Mask:0xFFFF0A00

11.1.3.1.93.3 Diagram



11.1.3.1.93.4 Fields

Field	Function
31-16 reserved_31_16	Reserved
15-12 CMDSTATUS	Command Status - 1: CmdErr: Indicates that the device controller encountered an error while processing the command. - 0: Indicates command success
11 reserved_11	Reserved
10 CMDACT	Command Active The software sets this bit to 1 to enable the device controller to execute the generic command. The device controller sets this bit to 0 after executing the command.
9	Reserved

Table continues on the next page...

Field	Function
reserved_9	
8 CMDIOC	Command Interrupt on Complete When this bit is set, the device controller issues a Generic Command Completion event after executing the command. Note that this interrupt is mapped to DCFG.IntrNum. Note: This field must not set to '1' if the DCTL.RunStop field is '0'.
7-0 CMDTYP	CMDTYP Generic Command Type Specifies the type of generic command the software driver is requesting the controller to perform. - 02h: Set Periodic Parameters - 04h: Set Scratchpad Buffer Array Address Lo - 05h: Set Scratchpad Buffer Array Address Hi - 07h: Transmit Device Notification - 09h: Selected FIFO Flush - 0Ah: All FIFO Flush - 0Ch: Set Endpoint NRDY - 10h: Run SoC Bus LoopBack Test - 11h: Restart After Disconnect All other values are reserved.

11.1.3.1.94 Device Active USB Endpoint Enable Register (DALEPENA)

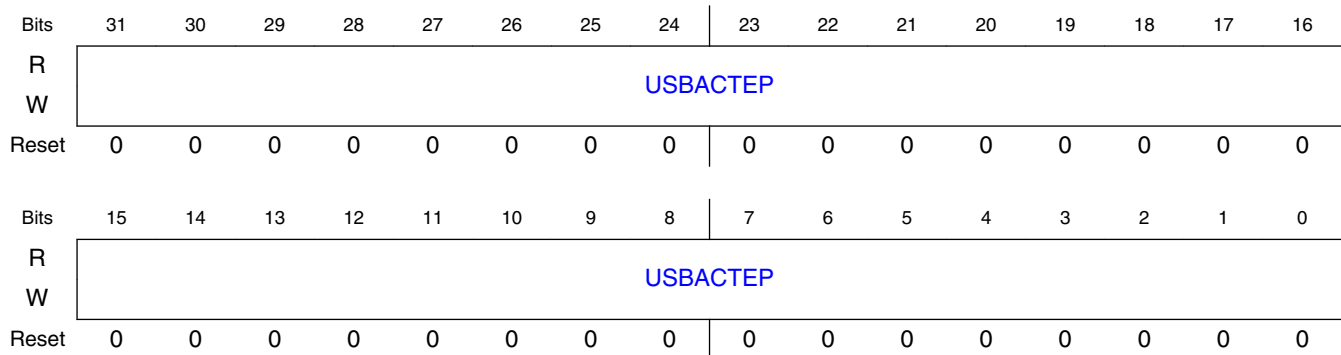
11.1.3.1.94.1 Offset

Register	Offset
DALEPENA	C720h

11.1.3.1.94.2 Function

This register indicates whether a USB endpoint is active in a given configuration or interface.

11.1.3.1.94.3 Diagram



11.1.3.1.94.4 Fields

Field	Function
31-0 USBACTEP	USB active endpoints This bit indicates if a USB endpoint is active in the current configuration and interface.

Field	Function
	Bit[0]: USB EP0-OUT Bit[1]: USB EP0-IN Bit[2]: USB EP1-OUT Bit[3]: USB EP1-IN Bit[4]: USB EP2-OUT Bit[5]: USB EP2-IN Bit[6]: USB EP3-OUT Bit[7]: USB EP3-IN The entity programming this register must set bits 0 and 1 because they enable control endpoints that map to physical endpoints (resources) after USB reset. Hardware clears these bits for all endpoints (other than EP0-OUT and EP0-IN) after detecting a USB reset event. After receiving SetConfiguration and SetInterface requests, the application must program endpoint registers accordingly and set these bits.

11.1.3.1.95 Device physical endpoint-n command parameter 2 register (DEPCMDPAR2)

11.1.3.1.95.1 Offset

Register	Offset
DEPCMDPAR2	C800h

11.1.3.1.95.2 Function

This register indicates the physical endpoint command Parameter 2.

11.1.3.1.95.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	PARAMETER															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PARAMETER															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.1.3.1.95.4 Fields

Field	Function
31-0 PARAMETER	PARAMETER

11.1.3.1.96 Device Physical Endpoint-n Command Parameter 1 Register (DEPCMDPAR1)**11.1.3.1.96.1 Offset**

Register	Offset
DEPCMDPAR1	C804h

11.1.3.1.96.2 Function

This register indicates the physical endpoint command parameter 1. It must be programmed before issuing the command.

11.1.3.1.96.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	PARAMETER															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PARAMETER															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.1.3.1.96.4 Fields

Field	Function
31-0 PARAMETER	PARAMETER

11.1.3.1.97 Device Physical Endpoint-n Command Parameter 0 Register (DEPCMDPAR0)

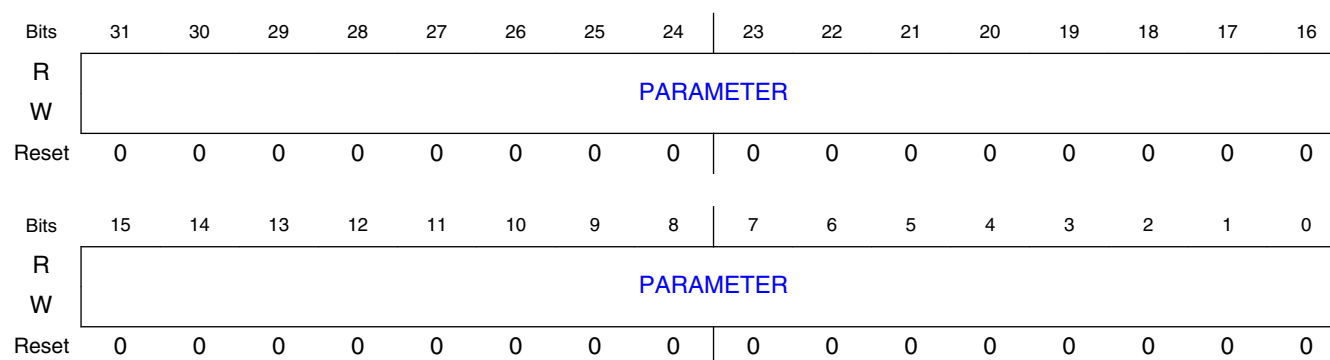
11.1.3.1.97.1 Offset

Register	Offset
DEPCMDPAR0	C808h

11.1.3.1.97.2 Function

This register indicates the physical endpoint command parameter 0. This must be programmed before or along with the command. For commands needing only one 32-bit parameter, this register must be programmed with the command register.

11.1.3.1.97.3 Diagram



11.1.3.1.97.4 Fields

Field	Function
31-0 PARAMETER	PARAMETER

11.1.3.1.98 Device Physical Endpoint-n Command Register (DEPCMD)

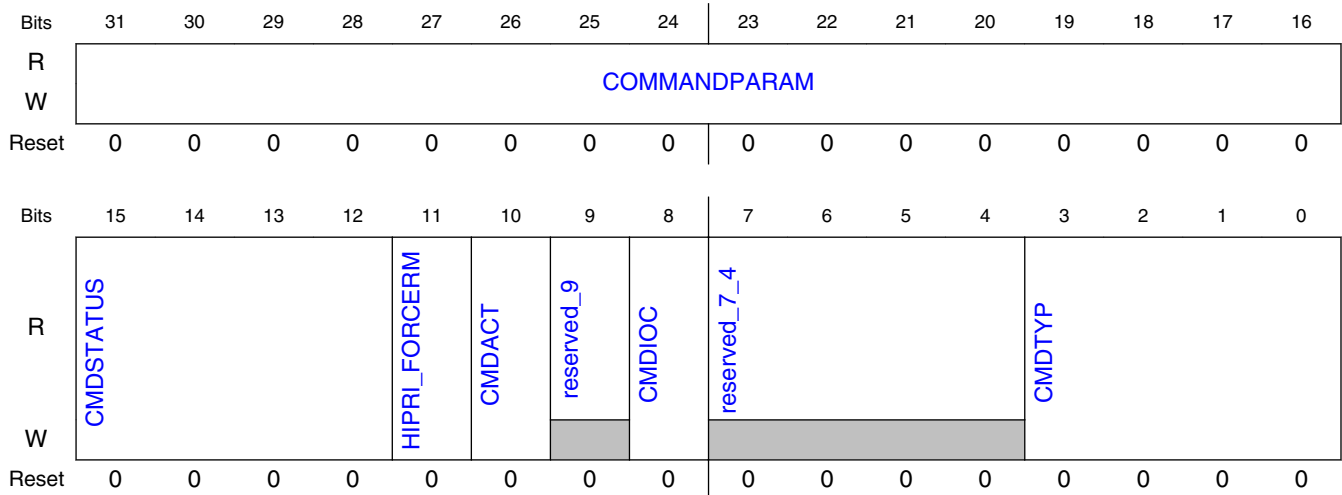
11.1.3.1.98.1 Offset

Register	Offset
DEPCMD	C80Ch

11.1.3.1.98.2 Function

This register enables software to issue physical endpoint-specific commands.

11.1.3.1.98.3 Diagram



11.1.3.1.98.4 Fields

Field	Function
31-16 COMMANDPARAM	Command Parameters or Event Parameters Command Parameters (CommandParam), when this register is written: For Start Transfer command: - [31:16]: StreamID.
15-12 CMDSTATUS	Command Completion Status (CmdStatus) Additional information about the completion of this command is available in this field.
11 HIPRI_FORCERM	HighPriority/ForceRM (HiPri_ForceRM) - HighPriority: Only valid for Start Transfer command - ForceRM: Only valid for End Transfer command - ClearPendIN: Only valid for Clear Stall command .
10 CMDACT	Command Active (CmdAct) Software sets this bit to 1 to enable the device endpoint controller to execute the generic command.
9 reserved_9	Reserved
8 CMDIOC	CMDIOC Command Interrupt on Complete (CmdIOC) When this bit is set, the device controller issues a generic Endpoint Command Complete event after executing the command.
7-4 reserved_7_4	Reserved

Table continues on the next page...

Field	Function
3-0 CMDTYP	Command Type Specifies the type of command the software driver is requesting the core to perform. 01h: Set Endpoint Configuration 64 or 96-bit Parameter

11.1.3.1.99 Device Interrupt Moderation Register (DEV_IMOD)

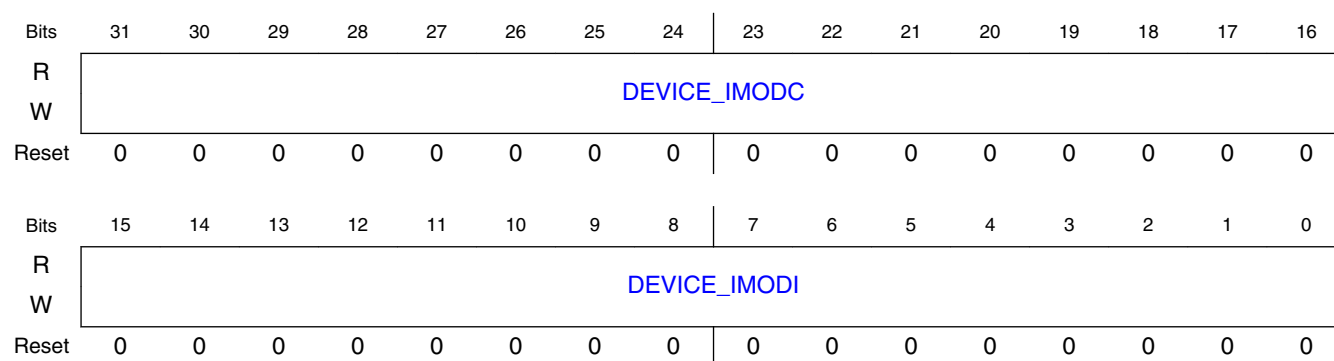
11.1.3.1.99.1 Offset

Register	Offset
DEV_IMOD	CA00h

11.1.3.1.99.2 Function

the Interrupt Moderation feature that allows the device software to throttle the interrupt rate.

11.1.3.1.99.3 Diagram



11.1.3.1.99.4 Fields

Field	Function
31-16 DEVICE_IMODC	Interrupt Moderation Down Counter Loaded with the DEVICE_IMODI value, whenever the hardware interrupt(n) line is de-asserted from the asserted state, counts down to 0, and stops.
15-0 DEVICE_IMODI	Moderation Interval (DEVICE_IMODI) This field holds the minimum inter-interrupt interval between events.

11.1.3.1.100 OTG Configuration Register (OCFG)

11.1.3.1.100.1 Offset

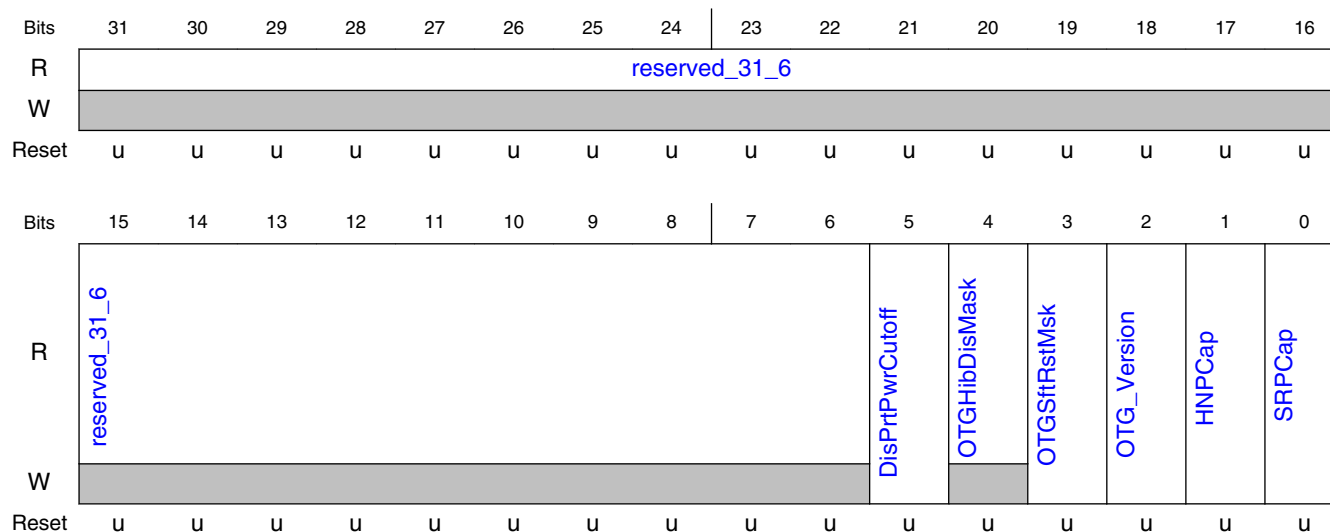
Register	Offset
OCFG	CC00h

11.1.3.1.100.2 Function

This register specifies the HNP and SRP capability of the DWC_usb3 core.

Reset Mask:0xFFFFFC0

11.1.3.1.100.3 Diagram



11.1.3.1.100.4 Fields

Field	Function
31-6 reserved_31_6	Reserved
5 DisPrtpwrCutoff	OTG Disable Port Power Cut Off (DisPrtpwrCutoff) - 0: The core automatically turns off the VBUS by clearing the OCTL.PrtpwrCtl after A_WAIT_BCON Timeout whenever the port is disconnected in disconnected state. If Hibernation is enabled and when a hibernation request is received in disconnected state, the core switches off VBUS instantly by clearing the OCTL.PrtpwrCtl. - 1: The core maintains VBUS ON even after A_WAIT_BCON Timeout when port is in disconnected state. The core is in a A_WAIT_BCON state continuously waiting for a Connect. If hibernation is enabled and when hibernation

Table continues on the next page...

Field	Function
	request is received in disconnected state, the core maintains VBUS ON and continues to drive VBUS even in hibernation.
4 OTGHibDisMask	OTG Hibernation Disable Mask. - 0 - Any change in PHY input signals relevant to OTG (ID, Vbus-valid, B-Valid) are masked from generating a corresponding event when the core receives Hibernation Save request from Host or Device Driver. The change in signals is masked until the Host or Device Run/Stop is programmed to 1. - 1 - The PHY input signals are not masked even after the core receives the Hibernation Save request from Host or Device Driver. For normal functionality, this bit must be 1'b0.
3 OTGSftRstMsk	OTG Soft Reset Mask. This bit is used to mask specific soft resets from affecting the OTG functionality of the core. When set, the xHCI-based USBCMD.HCRST in host mode and DCTL.CSftRst in device mode is masked from affecting reset signal outputs sent to the PHY, the OTG FSM logic of the core and also the resets to the VBUS filters inside the core. - 1'b0: The xHCI-based USBCMD.HCRST and DCTL.CSftRst resets the OTG logic of the core. - 1'b1: The xHCI-based USBCMD.HCRST and DCTL.CSftRst is masked from the OTG logic of the core. This bit can be programmed to allow existing xHCI flows (with USBCMD.HCRST programming) to function in OTG scenarios without any software changes. This bit must be programmed only when GCTL.PrtCapDir = 2'b11. Otherwise it must be set at 1'b0. Note: When using the core for OTG2 or OTG3 applications, it is not recommended to program USBCMD.HCRST during role switch.
2 OTG_Version	This is a debug bit and it must always be set to 1'b0.
1 HNPCap	RSP/HNP Capability. The terminology RSP is used when the core is operating in SS mode, and HNP is used when the core is operating in non-SS mode. The application uses this bit to control the DWC_usb3 core's RSP/HNP capabilities. - 1'b0: RSP/HNP capability is not enabled. - 1'b1: RSP/HNP capability is enabled. Note: This bit is writable only if RSP/HNP mode is specified for Mode of Operation in coreConsultant, that is when DWC_USB3_EN_OTG != 0 and DWC_USB3_MODE = DRD. If RSP/HNP mode is not specified, this bit is Read Only, and is set to 1'b0.
0 SRPCap	SRP capability The application uses this bit to control the USB 3.0 core's SRP capabilities. If this bit is not set for B-device, it cannot request the connected A-device (host) to activate VBUS and start a session. If this bit is not set for A-device, it cannot detect the SRP from B-device (device) to activate VBUS and start a session. NOTE: This bit is writable only if OTG is enabled. If OTG is not enabled, then this bit is Read Only and is set to 1'b0.

11.1.3.1.101 OTG Control Register (OCTL)

11.1.3.1.101.1 Offset

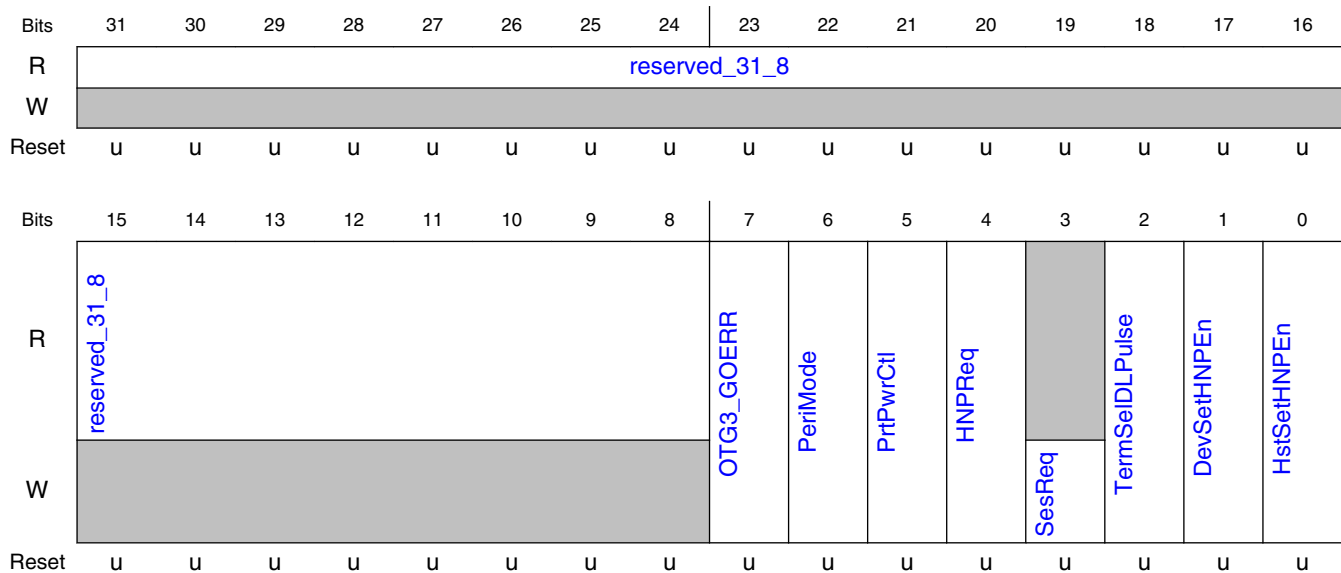
Register	Offset
OCTL	CC04h

11.1.3.1.101.2 Function

The OTG Control register controls the behavior of the OTG function of the core.

Reset Mask:0xFFFFFFFFFA0

11.1.3.1.101.3 Diagram



11.1.3.1.101.4 Fields

Field	Function
31-8 reserved_31_8	Reserved
7 OTG3_GOERR	OTG3_GOERR When set, this instructs the core's LTSSM to go to the error state during OTG 3.0 RSP. The software must set this bit when TRSP_ACK_ERR, TRSP_CNF_ERR, or TRSP_WRST_ERR timeout occurs. This bit is self-cleared by the core when the LTSSM moves to the error state.
6 PeriMode	Peripheral Mode The application uses this bit to program the core to work as a peripheral or as a host. - 1'b0: The OTG device acts as a host - 1'b1: The OTG device acts as a peripheral
5 PrtPwrCtl	Port Power Control Application sets this bit to initiate Vbus drive when it is an A-device. The application must clear this bit only if it wants to switch off the Vbus to B-device. The core clears this bit in the following conditions: Transition from any state to A-IDLE state defined in OTG2.0 state machine: - When AIDL_BDIS_TOUT occurs in A_SUSPEND; - When A_WAIT_BCON_TOUT occurs in A_WAIT_BCON; - Transition to any B- state defined in OTG2.0 state machine;
4 HNPReq	HNP Request - 1'b0: No HNP request - 1'b1: HNP request The application sets this bit to initiate a HNP request to the connected USB host. The application clears this bit by writing a 1'b0 when either of the following is detected: - OEVT.OTGBDevBHostEndEvt - OEVT.OTGBDevVBusChngEvt
3 SesReq	Session Request - 1'b0: No session request - 1'b1: Session request The application sets this bit to initiate a session request on the USB. Writing 1'b1 to this field triggers the core to send SRP (data line pulsing) on the PHY interface. In the absence of OEVT.OTGBDevSessVldDetEvt after a session request, the application must wait for atleast TB_SRP_FAIL time (6 secs) before initiating another session request. This field returns 1'b0 when read.
2 TermSelDLPulse	TermSel DLine Pulsing Selection This bit selects utmi_termselect to drive data line pulse during SRP. - 1'b0: Data line pulsing using utmi_txvalid (default). - 1'b1: Data line pulsing using utmi_termsel.

Table continues on the next page...

Field	Function
1 DevSetHNPEn	Device Set RSP/HNP Enable - 1'b0: RSP/HNP is not enabled in the application. - 1'b1: RSP/HNP is enabled in the application. The application sets this bit in the following scenario: - In HS/FS mode, when it successfully receives a SetFeature.SetHNPEnable command from the connected USB host. - In SS mode, when it has sent a b3_ntf_hst_rel to the A-device, or the A-device has sent an a3_ntf_host_req. Note: The terminology RSP is used when the core is operating in SS mode, and HNP is used when the core is operating in non-SS mode.
0 HstSetHNPEn	Host Set RSP/HNP Enable. - 1'b0: Host Set RSP/HNP is not enabled; - 1'b1: Host Set RSP/HNP is enabled; The application sets this bit in the following scenario: - In HS/FS mode, when it has successfully enabled HNP (using the SetFeature.SetHNPEnable command) from the connected device. - In SS mode, when it has successfully enabled the b3_rsp_enable feature in RSP-capable Device using SetFeature command while operating as an A-Host, or when it has received a b3_ntf_hst_rel through the SetFeature command while operating as A-peripheral. Note: The terminology RSP is used when the core is operating in SS mode, and HNP is used when the core is operating in non-SS mode.

11.1.3.1.102 OTG Events Register (OEVt)

11.1.3.1.102.1 Offset

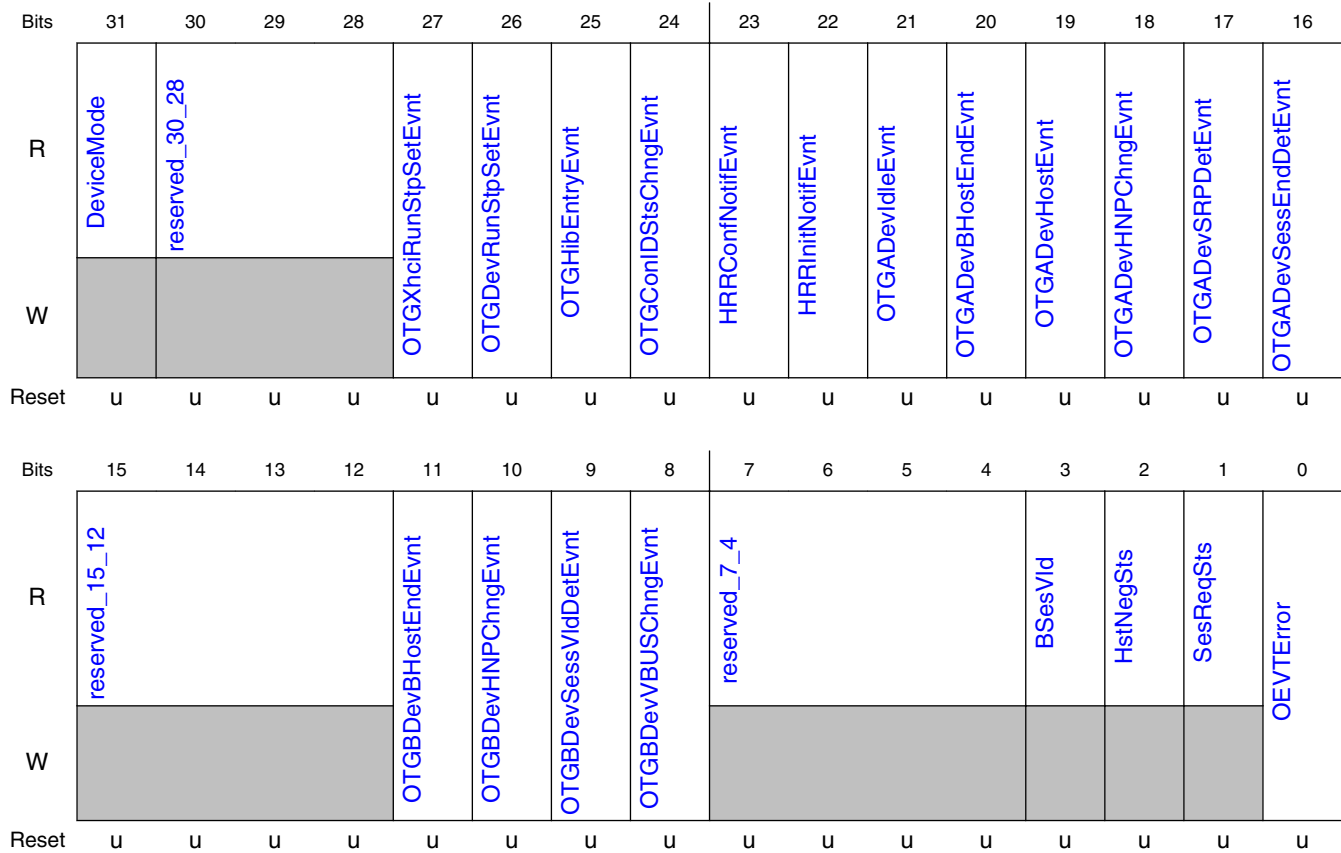
Register	Offset
OEVt	CC08h

11.1.3.1.102.2 Function

Any event set in this register will cause otg_interrupt signal to go high.

Reset Mask:0x7FDFFFF1

11.1.3.1.102.3 Diagram



11.1.3.1.102.4 Fields

Field	Function
31 DeviceMode	Device Mode Indicates whether the device is in A-device or B-device mode based on utmiotg_iddig - 1'b0: A-Device mode - 1'b1: B-Device mode The rest of the OTG Event Information bits (OTGxxxxEvtInfo) in OEVT register is based on the contents of this field.
30-28 reserved_30_28	Reserved
27 OTGXhciRunStpSetEvt	OTG Host Run Stop Set Event This event is set when the Host Driver programs the USBCMD.Run/Stop to 1'b1. Note: - During Hibernation Exit, upon receiving this event, the OTG Driver can start register accesses to the core. - This bit is applicable for OTG 2.0 and OTG 3.0 modes of operation.
26 OTGDevRunStpSetEvt	OTG Device Run Stop Set Event This event is set when the Device Driver programs the DCTL.Run/Stop to 1'b1. Note: During Hibernation Exit, upon receiving this event, the OTG Driver can start register accesses to the core.
25 OTGHibEntryEvt	OTG Hibernation Entry Event: - A-Device mode: This event is set when there is hibernation save request from the Host Driver by programming USBCMD.CSS bit to 1 when USBCMD.RunStp is 0. - B-Device mode: This event is set when there the Device Driver by programming DCTL.RunStp bit to 0. When this event is generated, the OTG Driver must disable all register accesses to the core.

Table continues on the next page...

Field	Function
24 OTGConIDStsChngEvnt	Connector ID Status Change Event Set in both A-Dev/B-Dev Mode. This event is generated when there is a change in connector ID status. When this bit is set, OEVT.OTGConIDStsChngEvnt is enabled. If not, the event is disabled.
23 HRRConfNotifEvnt	Host Role Request Confirm Notifier Event The core sets this bit after receiving a Host Role Request Device Notification TP with Confirm field set while operating as either an A-host or a B-host. Note: - This bit is applicable only when operating in SS mode. - This bit is applicable only for OTG 3.0 mode of operation.
22 HRRInitNotifEvt	Host Role Request Initiate Notifier Event The core sets this bit after receiving a Host Role Request Device Notification TP with Initiate field set while operating as either an A-host or a B-host.
21 OTGADevIdleEvnt	A-device A-IDLE Event Set in A-device Mode Only. The event is generated when A-device enters A-IDLE state. This event is set when the OTG 2.0 FSM of the core enters A-IDLE state from any other OTG state. A-device A-IDLE Event When this bit is set, OEVT.OTGADevIdleEvnt is enabled. If not, the event is disabled.
20 OTGADevBHostEndEvnt	A-device B-Host End Event Set in A-device Mode Only. The event is generated when B-device has completed its host role. Note: This bit is applicable for OTG 2.0 and OTG 3.0 modes of operation. When this bit is set, OEVT.OTGADevBHostEndEvnt is enabled. If not, the event is disabled.
19 OTGADevHostEvnt	A-device host event Set in A-device Mode Only. This event is generated when A-device enters host role. In HS/FS mode, it occurs after the initial connect to a B-device as A-host as well as when there is a role change from A-peripheral to A-host. Note: This bit is applicable only for OTG 2.0 mode of operation.
18 OTGADevHNPCChngEvnt	A-Dev HNP Change Event Set in A-device Mode Only. The event is generated when there is an HNP attempt. Note: This bit is applicable only for OTG 2.0 mode of operation. A-Dev HNP Change EventEn (OTGADevHNPCChngEvntEn) When this bit is set, OEVT.OTGADevHNPCChngEvnt is enabled. If not, the event is disabled.
17 OTGADevSRPDetEvnt	SRP Detect Event Set in A-device Mode Only. This event is asserted when a session request from the B-device is detected via SRP. Note: This bit is applicable for OTG 2.0 and OTG 3.0 modes of operation. A-Dev HNP Change EventEn (OTGADevHNPCChngEvntEn) When this bit is set, OEVT.OTGADevHNPCChngEvnt is enabled. If not, the event is disabled.
16 OTGADevSessEndDetEvnt	Session End Detected Event Set in A-device Mode Only. This event is asserted when the utmiotg_vbusvalid signal goes low indicating the end of a session. Note: This bit is applicable for OTG 2.0 and OTG 3.0 modes of operation. When this bit is set, OEVT.OTGADevSessEndEvnt is enabled. If not, the event is disabled.
15-12 reserved_15_12	Reserved
11 OTGBDevBHostEndEvnt	B-Device B-Host End Event Set in B-device Mode Only. This event is generated when B-device has completed its host role. Note: This bit is applicable for OTG 2.0 and OTG 3.0 modes of operation. When this bit is set, OEVT.OTGBDevHostEndEvnt is enabled. If not, the event is disabled.
10 OTGBDevHNPCChngEvnt	B-Dev HNP Change Event: Set in B-Device Mode only. This event is generated when there is a Success or Failure of an HNP attempt. Note: This bit is applicable for OTG 2.0 and OTG 3.0 modes of operation. When this bit is set, OEVT.OTGBDevHNPCChngEvnt is enabled. If not, the event is disabled.
9 OTGBDevSessVldDetEvnt	Session Valid Detected Event Set in B-device Mode Only. This event is asserted when there is a valid Vbus from A-device and B-device succeeds in starting a session. Note: This bit is applicable for OTG 2.0 and OTG 3.0 modes of operation.
8 OTGBDevVBUSChngEvnt	Vbus Change Event Set in B-device Mode Only. This event is asserted when the utmisrp_bvalid signal goes low (indicating the end of a session), or goes high. Note: This bit is applicable for OTG 2.0 and OTG 3.0 modes of operation. When this bit is set, OEVT.OTGBDevVBUSChngEvnt is enabled. If not, the event is disabled.

Table continues on the next page...

Field	Function
7-4 reserved_7_4	Reserved
3 BSesVld	Indicates the Device mode transceiver status. Indicates the Device mode transceiver status. The core updates this bit when: OEVTEN.OTGBDevVBUSChngEvnt is set. - 1'b0: B-session is not valid - 1'b1: B-session is valid Note: This bit is applicable for OTG 2.0 and OTG 3.0 modes of operation.
2 HstNegSts	Host Negotiation Status The core updates this bit when any of the following bits is set: - OEVTEN.OTGADevHNPChngEvnt - OEVTEN.OTGBDevHNPChngEvnt This bit indicates Host Negotiation Success or Failure. 1'b0: Host negotiation failure. - In A-device, for HS/FS, this indicates an imminent end of session indication from the core. - In B-device, for HS/LS, it indicates that the timer used to wait for an A-device to signal a connection (b_ase0_brst_tmout in OTG 2.0) timed out resulting in B-device staying as B-peripheral. 1'b1: Host negotiation success. - This indicates that the host negotiation was successful. Note: This bit is applicable only for OTG 2.0 mode of operation.
1 SesReqSts	Session Request Status Ignore this field. It is used only for Synopsys internal testing.
0 OEVTError	OTG Event Error There are no errors currently defined.

11.1.3.1.103 OTG Events Enable Register (OEVTEN)

11.1.3.1.103.1 Offset

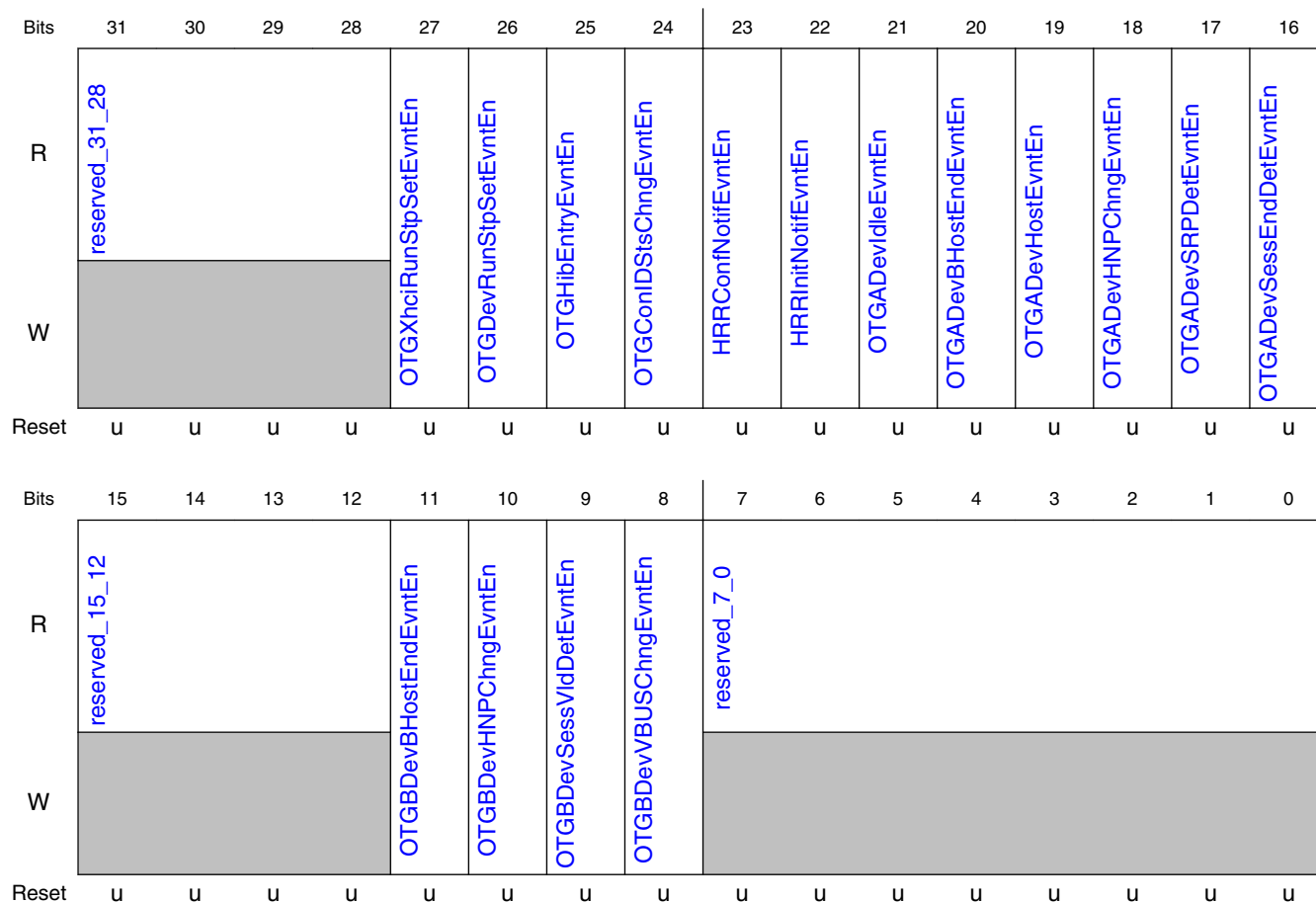
Register	Offset
OEVTEN	CC0Ch

11.1.3.1.103.2 Function

Setting a bit in this register enables the generation of corresponding events in OEVT and assertion of otg_interrupt due to this event.

Reset Mask:0xF000F0FF

11.1.3.1.103.3 Diagram



11.1.3.1.103.4 Fields

Field	Function
31-28 reserved_31_28	Reserved
27 OTGXhciRunStpSetEvtEn	OTG Host Run Stop Set Event Enable (OTGXhciRunStpSetEvtEn) When this bit is set, OEVT.OTGXhciRunStpSetEvt is enabled. If not, the event is disabled.
26 OTGDevRunStpSetEvtEn	OTG Device Run Stop Set Event Enable (OTGDevRunStpSetEvtEn) When this bit is set, OEVT.DevRunStpSet event is enabled. If not, that event is disabled.
25 OTGHibEntryEvtEn	OTG Hibernation Entry Event Enable (OTGHibEntryEn) When this bit is set, OEVT.HibEntryEvt is enabled. If not, the event is disabled.
24	OTGCommonEvtInfoEn[0] Connector ID Status Change Event (OTGConIDStsChngEvt) Set in both A-Dev/B-Dev Mode: This event is generated when there is a change in connector ID status

Table continues on the next page...

Universal Serial Bus Controller (USB)

Field	Function
OTGConIDStsChngEvntEn	
23 HRRConfNotifEvntEn	OTGCommonEvtInfoEn[2] HRRConfNotif Event Enable (HRRConfNotifEvntEn) When this bit is set, OEVT.HRRConfNotifEvnt is enabled. If not, the event is disabled.
22 HRRInitNotifEvntEn	OTGCommonEvtInfoEn[1] HRRInitNotif Event Enable (HRRInitNotifEvntEn) When this bit is set, OEVT.HRRInitNotifEvnt is enabled. If not, the event is disabled.
21 OTGDevIdleEvntEn	OTGDevEvtInfoEn[5] A-device A-IDLE Event (OTGDevIdleEvntEn) When this bit is set, OEVT.OTGDevIdleEvnt is enabled. If not, the event is disabled.
20 OTGDevBHostEndEvntEn	OTGDevEvtInfoEn[4] A-device B-Host End Event Enable (OTGDevBHostEndEvntEn) When this bit is set, OEVT.OTGDevBHostEndEvnt is enabled. If not, the event is disabled.
19 OTGDevHostEvntEn	OTGDevEvtInfoEn[3] A-device host event (OTGDevHostEvntEn) When this bit is set, OEVT.OTGDevHostEvnt is enabled. If not, the event is disabled.
18 OTGDevHNPCChngEvntEn	OTGDevEvtInfoEn[2] A-Dev HNP Change EventEn (OTGDevHNPCChngEvntEn) When this bit is set, OEVT.OTGDevHNPCChngEvnt is enabled. If not, the event is disabled.
17 OTGDevSRPDetEvntEn	OTGDevEvtInfoEn[1] SRP Detect Event Enable (OTGDevSRPDetEvntEn) When this bit is set, OEVT.OTGDevSRPDetEvnt is enabled. If not, the event is disabled.
16 OTGDevSessEndDetEvntEn	OTGDevEvtInfoEn[0] Session End Detected Event Enable (OTGDevSessEndDetEvntEn) When this bit is set, OEVT.OTGDevSessEndEvnt is enabled. Otherwise, the event is disabled.
15-12 reserved_15_12	Reserved
11 OTGBDevBHostEndEvntEn	OTGBDevEvtInfoEn[3] B-device B-Host End Event Enable (OTGBDevBHostEndEvntEn) When this bit is set, OEVT.OTGBDevHostEndEvnt is enabled. If not, the event is disabled
10 OTGBDevHNPCChngEvntEn	OTGBDevEvtInfoEn[2] B-Dev HNP Change Event Enable (OTGBDevHNPCChngEvntEn) When this bit is set, OEVT.OTGBDevHNPCChngEvnt is enabled. If not, the event is disabled
9 OTGBDevSessVldDetEvntEn	OTGBDevEvtInfoEn[1] Session Valid Detected Event Enable (OTGBDevSessVldDetEvntEn) Set in B-device mode only. This event is asserted when there is a valid VBUS from A-device and B-device succeeds in starting a session.
8 OTGBDevVBUSChngEvntEn	OTGBDevEvtInfoEn[0] VBUS Change Event Enable (OTGBDevVBUSChngEvntEn) When this bit is set, OEVT.OTGBDevVBUSChngEvnt is enabled. If not, the event is disabled.
7-0 reserved_7_0	Reserved

11.1.3.1.104 OTG Status Register (OSTS)

11.1.3.1.104.1 Offset

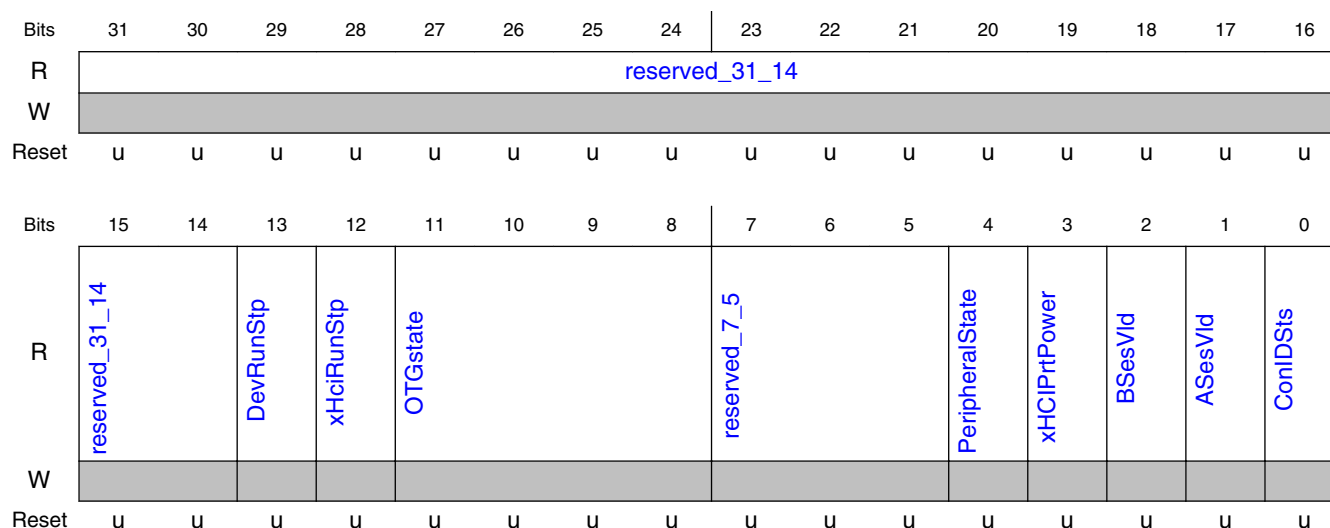
Register	Offset
OSTS	CC10h

11.1.3.1.104.2 Function

The OTG Status Register reflects the status of the OTG function of the core.

Reset Mask:0xFFFFC0E0

11.1.3.1.104.3 Diagram



11.1.3.1.104.4 Fields

Field	Function
31-14 reserved_31_14	Reserved
13 DevRunStp	This bit reflects the status of the Run/Stop bit in the DCTL Device register. - 0 DCTL.Run/Stop is set to 0. - 1 DCTL.Run/Stop is set to 1.
12 xHciRunStp	OTG Host Run Stop Set Event This event is set when the Host Driver programs the USBCMD.Run/Stop to 1'b1. Note: - During Hibernation Exit, upon receiving this event, the OTG Driver can start register accesses to the core. - This bit is applicable for OTG 2.0 and OTG 3.0 modes of operation.
11-8	This is a debug field indicating the current state of the OTG state machine. The encoding is as follows: - 4'b0000: A_IDLE - 4'b0010: A_WAIT_BCON - 4'b0011: A_WAIT_VFALL - 4'b0100: A_VBUS_ERR -

Table continues on the next page...

Field	Function
OTGstate	4'b0101: A_HOST - 4'b0110: A_SUSPEND - 4'b1111: A_PERIPHERAL - 4'b0111: A_WAIT_PPWR - 4'b1000: B_IDLE - 4'b1001: B_SRP_INIT - 4'b1010: B_PERIPHERAL - 4'b1011: B_WAIT_ACON - 4'b1100: B_HOST - 4'b1101: A_WAIT_SWITCH - 4'b1110: B_WAIT_SWITCH
7-5 reserved_7_5	Reserved
4 PeripheralState	OTG state Indicates whether the core is acting as a peripheral or host. - 1'b0: Host - 1'b1: Peripheral
3 xHCIPrtPower	This bit reflects the PORTSC.PP bit in the xHCI register.
2 BSesVld	B-Session Valid Indicates the Device mode transceiver status. In OTG mode, applications use this bit to determine if the device is connected. - 1'b0: B-session is not valid. - 1'b1: B-session is valid.
1 ASesVld	Indicates the Host mode transceiver status. - 1'b0: A-session is not valid - 1'b1: A-session is valid
0 ConIDSts	Connector ID Status: - Indicates the connector ID status - 1'b0: The DWC_usb3 core is in A-device mode - 1'b1: The DWC_usb3 core is in B-device mode Note: The reset value of this field depends on the power-on value of the IDDIG signal from the PHY.

11.1.3.1.105 ADP Configuration Register (ADPCFG)

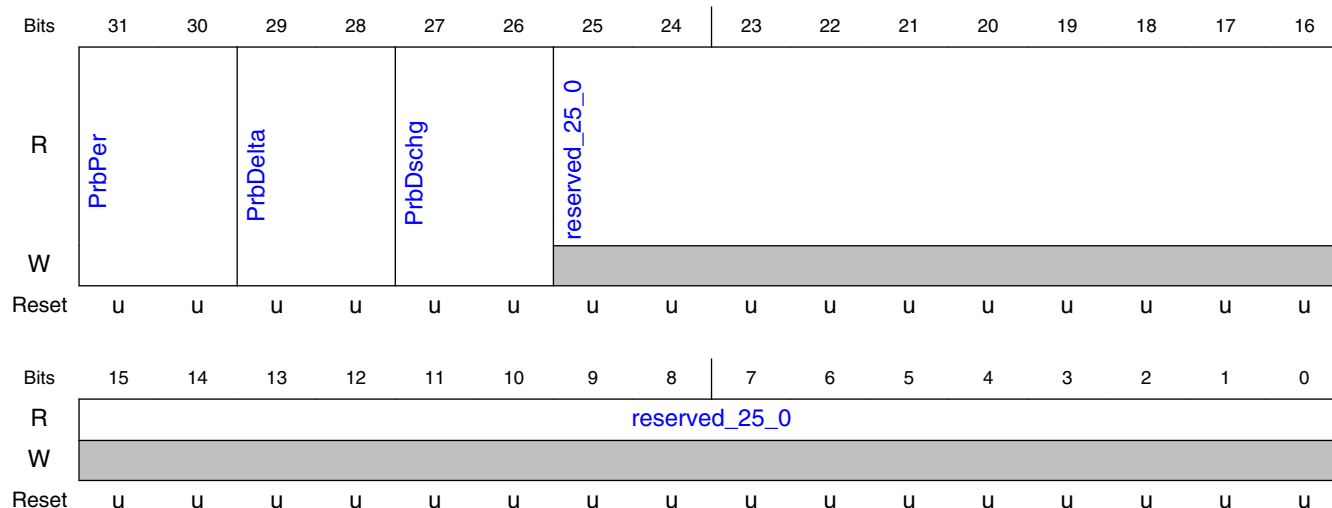
11.1.3.1.105.1 Offset

Register	Offset
ADPCFG	CC20h

11.1.3.1.105.2 Function

Reset Mask:0x3FFFFFFF

11.1.3.1.105.3 Diagram



11.1.3.1.105.4 Fields

Field	Function
31-30 PrbPer	Probe Period These bits set the T_ADP_PRB as follows: - 2'b00: 0.775 sec - 2'b01: 1.55 sec - 2'b10: 2.275 sec - 2'b11: Reserved The scaledown values for PrbPer are: - 2'b00: 12.5ms - 2'b10: 25 ms - 2'b11: 31.25 ms
29-28 PrbDelta	Probe Delta These bits set the resolution for RTIM value. They are defined in units of 32 kHz clock cycles as follows: - 2'b00: 1 cycles - 2'b01: 2 cycles - 2'b10: 3 cycles - 2'b11: 4 cycles For example, if this value is chosen to be 2'b01, it means that RTIM increments for every two 32 kHz clock cycles.
27-26 PrbDschg	Probe Discharge These bits set the time for TADP_DSCHG. They are defined as follows: - 2'b00: 4 msec - 2'b01: 8 msec - 2'b10: 16 msec - 2'b11: 32 msec The scaledown values for the PrbDschg are as follows: - 2'b00: 62.5 us - 2'b01: 125 us - 2'b10: 250 us - 2'b11: 500 us
25-0 reserved_25_0	Reserved

11.1.3.1.106 ADP Control Register (ADPCTL)

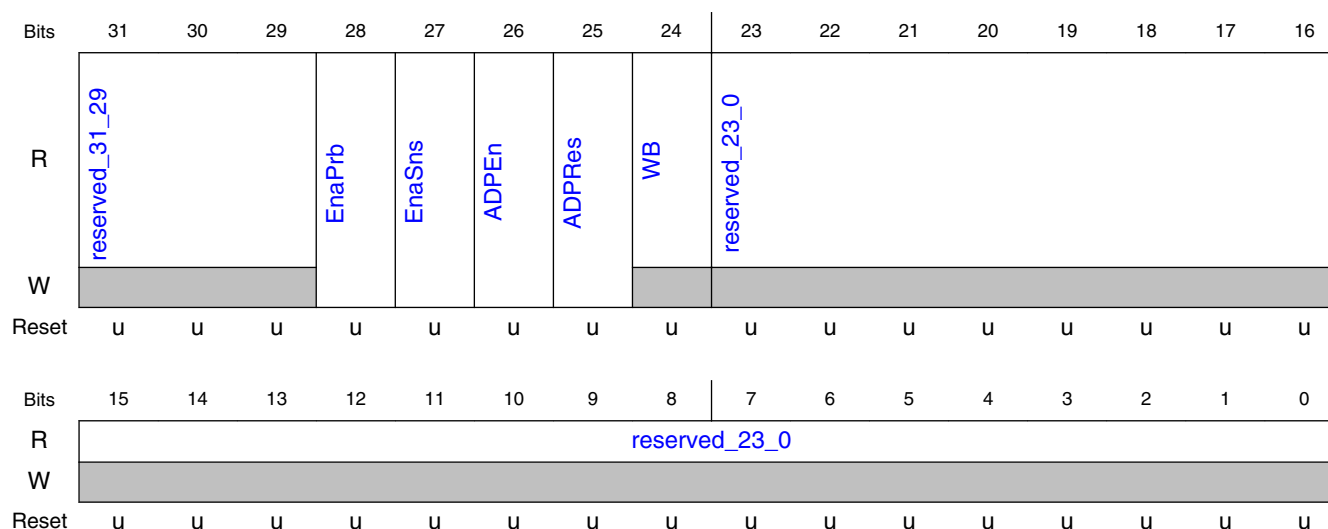
11.1.3.1.106.1 Offset

Register	Offset
ADPCTL	CC24h

11.1.3.1.106.2 Function

Reset Mask:0xE2FFFFFF

11.1.3.1.106.3 Diagram



11.1.3.1.106.4 Fields

Field	Function
31-29 reserved_31_29	Reserved
28 EnaPrb	Enable Probe: When set to 1'b1 along with ADPEn, the core performs a probe operation.
27 EnaSns	Enable Sense: When set to 1'b1 along with ADPEn, the core performs a sense operation.
26 ADPEn	ADP Enable: When set to 1'b1, the core performs either ADP probing or sensing based on EnaPrb and EnaSns. - ADPEn = 1'b0 gates the suspend clock for major portion of ADP related logic.
25 ADPRes	ADP Reset: When set to 1'b1, the ADP controller is reset. This bit is auto-cleared after the reset procedure is complete in the ADP controller.
24 WB	Write Busy - 1'b0: Write Completed - 1'b1: Write in Progress. The application can read or write ADPCFG and ADPCTL registers only if this field is cleared. The hardware sets this bit when the write is in progress in the Suspend clock domain.
23-0 reserved_23_0	Reserved

11.1.3.1.107 ADP Event Register (ADPEVT)

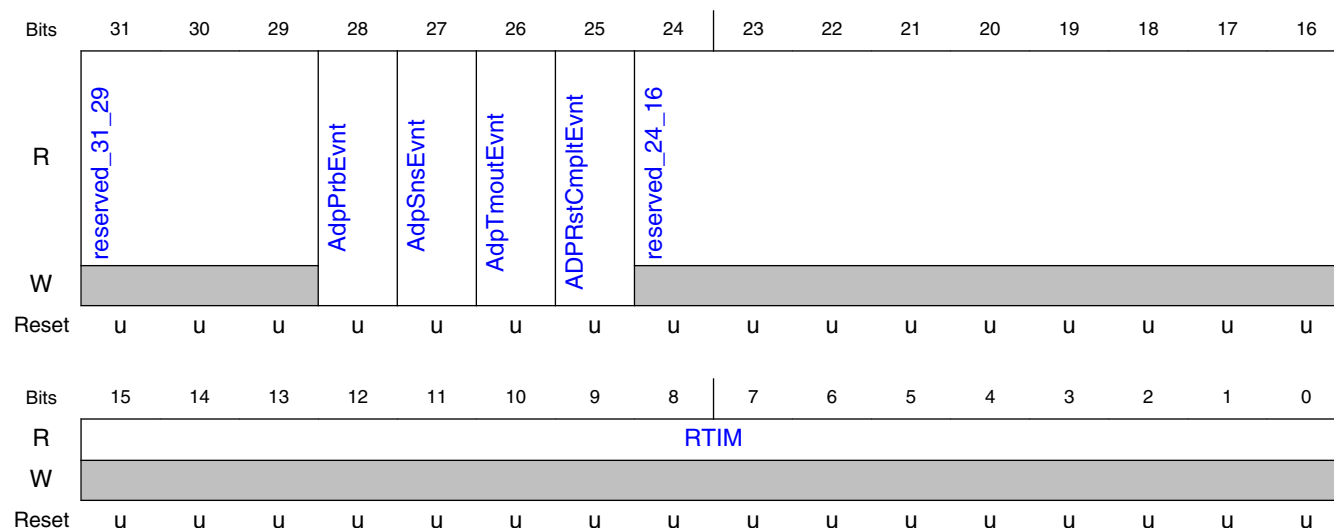
11.1.3.1.107.1 Offset

Register	Offset
ADPEVT	CC28h

11.1.3.1.107.2 Function

Reset Mask:0xFFFF0000

11.1.3.1.107.3 Diagram



11.1.3.1.107.4 Fields

Field	Function
31-29 reserved_31_29	Reserved
28 AdpPrbEvnt	ADP Probe Event When this event is set, it means that the Vbus voltage is greater than VadpPrb or VadpPrb is reached.
27 AdpSnsEvnt	ADP Sense Event When this event is set, it means that the Vbus voltage is greater than VadpSns or VadpSns is reached.
26 AdpTmoutEvnt	ADP Timeout Event This event is relevant when ADP probe command is executed. When this event is set, it means that the ramp time is completed (GADPCTL.RTIM has reached its terminal value of 0x7FF). This is a debug feature that allows software to read the ramp time after each cycle.
25 ADPRstCmpltEvnt	This event, when set, indicates that the ADP Reset command is successful.

Table continues on the next page...

Universal Serial Bus Controller (USB)

Field	Function
24-16 reserved_24_16	Reserved
15-0 RTIM	RAMP TIME: These bits capture the latest time it took for Vbus to ramp from VADP_SINK to VADP_PRB. The bits are defined in units of 32 kHz clock cycles as follows: - 0x000: 1 cycles - 0x001: 2 cycles - 0x002: 3 cycles and so on till, - 0xFFFF: 65536 cycles The maximum time of 65536 cycles corresponds to a time of 2.04 seconds. Note: For scaledown ramp_timeout: - PrbDelta = 2'b11 => 781.25 us - PrbDelta = 2'b10 => 1562.5 us - PrbDelta = 2'b01 => 3125 us - PrbDelta = 2'b00 => 6250 us

11.1.3.1.108 (ADPEVTEN)

11.1.3.1.108.1 Offset

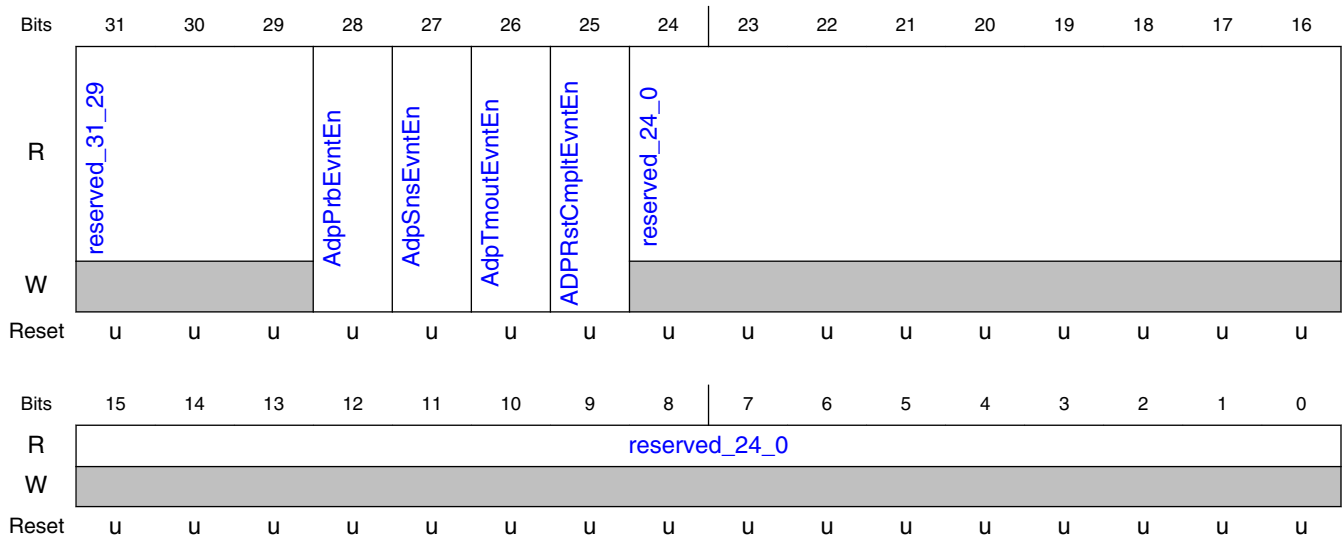
Register	Offset
ADPEVTEN	CC2Ch

11.1.3.1.108.2 Function

ADP Event Enable Register

Reset Mask:0xE1FFFFFF

11.1.3.1.108.3 Diagram



11.1.3.1.108.4 Fields

Field	Function
31-29 reserved_31_29	Reserved
28 AdpPrbEvntEn	ADP Probe Event Enable When this bit is set, ADPPrbEvnt in ADPEVT register is enabled.
27 AdpSnsEvntEn	ADP Sense Event Enable When this bit is set, AdpSnsEvntEn in ADPEVT register is enabled.
26 AdpTmoutEvntEn	ADP Timeout Event Enable When this bit is set, AdpTmoutEvntEn in ADPEVT register is enabled.
25 ADPRstCmplEvntEn	ADP Reset complete Event Enable When this bit is set, ADPRstCmplEvnt in ADPEVT register is enabled.
24-0 reserved_24_0	Reserved

11.1.3.1.109 BCFG (BCFG)

11.1.3.1.109.1 Offset

Register	Offset
BCFG	CC30h

11.1.3.1.109.2 Function

BC Configuration Register

Reset Mask:0xFFFFFFFFFC



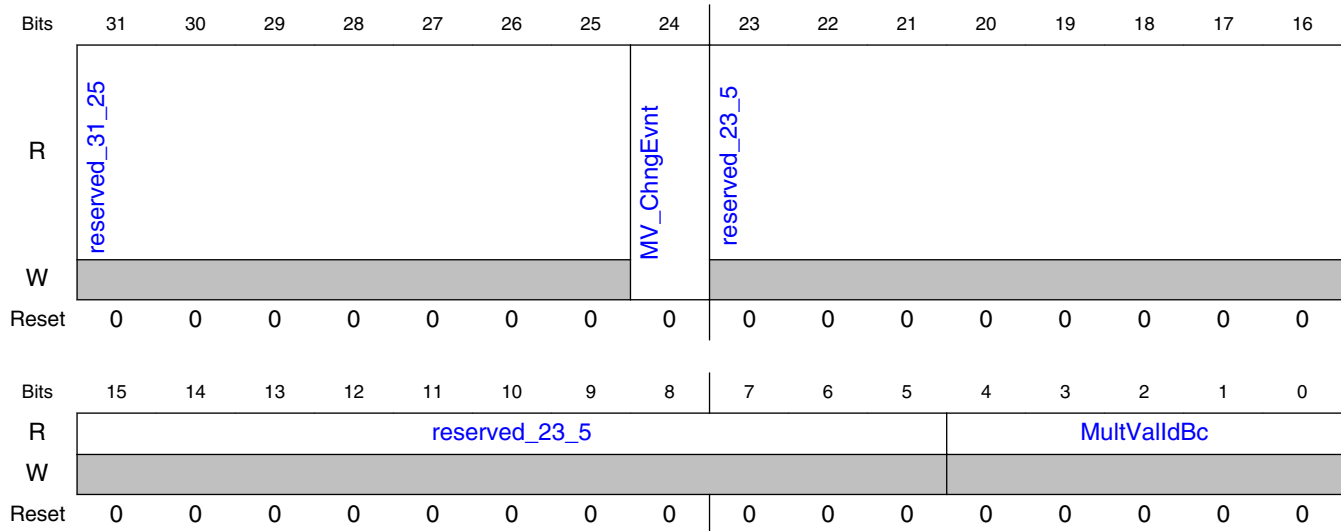
11.1.3.1.110 BCEVT (BCEVT)

11.1.3.1.110.1 Offset

11.1.3.1.110.2 Function

BC Event Register Writing 1 to the info bit in this register clears the register bit and the associated interrupt.

11.1.3.1.110.3 Diagram



11.1.3.1.110.4 Fields

Field	Function
31-25 reserved_31_25	Reserved
24 MV_ChngEvt	Multi-Valued input changed Event: This event indicates that there is a change in the value of at least one ACA pin value. This bit is present only if DWC_USB3_EN_BC = 1, otherwise it is Reserved.
23-5 reserved_23_5	Reserved
4-0 MultValdBc	Multi Valued ID pin Indicates the Battery Charger ACA inputs. - MultValdBc[4]: bc_rid_float - MultValdBc[3]: bc_rid_gnd - MultValdBc[2]: bc_rid_a - MultValdBc[1]: bc_rid_b - MultValdBc[0]: bc_rid_c

11.1.3.1.111 BCEVTEN (BCEVTEN)

11.1.3.1.111.1 Offset

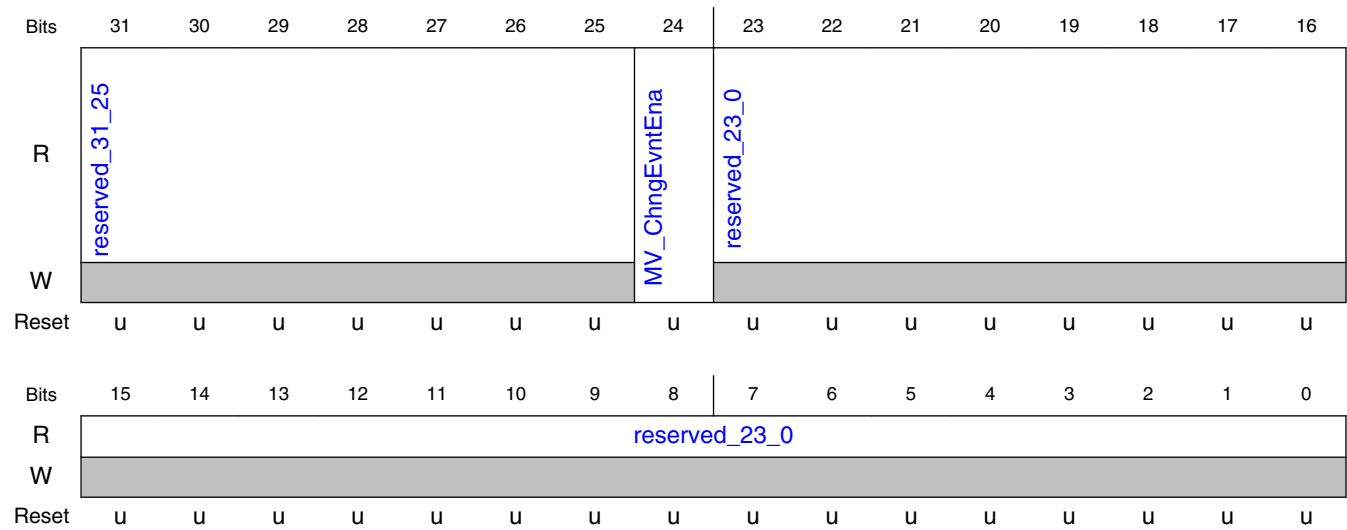
Register	Offset
BCEVTEN	CC3Ch

11.1.3.1.111.2 Function

BC Event Enable Register

Reset Mask:0xFEFFFFFFF

11.1.3.1.111.3 Diagram



11.1.3.1.111.4 Fields

Field	Function
31-25 reserved_31_25	Reserved
24 MV_ChngEvntEna	BCEvtInfoEna[0] Multi-Valued input changed Event Enable (MV_ChngEvntEn) When this bit is set, MV_ChngEvnt in BCEVT register is enabled.
23-0 reserved_23_0	Reserved

11.1.3.2 USB3_GLUE register descriptions

Between USB3.0 Core and USB PHY, there is a GLUE architecture to make the connection more configurable.

11.1.3.2.1 GLUE_usb3 memory map

Base Address for GLUE registers - USB1: 381F_0000, and USB2: 382F_0000

USB GLUE architecture makes the connection between Core and PHY more configurable. USB GLUE contains some configurable registers and status signal monitoring to Core/PHY. There are two GLUE blocks and each has its own configurable registers.

Offset	Register	Width (In bits)	Access	Reset value
0h	USB_CTL0_ADDR (USB_CTL0_ADDR)	32	RW	0020_F003h
20h	USB_STS0_ADDR (USB_STS0_ADDR)	32	RO	0000_07E8h
40h	PHY_CTL0_ADDR (PHY_CTL0_ADDR)	32	RW	3200_0CF0h
44h	PHY_CTL1_ADDR (PHY_CTL1_ADDR)	32	RW	0018_0006h
48h	PHY_CTL2_ADDR (PHY_CTL2_ADDR)	32	RW	0000_0100h
4Ch	PHY_CTL3_ADDR (PHY_CTL3_ADDR)	32	RW	94D4_E464h
50h	PHY_CTL4_ADDR (PHY_CTL4_ADDR)	32	RW	046C_7D09h
54h	PHY_CTL5_ADDR (PHY_CTL5_ADDR)	32	RW	0000_007Fh
80h	PHY_STS0_ADDR (PHY_STS0_ADDR)	32	W1C	0000_0000h

11.1.3.2.2 USB_CTL0_ADDR (USB_CTL0_ADDR)

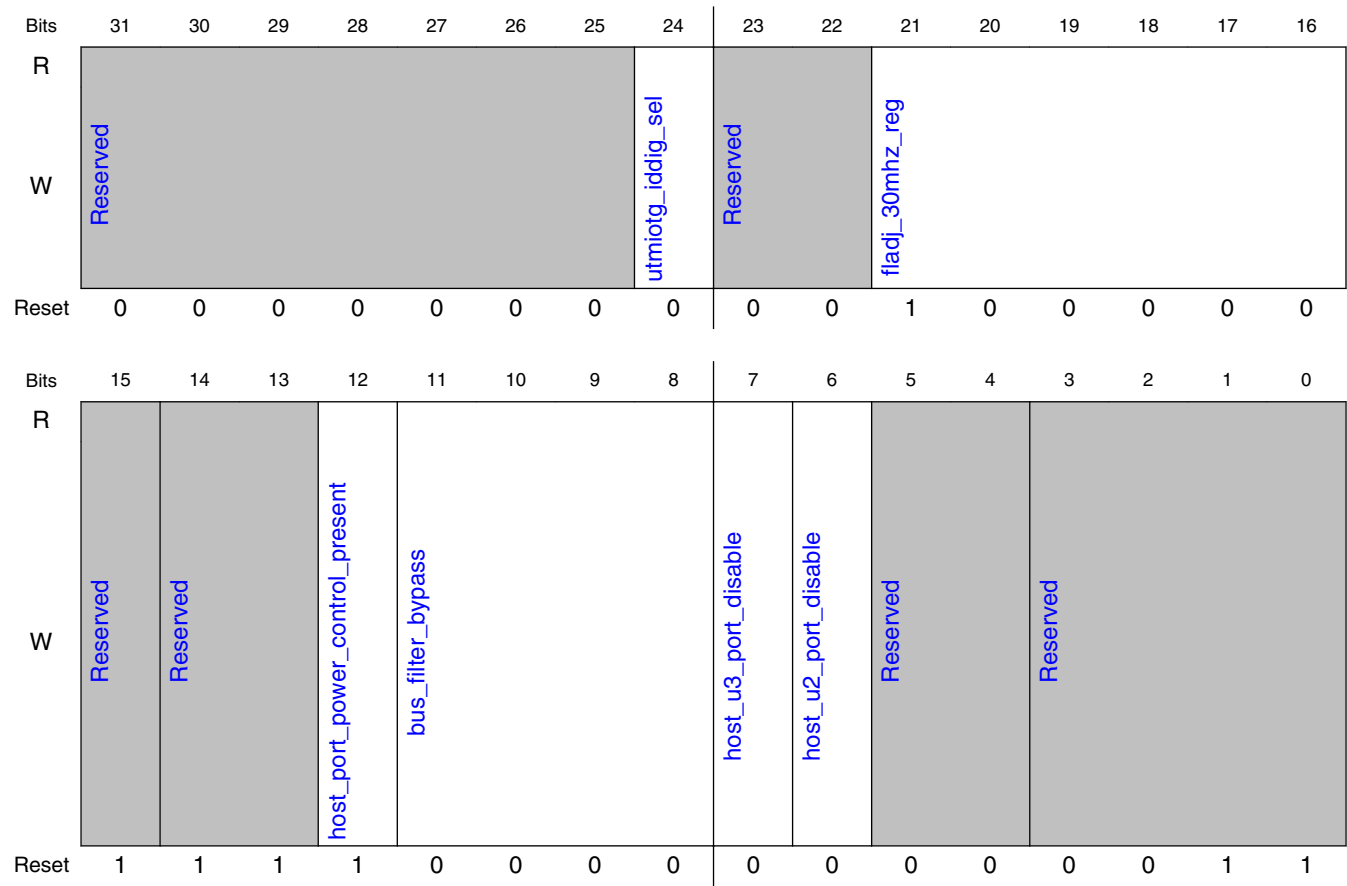
11.1.3.2.2.1 Offset

Register	Offset
USB_CTL0_ADDR	0h

11.1.3.2.2.2 Function

USB Core Configurable bits Register

11.1.3.2.2.3 Diagram



11.1.3.2.2.4 Fields

Field	Function
31-25 —	Reserved
24 utmiotg_iddig_sel	USB3.0 Core signal 0b - USB PHY ID0 1b - GPIO PAD
23-22 —	Reserved
21-16 fladj_30mhz_reg	USB3.0 Core signal
15 —	Reserved
14-13 —	Reserved

Table continues on the next page...

Field	Function
12 host_port_power _control_present	USB3.0 Core signal
11-8 bus_filter_bypas s	USB3.0 Core signal
7 host_u3_port_di sable	USB3.0 Core signal
6 host_u2_port_di sable	USB3.0 Core signal
5-4 —	Reserved
3-0 —	Reserved

11.1.3.2.3 USB_STS0_ADDR (USB_STS0_ADDR)

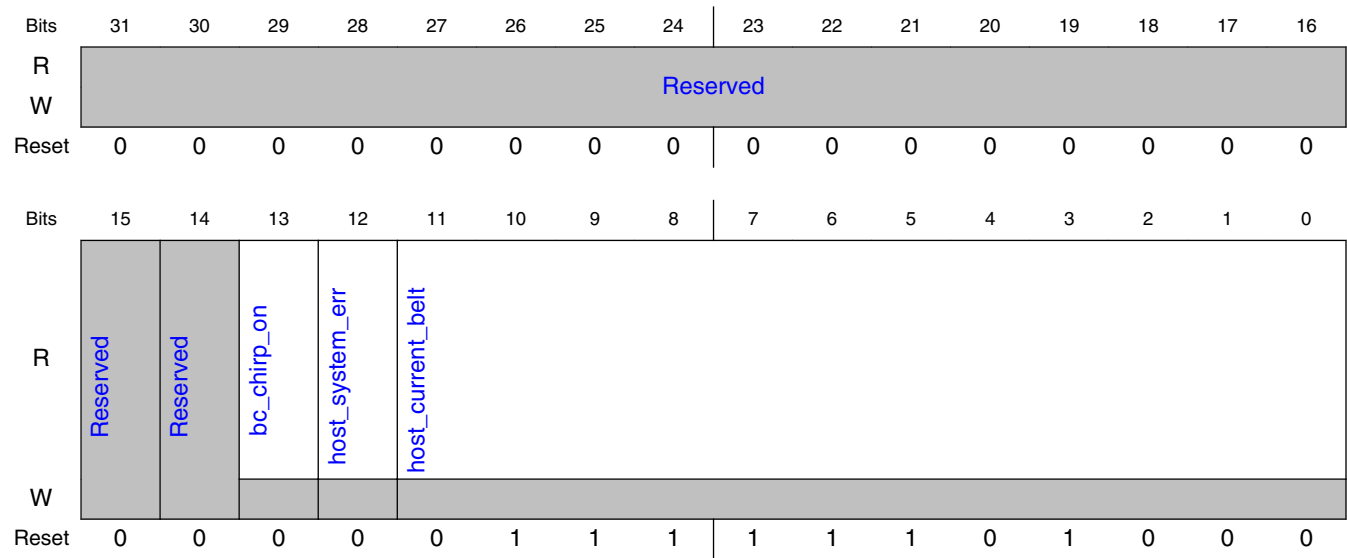
11.1.3.2.3.1 Offset

Register	Offset
USB_STS0_ADDR	20h

11.1.3.2.3.2 Function

USB3.0 Core Status bits Register

11.1.3.2.3.3 Diagram



11.1.3.2.3.4 Fields

Field	Function
31-15 —	Reserved
14 —	Reserved
13 bc_chirp_on	USB3.0 Core Signal
12 host_system_err	USB3.0 Core Signal
11-0 host_current_belt	USB3.0 Core Signal

11.1.3.2.4 PHY_CTL0_ADDR (PHY_CTL0_ADDR)

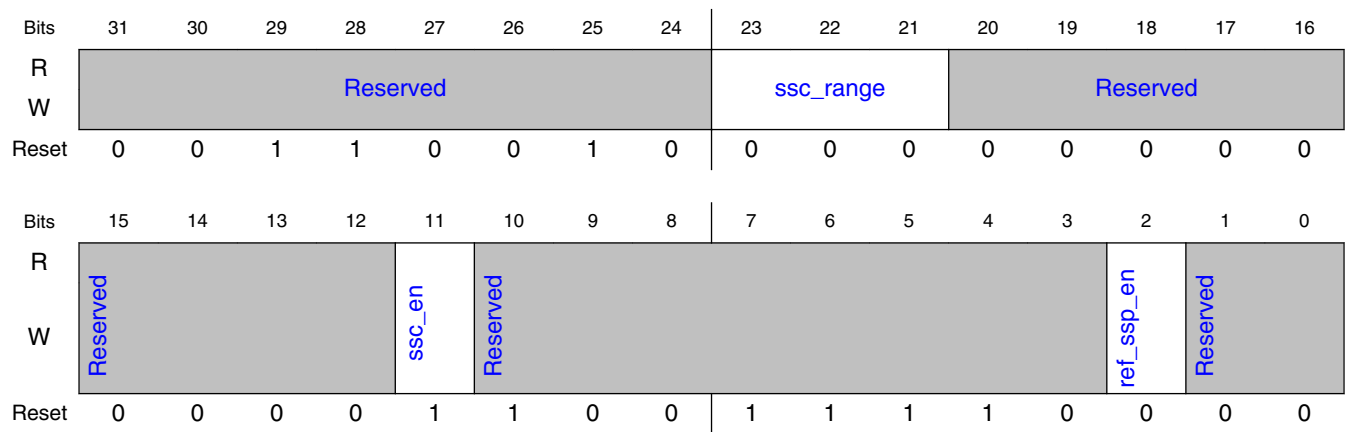
11.1.3.2.4.1 Offset

Register	Offset
PHY_CTL0_ADDR	40h

11.1.3.2.4.2 Function

USB3.0 PHY Status bits Register

11.1.3.2.4.3 Diagram



11.1.3.2.4.4 Fields

Field	Function
31-24 —	Reserved
23-21 ssc_range	USB3.0 PHY Signal
20-12 —	Reserved
11 ssc_en	USB3.0 PHY Signal
10-3 —	Reserved
2 ref_ssp_en	USB3.0 PHY Signal
1-0 —	Reserved

11.1.3.2.5 PHY_CTL1_ADDR (PHY_CTL1_ADDR)

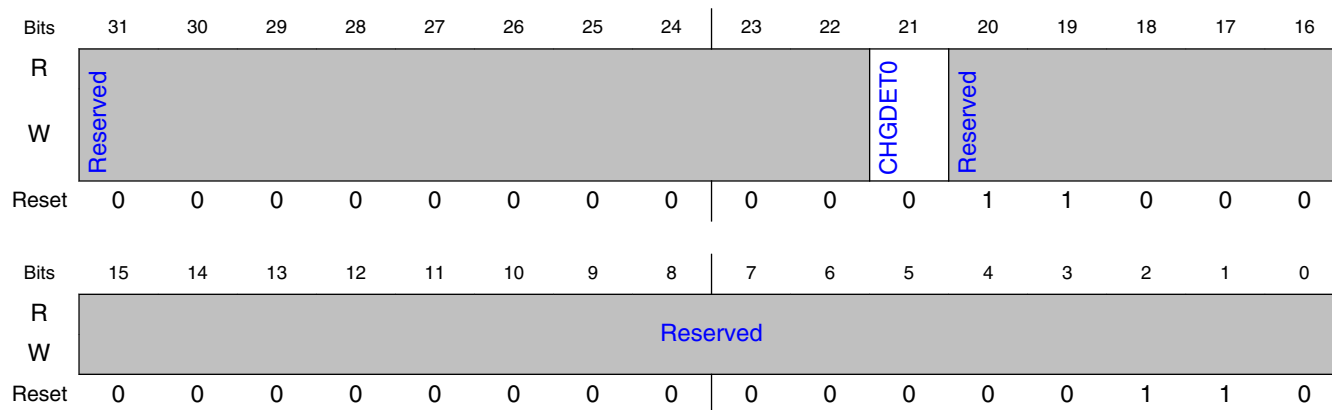
11.1.3.2.5.1 Offset

Register	Offset
PHY_CTL1_ADDR	44h

11.1.3.2.5.2 Function

USB3.0 PHY Status bits Register

11.1.3.2.5.3 Diagram



11.1.3.2.5.4 Fields

Field	Function
31-22 —	Reserved
21 CHGDET0	USB3.0 PHY Signal 0b - CHGDET0 will not generate USB interrupt 1b - CHGDET0 will generate USB interrupt when CHGDET0=1
20-0 —	Reserved

11.1.3.2.6 PHY_CTL2_ADDR (PHY_CTL2_ADDR)

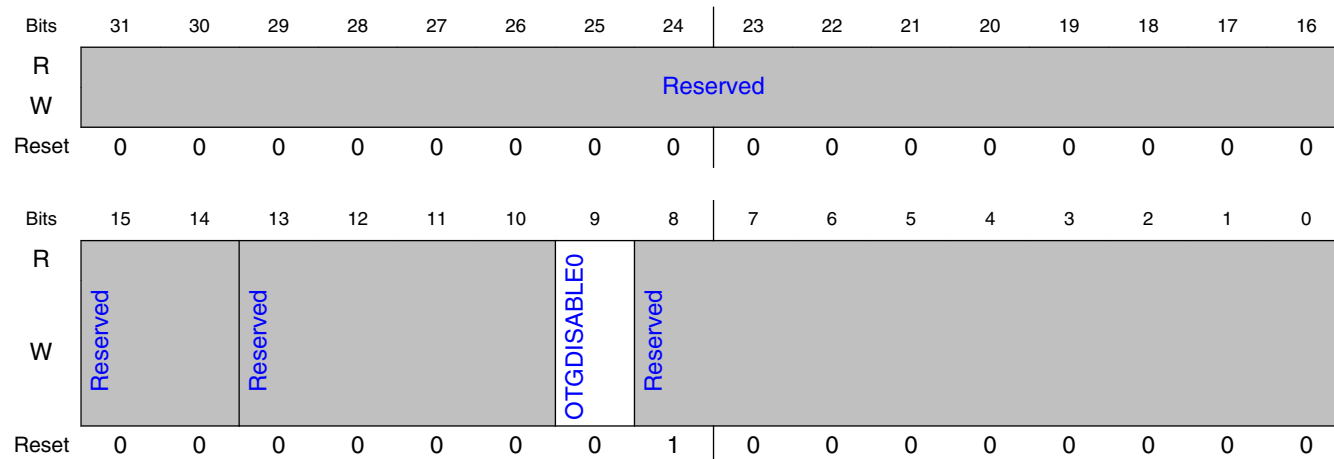
11.1.3.2.6.1 Offset

Register	Offset
PHY_CTL2_ADDR	48h

11.1.3.2.6.2 Function

USB3.0 PHY Status bits Register

11.1.3.2.6.3 Diagram



11.1.3.2.6.4 Fields

Field	Function
31-14 —	Reserved
13-10 —	Reserved
9 OTGDISABLE0	USB3.0 PHY Signal
8-0 —	Reserved

11.1.3.2.7 PHY_CTL3_ADDR (PHY_CTL3_ADDR)

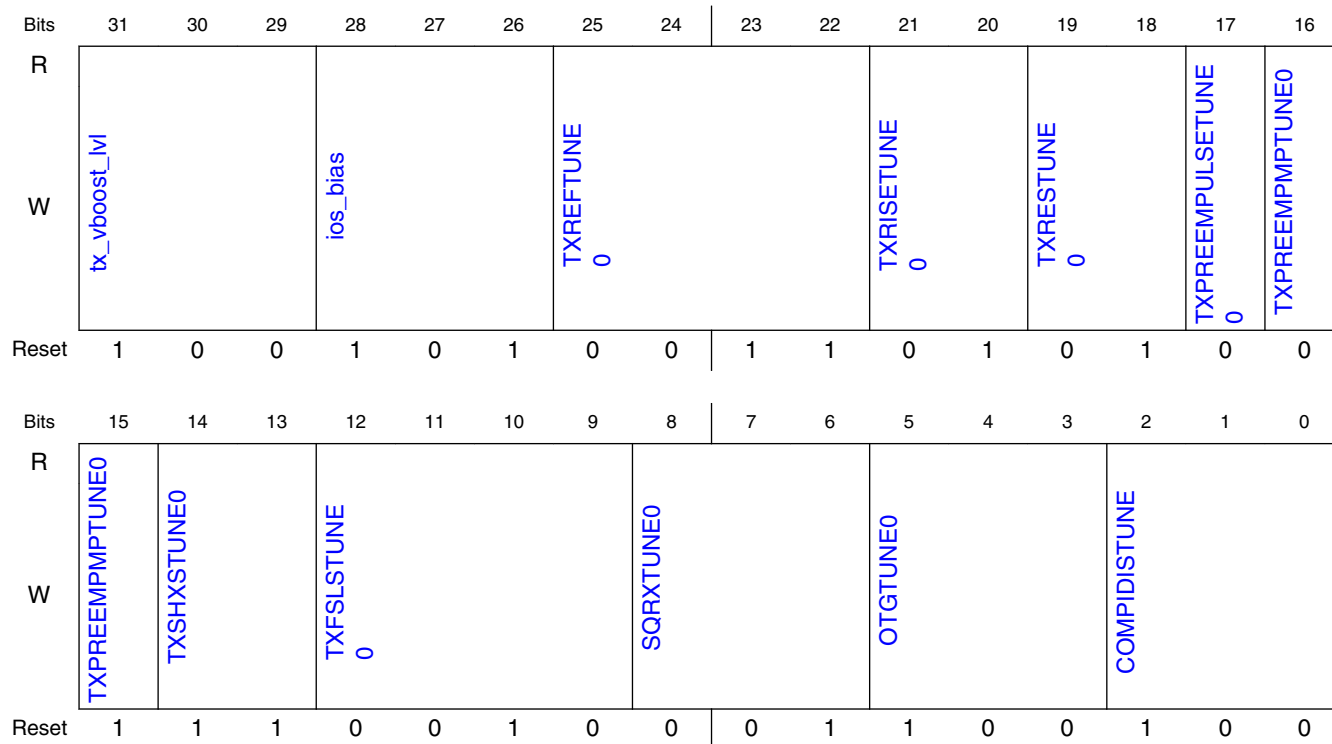
11.1.3.2.7.1 Offset

Register	Offset
PHY_CTL3_ADDR	4Ch

11.1.3.2.7.2 Function

USB3.0 PHY Status bits Register

11.1.3.2.7.3 Diagram



11.1.3.2.7.4 Fields

Field	Function
31-29 tx_vboost_lvl	USB3.0 PHY Signal
28-26 ios_bias	USB3.0 PHY Signal
25-22 TXREFTUNE0	USB3.0 PHY Signal
21-20 TXRISETUNE0	HS Transmitter Rise/Fall Time Adjustment Adjusts the rise/fall times of the high-speed waveform. To enable tuning at the board level, connect this bit to a register. NOTE: A positive binary bit setting change results in a –4% incremental change in the HS rise/fall time. A negative binary bit setting change results in a +4% incremental change in the HS rise/fall time.
19-18	USB Source Impedance Adjustment

Table continues on the next page...

Field	Function
TXRESTUNE0	Some applications require additional devices to be added on the USB, such as a series switch, which can add significant series resistance. This bus adjusts the driver source impedance to compensate for added series resistance on the USB. NOTE: Any setting other than the default can result in source impedance variation across process, voltage, and temperature conditions that does not meet USB 2.0 specification limits. If this bus is not used, leave it at the default setting 00b - Source impedance is increased by approximately 1.5 Ω 01b - Design default 10b - Source impedance is decreased by approximately 2 Ω 11b - Source impedance is decreased by approximately 4 Ω
17 TXPREEMPULSE TUNE0	HS Transmitter Pre-Emphasis Duration Control This signal controls the duration for which the HS pre-emphasis current is sourced onto DP or DM. The HS Transmitter pre-emphasis duration is defined in terms of unit amounts. One unit of pre-emphasis duration is approximately 580 ps and is defined as 1X pre-emphasis duration. This signal is valid only if either TXPREEMPMPPTUNE[1] or TXPREEMPMPPTUNE[0] is set to 1'b1. 0b - 2X (design default), long pre-emphasis current duration If TXPREEMPULSESETUNE is not used, set it to 1'b0 1b - 1X, short pre-emphasis current duration
16-15 TXPREEMPMP TUNE0	USB3.0 PHY Signal
14-13 TXSHXSTUNE0	USB3.0 PHY Signal
12-9 TXFSLSTUNE0	USB3.0 PHY Signal
8-6 SQRXTUNE0	USB3.0 PHY Signal
5-3 OTGTUNE0	USB3.0 PHY Signal
2-0 COMPIDISTUN E	USB3.0 PHY Signal

11.1.3.2.8 PHY_CTL4_ADDR (PHY_CTL4_ADDR)

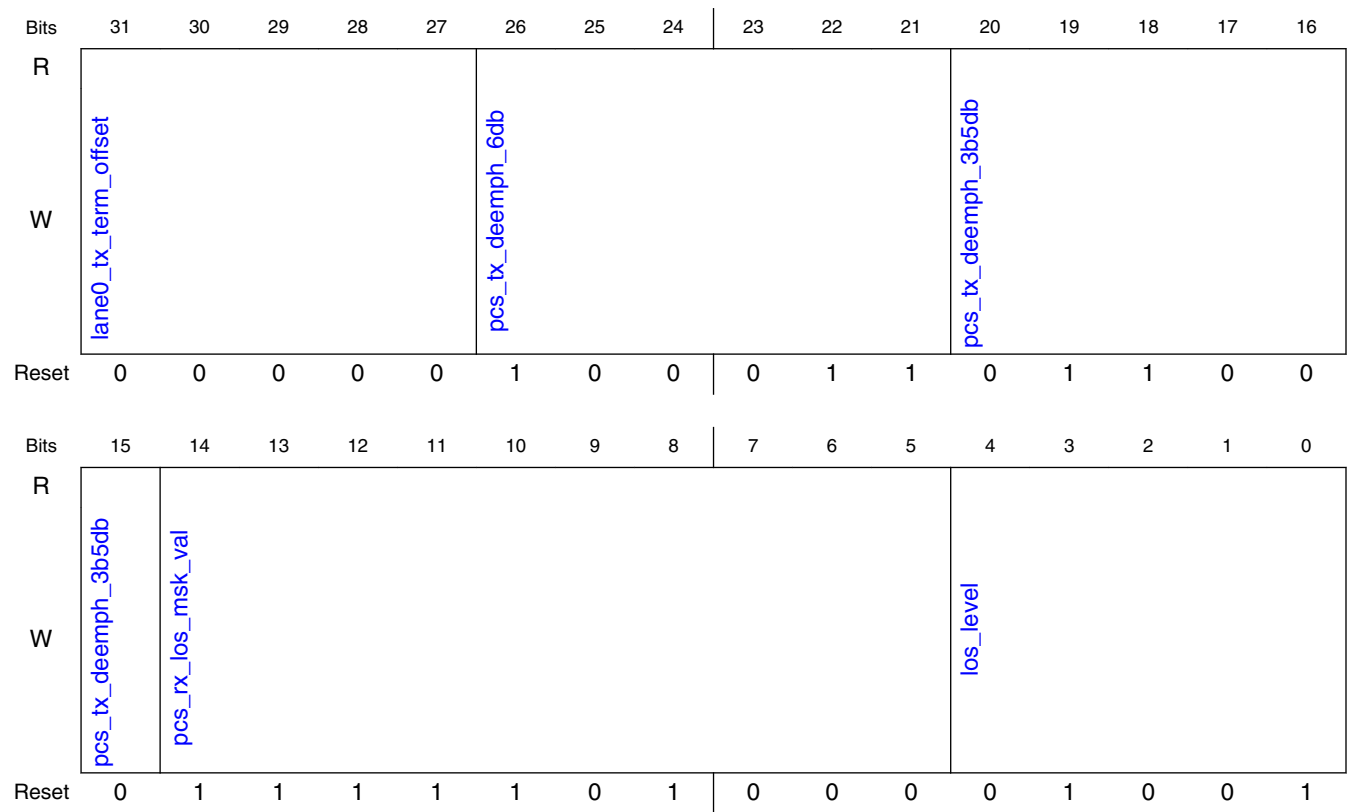
11.1.3.2.8.1 Offset

Register	Offset
PHY_CTL4_ADDR	50h

11.1.3.2.8.2 Function

USB3.0 PHY Status bits Register

11.1.3.2.8.3 Diagram



11.1.3.2.8.4 Fields

Field	Function
31-27 lane0_tx_term_o ffset	USB3.0 PHY Signal
26-21 pcs_tx_deemph _6db	USB3.0 PHY Signal
20-15 pcs_tx_deemph _3b5db	USB3.0 PHY Signal
14-5 pcs_rx_los_msk _val	USB3.0 PHY Signal
4-0 los_level	USB3.0 PHY Signal

11.1.3.2.9 PHY_CTL5_ADDR (PHY_CTL5_ADDR)

11.1.3.2.9.1 Offset

Register	Offset
PHY_CTL5_ADDR	54h

11.1.3.2.9.2 Function

USB3.0 PHY Status bits Register

11.1.3.2.9.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								pcs_tx_swing_full							
W																
Reset	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1

11.1.3.2.9.4 Fields

Field	Function
31-7 —	Reserved
6-0 pcs_tx_swing_full	USB3.0 PHY Signal

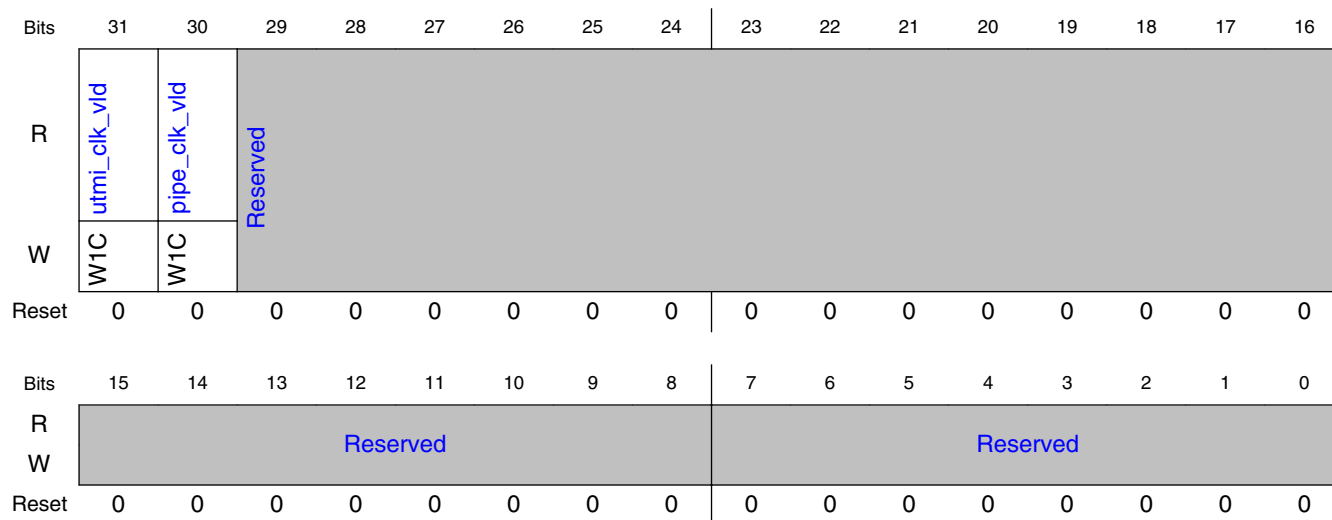
11.1.3.2.10 PHY_STS0_ADDR (PHY_STS0_ADDR)

11.1.3.2.10.1 Offset

Register	Offset
PHY_STS0_ADDR	80h

11.1.3.2.10.2 Function

USB3.0 PHY Status bits Register

11.1.3.2.10.3 Diagram**11.1.3.2.10.4 Fields**

Field	Function
31 utmi_clk_vld	USB3.0 PHY Signal synchronised by USB bus clock.After PHY and core reset pipe clock is stable if this bit is set.
30 pipe_clk_vld	USB3.0 PHY Signal synchronised by USB bus clock.After PHY and core reset pipe clock is stable if this bit is set.
29-8 —	Reserved
7-0 —	Reserved

11.2 Universal Serial Bus PHY (USB_PHY)

11.2.1 Overview

The USB 3.0 PHY is designed for USB 3.0 SuperSpeed applications connecting to a USB OTG controller. The USB 3.0 PHY supports the USB 3.0 SuperSpeed (5 Gbps) protocol and data rate and is backward compatible with USB 2.0 high-speed (480 Mbps), full-speed (12 Mbps), and low-speed (1.5 Mbps) protocols and data rates. This block guide provides an introduction to the USB 3.0 PHY and its features.

NOTE

The information within this block guide is Synopsys Proprietary. Used with permission.

11.2.1.1 Block Diagram

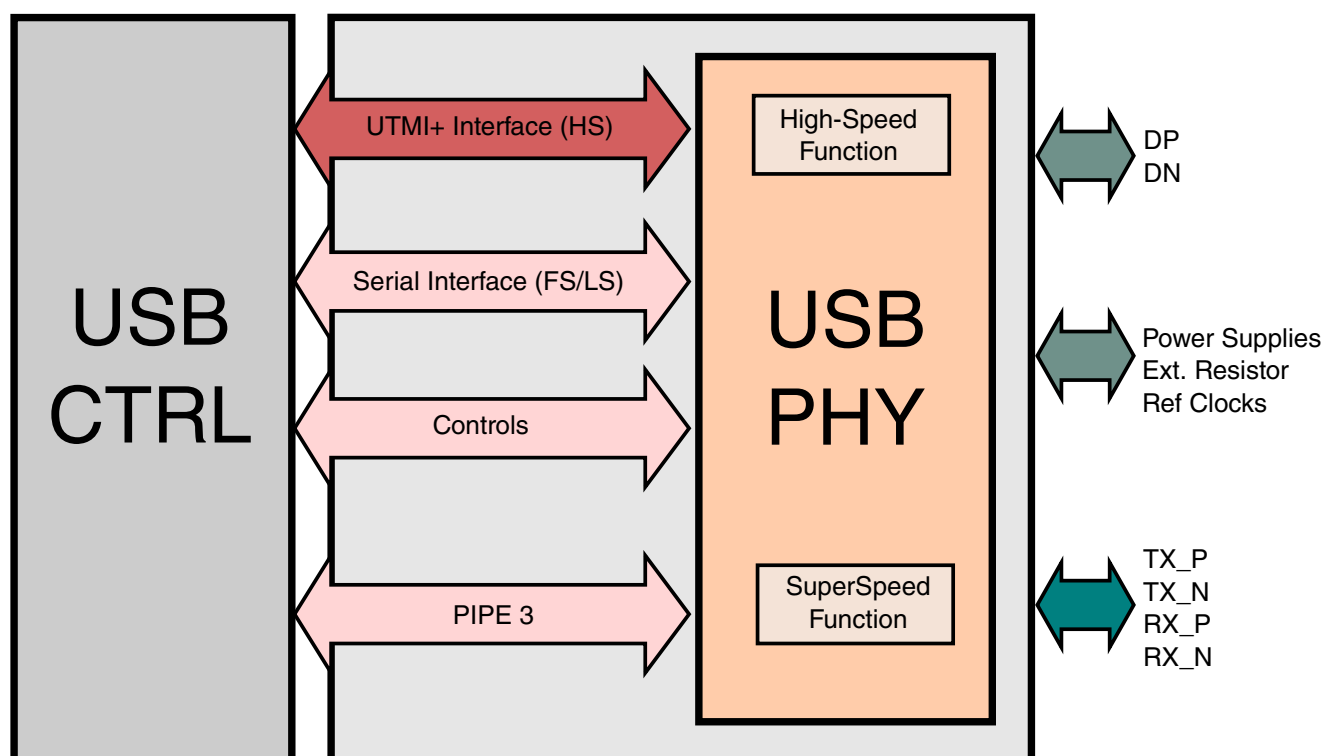


Figure 11-47. USB PHY Block Diagram

11.2.1.2 Features

The USB 3.0 PHY provides the following general features:

- Built-in VBUS threshold comparators;
- Single reference clock input for both USB 3.0 and USB 2.0 operating modes
- Integrated 3.3-V-to-1.8-V regulator in the PHY

The USB 3.0 PHY provides the following USB 3.0 features:

- 5-Gbps SuperSpeed data transmission rate through 3-m USB 3.0 cable
- PIPE 3-compliant SuperSpeed USB 3.0 Transceiver interface
- Integrated PHY includes transmitter, receiver, SSC generation, PLL, digital core, and ESD
- Adaptive Rx equalization
- Low-jitter PLL technology with excellent supply isolation

The USB 3.0 PHY provides the following USB 2.0 features:

- Built-in VBUS threshold comparators;
- 480-Mbps high-speed, 12-Mbps full-speed, and 1.5-Mbps low-speed serial (Host mode only) data transmission rates
- Supports high-speed power modes: suspend, resume, and remote wakeup
- 45- Ω termination, 1.5-k Ω pull-up and 15-k Ω pull-down resistors with support for independence control of the pull-down resistors
- 8/16-bit unidirectional parallel interfaces for HS, FS, and LS (Host mode only) modes of operation in accordance with the UTMI+ specification
- Dual (HS/FS) mode host/device support
- Data recovery from serial data on the USB connector
- SYNC/End-of-Packet (EOP) generation and checking
- Bit stuffing and unstuffing, and bit-stuffing error detection
- Non Return to Zero Invert (NRZI) encoding and decoding
- Bit serialization and deserialization
- VBUS threshold comparators

11.2.1.3 Standards Compliance

The USB 3.0 PHY is fully compliant with all the required features of the following standards:

- Universal Serial Bus 3.0 Specification, Revision 1.0
 - Supports 5.0-Gbps SuperSpeed (SS) through 3-m cable, 480-Mbps high-speed (HS), 12-Mbps fullspeed (FS), and 1.5-Mbps low-speed (LS) serial (Host mode only) data transmission rates, with a high-speed (HS) transmit-to-receive interpacket delay requirement of 38 bit-times as implied by the UTMI specification, version 1.05.

- Supports beaconing, receiver detection, and electrical idle
- Four-wire SuperSpeed connection (two Tx and two Rx unidirectional) with built-in Tx and Rx equalization
- PIPE 3-compliant USB 3.0 SuperSpeed interface
- On-The-Go Supplement to the USB 2.0 Specification, Revision 2.0
- UTMI+ Specification, Revision 1.0 (Level 3)
- Battery Charging Specification, Revision 1.2

11.2.2 Functional Description

11.2.2.1 PIPE Interface

The datapath of the USB 3.0 PHY is through the standard PIPE 3 interface. This interface can connect directly to any USB 3.0 controller. For more information about the PIPE 3 interface signals, refer to PHY Interface For the PCI Express and USB 3.0 Architectures.

11.2.2.2 Termination Tuning

The PHY uses an external resistor to calibrate the termination impedances of the PHY's high-speed inputs and outputs. The resistor is shared between the USB 2.0 high-speed outputs and the SuperSpeed I/O. Each time the PHY is taken out of a reset, a termination calibration is performed. The termination calibration is continuous for USB 2.0 operation except when the USB 2.0 port is suspended. The termination calibration is performed only when coming out of a reset in SuperSpeed operation.

A resistor calibration on the SuperSpeed link cannot be performed while the link is operational. At no point is a request for calibrating the SuperSpeed link using `rtune_req` required; this feature is provided only for completeness.

11.2.2.3 Battery Charging

The USB 3.0 PHY supports battery charger detection to enable devices to draw increased current from VBUS for charging and/or powering up from dedicated charging ports or charging downstream ports.

11.2.2.4 Dead Battery Provision

The dead battery provision enables a device to draw increased current (up to 100 mA) from VBUS in certain conditions. This function can be used only if all power supplies are stable and at their respective, normal, valid operating levels.

11.2.2.5 Data Contact Detect

Prior to a device performing battery charger detection, DCD is performed to determine if the D+ and D– pins in the device have made contact with those in the host or charger.

11.2.2.6 UART / Autoresume

The USB 3.0 PHY's single-ended receivers are used for UART and Autoresume data reception, while the full-speed drivers are used for data transmission. UART and Autoresume signaling levels are the same as signaling levels for USB 2.0 FS mode.

In this mode of operation, the driver source impedance is not calibrated; however, the driver retains the settings of the last calibration. In addition, the rise/fall time control is disabled, so changes in the state of DP and DN driven by this mode are not expected to meet the rise and fall times as defined in the USB 2.0 specification.

11.2.2.7 Serial Interface

The serial interface supports full-speed and low-speed data transmission rates to and from a USB 1.1 controller.

11.2.3 Clocks

The USB 3.0 PHY supports a wide range of input reference clocks from both external and on-chip clock sources. To support both SuperSpeed and high-speed operations, one of the following must be provided:

- A compatible, shared reference clock frequency
- Separate clock sources to support SuperSpeed operation and high-speed operation

11.2.4 External Signals

This section describes the USB 3.0 PHY signals that go to pins:

- External reference resistor: resref

- Receive pins: RX_N, RX_P
- Transmit pins: TX_N, TX_P
- USB 2.0 D+/D– lines: DN, DP
- USB mini-receptacle identifier: ID
- USB 5-V power supply pin: VBUS

11.3 PCI Express (PCIe)

11.3.1 Overview

This PCI Express dual mode (DM) controller provides a solution to implement a PCI Express port for a PCI Express root complex or endpoint application. A complete PCI Express port solution includes the controller, an analog PHY macro, and application logic to source and sink data.

NOTE

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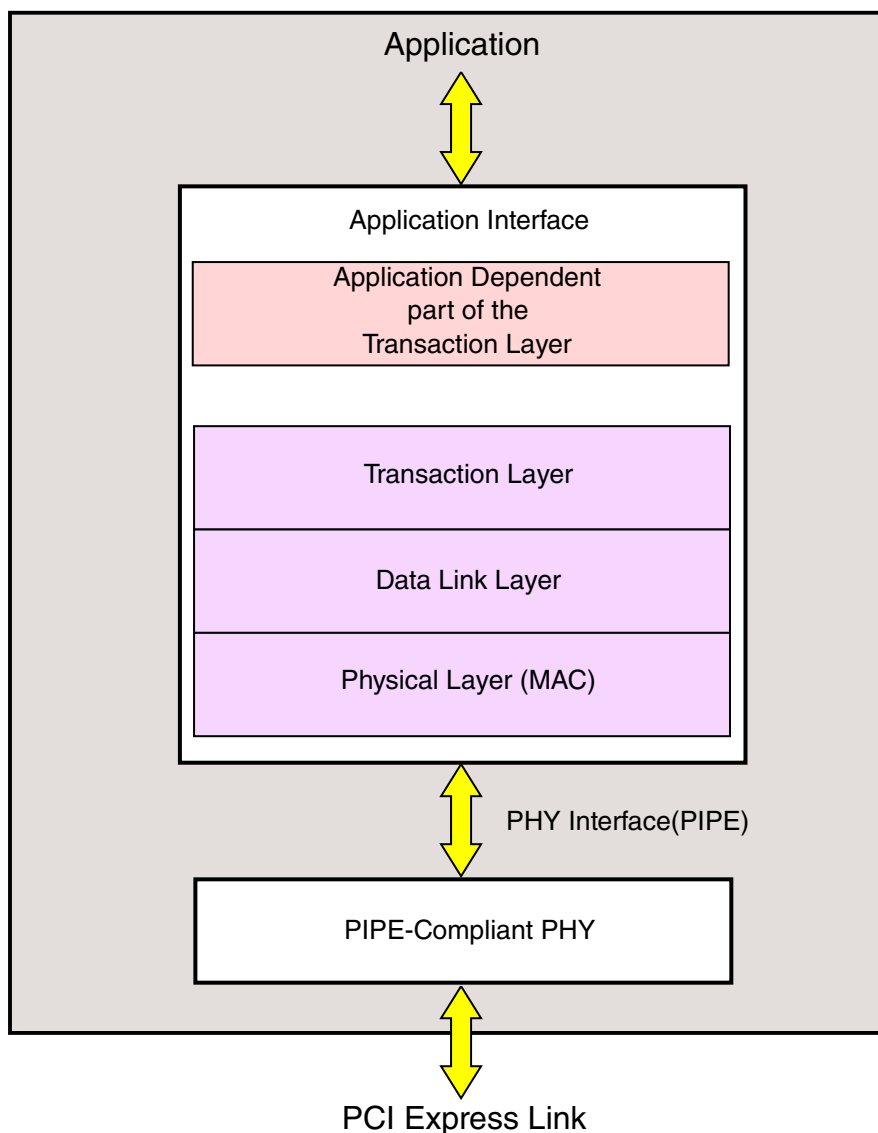


Figure 11-48. PCIe Application

11.3.1.1 Features

The PCI Express DM controller supports:

- Latency Tolerance Reporting (LTR)
- Address Translation Services (ATS) by your application
- Page Request Services (PRS) by your application
- Completion Timeout Ranges
- ID-Based Ordering (IDO) by your application
- M-PCIe
- PASID

- L1 Substates (L1SS)
- Extended Tag Support
- Resizable BAR (RBAR) with Expanded RBAR support,
- Separate Refclk with Independent Spread Spectrum Clocking (SRIS)
- Lightweight Notifications (LN)
- Readiness Notifications (RN)
- PCI Express Active State Power Management (ASPM)
- PCI Express Advanced Error Reporting (AER) with Multiple Header Logging
- PCI Express Advanced Error Reporting (AER)
- Vital Product Data (VPD)
- x1 lane
- Power Gating (UPF) Support
- Advanced Power and Clock Management
- Internal Address Translation Unit
- Internal MSI-X Generation Module
- AXI3
- Embedded DMA
- Upconfigure Support
- ECRC Generation and Checking
- Supports two Virtual Channels (VC)
- Max_Payload_Size (128 bytes)
- Configurable Filtering Rules for Posted, Non-posted, and Completion Traffic
- Configurable BAR Filtering, I/O Filtering, Configuration Filtering and Completion Lookup/Timeout
- MSI and MSI-X with Per-Vector Masking (PVM), Extended message data for MSI
- Type 1 Configuration Space
- Application-initiated Manual Lane Reversal/flip for situations where controller does not detect Lane 0
- Type 0 Configuration space

11.3.1.2 Standards

The PCIe controller implements the following standards:

- *PCI Express Base Specification, Revision 4.0, Version 0.7*
- Access to this specification requires membership in PCI-SIG.
- *PCI Local Bus Specification, Revision 3.0*
- Access to this specification requires membership in PCI-SIG.
- *PCI Bus Power Management Specification, Revision 1.2* Access to this specification requires membership in PCI-SIG.
- *PCI Express Card Electromechanical Specification, Revision 1.1* Access to this specification requires membership in PCI-SIG.

11.3.2 Functional Description

11.3.2.1 Link Establishment

The controller implements the LTSSM function according to the *PCI Express Base Specification, Revision 4.0, Version 0.7. .*

11.3.2.1.1 Transmit TLP Processing

The following section describe the flow of transmit TLPs through the controller.

NOTE

The controller does not check the TLP for errors.

The PCIe controller ensures transmitted TLP Payload sizes are less than or equal to the maximum payload size limit.

11.3.2.1.1.1 Transmit TLP Arbitration

TLPs and DLLPs have equal priorities during transmit arbitration as shown in the figure below. The priority of TLP types and DLLP types are shown in tables below.

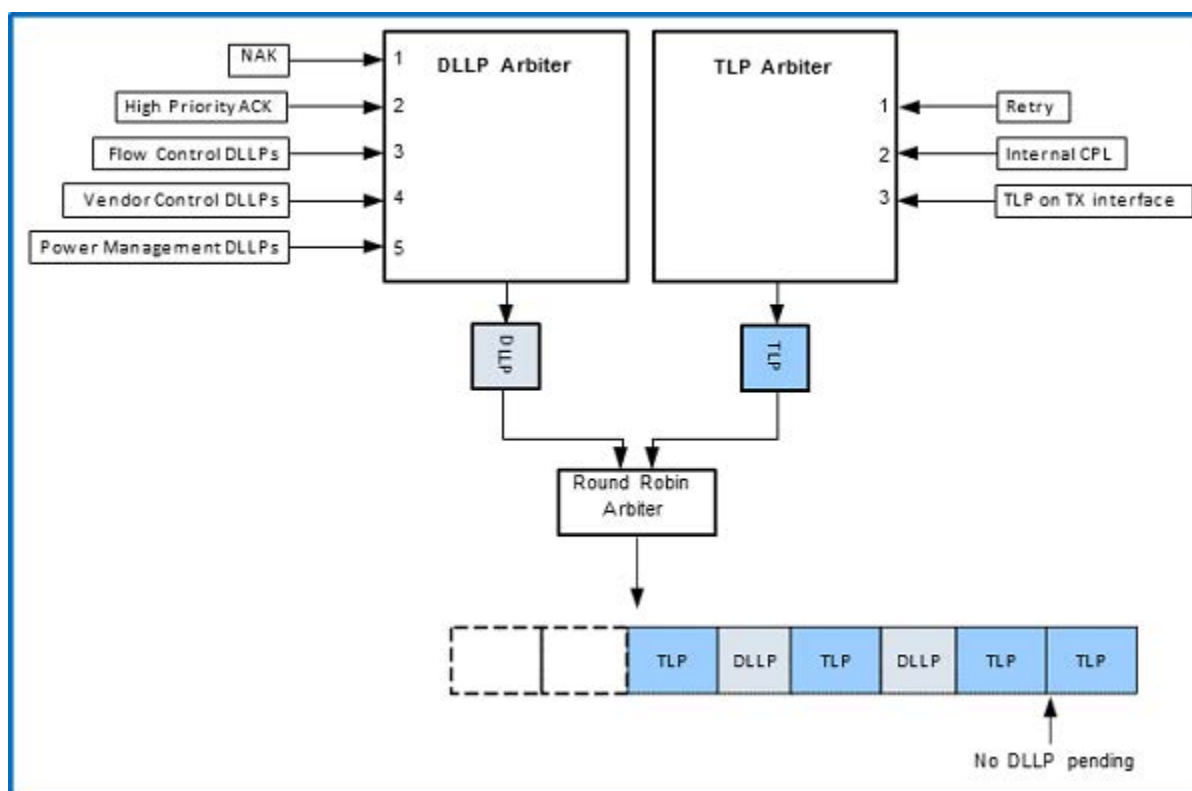


Figure 11-49. Transmit TLP/DLLP Arbitration

Table 11-25. TLP Arbitration Priority (1= Highest, 4= Lowest)

Priority	TLP Type
1	Completion of any TLP currently in progress
2	Retry buffer retransmissions
3	TLPs from the transaction layer (in the following order of priority): • Messages generated by the controller (including by your application through the MSI interface) • Upstream ports: Completions generated by the controller (including memory or I/O mapped application register space) for Type 0 configuration read and write requests, or responses to error conditions (unsupported requests) • Downstream ports: Completions generated by the controller for unsupported requests or completer aborts
4	TLPs on the AXI Master/Slave interface

Table 11-26. DLLP Arbitration Priority (1= Highest, 6 = Lowest)

Priority	DLLP Type
1	Completion of any DLLP currently in progress
2	NAK DLLP

Table continues on the next page...

Table 11-26. DLLP Arbitration Priority (1= Highest, 6 = Lowest) (continued)

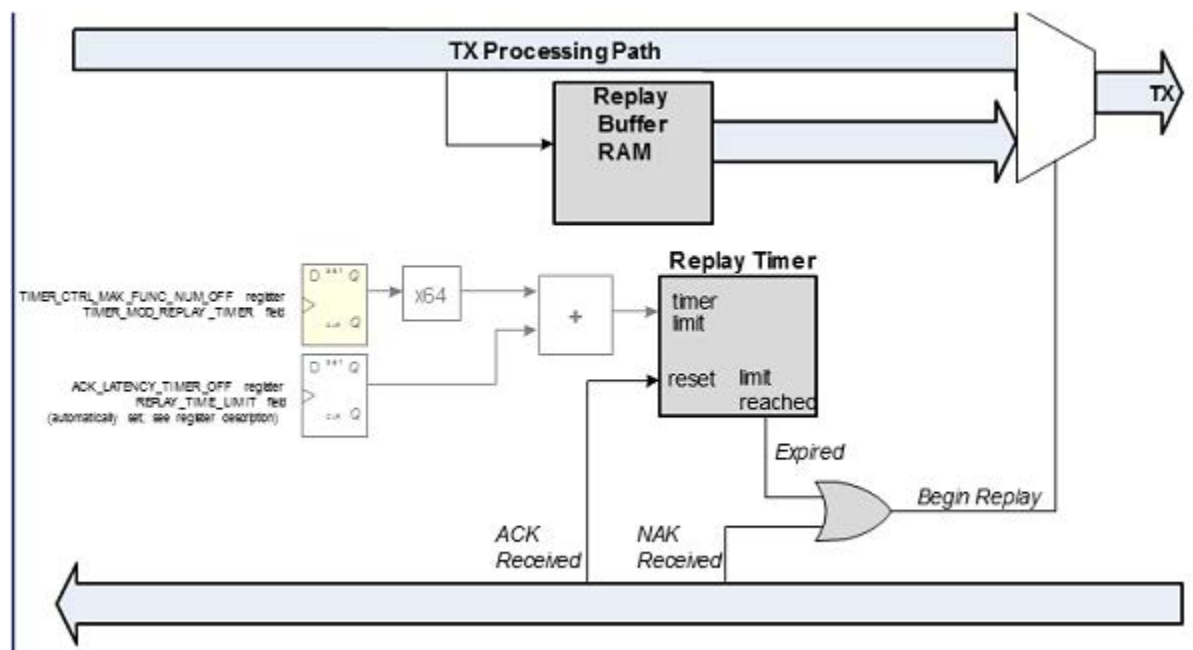
3	High Priority ACK DLLP
4	Flow Control DLLP
5	Vendor Specific Message DLLP
6	Power management or any other Low Priority DLLP

11.3.2.1.1.1 Effects of Flow Control Credits On Transmit Client Arbitration

The controller checks that enough flow control (FC) credits are available in the remote device for the specific type of transaction (posted, non-posted, completion) before allowing a transmission of a TLP. TLPs that passed the credit check are arbitrated according to the supported arbitration method. Internally generated completions and messages are also gated by the arbitration logic, though at highest priority, and must also pass the FC credit test before they are accepted for transmission.

11.3.2.1.1.2 Transmit Replay

You can modify the time-out value of the replay buffer as shown in the figure below. For a typical system, you do not have to modify the default settings, unless the remote device is executing unexpected replays.

Figure 11-50. Replay Buffer And Time (Representative)

11.3.2.1.2 Receive TLP Processing

This section describes the flow of received TLPs through the controller.

11.3.2.1.2.1 Receive Filtering

The controller contains a filter module that is responsible for the following tasks:

- Determine the status of a received TLP using filtering rules.
- Determine the destination of a received TLP based on the filtering status.
- Indicate the status of the received TLP using outputs.
- Report Errors to AER registers based on filter results. When more than one type of error is detected, Error Pollution, of the *PCI Express Base Specification, Revision 4.0, Version 0.7* is followed.
- The controller filters and routes received TLPs according to a set of rules determined by the TLP type based on the *PCI Express Base Specification, Revision 4.0, Version 0.7* and user-configurable filtering options.

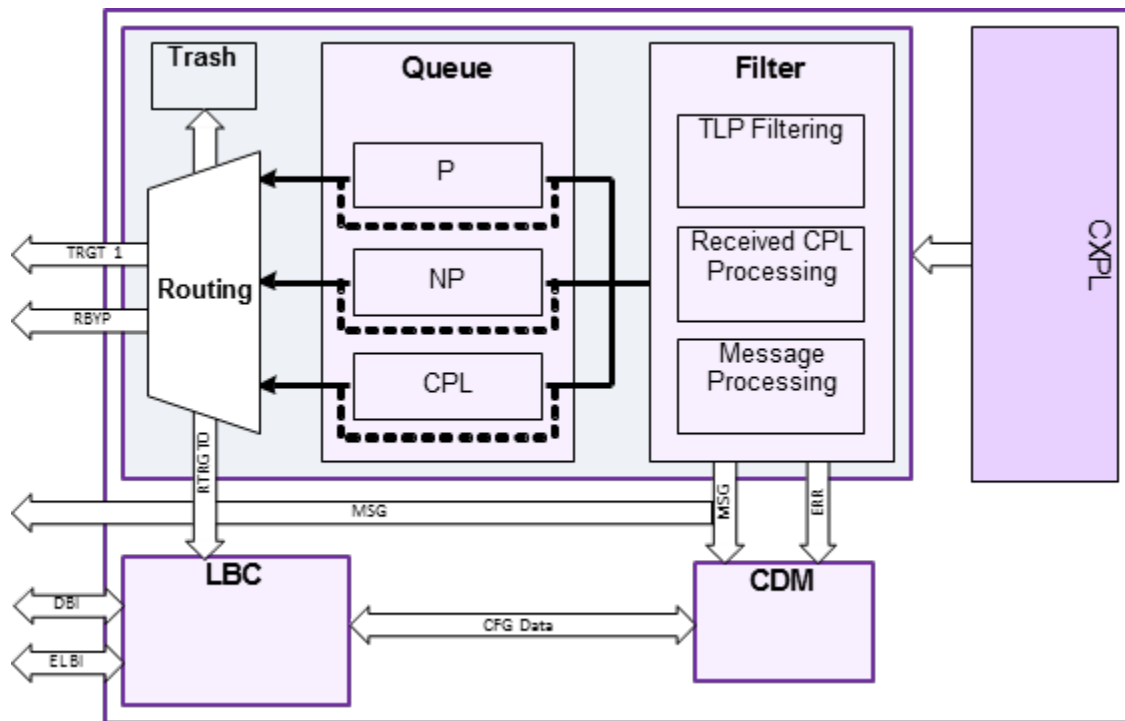


Figure 11-51. RADM Block Diagram

The following general rules apply to all incoming TLPs that are not malformed. For details on what happens to malformed TLPs, see "Error Detection for Received TLPs" By default:

- For a function in device power states D1, D2, and D3hot, the controller only accepts CFG and MSG requests TLPs for that function. All other incoming request types for that function are treated as unsupported requests (UR).
- When the controller detects an error in a received TLP, it normally performs the following:
 - Discards the TLP
 - Generates a completion (for non-posted requests) with the completion status set to CA or UR
 - Sets the status in the PCI-compatible Status register
 - Sets the status in the AER registers
 - Generates an error message (upstream port only)

All error-free MSG requests are decoded internally, signaled on the SII interface and then dropped. When you want to have the decoded message also sent to the application interface, then see “Routing of Received Messages”.

11.3.2.1.2.2 Receive Routing

This section discusses how the RADM routes different TLP types to different receive interfaces depending on TLP type and filter result. TRGT1 refers to the AXI interface. TRGT0 refers to the internal interface used to access the CDM registers or the ELBI in an upstream port.

11.3.2.1.2.2.1 EP Mode Routing Overview

The possible destinations of a posted or non-posted request TLP are TRGT1 interface, TRGT0 interface and Discard. By default:

- CFG requests are routed to TRGT0 and then to CDM through the LBC.
- BAR-matched MEM and I/O requests are routed to TRGT1.
- MSG requests are decoded internally, signaled on the SII interface, and then terminated.

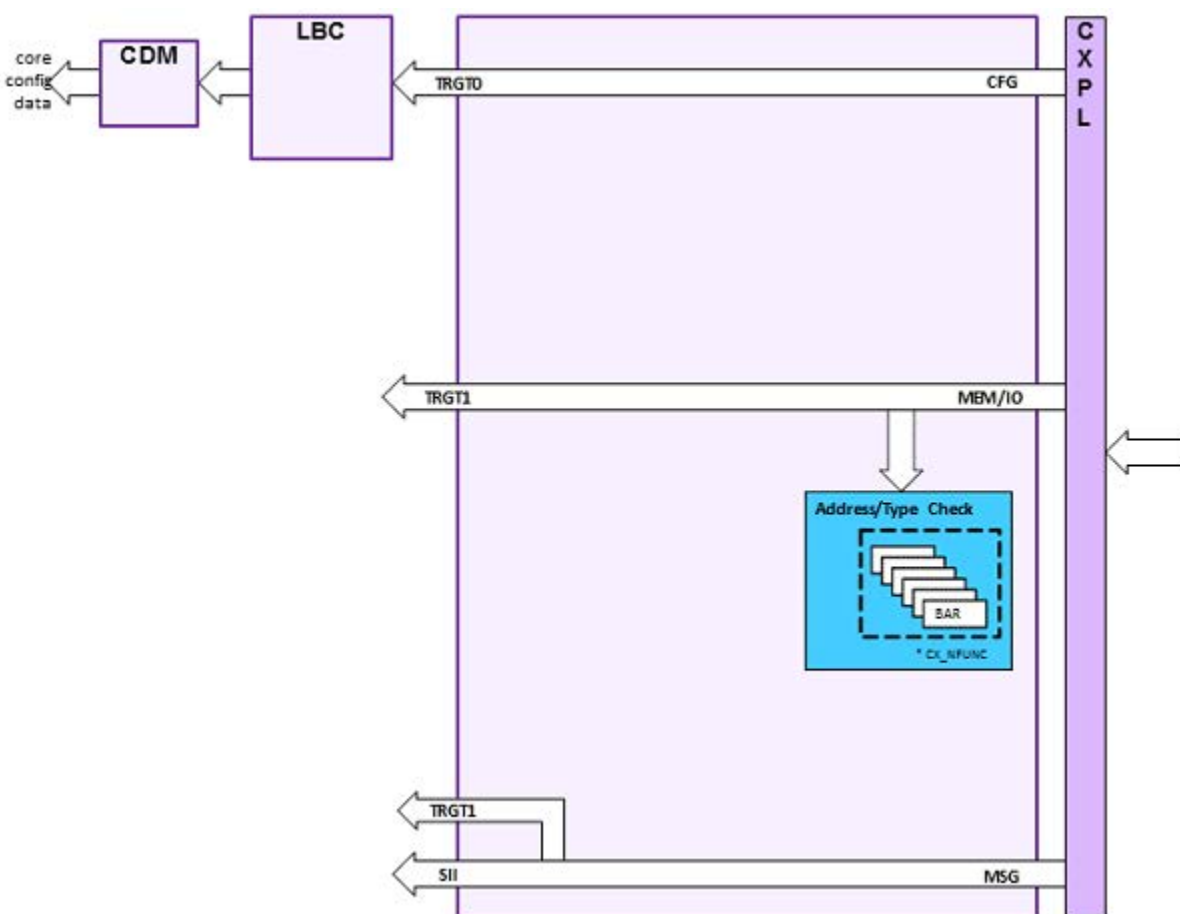


Figure 11-52. EP Mode Routing

11.3.2.1.2.2.2 RC Mode Routing Overview

The possible destinations of a posted or non-posted request are TRGT1 and Discard. By default:

- MEM requests outside of the memory range and prefetchable memory range as determined by the corresponding Base and Limit fields in the Type-1 header are routed to TRGT1.
- MSG requests are decoded internally, signaled on the SII interface, and then terminated.
- An RC does not expect to receive CFG or I/O requests.
- BARs should be disabled and not used.

The possible destinations of a completion TLP are TRGT1 and Discard.

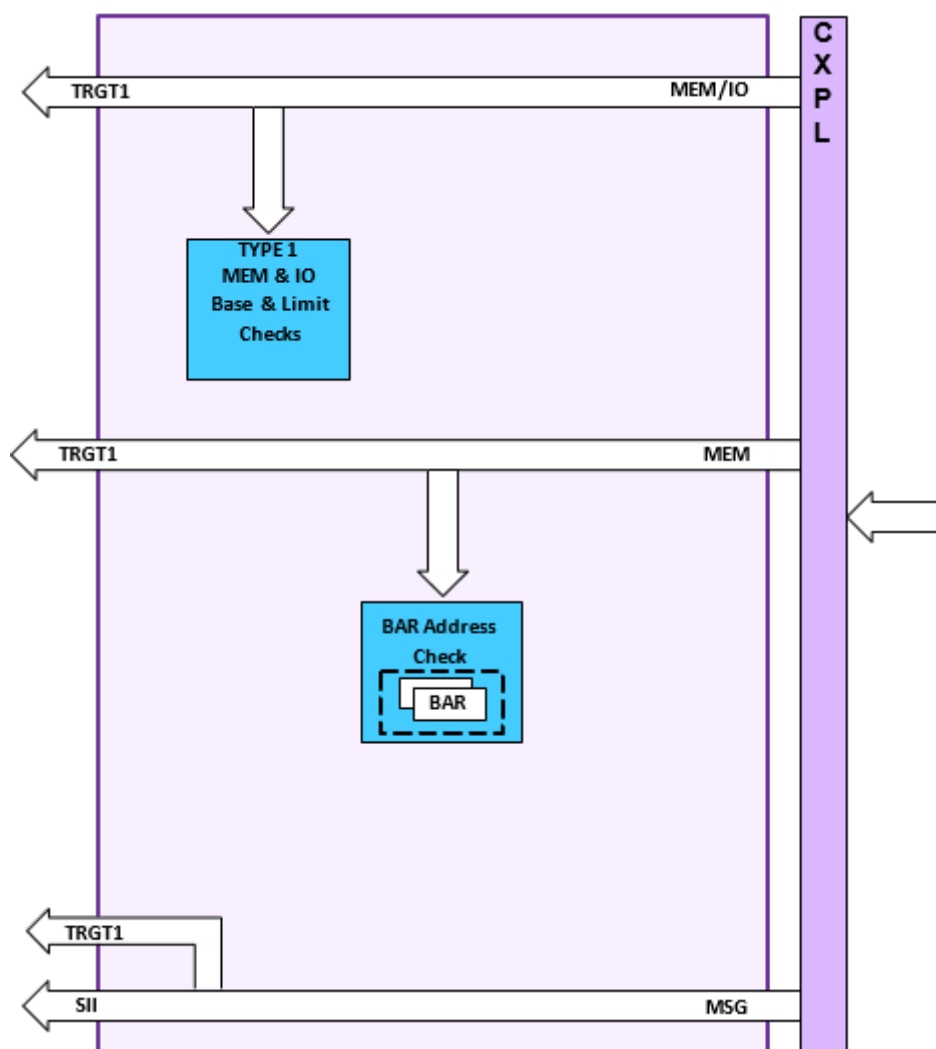


Figure 11-53. Default Request TLP Routing (Assuming no TLPs with CA/CRS/UR Error Status)

11.3.2.1.2.3 Error Handling

The controller supports the Baseline Capabilities, AER capabilities, and Advisory Non-Fatal Error Messaging as specified in section "Error Signaling and Logging" of the *PCI Express Base Specification, Revision 4.0, Version 0.7*. These include Correctable and Uncorrectable (Fatal and Non-Fatal) errors. For an RC port, the reporting of most errors is internal to the root port. No external error notifications are generated. One exception to this (for example) is unsupported request (UR) completion status.

11.3.2.1.2.3.1 Error Detection for Received TLPs

The controller performs all mandatory error detections, the error reporting mechanism based on the PCI Express Base Specification, Revision 4.0, Version 0.7. When the controller detects an error in a received TLP, it normally performs the following:

- Discards the TLP
- Generates a completion (for non-posted requests) with the completion status set to CA or UR
- Sets the status in the PCI-compatible status register
- Sets the status in the AER registers
- Generates an error MSG (upstream port only)

11.3.2.1.2.3.2 AER Multiple Header Logging

The controller supports the AER multiple header logging as specified in Section, "Multiple Error Handling (Advanced Error Reporting Capability)" of the *PCI Express Base Specification, Revision 4.0, Version 0.7*.

11.3.2.1.2.3.3 Physical Functions

By default, each physical function only logs the first header that caused an error. Per physical function, the controller implements a FIFO queue that interfaces with the AER Header Log registers (HDR_LOG_i_OFF). A valid queue entry is never overwritten. When the queue is full and your software does not read the header log registers fast enough, new header log information will be missed.

11.3.2.1.2.3.4 Virtual Functions

By default, each virtual function only logs the first header that caused an error. Each VF in each physical function has the exact same header log queue depth. Per VF, the controller implements a queue that interfaces with the AER Header Log registers (VF_HDR_LOG_i_OFF). A valid queue entry is never overwritten. When the queue is full and your software does not read the header log registers fast enough, new header log information will be missed.

The depth of each shared queue is identical for all PFs. The queue is not a FIFO, it is just a resource-saving mechanism, and each VF can randomly access its own header logs independently of other VFs.

Table 11-27. Shared VF Header Log Full Behavior

VF First Error Pointer Status	VF Shared Header Log Status	Core Actions
----------------------------------	--------------------------------	--------------

Table continues on the next page...

Table 11-27. Shared VF Header Log Full Behavior (continued)

Invalid	Full	• A read to the Header Log returns all 1's. • A read to the "TLP Prefix Log Present" bit indicates '0'. • The controller sets the Uncorrectable Error Status Register according to the First Error Pointer.
Valid	Full	• The controller sets the Uncorrectable Error Status Register according to the First Error Pointer that is associated with the rejected TLP. • The Header log can be cleared or updated to the next stored header log, if one exists, by writing to the bit of the Uncorrectable Status Error Register pointed to by the First Error Pointer. • This applies to both when a normally displayed TLP Header is present in the Header Log Register or when an overflow condition is shown on the Uncorrectable Status Error Register.

11.3.2.1.3 Register Module and Data Bus Interface (DBI)

11.3.2.1.3.1 Overview

The Local Bus Controller (LBC) module provides a mechanism for a link partner (in upstream mode) or a local CPU (through the DBI) to access:

- Internal registers (in the CDM)
- External application registers connected externally to the ELBI
- In RC mode:
 - There is no PCIe wire access to the CDM registers and ELBI.
 - There is full DBI access to the CDM registers and ELBI.
- The controller has 4096 bytes of register PCIe configuration and port logic space per function. The controller also has a common iATU and DMA configuration register space for all functions. The controller register space is fully accessible from the DBI without any restrictions.

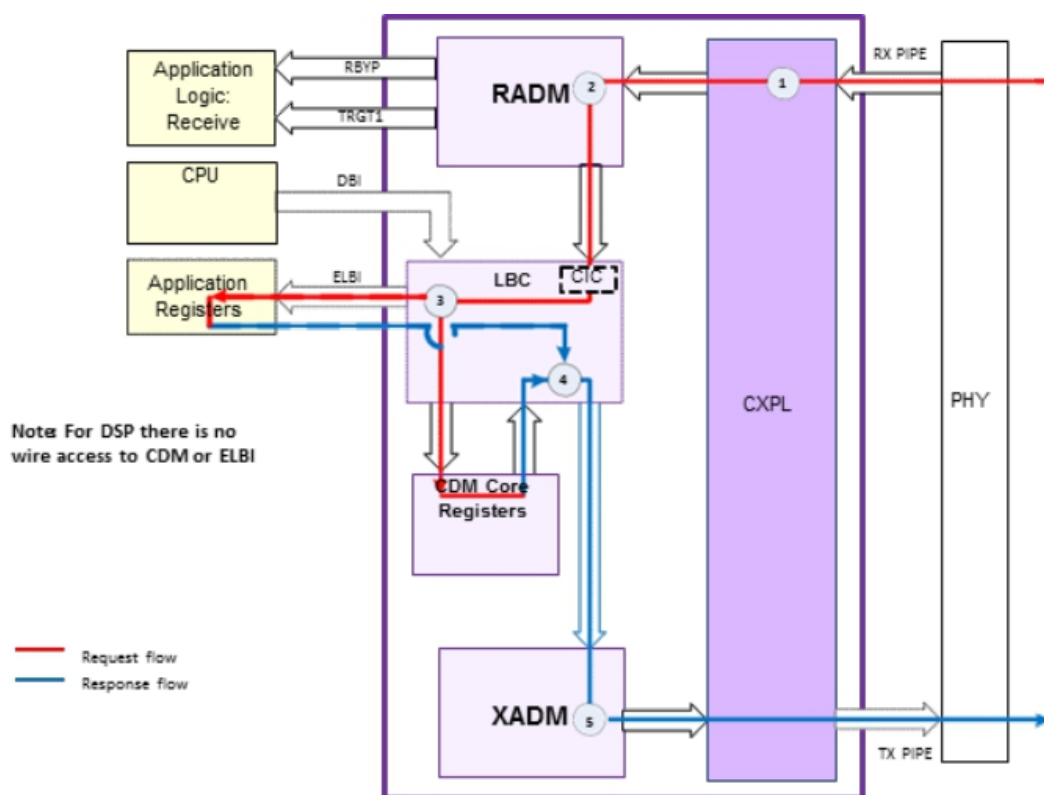
NOTE

Please refer to the Memory Map for the assigned space for the PCIe controller.

Table 11-28. EP Mode Port Region Access

Register Location		CDM					ELBI
Register Type		PCI-SIG		Port Logic			User
		Normal	Shadow	Misc	IATU	DMA	User
Access Details	CDM/ELBI Selector Bit ¹ Value	0	0	0	1	1	1
	CS2 Selector Bit Value	0	1	0	1	1	0
DBI/Wire Access Allowed							
DBI		Y	Y	Y	Y	Y	Y
Wire	CFG Request	Y	-	Y	-	-	Y
	BAR Matched MEM Request	-	-	Y ²	Y	Y	Y ¹

1. Selector bit locations are configuration-dependent.
2. IO request also supported.



PCIe Wire Access to CDM Registers or ELB(USP)

- ① Incoming request from PCIe remote link partner
- ② Request is filtered and routed by RADM through TRGT0 to LBC
- ③ LBC forwards the request to external registers(through ELBI) or internal registers in CDM
- ④ LBC forms completion TLPs with response received from ELBI or internal registers in CDM
- ⑤ PCIe controller transmits response completion to remote link partner

Figure 11-54. PCIe Wire Access to CDM Registers or ELB

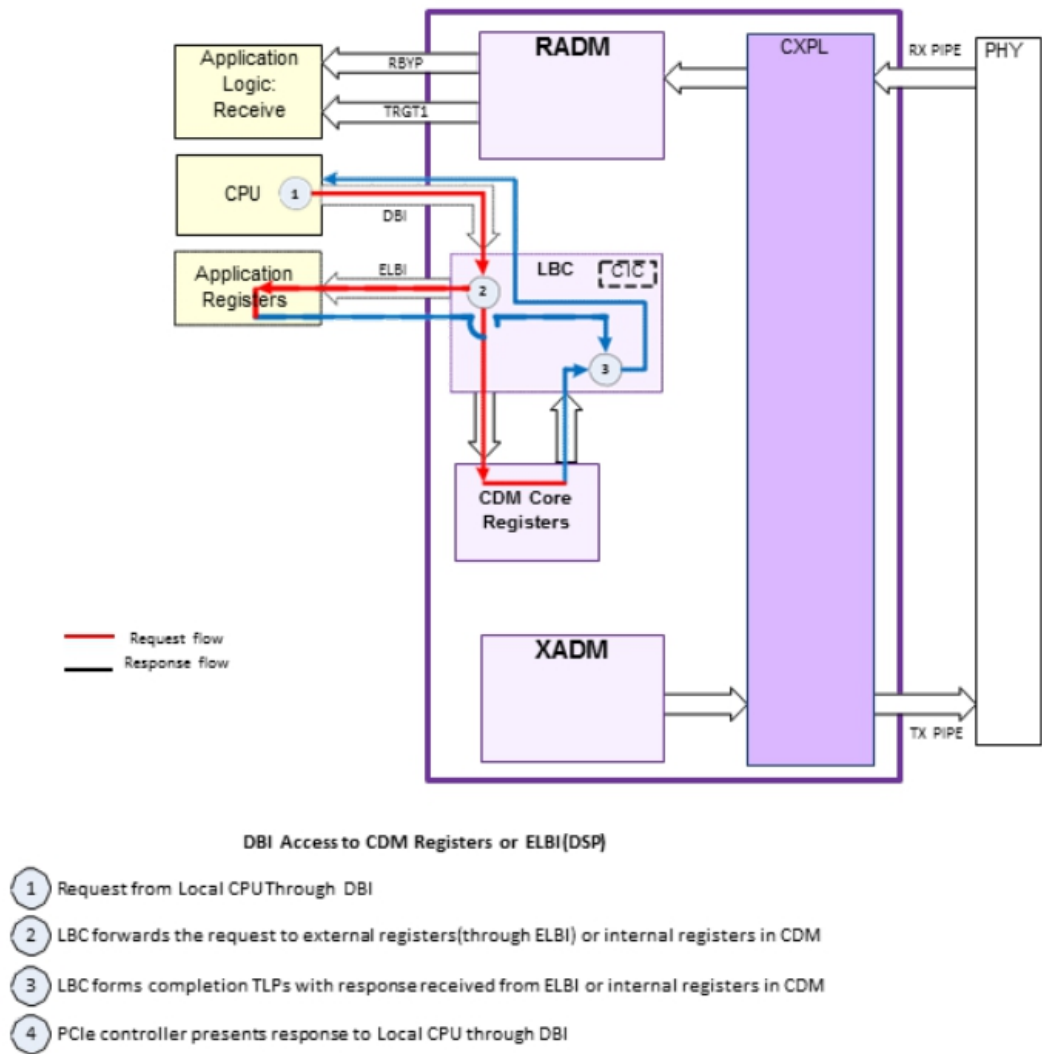


Figure 11-55. DBI Access to CDM Registers or ELBI

11.3.2.1.3.2 CDM Register Space Layout

11.3.2.1.3.2.1 Overview

The controller has 4096(0x1000) bytes of register PCIe configuration and port logic space per function. The controller also has a common iATU and DMA configuration register space for all functions. The controller register space is fully accessible from the DBI without any restrictions. Please see the Memory Map for the register base address assignment.

Table 11-29. Port Region Access

Register Location	CDM	ELBI
-------------------	-----	------

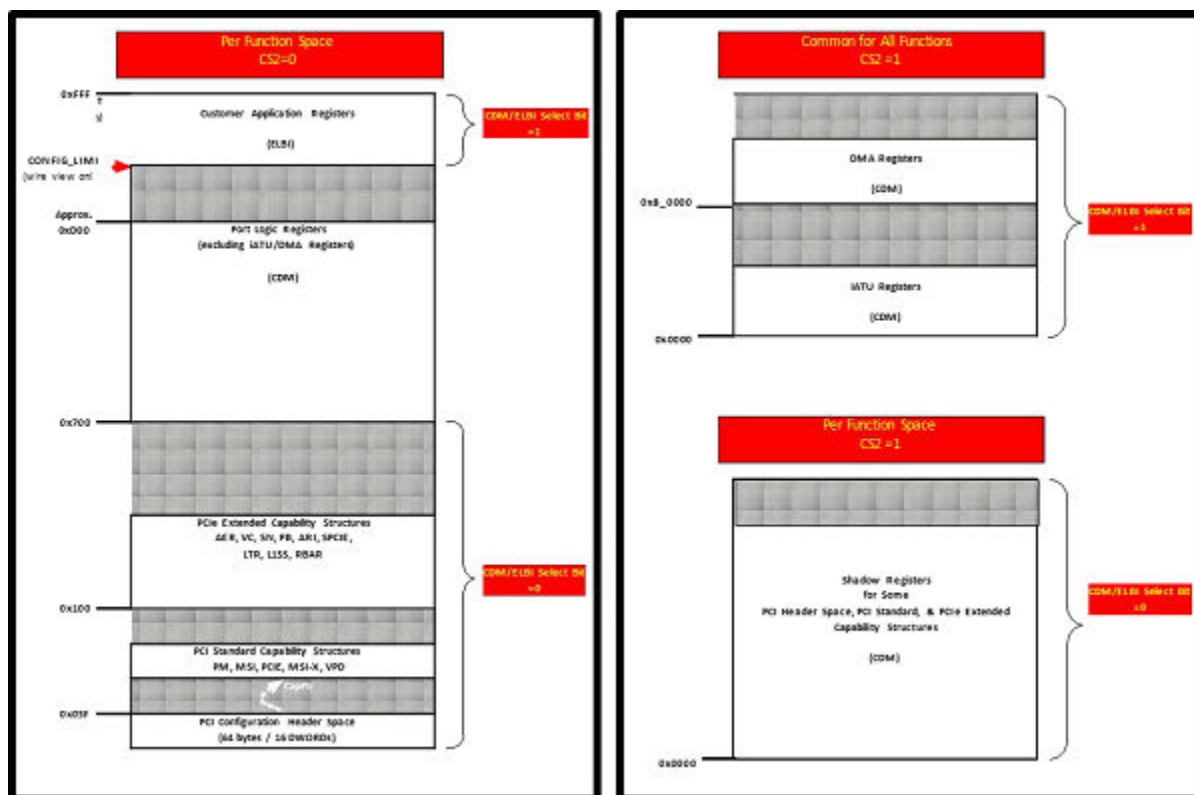
Table continues on the next page...

Table 11-29. Port Region Access (continued)

Register Type		PCI-SIG		Port Logic			User
		Normal	Shadow	Misc	IATU	DMA	User
Access Details	CDM/ELBI Selector Bit ¹ Value	0	0	0	1	1	1
	CS2 Selector Bit Value	0	1	0	1	1	0
DBI/Wire Access Allowed							
DBI		Y	Y	Y	Y	Y	Y
Wire	CFG Request	Y	-	Y	-	-	Y
	BAR Matched MEM Request	-	-	Y ²	Y	Y	Y ⁻¹

1. Selector bit locations are configuration-dependent.

2. IO request also supported.

**Figure 11-56. Controller Configuration Space Layout (EP Mode)**

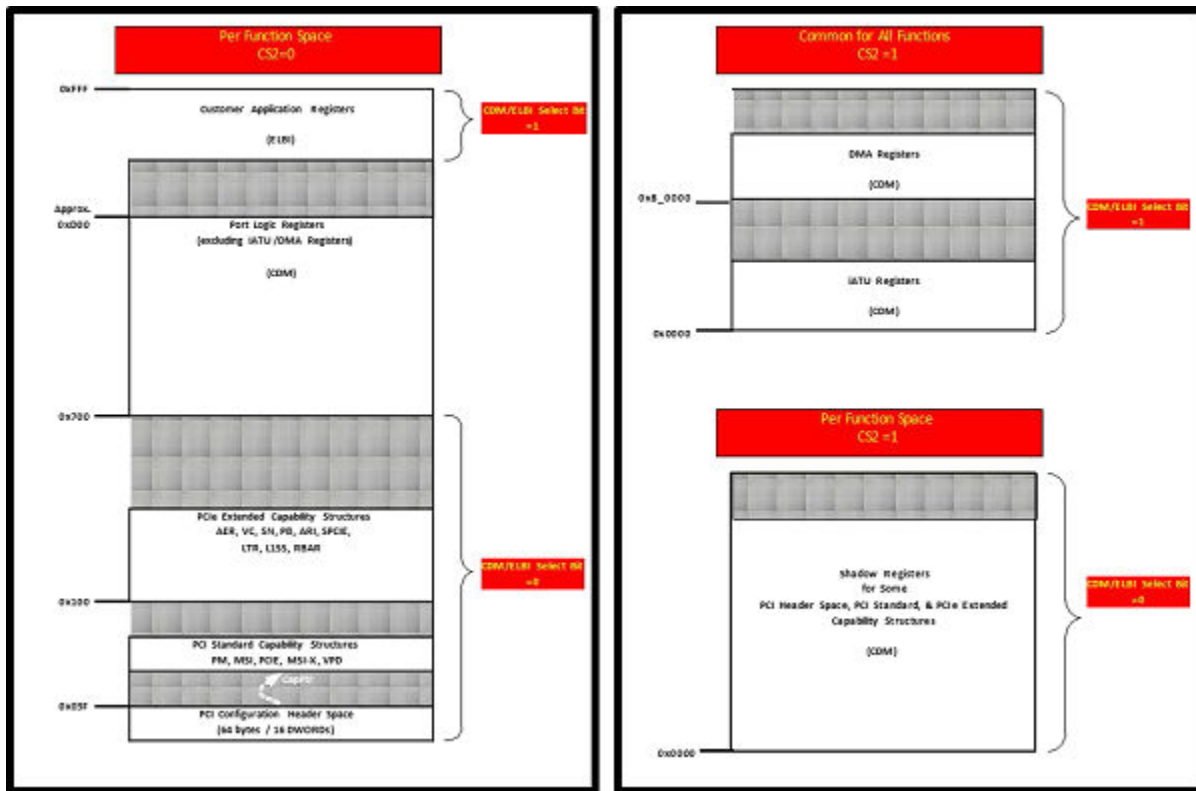


Figure 11-57. Controller Configuration Space Layout: (RC Mode)

Wire Access (Upstream Ports Only):

- Any CFG > CONFIG_LIMIT goes to TRGT1 or ELBI
- Any MEM/IO captured by a BAR whose interface target is TRGT0 goes to ELBI
- Any MEM captured by BAR_n ($n = \text{UNROLL_BAR_NUM}$) when
- ENABLE_MEM_MAP_UNROLL_{DMA|ATU}_REG=1, whose interface target is TRGT0, and whose address falls into the range defined in "Memory Mapping iATU and DMA Port Logic Registers" goes to the IATU/DMA registers.

DBI Access (Upstream and Downstream Ports):

- No concept of MEM/IO/CFG
- Uses the specified address bit to route the access to ELBI or CDM
- No restrictions regarding what spaces can be accessed or not

11.3.2.1.3.2.2 Core Configuration Space Registers

- CDM Registers
 - PCI Configuration Header and Capability Registers:
 - The *PCI configuration header and capability registers* in the previous diagrams are PCIe controller configuration registers specified by the *PCI Express Base Specification, Revision 4.0, Version 0.7*. Access from the PCIe wire is possible with CFG requests (upstream ports). These registers are fully accessible from the DBI without any restrictions.

- Port Logic Registers:
 - The port logic (PL) registers are configuration registers which are not specified by the *PCI Express Base Specification, Revision 4.0, Version 0.7*. Access from the PCIe wire is normally with CFG requests. These registers are fully accessible from the DBI without any restrictions.
- iATU and DMA Port Logic Registers:
 - This memory space is dedicated for iATU and DMA configuration registers. Access from the PCIe wire is with specific BAR matched MEM requests. These registers are fully accessible from the DBI without any restrictions.

11.3.2.1.3.3 Data Bus Interface (DBI)

11.3.2.1.3.3.1 Overview

The DBI can access all 4096 bytes (1024 DWORDs) of the PCI Express configuration space per function. DBI can also access iATU and DMA configuration space. This address space is fully accessible from the DBI without any restrictions.

11.3.2.1.3.4 Features and Limitations

The following limitations exist for the LBC.

- CDM Access Restrictions:
 - Maximum payload Length =1 DWORD.
 - PCIe MEM requests with Length > 1 are processed as completer abort.
 - PCIe MEM requests with Length =0 are processed as completer abort.
 - PCIe IO or CFG requests with Length !=1 are processed as malformed TLPs.
- ELBI Access Restrictions:
 - CS2 bit must be 1'b0.
 - PCIe MEM requests with Length =0 are processed as completer abort.
 - PCIe IO or CFG requests with Length !=1 are processed as malformed TLPs.
 - Does not support "Atomic Operations (AtomicOps)"
 - Does not support TPH. Core does not forward the TPH bits to the ELBI.
 - Data (Read)
- DBI Data/Address Port Widths (with no AXI bridge):
 - Data: 32 bit
 - Address: 32 bit

11.3.2.1.3.4.1 Additional AXI DBI Limitations

The following limitations exist when you configure the controller to use the AXI bridge module.

- The DBI address must be a DWORD address.
- AXI transfer widths of 8, 16, and 32 bits are supported. The AXI dedicated/shared DBI interface returns an error response to requests with width greater than 32 bits. The type of error response is programmable through `AMBA_ERROR_RESPONSE_DEFAULT_OFF` register1.
- For AHB you get an error response whenever the size is not equal to 32 bits.
- The CDM only supports 32-bit data access. The AXI dedicated/shared DBI interface returns an error response to multi beat requests. The type of error response is programmable through `AMBA_ERROR_RESPONSE_DEFAULT_OFF` register.
- Byte access within a DWORD for DBI requests is not possible for AHB.
- Byte access within a DWORD for DBI WRITE requests is possible for AXI (not for AHB)
 - Dedicated DBI: The write strobes `dbi_wstrb[3:0]` are sent to the LBC.
 - Shared DBI: The relevant four strobes from `slv_wstrb` are sent to the LBC

A zero-byte write over DBI is processed as a read and should never be issued. The AXI dedicated/shared DBI slave drops a zero-byte write before it reaches the internal (native controller) DBI and returns an error response. The type of error response is programmable through `AMBA_ERROR_RESPONSE_DEFAULT_OFF` register.

The type of error response returned by the AXI dedicated/shared DBI slave is the same as that programmed for UR completions in the `AMBA_ERROR_RESPONSE_DEFAULT_OFF` register.

11.3.2.1.4 Flow Control

The flow control mechanism is divided into two phases: initialization and update. The controller automatically performs both of these phases with minimal support required from your application. By default the RADM is responsible for returning flow control credits when it reads data out of the receive queues.

11.3.2.1.5 Completion Timeout Ranges

Timeout ranges are supported as defined in the *PCI Express Base Specification, Revision 4.0, Version 0.7*. The controller supports all ranges. The Device Control 2 Register has a default equal to the default in the specification: 0000b Default range: 50 us to 50 ms. When the default is used, then the timeout is in Range B: 0101b: 16 ms to 55 ms. This range was chosen for the default because the *PCI Express Base Specification, Revision*

4.0, *Version 0.7* states, "It is strongly recommended that the completion Timeout mechanism not expire in less than 10 ms". The table below illustrates the specification values versus the PCI Express controller values for the ranges.

NOTE

As the PCIe specification states, This mechanism is intended to be activated only when there is no reasonable expectation that the completion is returned, and should never occur under normal operating conditions.

Table 11-30. Comparison of PCIe Specification and PCIe Core Completion Timeout Ranges

Range	Encoding	Spec Minimum	Spec Maximum	PCIe controller Minimum	PCIe controller Maximum
Default	0000b	50 us	50 ms	28 ms	45 ms
A	0001b	50 us	100 us	65 us	99 ms
A	0010b	1 ms	10 ms	4.1 ms	6.2 ms
B	0101b	16 ms	55 ms	28 ms	44 ms
B	0110b	65 ms	210 ms	86 ms	131 ms
C	1001b	260 ms	900 ms	260 ms	390 ms
C	1010b	1s	3.5 s	1.8 s	2.8 s
D	1101b	4s	13 s	5.4 s	8.2 s
D	1110b	17s	64 s	38 s	58 s

11.3.2.1.6 Crosslink

Crosslink allows a downstream port to be connected to another downstream port or an upstream port to be connected to another upstream port. When a crosslink capable port negotiates a crosslink connection, the port changes its behavior accordingly. For this reason crosslink is supported in the DM controller because it contains the functionality for both upstream and downstream support.

Crosslink provides more than simply the ability to connect two like ports together. For instance, a DM operating in EP mode that negotiates a crosslink connection switches to RC mode and follows all of the *PCI Express Base Specification, Revision 4.0, Version 0.7* rules for an RC.

11.3.2.2 DMA Controller

This section describes the embedded multichannel DMA controller.

11.3.2.2.1 DMA Overview

The RC system CPU, or the EP application CPU, can offload the transferring of large blocks of data to the embedded DMA controller, leaving the CPU free to perform other tasks. It can simultaneously perform the following types of memory transactions (as shown in the figure 'System Level View of DMA'):

DMA write: Transfer (copy) of a block of data from local (application) memory to remote (link partner) memory.

DMA read: Transfer (copy) of a block of data from remote (link partner) memory to local (application) memory.

Therefore the DMA supports full duplex operation, processing read and write transfers at the same time, and in parallel with normal (non-DMA) traffic. Upon completion of a DMA transfer or an error, the DMA optionally interrupts the local CPU or sends an interrupt MWr (IMWr) to the remote CPU.

In linked list mode, the DMA fetches the transfer control information (called channel context) for each transfer (block), from a list of DMA elements that you have constructed in local memory.

11.3.2.2.2 DMA Architecture

This section describes the architecture of the DMA.

11.3.2.2.2.1 Architecture Overview

A DMA write and read channel operate independently to maximize the performance of the DMA read and write data transfers over the PCIe link. DMA with multiple read channels uses a weighted round robin (WRR) arbitration scheme to select the next read channel to be serviced. The same applies for multiple write channels.

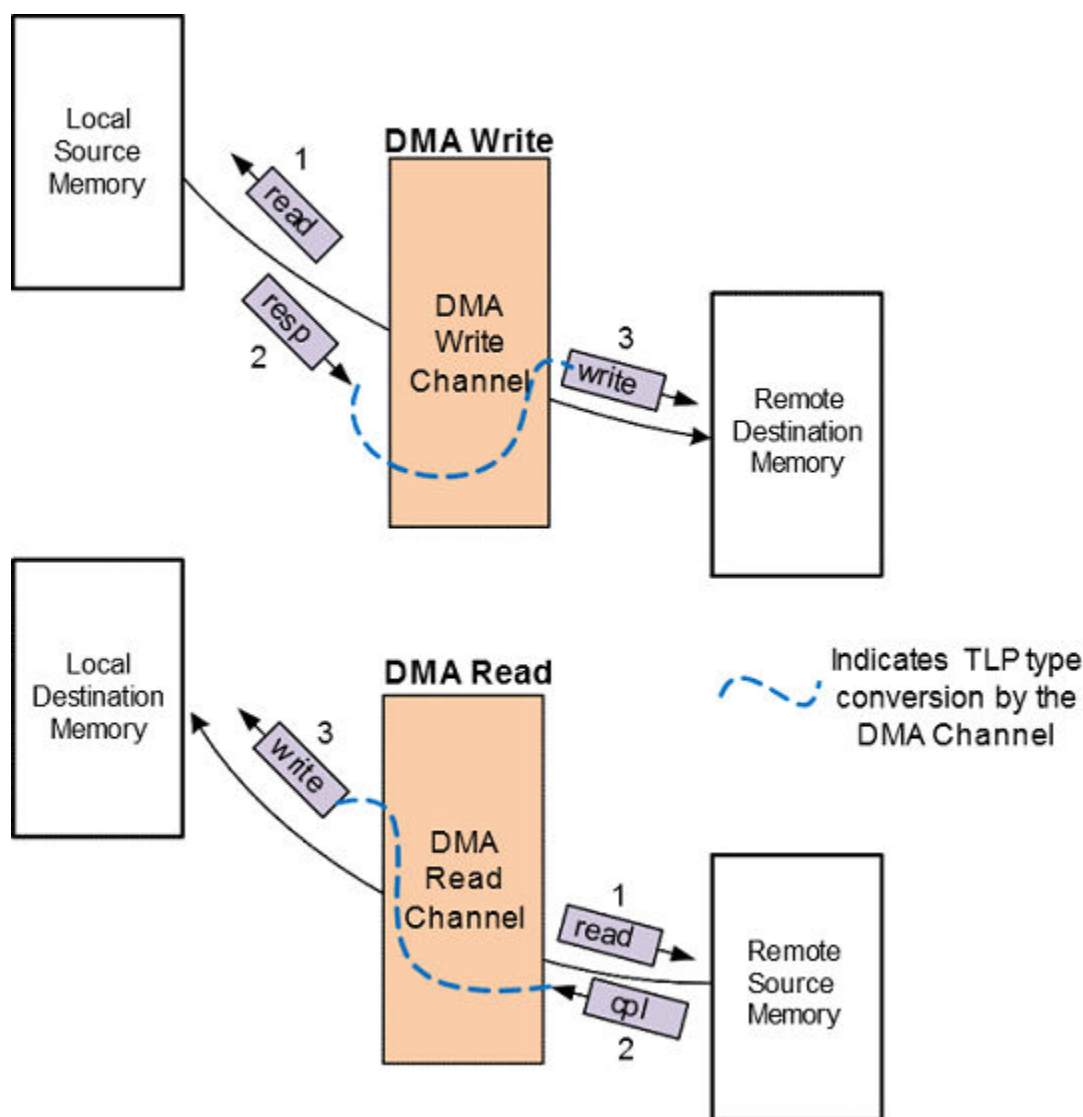


Figure 11-58. System Level View of DMA

11.3.2.2.2.1.1 DMA Read Transfer

The DMA injects multiple MRd requests of size less than or equal to the minimum of {Max Read Request Size, Max Payload Size} into the outbound request path, directed toward the remote link partner. The DMA converts the read responses into write requests, that it then transmits to the local application. When the DMA data transfer is complete, the CPU is notified. For a read transfer, the SAR is the address of the remote memory, and the DAR is the address of the local memory.

11.3.2.2.2.1.2 DMA Write Transfer

The DMA injects multiple MRd requests of size less than or equal to Max_Payload_Size into the inbound request path, directed toward the local application. The DMA converts the read responses into MWr TLPs, that it then transmits to the remote link partner. When the DMA data transfer is complete, the CPU is notified. For a write transfer, the SAR is the address of the local memory, and the DAR is the address of the remote memory.

11.3.2.2.2.2 Interrupts and Error Handling

11.3.2.2.2.2.1 Interrupt Handling

The DMA generates two interrupts per channel:

Done: The DMA successfully completes the transfer.

Abort: The DMA fails to complete the transfer, or an error occurs during the transfer.

The interrupt handling mechanism is different for linked list (LL) mode (than non LL mode), and there are also some differences between the read and write channels.

11.3.2.2.2.2.2 Non Linked List Mode

You enable the local and remote interrupts through the local and remote interrupt enable (LIE and RIE) bits:

DMA_CH_CONTROL1_OFF_WRCH_0.lie and
DMA_CH_CONTROL1_OFF_WRCH_0.rie.

In the write channel, there is only one error condition that results in an abort interrupt. For more details, see "Linked List Mode". You mask, clear, and read the status of each of the two interrupts (done and abort) through the DMA interrupt registers as indicated in figure 'Write Interrupt Generation - Non Linked List Mode - Shown For Write Channel #0'.

In the read channel, there are five error conditions that results in an abort interrupt. For more details, see "Linked List Mode". You mask and clear each of the two interrupts (done and abort) through the DMA interrupt registers as indicated in figure 'Read Interrupt Generation - Non Linked List Mode - Shown For Read Channel #0'. However, you can read the status of each of the five abort errors (that contribute to the abort interrupt) through DMA_READ_ERR_STATUS_LOW_OFF and DMA_READ_ERR_STATUS_HIGH_OFF.

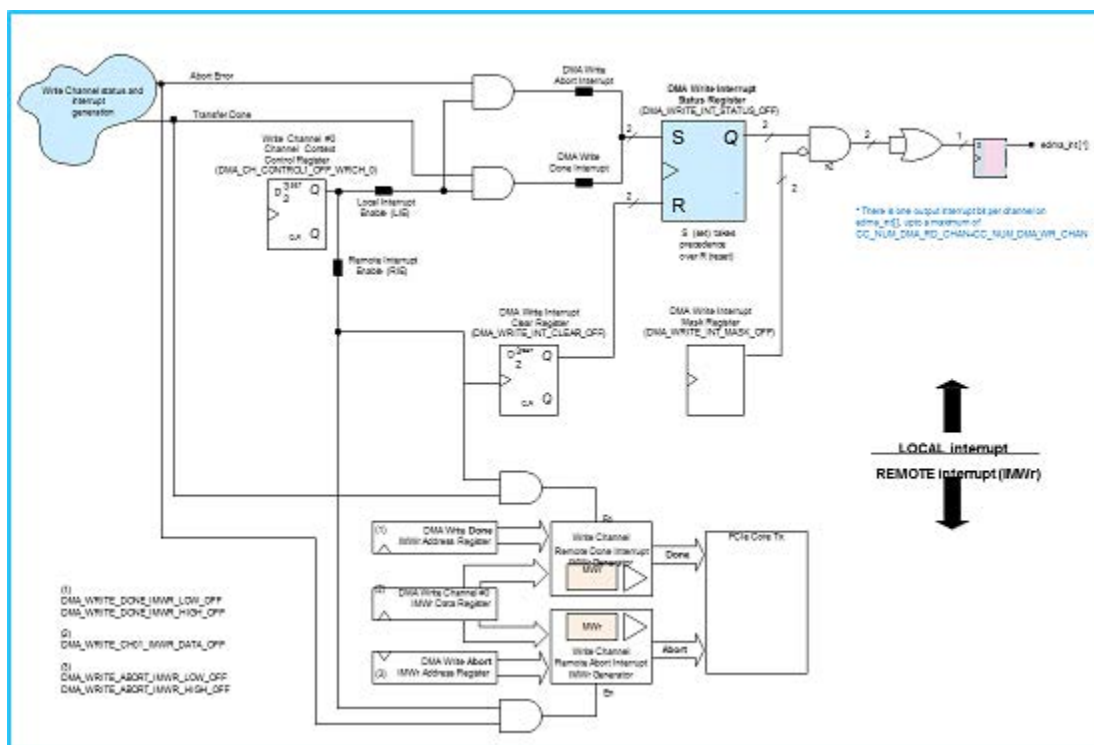


Figure 11-59. Write Interrupt Generation - Non Linked List Mode - Shown For Write Channel #0

If you want a remote interrupt and not a local interrupt then:

- Set LIE and RIE
- Mask the local interrupt using the mask register

This will allow you to poll the status register to distinguish a DONE from an ABORT

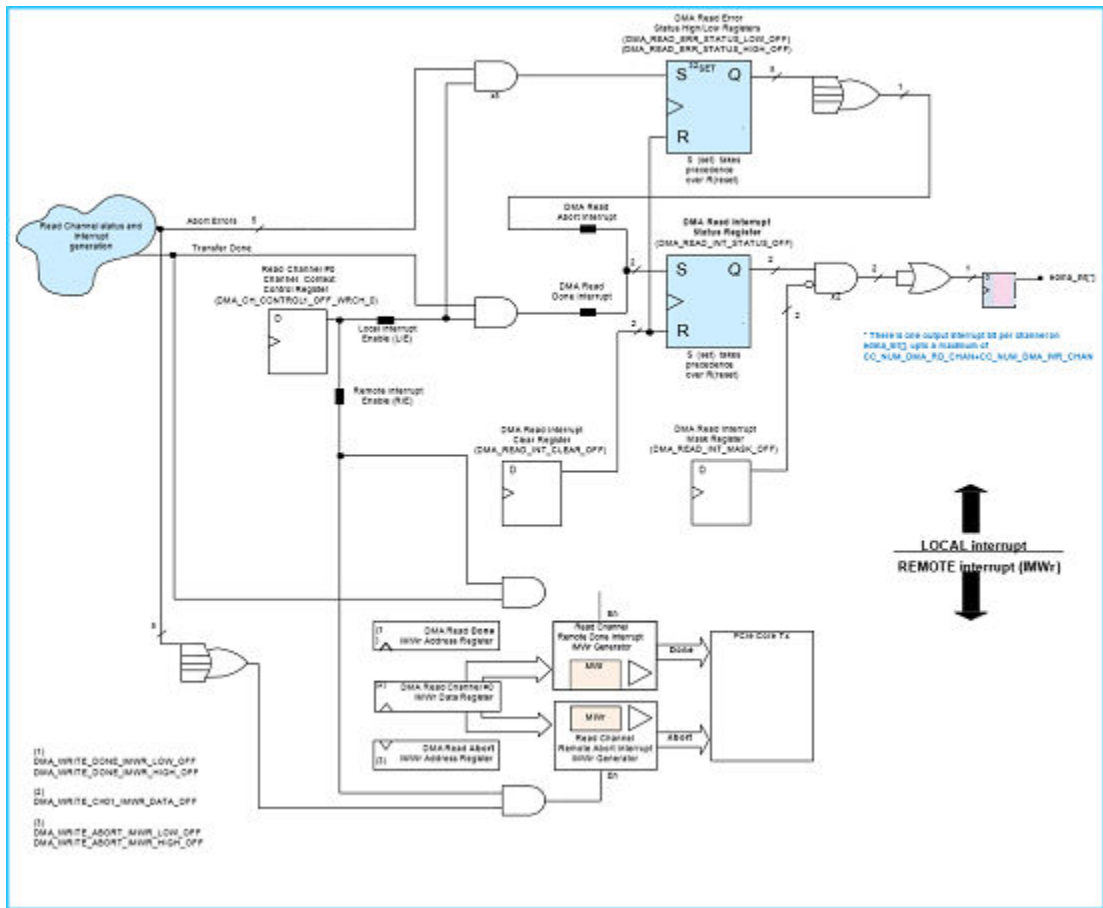


Figure 11-60. Read Interrupt Generation - Non Linked List Mode - Shown For Read Channel #0

11.3.2.2.2.2.3 *Linked List Mode*

The LIE and RIE bits in the LL element enable the channel done interrupts (local and remote). The LLLAIE and LLRAIE bits of the DMA_WRITE_LINKED_LIST_ERR_EN_OFF/DMA_READ_LINKED_LIST_ERR_EN_OFF registers enable the channel abort interrupts (local and remote). In the write channel, there are two error conditions that results in an abort interrupt. For more details, see "Linked List Mode". You mask and clear each of the two interrupts (done and abort) through the DMA interrupt registers as indicated in the figure below. You can read the status of each of the two abort errors (that contribute to the abort interrupt) through the DMA_WRITE_ERR_STATUS_OFF register.

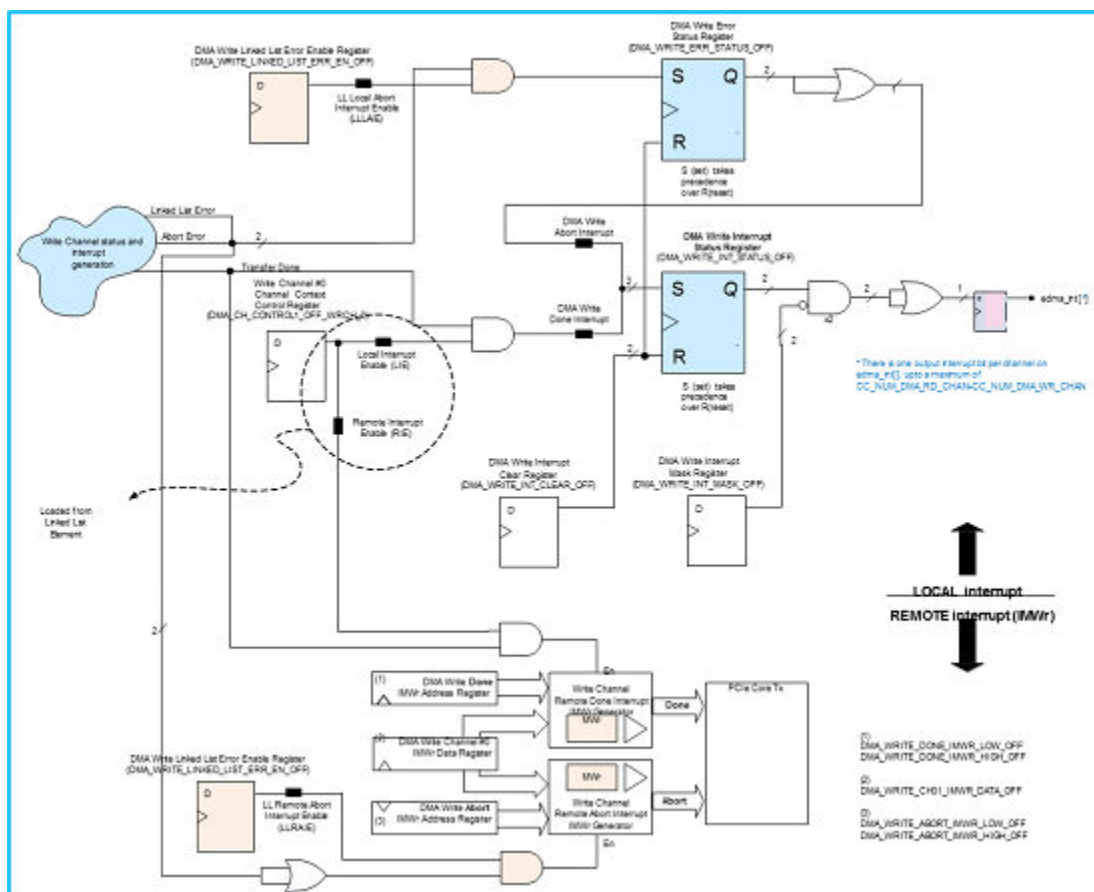


Figure 11-61. Write Interrupt Generation - Linked List Mode - Shown For Write Channel #0

In the read channel, there are six error conditions that results in an abort interrupt. For more details, see "Linked List Mode". You mask and clear each of the two interrupts (done and abort) through the DMA interrupt registers as indicated in the figure below. You can read the status of each of the six abort errors (that contribute to the abort interrupt) through the DMA_READ_ERR_STATUS_LOW_OFF and DMA_READ_ERR_STATUS_HIGH_OFF registers.

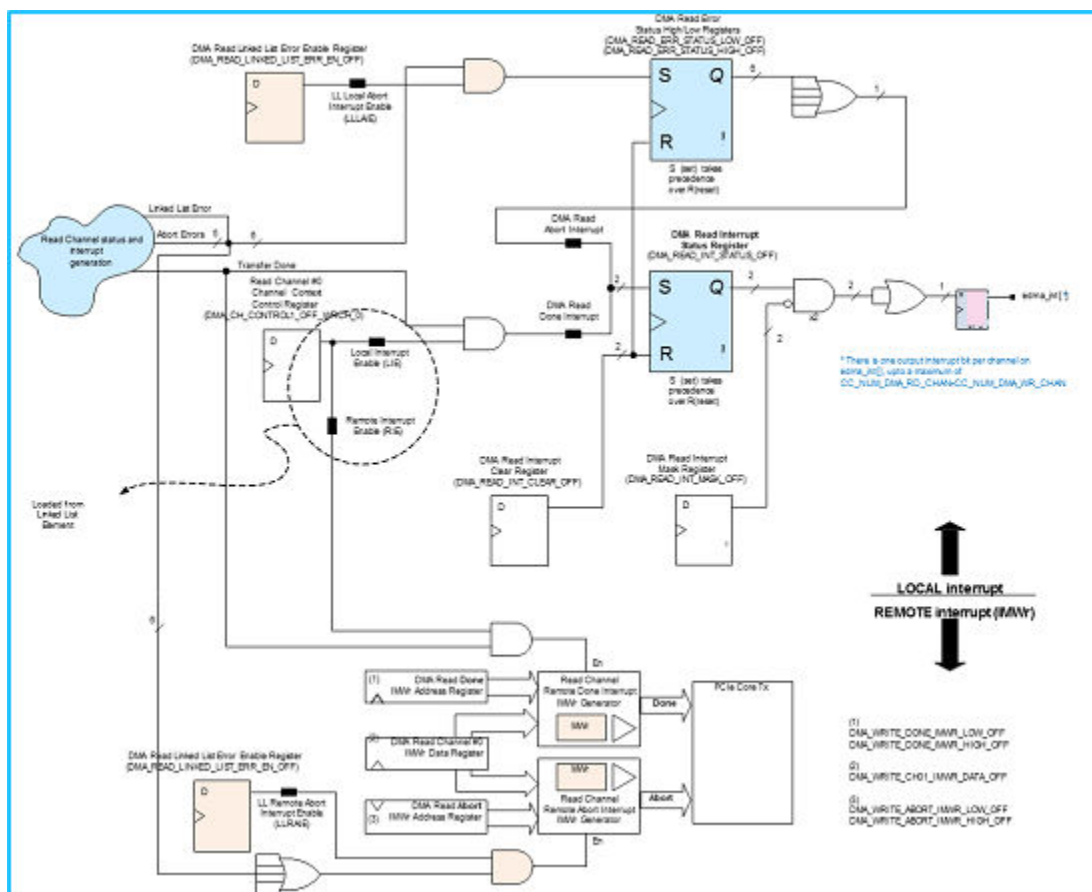


Figure 11-62. Read Interrupt Generation - Linked List Mode - Shown For Read Channel #0

NOTE

In non-linked list mode, LIE acts as a global switch. However when in linked list mode, LIE is just local to the current linked list element and the global switch is LLLAIE.

NOTE

If the DMA driver is running on the host and the interrupt service routine is reading local interrupts to determine if the transfer is successful, then you must set LIE and RIE in the same element and you should mask or ignore the local interrupt pin. Setting RIE and LIE in element A followed by RIE (only) in element B is not a verified usage scenario.

11.3.2.2.3 Linked List Mode

The DMA provides a linked list (LL) mode to efficiently move data from source to destination with minimal intervention from the local CPU. This mode provides an alternative to programming the DMA multiple times to transfer multiple blocks of data. The programming information (address, size, and so on) for each block of memory is pre-programmed by your software into a LL element (also known as a descriptor) in local memory. Each element (called a data element) in the LL structure (called a transfer list) can transfer up to 4 GB of data. You enable LL operation for a channel, by setting the LLE field of the DMA_CH_CONTROL1_OFF_WRCH_0 (Replace WRCH with RDCH for a read channel) register to 1. You can enable LL mode independently for each channel. When you enable LL for more than one channel, then you must have a separate LL structure in local memory for each channel. Your application must produce the LL element structure in local memory as shown in the figure below. Normally, all of the elements are contiguous (one after the other) in memory, and each element has six DWORDs containing the information about the block of data to be transferred. You program the channel context registers (DMA_LL_LOW_REG_WRCH_0 and DMA_LL_HIGH_OFF_WRCH_0) with the location of where you have placed the LL element structure in local memory.

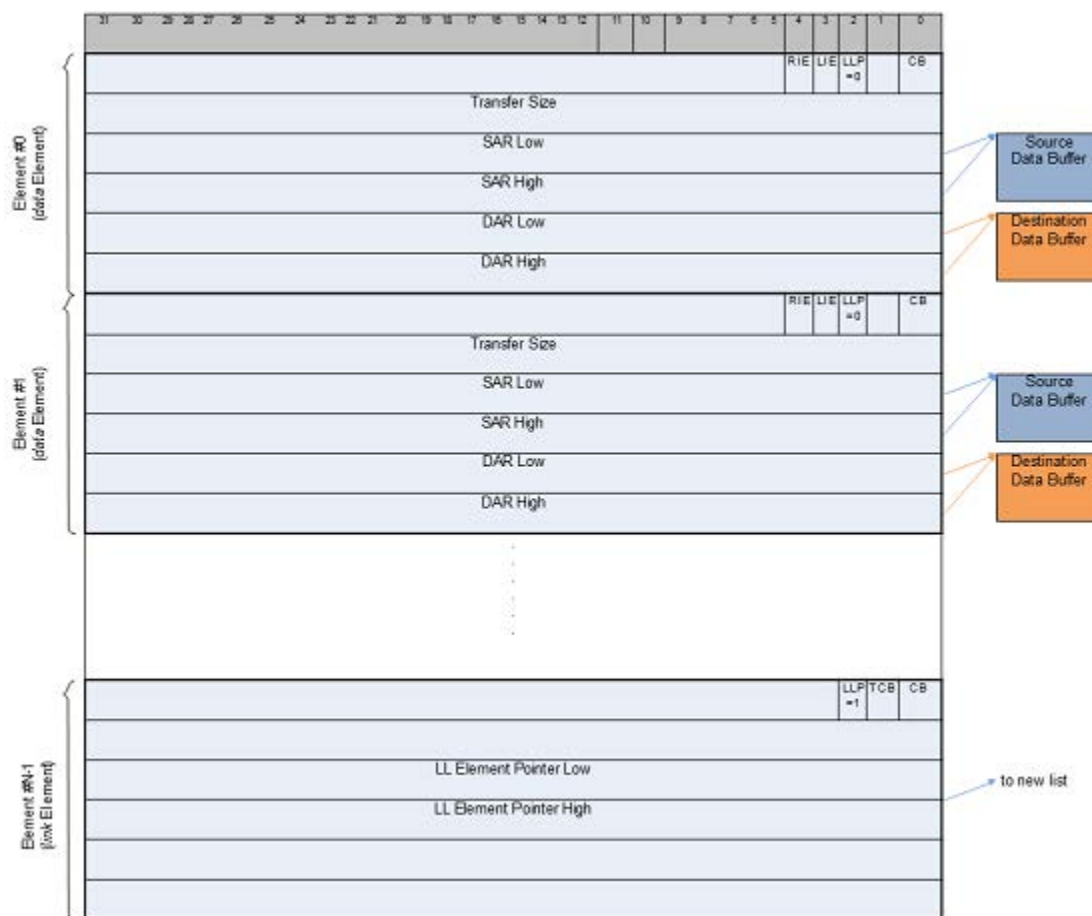


Figure 11-63. Linked List Element/Descriptor Structure in Local Memory with N Elements

When you start the DMA transfer (by writing to the DMA Write Doorbell Register `DMA_WRITE_DOORBELL_OFF` or DMA Read Doorbell Register `DMA_READ_DOORBELL_OFF`), the DMA reads (consumes) each element from local memory, and loads the information (SAR, DAR, size, and so on) from that element into the channel context registers in the DMA. These channel context registers determine the operation of the channel that the DMA controller is currently servicing. The DMA then proceeds to transfer the block of data (as defined by the element), and when it is finished, reads the next element from local memory. Normally, all of the elements are contiguous (one after the other) in memory, with the starting address defined in the channel context DMA Linked List Pointer Low Register `DMA_LL_P_LOW_OFF_WRCH_0`. When you want to jump in local memory to another element list (or recycle the consumed elements), then you set the LLP bit in the element (for example, link element #N-1 in figure shown)), specify the location of the next element structure using the LL Element Pointer DWORDs (as indicated in the figure), and, set TCB to 1 (for recycling) or to 0 (to jump to another list).

11.3.2.2.3.1 Relationship Between Element DWORDs and Channel Context Registers

Notice the similarity between a data element and the DMA Channel Context registers for each channel. Each element has six DWORDs as in figure 'Linked List Element/Descriptor Structure in Local Memory with N Elements'. There are eight channel context registers (DWORDs) for a channel. The DMA loads the six element DWORDs into the following channel context:

- CB, LLP, LIE, and RIE fields of the DMA Channel Control 1 register.
- DMA Transfer Size
- DMA SAR Low and DMA SAR High
- DMA DAR Low and DMA DAR High

The definitions of the element DWORD bit fields are the same as the DMA Channel Context registers described in the "DMA Software Register Map", with the exception of the LIE and RIE bits. The LIE and RIE bits in a LL element, only enable the done interrupt. In non-LL mode, the RIE and LIE bits (in the channel context registers) enable the done and abort interrupts.

When the LLP field of the first DWORD in the element is set to 1, the element is a link element. The DMA only loads the following information into the channel context registers:

- CB, TCB, and LLP bits of the first DWORD. The LIE and RIE bits are not defined in a link element.
- LL Element Pointer (3rd and 4th DWORDs) into the DMA Linked List Pointer registers (DMA_LL- P_LOW_REG_WRCH_0 and DMA_LL- P_HIGH_REG_WRCH_0).

Linked List Operation

- When you enable linked list operation (DMA_CH_CONTROL1_OFF_WRCH_0.LL = 1b1), then the DMA overwrites the following DMA channel context registers with the following information from linked list data elements:
 - The CB, LLP, LIE, and RIE fields of the DMA Channel Control 1 register.
 - DMA Transfer Size
 - DMA SAR Low & High
 - DMA DAR Low & High
- The structure of a link element is different to that of the data element. A data element has no TCB field. A link element has no LIE or RIE fields. It has no SAR, DAR, or Transfer Size DWORDs, but has LL Element Pointer DWORDs instead of the SAR DWORDs.
- The LIE and RIE bits in a LL element, only enable the done interrupt. In non-LL mode, the RIE and LIE bits enable the done and abort interrupts. For more information, see "Interrupts and Error Handling".

11.3.2.2.3 Using the DMA

This section describes how to use the DMA. It can simultaneously perform the following types of memory transactions:

- DMA write : Transfer (copy) of a block of data from local memory to remote memory
- DMA read : Transfer (copy) of a block of data from remote memory to local memory

Therefore the DMA supports full duplex operation, processing read and write transfers at the same time, and in parallel with normal (non-DMA) traffic. Upon completion of a DMA transfer or an error, the DMA sends an interrupt MWr (IMWr).

The channels are named read channel 7..0 and write channel 7..0. Each channel is programmed using a number of registers.

11.3.2.2.3.1 Source and Destination Address Registers (SAR, DAR)

The DMA channel context SAR and DAR registers (DMA_SAR_LOW_OFF_WRCH_01 / DMA_SAR_HIGH_OFF_WRCH_0 / DMA_DAR_LOW_OFF_WRCH_0 / DMA_DAR_HIGH_OFF_WRCH_0) provide support for remote-to-local, and local-to-remote PCIe address mapping. You program the start of the local and remote data buffers using these registers, and the DMA increments the SAR and DAR as the DMA transfer progresses. For a write transfer, the SAR is the address of the local memory, and the DAR is the address of the remote memory, as shown in Figure 'Write Transfer: SAR and DAR for Write Channel 0'.

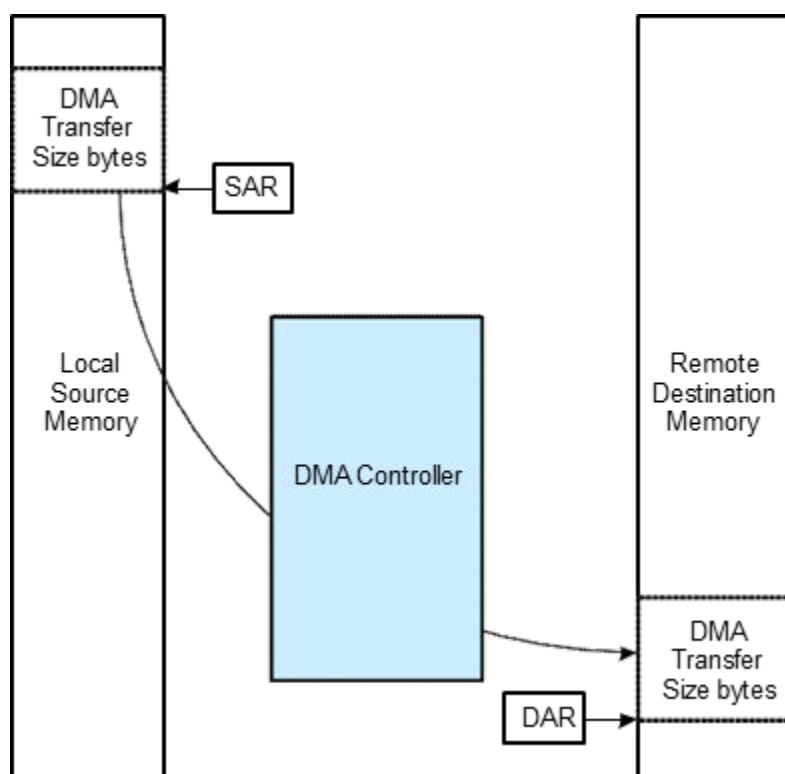


Figure 11-64. Write Transfer: SAR and DAR for Write Channel 0

For a read transfer, the SAR is the address of the remote memory, and the DAR is the address of the local memory, as shown in the figure below.

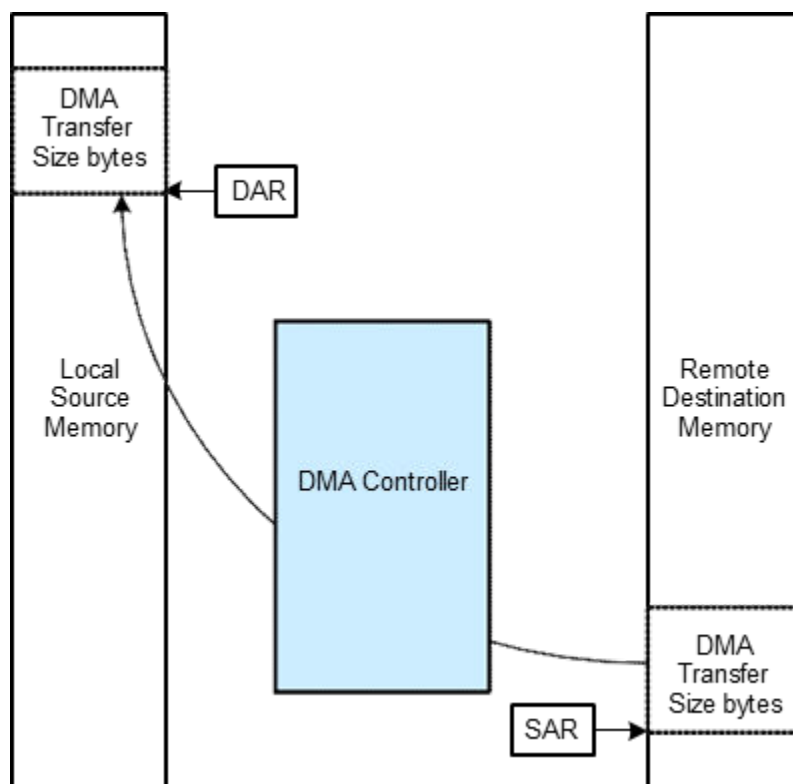


Figure 11-65. Read Transfer: SAR and DAR for Read Channel 0

The presence of the DMA controller does not affect:

- Normal filtering rules for inbound TLPs (for example, BAR checking in an Endpoint, as described in “Receive Filtering”)
- The operation of any internal or external address translation as described in “Internal Address Translation Unit (iATU)”.

NOTE

When you do not want the iATU to translate outbound requests that are generated by the internal DMA module, then you must implement one of the following approaches:

- Ensure that the combination of DMA channel programming and iATU control register programming, causes no translation of DMA traffic to be done in the iATU.
- Activate the DMA bypass mode to allow request TLPs which are initiated by the DMA controller to pass through the iATU untranslated. You can activate DMA bypass mode by setting the DMA Bypass field of the iATU Control 2 Register (IATU_REGION_CTRL_OFF_2_OUTBOUND_0).

11.3.2.2.3.2 DMA Transfer Size Register

You program the DMA transfer size using the DMA Transfer Size Register (DMA_TRANSFER_SIZE_OFF_WRCH_0). The DMA decrements the value in this register as the DMA transfer progresses. You can read this register at any time to determine the number of bytes remaining to be transferred. When all bytes are successfully transferred, the result is zero. The DMA transfer size can be of any value from one byte, up to a maximum of 4 GB.

11.3.2.2.3.3 Starting The DMA Transfer

After you program the DMA controller registers (including writing to the DMA Read Engine Enable or DMA Write Engine Enable register), you start a DMA transfer by writing the channel number to the Doorbell Number field of the DMA Write Doorbell Register (DMA_WRITE_DOORBELL_OFF) or DMA Read Doorbell Register (DMA_READ_DOORBELL_OFF). You can program and start both a read and a write transfer at the same time. The DMA supports full duplex operation, processing read and write transfers at the same time and in parallel with normal (non-DMA) traffic. You can program and start any number of channels in the DMA controller sequentially or simultaneously.

Sequentially: configure and start a channel, and then configure and start another channel

Simultaneously: configure multiple channels, and then start multiple channels

NOTE

You must not write to any of the context registers for a particular channel after you start the channel by writing the channel number to the Doorbell Number field of the DMA Write Doorbell Register (DMA_WRITE_DOORBELL_OFF) or DMA Read Doorbell Register (DMA_READ_DOORBELL_OFF).

11.3.2.2.3.4 Detecting the End of The DMA Transfer

If a DMA transfer proceeds without any errors, it stops automatically when finished.

11.3.2.2.3.4.1 Detecting End of Transfer without Errors

The normal end of a DMA transfer is detected by any of the following methods:

- Done local interrupt (pin) asserted.
- Done remote interrupt (IMWr) received.

- Channel status field of the Channel Control 1 register is Stopped, and the DMA Transfer Size register is 0x0.
- Polling of the DMA Write Interrupt Status Register (DMA_WRITE_INT_STATUS_OFF) or DMA Read Interrupt Status Register (DMA_READ_INT_STATUS_OFF).

11.3.2.2.3.4.2 Detecting End of Transfer with Errors

The abnormal end of a DMA transfer is detected by any of the following methods:

- Abort local interrupt (pin) asserted.
- Abort remote interrupt (IMWr) received.
- Polling of the DMA Write Interrupt Status Register (DMA_WRITE_INT_STATUS_OFF) or DMA Read Interrupt Status Register (DMA_READ_INT_STATUS_OFF).
- Channel Status field of the Channel Control 1 register is halted. When the DMA controller detects an error, it forces the DMA to stop issuing requests for the channel. It also sets the channel status field of the Channel Control 1 register to halted, generates an abort interrupt (if enabled), and sends an abort IMWr (if enabled). The DMA Transfer Size register indicates the remaining number bytes to be transferred, except when there is an AXI write error during a DMA read transfer. For more details, see table 'Possible Sources of an Error during a DMA Read Transfer'.
- Channel Status field of the Channel Control 1 register is Stopped, and the DMA Transfer Size register is not 0x0. You have prematurely stopped this channel as described in “Stopping the DMA Transfer (Software Stop)”.

11.3.2.2.3.5 Stopping the DMA Transfer (Software Stop)

You can manually abort (stop) the DMA transfer by writing the channel number to the Doorbell Number field and writing 1 to the Stop field in DMA_WRITE_DOORBELL_OFF. This causes the DMA to:

- Place the channel in a Stopped state.
- The channel Status field in DMA_CH_CONTROL1_OFF_WRCH_0 is Stopped and the value in DMA_TRANSFER_SIZE_OFF_WRCH_0 will not be 0x0.
- Wait for all outstanding pending transactions.
- Assert the abort interrupt (if it is enabled) in DMA_WRITE_INT_STATUS_OFF.

You might do this as part of error handling which is described in “Error Handling Assistance by Remote Software”, which is only necessary during software development if you incorrectly program the DMA write channel DAR. You might also do this as part of a function level reset (FLR). FLR does not directly affect the DMA transfer so you must manually stop the DMA transfer before initiating an FLR.

Use This Feature With Caution in Linked List Mode

You should not use the STOP feature in operational mode. It is only a debug feature and has associated hazards.

- Before setting the Stop bit, you must read the channel status field (CS) of the DMA Channel Control 1 register to ensure that the corresponding channel is Running (transferring data). To eliminate the possibility of a race condition between these two actions (read and write), you should confirm the presence of the abort bit in the DMA Write Interrupt Status Register (DMA_WRITE_INT_STATUS_OFF) or DMA Read Interrupt Status Register (DMA_READ_INT_STATUS_OFF) and check the status of any fatal error if some other event has occurred.
- After a Stop event, you cannot seamlessly resume the transfer again because the DMA will not carry on exactly from the point that it was stopped at. Therefore you must setup and start the complete channel context again, including the
 - linked list pointer
 - linked list structure in memory
 - related PCS and CCS values

11.3.2.2.3.6 TLP Generator Header Control

You must program bits [31:12] of the DMA Channel Control 1 Register (DMA_CH_CONTROL1_REG_WRCH_0/DMA_CH_CONTROL1_OFF_RDCH_0), so that the DMA TLP generator correctly sets the TC, RO, NS, AT, and FN TLP header fields for any TLPs that it generates. When SRIOV is enabled, then you must also program DMA_CH_CONTROL2_OFF_WRCH_0/DMA_CH_CONTROL2_OFF_RDCH_0.

11.3.2.2.3.7 DMA Software Register Map

This section discusses the grouping and access methods of the DMA registers. The following registers are present when the DMA is enabled.

The DMA Global Registers control the global settings

Table 11-31. DMA Global Registers

Name	Description
0x8_0000 +0x000	DMA Arbitration Scheme for TRGT1 Interface
+0x008	Number of DMA Channels
+0x00C	DMA Write Engine Enable
+0x010	DMA Write Doorbell
+0x018	DMA Write Engine Channel Arbitration Weight Low
+0x01C	DMA Write Engine Channel Arbitration Weight High
+0x02C	DMA Read Engine Enable
+0x030	DMA Read Doorbell
+0x034	DMA Global Read Control
+0x038	DMA Read Engine Channel Arbitration Weight Low
+0x03C	DMA Read Engine Channel Arbitration Weight High

The DMA Interrupt Registers control the generation of local interrupts and remote IMWr. The IMWr addresses and data are stored in RAM.

Table 11-32. DMA Interrupt Registers

Byte Offset	Description
0x8_0000 +0x4C	DMA Write Interrupt Status
+0x50	Not used (RsvdP).
+0x54	DMA Write Interrupt Mask
+0x58	DMA Write Interrupt Clear
+0x5C	DMA Write Error Status
+0x60	DMA Write Done IMWr Address Low
+0x64	DMA Write Done IMWr Address High
+0x68	DMA Write Abort IMWr Address Low
+0x6C	DMA Write Abort IMWr Address High
+0x70	DMA Write Channels 1 and 0 IMWr Data
+0x84	DMA Write Channels 3 and 2 IMWr Data
+0x88	DMA Write Channels 5 and 4 IMWr Data
+0x8C	DMA Write Channels 7 and 6 IMWr Data
+0x90	DMA Write Linked List Error Enable
up to +0x9C	Not used (RsvdP).
+0xA0	DMA Read Interrupt Status
+0xA4	Not used (RsvdP).
+0xA8	DMA Read Interrupt Mask
+0xAC	DMA Read Interrupt Clear
+0xB0	Not used (RsvdP).
+0xB4	DMA Read Error Status Low
+0xB8	DMA Read Error Status High

Table continues on the next page...

Table 11-32. DMA Interrupt Registers (continued)

up to +0xC0	Not used (RsvdP).
+0xC4	DMA Read Linked List Error Enable
+0xC8	Not used (RsvdP).
+0xCC	DMA Read Done IMWr Address Low
+0xD0	DMA Read Done IMWr Address High
+0xD4	DMA Read Abort IMWr Address Low
+0xD8	DMA Read Abort IMWr Address High
+0xDC	DMA Read Channels 1 and 0 IMWr Data
+0xE0	DMA Read Channels 3 and 2 IMWr Data
+0xE4	DMA Read Channels 5 and 4 IMWr Data
+0xE8	DMA Read Channels 7 and 6 IMWr Data

DMA Channel Context Registers: these registers (which are stored in RAM) are for the control and status information that is specific to each channel. There is a set of registers for each channel.

Table 11-33. DMA Channel Context Registers for Each Channel

Byte Offset	Description
DMA Channel Context	
$0x8_0000 + 0x200*i + 0x100*(WR RD)b$	DMA Channel Control 1 Register
$+0x004*i + 0x200*i + 0x100*(WR RD)$	DMA Channel Control 2 Register
$+0x008*i + 0x200*i + 0x100*(WR RD)$	DMA Transfer Size
$+0x00C*i + 0x200*i + 0x100*(WR RD)$	DMA SAR Low
$+0x010*i + 0x200*i + 0x100*(WR RD)$	DMA SAR High
$+0x014*i + 0x200*i + 0x100*(WR RD)$	DMA DAR Low
$+0x018*i + 0x200*i + 0x100*(WR RD)$	DMA DAR High
$+0x01C*i + 0x200*i + 0x100*(WR RD)$	DMA Linked List Pointer Low
$+0x020*i + 0x200*i + 0x100*(WR RD)$	DMA Linked List Pointer High

a. i is the number of channels

b. WR=0 | RD=1

11.3.2.3.7.1 Implementation of Channel Context Registers

- The channel context registers are stored in RAM whose contents are automatically initialized after power-on.
 - Therefore, you must program every “register” value as the default is undefined

- You must program all fields marked Reserved to 1'b0
- These registers are not affected by any of the reset signals
- You must not write to any of the context registers for a particular channel after you start the channel by writing the channel number to the Doorbell Number field of the DMA Write Doorbell Register (DMA_WRITE_DOORBELL_OFF) or DMA Read Doorbell Register (DMA_READ_DOORBELL_OFF).

Linked List Operation

When you enable linked list operation (Linked List Enable field of DMA Channel Control 1 Register (DMA_CH_CONTROL1_OFF_WRCH_0) to 1b1), the DMA overwrites the following DMA channel context registers with the following information from linked list data elements:

- The CB, LLP, LIE, and RIE fields of the DMA Channel Control 1 register.
- DMA Transfer Size
- DMA SAR Low
- DMA SAR High
- DMA DAR Low
- DMA DAR High

NOTE

- The structure of a link element is different to that of the data element. A data element has no TCB field. A link element has no LIE or RIE fields. It has no SAR, DAR, or Transfer Size DWORDs, but has LL Element Pointer DWORDs instead of the SAR DWORDs. For more details, see Relationship Between Element DWORDs and Channel Context Registers.
- The LIE and RIE bits in a LL element, only enable the Done interrupt. In non-LL mode, the RIE and LIE bits enable the Done and Abort interrupts. For more information, see Interrupts and Error Handling.

11.3.2.2.3.8 Programming Examples

This section provides a programming and operation flow for each of the following scenarios:

- 1 MB Write Transfer On Write Channel #0 Initiated by Local CPU (Non LL mode)
- 1 MB Read Transfer On Read Channel #0 Initiated by Local CPU (Non LL mode)
- 1 MB Write Transfer On Write Channel #5 Initiated by Remote CPU (Non LL mode)

- 1 MB Read Transfer On Read Channel #5 Initiated by Remote CPU (Non LL mode)
- Linked List Mode Programming Example (Write Channel #0)

NOTE

The order in which the registers in the following examples are programmed is not important, except for the Doorbell register, which initiates the DMA transfer. Therefore, to start both a "1 MB Write Transfer On Write Channel #0 Initiated by Local CPU (Non LL mode)" and a "1 MB Read Transfer On Read Channel #0 Initiated by Local CPU (Non LL mode)" by executing the commands on the table. If, as mentioned in "Starting The DMA Transfer", you want both channels to start at (approximately) the same time, then do not execute the last command (at 0x280) in the table 'DMA Register Setup' until after you have executed all the commands in table.

NOTE

You must enable the DMA before you start a DMA channel. Therefore you must write '1' to DMA Read Engine Enable or DMA Write Engine Enable register before you write '1' to the DMA Read Doorbell or DMA Write Doorbell register.

11.3.2.2.3.8.1 1 MB Write Transfer On Write Channel #0 Initiated by Local CPU (Non LL mode)

In this example, the IMWr generation is disabled, as the local CPU initiates the DMA transfer. The local CPU is interrupted using the edma_int bus. The SAR is the address of the local memory, and the DAR is the address of the remote memory, as shown in the figure below. The table below provides the programming details for this example transfer.

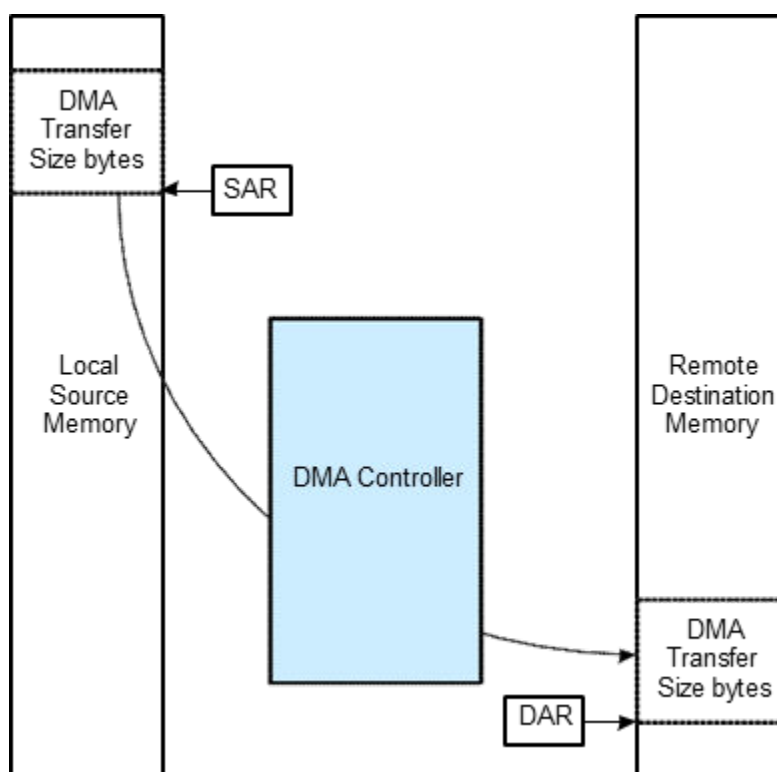


Figure 11-66. 1 MB Write DMA Transfer from 0xBEEF_BEE0 to 0xCAFE_CAF0

Table 11-34. 1 MB Write DMA Transfer from 0xBEEF_BEE0 to 0xCAFE_CAF0

Address (0x8_0000+)	Name	Value
0x00C	DMA Write Engine Enable You must not write 0 to this register. Even temporarily writing 0 to this register resets the DMA logic. For more details, see DMA Write Engine Enable Register (DMA_WRITE_ENGINE_EN_OFF).	0x1
0x054	DMA Write Interrupt Mask	0x0
0x200	DMA Channel Control 1 register - Local Interrupt Enable (LIE) =1 - Remote Interrupt Enable (RIE) =0 - AT, - RO, NS, TC, Function Number =0	0x04000008
0x208	DMA Transfer Size	0x00100000
0x20C	DMA SAR Low	0xBEEF_BEE0
0x210	DMA SAR High	0x0000_0000
0x214	DMA DAR Low	0xCAFE_CAF0
0x218	DMA DAR High	0x0000_0000
0x010	DMA Write Doorbell	0x0

11.3.2.2.3.8.2 1 MB Read Transfer On Read Channel #0 Initiated by Local CPU (Non LL mode)

In this example, the IMW_r generation is disabled, as the local CPU initiates the DMA transfer. The local CPU is interrupted using the edma_int bus. The SAR is the address of the remote memory, and the DAR is the address of the local memory, as shown in the figure below. The table below provides the programming details for this example transfer.

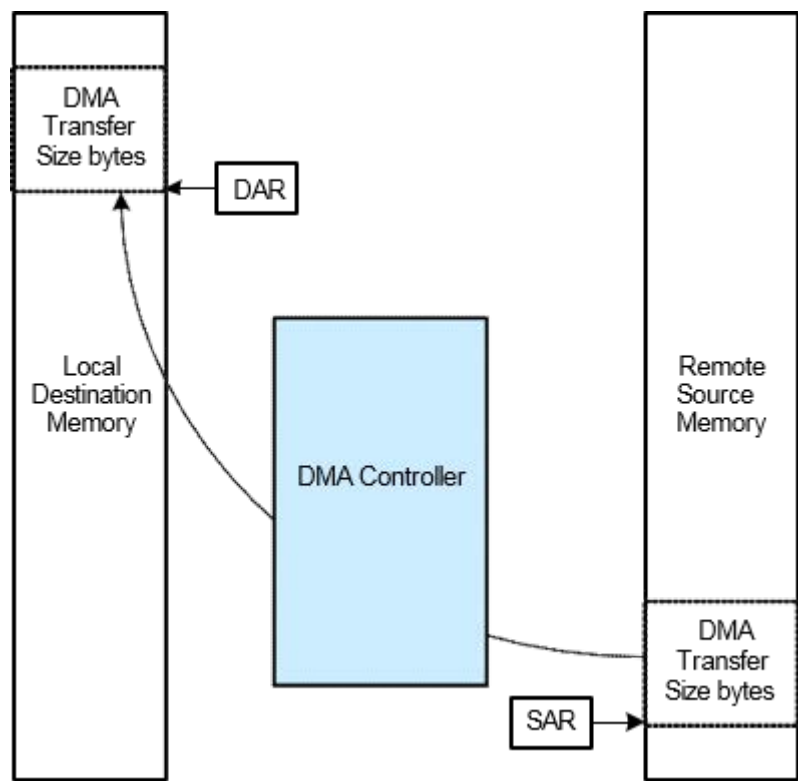


Figure 11-67. 1 MB Read DMA Transfer from 0xBEEF_BEE0 to 0xCAFE_CAF0

Table 11-35. DMA Register Setup

Address (0x8_0000+)	Name	Value
0x02C	DMA Read Engine Enable You must not write 0 to this register. Even temporarily writing 0 to this register resets the DMA logic. For more details, see DMA Read Engine Enable Register (DMA_READ_ENGINE_EN_OFF).	0x1
0x0A8	DMA Read Interrupt Mask	0x0
0x300	DMA Channel Control 1 register Local Interrupt Enable (LIE) =1	0x04000008

Table continues on the next page...

Table 11-35. DMA Register Setup (continued)

	• Remote Interrupt Enable (RIE) =0 • AT, • RO, NS, TC, Function Number =0	
0x308	DMA Transfer Size	0x00100000
0x30C	DMA SAR Low	0xBEEF_BEE0
0x310	DMA SAR High	0x0000_0000
0x314	DMA DAR Low	0xCAFE_CAF0
0x318	DMA DAR High	0x0000_0000
0x030	DMA Read Doorbell	0x0

11.3.2.2.3.8.3 1 MB Write Transfer On Write Channel #5 Initiated by Remote CPU (Non LL mode)

In this example, the local interrupt generation is disabled, as the remote CPU initiates the DMA transfer. The remote CPU is interrupted using an IMWr. The SAR is the address of the local memory, and the DAR is the address of the remote memory, as shown in the figure below. The table below provides the programming details for this example transfer.

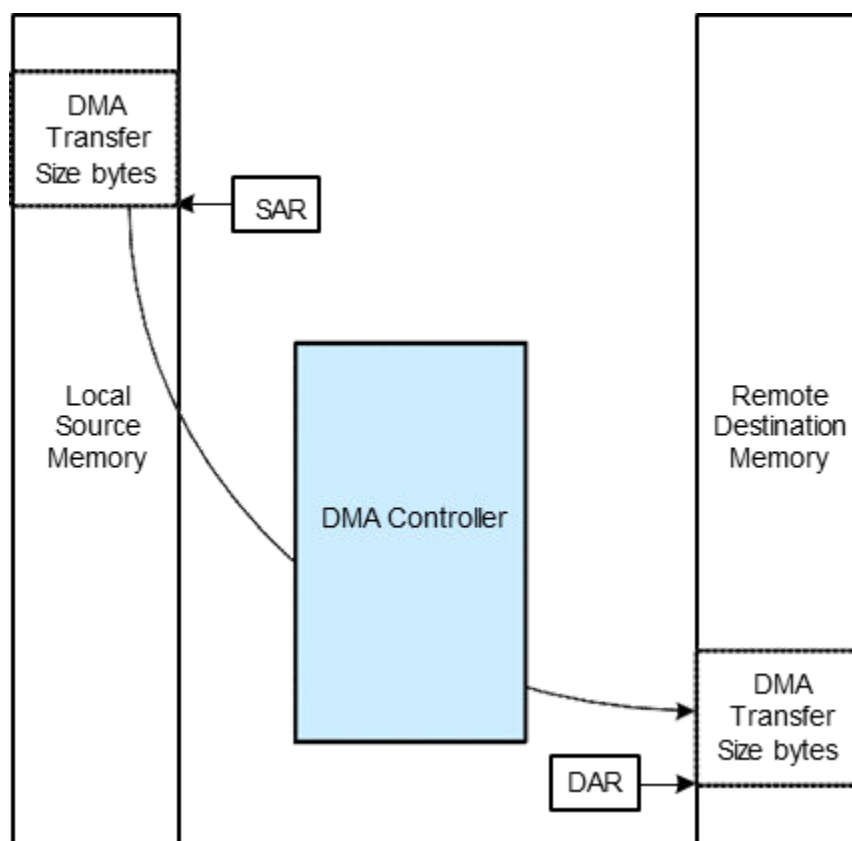
**Figure 11-68. 1 MB Write DMA Transfer from 0xBEEF_BEE0 to 0xCAFE_CAF0**

Table 11-36. DMA Register Setup

Address (0x8_0000+)	Name	Value
0x00C	DMA Write Engine Enable You must not write 0 to this register. Even temporarily writing 0 to this register resets the DMA logic. For more details, see DMA Write Engine Enable Register (DMA_WRITE_ENGINE_EN_OFF).	0x1
0x060 / 0x064	DMA Write Done IMWr Address Low and High	your IMWr Address #1
0x068 / 0x06C	DMA Write Abort IMWr Address Low and High	your IMWr Address #2
0x06C	DMA Write Channel 0 IMWr Data	your IMWr Data
0xA00	DMA Channel Control 1 register - Local Interrupt Enable (LIE) =0 - Remote Interrupt Enable (RIE) =1 - AT, - RO, NS, TC, Function Number =0	0x04000010
0xA08	DMA Transfer Size	0x00100000
0xA0C	DMA SAR Low	0xBEEF_BEE0
0xA10	DMA SAR High	0x0000_0000
0xA14	DMA DAR Low	0xCAFE_CAF0
0xA18	DMA DAR High	0x0000_0000

11.3.2.2.3.8.4 1 MB Read Transfer On Read Channel #5 Initiated by Remote CPU (Non LL mode)

In this example, the local interrupt generation is disabled, as the remote CPU initiates the DMA transfer. The remote CPU is interrupted using an IMWr. The SAR is the address of the remote memory, and the DAR is the address of the local memory, as shown in Figure below. The table below provides the programming details for this example transfer.

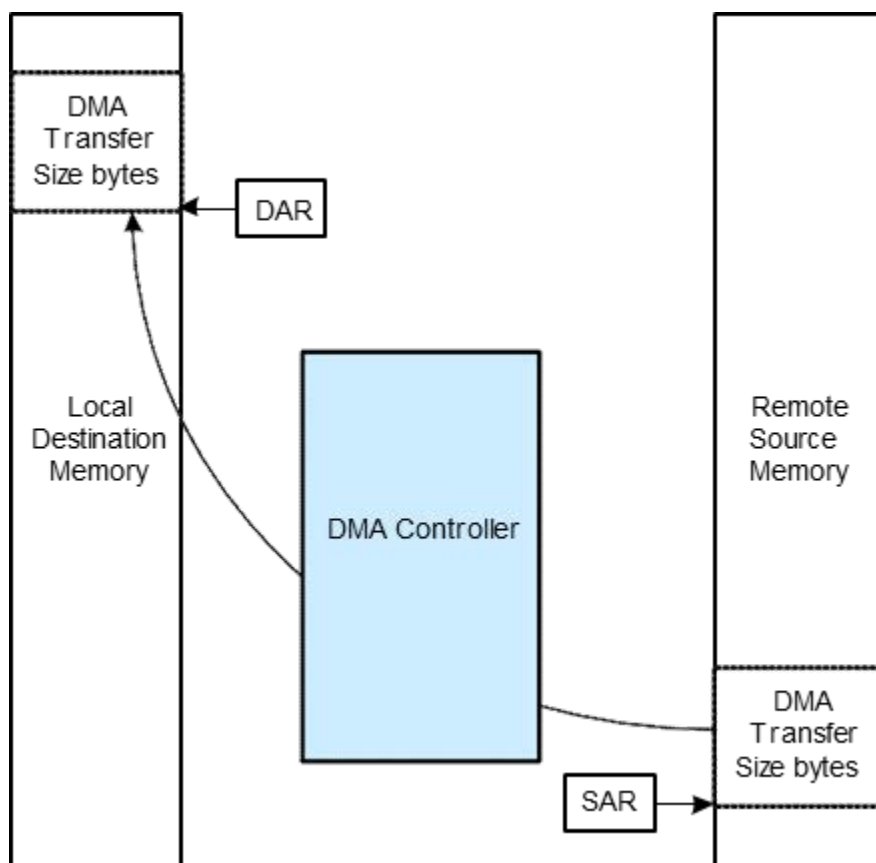


Figure 11-69. 1 MB Read DMA Transfer from 0xBEEF_BEE0 to 0xCAFE_CAF0

Table 11-37. DMA Register Setup

Address (0x700+)	Name	Value
0x29C	DMA Read Engine Enable You must not write “0” to this register. Even temporarily writing 0 to this register resets the DMA logic. For more details, see “DMA Read Engine Enable Register” (DMA_READ_ENGINE_EN).	0x1
0x33C / 0x340	DMA Read Done IMWr Address Low and high	your IMWr Address #1
0x344 / 0x348	DMA Read Abort IMWr Address Low and High	your IMWr Address #2
0x34C	DMA Read Channel 0 IMWr Data	your IMWr Data
0x36C	DMA Channel Context Index	0x8000_0005
0x370	DMA Channel Control 1 register - Local Interrupt Enable (LIE) =0 - Remote Interrupt Enable (RIE) =1 - AT, RO, NS, TC, Function Number =0 - TD =1	0x04000010
0x378	DMA Transfer Size	0x00100000
0x37C	DMA SAR Low	0xBEEF_BEE0
0x380	DMA SAR High	0x0000_0000
0x384	DMA DAR Low	0xCAFE_CAF0
0x388	DMA DAR High	0x0000_0000

Table continues on the next page...

Table 11-37. DMA Register Setup (continued)

0x2A0	DMA Read Doorbell	0x5
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11.3.2.2.3.8.5 *Linked List Mode Programming Example (Write Channel #0)*

This write channel example corresponds to figure 'Example Producer (Software) - Consumer (DMA) Synchronization Flow', with the following features:

- Recycled linked list (LL) of nine data elements and one link element.
- Transfer is initiated by local CPU.
- Each data element corresponds to a 1 MB (0x0010_0000) write transfer.
- The SARs for each element are separated by 2 MB (0x0020_0000). The DAR for each element are separated by 16 MB (0x0100_0000). The SAR is the address of the local memory, and the DAR is the address of the remote memory, as shown in the figure below.
- The LL transfer list is located at address 0x0000_0200 in local memory.
- No remote interrupt is generated so the IMWr generation is disabled, as the local CPU initiates the DMA transfer. The local CPU is interrupted using the edma_int bus.
- There is a Watermark interrupt at element #5, and an Empty interrupt at element #8. Both of these interrupts are done interrupts (as opposed to abort interrupts).

You should familiarize yourself with the Linked List Flow for Producer and Consumer as shown in Figure 'Linked List Flow for Producer and Consumer'. These are the main steps for this process in this example:

Step 1: Create TL in local memory:

- Create the 10 elements as per Table 'Linked List Element Initial Setup'.
 - For elements 0, 1, 2, 3, 4, 6, 7: RIE, LIE, LLP, CB =0,0,0,1
 - For element 5: RIE, LIE, LLP, CB =0,1,0,1 □ For element 8: RIE, LIE, LLP, CB =0,1,0,1 □ For element 9: LLP, TCB, CB =1,1,0
- For more details, see “PCS-CCS-CB-TCB Producer-Consumer Synchronization” and “Linked List Mode”

Step 2: Program and Start DMA:

- Configure and start the DMA registers as per Table 'DMA Register Setup'.

Step 4: Wait for done interrupt and recycle TL

- For more details, see “Using Interrupts for Linked List Producer-Consumer Synchronization”
- When recycling elements, consider “Element Recycling”

NOTE

If the DMA driver is running on the host and the interrupt service routine is reading local interrupts to determine if the transfer is successful, then you must set LIE and RIE in the same element and you should mask or ignore the local interrupt pin. Setting RIE and LIE in element A followed by RIE (only) in element B is not a verified usage scenario.

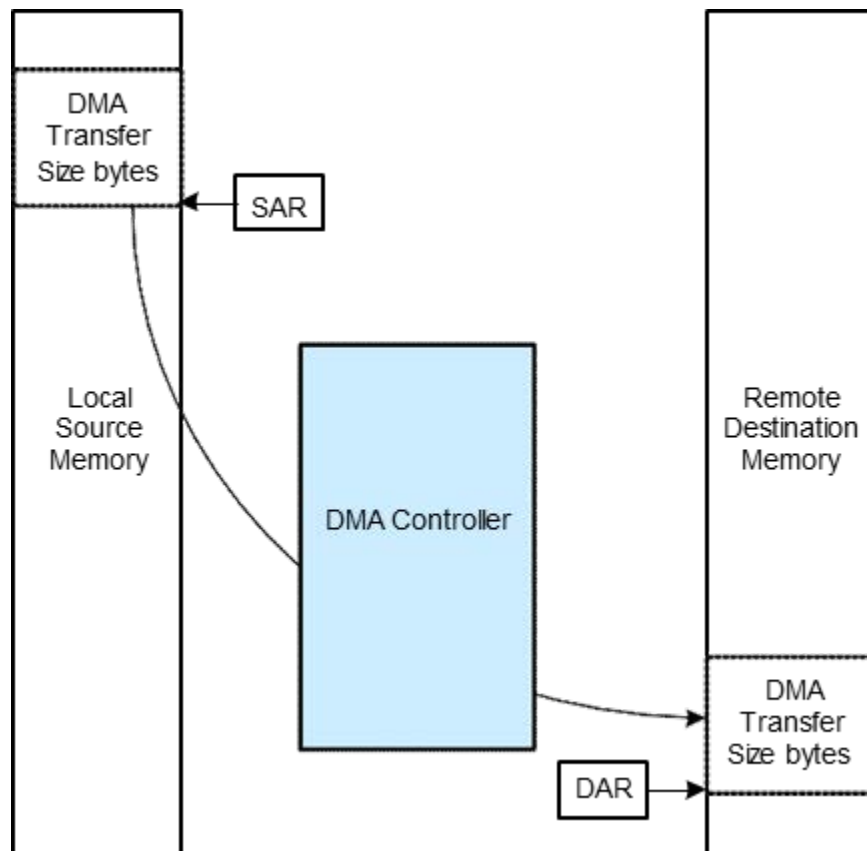


Figure 11-70. 1 MB Write DMA Transfer from SAR to DAR

Table 11-38. Linked List Element Initial Setup

Element #	Address	Data	Label
0	0x0000_0200	0x00000001	Channel Control
0	0x0000_0204	0x00100000	Transfer Size
0	0x0000_0208	0xBEEF_BEE0	SAR Low
0	0x0000_020C	0x0	SAR High
0	0x0000_0210	0xCAFE_CAF0	DAR Low
0	0x0000_0214	0x0	DAR High
1	0x0000_0218	0x00000001	Channel Control
1	0x0000_021C	0x00100000	Transfer Size
1	0x0000_0220	0xBF0F_BEE0	SAR Low

Table continues on the next page...

Table 11-38. Linked List Element Initial Setup (continued)

1	0x0000_0224	0x0	SAR High
1	0x0000_0228	0xCBFE_CAF0	DAR Low
1	0x0000_022C	0x0	DAR High
2	0x0000_0230	0x00000001	Channel Control
2	0x0000_0234	0x00100000	Transfer Size
2	0x0000_0238	0xBF2F_BEE0	SAR Low
2	0x0000_023C	0x0	SAR High
2	0x0000_0240	0xCCFE_CAF0	DAR Low
2	0x0000_0244	0x0	DAR High
3	0x0000_0248	0x00000001	Channel Control
3	0x0000_024C	0x00100000	Transfer Size
3	0x0000_0250	0xBF4F_BEE0	SAR Low
3	0x0000_0254	0x0	SAR High
3	0x0000_0258	0xCDFE_CAF0	DAR Low
3	0x0000_025C	0x0	DAR High
4	0x0000_0260	0x00000001	Channel Control
4	0x0000_0264	0x00100000	Transfer Size
4	0x0000_0268	0xBF6F_BEE0	SAR Low
4	0x0000_026C	0x0	SAR High
4	0x0000_0270	0xCEFE_CAF0	DAR Low
4	0x0000_0274	0x0	DAR High
5	0x0000_0278	0x00000009	Channel Control
5	0x0000_027C	0x00100000	Transfer Size
5	0x0000_0280	0xBF8F_BEE0	SAR Low
5	0x0000_0284	0x0	SAR High
5	0x0000_0288	0xCFFE_CAF0	DAR Low
5	0x0000_028C	0x0	DAR High
6	0x0000_0290	0x00000001	Channel Control
6	0x0000_0294	0x00100000	Transfer Size
6	0x0000_0298	0xBF9F_BEE0	SAR Low
6	0x0000_029C	0x0	SAR High
6	0x0000_0290	0xD0FE_CAF0	DAR Low
6	0x0000_0294	0x0	DAR High
7	0x0000_0298	0x00000001	Channel Control
7	0x0000_029C	0x00100000	Transfer Size
7	0x0000_02A0	0xBF9F_BEE0	SAR Low
7	0x0000_02A4	0x0	SAR High
7	0x0000_02A8	0xD1FE_CAF0	DAR Low
7	0x0000_02AC	0x0	DAR High
8	0x0000_02B0	0x00000009	Channel Control

Table continues on the next page...

Table 11-38. Linked List Element Initial Setup (continued)

8	0x0000_02B4	0x00100000	Transfer Size
8	0x0000_02B8	0xBFEB_BEE0	SAR Low
8	0x0000_02BC	0x0	SAR High
8	0x0000_02C0	0xD2FE_CAF0	DAR Low
8	0x0000_02C4	0x0	DAR High
9	0x0000_02C8	0x00000006	Channel Control
9	0x0000_02CC	0x0	Reserved
9	0x0000_02D0	0x0000_0200	Linked List Element Pointer Low
9	0x0000_02D4	0x0	Linked List Element Pointer High

Table 11-39. DMA Register Setup

Address (0x8_0000+)	Name	Value
0x030	DMA Write Engine Enable You must not write 0 to this register. Even temporarily writing 0 to this register resets the DMA logic. For more details, see DMA Write Engine Enable Register (DMA_WRITE_ENGINE_EN_OFF).	0x1
0x054	DMA Write Interrupt Mask	0x0
0x090	DMA Write Linked List Error Enable • LLLAIE =1	0x1
0x200	DMA Channel Control 1 register • Linked List Enable (LLE) =1 • Consumer Cycle Status (CCS) =PCS =1 • AT, • RO, NS, TC, Function Number =0	0x04000300
0x21C	DMA Linked List Pointer Low	0x0000_0200
0x220	DMA Linked List Pointer High	0x0000_0000
0x010	DMA Write Doorbell	0x0

11.3.2.2.4 Advanced DMA Information and Operation

This section discusses advanced features and operation of the DMA.

11.3.2.2.4.1 Overview

The DMA write and read channels operate independently to maximize the performance of the DMA read and write data transfers over the PCIe link.

The DMA can simultaneously perform the following types of memory transactions:

- DMA write: transfer (copy) of a block of data from local memory to remote memory
- DMA read: transfer (copy) of a block of data from remote memory to local memory

After you have programmed and started a DMA transfer (see "Using the DMA"), the DMA transfers the data as described in "**DMA Write Transfer**" and "**DMA Read Transfer**"

11.3.2.2.4.2 DMA Ordering

The DMA assumes that all DMA read channel data must be written to the local application memory in strict address order. For a DMA read transfer, each original outgoing request has a unique PCIe tag. The remote link partner or intermediate switch does not have to preserve the order of CPLs corresponding to such individual requests. When the DMA receives the CPLs out of order, it reorders that data in the "Read Buffer" as described in "**DMA Read Transfer**". DMA write transfer data is sourced from the local application memory and sent to the remote memory in address order.

11.3.2.2.4.3 Error Handling

The DMA supports full AXI and PCIe error handling in conjunction with the PCIe controller. The tables below list the possible sources of an error for DMA read and write transfers.

Table 11-40. Possible Sources of an Error during a DMA Read Transfer

Source	Type	Description
Application Write Error Detected	Fatal	The DMA read channel has received an error response from the AXI bus at the bridge master interface (or TRGT1 interface when the AXI bridge is not used) while writing posted data to it. This error is fatal. • You must restart the transfer from the beginning because the channel context is corrupted.

Table continues on the next page...

Table 11-40. Possible Sources of an Error during a DMA Read Transfer (continued)

		• The MWr is not rolled back. • The AXI bridge master does not report this error to software through error logging or MSG. NOTE: During linked list mode, the Application Write Error only corrupts the current element in the linked list. This is because the DMA does not fetch the next element from local memory until it has completely finished transferring the current block of data (as defined by the current linked list element). Finished implies that your application has successfully acknowledged (without ERROR) all of the MWr requests.
Unsupported Request (UR)	Non-fatal	The DMA read channel has received a PCIe UR CPL status from the remote device in response to the MRd request.
Completer Abort (CA)	Non-fatal	The DMA read channel has received a PCIe CA CPL status from the remote device in response to the MRd request.
CPL Time Out	Non-fatal	The DMA read channel has timed-out while waiting for the remote device to respond to the MRd request, or a malformed CplD has been received.
Data Poisoning	Fatal	The DMA read channel has detected data poisoning in the CPL from the remote device in response to the MRd request. The DMA read channel will drop the completion and then be halted.
Linked List Element Fetch Error Detected	Fatal	The DMA read channel has received an error response from the AXI bus (or TRGT1 interface when the AXI bridge is not used) while reading a linked list element from local memory.

Table 11-41. Possible Sources of an Error during a DMA Write Transfer

Source	Type	Description
Application Read Error Detected	Fatal	The DMA write channel has received an error response from the AXI bus at the bridge master interface (or TRGT1 interface when the AXI bridge is not used) while reading data from it. • The response TLP is discarded by the DMA and is not converted to an outbound MWr TLP.

Table continues on the next page...

Table 11-41. Possible Sources of an Error during a DMA Write Transfer (continued)

		• The UR or CA CPL status TLP generated by the bridge is discarded by the DMA. • The AXI bridge master does not report this error to software through error logging or MSG.
Linked List Element Fetch Error Detected	Fatal	The DMA write channel has received an error response from the AXI bus at the bridge master interface (or TRGT1 interface when the AXI bridge is not used) while reading a linked list element from local memory. • The UR or CA CPL status TLP generated by the bridge is discarded by the DMA. • The AXI bridge master1 does not report this (or for non-DMA traffic) error to software through error logging or MSG.

When the DMA detects an error, it performs the following actions (for the channel with the error):

- Stops issuing new requests to the remote link partner.
- Stops issuing new requests.
 - All data in the “Read Buffer”, that was received before the error was detected, is valid. In the case of a non-fatal error, this is reflected in the DMA Transfer Size Register DMA_TRANSFER_SIZE_OFF_WRCH_0.
 - All data in the read buffer (for that channel), that was received after the error was detected, is discarded.
- Places the DMA channel in a halted state (channel status field of the DMA Channel Control 1 Register is halted).
- Waits for all outstanding requests and CPLs to complete.
- Generates the abort interrupt or IMWr, if enabled, as outlined in “Interrupt Handling”.

For a DMA read transfer, you can read the DMA Read Error Status High Register (DMA_READ_ERR_STATUS_HIGH_OFF) and DMA Read Error Status Low Register (DMA_READ_ERR_STATUS_LOW_OFF) to identify the source of the error. For a DMA write transfer (in linked list mode only), you can read the DMA Write Error Status Register (DMA_WRITE_ERR_STATUS_OFF) to identify the source of the error. After your software has finished error and interrupt routine handling for Non-Fatal errors, you can request the DMA to continue processing (for the channel with the error) by:

1. Reprogramming the SAR based on the number of bytes remaining in the DMA Transfer Size register
2. Writing the channel number to the Doorbell Number field of the DMA Write Doorbell Register (DMA_WRITE_DOORBELL_OFF) or DMA Read Doorbell Register (DMA_READ_DOORBELL_REG).

NOTE

You cannot continue after a Fatal Error is detected, as the channel context is corrupted. Any actions that were in progress are not rolled back. To exit from this condition, you need to Soft Reset the complete DMA controller, as described in DMA Write Engine Enable Register (DMA_WRITE_ENGINE_EN_OFF) and DMA Read Engine Enable Register (DMA_READ_ENGINE_EN_OFF).

When an error occurs in linked list mode, the next LL element is not fetched from local memory. DMA Linked List Pointer Low Register (DMA_LL_LOW_OFF_WRCH_0) is not incremented by the DMA, but remains pointing to the element that caused the error.

11.3.2.2.4.3.1 Error Handling Assistance by Remote Software

During a DMA write transfer, the DMA is not aware of MWr errors that are detected in the remote device, because an MWr is a posted request and has no corresponding completion TLP. In these cases, your remote CPU must manually abort the DMA transfer by writing the channel number to the Doorbell Number field, and 1 to the Stop field of the DMA Write Doorbell Register (DMA_WRITE_DOORBELL_OFF). This causes the DMA to:

- Place the channel in a stopped state. The channel Status field of the DMA Channel Control 1 Register (DMA_CH_CONTROL1_OFF_WRCH_0) is stopped and the DMA Transfer Size register is not 0x0.
- Wait for all outstanding pending transactions.
- Assert the abort interrupt if it is enabled

11.3.2.2.4.4 Linked List Operation

This section describes the detailed operation of the DMA in linked list (LL) mode. It also discusses how the software should produce LL elements¹, and how the DMA and software maintain synchronization.

11.3.2.2.4.4.1 LL Operation Overview

You should first read "Linked List Mode" for an overview of LL mode. In this section, a normal LL operation called recycling2 is described in detail. The process is described for a write channel, but it is possible to have duplicate processes running in parallel for other write or read channels. In this process interrupts are used to trigger the producer (software) to recycle elements. Typically a Watermark interrupt (positioned near the middle of the transfer list (TL)) and an Empty interrupt (in the last data element) are used by setting LIE in these two elements to '1'.

The steps in this process are:

1. Software creates a LL element structure called a TL in local memory consisting of N-1 data elements and one link element.

The link elements "LL Element Pointer Low/High DWORDs"(as shown in Linked List Element figure in Linked List Mode topic) are programmed to point back to the beginning of the first data element.

One of the data elements (near the middle of the TL) is programmed to generate a "Watermark" interrupt.

The last data element is programmed to generate an "Empty" interrupt.

2. Software programs the DMA with the location of the TL. This is done by writing to the DMA Linked List Pointer Low Register (DMA_LL_P_LOW_OFF_WRCH_0) and DMA Linked List Pointer High Register (DMA_LL_P_HIGH_OFF_WRCH_0).
3. Software starts the DMA process by ringing the write channel Doorbell. This is done by writing to the DMA Write Doorbell Register (DMA_WRITE_DOORBELL_OFF).

The solid purple loop L1 in [Figure 11-71](#) corresponds to steps 1-3.

4. The DMA reads (**consumes**) each data element from local memory, and if the CB and CCS bits match, loads the information (SAR, DAR, size, and so on) from that data element into the channel context registers. These context registers determine the operation of the channel that the DMA is currently servicing. The DMA executes the check for the CB/CCS match after it has evaluated TCB and toggled CCS. The DMA always loads link elements into the channel context.

NOTE

If the DMA stops on a link element, then your software must set the DMA Linked List Pointer registers, and restart the DMA process by ringing the doorbell.

With recycling, the same LL structure is used over and over again. Another mode of operation would be to jump to a new LL element structure when the current one is consumed.

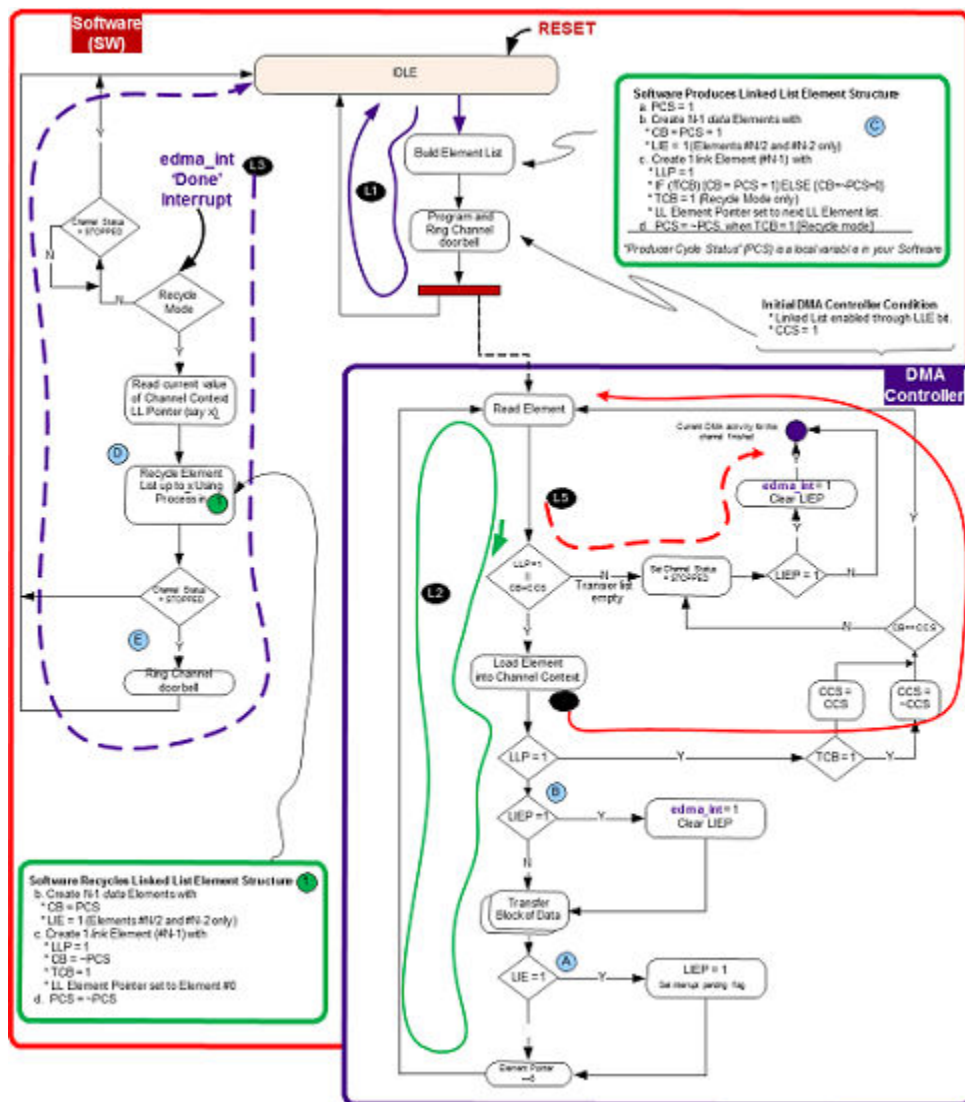


Figure 11-71. Linked List Flow for Producer and Consumer

5. The DMA then proceeds to transfer the block of data (as defined by the element), and when it is finished, reads the next element from local memory.

The solid green loop L2 in [Figure 11-71](#) corresponds to steps 4-5.

6. The last element in the list (called a link element, and indicated by having its LLP field set to 1) is not used to transfer a block of data.

It causes the DMA to effectively repeat the task in step 4.

Its TCB is typically set to 1. This causes the DMA to toggle its CCS bit, and the software to toggle its PCS bit. The software and DMA use the "PCS-CCS-CB-TCB" producer-consumer synchronization mechanism to ensure that:

Software does not recycle elements that have not yet being consumed by the DMA

DMA correctly recognizes and consumes recycled elements. The solid red loop L4 in [Figure 11-71](#) corresponds to step 6.

7. Upon reception of the interrupt (mentioned in step 1), the software starts to recycle the TL.

Software reprograms each data element with new DMA transfer information.

The software and DMA use the "PCS-CCS-CB-TCB" producer-consumer synchronization mechanism described later in this section.

The dashed purple loop L3 in [Figure 11-71](#) corresponds to step 7.

8. As some point, the software wants to terminate the complete DMA process, by not recycling any more elements. The DMA recognizes this condition when CB != CCS, and sets the channel status to stopped

The dashed red loop L5 in [Figure 11-71](#) corresponds to step 8.

11.3.2.2.4.4.2 Using Interrupts for Linked List Producer-Consumer Synchronization

When the DMA is finished with an element, the DMA checks the LIE bit, before reading the next LL element. For more details, see circled step A in [Figure 11-71](#). When the LIE is set, then the DMA does not assert the "done" interrupt immediately, but sets an internal interrupt pending flag (LIEP). It asserts the actual done interrupt after the next data element has been read from system memory. For more details, see circled step B in [Figure 11-71](#). This automatic internal process of delaying the interrupt avoids a race condition between the DMA and software when the LIE bit is set in an element.

Your software should set the "Watermark" interrupt early to schedule the recycling of the consumed elements. Placing the Watermark interrupt too far down in the list, combined with a slow element recycling process in your application; ensures that the DMA returns to the start of the LL before your software has recycled it. The DMA channel STOPS, and you have to restart it by writing to its Doorbell. Typically a "Watermark interrupt" (positioned near the middle of the TL) and an Empty interrupt (in the last data element) are used by setting LIE in these two elements to 1b1. Upon reception of the interrupt, the software starts to recycle the TL.

Operational interrupts are always "done" interrupts. For more details, see "Interrupts and Error Handling"

NOTE

The DMA processes the remote interrupt enable (RIE) bit in the LL element in the same way as the LIE. Therefore, everywhere that LIE and edma_int are mentioned in this section can be interpreted to mean LIE and edma_int, or RIE and IMWr.

11.3.2.2.4.4.3 PCS-CCS-CB-TCB Producer-Consumer Synchronization

The software and DMA use the "PCS-CCS-CB-TCB" producer-consumer synchronization mechanism to ensure that software does not recycle elements that have not yet being consumed by the DMA, and that the DMA correctly recognizes and consumes recycled elements. This process, which is shared between the DMA and the software, is illustrated through an example producer-consumer flow in the figure below.

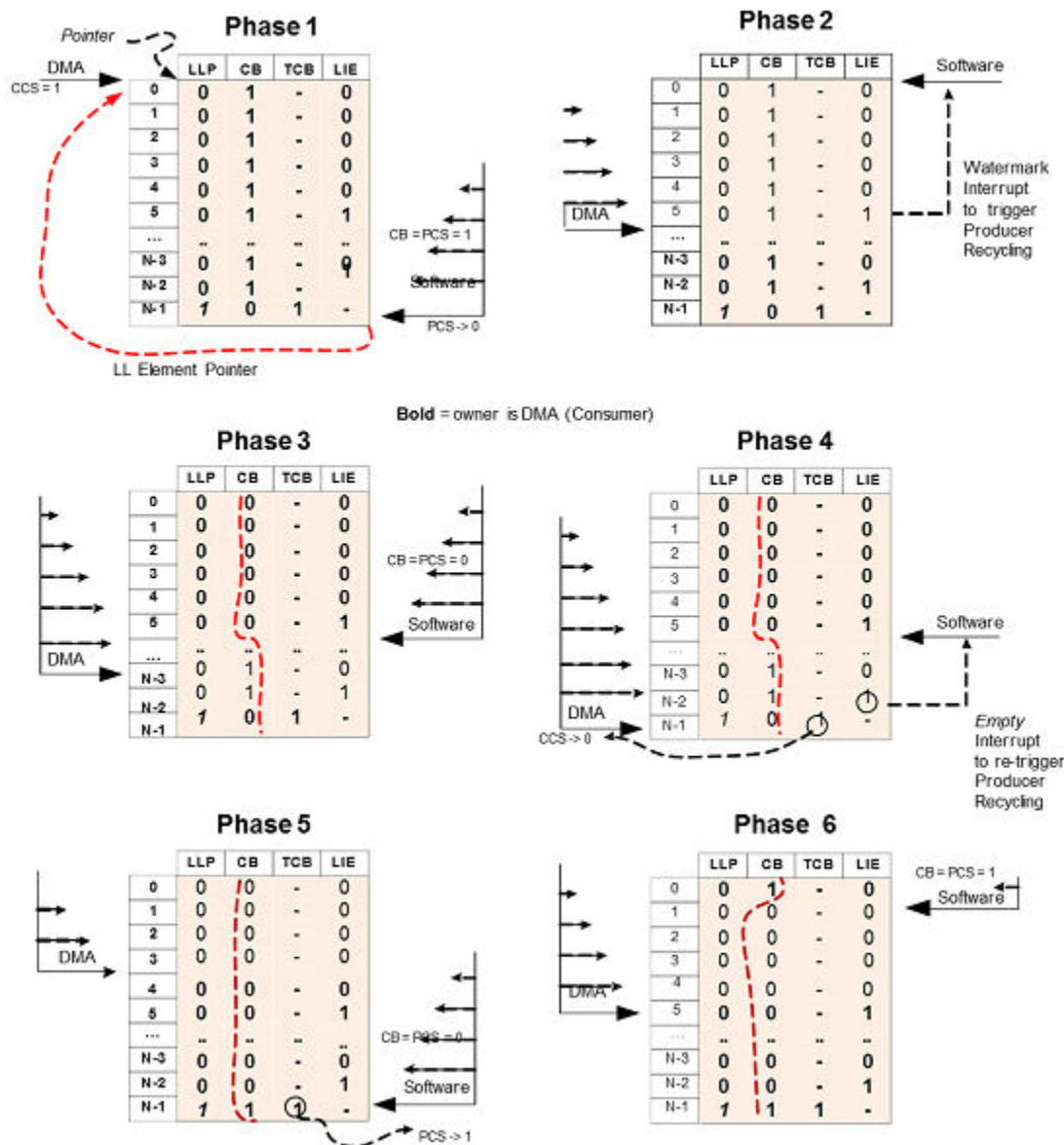


Figure 11-72. Example Producer (Software) - Consumer (DMA) Synchronization Flow

Before looking at the example in detail, it is useful to note that the DMA performs the following two tests as part of this process:

- Consumer-Owned Element, or Transfer List (TL) Empty Test

The solid red loop L4 in the [Figure 11-71](#) corresponds to this test. The Cycle Bit (CB) of DMA Channel Control 1 Register (DMA_CH_CONTROL1_OFF_WRCH_0) (which was loaded from the LL element) is tested against the DMA Consumer Cycle State (CCS) bit of the same register. When CB = CCS, the element is owned by the consumer (DMA) and the data transfer

is executed. When $CB \neq CCS$, the TL is empty and the DMA sets channel Status (CS) in DMA Channel Control 1 Register (DMA_CH_CONTROL1_OFF_WRCH_0) to "Stopped".

- Toggle Cycle Bit (TCB) Test

The Dashed Red Loop Marked L5 in [Figure 11-71](#) corresponds to this test. When the Toggle Cycle Bit (TCB) of the DMA Channel Control 1 Register (DMA_CH_CONTROL1_OFF_WRCH_0) (which was loaded from the LL element) is set to 1, the DMA toggles the CCS bit of the DMA Channel Control 1 Register (DMA_CH_CONTROL1_OFF_WRCH_0) to be $\sim CB$. The producer (software) also performs a similar action on its local variable "Producer Cycle Status" (PCS), as it sets $PCS = \sim PCS$. The TCB field is only valid in the last LL element (the link element). Toggling the PCS and CCS flags in this manner synchronizes the consumer to the producer.

The example in [Figure 11-72](#) has the following six phases:

1. The software builds the transfer list for the write (or read) channel, according to the steps outlined in the circled step C in [Figure 11-71](#). For each element, $CB = PCS = CCS = 1$. The exception to this is the link element where CB is set to the next iteration value because the link element is always loaded into the channel context.
 - The link element points to the start of the TL and its TCB bit is 1. Software toggles its PCS variable.
 - Software writes the location of the TL into the channel context LL Pointer register.
 - Software rings the write channel Doorbell.
2. The DMA starts processing elements. After processing element 5 (and fetching the next element), the DMA asserts a Watermark "done" interrupt to signal the software to start recycling elements.
3. The software is recycling elements. It reads the channel context DMA Linked List Pointer Low Register (DMA_LL_P_LOW_OFF_WRCH_0) and DMA Linked List Pointer High Register (DMA_LL_P_HIGH_OFF_WRCH_0), and recycles elements up to this location in the TL. It sets CB for each element to be PCS. The Dashed Purple Loop Marked L3 in [Figure 11-71](#) corresponds to this activity. The DMA continues processing (consuming) elements as before.
4. The DMA then asserts an Empty "done" interrupt (from the LIE bit in the last data element at the end of TL that was set in the previous phase) to indicate to the software, to start recycling elements again. The DMA reaches the end of the TL, and as the TCB is set to 1, it toggles its CCS bit to 0 ($\sim CB$). For more details, see circled step D in [Figure 11-71](#).
5. The software reads the channel context DMA Linked List Pointer Low Register (DMA_LL_P_LOW_REG_WRCH_0) and DMA Linked List Pointer High Register

(DMA_LL_P_HIGH_OFF_WRCH_0), and recycles elements up to this location in the TL. It sets CB for each element to be PCS. As it recycles the link element, it detects that TCB is set to 1. It toggles its internal variable PCS to 0, and the initial TL has been fully recycled. The DMA starts processing the recycled elements.

6. The DMA continues processing (consuming) elements as before. Software did not have to ring the Doorbell, as the first element was already recycled before the DMA started processing it. The software might continue recycling from the top of the list, if the DMA had already returned to the top of the list when the software read the channel context DMA Linked List Pointer Low Register (DMA_LL_P_LOW_OFF_WRCH_0) and DMA Linked List Pointer High Register (DMA_LL_P_HIGH_OFF_WRCH_0), in step 5. If not, it starts recycling again when the DMA processes element 4 and the DMA asserts a Watermark “done” interrupt.

11.3.2.2.4.4.4 *Element Recycling*

When your software is recycling elements, it should program the first DWORD (control bits) in the [Figure 11-63](#) only after the other DWORDs have been programmed. In addition, it should set the CB bit (to ~PCS) within this DWORD only after the other bits in this DWORD have been set. This programming sequence avoids the DMA incorrectly fetching an element that software is currently recycling.

When the software is finished recycling it must check the channel context status (CS). When CS is "Stopped", the software must ring the channel Doorbell again. For more details, see circled step E in [Figure 11-59](#). If your element recycling process was fast enough, then the DMA will seamlessly move from processing the last old element to the first new element, without the need for the software to ring the channel Doorbell again. To eliminate the possibility of a race condition between these two actions (read and write), you should first confirm the presence of the "done" interrupt bit in the DMA Write Interrupt Status Register (DMA_WRITE_INT_STATUS_OFF) or DMA Read Interrupt Status Register (DMA_READ_INT_STATUS_OFF) and also check the DMA Linked List Pointer Low Register (DMA_LL_P_LOW_REG_WRCH_0) to confirm that it is pointing to the correct location.

11.3.2.2.4.4.5 *Link Down Recovery*

During normal operation when the link goes down, the controller generates a "link reset" request. A

high-to-low transition on link_req_rst_not indicates that the PCIe controller is requesting external logic to reset the PCIe controller because the PHY link is down. For more details, see "Hot Reset". To determine the progress of the linked list operation when the

link went down, you can read the contents of the channel context DMA Linked List Pointer Low Register (DMA_LL_P_LOW_OFF_WRCH_0) and DMA Linked List Pointer High Register (DMA_LL_P_HIGH_OFF_WRCH_0).

11.3.2.3 M-PCIe

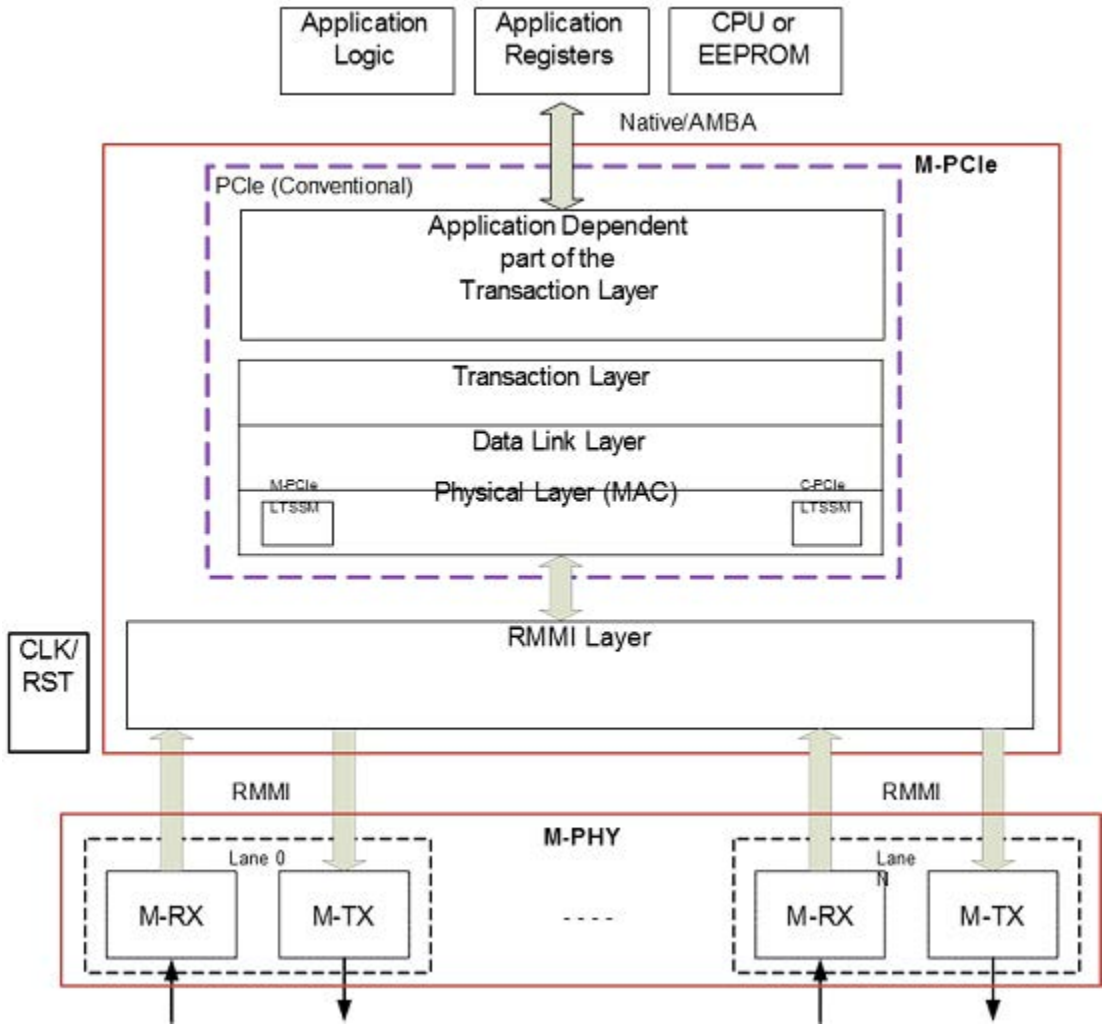
This section describes the optional M-PCIe feature.

11.3.2.3.1 Feature Description

The new M-PCIe specification enables the PCIe controller to operate over the low-power MIPI M-PHY physical layer technology, extending the benefits of PCIe to mobile devices including thin laptops, tablets and smartphones. The M-PCIe controller is compliant with the M-PCIe ECN1 specification. The interface between the controller and the PHY is compliant with the RMMI (Reference M-PHY MODULE Interface) specified in Annex A of the M-PHY Specification.

A complete M-PCIe port solution includes the controller, an analog PHY macro, and application logic to source and sink data. The physical layer is split across the RMMI such that the MAC functionality (LTSSM, lane-to-lane deskew) is in the controller and the PHY functionality is implemented in the RMMI-compliant PHY. The PHY module resides outside of the controller, interfacing through the standard RMMI. For more details, see Integrating with the PHY in the User Guide.

Figure 11-1 Complete M-PCIe Port Solution



a. HS=high-speed. LS=low-speed.

11.3.2.3.1.1 Features

This release supports the following features: **Table 11-1M- PCIe Features**

M-PCIe Feature	Supported
HSa-Gear Link Speeds	HS-Gear1 HS-Gear2 HS-Gear3
HS-Gear1 Base RATE-A Frequencies	31.2 MHz 62.4 MHz 124.8 MHz
Elastic Buffer Maximum Frequency Difference Compensation Between Link Partners • HS-Mode: • LS-Mode:	+/- 2000 ppm +/- 300 %

Table continues on the next page...

Dynamic Frequency Gear Speed Changing Mode	✓
Dynamic Width Gear Speed Changing Mode (RMMI Standard does not Support this)	✓
Controller and PHY can have Different Gear Speed Changing Modes	✓
x1, x2, x4, x8, x16 Link Width Support	✓
10-bit RMMI Width	✓
20-bit RMMI Width	✓
40-bit RMMI Width	✓
Controller and PHY can have Different RMMI Widths	✓
Runtime Selectable Mode of Operation Between M-PCIe and Conventional PCIe (Selectable PHY)	✓
Automatic RRAP Function (Off-loads Attribute Discovery/ Configuration From Your CPU In Downstream Port)	✓
LS-Gear Link Speeds Note: All bytes on the RMMI are valid in LS mode.	LS-Gear1
RATE-A / RATE-B Series Support	✓
Dynamic Asymmetric Link Width Adjustment	✓
Static (Compile Time) Asymmetric Link Width Adjustment	✗
Dynamic Link Reconfiguration Mechanism (Up Configure)	✓
Digital Remote Loopback	✓
Digital Local (RMMI) Loopback	✓

M-PCIe Feature	Supported
M-PHY Interoperability Support	✓
Third Party M-PHY Interoperability Support Through Custom PHY Flow	✓
Hardware Compliance Pattern Generator	✓
Internal Databus Width	32, 64, 128
Optional M-PCIe Features	
Manual Lane Flip	✓
Manual Polarity Change	✓
WAKE# Sideband Signal Support	✓
Optional System Features	
Advanced Power Management, Power Gating, and UPF Support	✓

Table continues on the next page...

PCI Express (PCIe)

Multifunction Support	✓
DMA Support	✓
ARI Support	✓
SR-IOV Support	✓
Latency Tolerance Reporting (LTR) Support	✓
OBFF Support	✓
Address Translation Services (ATS) Support	✓
Atomic Ops Support	✓
TLP Processing Hints (TPH) Support	✓
TLP Prefix Support (Including PASID)	✓
ID Based Ordering (IDO) Support	✓
Conventional PCIe Features Not Applicable in M-PCIe Mode	
CLKREQ# Sideband Signal and L1 Clock Power Management (CPM) Support	✗
Separate Refclk with Independent Spread Spectrum Clocking (SRIS)	✗
L1 Substates Support	✗
Dynamic Lane Reversal	✗
Dynamic Polarity Change	✗

a. HS=high-speed. LS=low-speed.

11.3.2.3.1.2 Limitations

Table 11-2 identifies the temporary limitations when using the M-PCIe module.

Table 11-2 M-PCIe Limitations

Limitation	Note
Limited Analog Loopback Support	Analog loopback requires the M-PHY to autonomously loop and re-transmit the Rx data. However, the RMMI specification does not define the entry mechanism for analog loopback. • If you are using the M-PHY (which has no input to accept the cores pa_rx_analog_loopback output), the analog loopback path must

Table continues on the next page...

	pass through the controller involving the elastic buffer. The Tx lanes in the controller use the Tx (and not the Rx) clock. The elastic buffer cannot operate indefinitely under this condition. If there is a frequency difference between the Rx and Tx clocks, there will be some data overflow or underflow. If you are using a third party M-PHY, the M-PHY enters analog loop-back mode when the controller asserts <code>pa_rx_analog_loopback</code>. If the M-PHY does not support such an input, you can instruct the M-PCIe controller to do loopback from Rx to Tx through the elastic buffer.
Verilog Testbench (VTB) VIP not Included	M-PCIe configurations currently have no M-PCIe VIP and connect instead to an RTL DUT of the opposite port type.
No Midstream De-assertion of TX_PhyDIRDY	The controller does not support midstream de-assertion of <code>phy_mac_tx_phydirty</code> during burst mode. This is not an M-PCIe or RMMI requirement. However, the <code>tx_symbolclk</code> is derived from the same clock that the M-PHY TX path uses, and there is no reason for the M-PHY to insert wait cycles.
Midstream De-assertion of RX_PhyDORDY	The <code>rx_symbolclk</code> is the recovered clock and there is no elastic buffer in the M-PHY. Therefore, there is no reason for the M-PHY to insert de-assertion of <code>mac_phy_rx_phydirty</code> during burst mode. If your M-PHY de-asserts <code>mac_phy_rx_phydirty</code> during burst mode, there is a risk of underflowing the elastic buffer in the controller.
TLP in 2K+ PPM System with Payload Greater than 2 KB not Supported.	The controller can handle a clock difference of 2K PPM only. When clock compensation greater than 2K PPM is required, the elastic buffer overflows or underflows on receiving a TLP with payload greater than 2 KB.
M-PHY RX Registers Cannot be set when LTSSM Exits L1 or L2.	The controller cannot set M-PHY RX registers when: - LTSSM exits L1 in a downstream or upstream port - LTSSM exits L2 in a downstream port.

11.3.2.3.1.3 Frequency, Speed, and Width Support

To set the M-PCIe frequency, datapath width, and speed modes of the controller, use the following configuration parameters in the coreConsultant GUI:

- Controller Base Frequency (CM_FREQ)
- PHY Base Frequency (CM_PHY_FREQ)
- Controller Gear2 and Gear3 Modes (CM_GEAR2_MODE and CM_GEAR3_MODE)
- PHY Gear2 and Gear3 Modes (CM_PHY_GEAR2_MODE and CM_PHY_GEAR3_MODE)
- Maximum Tx and Rx Link Width (CM_TXNL_GUI and CM_RXNL_GUI)
- $CX_NL = \max \{CM_TXNL_GUI, CM_RXNL_GUI\}$
- $CX_NB = (CM_FREQ == 124.8) ? 1 : (CM_FREQ == 62.4) ? 2 : 4$
- CM_FREQ_STEP_DOWN_EN. This is the enable for the “Frequency Step Module”

11.3.2.3.1.4 Gear2/Gear3 Modes

For Dynamic Frequency (DF) configurations, the number of active symbols on the RMMI is constant; the frequency of the controller doubles each time as the controller transitions from Gear1 -> Gear2 -> Gear3 rates. For Dynamic Width (DW) configurations, the frequency of the controller is constant; the number of active symbols on the RMMI doubles each time as the controller transitions from Gear1 -> Gear2 -> Gear3 rates. The controller cannot have different Gear2 and Gear3 modes. For example, you cannot configure the controller for Dynamic Width in Gear2 mode, and at the same time configure it for Dynamic Frequency in Gear3 mode. For more details, see Figure 11-3.

Figure 11-3 Gear2/Gear3 Speed Changing Paths

				Maxa Link Width							
Gear1 RATE A Frequency (CM_FREQ)	Gear2 Mode	Gear3 Mode	RMMI Widthc	For 32-bit datapath	For 64-bit datapathd	For 128-bit datapath	For 256-bit datapath	Maximum core_clk Frequency (x CM_FREQ)	Gear1 Speed Modee	Gear2 Speed Mode	Gear3 Speed Mode
124.8	Disable	Disable	10-bit	x4	x8	x16	-	x1	g1_1s		
62.4			20-bit	x2	x4	x8	x16	x1	g1_2s		

Table continues on the next page...

PCI Express (PCIe)

31.2			40-bit	x1	x2	x4	x8	x1	g1_4s		
124.8	DW	Disable	20-bit	x2	x4	x8	x16	x1	g1_1s	g2_2s_dw	
62.4	DW		40-bit	x1	x2	x4	x8	x1	g1_2s	g2_4s_dw	
124.8	DF		10-bit	x4	x8	x16	-	x2	g1_1s	g2_1s_df	
62.4	DF		20-bit	x2	x4	x8	x16	x2	g1_2s	g2_2s_df	
31.2	DF		40-bit	x1	x2	x4	x8	x2	g1_4s	g2_4s_df	
124.8	DW	DW	40-bit	x1	x2	x4	x8	x1	g1_1s	g2_2s_dw	g3_4s_dw
124.8	DF	DF	10-bit	x4	x8	x16	-	x4	g1_1s	g2_1s_df	g3_1s_df
62.4	DF	DF	20-bit	x2	x4	x8	x16	x4	g1_2s	g2_2s_df	g3_2s_df
31.2	DF	DF	40-bit	x1	x2	x4	x8	x4	g1_4s	g2_4s_df	g3_4s_df

a. The value in each cell indicates the maximum link width supported. You can configure the controller with a link width up to this value, and coreConsultant automatically calculates the datapath width.

b. Corresponding RATE B frequencies are 145.76, 72.88, and 36.44 MHz.

c. The actual physical RMMI interface I/O width is 10*CM_NB bits per lane. The coreConsultant tool automatically calculates this width from CM_GEAR2_MODE, CM_GEAR3_MODE, and CM_FREQ. In a DW configuration, the number of active bits in the RMMI interface increases as the controller transitions to a higher Gear speed.

d. A 64-bit datapath requires a 32-bit/64-bit license. You cannot create a 32-bit or 64-bit configuration with a 128-bit/256-bit license

e. The 'g' value indicates the speed mode. The 's' value indicates the number of 10-bit symbols processed per clock cycle per lane by the controller RMMI I/F, in the indicated speed mode. The 'dw/df' value indicates how the speed change to that mode is achieved

11.3.2.3.1.5 Controller PHY Compatibility

The following tables indicate which combinations of controller configurations from Table 11-3 and PHYs are supported. In Table 11-4, the triplet of values for each row indicates how the controller moves from Gear1 to Gear2 speed mode and from Gear2 to Gear3 speed mode. The triplet of values also identifies the particular controller configuration in Table 11-3. The controller always uses the same mechanism (DF or DW) to go from Gear1 to Gear2 and from Gear2 to Gear3.

Table 11-4 Gear3 Controller-PHY Combinations Supported

		PHY			
		g1_1s g2_2s_dw g3_4s_dw	g1_1s g2_1s_df g3_1s_df	g1_2s g2_2s_df g3_2s_df	g1_4s g2_4s_df g3_4s_df
CONTROLLER	g1_1s g2_2s_dw g3_4s_dw	✓	✗	✗	✗
	g1_1s g2_1s_df g3_1s_df	✗	✓	✗	✗
	g1_2s g2_2s_df g3_2s_df	✗	✓	✓	✗
	g1_4s g2_4s_df g3_4s_df	✗	✓	✓	✓

Table 11-5 Gear2 Controller-PHY Combinations Supported

		PHY				
		g1_1s g2_2s_dw	g1_2s g2_4s_dw	g1_1s g2_1s_df	g1_2s g2_2s_df	g1_4s g2_4s_df
CONTROLLER	g1_1s g2_2s_dw	✓	✗	✓	✗	✗
	g1_2s g2_4s_dw	✓	✓	✓	✓	✗
	g1_1s g2_1s_df	✗	✗	✓	✗	✗
	g1_2s g2_2s_df	✗	✗	✓	✓	✗
	g1_4s g2_4s_df	✗	✗	✓	✓	✓

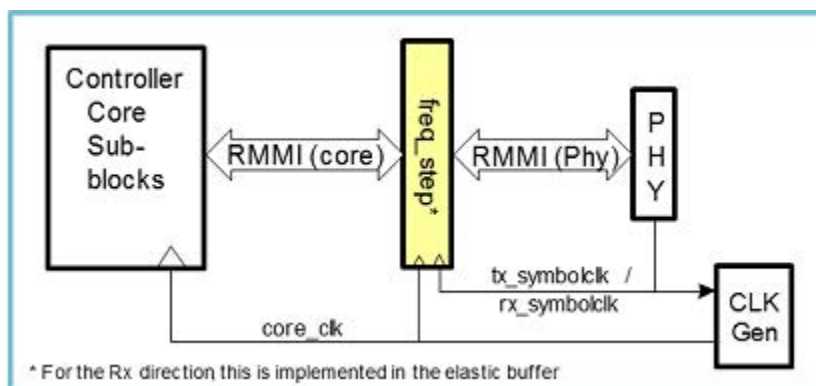
Table 11-6 Gear1 Controller-PHY Combinations Supported

		PHY		
		g1_1s	g1_2s	g1_4s
CONTROLLER	g1_1s	✓	✗	✗
	g1_2s	✓	✓	✗
	g1_4s	✓	✓	✓

11.3.2.3.1.6 Frequency Step Module

When the controller RMMI lane width (CM_NB) and M-PHY RMMI lane width (CM_PHY_NB) are different, or when the controller and the PHY have different Gear2 speed changing modes, the automatically-derived CM_FREQ_STEP_DOWN_EN parameter is defined, and a RMMI adapter module¹ is added between the controller RMMI and the PHY RMMI.

Figure 11-4 Location of freq_step RMMI Adapter Module



This module steps up/down the signals to/from the RMMI interface and adapts (x2, x4, x0.5, x0.25) the controller RMMI width to the PHY RMMI width when the controller clock frequency is slower or faster than the RMMI clock frequency.

11.3.3 Memory Map and Register Definition

This section includes the PCIe module memory map and detailed descriptions of all registers.

NOTE

Synopsys Proprietary. Used with permission.

11.3.3.1 Register Descriptions

11.3.3.1.1 PCIe Memory Map

pcie base address: 3380_0000h

Offset	Register	Width (In bits)	Access	Reset value
0h	Device ID and Vendor ID Register. (TYPE1_DEV_ID_VEND_ID_REG)	32	RW	ABCD_16C3h
4h	Status and Command Register. (TYPE1_STATUS_COMMAND_REG)	32	RW	0010_0000h
8h	Class Code and Revision ID Register. (TYPE1_CLASS_CODE_REV_ID_REG)	32	RW	0000_0001h
Ch	Header Type, Latency Timer, and Cache Line Size Register. (TYPE1_HDR_TYPE_LAT_CACHE_LINE_SIZE_REG)	32	RW	0001_0000h
18h	Secondary Latency Timer, Subordinate Bus Number, Secondary Bus Number, and Primary Bus Number Register. (SEC_LAT_TIMER_SUB_BUS_SEC_BUS_PRI_BUS_REG)	32	RW	0000_0000h
1Ch	Secondary Status, and I/O Limit and Base Register. (SEC_STAT_IO_LIMIT_IO_BASE_REG)	32	RW	0000_0000h
20h	Memory Limit and Base Register. (MEM_LIMIT_MEM_BASE_REG)	32	RW	0000_0000h
24h	Prefetchable Memory Limit and Base Register. (PREF_MEM_LIMIT_PREF_MEM_BASE_REG)	32	RW	0000_0000h
28h	Prefetchable Base Upper 32 Bits Register. (PREF_BASE_UPPER_REG)	32	RO	0000_0000h
2Ch	Prefetchable Limit Upper 32 Bits Register. (PREF_LIMIT_UPPER_REG)	32	RO	0000_0000h
30h	I/O Limit and Base Upper 16 Bits Register. (IO_LIMIT_UPPER_IO_BASE_UPPER_REG)	32	RO	0000_0000h
34h	Capabilities Pointer Register. (TYPE1_CAP_PTR_REG)	32	RW	0000_0040h
38h	Expansion ROM Base Address Register. (TYPE1_EXP_ROM_BASE_REG)	32	RW	0000_0000h
3Ch	Bridge Control, Interrupt Pin, and Interrupt Line Register. (BRIDGE_CTRL_INT_PIN_INT_LINE_REG)	32	RW	0000_01FFh
40h	Power Management Capabilities Register. (CAP_ID_NXT_PTR_REG)	32	RW	DBC3_5001h
44h	Power Management Control and Status Register. (CON_STATUS_REG)	32	RW	0000_0000h
50h	MSI Capability ID, Next Pointer, Capability/Control Registers. (PCI_MSI_CAP_ID_NEXT_CTRL_REG)	32	RW	0180_7005h
54h	MSI Message Lower Address Register. (MSI_CAP_OFF_04H_REG)	32	RW	0000_0000h
58h	For a 32 bit MSI Message, this register contains Data. (MSI_CAP_OFF_08H_REG)	32	RW	0000_0000h
5Ch	For a 64 bit MSI Message, this register contains Data. (MSI_CAP_OFF_0CH_REG)	32	RW	0000_0000h
60h	Used for MSI when Vector Masking Capable. (MSI_CAP_OFF_10H_REG)	32	RW	0000_0000h
64h	Used for MSI 64 bit messaging when Vector Masking Capable. (MSI_CAP_OFF_14H_REG)	32	RO	0000_0000h
70h	PCI Express Capabilities, ID, Next Pointer Register. (PCIE_CAP_ID_PCIE_NEXT_CAP_PTR_PCIE_CAP_REG)	32	RW	0042_0010h
74h	Device Capabilities Register. (DEVICE_CAPABILITIES_REG)	32	RW	0000_8000h

Table continues on the next page...

PCI Express (PCIe)

Offset	Register	Width (In bits)	Access	Reset value
78h	Device Control and Status Register. (DEVICE_CONTROL_DEVICE_STATUS)	32	RW	0000_2010h
7Ch	Link Capabilities Register. (LINK_CAPABILITIES_REG)	32	RW	0073_CC12h
80h	Link Control and Status Register. (LINK_CONTROL_LINK_STATUS_REG)	32	RW	1011_0000h
84h	Slot Capabilities Register. (SLOT_CAPABILITIES_REG)	32	RW	0000_0000h
88h	Slot Control and Status Register. (SLOT_CONTROL_SLOT_STATUS)	32	RW	0040_03C0h
8Ch	Root Control and Capabilities Register. (ROOT_CONTROL_ROOT_CAPABILITIES_REG)	32	RW	0001_0000h
90h	Root Status Register. (ROOT_STATUS_REG)	32	W1C	0000_0000h
94h	Device Capabilities 2 Register. (DEVICE_CAPABILITIES2_REG)	32	RO	0000_041Fh
98h	Device Control 2 and Status 2 Register. (DEVICE_CONTROL2_DEVICE_STATUS2_REG)	32	RW	0000_0000h
9Ch	Link Capabilities 2 Register. (LINK_CAPABILITIES2_REG)	32	RO	0000_0006h
A0h	Link Control 2 and Status 2 Register. (LINK_CONTROL2_LINK_STATUS2_REG)	32	RW	0001_0002h
100h	Advanced Error Reporting Extended Capability Header. (AER_EXT_CAP_HDR_OFF)	32	RW	1482_0001h
104h	Uncorrectable Error Status Register. (UNCORR_ERR_STATUS_OFF)	32	W1C	0000_0000h
108h	Uncorrectable Error Mask Register. (UNCORR_ERR_MASK_OFF)	32	RW	0040_0000h
10Ch	Uncorrectable Error Severity Register. (UNCORR_ERR_SEV_OFF)	32	RW	0046_2030h
110h	Correctable Error Status Register. (CORR_ERR_STATUS_OFF)	32	W1C	0000_0000h
114h	Correctable Error Mask Register. (CORR_ERR_MASK_OFF)	32	RW	0000_E000h
118h	Advanced Error Capabilities and Control Register. (ADV_ERR_CAP_CTRL_OFF)	32	RW	0000_00A0h
11Ch	Header Log Register 0. (HDR_LOG_0_OFF)	32	RO	0000_0000h
120h	Header Log Register 1. (HDR_LOG_1_OFF)	32	RO	0000_0000h
124h	Header Log Register 2. (HDR_LOG_2_OFF)	32	RO	0000_0000h
128h	Header Log Register 3. (HDR_LOG_3_OFF)	32	RO	0000_0000h
12Ch	Root Error Command Register. (ROOT_ERR_CMD_OFF)	32	RW	0000_0000h
130h	Root Error Status Register. (ROOT_ERR_STATUS_OFF)	32	RW	0000_0000h
134h	Error Source Identification Register. (ERR_SRC_ID_OFF)	32	RO	0000_0000h
138h	TLP Prefix Log Register 1. (TLP_PREFIX_LOG_1_OFF)	32	RO	0000_0000h
13Ch	TLP Prefix Log Register 2. (TLP_PREFIX_LOG_2_OFF)	32	RO	0000_0000h
140h	TLP Prefix Log Register 3. (TLP_PREFIX_LOG_3_OFF)	32	RO	0000_0000h
144h	TLP Prefix Log Register 4. (TLP_PREFIX_LOG_4_OFF)	32	RO	0000_0000h
148h	L1 Substates Extended Capability Header. (L1SUB_CAP_HEADER_REG)	32	RW	0001_001Eh
14Ch	L1 Substates Capability Register. (L1SUB_CAPABILITY_REG)	32	RW	0028_0A1Bh
150h	L1 Substates Control 1 Register. (L1SUB_CONTROL1_REG)	32	RW	0000_0A00h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
154h	L1 Substates Control 2 Register. (L1SUB_CONTROL2_REG)	32	RW	0000_0028h
700h	Ack Latency Timer and Replay Timer Register. (ACK_LATENCY_TIMER_OFF)	32	RW	1846_0817h
704h	Vendor Specific DLLP Register. (VENDOR_SPEC_DLLP_OFF)	32	RW	FFFF_FFFFh
708h	Port Force Link Register. (PORT_FORCE_OFF)	32	RW	0000_0004h
70Ch	Ack Frequency and L0-L1 ASPM Control Register. (ACK_F_ASPM_CTRL_OFF)	32	RW	1B2C_2C00h
710h	Port Link Control Register. (PORT_LINK_CTRL_OFF)	32	RW	0001_0120h
714h	Lane Skew Register. (LANE_SKEW_OFF)	32	RW	0000_0000h
718h	Timer Control and Max Function Number Register. (TIMER_CTRL_MAX_FUNC_NUM_OFF)	32	RW	0000_C000h
71Ch	Symbol Timer Register and Filter Mask 1 Register. (SYMBOL_TIMER_FILTER_1_OFF)	32	RW	0000_0280h
720h	Filter Mask 2 Register. (FILTER_MASK_2_OFF)	32	RW	0000_0000h
724h	AMBA Multiple Outbound Decomposed NP SubRequests Control Register. (AMBA_MUL_OB_DECOMP_NP_SUB_REQ_CTRL_OFF)	32	RW	0000_0001h
728h	Debug Register 0 (PL_DEBUG0_OFF)	32	RO	See description.
72Ch	Debug Register 1 (PL_DEBUG1_OFF)	32	RO	See description.
730h	Transmit Posted FC Credit Status (TX_P_FC_CREDIT_STATUS_OFF)	32	RO	0000_0000h
734h	Transmit Non-Posted FC Credit Status (TX_NP_FC_CREDIT_STATUS_OFF)	32	RO	0000_0000h
738h	Transmit Completion FC Credit Status (TX_CPL_FC_CREDIT_STATUS_OFF)	32	RO	0000_0000h
73Ch	Queue Status (QUEUE_STATUS_OFF)	32	RW	0000_0000h
740h	VC Transmit Arbitration Register 1 (VC_TX_ARBI_1_OFF)	32	RO	0000_000Fh
744h	VC Transmit Arbitration Register 2 (VC_TX_ARBI_2_OFF)	32	RO	0000_0000h
748h	Segmented-Buffer VC0 Posted Receive Queue Control. (VC0_P_RX_Q_CTRL_OFF)	32	RW	4520_C019h
74Ch	Segmented-Buffer VC0 Non-Posted Receive Queue Control. (VC0_NP_RX_Q_CTRL_OFF)	32	RW	0520_C003h
750h	Segmented-Buffer VC0 Completion Receive Queue Control. (VC0_CPL_RX_Q_CTRL_OFF)	32	RW	0580_0000h
80Ch	Link Width and Speed Change Control Register. (GEN2_CTRL_OFF)	32	RW	0002_012Ch
810h	PHY Status Register. (PHY_STATUS_OFF)	32	RO	See description.
814h	PHY Control Register. (PHY_CONTROL_OFF)	32	RW	0000_0000h
81Ch	Programmable Target Map Control Register. (TRGT_MAP_CTRL_OFF)	32	RW	0000_007Fh
820h	Integrated MSI Reception Module (iMRM) Address Register. (MSI_CTRL_ADDR_OFF)	32	RW	0000_0000h

Table continues on the next page...

PCI Express (PCIe)

Offset	Register	Width (In bits)	Access	Reset value
824h	Integrated MSI Reception Module Upper Address Register. (MSI_CTRL_UPPER_ADDR_OFF)	32	RW	0000_0000h
828h	Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_0_EN_OFF)	32	RW	0000_0000h
82Ch	Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_0_MASK_OFF)	32	RW	0000_0000h
830h	Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_0_STATUS_OFF)	32	W1C	0000_0000h
834h	Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_1_EN_OFF)	32	RW	0000_0000h
838h	Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_1_MASK_OFF)	32	RW	0000_0000h
83Ch	Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_1_STATUS_OFF)	32	W1C	0000_0000h
840h	Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_2_EN_OFF)	32	RW	0000_0000h
844h	Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_2_MASK_OFF)	32	RW	0000_0000h
848h	Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_2_STATUS_OFF)	32	W1C	0000_0000h
84Ch	Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_3_EN_OFF)	32	RW	0000_0000h
850h	Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_3_MASK_OFF)	32	RW	0000_0000h
854h	Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_3_STATUS_OFF)	32	W1C	0000_0000h
858h	Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_4_EN_OFF)	32	RW	0000_0000h
85Ch	Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_4_MASK_OFF)	32	RW	0000_0000h
860h	Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_4_STATUS_OFF)	32	W1C	0000_0000h
864h	Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_5_EN_OFF)	32	RW	0000_0000h
868h	Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_5_MASK_OFF)	32	RW	0000_0000h
86Ch	Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_5_STATUS_OFF)	32	W1C	0000_0000h
870h	Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_6_EN_OFF)	32	RW	0000_0000h
874h	Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_6_MASK_OFF)	32	RW	0000_0000h
878h	Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_6_STATUS_OFF)	32	W1C	0000_0000h
87Ch	Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_7_EN_OFF)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
880h	Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_7_MASK_OFF)	32	RW	0000_0000h
884h	Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_7_STATUS_OFF)	32	W1C	0000_0000h
888h	Integrated MSI Reception Module General Purpose IO Register. (MSI_GPIO_IO_OFF)	32	RW	0000_0000h
88Ch	RADM clock gating enable control register. (CLOCK_GATING_CTRL_OFF)	32	RW	0000_0001h
8B4h	Order Rule Control Register. (ORDER_RULE_CTRL_OFF)	32	RW	0000_0000h
8B8h	PIPE Loopback Control Register. (PIPE_LOOPBACK_CONTROL_OFF)	32	RW	0000_0001h
8BCh	DBI Read-Only Write Enable Register. (MISC_CONTROL_1_OFF)	32	RW	0000_0001h
8C0h	UpConfigure Multi-lane Control Register. (MULTI_LANE_CONTROL_OFF)	32	RW	0000_0000h
8C4h	PHY Interoperability Control Register. (PHY_INTEROP_CTRL_OFF)	32	RW	0000_0237h
8C8h	TRGT_CPL_LUT Delete Entry Control register. (TRGT_CPL_LUT_DELETE_ENTRY_OFF)	32	RW	0000_0000h
8CCh	Link Reset Request Flush Control Register. (LINK_FLUSH_CTRL_OFF)	32	RW	FF00_0001h
8D0h	AXI Bridge Slave Error Response Register. (AMBA_ERROR_RESPONSE_DEFAULT_OFF)	32	RW	0000_9C00h
8D4h	Link Down AXI Bridge Slave Timeout Register. (AMBA_LINK_TIMEOUT_OFF)	32	RW	0000_0032h
8D8h	AMBA Ordering Control. (AMBA_ORDERING_CTRL_OFF)	32	RW	0000_0000h
8E0h	ACE Cache Coherency Control Register 1 (COHERENCY_CTRL_1_OFF)	32	RW	0000_0000h
8E4h	ACE Cache Coherency Control Register 2 (COHERENCY_CTRL_2_OFF)	32	RW	0000_0000h
8E8h	ACE Cache Coherency Control Register 3 (COHERENCY_CTRL_3_OFF)	32	RW	0000_0000h
8F0h	Lower 20 bits of the programmable AXI address where Messages coming from wire are mapped to. (AXI_MSTR_MSG_ADDR_LOW_OFF)	32	RW	0000_0000h
8F4h	Upper 32 bits of the programmable AXI address where Messages coming from wire are mapped to. (AXI_MSTR_MSG_ADDR_HIGH_OFF)	32	RW	0000_0000h
8F8h	PCIe Controller IIP Release Version Number. (PCIE_VERSION_NUMBER_OFF)	32	RO	3530_302Ah
8FCh	PCIe Controller IIP Release Version Type. (PCIE_VERSION_TYPE_OFF)	32	RO	6761_2A2Ah
B40h	Auxiliary Clock Frequency Control Register. (AUX_CLK_FREQ_OFF)	32	RW	0000_000Ah
B44h	L1 Substates Timing Register. (L1_SUBSTATES_OFF)	32	RW	0000_00D2h
8000_0000h	iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_OUTBOUND_0)	32	RW	0000_0000h
8000_0004h	iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_OUTBOUND_0)	32	RW	0000_0000h

Table continues on the next page...

PCI Express (PCIe)

Offset	Register	Width (In bits)	Access	Reset value
8000_0008h	iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_OFF_OUTBOUND_0)	32	RW	0000_0000h
8000_000Ch	iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_0)	32	RW	0000_0000h
8000_0010h	iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_OUTBOUND_0)	32	RW	0000_FFFFh
8000_0014h	iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_0)	32	RW	0000_0000h
8000_0018h	iATU Upper Target Address Register. (IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_0)	32	RW	0000_0000h
8000_0100h	iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_INBOUND_0)	32	RW	0000_0000h
8000_0104h	iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_INBOUND_0)	32	RW	0000_0000h
8000_0108h	iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_OFF_INBOUND_0)	32	RW	0000_0000h
8000_010Ch	iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_INBOUND_0)	32	RW	0000_0000h
8000_0110h	iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_INBOUND_0)	32	RW	0000_FFFFh
8000_0114h	iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_INBOUND_0)	32	RW	0000_0000h
8000_0200h	iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_OUTBOUND_1)	32	RW	0000_0000h
8000_0204h	iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_OUTBOUND_1)	32	RW	0000_0000h
8000_0208h	iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_OFF_OUTBOUND_1)	32	RW	0000_0000h
8000_020Ch	iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_1)	32	RW	0000_0000h
8000_0210h	iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_OUTBOUND_1)	32	RW	0000_FFFFh
8000_0214h	iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_1)	32	RW	0000_0000h
8000_0218h	iATU Upper Target Address Register. (IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_1)	32	RW	0000_0000h
8000_0300h	iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_INBOUND_1)	32	RW	0000_0000h
8000_0304h	iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_INBOUND_1)	32	RW	0000_0000h
8000_0308h	iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_OFF_INBOUND_1)	32	RW	0000_0000h
8000_030Ch	iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_INBOUND_1)	32	RW	0000_0000h
8000_0310h	iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_INBOUND_1)	32	RW	0000_FFFFh

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8000_0314h	iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_INBOUND_1)	32	RW	0000_0000h
8000_0400h	iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_O UTBOUND_2)	32	RW	0000_0000h
8000_0404h	iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_O UTBOUND_2)	32	RW	0000_0000h
8000_0408h	iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_O FF_OUTBOUND_2)	32	RW	0000_0000h
8000_040Ch	iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_ OFF_OUTBOUND_2)	32	RW	0000_0000h
8000_0410h	iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_OUTB OUND_2)	32	RW	0000_FFFFh
8000_0414h	iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_ OFF_OUTBOUND_2)	32	RW	0000_0000h
8000_0418h	iATU Upper Target Address Register. (IATU_UPPER_TARGET_AD DR_OFF_OUTBOUND_2)	32	RW	0000_0000h
8000_0500h	iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_I NBOUND_2)	32	RW	0000_0000h
8000_0504h	iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_I NBOUND_2)	32	RW	0000_0000h
8000_0508h	iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_O FF_INBOUND_2)	32	RW	0000_0000h
8000_050Ch	iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_ OFF_INBOUND_2)	32	RW	0000_0000h
8000_0510h	iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_INBOUND_ 2)	32	RW	0000_FFFFh
8000_0514h	iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_ OFF_INBOUND_2)	32	RW	0000_0000h
8000_0600h	iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_O UTBOUND_3)	32	RW	0000_0000h
8000_0604h	iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_O UTBOUND_3)	32	RW	0000_0000h
8000_0608h	iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_O FF_OUTBOUND_3)	32	RW	0000_0000h
8000_060Ch	iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_ OFF_OUTBOUND_3)	32	RW	0000_0000h
8000_0610h	iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_OUTB OUND_3)	32	RW	0000_FFFFh
8000_0614h	iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_ OFF_OUTBOUND_3)	32	RW	0000_0000h
8000_0618h	iATU Upper Target Address Register. (IATU_UPPER_TARGET_AD DR_OFF_OUTBOUND_3)	32	RW	0000_0000h
8000_0700h	iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_I NBOUND_3)	32	RW	0000_0000h
8000_0704h	iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_I NBOUND_3)	32	RW	0000_0000h

Table continues on the next page...

PCI Express (PCIe)

Offset	Register	Width (In bits)	Access	Reset value
8000_0708h	iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_OFF_INBOUND_3)	32	RW	0000_0000h
8000_070Ch	iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_INBOUND_3)	32	RW	0000_0000h
8000_0710h	iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_INBOUND_3)	32	RW	0000_FFFFh
8000_0714h	iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_INBOUND_3)	32	RW	0000_0000h
8008_0000h	DMA Arbitration Scheme for TRGT1 Interface. (DMA_CTRL_DATA_ARB_PRIOR_OFF)	32	RW	0000_0688h
8008_0008h	DMA Number of Channels Register. (DMA_CTRL_OFF)	32	RW	0001_0001h
8008_000Ch	DMA Write Engine Enable Register. (DMA_WRITE_ENGINE_EN_OFF)	32	RW	0000_0000h
8008_0010h	DMA Write Doorbell Register. (DMA_WRITE_DOORBELL_OFF)	32	RW	0000_0000h
8008_0018h	DMA Write Engine Channel Arbitration Weight Low Register. (DMA_WRITE_CHANNEL_ARB_WEIGHT_LOW_OFF)	32	RW	0000_8421h
8008_001Ch	DMA Write Engine Channel Arbitration Weight High Register. (DMA_WRITE_CHANNEL_ARB_WEIGHT_HIGH_OFF)	32	RW	0000_8421h
8008_002Ch	DMA Read Engine Enable Register. (DMA_READ_ENGINE_EN_OFF)	32	RW	0000_0000h
8008_0030h	DMA Read Doorbell Register. (DMA_READ_DOORBELL_OFF)	32	RW	0000_0000h
8008_0038h	DMA Read Engine Channel Arbitration Weight Low Register. (DMA_READ_CHANNEL_ARB_WEIGHT_LOW_OFF)	32	RW	0000_8421h
8008_003Ch	DMA Read Engine Channel Arbitration Weight High Register. (DMA_READ_CHANNEL_ARB_WEIGHT_HIGH_OFF)	32	RW	0000_8421h
8008_004Ch	DMA Write Interrupt Status Register. (DMA_WRITE_INT_STATUS_OFF)	32	RW	0000_0000h
8008_0054h	DMA Write Interrupt Mask Register. (DMA_WRITE_INT_MASK_OFF)	32	RW	0001_0001h
8008_0058h	DMA Write Interrupt Clear Register. (DMA_WRITE_INT_CLEAR_OFF)	32	W1C	0000_0000h
8008_005Ch	DMA Write Error Status Register (DMA_WRITE_ERR_STATUS_OFF)	32	RO	0000_0000h
8008_0060h	DMA Write Done IMWr Address Low Register. (DMA_WRITE_DONE_IMWR_LOW_OFF)	32	RW	0000_0000h
8008_0064h	DMA Write Done IMWr Interrupt Address High Register. (DMA_WRITE_DONE_IMWR_HIGH_OFF)	32	RW	0000_0000h
8008_0068h	DMA Write Abort IMWr Address Low Register. (DMA_WRITE_ABORT_IMWR_LOW_OFF)	32	RW	0000_0000h
8008_006Ch	DMA Write Abort IMWr Address High Register. (DMA_WRITE_ABORT_IMWR_HIGH_OFF)	32	RW	0000_0000h
8008_0070h	DMA Write Channel 1 and 0 IMWr Data Register. (DMA_WRITE_CH01_IMWR_DATA_OFF)	32	RW	0000_0000h
8008_0074h	DMA Write Channel 3 and 2 IMWr Data Register. (DMA_WRITE_CH23_IMWR_DATA_OFF)	32	RW	0000_0000h
8008_0078h	DMA Write Channel 5 and 4 IMWr Data Register. (DMA_WRITE_CH45_IMWR_DATA_OFF)	32	RW	0000_0000h

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
8008_007Ch	DMA Write Channel 7 and 6 IMWr Data Register. (DMA_WRITE_CH67_IMWR_DATA_OFF)	32	RW	0000_0000h
8008_0090h	DMA Write Linked List Error Enable Register. (DMA_WRITE_LINKED_LIST_ERR_EN_OFF)	32	RW	0000_0000h
8008_00A0h	DMA Read Interrupt Status Register. (DMA_READ_INT_STATUS_OFF)	32	RW	0000_0000h
8008_00A8h	DMA Read Interrupt Mask Register. (DMA_READ_INT_MASK_OFF)	32	RW	0001_0001h
8008_00ACh	DMA Read Interrupt Clear Register. (DMA_READ_INT_CLEAR_OFF)	32	WO	0000_0000h
8008_00B4h	DMA Read Error Status Low Register. (DMA_READ_ERR_STATUS_LOW_OFF)	32	RO	0000_0000h
8008_00B8h	DMA Read Error Status High Register. (DMA_READ_ERR_STATUS_HIGH_OFF)	32	RO	0000_0000h
8008_00C4h	DMA Read Linked List Error Enable Register. (DMA_READ_LINKED_LIST_ERR_EN_OFF)	32	RW	0000_0000h
8008_00CCh	DMA Read Done IMWr Address Low Register. (DMA_READ_DONE_IMWR_LOW_OFF)	32	RW	0000_0000h
8008_00D0h	DMA Read Done IMWr Address High Register. (DMA_READ_DONE_IMWR_HIGH_OFF)	32	RW	0000_0000h
8008_00D4h	DMA Read Abort IMWr Address Low Register. (DMA_READ_ABORT_IMWR_LOW_OFF)	32	RW	0000_0000h
8008_00D8h	DMA Read Abort IMWr Address High Register. (DMA_READ_ABORT_IMWR_HIGH_OFF)	32	RW	0000_0000h
8008_00DCh	DMA Read Channel 1 and 0 IMWr Data Register. (DMA_READ_CH01_IMWR_DATA_OFF)	32	RW	0000_0000h
8008_00E0h	DMA Read Channel 3 and 2 IMWr Data Register. (DMA_READ_CH23_IMWR_DATA_OFF)	32	RW	0000_0000h
8008_00E4h	DMA Read Channel 5 and 4 IMWr Data Register. (DMA_READ_CH45_IMWR_DATA_OFF)	32	RW	0000_0000h
8008_00E8h	DMA Read Channel 7 and 6 IMWr Data Register. (DMA_READ_CH67_IMWR_DATA_OFF)	32	RW	0000_0000h
8008_0200h	DMA Write Channel Control 1 Register. (DMA_CH_CONTROL1_OFF_WRCH_0)	32	RW	0000_0000h
8008_0208h	DMA Write Transfer Size Register. (DMA_TRANSFER_SIZE_OFF_WRCH_0)	32	RW	0000_0000h
8008_020Ch	DMA Write SAR Low Register. (DMA_SAR_LOW_OFF_WRCH_0)	32	RW	0000_0000h
8008_0210h	DMA Write SAR High Register. (DMA_SAR_HIGH_OFF_WRCH_0)	32	RW	0000_0000h
8008_0214h	DMA Write DAR Low Register. (DMA_DAR_LOW_OFF_WRCH_0)	32	RW	0000_0000h
8008_0218h	DMA Write DAR High Register. (DMA_DAR_HIGH_OFF_WRCH_0)	32	RW	0000_0000h
8008_021Ch	DMA Write Linked List Pointer Low Register. (DMA_LLP_LOW_OFF_WRCH_0)	32	RW	0000_0000h
8008_0220h	DMA Write Linked List Pointer High Register. (DMA_LLP_HIGH_OFF_WRCH_0)	32	RW	0000_0000h
8008_0300h	DMA Read Channel Control 1 Register. (DMA_CH_CONTROL1_OFF_RDCH_0)	32	RW	0000_0000h

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PCI Express (PCIe)

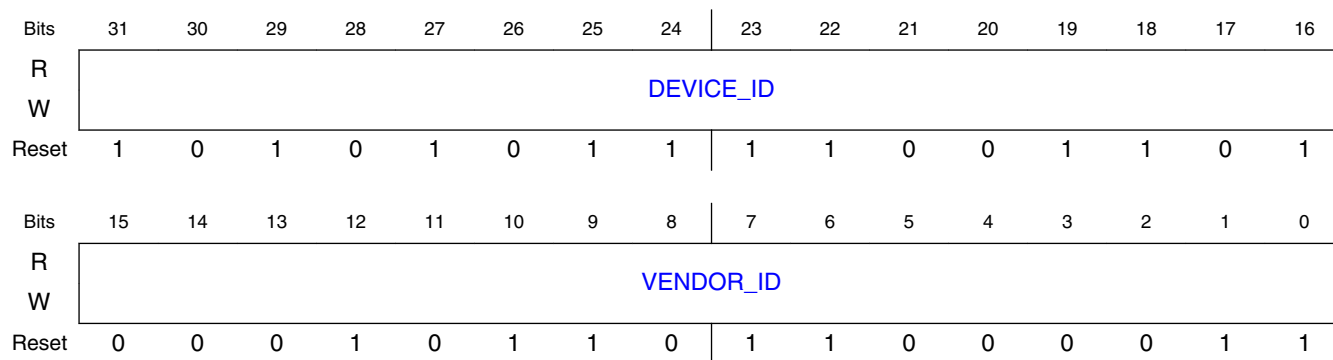
Offset	Register	Width (In bits)	Access	Reset value
8008_0308h	DMA Read Transfer Size Register. (DMA_TRANSFER_SIZE_OFF_RDCH_0)	32	RW	0000_0000h
8008_030Ch	DMA Read SAR Low Register. (DMA_SAR_LOW_OFF_RDCH_0)	32	RW	0000_0000h
8008_0310h	DMA Read SAR High Register. (DMA_SAR_HIGH_OFF_RDCH_0)	32	RW	0000_0000h
8008_0314h	DMA Read DAR Low Register. (DMA_DAR_LOW_OFF_RDCH_0)	32	RW	0000_0000h
8008_0318h	DMA Read DAR High Register. (DMA_DAR_HIGH_OFF_RDCH_0)	32	RW	0000_0000h
8008_031Ch	DMA Read Linked List Pointer Low Register. (DMA_LL_P_LOW_OFF_RDCH_0)	32	RW	0000_0000h
8008_0320h	DMA Read Linked List Pointer High Register. (DMA_LL_P_HIGH_OFF_RDCH_0)	32	RW	0000_0000h

11.3.3.1.2 Device ID and Vendor ID Register. (TYPE1_DEV_ID_VEND_ID_REG)

11.3.3.1.2.1 Offset

Register	Offset
TYPE1_DEV_ID_VEND_ID_REG	0h

11.3.3.1.2.2 Diagram



11.3.3.1.2.3 Fields

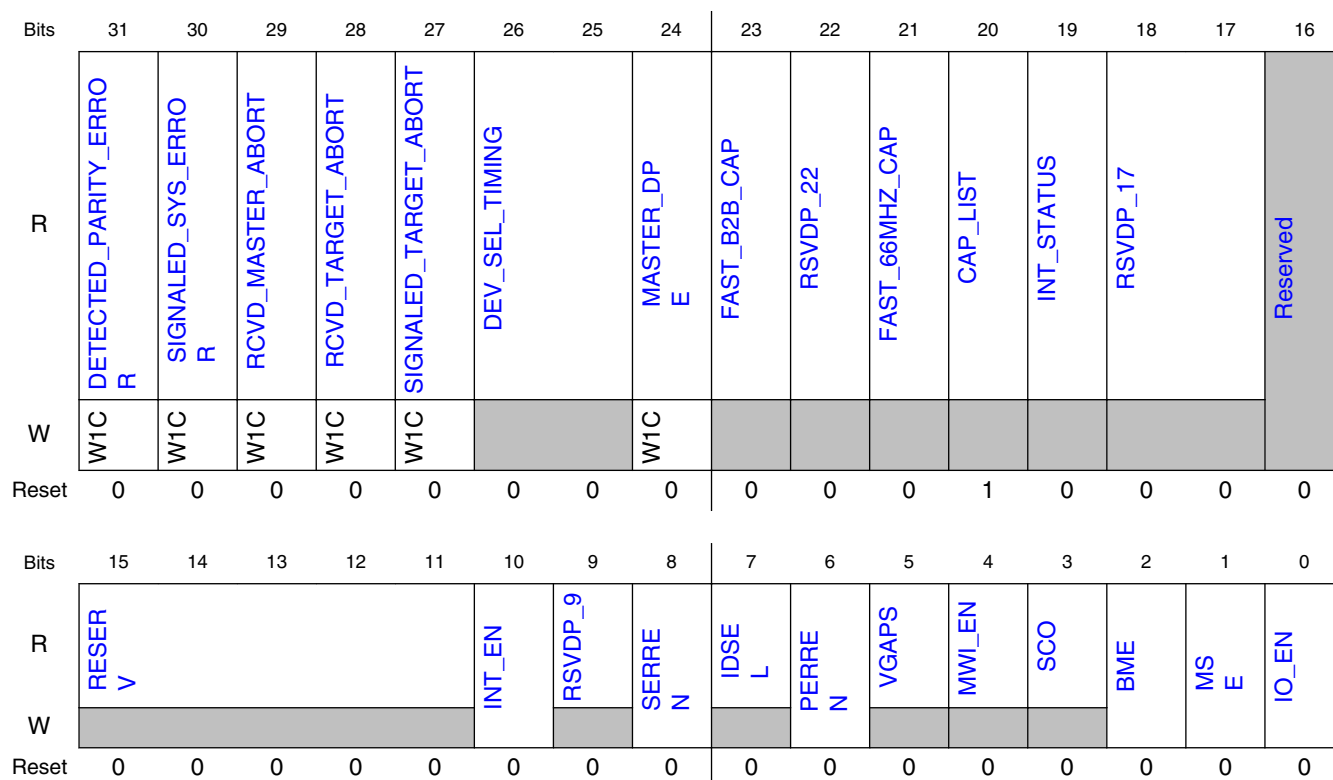
Field	Function
31-16 DEVICE_ID	Device ID. The Device ID register identifies the particular Function. This identifier is allocated by the vendor. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.
15-0 VENDOR_ID	Vendor ID. The Vendor ID register identifies the manufacturer of the Function. Valid vendor identifiers are allocated by the PCI-SIG to ensure uniqueness. It is not permitted to populate this register with a value of FFFFh, which is an invalid value for Vendor ID. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.

11.3.3.1.3 Status and Command Register. (TYPE1_STATUS_COMMAND_REG)

11.3.3.1.3.1 Offset

Register	Offset
TYPE1_STATUS_COMMAND_REG	4h

11.3.3.1.3.2 Diagram



11.3.3.1.3.3 Fields

Field	Function
31 DETECTED_PARITY_ERROR	Detected Parity Error. This bit is set by a Function whenever it receives a Poisoned TLP, regardless of the state the Parity Error Response bit in the Command register. The bit is set when the Poisoned TLP is received by a Function's primary side.
30 SIGNALLED_SYSTEM_ERROR	Signaled System Error. This bit is set when a Function sends an ERR_FATAL or ERR_NONFATAL Message, and the SERR# Enable bit in the Command register is 1b.
29 RCVD_MASTER_ABORT	Received Master Abort. This bit is set when a Requester receives a Completion with Unsupported Request Completion status. The bit is set when the Unsupported Request is received by a Function's primary side.
28 RCVD_TARGET_ABORT	Received Target Abort. This bit is set when a Requester receives a Completion with Completer Abort Completion status. The bit is set when the Completer Abort is received by a Function's primary side.
27 SIGNALLED_TARGET_ABORT	Signaled Target Abort. This bit is set when a Function completes a Posted or Non-Posted Request as a Completer Abort error. This applies to a Function when the Completer Abort was generated by its primary side.
26-25 DEV_SEL_TIMING	DEVSEL Timing. This field was originally described in the PCI Local Bus Specification. Its functionality does not apply to PCI Express. The controller hardwires it to 00b.
24 MASTER_DPE	Master Data Parity Error. This bit is set by a Function if the Parity Error Response bit in the Command register is 1b and either of the following two conditions occurs: - Port receives a Poisoned Completion going downstream - Port transmits a Poisoned Request upstream If the Parity Error Response bit is 0b, this bit is never set.
23 FAST_B2B_CAP	Fast Back-to-Back Transactions Capable. This bit was originally described in the PCI Local Bus Specification. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b.
22 RSVDP_22	Reserved for future use.
21 FAST_66MHZ_CAP	66 MHz Capable. This bit was originally described in the PCI Local Bus Specification. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b.
20 CAP_LIST	Capabilities List. Indicates the presence of an Extended Capability list item. Since all PCI Express device Functions are required to implement the PCI Express Capability structure, the controller hardwires this bit to 1b.
19 INT_STATUS	Interrupt Status. When set, indicates that an INTx emulation interrupt is pending internally in the Function. INTx emulation interrupts forwarded by Functions from the secondary side are not reflected in this bit. Setting the Interrupt Disable bit has no effect on the state of this bit. For Functions that do not generate INTx interrupts, the controller hardwires this bit to 0b.
18-17 RSVDP_17	Reserved for future use.
16 —	Reserved.
15-11	Reserved.

Table continues on the next page...

Field	Function
RESERV	
10 INT_EN	Interrupt Disable. Controls the ability of a Function to generate INTx emulation interrupts. When set, Functions are prevented from asserting INTx interrupts. Note: - Any INTx emulation interrupts already asserted by the Function must be deasserted when this bit is set. INTx interrupts use virtual wires that must, if asserted, be deasserted using the appropriate Deassert_INTx message(s) when this bit is set. - Only the INTx virtual wire interrupt(s) associated with the Function(s) for which this bit is set are affected. - For Functions that generate INTx interrupts on their own behalf, this bit is required. This bit has no effect on interrupts forwarded from the secondary side. For Functions that do not generate INTx interrupts on their own behalf this bit is optional. If this bit is not implemented, the controller hardwires it to 0b.
9 RSVDP_9	Reserved for future use.
8 SERREN	SERR# Enable. When set, this bit enables reporting upstream of Non-fatal and Fatal errors detected by the Function. Note: The errors are reported if enabled either through this bit or through the PCI Express specific bits in the Device Control register. For more details see the "Error Registers" section of the PCI Express Specification. In addition, this bit controls transmission by the primary interface of ERR_NONFATAL and ERR_FATAL error Messages forwarded from the secondary interface. This bit does not affect the transmission of forwarded ERR_COR messages.
7 IDSEL	IDSEL Stepping/Wait Cycle Control. This bit was originally described in the PCI Local Bus Specification. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b.
6 PERREN	Parity Error Response. This bit controls the logging of poisoned TLPs in the Master Data Parity Error bit in the Status register. For more details see the "Error Registers" section of the PCI Express Specification.
5 VGAPS	VGA Palette Snoop. This bit was originally described in the PCI Local Bus Specification and the PCI-to-PCI Bridge Architecture Specification. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b.
4 MWI_EN	Memory Write and Invalidate. This bit was originally described in the PCI Local Bus Specification and the PCI-to-PCI Bridge Architecture Specification. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b. For PCI Express to PCI/PCI-X Bridges, refer to the PCI Express to PCI/PCI-X Bridge Specification for requirements for this register.
3 SCO	Special Cycle Enable. This bit was originally described in the PCI Local Bus Specification. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b.
2 BME	Bus Master Enable. This bit controls forwarding of Memory or I/O requests by a port in the Upstream direction. When this bit is 0b, Memory and I/O Requests received at a Root Port must be handled as Unsupported Requests (UR) For Non-Posted Requests a Completion with UR completion status must be returned. This bit does not affect forwarding of Completions in either the Upstream or Downstream direction. The forwarding of Requests other than Memory or I/O Requests is not controlled by this bit.
1 MSE	Memory Space Enable. This bit controls a Function's response to Memory Space accesses received on its primary side. - When set, the Function is enabled to decode the address and further process Memory Space accesses. - When clear, all received Memory Space accesses are caused to be handled as Unsupported Requests. You cannot write to this register if your configuration has no MEM bars; that is, the internal signal has_mem_bar = 0. Note: The access attributes of this field are as follows: - Dbi: ! has_mem_bar ? RO : RW
0 IO_EN	IO Space Enable. This bit controls a Function's response to I/O Space accesses received on its primary side. - When set, the Function is enabled to decode the address and further process I/O Space accesses. - When clear, all received I/O accesses are caused to be handled as Unsupported Requests. You cannot write to this register if your configuration has no IO bars; that is, the internal signal has_io_bar = 0. Note: The access attributes of this field are as follows: - Dbi: ! has_io_bar ? RO : RW

11.3.3.1.4 Class Code and Revision ID Register. (TYPE1_CLASS_CODE_REV_ID_REG)

11.3.3.1.4.1 Offset

Register	Offset
TYPE1_CLASS_CODE_REV_ID_REG	8h

11.3.3.1.4.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	BASE_CLASS_CODE								SUBCLASS_CODE							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PROGRAM_INTERFACE								REVISION_ID							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

11.3.3.1.4.3 Fields

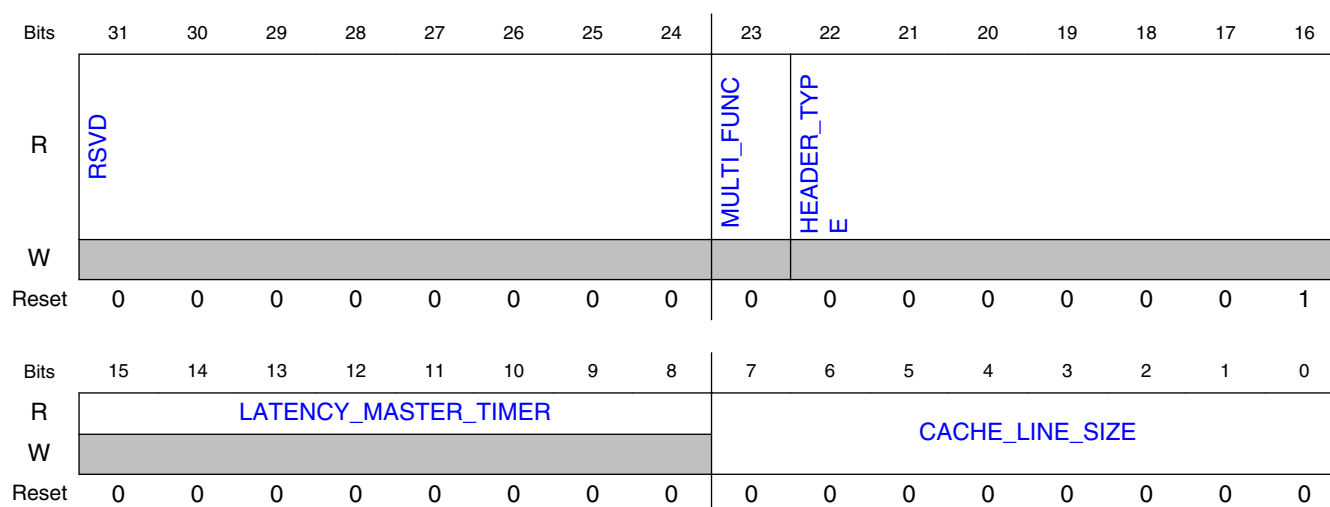
Field	Function
31-24 BASE_CLASS_CODE	Base Class Code. A code that broadly classifies the type of operation the Function performs. Encodings for base class, are provided in the PCI Code and ID Assignment Specification. All unspecified encodings are reserved. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.
23-16 SUBCLASS_CODE	Sub-Class Code. Specifies a base class sub-class, which identifies more specifically the operation of the Function. Encodings for sub-class are provided in the PCI Code and ID Assignment Specification. All unspecified encodings are reserved. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.
15-8 PROGRAM_INTERFACE	Programming Interface. This field identifies a specific register level programming interface (if any) so that device independent software can interact with the Function. Encodings for interface are provided in the PCI Code and ID Assignment Specification. All unspecified encodings are reserved. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.
7-0 REVISION_ID	Revision ID. The value of this field specifies a Function specific revision identifier. The value is chosen by the vendor. Zero is an acceptable value. The Revision ID should be viewed as a vendor defined extension to the Device ID. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.

11.3.3.1.5 Header Type, Latency Timer, and Cache Line Size Register. (TYPE1_HDR_TYPE_LAT_CACHE_LINE_SIZE_REG)

11.3.3.1.5.1 Offset

Register	Offset
TYPE1_HDR_TYPE_LAT_CACHE_LINE_SIZE_REG	Ch

11.3.3.1.5.2 Diagram



11.3.3.1.5.3 Fields

Field	Function
31-24 —	Reserved.
23 MULTI_FUNC	Multi-Function Device. - When set, indicates that the device may contain multiple Functions, but not necessarily. Software is permitted to probe for Functions other than Function 0. - When clear, software must not probe for Functions other than Function 0 unless explicitly indicated by another mechanism, such as an ARI or SR-IOV Capability structure. Except where stated otherwise, it is recommended that this bit be set if there are multiple Functions, and clear if there is only one Function. Note: This register field is sticky.
22-16 HEADER_TYPE	Header Layout. This field identifies the layout of the second part of the predefined header. The controller uses 000 0001b encoding. The encoding 000 0010b is reserved. This encoding was originally described in the PC Card Standard Electrical Specification and is used in previous versions of the programming model. Careful consideration should be given to any attempt to repurpose it.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
15-8 LATENCY_TIMER	Latency Timer. This register is also referred to as Primary Latency Timer. The Latency Timer was originally described in the PCI Local Bus Specification and the PCI-to-PCI Bridge Architecture Specification. Its functionality does not apply to PCI Express. The controller hardwires this register to 00h.
7-0 CACHE_LINE_SIZE	Cache Line Size. The Cache Line Size register is programmed by the system firmware or the operating system to system cache line size. However, legacy conventional PCI software may not always be able to program this register correctly especially in the case of Hot-Plug devices. This read-write register is implemented for legacy compatibility purposes but has no effect on any PCI Express device behavior.

11.3.3.1.6 Secondary Latency Timer, Subordinate Bus Number, Secondary Bus Number, and Primary Bus Number Register. (SEC_LAT_TIMER_SUB_BUS_SEC_BUS_PRI_BUS_REG)

11.3.3.1.6.1 Offset

Register	Offset
SEC_LAT_TIMER_SUB_BUS_SEC_BUS_PRI_BUS_REG	18h

11.3.3.1.6.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SEC_LAT_TIMER								SUB_BUS							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SEC_BUS								PRIM_BUS							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.6.3 Fields

Field	Function
31-24 SEC_LAT_TIMER	Secondary Latency Timer. This register does not apply to PCI Express. The controller hardwires it to 00h.
23-16	Subordinate Bus Number. The Subordinate Bus Number register is used to record the bus number of the highest numbered PCI bus segment which is behind (or subordinate to) the bridge. Configuration

Table continues on the next page...

Field	Function
SUB_BUS	software programs the value in this register. The bridge uses this register in conjunction with the Secondary Bus Number register to determine when to respond to and pass on a Type 1 configuration transaction on the primary interface to the secondary interface.
15-8 SEC_BUS	Secondary Bus Number. The Secondary Bus Number register is used to record the bus number of the PCI bus segment to which the secondary interface of the bridge is connected. Configuration software programs the value in this register. The bridge uses this register to determine when to respond to and convert a Type 1 configuration transaction on the primary interface into a Type 0 transaction on the secondary interface.
7-0 PRIM_BUS	Primary Bus Number. This register is not used by PCI Express Functions. It is implemented for compatibility with legacy software.

11.3.3.1.7 Secondary Status, and I/O Limit and Base Register. (SEC_STAT_IO_LIMIT_IO_BASE_REG)

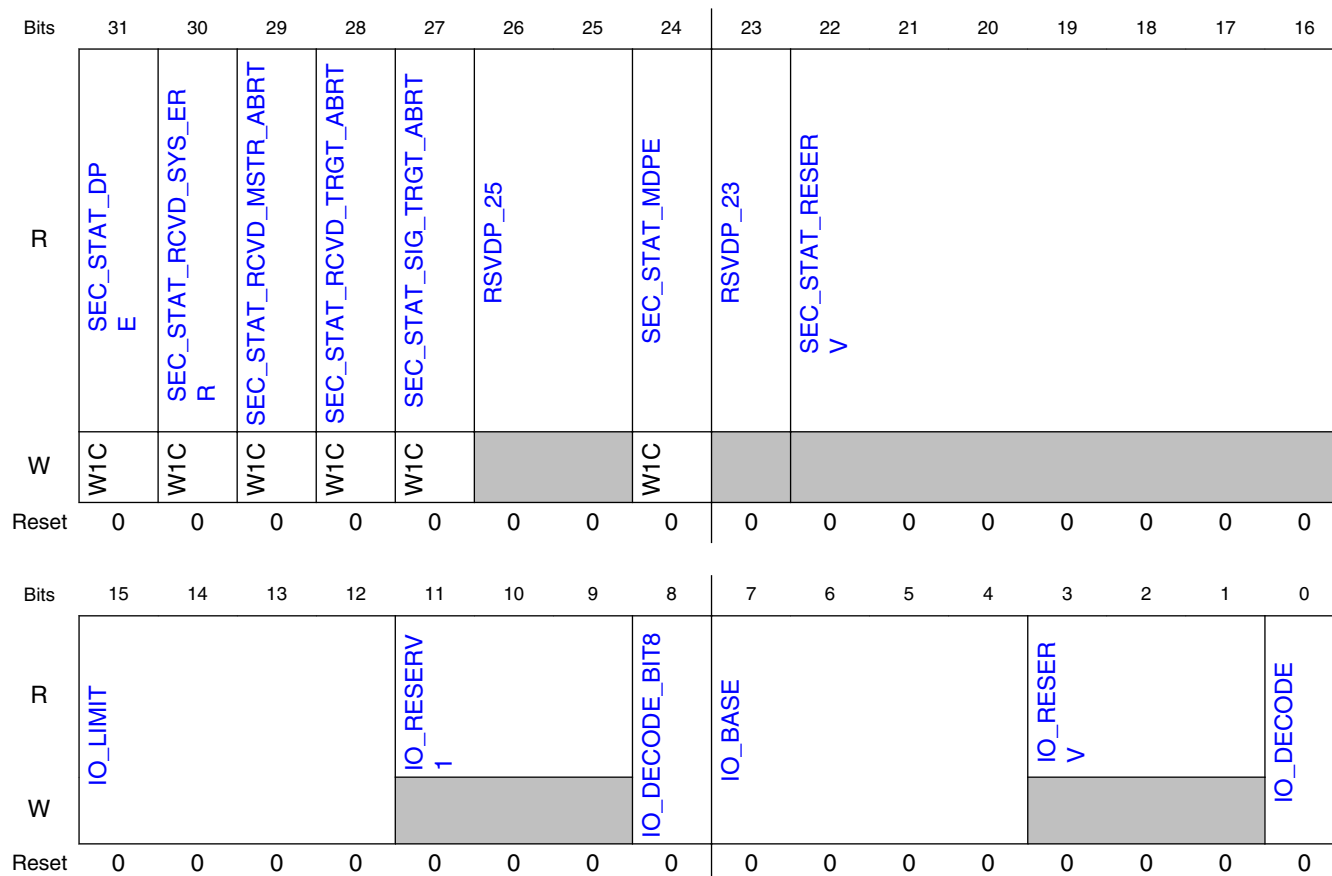
11.3.3.1.7.1 Offset

Register	Offset
SEC_STAT_IO_LIMIT_IO_BASE_REG	1Ch

11.3.3.1.7.2 Function

Secondary Status, and I/O Limit and Base Register. The I/O Limit and I/O Base registers are optional and define an address range that is used by the bridge to determine when to forward I/O transactions from one interface to the other. If a bridge does not implement an I/O address range, then both the I/O Limit and I/O Base registers must be implemented as read-only registers that return zero when read. If a bridge supports an I/O address range, then these registers must be initialized by configuration software so default states are not specified.

11.3.3.1.7.3 Diagram



11.3.3.1.7.4 Fields

Field	Function
31 SEC_STAT_DP E	Detected Parity Error. This bit is set by a Function when a Poisoned TLP is received by its secondary side, regardless of the state the Parity Error Response Enable bit in the Bridge Control register.
30 SEC_STAT_RC VD_SYS_ERR	Received System Error. This bit is set when the secondary side of a Function receives an ERR_FATAL or ERR_NONFATAL message.
29 SEC_STAT_RC VD_MSTR_ABR T	Received Master Abort. This bit is set when the secondary side of a Function (for requests initiated by the Type 1 header Function itself) receives a Completion with Unsupported Request Completion status.
28 SEC_STAT_RC VD_TRGT_ABR T	Received Target Abort. This bit is set when the secondary side of a Function (for requests initiated by the Type 1 header Function itself) receives a Completion with Completer Abort Completion status.

Table continues on the next page...

Field	Function
27 SEC_STAT_SIG_TRGT_ABRT	Signaled Target Abort. This bit is set when the secondary side of the Function (for Requests completed by the Type 1 header Function itself) completes a Posted or Non-Posted request as a Completer Abort error.
26-25 RSVDP_25	Reserved for future use.
24 SEC_STAT_MDPE	Master Data Parity Error. This bit is set by a Function if the Parity Error Response Enable bit in the Bridge Control register is set, and either of the following two conditions occurs: - Port receives a Poisoned Completion coming Upstream - Port transmits a Poisoned Request Downstream If the Parity Error Response Enable bit is clear, this bit is never set.
23 RSVDP_23	Reserved for future use.
22-16 SEC_STAT_RESERV	Reserved.
15-12 IO_LIMIT	I/O Limit Address. These bits correspond to the address[15:12] of IO address range. For the purpose of address decoding, the bridge assumes that the lower 12 address bits, address[11:0], of the I/O limit address (not implemented in the I/O Limit register) are FFFh. The I/O Limit register can be programmed to a smaller value than the I/O Base register, if there are no I/O addresses on the secondary side of the bridge. In this case, the bridge will not forward any I/O transactions from the primary bus to the secondary and will forward all I/O transactions from the secondary bus to the primary bus.
11-9 IO_RESERV1	Reserved.
8 IO_DECODE_BIT8	I/O Addressing Encode (IO Limit Address). This bit encodes the IO addressing capability of the bridge. IO_DECODE_BIT8 indicates the following: - 0h: The bridge supports only 16-bit I/O addressing (for ISA compatibility). For the purpose of address decoding, the bridge assumes that the upper 16 address bits, Address[31:16], of the I/O limit address (not implemented in I/O Limit register) are zero. Note: The bridge must still perform a full 32-bit decode of the I/O address (that is, check that Address[31:16] are 0000h). In this case, the I/O address range supported by the bridge will be restricted to the first 64 KB of I/O Space (0000 0000h to 0000 FFFFh). - 01h: The bridge supports 32-bit I/O address decoding, and the I/O Limit Upper 16 Bits hold the upper 16 bits, corresponding to Address[31:16], of the 32-bit Limit address. In this case, system configuration software is permitted to locate the I/O address range supported by the bridge anywhere in the 4-GB I/O Space. Note: The 4-KB alignment and granularity restrictions still apply when the bridge supports 32-bit I/O addressing. Note: The access attributes of this field are as follows: - Dbi: R
7-4 IO_BASE	I/O Base Address. These bits correspond to the address[15:12] of IO address range. For the purpose of address decoding, the bridge assumes that the lower 12 address bits, address[11:0], of the I/O base address (not implemented in the I/O Base register) are zero.
3-1 IO_RESERV	Reserved.
0 IO_DECODE	I/O Addressing Encode (IO Base Address) This bit encodes the IO addressing capability of the bridge. IO_DECODE indicates the following: - 0h: The bridge supports only 16-bit I/O addressing (for ISA compatibility). For the purpose of address decoding, the bridge assumes that the upper 16 address bits, Address[31:16], of the I/O base address (not implemented in I/O base register) are zero. Note: The bridge must still perform a full 32-bit decode of the I/O address (that is, check that Address[31:16] are 0000h). In this case, the I/O address range supported by the bridge will be restricted to the first 64 KB of I/O Space (0000 0000h to 0000 FFFFh). - 01h: The bridge supports 32-bit I/O address decoding, and the I/O Base Upper 16 Bits hold the upper 16 bits, corresponding to Address[31:16], of the 32-bit Base address. In this case, system configuration software is permitted to locate the I/O address range supported by the bridge anywhere in the 4-GB I/O Space. Note: The 4-KB alignment and granularity restrictions still apply when the bridge supports 32-bit I/O addressing. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R

11.3.3.1.8 Memory Limit and Base Register. (MEM_LIMIT_MEM_BASE_REG)

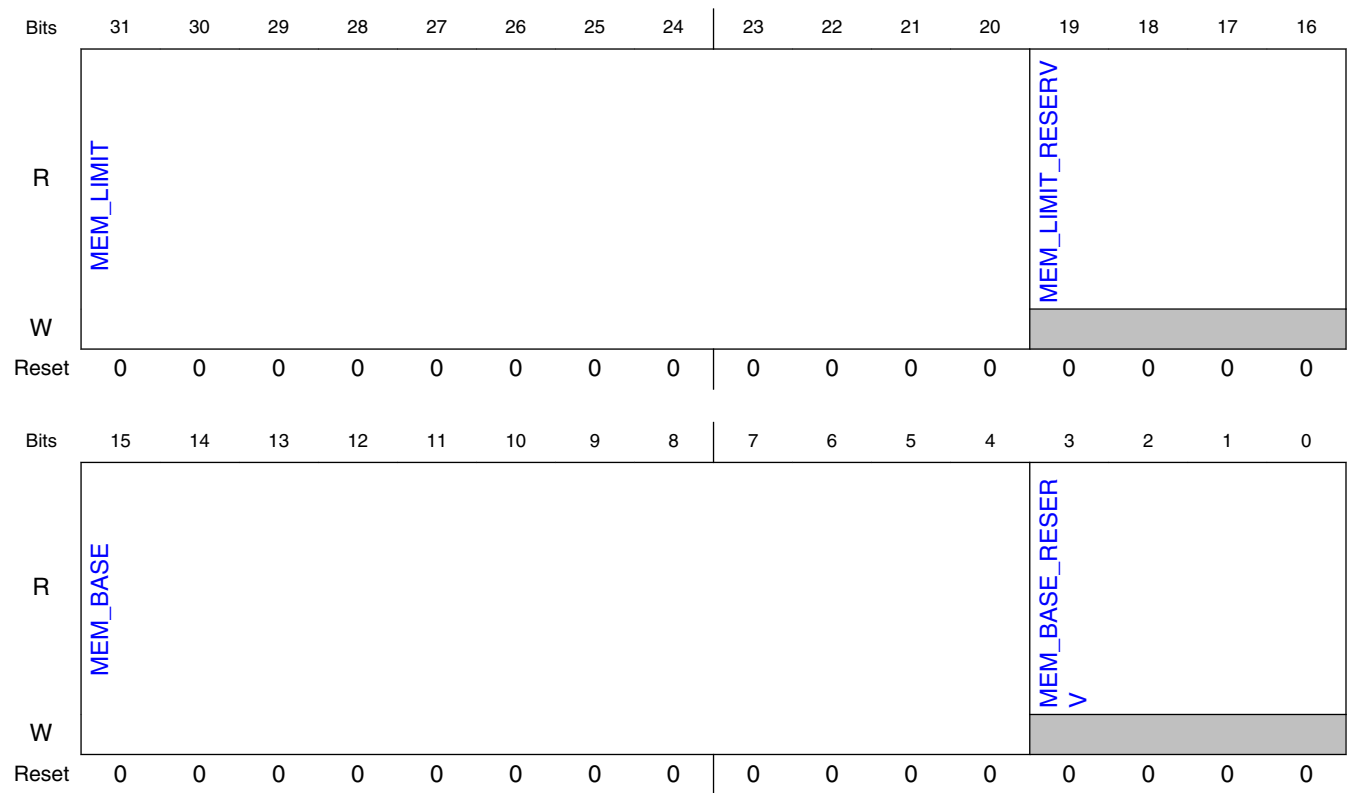
11.3.3.1.8.1 Offset

Register	Offset
MEM_LIMIT_MEM_BASE_REG	20h

11.3.3.1.8.2 Function

Memory Limit and Base Register. The Memory Limit and Memory Base registers define a memory mapped address range which is used by the bridge to determine when to forward memory transactions from one interface to the other. If there is no prefetchable memory space, and there is no memory-mapped space on the secondary side of the bridge, then the bridge will not forward any memory transactions from the primary bus to the secondary bus and will forward all memory transactions from the secondary bus to the primary bus.

11.3.3.1.8.3 Diagram



11.3.3.1.8.4 Fields

Field	Function
31-20 MEM_LIMIT	Memory Limit Address. These bits correspond to the upper 12 address bits, Address[31:20], of 32-bit addresses. For the purpose of address decoding, the bridge assumes that the lower 20 address bits, Address[19:0], of the memory limit address (not implemented in the Memory Limit register) are F FFFFh. The Memory Limit register must be programmed to a smaller value than the Memory Base register if there is no memory-mapped address space on the secondary side of the bridge.
19-16 MEM_LIMIT_RESERVED	Reserved.
15-4 MEM_BASE	Memory Base Address. These bits correspond to the upper 12 address bits, Address[31:20], of 32-bit addresses. For the purpose of address decoding, the bridge assumes that the lower 20 address bits, Address[19:0], of the memory base address (not implemented in the Memory Base register) are zero.
3-0 MEM_BASE_RESERVED	Reserved.

11.3.3.1.9 Prefetchable Memory Limit and Base Register. (PREF_MEM_LIMIT_PREF_MEM_BASE_REG)

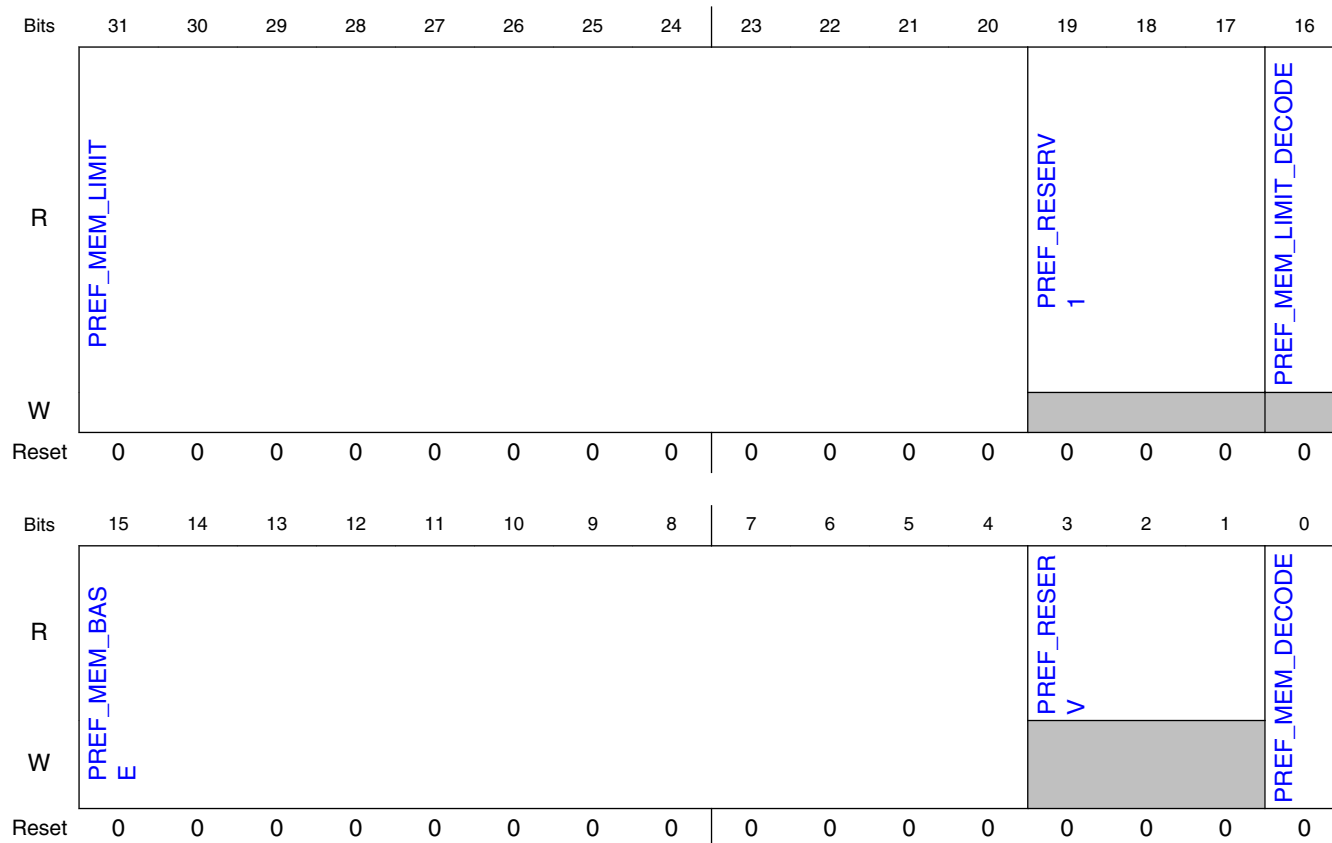
11.3.3.1.9.1 Offset

Register	Offset
PREF_MEM_LIMIT_PREF_MEM_BASE_REG	24h

11.3.3.1.9.2 Function

Prefetchable Memory Limit and Base Register. The Prefetchable Memory Limit and Prefetchable Memory Base registers must indicate that 64-bit addresses are supported, as defined in PCI-to-PCI Bridge Architecture Specification. The Prefetchable Memory Limit and Prefetchable Memory Base registers are optional. They define a prefetchable memory address range which is used by the bridge to determine when to forward memory transactions from one interface to the other (see the PCI-to-PCI Bridge Architecture Specification for additional details).

11.3.3.1.9.3 Diagram



11.3.3.1.9.4 Fields

Field	Function
31-20 PREF_MEM_LIMIT	Prefetchable Memory Limit Address. If the Prefetchable Memory Limit register indicates support for 32-bit addressing, then the Prefetchable Limit Upper 32 Bits register is implemented as a read-only register that returns zero when read. If the Prefetchable Memory Limit registers indicate support for 64-bit addressing, then the Prefetchable Limit Upper 32 Bits register is implemented as a read/write register which must be initialized by configuration software. If a 64-bit prefetchable memory address range is supported, the Prefetchable Limit Upper 32 Bits register specifies the upper 32 bits, corresponding to Address[63:32], of the 64-bit limit addresses which specify the prefetchable memory address range.
19-17 PREF_RESERV_1	Reserved.
16 PREF_MEM_LIMIT_DECODE	Prefetchable Memory Limit Decode. This bit encodes whether or not the bridge supports 64-bit addresses. The value of PREF_MEM_LIMIT_DECODE indicates the following: - 0b: Indicates that the bridge supports only 32 bit addresses - 1b: Indicates that the bridge supports 64 bit addresses. Prefetchable Limit Upper 32 Bits registers holds the rest of the 64-bit prefetchable limit address.
15-4 PREF_MEM_BASE	Prefetchable Memory Base Address. If the Prefetchable Memory Base register indicates support for 32-bit addressing, then the Prefetchable Base Upper 32 Bits register is implemented as a read-only register that returns zero when read. If the Prefetchable Memory Base register indicates support for 64-bit addressing, then the Prefetchable Limit Upper 32 Bits register is implemented as a read/write register

Table continues on the next page...

Field	Function
	which must be initialized by configuration software. If a 64-bit prefetchable memory address range is supported, the Prefetchable Base Upper 32 Bits register specifies the upper 32 bits, corresponding to Address[63:32], of the 64-bit base addresses which specify the prefetchable memory address range.
3-1 PREF_RESERV	Reserved.
0 PREF_MEM_DECODE	Prefetchable Memory Base Decode. This bit encodes whether or not the bridge supports 64-bit addresses. The value of PREF_MEM_DECODE indicates the following: - 0b: Indicates that the bridge supports only 32 bit addresses. - 1b: Indicates that the bridge supports 64 bit addresses. Prefetchable Base Upper 32 Bits registers holds the rest of the 64-bit prefetchable base address. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.

11.3.3.1.10 Prefetchable Base Upper 32 Bits Register. (PREF_BASE_UPPER_REG)

11.3.3.1.10.1 Offset

Register	Offset
PREF_BASE_UPPER_REG	28h

11.3.3.1.10.2 Function

Prefetchable Base Upper 32 Bits Register. The Prefetchable Base Upper 32 Bits register is an optional extension to the Prefetchable Memory Base register.

11.3.3.1.10.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	PREF_MEM_BASE_UPPER															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PREF_MEM_BASE_UPPER															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.10.4 Fields

Field	Function
31-0 PREF_MEM_BASE_UPPER	Prefetchable Base Upper 32 Bit. If the Prefetchable Memory Base register indicates support for 32-bit addressing, then this register is implemented as read-only register that returns zero when read. If the Prefetchable Memory Base register indicate support for 64-bit addressing, then this register is implemented as read/write register which must be initialized by configuration software. This register specifies the upper 32 bits, corresponding to Address[63:32], of the 64-bit base addresses which specify the prefetchable memory address range. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R

11.3.3.1.11 Prefetchable Limit Upper 32 Bits Register. (PREF_LIMIT_UPPER_REG)

11.3.3.1.11.1 Offset

Register	Offset
PREF_LIMIT_UPPER_REG	2Ch

11.3.3.1.11.2 Function

Prefetchable Limit Upper 32 Bits Register. The Prefetchable Limit Upper 32 Bits register is an optional extension to the Prefetchable Memory Limit register.

11.3.3.1.11.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	PREF_MEM_LIMIT_UPPER															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PREF_MEM_LIMIT_UPPER															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.11.4 Fields

Field	Function
31-0 PREF_MEM_LI MIT_UPPER	Prefetchable Limit Upper 32 Bit. If the Prefetchable Memory Limit register indicate support for 64-bit addressing, then this register is implemented as read/write register which must be initialized by configuration software. This register specifies the upper 32 bits, corresponding to Address[63:32], of the 64-bit base addresses which specify the prefetchable memory address range. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R

11.3.3.1.12 I/O Limit and Base Upper 16 Bits Register. (IO_LIMIT_UPPER_I O_BASE_UPPER_REG)

11.3.3.1.12.1 Offset

Register	Offset
IO_LIMIT_UPPER_IO_ BASE_UPPER_REG	30h

11.3.3.1.12.2 Function

I/O Limit and Base Upper 16 Bits Register. The I/O Limit Upper 16 Bits and I/O Base Upper 16 Bits registers are optional extensions to the I/O Limit and I/O Base registers.

11.3.3.1.12.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	IO_LIMIT_UPPER															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	IO_BASE_UPPER															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.12.4 Fields

Field	Function
31-16 IO_LIMIT_UPPE R	I/O Limit Upper 16 Bits. If the I/O Limit register indicates support for 16-bit I/O address decoding, then this register is implemented as a read-only register which return zero when read. If the I/O Limit register indicates support for 32-bit I/O addressing, then this register must be initialized by configuration software.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
	If 32-bit I/O address decoding is supported, this register specifies the upper 16 bits, corresponding to Address[31:16], of the 32-bit limit address, that specify the I/O address range. See the PCI-to-PCI Bridge Architecture Specification for additional details). Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
15-0 IO_BASE_UPPER	I/O Base Upper 16 Bits. If the I/O Base register indicates support for 16-bit I/O address decoding, then this register is implemented as a read-only register which return zero when read. If the I/O base register indicates support for 32-bit I/O addressing, then this register must be initialized by configuration software. If 32-bit I/O address decoding is supported, this register specifies the upper 16 bits, corresponding to Address[31:16], of the 32-bit base address, that specify the I/O address range. See the PCI-to-PCI Bridge Architecture Specification for additional details. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R

11.3.3.1.13 Capabilities Pointer Register. (TYPE1_CAP_PTR_REG)

11.3.3.1.13.1 Offset

Register	Offset
TYPE1_CAP_PTR_REG	34h

11.3.3.1.13.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	RSVDP_8															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	RSVDP_8								CAP_POINTER							
W																
Reset	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

11.3.3.1.13.3 Fields

Field	Function
31-8 RSVDP_8	Reserved for future use.
7-0 CAP_POINTER	Capabilities Pointer. This register is used to point to a linked list of capabilities implemented by this Function. Since all PCI Express Functions are required to implement the PCI Express Capability structure, this register must point to a valid capability structure and either this structure is the PCI Express Capability structure, or a subsequent list item points to the PCI Express Capability structure. The bottom

Field	Function
	two bits are Reserved and must be set to 00b. Software must mask these bits off before using this register as a pointer in Configuration Space to the first entry of a linked list of new capabilities. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.

11.3.3.1.14 Expansion ROM Base Address Register. (TYPE1_EXP_ROM_BASE_REG)

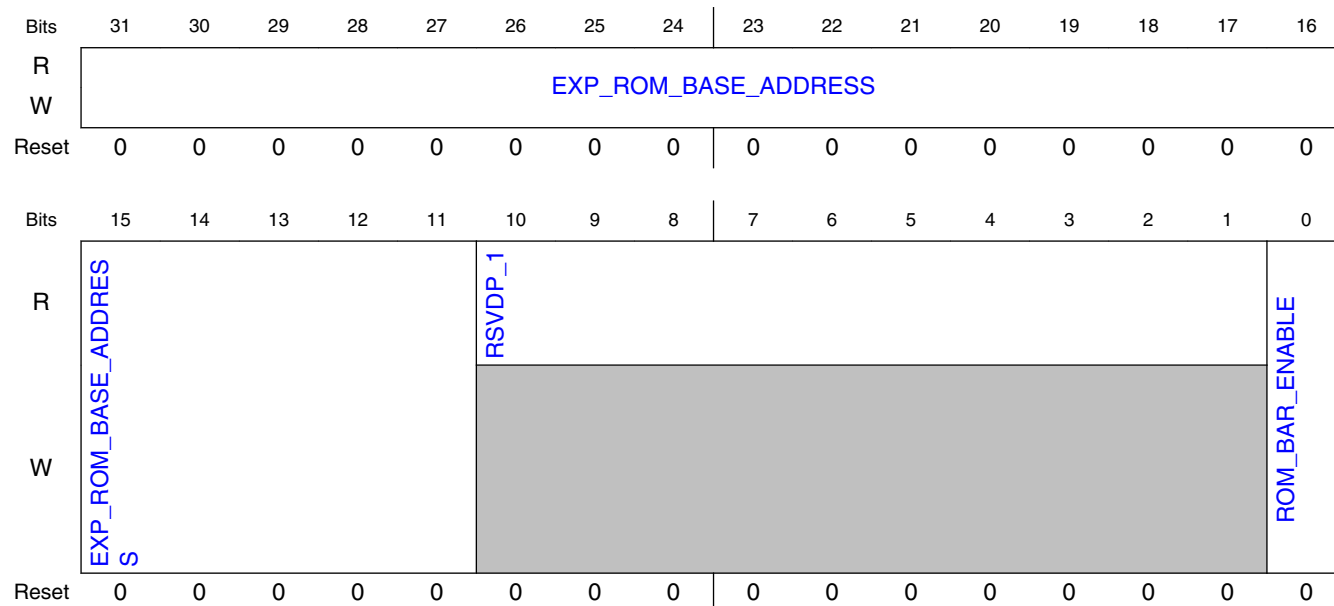
11.3.3.1.14.1 Offset

Register	Offset
TYPE1_EXP_ROM_BASE_REG	38h

11.3.3.1.14.2 Function

Expansion ROM Base Address Register. This register is defined to handle the base address and size information for this expansion ROM. The mask for this ROM BAR exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address.

11.3.3.1.14.3 Diagram



11.3.3.1.14.4 Fields

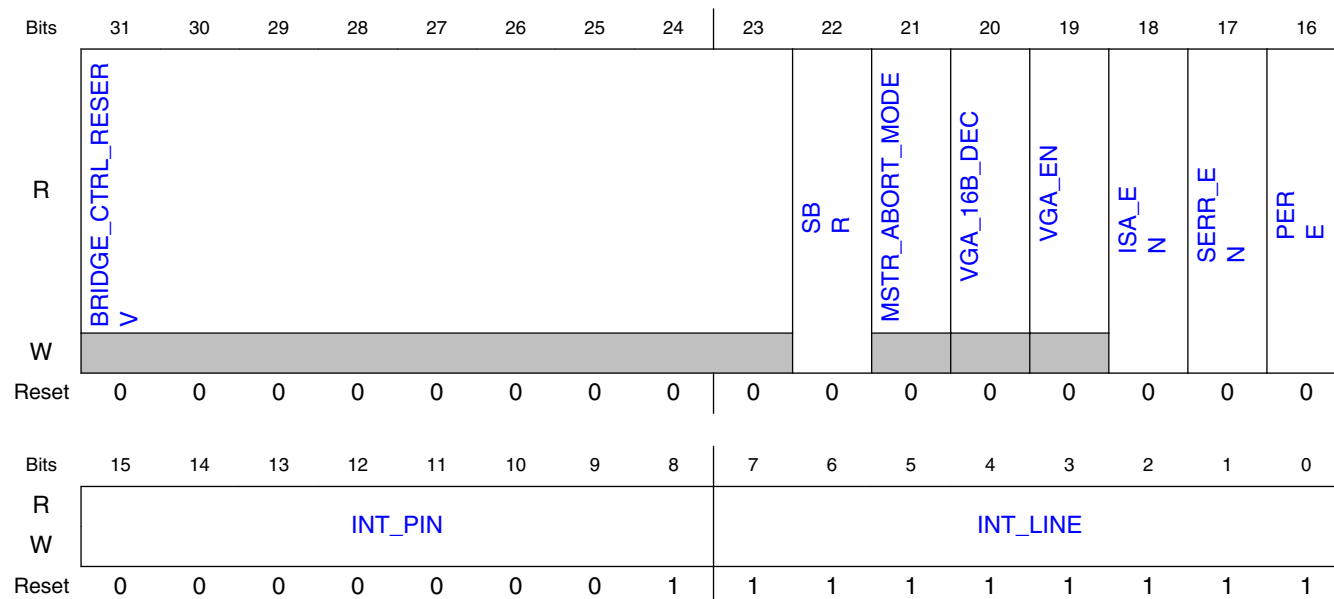
Field	Function
31-11 EXP_ROM_BASE_ADDRESS	Expansion ROM Base Address. Upper 21 bits of the Expansion ROM base address. The number of bits (out of these 21) that a Function actually implements depends on how much address space the Function requires. The mask for this ROM BAR exists (if implemented) as a shadow register at this address. The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the second register at this address. Note: The access attributes of this field are as follows: - Dbi: R/W
10-1 RSVDP_1	Reserved for future use.
0 ROM_BAR_ENABLE	Expansion ROM Enable. This bit controls whether or not the Function accepts accesses to its expansion ROM. When this bit is 0b, the Function's expansion ROM address space is disabled. When the bit is 1b, address decoding is enabled using the parameters in the other part of the Expansion ROM Base Address register. The Memory Space Enable bit in the Command register has precedence over the Expansion ROM Enable bit. A Function must claim accesses to its expansion ROM only if both the Memory Space Enable bit and the Expansion ROM Enable bit are set. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.15 Bridge Control, Interrupt Pin, and Interrupt Line Register. (BRIDGE_CTRL_INT_PIN_INT_LINE_REG)

11.3.3.1.15.1 Offset

Register	Offset
BRIDGE_CTRL_INT_PIN_INT_LINE_REG	3Ch

11.3.3.1.15.2 Diagram



11.3.3.1.15.3 Fields

Field	Function
31-23 BRIDGE_CTRL_RESERV	Reserved.
22 SBR	Secondary Bus Reset. Setting this bit triggers a hot reset on the corresponding PCI Express Port. Software must ensure a minimum reset duration (Trst) as defined in the PCI Local Bus Specification. Software and systems must honor first-access-following-reset timing requirements, unless the Readiness Notifications mechanism is used or if the Immediate Readiness bit in the relevant Function's Status Register register is set. Port configuration registers must not be changed, except as required to update Port status.
21 MSTR_ABORT_MODE	Master Abort Mode. This bit was originally described in the PCI-to-PCI Bridge Architecture Specification. Its functionality does not apply to PCI Express. The controller hardwires this bit to 0b. Note: The access attributes of this field are as follows: - Dbi: R/W
20 VGA_16B_DEC	VGA 16 bit decode. This bit only has meaning if VGA Enable bit is set. This bit enables system configuration software to select between 10-bit and 16-bit I/O address decoding for all VGA I/O register accesses that are forwarded from primary to secondary. The following actions are taken based on the value of the VGA_16B_DEC bit: - 0b: Execute 10-bit address decodes on VGA I/O accesses - 1b: Execute 16-bit address decodes on VGA I/O accesses For Functions that do not support VGA, the controller hardwires this bit to 0b. Note: The access attributes of this field are as follows: - Dbi: R
19 VGA_EN	VGA Enable. Modifies the response by the bridge to VGA compatible addresses. If the VGA Enable bit is set, the bridge will positively decode and forward the following accesses on the primary interface to the secondary interface (and, conversely, block the forwarding of these addresses from the secondary to primary interface): - Memory accesses in the range 000A 0000h to 000B FFFFh - I/O addresses in the first 64 KB of the I/O address space (Address[31:16] are 0000h) where Address[9:0] are in the ranges 3B0h to 3BBh and 3C0h to 3DFh (inclusive of ISA address aliases determined by the setting of VGA 16-bit Decode) If the VGA Enable bit is set, forwarding of these accesses is independent of the I/O address range and memory address ranges defined by the I/O Base and Limit registers, the Memory Base and

Table continues on the next page...

Field	Function
	Limit registers, and the Prefetchable Memory Base and Limit registers of the bridge. (Forwarding of these accesses is also independent of the setting of the ISA Enable bit (in the Bridge Control register) when the VGA Enable bit is set. Forwarding of these accesses is qualified by the I/O Space Enable and Memory Space Enable bits in the Command register.) The following actions are taken based on the value of the VGA_EN bit: - 0b: Do not forward VGA compatible memory and I/O addresses from the primary to the secondary interface (addresses defined above) unless they are enabled for forwarding by the defined I/O and memory address ranges - 1b: Forward VGA compatible memory and I/O addresses (addresses defined above) from the primary interface to the secondary interface (if the I/O Space Enable and Memory Space Enable bits are set) independent of the I/O and memory address ranges and independent of the ISA Enable bit For Functions that do not support VGA, the controller hardwires this bit to 0b. Note: The access attributes of this field are as follows: - Dbi: R
18 ISA_EN	ISA Enable. Modifies the response by the bridge to ISA I/O addresses. This applies only to I/O addresses that are enabled by the I/O Base and I/O Limit registers and are in the first 64 KB of I/O address space (0000 0000h to 0000 FFFFh). If this bit is set, the bridge will block any forwarding from primary to secondary of I/O transactions addressing the last 768 bytes in each 1-KB block. In the opposite direction (secondary to primary), I/O transactions will be forwarded if they address the last 768 bytes in each 1-KB block. The following actions are taken based on the value of the ISA_EN bit: - 0b: Forward downstream all I/O addresses in the address range defined by the I/O Base and I/O Limit registers - 1b: Forward upstream ISA I/O addresses in the address range defined by the I/O Base and I/O Limit registers that are in the first 64 KB of PCI I/O address space (top 768 bytes of each 1-KB block.
17 SERR_EN	SERR# Enable. This bit controls forwarding of ERR_COR, ERR_NONFATAL and ERR_FATAL from secondary to primary.
16 PERE	Parity Error Response Enable. This bit controls the logging of poisoned TLPs in the Master Data Parity Error bit in the Secondary Status register.
15-8 INT_PIN	Interrupt PIN. The Interrupt Pin register register that identifies the legacy interrupt Message(s) the Function uses. Valid values are: - 01h, 02h, 03h, and 04h: map to legacy interrupt Messages for INTA, INTB, INTC, and INTD respectively. - 00h: indicates that the Function uses no legacy interrupt Message(s). - 05h through FFh: Reserved. PCI Express defines one legacy interrupt Message for a single Function device and up to four legacy interrupt Messages for a multi-Function device. For a single Function device, only INTA may be used. Any Function on a multi-Function device can use any of the INTx Messages. If a device implements a single legacy interrupt Message, it must be INTA; if it implements two legacy interrupt Messages, they must be INTA and INTB; and so forth. For a multi-Function device, all Functions may use the same INTx Message or each may have its own (up to a maximum of four Functions) or any combination thereof. A single Function can never generate an interrupt request on more than one INTx Message. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
7-0 INT_LINE	Interrupt Line. The Interrupt Line register communicates interrupt line routing information. The register must be implemented by any Function that uses an interrupt pin. Values in this register are programmed by system software and are system architecture specific. The Function itself does not use this value; rather the value in this register is used by device drivers and operating systems.

11.3.3.1.16 Power Management Capabilities Register. (CAP_ID_NXT_PTR_REG)

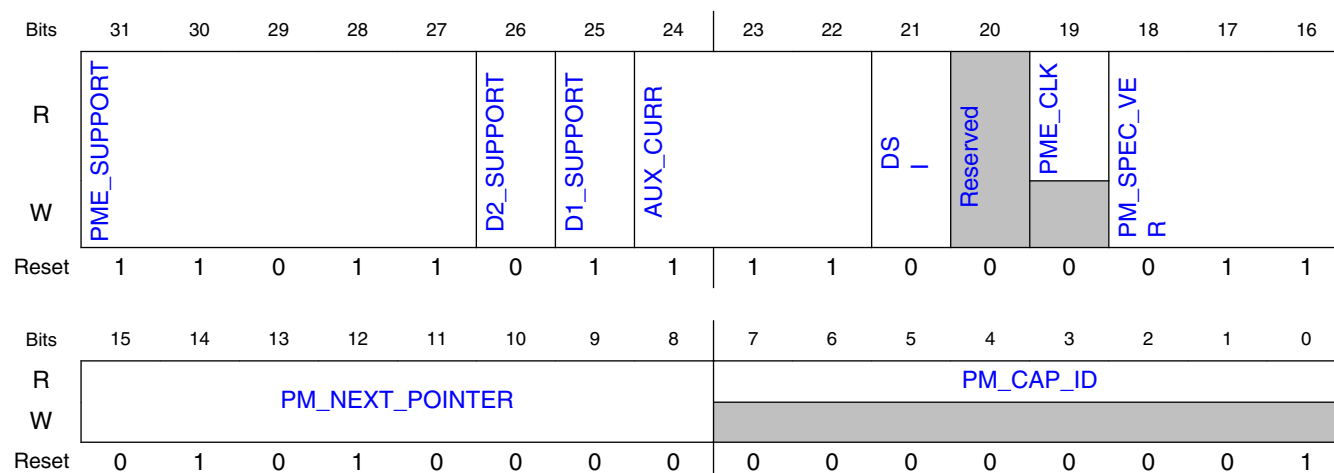
11.3.3.1.16.1 Offset

Register	Offset
CAP_ID_NXT_PTR_REG	40h

11.3.3.1.16.2 Function

Power Management Capabilities Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.16.3 Diagram



11.3.3.1.16.4 Fields

Field	Function
31-27 PME_SUPPORT	Power Management Event Support. For a description of this standard PCIe register field, see the PCI Express Specification. The read value from this field is the write value && (sys_aux_pwr_det, 1'b1, D2_SUPPORT, D1_SUPPORT, 1'b1), where D1_SUPPORT and D2_SUPPORT are fields in this register. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
26 D2_SUPPORT	D2 State Support. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
25 D1_SUPPORT	D1 State Support. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
24-22 AUX_CURR	Auxiliary Current Requirements. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
21 DSI	Device Specific Initialization Bit. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
20 —	Reserved.
19 PME_CLK	PCI Clock Requirement. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
18-16 PM_SPEC_VER	Power Management Spec Version. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
15-8 PM_NEXT_POINTER	Next Capability Pointer. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
7-0 PM_CAP_ID	Power Management Capability ID. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.17 Power Management Control and Status Register. (CON_STATUS_REG)

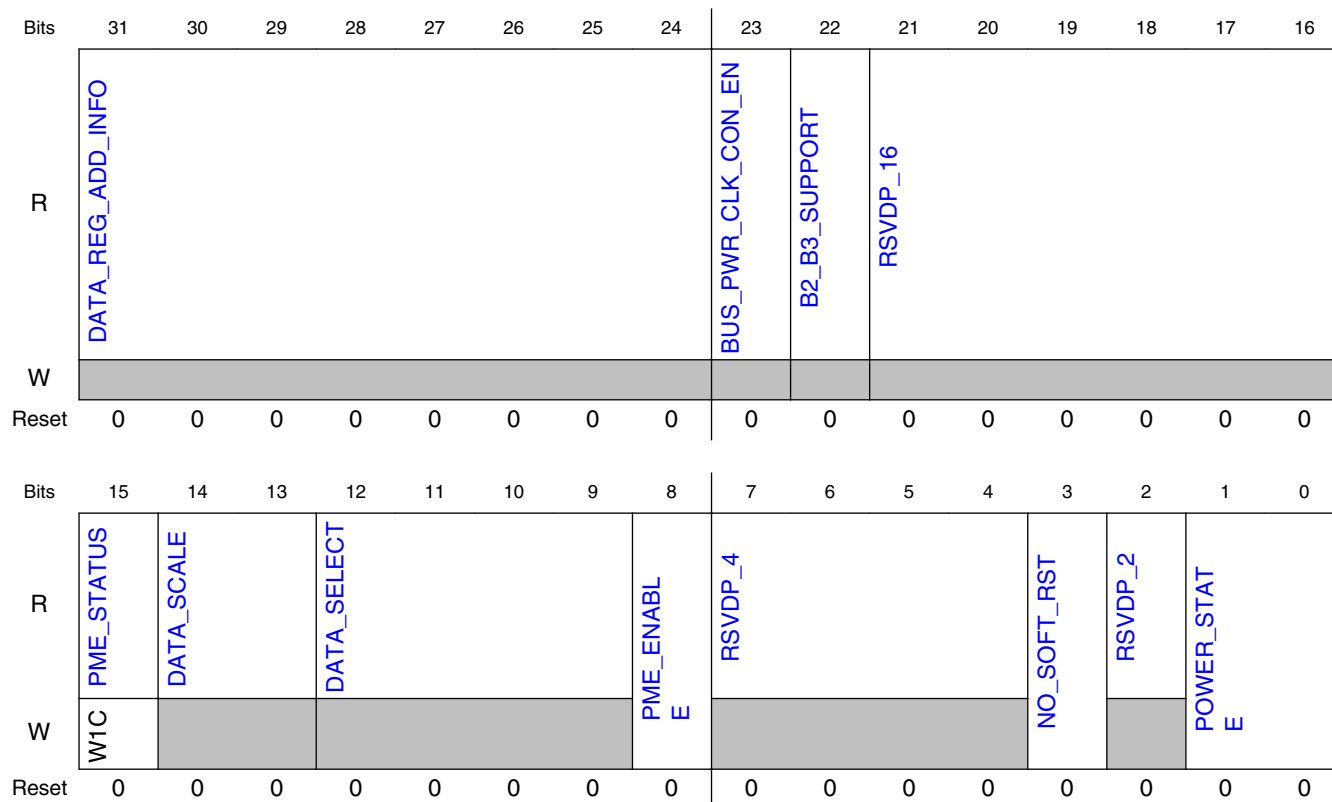
11.3.3.1.17.1 Offset

Register	Offset
CON_STATUS_REG	44h

11.3.3.1.17.2 Function

Power Management Control and Status Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.17.3 Diagram



11.3.3.1.17.4 Fields

Field	Function
31-24 DATA_REG_AD D_INFO	Power Data Information Register. For a description of this standard PCIe register field, see the PCI Express Specification.
23 BUS_PWR_CLK _CON_EN	Bus Power/Clock Control Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
22 B2_B3_SUPPO RT	B2B3 Support for D3hot. For a description of this standard PCIe register field, see the PCI Express Specification.
21-16 RSVDP_16	Reserved for future use.
15 PME_STATUS	PME Status. For a description of this standard PCIe register field, see the PCI Express Specification.
14-13 DATA_SCALE	Data Scaling Factor. For a description of this standard PCIe register field, see the PCI Express Specification.
12-9	Data Select. For a description of this standard PCIe register field, see the PCI Express Specification.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
DATA_SELECT	
8 PME_ENABLE	PME Enable. For a description of this standard PCIe register field, see the PCI Express Specification. The PMC registers this value under aux power. Sometimes it might remember the old value, even if you try to clear it by writing '0'. Note: This register field is sticky.
7-4 RSVDP_4	Reserved for future use.
3 NO_SOFT_RST	No soft Reset. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
2 RSVDP_2	Reserved for future use.
1-0 POWER_STATE	Power State. For a description of this standard PCIe register field, see the PCI Express Specification. You can write to this register. However, the read-back value is the actual power state, not the write value. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.18 MSI Capability ID, Next Pointer, Capability/Control Registers. (PCI_MSI_CAP_ID_NEXT_CTRL_REG)

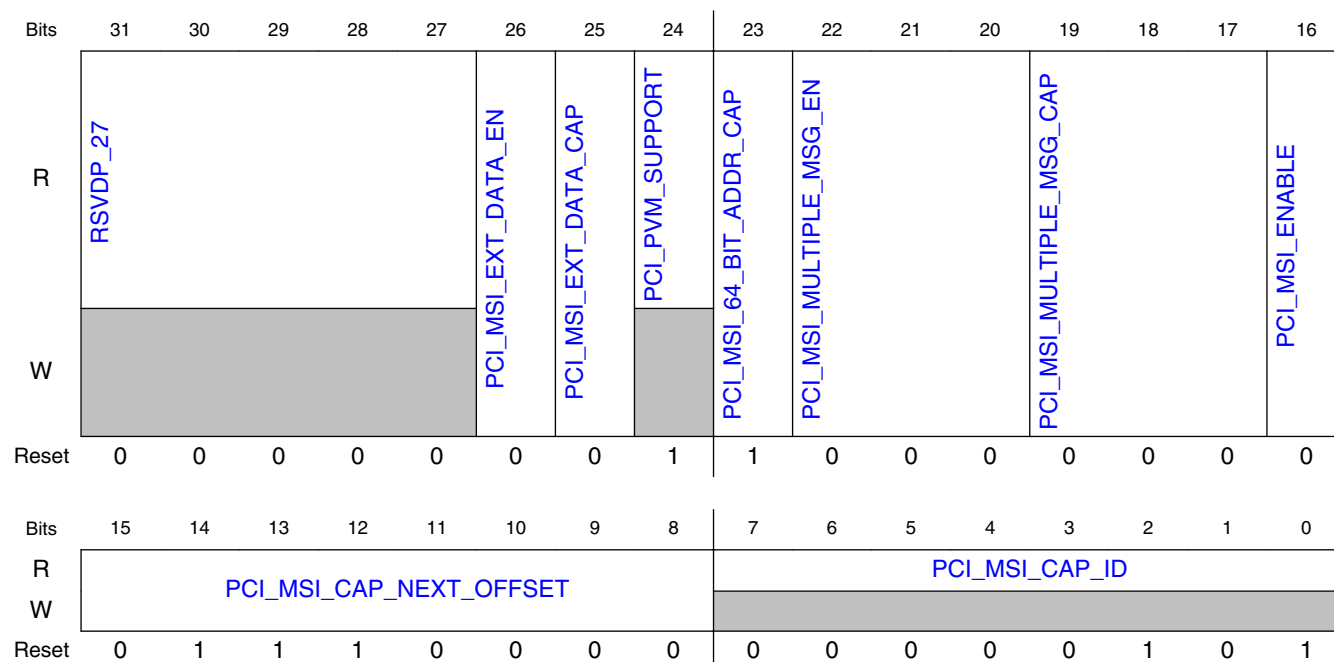
11.3.3.1.18.1 Offset

Register	Offset
PCI_MSI_CAP_ID_NEXT_CTRL_REG	50h

11.3.3.1.18.2 Function

MSI Capability ID, Next Pointer, Capability/Control Registers. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.18.3 Diagram



11.3.3.1.18.4 Fields

Field	Function
31-27 RSVDP_27	Reserved for future use.
26 PCI_MSI_EXT_DATA_EN	Extended Message Data Enable. For a description of this standard PCIe register, see the PCI-SIG ECN for Extended MSI Data, Feb 24, 2016, affecting PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: PCI_MSI_CAP_ID_NEXT_CTRL_REG.PCI_MSI_EXT_DATA_CAP ? RW : RO
25 PCI_MSI_EXT_DATA_CAP	Extended Message Data Capable. For a description of this standard PCIe register, see the PCI-SIG ECN for Extended MSI Data, Feb 24, 2016, affecting PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
24 PCI_PVM_SUPPORT	MSI Per Vector Masking Capable. For a description of this standard PCIe register field, see the PCI Express Specification.
23 PCI_MSI_64_BIT_ADDR_CAP	MSI 64-bit Address Capable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
22-20 PCI_MSI_MULTIPLE_MSG_EN	MSI Multiple Message Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
19-17	MSI Multiple Message Capable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
PCI_MSI_MULTIPLE_MSG_CAP	
16 PCI_MSI_ENABLE	MSI Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
15-8 PCI_MSI_CAP_NEXT_OFFSET	MSI Capability Next Pointer. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
7-0 PCI_MSI_CAP_ID	MSI Capability ID. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.19 MSI Message Lower Address Register. (MSI_CAP_OFF_04H_REG)

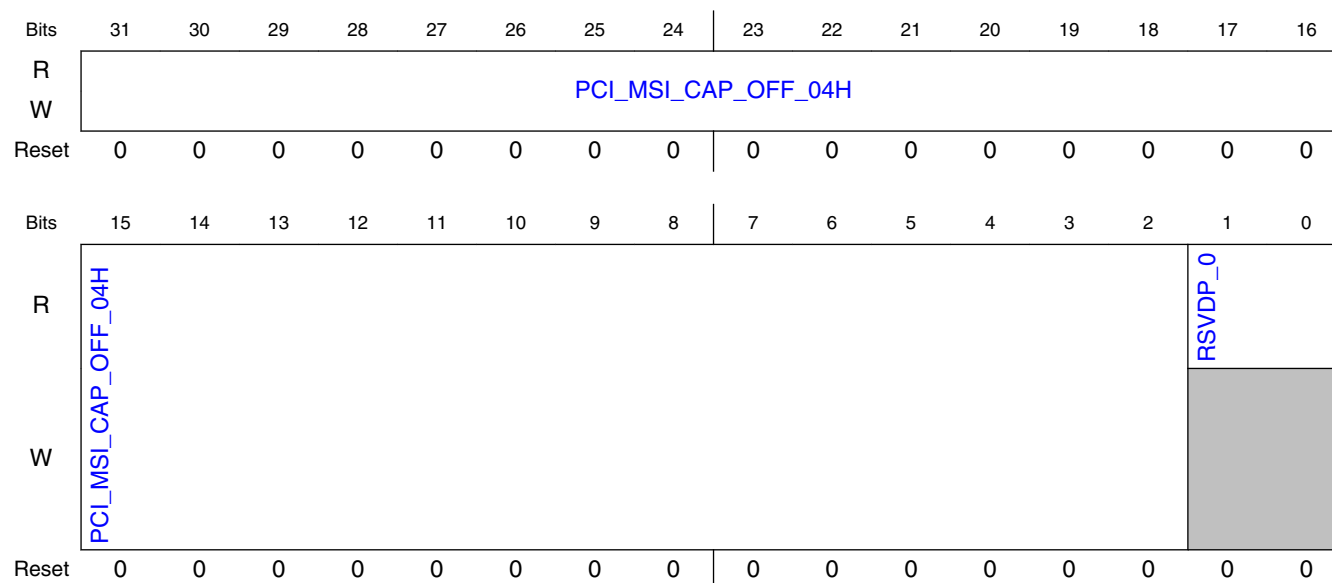
11.3.3.1.19.1 Offset

Register	Offset
MSI_CAP_OFF_04H_REG	54h

11.3.3.1.19.2 Function

MSI Message Lower Address Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.19.3 Diagram



11.3.3.1.19.4 Fields

Field	Function
31-2 PCI_MSI_CAP_OFF_04H	MSI Message Lower Address Field. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
1-0 RSVDP_0	Reserved for future use.

11.3.3.1.20 For a 32 bit MSI Message, this register contains Data. (MSI_CAP_OFF_08H_REG)

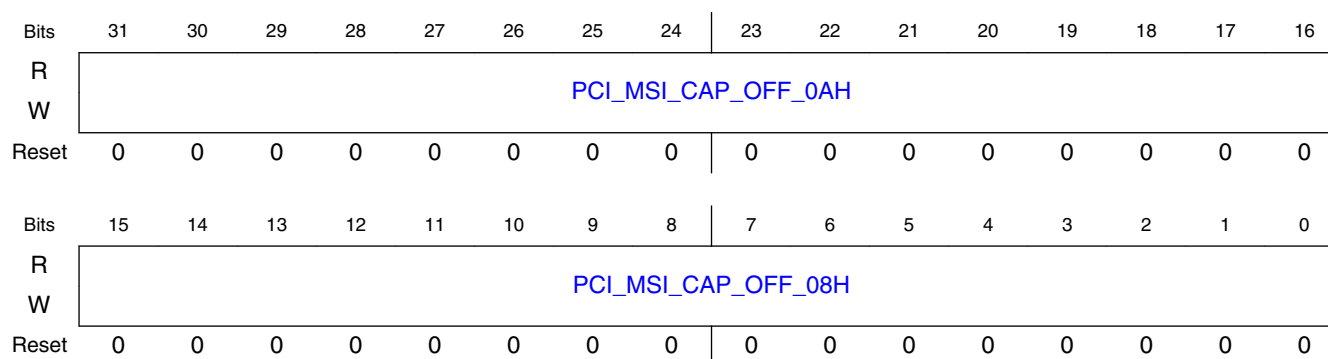
11.3.3.1.20.1 Offset

Register	Offset
MSI_CAP_OFF_08H_REG	58h

11.3.3.1.20.2 Function

For a 32 bit MSI Message, this register contains Data. For 64 bit it contains the Upper Address. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.20.3 Diagram



11.3.3.1.20.4 Fields

Field	Function
31-16 PCI_MSI_CAP_OFF_0AH	For a 32 bit MSI Message, this field contains Ext MSI Data. For 64-bit it contains upper 16 bits of the Upper Address. For a description of this standard PCIe register field, see the PCI Express Specification Note: The access attributes of this field are as follows: - Dbi: PCI_MSI_64_BIT_ADDR_CAP `DEFAULT_EXT_MSI_DATA_CAPABLE ? R/W : R
15-0 PCI_MSI_CAP_OFF_08H	For a 32-bit MSI Message, this field contains Data. For 64-bit it contains lower 16 bits of the Upper Address. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: PCI_MSI_64_BIT_ADDR_CAP ? R/W : R

11.3.3.1.21 For a 64 bit MSI Message, this register contains Data. (MSI_CAP_OFF_0CH_REG)

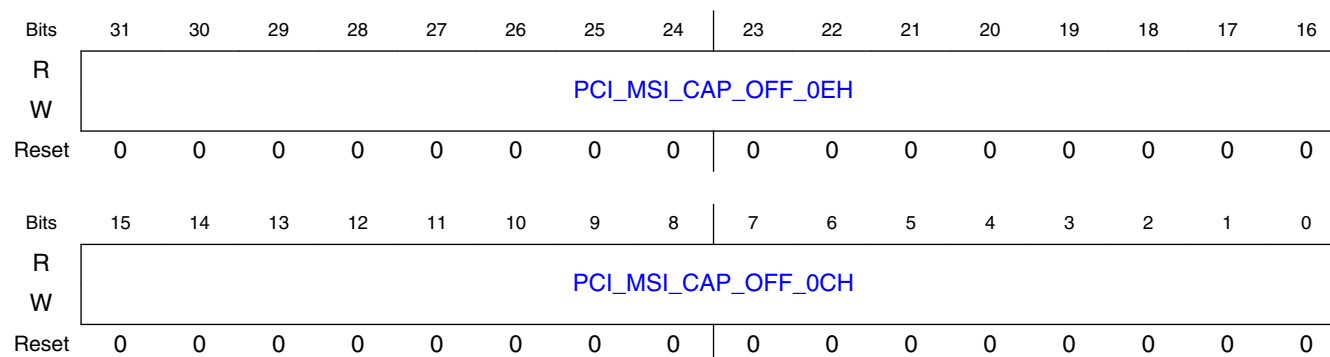
11.3.3.1.21.1 Offset

Register	Offset
MSI_CAP_OFF_0CH_REG	5Ch

11.3.3.1.21.2 Function

For a 64 bit MSI Message, this register contains Data. For 32 bit, it contains Mask Bits if PVM enabled. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.21.3 Diagram



11.3.3.1.21.4 Fields

Field	Function
31-16 PCI_MSI_CAP_OFF_0EH	For a 64-bit MSI Message, this field contains Data. For 32-bit, it contains the upper Mask Bits if PVM is enabled. For a description of this standard PCIe register field, see the PCI Express Specification Note: The access attributes of this field are as follows: - Dbi: (!MSI_64_EN && MSI_PVM_EN_VALUE) ? RW: MSI_64_EN && DEFAULT_EXT_MSI_DATA_CAPABLE ? RW : RO
15-0 PCI_MSI_CAP_OFF_0CH	For a 64-bit MSI Message, this field contains Data. For 32-bit, it contains the lower Mask Bits if PVM is enabled. For a description of this standard PCIe register field, see the PCI Express Specification Note: The access attributes of this field are as follows: - Dbi: PCI_MSI_64_BIT_ADDR_CAP MSI_PVM_EN ? R/W : R

11.3.3.1.22 Used for MSI when Vector Masking Capable. (MSI_CAP_OFF_10H_REG)

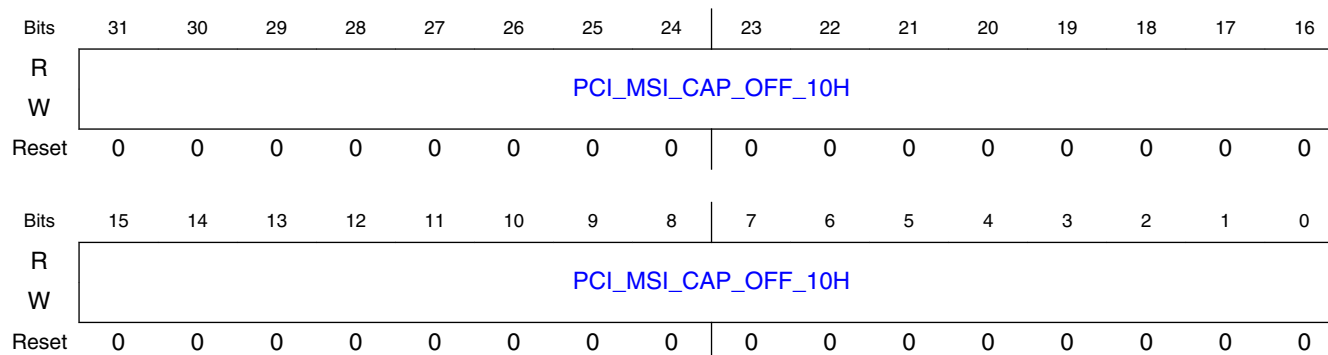
11.3.3.1.22.1 Offset

Register	Offset
MSI_CAP_OFF_10H_REG	60h

11.3.3.1.22.2 Function

Used for MSI when Vector Masking Capable. For 32 bit contains Pending Bits. For 64 bit, contains Mask Bits. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.22.3 Diagram



11.3.3.1.22.4 Fields

Field	Function
31-0 PCI_MSI_CAP_OFF_10H	Used for MSI when Vector Masking Capable. For 32-bit contains Pending Bits. For 64-bit, contains Mask Bits. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: PCI_MSI_64_BIT_ADDR_CAP && MSI_PVM_EN ? R/W : R

11.3.3.1.23 Used for MSI 64 bit messaging when Vector Masking Capable. (MSI_CAP_OFF_14H_REG)

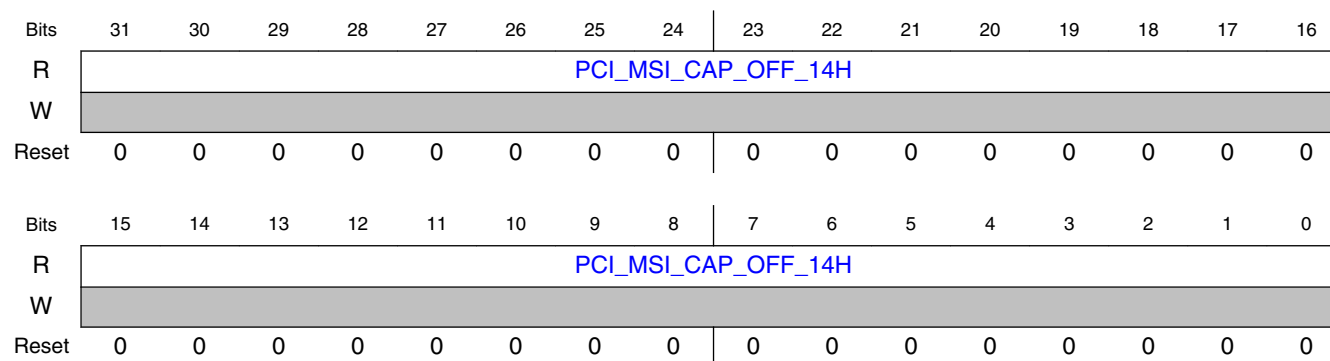
11.3.3.1.23.1 Offset

Register	Offset
MSI_CAP_OFF_14H_REG	64h

11.3.3.1.23.2 Function

Used for MSI 64 bit messaging when Vector Masking Capable. Contains Pending Bits. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.23.3 Diagram



11.3.3.1.23.4 Fields

Field	Function
31-0 PCI_MSI_CAP_OFF_14H	Used for MSI 64-bit messaging when Vector Masking Capable. Contains Pending Bits. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.24 PCI Express Capabilities, ID, Next Pointer Register. (PCIE_CAP_ID_PCIE_NEXT_CAP_PTR_PCIE_CAP_REG)

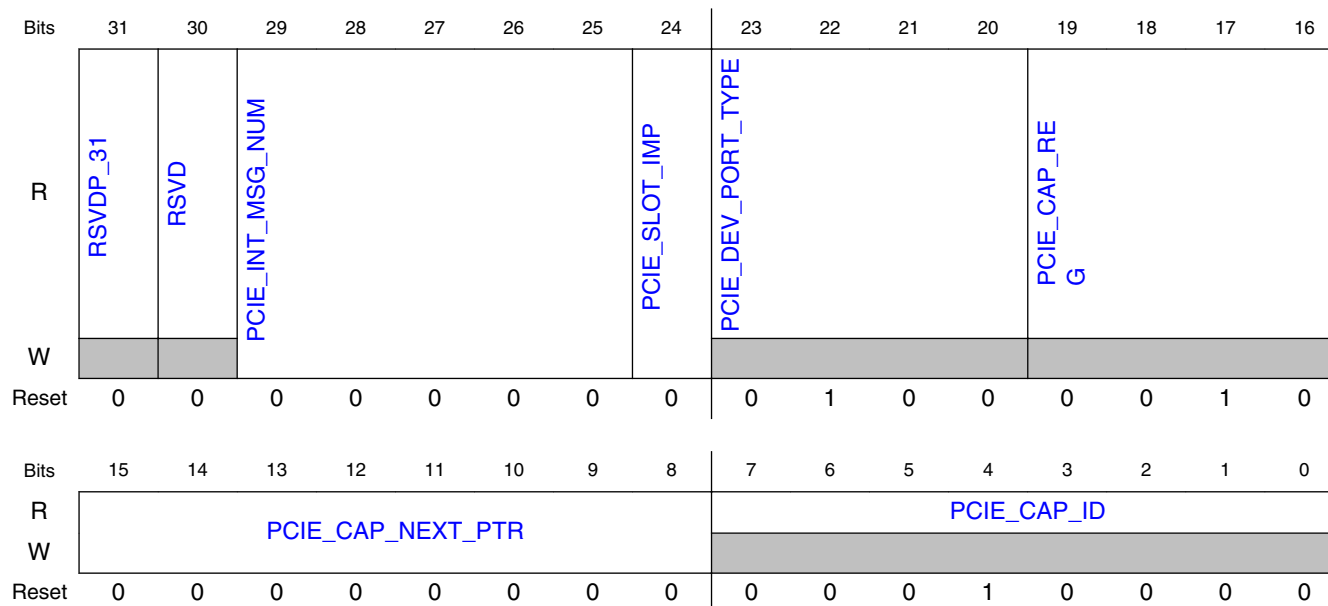
11.3.3.1.24.1 Offset

Register	Offset
PCIE_CAP_ID_PCIE_NEXT_CAP_PTR_PCIE_CAP_REG	70h

11.3.3.1.24.2 Function

PCI Express Capabilities, ID, Next Pointer Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.24.3 Diagram



11.3.3.1.24.4 Fields

Field	Function
31 RSVDP_31	Reserved for future use.
30 RSVD	Reserved. For a description of this standard PCIe register field, see the PCI Express Specification.
29-25 PCIE_INT_MSG_NUM	PCIe Interrupt Message Number. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
24 PCIE_SLOT_IMP	PCIe Slot Implemented Valid. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
23-20 PCIE_DEV_PORT_TYPE	PCIe Device/PortType. For a description of this standard PCIe register field, see the PCI Express Specification.
19-16 PCIE_CAP_REG	PCIe Capability Version Number. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
15-8 PCIE_CAP_NEXT_PTR	PCIe Next Capability Pointer. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
7-0 PCIE_CAP_ID	PCIe Capability ID. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.25 Device Capabilities Register. (DEVICE_CAPABILITIES_REG)

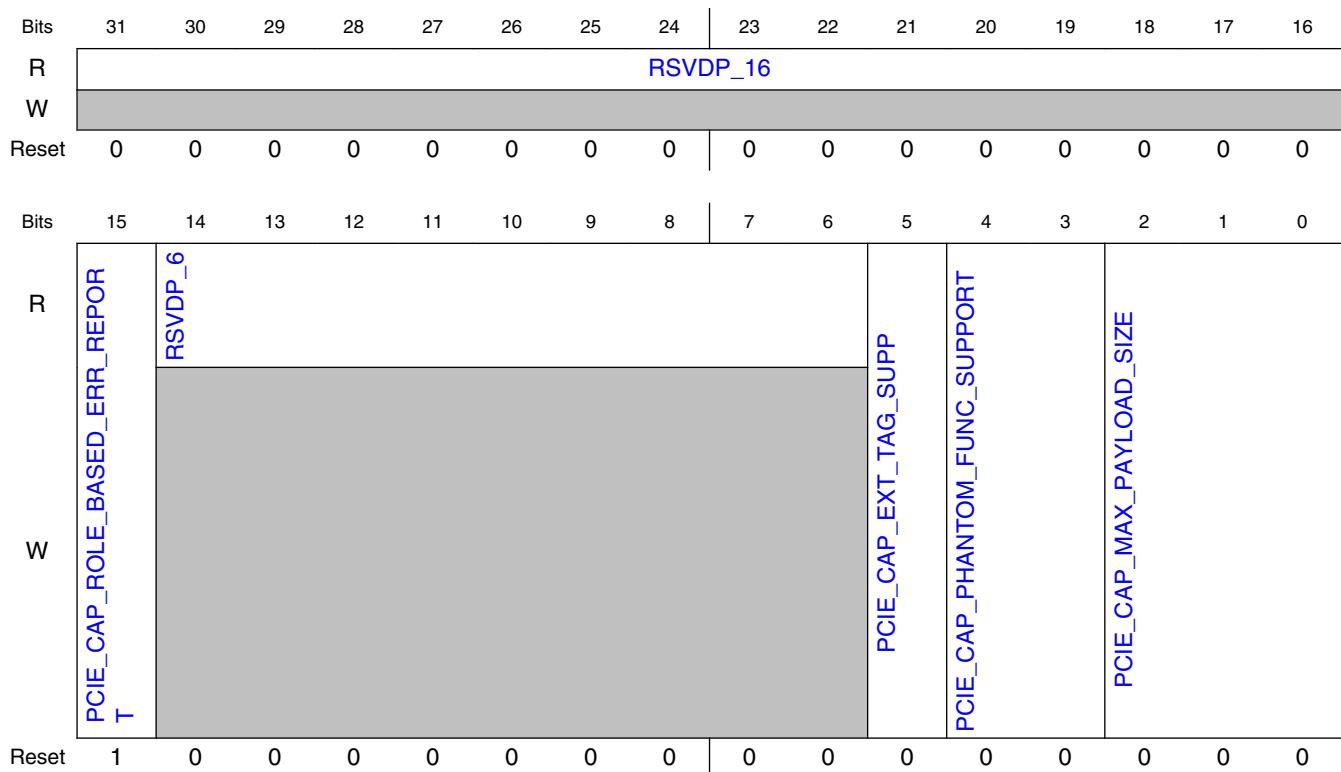
11.3.3.1.25.1 Offset

Register	Offset
DEVICE_CAPABILITIES_REG	74h

11.3.3.1.25.2 Function

Device Capabilities Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.25.3 Diagram



11.3.3.1.25.4 Fields

Field	Function
31-16	Reserved for future use.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
RSVDP_16	
15 PCIE_CAP_ROLE_BASED_ERROR_REPORT	Role-based Error Reporting Implemented. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
14-6 RSVDP_6	Reserved for future use.
5 PCIE_CAP_EXTENDED_TAG_SUPPORT	Extended Tag Field Supported. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
4-3 PCIE_CAP_PHANTOM_FUNCTION_SUPPORT	Phantom Functions Supported. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
2-0 PCIE_CAP_MAX_PAYLOAD_SIZE	Max Payload Size Supported. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.

11.3.3.1.26 Device Control and Status Register. (DEVICE_CONTROL_DEVICE_STATUS)

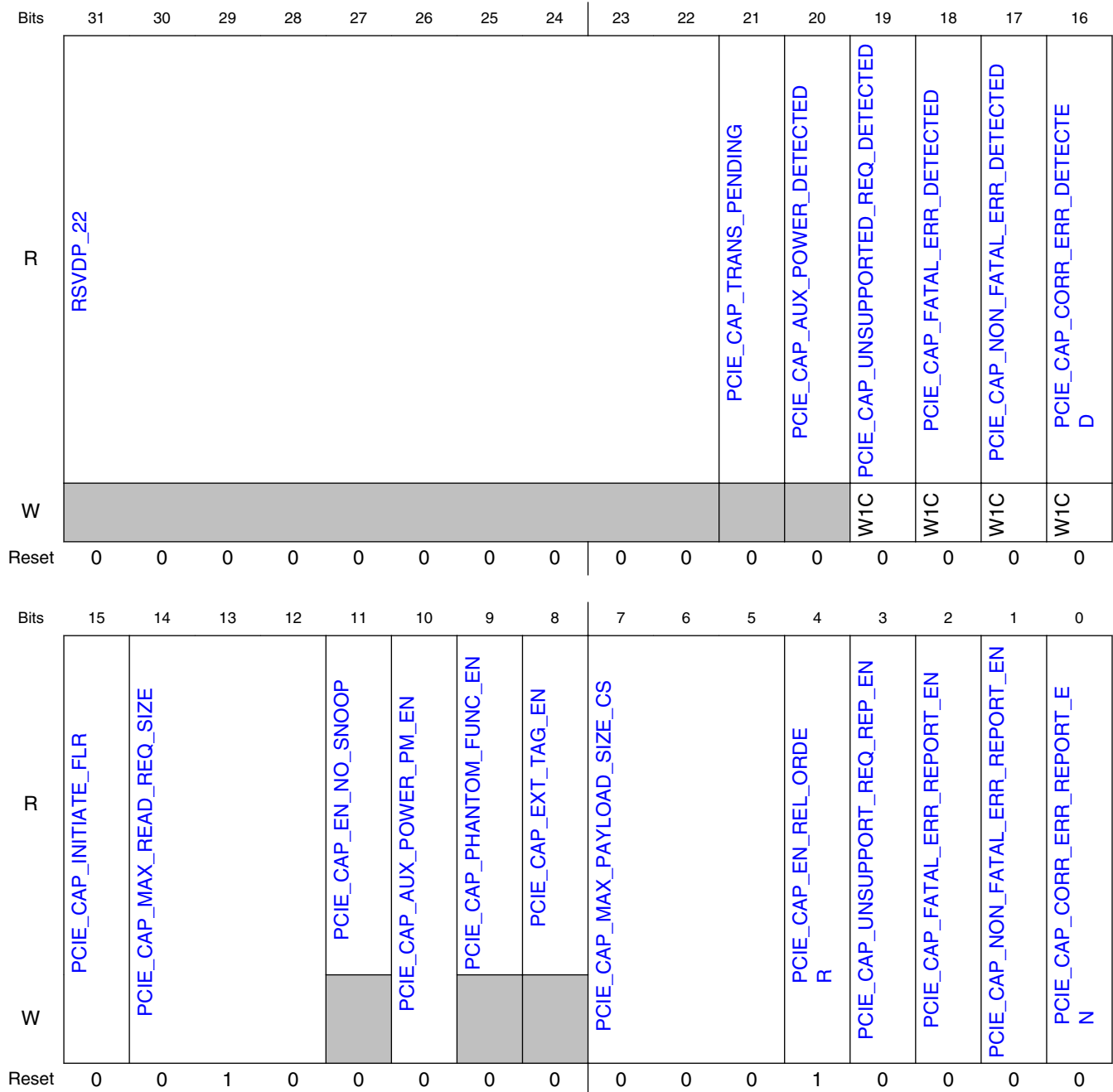
11.3.3.1.26.1 Offset

Register	Offset
DEVICE_CONTROL_DEVICE_STATUS	78h

11.3.3.1.26.2 Function

Device Control and Status Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.26.3 Diagram



11.3.3.1.26.4 Fields

Field	Function
31-22 RSVDP_22	Reserved for future use.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
21 PCIE_CAP_TRANS_PENDING	Transactions Pending Status. For a description of this standard PCIe register field, see the PCI Express Specification.
20 PCIE_CAP_AUX_POWER_DETECTED	Aux Power Detected Status. For a description of this standard PCIe register field, see the PCI Express Specification. This bit is derived by sampling the sys_aux_pwr_det input.
19 PCIE_CAP_UNSUPPORTED_REQ_DETECTED	Unsupported Request Detected Status. For a description of this standard PCIe register field, see the PCI Express Specification.
18 PCIE_CAP_FATAL_ERR_DETECTED	Fatal Error Detected Status. For a description of this standard PCIe register field, see the PCI Express Specification.
17 PCIE_CAP_NON_FATAL_ERR_DETECTED	Non-Fatal Error Detected Status. For a description of this standard PCIe register field, see the PCI Express Specification.
16 PCIE_CAP_CORR_ERR_DETECTED	Correctable Error Detected Status. For a description of this standard PCIe register field, see the PCI Express Specification.
15 PCIE_CAP_INITIATE_FLR	Initiate Function Level Reset (for endpoints). For a description of this standard PCIe register field, see the PCI Express Specification.
14-12 PCIE_CAP_MAX_READ_REQ_SIZE	Max Read Request Size. For a description of this standard PCIe register field, see the PCI Express Specification.
11 PCIE_CAP_ENABLE_NO_SNOOP	Enable No Snoop. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R
10 PCIE_CAP_AUX_POWER_PM_ENABLE	Aux Power PM Enable. For a description of this standard PCIe register field, see the PCI Express Specification. This bit is derived by sampling the sys_aux_pwr_det input. Note: This register field is sticky.
9 PCIE_CAP_PHANTOM_FUNC_ENABLE	Phantom Functions Enable. For a description of this standard PCIe register field, see the PCI Express Specification. The write value is gated with the PCIE_CAP_PHANTOM_FUNC_SUPPORT field of DEVICE_CAPABILITIES_REG. Note: The access attributes of this field are as follows: - Dbi: DEVICE_CAPABILITIES_REG.PCIE_CAP_PHANTOM_FUNC_SUPPORT ? RW : RO
8 PCIE_CAP_EXTENDED_TAG_ENABLE	Extended Tag Field Enable. For a description of this standard PCIe register field, see the PCI Express Specification. The write value is gated with the PCIE_CAP_EXT_TAG_SUPP field of DEVICE_CAPABILITIES_REG. Note: The access attributes of this field are as follows: - Dbi: DEVICE_CAPABILITIES_REG.PCIE_CAP_EXT_TAG_SUPP ? RW : RO

Table continues on the next page...

Field	Function
7-5 PCIE_CAP_MAX_PAYLOAD_SIZE_CS	Max Payload Size. Max_Payload_Size . This field sets maximum TLP payload size for the Function. Permissible values that can be programmed are indicated by the Max_Payload_Size Supported field (PCIE_CAP_MAX_PAYLOAD_SIZE) in the Device Capabilities register (DEVICE_CAPABILITIES_REG).
4 PCIE_CAP_ENABLE_REL_ORDER	Enable Relaxed Ordering. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
3 PCIE_CAP_UNSUPPORTED_REQUEST_REPORTING_ENABLE	Unsupported Request Reporting Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
2 PCIE_CAP_FATAL_ERROR_REPORTING_ENABLE	Fatal Error Reporting Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
1 PCIE_CAP_NON_FATAL_ERROR_REPORTING_ENABLE	Non-fatal Error Reporting Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
0 PCIE_CAP_CORRECTABLE_ERROR_REPORTING_ENABLE	Correctable Error Reporting Enable. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.27 Link Capabilities Register. (LINK_CAPABILITIES_REG)

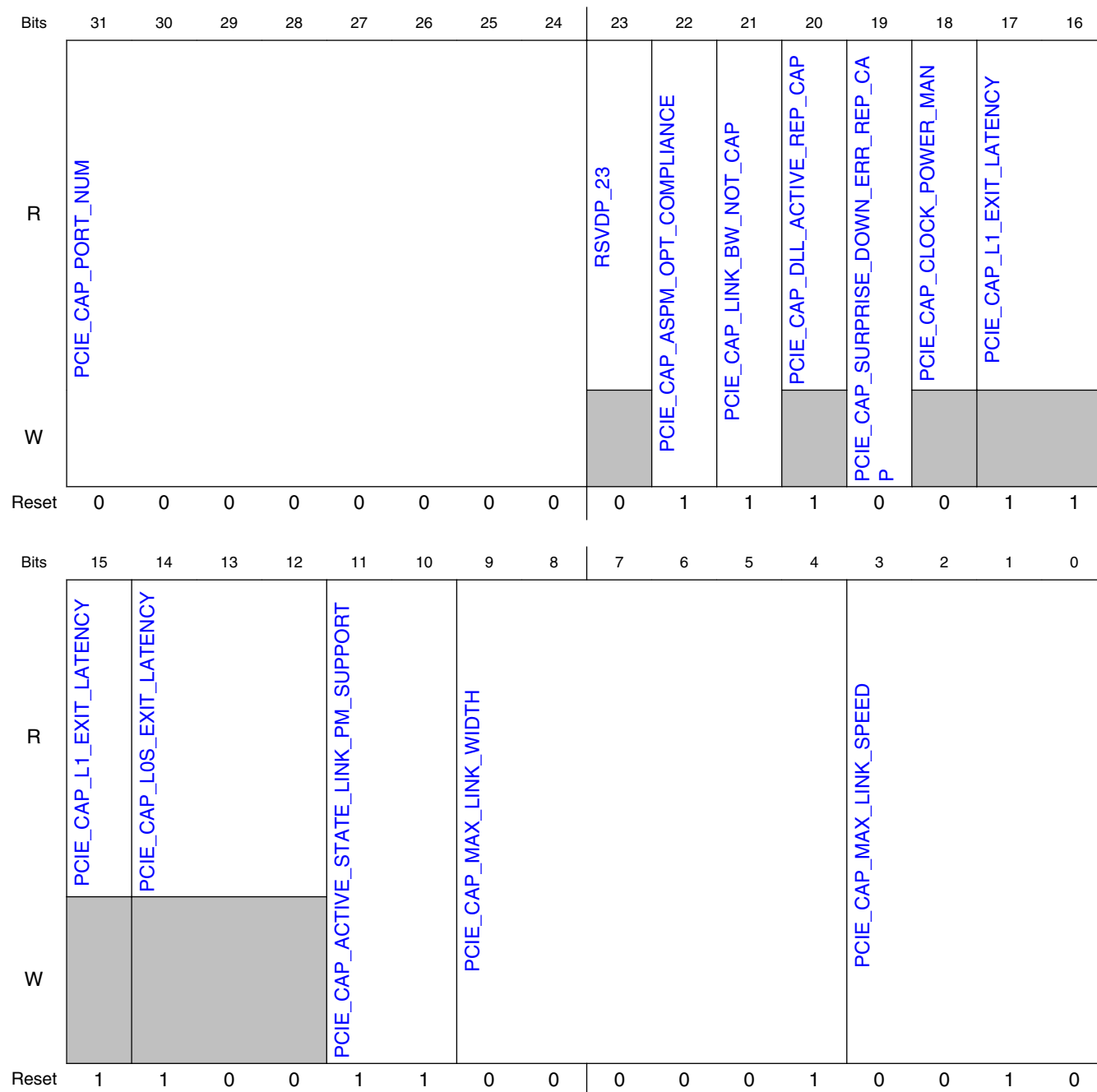
11.3.3.1.27.1 Offset

Register	Offset
LINK_CAPABILITIES_REGISTER	7Ch

11.3.3.1.27.2 Function

Link Capabilities Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.27.3 Diagram



11.3.3.1.27.4 Fields

Field	Function
31-24 PCIE_CAP_PORT_NUM	Port Number. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R

Table continues on the next page...

Field	Function
23 RSVDP_23	Reserved for future use.
22 PCIE_CAP_AS PM_OPT_COM PLIANCE	ASPM Optionality Compliance. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
21 PCIE_CAP_LIN K_BW_NOT_CA P	Link Bandwidth Notification Capable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
20 PCIE_CAP_DLL _ACTIVE_REP_ CAP	Data Link Layer Link Active Reporting Capable. For a description of this standard PCIe register field, see the PCI Express Specification.
19 PCIE_CAP_SU RPRISE_DOWN _ERR_REP_CA P	Surprise Down Error Reporting Capable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
18 PCIE_CAP_CL OCK_POWER_ MAN	Clock Power Management. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
17-15 PCIE_CAP_L1_ EXIT_LATENCY	L1 Exit Latency. For a description of this standard PCIe register field, see the PCI Express Specification. There are two each of these register fields, this one and a shadow one at the same address. The Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG) determines which one is used by the controller and which one is accessed by a read request. Common Clock operation is enabled in the controller when you set the Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG). The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the shadow field at this location. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
14-12 PCIE_CAP_L0S _EXIT_LATENC Y	L0s Exit Latency. For a description of this standard PCIe register field, see the PCI Express Specification. There are two each of these register fields, this one and a shadow one at the same address. The Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG) determines which one is used by the controller and which one is accessed by a read request. Common Clock operation is enabled in the controller when you set the Common Clock bit (PCIE_CAP_COMMON_CLK_CONFIG) of the Link Control Register (LINK_CONTROL_LINK_STATUS_REG). The assertion of CS2 (that is, assert the dbi_cs2 input, or the CS2 address bit for the AXI bridge) is required to write to the shadow field at this location. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
11-10 PCIE_CAP_AC TIVE_STATE_LI NK_PM_SUPP ORT	Level of ASPM (Active State Power Management) Support. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
9-4	Maximum Link Width. For a description of this standard PCIe register field, see the PCI Express Specification. In M-PCIe mode, the reset and dynamic values of this field are calculated by the controller.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
PCIE_CAP_MAX_LINK_WIDTH	Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.
3-0 PCIE_CAP_MAX_LINK_SPEED	Maximum Link Speed. For a description of this standard PCIe register field, see the PCI Express Specification. In M-PCIe mode, the reset and dynamic values of this field are calculated by the controller. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R Note: This register field is sticky.

11.3.3.1.28 Link Control and Status Register. (LINK_CONTROL_LINK_STATUS_REG)

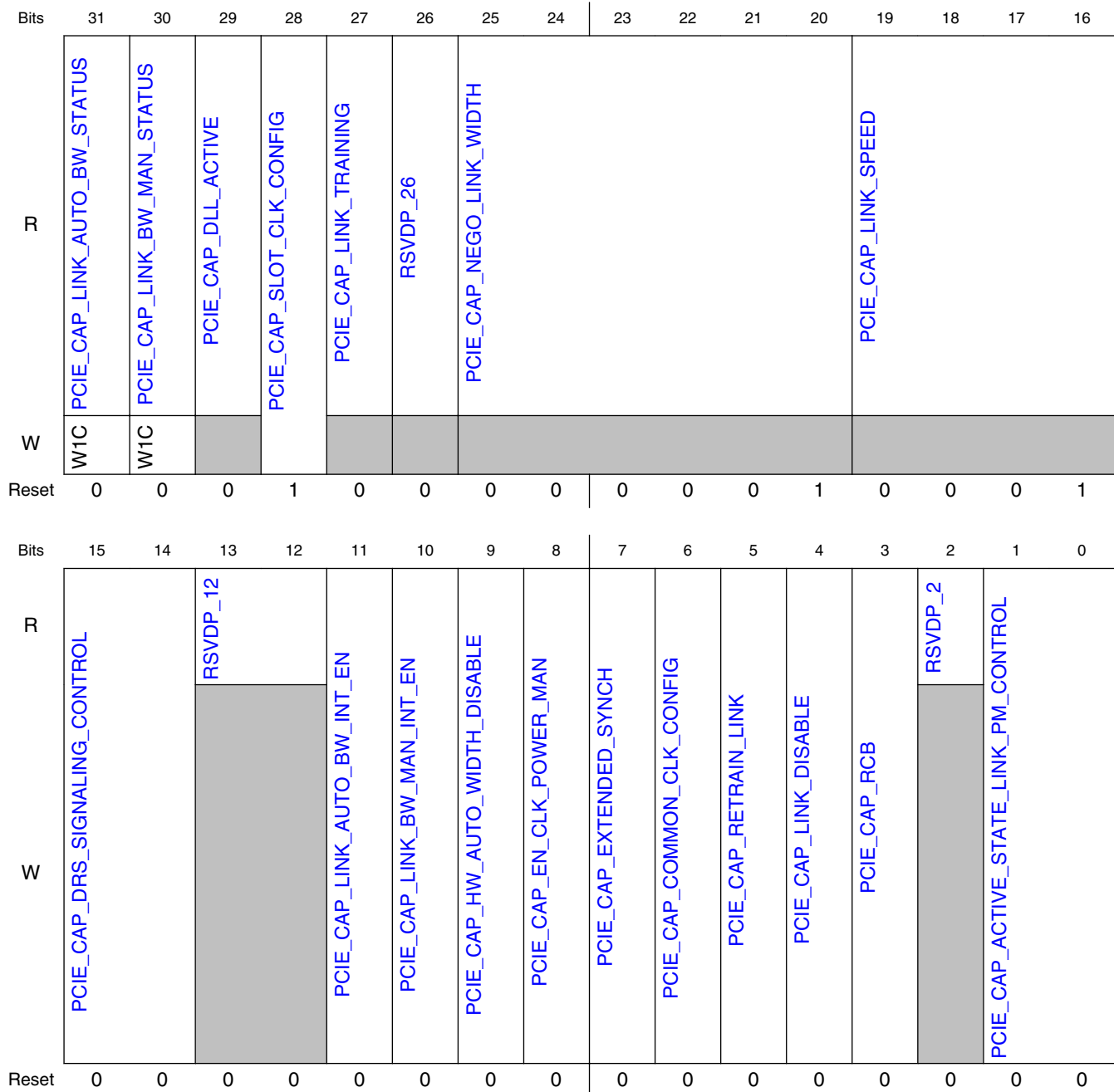
11.3.3.1.28.1 Offset

Register	Offset
LINK_CONTROL_LINK_STATUS_REG	80h

11.3.3.1.28.2 Function

Link Control and Status Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.28.3 Diagram



11.3.3.1.28.4 Fields

Field	Function
31	Link Autonomous Bandwidth Status. For a description of this standard PCIe register field, see the PCI Express Specification. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
PCIE_CAP_LINK_AUTO_BW_STATUS	
30 PCIE_CAP_LINK_BW_MAN_STATUS	Link Bandwidth Management Status. For a description of this standard PCIe register field, see the PCI Express Specification. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG.
29 PCIE_CAP_DLL_ACTIVE	Data Link Layer Active. For a description of this standard PCIe register field, see the PCI Express Specification.
28 PCIE_CAP_SLOT_CLK_CONFIG	Slot Clock Configuration. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
27 PCIE_CAP_LINK_TRAINING	LTSSM is in Configuration or Recovery State. For a description of this standard PCIe register field, see the PCI Express Specification.
26 RSVDP_26	Reserved for future use.
25-20 PCIE_CAP_NEGOTIATED_LINK_WIDTH	Negotiated Link Width. For a description of this standard PCIe register field, see the PCI Express Specification.
19-16 PCIE_CAP_LINK_SPEED	Current Link Speed. For a description of this standard PCIe register field, see the PCI Express Specification.
15-14 PCIE_CAP_DRS_SIGNALING_CONTROL	DRS Signaling Control. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: LINK_CAPABILITIES2_REG.DRS_SUPPORTED ? RW : RO
13-12 RSVDP_12	Reserved for future use.
11 PCIE_CAP_LINK_AUTO_BW_INTERRUPT_ENABLE	Link Autonomous Bandwidth Management Interrupt Enable. For a description of this standard PCIe register field, see the PCI Express Specification. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG. Note: The access attributes of this field are as follows: - Dbi: LINK_CAPABILITIES_REG.PCIE_CAP_LINK_BW_NOT_CAP ? RW : RO
10 PCIE_CAP_LINK_BW_MAN_INTERRUPT_ENABLE	Link Bandwidth Management Interrupt Enable. For a description of this standard PCIe register field, see the PCI Express Specification. The write value is gated with the PCIE_CAP_LINK_BW_NOT_CAP field in LINK_CAPABILITIES_REG. Note: The access attributes of this field are as follows: - Dbi: LINK_CAPABILITIES_REG.PCIE_CAP_LINK_BW_NOT_CAP ? RW : RO
9 PCIE_CAP_HW_AUTO_WIDTH_DISABLE	Hardware Autonomous Width Disable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
8	Enable Clock Power Management. For a description of this standard PCIe register field, see the PCI Express Specification. The write value is gated with the PCIE_CAP_CLOCK_POWER_MAN field in

Table continues on the next page...

Field	Function
PCIE_CAP_EN_CLK_POWER_MAN	LINK_CAPABILITIES_REG. Note: The access attributes of this field are as follows: - Dbi: LINK_CAPABILITIES_REG.PCIE_CAP_CLOCK_POWER_MAN ? RWS : ROS Note: This register field is sticky.
7 PCIE_CAP_EXTENDED_SYNC_H	Extended Synch. For a description of this standard PCIe register field, see the PCI Express Specification.
6 PCIE_CAP_COMMON_CLK_CONFIG	Common Clock Configuration. For a description of this standard PCIe register field, see the PCI Express Specification.
5 PCIE_CAP_RETRAIN_LINK	Initiate Link Retrain. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: see description
4 PCIE_CAP_LINK_DISABLE	Initiate Link Disable. For a description of this standard PCIe register field, see the PCI Express Specification. In a DSP that supports crosslink, the controller gates the write value with the CROSS_LINK_EN field in PORT_LINK_CTRL_OFF.
3 PCIE_CAP_RCB	Read Completion Boundary (RCB). Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
2 RSVDP_2	Reserved for future use.
1-0 PCIE_CAP_ACTIVE_STATE_LINK_PM_CONTROL	Active State Power Management (ASPM) Control. Software must not enable L0s in either direction on a given Link unless components on both sides of the Link each support L0s; otherwise, the result is undefined. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.29 Slot Capabilities Register. (SLOT_CAPABILITIES_REG)

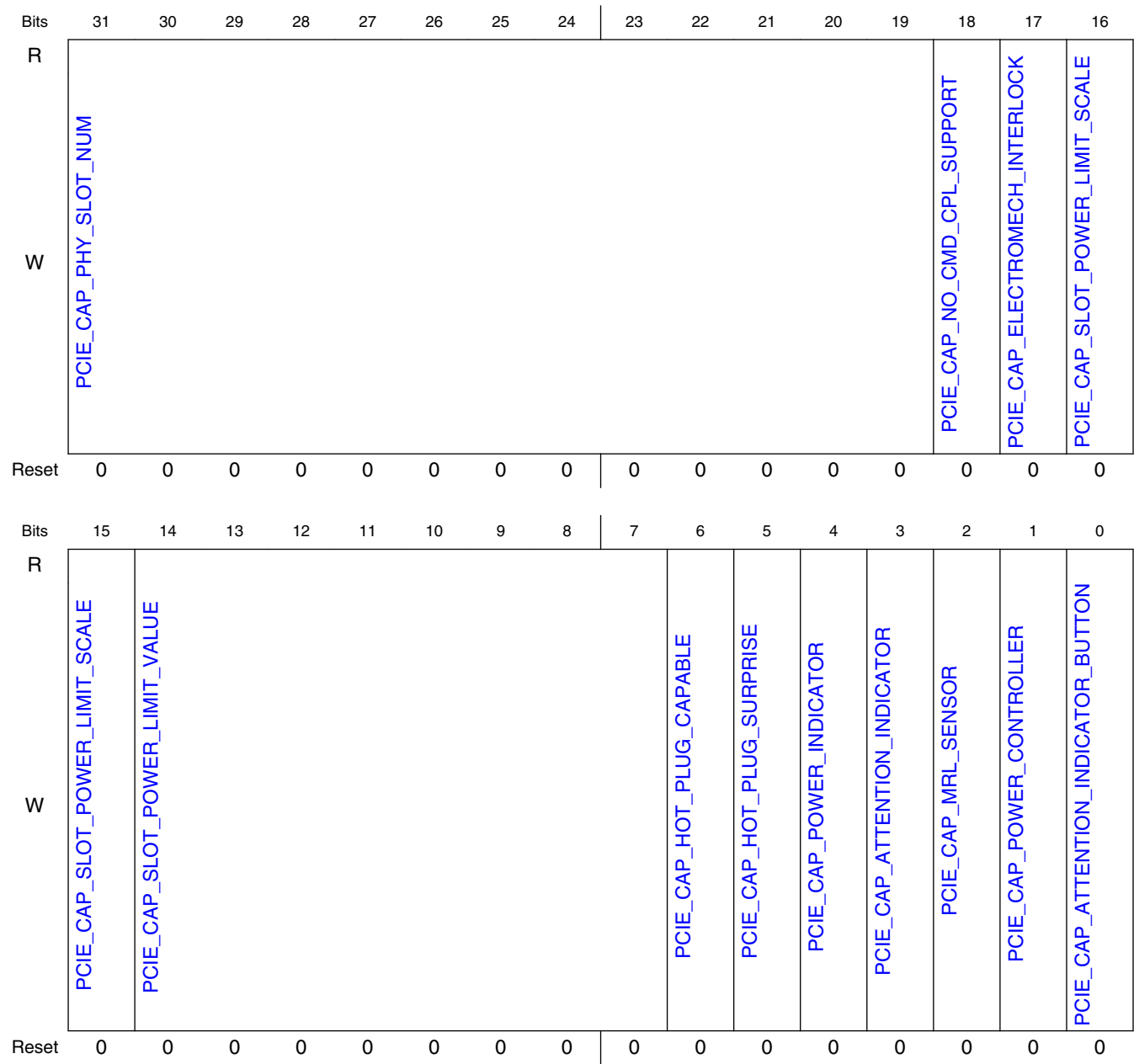
11.3.3.1.29.1 Offset

Register	Offset
SLOT_CAPABILITIES_REG	84h

11.3.3.1.29.2 Function

Slot Capabilities Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.29.3 Diagram



11.3.3.1.29.4 Fields

Field	Function
31-19 PCIE_CAP_PHY_SLOT_NUM	Physical Slot Number. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R

Table continues on the next page...

Field	Function
18 PCIE_CAP_NO_CMD_CPL_SUPPORT	No Command Completed Support. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
17 PCIE_CAP_ELECTROMECHANICAL_INTERLOCK	Electromechanical Interlock Present. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
16-15 PCIE_CAP_SLOT_POWER_LIMIT_SCALE	Slot Power Limit Scale. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
14-7 PCIE_CAP_SLOT_POWER_LIMIT_VALUE	Slot Power Limit Value. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
6 PCIE_CAP_HOT_PLUG_CAPABLE	Hot Plug Capable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
5 PCIE_CAP_HOT_PLUG_SURPRISE	Hot Plug Surprise possible. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
4 PCIE_CAP_POWER_INDICATOR	Power Indicator Present. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
3 PCIE_CAP_ATTENTION_INDICATOR	Attention Indicator Present. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
2 PCIE_CAP_MRL_SENSOR	MRL Present. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
1 PCIE_CAP_POWER_CONTROLLER	Power Controller Present. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R
0 PCIE_CAP_ATTENTION_INDICATOR_BUTTON	Attention Button Present. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W else R

11.3.3.1.30 Slot Control and Status Register. (SLOT_CONTROL_SLOT_STATUS)

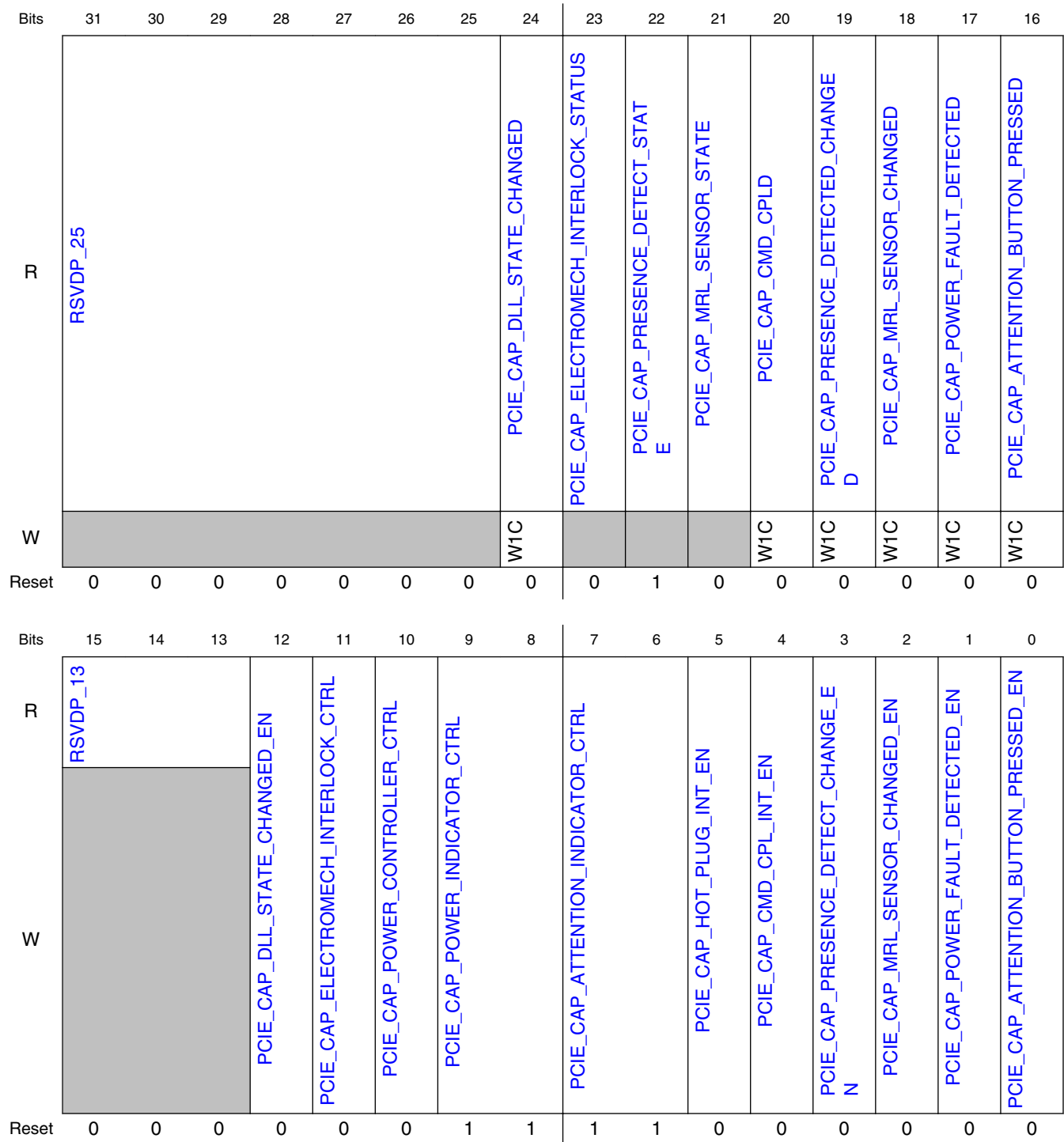
11.3.3.1.30.1 Offset

Register	Offset
SLOT_CONTROL_SLOT_STATUS	88h

11.3.3.1.30.2 Function

Slot Control and Status Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.30.3 Diagram



11.3.3.1.30.4 Fields

Field	Function
31-25	Reserved for future use.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
RSVDP_25	
24 PCIE_CAP_DLL _STATE_CHAN GED	Data Link Layer State Changed. For a description of this standard PCIe register field, see the PCI Express Specification.
23 PCIE_CAP_ELE CTROMECH_IN TERLOCK_STA TUS	Electromechanical Interlock Status. For a description of this standard PCIe register field, see the PCI Express Specification.
22 PCIE_CAP_PR ESENCE_DETE CT_STATE	Presence Detect State. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R
21 PCIE_CAP_MR L_SENSOR_ST ATE	MRL Sensor State. For a description of this standard PCIe register field, see the PCI Express Specification.
20 PCIE_CAP_CM D_CPLD	Command Completed. For a description of this standard PCIe register field, see the PCI Express Specification.
19 PCIE_CAP_PR ESENCE_DETE CTED_CHANG ED	Presence Detect Changed. For a description of this standard PCIe register field, see the PCI Express Specification.
18 PCIE_CAP_MR L_SENSOR_CH ANGED	MRL Sensor Changed. For a description of this standard PCIe register field, see the PCI Express Specification.
17 PCIE_CAP_PO WER_FAULT_D ETECTED	Power Fault Detected. For a description of this standard PCIe register field, see the PCI Express Specification.
16 PCIE_CAP_ATT ENTION_BUTT ON_PRESSED	Attention Button Pressed. For a description of this standard PCIe register field, see the PCI Express Specification.
15-13 RSVDP_13	Reserved for future use.
12 PCIE_CAP_DLL _STATE_CHAN GED_EN	Data Link Layer State Changed Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
11	Electromechanical Interlock Control. For a description of this standard PCIe register field, see the PCI Express Specification.

Table continues on the next page...

Field	Function
PCIE_CAP_ELECTROMECHANICAL_LOCK_CTRL	
10 PCIE_CAP_POWER_CONTROLLER_CTRL	Power Controller Control. For a description of this standard PCIe register field, see the PCI Express Specification.
9-8 PCIE_CAP_POWER_INDICATOR_CTRL	Power Indicator Control. For a description of this standard PCIe register field, see the PCI Express Specification.
7-6 PCIE_CAP_ATTENTION_INDICATOR_CTRL	Attention Indicator Control. For a description of this standard PCIe register field, see the PCI Express Specification.
5 PCIE_CAP_HOT_PLUG_INTERRUPT_ENABLE	Hot Plug Interrupt Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
4 PCIE_CAP_COMMAND_COMPLETED_INTERRUPT_ENABLE	Command Completed Interrupt Enable. For a description of this standard PCIe register field, see the PCI Express Specification. Write value is gated with PCIE_CAP_NO_CMD_CPL_SUPPORT field in SLOT_CAPABILITIES_REG. Note: The access attributes of this field are as follows: - Dbi: SLOT_CAPABILITIES_REG.PCIE_CAP_NO_CMD_CPL_SUPPORT ? RO : RW
3 PCIE_CAP_PRESENCE_DETECT_CHANGE_ENABLE	Presence Detect Changed Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
2 PCIE_CAP_MRL_SENSOR_CHANGED_ENABLE	MRL Sensor Changed Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
1 PCIE_CAP_POWER_FAULT_DETECTED_ENABLE	Power Fault Detected Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
0 PCIE_CAP_ATTENTION_BUTTON_PRESSED_ENABLE	Attention Button Pressed Enable. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.31 Root Control and Capabilities Register. (ROOT_CONTROL_ROOT_CAPABILITIES_REG)

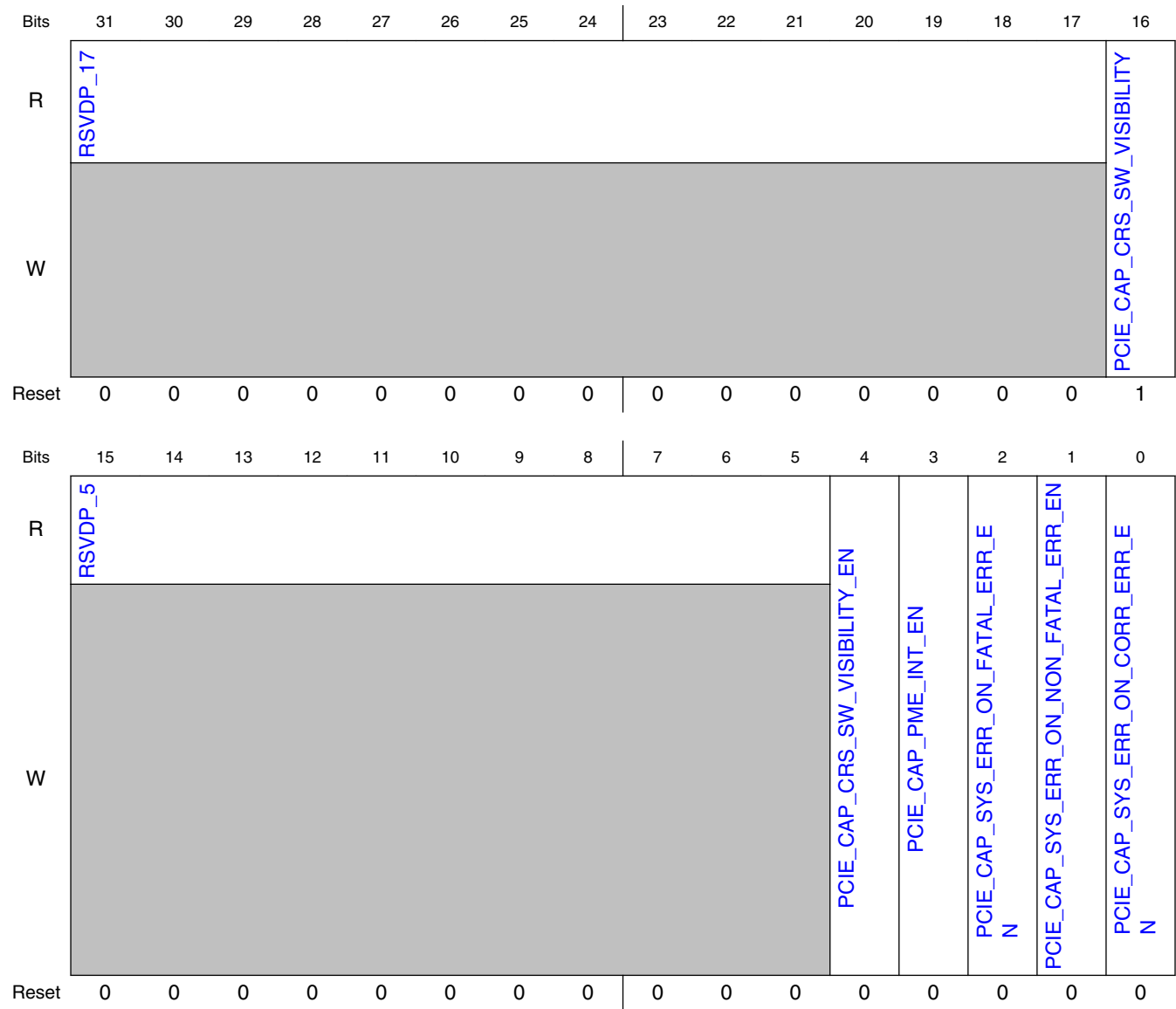
11.3.3.1.31.1 Offset

Register	Offset
ROOT_CONTROL_ROOT_CAPABILITIES_REG	8Ch

11.3.3.1.31.2 Function

Root Control and Capabilities Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.31.3 Diagram



11.3.3.1.31.4 Fields

Field	Function
31-17 RSVDP_17	Reserved for future use.
16 PCIE_CAP_CR S_SW_VISIBILI TY	CRS Software Visibility Capable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W (Sticky) else R (Sticky) Note: This register field is sticky.
15-5 RSVDP_5	Reserved for future use.
4 PCIE_CAP_CR S_SW_VISIBILI TY_EN	Configuration Request Retry Status (CRS) Software Visibility Enable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: ROOT_CONTROL_ROOT_CAPABILITIES_REG.PCIE_CAP_CR_S_SW_VISIBILITY ? RW : RO
3 PCIE_CAP_PM E_INT_EN	PME Interrupt Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
2 PCIE_CAP_SY S_ERR_ON_FA TAL_ERR_EN	System Error on Fatal Error Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
1 PCIE_CAP_SY S_ERR_ON_NO N_FATAL_ERR _EN	System Error on Non-fatal Error Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
0 PCIE_CAP_SY S_ERR_ON_CO RR_ERR_EN	System Error on Correctable Error Enable. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.32 Root Status Register. (ROOT_STATUS_REG)

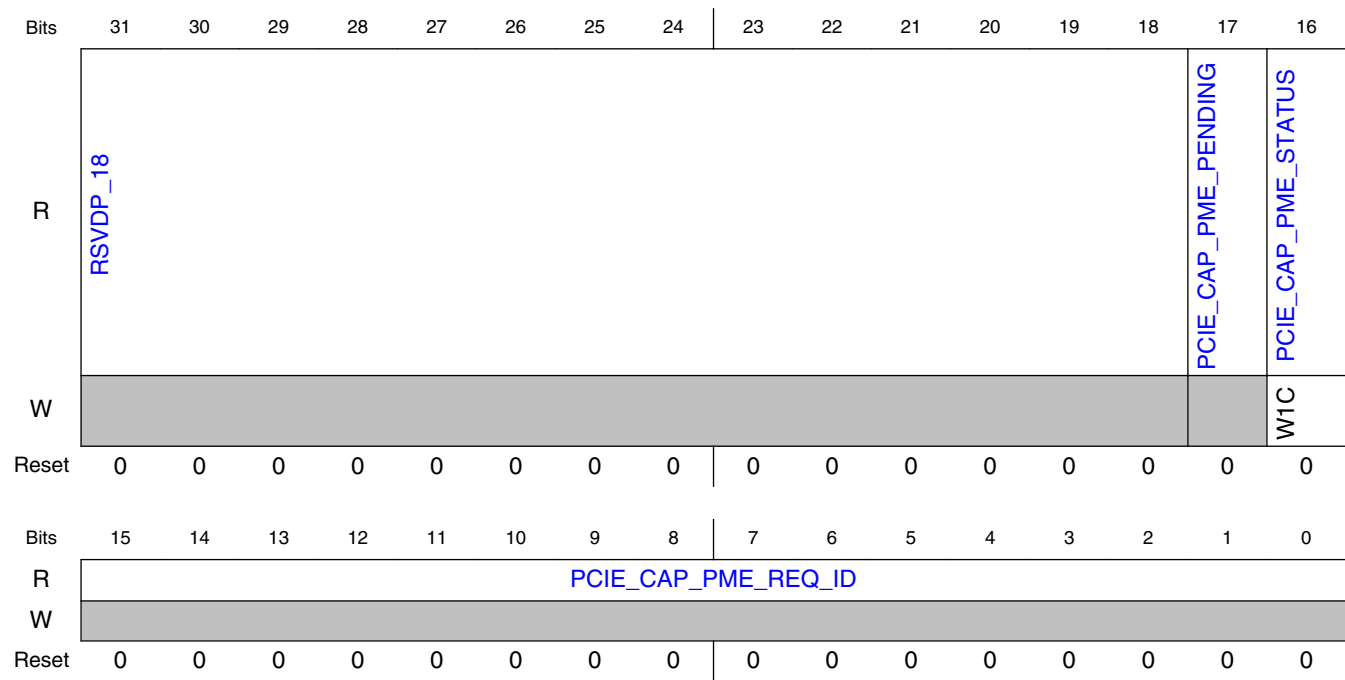
11.3.3.1.32.1 Offset

Register	Offset
ROOT_STATUS_REG	90h

11.3.3.1.32.2 Function

Root Status Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.32.3 Diagram



11.3.3.1.32.4 Fields

Field	Function
31-18 RSVDP_18	Reserved for future use.
17 PCIE_CAP_PME_PENDING	PME Pending. For a description of this standard PCIe register field, see the PCI Express Specification.
16 PCIE_CAP_PME_STATUS	PME Status. For a description of this standard PCIe register field, see the PCI Express Specification.
15-0 PCIE_CAP_PME_REQ_ID	PME Requester ID. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.33 Device Capabilities 2 Register. (DEVICE_CAPABILITIES2_REG)

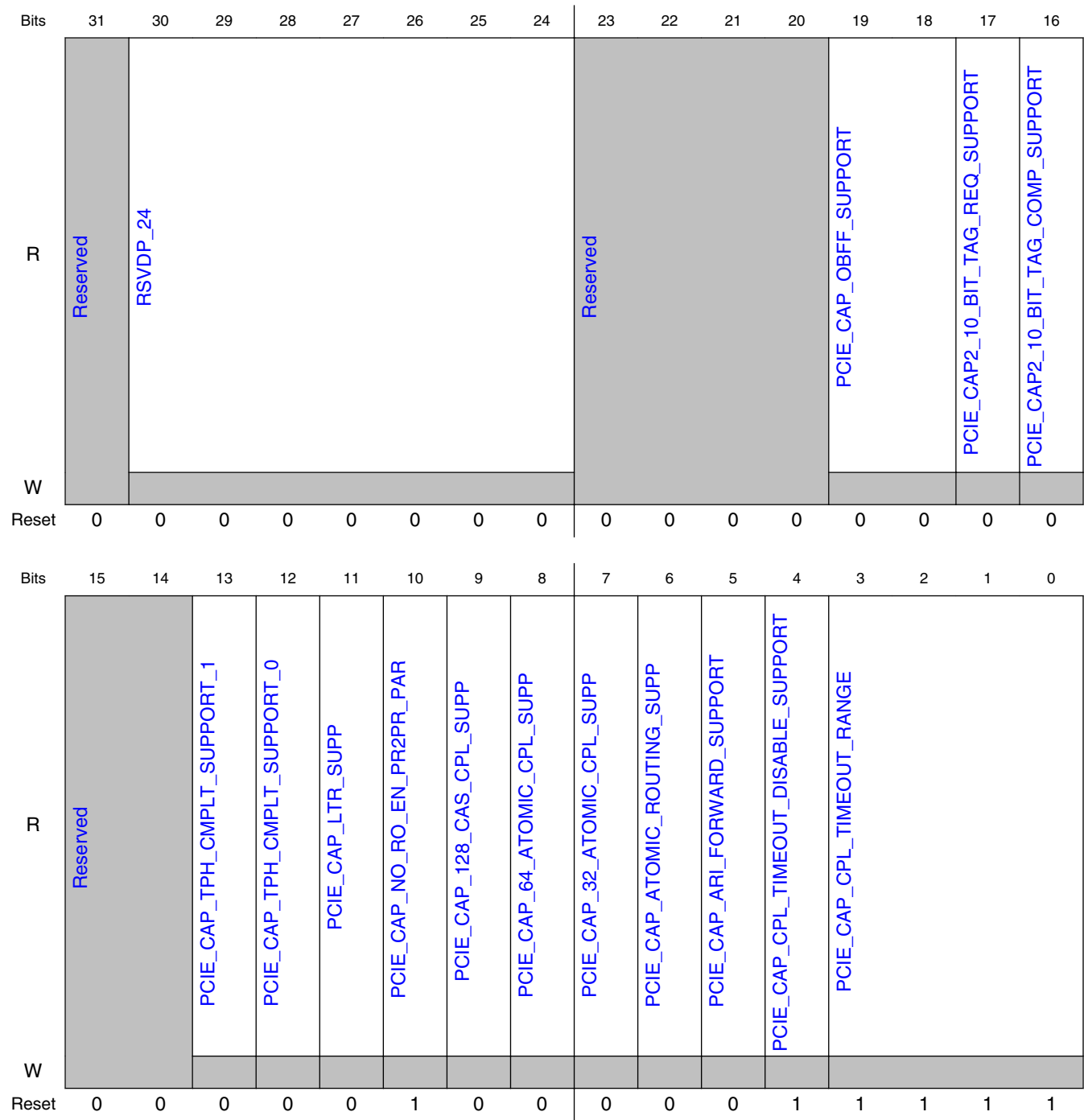
11.3.3.1.33.1 Offset

Register	Offset
DEVICE_CAPABILITIES2_REG	94h

11.3.3.1.33.2 Function

Device Capabilities 2 Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.33.3 Diagram



11.3.3.1.33.4 Fields

Field	Function
31	Reserved.

Table continues on the next page...

Field	Function
—	
30-24 RSVDP_24	Reserved for future use.
23-20 —	Reserved.
19-18 PCIE_CAP_OB FF_SUPPORT	(OBFF) Optimized Buffer Flush/fill Supported. For a description of this standard PCIe register field, see the PCI Express Specification.
17 PCIE_CAP2_10 _BIT_TAG_REQ _SUPPORT	10-Bit Tag Requester Supported. For a description of this standard PCIe register field, see the PCI Express Base Specification 4.0.
16 PCIE_CAP2_10 _BIT_TAG_CO MP_SUPPORT	10-Bit Tag Completer Supported. For a description of this standard PCIe register field, see the PCI Express Base Specification 4.0.
15-14 —	Reserved.
13 PCIE_CAP_TP H_CMPLT_SUP PORT_1	TPH Completer Supported Bit 1. For a description of this standard PCIe register field, see the PCI Express Specification.
12 PCIE_CAP_TP H_CMPLT_SUP PORT_0	TPH Completer Supported Bit 0. For a description of this standard PCIe register field, see the PCI Express Specification.
11 PCIE_CAP_LTR _SUPP	LTR Mechanism Supported. For a description of this standard PCIe register field, see the PCI Express Specification.
10 PCIE_CAP_NO _RO_EN_PR2P R_PAR	No Relaxed Ordering Enabled PR-PR Passing. For a description of this standard PCIe register field, see the PCI Express Specification.
9 PCIE_CAP_128 _CAS_CPL_SU PP	128 Bit CAS Completer Supported. For a description of this standard PCIe register field, see the PCI Express Specification.
8 PCIE_CAP_64_ ATOMIC_CPL_ SUPP	64 Bit AtomicOp Completer Supported. For a description of this standard PCIe register field, see the PCI Express Specification.
7 PCIE_CAP_32_ ATOMIC_CPL_ SUPP	32 Bit AtomicOp Completer Supported. For a description of this standard PCIe register field, see the PCI Express Specification.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
6 PCIE_CAP_ATOMIC_ROUTING_SUPPORTED	Atomic Operation Routing Supported. For a description of this standard PCIe register field, see the PCI Express Specification.
5 PCIE_CAP_ARI_FORWARD_SUPPORT	ARI Forwarding Supported. For a description of this standard PCIe register field, see the PCI Express Specification.
4 PCIE_CAP_COMPLETION_TIMEOUT_DISABLE_SUPPORTED	Completion Timeout Disable Supported. For a description of this standard PCIe register field, see the PCI Express Specification.
3-0 PCIE_CAP_COMPLETION_TIMEOUT_RANGES_SUPPORTED	Completion Timeout Ranges Supported. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.34 Device Control 2 and Status 2 Register. (DEVICE_CONTROL2_DEVICE_STATUS2_REG)

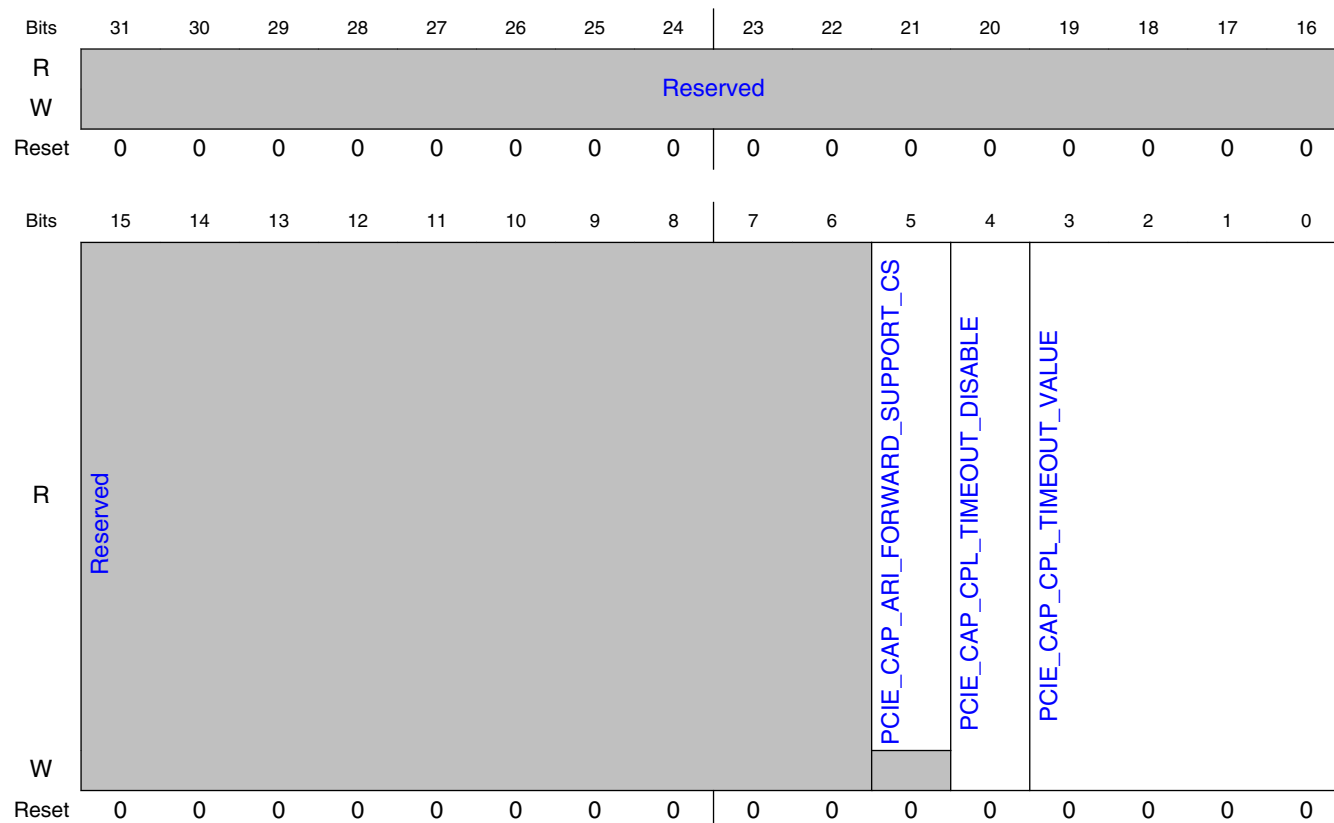
11.3.3.1.34.1 Offset

Register	Offset
DEVICE_CONTROL2_DEVICE_STATUS2_REG	98h

11.3.3.1.34.2 Function

Device Control 2 and Status 2 Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.34.3 Diagram



11.3.3.1.34.4 Fields

Field	Function
31-6 —	Reserved.
5 PCIE_CAP_ARI_FORWARD_SUPPORT_CS	ARI Forwarding Enable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
4 PCIE_CAP_CPL_TIMEOUT_DISABLE	Completion Timeout Disable. For a description of this standard PCIe register field, see the PCI Express Specification.
3-0 PCIE_CAP_CPL_TIMEOUT_VALUE	Completion Timeout Value. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.35 Link Capabilities 2 Register. (LINK_CAPABILITIES2_REG)

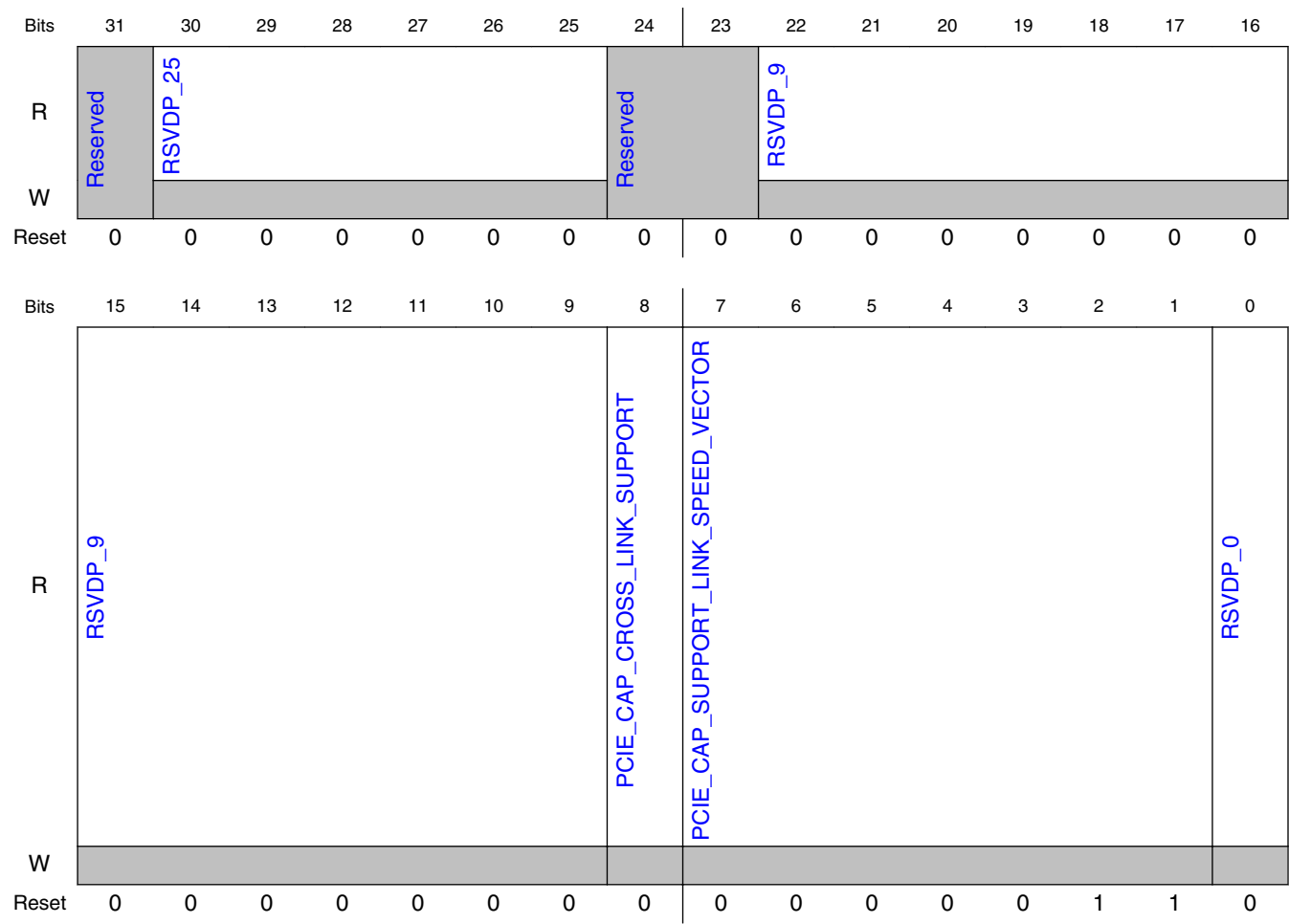
11.3.3.1.35.1 Offset

Register	Offset
LINK_CAPABILITIES2_REG	9Ch

11.3.3.1.35.2 Function

Link Capabilities 2 Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.35.3 Diagram



11.3.3.1.35.4 Fields

Field	Function
31 —	Reserved.
30-25 RSVDP_25	Reserved for future use.
24-23 —	Reserved.
22-9 RSVDP_9	Reserved for future use.
8 PCIE_CAP_CROSS_LINK_SUPPORT	Cross Link Supported. For a description of this standard PCIe register field, see the PCI Express Specification.
7-1 PCIE_CAP_SUPPORTED_LINK_SPEED_VECTOR	Supported Link Speeds Vector. For a description of this standard PCIe register field, see the PCI Express Specification. This field has a default of (PCIE_CAP_MAX_LINK_SPEED == 0100) ? 0001111 : (PCIE_CAP_MAX_LINK_SPEED == 0011) ? 0000111 : (PCIE_CAP_MAX_LINK_SPEED == 0010) ? 0000011 : 0000001 where PCIE_CAP_MAX_LINK_SPEED is a field in the LINK_CAPABILITIES_REG register.
0 RSVDP_0	Reserved for future use.

11.3.3.1.36 Link Control 2 and Status 2 Register. (LINK_CONTROL2_LINK_STATUS2_REG)

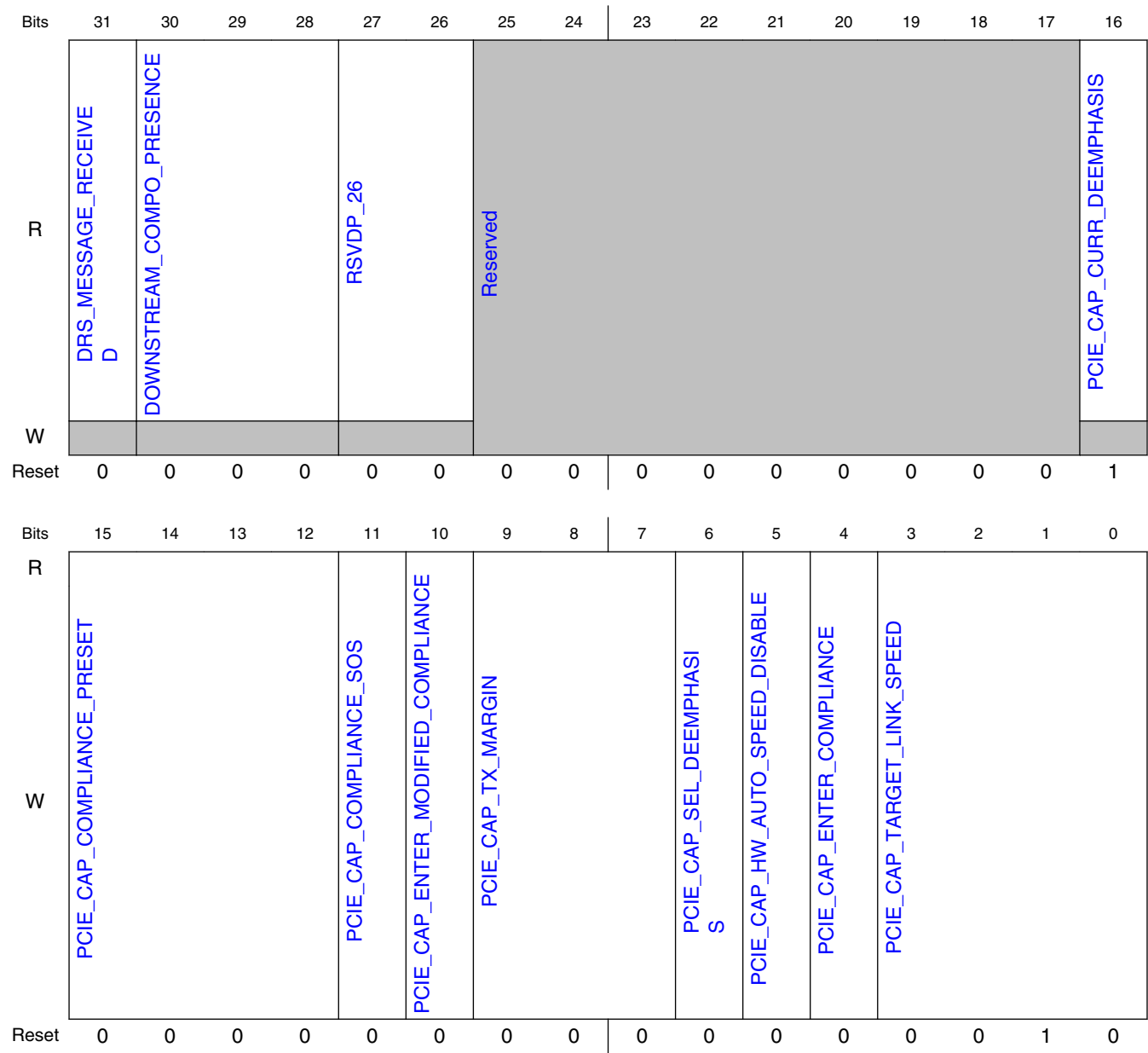
11.3.3.1.36.1 Offset

Register	Offset
LINK_CONTROL2_LINK_STATUS2_REG	A0h

11.3.3.1.36.2 Function

Link Control 2 and Status 2 Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.36.3 Diagram



11.3.3.1.36.4 Fields

Field	Function
31 DRS_MESSAGE_RECEIVED	DRS Message Received. For a description of this standard PCIe register field, see the PCI Express Base Specification 4.0. Note: The access attributes of this field are as follows: - Dbi: RW1C
30-28	Downstream Component Presence. For a description of this standard PCIe register field, see the PCI Express Base Specification 4.0.

Table continues on the next page...

Field	Function
DOWNSTREAM_COMPO_PRESENCE	
27-26 RSVDP_26	Reserved for future use.
25-17 —	Reserved.
16 PCIE_CAP_CURR_DEEMPHASIS	Current De-emphasis Level. For a description of this standard PCIe register field, see the PCI Express Specification. In M-PCIe mode this register is always 0x0. In C-PCIe mode, its contents are derived by sampling the PIPE
15-12 PCIE_CAP_COMPLIANCE_PRESSET	Sets Compliance Preset/De-emphasis for 5 GT/s and 8 GT/s. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
11 PCIE_CAP_COMPLIANCE_SOS	Sets Compliance Skip Ordered Sets transmission. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
10 PCIE_CAP_ENTER_MODIFIED_COMPLIANCE	Enter Modified Compliance. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
9-7 PCIE_CAP_TX_MARGIN	Controls Transmit Margin for Debug or Compliance. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
6 PCIE_CAP_SELECT_DEEMPHASIS	Controls Selectable De-emphasis for 5 GT/s. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.
5 PCIE_CAP_HW_AUTO_SPEED_DISABLE	Hardware Autonomous Speed Disable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
4 PCIE_CAP_ENTER_COMPLIANCE	Enter Compliance Mode. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
3-0 PCIE_CAP_TARGET_LINK_SPEED	Target Link Speed. For a description of this standard PCIe register field, see the PCI Express Specification. In M-PCIe mode, the contents of this field are derived from other registers. Note: This register field is sticky.

11.3.3.1.37 Advanced Error Reporting Extended Capability Header. (AER_EXT_CAP_HDR_OFF)

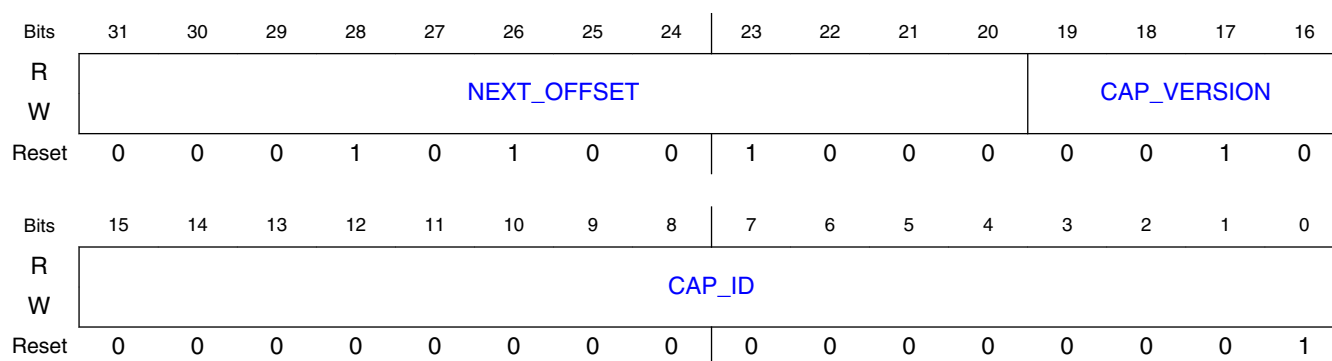
11.3.3.1.37.1 Offset

Register	Offset
AER_EXT_CAP_HDR_OFF	100h

11.3.3.1.37.2 Function

Advanced Error Reporting Extended Capability Header. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.37.3 Diagram



11.3.3.1.37.4 Fields

Field	Function
31-20 NEXT_OFFSET	Next Capability Offset. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.
19-16 CAP_VERSION	Capability Version. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.
15-0 CAP_ID	AER Extended Capability ID. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.

11.3.3.1.38 Uncorrectable Error Status Register. (UNCORR_ERR_STATU S_OFF)

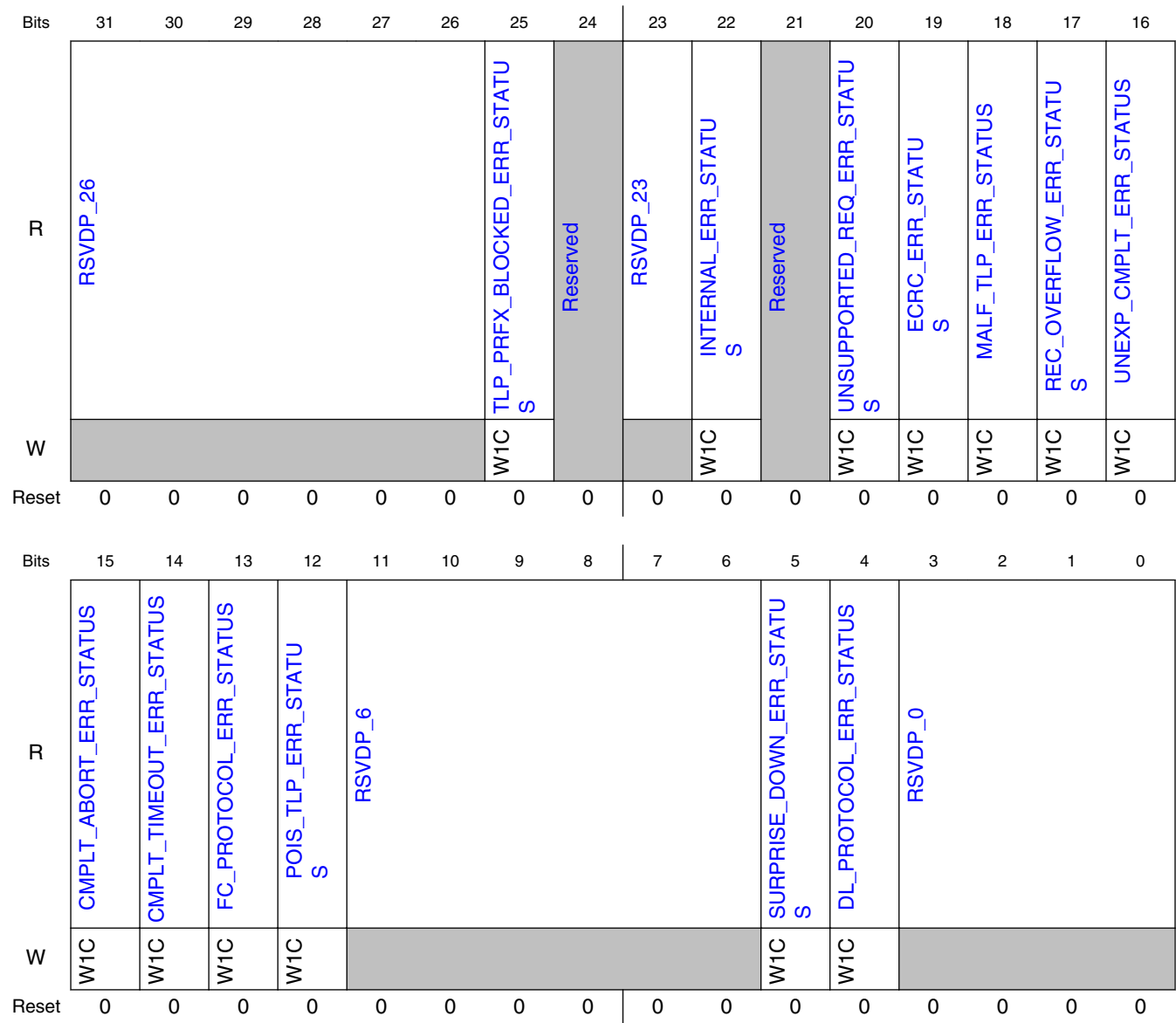
11.3.3.1.38.1 Offset

Register	Offset
UNCORR_ERR_STATU S_OFF	104h

11.3.3.1.38.2 Function

Uncorrectable Error Status Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.38.3 Diagram



11.3.3.1.38.4 Fields

Field	Function
31-26 RSVDP_26	Reserved for future use.
25 TLP_PRFX_BLOCKED_ERR_STATU	TLP Prefix Blocked Error Status. For a description of this standard PCIe register field, see the PCI Express Specification. Note: Not supported.
24	Reserved.

Table continues on the next page...

Field	Function
—	
23 RSVDP_23	Reserved for future use.
22 INTERNAL_ER R_STATUS	Uncorrectable Internal Error Status. For a description of this standard PCIe register field, see the PCI Express Specification. The controller sets this bit when your application asserts app_err_bus[9]. It does not set this bit when it detects internal uncorrectable internal errors such as parity and ECC failures. You should use the outputs from these errors to drive the app_err_bus[9] input. For more details, see the "Data Integrity (Wire, Datapath, and RAM Protection)" section in the Databook.
21 —	Reserved.
20 UNSUPPORTED_REQ_ERR_STATUS	Unsupported Request Error Status. For a description of this standard PCIe register field, see the PCI Express Specification.
19 ECRC_ERR_STATUS	ECRC Error Status. For a description of this standard PCIe register field, see the PCI Express Specification.
18 MALF_TLP_ERR_STATUS	Malformed TLP Status. For a description of this standard PCIe register field, see the PCI Express Specification.
17 REC_OVERFLOW_ERR_STATUS	Receiver Overflow Status. For a description of this standard PCIe register field, see the PCI Express Specification.
16 UNEXP_CMPLT_ERR_STATUS	Unexpected Completion Status. For a description of this standard PCIe register field, see the PCI Express Specification.
15 CMPLT_ABORT_ERR_STATUS	Completer Abort Status. For a description of this standard PCIe register field, see the PCI Express Specification.
14 CMPLT_TIMEOUT_ERR_STATUS	Completion Timeout Status. For a description of this standard PCIe register field, see the PCI Express Specification.
13 FC_PROTOCOL_ERR_STATUS	Flow Control Protocol Error Status. For a description of this standard PCIe register field, see the PCI Express Specification.
12 POIS_TLP_ERR_STATUS	Poisoned TLP Status. For a description of this standard PCIe register field, see the PCI Express Specification.
11-6 RSVDP_6	Reserved for future use.
5	Surprise Down Error Status (Optional). For a description of this standard PCIe register field, see the PCI Express Specification.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
SURPRISE_DOWN_ERR_STATUS	
4 DL_PROTOCOL_ERR_STATUS	Data Link Protocol Error Status. For a description of this standard PCIe register field, see the PCI Express Specification.
3-0 RSVDP_0	Reserved for future use.

11.3.3.1.39 Uncorrectable Error Mask Register. (UNCORR_ERR_MASK_OFF)

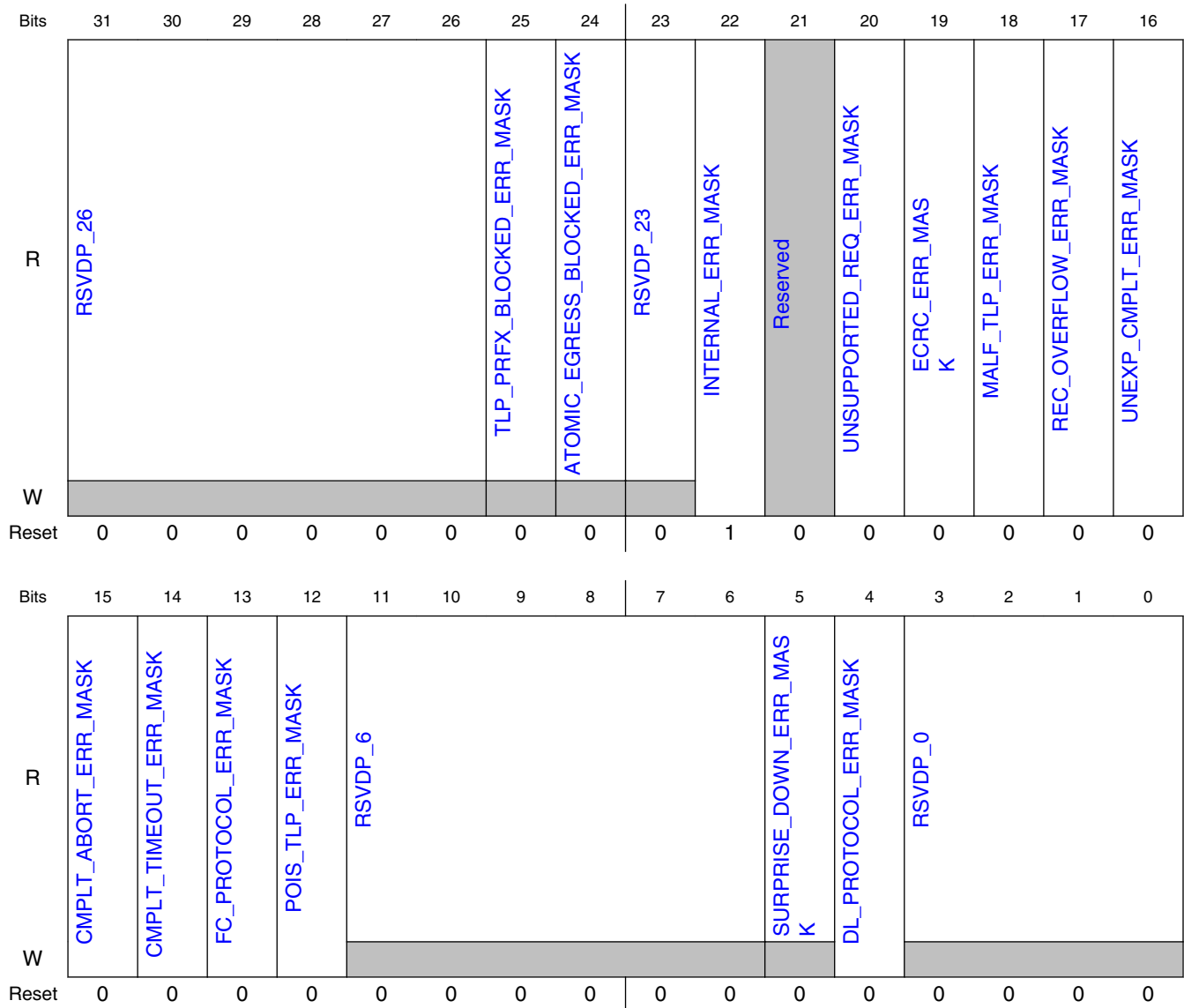
11.3.3.1.39.1 Offset

Register	Offset
UNCORR_ERR_MASK_OFF	108h

11.3.3.1.39.2 Function

Uncorrectable Error Mask Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.39.3 Diagram



11.3.3.1.39.4 Fields

Field	Function
31-26 RSVDP_26	Reserved for future use.
25 TLP_PRFX_BLOCKED_ERR_MASK	TLP Prefix Blocked Error Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: Not supported. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
24 ATOMIC_EGRESS_BLOCKED_ERR_MASK	AtomicOp Egress Block Mask (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
23 RSVDP_23	Reserved for future use.
22 INTERNAL_ERROR_MASK	Uncorrectable Internal Error Mask (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
21 —	Reserved.
20 UNSUPPORTED_REQ_ERR_MASK	Unsupported Request Error Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
19 ECRC_ERR_MASK	ECRC Error Mask (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
18 MALF_TLP_ERR_MASK	Malformed TLP Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
17 REC_OVERFLOW_ERR_MASK	Receiver Overflow Mask (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
16 UNEXP_CMPLT_ERR_MASK	Unexpected Completion Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
15 CMPLT_ABORT_ERR_MASK	Completer Abort Error Mask (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
14 CMPLT_TIMEOUT_ERR_MASK	Completion Timeout Error Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
13 FC_PROTOCOL_ERR_MASK	Flow Control Protocol Error Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
12 POIS_TLP_ERR_MASK	Poisoned TLP Error Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
11-6 RSVDP_6	Reserved for future use.
5	Surprise Down Error Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi:

Table continues on the next page...

Field	Function
SURPRISE_DOWN_ERR_MASK	LINK_CAPABILITIES_REG.PCIE_CAP_SURPRISE_DOWN_ERR_REP_CAP ? RW : RO Note: This register field is sticky.
4 DL_PROTOCOL_ERR_MASK	Data Link Protocol Error Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
3-0 RSVDP_0	Reserved for future use.

11.3.3.1.40 Uncorrectable Error Severity Register. (UNCORR_ERR_SEV_OFFSET)

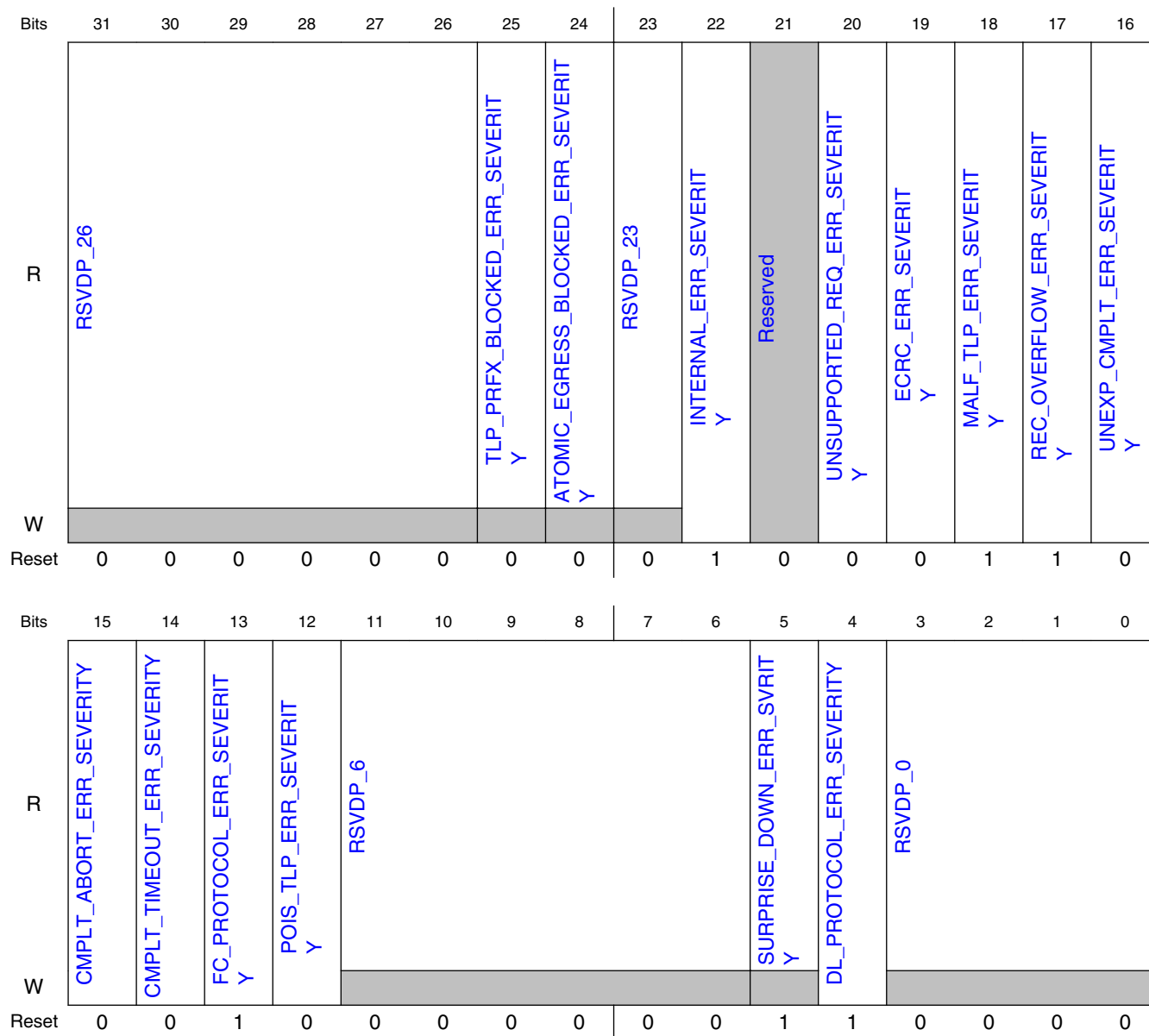
11.3.3.1.40.1 Offset

Register	Offset
UNCORR_ERR_SEV_OFFSET	10Ch

11.3.3.1.40.2 Function

Uncorrectable Error Severity Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.40.3 Diagram



11.3.3.1.40.4 Fields

Field	Function
31-26 RSVDP_26	Reserved for future use.
25 TLP_PRFX_BLK OCKED_ERR_S EVERITY	TLP Prefix Blocked Error Severity (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: Not supported. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.

Table continues on the next page...

Field	Function
24 ATOMIC_EGRESS_BLOCKED_ERROR_SEVERITY	AtomicOp Egress Blocked Severity (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
23 RSVDP_23	Reserved for future use.
22 INTERNAL_ERROR_SEVERITY	Uncorrectable Internal Error Severity (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
21 —	Reserved.
20 UNSUPPORTED_REQUEST_ERROR_SEVERITY	Unsupported Request Error Severity. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
19 ECRC_ERROR_SEVERITY	ECRC Error Severity (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
18 MALFORMED_TLP_ERROR_SEVERITY	Malformed TLP Severity. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
17 RECEIVER_OVERFLOW_ERROR_SEVERITY	Receiver Overflow Error Severity (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
16 UNEXPECTED_COMPLETION_ERROR_SEVERITY	Unexpected Completion Error Severity. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
15 COMPLETER_ABORT_ERROR_SEVERITY	Completer Abort Error Severity (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
14 COMPLETION_TIMEOUT_ERROR_SEVERITY	Completion Timeout Error Severity. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
13 FLOW_CONTROL_PROTOCOL_ERROR_SEVERITY	Flow Control Protocol Error Severity (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
12 POISONED_TLP_ERROR_SEVERITY	Poisoned TLP Severity. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
11-6 RSVDP_6	Reserved for future use.
5 SURPRISE_DOWN_ERR_SEVERITY	Surprise Down Error Severity (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: LINK_CAPABILITIES_REG.PCIE_CAP_SURPRISE_DOWN_ERR_REP_CAP ? RW : RO Note: This register field is sticky.
4 DL_PROTOCOL_ERR_SEVERITY	Data Link Protocol Error Severity. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
3-0 RSVDP_0	Reserved for future use.

11.3.3.1.41 Correctable Error Status Register. (CORR_ERR_STATUS_OFF)

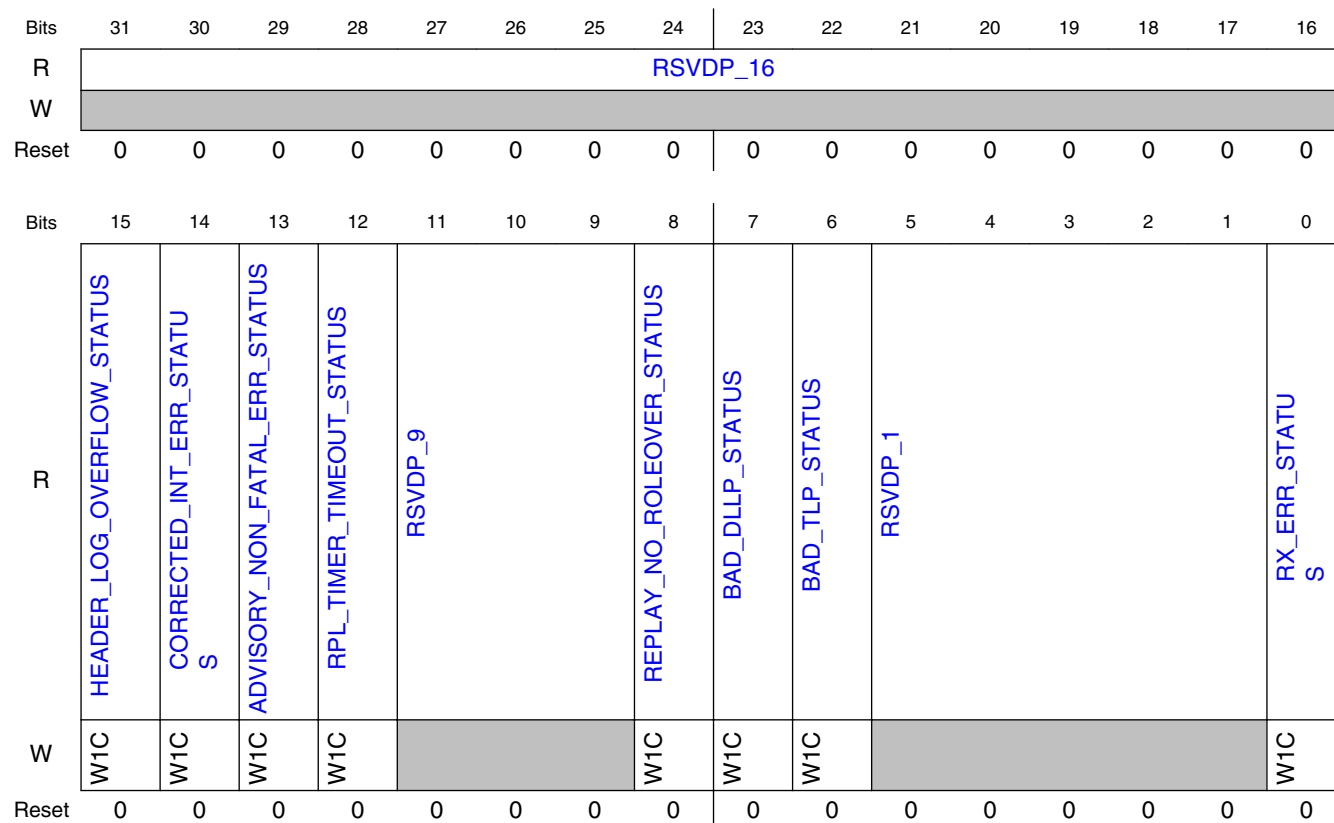
11.3.3.1.41.1 Offset

Register	Offset
CORR_ERR_STATUS_OFF	110h

11.3.3.1.41.2 Function

Correctable Error Status Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.41.3 Diagram



11.3.3.1.41.4 Fields

Field	Function
31-16 RSVDP_16	Reserved for future use.
15 HEADER_LOG_OVERFLOW_STATUS	Header Log Overflow Error Status (Optional). For a description of this standard PCIe register field, see the PCI Express Specification.
14 CORRECTED_INT_ERR_STATUS	Corrected Internal Error Status (Optional). For a description of this standard PCIe register field, see the PCI Express Specification.
13 ADVISORY_NON_FATAL_ERR_STATUS	Advisory Non-Fatal Error Status. For a description of this standard PCIe register field, see the PCI Express Specification.
12	Replay Timer Timeout Status. For a description of this standard PCIe register field, see the PCI Express Specification.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
RPL_TIMER_TIMEOUT_STATUSES	
11-9 RSVDP_9	Reserved for future use.
8 REPLAY_NO_ROLLOVER_STATUS	REPLAY_NUM Rollover Status. For a description of this standard PCIe register field, see the PCI Express Specification.
7 BAD_DLLP_STATUS	Bad DLLP Status. For a description of this standard PCIe register field, see the PCI Express Specification.
6 BAD_TLP_STATUS	Bad TLP Status. For a description of this standard PCIe register field, see the PCI Express Specification.
5-1 RSVDP_1	Reserved for future use.
0 RX_ERR_STATUS	Receiver Error Status (Optional). For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.42 Correctable Error Mask Register. (CORR_ERR_MASK_OFF)

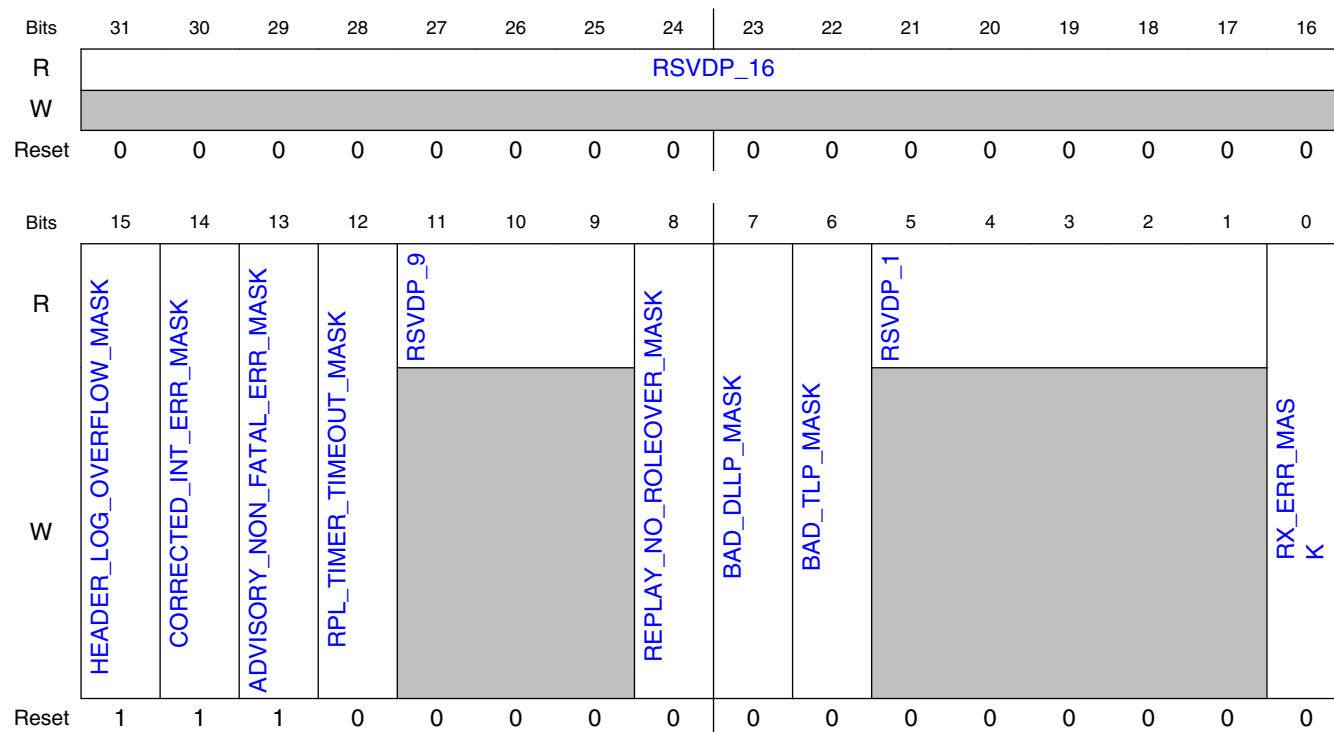
11.3.3.1.42.1 Offset

Register	Offset
CORR_ERR_MASK_OFF	114h

11.3.3.1.42.2 Function

Correctable Error Mask Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.42.3 Diagram



11.3.3.1.42.4 Fields

Field	Function
31-16 RSVDP_16	Reserved for future use.
15 HEADER_LOG_OVERFLOW_MASK	Header Log Overflow Error Mask (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
14 CORRECTED_INT_ERR_MASK	Corrected Internal Error Mask (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
13 ADVISORY_NON_FATAL_ERR_MASK	Advisory Non-Fatal Error Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
12 RPL_TIMER_TIMEOUT_MASK	Replay Timer Timeout Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
11-9 RSVDP_9	Reserved for future use.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
8 REPLAY_NO_ROLLOVER_MASK	REPLAY_NUM Rollover Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
7 BAD_DLLP_MASK	Bad DLLP Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
6 BAD_TLP_MASK	Bad TLP Mask. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
5-1 RSVDP_1	Reserved for future use.
0 RX_ERR_MASK	Receiver Error Mask (Optional). For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

11.3.3.1.43 Advanced Error Capabilities and Control Register. (ADV_ERR_CAP_CTRL_OFF)

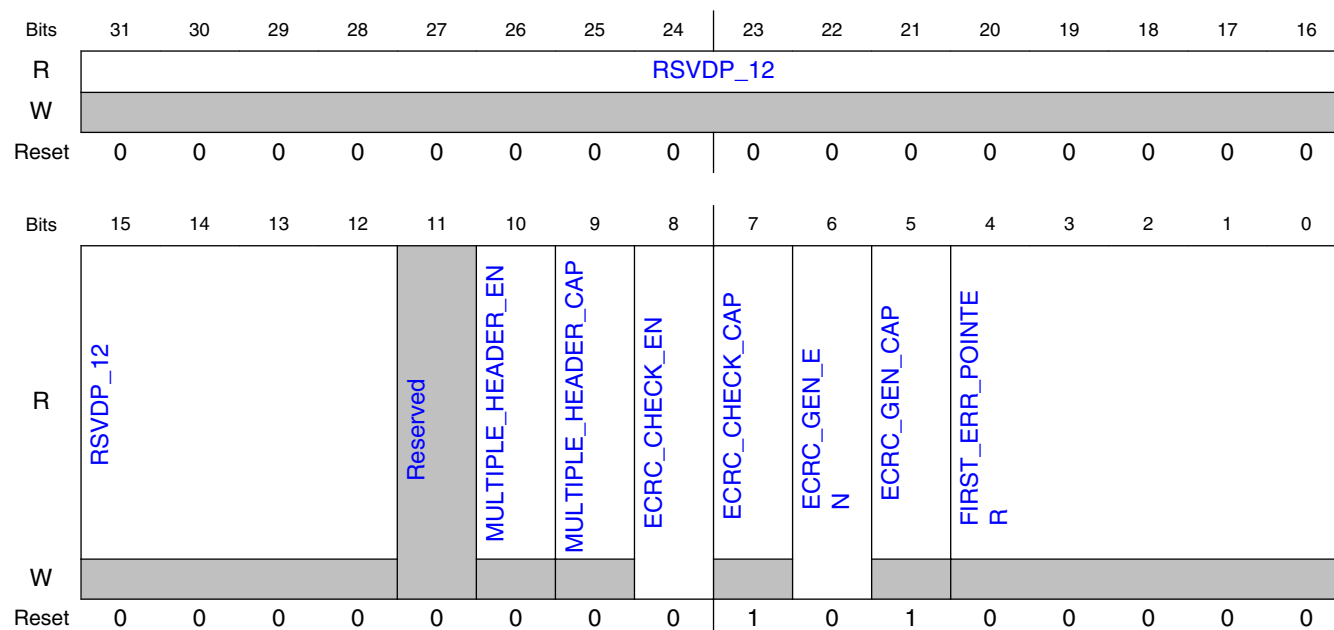
11.3.3.1.43.1 Offset

Register	Offset
ADV_ERR_CAP_CTRL_OFF	118h

11.3.3.1.43.2 Function

Advanced Error Capabilities and Control Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.43.3 Diagram



11.3.3.1.43.4 Fields

Field	Function
31-12 RSVDP_12	Reserved for future use.
11 —	Reserved.
10 MULTIPLE_HEADER_EN	Multiple Header Recording Enable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
9 MULTIPLE_HEADER_CAP	Multiple Header Recording Capable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
8 ECRC_CHECK_EN	ECRC Check Enable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
7 ECRC_CHECK_CAP	ECRC Check Capable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
6 ECRC_GEN_EN	ECRC Generation Enable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
5	ECRC Generation Capable. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
ECRC_GEN_C AP	
4-0 FIRST_ERR_P OINTER	First Error Pointer. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

11.3.3.1.44 Header Log Register 0. (HDR_LOG_0_OFF)

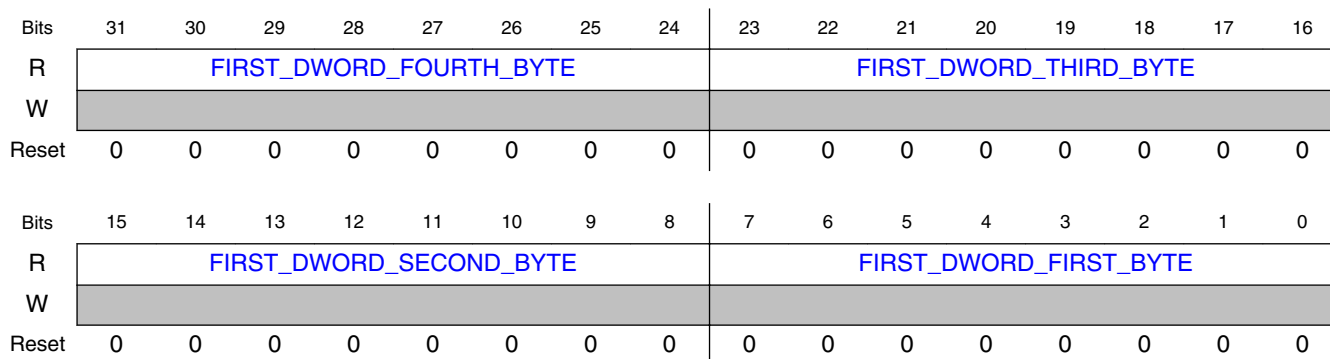
11.3.3.1.44.1 Offset

Register	Offset
HDR_LOG_0_OFF	11Ch

11.3.3.1.44.2 Function

Header Log Register 0. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.44.3 Diagram



11.3.3.1.44.4 Fields

Field	Function
31-24 FIRST_DWORD _FOURTH_BYT E	Byte 3 of Header log register of First 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

Table continues on the next page...

Field	Function
23-16 FIRST_DWORD_THIRD_BYTE	Byte 2 of Header log register of First 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
15-8 FIRST_DWORD_SECOND_BYTE	Byte 1 of Header log register of First 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
7-0 FIRST_DWORD_FIRST_BYTE	Byte 0 of Header log register of First 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

11.3.3.1.45 Header Log Register 1. (HDR_LOG_1_OFF)

11.3.3.1.45.1 Offset

Register	Offset
HDR_LOG_1_OFF	120h

11.3.3.1.45.2 Function

Header Log Register 1. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.45.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SECOND_DWORD_FOURTH_BYTE								SECOND_DWORD_THIRD_BYTE							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SECOND_DWORD_SECOND_BYTE								SECOND_DWORD_FIRST_BYTE							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.45.4 Fields

Field	Function
31-24 SECOND_DWORD_FOURTH_BYTE	Byte 3 of Header log register of Second 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
23-16 SECOND_DWORD_THIRD_BYTE	Byte 2 of Header log register of Second 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
15-8 SECOND_DWORD_SECOND_BYTE	Byte 1 of Header log register of Second 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
7-0 SECOND_DWORD_FIRST_BYTE	Byte 0 of Header log register of Second 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

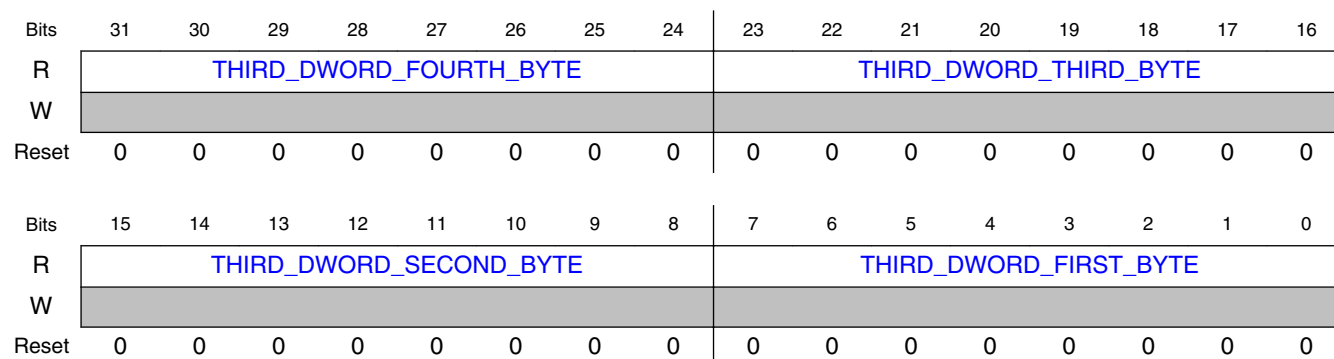
11.3.3.1.46 Header Log Register 2. (HDR_LOG_2_OFF)**11.3.3.1.46.1 Offset**

Register	Offset
HDR_LOG_2_OFF	124h

11.3.3.1.46.2 Function

Header Log Register 2. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.46.3 Diagram



11.3.3.1.46.4 Fields

Field	Function
31-24 THIRD_DWORD_FOURTH_BYTE	Byte 3 of Header log register of Third 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
23-16 THIRD_DWORD_THIRD_BYTE	Byte 2 of Header log register of Third 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
15-8 THIRD_DWORD_SECOND_BYTE	Byte 1 of Header log register of Third 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
7-0 THIRD_DWORD_FIRST_BYTE	Byte 0 of Header log register of Third 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

11.3.3.1.47 Header Log Register 3. (HDR_LOG_3_OFF)

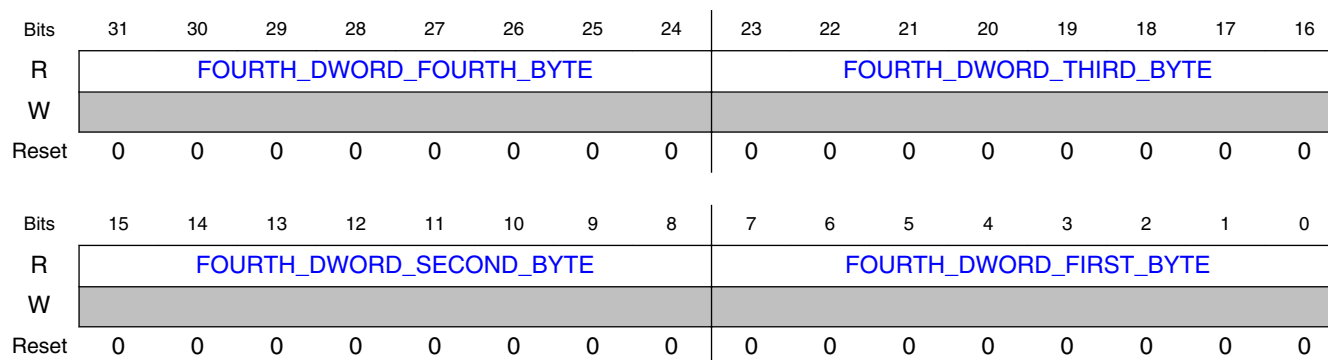
11.3.3.1.47.1 Offset

Register	Offset
HDR_LOG_3_OFF	128h

11.3.3.1.47.2 Function

Header Log Register 3. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.47.3 Diagram



11.3.3.1.47.4 Fields

Field	Function
31-24 FOURTH_DWORD_FOURTH_BYTE	Byte 3 of Header log register of Fourth 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
23-16 FOURTH_DWORD_THIRD_BYTE	Byte 2 of Header log register of Fourth 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
15-8 FOURTH_DWORD_SECOND_BYTE	Byte 1 of Header log register of Fourth 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
7-0 FOURTH_DWORD_FIRST_BYTE	Byte 0 of Header log register of Fourth 32 bit Data Word. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

11.3.3.1.48 Root Error Command Register. (ROOT_ERR_CMD_OFF)

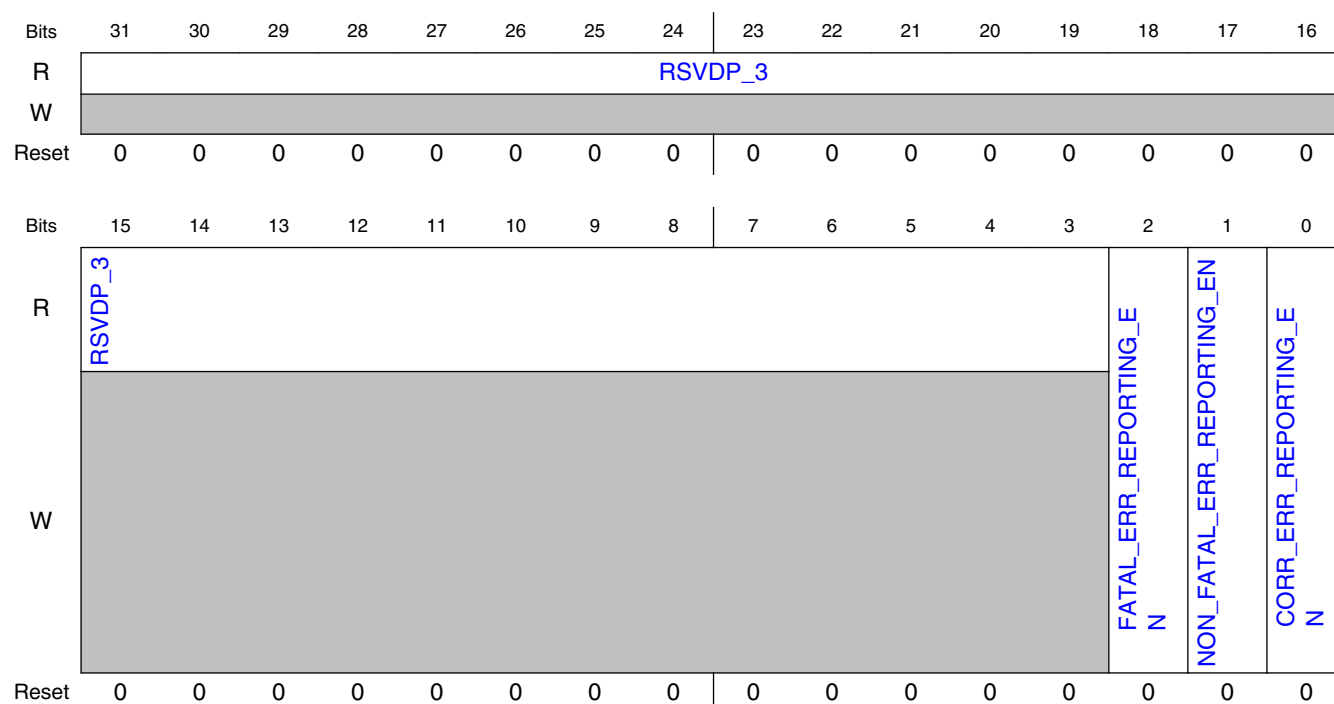
11.3.3.1.48.1 Offset

Register	Offset
ROOT_ERR_CMD_OFF	12Ch

11.3.3.1.48.2 Function

Root Error Command Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.48.3 Diagram



11.3.3.1.48.4 Fields

Field	Function
31-3 RSVDP_3	Reserved for future use.
2 FATAL_ERR_R EPORTING_EN	Fatal Error Reporting Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
1 NON_FATAL_E RR_REPORTIN G_EN	Non-Fatal Error Reporting Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
0 CORR_ERR_R EPORTING_EN	Correctable Error Reporting Enable. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.49 Root Error Status Register. (ROOT_ERR_STATUS_OFF)

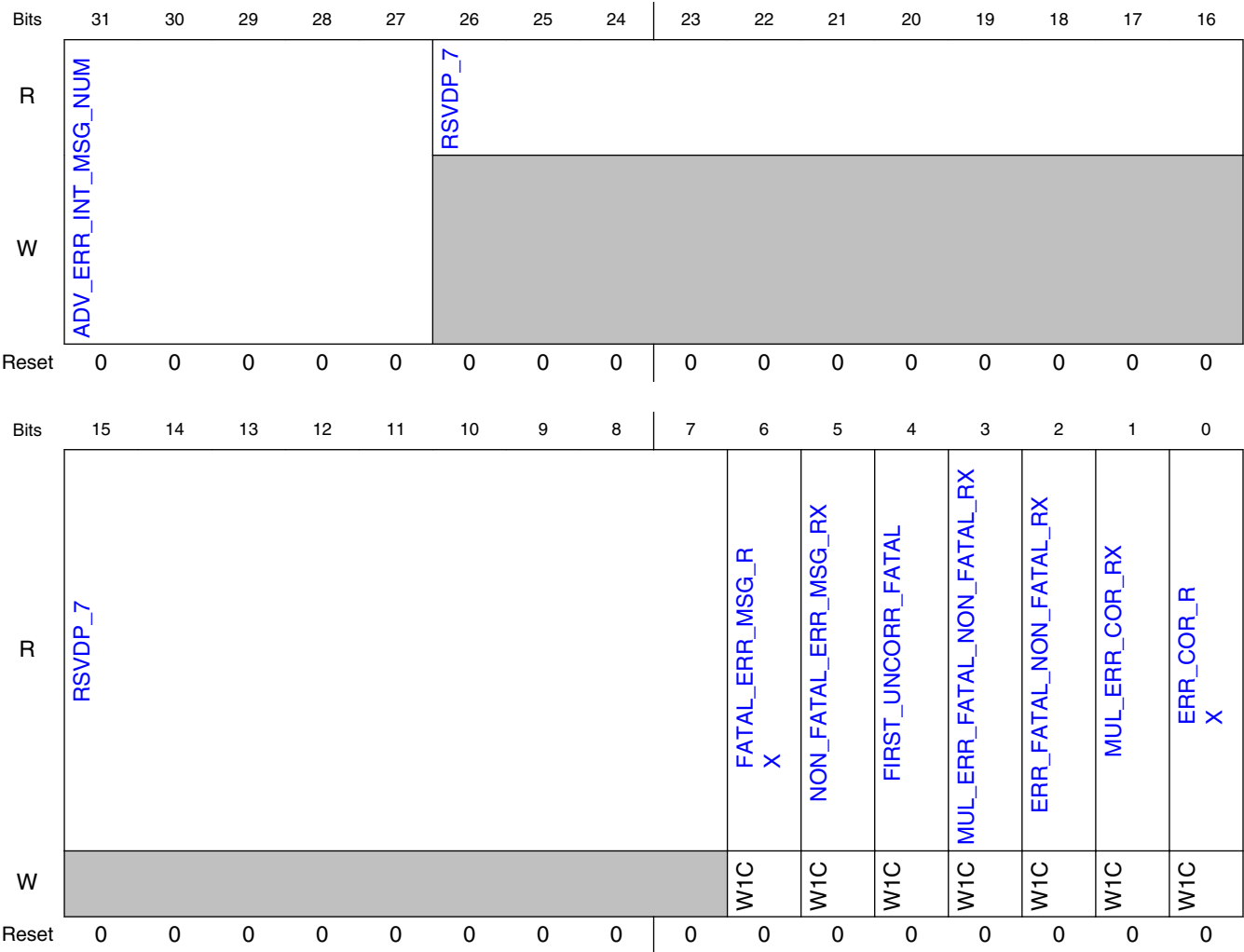
11.3.3.1.49.1 Offset

Register	Offset
ROOT_ERR_STATUS_OFF	130h

11.3.3.1.49.2 Function

Root Error Status Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.49.3 Diagram



11.3.3.1.49.4 Fields

Field	Function
31-27 ADV_ERR_INT_MSG_NUM	Advanced Error Interrupt Message Number. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.
26-7 RSVDP_7	Reserved for future use.
6 FATAL_ERR_MSG_RX	One or more Fatal Error Messages Received. For a description of this standard PCIe register field, see the PCI Express Specification.
5 NON_FATAL_ERR_MSG_RX	One or more Non-Fatal Error Messages Received. For a description of this standard PCIe register field, see the PCI Express Specification.
4 FIRST_UNCORR_FATAL	First Uncorrectable Error is Fatal. For a description of this standard PCIe register field, see the PCI Express Specification.
3 MUL_ERR_FATAL_NON_FATAL_RX	Multiple Fatal or Non-Fatal Errors Received. For a description of this standard PCIe register field, see the PCI Express Specification.
2 ERR_FATAL_NON_FATAL_RX	Fatal or Non-Fatal Error Received. For a description of this standard PCIe register field, see the PCI Express Specification.
1 MUL_ERR_CORR_RX	Multiple Correctable Errors Received. For a description of this standard PCIe register field, see the PCI Express Specification.
0 ERR_CORR_RX	Correctable Error Received. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.50 Error Source Identification Register. (ERR_SRC_ID_OFF)

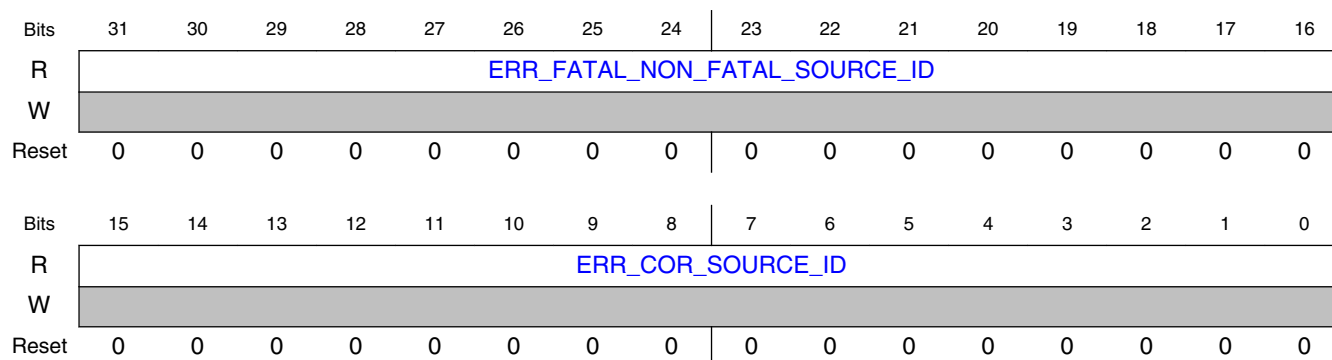
11.3.3.1.50.1 Offset

Register	Offset
ERR_SRC_ID_OFF	134h

11.3.3.1.50.2 Function

Error Source Identification Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.50.3 Diagram



11.3.3.1.50.4 Fields

Field	Function
31-16 ERR_FATAL_NON_FATAL_SOURCE_ID	Source of Fatal/Non-Fatal Error. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
15-0 ERR_COR_SOURCE_ID	Source of Correctable Error. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

11.3.3.1.51 TLP Prefix Log Register 1. (TLP_PREFIX_LOG_1_OFF)

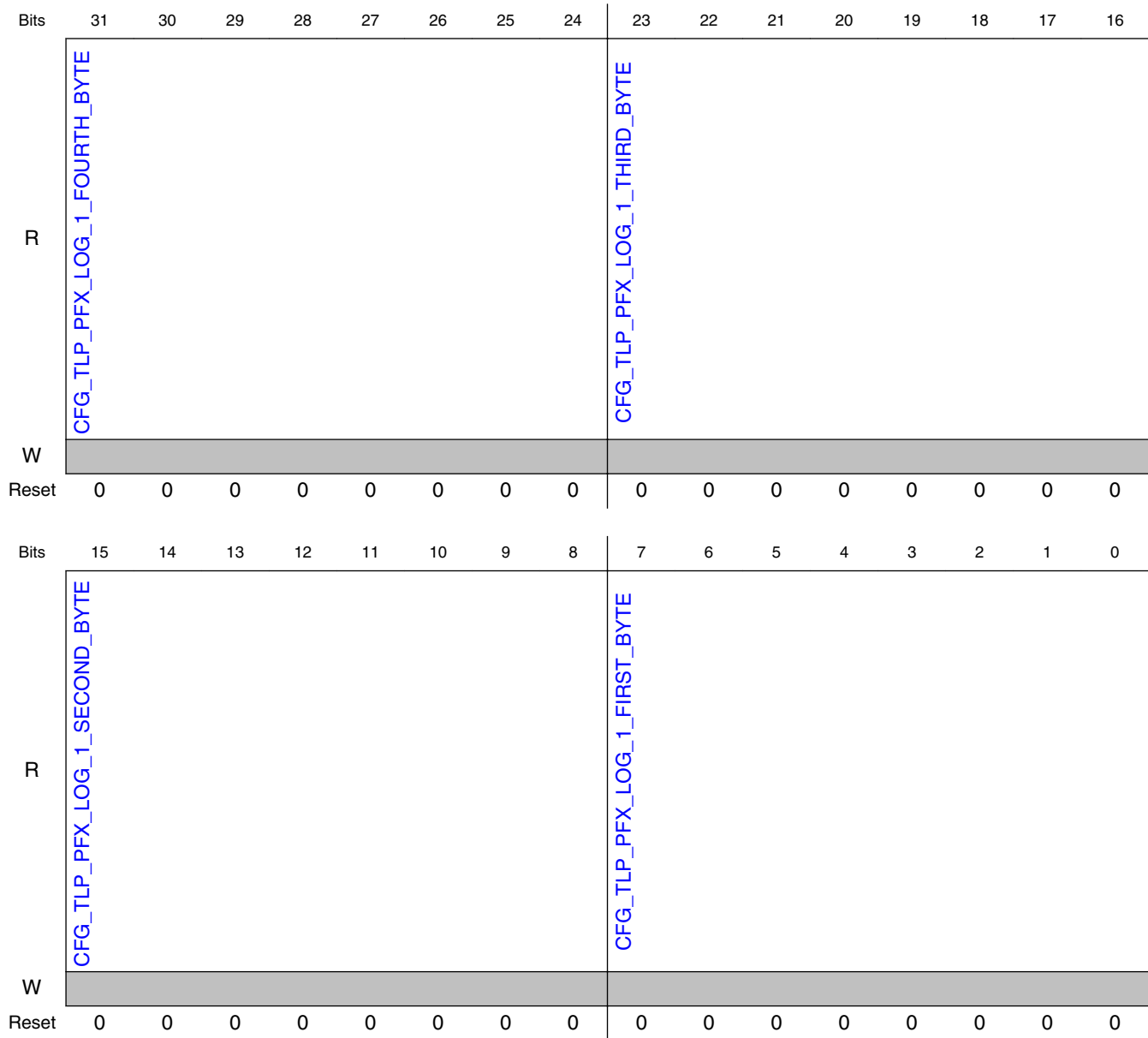
11.3.3.1.51.1 Offset

Register	Offset
TLP_PREFIX_LOG_1_OFF	138h

11.3.3.1.51.2 Function

TLP Prefix Log Register 1. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.51.3 Diagram



11.3.3.1.51.4 Fields

Field	Function
31-24 CFG_TLP_PFX_LOG_1_FOURTH_BYTE	Byte 3 of Error TLP Prefix Log 1. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
23-16	Byte 2 of Error TLP Prefix Log 1. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
CFG_TLP_PFX_LOG_1_THIRD_BYTE	
15-8 CFG_TLP_PFX_LOG_1_SECONDBYTE	Byte 1 of Error TLP Prefix Log 1. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
7-0 CFG_TLP_PFX_LOG_1_FIRST_BYTE	Byte 0 of Error TLP Prefix Log 1. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

11.3.3.1.52 TLP Prefix Log Register 2. (TLP_PREFIX_LOG_2_OFF)

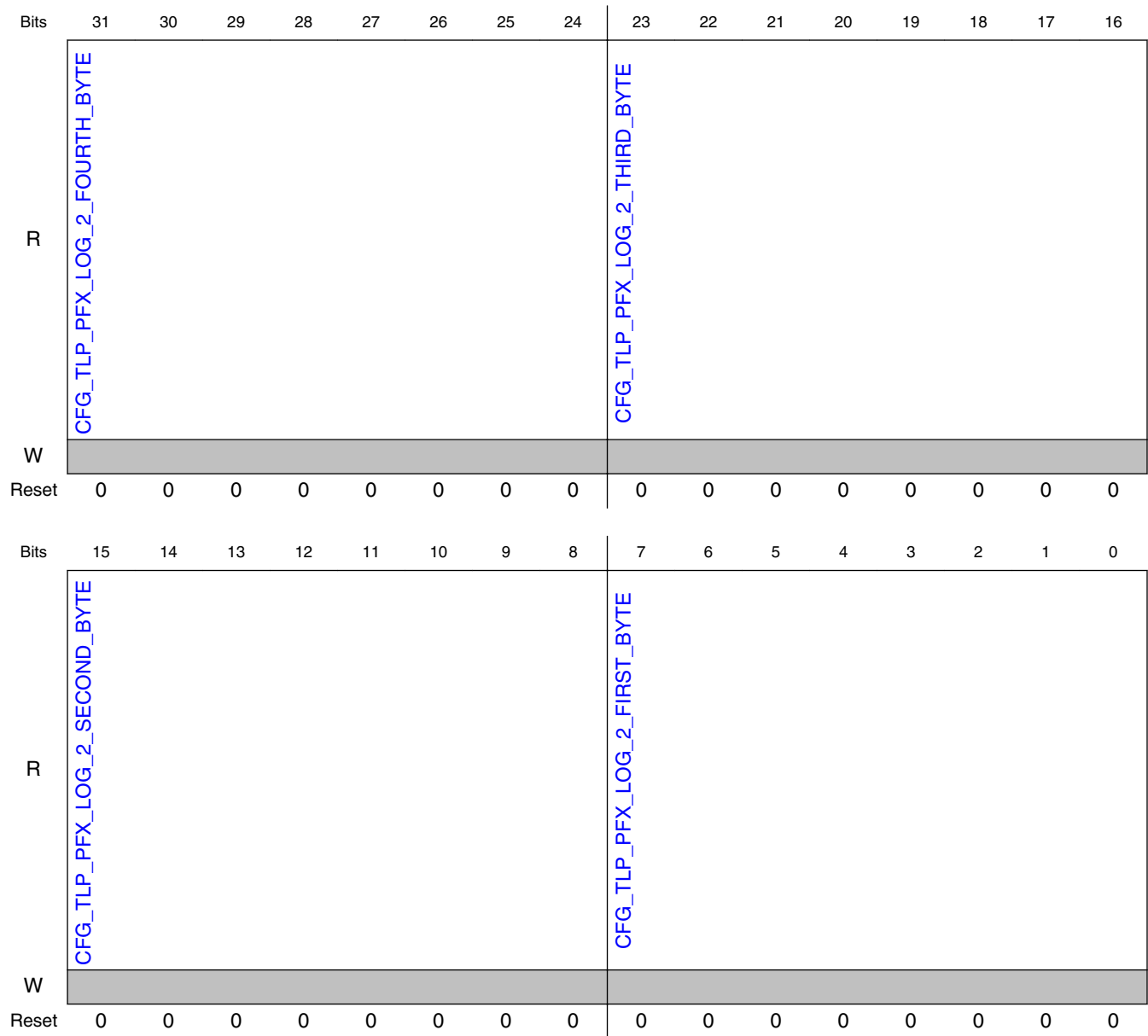
11.3.3.1.52.1 Offset

Register	Offset
TLP_PREFIX_LOG_2_OFF	13Ch

11.3.3.1.52.2 Function

TLP Prefix Log Register 2. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.52.3 Diagram



11.3.3.1.52.4 Fields

Field	Function
31-24 CFG_TLP_PFX_LOG_2_FOURTH_BYTE	Byte 3 Error TLP Prefix Log 2. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
23-16	Byte 2 Error TLP Prefix Log 2. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
CFG_TLP_PFX_LOG_2_THIRD_BYTE	
15-8 CFG_TLP_PFX_LOG_2_SECONDBYTE	Byte 1 Error TLP Prefix Log 2. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
7-0 CFG_TLP_PFX_LOG_2_FIRST_BYTE	Byte 0 Error TLP Prefix Log 2. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

11.3.3.1.53 TLP Prefix Log Register 3. (TLP_PREFIX_LOG_3_OFF)

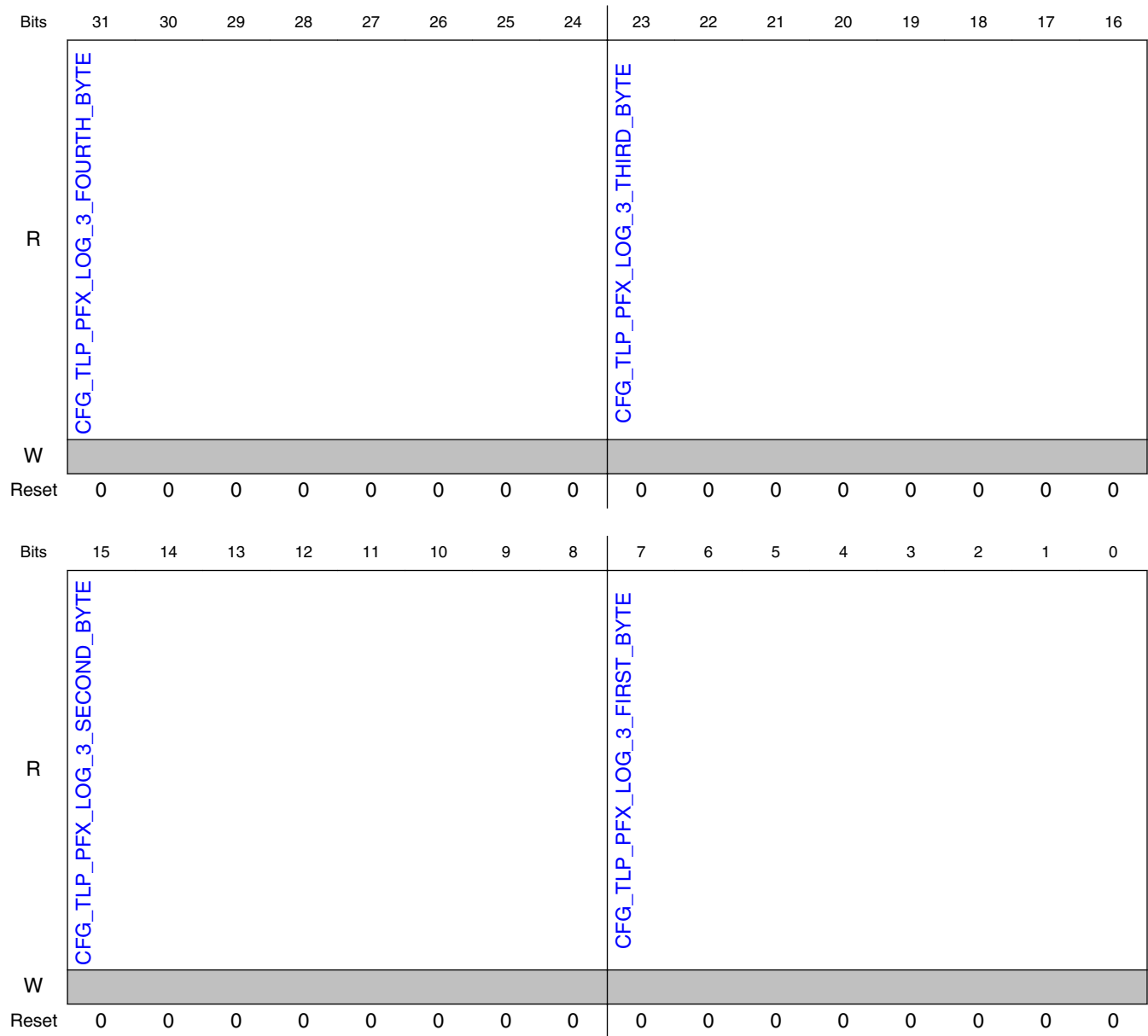
11.3.3.1.53.1 Offset

Register	Offset
TLP_PREFIX_LOG_3_OFF	140h

11.3.3.1.53.2 Function

TLP Prefix Log Register 3. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.53.3 Diagram



11.3.3.1.53.4 Fields

Field	Function
31-24 CFG_TLP_PFX_LOG_3_FOURTH_BYTE	Byte 3 Error TLP Prefix Log 3. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
23-16	Byte 2 Error TLP Prefix Log 3. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
CFG_TLP_PFX_LOG_3_THIRD_BYTE	
15-8 CFG_TLP_PFX_LOG_3_SECONDBYTE	Byte 1 Error TLP Prefix Log 3. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
7-0 CFG_TLP_PFX_LOG_3_FIRST_BYTE	Byte 0 Error TLP Prefix Log 3. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

11.3.3.1.54 TLP Prefix Log Register 4. (TLP_PREFIX_LOG_4_OFF)

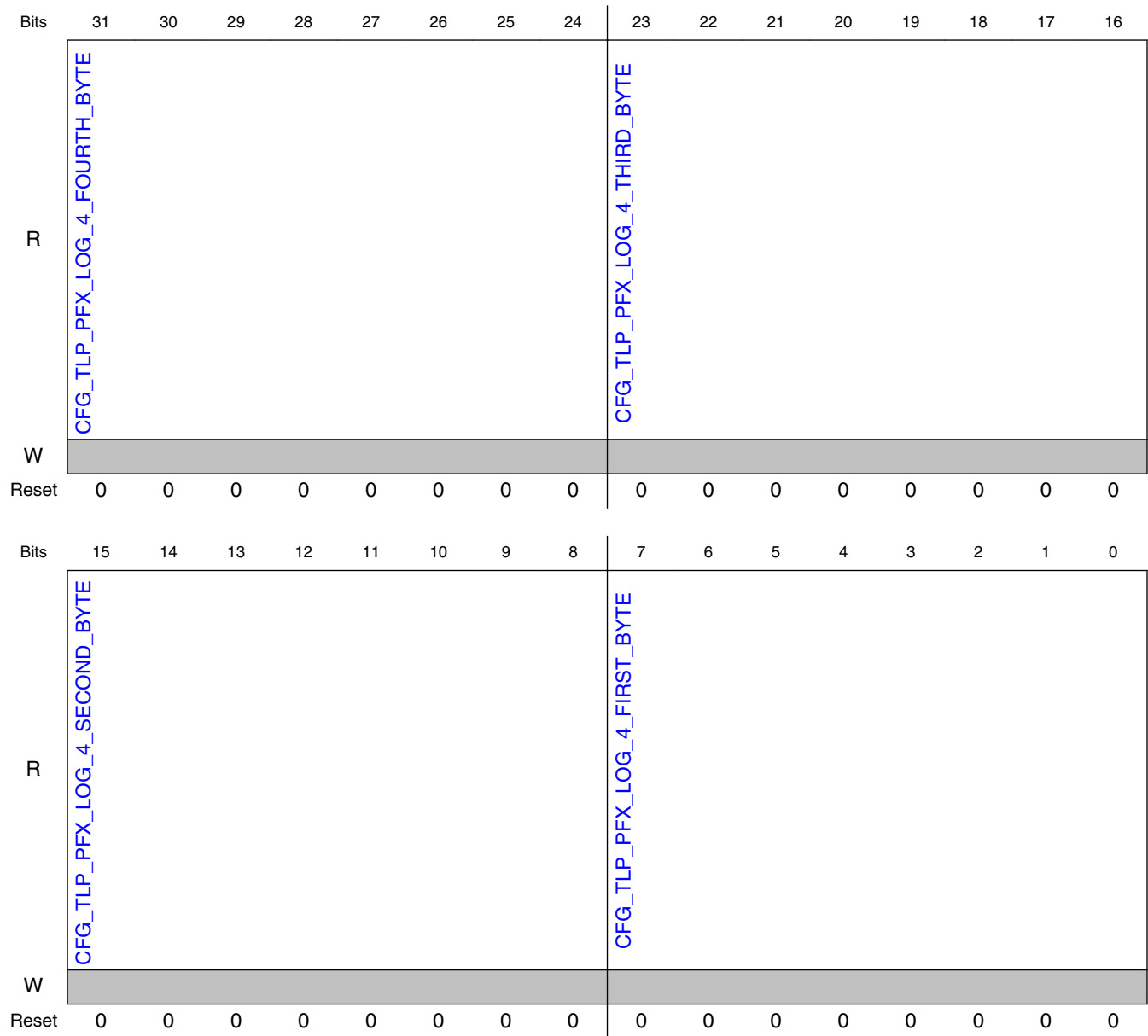
11.3.3.1.54.1 Offset

Register	Offset
TLP_PREFIX_LOG_4_OFF	144h

11.3.3.1.54.2 Function

TLP Prefix Log Register 4. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.54.3 Diagram



11.3.3.1.54.4 Fields

Field	Function
31-24 CFG_TLP_PFX_LOG_4_FOURTH_BYTE	Byte 3 Error TLP Prefix Log 4. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
23-16	Byte 2 Error TLP Prefix Log 4. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
CFG_TLP_PFX_LOG_4_THIRD_BYTE	
15-8 CFG_TLP_PFX_LOG_4_SECOND_BYTE	Byte 1 Error TLP Prefix Log 4. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.
7-0 CFG_TLP_PFX_LOG_4_FIRST_BYTE	Byte 0 Error TLP Prefix Log 4. For a description of this standard PCIe register field, see the PCI Express Specification. Note: This register field is sticky.

11.3.3.1.55 L1 Substates Extended Capability Header. (L1SUB_CAP_HEADER_REG)

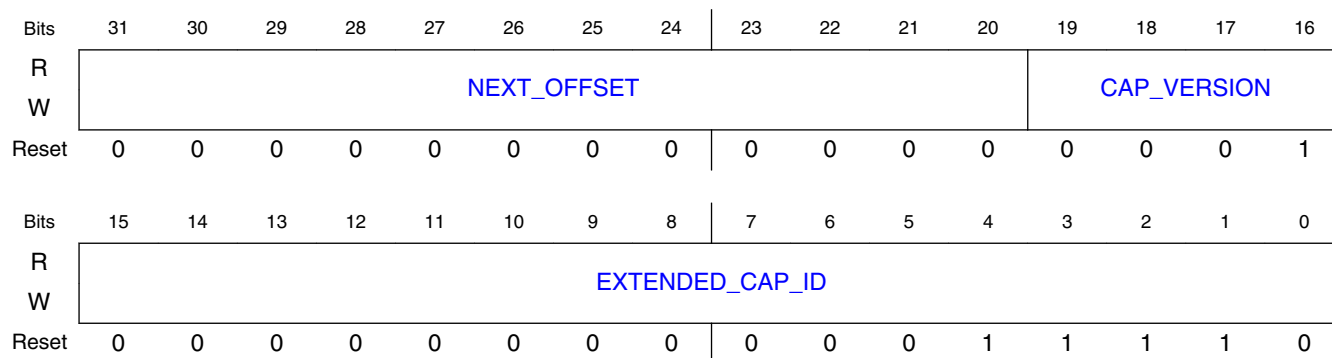
11.3.3.1.55.1 Offset

Register	Offset
L1SUB_CAP_HEADER_REG	148h

11.3.3.1.55.2 Function

L1 Substates Extended Capability Header. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.55.3 Diagram



11.3.3.1.55.4 Fields

Field	Function
31-20 NEXT_OFFSET	Next Capability Offset. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.
19-16 CAP_VERSION	Capability Version. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.
15-0 EXTENDED_CAP_ID	L1SUB Extended Capability ID. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: if (DBI_RO_WR_EN == 1) then R/W(sticky) else R(sticky) Note: This register field is sticky.

11.3.3.1.56 L1 Substates Capability Register. (L1SUB_CAPABILITY_REG)

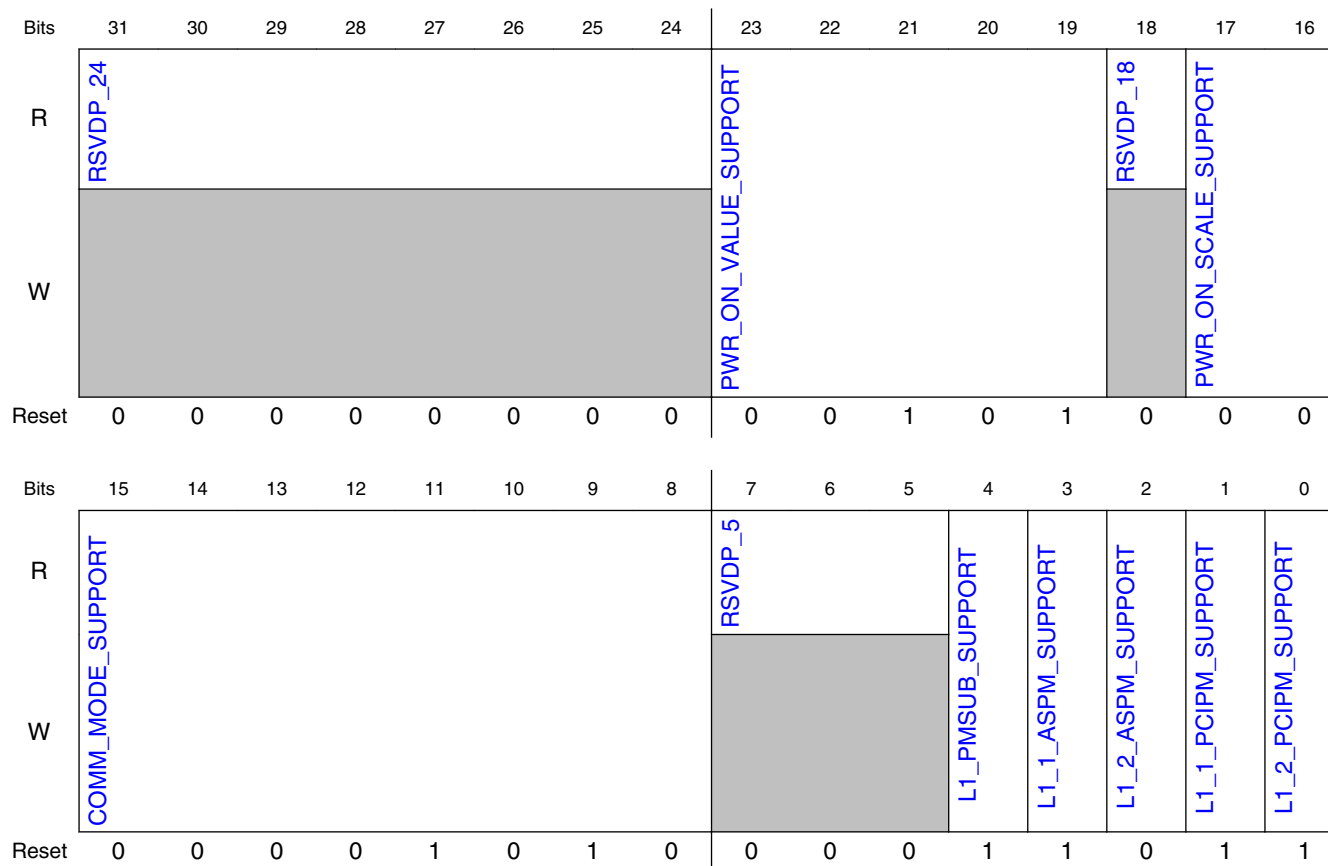
11.3.3.1.56.1 Offset

Register	Offset
L1SUB_CAPABILITY_REG	14Ch

11.3.3.1.56.2 Function

L1 Substates Capability Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.56.3 Diagram



11.3.3.1.56.4 Fields

Field	Function
31-24 RSVDP_24	Reserved for future use.
23-19 PWR_ON_VALUE_SUPPORT	Port T Power On Value. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
18 RSVDP_18	Reserved for future use.
17-16 PWR_ON_SCALE_SUPPORT	Port T Power On Scale. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
15-8 COMM_MODE_SUPPORT	Port Common Mode Restore Time. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
7-5	Reserved for future use.

Table continues on the next page...

Field	Function
RSVDP_5	
4 L1_PMSUB_SU PPORT	L1 PM Substates ECN Supported. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
3 L1_1_ASPM_S UPPORT	ASPM L11 Supported. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
2 L1_2_ASPM_S UPPORT	ASPM L12 Supported. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
1 L1_1_PCIPM_S UPPORT	PCI-PM L11 Supported. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
0 L1_2_PCIPM_S UPPORT	PCI-PM L12 Supported. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.57 L1 Substates Control 1 Register. (L1SUB_CONTROL1_REG)

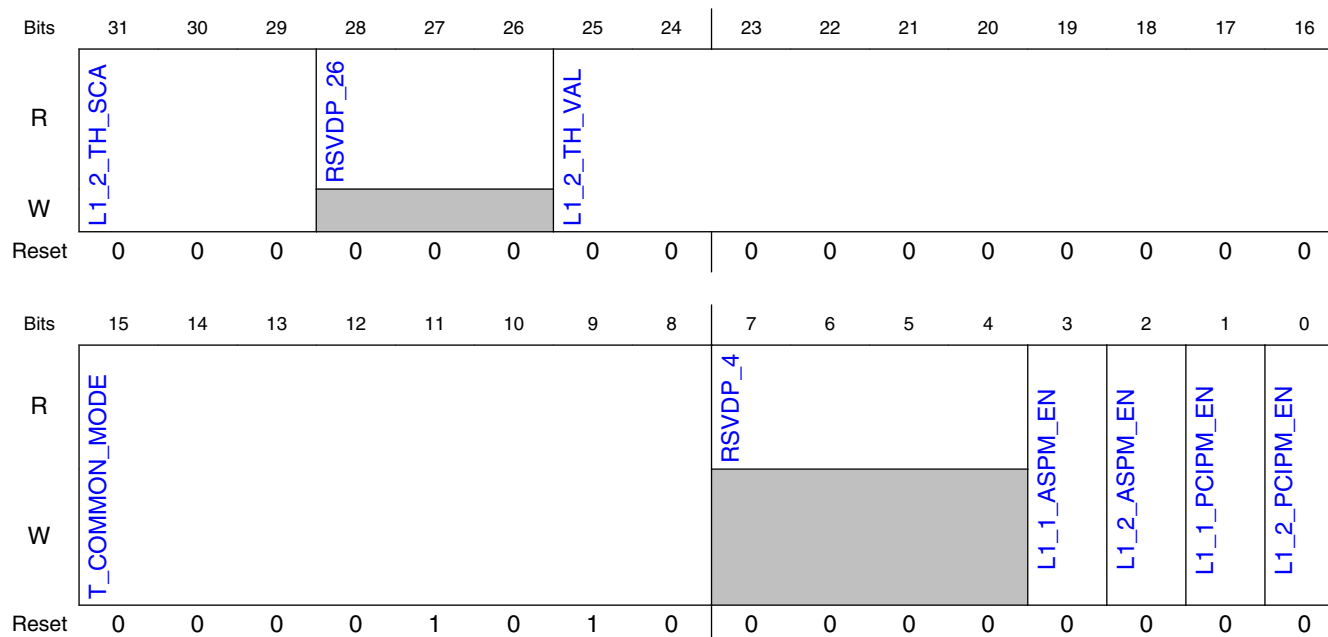
11.3.3.1.57.1 Offset

Register	Offset
L1SUB_CONTROL1_R EG	150h

11.3.3.1.57.2 Function

L1 Substates Control 1 Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.57.3 Diagram



11.3.3.1.57.4 Fields

Field	Function
31-29 L1_2_TH_SCA	LTR L12 Threshold Scale. For a description of this standard PCIe register field, see the PCI Express Specification.
28-26 RSVDP_26	Reserved for future use.
25-16 L1_2_TH_VAL	LTR L12 Threshold Value. For a description of this standard PCIe register field, see the PCI Express Specification.
15-8 T_COMMON_M ODE	Common Mode Restore Time. For a description of this standard PCIe register field, see the PCI Express Specification. Note: The access attributes of this field are as follows: - Dbi: R/W
7-4 RSVDP_4	Reserved for future use.
3 L1_1_ASPM_E N	ASPM L11 Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
2 L1_2_ASPM_E N	ASPM L12 Enable. For a description of this standard PCIe register field, see the PCI Express Specification.
1 L1_1_PCIPM_E N	PCI-PM L11 Enable. For a description of this standard PCIe register field, see the PCI Express Specification.

Table continues on the next page...

Field	Function
0 L1_2_PCIPM_EN	PCI-PM L12 Enable. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.58 L1 Substates Control 2 Register. (L1SUB_CONTROL2_REG)

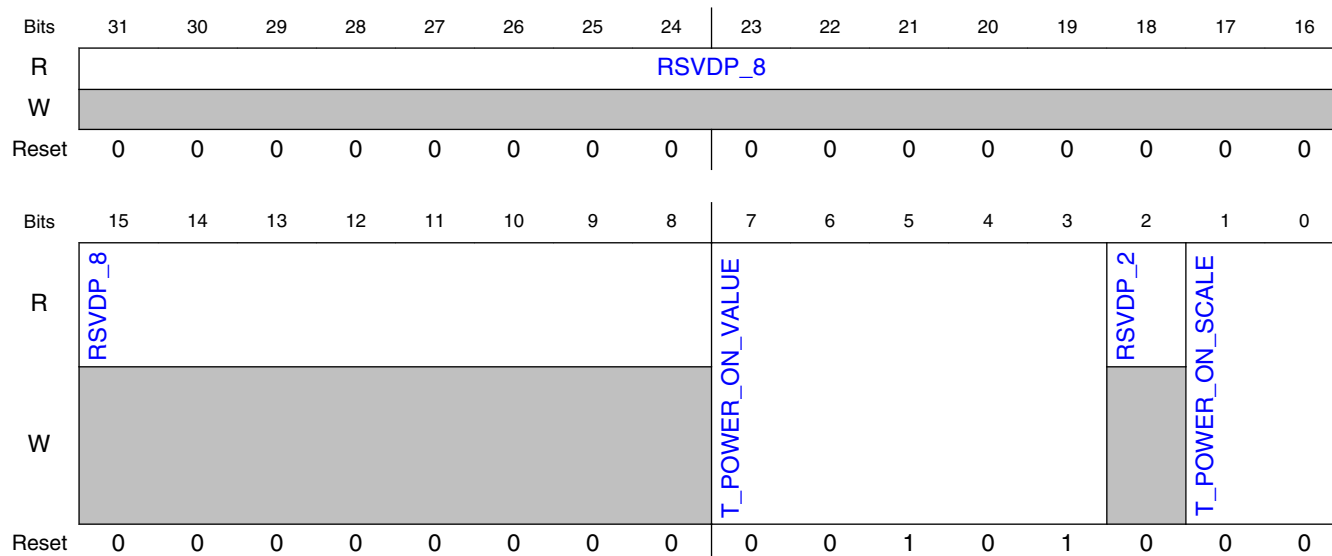
11.3.3.1.58.1 Offset

Register	Offset
L1SUB_CONTROL2_REG	154h

11.3.3.1.58.2 Function

L1 Substates Control 2 Register. For a description of this standard PCIe register, see the PCI Express Specification.

11.3.3.1.58.3 Diagram



11.3.3.1.58.4 Fields

Field	Function
31-8	Reserved for future use.

Table continues on the next page...

PCI Express (PCIe)

Field	Function
RSVDP_8	
7-3 T_POWER_ON_VALUE	T Power On Value. For a description of this standard PCIe register field, see the PCI Express Specification.
2 RSVDP_2	Reserved for future use.
1-0 T_POWER_ON_SCALE	T Power On Scale. For a description of this standard PCIe register field, see the PCI Express Specification.

11.3.3.1.59 Ack Latency Timer and Replay Timer Register. (ACK_LATENCY_TIMER_OFF)

11.3.3.1.59.1 Offset

Register	Offset
ACK_LATENCY_TIMER_OFF	700h

11.3.3.1.59.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	REPLAY_TIME_LIMIT															
W																
Reset	0	0	0	1	1	0	0	0	0	1	0	0	0	1	1	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ROUND_TRIP_LATENCY_TIME_LIMIT															
W																
Reset	0	0	0	0	1	0	0	0	0	0	0	1	0	1	1	1

11.3.3.1.59.3 Fields

Field	Function
31-16 REPLAY_TIME_LIMIT	Replay Timer Limit. The replay timer expires when it reaches this limit. The controller initiates a replay upon reception of a NAK or when the replay timer expires. For more details, see "Transmit Replay". You can modify the effective timer limit with the TIMER_MOD_REPLAY_TIMER field of the TIMER_CTRL_MAX_FUNC_NUM_OFF register. After reset, the controller updates the default according to the Negotiated Link Width, Max_Payload_Size, and speed. The value is determined from Tables 3-4,

Table continues on the next page...

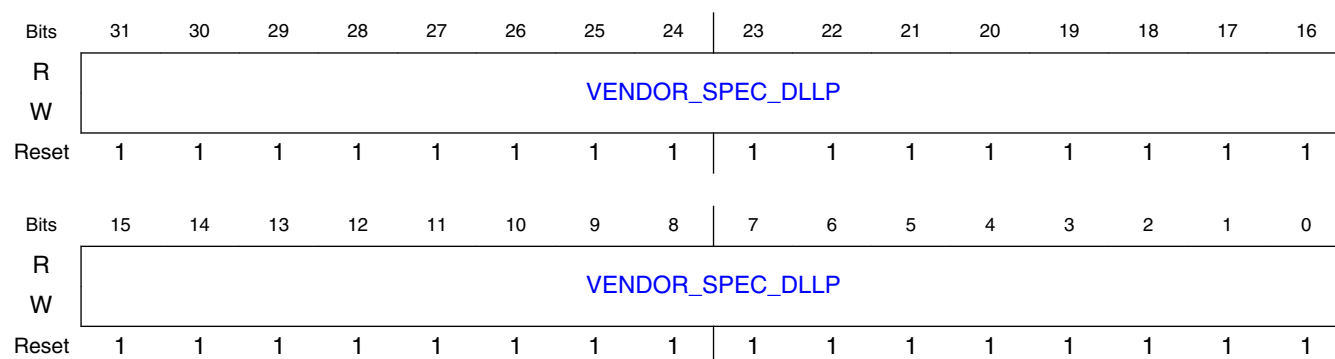
Field	Function
	3-5, and 3-6 of the PCIe 3.0 specification. If there is a change in the payload size or link speed, the controller will override any value that you have written to this register field, and reset the field back to the specification-defined value. It will not change the value in the TIMER_MOD_REPLAY_TIMER field of the TIMER_CTRL_MAX_FUNC_NUM_OFF register.
15-0 ROUND_TRIP_LATENCY_TIMER_LIMIT	Ack Latency Timer Limit. The Ack latency timer expires when it reaches this limit. For more details, see "Ack Scheduling". You can modify the effective timer limit with the TIMER_MOD_ACK_NAK field of the TIMER_CTRL_MAX_FUNC_NUM_OFF register. After reset, the controller updates the default according to the Negotiated Link Width, Max_Payload_Size, and speed. The value is determined from Tables 3-7, 3-8, and 3-9 of the PCIe 3.0 specification. The limit must reflect the round trip latency from requester to completer. If there is a change in the payload size or link width, the controller will override any value that you have written to this register field, and reset the field back to the specification-defined value. It will not change the value in the TIMER_MOD_ACK_NAK field of the TIMER_CTRL_MAX_FUNC_NUM_OFF register.

11.3.3.1.60 Vendor Specific DLLP Register. (VENDOR_SPEC_DLLP_OFF)

11.3.3.1.60.1 Offset

Register	Offset
VENDOR_SPEC_DLLP_OFF	704h

11.3.3.1.60.2 Diagram



11.3.3.1.60.3 Fields

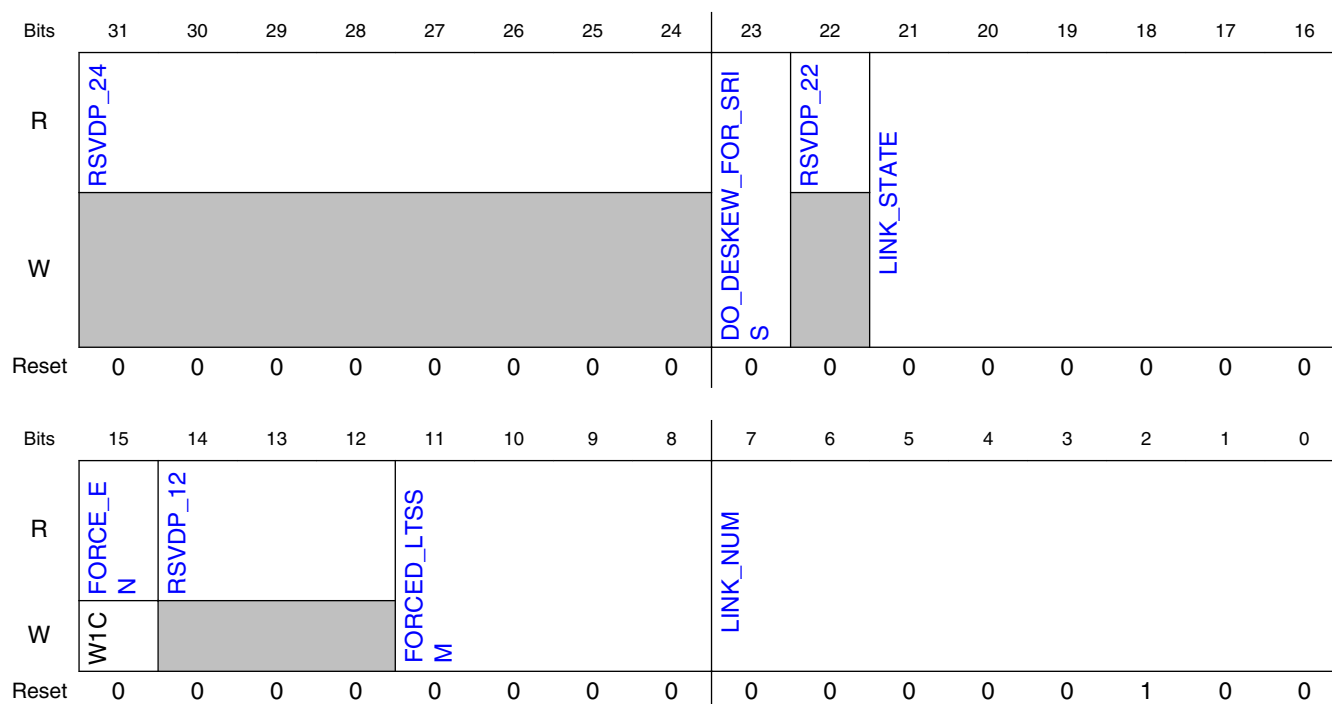
Field	Function
31-0 VENDOR_SPEC_DLLP	Vendor Specific DLLP Register. Used to send a specific PCI Express DLLP. Your application writes the 8-bit DLLP Type and 24-bits of Payload data into this register, then sets the field VENDOR_SPECIFIC_DLLP_REQ of PORT_LINK_CTRL_OFF to send the DLLP. - [7:0] = Type - [31:8] = Payload (24 bits) The dllp type is in bits [7:0] while the remainder is the vendor defined payload. Note: This register field is sticky.

11.3.3.1.61 Port Force Link Register. (PORT_FORCE_OFF)

11.3.3.1.61.1 Offset

Register	Offset
PORT_FORCE_OFF	708h

11.3.3.1.61.2 Diagram



11.3.3.1.61.3 Fields

Field	Function
31-24 RSVDP_24	Reserved for future use.
23 DO_DESKEW_FOR_SRIS	Use the transitions from TS2 to Logical Idle Symbol, SKP OS to Logical Idle Symbol, and FTS Sequence to SKP OS to do deskew for SRIS instead of using received SKP OS if DO_DESKEW_FOR_SRIS is set to 1. Note: This register field is sticky.
22 RSVDP_22	Reserved for future use.

Table continues on the next page...

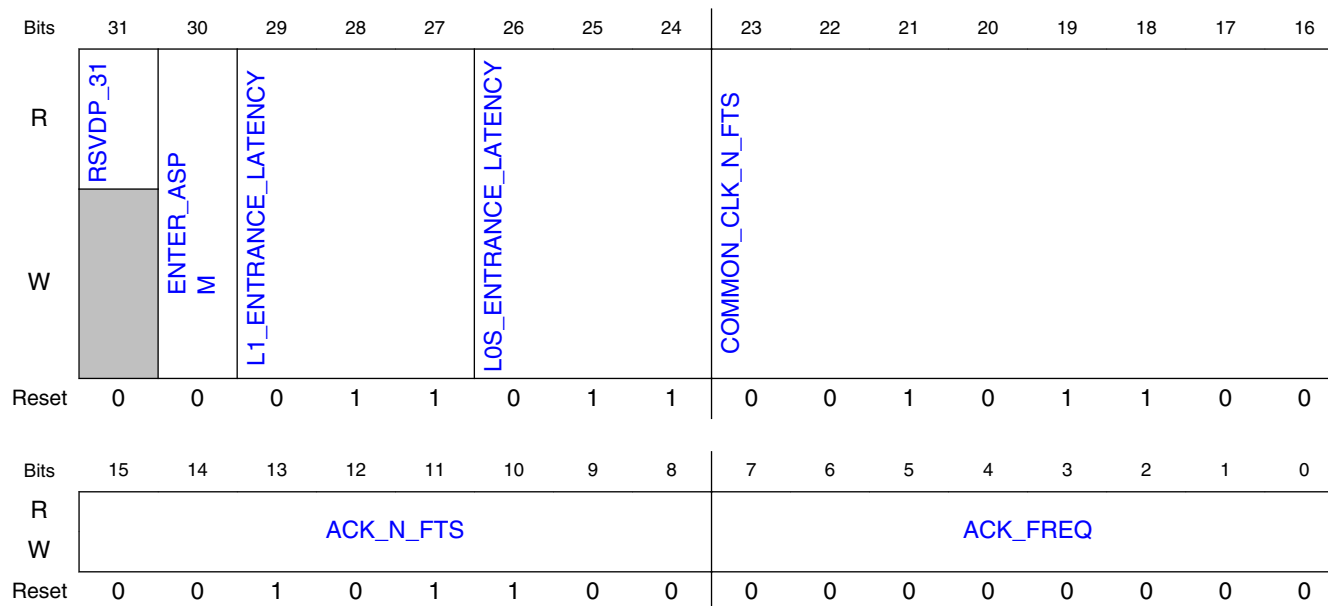
Field	Function
21-16 LINK_STATE	Forced LTSSM State. The LTSSM state that the controller is forced to when you set the FORCE_EN bit (Force Link). LTSSM state encoding is defined by the lts_state variable in workspace/src/Layer1/smlh_ltssm.v. Note: This register field is sticky.
15 FORCE_EN	Force Link. The controller supports a testing and debug capability to allow your software to force the LTSSM state machine into a specific state, and to force the controller to transmit a specific Link Command. Asserting this bit triggers the following actions: - Forces the LTSSM to the state specified by the Forced LTSSM State field. - Forces the controller to transmit the command specified by the Forced Link Command field. This is a self-clearing register field. Reading from this register field always returns a "0".
14-12 RSVDP_12	Reserved for future use.
11-8 FORCED_LTSSM	Forced Link Command. The link command that the controller is forced to transmit when you set FORCE_EN bit (Force Link). Link command encoding is defined by the ltssm_cmd variable in workspace/src/Layer1/smlh_ltssm.v. Note: This register field is sticky.
7-0 LINK_NUM	Link Number. Not used for endpoint. Not used for M-PCIe. Note: This register field is sticky.

11.3.3.1.62 Ack Frequency and L0-L1 ASPM Control Register. (ACK_F_AS PM_CTRL_OFF)

11.3.3.1.62.1 Offset

Register	Offset
ACK_F_AS PM_CTRL_OFF	70Ch

11.3.3.1.62.2 Diagram



11.3.3.1.62.3 Fields

Field	Function
31 RSVDP_31	Reserved for future use.
30 ENTER_ASPM	ASPM L1 Entry Control. - 1: Core enters ASPM L1 after a period in which it has been idle. - 0: Core enters ASPM L1 only after idle period during which both receive and transmit are in L0s. Note: This register field is sticky.
29-27 L1_ENTRANCE_LATENCY	L1 Entrance Latency. Value range is: - 000: 1 us - 001: 2 us - 010: 4 us - 011: 8 us - 100: 16 us - 101: 32 us - 110 or 111: 64 us Note: Programming this timer with a value greater than 32us has no effect unless extended sync is used, or all of the credits are infinite. Note: This register field is sticky.
26-24 L0S_ENTRANCE_LATENCY	L0s Entrance Latency. Values correspond to: - 000: 1 us - 001: 2 us - 010: 3 us - 011: 4 us - 100: 5 us - 101: 6 us - 110 or 111: 7 us This field is applicable to STALL while in L0 for M-PCIe. Note: This register field is sticky.
23-16 COMMON_CLK_N_FTS	Common Clock N_FTS. This is the N_FTS when common clock is used. The number of Fast Training Sequence ordered sets to be transmitted when transitioning from L0s to L0. The maximum number of FTS ordered-sets that a component can request is 255. The controller does not support a value of zero; a value of zero can cause the LTSSM to go into the recovery state when exiting from L0s. This field is reserved (fixed to '0') for M-PCIe. Note: The access attributes of this field are as follows: - Dbi: R
15-8 ACK_N_FTS	N_FTS. The number of Fast Training Sequence ordered sets to be transmitted when transitioning from L0s to L0. The maximum number of FTS ordered-sets that a component can request is 255. The controller does not support a value of zero; a value of zero can cause the LTSSM to go into the recovery state when exiting from L0s. This field is reserved (fixed to '0') for M-PCIe. Note: This register field is sticky.
7-0 ACK_FREQ	Ack Frequency. The controller accumulates the number of pending ACKs specified here (up to 255) before sending an ACK DLLP. - 0: Indicates that this Ack frequency control feature is turned off. The controller schedules a low-priority ACK DLLP for every TLP that it receives. - 1-255: Indicates that the

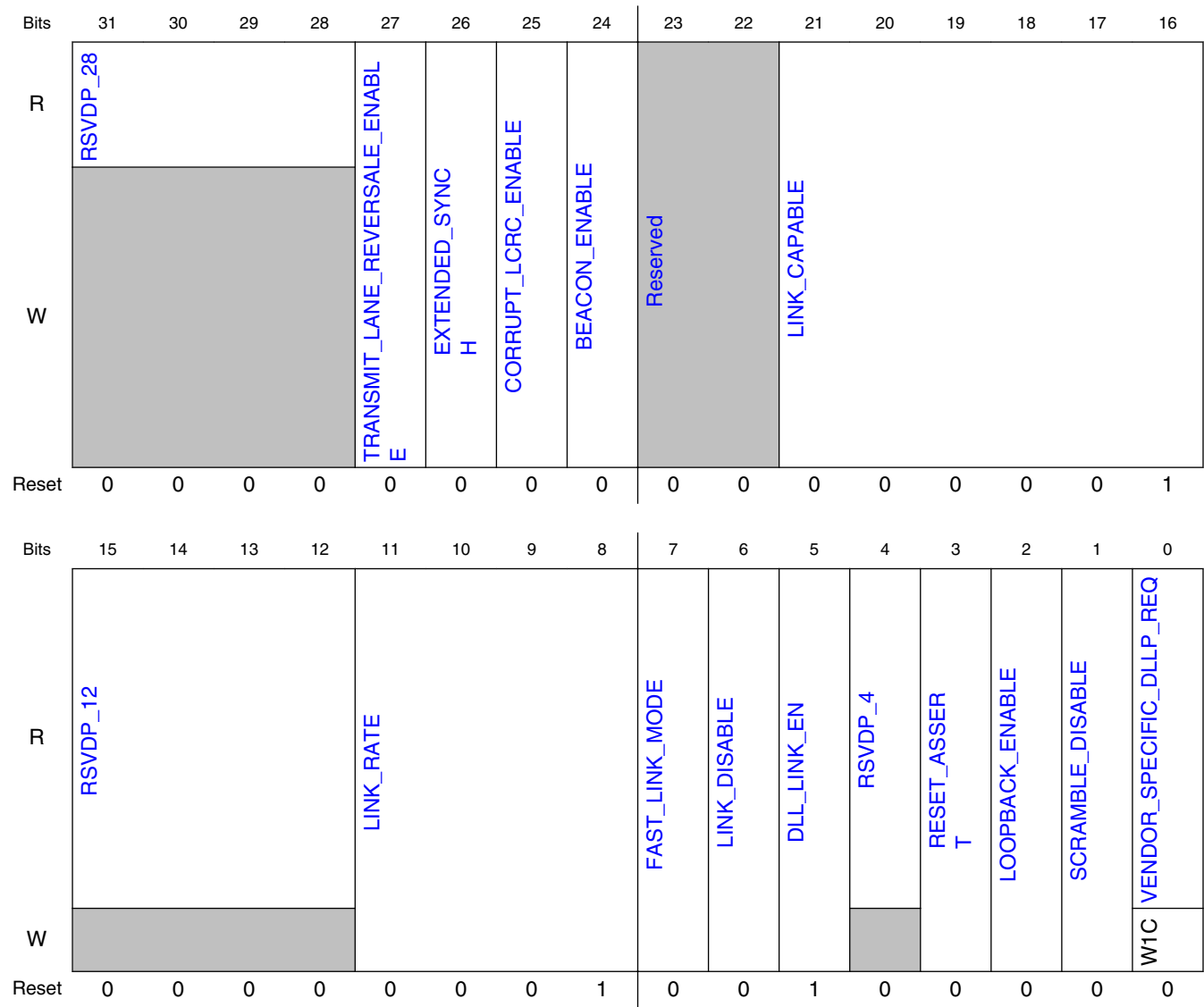
Field	Function
	controller will schedule a high-priority ACK after receiving this number of TLPs. It might schedule the ACK before receiving this number of TLPs, but never later. For a typical system, you do not have to modify the default setting. For more details, see "ACK/NAK Scheduling". Note: This register field is sticky.

11.3.3.1.63 Port Link Control Register. (PORT_LINK_CTRL_OFF)

11.3.3.1.63.1 Offset

Register	Offset
PORT_LINK_CTRL_OFF	710h

11.3.3.1.63.2 Diagram



11.3.3.1.63.3 Fields

Field	Function
31-28 RSVDP_28	Reserved for future use.
27 TRANSMIT_LANE_REVERSALE_ENABLE	TRANSMIT_LANE_REVERSALE_ENABLE is an internally reserved field. Do not use. Note: This register field is sticky.
26	EXTENDED_SYNC is an internally reserved field. Do not use. Note: This register field is sticky.

Table continues on the next page...

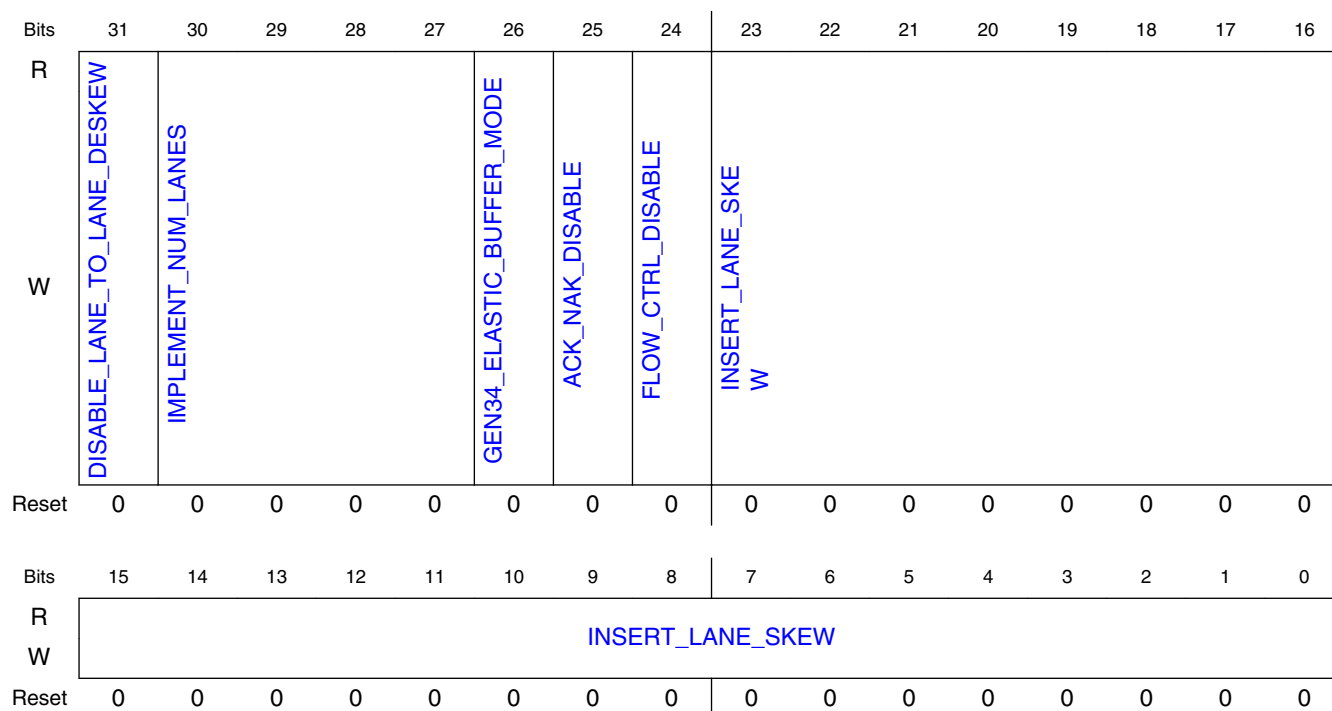
Field	Function
EXTENDED_SY NCH	
25 CORRUPT_LCR C_ENABLE	CORRUPT_LCRC_ENABLE is an internally reserved field. Do not use. Note: This register field is sticky.
24 BEACON_ENA BLE	BEACON_ENABLE is an internally reserved field. Do not use. Note: This register field is sticky.
23-22 —	Reserved.
21-16 LINK_CAPABLE	Link Mode Enable. Sets the number of lanes in the link that you want to connect to the link partner. When you have unused lanes in your system, then you must change the value in this register to reflect the number of lanes. You must also change the value in the "Predetermined Number of Lanes" field of the "Link Width and Speed Change Control Register". For more information, see "How to Tie Off Unused Lanes". For information on upsizing and downsizing the link width, see "Link Establishment". - 000001: x1 - 000011: x2 - 000111: x4 - 001111: x8 - 011111: x16 - 111111: x32 (not supported) This field is reserved (fixed to '0') for M-PCIe. Note: This register field is sticky.
15-12 RSVDP_12	Reserved for future use.
11-8 LINK_RATE	LINK_RATE is an internally reserved field. Do not use. Note: This register field is sticky.
7 FAST_LINK_M ODE	Fast Link Mode. Sets all internal LTSSM millisecond timers to Fast Mode for speeding up simulation. Forces the LTSSM training (link initialization) to use shorter time-outs and to link up faster. The default scaling factor can be changed through the FAST_LINK_SCALING_FACTOR field in the TIMER_CTRL_MAX_FUNC_NUM_OFF register. If this bit is set to '1', tRRAPInitiatorResponse is set to 1.88 ms(60 ms/32). Note: This register field is sticky.
6 LINK_DISABLE	LINK_DISABLE is an internally reserved field. Do not use. Note: This register field is sticky.
5 DLL_LINK_EN	DLL Link Enable. Enables link initialization. When DLL Link Enable =0, the controller does not transmit InitFC DLLPs and does not establish a link. Note: This register field is sticky.
4 RSVDP_4	Reserved for future use.
3 RESET_ASSER T	Reset Assert. Triggers a recovery and forces the LTSSM to the hot reset state (downstream port only). Note: This register field is sticky.
2 LOOPBACK_EN ABLE	Loopback Enable. Turns on loopback. For more details, see "Loopback". For M-PCIe, to force the master to enter Digital Loopback mode, you must set this field to "1" during Configuration.start state(initial discovery/configuration). M-PCIe doesn't support loopback mode from L0 state - only from Configuration.start. Note: This register field is sticky.
1 SCRAMBLE_DI SABLE	Scramble Disable. Turns off data scrambling. Note: This register field is sticky.
0 VENDOR_SPE CIFIC_DLLP_R EQ	Vendor Specific DLLP Request. When software writes a '1' to this bit, the controller transmits the DLLP contained in the VENDOR_SPEC_DLLP field of VENDOR_SPEC_DLLP_OFF. Reading from this self-clearing register field always returns a '0'.

11.3.3.1.64 Lane Skew Register. (LANE_SKEW_OFF)

11.3.3.1.64.1 Offset

Register	Offset
LANE_SKEW_OFF	714h

11.3.3.1.64.2 Diagram



11.3.3.1.64.3 Fields

Field	Function
31 DISABLE_LANE_TO_LANE_DESKEW	Disable Lane-to-Lane Deskew. Causes the controller to disable the internal Lane-to-Lane deskew logic. Note: This register field is sticky.
30-27 IMPLEMENT_NUM_LANES	Implementation-specific Number of Lanes. Set the implementation-specific number of lanes. Allowed values are: - 4'b0000: 1 lane - 4'b0001: 2 lanes - 4'b0011: 4 lanes - 4'b0111: 8 lanes - 4'b1111: 16 lanes The number of lanes to be used when in Loopback Master. The number of lanes programmed must be equal to or less than the valid number of lanes set in LINK_CAPABLE field. You must configure this field before initiating Loopback by writing in the LOOPBACK_ENABLE field. The controller will transition from Loopback.Entry to Loopback.Active after receiving two consecutive TS1 Ordered Sets with the Loopback

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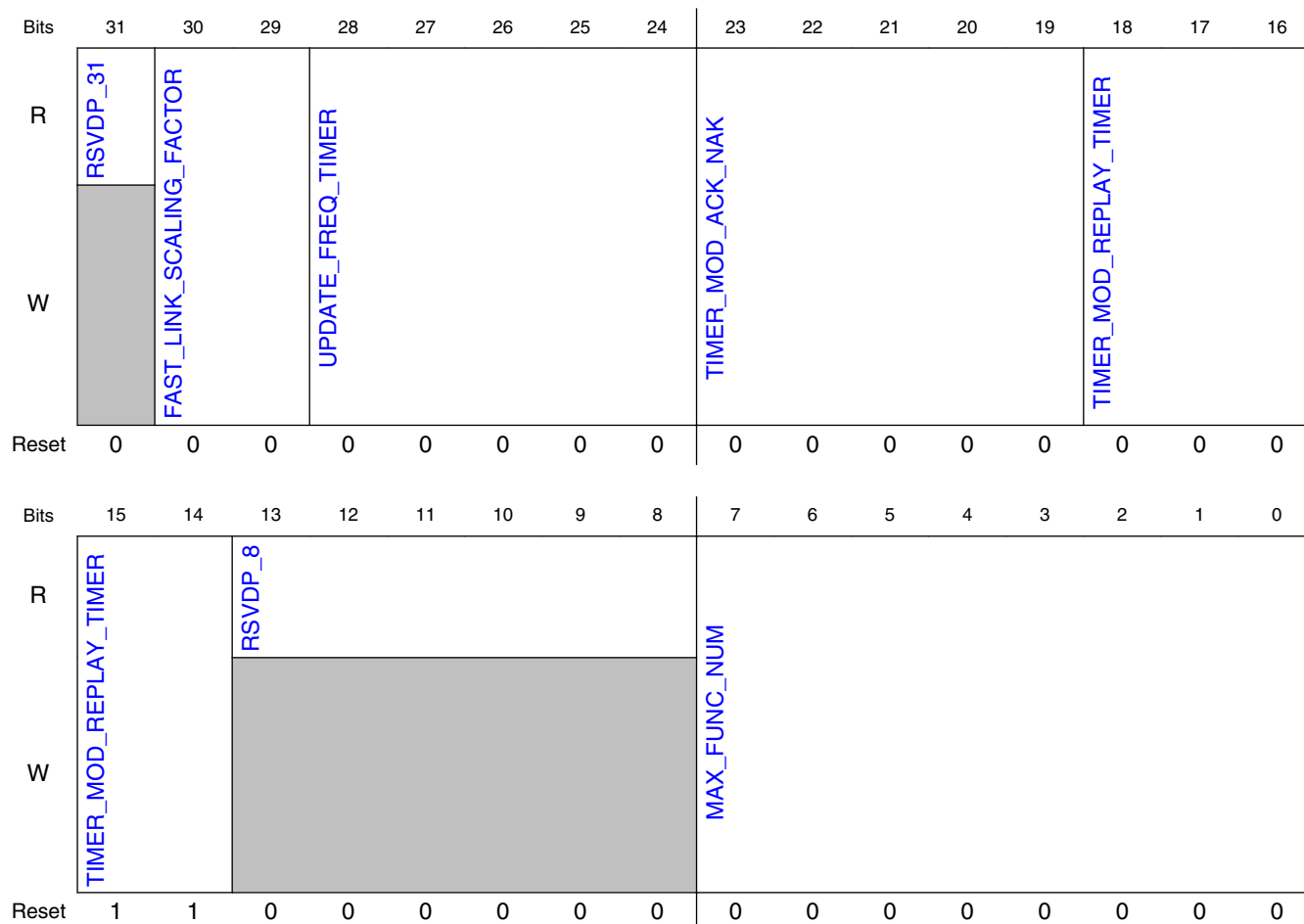
Field	Function
	bit asserted on the implementation specific number of lanes configured in this field. Note: This register field is sticky.
26 GEN34_ELASTIC_BUFFER_MODE	Selects Elasticity Buffer operating mode in Gen3 or Gen4 rate: 0: Nominal Half Full Buffer mode 1: Nominal Empty Buffer Mode This register bit only affects Gen3 or Gen4 operating rate. For Gen1 or Gen2 operating rate the Elasticity Buffer operating mode is always the Nominal Half Full Buffer mode. Note: This register field is sticky.
25 ACK_NAK_DISABLE	Ack/Nak Disable. Prevents the controller from sending ACK and NAK DLLPs. Note: This register field is sticky.
24 FLOW_CTRL_DISABLE	Flow Control Disable. Prevents the controller from sending FC DLLPs. Note: This register field is sticky.
23-0 INSERT_LANE_SKEW	INSERT_LANE_SKEW is an internally reserved field. Do not use. Note: This register field is sticky.

11.3.3.1.65 Timer Control and Max Function Number Register. (TIMER_CTRL_MAX_FUNC_NUM_OFF)

11.3.3.1.65.1 Offset

Register	Offset
TIMER_CTRL_MAX_FUNC_NUM_OFF	718h

11.3.3.1.65.2 Diagram



11.3.3.1.65.3 Fields

Field	Function
31 RSVDP_31	Reserved for future use.
30-29 FAST_LINK_SCALING_FACTOR	Fast Link Timer Scaling Factor. Sets the scaling factor of LTSSM timer when FAST_LINK_MODE field in PORT_LINK_CTRL_OFF is set to '1'. - 0: Scaling Factor is 1024 (1ms is 1us) - 1: Scaling Factor is 256 (1ms is 4us) - 2: Scaling Factor is 64 (1ms is 16us) - 3: Scaling Factor is 16 (1ms is 64us). Not used for M-PCIe. Note: This register field is sticky.
28-24 UPDATE_FREQ_TIMER	UPDATE_FREQ_TIMER is an internally reserved field. Do not use. Note: This register field is sticky.
23-19 TIMER_MOD_ACK_NAK	Ack Latency Timer Modifier. Increases the timer value for the Ack latency timer in increments of 64 clock cycles. A value of "0" represents no modification to the timer value. For more details, see the ROUND_TRIP_LATENCY_TIME_LIMIT field of the ACK_LATENCY_TIMER_OFF register. Note: This register field is sticky.

Table continues on the next page...

Field	Function
18-14 TIMER_MOD_REPLAY_TIMER	Replay Timer Limit Modifier. Increases the time-out value for the replay timer in increments of 64 clock cycles at Gen1 or Gen2 speed, and in increments of 256 clock cycles at Gen3 speed. A value of "0" represents no modification to the timer limit. For more details, see the REPLAY_TIME_LIMIT field of the ACK_LATENCY_TIMER_OFF register. At Gen3 speed, the controller automatically changes the value of this field to DEFAULT_GEN3_REPLAY_ADJ. For M-PCIe, this field increases the time-out value for the replay timer in increments of 64 clock cycles at HS-Gear1, HS-Gear2, or HS-Gear3 speed. Note: This register field is sticky.
13-8 RSVDP_8	Reserved for future use.
7-0 MAX_FUNC_NUM	Maximum function number that can be used in a request. Configuration requests targeted at function numbers above this value are returned with UR (unsupported request). Note: This register field is sticky.

11.3.3.1.66 Symbol Timer Register and Filter Mask 1 Register. (SYMBOL_TIMER_FILTER_1_OFF)

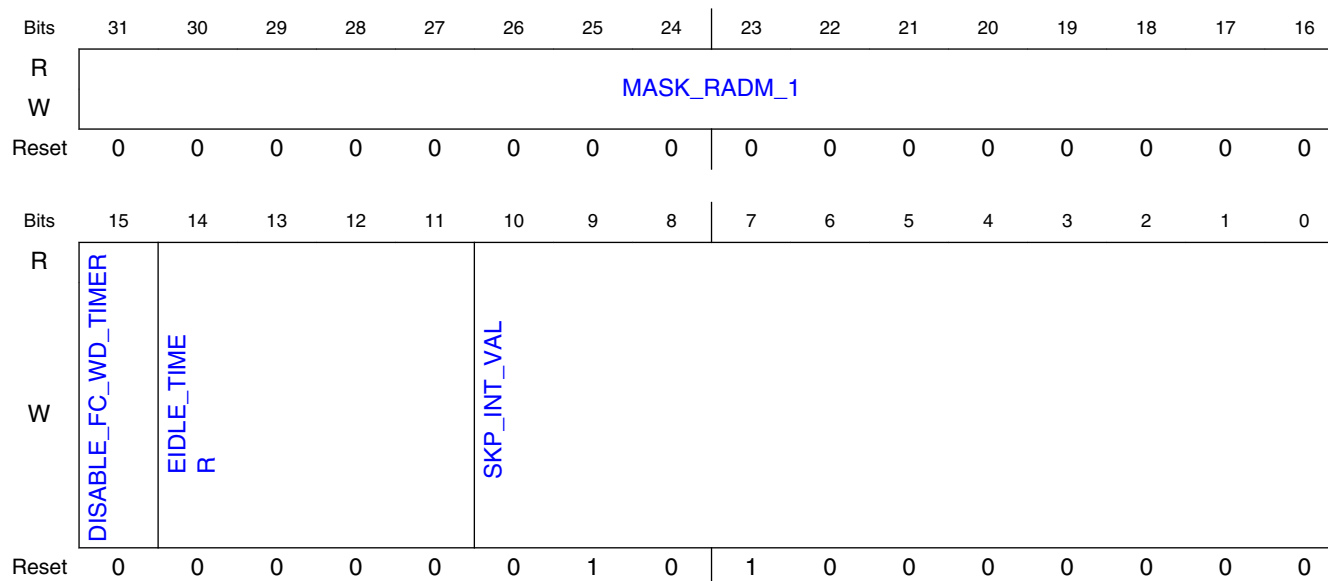
11.3.3.1.66.1 Offset

Register	Offset
SYMBOL_TIMER_FILTER_1_OFF	71Ch

11.3.3.1.66.2 Function

Symbol Timer Register and Filter Mask 1 Register. The Filter Mask 1 Register modifies the RADM filtering and error handling rules. For more details, see the "Receive Filtering" section. In each case, '0' applies the associated filtering rule and '1' masks the associated filtering rule.

11.3.3.1.66.3 Diagram



11.3.3.1.66.4 Fields

Field	Function
31-16 MASK_RADM_1	Filter Mask 1. The Filter Mask 1 Register modifies the RADM filtering and error handling rules. For more details, see the "Receive Filtering" section. In each case, '0' applies the associated filtering rule and '1' masks the associated filtering rule. Note: This register field is sticky.
15 DISABLE_FC_WD_TIMER	Disable FC Watchdog Timer. Note: This register field is sticky.
14-11 EIDLE_TIMER	EIDLE_TIMER is an internally reserved field. Do not use. Note: This register field is sticky.
10-0 SKP_INT_VAL	SKP Interval Value. The number of symbol times to wait between transmitting SKP ordered sets. Note that the controller actually waits the number of symbol times in this register plus 1 between transmitting SKP ordered sets. Your application must program this register accordingly. For example, if 1536 were programmed into this register (in a 250 MHz controller), then the controller actually transmits SKP ordered sets once every 1537 symbol times. The value programmed to this register is actually clock ticks and not symbol times. In a 125 MHz controller, programming the value programmed to this register should be scaled down by a factor of 2 (because one clock tick = two symbol times in this case). Note: This value is not used at Gen3 speed; the skip interval is hardcoded to 370 blocks. Note: This register field is sticky.

11.3.3.1.67 Filter Mask 2 Register. (FILTER_MASK_2_OFF)

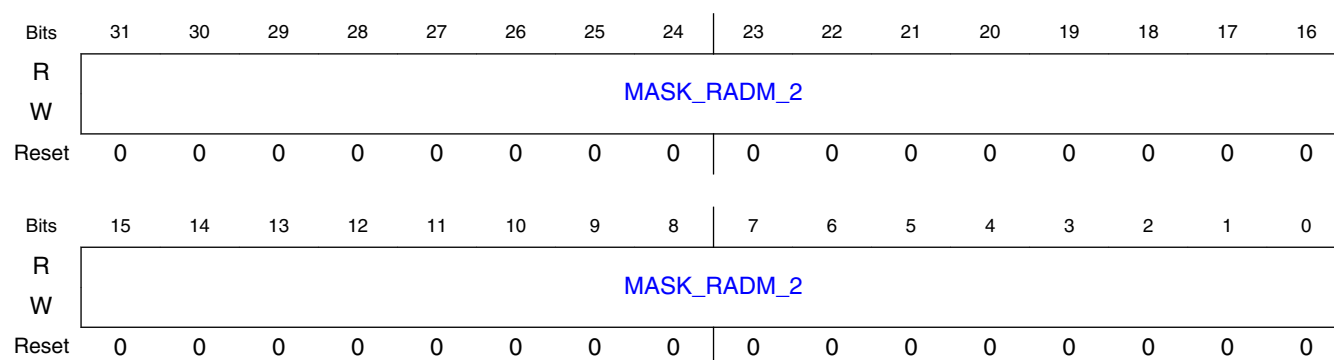
11.3.3.1.67.1 Offset

Register	Offset
FILTER_MASK_2_OFF	720h

11.3.3.1.67.2 Function

Filter Mask 2 Register. This register modifies the RADM filtering and error handling rules. For more details, see the "Receive Filtering" section. In each case, '0' applies the associated filtering rule and '1' masks the associated filtering rule.

11.3.3.1.67.3 Diagram



11.3.3.1.67.4 Fields

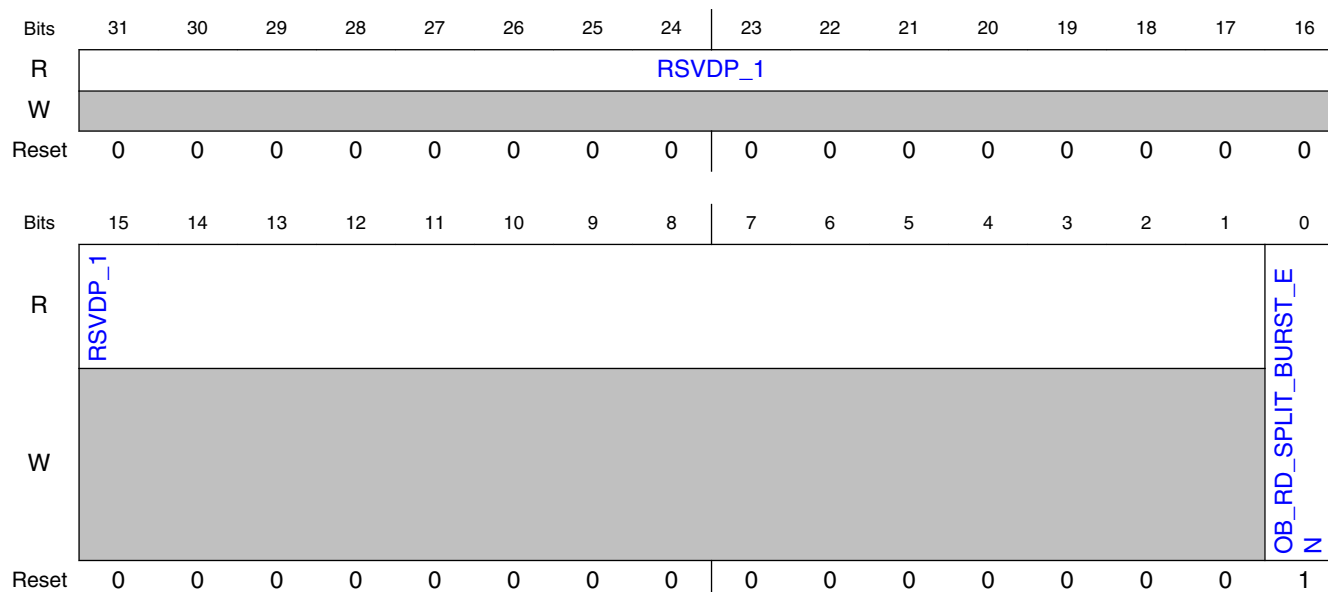
Field	Function
31-0 MASK_RADM_2	Filter Mask 2. This field modifies the RADM filtering and error handling rules. For more details, see the "Receive Filtering" section. In each case, '0' applies the associated filtering rule and '1' masks the associated filtering rule. Note: This register field is sticky.

11.3.3.1.68 AMBA Multiple Outbound Decomposed NP SubRequests Control Register. (AMBA_MUL_OB_DECOMP_NP_SUB_REQ_CTRL_OFF)

11.3.3.1.68.1 Offset

Register	Offset
AMBA_MUL_OB_DECOMP_NP_SUB_REQ_CTRL_OFF	724h

11.3.3.1.68.2 Diagram



11.3.3.1.68.3 Fields

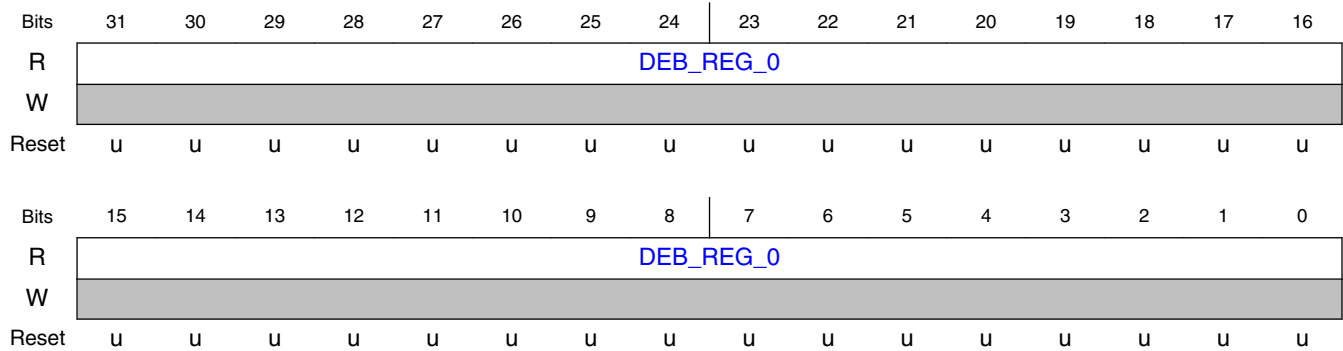
Field	Function
31-1 RSVDP_1	Reserved for future use.
0 OB_RD_SPLIT_BURST_EN	Enable AMBA Multiple Outbound Decomposed NP SubRequests. This bit when set to "0" disables the possibility of having multiple outstanding non-posted requests that were derived from decomposition of an outbound AMBA request. For more details, see "AXI Bridge Ordering" in the AXI chapter of the Databook. You should not clear this register unless your application master is requesting an amount of read data greater than Max_Read_Request_Size, and the remote device (or switch) is reordering completions that have different tags. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.

11.3.3.1.69 Debug Register 0 (PL_DEBUG0_OFF)

11.3.3.1.69.1 Offset

Register	Offset
PL_DEBUG0_OFF	728h

11.3.3.1.69.2 Diagram



11.3.3.1.69.3 Fields

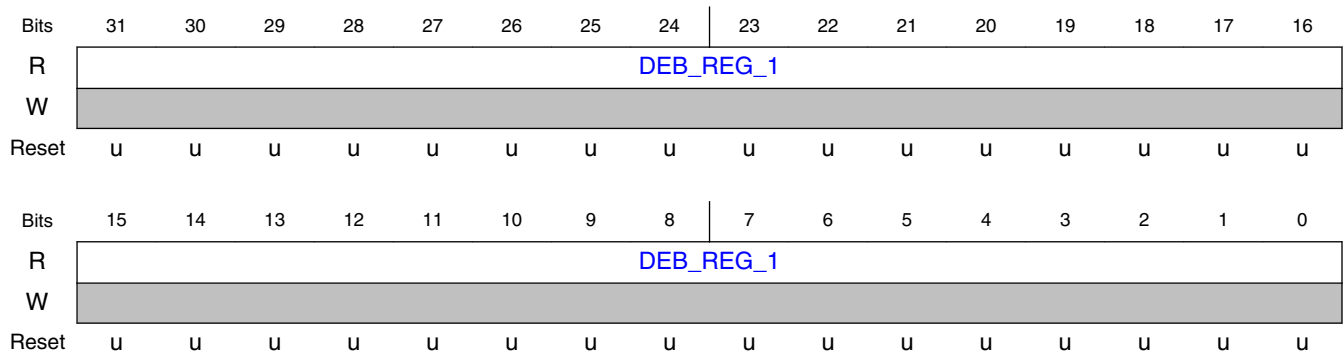
Field	Function
31-0 DEB_REG_0	The value on cxpl_debug_info[31:0].

11.3.3.1.70 Debug Register 1 (PL_DEBUG1_OFF)

11.3.3.1.70.1 Offset

Register	Offset
PL_DEBUG1_OFF	72Ch

11.3.3.1.70.2 Diagram



11.3.3.1.70.3 Fields

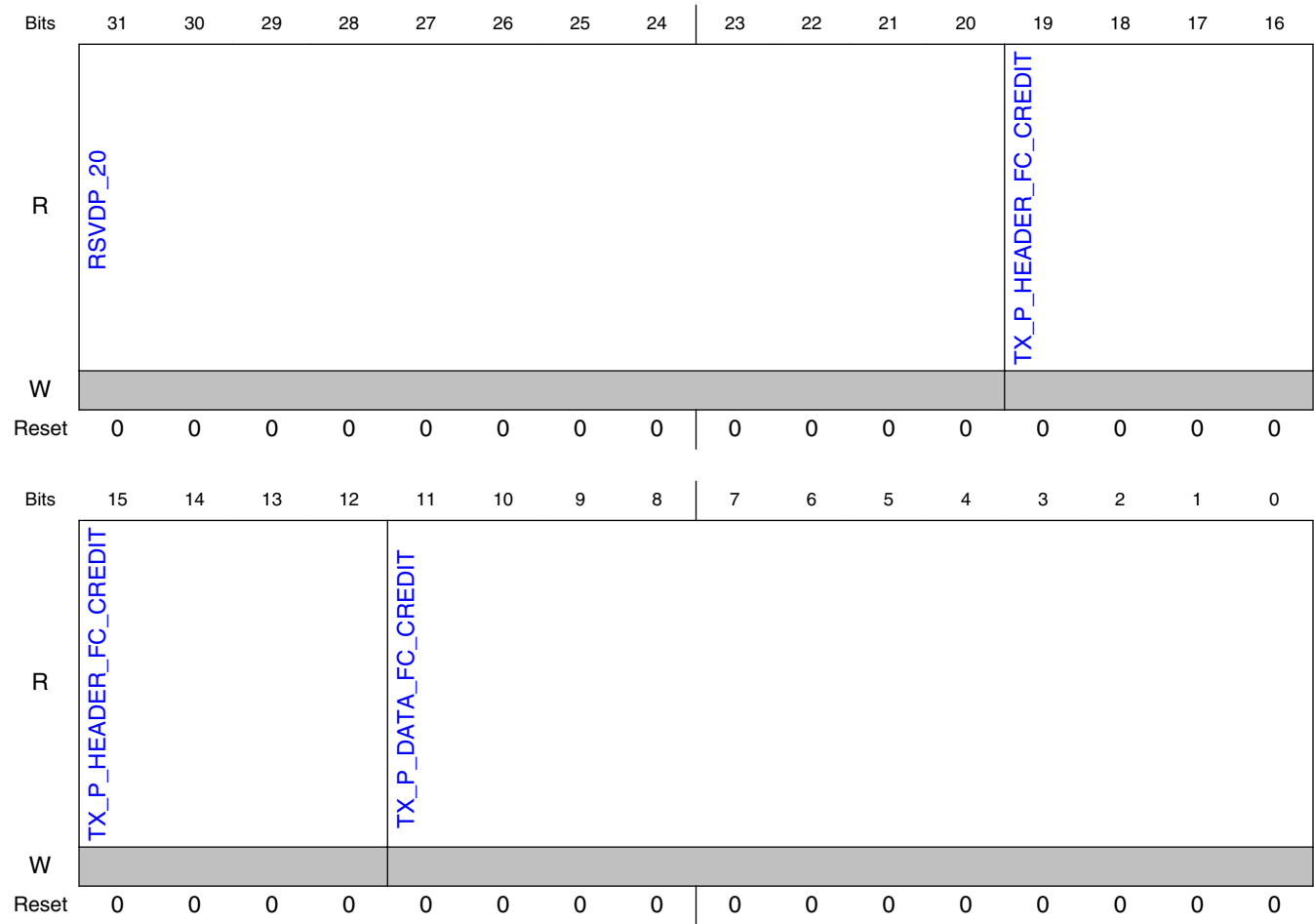
Field	Function
31-0 DEB_REG_1	The value on cxpl_debug_info[63:32].

11.3.3.1.71 Transmit Posted FC Credit Status (TX_P_FC_CREDIT_STATUS_OFF)

11.3.3.1.71.1 Offset

Register	Offset
TX_P_FC_CREDIT_STATUS_OFF	730h

11.3.3.1.71.2 Diagram



11.3.3.1.71.3 Fields

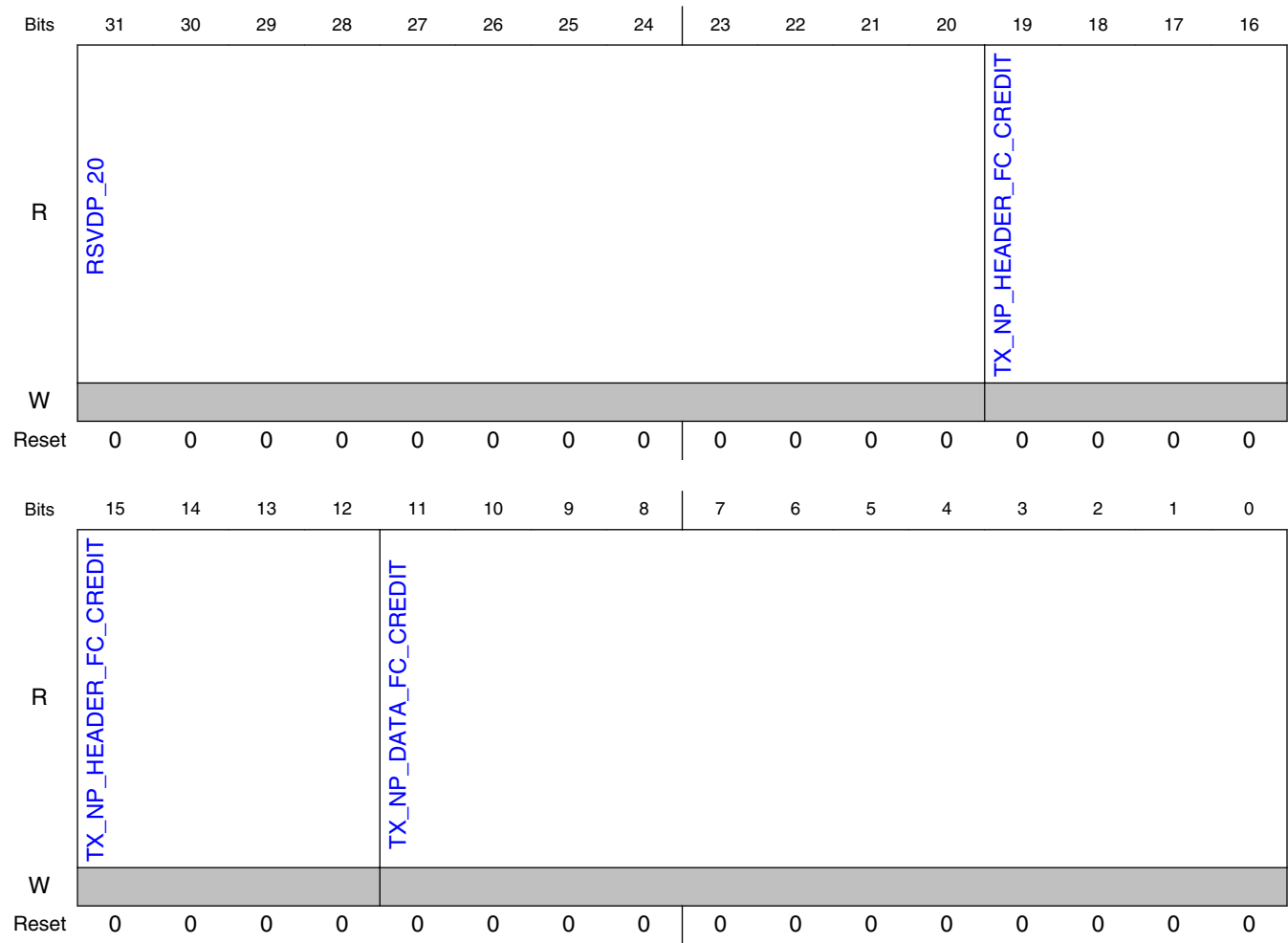
Field	Function
31-20 RSVDP_20	Reserved for future use.
19-12 TX_P_HEADER_FC_CREDIT	Transmit Posted Header FC Credits. The posted Header credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. Default value depends on the number of advertised credits for header and data [12'b0, xtlh_xadm_ph_cdts, xtlh_xadm_pd_cdts]; When the number of advertised completion credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF].
11-0 TX_P_DATA_FC_CREDIT	Transmit Posted Data FC Credits. The posted Data credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. Default value depends on the number of advertised credits for header and data [12'b0, xtlh_xadm_ph_cdts, xtlh_xadm_pd_cdts]; When the number of advertised completion credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF].

11.3.3.1.72 Transmit Non-Posted FC Credit Status (TX_NP_FC_CREDIT_STATUS_OFF)

11.3.3.1.72.1 Offset

Register	Offset
TX_NP_FC_CREDIT_STATUS_OFF	734h

11.3.3.1.72.2 Diagram



11.3.3.1.72.3 Fields

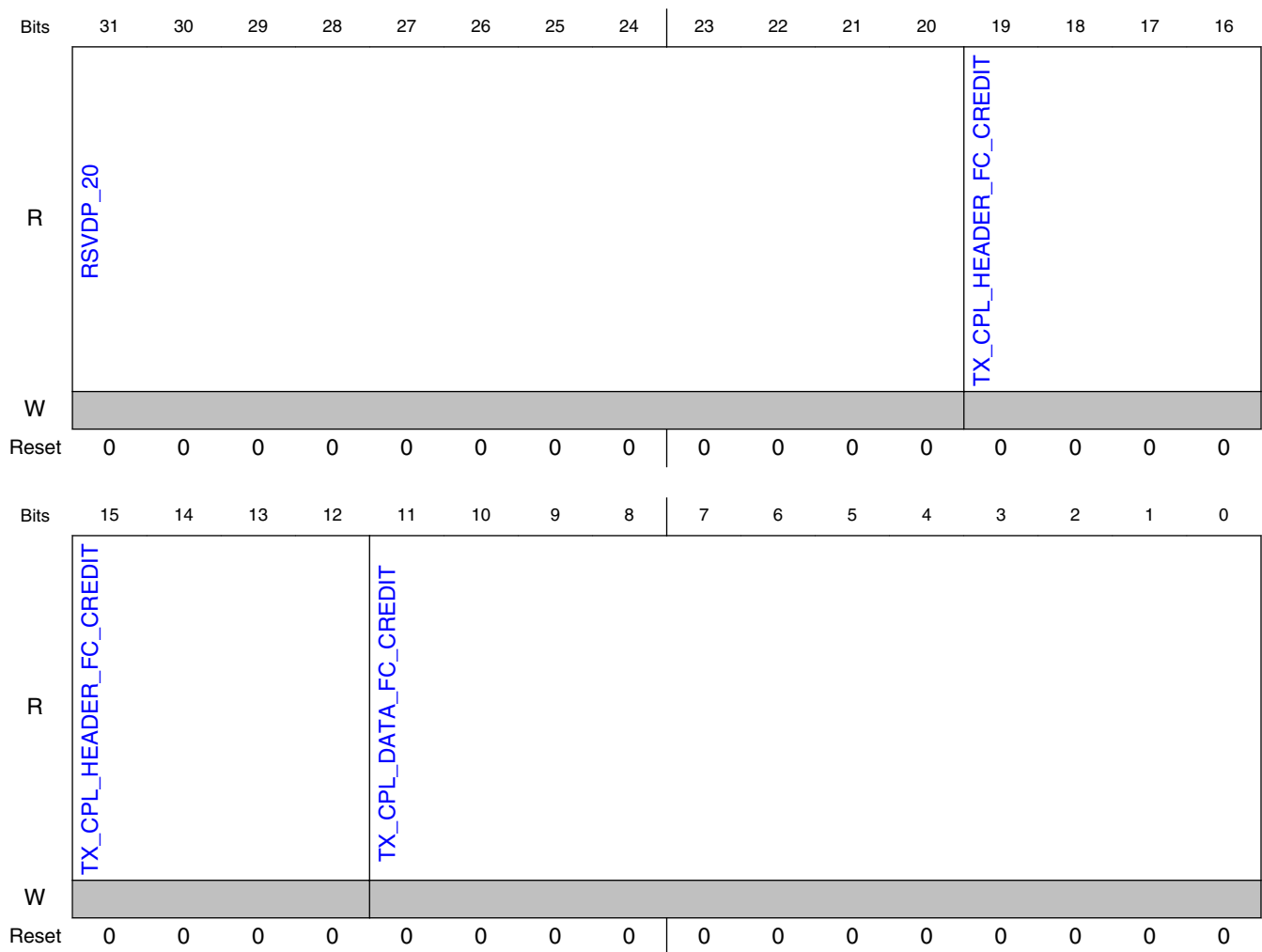
Field	Function
31-20 RSVDP_20	Reserved for future use.
19-12 TX_NP_HEADE R_FC_CREDIT	Transmit Non-Posted Header FC Credits. The non-posted Header credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. Default value depends on the number of advertised credits for header and data [12'b0, xtlh_xadm_nph_cdts, xtlh_xadm_npd_cdts]; When the number of advertised completion credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF].
11-0 TX_NP_DATA_ FC_CREDIT	Transmit Non-Posted Data FC Credits. The non-posted Data credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. Default value depends on the number of advertised credits for header and data [12'b0, xtlh_xadm_nph_cdts, xtlh_xadm_npd_cdts]; When the number of advertised completion credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF].

11.3.3.1.73 Transmit Completion FC Credit Status (TX_CPL_FC_CREDIT_STATUS_OFF)

11.3.3.1.73.1 Offset

Register	Offset
TX_CPL_FC_CREDIT_STATUS_OFF	738h

11.3.3.1.73.2 Diagram



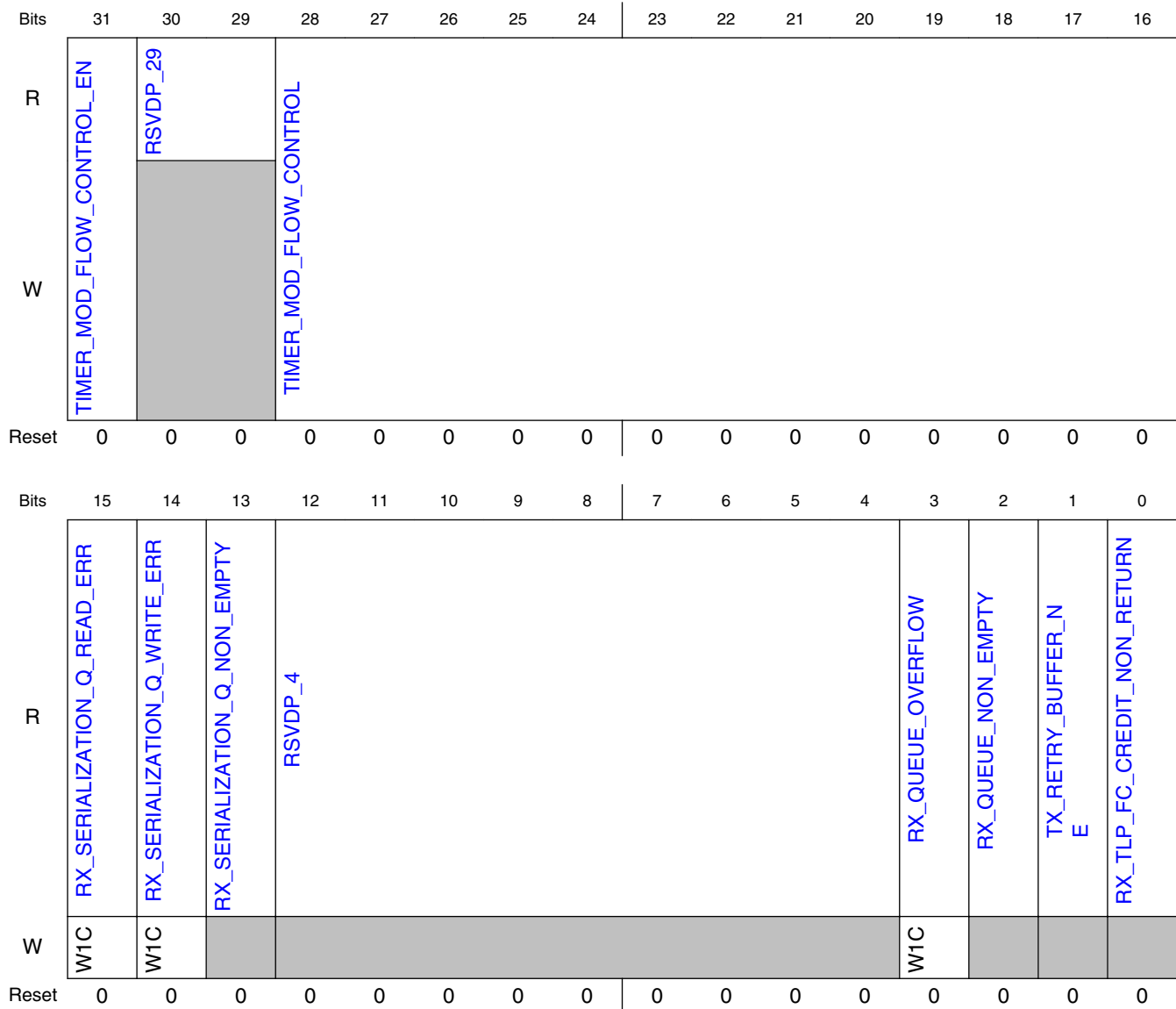
11.3.3.1.73.3 Fields

Field	Function
31-20 RSVDP_20	Reserved for future use.
19-12 TX_CPL_HEADER_FC_CREDIT	Transmit Completion Header FC Credits. The Completion Header credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. Default value depends on the number of advertised credits for header and data [12'b0, xtlh_xadm_cplh_cdts, xtlh_xadm_cpId_cdts]; When the number of advertised completion credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF].
11-0 TX_CPL_DATA_FC_CREDIT	Transmit Completion Data FC Credits. The Completion Data credits advertised by the receiver at the other end of the link, updated with each UpdateFC DLLP. Default value depends on the number of advertised credits for header and data [12'b0, xtlh_xadm_cplh_cdts, xtlh_xadm_cpId_cdts]; When the number of advertised completion credits (both header and data) are infinite, then the default would be [12'b0, 8'hFF, 12'hFFF].

11.3.3.1.74 Queue Status (QUEUE_STATUS_OFF)**11.3.3.1.74.1 Offset**

Register	Offset
QUEUE_STATUS_OFF	73Ch

11.3.3.1.74.2 Diagram



11.3.3.1.74.3 Fields

Field	Function
31 TIMER_MOD_FLOW_CONTROL_EN	FC Latency Timer Override Enable. When this bit is set, the value from the "FC Latency Timer Override Value" field in this register will override the FC latency timer value that the controller calculates according to the PCIe specification. Note: This register field is sticky.
30-29 RSVDP_29	Reserved for future use.

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PCI Express (PCIe)

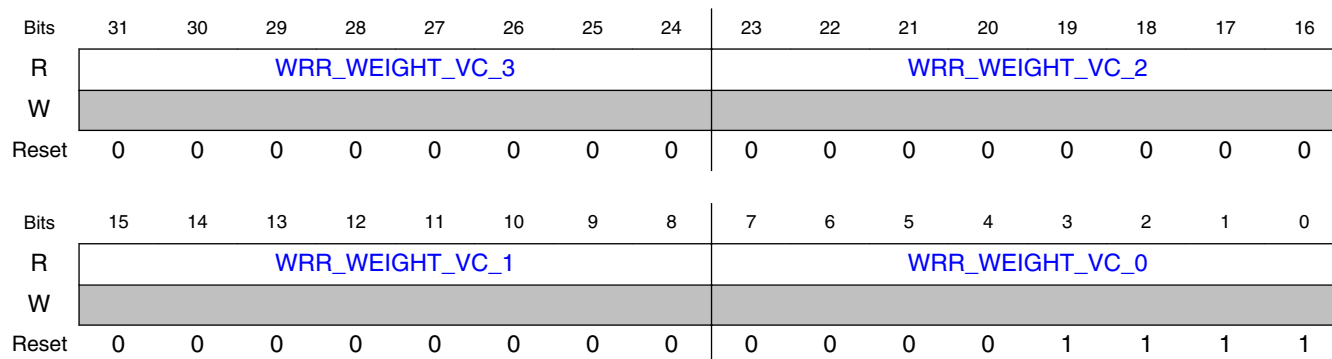
Field	Function
28-16 TIMER_MOD_F LOW_CONTR OL	FC Latency Timer Override Value. When you set the "FC Latency Timer Override Enable" in this register, the value in this field will override the FC latency timer value that the controller calculates according to the PCIe specification. For more details, see "Flow Control". Note: This register field is sticky.
15 RX_SERIALI ZATION_Q_R EAD_ _ERR	Receive Serialization Read Error. Indicates the serialization queue has attempted to read an incorrectly formatted TLP.
14 RX_SERIALI ZATION_Q_W RIT E_ERR	Receive Serialization Queue Write Error. Indicates insufficient buffer space available to write to the serialization queue.
13 RX_SERIALI ZATION_Q_N ON_ EMPTY	Receive Serialization Queue Not Empty. Indicates there is data in the serialization queue.
12-4 RSVDP_4	Reserved for future use.
3 RX_QUEUE_ O VERFLOW	Receive Credit Queue Overflow. Indicates insufficient buffer space available to write to the P/NP/CPL credit queue.
2 RX_QUEUE_ N ON_EMPTY	Receive Credit Queue Not Empty. Indicates there is data in one or more of the receive buffers.
1 TX_RETRY_ BU FFER_NE	Transmit Retry Buffer Not Empty. Indicates that there is data in the transmit retry buffer.
0 RX_TLP_FC_ C REDIT_N ON_R ETURN	Received TLP FC Credits Not Returned. Indicates that the controller has received a TLP but has not yet sent an UpdateFC DLLP indicating that the credits for that TLP have been restored by the receiver at the other end of the link.

11.3.3.1.75 VC Transmit Arbitration Register 1 (VC_TX_ARBI_1_OFF)

11.3.3.1.75.1 Offset

Register	Offset
VC_TX_ARBI_1_OFF	740h

11.3.3.1.75.2 Diagram



11.3.3.1.75.3 Fields

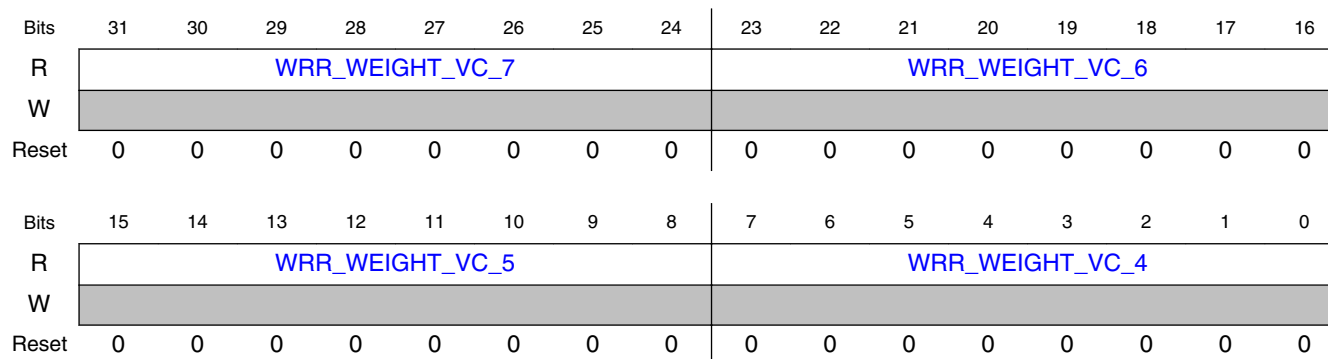
Field	Function
31-24 WRR_WEIGHT_VC_3	WRR Weight for VC3. Note: The access attributes of this field are as follows: - Dbi: R
23-16 WRR_WEIGHT_VC_2	WRR Weight for VC2. Note: The access attributes of this field are as follows: - Dbi: R
15-8 WRR_WEIGHT_VC_1	WRR Weight for VC1. Note: The access attributes of this field are as follows: - Dbi: R
7-0 WRR_WEIGHT_VC_0	WRR Weight for VC0. Note: The access attributes of this field are as follows: - Dbi: R

11.3.3.1.76 VC Transmit Arbitration Register 2 (VC_TX_ARBI_2_OFF)

11.3.3.1.76.1 Offset

Register	Offset
VC_TX_ARBI_2_OFF	744h

11.3.3.1.76.2 Diagram



11.3.3.1.76.3 Fields

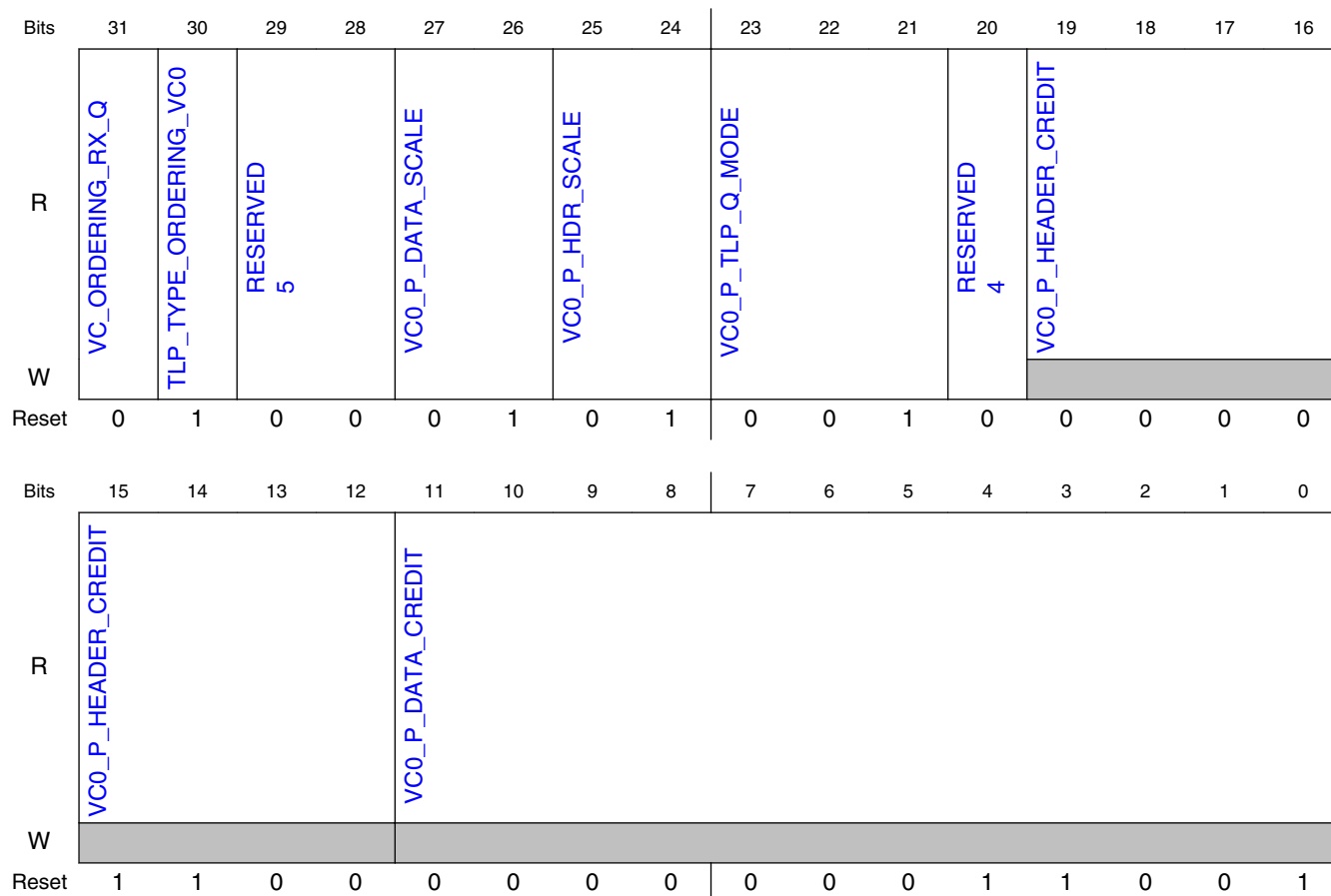
Field	Function
31-24 WRR_WEIGHT_VC_7	WRR Weight for VC7. Note: The access attributes of this field are as follows: - Dbi: R
23-16 WRR_WEIGHT_VC_6	WRR Weight for VC6. Note: The access attributes of this field are as follows: - Dbi: R
15-8 WRR_WEIGHT_VC_5	WRR Weight for VC5. Note: The access attributes of this field are as follows: - Dbi: R
7-0 WRR_WEIGHT_VC_4	WRR Weight for VC4. Note: The access attributes of this field are as follows: - Dbi: R

11.3.3.1.77 Segmented-Buffer VC0 Posted Receive Queue Control. (VC0_P_RX_Q_CTRL_OFF)

11.3.3.1.77.1 Offset

Register	Offset
VC0_P_RX_Q_CTRL_OFF	748h

11.3.3.1.77.2 Diagram



11.3.3.1.77.3 Fields

Field	Function
31 VC_ORDERING_RX_Q	VC Ordering for Receive Queues. Determines the VC ordering rule for the receive queues, used only in the segmented-buffer configuration: - 1: Strict ordering, higher numbered VCs have higher priority - 0: Round robin Note: This register field is sticky.
30 TLP_TYPE_ORDERING_VC0	TLP Type Ordering for VC0. Determines the TLP type ordering rule for VC0 receive queues, used only in the segmented-buffer configuration: - 1: PCIe ordering rules (recommended) - 0: Strict ordering: posted, completion, then non-posted Note: This register field is sticky.
29-28 RESERVED5	Reserved. Note: This register field is sticky.
27-26 VC0_P_DATA_SCALE	VC0 Scale Posted Data Credits. Note: This register field is sticky.
25-24 VC0_P_HDR_SCALE	VC0 Scale Posted Header Credits. Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

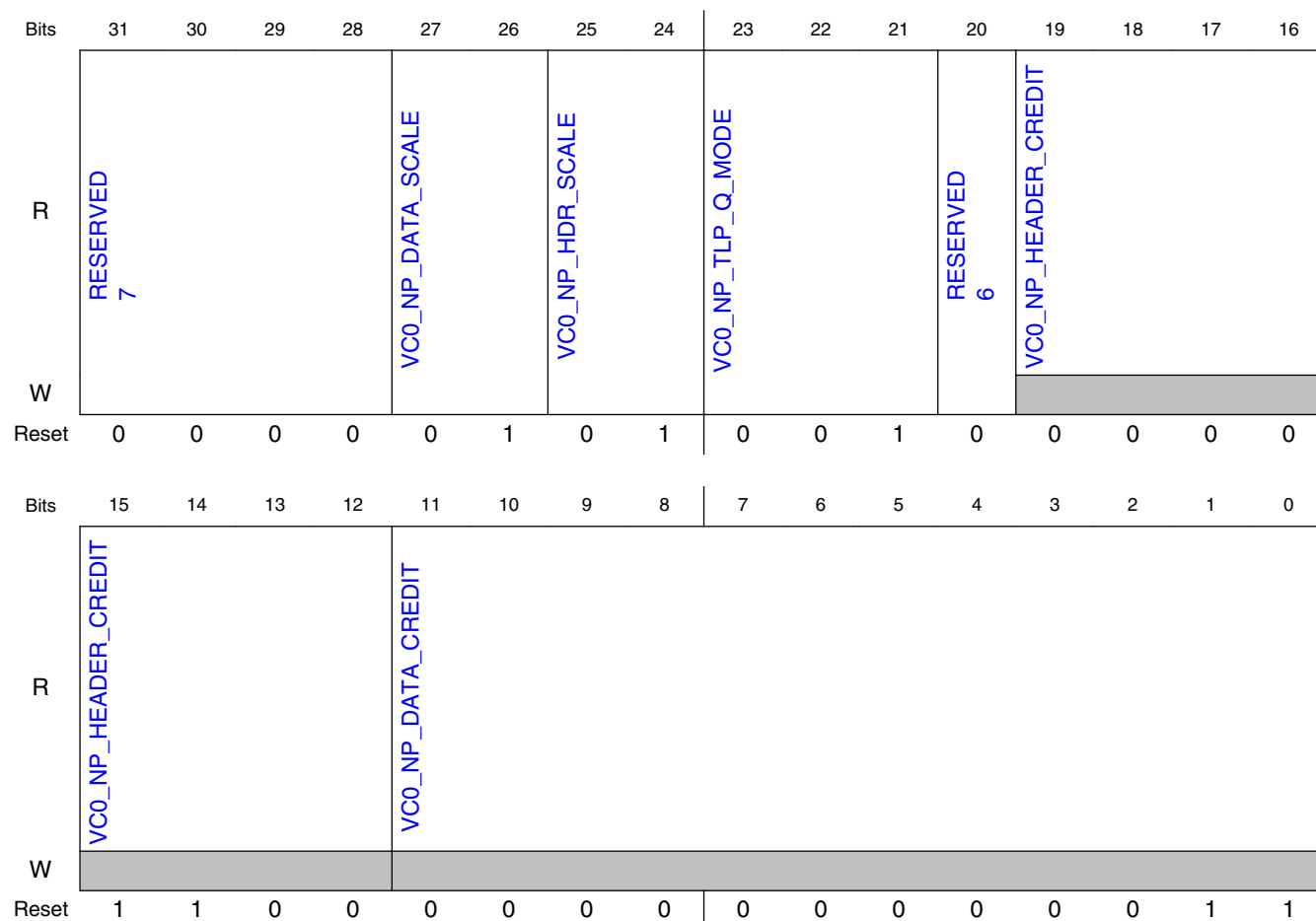
Field	Function
23-21 VC0_P_TLP_Q_MODE	Reserved. Note: This register field is sticky.
20 RESERVED4	Reserved. Note: This register field is sticky.
19-12 VC0_P_HEADER_CREDIT	VC0 Posted Header Credits. The number of initial posted header credits for VC0, used only in the segmented-buffer configuration. Note: The access attributes of this field are as follows: - Dbi: R (sticky) Note: This register field is sticky.
11-0 VC0_P_DATA_CREDIT	VC0 Posted Data Credits. The number of initial posted data credits for VC0, used only in the segmented-buffer configuration. Note: The access attributes of this field are as follows: - Dbi: R (sticky) Note: This register field is sticky.

11.3.3.1.78 Segmented-Buffer VC0 Non-Posted Receive Queue Control. (VC0_NP_RX_Q_CTRL_OFF)

11.3.3.1.78.1 Offset

Register	Offset
VC0_NP_RX_Q_CTRL_OFF	74Ch

11.3.3.1.78.2 Diagram



11.3.3.1.78.3 Fields

Field	Function
31-28 RESERVED7	Reserved. Note: This register field is sticky.
27-26 VC0_NP_DATA_SCALE	VC0 Scale Non-Posted Data Credits. Note: This register field is sticky.
25-24 VC0_NP_HDR_SCALE	VC0 Scale Non-Posted Header Credits. Note: This register field is sticky.
23-21 VC0_NP_TLP_Q_MODE	Reserved. Note: This register field is sticky.
20 RESERVED6	Reserved. Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

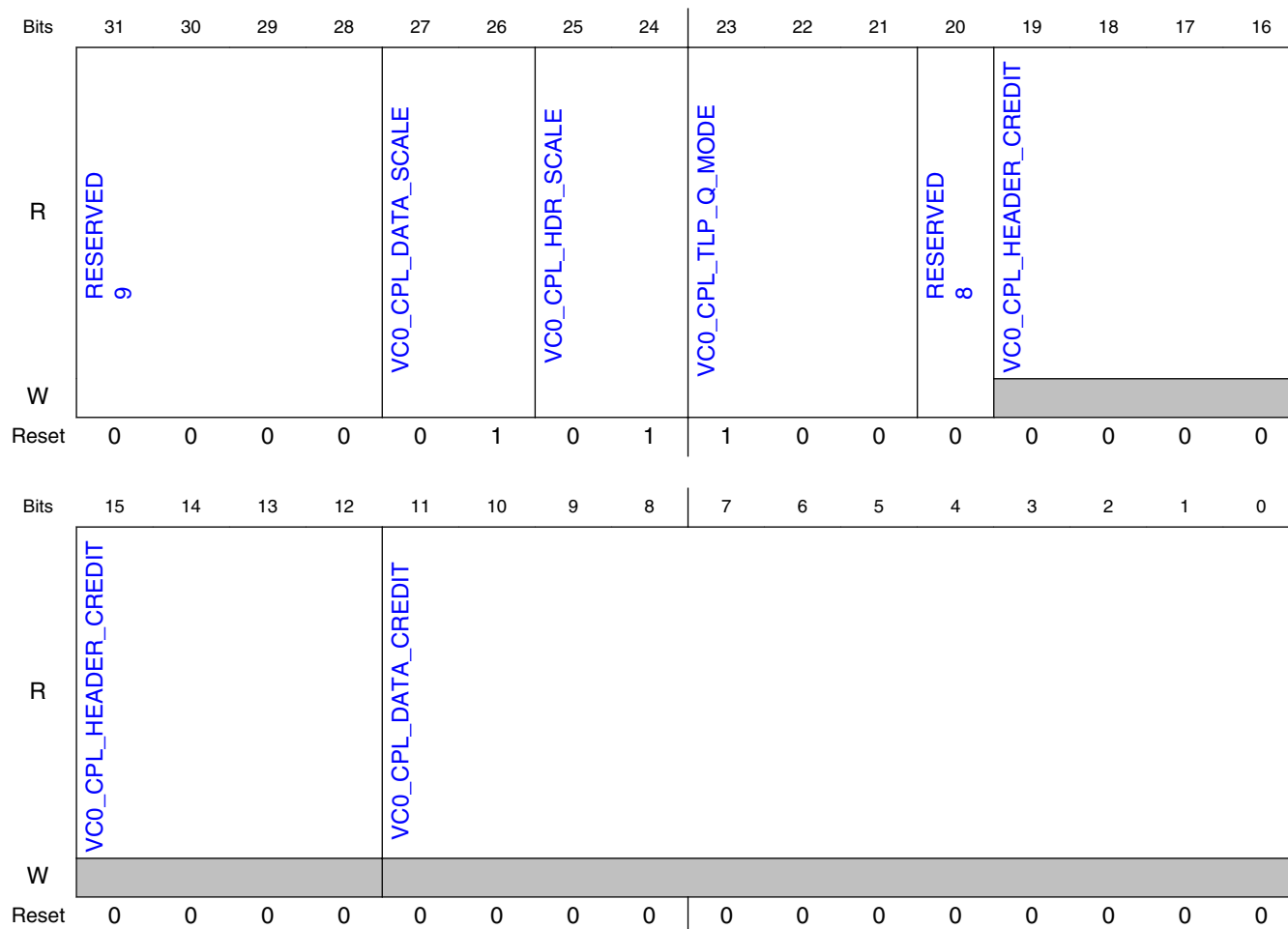
Field	Function
19-12 VC0_NP_HEADER_CREDIT	VC0 Non-Posted Header Credits. The number of initial non-posted header credits for VC0, used only in the segmented-buffer configuration. Note: The access attributes of this field are as follows: - Dbi: R (sticky) Note: This register field is sticky.
11-0 VC0_NP_DATA_CREDIT	VC0 Non-Posted Data Credits. The number of initial non-posted data credits for VC0, used only in the segmented-buffer configuration. Note: The access attributes of this field are as follows: - Dbi: R (sticky) Note: This register field is sticky.

11.3.3.1.79 Segmented-Buffer VC0 Completion Receive Queue Control. (VC0_CPL_RX_Q_CTRL_OFF)

11.3.3.1.79.1 Offset

Register	Offset
VC0_CPL_RX_Q_CTRL_OFF	750h

11.3.3.1.79.2 Diagram



11.3.3.1.79.3 Fields

Field	Function
31-28 RESERVED9	Reserved. Note: This register field is sticky.
27-26 VC0_CPL_DATA_SCALE	VC0 Scale CPL Data Credits. Note: This register field is sticky.
25-24 VC0_CPL_HDR_SCALE	VC0 Scale CPL Header Credits. Note: This register field is sticky.
23-21 VC0_CPL_TLP_Q_MODE	Reserved. Note: This register field is sticky.
20	Reserved. Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

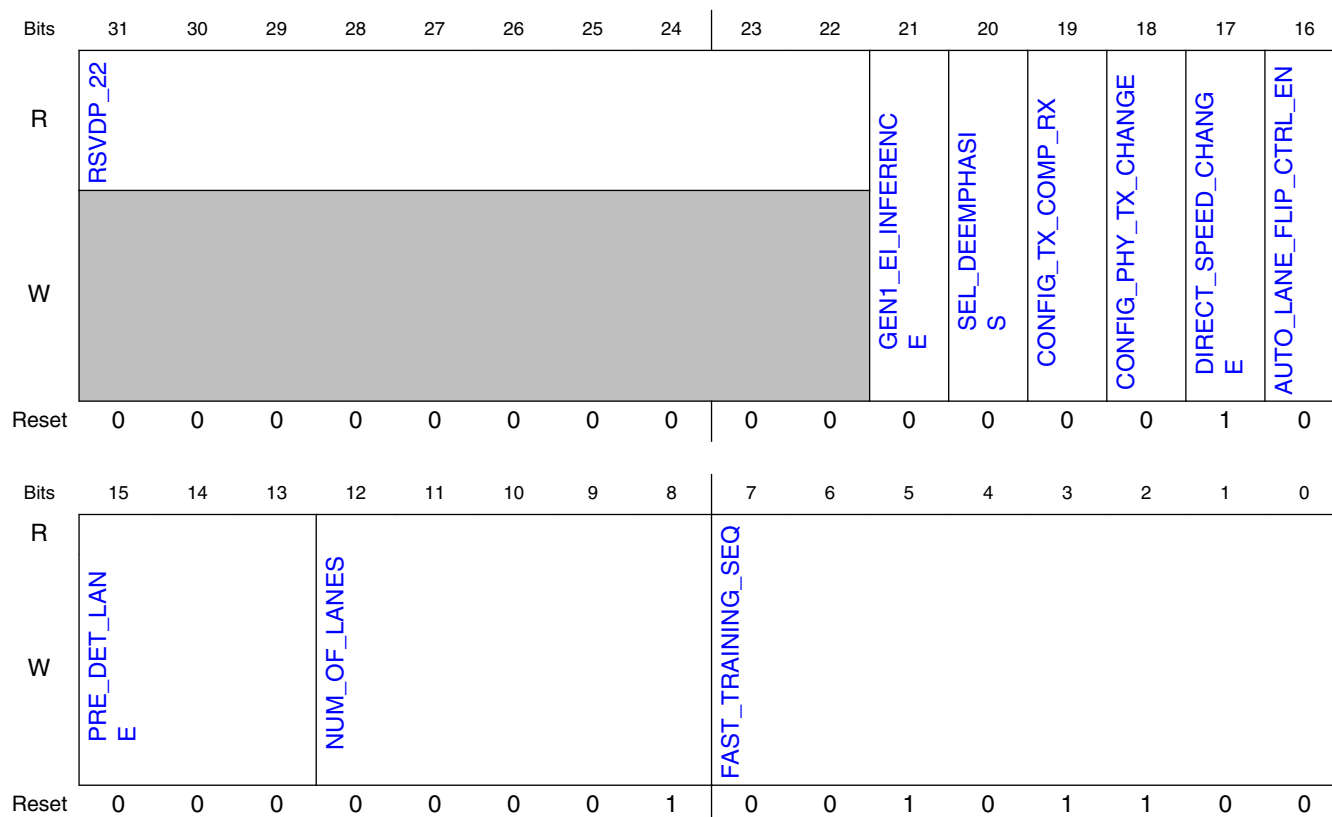
Field	Function
RESERVED8	
19-12 VC0_CPL_HEA DER_CREDIT	VC0 Completion Header Credits. The number of initial Completion header credits for VC0, used only in the segmented-buffer configuration. Note: The access attributes of this field are as follows: - Dbi: R (sticky) Note: This register field is sticky.
11-0 VC0_CPL_DAT A_CREDIT	VC0 Completion Data Credits. The number of initial Completion data credits for VC0, used only in the segmented-buffer configuration. Note: The access attributes of this field are as follows: - Dbi: R (sticky) Note: This register field is sticky.

11.3.3.1.80 Link Width and Speed Change Control Register. (GEN2_CTRL_OFF)

11.3.3.1.80.1 Offset

Register	Offset
GEN2_CTRL_OFF	80Ch

11.3.3.1.80.2 Diagram



11.3.3.1.80.3 Fields

Field	Function
31-22 RSVDP_22	Reserved for future use.
21 GEN1_EI_INFERENCE	Electrical Idle Inference Mode at Gen1 Rate. Programmable mode to determine inferred electrical idle (EI) in Recovery.Speed or Loopback.Active (as slave) state at Gen1 speed by looking for a "1" value on RxElecIdle instead of looking for a "0" on RxValid. If the PHY fails to deassert the RxValid signal in Recovery.Speed or Loopback.Active (because of corrupted EIOS for example), then EI cannot be inferred successfully in the controller by just detecting the condition RxValid=0. - 0: Use RxElecIdle signal to infer Electrical Idle - 1: Use RxValid signal to infer Electrical Idle Note: This register field is sticky.
20 SEL_DEEMPHASIS	Used to set the de-emphasis level for upstream ports. This bit selects the level of de-emphasis the link operates at. - 0: -6 dB - 1: -3.5 dB This field is reserved (fixed to '0') for M-PCIe. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
19 CONFIG_TX_COMP_RX	Config Tx Compliance Receive Bit. When set to 1, signals LTSSM to transmit TS ordered sets with the compliance receive bit assert (equal to "1"). This field is reserved (fixed to '0') for M-PCIe. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
18 CONFIG_PHY_TX_CHANGE	Config PHY Tx Swing. Controls the PHY transmitter voltage swing level. The controller drives the mac_phy_txswing output from this register bit field. - 0: Full Swing - 1: Low Swing This field is reserved (fixed to '0') for M-PCIe. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
17 DIRECT_SPEED_CHANGE	Directed Speed Change. Writing "1" to this field instructs the LTSSM to initiate a speed change to Gen2 or Gen3 after the link is initialized at Gen1 speed. When the speed change occurs, the controller will clear the contents of this field; and a read to this field by your software will return a "0". To manually initiate the speed change: - Write to LINK_CONTROL2_LINK_STATUS2_REG[PCIE_CAP_TARGET_LINK_SPEED] in the local device, deassert this field, and then assert this field. This field is reserved (fixed to '0') for M-PCIe. Note: The access attributes of this field are as follows: - Dbi: R/W
16 AUTO_LANE_FLIP_CTRL_EN	Enable Auto flipping of the lanes. This field is reserved (fixed to '0') for M-PCIe. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
15-13 PRE_DET_LANE	Predetermined Lane for Auto Flip. This field defines which physical lane is connected to logical Lane0 by the flip operation performed in Detect. Allowed values are: - 3'b000: Connect logical Lane0 to physical lane 0 - 3'b001: Connect logical Lane0 to physical lane 1 - 3'b010: Connect logical Lane0 to physical lane 3 - 3'b011: Connect logical Lane0 to physical lane 7 - 3'b100: Connect logical Lane0 to physical lane 15 This field is used to restrict the receiver detect procedure to a particular lane when the default detect and polling procedure performed on all lanes cannot be successful. A notable example of when it is useful to program this field to a value different from the default, is when a lane is asymmetrically broken, that is, it is detected in Detect LTSSM state but it cannot exit Electrical Idle in Polling LTSSM state. Note: This field is reserved (fixed to '0') for M-PCIe. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
12-8 NUM_OF_LANES	Predetermined Number of Lanes. Defines the number of lanes which are connected and not bad. Used to limit the effective link width to ignore "broken" or "unused" lanes that detect a receiver. Indicates the number of lanes to check for exit from Electrical Idle in Polling.Active and L2.Idle. It is possible that the LTSSM might detect a receiver on a bad or broken lane during the Detect Substate. However, it is also possible that such a lane might also fail to exit Electrical Idle and therefore prevent a valid link from being configured. This value is referred to as the "Predetermined Number of Lanes" in section 4.2.6.2.1 of the PCI Express Base 3.0 Specification, revision 1.0. Encoding is as follows: - 0x01: 1 lane - 0x02: 2 lanes - 0x03: 3 lanes - .. When you have unused lanes in your system, then you must change the value in this register to reflect the number of lanes. You must also change the value in the "Link Mode Enable" field of PORT_LINK_CTRL_OFF. The value in this register is normally the same as the encoded value in

Table continues on the next page...

PCI Express (PCIe)

Field	Function
	PORT_LINK_CTRL_OFF. If you find that one of your used lanes is bad then you must reduce the value in this register. For more information, see "How to Tie Off Unused Lanes." For information on upsizing and downsizing the link width, see "Link Establishment." This field is reserved (fixed to '0') for M-PCIe. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.
7-0 FAST_TRAINING_SEQ	Sets the Number of Fast Training Sequences (N_FTS) that the controller advertises as its N_FTS during Gen2 or Gen3 link training. This value is used to inform the link partner about the PHY's ability to recover synchronization after a low power state. The number should be provided by the PHY vendor. Do not set N_FTS to zero; doing so can cause the LTSSM to go into the recovery state when exiting from L0s. This field is reserved (fixed to '0') for M-PCIe. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.

11.3.3.1.81 PHY Status Register. (PHY_STATUS_OFF)

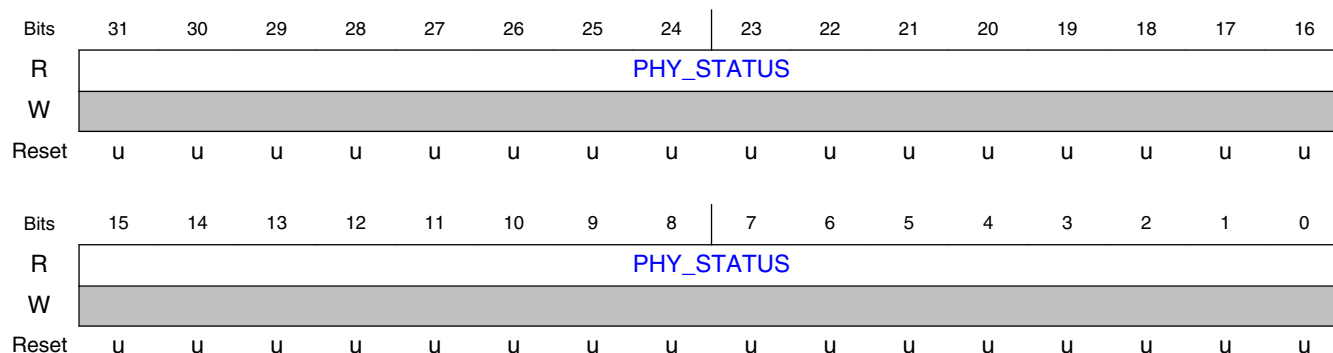
11.3.3.1.81.1 Offset

Register	Offset
PHY_STATUS_OFF	810h

11.3.3.1.81.2 Function

PHY Status Register. Memory mapped register from phy_cfg_status GPIO input pins.

11.3.3.1.81.3 Diagram



11.3.3.1.81.4 Fields

Field	Function
31-0 PHY_STATUS	PHY Status. Data received directly from the phy_cfg_status bus. This is a GPIO register reflecting the values on the static phy_cfg_status input signals. The usage is left completely to the user and does not in any way influence controller functionality. You can use it for any static sideband status signalling

Field	Function
	requirements that you have for your PHY. This field is reserved (fixed to '0') for M-PCle. Note: This register field is sticky.

11.3.3.1.82 PHY Control Register. (PHY_CONTROL_OFF)

11.3.3.1.82.1 Offset

Register	Offset
PHY_CONTROL_OFF	814h

11.3.3.1.82.2 Function

PHY Control Register. Memory mapped register to cfg_phy_control GPIO output pins.

11.3.3.1.82.3 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	PHY_CONTROL															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PHY_CONTROL															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.82.4 Fields

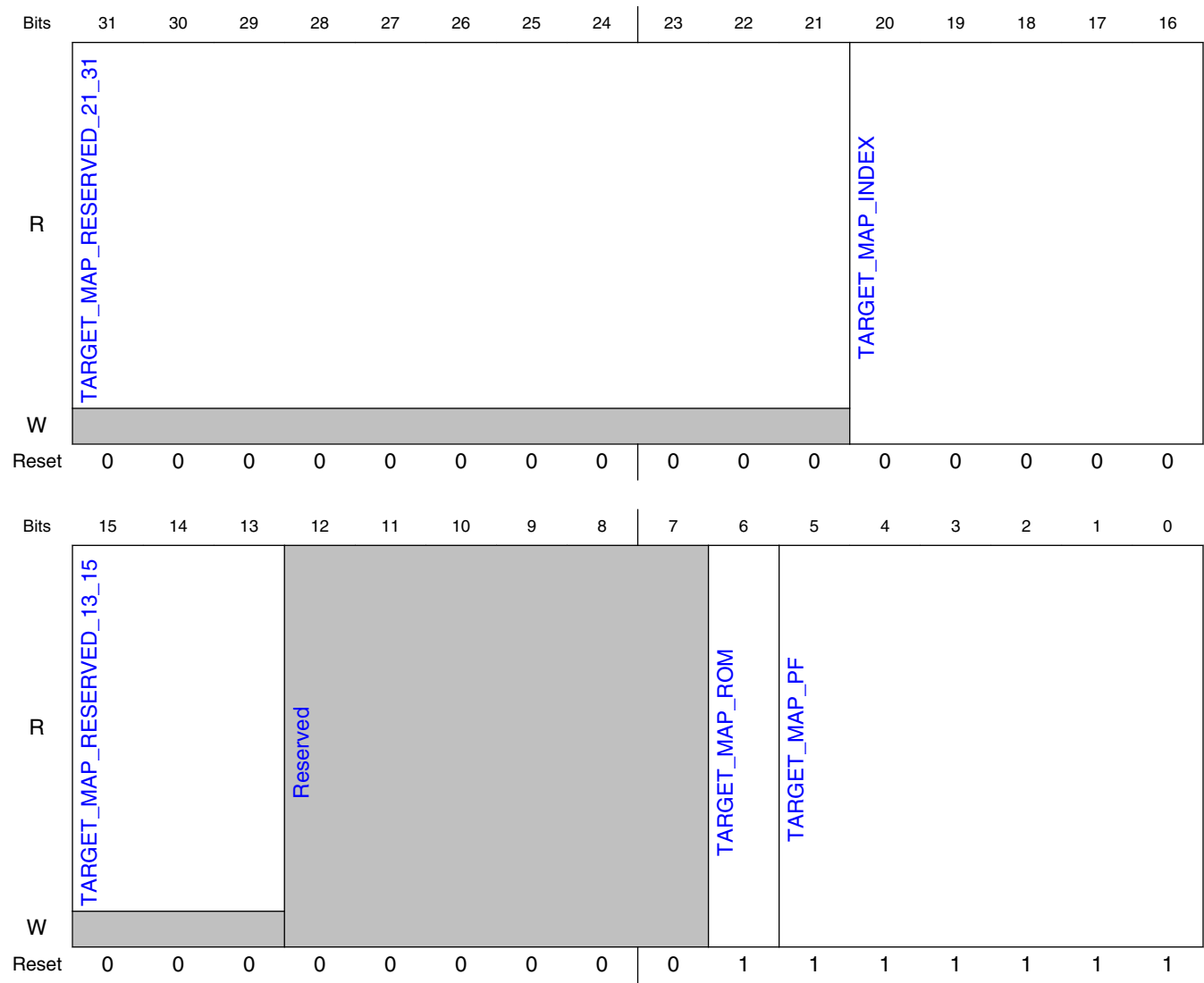
Field	Function
31-0 PHY_CONTROL	PHY Control. Data sent directly to the cfg_phy_control bus. These is a GPIO register driving the values on the static cfg_phy_control output signals. The usage is left completely to the user and does not in any way influence controller functionality. You can use it for any static sideband control signalling requirements that you have for your PHY. This field is reserved (fixed to '0') for M-PCle. Note: This register field is sticky.

11.3.3.1.83 Programmable Target Map Control Register. (TRGT_MAP_CTRL_L_OFF)

11.3.3.1.83.1 Offset

Register	Offset
TRGT_MAP_CTRL_OFF	81Ch

11.3.3.1.83.2 Diagram



11.3.3.1.83.3 Fields

Field	Function
31-21 TARGET_MAP_RESERVED_21_31	Reserved. Note: The access attributes of this field are as follows: - Dbi: R (sticky)
20-16 TARGET_MAP_INDEX	The number of the PF Function on which the Target Values are set. This register does not respect the Byte Enable setting. any write will affect all register bits.
15-13 TARGET_MAP_RESERVED_13_15	Reserved. Note: The access attributes of this field are as follows: - Dbi: R (sticky)
12-7 —	Reserved.
6 TARGET_MAP_ROM	Target Value for the ROM page of the PF Function selected by the index number. This register does not respect the Byte Enable setting. any write will affect all register bits.
5-0 TARGET_MAP_PF	Target Values for each BAR on the PF Function selected by the index number. This register does not respect the Byte Enable setting. any write will affect all register bits.

11.3.3.1.84 Integrated MSI Reception Module (iMRM) Address Register. (MSI_CTRL_ADDR_OFF)

11.3.3.1.84.1 Offset

Register	Offset
MSI_CTRL_ADDR_OFF	820h

11.3.3.1.84.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_ADDR															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_ADDR															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.84.3 Fields

Field	Function
31-0 MSI_CTRL_ADDR	Integrated MSI Reception Module Address. System specified address for MSI memory write transaction termination. Within the AXI Bridge, every received Memory Write request is examined to see if it targets the MSI Address that has been specified in this register; and also to see if it satisfies the definition of an MSI interrupt request. When these conditions are satisfied the Memory Write request is marked as an MSI request. Note: This register field is sticky.

11.3.3.1.85 Integrated MSI Reception Module Upper Address Register. (MSI_CTRL_UPPER_ADDR_OFF)

11.3.3.1.85.1 Offset

Register	Offset
MSI_CTRL_UPPER_ADDR_OFF	824h

11.3.3.1.85.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_UPPER_ADDR															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_UPPER_ADDR															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.85.3 Fields

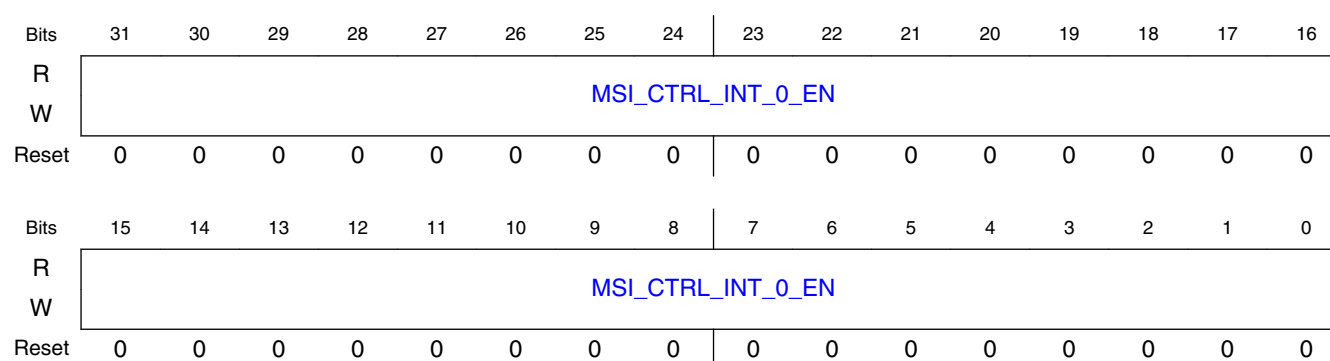
Field	Function
31-0 MSI_CTRL_UPPER_ADDR	Integrated MSI Reception Module Upper Address. System specified upper address for MSI memory write transaction termination. Allows functions to support a 64-bit MSI address. Note: This register field is sticky.

11.3.3.1.86 Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_0_EN_OFF)

11.3.3.1.86.1 Offset

Register	Offset
MSI_CTRL_INT_0_EN_OFF	828h

11.3.3.1.86.2 Diagram



11.3.3.1.86.3 Fields

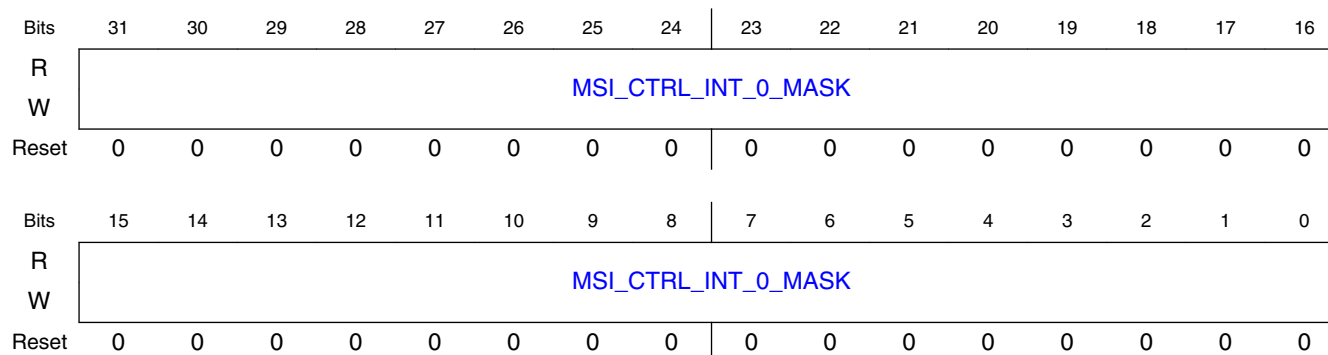
Field	Function
31-0 MSI_CTRL_INT_0_EN	MSI Interrupt#i Enable. Specifies which interrupts are enabled. When an MSI is received from a disabled interrupt, no status bit gets set in MSI controller interrupt status register. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.87 Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_0_MASK_OFF)

11.3.3.1.87.1 Offset

Register	Offset
MSI_CTRL_INT_0_MASK_OFF	82Ch

11.3.3.1.87.2 Diagram



11.3.3.1.87.3 Fields

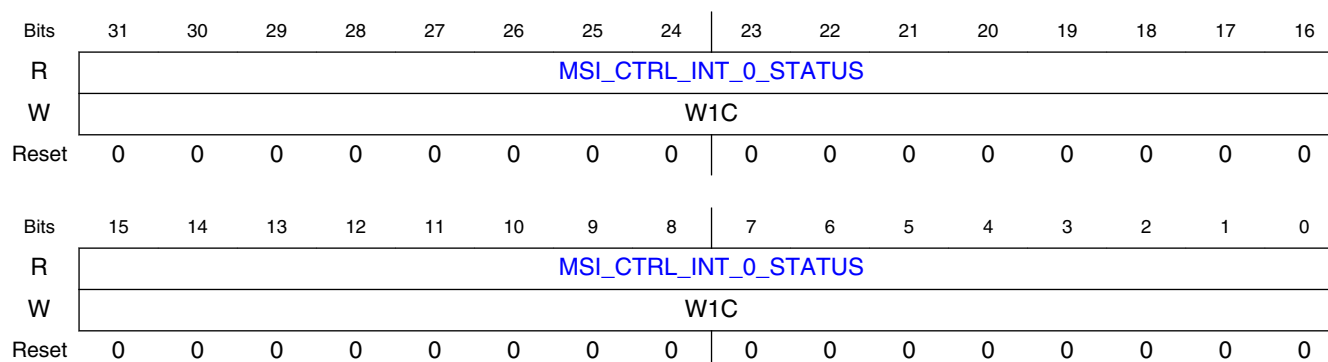
Field	Function
31-0 MSI_CTRL_INT_0_MASK	MSI Interrupt#i Mask. Allows enabled interrupts to be masked. When an MSI is received for a masked interrupt, the corresponding status bit gets set in the interrupt status register but the msi_ctrl_int output is not set HIGH. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.88 Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_0_STATUS_OFF)

11.3.3.1.88.1 Offset

Register	Offset
MSI_CTRL_INT_0_STATUS_OFF	830h

11.3.3.1.88.2 Diagram



11.3.3.1.88.3 Fields

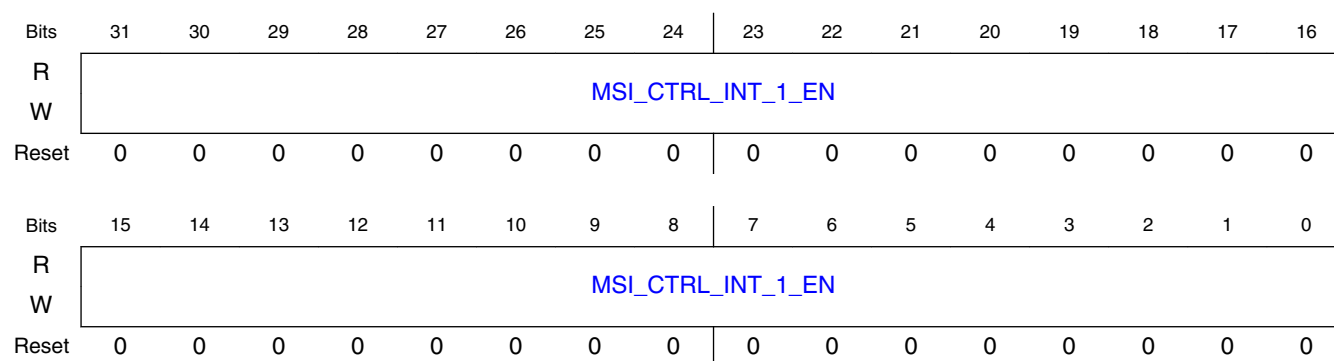
Field	Function
31-0 MSI_CTRL_INT_0_STATUS	MSI Interrupt#i Status. When an MSI is detected for EP#i, one bit in this register is set. The decoding of the data payload of the MSI Memory Write request determines which bit gets set. A status bit is cleared by writing a 1 to the bit. Each bit corresponds to a single MSI Interrupt Vector.

11.3.3.1.89 Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_1_EN_OFF)

11.3.3.1.89.1 Offset

Register	Offset
MSI_CTRL_INT_1_EN_OFF	834h

11.3.3.1.89.2 Diagram



11.3.3.1.89.3 Fields

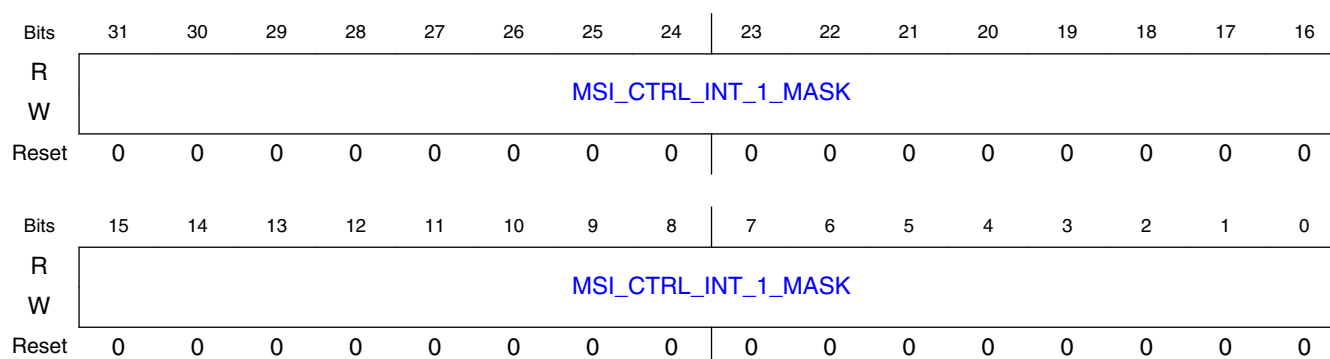
Field	Function
31-0 MSI_CTRL_INT_1_EN	MSI Interrupt#i Enable. Specifies which interrupts are enabled. When an MSI is received from a disabled interrupt, no status bit gets set in MSI controller interrupt status register. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.90 Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_1_MASK_OFF)

11.3.3.1.90.1 Offset

Register	Offset
MSI_CTRL_INT_1_MASK_OFF	838h

11.3.3.1.90.2 Diagram



11.3.3.1.90.3 Fields

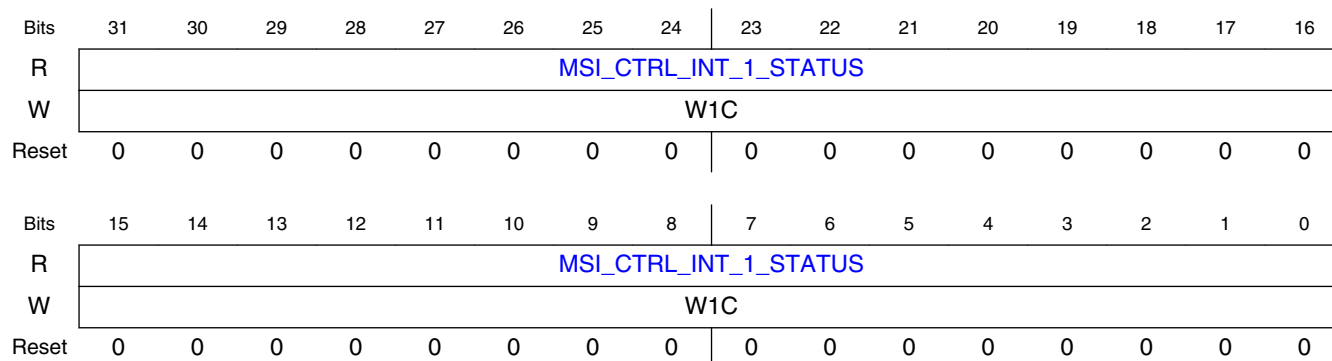
Field	Function
31-0 MSI_CTRL_INT_1_MASK	MSI Interrupt#i Mask. Allows enabled interrupts to be masked. When an MSI is received for a masked interrupt, the corresponding status bit gets set in the interrupt status register but the msi_ctrl_int output is not set HIGH. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.91 Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_1_STATUS_OFF)

11.3.3.1.91.1 Offset

Register	Offset
MSI_CTRL_INT_1_STATUS_OFF	83Ch

11.3.3.1.91.2 Diagram



11.3.3.1.91.3 Fields

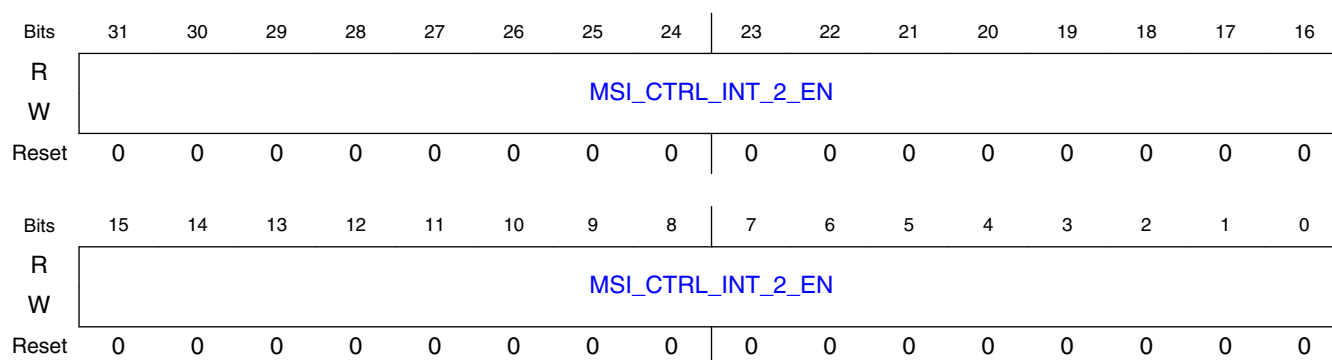
Field	Function
MSI_CTRL_INT_1_STATUS	MSI Interrupt#i Status. When an MSI is detected for EP#i, one bit in this register is set. The decoding of the data payload of the MSI Memory Write request determines which bit gets set. A status bit is cleared by writing a 1 to the bit. Each bit corresponds to a single MSI Interrupt Vector.

11.3.3.1.92 Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_2_EN_OFF)

11.3.3.1.92.1 Offset

Register	Offset
MSI_CTRL_INT_2_EN_OFF	840h

11.3.3.1.92.2 Diagram



11.3.3.1.92.3 Fields

Field	Function
31-0 MSI_CTRL_INT_2_EN	MSI Interrupt#i Enable. Specifies which interrupts are enabled. When an MSI is received from a disabled interrupt, no status bit gets set in MSI controller interrupt status register. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.93 Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_2_MASK_OFF)

11.3.3.1.93.1 Offset

Register	Offset
MSI_CTRL_INT_2_MASK_OFF	844h

11.3.3.1.93.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_2_MASK															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_2_MASK															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.93.3 Fields

Field	Function
31-0 MSI_CTRL_INT_2_MASK	MSI Interrupt#i Mask. Allows enabled interrupts to be masked. When an MSI is received for a masked interrupt, the corresponding status bit gets set in the interrupt status register but the msi_ctrl_int output is not set HIGH. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.94 Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_2_STATUS_OFF)

11.3.3.1.94.1 Offset

Register	Offset
MSI_CTRL_INT_2_STATUS_OFF	848h

11.3.3.1.94.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_2_STATUS															
W	W1C															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_2_STATUS															
W	W1C															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.94.3 Fields

Field	Function
31-0 MSI_CTRL_INT_2_STATUS	MSI Interrupt#i Status. When an MSI is detected for EP#i, one bit in this register is set. The decoding of the data payload of the MSI Memory Write request determines which bit gets set. A status bit is cleared by writing a 1 to the bit. Each bit corresponds to a single MSI Interrupt Vector.

11.3.3.1.95 Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_3_EN_OFF)

11.3.3.1.95.1 Offset

Register	Offset
MSI_CTRL_INT_3_EN_OFF	84Ch

11.3.3.1.95.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_3_EN															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_3_EN															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.95.3 Fields

Field	Function
31-0 MSI_CTRL_INT_3_EN	MSI Interrupt#i Enable. Specifies which interrupts are enabled. When an MSI is received from a disabled interrupt, no status bit gets set in MSI controller interrupt status register. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.96 Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_3_MASK_OFF)

11.3.3.1.96.1 Offset

Register	Offset
MSI_CTRL_INT_3_MASK_OFF	850h

11.3.3.1.96.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_3_MASK															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_3_MASK															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.96.3 Fields

Field	Function
31-0 MSI_CTRL_INT_3_MASK	MSI Interrupt#i Mask. Allows enabled interrupts to be masked. When an MSI is received for a masked interrupt, the corresponding status bit gets set in the interrupt status register but the msi_ctrl_int output is not set HIGH. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.97 Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_3_STATUS_OFF)

11.3.3.1.97.1 Offset

Register	Offset
MSI_CTRL_INT_3_STATUS_OFF	854h

11.3.3.1.97.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_3_STATUS															
W	W1C															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_3_STATUS															
W	W1C															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.97.3 Fields

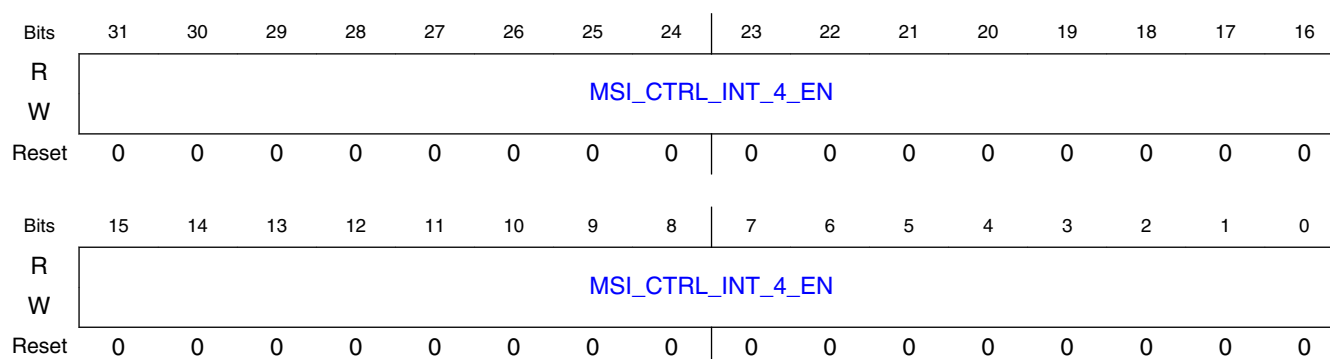
Field	Function
31-0 MSI_CTRL_INT_3_STATUS	MSI Interrupt#i Status. When an MSI is detected for EP#i, one bit in this register is set. The decoding of the data payload of the MSI Memory Write request determines which bit gets set. A status bit is cleared by writing a 1 to the bit. Each bit corresponds to a single MSI Interrupt Vector.

11.3.3.1.98 Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_4_EN_OFF)

11.3.3.1.98.1 Offset

Register	Offset
MSI_CTRL_INT_4_EN_OFF	858h

11.3.3.1.98.2 Diagram



11.3.3.1.98.3 Fields

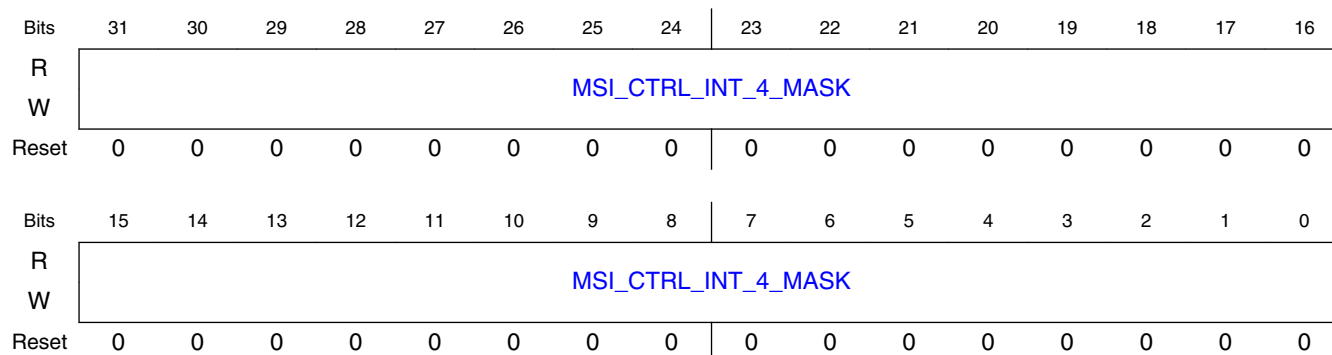
Field	Function
31-0 MSI_CTRL_INT_4_EN	MSI Interrupt#i Enable. Specifies which interrupts are enabled. When an MSI is received from a disabled interrupt, no status bit gets set in MSI controller interrupt status register. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.99 Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_4_MASK_OFF)

11.3.3.1.99.1 Offset

Register	Offset
MSI_CTRL_INT_4_MASK_OFF	85Ch

11.3.3.1.99.2 Diagram



11.3.3.1.99.3 Fields

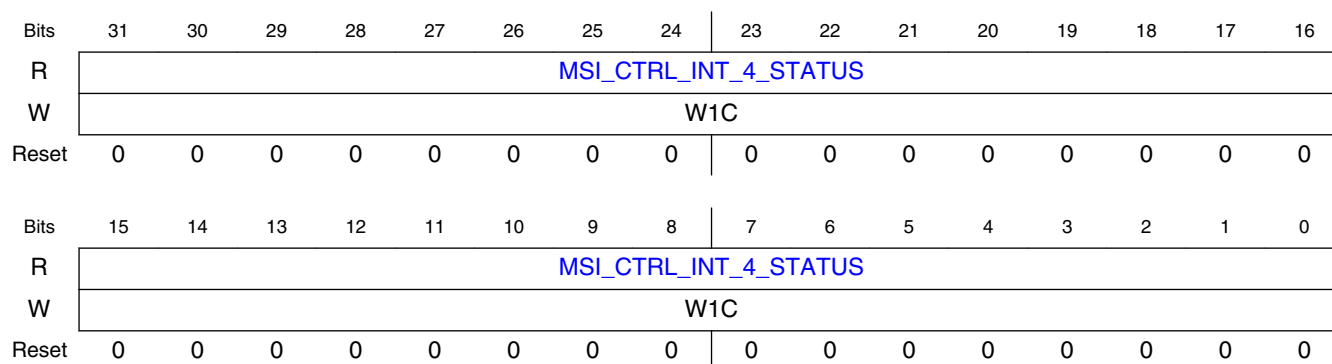
Field	Function
31-0 MSI_CTRL_INT_4_MASK	MSI Interrupt#i Mask. Allows enabled interrupts to be masked. When an MSI is received for a masked interrupt, the corresponding status bit gets set in the interrupt status register but the msi_ctrl_int output is not set HIGH. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.100 Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_4_STATUS_OFF)

11.3.3.1.100.1 Offset

Register	Offset
MSI_CTRL_INT_4_STATUS_OFF	860h

11.3.3.1.100.2 Diagram



11.3.3.1.100.3 Fields

Field	Function
31-0 MSI_CTRL_INT_4_STATUS	MSI Interrupt#i Status. When an MSI is detected for EP#i, one bit in this register is set. The decoding of the data payload of the MSI Memory Write request determines which bit gets set. A status bit is cleared by writing a 1 to the bit. Each bit corresponds to a single MSI Interrupt Vector.

11.3.3.1.101 Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_5_EN_OFF)

11.3.3.1.101.1 Offset

Register	Offset
MSI_CTRL_INT_5_EN_OFF	864h

11.3.3.1.101.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_5_EN															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_5_EN															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.101.3 Fields

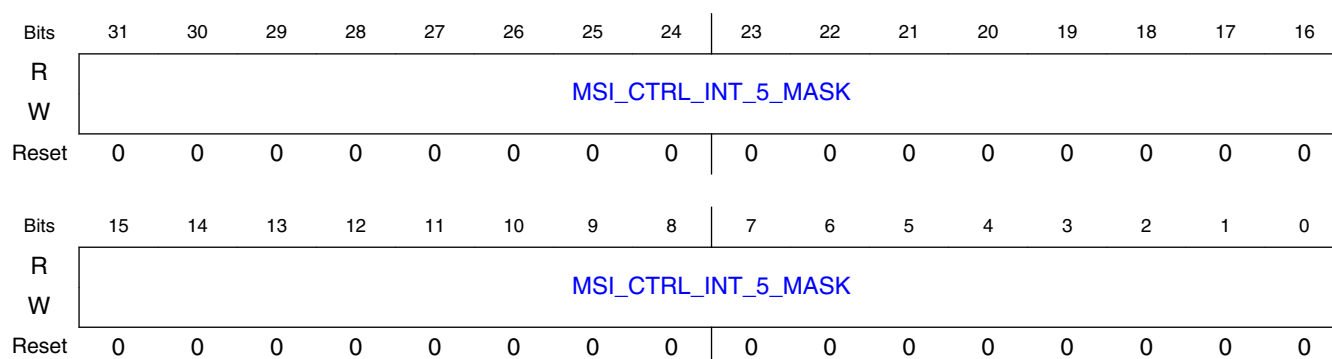
Field	Function
31-0 MSI_CTRL_INT_5_EN	MSI Interrupt#i Enable. Specifies which interrupts are enabled. When an MSI is received from a disabled interrupt, no status bit gets set in MSI controller interrupt status register. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.102 Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_5_MASK_OFF)

11.3.3.1.102.1 Offset

Register	Offset
MSI_CTRL_INT_5_MASK_OFF	868h

11.3.3.1.102.2 Diagram



11.3.3.1.102.3 Fields

Field	Function
31-0 MSI_CTRL_INT_5_MASK	MSI Interrupt#i Mask. Allows enabled interrupts to be masked. When an MSI is received for a masked interrupt, the corresponding status bit gets set in the interrupt status register but the msi_ctrl_int output is not set HIGH. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.103 Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_5_STATUS_OFF)

11.3.3.1.103.1 Offset

Register	Offset
MSI_CTRL_INT_5_STATUS_OFF	86Ch

11.3.3.1.103.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_5_STATUS															
W	W1C															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_5_STATUS															
W	W1C															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.103.3 Fields

Field	Function
MSI_CTRL_INT_5_STATUS	MSI Interrupt#i Status. When an MSI is detected for EP#i, one bit in this register is set. The decoding of the data payload of the MSI Memory Write request determines which bit gets set. A status bit is cleared by writing a 1 to the bit. Each bit corresponds to a single MSI Interrupt Vector.

11.3.3.1.104 Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_6_EN_OFF)

11.3.3.1.104.1 Offset

Register	Offset
MSI_CTRL_INT_6_EN_OFF	870h

11.3.3.1.104.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_6_EN															
W	MSI_CTRL_INT_6_EN															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_6_EN															
W	MSI_CTRL_INT_6_EN															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.104.3 Fields

Field	Function
31-0 MSI_CTRL_INT_6_EN	MSI Interrupt#i Enable. Specifies which interrupts are enabled. When an MSI is received from a disabled interrupt, no status bit gets set in MSI controller interrupt status register. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.105 Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_6_MASK_OFF)

11.3.3.1.105.1 Offset

Register	Offset
MSI_CTRL_INT_6_MASK_OFF	874h

11.3.3.1.105.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_6_MASK															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_6_MASK															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.105.3 Fields

Field	Function
31-0 MSI_CTRL_INT_6_MASK	MSI Interrupt#i Mask. Allows enabled interrupts to be masked. When an MSI is received for a masked interrupt, the corresponding status bit gets set in the interrupt status register but the msi_ctrl_int output is not set HIGH. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.106 Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_6_STATUS_OFF)

11.3.3.1.106.1 Offset

Register	Offset
MSI_CTRL_INT_6_STATUS_OFF	878h

11.3.3.1.106.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_6_STATUS															
W	W1C															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_6_STATUS															
W	W1C															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.106.3 Fields

Field	Function
31-0 MSI_CTRL_INT_6_STATUS	MSI Interrupt#i Status. When an MSI is detected for EP#i, one bit in this register is set. The decoding of the data payload of the MSI Memory Write request determines which bit gets set. A status bit is cleared by writing a 1 to the bit. Each bit corresponds to a single MSI Interrupt Vector.

11.3.3.1.107 Integrated MSI Reception Module Interrupt#i Enable Register. (MSI_CTRL_INT_7_EN_OFF)

11.3.3.1.107.1 Offset

Register	Offset
MSI_CTRL_INT_7_EN_OFF	87Ch

11.3.3.1.107.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_7_EN															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_7_EN															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.107.3 Fields

Field	Function
31-0 MSI_CTRL_INT_7_EN	MSI Interrupt#i Enable. Specifies which interrupts are enabled. When an MSI is received from a disabled interrupt, no status bit gets set in MSI controller interrupt status register. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.108 Integrated MSI Reception Module Interrupt#i Mask Register. (MSI_CTRL_INT_7_MASK_OFF)

11.3.3.1.108.1 Offset

Register	Offset
MSI_CTRL_INT_7_MASK_OFF	880h

11.3.3.1.108.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_7_MASK															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_7_MASK															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.108.3 Fields

Field	Function
31-0 MSI_CTRL_INT_7_MASK	MSI Interrupt#i Mask. Allows enabled interrupts to be masked. When an MSI is received for a masked interrupt, the corresponding status bit gets set in the interrupt status register but the msi_ctrl_int output is not set HIGH. Each bit corresponds to a single MSI Interrupt Vector. Note: This register field is sticky.

11.3.3.1.109 Integrated MSI Reception Module Interrupt#i Status Register. (MSI_CTRL_INT_7_STATUS_OFF)

11.3.3.1.109.1 Offset

Register	Offset
MSI_CTRL_INT_7_STATUS_OFF	884h

11.3.3.1.109.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	MSI_CTRL_INT_7_STATUS															
W	W1C															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MSI_CTRL_INT_7_STATUS															
W	W1C															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.109.3 Fields

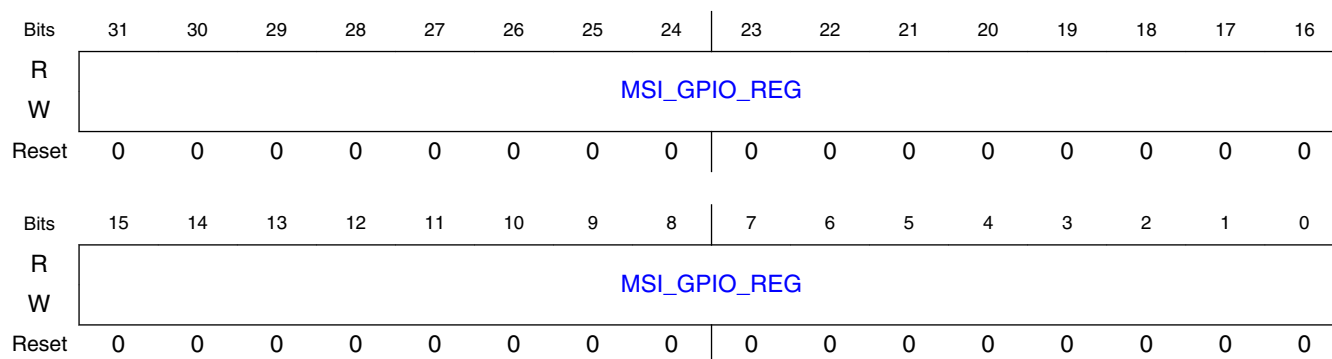
Field	Function
31-0 MSI_CTRL_INT_7_STATUS	MSI Interrupt#i Status. When an MSI is detected for EP#i, one bit in this register is set. The decoding of the data payload of the MSI Memory Write request determines which bit gets set. A status bit is cleared by writing a 1 to the bit. Each bit corresponds to a single MSI Interrupt Vector.

11.3.3.1.110 Integrated MSI Reception Module General Purpose IO Register. (MSI_GPIO_IO_OFF)

11.3.3.1.110.1 Offset

Register	Offset
MSI_GPIO_IO_OFF	888h

11.3.3.1.110.2 Diagram



11.3.3.1.110.3 Fields

Field	Function
31-0 MSI_GPIO_REG	MSI GPIO Register. The contents of this register drives the top-level GPIO msi_ctrl_io[31:0] Note: This register field is sticky.

11.3.3.1.111 RADM clock gating enable control register. (CLOCK_GATING_CTRL_OFF)

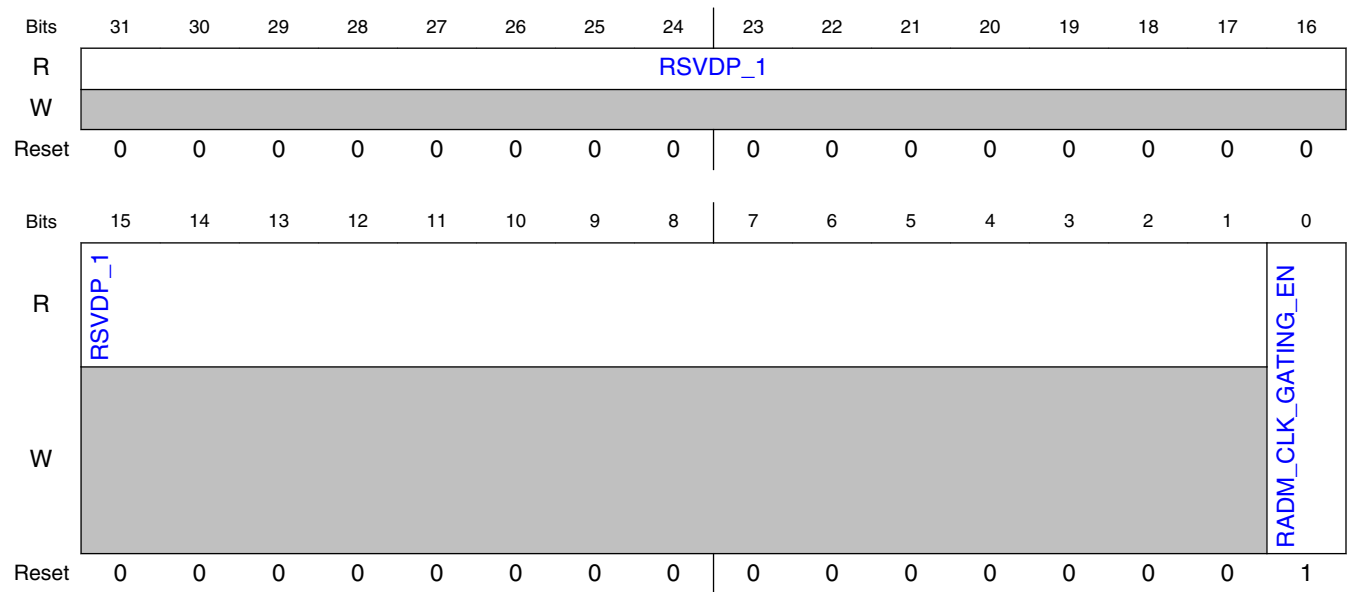
11.3.3.1.111.1 Offset

Register	Offset
CLOCK_GATING_CTRL_OFF	88Ch

11.3.3.1.111.2 Function

RADM clock gating enable control register. Using this register you can disable the RADM clock gating feature. The DWC_pcie_clk_rst.v modules uses the en_radm_clk_g output to gate core_clk and create the radm_clk_g clock input. The controller de-asserts the en_radm_clk_g output when there is no Rx traffic, Rx queues and pre/post-queue pipelines are empty, RADM completion LUT is empty, and there are no FLR actions pending. You must set the RADM_CLK_GATING_EN field to enable this functionality; otherwise the en_radm_clk_g output will always be set to '1'.

11.3.3.1.111.3 Diagram



11.3.3.1.111.4 Fields

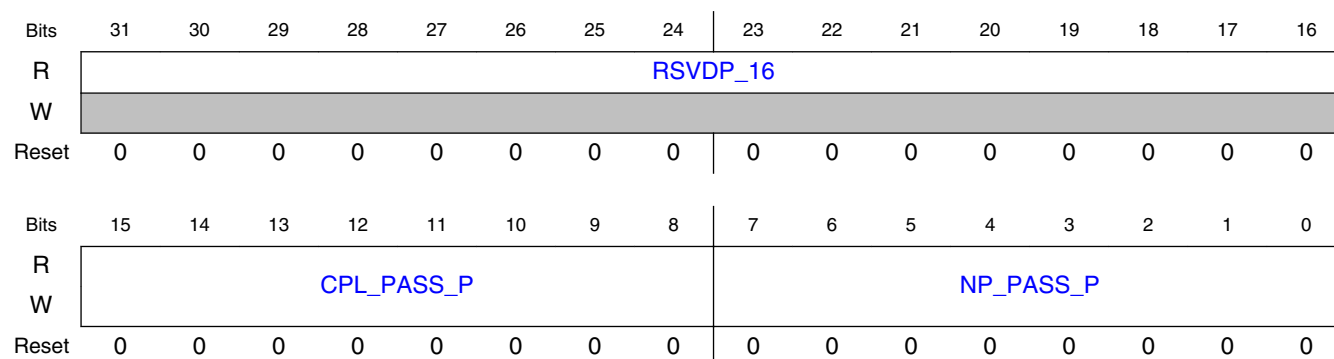
Field	Function
31-1 RSVDP_1	Reserved for future use.
0 RADM_CLK_GATING_EN	Enable Radm clock gating feature. - 0: Disable - 1: Enable(default)

11.3.3.1.112 Order Rule Control Register. (ORDER_RULE_CTRL_OFF)

11.3.3.1.112.1 Offset

Register	Offset
ORDER_RULE_CTRL_OFF	8B4h

11.3.3.1.112.2 Diagram



11.3.3.1.112.3 Fields

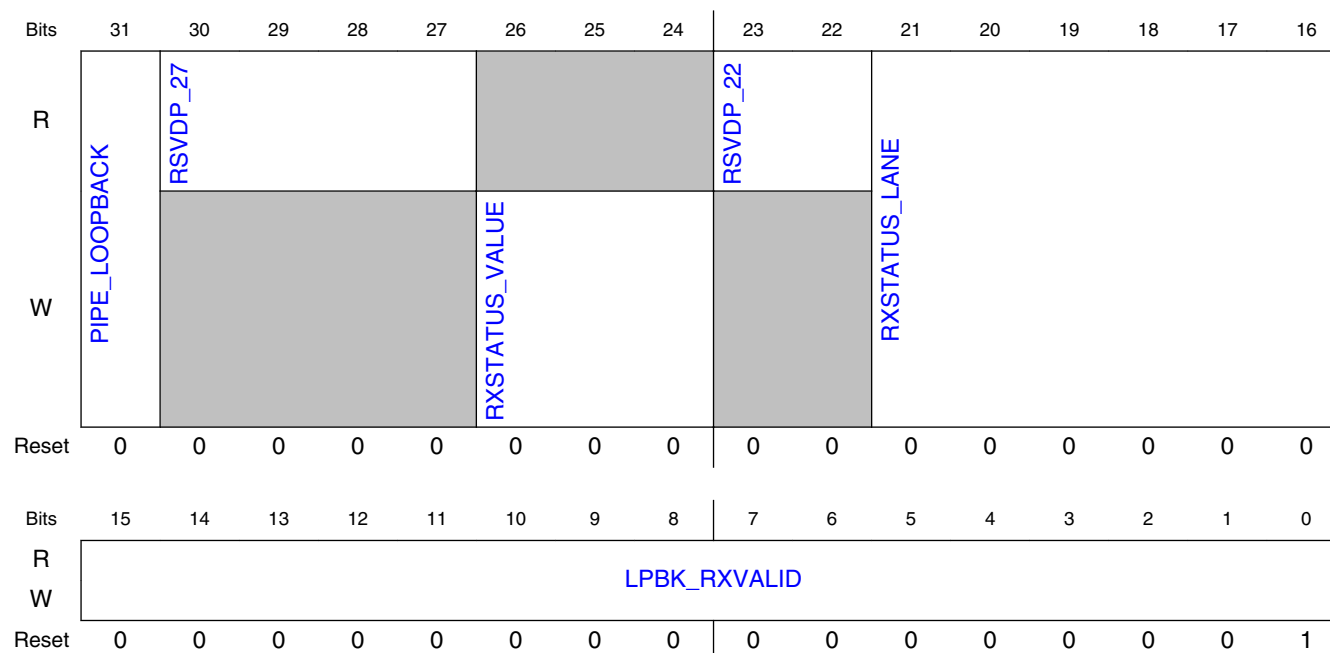
Field	Function
31-16 RSVDP_16	Reserved for future use.
15-8 CPL_PASS_P	Completion Passing Posted Ordering Rule Control. Determines if CPL can pass halted P queue. - 0: CPL can not pass P (recommended) - 1: CPL can pass P
7-0 NP_PASS_P	Non-Posted Passing Posted Ordering Rule Control. Determines if NP can pass halted P queue. - 0 : NP can not pass P (recommended). - 1 : NP can pass P

11.3.3.1.113 PIPE Loopback Control Register. (PIPE_LOOPBACK_CONTROL_OFF)

11.3.3.1.113.1 Offset

Register	Offset
PIPE_LOOPBACK_CONTROL_OFF	8B8h

11.3.3.1.113.2 Diagram



11.3.3.1.113.3 Fields

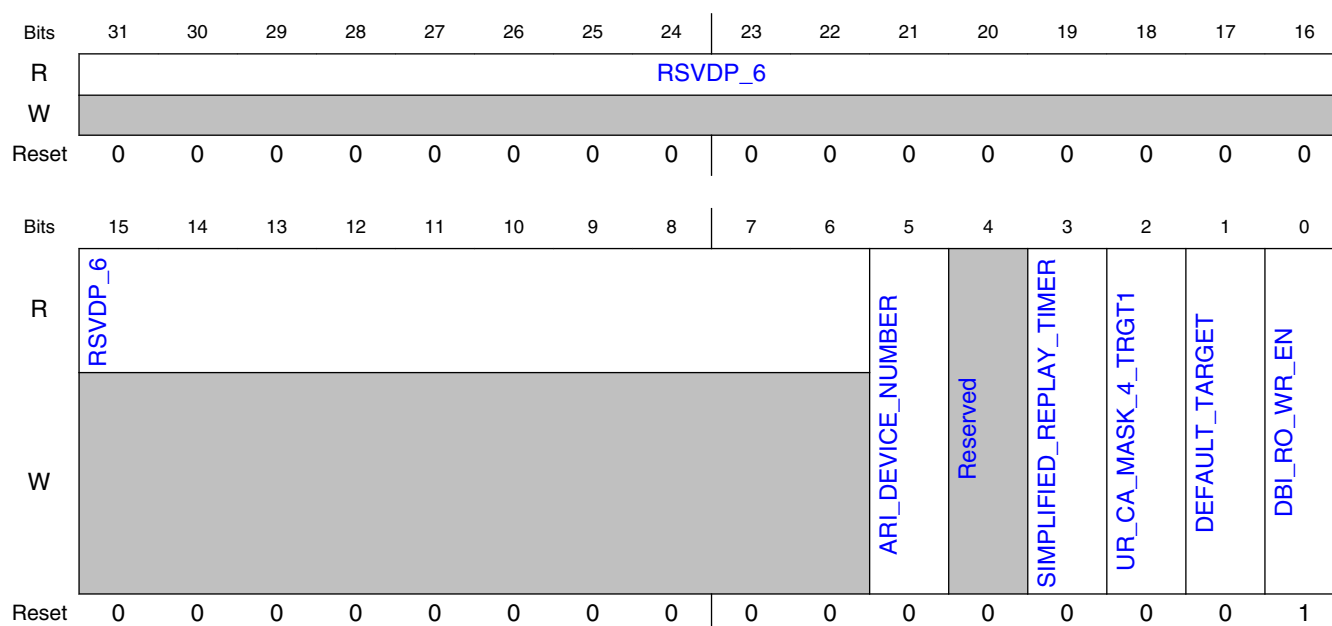
Field	Function
31 PIPE_LOOPBACK	PIPE Loopback Enable. Indicates RMMI Loopback if M-PCIe. Note: This register field is sticky.
30-27 RSVDP_27	Reserved for future use.
26-24 RXSTATUS_VALUE	RXSTATUS_VALUE is an internally reserved field. Do not use.
23-22 RSVDP_22	Reserved for future use.
21-16 RXSTATUS_LANE	RXSTATUS_LANE is an internally reserved field. Do not use. Note: This register field is sticky.
15-0 LPBK_RXVALID	LPBK_RXVALID is an internally reserved field. Do not use. Note: This register field is sticky.

11.3.3.1.114 DBI Read-Only Write Enable Register. (MISC_CONTROL_1_0 FF)

11.3.3.1.114.1 Offset

Register	Offset
MISC_CONTROL_1_0 FF	8BCh

11.3.3.1.114.2 Diagram



11.3.3.1.114.3 Fields

Field	Function
31-6 RSVDP_6	Reserved for future use.
5 ARI_DEVICE_NUMBER	When ARI is enabled, this field enables use of the device ID. Note: This register field is sticky.
4 —	Reserved.
3	Enables Simplified Replay Timer (Gen4). For more details, see "Transmit Replay" in the Controller Operations chapter of the Databook. Simplified Replay Timer Values are: - A value from 24,000 to 31,000

Table continues on the next page...

Field	Function
SIMPLIFIED_REPLY_TIMER	Symbol Times when Extended Synch is 0b. - A value from 80,000 to 100,000 Symbol Times when Extended Synch is 1b. Must not be changed while link is in use. Note: This register field is sticky.
2 UR_CA_MASK_4_TRGT1	This field only applies to request TLPs (with UR filtering status) that you have chosen to forward to the application (when you set DEFAULT_TARGET in this register). - When you set this field to '1', the core suppresses error logging, Error Message generation, and CPL generation (for non-posted requests). Note: This register field is sticky.
1 DEFAULT_TARGET	Default target a received IO or MEM request with UR/CA/CRS is sent to by the controller. - 0: The controller drops all incoming I/O or MEM requests (after corresponding error reporting). A completion with UR status will be generated for non-posted requests. - 1: The controller forwards all incoming I/O or MEM requests with UR/CA/CRS status to your application. Note: This register field is sticky.
0 DBI_RO_WR_EN	Write to RO Registers Using DBI. When you set this field to "1", then some RO and HwInit bits are writable from the local application through the DBI. Note: This register field is sticky.

11.3.3.1.115 UpConfigure Multi-lane Control Register. (MULTI_LANE_CONTROL_OFF)

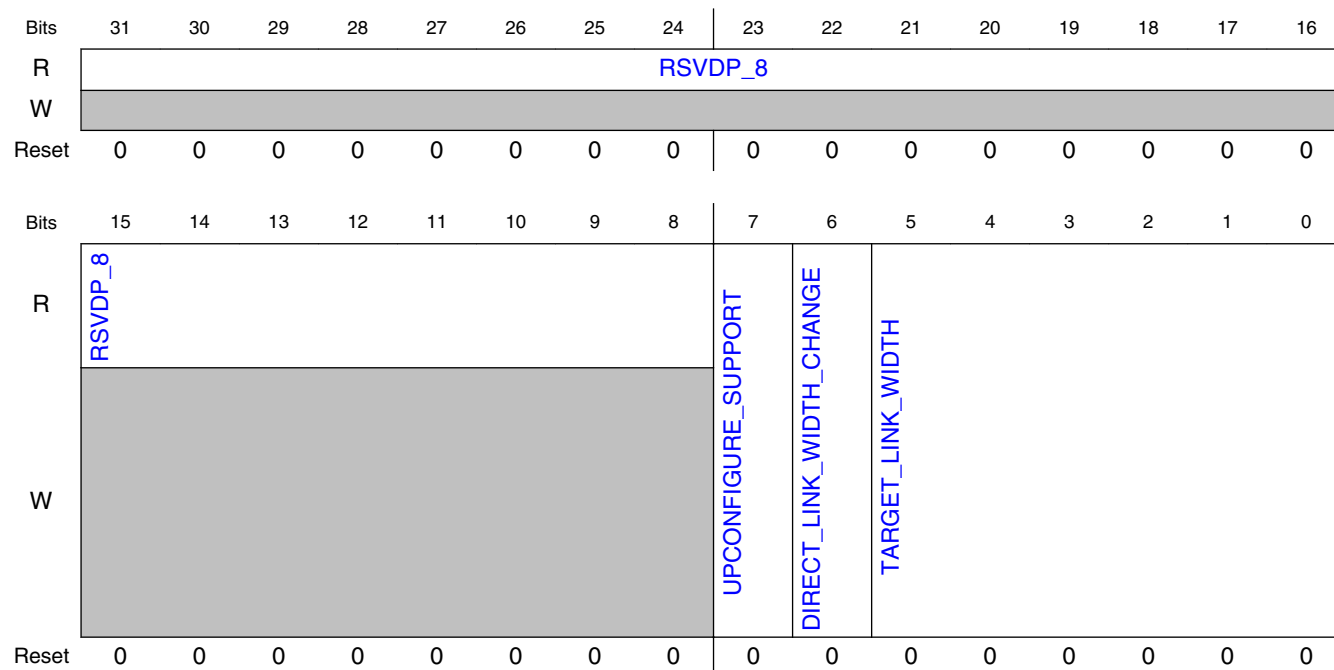
11.3.3.1.115.1 Offset

Register	Offset
MULTI_LANE_CONTROL_OFF	8C0h

11.3.3.1.115.2 Function

UpConfigure Multi-lane Control Register. Used when upsizing or downsizing the link width through Configuration state without bringing the link down. For more details, see the "Link Establishment" section in the "Controller Operations" chapter of the Databook.

11.3.3.1.115.3 Diagram



11.3.3.1.115.4 Fields

Field	Function
31-8 RSVDP_8	Reserved for future use.
7 UPCONFIGURE_SUPPORT	Upconfigure Support. The controller sends this value as the Link Upconfigure Capability in TS2 Ordered Sets in Configuration.Complete state. This field is reserved (fixed to '0') for M-PCIe. Note: This register field is sticky.
6 DIRECT_LINK_WIDTH_CHANGE	Directed Link Width Change. The controller always moves to Configuration state through Recovery state when this bit is set to '1'. - If the upconfigure_capable variable is '1' and the PCIE_CAP_HW_AUTO_WIDTH_DISABLE bit in LINK_CONTROL_LINK_STATUS_REG is '0', the controller starts upconfigure or autonomous width downsizing (to the TARGET_LINK_WIDTH value) in the Configuration state. - If TARGET_LINK_WIDTH value is 0x0, the controller does not start upconfigure or autonomous width downsizing in the Configuration state. The controller self-clears this field when the controller accepts this request. This field is reserved (fixed to '0') for M-PCIe.
5-0 TARGET_LINK_WIDTH	Target Link Width. Values correspond to: - 6'b000000: Core does not start upconfigure or autonomous width downsizing in the Configuration state. - 6'b000001: x1 - 6'b000010: x2 - 6'b000100: x4 - 6'b001000: x8 - 6'b010000: x16 - 6'b100000: x32 This field is reserved (fixed to '0') for M-PCIe.

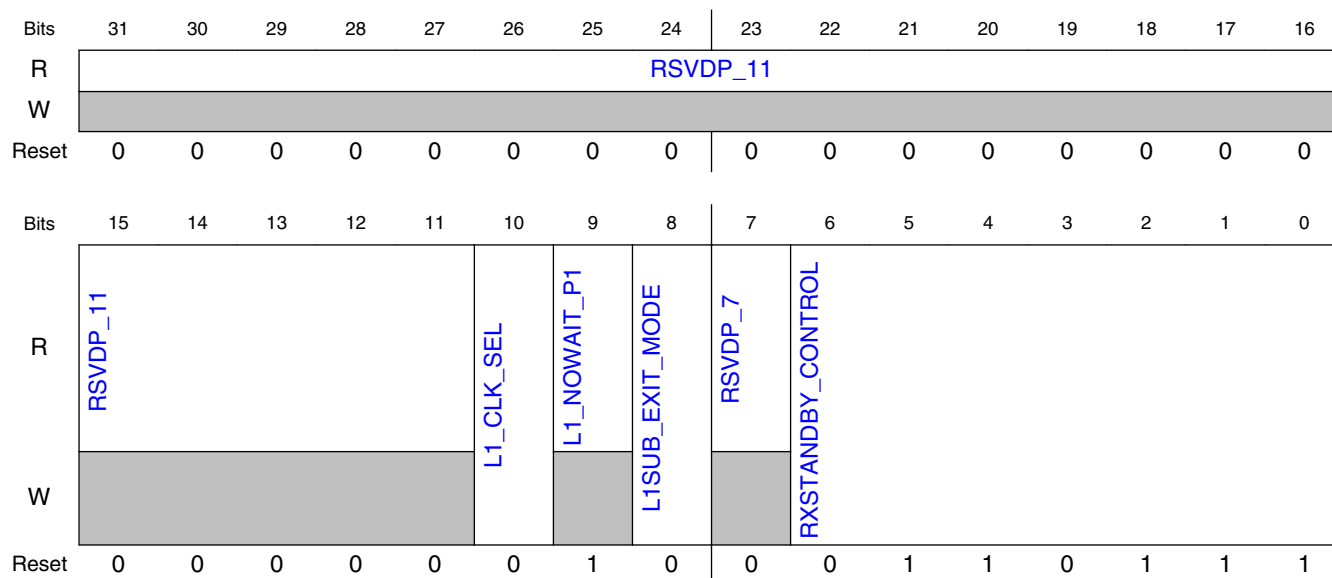
11.3.3.1.116 PHY Interoperability Control Register. (PHY_INTEROP_CTRL_OFF)

11.3.3.1.116.1 Offset

Register	Offset
PHY_INTEROP_CTRL_OFF	8C4h

11.3.3.1.116.2 Function

PHY Interoperability Control Register. This register is reserved for internal use. You should not write to this register and change the default unless specifically instructed by Synopsys support.

11.3.3.1.116.3 Diagram**11.3.3.1.116.4 Fields**

Field	Function
31-11 RSVDP_11	Reserved for future use.
10 L1_CLK_SEL	L1 Clock control bit. - 1: Controller does not request aux_clk switch and core_clk gating in L1. - 0: Controller requests aux_clk switch and core_clk gating in L1. Note: This register field is sticky.
9 L1_NOWAIT_P1	L1 entry control bit. - 1: Core does not wait for PHY to acknowledge transition to P1 before entering L1. - 0: Core waits for the PHY to acknowledge transition to P1 before entering L1. Note: The access attributes of this field are as follows: - Dbi: R/W (sticky) Note: This register field is sticky.

Table continues on the next page...

Field	Function
8 L1SUB_EXIT_MODE	L1 Exit Control Using phy_mac_pclckack_n. - 1: Core exits L1 without waiting for the PHY to assert phy_mac_pclckack_n. - 0: Core waits for the PHY to assert phy_mac_pclckack_n before exiting L1. Note: This register field is sticky.
7 RSVDP_7	Reserved for future use.
6-0 RXSTANDBY_CONTROL	Rxstandby Control. Bits 0..5 determine if the controller asserts the RxStandby signal (mac_phy_rxstandby) in the indicated condition. Bit 6 enables the controller to perform the RxStandby/RxStandbyStatus handshake. - [0]: Rx EIOS and subsequent TX-IDLE-MIN - [1]: Rate Change - [2]: Inactive lane for upconfigure/downconfigure - [3]: PowerDown=P1orP2 - [4]: RxL0s.Idle - [5]: EI Infer in L0 - [6]: Execute RxStandby/RxStandbyStatus Handshake This field is reserved (fixed to '0') for M-PCIE. Note: This register field is sticky.

11.3.3.1.117 TRGT_CPL_LUT Delete Entry Control register. (TRGT_CPL_LUT_DELETE_ENTRY_OFF)

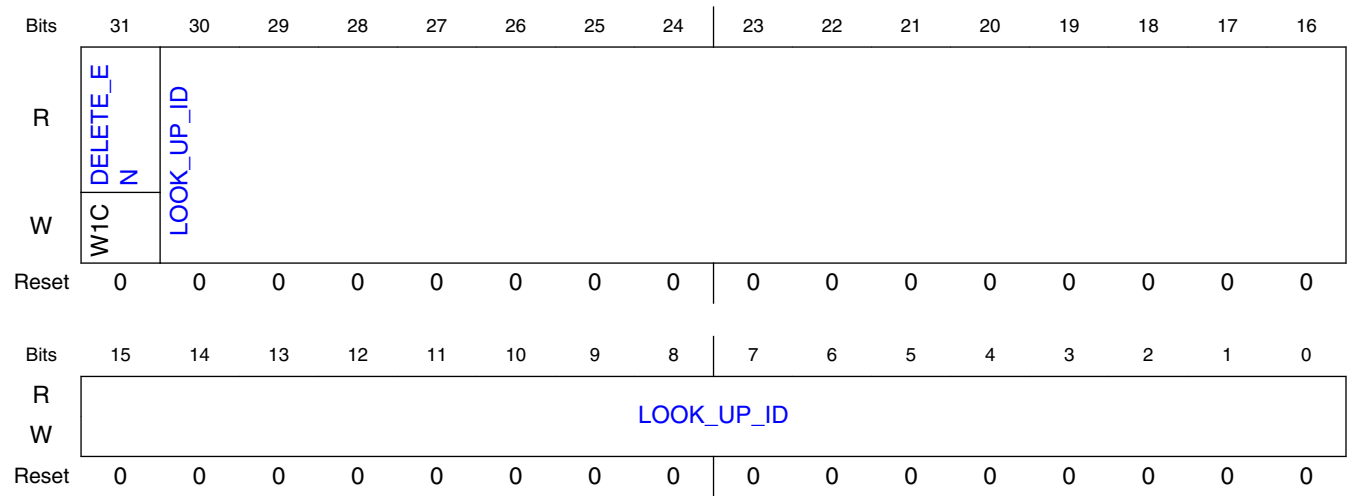
11.3.3.1.117.1 Offset

Register	Offset
TRGT_CPL_LUT_DELETE_ENTRY_OFF	8C8h

11.3.3.1.117.2 Function

TRGT_CPL_LUT Delete Entry Control register. Using this register you can delete one entry in the target completion LUT. You should only use this register when you know that your application will never send the completion because of an FLR or any other reason. Note:: The target completion LUT (and associated target completion timeout event) is watching for completions (from your application on XALI0/1/2 or AXI master read channel) corresponding to previously received non-posted requests from the PCIe wire.

11.3.3.1.117.3 Diagram



11.3.3.1.117.4 Fields

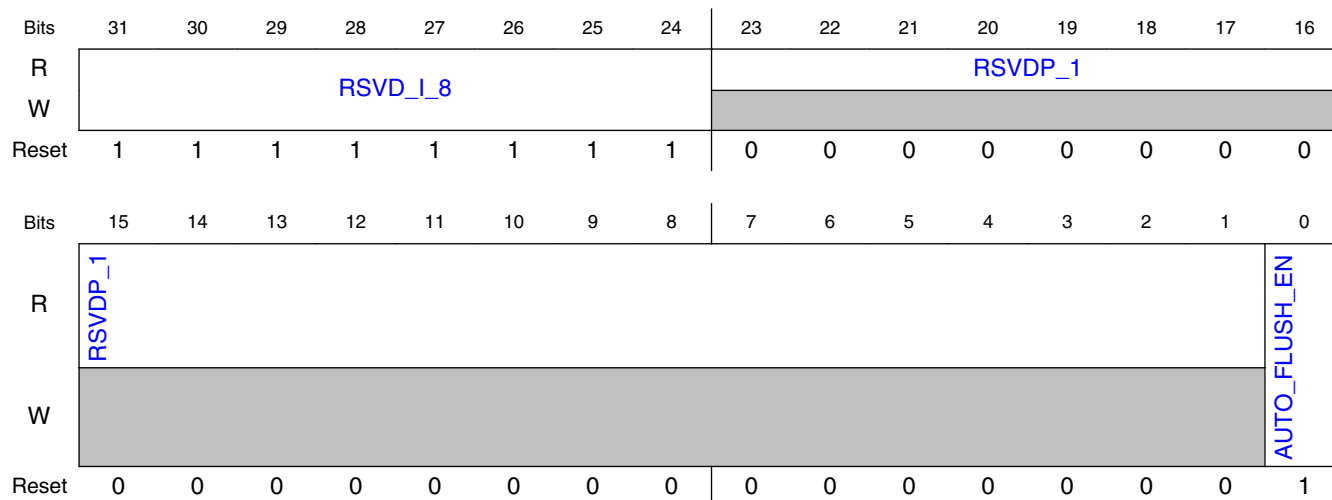
Field	Function
31 DELETE_EN	This is a one-shot bit. A '1' write to this bit triggers the deletion of the target completion LUT entry that is specified in the LOOK_UP_ID field. This is a self-clearing register field. Reading from this register field always returns a '0'.
30-0 LOOK_UP_ID	This number selects one entry to delete of the TRGT_CPL_LUT.

11.3.3.1.118 Link Reset Request Flush Control Register. (LINK_FLUSH_CONTROL_OFF)

11.3.3.1.118.1 Offset

Register	Offset
LINK_FLUSH_CONTROL_OFF	8CCh

11.3.3.1.118.2 Diagram



11.3.3.1.118.3 Fields

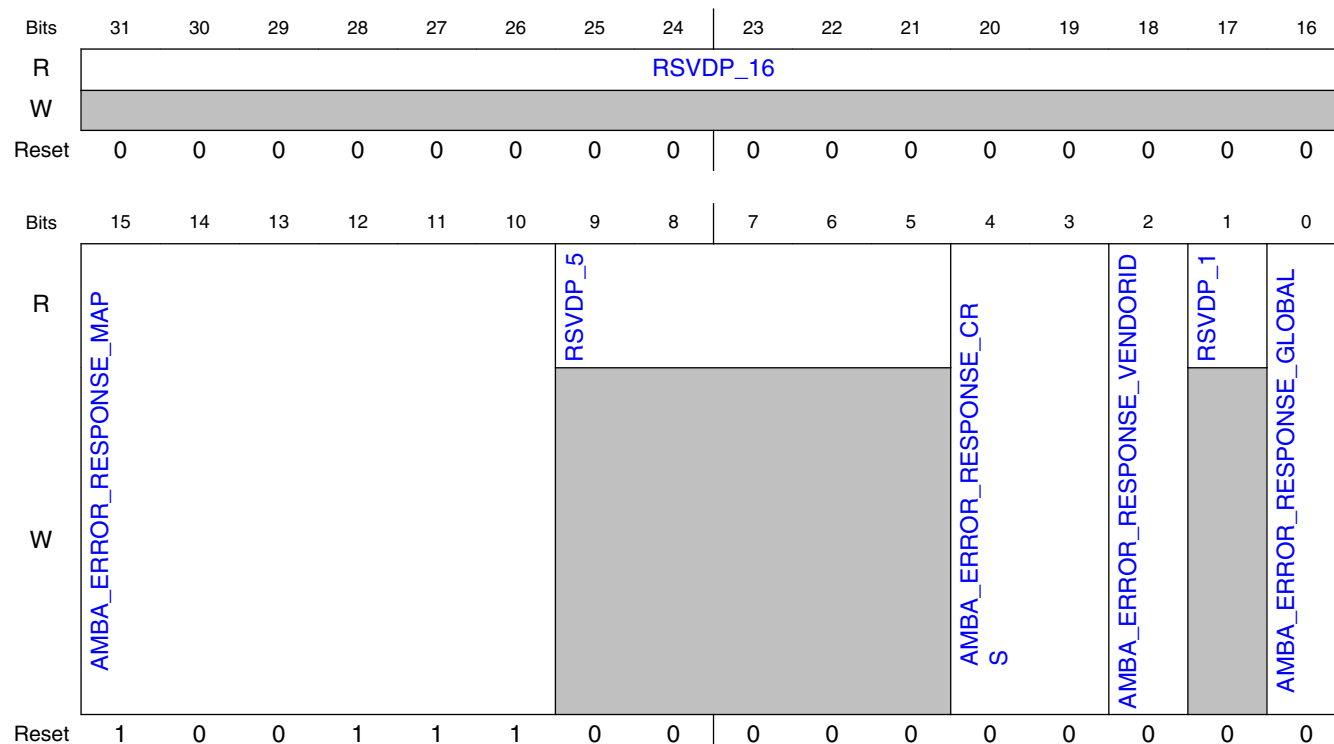
Field	Function
31-24 RSVD_I_8	This is an internally reserved field. Do not use. Note: This register field is sticky.
23-1 RSVDP_1	Reserved for future use.
0 AUTO_FLUSH_EN	Enables automatic flushing of pending requests before sending the reset request to the application logic to reset the PCIe controller and the AXI Bridge. The flushing process is initiated if any of the following events occur: - Hot reset request. A downstream port (DSP) can "hot reset" an upstream port (USP) by sending two consecutive TS1 ordered sets with the hot reset bit asserted. - Warm (Soft) reset request. Generated when exiting from D3 to D0 and <code>cfg_pm_no_soft_rst=0</code> . - Link down reset request. A high to low transition on <code>smlh_req_rst_not</code> indicates the link has gone down and the controller is requesting a reset. If you disable automatic flushing, your application is responsible for resetting the PCIe controller and the AXI Bridge. For more details see "Warm and Hot Resets" section in the Architecture chapter of the Databook. Note: This register field is sticky.

11.3.3.1.119 AXI Bridge Slave Error Response Register. (AMBA_ERROR_RESPONSE_DEFAULT_OFF)

11.3.3.1.119.1 Offset

Register	Offset
AMBA_ERROR_RESPONSE_DEFAULT_OFF	8D0h

11.3.3.1.119.2 Diagram



11.3.3.1.119.3 Fields

Field	Function
31-16 RSVDP_16	Reserved for future use.
15-10 AMBA_ERROR_RESPONSE_MAP	AXI Slave Response Error Map. Allows you to selectively map the errors received from the PCIe completion (for non-posted requests) to the AXI slave responses, slv_resp or slv_bresp. The recommended setting is SLVERR. CRS is always mapped to OKAY. - [0] -- 0: UR (unsupported request) -> DECERR -- 1: UR (unsupported request) -> SLVERR - [1] -- 0: CRS (configuration retry status) -> DECERR -- 1: CRS (configuration retry status) -> SLVERR - [2] -- 0: CA (completer abort) -> DECERR -- 1: CA (completer abort) -> SLVERR - [3]: Reserved - [4]: Reserved - [5]: -- 0: Completion Timeout -> DECERR -- 1: Completion Timeout -> SLVERR The AXI bridge internally drops (processes internally but not passed to your application) a completion that has been marked by the Rx filter as UC or MLF, and does not pass its status directly down to the slave interface. It waits for a timeout and then signals "Completion Timeout" to the slave interface. The controller sets the AXI slave read databus to 0xFFFF for all error responses. Note: This register field is sticky.
9-5 RSVDP_5	Reserved for future use.
4-3 AMBA_ERROR_RESPONSE_CR	CRS Slave Error Response Mapping. Determines the AXI slave response for CRS completions. For more details see "Error Handling" in the AXI chapter of the Databook. AHB: - always returns OKAY AXI: - 00: OKAY - 01: OKAY with all FFFF_FFFF data for all CRS completions - 10: OKAY with FFFF_0001 data for CRS completions to vendor ID read requests, OKAY with FFFF_FFFF data for all other CRS completions - 11: SLVERR/DECERR (the AXI_ERROR_RESPONSE_MAP field determines the PCIe-to-AXI Slave error response mapping) Note: This register field is sticky.

Table continues on the next page...

Field	Function
2 AMBA_ERROR_RESPONSE_VENDORID	Vendor ID Non-existent Slave Error Response Mapping. Determines the AXI slave response for errors on reads to non-existent Vendor ID register. For more details see "Error Handling" in the AXI chapter of the Databook. AHB: - 0: OKAY (with FFFF data). The controller ignores the setting in the bit when bit 0 of this register is '0'. - 1: ERROR AXI: - 0: OKAY (with FFFF data). - 1: SLVERR/DECERR (the AXI_ERROR_RESPONSE_MAP field determines the PCIe-to-AXI Slave error response mapping) Note: This register field is sticky.
1 RSVDP_1	Reserved for future use.
0 AMBA_ERROR_RESPONSE_GLOBAL	Global Slave Error Response Mapping. Determines the AXI slave response for all error scenarios on non-posted requests. For more details see "Error Handling" in the AXI chapter of the Databook. AHB: - 0: OKAY (with FFFF data for non-posted requests) and ignore the setting in bit [2] of this register. - 1: ERROR for normal link (data) accesses and look at bit [2] for other scenarios. AXI: - 0: OKAY (with FFFF data for non-posted requests) - 1: SLVERR/DECERR (the AXI_ERROR_RESPONSE_MAP field determines the PCIe-to-AXI Slave error response mapping) The error response mapping is not applicable to Non-existent Vendor ID register reads. Note: This register field is sticky.

11.3.3.1.120 Link Down AXI Bridge Slave Timeout Register. (AMBA_LINK_TIMEOUT_OFF)

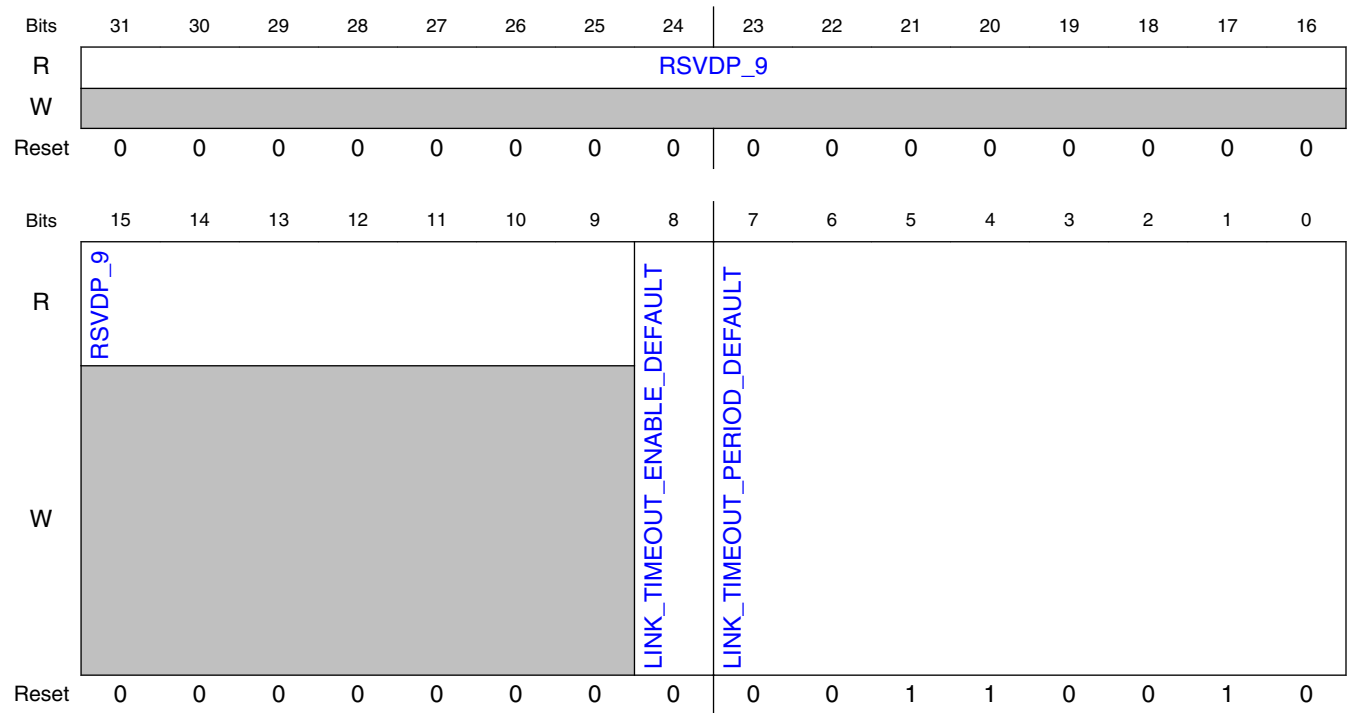
11.3.3.1.120.1 Offset

Register	Offset
AMBA_LINK_TIMEOUT_OFF	8D4h

11.3.3.1.120.2 Function

Link Down AXI Bridge Slave Timeout Register. If your application AXI master issues outbound requests to the AXI bridge slave interface before the PCIe link is operational, the controller starts a "flush" timer. The timeout value of the timer is set by this register. The timer will timeout and then flush the bridge TX request queues after this amount of time. The timer counts when there are pending outbound AXI slave interface (or DMA) requests and the PCIe TX link is not transmitting any of these requests. For more details, see the "AXI Bridge Initialization, Clocking and Reset" section in the AXI chapter of the Databook.

11.3.3.1.120.3 Diagram



11.3.3.1.120.4 Fields

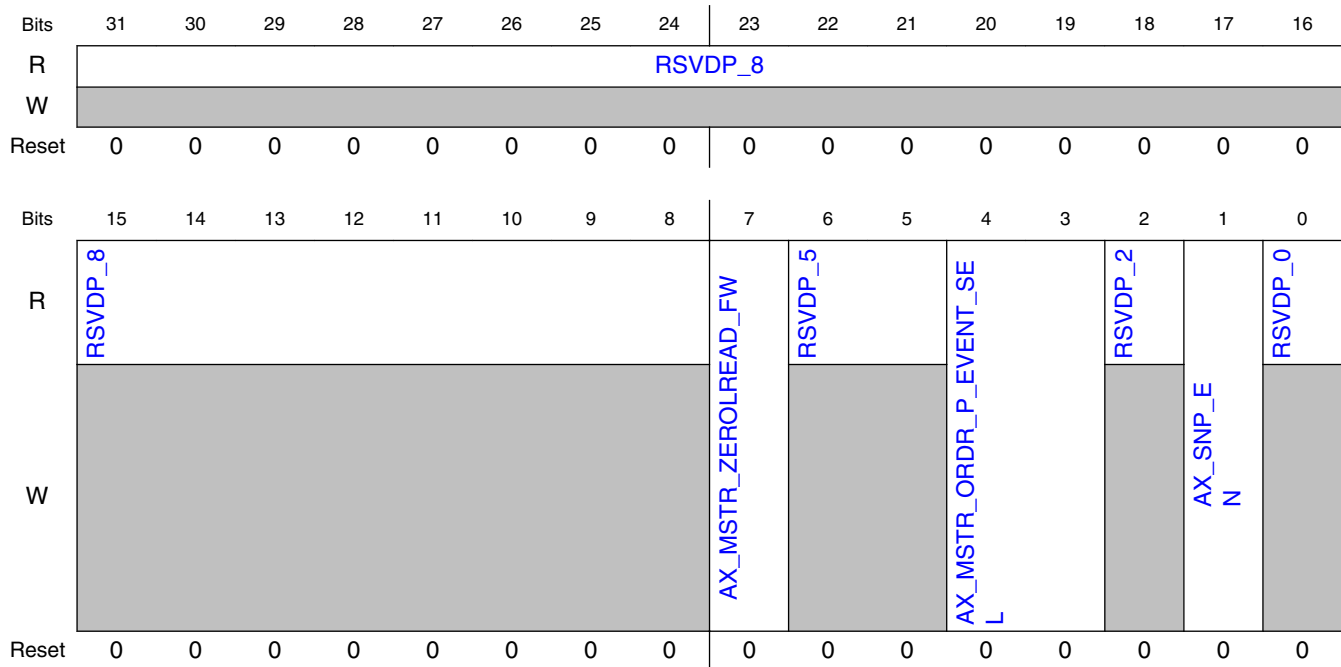
Field	Function
31-9 RSVDP_9	Reserved for future use.
8 LINK_TIMEOUT_ENABLE_DEFAULT	Disable Flush. You can disable the flush feature by setting this field to "1". Note: This register field is sticky.
7-0 LINK_TIMEOUT_PERIOD_DEFAULT	Timeout Value (ms). The timer will timeout and then flush the bridge TX request queues after this amount of time. The timer counts when there are pending outbound AXI slave interface requests and the PCIe TX link is not transmitting any of these requests. The timer is clocked by core_clk. For an M-PCIe configuration: - Time unit of this field is 4 ms. - Margin of error for RateA clock is < 1%. - Margin of error for RateB clock is between 16% and 17%. Note: This register field is sticky.

11.3.3.1.121 AMBA Ordering Control. (AMBA_ORDERING_CTRL_OFF)

11.3.3.1.121.1 Offset

Register	Offset
AMBA_ORDERING_CTL_OFF	8D8h

11.3.3.1.121.2 Diagram



11.3.3.1.121.3 Fields

Field	Function
31-8 RSVDP_8	Reserved for future use.
7 AX_MSTR_ZEROLREAD_FW	AXI Master Zero Length Read Forward to the application. The DW PCIe controller AXI bridge is able to terminate in order with the Posted transactions the zero length read, implementing the PCIe express flush semantics of the Posted transactions. - 0x0: The zero length Read is terminated at the DW PCIe AXI bridge master - 0x1: The zero length Read is forward to the application.
6-5 RSVDP_5	Reserved for future use.
4-3 AX_MSTR_ORDR_P_EVENT_SELECTOR	AXI Master Posted Ordering Event Selector. This field selects how the master interface determines when a P write is completed when enforcing the PCIe ordering rule, "NP must not pass P" at the AXI Master Interface. The AXI protocol does not support ordering between channels. Therefore, NP reads can pass P on your AXI bus fabric. This can result in an ordering violation when the read overtakes a P that is going to the same address. Therefore, the bridge master does not issue any NP requests until all outstanding P writes reach their destination. It does this by waiting for the all of the write responses on

Table continues on the next page...

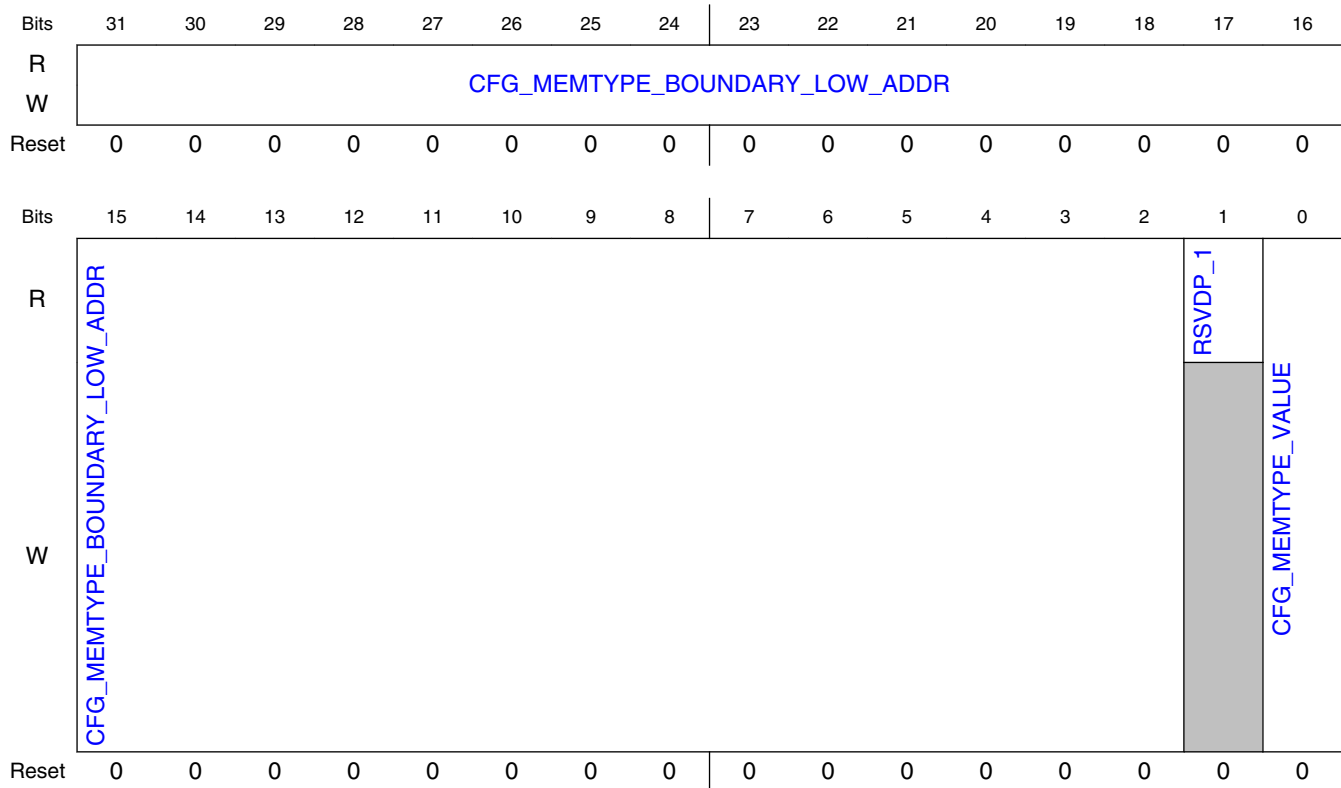
Field	Function
	the B channel. This can affect the performance of the master read channel. For scenarios where the interconnect serializes the AXI master "AW", "W" and "AR" channels, you can increase the performance by reducing the need to wait until the complete Posted transaction has effectively reached the application slave. - 00: B'last event: wait for the all of the write responses on the B channel thereby ensuring that the complete Posted transaction has effectively reached the application slave (default). - 01: AW'last event: wait until the complete Posted transaction has left the AXI address channel at the bridge master. - 10: W'last event: wait until the complete Posted transaction has left the AXI data channel at the bridge master. - 11: Reserved Note 2: This setting will not affect: - MSI interrupt catcher and P data ordering. This is always driven by the B'last event. - DMA read engine TLP ordering. This is always driven by the B'last event. - NP write transactions which are always serialized with P write transactions.
2 RSVDP_2	Reserved for future use.
1 AX_SNP_EN	AXI Serialize Non-Posted Requests Enable. This field enables the AXI Bridge to serialize same ID Non-Posted Read/Write Requests on the wire. Serialization implies one outstanding same ID NP Read or Write on the wire and used to avoid AXI RAR and WAW hazards at the remote link partner. For more details, see the "Optional Serialization of AXI Slave Non-posted Requests" section in the AXI chapter of the Databook.
0 RSVDP_0	Reserved for future use.

11.3.3.1.122 ACE Cache Coherency Control Register 1 (COHERENCY_CONTROL_1_OFF)

11.3.3.1.122.1 Offset

Register	Offset
COHERENCY_CONTROL_1_OFF	8E0h

11.3.3.1.122.2 Diagram



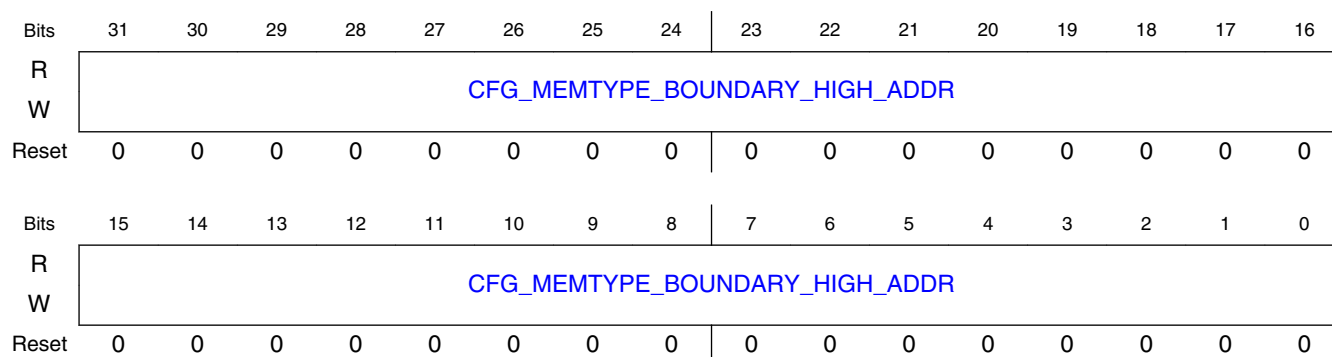
11.3.3.1.122.3 Fields

Field	Function
31-2 CFG_MEMTYPE_BOUNDARY_LOW_ADDR	Boundary Lower Address For Memory Type. Bits [31:0] of dword-aligned address of the boundary for Memory type. The two lower address LSBs are "00". Addresses up to but not including this value are in the lower address space region; addresses equal or greater than this value are in the upper address space region. Note: This register field is sticky.
1 RSVDP_1	Reserved for future use.
0 CFG_MEMTYPE_VALUE	Sets the memory type for the lower and upper parts of the address space: - 0: lower = Peripheral; upper = Memory - 1: lower = Memory type; upper = Peripheral Note: This register field is sticky.

11.3.3.1.123 ACE Cache Coherency Control Register 2 (COHERENCY_CONTROL_2_OFF)

11.3.3.1.123.1 Offset

Register	Offset
COHERENCY_CONTROL_2_OFF	8E4h

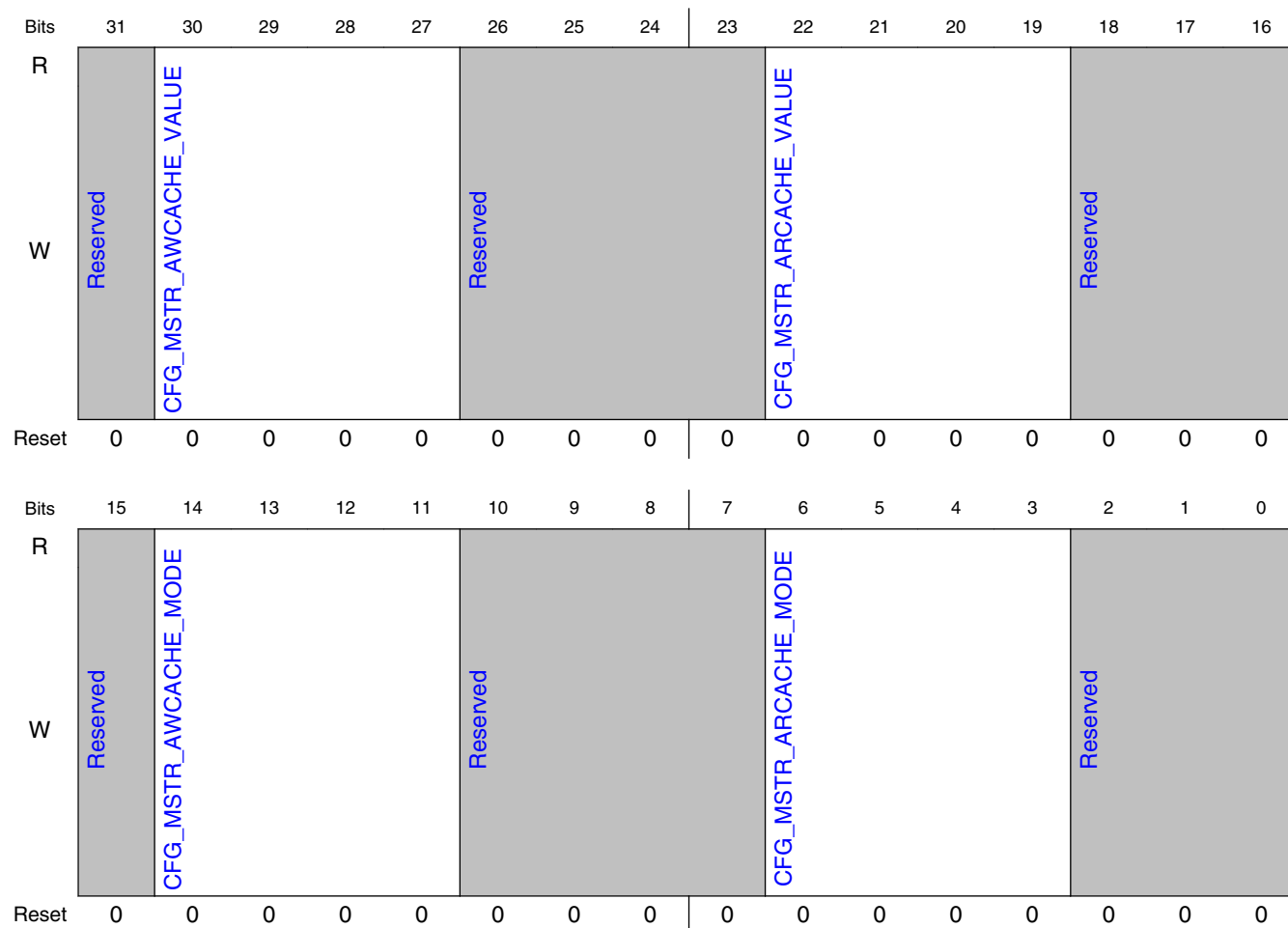
11.3.3.1.123.2 Diagram**11.3.3.1.123.3 Fields**

Field	Function
CFG_MEMTYPE_BOUNDARY_HIGH_ADDR	Boundary Upper Address For Memory Type. Bits [63:32] of the 64-bit dword-aligned address of the boundary for Memory type. Note: This register field is sticky.

11.3.3.1.124 ACE Cache Coherency Control Register 3 (COHERENCY_CONTROL_3_OFF)**11.3.3.1.124.1 Offset**

Register	Offset
COHERENCY_CONTROL_3_OFF	8E8h

11.3.3.1.124.2 Diagram



11.3.3.1.124.3 Fields

Field	Function
31 —	Reserved.
30-27 CFG_MSTR_AWCACHE_VALUE	Master Write CACHE Signal Value. Value of the individual bits in mstr_awcache when CFG_MSTR_AWCACHE_MODE is '1'. Note: not applicable to message requests; for message requests the value of mstr_awcache is always "0000" Note: This register field is sticky.
26-23 —	Reserved.
22-19 CFG_MSTR_ARCACHE_VALUE	Master Read CACHE Signal Value. Value of the individual bits in mstr_arcache when CFG_MSTR_ARCACHE_MODE is '1'. Note: This register field is sticky.
18-15	Reserved.

Table continues on the next page...

Field	Function
—	
14-11 CFG_MSTR_AWCACHE_MODE	Master Write CACHE Signal Behavior. Defines how the individual bits in mstr_awcache are controlled: - 0: set automatically by the AXI master - 1: set by the value of the corresponding bit of the CFG_MSTR_AWCACHE_VALUE field Note: for message requests the value of mstr_awcache is always "0000" regardless of the value of this bit Note: This register field is sticky.
10-7 —	Reserved.
6-3 CFG_MSTR_ARCACHE_MODE	Master Read CACHE Signal Behavior. Defines how the individual bits in mstr_arcache are controlled: - 0: set automatically by the AXI master - 1: set by the value of the corresponding bit of the CFG_MSTR_ARCACHE_VALUE field Note: This register field is sticky.
2-0 —	Reserved.

11.3.3.1.125 Lower 20 bits of the programmable AXI address where Messages coming from wire are mapped to. (AXI_MSTR_MSG_ADDR_LOW_OFF)

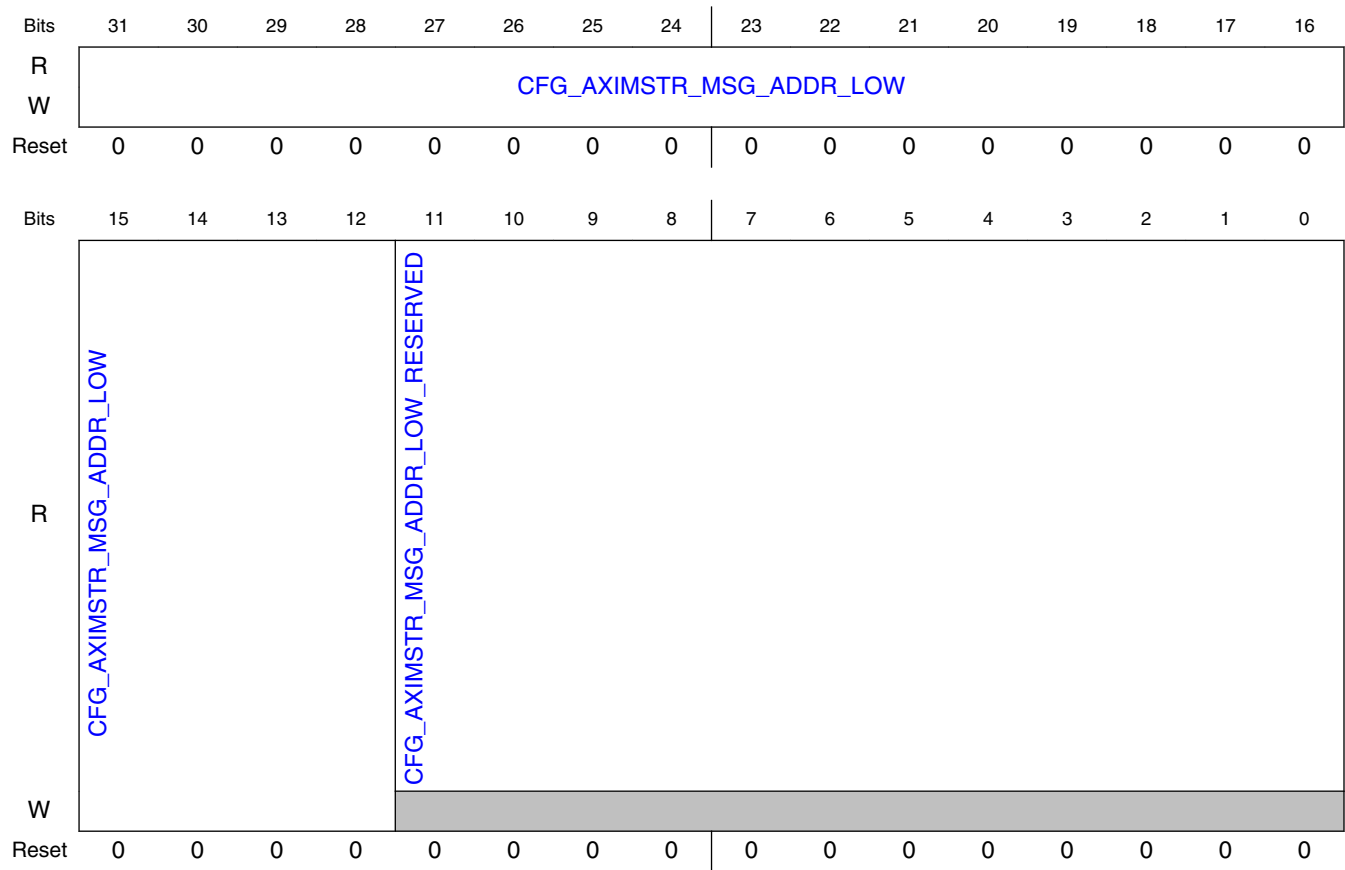
11.3.3.1.125.1 Offset

Register	Offset
AXI_MSTR_MSG_ADDR_LOW_OFF	8F0h

11.3.3.1.125.2 Function

Lower 20 bits of the programmable AXI address where Messages coming from wire are mapped to. Bits [11:0] of the register are tied to zero for the address to be 4k-aligned. In previous releases, the third and fourth DWORDs of a message (Msg/MsgD) TLP header were delivered though the AXI master address bus (mstr_awaddr). These DWORDS are now supplied through the mstr_awmisc_info_hdr_34dw[63:0] output; and the value on mstr_awaddr is driven to the value you have programmed into the AXI_MSTR_MSG_ADDR_LOW_OFF and AXI_MSTR_MSG_ADDR_HIGH_OFF registers.

11.3.3.1.125.3 Diagram



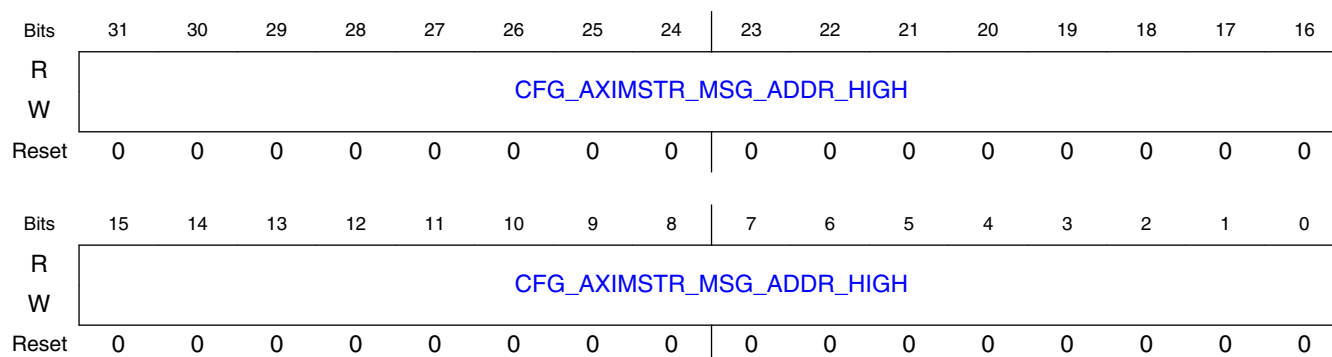
11.3.3.1.125.4 Fields

Field	Function
31-12 CFG_AXIMSTR_MSG_ADDR_LOW	Lower 20 bits of the programmable AXI address for Messages. Note: This register field is sticky.
11-0 CFG_AXIMSTR_MSG_ADDR_LOW_RESERVED	Reserved for future use. Note: This register field is sticky.

11.3.3.1.126 Upper 32 bits of the programmable AXI address where Messages coming from wire are mapped to. (AXI_MSTR_MSG_ADDR_HIGH_OFF)

11.3.3.1.126.1 Offset

Register	Offset
AXI_MSTR_MSG_ADDR_HIGH_OFF	8F4h

11.3.3.1.126.2 Diagram**11.3.3.1.126.3 Fields**

Field	Function
31-0 CFG_AXIMSTR_MSG_ADDR_HIGH	Upper 32 bits of the programmable AXI address for Messages. Note: This register field is sticky.

11.3.3.1.127 PCIe Controller IIP Release Version Number. (PCIE_VERSION_NUMBER_OFF)**11.3.3.1.127.1 Offset**

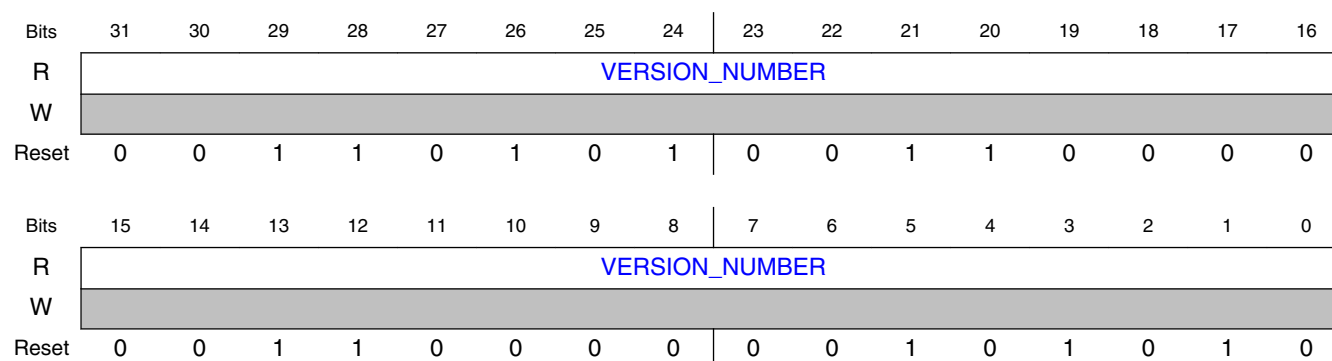
Register	Offset
PCIE_VERSION_NUMBER_OFF	8F8h

11.3.3.1.127.2 Function

PCIe Controller IIP Release Version Number. The version number is given in hex format. You should convert each pair of hex characters to ASCII to interpret. Using 4.70a (GA)

as an example: - VERSION_NUMBER = 0x3437302a which translates to 470* -
 VERSION_TYPE = 0x67612a2a which translates to ga** Using 4.70a-ea01 as an
 example: - VERSION_NUMBER = 0x3437302a which translates to 470* -
 VERSION_TYPE = 0x65613031 which translates to ea01 GA is a general release

11.3.3.1.127.3 Diagram



11.3.3.1.127.4 Fields

Field	Function
31-0 VERSION_NUMBER	Version Number.

11.3.3.1.128 PCIe Controller IIP Release Version Type. (PCIE_VERSION_TYPE_OFF)

11.3.3.1.128.1 Offset

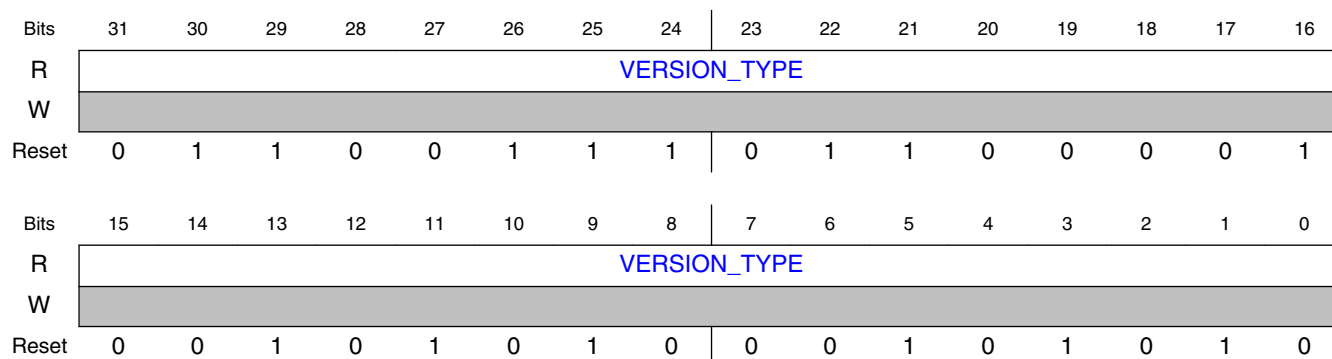
Register	Offset
PCIE_VERSION_TYPE_OFF	8FCh

11.3.3.1.128.2 Function

PCIe Controller IIP Release Version Type. The type is given in hex format. You should convert each pair of hex characters to ASCII to interpret. Using 4.70a (GA) as an example: - VERSION_NUMBER = 0x3437302a which translates to 470* -
 VERSION_TYPE = 0x67612a2a which translates to ga** Using 4.70a-ea01 as an

example: - VERSION_NUMBER = 0x3437302a which translates to 470* -
VERSION_TYPE = 0x65613031 which translates to ea01 GA is a general release.

11.3.3.1.128.3 Diagram



11.3.3.1.128.4 Fields

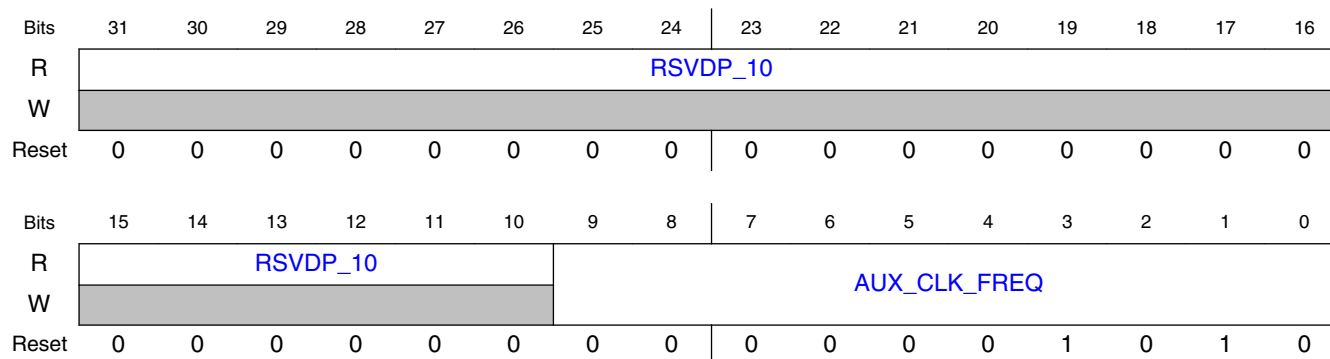
Field	Function
31-0 VERSION_TYPE	Version Type.

11.3.3.1.129 Auxiliary Clock Frequency Control Register. (AUX_CLK_FREQ_OFF)

11.3.3.1.129.1 Offset

Register	Offset
AUX_CLK_FREQ_OFF	B40h

11.3.3.1.129.2 Diagram



11.3.3.1.129.3 Fields

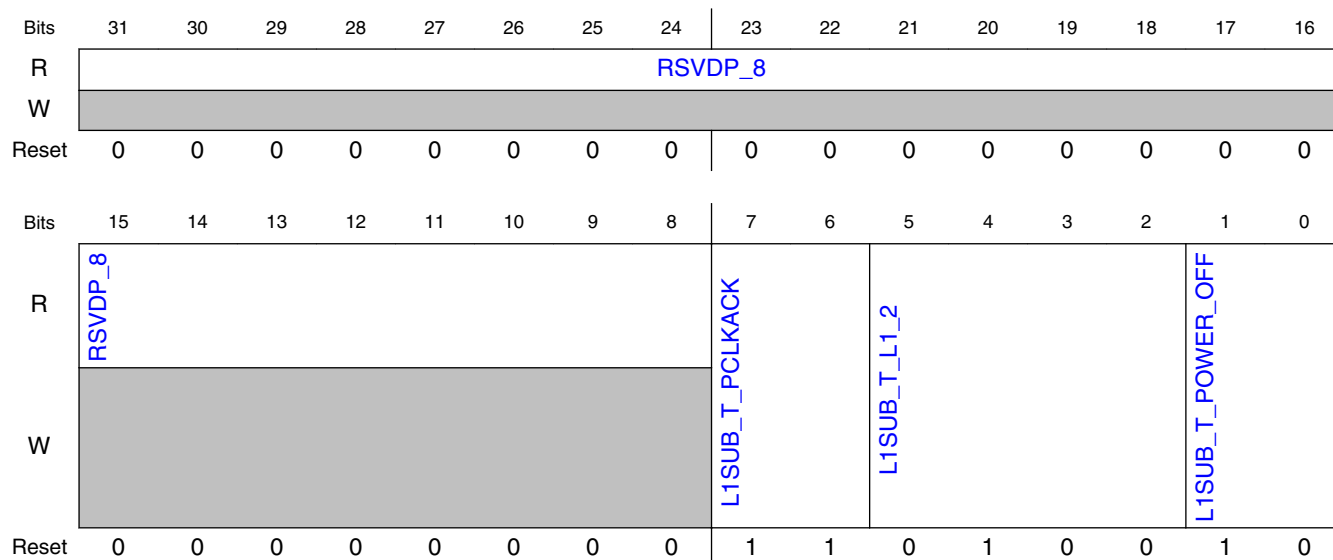
Field	Function
31-10 RSVDP_10	Reserved for future use.
9-0 AUX_CLK_FREQ	The aux_clk frequency in MHz. This value is used to provide a 1 us reference for counting time during low-power states with aux_clk when the PHY has removed the pipe_clk. Frequencies lower than 1 MHz are possible but with a loss of accuracy in the time counted. If the actual frequency (f) of aux_clk does not exactly match the programmed frequency (f_prog), then there is an error in the time counted by the controller that can be expressed in percentage as: $err\% = (f_prog/f - 1) * 100$. For example if f=2.5 MHz and f_prog=3 MHz, then $err\% = (3/2.5 - 1) * 100 = 20\%$, meaning that the time counted by the controller on aux_clk will be 20% greater than the time in us programmed in the corresponding time register (for example T_POWER_ON). Note: This register field is sticky.

11.3.3.1.130 L1 Substates Timing Register. (L1_SUBSTATES_OFF)

11.3.3.1.130.1 Offset

Register	Offset
L1_SUBSTATES_OFF	B44h

11.3.3.1.130.2 Diagram



11.3.3.1.130.3 Fields

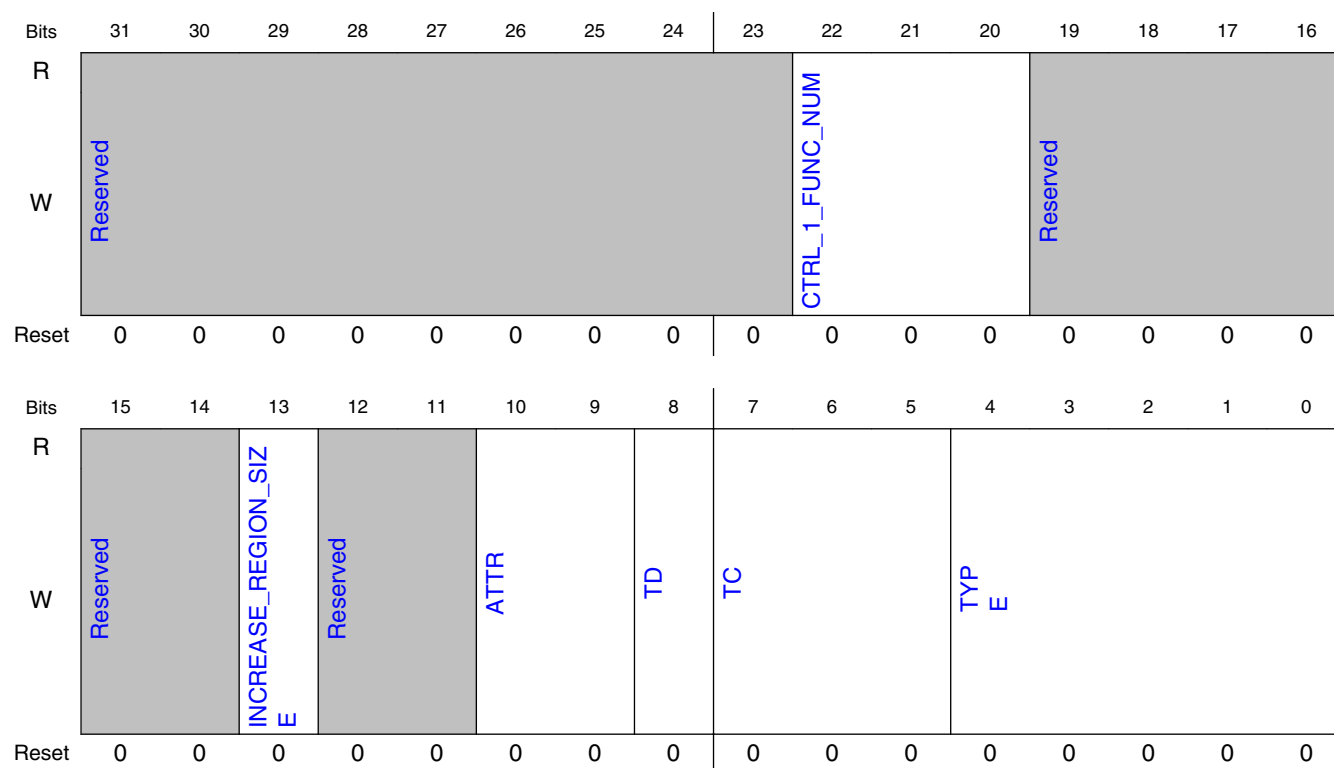
Field	Function
31-8 RSVDP_8	Reserved for future use.
7-6 L1SUB_T_PCLKACK	Max delay (in 1us units) between a MAC request to remove the clock on mac_phy_pclkreq_n and a PHY response on phy_mac_pclckack_n. If the PHY does not respond within this time the request is aborted. Range is 0..3 Note: This register field is sticky.
5-2 L1SUB_T_L1_2	Duration (in 1us units) of L1.2. Range is 0.15. Note: The timeout value can vary by 50%. Note: This register field is sticky.
1-0 L1SUB_T_POWER_OFF	Duration (in 1us units) of L1.2.Entry. Range is 0.3. Note: The timeout value can vary by 50%. Note: This register field is sticky.

11.3.3.1.131 iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_OUTBOUND_0)

11.3.3.1.131.1 Offset

Register	Offset
IATU_REGION_CTRL_1_OFF_OUTBOUND_0	8000_0000h

11.3.3.1.131.2 Diagram



11.3.3.1.131.3 Fields

Field	Function
31-23 —	Reserved.
22-20 CTRL_1_FUNC_NUM	Function Number. - When the address of an outbound TLP is matched to this region and the FUNC_BYPASS field in the "iATU Control 2 Register" is '0', then the function number used in generating the function part of the requester ID (RID) field of the TLP is taken from this 5-bit register. When you are using the AXI Bridge, then this field is swapped before AXI decomposition occurs so that the correct "Max_Read_Request_Size" and "Max_Payload_Size" values are used. Note: This register field is sticky.
19-14 —	Reserved.
13 INCREASE_REGION_SIZE	Increase the maximum ATU Region size. When set, the size is determined by the Maximum Size of iATU Region. When clear, the maximum ATU Region size is 4 GB (default). Note: This register field is sticky.
12-11 —	Reserved.
10-9 ATTR	When the address of an outbound TLP is matched to this region, then the ATTR field of the TLP is changed to the value in this register. Note: This register field is sticky.

Table continues on the next page...

PCI Express (PCIe)

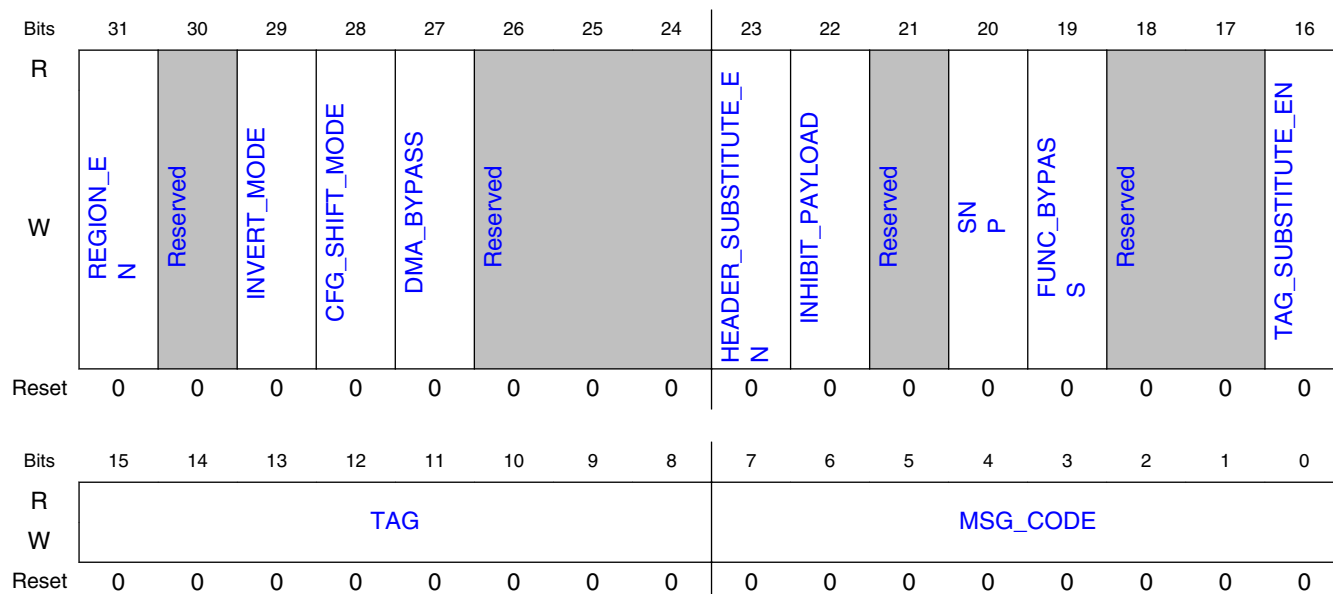
Field	Function
8 TD	When the address of an outbound TLP is matched to this region, then the TD field of the TLP is changed to the value in this register. Note: This register field is sticky.
7-5 TC	When the address of an outbound TLP is matched to this region, then the TC field of the TLP is changed to the value in this register. Note: This register field is sticky.
4-0 TYPE	When the address of an outbound TLP is matched to this region, then the TYPE field of the TLP is changed to the value in this register. Note: This register field is sticky.

11.3.3.1.132 iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_OUTBOUND_0)

11.3.3.1.132.1 Offset

Register	Offset
IATU_REGION_CTRL_2_OFF_OUTBOUND_0	8000_0004h

11.3.3.1.132.2 Diagram



11.3.3.1.132.3 Fields

Field	Function
31 REGION_EN	Region Enable. This bit must be set to '1' for address translation to take place. Note: This register field is sticky.
30 —	Reserved.
29 INVERT_MODE	Invert Mode. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). Note: This register field is sticky.
28 CFG_SHIFT_M ODE	CFG Shift Mode. The iATU uses bits [27:12] of the untranslated address (on the XALI0/1/2 interface or AXI slave interface address) to form the BDF number of the outgoing CFG TLP. This supports the Enhanced Configuration Address Mapping (ECAM) mechanism (Section 7.2.2 of the PCI Express Base 3.1 Specification, revision 1.0) by allowing all outgoing I/O and MEM TLPs (that have been translated to CFG) to be mapped from memory space into any 256 MB region of the PCIe configuration space. Note: This register field is sticky.
27 DMA_BYPASS	DMA Bypass Mode. Allows request TLPs which are initiated by the DMA controller to pass through the iATU untranslated. Note: This field is reserved for the SW product. You must set it to '0'. Note: This register field is sticky.
26-24 —	Reserved.
23 HEADER_SUBS TITUTE_EN	Header Substitute Enable. When enabled and region address is matched, the iATU fully substitutes bytes 8-11 (for 3 DWORD header) or bytes 12-15 (for 4 DWORD header) of the outbound TLP header with the contents of the LWR_TARGET_RW field in IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i. - 1: LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i register is used to fill bytes 8-to-11 (for 3 DWORD header) or bytes 12-to-15 (for 4 DWORD header) of the translated TLP header. - 0: LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i register forms the new address of the translated region. Note: This register field is sticky.
22 INHIBIT_PAYLO AD	Inhibit TLP Payload Data for TLP's in Matched Region; assign iATU region to be TLP without data. When enabled and region address is matched, the iATU marks all TLPs as having no payload data by forcing the TLP header Fmt[1] bit =0, regardless of the application inputs such as slv_wstrb. - 1: Fmt[1] =0 so that only TLP type without data is sent. For example, a Msg instead of MsgD will be sent. - 0: Fmt[1] =0/1 so that TLPs with or without data can be sent. Note: This register field is sticky.
21 —	Reserved.
20 SNP	Serialize Non-Posted Requests. In this mode, when the AXI Bridge is populated, same AXI ID Non-Posted Read/Write Requests are transmitted on the wire if there are no other same ID Non-Posted Requests outstanding. Note: This register field is sticky.
19 FUNC_BYPASS	Function Number Translation Bypass. In this mode, the function number of the translated TLP is taken from your application transmit interface and not from the CTRL_1_FUNC_NUM field of the "iATU Control 1 Register" or the VF_NUMBER field of the "iATU Control 3 Register." Note: This register field is sticky.
18-17 —	Reserved.
16 TAG_SUBSTIT UTE_EN	TAG Substitute Enable. When enabled and region address is matched, the iATU substitutes the TAG field of the outbound TLP header with the contents of the TAG field in this register. The expected usage scenario is translation from AXI MWr to Vendor Defined Msg/MsgD. Note: This register field is sticky.
15-8 TAG	TAG. The substituted TAG field (byte 6) in the outgoing TLP header when TAG_SUBSTITUTE_EN is set. Note: This register field is sticky.

Table continues on the next page...

Field	Function
7-0 MSG_CODE	MSG TLPs (Message Code). When the address of an outbound TLP is matched to this region, and the translated TLP TYPE field is Msg or MsgD; then the message field of the TLP is changed to the value in this register. Memory TLPs: (ST;Steering Tag). When the ST field of an outbound TLP is matched to this region, and the translated TLP TYPE field targets memory space; then the ST field of the TLP is changed to the value in this register. Note: This register field is sticky.

11.3.3.1.133 iATU Lower Base Address Register. (IATU_LWR_BASE_ADDRESS_OFFSET_OUTBOUND_0)

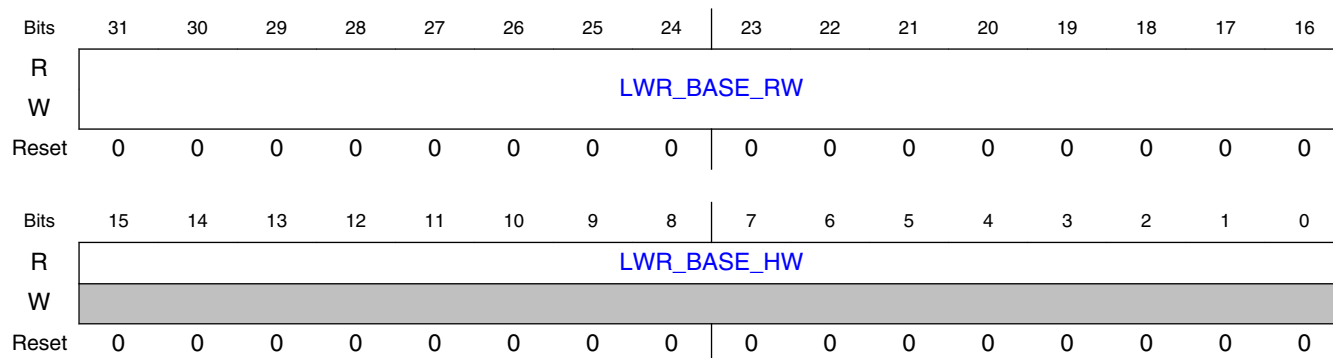
11.3.3.1.133.1 Offset

Register	Offset
IATU_LWR_BASE_ADDRESS_OFFSET_OUTBOUND_0	8000_0008h

11.3.3.1.133.2 Function

iATU Lower Base Address Register. The Minimum Size of iATU Region (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{Minimum Size of iATU Region})$ bits are zero.

11.3.3.1.133.3 Diagram



11.3.3.1.133.4 Fields

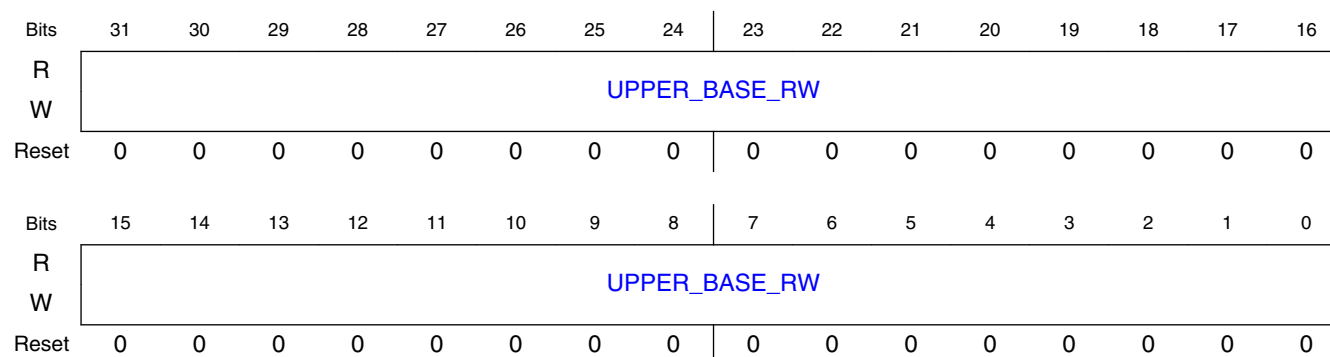
Field	Function
31-16 LWR_BASE_RW	Forms bits [31:n] of the start address of the address region to be translated. n is log2(Minimum Size of iATU Region) Note: This register field is sticky.
15-0 LWR_BASE_HW	Forms bits [n-1:0] of the start address of the address region to be translated. The start address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is log2(Minimum Size of iATU Region)

11.3.3.1.134 iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_0)

11.3.3.1.134.1 Offset

Register	Offset
IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_0	8000_000Ch

11.3.3.1.134.2 Diagram



11.3.3.1.134.3 Fields

Field	Function
31-0 UPPER_BASE_RW	Forms bits [63:32] of the start (and end) address of the address region to be translated. In systems with a 32-bit address space, this register is not used and therefore writing to this register has no effect. Note: This register field is sticky.

11.3.3.1.135 iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_OUTBOUND_0)

11.3.3.1.135.1 Offset

Register	Offset
IATU_LIMIT_ADDR_OFF_OUTBOUND_0	8000_0010h

11.3.3.1.135.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	LIMIT_ADDR_RW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	LIMIT_ADDR_HW															
W																
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

11.3.3.1.135.3 Fields

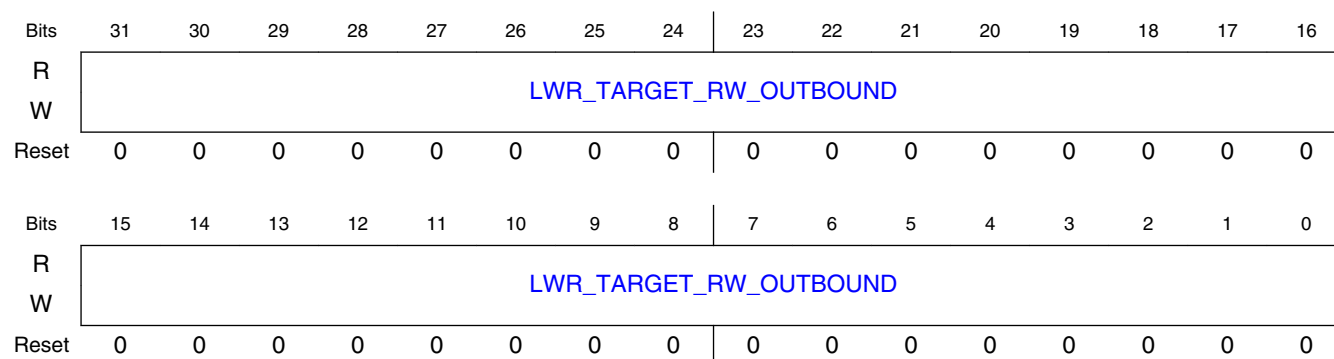
Field	Function
31-16 LIMIT_ADDR_RW	Forms upper bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. Note: This register field is sticky.
15-0 LIMIT_ADDR_HW	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller.

11.3.3.1.136 iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_0)

11.3.3.1.136.1 Offset

Register	Offset
IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_0	8000_0014h

11.3.3.1.136.2 Diagram



11.3.3.1.136.3 Fields

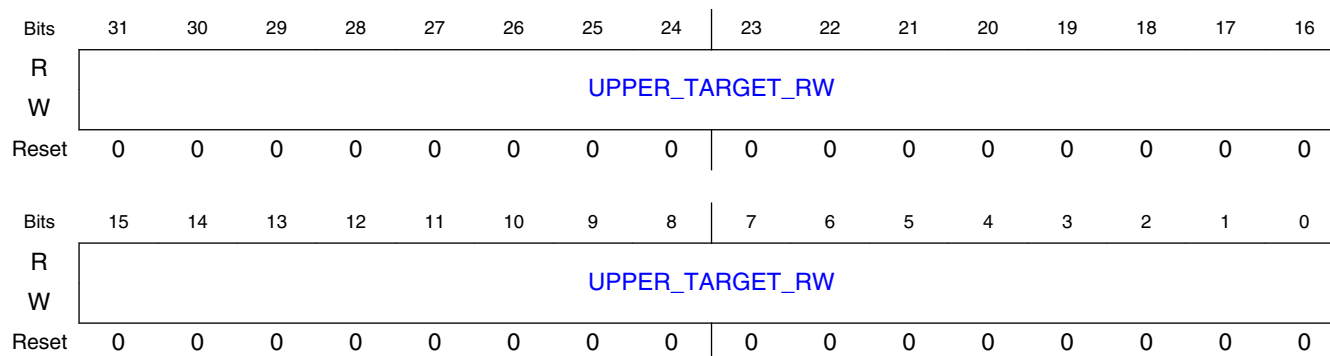
Field	Function
LWR_TARGET_RW_OUTBOUND	31-0 When HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_OFF_OUTBOUND_ is '0' (normal operation): - LWR_TARGET_RW[31:n] forms MSB's of the Lower Target part of the new address of the translated region; - LWR_TARGET_RW[n-1:0] are not used. (The start address must be aligned to the Minimum Size of iATU Region kB boundary, so the lower bits of the start address of the new address of the translated region (bits n-1:0) are always '0'). - n is log2(Minimum Size of iATU Region). When HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_OFF_OUTBOUND_i is '1': - LWR_TARGET_RW[31:0] forms bytes 8-11 (for 3 dword header) or bytes 12-15 (for 4 dword header) of the outbound TLP header. Usage scenarios include the transmission of Vendor Defined Messages where the controller determines the content of bytes 12 to 15 of the TLP header. Note: This register field is sticky.

11.3.3.1.137 iATU Upper Target Address Register. (IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_0)

11.3.3.1.137.1 Offset

Register	Offset
IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_0	8000_0018h

11.3.3.1.137.2 Diagram



11.3.3.1.137.3 Fields

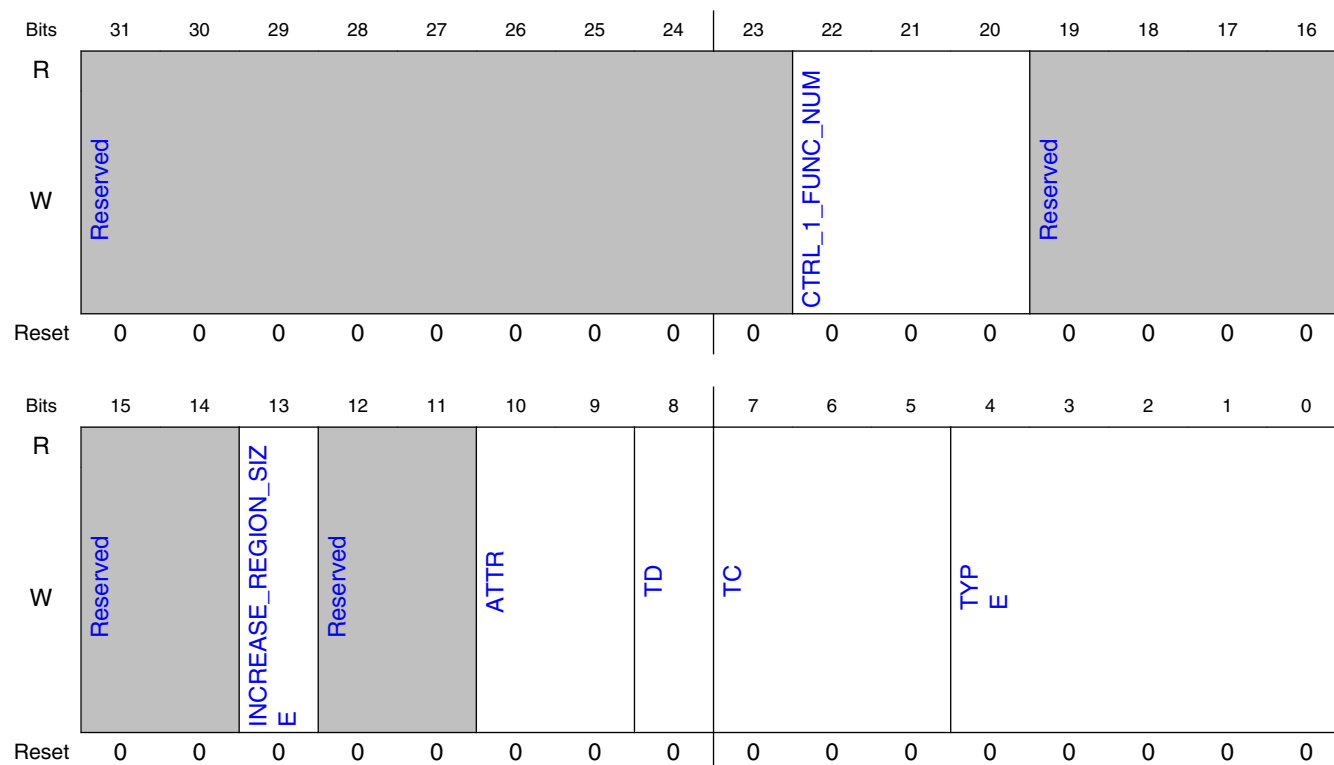
Field	Function
31-0 UPPER_TARGET_RW	Forms bits [63:32] of the start address (Upper Target part) of the new address of the translated region. Note: This register field is sticky.

11.3.3.1.138 iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_INBOUND_0)

11.3.3.1.138.1 Offset

Register	Offset
IATU_REGION_CTRL_1_OFF_INBOUND_0	8000_0100h

11.3.3.1.138.2 Diagram



11.3.3.1.138.3 Fields

Field	Function
31-23 —	Reserved.
22-20 CTRL_1_FUNC_NUM	Function Number. - MEM-I/O: When the Address and BAR matching logic in the controller indicate that a MEM-I/O transaction matches a BAR in the function corresponding to this value, then address translation proceeds. This check is only performed if the "Function Number Match Enable" bit of the "iATU Control 2 Register" is set. - CFG0/CFG1: When the destination function number as specified in the routing ID of the TLP header matches the function, then address translation proceeds. This check is only performed if the "Function Number Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
19-14 —	Reserved.
13 INCREASE_REGION_SIZE	Increase the maximum ATU Region size. When set, the size is determined by the Maximum Size of iATU Region. When clear, the maximum ATU Region size is 4 GB (default). Note: This register field is sticky.
12-11 —	Reserved.
10-9 ATTR	When the ATTR field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "ATTR Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.

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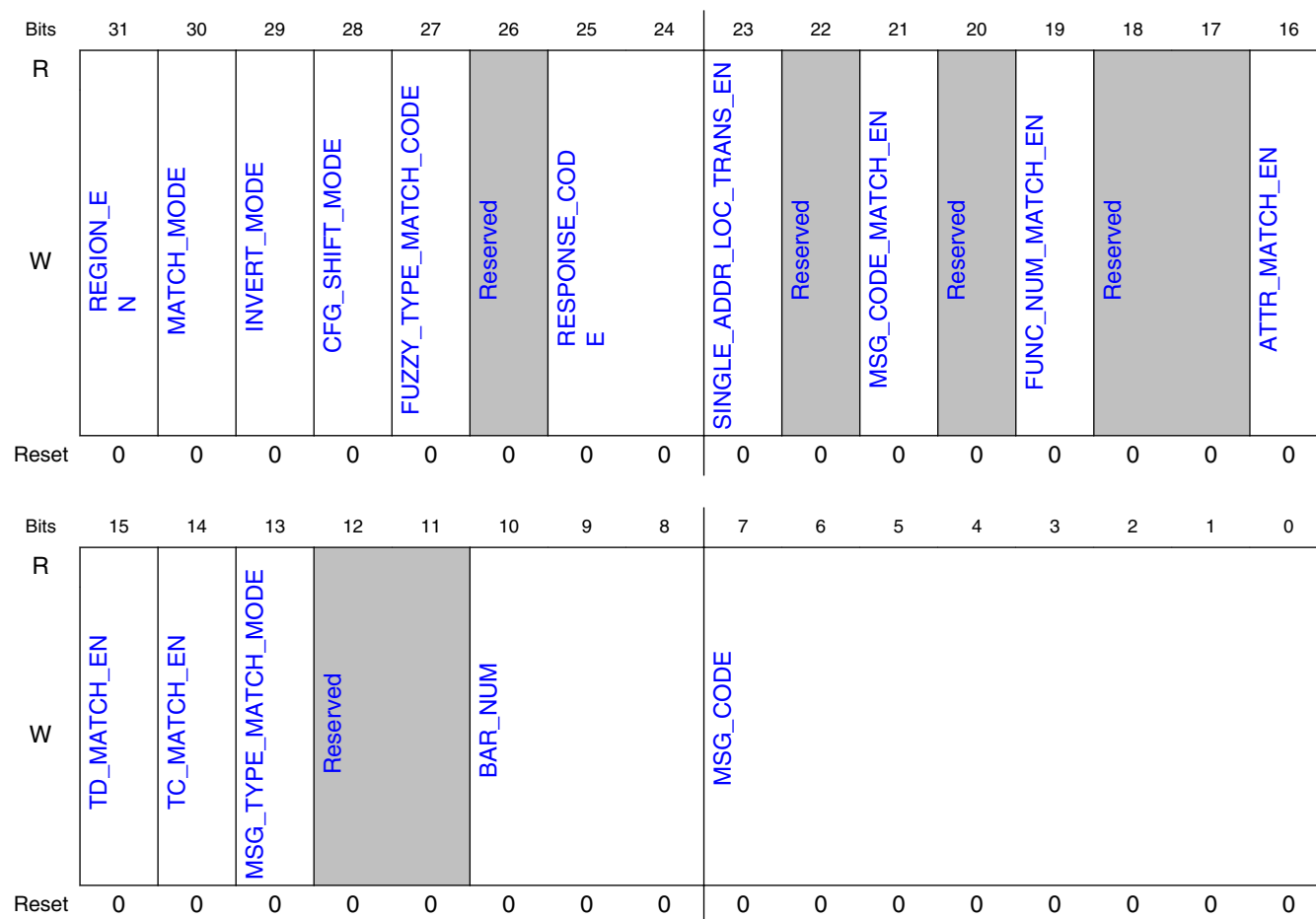
Field	Function
8 TD	When the TD field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TD Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
7-5 TC	When the TC field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TC Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
4-0 TYPE	When the TYPE field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). Note: This register field is sticky.

11.3.3.1.139 iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_INBOUND_0)

11.3.3.1.139.1 Offset

Register	Offset
IATU_REGION_CTRL_2_OFF_INBOUND_0	8000_0104h

11.3.3.1.139.2 Diagram



11.3.3.1.139.3 Fields

Field	Function
31 REGION_EN	Region Enable. This bit must be set to '1' for address translation to take place. Note: This register field is sticky.
30 MATCH_MODE	Match Mode. Determines Inbound matching mode for TLPs. The mode depends on the type of TLP that is received as follows: For MEM-I/O TLPs, this field is interpreted as follows: - 0: Address Match Mode. The iATU operates using addresses as in the outbound direction. The Region Base and Limit Registers must be setup. - 1: BAR Match Mode. BAR matching is used. The "BAR Number" field is relevant. Not used for RC. For CFG0 TLPs, this field is interpreted as follows: - 0: Routing ID Match Mode. The iATU interprets the Routing ID (Bytes 8 to 11 of TLP header) as an address. This corresponds to the upper 16 bits of the address in MEM-I/O transactions. The Routing ID of the TLP must be within the base and limit of the iATU region for matching to proceed. - 1: Accept Mode. The iATU accepts all CFG0 transactions as address matches. The routing ID in the CFG0 TLP is ignored. This is useful as all received CFG0 TLPs should be processed regardless of the Bus number. For MSG/MSGD TLPs, this field is interpreted as follows: - 0: Address Match Mode. The iATU treats the third dword and fourth dword of the inbound MSG/MSGD TLP as an address and it is matched against the Region Base and Limit Registers. - 1: Vendor ID Match Mode. This mode is relevant for ID-routed Vendor Defined Messages. The iATU ignores the Routing ID (Bus, Device, Function) in bits [31:16] of the third dword of the TLP header, but matches

Table continues on the next page...

Field	Function
	against the Vendor ID in bits [15:0] of the third dword of the TLP header. Bits [15:0] of the Region Upper Base register should be programmed with the required Vendor ID. The lower Base and Limit Register should be programmed to translate TLPs based on vendor specific information in the fourth dword of the TLP header. - If SINGLE_ADDRESS_LOCATION_TRANSLATE_EN = 1 AND MSG_TYPE_MATCH_MODE = 1, then Match Mode is ignored. Note: This register field is sticky.
29 INVERT_MODE	Invert Mode. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). Note: This register field is sticky.
28 CFG_SHIFT_MODE	CFG Shift Mode. This is useful for CFG transactions where the PCIe configuration mechanism maps bits [27:12] of the address to the bus/device and function number. This allows a CFG configuration space to be located in any 256MB window of your application memory space using a 28-bit effective address. Shifts bits [31:16] of the untranslated address to form bits [27:12] of the translated address. Note: This register field is sticky.
27 FUZZY_TYPE_MATCH_CODE	Fuzzy Type Match Mode. When enabled, the iATU relaxes the matching of the TLP TYPE field against the expected TYPE field so that - CfgRd0 and CfgRd1 TLPs are seen as identical. Similarly with CfgWr0 and CfgWr1. - MWr, MRd and MRdLk TLPs are seen as identical - The Routing field of Msg/MsgD TLPs is ignored - FetchAdd, Swap and CAS are seen as identical. For example, CFG0 in the TYPE field in the "iATU Control 1 Register" matches against an inbound CfgRd0, CfgRd1, CfgWr0 or CfgWr1 TLP. Note: This register field is sticky.
26 —	Reserved.
25-24 RESPONSE_CODE	Response Code. Defines the type of response to give for accesses matching this region. This overrides the normal RADM filter response. Note that this feature is not available for any region where Single Address Location Translate is enabled. - 00 - Normal RADM filter response is used. - 01 - Unsupported request (UR) - 10 - Completer abort (CA) - 11 - Not used / undefined / reserved. Note: This register field is sticky.
23 SINGLE_ADDR_LOC_TRANSLATE_EN	Single Address Location Translate Enable. When enabled, Rx TLPs can be translated to a single address location as determined by the target address register of the iATU region. The main usage scenario is translation of Messages (such as Vendor Defined or ATS Messages) to MWr TLPs when the AXI bridge is enabled. Note: This register field is sticky.
22 —	Reserved.
21 MSG_CODE_MATCH_EN	Message Code Match Enable (Msg TLPS). Ensures that a successful message Code TLP field comparison match (see Message Code field of the "iATU Control 2 Register") occurs (in MSG transactions) for address translation to proceed. ST Match Enable (Mem TLPS). Ensures that a successful ST TLP field comparison match (see ST field of the "iATU Control 2 Register") occurs (in MEM transactions) for address translation to proceed. Note: This register field is sticky.
20 —	Reserved.
19 FUNC_NUM_MATCH_EN	Function Number Match Enable. Ensures that a successful Function Number TLP field comparison match (see Function Number field of the "iATU Control 1 Register") occurs (in MEM-I/O and CFG0/CFG1 transactions) for address translation to proceed. Note: This register field is sticky.
18-17 —	Reserved.
16 ATTR_MATCH_EN	ATTR Match Enable. Ensures that a successful ATTR TLP field comparison match (see ATTR field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.

Table continues on the next page...

Field	Function
15 TD_MATCH_EN	TD Match Enable. Ensures that a successful TD TLP field comparison match (see TD field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.
14 TC_MATCH_EN	TC Match Enable. Ensures that a successful TC TLP field comparison match (see TC field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.
13 MSG_TYPE_MATCH_MODE	Message Type Match Mode. When enabled, and if single address location translate enable is set, then inbound TLPs of type MSG/MSGd which match the type field of the iatu_region_ctrl_1_OFF_inbound register ($\Rightarrow$ TYPE[4:3]=2'b10) will be translated. Message type match mode overrides any value of MATCH_MODE field in this register. Usage scenarios for this are translation of VDM or ATS messages when AXI bridge is configured on client interface. Note: This register field is sticky.
12-11 —	Reserved.
10-8 BAR_NUM	BAR Number. When the BAR number of an inbound MEM or IO TLP " that is matched by the normal internal BAR address matching mechanism " is the same as this field, address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Match Mode" bit of the "iATU Control 2 Register" is set. - 000b - BAR0 - 001b - BAR1 - 010b - BAR2 - 011b - BAR3 - 100b - BAR4 - 101b - BAR5 - 110b - ROM - 111b - reserved - IO translation would require either 00100b or 00101b in the inbound TLP TYPE; the BAR Number set in the range 000b - 101b and that BAR configured as an IO BAR. Note: This register field is sticky.
7-0 MSG_CODE	MSG TLPs: (Message Code). When the TYPE field of an inbound Msg/MsgD TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Message Code Match Enable" bit of the "iATU Control 2 Register" is set. Memory TLPs: (ST;Steering Tag). When the ST field of an inbound TLP is matched to this value, then address translation proceeds. This check is only performed if the "ST Match Enable" bit of the "iATU Control2 Register" is set. The setting is independent of the setting of the TH field. Note: This register field is sticky.

11.3.3.1.140 iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_OFF_INBOUND_0)

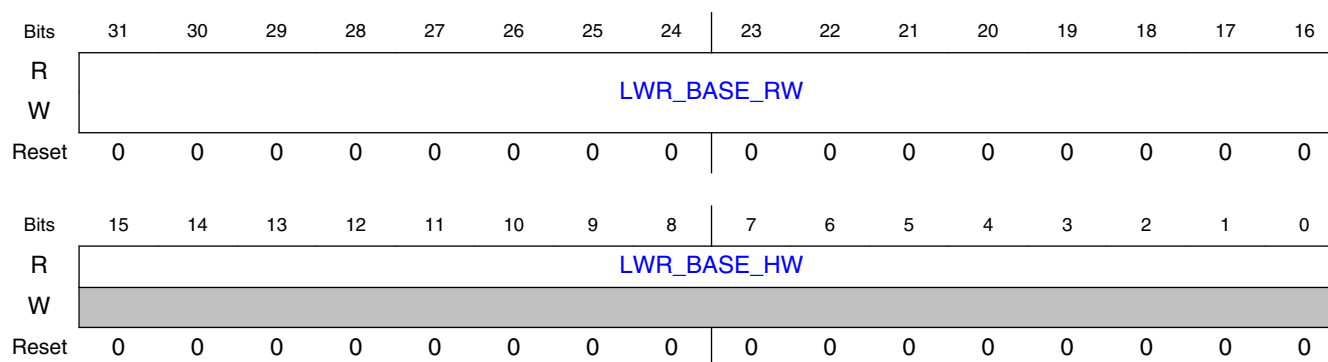
11.3.3.1.140.1 Offset

Register	Offset
IATU_LWR_BASE_ADDR_OFF_INBOUND_0	8000_0108h

11.3.3.1.140.2 Function

iATU Lower Base Address Register. The Minimum Size of iATU Region (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{Minimum Size of iATU Region})$ bits are zero.

11.3.3.1.140.3 Diagram



11.3.3.1.140.4 Fields

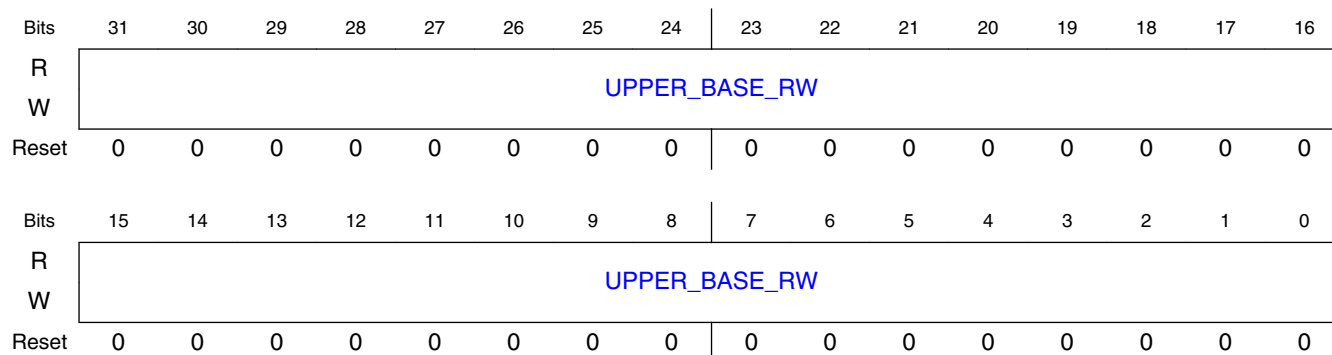
Field	Function
31-16 LWR_BASE_R W	Forms bits [31:n] of the start address of the address region to be translated. n is log2(Minimum Size of iATU Region) Note: This register field is sticky.
15-0 LWR_BASE_H W	Forms bits [n-1:0] of the start address of the address region to be translated. The start address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is log2(Minimum Size of iATU Region)

11.3.3.1.141 iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_INBOUND_0)

11.3.3.1.141.1 Offset

Register	Offset
IATU_UPPER_BASE_ADDR_OFF_INBOUND_0	8000_010Ch

11.3.3.1.141.2 Diagram



11.3.3.1.141.3 Fields

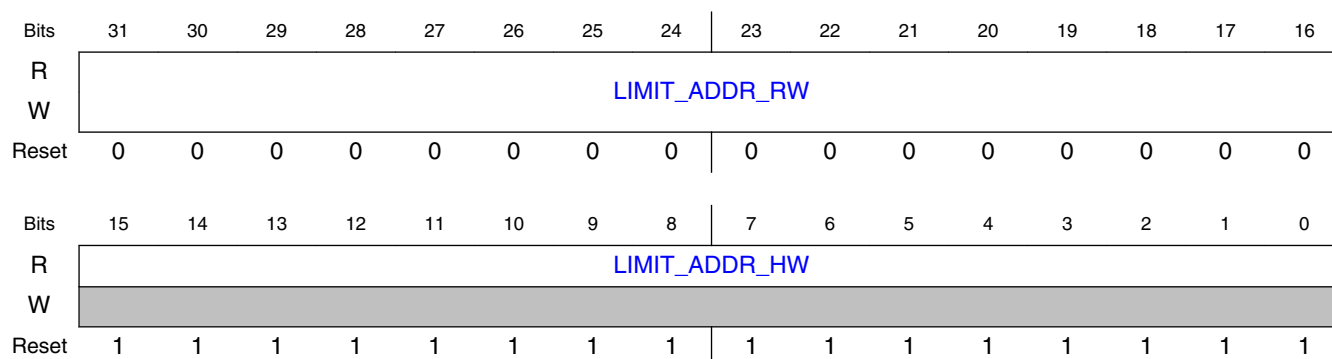
Field	Function
31-0 UPPER_BASE_ RW	Forms bits [63:32] of the start (and end) address of the address region to be translated. Note: This register field is sticky.

11.3.3.1.142 iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_INBOUND_0)

11.3.3.1.142.1 Offset

Register	Offset
IATU_LIMIT_ADDR_OFF_INBOUND_0	8000_0110h

11.3.3.1.142.2 Diagram



11.3.3.1.142.3 Fields

Field	Function
31-16 LIMIT_ADDR_R W	Forms upper bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. Note: This register field is sticky.
15-0 LIMIT_ADDR_H W	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller.

11.3.3.1.143 iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_INBOUND_0)

11.3.3.1.143.1 Offset

Register	Offset
IATU_LWR_TARGET_ADDR_OFF_INBOUND_0	8000_0114h

11.3.3.1.143.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	LWR_TARGET_RW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	LWR_TARGET_HW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.143.3 Fields

Field	Function
31-16 LWR_TARGET_R W	Forms MSB's of the Lower Target part of the new address of the translated region. These bits are always '0'. - Field size depends on log2(Minimum Size of iATU Region) in address match mode. - Field size depends on log2(BAR_MASK+1) in BAR match mode. Note: This register field is sticky.

Table continues on the next page...

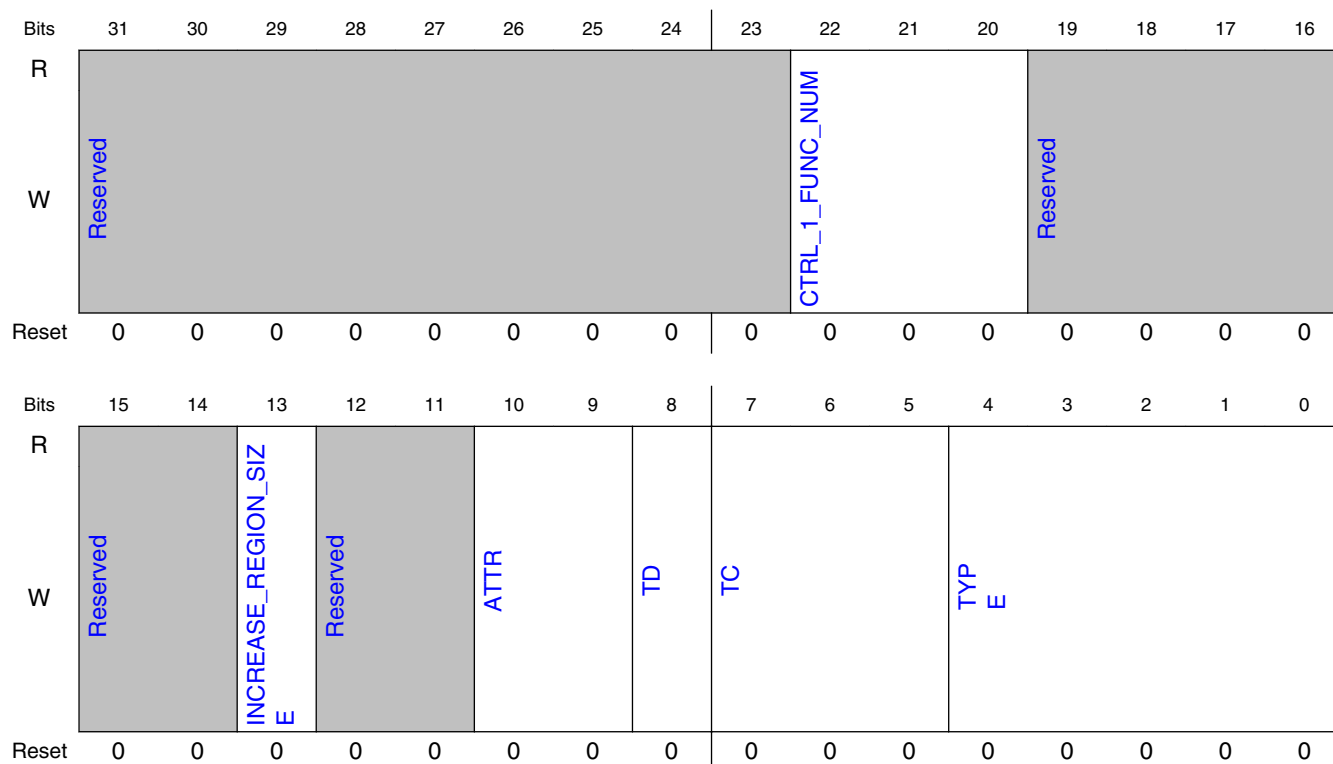
Field	Function
15-0 LWR_TARGET_HW	Forms the LSB's of the Lower Target part of the new address of the translated region. The start address must be aligned to the Minimum Size of iATU Region kB boundary (in address match mode); and to the Bar size boundary (in BAR match mode) so that these bits are always '0'. If the BAR is smaller than the iATU region size, then the iATU target address must align to the iATU region size; otherwise it must align to the BAR size. A write to this location is ignored by the PCIe controller. - Field size depends on $\log_2(\text{Minimum Size of iATU Region})$ in address match mode. - Field size depends on $\log_2(\text{BAR_MASK} + 1)$ in BAR match mode.

11.3.3.1.144 iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_OUTBOUND_1)

11.3.3.1.144.1 Offset

Register	Offset
IATU_REGION_CTRL_1_OFF_OUTBOUND_1	8000_0200h

11.3.3.1.144.2 Diagram



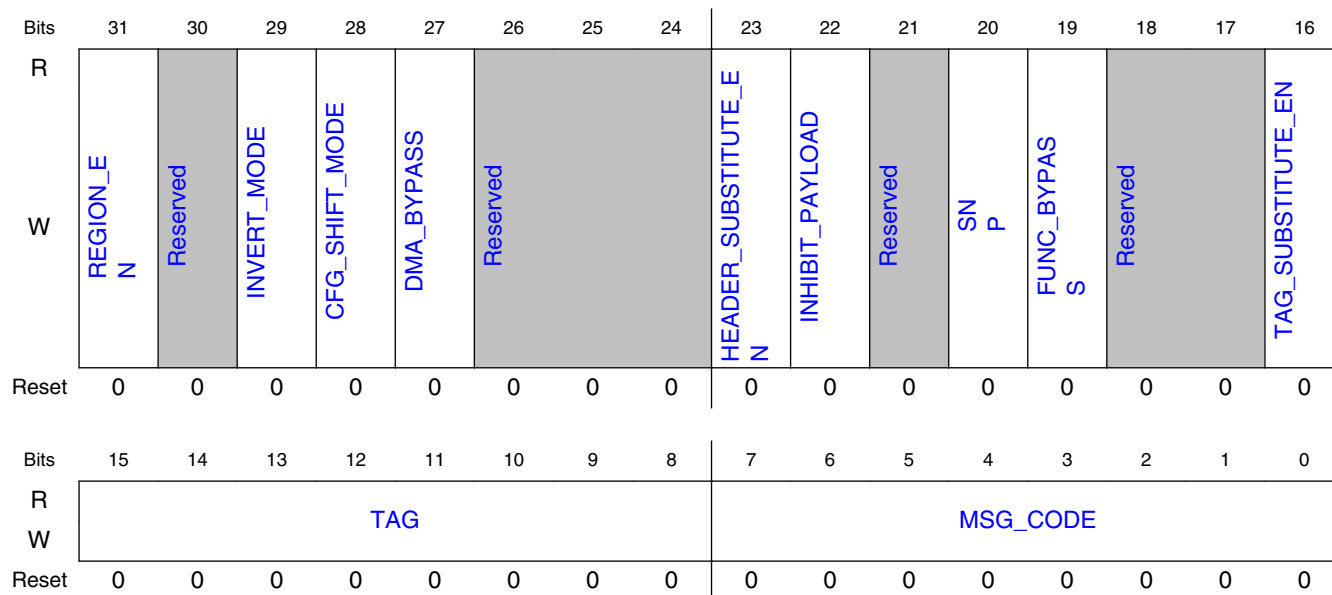
11.3.3.1.144.3 Fields

Field	Function
31-23 —	Reserved.
22-20 CTRL_1_FUNC_NUM	Function Number. - When the address of an outbound TLP is matched to this region and the FUNC_BYPASS field in the "iATU Control 2 Register" is '0', then the function number used in generating the function part of the requester ID (RID) field of the TLP is taken from this 5-bit register. When you are using the AXI Bridge, then this field is swapped before AXI decomposition occurs so that the correct "Max_Read_Request_Size" and "Max_Payload_Size" values are used. Note: This register field is sticky.
19-14 —	Reserved.
13 INCREASE_REGION_SIZE	Increase the maximum ATU Region size. When set, the size is determined by the Maximum Size of iATU Region. When clear, the maximum ATU Region size is 4 GB (default). Note: This register field is sticky.
12-11 —	Reserved.
10-9 ATTR	When the address of an outbound TLP is matched to this region, then the ATTR field of the TLP is changed to the value in this register. Note: This register field is sticky.
8 TD	When the address of an outbound TLP is matched to this region, then the TD field of the TLP is changed to the value in this register. Note: This register field is sticky.
7-5 TC	When the address of an outbound TLP is matched to this region, then the TC field of the TLP is changed to the value in this register. Note: This register field is sticky.
4-0 TYPE	When the address of an outbound TLP is matched to this region, then the TYPE field of the TLP is changed to the value in this register. Note: This register field is sticky.

11.3.3.1.145 iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_OUTBOUND_1)**11.3.3.1.145.1 Offset**

Register	Offset
IATU_REGION_CTRL_2_OFF_OUTBOUND_1	8000_0204h

11.3.3.1.145.2 Diagram



11.3.3.1.145.3 Fields

Field	Function
31 REGION_EN	Region Enable. This bit must be set to '1' for address translation to take place. Note: This register field is sticky.
30 —	Reserved.
29 INVERT_MODE	Invert Mode. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). Note: This register field is sticky.
28 CFG_SHIFT_MODE	CFG Shift Mode. The iATU uses bits [27:12] of the untranslated address (on the XALI0/1/2 interface or AXI slave interface address) to form the BDF number of the outgoing CFG TLP. This supports the Enhanced Configuration Address Mapping (ECAM) mechanism (Section 7.2.2 of the PCI Express Base 3.1 Specification, revision 1.0) by allowing all outgoing I/O and MEM TLPs (that have been translated to CFG) to be mapped from memory space into any 256 MB region of the PCIe configuration space. Note: This register field is sticky.
27 DMA_BYPASS	DMA Bypass Mode. Allows request TLPs which are initiated by the DMA controller to pass through the iATU untranslated. Note: This field is reserved for the SW product. You must set it to '0'. Note: This register field is sticky.
26-24 —	Reserved.
23 HEADER_SUBSTITUTE_EN	Header Substitute Enable. When enabled and region address is matched, the iATU fully substitutes bytes 8-11 (for 3 DWORD header) or bytes 12-15 (for 4 DWORD header) of the outbound TLP header with the contents of the LWR_TARGET_RW field in IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i. - 1: LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i register is used to fill bytes 8-to-11 (for 3 DWORD header) or bytes 12-to-15 (for 4 DWORD header) of the translated TLP header. - 0: LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i register forms the new address of the translated region. Note: This register field is sticky.

Table continues on the next page...

Field	Function
22 INHIBIT_PAYLOAD_AD	Inhibit TLP Payload Data for TLP's in Matched Region; assign iATU region to be TLP without data. When enabled and region address is matched, the iATU marks all TLPs as having no payload data by forcing the TLP header Fmt[1] bit =0, regardless of the application inputs such as slv_wstrb. - 1: Fmt[1] =0 so that only TLP type without data is sent. For example, a Msg instead of MsgD will be sent. - 0: Fmt[1] =0/1 so that TLPs with or without data can be sent. Note: This register field is sticky.
21 —	Reserved.
20 SNP	Serialize Non-Posted Requests. In this mode, when the AXI Bridge is populated, same AXI ID Non-Posted Read/Write Requests are transmitted on the wire if there are no other same ID Non-Posted Requests outstanding. Note: This register field is sticky.
19 FUNC_BYPASS	Function Number Translation Bypass. In this mode, the function number of the translated TLP is taken from your application transmit interface and not from the CTRL_1_FUNC_NUM field of the "iATU Control 1 Register" or the VF_NUMBER field of the "iATU Control 3 Register." Note: This register field is sticky.
18-17 —	Reserved.
16 TAG_SUBSTITUTE_EN	TAG Substitute Enable. When enabled and region address is matched, the iATU substitutes the TAG field of the outbound TLP header with the contents of the TAG field in this register. The expected usage scenario is translation from AXI MWr to Vendor Defined Msg/MsgD. Note: This register field is sticky.
15-8 TAG	TAG. The substituted TAG field (byte 6) in the outgoing TLP header when TAG_SUBSTITUTE_EN is set. Note: This register field is sticky.
7-0 MSG_CODE	MSG TLPs (Message Code). When the address of an outbound TLP is matched to this region, and the translated TLP TYPE field is Msg or MsgD; then the message field of the TLP is changed to the value in this register. Memory TLPs: (ST;Steering Tag). When the ST field of an outbound TLP is matched to this region, and the translated TLP TYPE field targets memory space; then the ST field of the TLP is changed to the value in this register. Note: This register field is sticky.

11.3.3.1.146 iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_OFF_OUTBOUND_1)

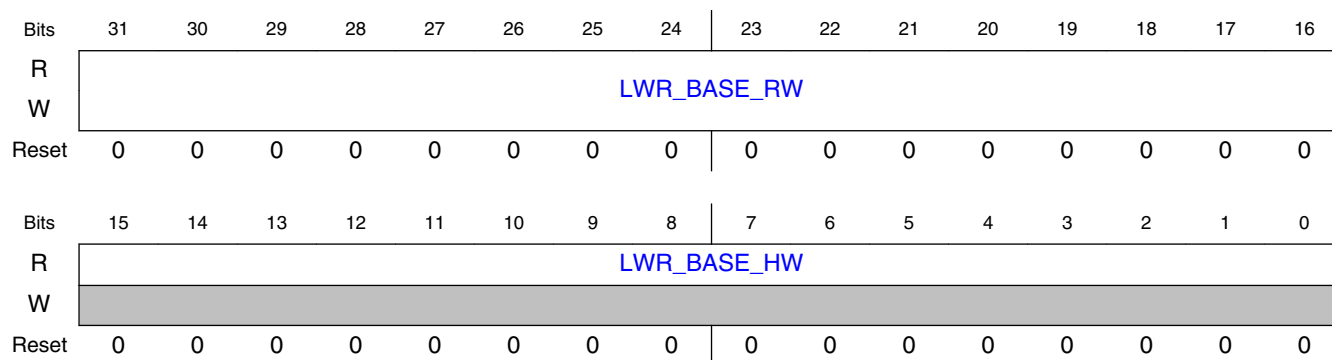
11.3.3.1.146.1 Offset

Register	Offset
IATU_LWR_BASE_ADDR_OFF_OUTBOUND_1	8000_0208h

11.3.3.1.146.2 Function

iATU Lower Base Address Register. The Minimum Size of iATU Region (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{Minimum Size of iATU Region})$ bits are zero.

11.3.3.1.146.3 Diagram



11.3.3.1.146.4 Fields

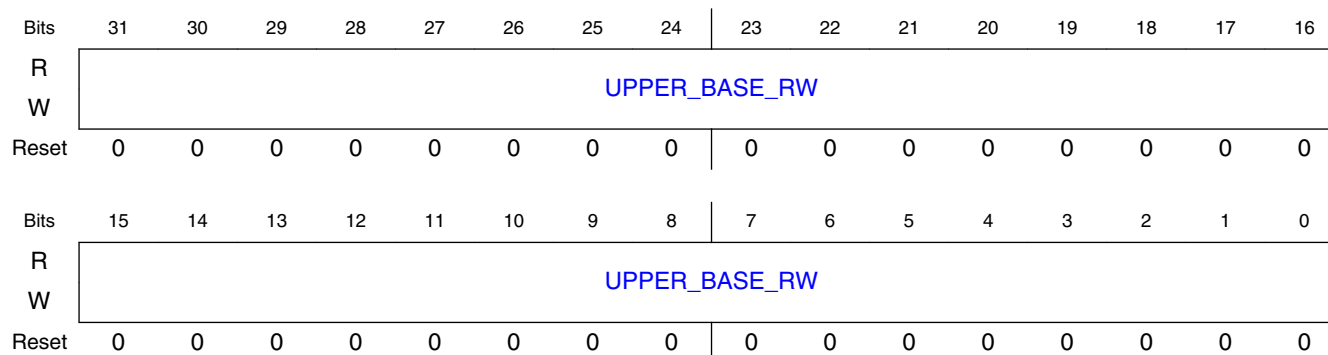
Field	Function
31-16 LWR_BASE_R W	Forms bits [31:n] of the start address of the address region to be translated. n is log2(Minimum Size of iATU Region) Note: This register field is sticky.
15-0 LWR_BASE_H W	Forms bits [n-1:0] of the start address of the address region to be translated. The start address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is log2(Minimum Size of iATU Region)

11.3.3.1.147 iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_1)

11.3.3.1.147.1 Offset

Register	Offset
IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_1	8000_020Ch

11.3.3.1.147.2 Diagram



11.3.3.1.147.3 Fields

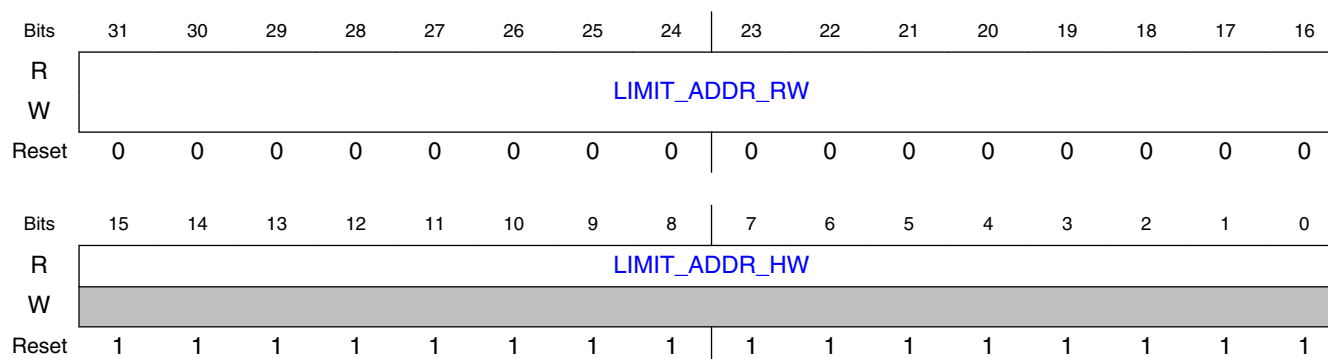
Field	Function
31-0 UPPER_BASE_ RW	Forms bits [63:32] of the start (and end) address of the address region to be translated. In systems with a 32-bit address space, this register is not used and therefore writing to this register has no effect. Note: This register field is sticky.

11.3.3.1.148 iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_OUTBOUND_1)

11.3.3.1.148.1 Offset

Register	Offset
IATU_LIMIT_ADDR_OFF_OUTBOUND_1	8000_0210h

11.3.3.1.148.2 Diagram



11.3.3.1.148.3 Fields

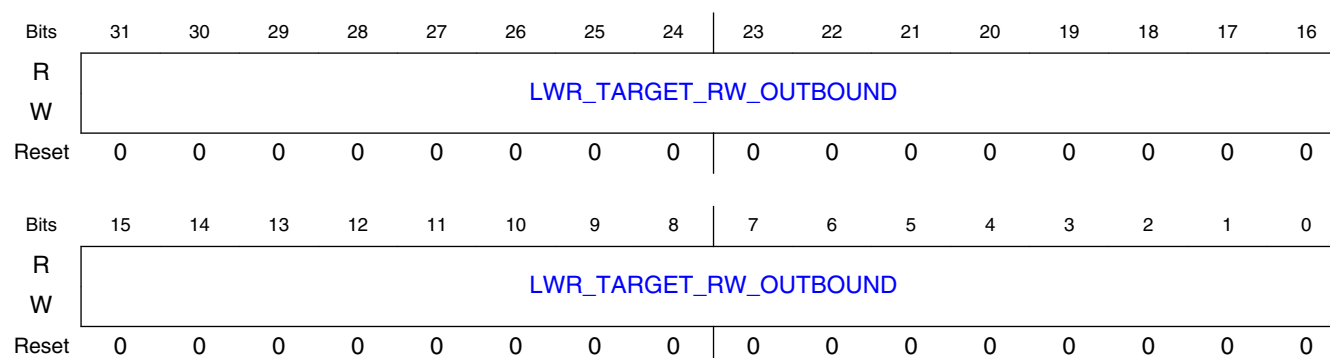
Field	Function
31-16 LIMIT_ADDR_R W	Forms upper bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. Note: This register field is sticky.
15-0 LIMIT_ADDR_H W	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller.

11.3.3.1.149 iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_1)

11.3.3.1.149.1 Offset

Register	Offset
IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_1	8000_0214h

11.3.3.1.149.2 Diagram



11.3.3.1.149.3 Fields

Field	Function
31-0 LWR_TARGET_RW_OUTBOUND	When HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_OFF_OUTBOUND_ is '0' (normal operation): - LWR_TARGET_RW[31:n] forms MSB's of the Lower Target part of the new address of the translated region; - LWR_TARGET_RW[n-1:0] are not used. (The start address must be aligned to the Minimum Size of iATU Region kB boundary, so the lower bits of the start address of the new address of the translated region (bits n-1:0) are always '0'). - n is log2(Minimum Size of iATU Region). When

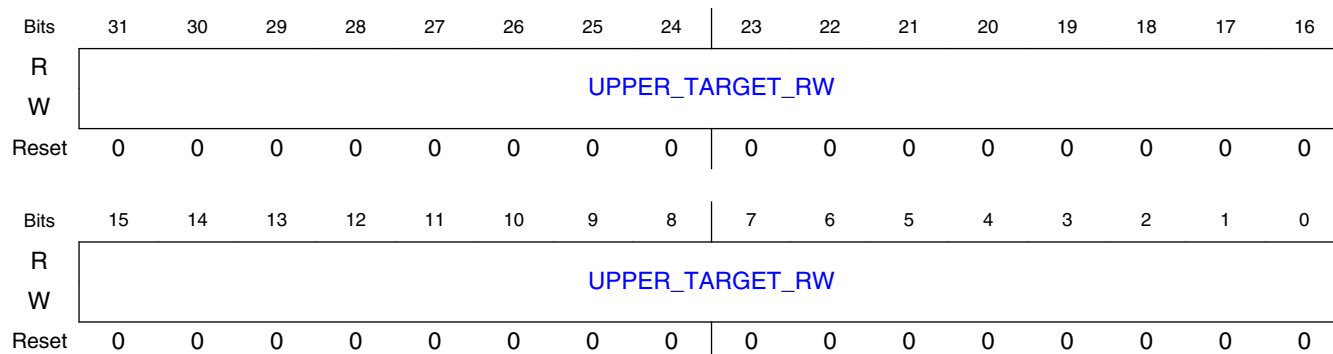
Field	Function
	HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_OFF_OUTBOUND_i is '1': - LWR_TARGET_RW[31:0] forms bytes 8-11 (for 3 dword header) or bytes 12-15 (for 4 dword header) of the outbound TLP header. Usage scenarios include the transmission of Vendor Defined Messages where the controller determines the content of bytes 12 to 15 of the TLP header. Note: This register field is sticky.

11.3.3.1.150 iATU Upper Target Address Register. (IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_1)

11.3.3.1.150.1 Offset

Register	Offset
IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_1	8000_0218h

11.3.3.1.150.2 Diagram



11.3.3.1.150.3 Fields

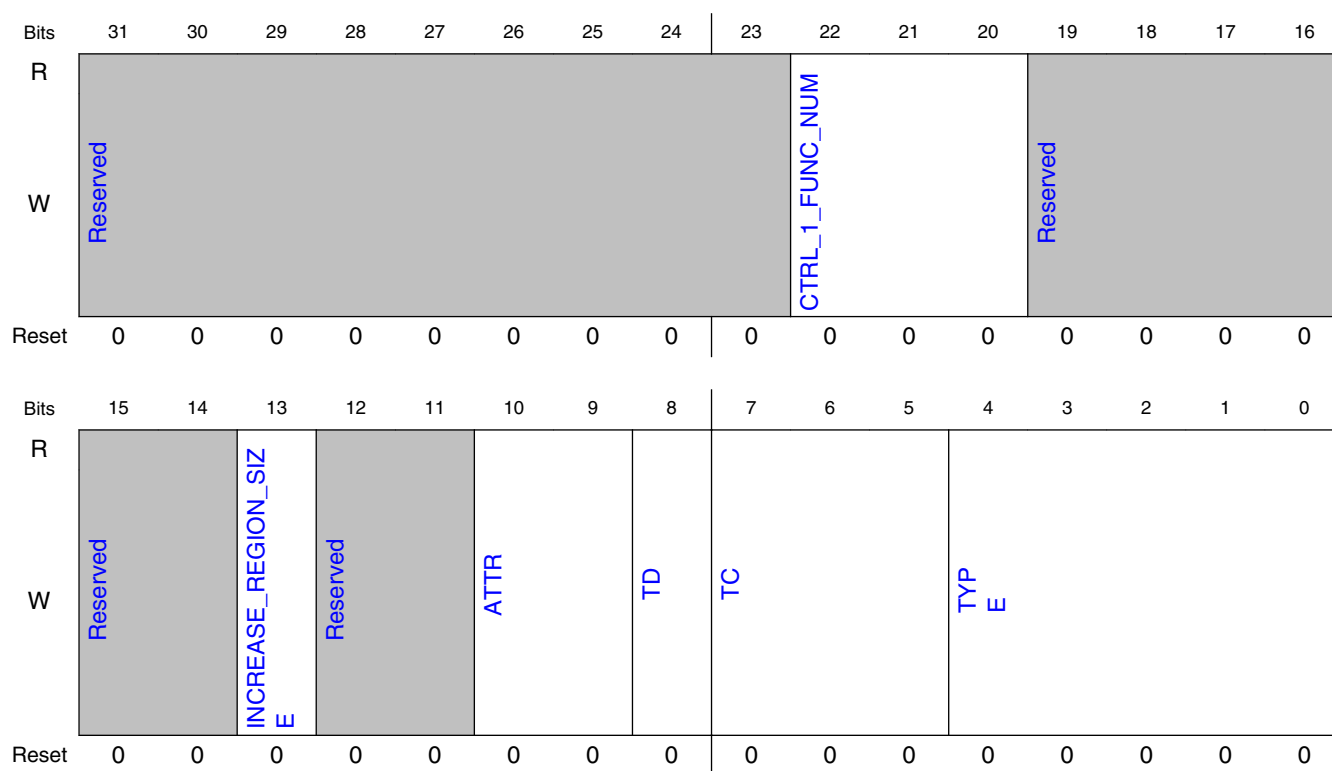
Field	Function
31-0 UPPER_TARGET_RW	Forms bits [63:32] of the start address (Upper Target part) of the new address of the translated region. Note: This register field is sticky.

11.3.3.1.151 iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFFSET_INBOUND_1)

11.3.3.1.151.1 Offset

Register	Offset
IATU_REGION_CTRL_1_OFFSET_INBOUND_1	8000_0300h

11.3.3.1.151.2 Diagram



11.3.3.1.151.3 Fields

Field	Function
31-23 —	Reserved.
22-20 CTRL_1_FUNC_NUM	Function Number. - MEM-I/O: When the Address and BAR matching logic in the controller indicate that a MEM-I/O transaction matches a BAR in the function corresponding to this value, then address translation proceeds. This check is only performed if the "Function Number Match Enable" bit of the "iATU Control 2 Register" is set. - CFG0/CFG1: When the destination function number as specified in the routing ID of the TLP header matches the function, then address translation proceeds. This check is only performed if the

Table continues on the next page...

PCI Express (PCIe)

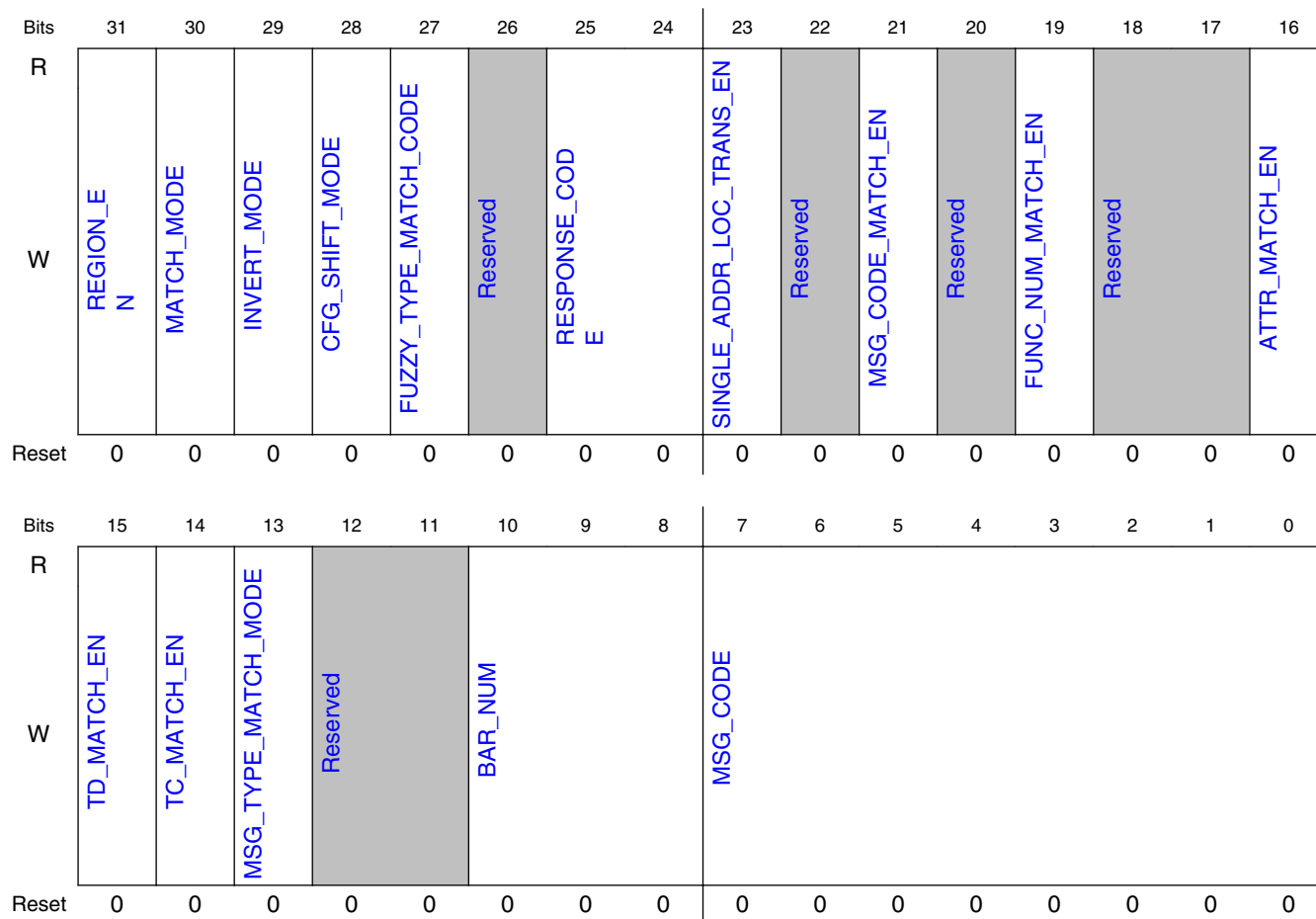
Field	Function
	"Function Number Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
19-14 —	Reserved.
13 INCREASE_REGION_SIZE	Increase the maximum ATU Region size. When set, the size is determined by the Maximum Size of iATU Region. When clear, the maximum ATU Region size is 4 GB (default). Note: This register field is sticky.
12-11 —	Reserved.
10-9 ATTR	When the ATTR field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "ATTR Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
8 TD	When the TD field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TD Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
7-5 TC	When the TC field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TC Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
4-0 TYPE	When the TYPE field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). Note: This register field is sticky.

11.3.3.1.152 iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_INBOUND_1)

11.3.3.1.152.1 Offset

Register	Offset
IATU_REGION_CTRL_2_OFF_INBOUND_1	8000_0304h

11.3.3.1.152.2 Diagram



11.3.3.1.152.3 Fields

Field	Function
31 REGION_EN	Region Enable. This bit must be set to '1' for address translation to take place. Note: This register field is sticky.
30 MATCH_MODE	Match Mode. Determines Inbound matching mode for TLPs. The mode depends on the type of TLP that is received as follows: For MEM-I/O TLPs, this field is interpreted as follows: - 0: Address Match Mode. The iATU operates using addresses as in the outbound direction. The Region Base and Limit Registers must be setup. - 1: BAR Match Mode. BAR matching is used. The "BAR Number" field is relevant. Not used for RC. For CFG0 TLPs, this field is interpreted as follows: - 0: Routing ID Match Mode. The iATU interprets the Routing ID (Bytes 8 to 11 of TLP header) as an address. This corresponds to the upper 16 bits of the address in MEM-I/O transactions. The Routing ID of the TLP must be within the base and limit of the iATU region for matching to proceed. - 1: Accept Mode. The iATU accepts all CFG0 transactions as address matches. The routing ID in the CFG0 TLP is ignored. This is useful as all received CFG0 TLPs should be processed regardless of the Bus number. For MSG/MSGD TLPs, this field is interpreted as follows: - 0: Address Match Mode. The iATU treats the third dword and fourth dword of the inbound MSG/MSGD TLP as an address and it is matched against the Region Base and Limit Registers. - 1: Vendor ID Match Mode. This mode is relevant for ID-routed Vendor Defined Messages. The iATU ignores the Routing ID (Bus, Device, Function) in bits [31:16] of the third dword of the TLP header, but matches

Table continues on the next page...

Field	Function
	against the Vendor ID in bits [15:0] of the third dword of the TLP header. Bits [15:0] of the Region Upper Base register should be programmed with the required Vendor ID. The lower Base and Limit Register should be programmed to translate TLPs based on vendor specific information in the fourth dword of the TLP header. - If SINGLE_ADDRESS_LOCATION_TRANSLATE_EN = 1 AND MSG_TYPE_MATCH_MODE = 1, then Match Mode is ignored. Note: This register field is sticky.
29 INVERT_MODE	Invert Mode. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). Note: This register field is sticky.
28 CFG_SHIFT_MODE	CFG Shift Mode. This is useful for CFG transactions where the PCIe configuration mechanism maps bits [27:12] of the address to the bus/device and function number. This allows a CFG configuration space to be located in any 256MB window of your application memory space using a 28-bit effective address. Shifts bits [31:16] of the untranslated address to form bits [27:12] of the translated address. Note: This register field is sticky.
27 FUZZY_TYPE_MATCH_CODE	Fuzzy Type Match Mode. When enabled, the iATU relaxes the matching of the TLP TYPE field against the expected TYPE field so that - CfgRd0 and CfgRd1 TLPs are seen as identical. Similarly with CfgWr0 and CfgWr1. - MWr, MRd and MRdLk TLPs are seen as identical - The Routing field of Msg/MsgD TLPs is ignored - FetchAdd, Swap and CAS are seen as identical. For example, CFG0 in the TYPE field in the "iATU Control 1 Register" matches against an inbound CfgRd0, CfgRd1, CfgWr0 or CfgWr1 TLP. Note: This register field is sticky.
26 —	Reserved.
25-24 RESPONSE_CODE	Response Code. Defines the type of response to give for accesses matching this region. This overrides the normal RADM filter response. Note that this feature is not available for any region where Single Address Location Translate is enabled. - 00 - Normal RADM filter response is used. - 01 - Unsupported request (UR) - 10 - Completer abort (CA) - 11 - Not used / undefined / reserved. Note: This register field is sticky.
23 SINGLE_ADDR_LOC_TRANSLATE_EN	Single Address Location Translate Enable. When enabled, Rx TLPs can be translated to a single address location as determined by the target address register of the iATU region. The main usage scenario is translation of Messages (such as Vendor Defined or ATS Messages) to MWr TLPs when the AXI bridge is enabled. Note: This register field is sticky.
22 —	Reserved.
21 MSG_CODE_MATCH_EN	Message Code Match Enable (Msg TLPS). Ensures that a successful message Code TLP field comparison match (see Message Code field of the "iATU Control 2 Register") occurs (in MSG transactions) for address translation to proceed. ST Match Enable (Mem TLPS). Ensures that a successful ST TLP field comparison match (see ST field of the "iATU Control 2 Register") occurs (in MEM transactions) for address translation to proceed. Note: This register field is sticky.
20 —	Reserved.
19 FUNC_NUM_MATCH_EN	Function Number Match Enable. Ensures that a successful Function Number TLP field comparison match (see Function Number field of the "iATU Control 1 Register") occurs (in MEM-I/O and CFG0/CFG1 transactions) for address translation to proceed. Note: This register field is sticky.
18-17 —	Reserved.
16 ATTR_MATCH_EN	ATTR Match Enable. Ensures that a successful ATTR TLP field comparison match (see ATTR field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.

Table continues on the next page...

Field	Function
15 TD_MATCH_EN	TD Match Enable. Ensures that a successful TD TLP field comparison match (see TD field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.
14 TC_MATCH_EN	TC Match Enable. Ensures that a successful TC TLP field comparison match (see TC field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.
13 MSG_TYPE_MATCH_MODE	Message Type Match Mode. When enabled, and if single address location translate enable is set, then inbound TLPs of type MSG/MSGd which match the type field of the iatu_region_ctrl_1_OFF_inbound register ($\Rightarrow$ TYPE[4:3]=2'b10) will be translated. Message type match mode overrides any value of MATCH_MODE field in this register. Usage scenarios for this are translation of VDM or ATS messages when AXI bridge is configured on client interface. Note: This register field is sticky.
12-11 —	Reserved.
10-8 BAR_NUM	BAR Number. When the BAR number of an inbound MEM or IO TLP " that is matched by the normal internal BAR address matching mechanism " is the same as this field, address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Match Mode" bit of the "iATU Control 2 Register" is set. - 000b - BAR0 - 001b - BAR1 - 010b - BAR2 - 011b - BAR3 - 100b - BAR4 - 101b - BAR5 - 110b - ROM - 111b - reserved - IO translation would require either 00100b or 00101b in the inbound TLP TYPE; the BAR Number set in the range 000b - 101b and that BAR configured as an IO BAR. Note: This register field is sticky.
7-0 MSG_CODE	MSG TLPs: (Message Code). When the TYPE field of an inbound Msg/MsgD TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Message Code Match Enable" bit of the "iATU Control 2 Register" is set. Memory TLPs: (ST;Steering Tag). When the ST field of an inbound TLP is matched to this value, then address translation proceeds. This check is only performed if the "ST Match Enable" bit of the "iATU Control2 Register" is set. The setting is independent of the setting of the TH field. Note: This register field is sticky.

11.3.3.1.153 iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_OFF_INBOUND_1)

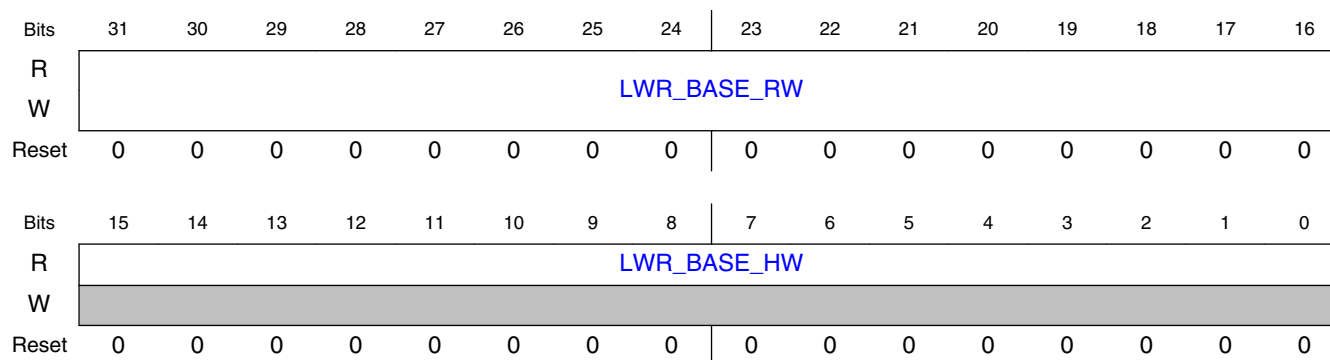
11.3.3.1.153.1 Offset

Register	Offset
IATU_LWR_BASE_ADDR_OFF_INBOUND_1	8000_0308h

11.3.3.1.153.2 Function

iATU Lower Base Address Register. The Minimum Size of iATU Region (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{Minimum Size of iATU Region})$ bits are zero.

11.3.3.1.153.3 Diagram



11.3.3.1.153.4 Fields

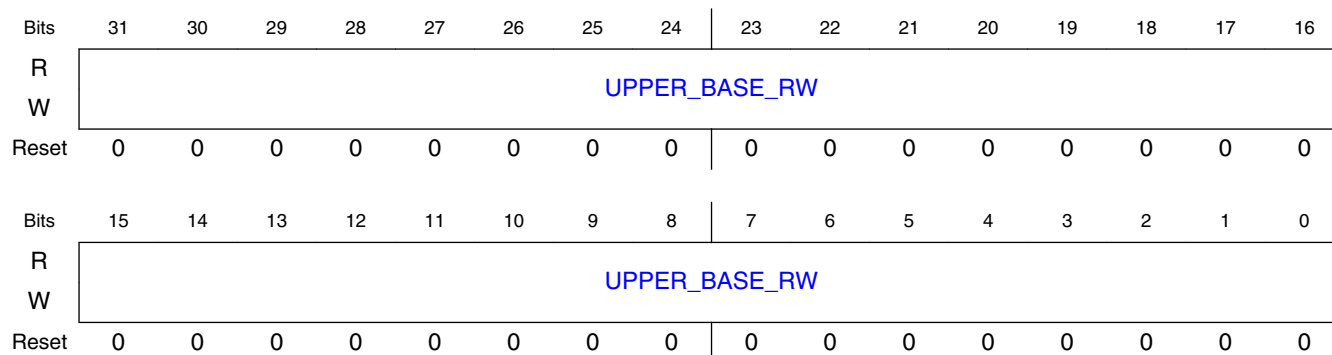
Field	Function
31-16 LWR_BASE_R W	Forms bits [31:n] of the start address of the address region to be translated. n is log2(Minimum Size of iATU Region) Note: This register field is sticky.
15-0 LWR_BASE_H W	Forms bits [n-1:0] of the start address of the address region to be translated. The start address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is log2(Minimum Size of iATU Region)

11.3.3.1.154 iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_INBOUND_1)

11.3.3.1.154.1 Offset

Register	Offset
IATU_UPPER_BASE_ADDR_OFF_INBOUND_1	8000_030Ch

11.3.3.1.154.2 Diagram



11.3.3.1.154.3 Fields

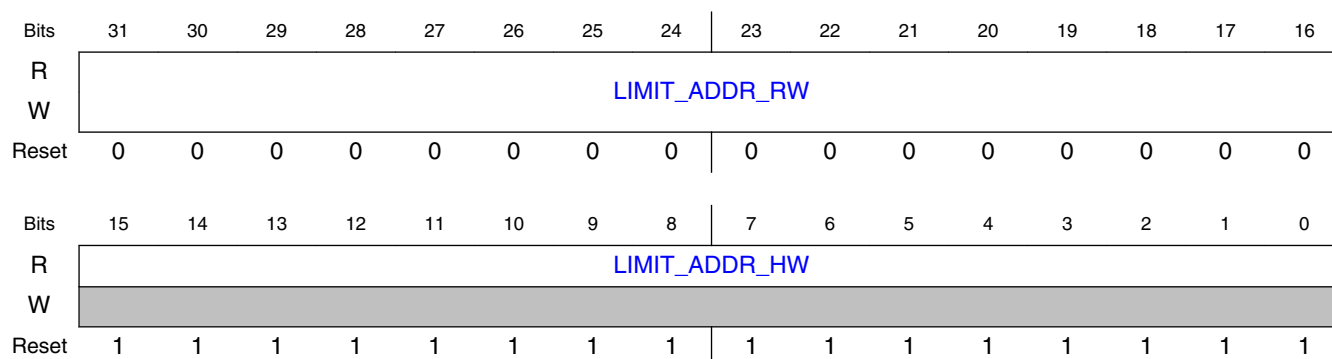
Field	Function
31-0 UPPER_BASE_ RW	Forms bits [63:32] of the start (and end) address of the address region to be translated. Note: This register field is sticky.

11.3.3.1.155 iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_INBOUND_1)

11.3.3.1.155.1 Offset

Register	Offset
IATU_LIMIT_ADDR_OFF_INBOUND_1	8000_0310h

11.3.3.1.155.2 Diagram



11.3.3.1.155.3 Fields

Field	Function
31-16 LIMIT_ADDR_R W	Forms upper bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. Note: This register field is sticky.
15-0 LIMIT_ADDR_H W	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller.

11.3.3.1.156 iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_INBOUND_1)

11.3.3.1.156.1 Offset

Register	Offset
IATU_LWR_TARGET_ADDR_OFF_INBOUND_1	8000_0314h

11.3.3.1.156.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	LWR_TARGET_RW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	LWR_TARGET_HW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.156.3 Fields

Field	Function
31-16 LWR_TARGET_RW	Forms MSB's of the Lower Target part of the new address of the translated region. These bits are always '0'. - Field size depends on log2(Minimum Size of iATU Region) in address match mode. - Field size depends on log2(BAR_MASK+1) in BAR match mode. Note: This register field is sticky.

Table continues on the next page...

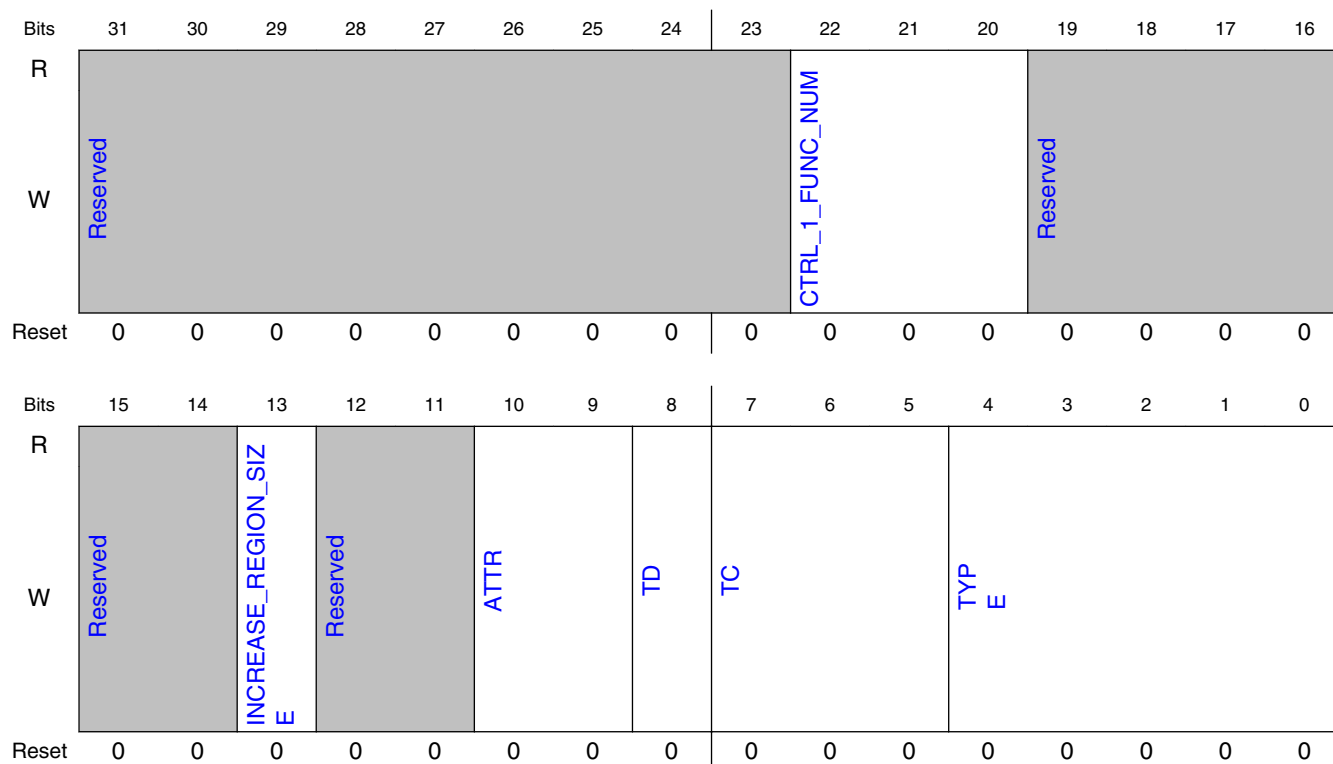
Field	Function
15-0 LWR_TARGET_HW	Forms the LSB's of the Lower Target part of the new address of the translated region. The start address must be aligned to the Minimum Size of iATU Region kB boundary (in address match mode); and to the Bar size boundary (in BAR match mode) so that these bits are always '0'. If the BAR is smaller than the iATU region size, then the iATU target address must align to the iATU region size; otherwise it must align to the BAR size. A write to this location is ignored by the PCIe controller. - Field size depends on log2(Minimum Size of iATU Region) in address match mode. - Field size depends on log2(BAR_MASK +1) in BAR match mode.

11.3.3.1.157 iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_OUTBOUND_2)

11.3.3.1.157.1 Offset

Register	Offset
IATU_REGION_CTRL_1_OFF_OUTBOUND_2	8000_0400h

11.3.3.1.157.2 Diagram



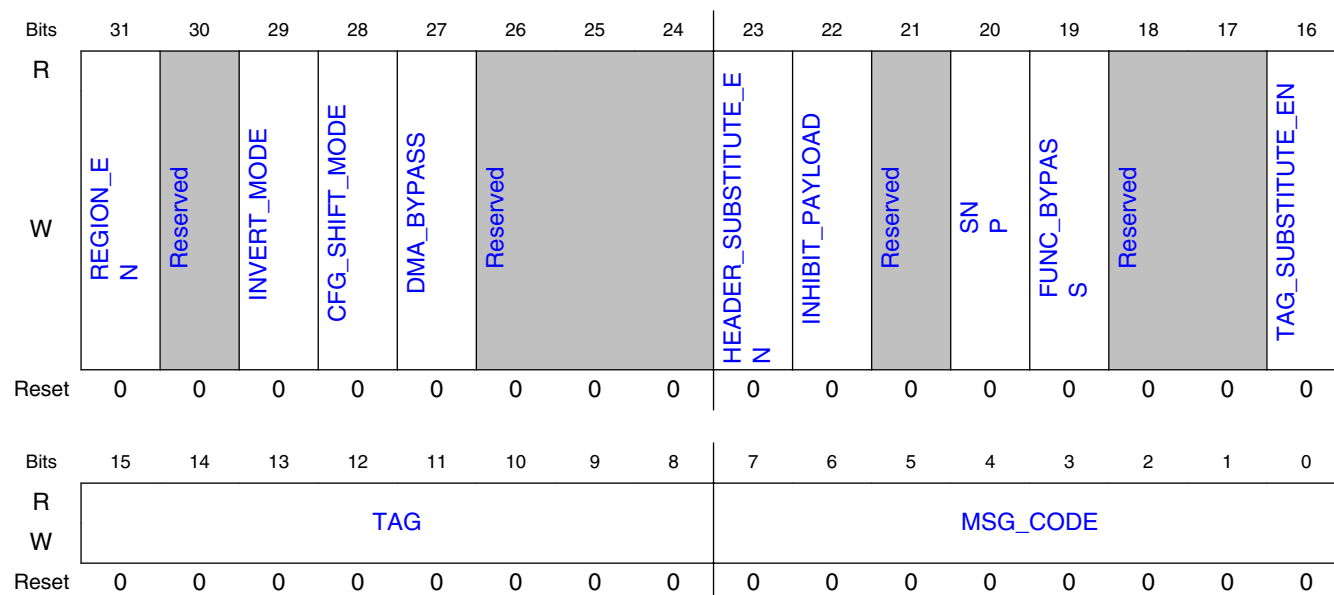
11.3.3.1.157.3 Fields

Field	Function
31-23 —	Reserved.
22-20 CTRL_1_FUNC_NUM	Function Number. - When the address of an outbound TLP is matched to this region and the FUNC_BYPASS field in the "iATU Control 2 Register" is '0', then the function number used in generating the function part of the requester ID (RID) field of the TLP is taken from this 5-bit register. When you are using the AXI Bridge, then this field is swapped before AXI decomposition occurs so that the correct "Max_Read_Request_Size" and "Max_Payload_Size" values are used. Note: This register field is sticky.
19-14 —	Reserved.
13 INCREASE_REGION_SIZE	Increase the maximum ATU Region size. When set, the size is determined by the Maximum Size of iATU Region. When clear, the maximum ATU Region size is 4 GB (default). Note: This register field is sticky.
12-11 —	Reserved.
10-9 ATTR	When the address of an outbound TLP is matched to this region, then the ATTR field of the TLP is changed to the value in this register. Note: This register field is sticky.
8 TD	When the address of an outbound TLP is matched to this region, then the TD field of the TLP is changed to the value in this register. Note: This register field is sticky.
7-5 TC	When the address of an outbound TLP is matched to this region, then the TC field of the TLP is changed to the value in this register. Note: This register field is sticky.
4-0 TYPE	When the address of an outbound TLP is matched to this region, then the TYPE field of the TLP is changed to the value in this register. Note: This register field is sticky.

11.3.3.1.158 iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_OUTBOUND_2)**11.3.3.1.158.1 Offset**

Register	Offset
IATU_REGION_CTRL_2_OFF_OUTBOUND_2	8000_0404h

11.3.3.1.158.2 Diagram



11.3.3.1.158.3 Fields

Field	Function
31 REGION_EN	Region Enable. This bit must be set to '1' for address translation to take place. Note: This register field is sticky.
30 —	Reserved.
29 INVERT_MODE	Invert Mode. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). Note: This register field is sticky.
28 CFG_SHIFT_MODE	CFG Shift Mode. The iATU uses bits [27:12] of the untranslated address (on the XALI0/1/2 interface or AXI slave interface address) to form the BDF number of the outgoing CFG TLP. This supports the Enhanced Configuration Address Mapping (ECAM) mechanism (Section 7.2.2 of the PCI Express Base 3.1 Specification, revision 1.0) by allowing all outgoing I/O and MEM TLPs (that have been translated to CFG) to be mapped from memory space into any 256 MB region of the PCIe configuration space. Note: This register field is sticky.
27 DMA_BYPASS	DMA Bypass Mode. Allows request TLPs which are initiated by the DMA controller to pass through the iATU untranslated. Note: This field is reserved for the SW product. You must set it to '0'. Note: This register field is sticky.
26-24 —	Reserved.
23 HEADER_SUBSTITUTE_EN	Header Substitute Enable. When enabled and region address is matched, the iATU fully substitutes bytes 8-11 (for 3 DWORD header) or bytes 12-15 (for 4 DWORD header) of the outbound TLP header with the contents of the LWR_TARGET_RW field in IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i. - 1: LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i register is used to fill bytes 8-to-11 (for 3 DWORD header) or bytes 12-to-15 (for 4 DWORD header) of the translated TLP header. - 0: LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i register forms the new address of the translated region. Note: This register field is sticky.

Table continues on the next page...

Field	Function
22 INHIBIT_PAYLOAD_AD	Inhibit TLP Payload Data for TLP's in Matched Region; assign iATU region to be TLP without data. When enabled and region address is matched, the iATU marks all TLPs as having no payload data by forcing the TLP header Fmt[1] bit =0, regardless of the application inputs such as slv_wstrb. - 1: Fmt[1] =0 so that only TLP type without data is sent. For example, a Msg instead of MsgD will be sent. - 0: Fmt[1] =0/1 so that TLPs with or without data can be sent. Note: This register field is sticky.
21 —	Reserved.
20 SNP	Serialize Non-Posted Requests. In this mode, when the AXI Bridge is populated, same AXI ID Non-Posted Read/Write Requests are transmitted on the wire if there are no other same ID Non-Posted Requests outstanding. Note: This register field is sticky.
19 FUNC_BYPASS	Function Number Translation Bypass. In this mode, the function number of the translated TLP is taken from your application transmit interface and not from the CTRL_1_FUNC_NUM field of the "iATU Control 1 Register" or the VF_NUMBER field of the "iATU Control 3 Register." Note: This register field is sticky.
18-17 —	Reserved.
16 TAG_SUBSTITUTE_EN	TAG Substitute Enable. When enabled and region address is matched, the iATU substitutes the TAG field of the outbound TLP header with the contents of the TAG field in this register. The expected usage scenario is translation from AXI MWr to Vendor Defined Msg/MsgD. Note: This register field is sticky.
15-8 TAG	TAG. The substituted TAG field (byte 6) in the outgoing TLP header when TAG_SUBSTITUTE_EN is set. Note: This register field is sticky.
7-0 MSG_CODE	MSG TLPs (Message Code). When the address of an outbound TLP is matched to this region, and the translated TLP TYPE field is Msg or MsgD; then the message field of the TLP is changed to the value in this register. Memory TLPs: (ST;Steering Tag). When the ST field of an outbound TLP is matched to this region, and the translated TLP TYPE field targets memory space; then the ST field of the TLP is changed to the value in this register. Note: This register field is sticky.

11.3.3.1.159 iATU Lower Base Address Register. (IATU_LWR_BASE_ADDRESS_OFFSET_OUTBOUND_2)

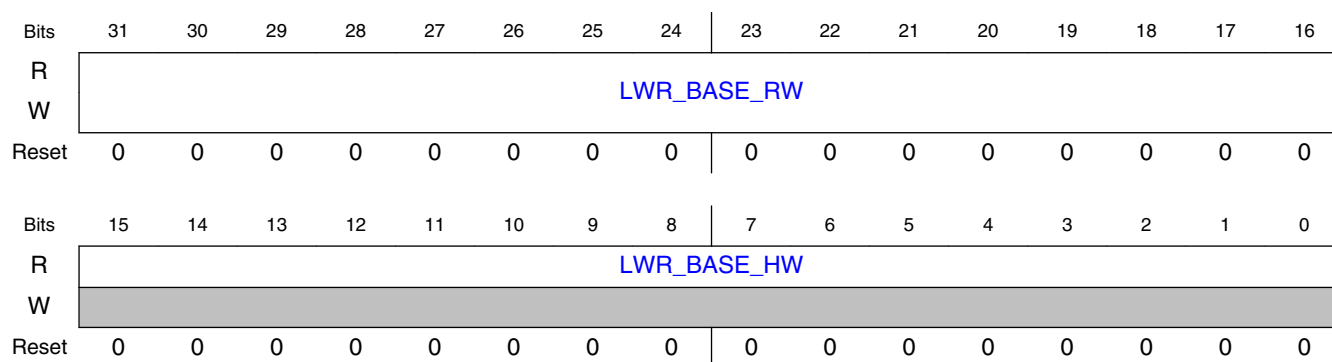
11.3.3.1.159.1 Offset

Register	Offset
IATU_LWR_BASE_ADDRESS_OFFSET_OUTBOUND_2	8000_0408h

11.3.3.1.159.2 Function

iATU Lower Base Address Register. The Minimum Size of iATU Region (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{Minimum Size of iATU Region})$ bits are zero.

11.3.3.1.159.3 Diagram



11.3.3.1.159.4 Fields

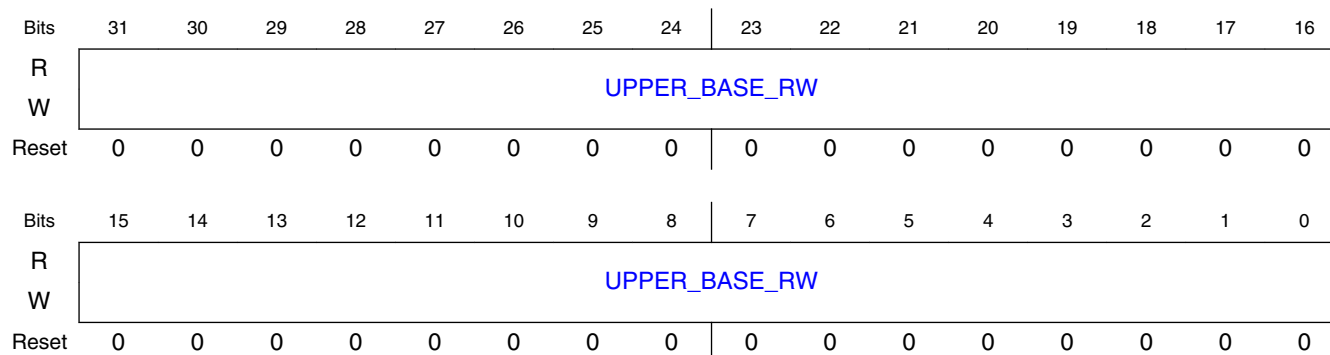
Field	Function
31-16 LWR_BASE_R W	Forms bits [31:n] of the start address of the address region to be translated. n is log2(Minimum Size of iATU Region) Note: This register field is sticky.
15-0 LWR_BASE_H W	Forms bits [n-1:0] of the start address of the address region to be translated. The start address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is log2(Minimum Size of iATU Region)

11.3.3.1.160 iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_2)

11.3.3.1.160.1 Offset

Register	Offset
IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_2	8000_040Ch

11.3.3.1.160.2 Diagram



11.3.3.1.160.3 Fields

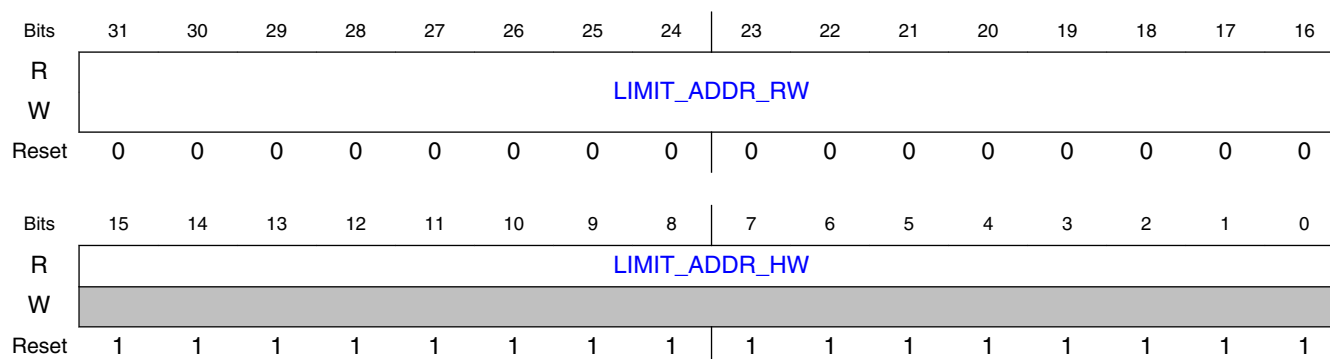
Field	Function
31-0 UPPER_BASE_ RW	Forms bits [63:32] of the start (and end) address of the address region to be translated. In systems with a 32-bit address space, this register is not used and therefore writing to this register has no effect. Note: This register field is sticky.

11.3.3.1.161 iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_OUTBOUND_2)

11.3.3.1.161.1 Offset

Register	Offset
IATU_LIMIT_ADDR_OFF_OUTBOUND_2	8000_0410h

11.3.3.1.161.2 Diagram



11.3.3.1.161.3 Fields

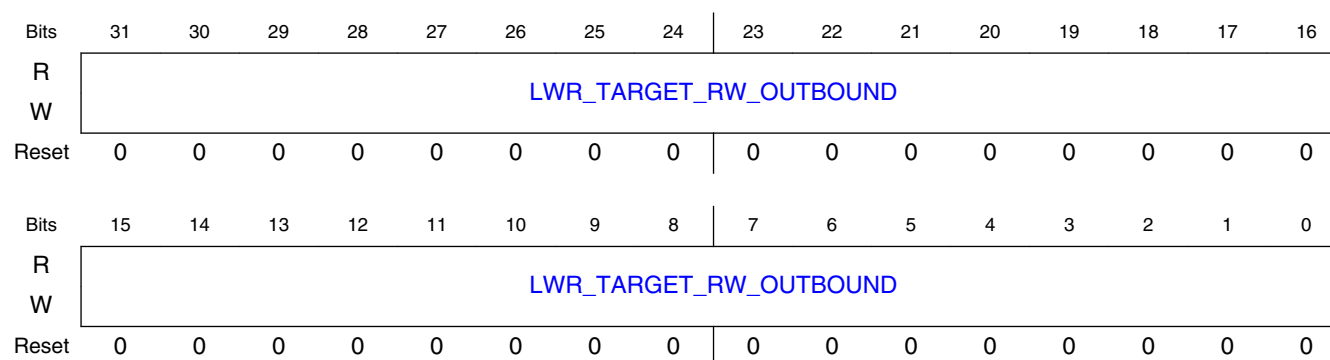
Field	Function
31-16 LIMIT_ADDR_R W	Forms upper bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. Note: This register field is sticky.
15-0 LIMIT_ADDR_H W	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller.

11.3.3.1.162 iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_2)

11.3.3.1.162.1 Offset

Register	Offset
IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_2	8000_0414h

11.3.3.1.162.2 Diagram



11.3.3.1.162.3 Fields

Field	Function
31-0 LWR_TARGET_RW_OUTBOUND	When HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_OFF_OUTBOUND_ is '0' (normal operation): - LWR_TARGET_RW[31:n] forms MSB's of the Lower Target part of the new address of the translated region; - LWR_TARGET_RW[n-1:0] are not used. (The start address must be aligned to the Minimum Size of iATU Region kB boundary, so the lower bits of the start address of the new address of the translated region (bits n-1:0) are always '0'). - n is log2(Minimum Size of iATU Region). When

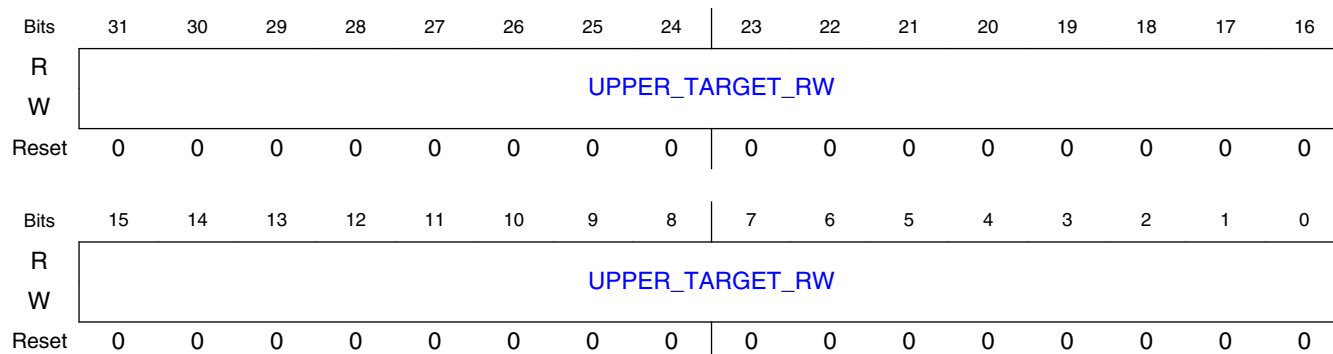
Field	Function
	HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_OFF_OUTBOUND_i is '1': - LWR_TARGET_RW[31:0] forms bytes 8-11 (for 3 dword header) or bytes 12-15 (for 4 dword header) of the outbound TLP header. Usage scenarios include the transmission of Vendor Defined Messages where the controller determines the content of bytes 12 to 15 of the TLP header. Note: This register field is sticky.

11.3.3.1.163 iATU Upper Target Address Register. (IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_2)

11.3.3.1.163.1 Offset

Register	Offset
IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_2	8000_0418h

11.3.3.1.163.2 Diagram



11.3.3.1.163.3 Fields

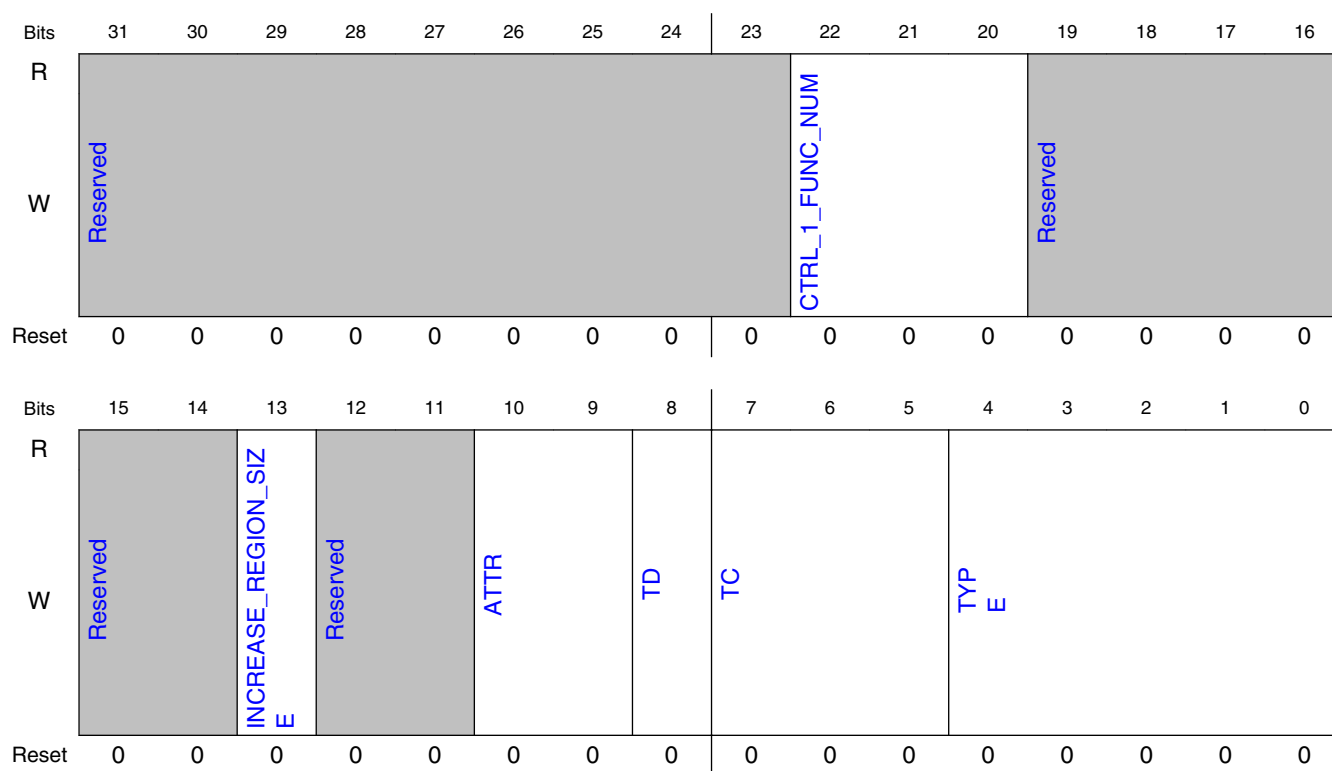
Field	Function
31-0 UPPER_TARGET_RW	Forms bits [63:32] of the start address (Upper Target part) of the new address of the translated region. Note: This register field is sticky.

11.3.3.1.164 iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFFSET_INBOUND_2)

11.3.3.1.164.1 Offset

Register	Offset
IATU_REGION_CTRL_1_OFFSET_INBOUND_2	8000_0500h

11.3.3.1.164.2 Diagram



11.3.3.1.164.3 Fields

Field	Function
31-23 —	Reserved.
22-20 CTRL_1_FUNC_NUM	Function Number. - MEM-I/O: When the Address and BAR matching logic in the controller indicate that a MEM-I/O transaction matches a BAR in the function corresponding to this value, then address translation proceeds. This check is only performed if the "Function Number Match Enable" bit of the "iATU Control 2 Register" is set. - CFG0/CFG1: When the destination function number as specified in the routing ID of the TLP header matches the function, then address translation proceeds. This check is only performed if the

Table continues on the next page...

PCI Express (PCIe)

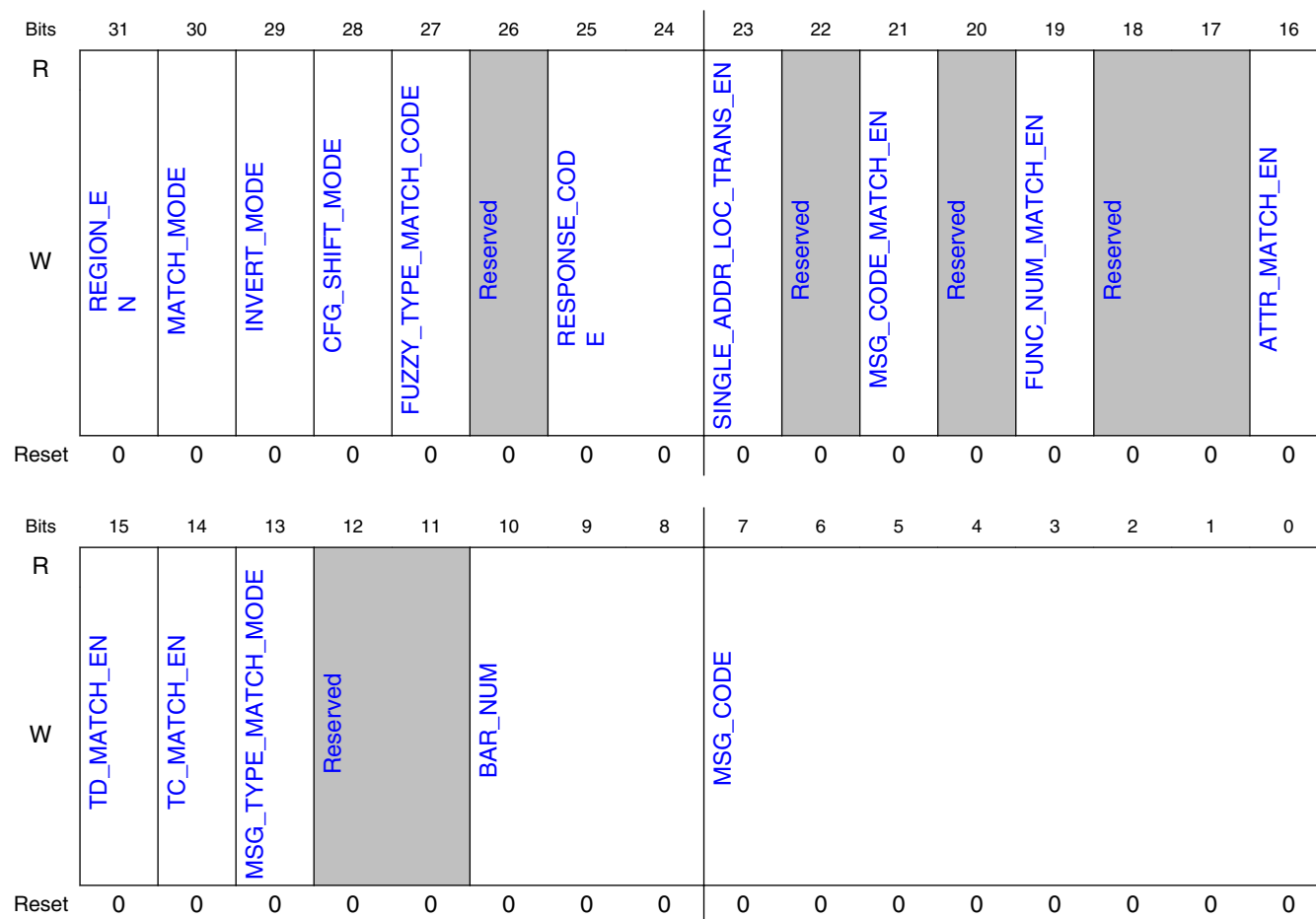
Field	Function
	"Function Number Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
19-14 —	Reserved.
13 INCREASE_REGION_SIZE	Increase the maximum ATU Region size. When set, the size is determined by the Maximum Size of iATU Region. When clear, the maximum ATU Region size is 4 GB (default). Note: This register field is sticky.
12-11 —	Reserved.
10-9 ATTR	When the ATTR field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "ATTR Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
8 TD	When the TD field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TD Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
7-5 TC	When the TC field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TC Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
4-0 TYPE	When the TYPE field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). Note: This register field is sticky.

11.3.3.1.165 iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_INBOUND_2)

11.3.3.1.165.1 Offset

Register	Offset
IATU_REGION_CTRL_2_OFF_INBOUND_2	8000_0504h

11.3.3.1.165.2 Diagram



11.3.3.1.165.3 Fields

Field	Function
31 REGION_EN	Region Enable. This bit must be set to '1' for address translation to take place. Note: This register field is sticky.
30 MATCH_MODE	Match Mode. Determines Inbound matching mode for TLPs. The mode depends on the type of TLP that is received as follows: For MEM-I/O TLPs, this field is interpreted as follows: - 0: Address Match Mode. The iATU operates using addresses as in the outbound direction. The Region Base and Limit Registers must be setup. - 1: BAR Match Mode. BAR matching is used. The "BAR Number" field is relevant. Not used for RC. For CFG0 TLPs, this field is interpreted as follows: - 0: Routing ID Match Mode. The iATU interprets the Routing ID (Bytes 8 to 11 of TLP header) as an address. This corresponds to the upper 16 bits of the address in MEM-I/O transactions. The Routing ID of the TLP must be within the base and limit of the iATU region for matching to proceed. - 1: Accept Mode. The iATU accepts all CFG0 transactions as address matches. The routing ID in the CFG0 TLP is ignored. This is useful as all received CFG0 TLPs should be processed regardless of the Bus number. For MSG/MSGD TLPs, this field is interpreted as follows: - 0: Address Match Mode. The iATU treats the third dword and fourth dword of the inbound MSG/MSGD TLP as an address and it is matched against the Region Base and Limit Registers. - 1: Vendor ID Match Mode. This mode is relevant for ID-routed Vendor Defined Messages. The iATU ignores the Routing ID (Bus, Device, Function) in bits [31:16] of the third dword of the TLP header, but matches

Table continues on the next page...

Field	Function
	against the Vendor ID in bits [15:0] of the third dword of the TLP header. Bits [15:0] of the Region Upper Base register should be programmed with the required Vendor ID. The lower Base and Limit Register should be programmed to translate TLPs based on vendor specific information in the fourth dword of the TLP header. - If SINGLE_ADDRESS_LOCATION_TRANSLATE_EN = 1 AND MSG_TYPE_MATCH_MODE = 1, then Match Mode is ignored. Note: This register field is sticky.
29 INVERT_MODE	Invert Mode. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). Note: This register field is sticky.
28 CFG_SHIFT_MODE	CFG Shift Mode. This is useful for CFG transactions where the PCIe configuration mechanism maps bits [27:12] of the address to the bus/device and function number. This allows a CFG configuration space to be located in any 256MB window of your application memory space using a 28-bit effective address. Shifts bits [31:16] of the untranslated address to form bits [27:12] of the translated address. Note: This register field is sticky.
27 FUZZY_TYPE_MATCH_CODE	Fuzzy Type Match Mode. When enabled, the iATU relaxes the matching of the TLP TYPE field against the expected TYPE field so that - CfgRd0 and CfgRd1 TLPs are seen as identical. Similarly with CfgWr0 and CfgWr1. - MWr, MRd and MRdLk TLPs are seen as identical - The Routing field of Msg/MsgD TLPs is ignored - FetchAdd, Swap and CAS are seen as identical. For example, CFG0 in the TYPE field in the "iATU Control 1 Register" matches against an inbound CfgRd0, CfgRd1, CfgWr0 or CfgWr1 TLP. Note: This register field is sticky.
26 —	Reserved.
25-24 RESPONSE_CODE	Response Code. Defines the type of response to give for accesses matching this region. This overrides the normal RADM filter response. Note that this feature is not available for any region where Single Address Location Translate is enabled. - 00 - Normal RADM filter response is used. - 01 - Unsupported request (UR) - 10 - Completer abort (CA) - 11 - Not used / undefined / reserved. Note: This register field is sticky.
23 SINGLE_ADDR_LOC_TRANSLATE_EN	Single Address Location Translate Enable. When enabled, Rx TLPs can be translated to a single address location as determined by the target address register of the iATU region. The main usage scenario is translation of Messages (such as Vendor Defined or ATS Messages) to MWr TLPs when the AXI bridge is enabled. Note: This register field is sticky.
22 —	Reserved.
21 MSG_CODE_MATCH_EN	Message Code Match Enable (Msg TLPS). Ensures that a successful message Code TLP field comparison match (see Message Code field of the "iATU Control 2 Register") occurs (in MSG transactions) for address translation to proceed. ST Match Enable (Mem TLPS). Ensures that a successful ST TLP field comparison match (see ST field of the "iATU Control 2 Register") occurs (in MEM transactions) for address translation to proceed. Note: This register field is sticky.
20 —	Reserved.
19 FUNC_NUM_MATCH_EN	Function Number Match Enable. Ensures that a successful Function Number TLP field comparison match (see Function Number field of the "iATU Control 1 Register") occurs (in MEM-I/O and CFG0/CFG1 transactions) for address translation to proceed. Note: This register field is sticky.
18-17 —	Reserved.
16 ATTR_MATCH_EN	ATTR Match Enable. Ensures that a successful ATTR TLP field comparison match (see ATTR field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.

Table continues on the next page...

Field	Function
15 TD_MATCH_EN	TD Match Enable. Ensures that a successful TD TLP field comparison match (see TD field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.
14 TC_MATCH_EN	TC Match Enable. Ensures that a successful TC TLP field comparison match (see TC field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.
13 MSG_TYPE_MATCH_MODE	Message Type Match Mode. When enabled, and if single address location translate enable is set, then inbound TLPs of type MSG/MSGd which match the type field of the iatu_region_ctrl_1_OFF_inbound register ($\Rightarrow$ TYPE[4:3]=2'b10) will be translated. Message type match mode overrides any value of MATCH_MODE field in this register. Usage scenarios for this are translation of VDM or ATS messages when AXI bridge is configured on client interface. Note: This register field is sticky.
12-11 —	Reserved.
10-8 BAR_NUM	BAR Number. When the BAR number of an inbound MEM or IO TLP " that is matched by the normal internal BAR address matching mechanism " is the same as this field, address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Match Mode" bit of the "iATU Control 2 Register" is set. - 000b - BAR0 - 001b - BAR1 - 010b - BAR2 - 011b - BAR3 - 100b - BAR4 - 101b - BAR5 - 110b - ROM - 111b - reserved - IO translation would require either 00100b or 00101b in the inbound TLP TYPE; the BAR Number set in the range 000b - 101b and that BAR configured as an IO BAR. Note: This register field is sticky.
7-0 MSG_CODE	MSG TLPs: (Message Code). When the TYPE field of an inbound Msg/MsgD TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Message Code Match Enable" bit of the "iATU Control 2 Register" is set. Memory TLPs: (ST;Steering Tag). When the ST field of an inbound TLP is matched to this value, then address translation proceeds. This check is only performed if the "ST Match Enable" bit of the "iATU Control2 Register" is set. The setting is independent of the setting of the TH field. Note: This register field is sticky.

11.3.3.1.166 iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_OFF_INBOUND_2)

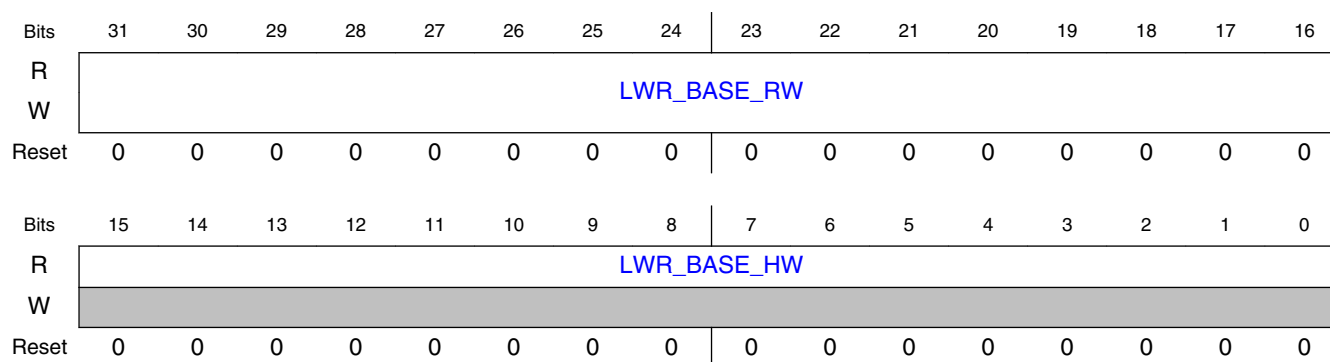
11.3.3.1.166.1 Offset

Register	Offset
IATU_LWR_BASE_ADDR_OFF_INBOUND_2	8000_0508h

11.3.3.1.166.2 Function

iATU Lower Base Address Register. The Minimum Size of iATU Region (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{Minimum Size of iATU Region})$ bits are zero.

11.3.3.1.166.3 Diagram



11.3.3.1.166.4 Fields

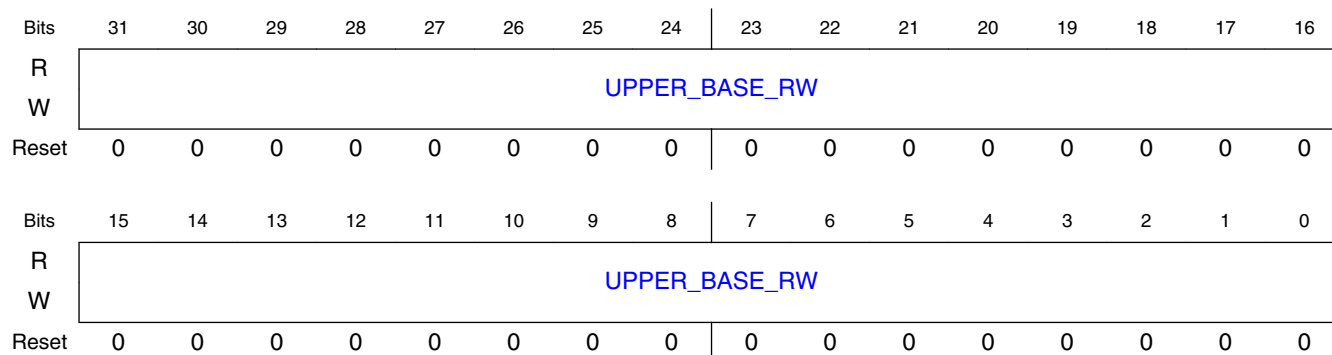
Field	Function
31-16 LWR_BASE_R W	Forms bits [31:n] of the start address of the address region to be translated. n is log2(Minimum Size of iATU Region) Note: This register field is sticky.
15-0 LWR_BASE_H W	Forms bits [n-1:0] of the start address of the address region to be translated. The start address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is log2(Minimum Size of iATU Region)

11.3.3.1.167 iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_INBOUND_2)

11.3.3.1.167.1 Offset

Register	Offset
IATU_UPPER_BASE_ADDR_OFF_INBOUND_2	8000_050Ch

11.3.3.1.167.2 Diagram



11.3.3.1.167.3 Fields

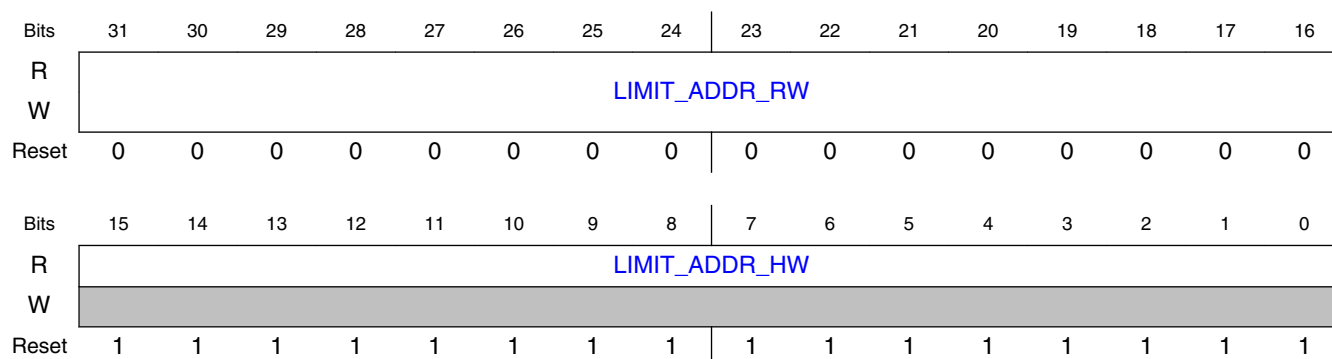
Field	Function
31-0 UPPER_BASE_ RW	Forms bits [63:32] of the start (and end) address of the address region to be translated. Note: This register field is sticky.

11.3.3.1.168 iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_INBOUND_2)

11.3.3.1.168.1 Offset

Register	Offset
IATU_LIMIT_ADDR_OFF_INBOUND_2	8000_0510h

11.3.3.1.168.2 Diagram



11.3.3.1.168.3 Fields

Field	Function
31-16 LIMIT_ADDR_R W	Forms upper bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. Note: This register field is sticky.
15-0 LIMIT_ADDR_H W	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller.

11.3.3.1.169 iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_INBOUND_2)

11.3.3.1.169.1 Offset

Register	Offset
IATU_LWR_TARGET_ADDR_OFF_INBOUND_2	8000_0514h

11.3.3.1.169.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	LWR_TARGET_RW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	LWR_TARGET_HW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.169.3 Fields

Field	Function
31-16 LWR_TARGET_R W	Forms MSB's of the Lower Target part of the new address of the translated region. These bits are always '0'. - Field size depends on log2(Minimum Size of iATU Region) in address match mode. - Field size depends on log2(BAR_MASK+1) in BAR match mode. Note: This register field is sticky.

Table continues on the next page...

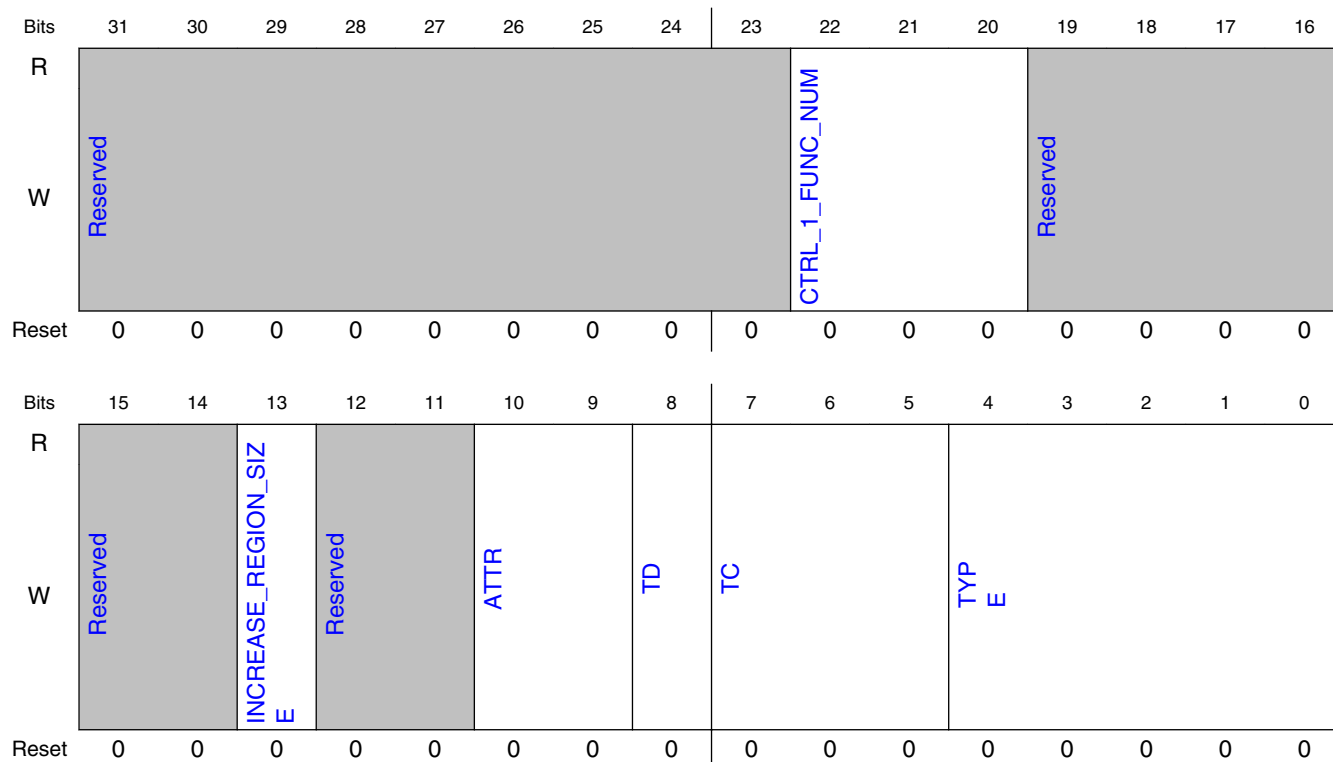
Field	Function
15-0 LWR_TARGET_HW	Forms the LSB's of the Lower Target part of the new address of the translated region. The start address must be aligned to the Minimum Size of iATU Region kB boundary (in address match mode); and to the Bar size boundary (in BAR match mode) so that these bits are always '0'. If the BAR is smaller than the iATU region size, then the iATU target address must align to the iATU region size; otherwise it must align to the BAR size. A write to this location is ignored by the PCIe controller. - Field size depends on $\log_2(\text{Minimum Size of iATU Region})$ in address match mode. - Field size depends on $\log_2(\text{BAR_MASK} + 1)$ in BAR match mode.

11.3.3.1.170 iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_OUTBOUND_3)

11.3.3.1.170.1 Offset

Register	Offset
IATU_REGION_CTRL_1_OFF_OUTBOUND_3	8000_0600h

11.3.3.1.170.2 Diagram



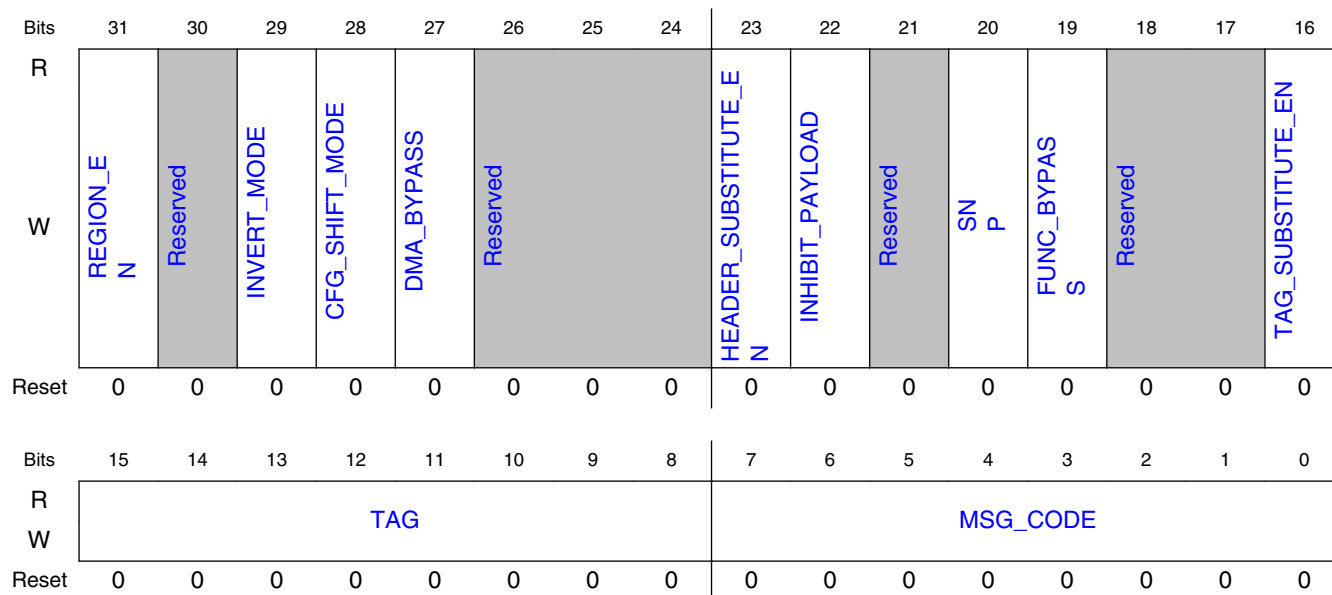
11.3.3.1.170.3 Fields

Field	Function
31-23 —	Reserved.
22-20 CTRL_1_FUNC_NUM	Function Number. - When the address of an outbound TLP is matched to this region and the FUNC_BYPASS field in the "iATU Control 2 Register" is '0', then the function number used in generating the function part of the requester ID (RID) field of the TLP is taken from this 5-bit register. When you are using the AXI Bridge, then this field is swapped before AXI decomposition occurs so that the correct "Max_Read_Request_Size" and "Max_Payload_Size" values are used. Note: This register field is sticky.
19-14 —	Reserved.
13 INCREASE_REGION_SIZE	Increase the maximum ATU Region size. When set, the size is determined by the Maximum Size of iATU Region. When clear, the maximum ATU Region size is 4 GB (default). Note: This register field is sticky.
12-11 —	Reserved.
10-9 ATTR	When the address of an outbound TLP is matched to this region, then the ATTR field of the TLP is changed to the value in this register. Note: This register field is sticky.
8 TD	When the address of an outbound TLP is matched to this region, then the TD field of the TLP is changed to the value in this register. Note: This register field is sticky.
7-5 TC	When the address of an outbound TLP is matched to this region, then the TC field of the TLP is changed to the value in this register. Note: This register field is sticky.
4-0 TYPE	When the address of an outbound TLP is matched to this region, then the TYPE field of the TLP is changed to the value in this register. Note: This register field is sticky.

11.3.3.1.171 iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_OUTBOUND_3)**11.3.3.1.171.1 Offset**

Register	Offset
IATU_REGION_CTRL_2_OFF_OUTBOUND_3	8000_0604h

11.3.3.1.171.2 Diagram



11.3.3.1.171.3 Fields

Field	Function
31 REGION_EN	Region Enable. This bit must be set to '1' for address translation to take place. Note: This register field is sticky.
30 —	Reserved.
29 INVERT_MODE	Invert Mode. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). Note: This register field is sticky.
28 CFG_SHIFT_MODE	CFG Shift Mode. The iATU uses bits [27:12] of the untranslated address (on the XALI0/1/2 interface or AXI slave interface address) to form the BDF number of the outgoing CFG TLP. This supports the Enhanced Configuration Address Mapping (ECAM) mechanism (Section 7.2.2 of the PCI Express Base 3.1 Specification, revision 1.0) by allowing all outgoing I/O and MEM TLPs (that have been translated to CFG) to be mapped from memory space into any 256 MB region of the PCIe configuration space. Note: This register field is sticky.
27 DMA_BYPASS	DMA Bypass Mode. Allows request TLPs which are initiated by the DMA controller to pass through the iATU untranslated. Note: This field is reserved for the SW product. You must set it to '0'. Note: This register field is sticky.
26-24 —	Reserved.
23 HEADER_SUBSTITUTE_EN	Header Substitute Enable. When enabled and region address is matched, the iATU fully substitutes bytes 8-11 (for 3 DWORD header) or bytes 12-15 (for 4 DWORD header) of the outbound TLP header with the contents of the LWR_TARGET_RW field in IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i. - 1: LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i register is used to fill bytes 8-to-11 (for 3 DWORD header) or bytes 12-to-15 (for 4 DWORD header) of the translated TLP header. - 0: LWR_TARGET_RW in the iATU_LWR_TARGET_ADDR_OFF_OUTBOUND_i register forms the new address of the translated region. Note: This register field is sticky.

Table continues on the next page...

Field	Function
22 INHIBIT_PAYLOAD_AD	Inhibit TLP Payload Data for TLP's in Matched Region; assign iATU region to be TLP without data. When enabled and region address is matched, the iATU marks all TLPs as having no payload data by forcing the TLP header Fmt[1] bit =0, regardless of the application inputs such as slv_wstrb. - 1: Fmt[1] =0 so that only TLP type without data is sent. For example, a Msg instead of MsgD will be sent. - 0: Fmt[1] =0/1 so that TLPs with or without data can be sent. Note: This register field is sticky.
21 —	Reserved.
20 SNP	Serialize Non-Posted Requests. In this mode, when the AXI Bridge is populated, same AXI ID Non-Posted Read/Write Requests are transmitted on the wire if there are no other same ID Non-Posted Requests outstanding. Note: This register field is sticky.
19 FUNC_BYPASS	Function Number Translation Bypass. In this mode, the function number of the translated TLP is taken from your application transmit interface and not from the CTRL_1_FUNC_NUM field of the "iATU Control 1 Register" or the VF_NUMBER field of the "iATU Control 3 Register." Note: This register field is sticky.
18-17 —	Reserved.
16 TAG_SUBSTITUTE_EN	TAG Substitute Enable. When enabled and region address is matched, the iATU substitutes the TAG field of the outbound TLP header with the contents of the TAG field in this register. The expected usage scenario is translation from AXI MWr to Vendor Defined Msg/MsgD. Note: This register field is sticky.
15-8 TAG	TAG. The substituted TAG field (byte 6) in the outgoing TLP header when TAG_SUBSTITUTE_EN is set. Note: This register field is sticky.
7-0 MSG_CODE	MSG TLPs (Message Code). When the address of an outbound TLP is matched to this region, and the translated TLP TYPE field is Msg or MsgD; then the message field of the TLP is changed to the value in this register. Memory TLPs: (ST;Steering Tag). When the ST field of an outbound TLP is matched to this region, and the translated TLP TYPE field targets memory space; then the ST field of the TLP is changed to the value in this register. Note: This register field is sticky.

11.3.3.1.172 iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_OFF_OUTBOUND_3)

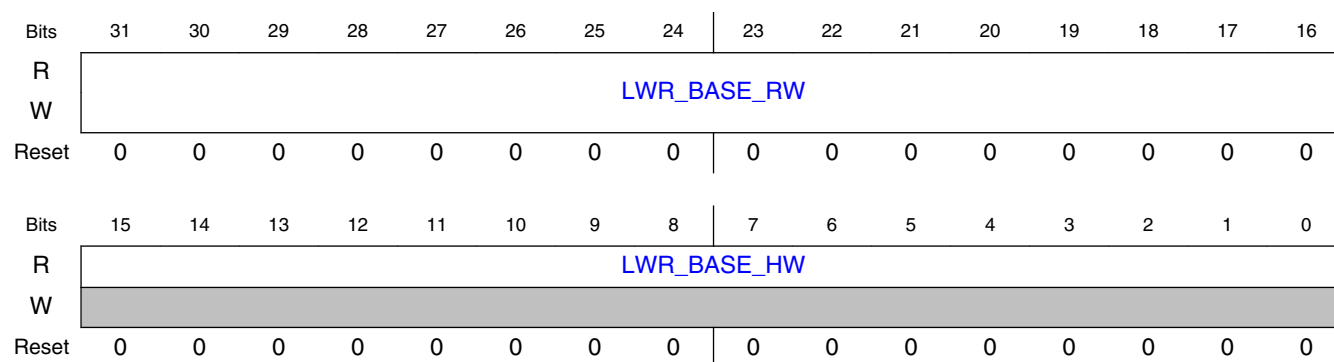
11.3.3.1.172.1 Offset

Register	Offset
IATU_LWR_BASE_ADDR_OFF_OUTBOUND_3	8000_0608h

11.3.3.1.172.2 Function

iATU Lower Base Address Register. The Minimum Size of iATU Region (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{Minimum Size of iATU Region})$ bits are zero.

11.3.3.1.172.3 Diagram



11.3.3.1.172.4 Fields

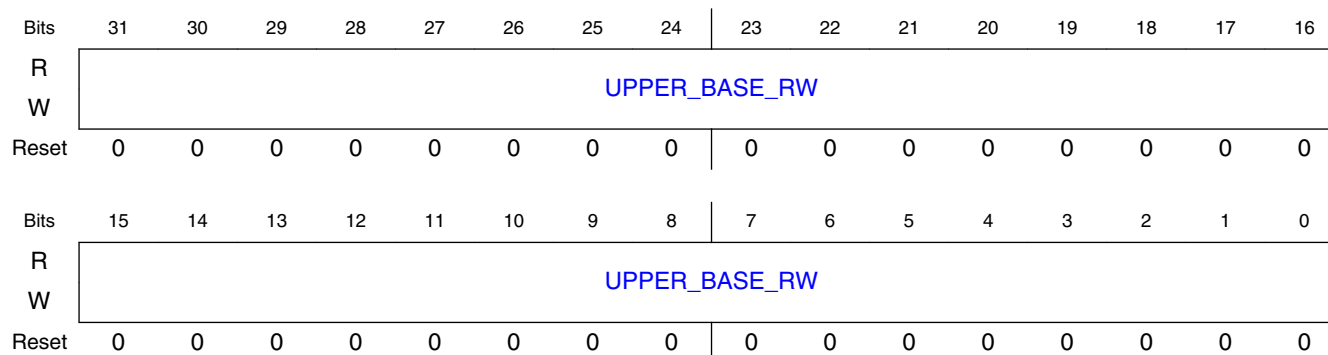
Field	Function
31-16 LWR_BASE_R W	Forms bits [31:n] of the start address of the address region to be translated. n is log2(Minimum Size of iATU Region) Note: This register field is sticky.
15-0 LWR_BASE_H W	Forms bits [n-1:0] of the start address of the address region to be translated. The start address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is log2(Minimum Size of iATU Region)

11.3.3.1.173 iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_3)

11.3.3.1.173.1 Offset

Register	Offset
IATU_UPPER_BASE_ADDR_OFF_OUTBOUND_3	8000_060Ch

11.3.3.1.173.2 Diagram



11.3.3.1.173.3 Fields

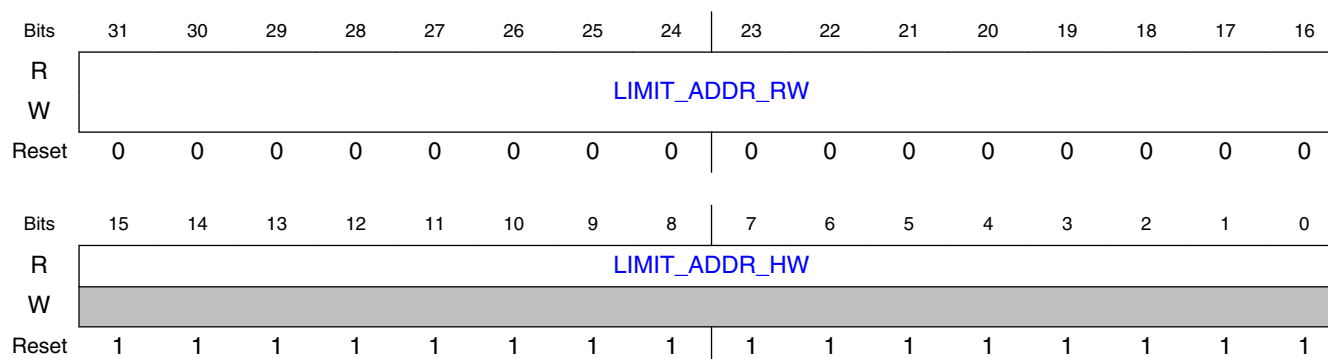
Field	Function
31-0 UPPER_BASE_ RW	Forms bits [63:32] of the start (and end) address of the address region to be translated. In systems with a 32-bit address space, this register is not used and therefore writing to this register has no effect. Note: This register field is sticky.

11.3.3.1.174 iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_OUTBOUND_3)

11.3.3.1.174.1 Offset

Register	Offset
IATU_LIMIT_ADDR_OFF_OUTBOUND_3	8000_0610h

11.3.3.1.174.2 Diagram



11.3.3.1.174.3 Fields

Field	Function
31-16 LIMIT_ADDR_R W	Forms upper bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. Note: This register field is sticky.
15-0 LIMIT_ADDR_H W	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller.

11.3.3.1.175 iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_3)

11.3.3.1.175.1 Offset

Register	Offset
IATU_LWR_TARGET_ADDR_OFF_OUTBOUND_3	8000_0614h

11.3.3.1.175.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	LWR_TARGET_RW_OUTBOUND															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	LWR_TARGET_RW_OUTBOUND															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.175.3 Fields

Field	Function
31-0 LWR_TARGET_RW_OUTBOUND	When HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_OFF_OUTBOUND_ is '0' (normal operation): - LWR_TARGET_RW[31:n] forms MSB's of the Lower Target part of the new address of the translated region; - LWR_TARGET_RW[n-1:0] are not used. (The start address must be aligned to the Minimum Size of iATU Region kB boundary, so the lower bits of the start address of the new address of the translated region (bits n-1:0) are always '0'). - n is log2(Minimum Size of iATU Region). When

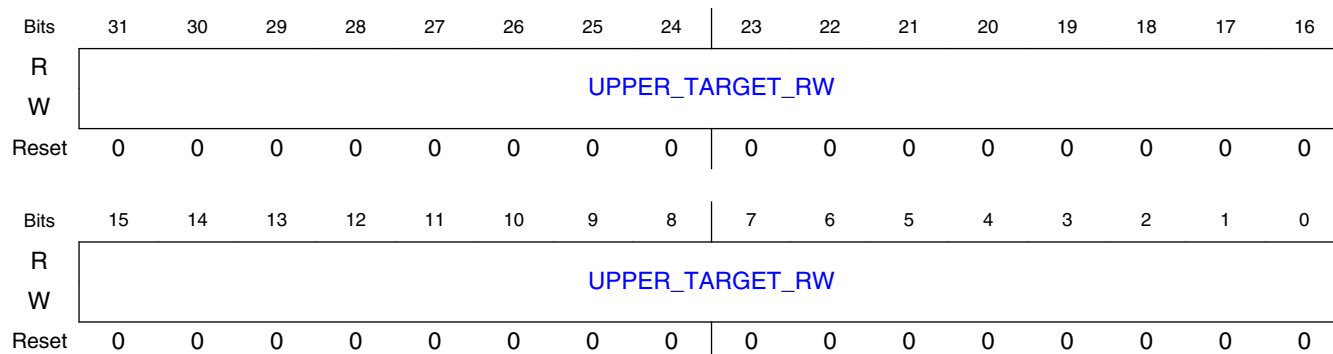
Field	Function
	HEADER_SUBSTITUTE_EN in IATU_REGION_CTRL_2_OFF_OUTBOUND_i is '1': - LWR_TARGET_RW[31:0] forms bytes 8-11 (for 3 dword header) or bytes 12-15 (for 4 dword header) of the outbound TLP header. Usage scenarios include the transmission of Vendor Defined Messages where the controller determines the content of bytes 12 to 15 of the TLP header. Note: This register field is sticky.

11.3.3.1.176 iATU Upper Target Address Register. (IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_3)

11.3.3.1.176.1 Offset

Register	Offset
IATU_UPPER_TARGET_ADDR_OFF_OUTBOUND_3	8000_0618h

11.3.3.1.176.2 Diagram



11.3.3.1.176.3 Fields

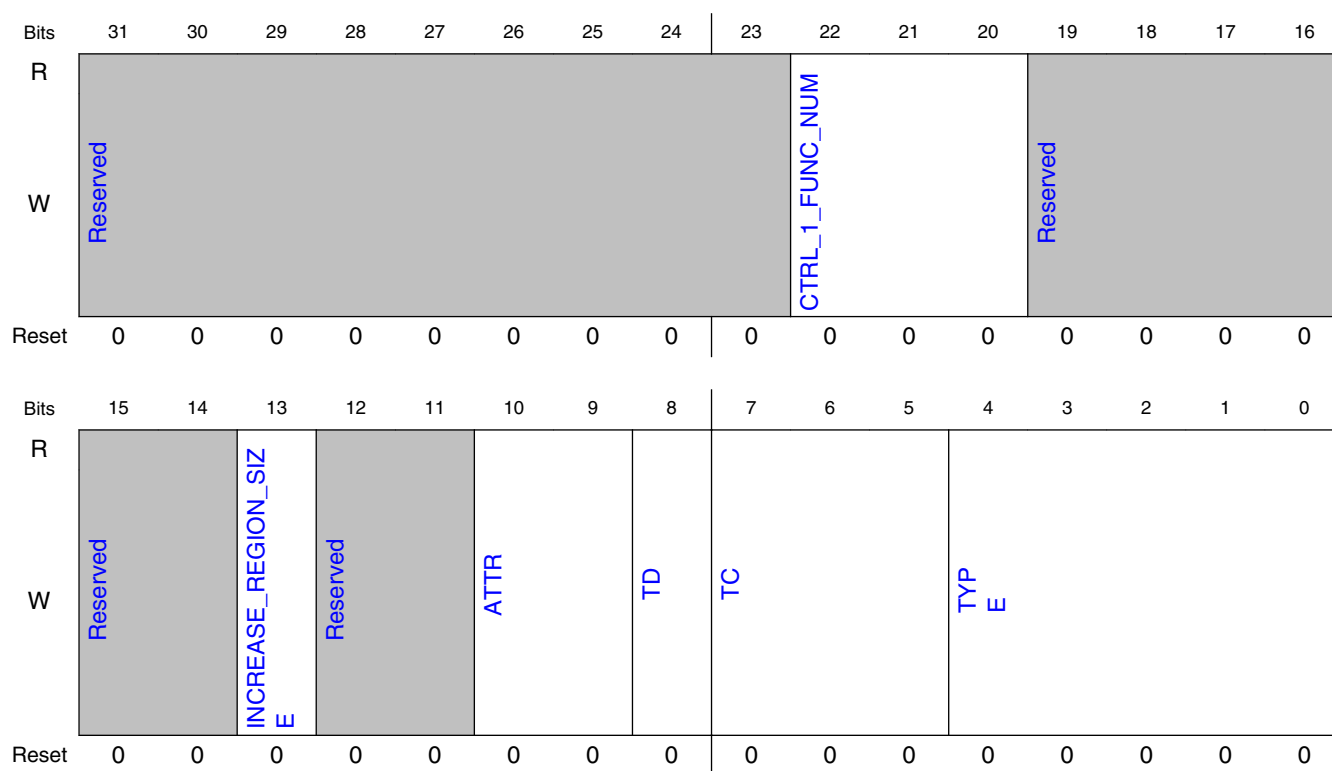
Field	Function
31-0 UPPER_TARGET_RW	Forms bits [63:32] of the start address (Upper Target part) of the new address of the translated region. Note: This register field is sticky.

11.3.3.1.177 iATU Region Control 1 Register. (IATU_REGION_CTRL_1_OFF_INBOUND_3)

11.3.3.1.177.1 Offset

Register	Offset
IATU_REGION_CTRL_1_OFF_INBOUND_3	8000_0700h

11.3.3.1.177.2 Diagram



11.3.3.1.177.3 Fields

Field	Function
31-23 —	Reserved.
22-20 CTRL_1_FUNC_NUM	Function Number. - MEM-I/O: When the Address and BAR matching logic in the controller indicate that a MEM-I/O transaction matches a BAR in the function corresponding to this value, then address translation proceeds. This check is only performed if the "Function Number Match Enable" bit of the "iATU Control 2 Register" is set. - CFG0/CFG1: When the destination function number as specified in the routing ID of the TLP header matches the function, then address translation proceeds. This check is only performed if the

Table continues on the next page...

PCI Express (PCIe)

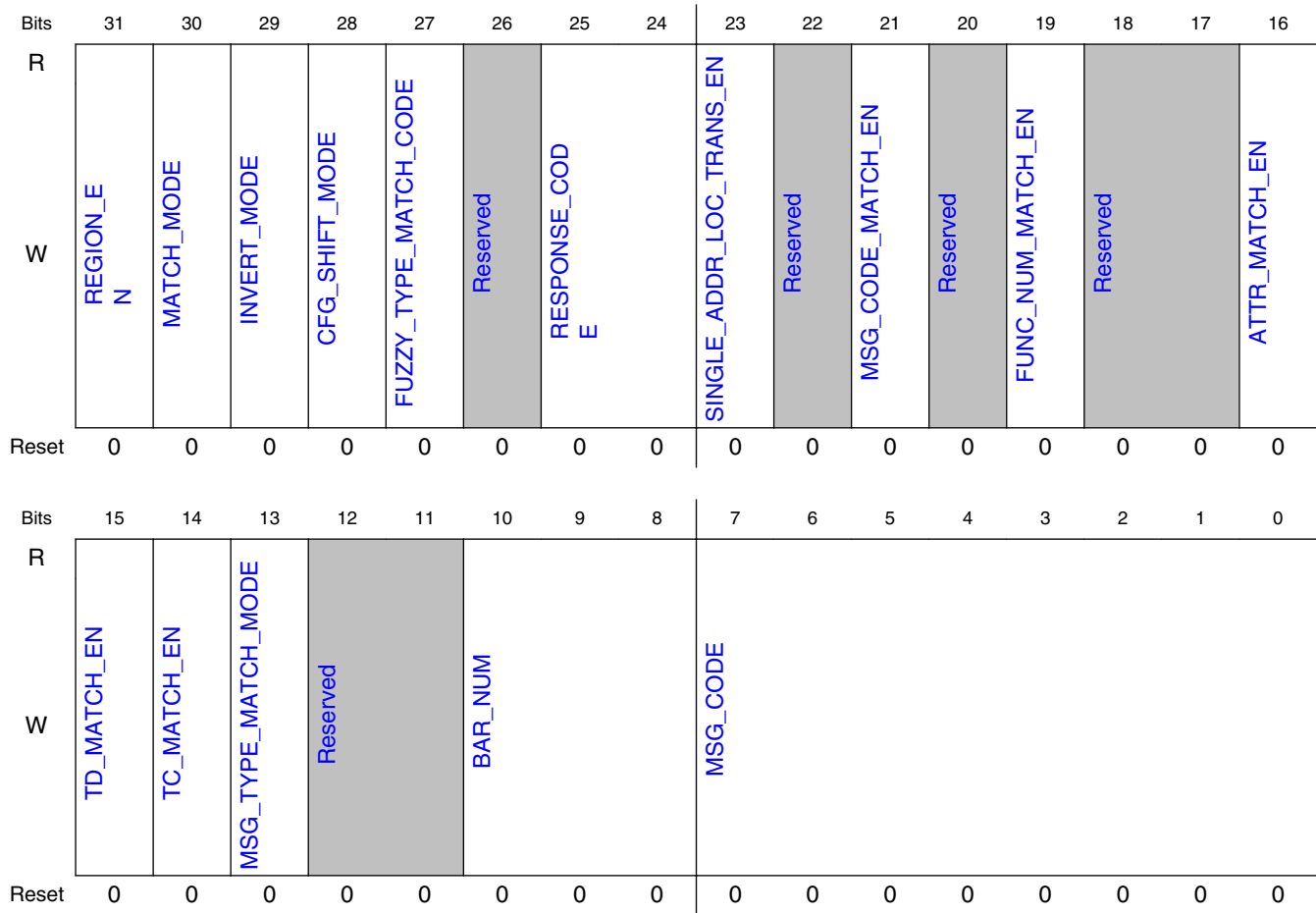
Field	Function
	"Function Number Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
19-14 —	Reserved.
13 INCREASE_REGION_SIZE	Increase the maximum ATU Region size. When set, the size is determined by the Maximum Size of iATU Region. When clear, the maximum ATU Region size is 4 GB (default). Note: This register field is sticky.
12-11 —	Reserved.
10-9 ATTR	When the ATTR field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "ATTR Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
8 TD	When the TD field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TD Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
7-5 TC	When the TC field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "TC Match Enable" bit of the "iATU Control 2 Register" is set. Note: This register field is sticky.
4-0 TYPE	When the TYPE field of an inbound TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). Note: This register field is sticky.

11.3.3.1.178 iATU Region Control 2 Register. (IATU_REGION_CTRL_2_OFF_INBOUND_3)

11.3.3.1.178.1 Offset

Register	Offset
IATU_REGION_CTRL_2_OFF_INBOUND_3	8000_0704h

11.3.3.1.178.2 Diagram



11.3.3.1.178.3 Fields

Field	Function
31 REGION_EN	Region Enable. This bit must be set to '1' for address translation to take place. Note: This register field is sticky.
30 MATCH_MODE	Match Mode. Determines Inbound matching mode for TLPs. The mode depends on the type of TLP that is received as follows: For MEM-I/O TLPs, this field is interpreted as follows: - 0: Address Match Mode. The iATU operates using addresses as in the outbound direction. The Region Base and Limit Registers must be setup. - 1: BAR Match Mode. BAR matching is used. The "BAR Number" field is relevant. Not used for RC. For CFG0 TLPs, this field is interpreted as follows: - 0: Routing ID Match Mode. The iATU interprets the Routing ID (Bytes 8 to 11 of TLP header) as an address. This corresponds to the upper 16 bits of the address in MEM-I/O transactions. The Routing ID of the TLP must be within the base and limit of the iATU region for matching to proceed. - 1: Accept Mode. The iATU accepts all CFG0 transactions as address matches. The routing ID in the CFG0 TLP is ignored. This is useful as all received CFG0 TLPs should be processed regardless of the Bus number. For MSG/MSGD TLPs, this field is interpreted as follows: - 0: Address Match Mode. The iATU treats the third dword and fourth dword of the inbound MSG/MSGD TLP as an address and it is matched against the Region Base and Limit Registers. - 1: Vendor ID Match Mode. This mode is relevant for ID-routed Vendor Defined Messages. The iATU ignores the Routing ID (Bus, Device, Function) in bits [31:16] of the third dword of the TLP header, but matches

Table continues on the next page...

Field	Function
	against the Vendor ID in bits [15:0] of the third dword of the TLP header. Bits [15:0] of the Region Upper Base register should be programmed with the required Vendor ID. The lower Base and Limit Register should be programmed to translate TLPs based on vendor specific information in the fourth dword of the TLP header. - If SINGLE_ADDRESS_LOCATION_TRANSLATE_EN = 1 AND MSG_TYPE_MATCH_MODE = 1, then Match Mode is ignored. Note: This register field is sticky.
29 INVERT_MODE	Invert Mode. When set the address matching region is inverted. Therefore, an address match occurs when the untranslated address is in the region outside the defined range (Base Address to Limit Address). Note: This register field is sticky.
28 CFG_SHIFT_M ODE	CFG Shift Mode. This is useful for CFG transactions where the PCIe configuration mechanism maps bits [27:12] of the address to the bus/device and function number. This allows a CFG configuration space to be located in any 256MB window of your application memory space using a 28-bit effective address. Shifts bits [31:16] of the untranslated address to form bits [27:12] of the translated address. Note: This register field is sticky.
27 FUZZY_TYPE_ MATCH_CODE	Fuzzy Type Match Mode. When enabled, the iATU relaxes the matching of the TLP TYPE field against the expected TYPE field so that - CfgRd0 and CfgRd1 TLPs are seen as identical. Similarly with CfgWr0 and CfgWr1. - MWr, MRd and MRdLk TLPs are seen as identical - The Routing field of Msg/MsgD TLPs is ignored - FetchAdd, Swap and CAS are seen as identical. For example, CFG0 in the TYPE field in the "iATU Control 1 Register" matches against an inbound CfgRd0, CfgRd1, CfgWr0 or CfgWr1 TLP. Note: This register field is sticky.
26 —	Reserved.
25-24 RESPONSE_C ODE	Response Code. Defines the type of response to give for accesses matching this region. This overrides the normal RADM filter response. Note that this feature is not available for any region where Single Address Location Translate is enabled. - 00 - Normal RADM filter response is used. - 01 - Unsupported request (UR) - 10 - Completer abort (CA) - 11 - Not used / undefined / reserved. Note: This register field is sticky.
23 SINGLE_ADDR_ LOC_TRANS_ EN	Single Address Location Translate Enable. When enabled, Rx TLPs can be translated to a single address location as determined by the target address register of the iATU region. The main usage scenario is translation of Messages (such as Vendor Defined or ATS Messages) to MWr TLPs when the AXI bridge is enabled. Note: This register field is sticky.
22 —	Reserved.
21 MSG_CODE_M ATCH_EN	Message Code Match Enable (Msg TLPS). Ensures that a successful message Code TLP field comparison match (see Message Code field of the "iATU Control 2 Register") occurs (in MSG transactions) for address translation to proceed. ST Match Enable (Mem TLPS). Ensures that a successful ST TLP field comparison match (see ST field of the "iATU Control 2 Register") occurs (in MEM transactions) for address translation to proceed. Note: This register field is sticky.
20 —	Reserved.
19 FUNC_NUM_M ATCH_EN	Function Number Match Enable. Ensures that a successful Function Number TLP field comparison match (see Function Number field of the "iATU Control 1 Register") occurs (in MEM-I/O and CFG0/CFG1 transactions) for address translation to proceed. Note: This register field is sticky.
18-17 —	Reserved.
16 ATTR_MATCH_ EN	ATTR Match Enable. Ensures that a successful ATTR TLP field comparison match (see ATTR field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.

Table continues on the next page...

Field	Function
15 TD_MATCH_EN	TD Match Enable. Ensures that a successful TD TLP field comparison match (see TD field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.
14 TC_MATCH_EN	TC Match Enable. Ensures that a successful TC TLP field comparison match (see TC field of the "iATU Control 1 Register") occurs for address translation to proceed. Note: This register field is sticky.
13 MSG_TYPE_MATCH_MODE	Message Type Match Mode. When enabled, and if single address location translate enable is set, then inbound TLPs of type MSG/MSGd which match the type field of the iatu_region_ctrl_1_OFF_inbound register ($\Rightarrow$ TYPE[4:3]=2'b10) will be translated. Message type match mode overrides any value of MATCH_MODE field in this register. Usage scenarios for this are translation of VDM or ATS messages when AXI bridge is configured on client interface. Note: This register field is sticky.
12-11 —	Reserved.
10-8 BAR_NUM	BAR Number. When the BAR number of an inbound MEM or IO TLP " that is matched by the normal internal BAR address matching mechanism " is the same as this field, address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Match Mode" bit of the "iATU Control 2 Register" is set. - 000b - BAR0 - 001b - BAR1 - 010b - BAR2 - 011b - BAR3 - 100b - BAR4 - 101b - BAR5 - 110b - ROM - 111b - reserved - IO translation would require either 00100b or 00101b in the inbound TLP TYPE; the BAR Number set in the range 000b - 101b and that BAR configured as an IO BAR. Note: This register field is sticky.
7-0 MSG_CODE	MSG TLPs: (Message Code). When the TYPE field of an inbound Msg/MsgD TLP is matched to this value, then address translation proceeds (when all other enabled field-matches are successful). This check is only performed if the "Message Code Match Enable" bit of the "iATU Control 2 Register" is set. Memory TLPs: (ST;Steering Tag). When the ST field of an inbound TLP is matched to this value, then address translation proceeds. This check is only performed if the "ST Match Enable" bit of the "iATU Control2 Register" is set. The setting is independent of the setting of the TH field. Note: This register field is sticky.

11.3.3.1.179 iATU Lower Base Address Register. (IATU_LWR_BASE_ADDR_OFF_INBOUND_3)

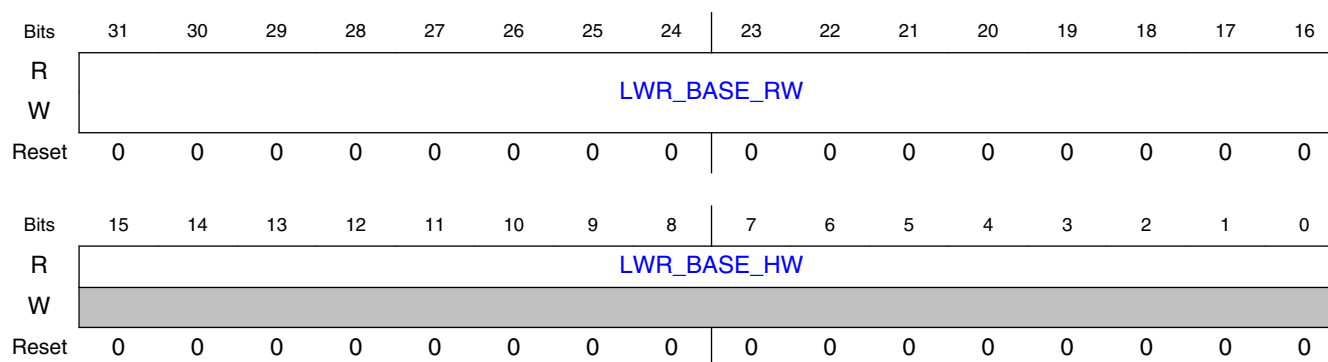
11.3.3.1.179.1 Offset

Register	Offset
IATU_LWR_BASE_ADDR_OFF_INBOUND_3	8000_0708h

11.3.3.1.179.2 Function

iATU Lower Base Address Register. The Minimum Size of iATU Region (Value Range: 4 kB, 8 kB, 16 kB, 32 kB, 64 kB defaults to 64 kB) specifies the minimum size of an address translation region. For example, if set to 64 kB; the lower 16 bits of the Base, Limit and Target registers are zero and all address regions are aligned on 64 kB boundaries. More precisely, the lower $\log_2(\text{Minimum Size of iATU Region})$ bits are zero.

11.3.3.1.179.3 Diagram



11.3.3.1.179.4 Fields

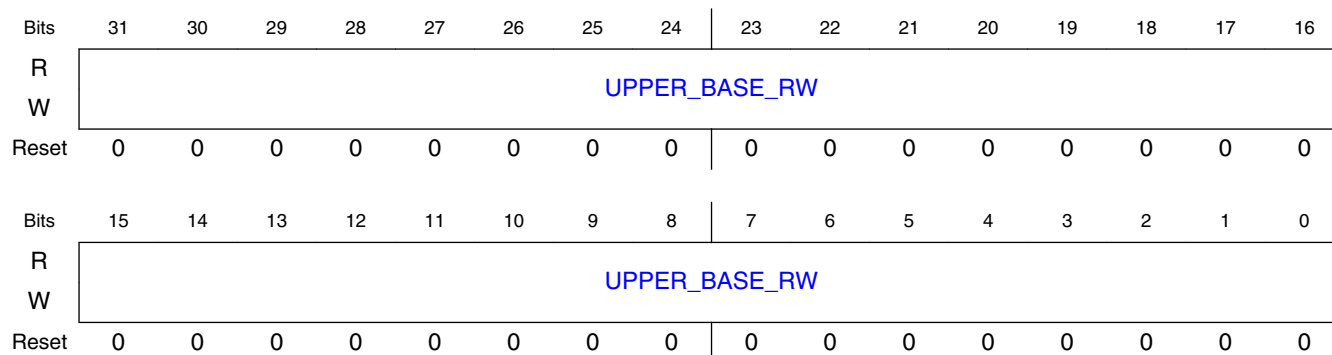
Field	Function
31-16 LWR_BASE_R W	Forms bits [31:n] of the start address of the address region to be translated. n is log2(Minimum Size of iATU Region) Note: This register field is sticky.
15-0 LWR_BASE_H W	Forms bits [n-1:0] of the start address of the address region to be translated. The start address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. n is log2(Minimum Size of iATU Region)

11.3.3.1.180 iATU Upper Base Address Register. (IATU_UPPER_BASE_ADDR_OFF_INBOUND_3)

11.3.3.1.180.1 Offset

Register	Offset
IATU_UPPER_BASE_ADDR_OFF_INBOUND_3	8000_070Ch

11.3.3.1.180.2 Diagram



11.3.3.1.180.3 Fields

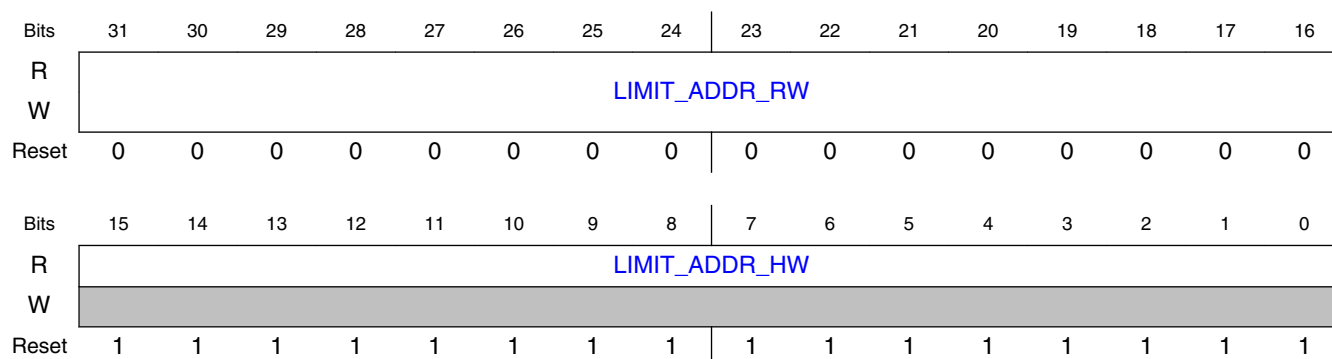
Field	Function
31-0 UPPER_BASE_ RW	Forms bits [63:32] of the start (and end) address of the address region to be translated. Note: This register field is sticky.

11.3.3.1.181 iATU Limit Address Register. (IATU_LIMIT_ADDR_OFF_INBOUND_3)

11.3.3.1.181.1 Offset

Register	Offset
IATU_LIMIT_ADDR_OFF_INBOUND_3	8000_0710h

11.3.3.1.181.2 Diagram



11.3.3.1.181.3 Fields

Field	Function
31-16 LIMIT_ADDR_R W	Forms upper bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller. Note: This register field is sticky.
15-0 LIMIT_ADDR_H W	Forms lower bits of the end address of the address region to be translated. The end address must be aligned to the Minimum Size of iATU Region kB boundary, so these bits are always 0. A write to this location is ignored by the PCIe controller.

11.3.3.1.182 iATU Lower Target Address Register. (IATU_LWR_TARGET_ADDR_OFF_INBOUND_3)

11.3.3.1.182.1 Offset

Register	Offset
IATU_LWR_TARGET_ADDR_OFF_INBOUND_3	8000_0714h

11.3.3.1.182.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	LWR_TARGET_RW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	LWR_TARGET_HW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.182.3 Fields

Field	Function
31-16 LWR_TARGET_R W	Forms MSB's of the Lower Target part of the new address of the translated region. These bits are always '0'. - Field size depends on log2(Minimum Size of iATU Region) in address match mode. - Field size depends on log2(BAR_MASK+1) in BAR match mode. Note: This register field is sticky.

Table continues on the next page...

Field	Function
15-0 LWR_TARGET_HW	Forms the LSB's of the Lower Target part of the new address of the translated region. The start address must be aligned to the Minimum Size of iATU Region kB boundary (in address match mode); and to the Bar size boundary (in BAR match mode) so that these bits are always '0'. If the BAR is smaller than the iATU region size, then the iATU target address must align to the iATU region size; otherwise it must align to the BAR size. A write to this location is ignored by the PCIe controller. - Field size depends on log2(Minimum Size of iATU Region) in address match mode. - Field size depends on log2(BAR_MASK +1) in BAR match mode.

11.3.3.1.183 DMA Arbitration Scheme for TRGT1 Interface. (DMA_CTRL_DATA_ARB_PRIOR_OFF)

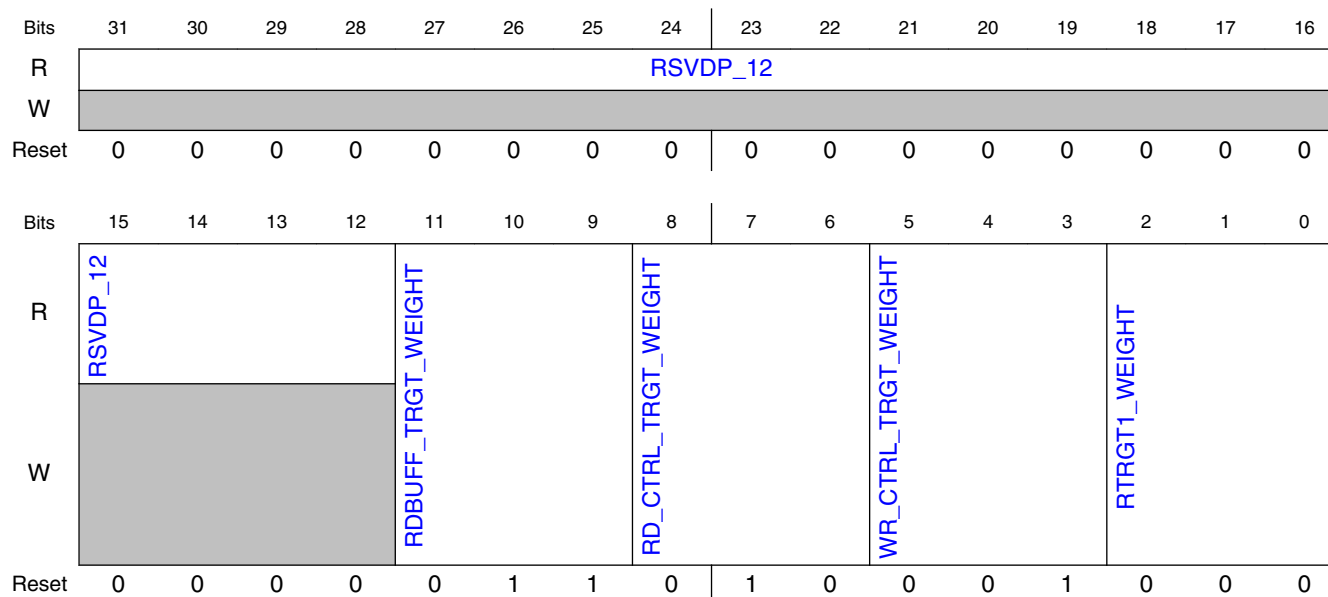
11.3.3.1.183.1 Offset

Register	Offset
DMA_CTRL_DATA_ARB_PRIOR_OFF	8008_0000h

11.3.3.1.183.2 Function

DMA Arbitration Scheme for TRGT1 Interface. This register is used to control traffic priorities among various sources that are delivered to your application through TRGT1 where 0x0 represents the highest priority. - Non-DMA Rx Requests - DMA Write Channel MRd Requests (DMA data requests and LL element/descriptor access) - DMA Read Channel MRd Requests (LL element/descriptor access) - DMA Read Channel MWr Requests Concurrent traffic from channels with same priority are sorted according to Round-Robin arbitration rules. The arbitration priority defaults to Non-DMA requests (highest), Write Channel MRd, Read Channel MRd, Read Channel MWr.

11.3.3.1.183.3 Diagram



11.3.3.1.183.4 Fields

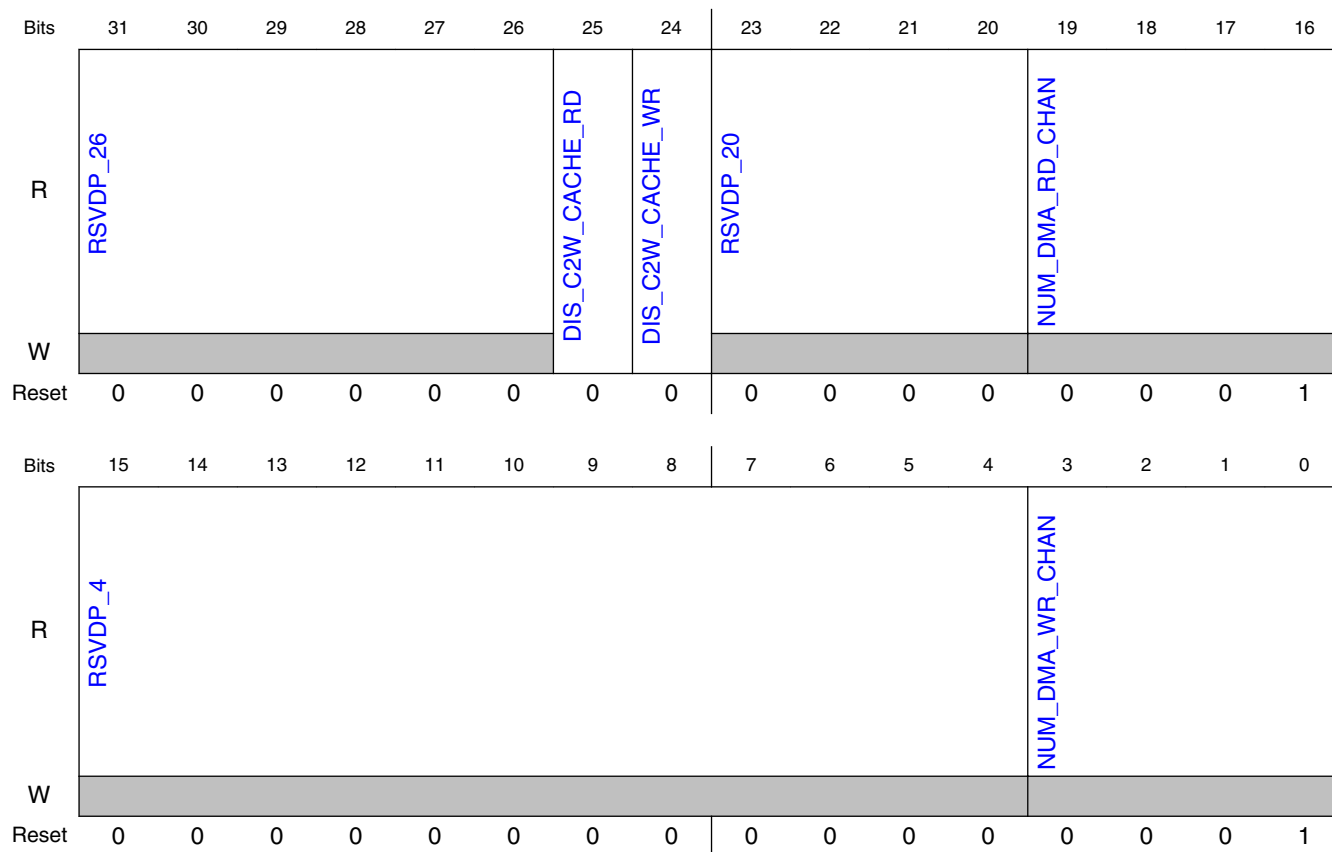
Field	Function
31-12 RSVDP_12	Reserved for future use.
11-9 RDBUFF_TRGT_WEIGHT	DMA Read Channel MWr Requests. Note: The access attributes of this field are as follows: - Dbi: R/W
8-6 RD_CTRL_TRGT_WEIGHT	DMA Read Channel MRd Requests. For LL element/descriptor access. Note: The access attributes of this field are as follows: - Dbi: R/W
5-3 WR_CTRL_TRGT_WEIGHT	DMA Write Channel MRd Requests. For DMA data requests and LL element/descriptor access. Note: The access attributes of this field are as follows: - Dbi: R/W
2-0 RTRGT1_WEIGHT	Non-DMA Rx Requests. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.184 DMA Number of Channels Register. (DMA_CTRL_OFF)

11.3.3.1.184.1 Offset

Register	Offset
DMA_CTRL_OFF	8008_0008h

11.3.3.1.184.2 Diagram



11.3.3.1.184.3 Fields

Field	Function
31-26 RSVDP_26	Reserved for future use.
25 DIS_C2W_CAC HE_RD	Disable DMA Read Channels "completion to memory write" context cache pre-fetch function. Note: For internal debugging only. Note: The access attributes of this field are as follows: - Dbi: R/W
24 DIS_C2W_CAC HE_WR	Disable DMA Write Channels "completion to memory write" context cache pre-fetch function. Note: For internal debugging only. Note: The access attributes of this field are as follows: - Dbi: R/W

Table continues on the next page...

PCI Express (PCIe)

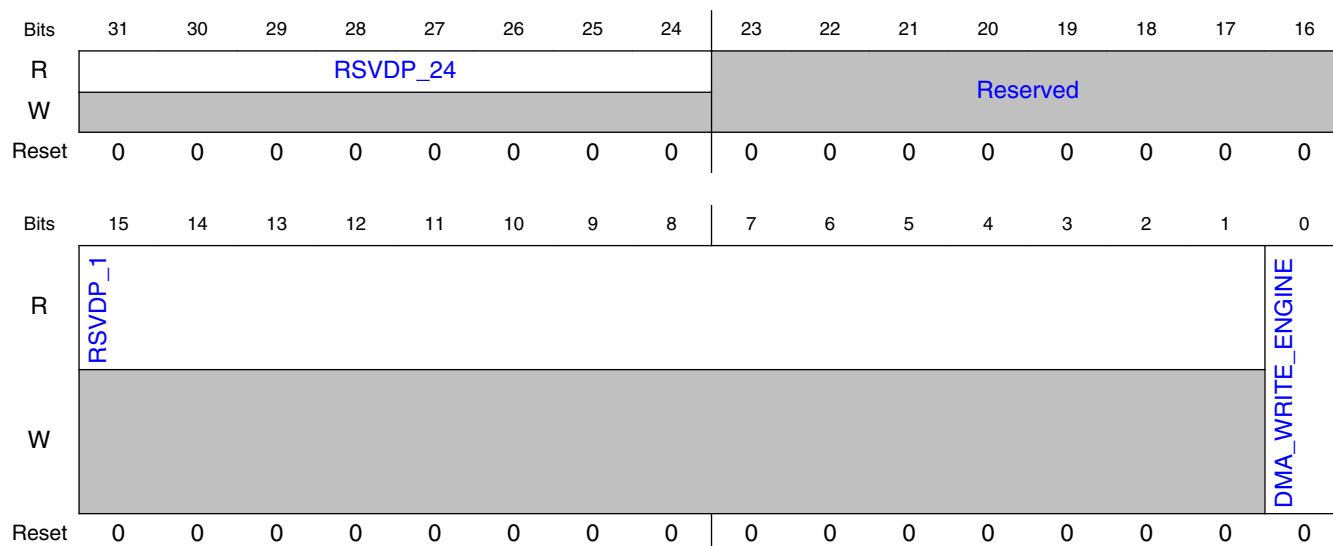
Field	Function
23-20 RSVDP_20	Reserved for future use.
19-16 NUM_DMA_RD_CHAN	Number of Read Channels. You can read this register to determine the number of read channels the DMA controller has been configured to support.
15-4 RSVDP_4	Reserved for future use.
3-0 NUM_DMA_WR_CHAN	Number of Write Channels. You can read this register to determine the number of write channels the DMA controller has been configured to support.

11.3.3.1.185 DMA Write Engine Enable Register. (DMA_WRITE_ENGINE_EN_OFF)

11.3.3.1.185.1 Offset

Register	Offset
DMA_WRITE_ENGINE_EN_OFF	8008_000Ch

11.3.3.1.185.2 Diagram



11.3.3.1.185.3 Fields

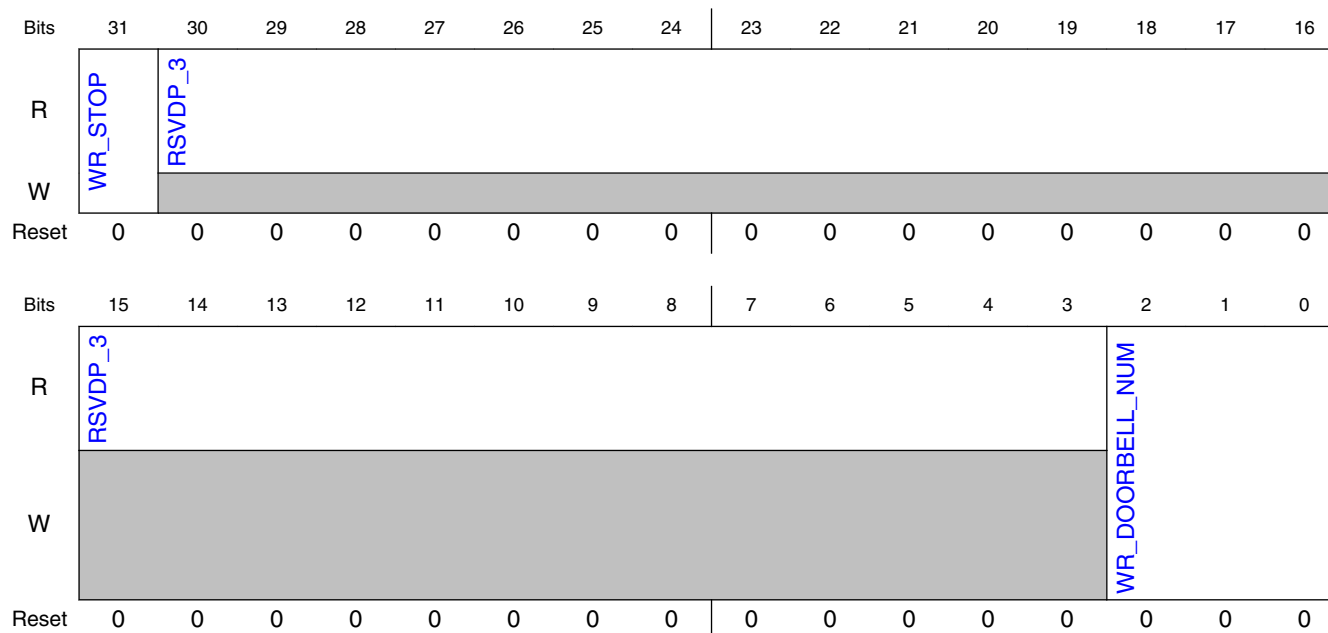
Field	Function
31-24 RSVDP_24	Reserved for future use.
23-16 —	Reserved.
15-1 RSVDP_1	Reserved for future use.
0 DMA_WRITE_ENGINE	DMA Write Engine Enable. - 1: Enable - 0: Disable (Soft Reset) For normal operation, you must initially set this bit to "1", before any other software setup actions. You do not need to toggle or rewrite to this bit during normal operation. You should set this bit to "0" when you want to "Soft Reset" the DMA controller write logic. There are three possible reasons for resetting the DMA controller write logic: - The "Abort Interrupt Status" bit is set (in the "DMA Write Interrupt Status Register" (DMA_WRITE_INT_STATUS_OFF), and any of the bits in the "DMA Write Error Status Register" (DMA_WRITE_ERR_STATUS_OFF) are set. Resetting the DMA controller write logic re-initializes the control logic, ensuring that the next DMA write transfer is executed successfully. - You have executed the procedure outlined in "Stop Bit", after which, the "Abort Interrupt Status" bit is set and the Channel Status field (CS) of the DMA write "DMA Channel Control 1 Register" (DMA_CH_CONTROL1_OFF_WRCH_0) is set to "Stopped." Resetting the DMA controller write logic re-initializes the control logic ensuring that the next DMA write transfer is executed successfully. - During software development, when you incorrectly program the DMA write engine. To "Soft Reset" the DMA controller write logic, you must: - De-assert the DMA write engine enable bit. - Wait for the DMA to complete any in-progress TLP transfer, by waiting until a read on the DMA write engine enable bit returns a "0". - Assert the DMA write engine enable bit. This "Soft Reset" does not clear the DMA configuration registers. The DMA write transfer does not start until you write to the "DMA Write Doorbell Register" (DMA_WRITE_DOORBELL_OFF). Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.186 DMA Write Doorbell Register. (DMA_WRITE_DOORBELL_OFF)

11.3.3.1.186.1 Offset

Register	Offset
DMA_WRITE_DOORBELL_OFF	8008_0010h

11.3.3.1.186.2 Diagram



11.3.3.1.186.3 Fields

Field	Function
31 WR_STOP	Stop. Set in conjunction with the Doorbell Number field. The DMA write channel stops issuing requests, sets the channel status to "Stopped", and asserts the "Abort" interrupt if it is enabled. Before setting the Stop bit, you must read the channel Status field (CS) of the "DMA Channel Control 1 Register" (DMA_CH_CONTROL1_OFF_WRCH_0) to ensure that the write channel is "Running" (transferring data). For more information, see "Stopping the DMA Transfer (Software Stop)." Note: The access attributes of this field are as follows: - Dbi: R/W
30-3 RSVDP_3	Reserved for future use.
2-0 WR_DOORBELL_NUM	Doorbell Number. You must write the channel number to this register to start the DMA write transfer for that channel. The DMA detects a write to this register field even if the value of this field does not change. You do not need to toggle or write any other value to this register to start a new transfer. The range of this field is 0x0 to 0x7, and 0x0 corresponds to channel 0. Also note that a write to this field triggers the controller to exit L1 substates. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.187 DMA Write Engine Channel Arbitration Weight Low Register.
(DMA_WRITE_CHANNEL_ARB_WEIGHT_LOW_OFF)

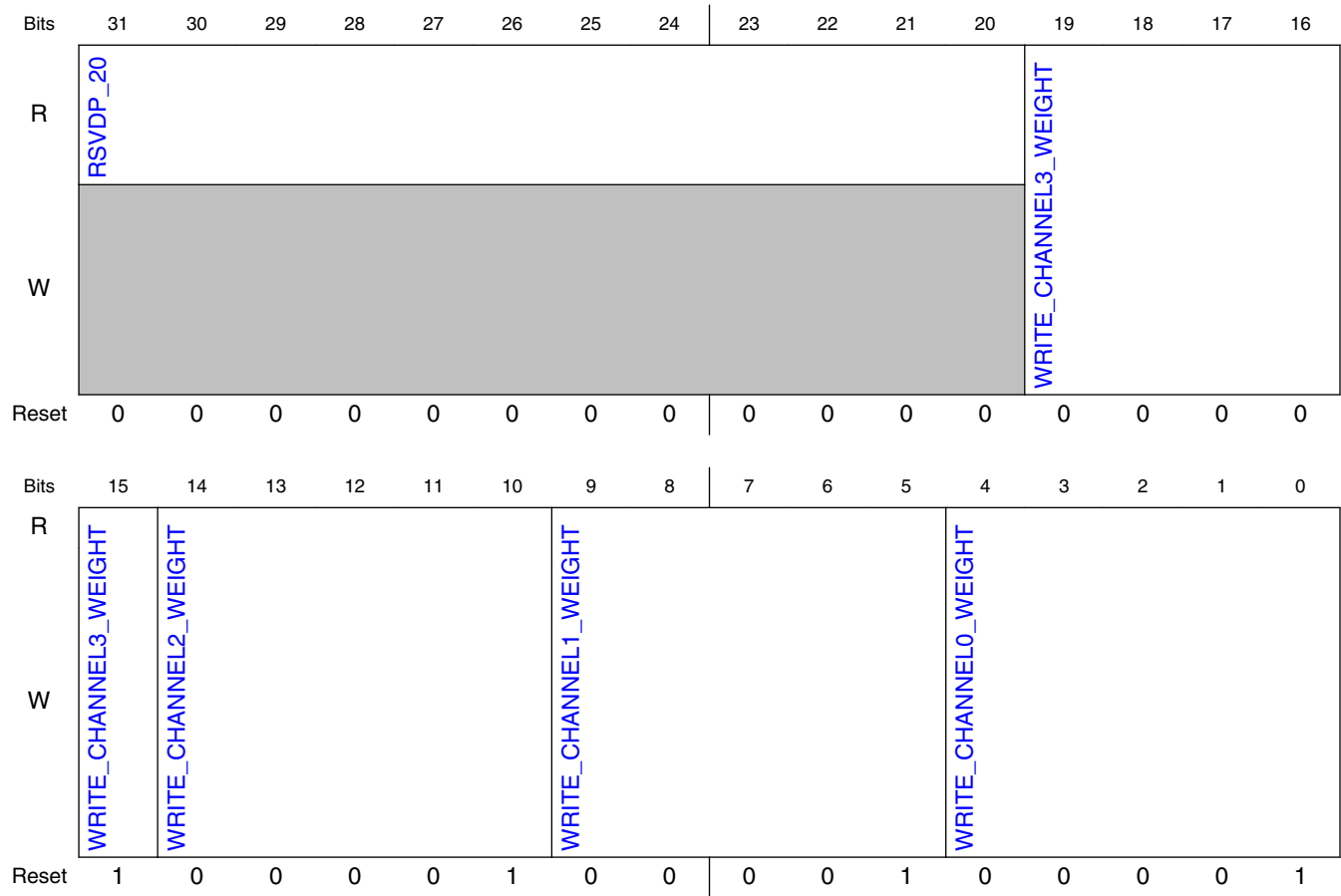
11.3.3.1.187.1 Offset

Register	Offset
DMA_WRITE_CHANNE L_ARB_WEIGHT_LOW_ OFF	8008_0018h

11.3.3.1.187.2 Function

DMA Write Engine Channel Arbitration Weight Low Register. The 5-bit channel weight (for write channels 0-3) specifies the maximum number of TLP requests that the DMA can issue for that channel before it must return to the arbitration routine. When the channel weight count is reached or DMA channel request transfer size reaches zero, the WWR arbiter selects the next channel to be processed. Your software must initialize this register before ringing the doorbell. For more details, see "Multichannel Arbitration". Value range is (0-0x1F) corresponding to (1-32) transaction requests.

11.3.3.1.187.3 Diagram



11.3.3.1.187.4 Fields

Field	Function
31-20 RSVDP_20	Reserved for future use.
19-15 WRITE_CHANN EL3_WEIGHT	Channel 3 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. Note: The access attributes of this field are as follows: - Dbi: R/W
14-10 WRITE_CHANN EL2_WEIGHT	Channel 2 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. Note: The access attributes of this field are as follows: - Dbi: R/W
9-5 WRITE_CHANN EL1_WEIGHT	Channel 1 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. Note: The access attributes of this field are as follows: - Dbi: R/W
4-0 WRITE_CHANN EL0_WEIGHT	Channel 0 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.188 DMA Write Engine Channel Arbitration Weight High Register. (DMA_WRITE_CHANNEL_ARB_WEIGHT_HIGH_OFF)

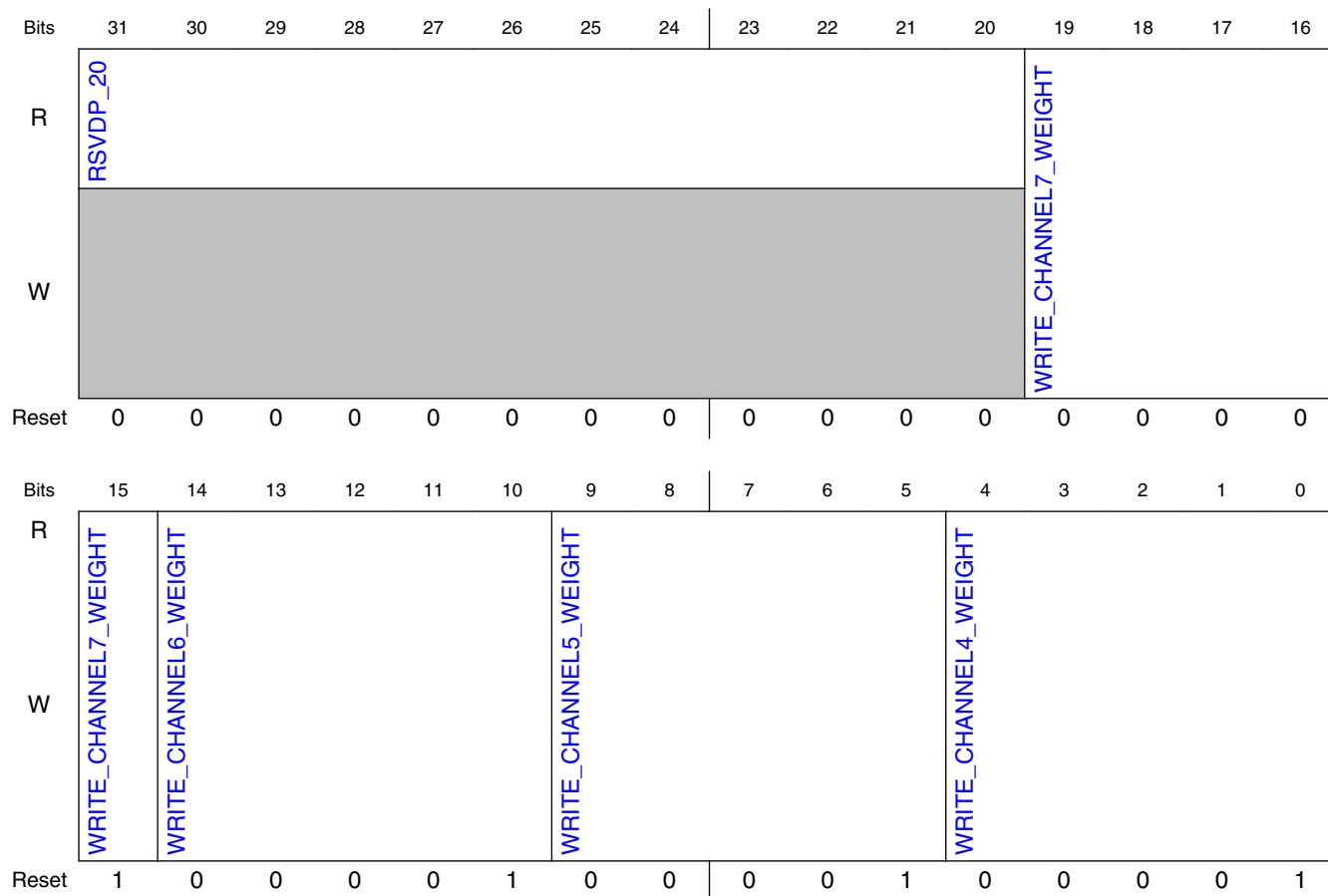
11.3.3.1.188.1 Offset

Register	Offset
DMA_WRITE_CHANNE L_ARB_WEIGHT_HIGH_ OFF	8008_001Ch

11.3.3.1.188.2 Function

DMA Write Engine Channel Arbitration Weight High Register. The 5-bit channel weight (for write channels 4-7) specifies the maximum number of TLP requests that the DMA can issue for that channel before it must return to the arbitration routine. When the channel weight count is reached or DMA channel request transfer size reaches zero, the WWR arbiter selects the next channel to be processed. Your software must initialize this register before ringing the doorbell. For more details, see "Multichannel Arbitration". Value range is (0-0x1F) corresponding to (1-32) transaction requests.

11.3.3.1.188.3 Diagram



11.3.3.1.188.4 Fields

Field	Function
31-20 RSVDP_20	Reserved for future use.
19-15 WRITE_CHANNEL7_WEIGHT	Channel 7 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. Note: The access attributes of this field are as follows: - Dbi: R/W
14-10 WRITE_CHANNEL6_WEIGHT	Channel 6 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. Note: The access attributes of this field are as follows: - Dbi: R/W
9-5 WRITE_CHANNEL5_WEIGHT	Channel 5 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means that one TLP is issued before moving to the next channel. Note: The access attributes of this field are as follows: - Dbi: R/W
4-0	Channel 4 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. A value of '0' means

PCI Express (PCIe)

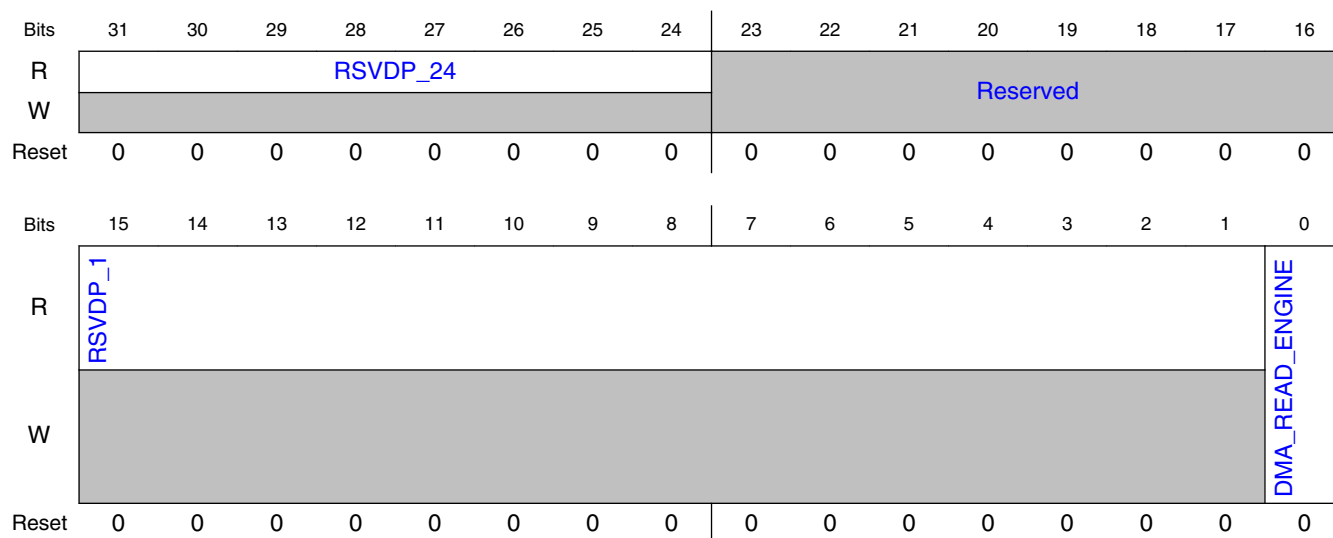
Field	Function
WRITE_CHANN EL4_WEIGHT	that one TLP is issued before moving to the next channel. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.189 DMA Read Engine Enable Register. (DMA_READ_ENGINE_EN_OFF)

11.3.3.1.189.1 Offset

Register	Offset
DMA_READ_ENGINE_ EN_OFF	8008_002Ch

11.3.3.1.189.2 Diagram



11.3.3.1.189.3 Fields

Field	Function
31-24 RSVDP_24	Reserved for future use.
23-16 —	Reserved.
15-1 RSVDP_1	Reserved for future use.

Table continues on the next page...

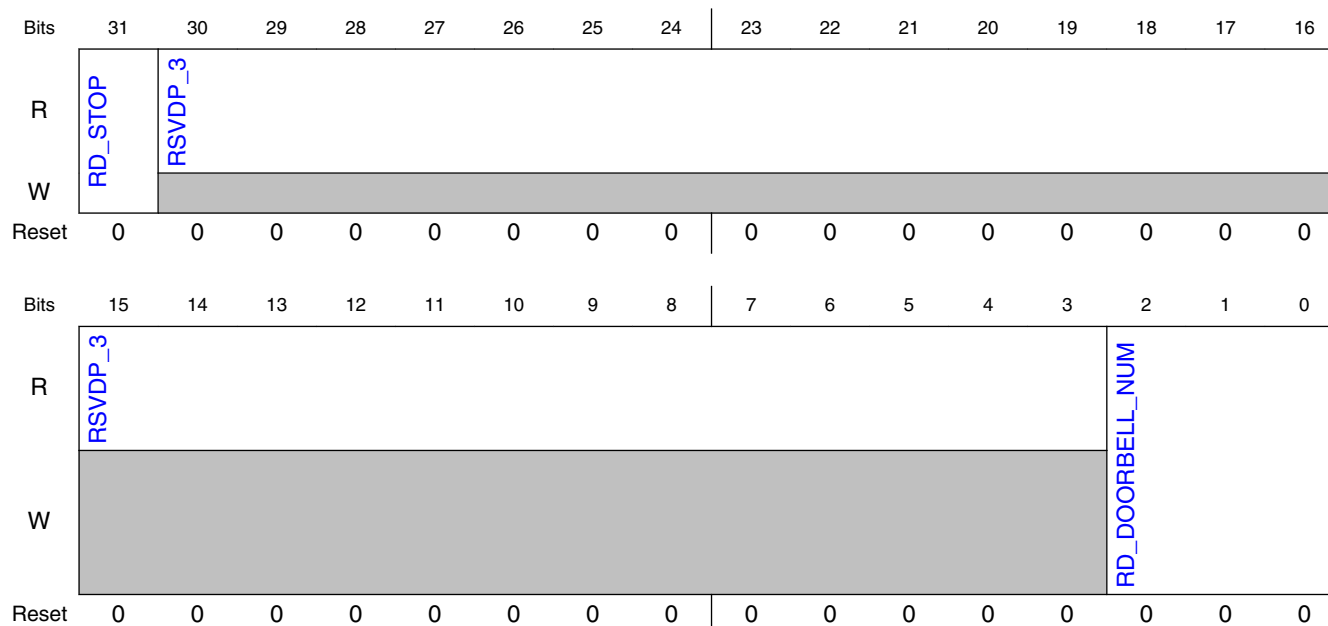
Field	Function
0 DMA_READ_ENGINE	DMA Read Engine Enable. - 1: Enable - 0: Disable (Soft Reset) For normal operation, you must initially set this bit to "1", before any other software setup actions. You do not need to toggle or rewrite to this bit during normal operation. You should set this field to "0" when you want to "Soft Reset" the DMA controller read logic. There are three possible reasons for resetting the DMA controller read logic: - The "Abort Interrupt Status" bit is set (in the "DMA Read Interrupt Status Register" (DMA_READ_INT_STATUS_OFF), and any of the bits in the "DMA Read Error Status Low Register" (DMA_READ_ERR_STATUS_LOW_OFF) is set. Resetting the DMA controller read logic re-initializes the control logic, ensuring that the next DMA read transfer is executed successfully. - You have executed the procedure outlined in "Stop Bit", after which, the "Abort Interrupt Status" bit is set and the channel Status field (CS) of the DMA read "DMA Channel Control 1 Register " (DMA_CH_CONTROL1_OFF_WRCH_0) is set to "Stopped". Resetting the DMA controller read logic re-initializes the control logic ensuring that the next DMA read transfer is executed successfully. - During software development, when you incorrectly program the DMA read engine. To "Soft Reset" the DMA controller read logic, you must: - De-assert the DMA read engine enable bit. - Wait for the DMA to complete any in-progress TLP transfer, by waiting until a read on the DMA read engine enable bit returns a "0". - Assert the DMA read engine enable bit. This "Soft Reset" does not clear the DMA configuration registers. The DMA read transfer does not start until you write to the "DMA Read Doorbell Register" (DMA_READ_DOORBELL_OFF). Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.190 DMA Read Doorbell Register. (DMA_READ_DOORBELL_OFF)

11.3.3.1.190.1 Offset

Register	Offset
DMA_READ_DOORBELL_OFF	8008_0030h

11.3.3.1.190.2 Diagram



11.3.3.1.190.3 Fields

Field	Function
31 RD_STOP	Stop. Set in conjunction with the Doorbell Number field. The DMA read channel stops issuing requests, sets the channel status to "Stopped", and asserts the "Abort" interrupt if it is enabled. Before setting the Stop bit, you must read the channel Status field (CS) of the "DMA Channel Control 1 Register" (DMA_CH_CONTROL1_OFF_RDCH_0) to ensure that the read channel is "Running" (transferring data). For more information, see "Stopping the DMA Transfer (Software Stop)". Note: The access attributes of this field are as follows: - Dbi: R/W
30-3 RSVDP_3	Reserved for future use.
2-0 RD_DOORBELL_NUM	Doorbell Number. You must write 0x0 to this register to start the DMA read transfer for that channel. The DMA detects a write to this register field even if the value of this field does not change. The range of this field is 0x0 to 0x7, and 0x0 corresponds to channel 0. Also note that a write to this field triggers the controller to exit L1 substates. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.191 DMA Read Engine Channel Arbitration Weight Low Register. (DMA_READ_CHANNEL_ARB_WEIGHT_LOW_OFF)

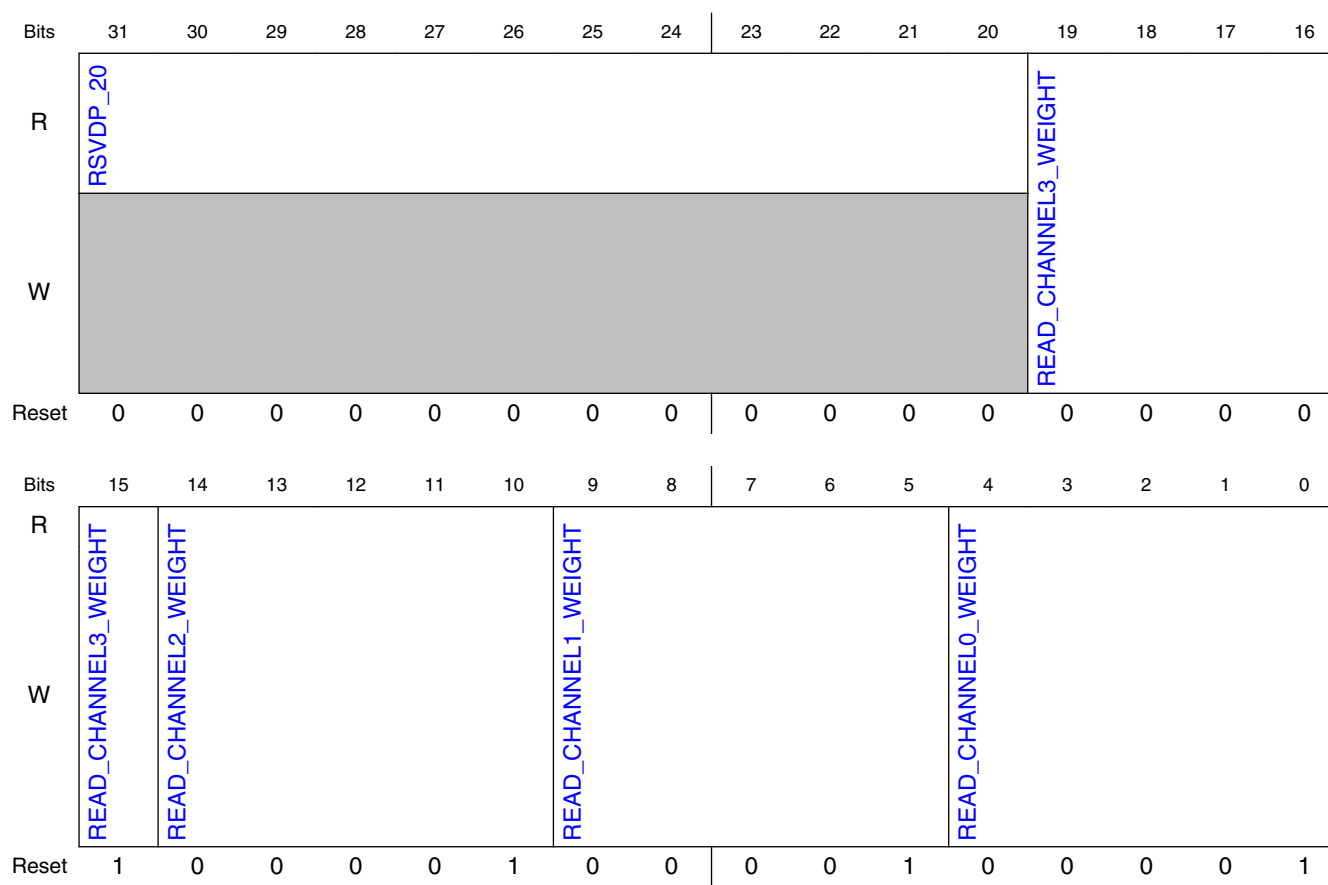
11.3.3.1.191.1 Offset

Register	Offset
DMA_READ_CHANNEL_ ARB_WEIGHT_LOW_ OFF	8008_0038h

11.3.3.1.191.2 Function

DMA Read Engine Channel Arbitration Weight Low Register. The 5-bit channel weight (for read channels 0-3) specifies the maximum number of TLP requests that the DMA can issue for that channel before it must return to the arbitration routine. When the channel weight count is reached or DMA channel request transfer size reaches zero, the WWR arbiter selects the next channel to be processed. Your software must initialize this register before ringing the doorbell. For more details, see "Multichannel Arbitration". Value range is (0-0x1F) corresponding to (1-32) transaction requests.

11.3.3.1.191.3 Diagram



11.3.3.1.191.4 Fields

Field	Function
31-20 RSVDP_20	Reserved for future use.
19-15 READ_CHANN EL3_WEIGHT	Channel 3 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. Note: The access attributes of this field are as follows: - Dbi: R/W
14-10 READ_CHANN EL2_WEIGHT	Channel 2 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. Note: The access attributes of this field are as follows: - Dbi: R/W
9-5 READ_CHANN EL1_WEIGHT	Channel 1 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. Note: The access attributes of this field are as follows: - Dbi: R/W
4-0 READ_CHANN EL0_WEIGHT	Channel 0 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. Note: The access attributes of this field are as follows: - Dbi: R/W

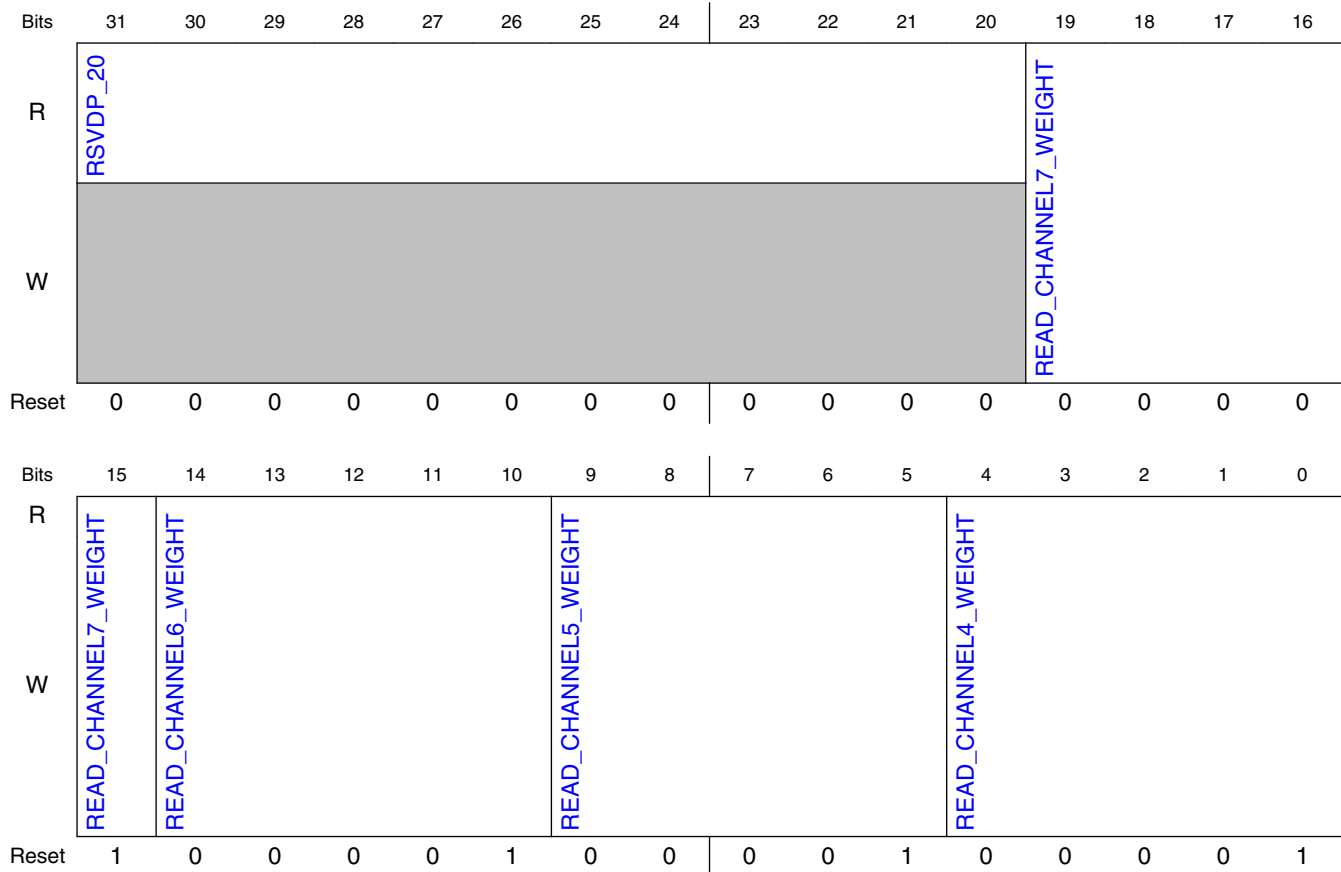
11.3.3.1.192 DMA Read Engine Channel Arbitration Weight High Register. (DMA_READ_CHANNEL_ARB_WEIGHT_HIGH_OFF)**11.3.3.1.192.1 Offset**

Register	Offset
DMA_READ_CHANNEL_ ARB_WEIGHT_HIGH_ OFF	8008_003Ch

11.3.3.1.192.2 Function

DMA Read Engine Channel Arbitration Weight High Register. The 5-bit channel weight (for read channels 4-7) specifies the maximum number of TLP requests that the DMA can issue for that channel before it must return to the arbitration routine. When the channel weight count is reached or DMA channel request transfer size reaches zero, the WWR arbiter selects the next channel to be processed. Your software must initialize this register before ringing the doorbell. For more details, see "Multichannel Arbitration". Value range is (0-0x1F) corresponding to (1-32) transaction requests.

11.3.3.1.192.3 Diagram



11.3.3.1.192.4 Fields

Field	Function
31-20 RSVDP_20	Reserved for future use.
19-15 READ_CHANN EL7_WEIGHT	Channel 7 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. Note: The access attributes of this field are as follows: - Dbi: R/W
14-10 READ_CHANN EL6_WEIGHT	Channel 6 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. Note: The access attributes of this field are as follows: - Dbi: R/W
9-5 READ_CHANN EL5_WEIGHT	Channel 5 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. Note: The access attributes of this field are as follows: - Dbi: R/W
4-0 READ_CHANN EL4_WEIGHT	Channel 4 Weight. The weight is initialized by software before ringing the doorbell. The value is used by the channel weighted round robin arbiter to select the next channel read request. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.193 DMA Write Interrupt Status Register. (DMA_WRITE_INT_ST ATUS_OFF)

11.3.3.1.193.1 Offset

Register	Offset
DMA_WRITE_INT_ST ATUS_OFF	8008_004Ch

11.3.3.1.193.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	RSVDP_24								WR_ABORT_INT_STATUS							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	RSVDP_8								WR_DONE_INT_STATUS							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.193.3 Fields

Field	Function
31-24 RSVDP_24	Reserved for future use.
23-16 WR_ABORT_IN T_STATUS	Abort Interrupt Status. The DMA write channel has detected an error, or you manually stopped the transfer as described in "Error Handling Assistance by Remote Software". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA write interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the DMA write interrupt Clear register to clear this interrupt bit. Note: You can write to this register to emulate interrupt generation, during software or hardware testing. A write to the address triggers an interrupt, but the DMA does not set the Done or Abort bits in this register. Note: The access attributes of this field are as follows: - Dbi: R/W
15-8 RSVDP_8	Reserved for future use.
7-0 WR_DONE_INT _STATUS	Done Interrupt Status. The DMA write channel has successfully completed the DMA transfer. For more details, see "Interrupts and Error Handling". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA write interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the DMA write interrupt Clear register to clear this interrupt bit. Note: You can write to this register to emulate interrupt generation, during software or hardware testing. A write to

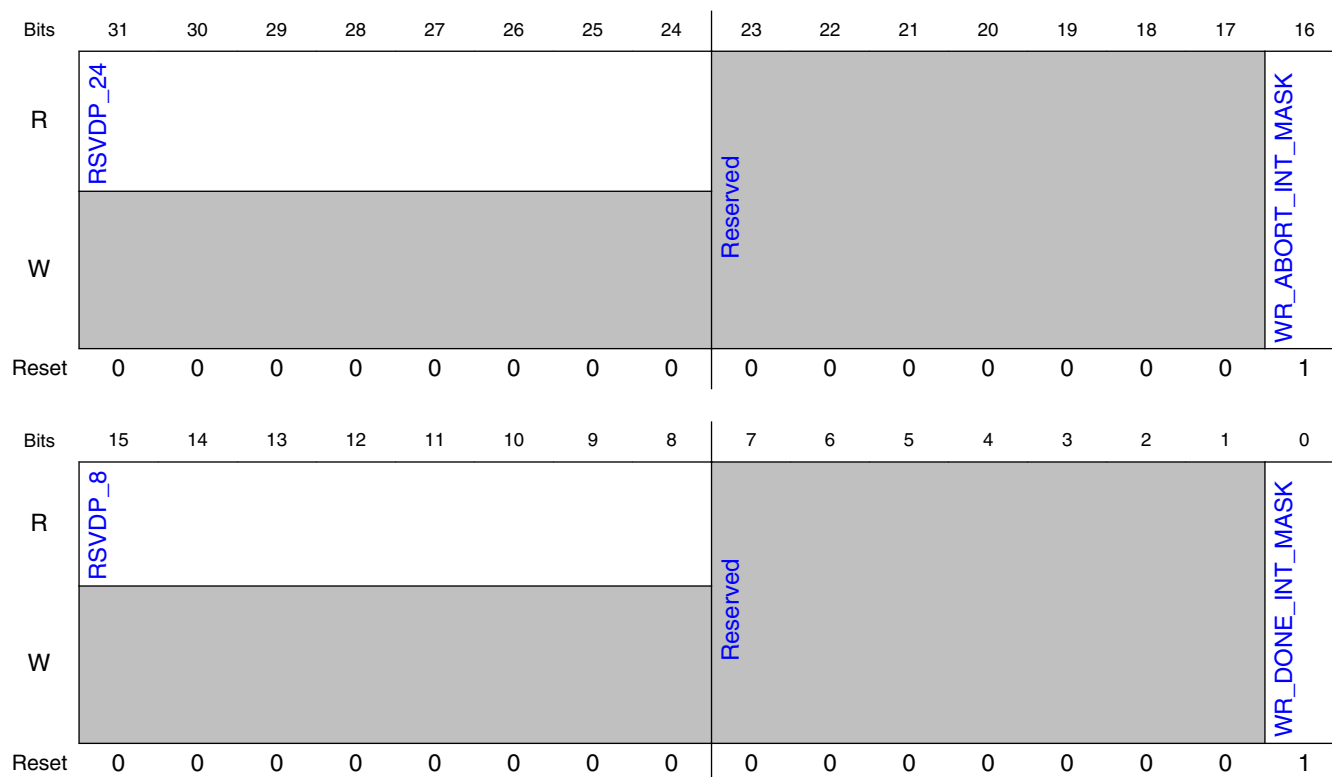
Field	Function
	the address triggers an interrupt, but the DMA does not set the Done or Abort bits in this register. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.194 DMA Write Interrupt Mask Register. (DMA_WRITE_INT_MASK_OFF)

11.3.3.1.194.1 Offset

Register	Offset
DMA_WRITE_INT_MASK_OFF	8008_0054h

11.3.3.1.194.2 Diagram



11.3.3.1.194.3 Fields

Field	Function
31-24	Reserved for future use.

Table continues on the next page...

PCI Express (PCIe)

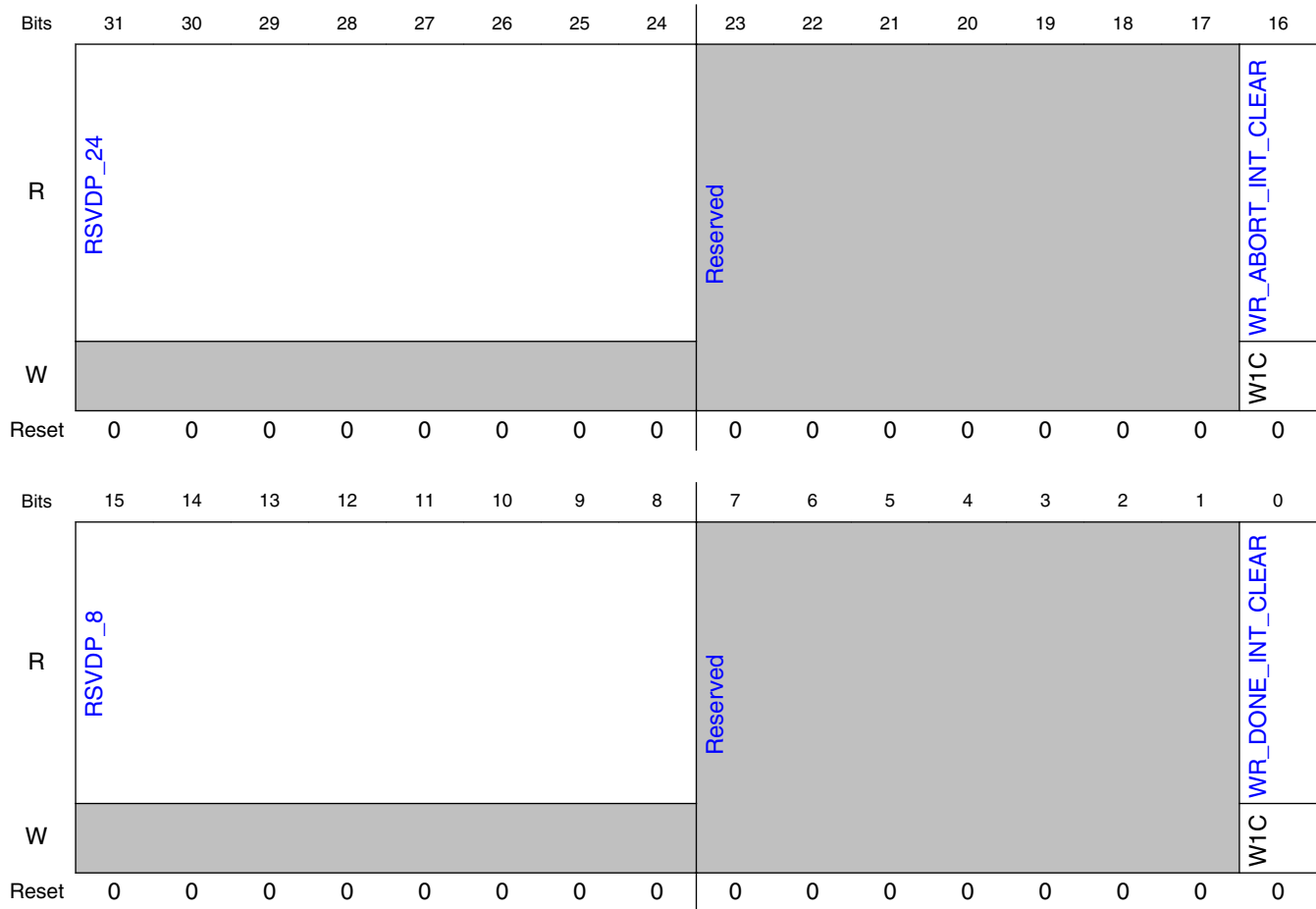
Field	Function
RSVDP_24	
23-17 —	Reserved.
16 WR_ABORT_INT_MASK	Abort Interrupt Mask. Prevents the Abort interrupt status field in the DMA write interrupt status register from asserting the edma_int output. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: The access attributes of this field are as follows: - Dbi: R/W
15-8 RSVDP_8	Reserved for future use.
7-1 —	Reserved.
0 WR_DONE_INT_MASK	Done Interrupt Mask. Prevents the Done interrupt status field in the DMA write interrupt status register from asserting the edma_int output. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.195 DMA Write Interrupt Clear Register. (DMA_WRITE_INT_CLEAR_OFF)

11.3.3.1.195.1 Offset

Register	Offset
DMA_WRITE_INT_CLEAR_OFF	8008_0058h

11.3.3.1.195.2 Diagram



11.3.3.1.195.3 Fields

Field	Function
31-24 RSVDP_24	Reserved for future use.
23-17 —	Reserved.
16 WR_ABORT_IN T_CLEAR	Abort Interrupt Clear. You must write a 1'b1 to clear the corresponding bit in the Abort interrupt status field of the DMA write interrupt status register. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: Reading from this self-clearing register field always returns a "0".
15-8 RSVDP_8	Reserved for future use.
7-1 —	Reserved.

Table continues on the next page...

PCI Express (PCIe)

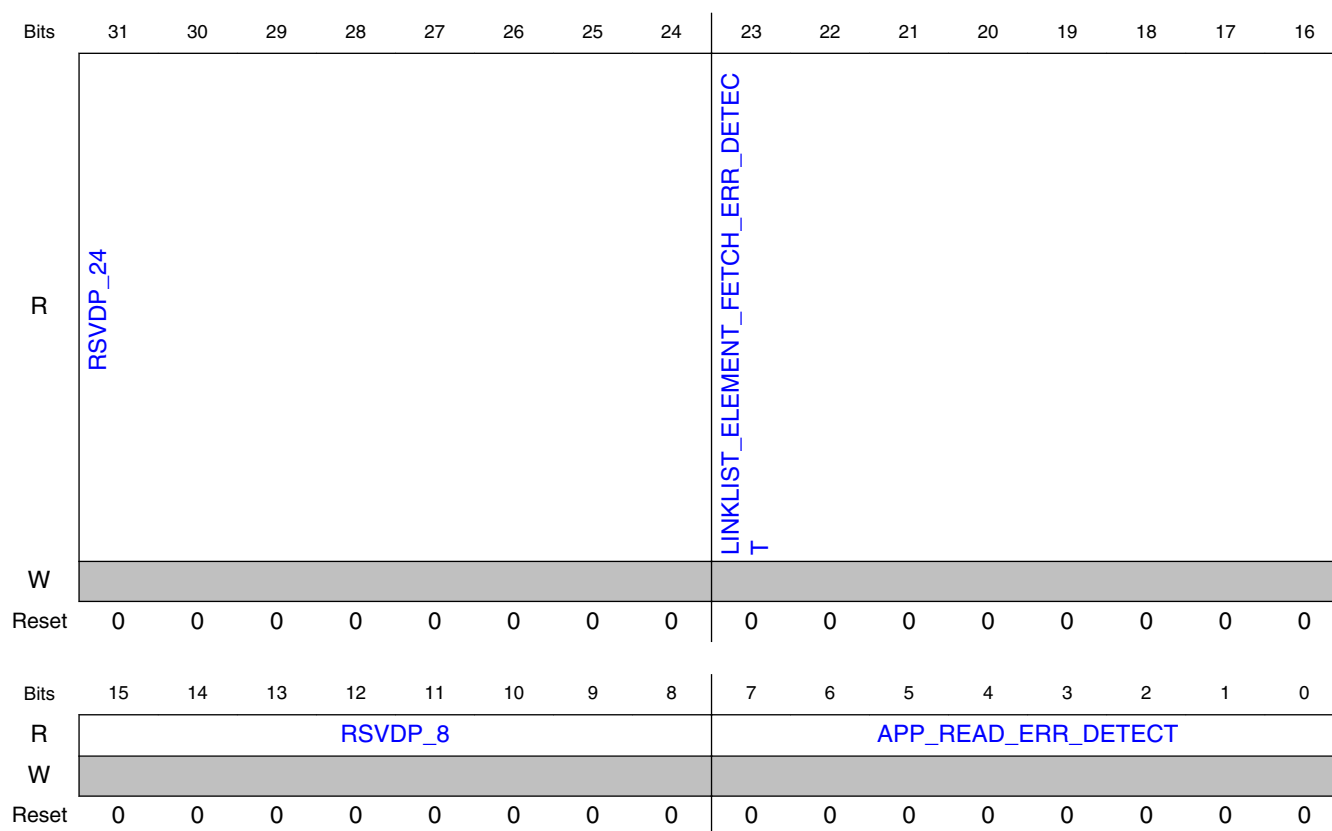
Field	Function
0 WR_DONE_INT_CLEAR	Done Interrupt Clear. You must write a 1'b1 to clear the corresponding bit in the Done interrupt status field of the DMA write interrupt status register. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: Reading from this self-clearing register field always returns a "0".

11.3.3.1.196 DMA Write Error Status Register (DMA_WRITE_ERR_STATUS_OFF)

11.3.3.1.196.1 Offset

Register	Offset
DMA_WRITE_ERR_STATUS_OFF	8008_005Ch

11.3.3.1.196.2 Diagram



11.3.3.1.196.3 Fields

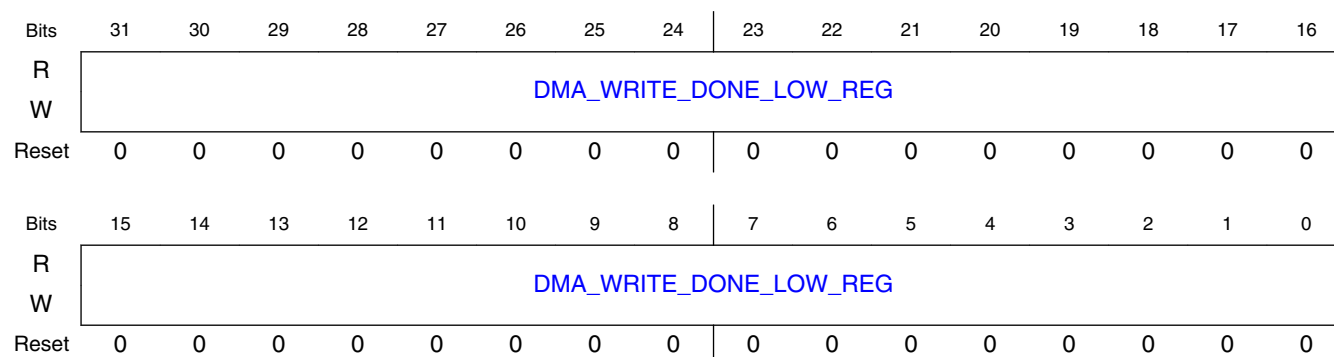
Field	Function
31-24 RSVDP_24	Reserved for future use.
23-16 LINKLIST_ELEMENT_FETCH_ERR_DETECT	Linked List Element Fetch Error Detected. The DMA write channel has received an error response from the AXI bus (or TRGT1 interface when the AXI Bridge is not used) while reading a linked list element from local memory. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA write interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the "DMA Write Interrupt Clear Register" (DMA_WRITE_INT_CLEAR_OFF) to clear this error bit.
15-8 RSVDP_8	Reserved for future use.
7-0 APP_READ_ERR_DETECT	Application Read Error Detected. The DMA write channel has received an error response from the AXI bus (or TRGT1 interface when the AXI Bridge is not used) while reading data from it. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA write interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the "DMA Write Interrupt Clear Register" (DMA_WRITE_INT_CLEAR_OFF) to clear this error bit.

11.3.3.1.197 DMA Write Done IMWr Address Low Register. (DMA_WRITE_DONE_IMWR_LOW_OFF)

11.3.3.1.197.1 Offset

Register	Offset
DMA_WRITE_DONE_IMWR_LOW_OFF	8008_0060h

11.3.3.1.197.2 Diagram



11.3.3.1.197.3 Fields

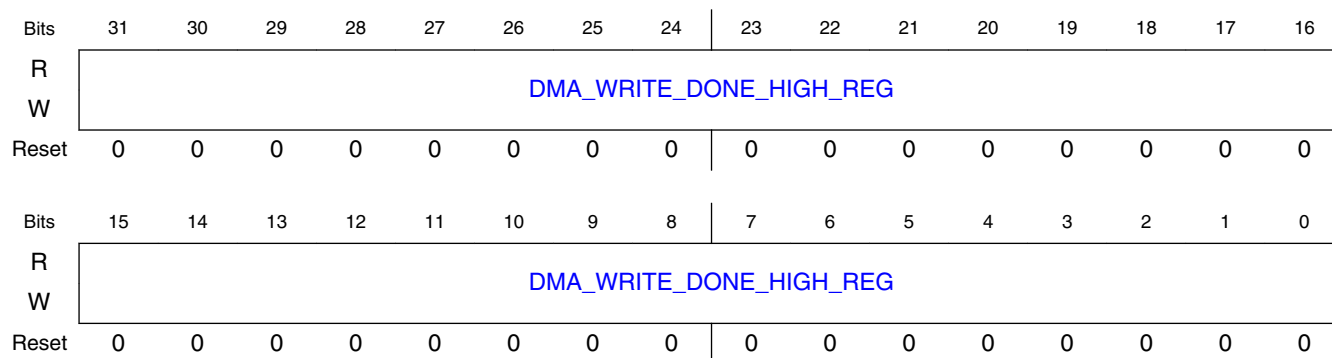
Field	Function
DMA_WRITE_DONE_LOW_REG 31-0	The DMA uses this field to generate bits [31:0] of the address field for the Done IMWr TLP. Bits [1:0] must be "00" as this address must be dword aligned. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.198 DMA Write Done IMWr Interrupt Address High Register. (DMA_WRITE_DONE_IMWR_HIGH_OFF)

11.3.3.1.198.1 Offset

Register	Offset
DMA_WRITE_DONE_IMWR_HIGH_OFF	8008_0064h

11.3.3.1.198.2 Diagram



11.3.3.1.198.3 Fields

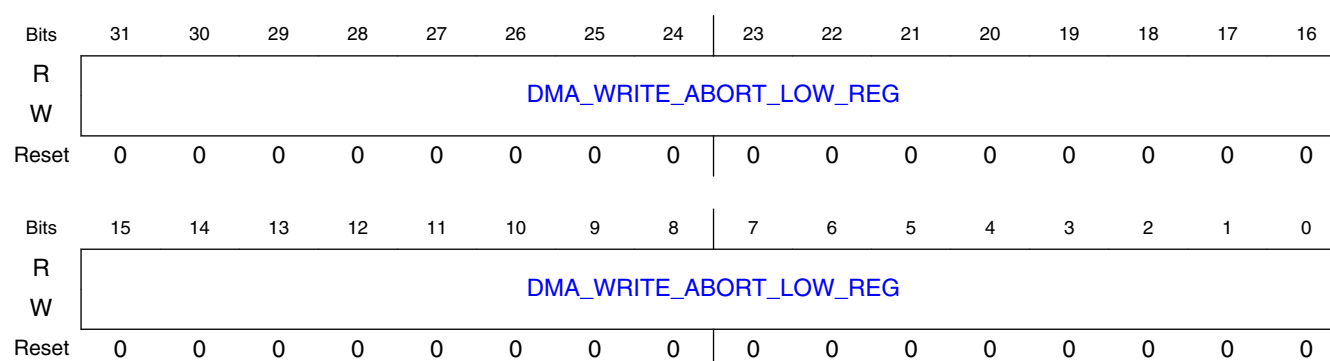
Field	Function
DMA_WRITE_DONE_HIGH_REG 31-0	The DMA uses this field to generate bits [63:32] of the address field for the Done IMWr TLP. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.199 DMA Write Abort IMWr Address Low Register. (DMA_WRITE_ABORT_IMWR_LOW_OFF)

11.3.3.1.199.1 Offset

Register	Offset
DMA_WRITE_ABORT_IMWR_LOW_OFF	8008_0068h

11.3.3.1.199.2 Diagram



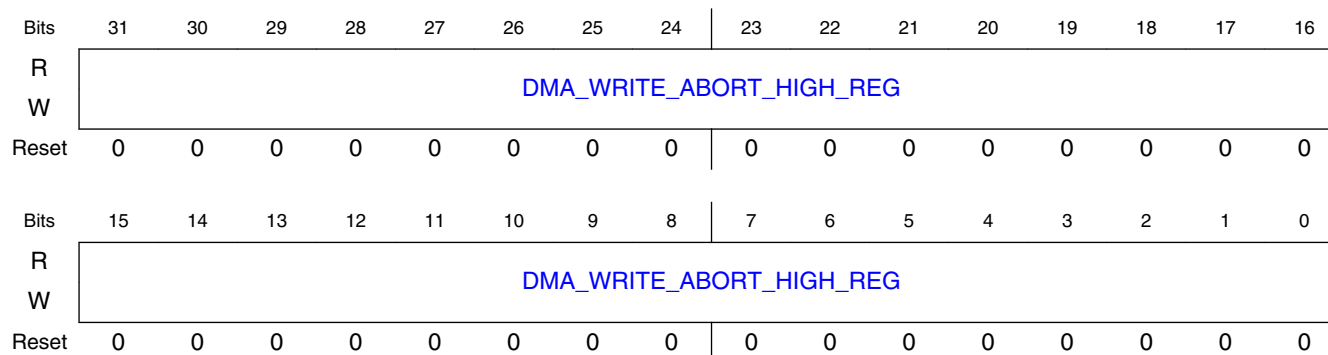
11.3.3.1.199.3 Fields

Field	Function
31-0 DMA_WRITE_ABORT_LOW_REG	The DMA uses this field to generate bits [31:0] of the address field for the Abort IMWr TLP it generates. Bits [1:0] must be "00" as this address must be dword aligned. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.200 DMA Write Abort IMWr Address High Register. (DMA_WRITE_ABORT_IMWR_HIGH_OFF)

11.3.3.1.200.1 Offset

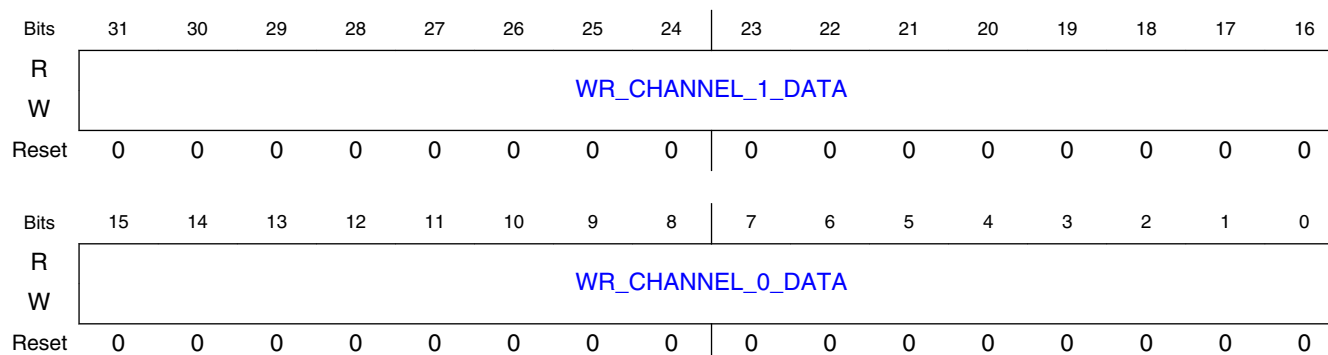
Register	Offset
DMA_WRITE_ABORT_IMWR_HIGH_OFF	8008_006Ch

11.3.3.1.200.2 Diagram**11.3.3.1.200.3 Fields**

Field	Function
31-0 DMA_WRITE_ABORT_HIGH_REG	The DMA uses this field to generate bits [63:32] of the address field for the Abort IMWr TLP. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.201 DMA Write Channel 1 and 0 IMWr Data Register. (DMA_WRITE_CH01_IMWR_DATA_OFF)**11.3.3.1.201.1 Offset**

Register	Offset
DMA_WRITE_CH01_IMWR_DATA_OFF	8008_0070h

11.3.3.1.201.2 Diagram

11.3.3.1.201.3 Fields

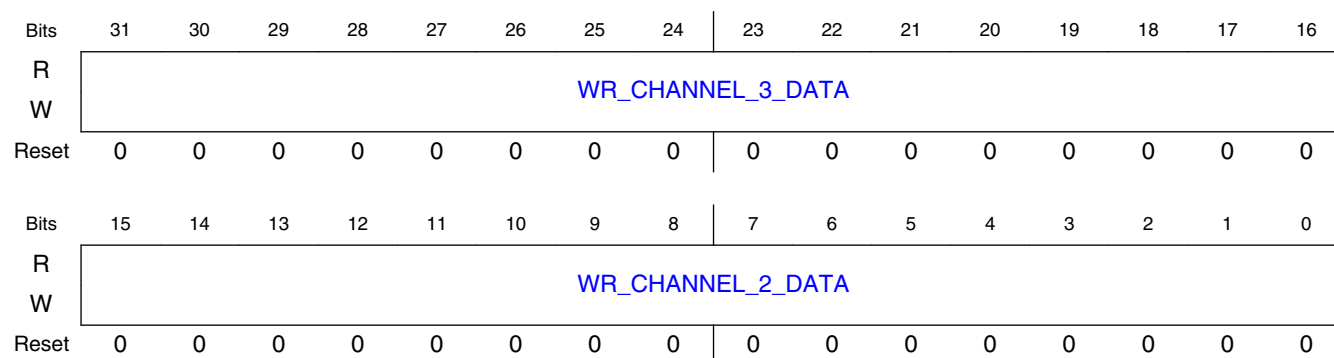
Field	Function
31-16 WR_CHANNEL_1_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 1. Note: The access attributes of this field are as follows: - Dbi: R/W
15-0 WR_CHANNEL_0_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 0. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.202 DMA Write Channel 3 and 2 IMWr Data Register. (DMA_WRITE_CH23_IMWR_DATA_OFF)

11.3.3.1.202.1 Offset

Register	Offset
DMA_WRITE_CH23_IMWR_DATA_OFF	8008_0074h

11.3.3.1.202.2 Diagram



11.3.3.1.202.3 Fields

Field	Function
31-16 WR_CHANNEL_3_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 3. Note: The access attributes of this field are as follows: - Dbi: R/W

Table continues on the next page...

PCI Express (PCIe)

Field	Function
15-0 WR_CHANNEL_2_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 2. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.203 DMA Write Channel 5 and 4 IMWr Data Register. (DMA_WRITE_CH45_IMWR_DATA_OFF)

11.3.3.1.203.1 Offset

Register	Offset
DMA_WRITE_CH45_IMWR_DATA_OFF	8008_0078h

11.3.3.1.203.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	WR_CHANNEL_5_DATA															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	WR_CHANNEL_4_DATA															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.203.3 Fields

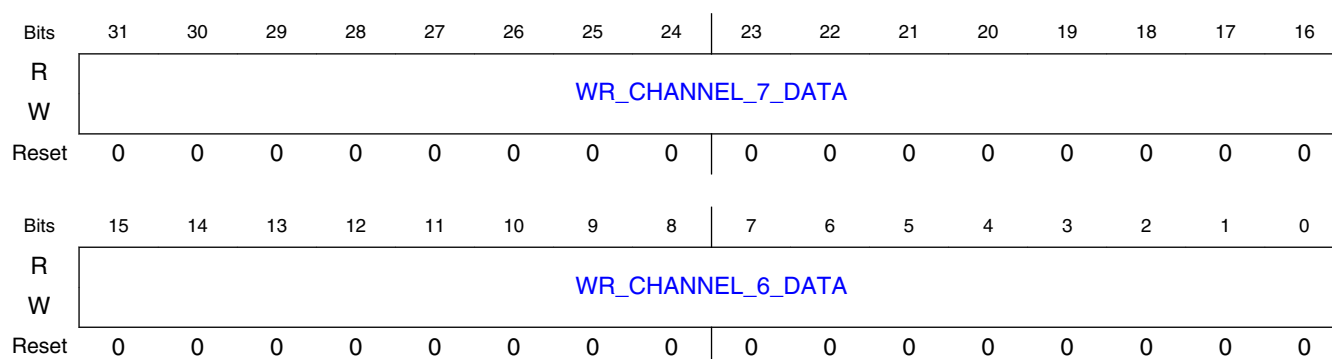
Field	Function
31-16 WR_CHANNEL_5_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 5. Note: The access attributes of this field are as follows: - Dbi: R/W
15-0 WR_CHANNEL_4_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 4. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.204 DMA Write Channel 7 and 6 IMWr Data Register. (DMA_WRITE_CH67_IMWR_DATA_OFF)

11.3.3.1.204.1 Offset

Register	Offset
DMA_WRITE_CH67_IMWR_DATA_OFF	8008_007Ch

11.3.3.1.204.2 Diagram



11.3.3.1.204.3 Fields

Field	Function
31-16 WR_CHANNEL_7_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 7. Note: The access attributes of this field are as follows: - Dbi: R/W
15-0 WR_CHANNEL_6_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for write channel 6. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.205 DMA Write Linked List Error Enable Register. (DMA_WRITE_LINKED_LIST_ERR_EN_OFF)

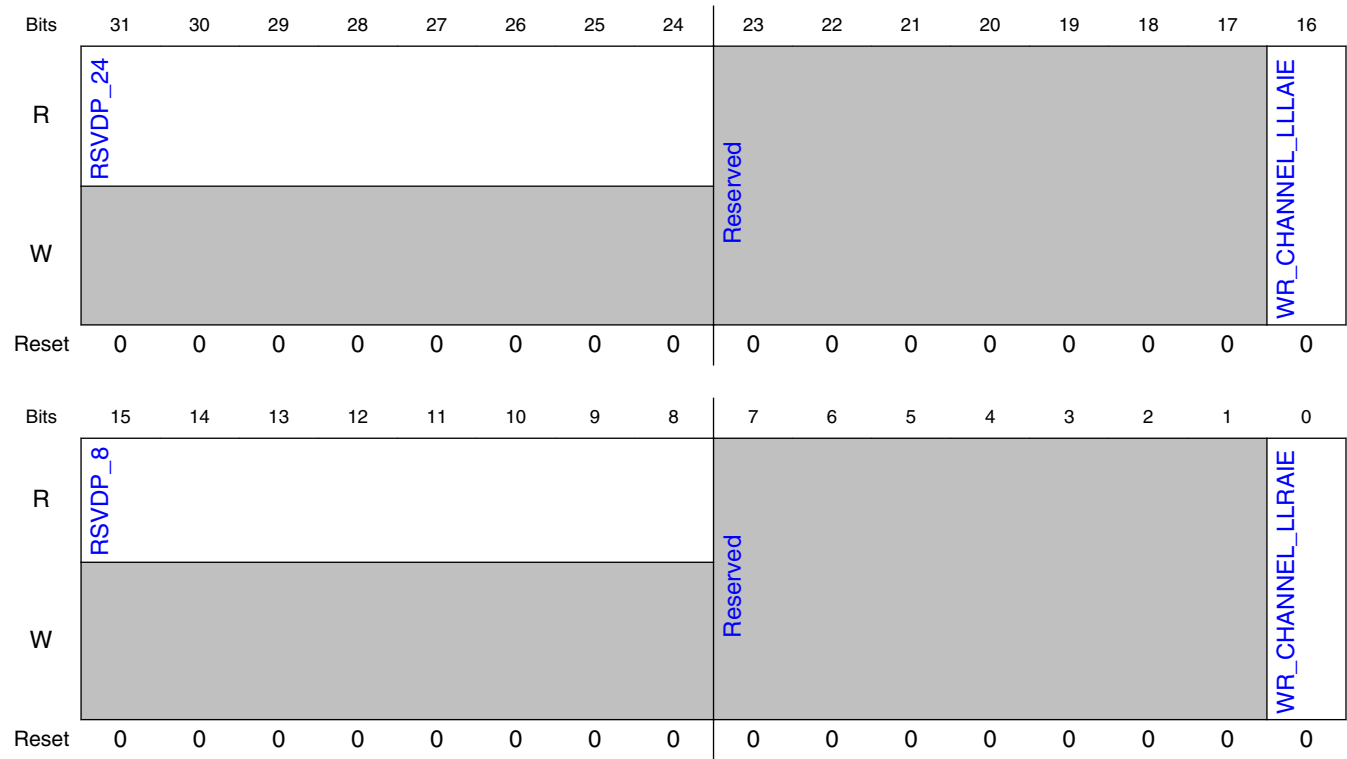
11.3.3.1.205.1 Offset

Register	Offset
DMA_WRITE_LINKED_LIST_ERR_EN_OFF	8008_0090h

11.3.3.1.205.2 **Function**

DMA Write Linked List Error Enable Register. The LIE and RIE bits in the LL element enable the channel "done" interrupts (local and remote). The LLLAIE and LLRAIE bits enable the channel "abort" interrupts (local and remote).

11.3.3.1.205.3 **Diagram**



11.3.3.1.205.4 **Fields**

Field	Function
31-24 RSVDP_24	Reserved for future use.
23-17 —	Reserved.
16 WR_CHANNEL_LLLAIE	Write Channel LL Local Abort Interrupt Enable (LLLAIE). You enable the write channel local abort interrupt through this bit. The LIE and RIE bits in the LL element enable the write channel done interrupts. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Used in linked list mode only. For more details, see "Interrupt Handling". Note: The access attributes of this field are as follows: - Dbi: R/W
15-8	Reserved for future use.

Table continues on the next page...

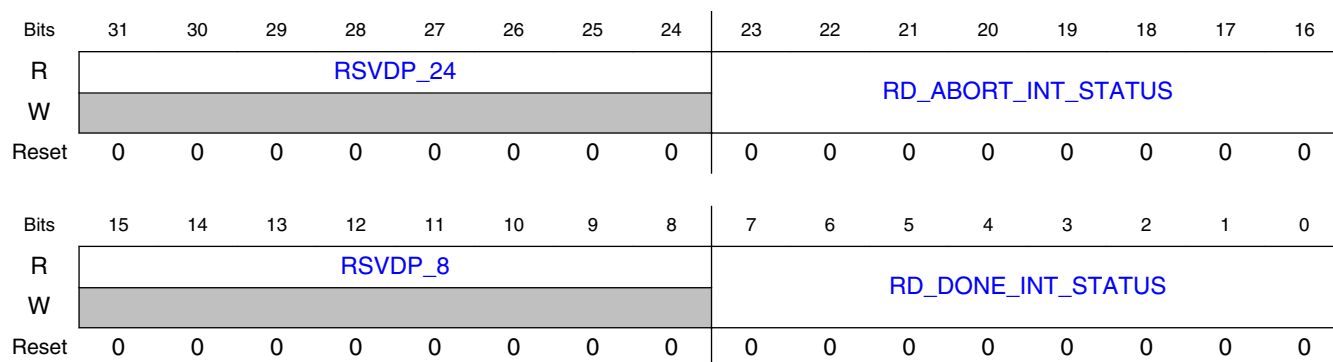
Field	Function
RSVDP_8	
7-1 —	Reserved.
0 WR_CHANNEL_LLRAIE	Write Channel LL Remote Abort Interrupt Enable (LLRAIE). You enable the write channel remote abort interrupt through this bit. The LIE and RIE bits in the LL element enable the write channel done interrupts. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Used in linked list mode only. For more details, see "Interrupt Handling". Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.206 DMA Read Interrupt Status Register. (DMA_READ_INT_STATUS_OFF)

11.3.3.1.206.1 Offset

Register	Offset
DMA_READ_INT_STATUS_OFF	8008_00A0h

11.3.3.1.206.2 Diagram



11.3.3.1.206.3 Fields

Field	Function
31-24 RSVDP_24	Reserved for future use.
23-16 RD_ABORT_INT_STATUS	Abort Interrupt Status. The DMA read channel has detected an error, or you manually stopped the transfer as described in "Stopping the DMA Transfer (Software Stop)". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. You can read the "DMA Read Error Status Low Register" (DMA_READ_ERR_STATUS_LOW_OFF) and "DMA Read Error Status High Register"

Table continues on the next page...

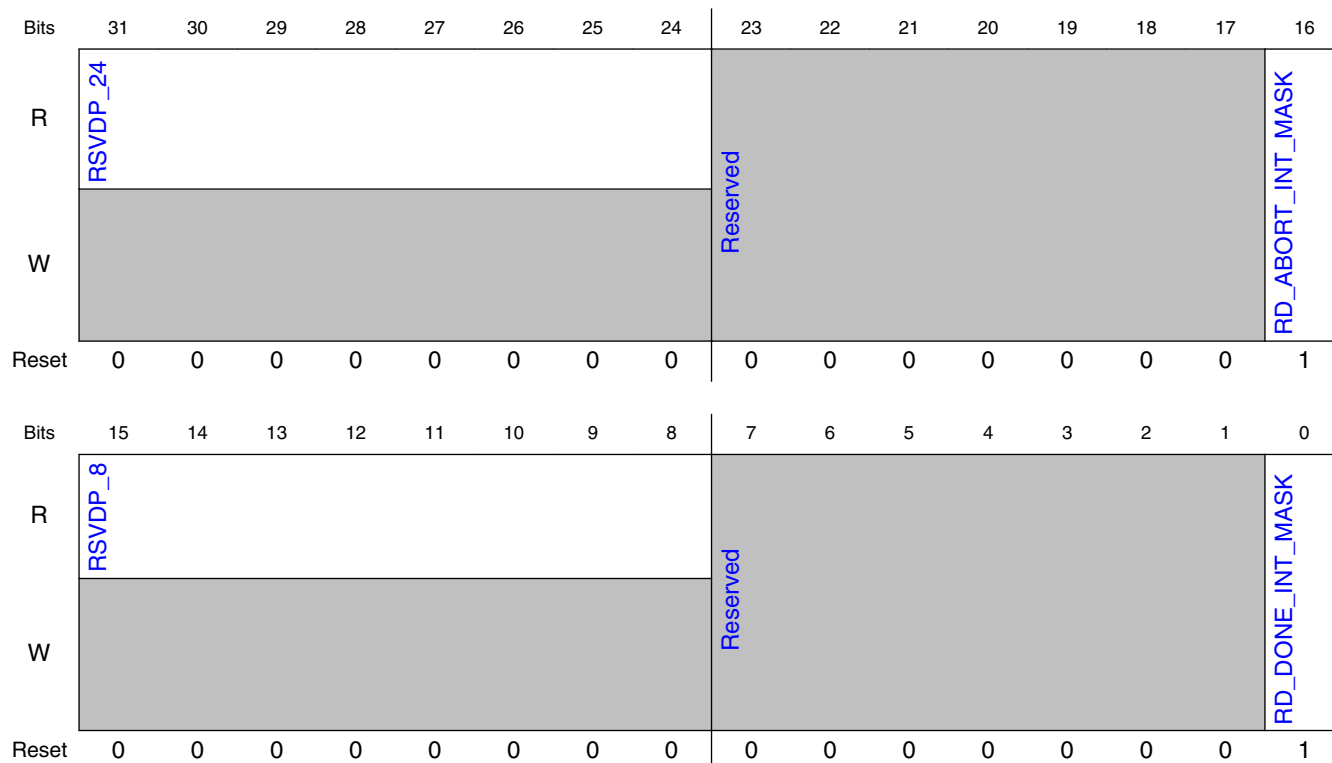
Field	Function
	(DMA_READ_ERR_STATUS_HIGH_OFF) to determine the source of the error. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA read interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the DMA read interrupt Clear register to clear this interrupt bit. Note: You can write to this register to emulate interrupt generation, during software or hardware testing. A write to the address triggers an interrupt, but the DMA does not set the Done or Abort bits in this register. Note: The access attributes of this field are as follows: - Dbi: R/W
15-8 RSVDP_8	Reserved for future use.
7-0 RD_DONE_INT_STATUS	Done Interrupt Status. The DMA read channel has successfully completed the DMA read transfer. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA read interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the DMA read interrupt Clear register to clear this interrupt bit. Note: You can write to this register to emulate interrupt generation, during software or hardware testing. A write to the address triggers an interrupt, but the DMA does not set the Done or Abort bits in this register. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.207 DMA Read Interrupt Mask Register. (DMA_READ_INT_MASK_OFF)

11.3.3.1.207.1 Offset

Register	Offset
DMA_READ_INT_MASK_OFF	8008_00A8h

11.3.3.1.207.2 Diagram



11.3.3.1.207.3 Fields

Field	Function
31-24 RSVDP_24	Reserved for future use.
23-17 —	Reserved.
16 RD_ABORT_INT_MASK	Abort Interrupt Mask. Prevents the Abort interrupt status field in the DMA read interrupt status register from asserting the edma_int output. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: The access attributes of this field are as follows: - Dbi: R/W
15-8 RSVDP_8	Reserved for future use.
7-1 —	Reserved.
0 RD_DONE_INT_MASK	Done Interrupt Mask. Prevents the Done interrupt status field in the DMA read interrupt status register from asserting the edma_int output. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.208 DMA Read Interrupt Clear Register. (DMA_READ_INT_CLEAR_OFF)

11.3.3.1.208.1 Offset

Register	Offset
DMA_READ_INT_CLEAR_OFF	8008_00ACh

11.3.3.1.208.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	RSVDP_24															
W									RD_ABORT_INT_CLEAR							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	RSVDP_8															
W									RD_DONE_INT_CLEAR							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.208.3 Fields

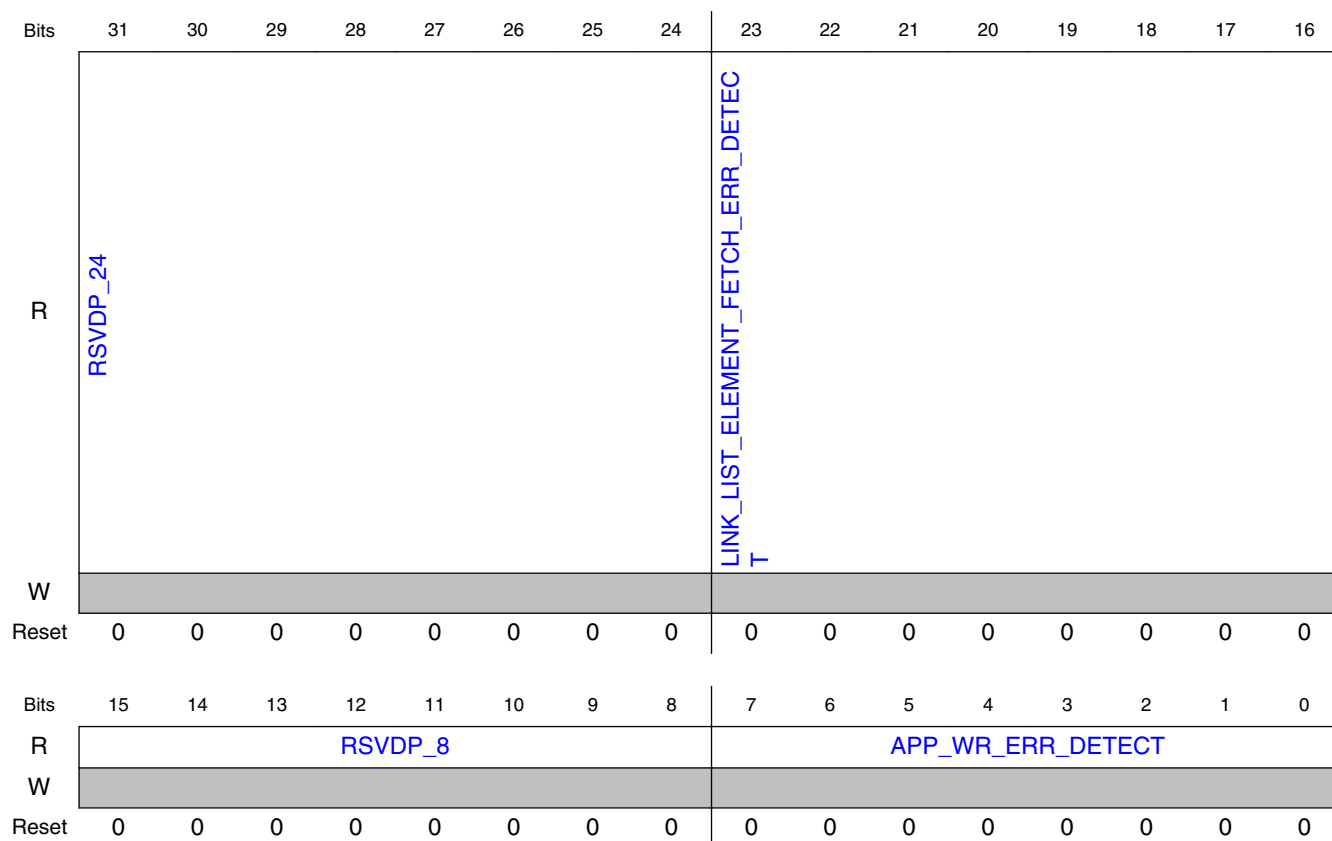
Field	Function
31-24 RSVDP_24	Reserved for future use.
23-16 RD_ABORT_INT_CLEAR	Abort Interrupt Clear. You must write a 1'b1 to clear the corresponding bit in the Abort interrupt status field of the DMA read interrupt status register. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: Reading from this self-clearing register field always returns a "0".
15-8 RSVDP_8	Reserved for future use.
7-0 RD_DONE_INT_CLEAR	Done Interrupt Clear. You must write a 1'b1 to clear the corresponding bit in the Done interrupt status field of the DMA read interrupt status register. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Note: Reading from this self-clearing register field always returns a "0".

11.3.3.1.209 DMA Read Error Status Low Register. (DMA_READ_ERR_STATUS_LOW_OFF)

11.3.3.1.209.1 Offset

Register	Offset
DMA_READ_ERR_STATUS_LOW_OFF	8008_00B4h

11.3.3.1.209.2 Diagram



11.3.3.1.209.3 Fields

Field	Function
31-24 RSVDP_24	Reserved for future use.
23-16 LINK_LIST_ELEMENT_FETCH_ERR_DETECT	Linked List Element Fetch Error Detected. - The DMA read channel has received an error response from the AXI bus while reading a linked list element from local memory. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA read interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the "DMA Read Interrupt Clear Register" (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this clears all bits in this register, and also the DMA Read Error Status High register (DMA_READ_ERR_STATUS_HIGH_OFF).

Table continues on the next page...

PCI Express (PCIe)

Field	Function
15-8 RSVDP_8	Reserved for future use.
7-0 APP_WR_ERR_DETECT	Application Write Error Detected. The DMA read channel has received an error response from the AXI bus (or TRGT1 interface when the AXI Bridge is not used) while writing data to it. This error is fatal. You must restart the transfer from the beginning, as the channel context is corrupted, and the transfer is not rolled back. For more details, see "Linked List Mode". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA read interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the "DMA Read Interrupt Clear Register" (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this clears all bits in this register, and also the DMA Read Error Status High register (DMA_READ_ERR_STATUS_HIGH_OFF).

11.3.3.1.210 DMA Read Error Status High Register. (DMA_READ_ERR_STATUS_HIGH_OFF)

11.3.3.1.210.1 Offset

Register	Offset
DMA_READ_ERR_STATUS_HIGH_OFF	8008_00B8h

11.3.3.1.210.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DATA_POISONING								CPL_TIMEOUT							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CPL_ABORT								UNSUPPORTED_REQ							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.210.3 Fields

Field	Function
31-24 DATA_POISONING	Data Poisoning. The DMA read channel has detected data poisoning in the completion from the remote device (in response to the MRd request). The DMA read channel will drop the completion and then be halted. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA read interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field

Table continues on the next page...

Field	Function
	of the "DMA Read Interrupt Clear Register" (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this also clears the other error bits for the same channel in this register and in the DMA Read Error Status Low register.
23-16 CPL_TIMEOUT	Completion Time Out. The DMA read channel has timed-out while waiting for the remote device to respond to the MRd request, or a malformed CplD has been received. For more details, see "Linked List Mode". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA read interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the "DMA Read Interrupt Clear Register" (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this also clears the other error bits for the same channel in this register and in the DMA Read Error Status Low register.
15-8 CPL_ABORT	Completer Abort. The DMA read channel has received a PCIe completer abort completion status from the remote device in response to the MRd request. For more details, see "Linked List Mode". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA read interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the "DMA Read Interrupt Clear Register" (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this also clears the other error bits for the same channel in this register and in the DMA Read Error Status Low register.
7-0 UNSUPPORTED_REQ	Unsupported Request. The DMA read channel has received a PCIe unsupported request completion status from the remote device in response to the MRd request. For more details, see "Linked List Mode". Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. - Enabling: For details, see "Interrupts and Error Handling". - Masking: The DMA read interrupt Mask register has no effect on this register. - Clearing: You must write a 1'b1 to the corresponding channel bit in the Abort interrupt field of the "DMA Read Interrupt Clear Register" (DMA_READ_INT_CLEAR_OFF) to clear this error bit. Note, this also clears the other error bits for the same channel in this register and in the DMA Read Error Status Low register.

11.3.3.1.211 DMA Read Linked List Error Enable Register. (DMA_READ_LINKED_LIST_ERR_EN_OFF)

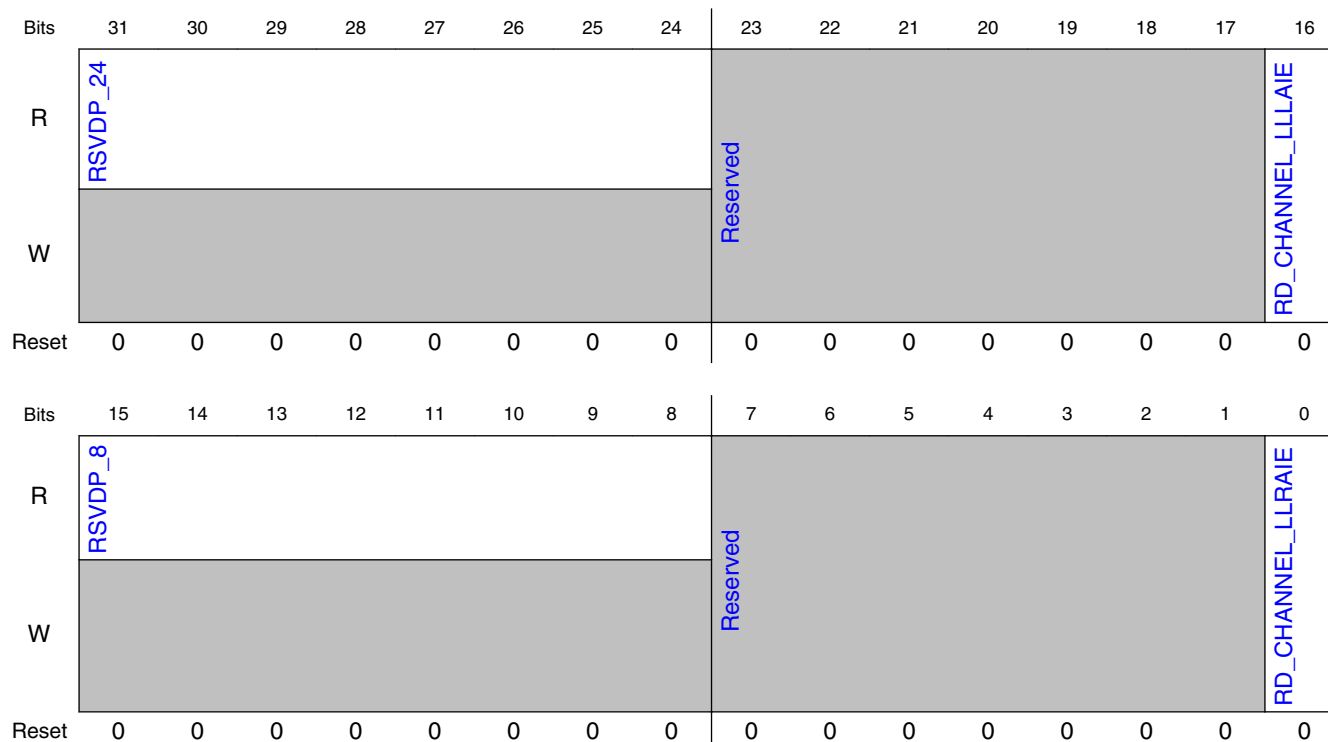
11.3.3.1.211.1 Offset

Register	Offset
DMA_READ_LINKED_LIST_ERR_EN_OFF	8008_00C4h

11.3.3.1.211.2 Function

DMA Read Linked List Error Enable Register. The LIE and RIE bits in the LL element enable the channel "done" interrupts (local and remote). The LLLAIE and LLRAIE bits enable the channel "abort" interrupts (local and remote).

11.3.3.1.211.3 Diagram



11.3.3.1.211.4 Fields

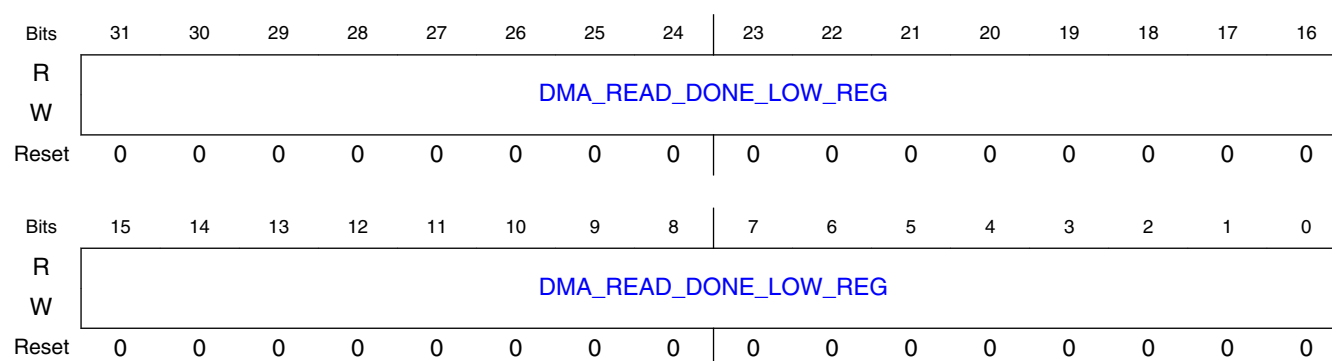
Field	Function
31-24 RSVDP_24	Reserved for future use.
23-17 —	Reserved.
16 RD_CHANNEL_LLLAIE	Read Channel LL Local Abort Interrupt Enable (LLLAIE). You enable the read channel Local Abort interrupt through this bit. The LIE and RIE bits in the LL element enable the read channel done interrupts. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Used in linked list mode only. For more details, see "Interrupt Handling". Note: The access attributes of this field are as follows: - Dbi: R/W
15-8 RSVDP_8	Reserved for future use.
7-1 —	Reserved.
0 RD_CHANNEL_LLRAIE	Read Channel LL Remote Abort Interrupt Enable (LLRAIE). You enable the read channel Remote Abort interrupt through this bit. The LIE and RIE bits in the LL element enable the read channel done interrupts. Each bit corresponds to a DMA channel. Bit [0] corresponds to channel 0. Used in linked list mode only. For more details, see "Interrupt Handling". Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.212 DMA Read Done IMWr Address Low Register. (DMA_READ_DONE_IMWR_LOW_OFF)

11.3.3.1.212.1 Offset

Register	Offset
DMA_READ_DONE_IMWR_LOW_OFF	8008_00CCh

11.3.3.1.212.2 Diagram



11.3.3.1.212.3 Fields

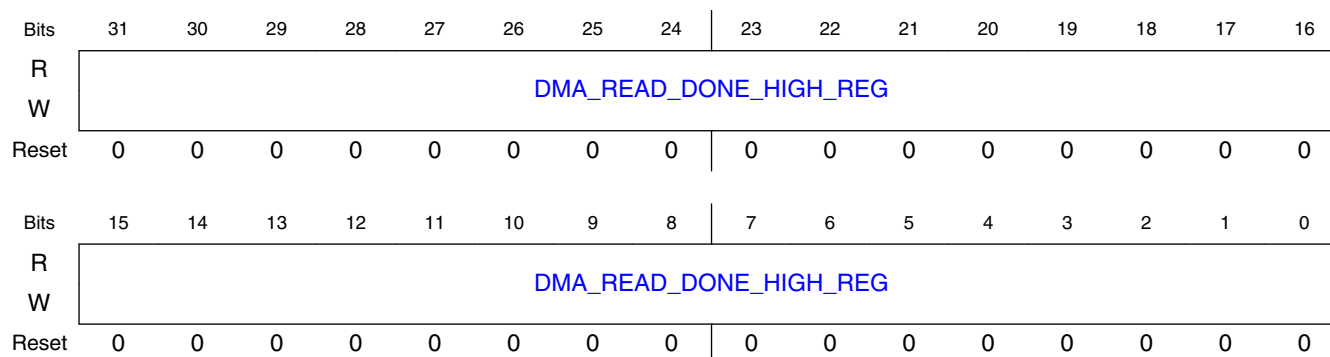
Field	Function
31-0 DMA_READ_DONE_LOW_REG	The DMA uses this field to generate bits [31:0] of the address field for the Done IMWr TLP. Bits [1:0] must be "00" as this address must be dword aligned. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.213 DMA Read Done IMWr Address High Register. (DMA_READ_DONE_IMWR_HIGH_OFF)

11.3.3.1.213.1 Offset

Register	Offset
DMA_READ_DONE_IMWR_HIGH_OFF	8008_00D0h

11.3.3.1.213.2 Diagram



11.3.3.1.213.3 Fields

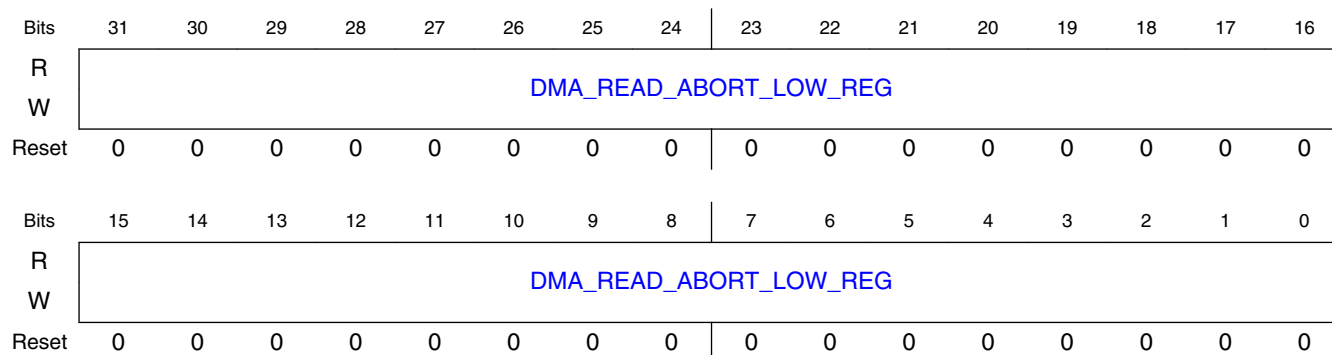
Field	Function
31-0 DMA_READ_D ONE_HIGH_RE G	The DMA uses this field to generate bits [63:32] of the address field for the Done IMWr TLP. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.214 DMA Read Abort IMWr Address Low Register. (DMA_READ_ABORT_IMWR_LOW_OFF)

11.3.3.1.214.1 Offset

Register	Offset
DMA_READ_ABORT_I MWR_LOW_OFF	8008_00D4h

11.3.3.1.214.2 Diagram



11.3.3.1.214.3 Fields

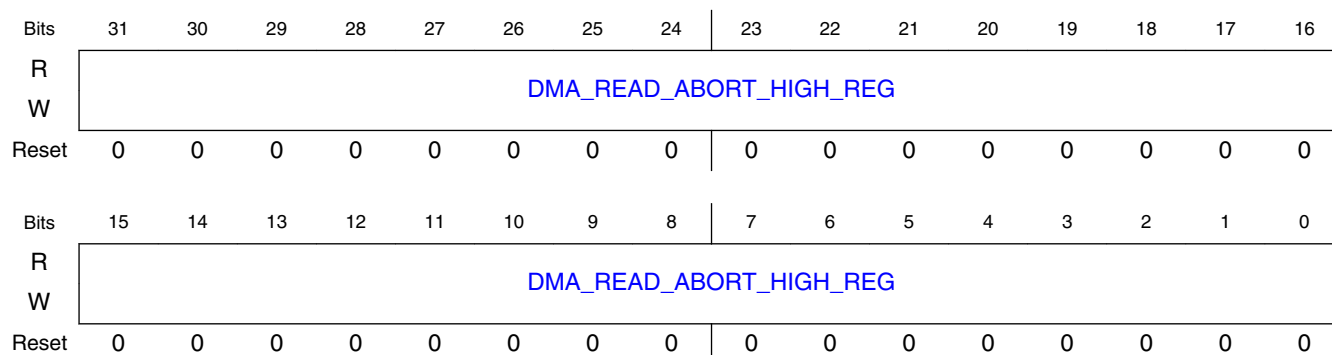
Field	Function
31-0 DMA_READ_ABORT_LOW_REG	The DMA uses this field to generate bits [31:0] of the address field for the Abort IMWr TLP. Bits [1:0] must be "00" as this address must be dword aligned. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.215 DMA Read Abort IMWr Address High Register. (DMA_READ_ABORT_IMWR_HIGH_OFF)

11.3.3.1.215.1 Offset

Register	Offset
DMA_READ_ABORT_IMWR_HIGH_OFF	8008_00D8h

11.3.3.1.215.2 Diagram



11.3.3.1.215.3 Fields

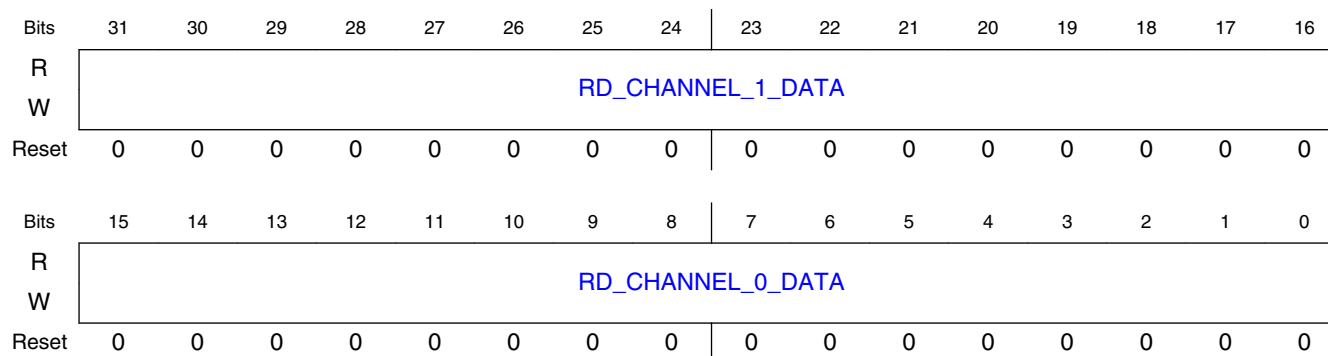
Field	Function
31-0 DMA_READ_ABORT_HIGH_REGISTER	The DMA uses this field to generate bits [63:32] of the address field for the Abort IMWr TLP. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.216 DMA Read Channel 1 and 0 IMWr Data Register. (DMA_READ_CH01_IMWR_DATA_OFF)

11.3.3.1.216.1 Offset

Register	Offset
DMA_READ_CH01_IMWR_DATA_OFF	8008_00DCh

11.3.3.1.216.2 Diagram



11.3.3.1.216.3 Fields

Field	Function
31-16 RD_CHANNEL_1_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 1. Note: The access attributes of this field are as follows: - Dbi: R/W
15-0 RD_CHANNEL_0_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 0. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.217 DMA Read Channel 3 and 2 IMWr Data Register. (DMA_READ_CH23_IMWR_DATA_OFF)

11.3.3.1.217.1 Offset

Register	Offset
DMA_READ_CH23_IMWR_DATA_OFF	8008_00E0h

11.3.3.1.217.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	RD_CHANNEL_3_DATA															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	RD_CHANNEL_2_DATA															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

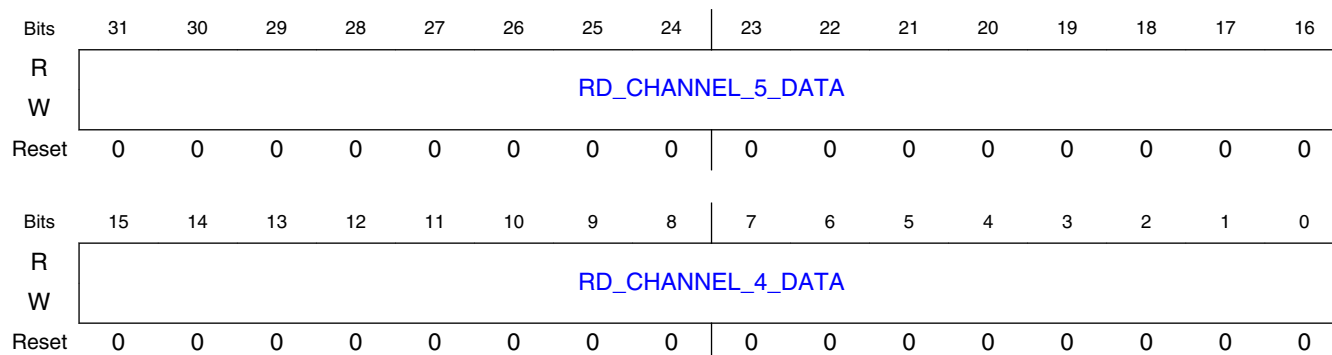
11.3.3.1.217.3 Fields

Field	Function
31-16 RD_CHANNEL_3_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 3. Note: The access attributes of this field are as follows: - Dbi: R/W
15-0 RD_CHANNEL_2_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 2. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.218 DMA Read Channel 5 and 4 IMWr Data Register. (DMA_READ_CH45_IMWR_DATA_OFF)

11.3.3.1.218.1 Offset

Register	Offset
DMA_READ_CH45_IMWR_DATA_OFF	8008_00E4h

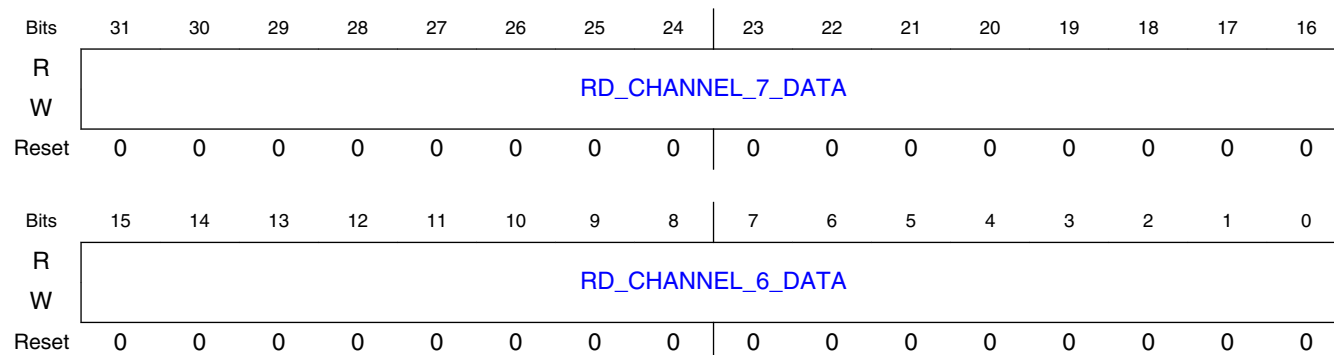
11.3.3.1.218.2 Diagram**11.3.3.1.218.3 Fields**

Field	Function
31-16 RD_CHANNEL_5_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 5. Note: The access attributes of this field are as follows: - Dbi: R/W
15-0 RD_CHANNEL_4_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 4. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.219 DMA Read Channel 7 and 6 IMWr Data Register. (DMA_READ_CH67_IMWR_DATA_OFF)**11.3.3.1.219.1 Offset**

Register	Offset
DMA_READ_CH67_IMWR_DATA_OFF	8008_00E8h

11.3.3.1.219.2 Diagram



11.3.3.1.219.3 Fields

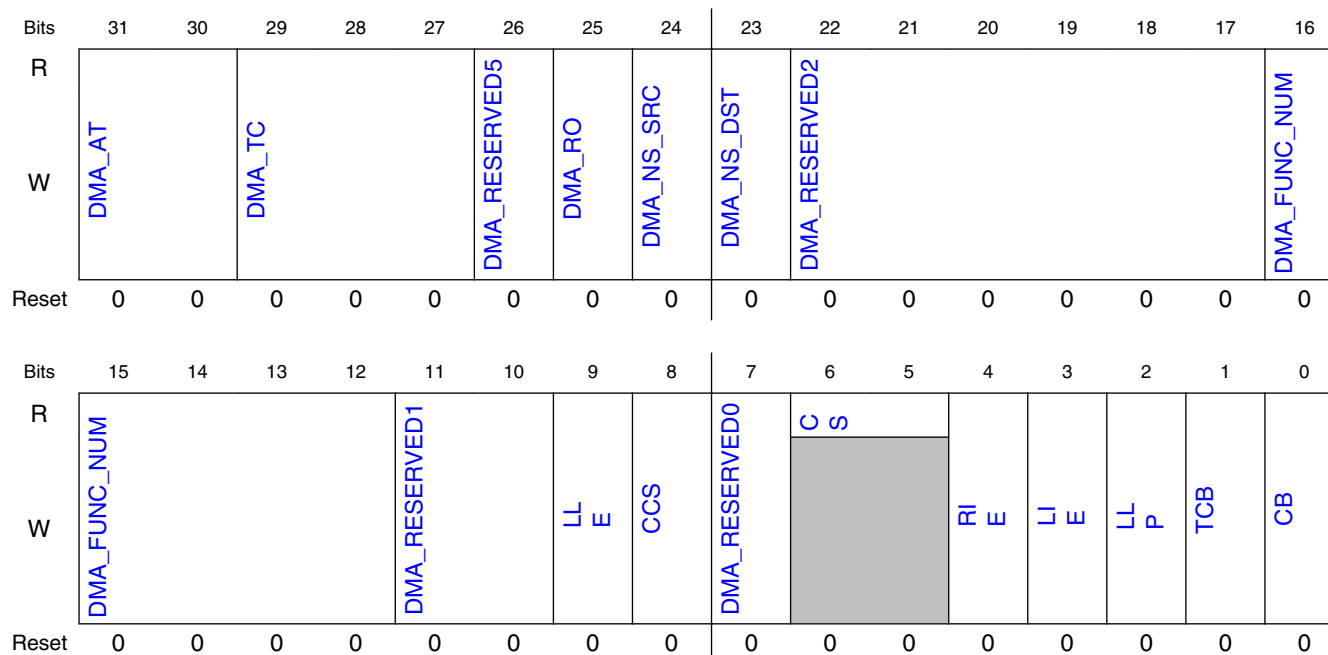
Field	Function
31-16 RD_CHANNEL_7_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 7. Note: The access attributes of this field are as follows: - Dbi: R/W
15-0 RD_CHANNEL_6_DATA	The DMA uses this field to generate the data field for the Done or Abort IMWr TLPs it generates for read channel 6. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.220 DMA Write Channel Control 1 Register. (DMA_CH_CONTROL1_OFF_WRCH_0)

11.3.3.1.220.1 Offset

Register	Offset
DMA_CH_CONTROL1_OFF_WRCH_0	8008_0200h

11.3.3.1.220.2 Diagram



11.3.3.1.220.3 Fields

Field	Function
31-30 DMA_AT	Address Translation TLP Header Bit (AT) The DMA uses this TLP header field when generating MRd/MWr (not IMWr) TLPs. Note: The access attributes of this field are as follows: - Dbi: R/W
29-27 DMA_TC	Traffic Class TLP Header Bit (TC) The DMA uses this TLP header field when generating MRd/MWr (not IMWr) TLPs. Note: The access attributes of this field are as follows: - Dbi: R/W
26 DMA_RESERVED5	Reserved. Note: The access attributes of this field are as follows: - Dbi: R/W
25 DMA_RO	Relaxed Ordering TLP Header Bit (RO) The DMA uses this TLP header field when generating MRd/MWr (not IMWr) TLPs. Note: The access attributes of this field are as follows: - Dbi: R/W
24 DMA_NS_SRC	Source No Snoop TLP Header Bit (DMA_NS_SRC). The DMA uses this TLP header field when generating MRd (SAR addressing space) (not IMWr) TLPs. Note: The access attributes of this field are as follows: - Dbi: R/W
23 DMA_NS_DST	Destination No Snoop TLP Header Bit (DMA_NS_DST). The DMA uses this TLP header field when generating MWr (DAR addressing space) (not IMWr) TLPs. Note: The access attributes of this field are as follows: - Dbi: R/W
22-17 DMA_RESERVED2	Reserved. Note: The access attributes of this field are as follows: - Dbi: R/W
16-12	Function Number (FN). The controller uses this field when generating the requester ID for the MRd/MWr DMA TLP. When you have enabled SR-IOV, then this field is ignored if you have set the VFE field in the

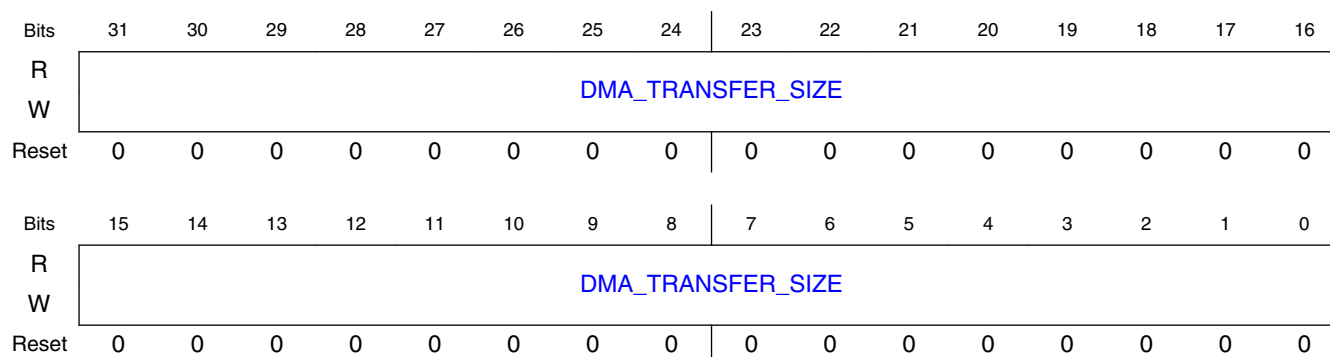
Table continues on the next page...

Field	Function
DMA_FUNC_NUM	"DMA Write Channel Control 2 Register" (DMA_CH_CONTROL2_OFF_WRCH_0). Note: The access attributes of this field are as follows: - Dbi: R/W
11-10 DMA_RESERVED1	Reserved. Note: The access attributes of this field are as follows: - Dbi: R/W
9 LLE	Linked List Enable (LLE). - 0: Disable linked list operation - 1: Enable linked list operation Note: The access attributes of this field are as follows: - Dbi: R/W
8 CCS	Consumer Cycle State (CCS). Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more details, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". You must initialize this bit. The DMA updates this bit during linked list operation. Note: The access attributes of this field are as follows: - Dbi: R/W
7 DMA_RESERVED0	Reserved. Note: The access attributes of this field are as follows: - Dbi: R/W
6-5 CS	Channel Status (CS). The channel status bits identify the current operational state of the DMA channel. The operation state encoding for each DMA channel is as follows: - 00: Reserved - 01: Running. This channel is active and transferring data. - 10: Halted. An error condition has been detected, and the DMA has stopped this channel. - 11: Stopped. The DMA has transferred all data for this channel, or you have prematurely stopped this channel by writing to the Stop field of the "DMA Write Doorbell Register" (DMA_WRITE_DOORBELL_OFF) or "DMA Read Doorbell Register" (DMA_READ_DOORBELL_OFF).
4 RIE	Remote Interrupt Enable (RIE). You must set this bit to enable the generation of the Done or Abort Remote interrupts. For more details, see "Interrupts and Error Handling". In LL mode, the DMA overwrites this with the RIE of the LL element. The RIE bit in a LL element only enables the Done interrupt. In non-LL mode, the RIE bit enables the Done and Abort interrupts. This field is not defined in a link LL element. Note: The access attributes of this field are as follows: - Dbi: R/W
3 LIE	Local Interrupt Enable (LIE). You must set this bit to enable the generation of the Done or Abort Local interrupts. For more details, see "Interrupts and Error Handling". In LL mode, the DMA overwrites this with the LIE of the LL element. The LIE bit in a LL element only enables the Done interrupt. In non-LL mode, the LIE bit enables the Done and Abort interrupts. This field is not defined in a link LL element. Note: The access attributes of this field are as follows: - Dbi: R/W
2 LLP	Load Link Pointer (LLP). Used in linked list mode only. Indicates that this linked list element is a link element, and its LL element pointer dwords are pointing to the next (non-contiguous) element. The DMA loads this field with the LLP of the linked list element. Note: The access attributes of this field are as follows: - Dbi: R/W
1 TCB	Toggle Cycle Bit (TCB). Indicates to the DMA to toggle its interpretation of the CB. Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more details, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". The DMA loads this field with the TCB of the linked list element. this field is not defined in a data LL element. Note: The access attributes of this field are as follows: - Dbi: R/W
0 CB	Cycle Bit (CB). Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more details, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". The DMA loads this field with the CB of the linked list element. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.221 DMA Write Transfer Size Register. (DMA_TRANSFER_SIZE_OFF_WRCH_0)

11.3.3.1.221.1 Offset

Register	Offset
DMA_TRANSFER_SIZE_OFF_WRCH_0	8008_0208h

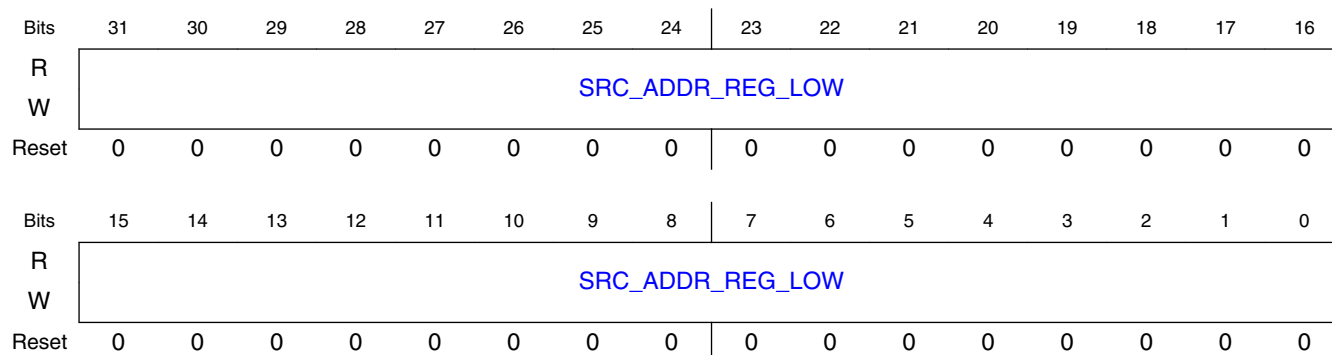
11.3.3.1.221.2 Diagram**11.3.3.1.221.3 Fields**

Field	Function
31-0 DMA_TRANSFER_SIZE	DMA Transfer Size. You program this register with the size of the DMA transfer. The maximum DMA transfer size is 4Gbytes. The minimum transfer size is one byte (0x1). This field is automatically decremented by the DMA as the DMA write channel transfer progresses. This field indicates the number bytes remaining to be transferred. When all bytes are successfully transferred the current transfer size is zero. In LL mode, the DMA overwrites this register with the corresponding dword of the LL element. You can read this register to monitor the transfer progress, however in some scenarios this register is updated after a delay. For example, when less than 3 channels are doorbelled, this register is updated only after a descriptor finishes(linked list mode), or the transfer ends (non-linked list mode). Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.222 DMA Write SAR Low Register. (DMA_SAR_LOW_OFF_WRCH_0)**11.3.3.1.222.1 Offset**

Register	Offset
DMA_SAR_LOW_OFF_WRCH_0	8008_020Ch

11.3.3.1.222.2 Diagram



11.3.3.1.222.3 Fields

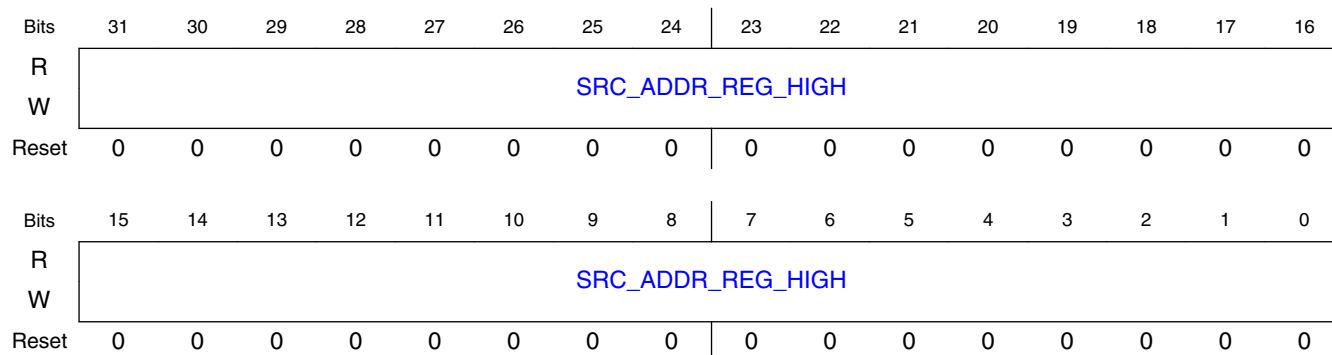
Field	Function
31-0 SRC_ADDR_REG_LOW	Source Address Register (Lower 32 bits). Indicates the next address to be read from. The DMA increments the SAR as the DMA transfer progresses. In LL mode, the DMA overwrites this with the corresponding dword of the LL element. - DMA Read: The SAR is the address of the remote memory. - DMA Write: The SAR is the address of the local memory. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.223 DMA Write SAR High Register. (DMA_SAR_HIGH_OFF_WRC H_0)

11.3.3.1.223.1 Offset

Register	Offset
DMA_SAR_HIGH_OFF_WRC H_0	8008_0210h

11.3.3.1.223.2 Diagram



11.3.3.1.223.3 Fields

Field	Function
31-0 SRC_ADDR_REG_HIGH	Source Address Register (Higher 32 bits). In LL mode, the DMA overwrites this with the corresponding dword of the LL element. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.224 DMA Write DAR Low Register. (DMA_DAR_LOW_OFF_WRCH_0)

11.3.3.1.224.1 Offset

Register	Offset
DMA_DAR_LOW_OFF_WRCH_0	8008_0214h

11.3.3.1.224.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DST_ADDR_REG_LOW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DST_ADDR_REG_LOW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.224.3 Fields

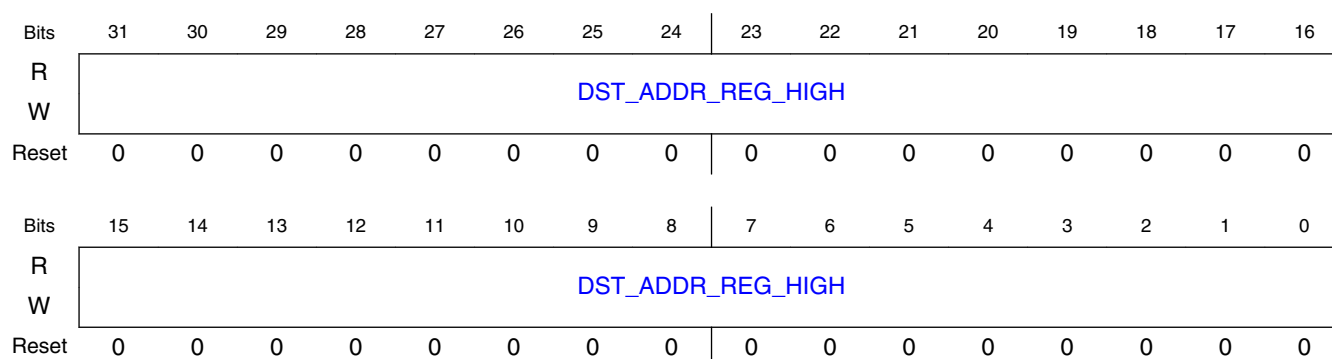
Field	Function
31-0 DST_ADDR_REG_LOW	Destination Address Register (Lower 32 bits). Indicates the next address to be written to. The DMA increments the DAR as the DMA transfer progresses. In LL mode, the DMA overwrites this with the corresponding dword of the LL element. - DMA Read: The DAR is the address of the local memory. - DMA Write: The DAR is the address of the remote memory. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.225 DMA Write DAR High Register. (DMA_DAR_HIGH_OFF_WRC_H_0)

11.3.3.1.225.1 Offset

Register	Offset
DMA_DAR_HIGH_OFF_WRC_H_0	8008_0218h

11.3.3.1.225.2 Diagram



11.3.3.1.225.3 Fields

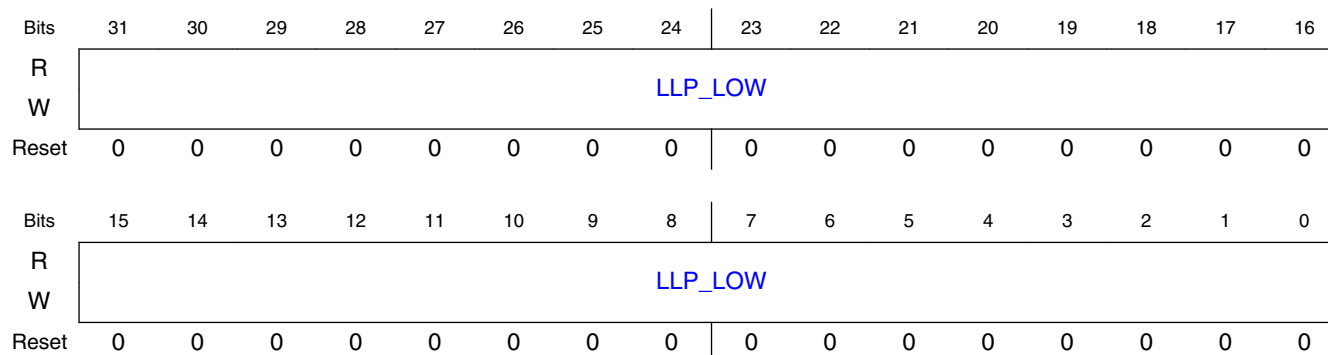
Field	Function
31-0 DST_ADDR_REG_HIGH	Destination Address Register (Higher 32 bits). In LL mode, the DMA overwrites this with the corresponding dword of the LL element. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.226 DMA Write Linked List Pointer Low Register. (DMA_LL_LOW_OFF_WRC_H_0)

11.3.3.1.226.1 Offset

Register	Offset
DMA_LL_LOW_OFF_WRC_H_0	8008_021Ch

11.3.3.1.226.2 Diagram



11.3.3.1.226.3 Fields

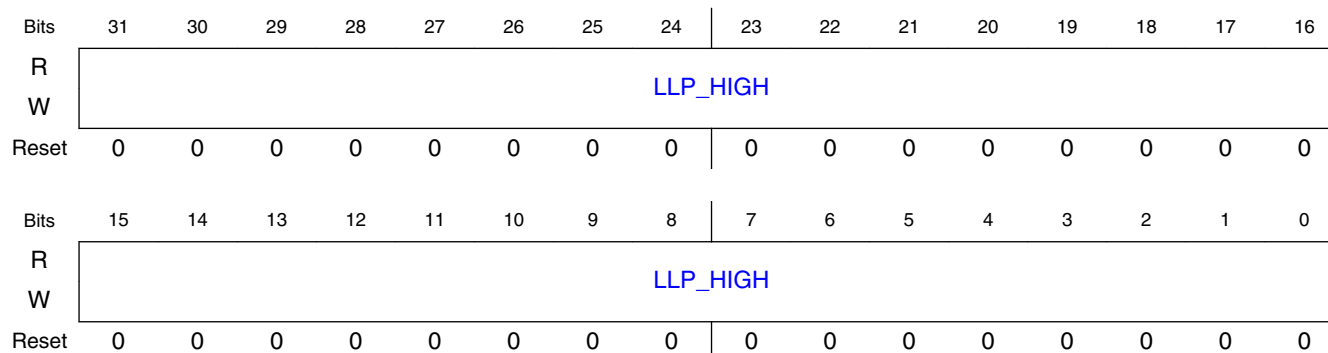
Field	Function
31-0 LLP_LOW	Lower bits of the address of the linked list transfer list in local memory. Used in linked list mode only. Updated by the DMA to point to the next element in the transfer list after the previous element is consumed. - When the current element is a data element; this field is incremented by 6. - When the current element is a link element; this field is overwritten by the LL Element Pointer of the element. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.227 DMA Write Linked List Pointer High Register. (DMA_LLPHIGH_OFF_WRCH_0)

11.3.3.1.227.1 Offset

Register	Offset
DMA_LLPHIGH_OFF_WRCH_0	8008_0220h

11.3.3.1.227.2 Diagram



11.3.3.1.227.3 Fields

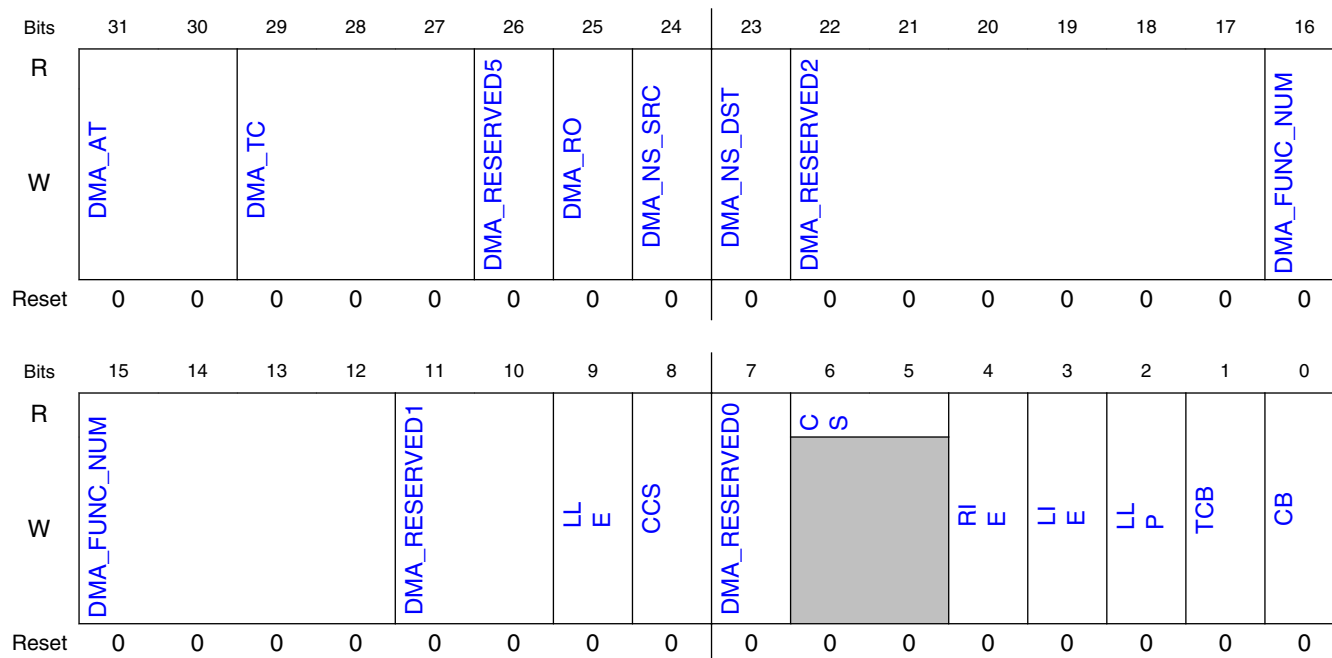
Field	Function
31-0 LLP_HIGH	Upper 32 bits of the address of the linked list transfer list in local memory. Used in linked list mode only. Updated by the DMA to point to the next element in the transfer list as elements are consumed. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.228 DMA Read Channel Control 1 Register. (DMA_CH_CONTR OL1_OFF_RDCH_0)

11.3.3.1.228.1 Offset

Register	Offset
DMA_CH_CONTROL1_ OFF_RDCH_0	8008_0300h

11.3.3.1.228.2 Diagram



11.3.3.1.228.3 Fields

Field	Function
31-30 DMA_AT	Address Translation TLP Header Bit (AT) The DMA uses this TLP header field when generating MRd/MWr (not IMWr) TLPs. Note: The access attributes of this field are as follows: - Dbi: R/W
29-27 DMA_TC	Traffic Class TLP Header Bit (TC) The DMA uses this TLP header field when generating MRd/MWr (not IMWr) TLPs. Note: The access attributes of this field are as follows: - Dbi: R/W
26 DMA_RESERV ED5	Reserved. Note: The access attributes of this field are as follows: - Dbi: R/W
25 DMA_RO	Relaxed Ordering TLP Header Bit (RO) The DMA uses this TLP header field when generating MRd/MWr (not IMWr) TLPs. Note: The access attributes of this field are as follows: - Dbi: R/W
24 DMA_NS_SRC	Source No Snoop TLP Header Bit (DMA_NS_SRC). The DMA uses this TLP header field when generating MRd (SAR addressing space) (not IMWr) TLPs. Note: The access attributes of this field are as follows: - Dbi: R/W
23 DMA_NS_DST	Destination No Snoop TLP Header Bit (DMA_NS_DST). The DMA uses this TLP header field when generating MWr (DAR addressing space) (not IMWr) TLPs. Note: The access attributes of this field are as follows: - Dbi: R/W
22-17 DMA_RESERV ED2	Reserved. Note: The access attributes of this field are as follows: - Dbi: R/W
16-12 DMA_FUNC_N UM	Function Number (FN). The controller uses this field when generating the requester ID for the MRd/MWr DMA TLP. When you have enabled SR-IOV, then this field is ignored if you have set the VFE field in the "DMA Read Channel Control 2 Register" (DMA_CH_CONTROL2_OFF_RDCH_0). Note: The access attributes of this field are as follows: - Dbi: R/W
11-10 DMA_RESERV ED1	Reserved. Note: The access attributes of this field are as follows: - Dbi: R/W
9 LLE	Linked List Enable (LLE). - 0: Disable linked list operation - 1: Enable linked list operation Note: The access attributes of this field are as follows: - Dbi: R/W
8 CCS	Consumer Cycle State (CCS). Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more details, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". You must initialize this bit. The DMA updates this bit during linked list operation. Note: The access attributes of this field are as follows: - Dbi: R/W
7 DMA_RESERV ED0	Reserved. Note: The access attributes of this field are as follows: - Dbi: R/W
6-5 CS	Channel Status (CS). The channel status bits identify the current operational state of the DMA channel. The operation state encoding for each DMA channel is as follows: - 00: Reserved - 01: Running. This channel is active and transferring data. - 10: Halted. An error condition has been detected, and the DMA has stopped this channel. - 11: Stopped. The DMA has transferred all data for this channel, or you have prematurely stopped this channel by writing to the Stop field of the "DMA Read Doorbell Register" (DMA_WRITE_DOORBELL_OFF) or "DMA Read Doorbell Register" (DMA_READ_DOORBELL_OFF).
4 RIE	Remote Interrupt Enable (RIE). You must set this bit to enable the generation of the Done or Abort Remote interrupts. For more details, see "Interrupts and Error Handling". In LL mode, the DMA overwrites this with the RIE of the LL element. The RIE bit in a LL element only enables the Done interrupt. In non-LL mode, the RIE bit enables the Done and Abort interrupts. This field is not defined in a link LL element. Note: The access attributes of this field are as follows: - Dbi: R/W

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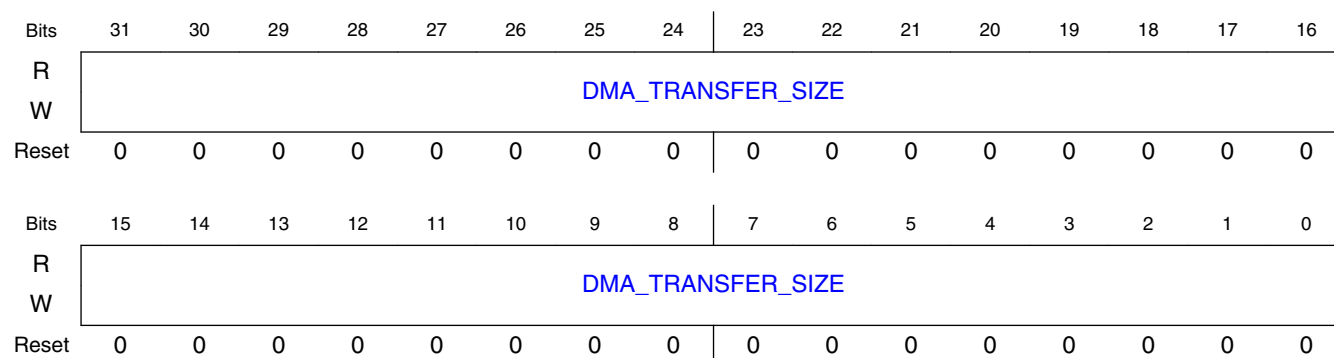
Field	Function
3 LIE	Local Interrupt Enable (LIE). You must set this bit to enable the generation of the Done or Abort Local interrupts. For more details, see "Interrupts and Error Handling". In LL mode, the DMA overwrites this with the LIE of the LL element. The LIE bit in a LL element only enables the Done interrupt. In non-LL mode, the LIE bit enables the Done and Abort interrupts. This field is not defined in a link LL element. Note: The access attributes of this field are as follows: - Dbi: R/W
2 LLP	Load Link Pointer (LLP). Used in linked list mode only. Indicates that this linked list element is a link element, and its LL element pointer dwords are pointing to the next (non-contiguous) element. The DMA loads this field with the LLP of the linked list element. Note: The access attributes of this field are as follows: - Dbi: R/W
1 TCB	Toggle Cycle Bit (TCB). Indicates to the DMA to toggle its interpretation of the CB. Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more details, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". The DMA loads this field with the TCB of the linked list element. this field is not defined in a data LL element. Note: The access attributes of this field are as follows: - Dbi: R/W
0 CB	Cycle Bit (CB). Used in linked list mode only. It is used to synchronize the producer (software) and the consumer (DMA). For more details, see "PCS-CCS-CB-TCB Producer-Consumer Synchronization". The DMA loads this field with the CB of the linked list element. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.229 DMA Read Transfer Size Register. (DMA_TRANSFER_SIZE_OFF_RDCH_0)

11.3.3.1.229.1 Offset

Register	Offset
DMA_TRANSFER_SIZE_OFF_RDCH_0	8008_0308h

11.3.3.1.229.2 Diagram



11.3.3.1.229.3 Fields

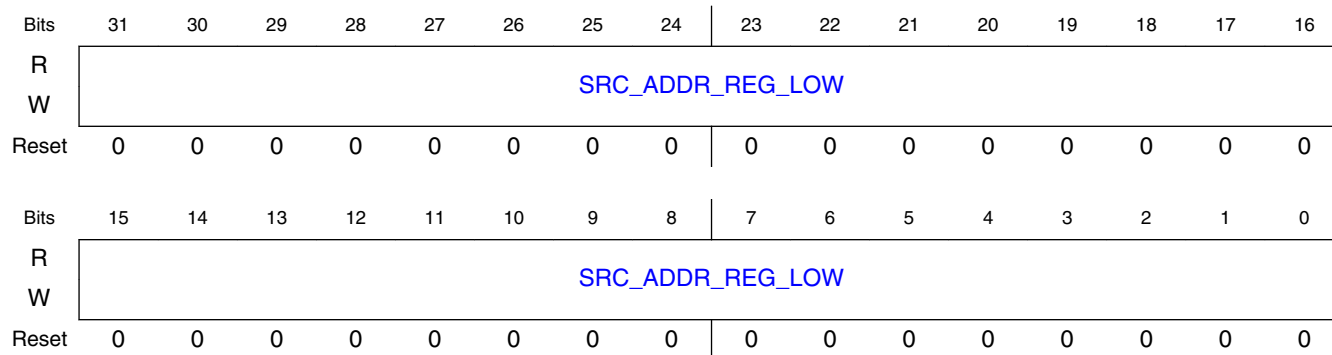
Field	Function
31-0 DMA_TRANSFERR_SIZE	DMA Transfer Size. You program this register with the size of the DMA transfer. The maximum DMA transfer size is 4Gbytes. The minimum transfer size is one byte (0x1). This field is automatically decremented by the DMA as the DMA read channel transfer progresses. This field indicates the number bytes remaining to be transferred. When all bytes are successfully transferred the current transfer size is zero. In LL mode, the DMA overwrites this register with the corresponding dword of the LL element. You can read this register to monitor the transfer progress, however in some scenarios this register is updated after a delay. For example, when less than 3 channels are doorbelled, this register is updated only after a descriptor finishes(linked list mode), or the transfer ends (non-linked list mode). Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.230 DMA Read SAR Low Register. (DMA_SAR_LOW_OFF_RDCH_0)

11.3.3.1.230.1 Offset

Register	Offset
DMA_SAR_LOW_OFF_RDCH_0	8008_030Ch

11.3.3.1.230.2 Diagram



11.3.3.1.230.3 Fields

Field	Function
31-0 SRC_ADDR_REG_LOW	Source Address Register (Lower 32 bits). Indicates the next address to be read from. The DMA increments the SAR as the DMA transfer progresses. In LL mode, the DMA overwrites this with the corresponding dword of the LL element. - DMA Read: The SAR is the address of the remote memory. - DMA Read: The SAR is the address of the local memory. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.231 DMA Read SAR High Register. (DMA_SAR_HIGH_OFF_RDCH_0)

11.3.3.1.231.1 Offset

Register	Offset
DMA_SAR_HIGH_OFF_RDCH_0	8008_0310h

11.3.3.1.231.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SRC_ADDR_REG_HIGH															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SRC_ADDR_REG_HIGH															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.231.3 Fields

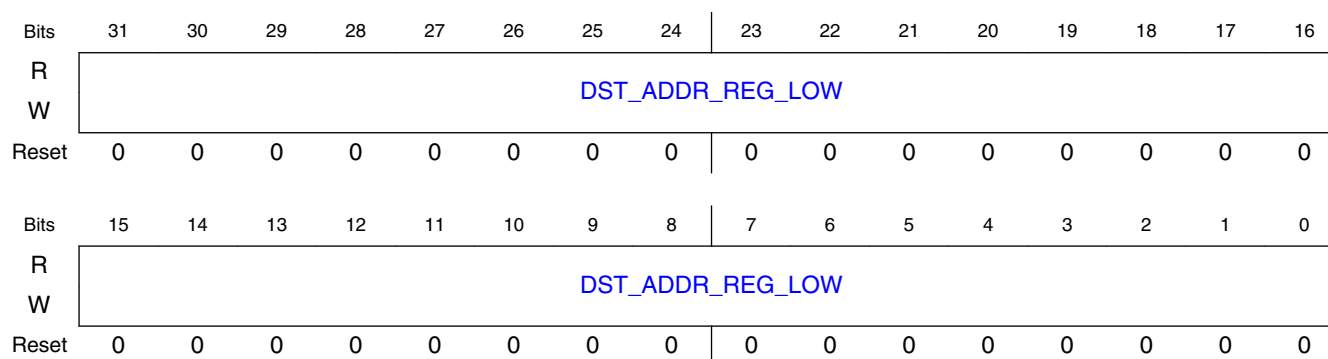
Field	Function
31-0 SRC_ADDR_REG_HIGH	Source Address Register (Higher 32 bits). In LL mode, the DMA overwrites this with the corresponding dword of the LL element. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.232 DMA Read DAR Low Register. (DMA_DAR_LOW_OFF_RDCH_0)

11.3.3.1.232.1 Offset

Register	Offset
DMA_DAR_LOW_OFF_RDCH_0	8008_0314h

11.3.3.1.232.2 Diagram



11.3.3.1.232.3 Fields

Field	Function
31-0 DST_ADDR_REG_LOW	Destination Address Register (Lower 32 bits). Indicates the next address to be written to. The DMA increments the DAR as the DMA transfer progresses. In LL mode, the DMA overwrites this with the corresponding dword of the LL element. - DMA Read: The DAR is the address of the local memory. - DMA Read: The DAR is the address of the remote memory. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.233 DMA Read DAR High Register. (DMA_DAR_HIGH_OFF_RDC H_0)

11.3.3.1.233.1 Offset

Register	Offset
DMA_DAR_HIGH_OFF_RDCH_0	8008_0318h

11.3.3.1.233.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DST_ADDR_REG_HIGH															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DST_ADDR_REG_HIGH															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.233.3 Fields

Field	Function
31-0 DST_ADDR_REG_HIGH	Destination Address Register (Higher 32 bits). In LL mode, the DMA overwrites this with the corresponding dword of the LL element. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.234 DMA Read Linked List Pointer Low Register. (DMA_LL_P_LOW_OFF_RDCH_0)

11.3.3.1.234.1 Offset

Register	Offset
DMA_LL_P_LOW_OFF_RDCH_0	8008_031Ch

11.3.3.1.234.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	LLP_LOW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	LLP_LOW															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.234.3 Fields

Field	Function
31-0 LLP_LOW	Lower bits of the address of the linked list transfer list in local memory. Used in linked list mode only. Updated by the DMA to point to the next element in the transfer list after the previous element is consumed. - When the current element is a data element; this field is incremented by 6. - When the current element is a link element; this field is overwritten by the LL Element Pointer of the element. Note: The access attributes of this field are as follows: - Dbi: R/W

11.3.3.1.235 DMA Read Linked List Pointer High Register. (DMA_LL_P_HIGH_OFF_RDCH_0)

11.3.3.1.235.1 Offset

Register	Offset
DMA_LL_P_HIGH_OFF_RDCH_0	8008_0320h

11.3.3.1.235.2 Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	LLP_HIGH															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	LLP_HIGH															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.3.3.1.235.3 Fields

Field	Function
31-0 LLP_HIGH	Upper 32 bits of the address of the linked list transfer list in local memory. Used in linked list mode only. Updated by the DMA to point to the next element in the transfer list as elements are consumed. Note: The access attributes of this field are as follows: - Dbi: R/W

11.4 PCI Express PHY (PCle_PHY)

11.4.1 Overview

The PCIe 2 PHY implements the complete PCI Express 2.0 physical layer capability for connecting a root complex or endpoint to a PCI Express system. The PCIe 2 PHY supports the 5-Gbps data rate of the PCI Express Gen 2 specification and is backward compatible with the 2.5-Gbps Gen 1.1 specification. This chapter provides an introduction to the PCIe 2 PHY and its features.

11.4.1.1 Block Diagram

This section provides the high-level block diagram of the PCIe 2 PHY.

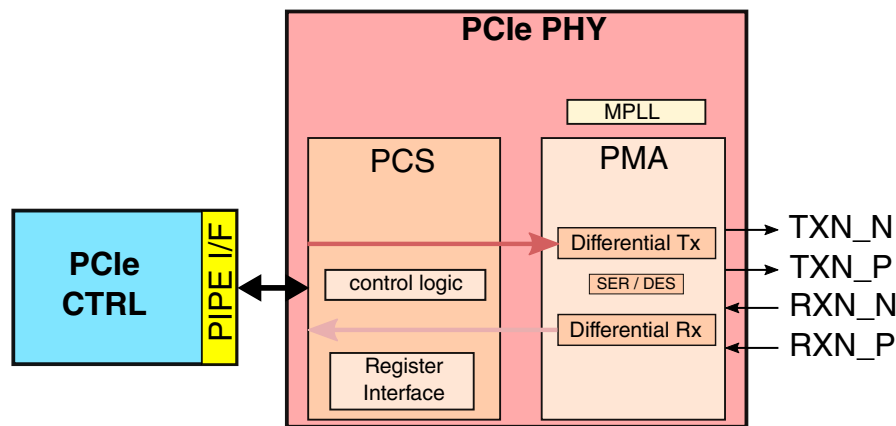


Figure 11-73. PCIe2 PHY Block Diagram

11.4.1.2 Features

The PCIe 2 PHY provides the following features:

- PIPE3-compliant transceiver interface
 - 5-Gbps data transmission rate PIPE3-compliant transceiver interface
- Integrated PHY includes transmitter, receiver, PLL, and digital core
- Programmable RX equalization
- Integrated regulator to support either 3.3-V or 1.8-V I/O power supply Low-jitter PLL
- IEEE 1149.6 (JTAG) boundary scan

11.4.2 Functional Descriptions

This section includes the relevant sub sections that detail the PCIe 2 PHY operation.

11.4.2.1 PIPE Interface

The data path of the PCIe 2 PHY is through the standard PIPE 3 interface. It supports one lane.

11.4.2.2 Detection of Electrical Idle Using PHY Loss of Signal Circuitry

The PCIe 2 PHY contains a loss-of-signal (LOS) detector that informs the controller when signaling is present on the high-speed input lines. The PHY deasserts the electrical idle signal to inform the PIPE interface when an electrical idle condition has been broken. Due to the LOS detector's bandwidth limitations, the detector can detect incoming data of up to 1.5 Gbps. At high transition density during normal data transmission, the detector might not detect signals on the line. Therefore, care must be applied to the controller to prevent false detection of electrical idle or beacon signaling.

11.4.2.3 Low-Power States Entry and Exit

Whenever the controller exits from a low-power state (or aborts entry into a low-power state), the controller waits for the PIPE 3 interface to fully complete the entry into the specific low-power state, then executes the exit from this low-power state.

11.4.2.4 Clocks

The PCIe 2 PHY supports a 100-MHz input to the MPLL. The PHY also provides a feature that halves the input clock. Therefore, the PHY supports either 100-MHz or 200-MHz reference clocks. The PIPE clock domain is the primary clock domain in the PIPE PCS, and because most PIPE signals are synchronous with PCLK, this clock domain is also a primary clock domain in any higher-level logic that communicates with the PHY via the PIPE interface.

PCLK depends on the selected PIPE interface width sourced from the PHY's MPLL. The PCIe 2 PHY does not support dynamic width selection. This clock domain should be considered synchronous to the PCS clock domain. The PCS clock domain or transmit clock domain runs at a fixed 250 MHz.

11.4.2.5 Reset

there is one active-high and one active-low asynchronous reset signals within the PCIe2 wrapper. The active-low reset is for the PCS and PIPE3 interface, and the active-high reset is for the power on reset for the PHY (asserted during power-on for all modes of operation).

11.4.3 PCIe PHY Memory Map/Register Definition

11.4.3.1 PCIe PHY register descriptions

11.4.3.1.1 PCIe PHY memory map

Offset	Register	Width (In bits)	Access	Reset value
0h	Low 16 bits of IDCODE (SUP_DIG_IDCODE_LO)	16	RO	64CDh
2h	High 16 bits of IDCODE (SUP_DIG_IDCODE_HI)	16	RO	A21Ch
4h	Debug controls (SUP_DIG_DEBUG)	16	RW	Table 11-41
6h	Resistor tuning debug controls (SUP_DIG_RTUNE_DEBUG)	16	RW	Table 11-41
8h	Resistor tuning register status (SUP_DIG_RTUNE_STAT)	16	RO	Table 11-41
Ah	Current MPLL phase selector value (SUP_DIG_SS_PHASE)	16	RW	8000h
Ch	Frequency Control for Spread Spectrum (SUP_DIG_SS_FREQ)	16	RW	Table 11-41
Eh	Current HSP_PMI SSC phase sel value (SUP_DIG_HSP_SS_PHASE)	16	RW	Table 11-41
10h	Control for fixed or variable MPLL Count Selection (SUP_DIG_MPLL_CNT_CTL)	16	RW	Table 11-41
20h	Override values for incoming controls from ASIC (SUP_DIG_ATEOVRD)	16	RW	Table 11-41
22h	Override values for incoming RTUNE and MPLL controls from ASIC (SUP_DIG_MPLL_OVRD_IN_LO)	16	RW	004Ch
24h	Override values for incoming VBOOST LVL and MPLL controls from ASIC (SUP_DIG_MPLL_OVRD_IN_HI)	16	RW	0000h

Table continues on the next page...

PCI Express PHY (PCIe_PHY)

Offset	Register	Width (In bits)	Access	Reset value
26h	Override values for incoming SSC controls from ASIC (SUP_DIG_SSC_OVRD_IN)	16	RW	Table 11-41
28h	Override values for incoming BS controls from ASIC (SUP_DIG_BS_OVRD_IN)	16	RW	Table 11-41
2Ah	Override values for acjt_level and los_level (SUP_DIG_LEVEL_OVRD_IN)	16	RW	Table 11-41
2Ch	Override values for support block (SUP_DIG_SUP_OVRD_OUT)	16	RW	Table 11-41
2Eh	Current values for incoming MPLL controls from ASIC (SUP_DIG_MPLL_ASIC_IN)	16	RO	Table 11-41
30h	Current values for incoming boundary scan controls from ASIC (SUP_DIG_BS_ASIC_IN)	16	RO	Table 11-41
32h	Current values for incoming los/acjt level, MPLL half rate mode and vreg bypass from ASIC (SUP_DIG_LEVEL_ASIC_IN)	16	RO	Table 11-41
34h	Current values for incoming SSC controls from ASIC (SUP_DIG_SSC_ASIC_IN)	16	RO	Table 11-41
36h	Current values for outgoing support block controls to ASIC (SUP_DIG_SUP_ASIC_OUT)	16	RO	Table 11-41
38h	SSP_SUP_DIG_ATEOVRD_STATUS (SUP_DIG_ATEOVRD_STATUS)	16	RO	Table 11-41
3Ah	Override values for HSP (SUP_DIG_HSP_OVRD_IN)	16	RW	Table 11-41
3Ch	Current values for incoming HSP & REF XO controls from ASIC (SUP_DIG_HSP_ASIC_IN)	16	RO	Table 11-41
40h	Scope Counter configuration (SUP_DIG_SCOPE_ENABLES)	16	RW	Table 11-41
42h	Number of samples to count (SUP_DIG_SCOPE_SAMPLES)	16	RW	0100h
44h	Results of scope counting (SUP_DIG_SCOPE_COUNT)	16	RO	0000h
46h	Masked scope counting Start and Status (SUP_DIG_SCOPE_CTL)	16	RW	Table 11-41
48h	Current count value for mask_000 (SUP_DIG_SCOPE_MASK_000)	16	RW	0000h
4Ah	Current count value for mask_001 (SUP_DIG_SCOPE_MASK_001)	16	RW	0000h
4Ch	Current count value for mask_010 (SUP_DIG_SCOPE_MASK_010)	16	RW	0000h
4Eh	Current count value for mask_011 (SUP_DIG_SCOPE_MASK_011)	16	RW	0000h
50h	Current count value for mask_100 (SUP_DIG_SCOPE_MASK_100)	16	RW	0000h
52h	Current count value for mask_101 (SUP_DIG_SCOPE_MASK_101)	16	RW	0000h
54h	Current count value for mask_110 (SUP_DIG_SCOPE_MASK_110)	16	RW	0000h
56h	Current count value for mask_111 (SUP_DIG_SCOPE_MASK_111)	16	RW	0000h
60h	MPLL loop control bits (SUP_ANA_MPLL_LOOP_CTL)	16	RW	Table 11-41
62h	ATB Select bits (SUP_ANA_MPLL_ATB_MEAS1)	16	RW	Table 11-41
64h	ATB Select bits (SUP_ANA_MPLL_ATB_MEAS2)	16	RW	Table 11-41
66h	Local overrides (SUP_ANA_MPLL_OVRD)	16	RW	Table 11-41
68h	RTUNE control bits (SUP_ANA_RTUNE_RTUNE_CTRL)	16	RW	Table 11-41
6Ah	ATB_SWITCHYARD control bits (SUP_ANA_ATB_SWITCHYARD_CTRL)	16	RW	Table 11-41
6Ch	controls sscf reference clock rate (SUP_ANA_SSC_CLK_CNTRL)	16	RW	Table 11-41

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
2000h	Override values for incoming TX controls from ASIC (LANE0_DIG_TX_OVRD_IN_LO)	16	RW	Table 11-41
2002h	Override values for incoming TX controls from ASIC (LANE0_DIG_TX_OVRD_IN_HI)	16	RW	Table 11-41
2004h	Override values for incoming TX drive controls from ASIC (LANE0_DIG_TX_OVRD_DRV_LO)	16	RW	Table 11-41
2006h	Override values for incoming TX drive controls from ASIC (LANE0_DIG_TX_OVRD_DRV_HI)	16	RW	Table 11-41
2008h	Override values for outgoing TX controls to ASIC (LANE0_DIG_TX_OVRD_OUT)	16	RW	Table 11-41
200Ah	Override values for incoming RX controls from ASIC (LANE0_DIG_RX_OVRD_IN_LO)	16	RW	Table 11-41
200Ch	Override values for incoming RX controls from ASIC (LANE0_DIG_RX_OVRD_IN_HI)	16	RW	Table 11-41
200Eh	Override values for outgoing RX controls to ASIC (LANE0_DIG_RX_OVRD_OUT)	16	RW	Table 11-41
2010h	Current values for incoming TX controls from ASIC (LANE0_DIG_TX_ASIC_IN)	16	RO	Table 11-41
2012h	Current values for incoming TX level controls from ASIC (LANE0_DIG_TX_ASIC_DRV_LO)	16	RO	Table 11-41
2014h	Current values for incoming TX level controls from ASIC (LANE0_DIG_TX_ASIC_DRV_HI)	16	RO	Table 11-41
2016h	Current values for outgoing TX status controls from PHY (LANE0_DIG_TX_ASIC_OUT)	16	RO	Table 11-41
2018h	Current values for incoming RX controls from ASIC (LANE0_DIG_RX_ASIC_IN)	16	RO	Table 11-41
201Ah	Current values for outgoing RX status controls from PHY (LANE0_DIG_RX_ASIC_OUT)	16	RO	Table 11-41
2020h	Transmit debug controls (LANE0_DIG_TX_DEBUG)	16	RW	Table 11-41
2022h	SSP_LANE0_DIG_TX_CM_WAIT_TIME_OVRD (LANE0_DIG_TX_CM_WAIT_TIME_OVRD)	16	RW	Table 11-41
2024h	Calculated values for N_USE and N_TRISTATE. (LANE0_DIG_TX_VMD_FSM_TX_VCM_0)	16	RO	Table 11-41
2026h	Calculated values for N_FIXED and N_TRAILER. (LANE0_DIG_TX_VMD_FSM_TX_VCM_1)	16	RO	Table 11-41
2028h	Access the TX_ANA VMD Control Shift Register (LANE0_DIG_TX_VMD_FSM_TX_VCM_DEBUG_IN)	16	RW	Table 11-41
202Ah	Pattern Generator controls (LANE0_DIG_TX_LBERT_CTL)	16	RW	Table 11-41
202Ch	Pattern Matcher controls (LANE0_DIG_RX_LBERT_CTL)	16	RW	Table 11-41
202Eh	Pattern match error counter (LANE0_DIG_RX_LBERT_ERR)	16	RW	0000h
2030h	Control bits for per-transceiver scope portion (LANE0_DIG_RX_SCOPE_CTL)	16	RW	Table 11-41
2032h	Scope sampling phase. (LANE0_DIG_RX_SCOPE_PHASE)	16	RW	Table 11-41
2034h	Current frequency integrator value. (LANE0_DIG_RX_DPLL_FREQ)	16	RW	Table 11-41

Table continues on the next page...

PCI Express PHY (PCIe_PHY)

Offset	Register	Width (In bits)	Access	Reset value
2036h	Control bits for receiver in recovered domain (LANE0_DIG_RX_CDR_CTL)	16	RW	000Fh
2038h	Current values for CDR and Adaptation FSM. (LANE0_DIG_RX_CDR_CDR_FSM_DEBUG)	16	RO	Table 11-41
203Ah	Control bits for receiver in recovered domain (LANE0_DIG_RX_CDR_LOCK_VEC_OVRD)	16	RW	8000h
203Ch	Current values for the locked vector. (LANE0_DIG_RX_CDR_LOCK_VEC)	16	RO	Table 11-41
203Eh	Observation of adaptation FSM register. (LANE0_DIG_RX_CDR_ADAP_FSM)	16	RO	Table 11-41
2040h	ATB Control Register (LANE0_ANA_RX_ATB0)	16	RW	Table 11-41
2042h	ATB and Misc Control Register (LANE0_ANA_RX_ATB1)	16	RW	Table 11-41
2044h	RX State Control Register (LANE0_ANA_RX_ENPWR0)	16	RW	Table 11-41
2046h	Phase select Value (LANE0_ANA_RX_PMI_X_PHASE)	16	RW	Table 11-41
2048h	Additional RX State Control Register (LANE0_ANA_RX_ENPWR1)	16	RW	Table 11-41
204Ah	Yet another RX State Control Register (LANE0_ANA_RX_ENPWR2)	16	RW	Table 11-41
204Ch	RX Scope controls and clock DCC control voltage probe (LANE0_ANA_RX_SCOPE_VDCC)	16	RW	Table 11-41
2056h	Register 3 (LANE0_ANA_TX_TXDRV_CNTRL)	16	RW	Table 11-41
2058h	Register 4 (LANE0_ANA_TX_POWER_CTL)	16	RW	Table 11-41
205Ah	Register 5 (LANE0_ANA_TX_ALT_BLOCK)	16	RW	Table 11-41
205Ch	Register 6 (LANE0_ANA_TX_ALT_AND_LOOPBACK)	16	RW	Table 11-41
205Eh	Register 7 (LANE0_ANA_TX_TX_ATB_REG)	16	RW	Table 11-41
2060h	Register 8 (LANE0_ANA_TX_TX_BOOST)	16	RW	Table 11-41
2080h	Control bits for CDR init phase portion (LANE0_DIG_RX_CDR_FSM_INIT_CTL)	16	RW	Table 11-41
2082h	Control bits for CDR track phase portion (LANE0_DIG_RX_CDR_FSM_TRACK_CTL)	16	RW	Table 11-41
2084h	Control bits for CDR lock phase portion (LANE0_DIG_RX_CDR_FSM_LOCK_CTL)	16	RW	Table 11-41
1_0000h	Low 16 bits of IDCODE (SUPX_DIG_IDCODE_LO)	16	RO	04CDh
1_0002h	High 16 bits of IDCODE (SUPX_DIG_IDCODE_HI)	16	RO	3006h
1_0004h	Debug controls (SUPX_DIG_DEBUG)	16	RW	Table 11-41
1_0006h	Resistor tuning debug controls (SUPX_DIG_RTUNE_DEBUG)	16	RW	Table 11-41
1_0008h	Resistor tuning register status (SUPX_DIG_RTUNE_STAT)	16	RO	Table 11-41
1_000Ah	Current MPLL phase selector value (SUPX_DIG_SS_PHASE)	16	RW	8000h
1_000Ch	Frequency Control for Spread Spectrum (SUPX_DIG_SS_FREQ)	16	RW	Table 11-41
1_000Eh	Current HSP_PMI_X SSC phase sel value (SUPX_DIG_HSP_SS_PHASE)	16	RW	Table 11-41
1_0010h	Control for fixed or variable MPLL Count Selection (SUPX_DIG_MPLL_CNT_CTL)	16	RW	Table 11-41
1_0020h	Override values for incoming controls from ASIC (SUPX_DIG_ATEOVRD)	16	RW	Table 11-41

Table continues on the next page...

Offset	Register	Width (In bits)	Access	Reset value
1_0022h	Override values for incoming RTUNE and MPLL controls from ASIC (SUPX_DIG_MPLL_OVRD_IN_LO)	16	RW	004Ch
1_0024h	Override values for incoming VBOOST LVL and MPLL controls from ASIC (SUPX_DIG_MPLL_OVRD_IN_HI)	16	RW	0000h
1_0026h	Override values for incoming SSC controls from ASIC (SUPX_DIG_SSC_OVRD_IN)	16	RW	Table 11-41
1_0028h	Override values for incoming BS controls from ASIC (SUPX_DIG_BS_OVRD_IN)	16	RW	Table 11-41
1_002Ah	Override values for acjt_level and los_level (SUPX_DIG_LEVEL_OVRD_IN)	16	RW	Table 11-41
1_002Ch	Override values for support block (SUPX_DIG_SUP_OVRD_OUT)	16	RW	Table 11-41
1_002Eh	Current values for incoming MPLL controls from ASIC (SUPX_DIG_MPLL_ASIC_IN)	16	RO	Table 11-41
1_0030h	Current values for incoming boundary scan controls from ASIC (SUPX_DIG_BS_ASIC_IN)	16	RO	Table 11-41
1_0032h	Current values for incoming los/acjt level, MPLL half rate mode and vreg bypass from ASIC (SUPX_DIG_LEVEL_ASIC_IN)	16	RO	Table 11-41
1_0034h	Current values for incoming SSC controls from ASIC (SUPX_DIG_SSC_ASIC_IN)	16	RO	Table 11-41
1_0036h	Current values for outgoing support block controls to ASIC (SUPX_DIG_SUP_ASIC_OUT)	16	RO	Table 11-41
1_0038h	SSP_SUPX_DIG_ATEOVRD_STATUS (SUPX_DIG_ATEOVRD_STATUS)	16	RO	Table 11-41
1_003Ah	Override values for HSP (SUPX_DIG_HSP_OVRD_IN)	16	RW	Table 11-41
1_003Ch	Current values for incoming HSP & REF XO controls from ASIC (SUPX_DIG_HSP_ASIC_IN)	16	RO	Table 11-41
1_0040h	Scope Counter configuration (SUPX_DIG_SCOPE_ENABLES)	16	RW	Table 11-41
1_0042h	Number of samples to count (SUPX_DIG_SCOPE_SAMPLES)	16	RW	0100h
1_0044h	Results of scope counting (SUPX_DIG_SCOPE_COUNT)	16	RO	0000h
1_0046h	Masked scope counting Start and Status (SUPX_DIG_SCOPE_CTL)	16	RW	Table 11-41
1_0048h	Current count value for mask_000 (SUPX_DIG_SCOPE_MASK_000)	16	RW	0000h
1_004Ah	Current count value for mask_001 (SUPX_DIG_SCOPE_MASK_001)	16	RW	0000h
1_004Ch	Current count value for mask_010 (SUPX_DIG_SCOPE_MASK_010)	16	RW	0000h
1_004Eh	Current count value for mask_011 (SUPX_DIG_SCOPE_MASK_011)	16	RW	0000h
1_0050h	Current count value for mask_100 (SUPX_DIG_SCOPE_MASK_100)	16	RW	0000h
1_0052h	Current count value for mask_101 (SUPX_DIG_SCOPE_MASK_101)	16	RW	0000h
1_0054h	Current count value for mask_110 (SUPX_DIG_SCOPE_MASK_110)	16	RW	0000h
1_0056h	Current count value for mask_111 (SUPX_DIG_SCOPE_MASK_111)	16	RW	0000h
1_0060h	MPLL loop control bits (SUPX_ANA_MPLL_LOOP_CTL)	16	RW	Table 11-41
1_0062h	ATB Select bits (SUPX_ANA_MPLL_ATB_MEAS1)	16	RW	Table 11-41
1_0064h	ATB Select bits (SUPX_ANA_MPLL_ATB_MEAS2)	16	RW	Table 11-41
1_0066h	Local overrides (SUPX_ANA_MPLL_OVRD)	16	RW	Table 11-41
1_0068h	RTUNE control bits (SUPX_ANA_RTUNE_RTUNE_CTRL)	16	RW	Table 11-41

Table continues on the next page...

PCI Express PHY (PCle_PHY)

Offset	Register	Width (In bits)	Access	Reset value
1_006Ah	ATB_SWITCHYARD control bits (SUPX_ANA_ATB_SWITCHYARD_CTRL)	16	RW	Table 11-41
1_006Ch	controls sscf reference clock rate (SUPX_ANA_SSC_CLK_CNTRL)	16	RW	Table 11-41
1_2000h	Override values for incoming TX controls from ASIC (LANEX_DIG_TX_OVRD_IN_LO)	16	RW	Table 11-41
1_2002h	Override values for incoming TX controls from ASIC (LANEX_DIG_TX_OVRD_IN_HI)	16	RW	Table 11-41
1_2004h	Override values for incoming TX drive controls from ASIC (LANEX_DIG_TX_OVRD_DRV_LO)	16	RW	Table 11-41
1_2006h	Override values for incoming TX drive controls from ASIC (LANEX_DIG_TX_OVRD_DRV_HI)	16	RW	Table 11-41
1_2008h	Override values for outgoing TX controls to ASIC (LANEX_DIG_TX_OVRD_OUT)	16	RW	Table 11-41
1_200Ah	Override values for incoming RX controls from ASIC (LANEX_DIG_RX_OVRD_IN_LO)	16	RW	Table 11-41
1_200Ch	Override values for incoming RX controls from ASIC (LANEX_DIG_RX_OVRD_IN_HI)	16	RW	Table 11-41
1_200Eh	Override values for outgoing RX controls to ASIC (LANEX_DIG_RX_OVRD_OUT)	16	RW	Table 11-41
1_2010h	Current values for incoming TX controls from ASIC (LANEX_DIG_TX_ASIC_IN)	16	RO	Table 11-41
1_2012h	Current values for incoming TX level controls from ASIC (LANEX_DIG_TX_ASIC_DRV_LO)	16	RO	Table 11-41
1_2014h	Current values for incoming TX level controls from ASIC (LANEX_DIG_TX_ASIC_DRV_HI)	16	RO	Table 11-41
1_2016h	Current values for outgoing TX status controls from PHY (LANEX_DIG_TX_ASIC_OUT)	16	RO	Table 11-41
1_2018h	Current values for incoming RX controls from ASIC (LANEX_DIG_RX_ASIC_IN)	16	RO	Table 11-41
1_201Ah	Current values for outgoing RX status controls from PHY (LANEX_DIG_RX_ASIC_OUT)	16	RO	Table 11-41
1_2020h	Transmit debug controls (LANEX_DIG_TX_DEBUG)	16	RW	Table 11-41
1_2022h	SSP_LANEX_DIG_TX_CM_WAIT_TIME_OVRD (LANEX_DIG_TX_CM_WAIT_TIME_OVRD)	16	RW	Table 11-41
1_2024h	Calculated values for N_USE and N_TRISTATE. (LANEX_DIG_TX_VMD_FSM_TX_VCM_0)	16	RO	Table 11-41
1_2026h	Calculated values for N_FIXED and N_TRAILER. (LANEX_DIG_TX_VMD_FSM_TX_VCM_1)	16	RO	Table 11-41
1_2028h	Access the TX_ANA VMD Control Shift Register (LANEX_DIG_TX_VMD_FSM_TX_VCM_DEBUG_IN)	16	RW	Table 11-41
1_202Ah	Pattern Generator controls (LANEX_DIG_TX_LBERT_CTL)	16	RW	Table 11-41
1_202Ch	Pattern Matcher controls (LANEX_DIG_RX_LBERT_CTL)	16	RW	Table 11-41
1_202Eh	Pattern match error counter (LANEX_DIG_RX_LBERT_ERR)	16	RW	0000h
1_2030h	Control bits for per-transceiver scope portion (LANEX_DIG_RX_SCOPE_CTL)	16	RW	Table 11-41

Table continues on the next page...

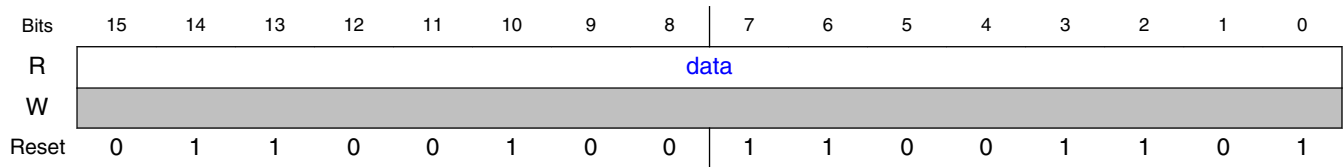
Offset	Register	Width (In bits)	Access	Reset value
1_2032h	Scope sampling phase. (LANEX_DIG_RX_SCOPE_PHASE)	16	RW	Table 11-41
1_2034h	Current frequency integrator value. (LANEX_DIG_RX_DPLL_FREQ)	16	RW	Table 11-41
1_2036h	Control bits for receiver in recovered domain (LANEX_DIG_RX_CDR_CTL)	16	RW	000Fh
1_2038h	Current values for CDR and Adaptation FSM. (LANEX_DIG_RX_CDR_CDR_FSM_DEBUG)	16	RO	Table 11-41
1_203Ah	Control bits for receiver in recovered domain (LANEX_DIG_RX_CDR_LOCK_VEC_OVRD)	16	RW	8000h
1_203Ch	Current values for the locked vector. (LANEX_DIG_RX_CDR_LOCK_VEC)	16	RO	Table 11-41
1_203Eh	Observation of adaptation FSM register. (LANEX_DIG_RX_CDR_ADAP_FSM)	16	RO	Table 11-41
1_2040h	ATB Control Register (LANEX_ANA_RX_ATB0)	16	RW	Table 11-41
1_2042h	ATB and Misc Control Register (LANEX_ANA_RX_ATB1)	16	RW	Table 11-41
1_2044h	RX State Control Register (LANEX_ANA_RX_ENPWR0)	16	RW	Table 11-41
1_2046h	Phase select Value (LANEX_ANA_RX_PMI_X_PHASE)	16	RW	Table 11-41
1_2048h	Additional RX State Control Register (LANEX_ANA_RX_ENPWR1)	16	RW	Table 11-41
1_204Ah	Yet another RX State Control Register (LANEX_ANA_RX_ENPWR2)	16	RW	Table 11-41
1_204Ch	RX Scope controls and clock DCC control voltage probe (LANEX_ANA_RX_SCOPE_VDCC)	16	RW	Table 11-41
1_2056h	Register 3 (LANEX_ANA_TX_TXDRV_CNTRL)	16	RW	Table 11-41
1_2058h	Register 4 (LANEX_ANA_TX_POWER_CTL)	16	RW	Table 11-41
1_205Ah	Register 5 (LANEX_ANA_TX_ALT_BLOCK)	16	RW	Table 11-41
1_205Ch	Register 6 (LANEX_ANA_TX_ALT_AND_LOOPBACK)	16	RW	Table 11-41
1_205Eh	Register 7 (LANEX_ANA_TX_TX_ATB_REG)	16	RW	Table 11-41
1_2060h	Register 8 (LANEX_ANA_TX_TX_BOOST)	16	RW	Table 11-41
1_2080h	Control bits for CDR init phase portion (LANEX_DIG_RX_CDR_FSM_INIT_CTL)	16	RW	Table 11-41
1_2082h	Control bits for CDR track phase portion (LANEX_DIG_RX_CDR_FSM_TRACK_CTL)	16	RW	Table 11-41
1_2084h	Control bits for CDR lock phase portion (LANEX_DIG_RX_CDR_FSM_LOCK_CTL)	16	RW	Table 11-41

11.4.3.1.2 Low 16 bits of IDCODE (SUP_DIG_IDCODE_LO)

11.4.3.1.2.1 Offset

Register	Offset
SUP_DIG_IDCODE_LO	0h

11.4.3.1.2.2 Diagram



11.4.3.1.2.3 Fields

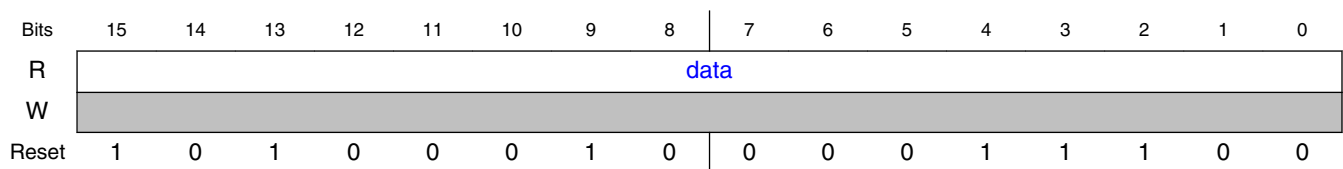
Field	Function
15-0 data	data

11.4.3.1.3 High 16 bits of IDCODE (SUP_DIG_IDCODE_HI)

11.4.3.1.3.1 Offset

Register	Offset
SUP_DIG_IDCODE_HI	2h

11.4.3.1.3.2 Diagram



11.4.3.1.3.3 Fields

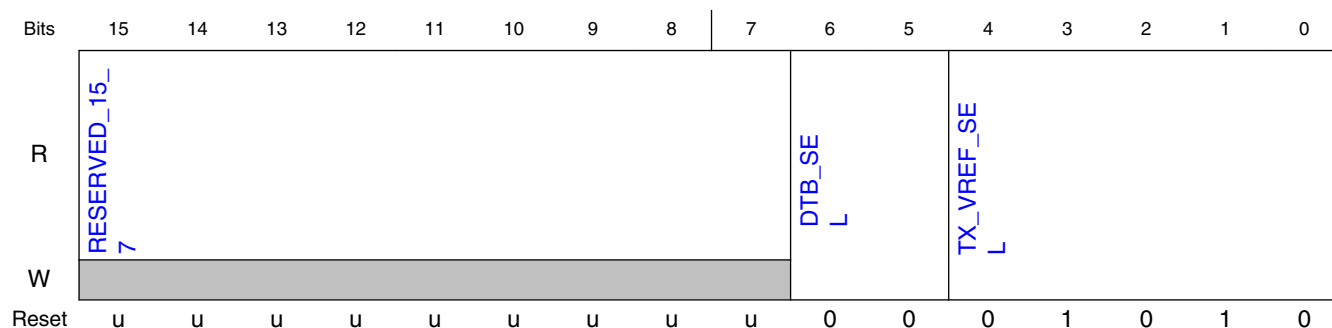
Field	Function
15-0 data	data

11.4.3.1.4 Debug controls (SUP_DIG_DEBUG)

11.4.3.1.4.1 Offset

Register	Offset
SUP_DIG_DEBUG	4h

11.4.3.1.4.2 Diagram



11.4.3.1.4.3 Fields

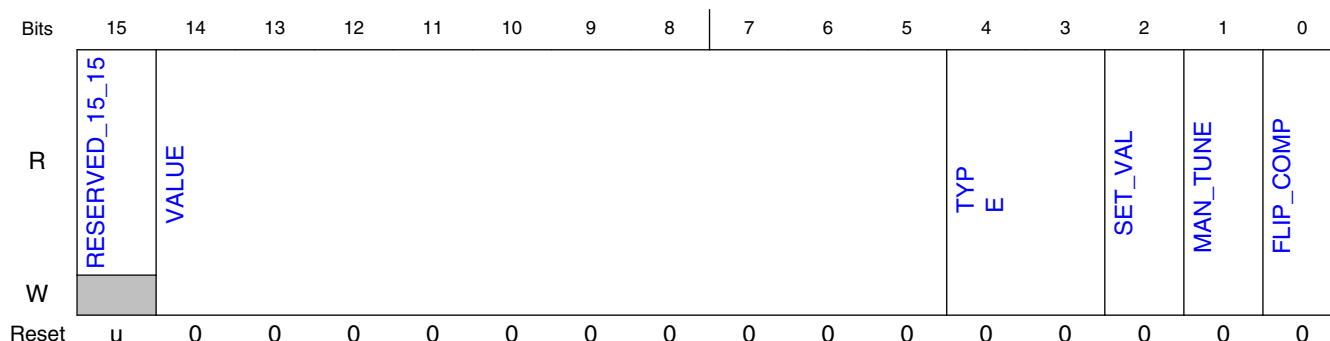
Field	Function
15-7 RESERVED_15_7	Reserved for Future use
6-5 DTB_SEL	The lane DTB's are OR'd together with the support DTB signals selected with the below encodings 0 - none 1 - reset_ctl DTB output 2 - scope DTB output 3 - rtune DTB output
4-0 TX_VREF_SEL	TX_VREF_SEL

11.4.3.1.5 Resistor tuning debug controls (SUP_DIG_RTUNE_DEBUG)

11.4.3.1.5.1 Offset

Register	Offset
SUP_DIG_RTUNE_DEB UG	6h

11.4.3.1.5.2 Diagram



11.4.3.1.5.3 Fields

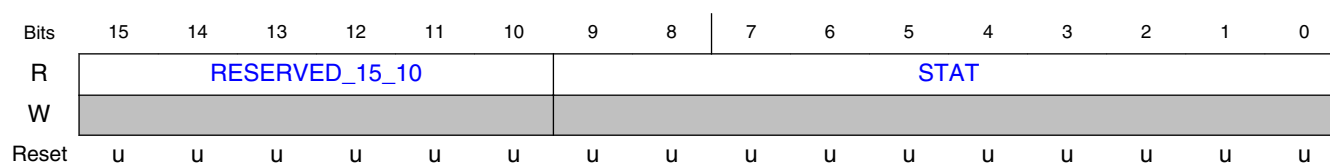
Field	Function
15 RESERVED_15_15	Reserved for Future use
14-5 VALUE	Value to use when triggering SET_VAL field only the 6 LSB's are used when setting rx or tx cal values (USER2 - read-write and custom)
4-3 TYPE	Type of manual tuning or register read/write to execute 0 - ADC, or read/write rt_value 1 - Rx tune, or read/write rx_cal_val (only 6 bits) 2 - Tx tune, or read/write tx_cal_val (only 6 bits) 3 - Resref detect (no affect when triggering SET_VAL field) (USER2 - read-write and custom)
2 SET_VAL	Set value Write to a 1 to manually write the register specified by the TYPE field to the value in the VALUE field (USER2 - read-write and custom)
1 MAN_TUNE	Write to a 1 to do a manual tuning specified by TYPE field starting a manual tune while a tune is currently running can cause unpredictable results. Write to a 1 to do a manual tuning specified by TYPE field starting a manual tune while a tune is currently running can cause unpredictable results. For use only when you know the state of the rtune circuit. (USER2 - read-write and custom)
0 FLIP_COMP	Invert analog comparator output (USER2 - read-write and custom)

11.4.3.1.6 Resistor tuning register status (SUP_DIG_RTUNE_STAT)

11.4.3.1.6.1 Offset

Register	Offset
SUP_DIG_RTUNE_STAT	8h

11.4.3.1.6.2 Diagram



11.4.3.1.6.3 Fields

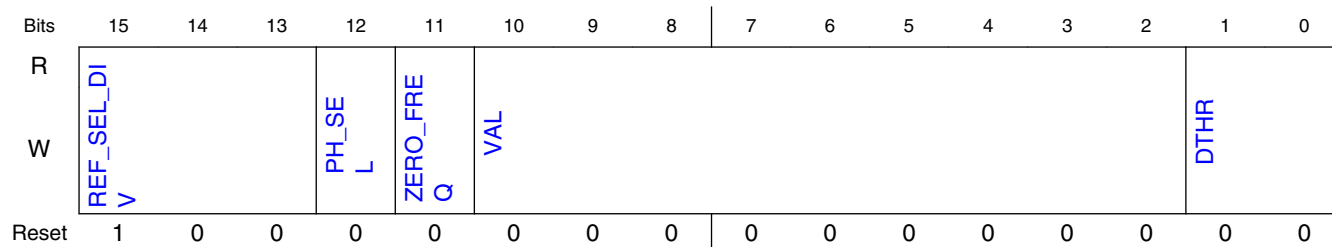
Field	Function
15-10 RESERVED_15_10	Reserved for Future use
9-0 STAT	Current value of the register specified by the TYPE field in the debug control register. Current value of the register specified by the TYPE field in the debug control register. Set the TYPE field in the DEBUG register, and then do a read of this register. (RU - read-only volatile)

11.4.3.1.7 Current MPLL phase selector value (SUP_DIG_SS_PHASE)

11.4.3.1.7.1 Offset

Register	Offset
SUP_DIG_SS_PHASE	Ah

11.4.3.1.7.2 Diagram



11.4.3.1.7.3 Fields

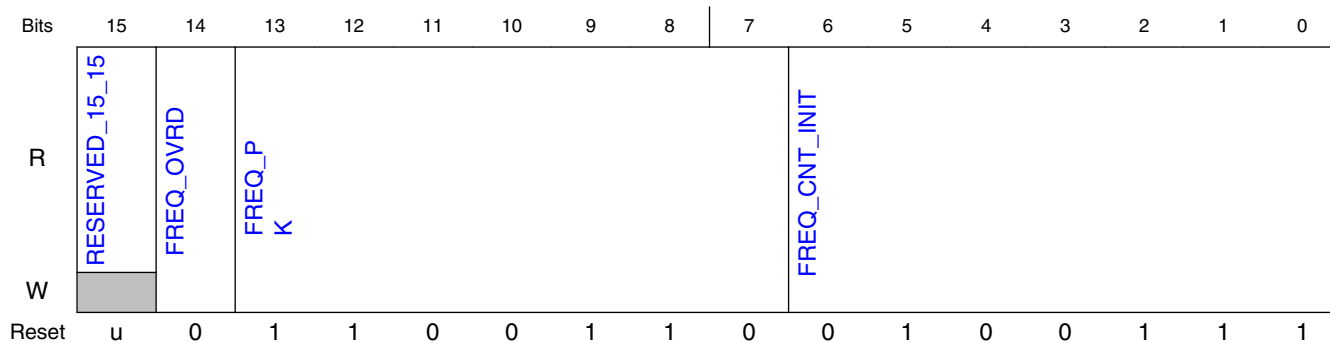
Field	Function
15-13 REF_SEL_DIV	Reference Divisor to get better resolution and lower PPM errors when synthesizing from a non-integer reference clock. Reference Divisor to get better resolution and lower PPM errors when synthesizing from a non-integer reference clock. NOTE: Default value is 4. One is added in hardware to give a range of 1 to 8. (volatile and 2 reads needed to read value)
12 PH_SEL	Phase Mixer Clock select. Phase Mixer Clock select. 1: is pmix. 0 is VCO. (volatile and 2 reads needed to read value)
11 ZERO_FREQ	Zero frequency register. Zero frequency register. NOTES: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value)
10-2 VAL	Phase value from zero reference (volatile and 2 reads needed to read value)
1-0 DTHR	Bits below the useful resolution (volatile and 2 reads needed to read value)

11.4.3.1.8 Frequency Control for Spread Spectrum (SUP_DIG_SS_FREQ)

11.4.3.1.8.1 Offset

Register	Offset
SUP_DIG_SS_FREQ	Ch

11.4.3.1.8.2 Diagram



11.4.3.1.8.3 Fields

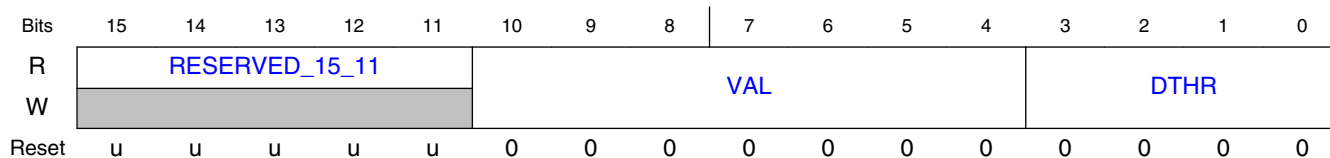
Field	Function
15 RESERVED_15_15	Reserved for Future use
14 FREQ_OVRD	Frequency Register Override. Frequency Register Override. NOTE: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value)
13-7 FREQ_PK	Peak Frequency Value (for changing direction) (volatile and 2 reads needed to read value)
6-0 FREQ_CNT_INIT	Initial Frequency Counter Value (volatile and 2 reads needed to read value)

11.4.3.1.9 Current HSP_PMI SS phase sel value (SUP_DIG_HSP_SS_PHASE)

11.4.3.1.9.1 Offset

Register	Offset
SUP_DIG_HSP_SS_PHASE	Eh

11.4.3.1.9.2 Diagram



11.4.3.1.9.3 Fields

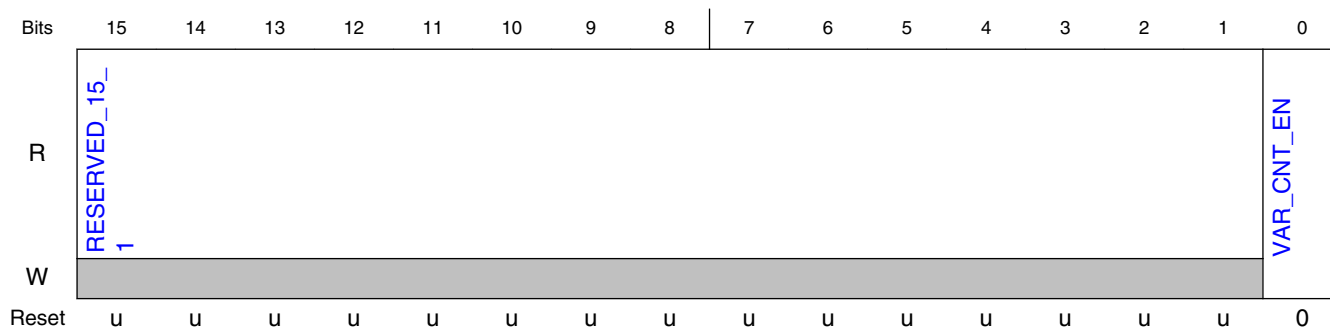
Field	Function
15-11 RESERVED_15_11	Reserved for Future use
10-4 VAL	SSC phase value from zero reference (volatile and 2 reads needed to read value)
3-0 DTHR	Bits below the useful resolution (volatile and 2 reads needed to read value)

11.4.3.1.10 Control for fixed or variable MPLL Count Selection (SUP_DIG_MPLL_CNT_CTL)

11.4.3.1.10.1 Offset

Register	Offset
SUP_DIG_MPLL_CNT_CTL	10h

11.4.3.1.10.2 Diagram



11.4.3.1.10.3 Fields

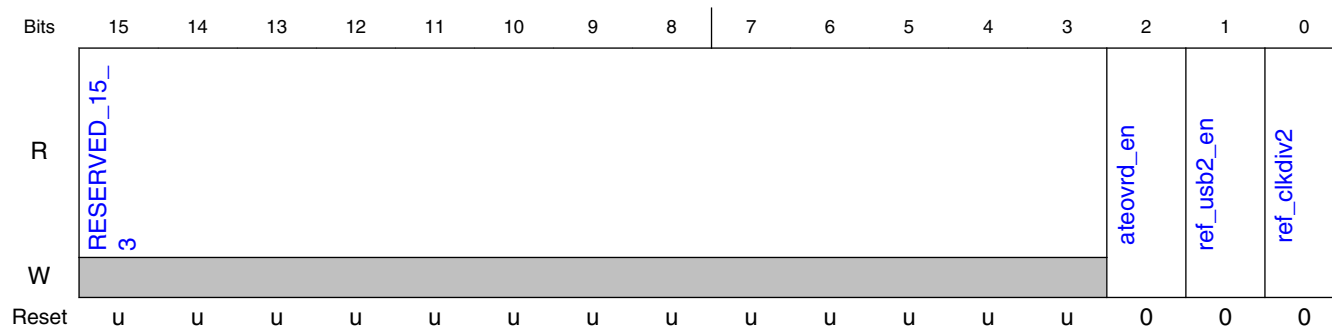
Field	Function
15-1 RESERVED_15_1	Reserved for Future use
0 VAR_CNT_EN	Selects variable count when enabled (Default = fixed count)

11.4.3.1.11 Override values for incoming controls from ASIC (SUP_DIG_ATEOVRD)

11.4.3.1.11.1 Offset

Register	Offset
SUP_DIG_ATEOVRD	20h

11.4.3.1.11.2 Diagram



11.4.3.1.11.3 Fields

Field	Function
15-3 RESERVED_15_3	Reserved for Future use
2 ateovrd_en	Override enable for ATE signals. Override enable for ATE signals. NOTE: This register is only accessible through cregs. Does not retain state during reset or ATPG. (USER2 - read-write and custom)

Table continues on the next page...

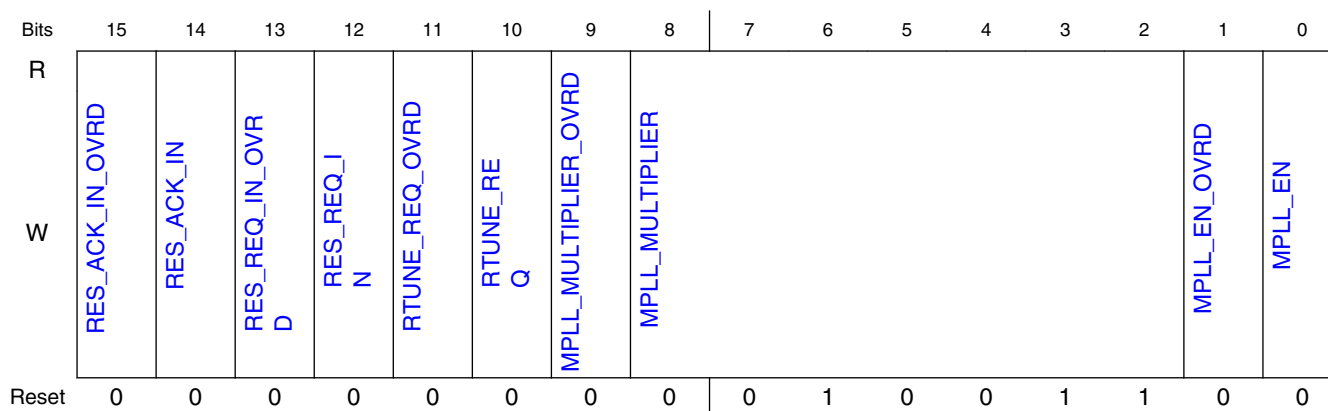
Field	Function
1 ref_usb2_en	Override value for USB HSP PHY ref_clk enable. Override value for USB HSP PHY ref_clk enable. (USER2 - read-write and custom)
0 ref_clkdiv2	Override value for SSP ref_clk pre-scaler. Override value for SSP ref_clk pre-scaler. (USER2 - read-write and custom)

11.4.3.1.12 Override values for incoming RTUNE and MPLL controls from ASIC (SUP_DIG_MPLL_OVRD_IN_LO)

11.4.3.1.12.1 Offset

Register	Offset
SUP_DIG_MPLL_OVRD_IN_LO	22h

11.4.3.1.12.2 Diagram



11.4.3.1.12.3 Fields

Field	Function
15 RES_ACK_IN_OVRD	Override enable for res_ack_in
14 RES_ACK_IN	Override value for res_ack_in
13	Override enable for res_req_in

Table continues on the next page...

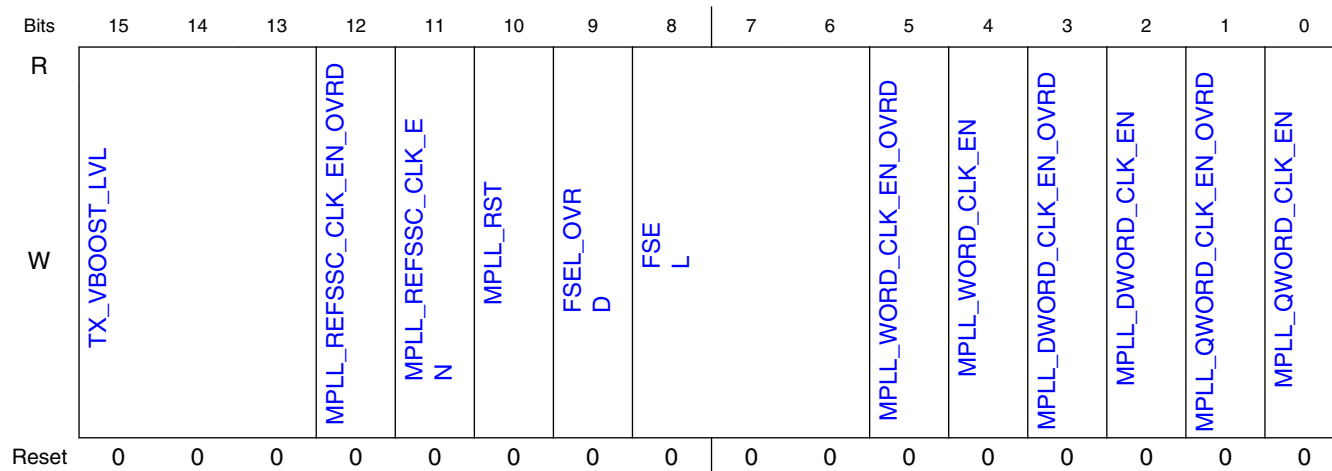
Field	Function
RES_REQ_IN_OVRD	
12 RES_REQ_IN	Override value for res_req_in
11 RTUNE_REQ_OVRD	Override enable for rtune_req
10 RTUNE_REQ	Override value for rtune_req
9 MPLL_MULTIPLIER_OVRD	Override enable for mpll_multipler
8-2 MPLL_MULTIPLIER	Override value for mpll_multipler
1 MPLL_EN_OVRD	Override enable for mpll_en
0 MPLL_EN	Override value for mpll_en

11.4.3.1.13 Override values for incoming VBOOST LVL and MPLL controls from ASIC (SUP_DIG_MPLL_OVRD_IN_HI)

11.4.3.1.13.1 Offset

Register	Offset
SUP_DIG_MPLL_OVRD_IN_HI	24h

11.4.3.1.13.2 Diagram



11.4.3.1.13.3 Fields

Field	Function
15-13 TX_VBOOST_LVL	Override value for tx_vboost_lvl. Override value for tx_vboost_lvl. Override enabled using LEVEL_EN from LEVEL_OVRD_IN register.
12 MPLL_REFSSC_CLK_EN_OVRD	Override enable for mpll_refssc_clk_en
11 MPLL_REFSSC_CLK_EN	Override value for mpll_refssc_clk_en
10 MPLL_RST	Reset the MPLL state machine Writing the register with this bit set will reset the MPLL power up/down FSM, regardless of the current state of the register bit.
9 FSEL_OVRD	Override enable for fsel[2:0]
8-6 FSEL	Override value for fsel [2:0]
5 MPLL_WORD_CLK_EN_OVRD	Override enable for mpll_word_clk_en
4 MPLL_WORD_CLK_EN	Override value for mpll_word_clk_en
3	Override enable for mpll_dword_clk_en

Table continues on the next page...

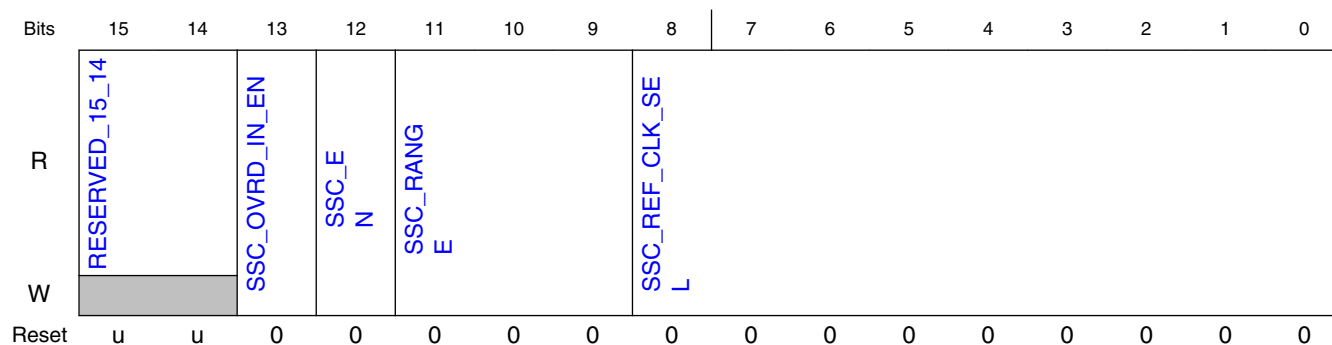
Field	Function
MPLL_DWORD_CLK_EN_OVRD	
2 MPLL_DWORD_CLK_EN	Override value for mpll_dword_clk_en
1 MPLL_QWORD_CLK_EN_OVRD	Override enable for mpll_qword_clk_en
0 MPLL_QWORD_CLK_EN	Override value for mpll_qword_clk_en

11.4.3.1.14 Override values for incoming SSC controls from ASIC (SUP_DIG_SSC_OVRD_IN)

11.4.3.1.14.1 Offset

Register	Offset
SUP_DIG_SSC_OVRD_IN	26h

11.4.3.1.14.2 Diagram



11.4.3.1.14.3 Fields

Field	Function
15-14	Reserved for Future use

Table continues on the next page...

PCI Express PHY (PCIe_PHY)

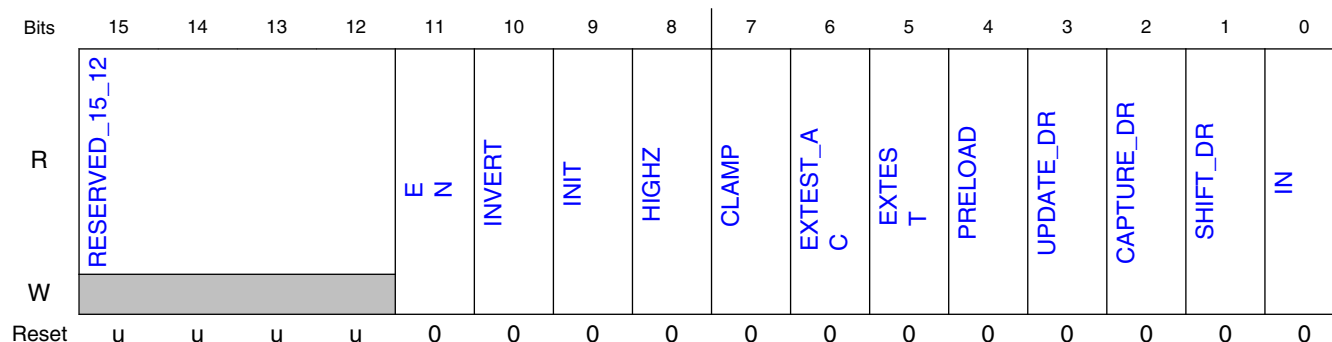
Field	Function
RESERVED_15_14	
13 SSC_OVRD_IN_EN	Override enable for Spread Spectrum generator.
12 SSC_EN	Override value for SSC enable.
11-9 SSC_RANGE	Override value for SSC modulation range.
8-0 SSC_REF_CLK_SEL	Override value for reference clock scaling

11.4.3.1.15 Override values for incoming BS controls from ASIC (SUP_DIG_BS_OVRD_IN)

11.4.3.1.15.1 Offset

Register	Offset
SUP_DIG_BS_OVRD_IN	28h

11.4.3.1.15.2 Diagram



11.4.3.1.15.3 Fields

Field	Function
15-12	Reserved for Future use

Table continues on the next page...

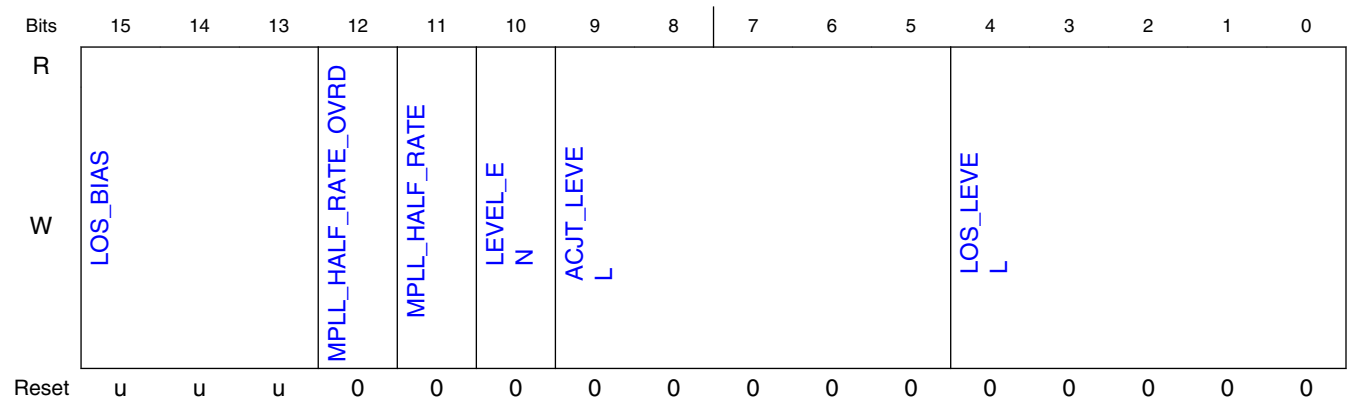
Field	Function
RESERVED_15_12	
11 EN	Enable override values for all inputs controlled by this register
10 INVERT	Override value for bs_invert
9 INIT	Override value for bs_init
8 HIGHZ	Override value for bs_highz
7 CLAMP	Override value for bs_clamp
6 EXTEST_AC	Override value for bs_extest_ac
5 EXTEST	Override value for bs_extest
4 PRELOAD	Override value for bs_preload
3 UPDATE_DR	Override value for bs_update_dr
2 CAPTURE_DR	Override value for bs_capture_dr
1 SHIFT_DR	Override value for bs_shift_dr
0 IN	Override value for bs_in

11.4.3.1.16 Override values for acjt_level and los_level (SUP_DIG_LEVEL_OVRD_IN)

11.4.3.1.16.1 Offset

Register	Offset
SUP_DIG_LEVEL_OVRD_IN	2Ah

11.4.3.1.16.2 Diagram



11.4.3.1.16.3 Fields

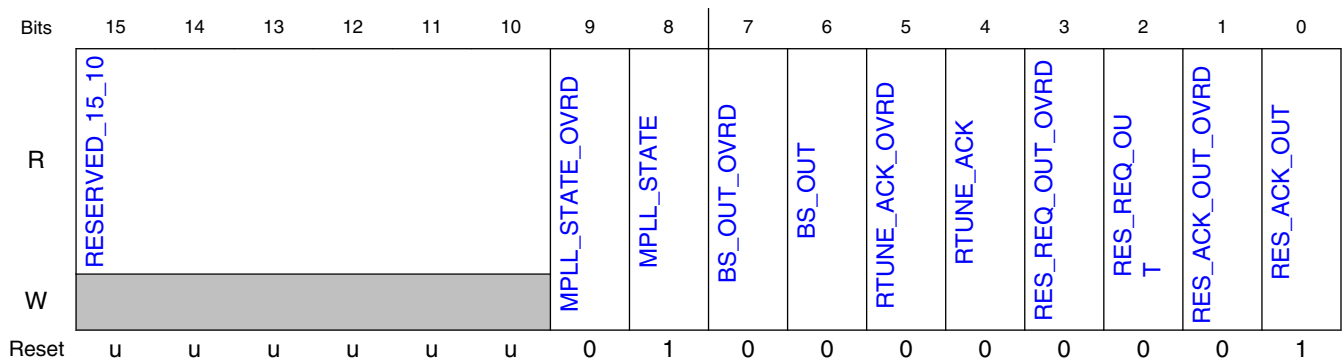
Field	Function
15-13 LOS_BIAS	Override value for los_bias
12 MPLL_HALF_RATE_OVRD	Enable override value for mpll_half_rate
11 MPLL_HALF_RATE	Override value for mpll_half_rate
10 LEVEL_EN	Enable override values for tx_vboost_lvl/los_bias/los_level/acjt_level inputs controlled by this and MPLL_OVRD_IN_HI registers
9-5 ACJT_LEVEL	Override value for acjt_level
4-0 LOS_LEVEL	Override value for los_level

11.4.3.1.17 Override values for support block (SUP_DIG_SUP_OVRD_OUT)

11.4.3.1.17.1 Offset

Register	Offset
SUP_DIG_SUP_OVRD_OUT	2Ch

11.4.3.1.17.2 Diagram



11.4.3.1.17.3 Fields

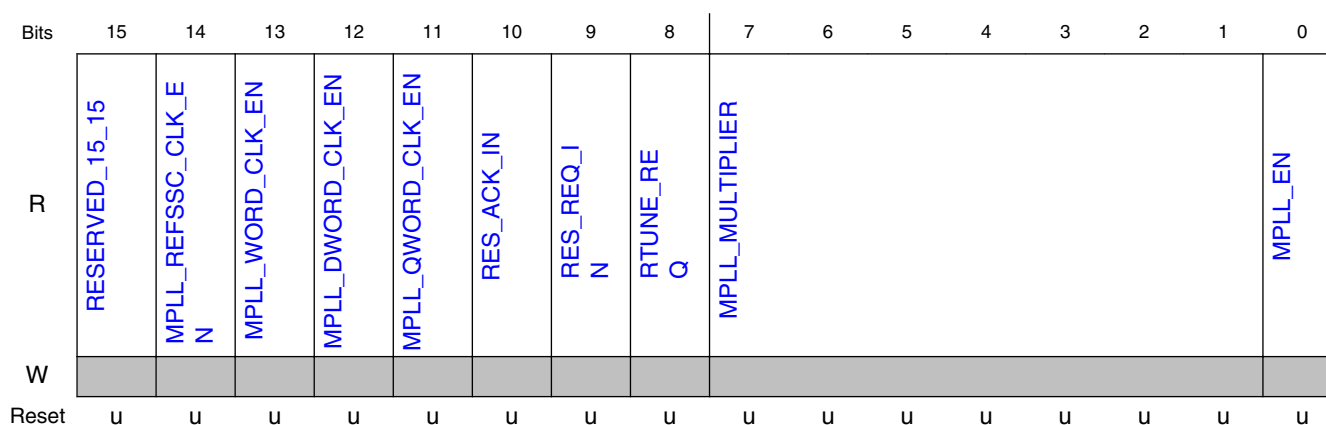
Field	Function
15-10 RESERVED_15_10	Reserved for Future use
9 MPLL_STATE_OVRD	Override enable for mpll_state output
8 MPLL_STATE	Override value for mpll_state output
7 BS_OUT_OVRD	Override enable for bs_out output
6 BS_OUT	Override value for bs_out output
5 RTUNE_ACK_OVRD	Override enable for rtune_ack output
4 RTUNE_ACK	Override value for rtune_ack output
3 RES_REQ_OUT_OVRD	Override enable for res_req_out output
2 RES_REQ_OUT	Override value for res_req_out output
1 RES_ACK_OUT_OVRD	Override enable for res_ack_out output
0 RES_ACK_OUT	Override value for res_ack_out output

11.4.3.1.18 Current values for incoming MPLL controls from ASIC (SUP_DIG_MPLL_ASIC_IN)

11.4.3.1.18.1 Offset

Register	Offset
SUP_DIG_MPLL_ASIC_IN	2Eh

11.4.3.1.18.2 Diagram



11.4.3.1.18.3 Fields

Field	Function
15 RESERVED_15_15	Reserved for Future use
14 MPLL_REFSSC_CLK_EN	Value from ASIC for mpll_refssc_clk_en (RU - read-only volatile)
13 MPLL_WORD_CLK_EN	Value from ASIC for mpll_word_clk_en (RU - read-only volatile)
12 MPLL_DWORD_CLK_EN	Value from ASIC for mpll_dword_clk_en (RU - read-only volatile)
11	Value from ASIC for mpll_qword_clk_en (RU - read-only volatile)

Table continues on the next page...

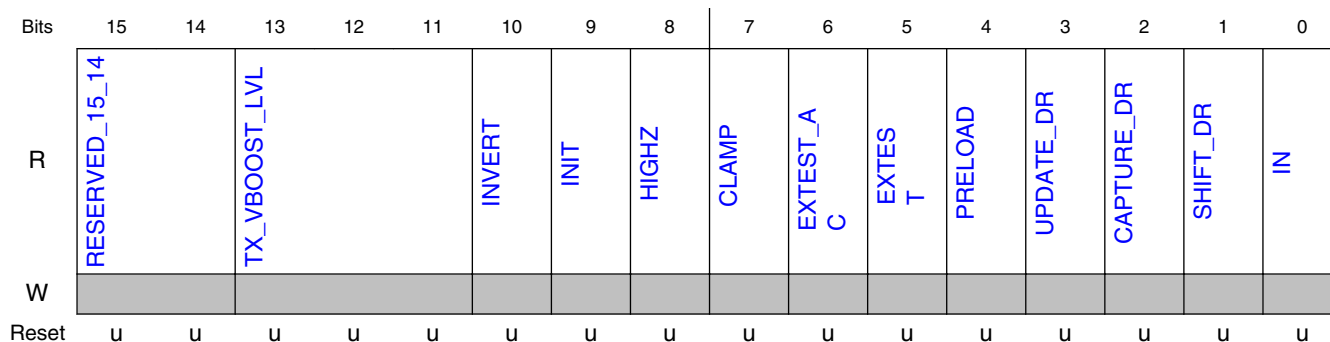
Field	Function
MPLL_QWORD_CLK_EN	
10 RES_ACK_IN	Value from ASIC for res_ack_in (RU - read-only volatile)
9 RES_REQ_IN	Value from ASIC for res_req_in (RU - read-only volatile)
8 RTUNE_REQ	Value from ASIC for rtune_req (RU - read-only volatile)
7-1 MPLL_MULTIPLIER	Value from ASIC for mpll_multiplier (RU - read-only volatile)
0 MPLL_EN	Value from ASIC for mpll_en (RU - read-only volatile)

11.4.3.1.19 Current values for incoming boundary scan controls from ASIC (SUP_DIG_BS_ASIC_IN)

11.4.3.1.19.1 Offset

Register	Offset
SUP_DIG_BS_ASIC_IN	30h

11.4.3.1.19.2 Diagram



11.4.3.1.19.3 Fields

Field	Function
15-14	Reserved for Future use

Table continues on the next page...

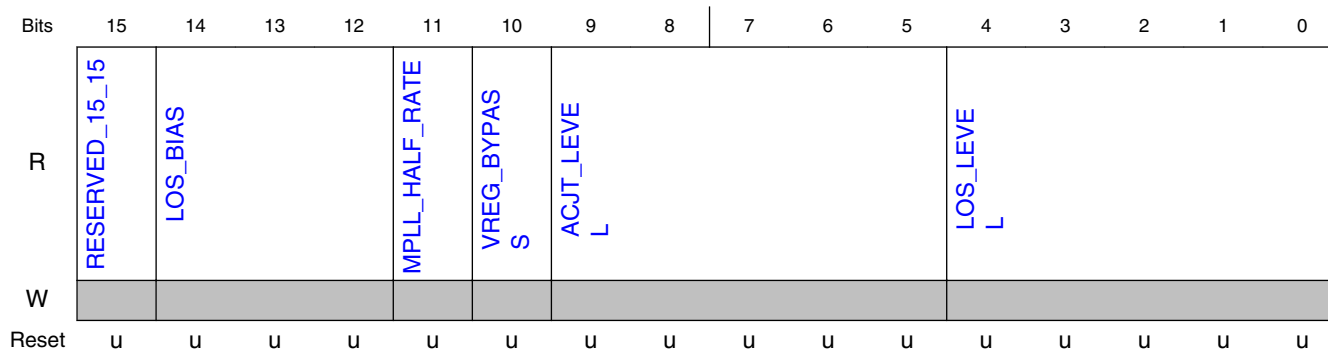
Field	Function
RESERVED_15_14	
13-11 TX_VBOOST_LVL	Value from ASIC for tx_vboost_lvl -- Not as bs_* signal, but we're out of bits. Value from ASIC for tx_vboost_lvl -- Not as bs_* signal, but we're out of bits. (RU - read-only volatile)
10 INVERT	Value from ASIC for bs_invert (RU - read-only volatile)
9 INIT	Value from ASIC for bs_init (RU - read-only volatile)
8 HIGHZ	Value from ASIC for bs_highz (RU - read-only volatile)
7 CLAMP	Value from ASIC for bs_clamp (RU - read-only volatile)
6 EXTEST_AC	Value from ASIC for bs_extest_ac (RU - read-only volatile)
5 EXTEST	Value from ASIC for bs_extest (RU - read-only volatile)
4 PRELOAD	Value from ASIC for bs_preload (RU - read-only volatile)
3 UPDATE_DR	Value from ASIC for bs_update_dr (RU - read-only volatile)
2 CAPTURE_DR	Value from ASIC for bs_capture_dr (RU - read-only volatile)
1 SHIFT_DR	Value from ASIC for bs_shift_dr (RU - read-only volatile)
0 IN	Value from ASIC for bs_in (RU - read-only volatile)

11.4.3.1.20 Current values for incoming los/acjt level, MPLL half rate mode and vreg bypass from ASIC (SUP_DIG_LEVEL_ASIC_IN)

11.4.3.1.20.1 Offset

Register	Offset
SUP_DIG_LEVEL_ASIC_IN	32h

11.4.3.1.20.2 Diagram



11.4.3.1.20.3 Fields

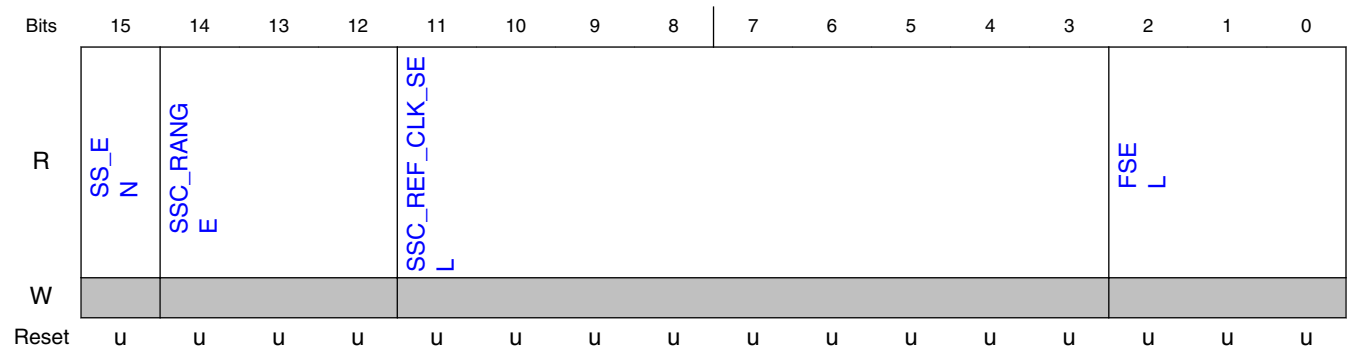
Field	Function
15 RESERVED_15_15	Reserved for Future use
14-12 LOS_BIAS	Value from ASIC for los_bias (RU - read-only volatile)
11 MPLL_HALF_RATE	Value from ASIC for mpll_half_rate (RU - read-only volatile)
10 VREG_BYPASS	Value from ASIC for vreg_bypass (RU - read-only volatile)
9-5 ACJT_LEVEL	Value from ASIC for acjt_level (RU - read-only volatile)
4-0 LOS_LEVEL	Value from ASIC for los_level (RU - read-only volatile)

11.4.3.1.21 Current values for incoming SSC controls from ASIC (SUP_DIG_SSC_ASIC_IN)

11.4.3.1.21.1 Offset

Register	Offset
SUP_DIG_SSC_ASIC_IN	34h

11.4.3.1.21.2 Diagram



11.4.3.1.21.3 Fields

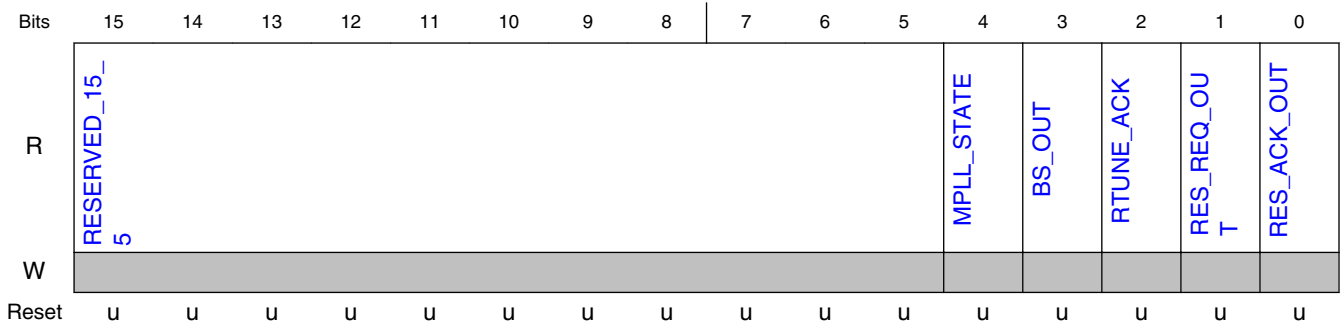
Field	Function
15 SS_EN	Value from ASIC for ssc_en (RU - read-only volatile)
14-12 SSC_RANGE	Value from ASIC for ssc_range (RU - read-only volatile)
11-3 SSC_REF_CLK_SEL	Value from ASIC for ssc_ref_clk_sel (RU - read-only volatile)
2-0 FSEL	Value from ASIC for fsel (RU - read-only volatile)

11.4.3.1.22 Current values for outgoing support block controls to ASIC (SUP_DIG_SUP_ASIC_OUT)

11.4.3.1.22.1 Offset

Register	Offset
SUP_DIG_SUP_ASIC_OUT	36h

11.4.3.1.22.2 Diagram



11.4.3.1.22.3 Fields

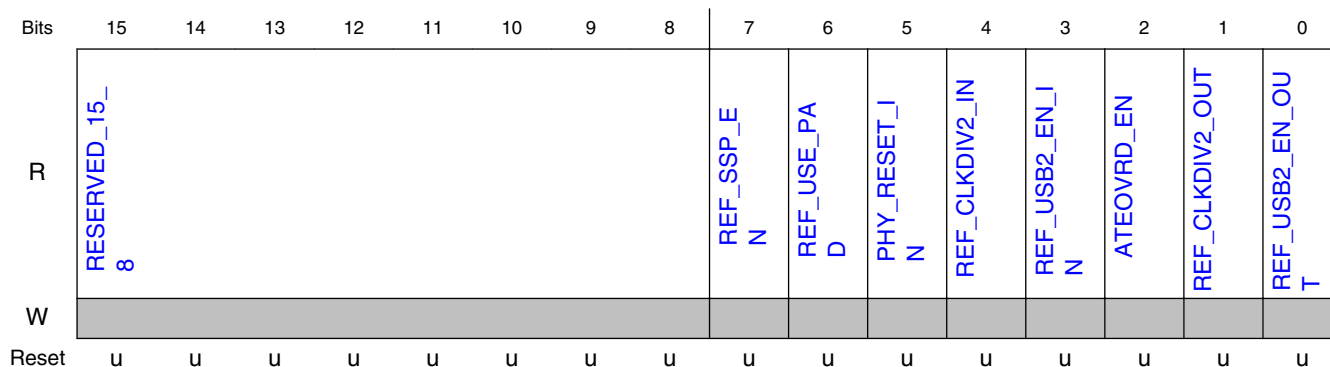
Field	Function
15-5 RESERVED_15_5	Reserved for Future use
4 MPLL_STATE	Value from PHY for mpll_state output (RU - read-only volatile)
3 BS_OUT	Value from PHY for bs_out output (RU - read-only volatile)
2 RTUNE_ACK	Value from PHY for rtune_ack output (RU - read-only volatile)
1 RES_REQ_OUT	Value from PHY for res_req_out output (RU - read-only volatile)
0 RES_ACK_OUT	Value from PHY for res_ack_out output (RU - read-only volatile)

11.4.3.1.23 SSP_SUP_DIG_ATEOVRD_STATUS (SUP_DIG_ATEOVRD_STATUS)

11.4.3.1.23.1 Offset

Register	Offset
SUP_DIG_ATEOVRD_STATUS	38h

11.4.3.1.23.2 Diagram



11.4.3.1.23.3 Fields

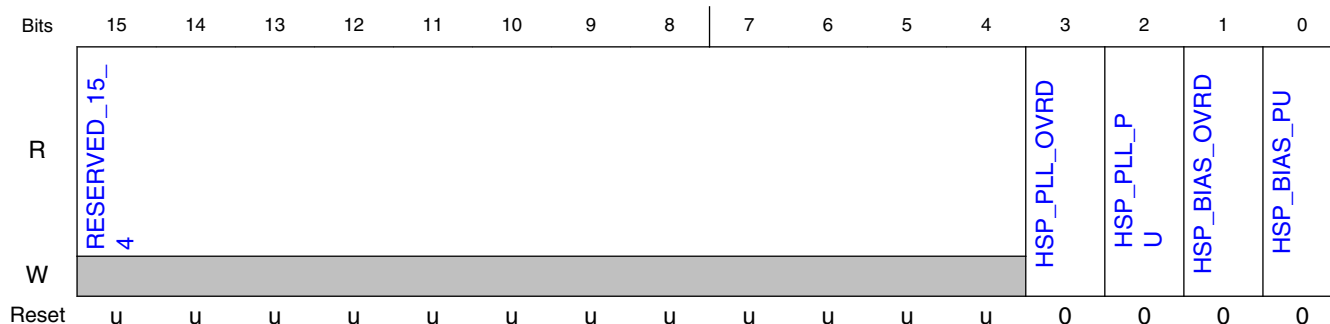
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 REF_SSP_EN	Value from ASIC for ref_ssp_en. Value from ASIC for ref_ssp_en. (RU - read-only volatile)
6 REF_USE_PAD	Value from ASIC for ref_use_pad. Value from ASIC for ref_use_pad. (RU - read-only volatile)
5 PHY_RESET_IN	Value from ASIC for phy_reset (RU - read-only volatile)
4 REF_CLKDIV2_IN	Value from ASIC for ref_clkdiv2 (RU - read-only volatile)
3 REF_USB2_EN_IN	Value from ASIC for ref_usb2_en (RU - read-only volatile)
2 ATEOVRD_EN	When set, values from ATEOVRD register are being sent to PHY (RU - read-only volatile)
1 REF_CLKDIV2_OUT	Value from ATEOVRD for ref_clkdiv2 (RU - read-only volatile)
0 REF_USB2_EN_OUT	Value from ATEOVRD for ref_usb2_en (RU - read-only volatile)

11.4.3.1.24 Override values for HSP (SUP_DIG_HSP_OVRD_IN)

11.4.3.1.24.1 Offset

Register	Offset
SUP_DIG_HSP_OVRD_IN	3Ah

11.4.3.1.24.2 Diagram



11.4.3.1.24.3 Fields

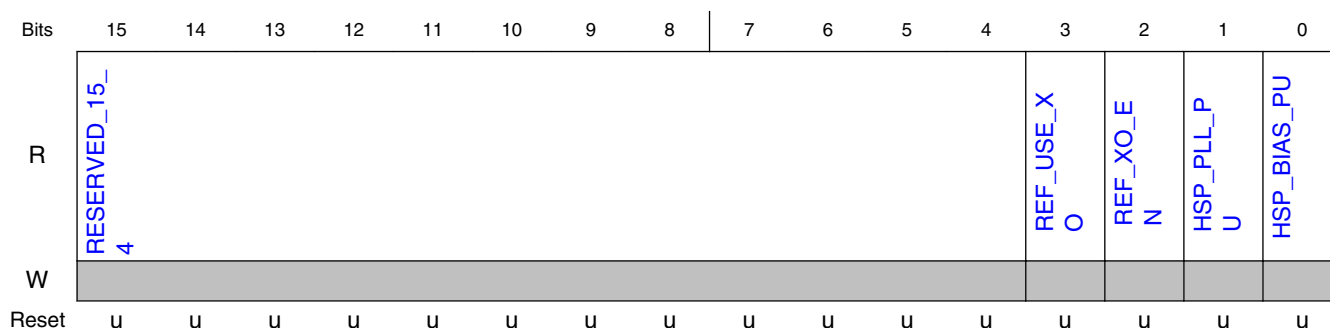
Field	Function
15-4 RESERVED_15_4	Reserved for Future use
3 HSP_PLL_OVRD	Override enable for HSP pll_pu signal
2 HSP_PLL_PU	Override value for HSP pll_pu signal
1 HSP_BIAS_OVRD	Override enable for HSP bias_pu signal
0 HSP_BIAS_PU	Override value for HSP bias_pu signal

11.4.3.1.25 Current values for incoming HSP & REF XO controls from ASIC (SUP_DIG_HSP_ASIC_IN)

11.4.3.1.25.1 Offset

Register	Offset
SUP_DIG_HSP_ASIC_IN	3Ch

11.4.3.1.25.2 Diagram



11.4.3.1.25.3 Fields

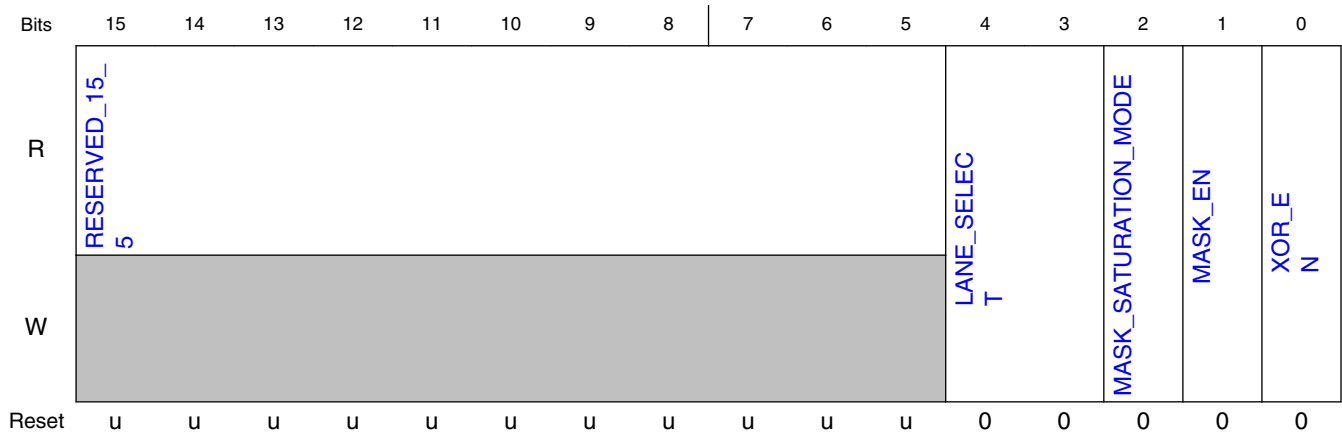
Field	Function
15-4 RESERVED_15_4	Reserved for Future use
3 REF_USE_XO	Value from ASIC for ref_use_xo (RU - read-only volatile)
2 REF_XO_EN	Value from ASIC for ref_xo_en (RU - read-only volatile)
1 HSP_PLL_PU	Value from ASIC for hsp_pll_pu (RU - read-only volatile)
0 HSP_BIAS_PU	Value from ASIC for hsp_bias_pu (RU - read-only volatile)

11.4.3.1.26 Scope Counter configuration (SUP_DIG_SCOPE_ENABLES)

11.4.3.1.26.1 Offset

Register	Offset
SUP_DIG_SCOPE_ENABLES	40h

11.4.3.1.26.2 Diagram



11.4.3.1.26.3 Fields

Field	Function
15-5 RESERVED_15_5	Reserved for Future use
4-3 LANE_SELECT	Scope Counter Input select
2 MASK_SATURATION_MODE	Method of mask saturation. Method of mask saturation. When 1 Saturate when the first mask_counter reaches sample_limit. When 0 Saturate when all mask_counters have reached sample_limit.
1 MASK_EN	Enable scope_mask input for tracking count values. Enable scope_mask input for tracking count values. Clears registers when deasserted.
0 XOR_EN	Use scope_xor input for count values.

11.4.3.1.27 Number of samples to count (SUP_DIG_SCOPE_SAMPLES)

11.4.3.1.27.1 Offset

Register	Offset
SUP_DIG_SCOPE_SAMPLES	42h

11.4.3.1.27.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SAMPLES															
W																
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

11.4.3.1.27.3 Fields

Field	Function
15-0 SAMPLES	Number of Sample to count.

11.4.3.1.28 Results of scope counting (SUP_DIG_SCOPE_COUNT)**11.4.3.1.28.1 Offset**

Register	Offset
SUP_DIG_SCOPE_COUNT	44h

11.4.3.1.28.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	COUNT															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.4.3.1.28.3 Fields

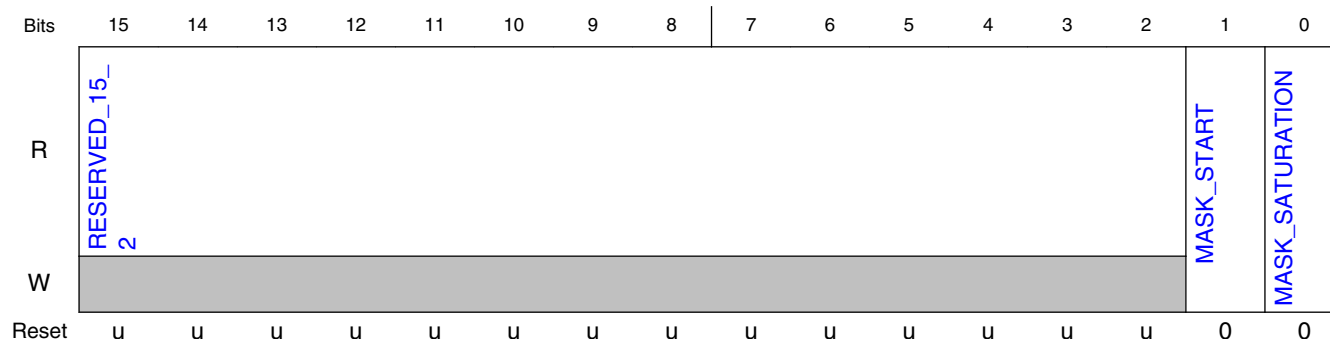
Field	Function
15-0 COUNT	A write to this register will start the counting process The value of FFFF indicates counting still in progress. A write to this register will start the counting process The value of FFFF indicates counting still in progress. Asserting MASK_EN will also start the counting if in MASK mode. (RU - read-only volatile)

11.4.3.1.29 Masked scope counting Start and Status (SUP_DIG_SCOPE_CTL)

11.4.3.1.29.1 Offset

Register	Offset
SUP_DIG_SCOPE_CTL	46h

11.4.3.1.29.2 Diagram



11.4.3.1.29.3 Fields

Field	Function
15-2 RESERVED_15_2	Reserved for Future use
1 MASK_START	A write to this register will start the counting process The value of FFFF indicates counting still in progress Write will trigger mask counting. A write to this register will start the counting process

Table continues on the next page...

Field	Function
	The value of FFFF indicates counting still in progress Write will trigger mask counting. (USER1 - read-write volatile)
0	When assert mask registers have saturated.
MASK_SATURATION	When assert mask registers have saturated. (USER1 - read-write volatile)

11.4.3.1.30 Current count value for mask_000 (SUP_DIG_SCOPE_MASK_000)

11.4.3.1.30.1 Offset

Register	Offset
SUP_DIG_SCOPE_MASK_000	48h

11.4.3.1.30.2 Diagram



11.4.3.1.30.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_000	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.31 Current count value for mask_001 (SUP_DIG_SCOPE_MASK_001)

11.4.3.1.31.1 Offset

Register	Offset
SUP_DIG_SCOPE_MASK_001	4Ah

11.4.3.1.31.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MASK_VAL_001															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.4.3.1.31.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_001	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.32 Current count value for mask_010 (SUP_DIG_SCOPE_MASK_010)

11.4.3.1.32.1 Offset

Register	Offset
SUP_DIG_SCOPE_MASK_010	4Ch

11.4.3.1.32.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MASK_VAL_010															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.4.3.1.32.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_01 0	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.33 Current count value for mask_011 (SUP_DIG_SCOPE_MASK_011)**11.4.3.1.33.1 Offset**

Register	Offset
SUP_DIG_SCOPE_MASK_011	4Eh

11.4.3.1.33.2 Diagram**11.4.3.1.33.3 Fields**

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_01 1	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.34 Current count value for mask_100 (SUP_DIG_SCOPE_MASK_100)

11.4.3.1.34.1 Offset

Register	Offset
SUP_DIG_SCOPE_MASK_100	50h

11.4.3.1.34.2 Diagram



11.4.3.1.34.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_100	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.35 Current count value for mask_101 (SUP_DIG_SCOPE_MASK_101)

11.4.3.1.35.1 Offset

Register	Offset
SUP_DIG_SCOPE_MASK_101	52h

11.4.3.1.35.2 Diagram



11.4.3.1.35.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_10 1	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.36 Current count value for mask_110 (SUP_DIG_SCOPE_MASK_110)**11.4.3.1.36.1 Offset**

Register	Offset
SUP_DIG_SCOPE_MAS K_110	54h

11.4.3.1.36.2 Diagram**11.4.3.1.36.3 Fields**

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_11 0	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.37 Current count value for mask_111 (SUP_DIG_SCOPE_MASK_111)

11.4.3.1.37.1 Offset

Register	Offset
SUP_DIG_SCOPE_MASK_111	56h

11.4.3.1.37.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MASK_VAL_111															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.4.3.1.37.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_111	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.38 MPLL loop control bits (SUP_ANA_MPLL_LOOP_CTL)

11.4.3.1.38.1 Offset

Register	Offset
SUP_ANA_MPLL_LOOP_CTL	60h

11.4.3.1.38.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	RESERVED_15_8								PROP_CNTRL				INT_CNTRL		GD	VMB
W																
Reset	u	u	u	u	u	u	u	u	1	1	0	0	0	0	0	0

11.4.3.1.38.3 Fields

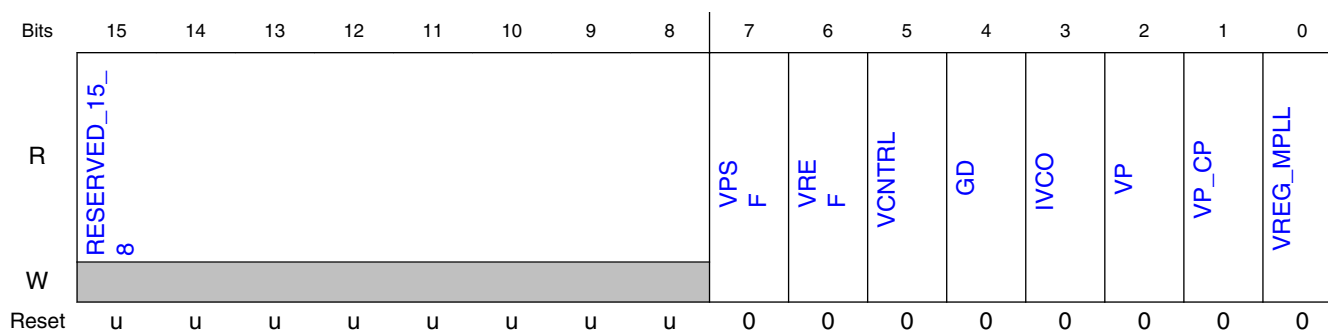
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7-4 PROP_CNTRL	Charge pump proportional current setting
3-2 INT_CNTRL	Charge pump integrating current setting
1 GD	Measure MPLL local GD
0 VMB	Measure MPLL Master Bias Voltage

11.4.3.1.39 ATB Select bits (SUP_ANA_MPLL_ATB_MEAS1)

11.4.3.1.39.1 Offset

Register	Offset
SUP_ANA_MPLL_ATB_MEAS1	62h

11.4.3.1.39.2 Diagram



11.4.3.1.39.3 Fields

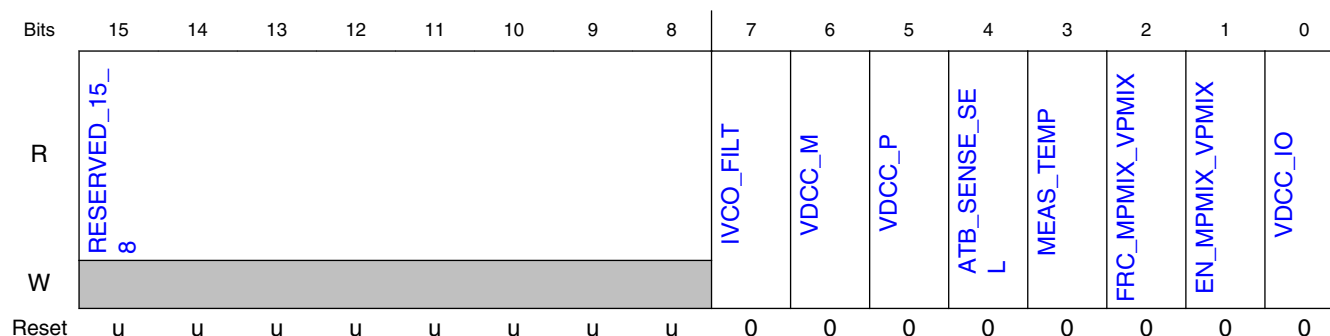
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 VPSF	puts vpsf on atb_s_p
6 VREF	puts vref from charge pump on atb_s_p
5 VCNTRL	puts vcntrl on atb_s_m
4 GD	puts gd on atb_s_m
3 IVCO	puts VCO current/93 on atb_f_m
2 VP	puts vpa on atb_s_m
1 VP_CP	puts vp_cp on atb_s_p
0 VREG_MPLL	puts vreg_mpll on atb_s_p

11.4.3.1.40 ATB Select bits (SUP_ANA_MPLL_ATB_MEAS2)

11.4.3.1.40.1 Offset

Register	Offset
SUP_ANA_MPLL_ATB_MEAS2	64h

11.4.3.1.40.2 Diagram



11.4.3.1.40.3 Fields

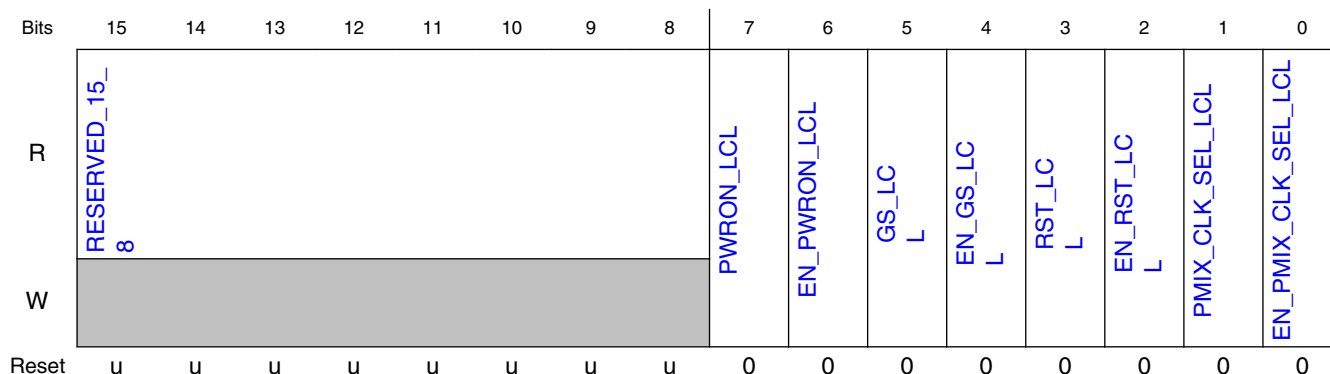
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 IVCO_FILT	puts filtered version of ivco on atb_s_p
6 VDCC_M	puts dcc output vdcc_p on atb_s_m
5 VDCC_P	puts dcc output vdcc_m on atb_s_p
4 ATB_SENSE_SE_L	connects internal atb sense bus to external bus
3 MEAS_TEMP	tells POR block to measure the temperature
2 FRC_MPMIX_VPMIX	Force mpll_pmix_vreg to use atb_s_m instead of vbg as its input
1 EN_MPMIX_VPMIX	Put vreg_pmix on atb_s_p
0 VDCC_IO	when asserted, bit 5, 6 of this register measures i phase clock DCC voltage, otherwise, q phase DCC voltage is measured

11.4.3.1.41 Local overrides (SUP_ANA_MPLL_OVRD)

11.4.3.1.41.1 Offset

Register	Offset
SUP_ANA_MPLL_OVRD	66h

11.4.3.1.41.2 Diagram



11.4.3.1.41.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 PWRON_LCL	local power_on value
6 EN_PWRON_LCL	Enable local control of power_on
5 GS_LCL	local gear_shift value
4 EN_GS_LCL	Enable local control of gear_shift
3 RST_LCL	local reset value
2 EN_RST_LCL	enable local control of reset
1 PMIX_CLK_SEL_LCL	local pmix_clk_sel value
0	enable local control of pmix_clk_sel

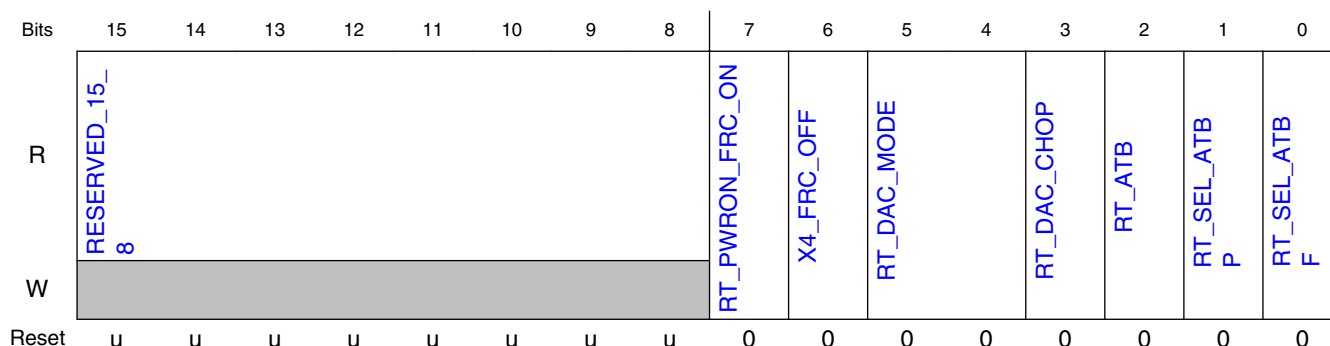
Field	Function
EN_PMI_X_CLK_SEL_LCL	

11.4.3.1.42 RTUNE control bits (SUP_ANA_RTUNE_RTUNE_CTRL)

11.4.3.1.42.1 Offset

Register	Offset
SUP_ANA_RTUNE_RTUNE_CTRL	68h

11.4.3.1.42.2 Diagram



11.4.3.1.42.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 RT_PWRON_FRC_ON	When set, forces RTUNE block to be on
6 X4_FRC_OFF	When set, do not multiply test current by 4
5-4 RT_DAC_MODE	Margin DAC mode control bits(00=powerdown, 01=DAC drives atb_s_p/m directly, 10=DAC drives atb_s_p/m to the RX in margining mode, 11=illegal state)
3	Margin DAC chop control bit.

Table continues on the next page...

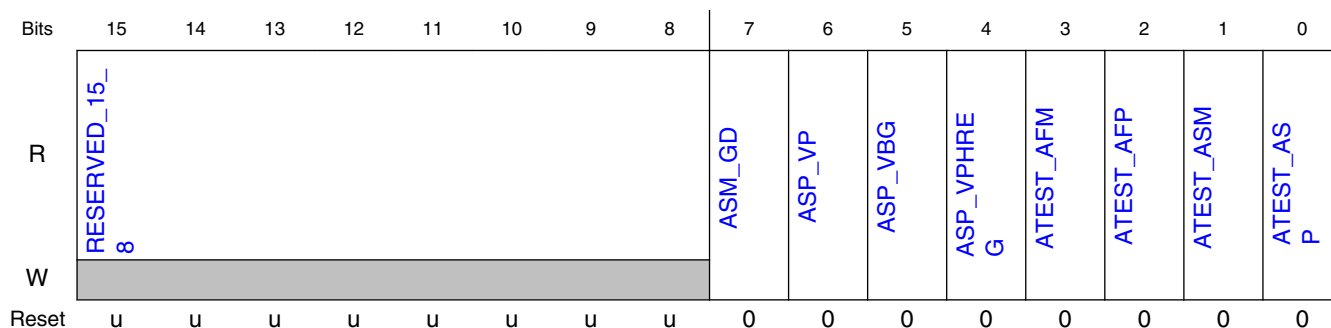
Field	Function
RT_DAC_CHOP	Margin DAC chop control bit. ref to ssp analog engineering doc 8.5
2 RT_ATB	RTUNE ATB mode control bit (1=RTUNE performs ADC on ATB input, 0=not accessing ATB. refer to ssp analog engineering specification 8.1.4)
1 RT_SEL_ATBP	RTUNE ATB sense input select bit (1=atb_s_p, 0=atb_s_m)
0 RT_SEL_ATBF	RTUNE ATB input select bit (1=atb_fm, 0=atb_s_p/m)

11.4.3.1.43 ATB_SWITCHYARD control bits (SUP_ANA_ATB_SWITCHYARD_CTRL)

11.4.3.1.43.1 Offset

Register	Offset
SUP_ANA_ATB_SWITCHYARD_CTRL	6Ah

11.4.3.1.43.2 Diagram



11.4.3.1.43.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 ASM_GD	Connect gd to atb_s_m

Table continues on the next page...

PCI Express PHY (PCle_PHY)

Field	Function
6 ASP_VP	Connect vp to atb_s_p
5 ASP_VBG	Connect vbg to atb_s_p, which is divided down value from a resistor stack
4 ASP_VPHREG	Connect vphreg to atb_s_p
3 ATEST_AFM	drive ATEST from atb_f_m
2 ATEST_AFP	drive ATEST from atb_f_p
1 ATEST_ASM	Connet ATEST to atb_s_m
0 ATEST_ASP	Connect ATEST to atb_s_p

11.4.3.1.44 controls sscf reference clock rate (SUP_ANA_SSC_CLK_CNTRL)

11.4.3.1.44.1 Offset

Register	Offset
SUP_ANA_SSC_CLK_CNTRL	6Ch

11.4.3.1.44.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	RESERVED_15_8								NC	SSC_CLK_DIV125						
W																
Reset	u	u	u	u	u	u	u	u	0	1	1	1	1	1	0	1

11.4.3.1.44.3 Fields

Field	Function
15-8	Reserved for Future use

Table continues on the next page...

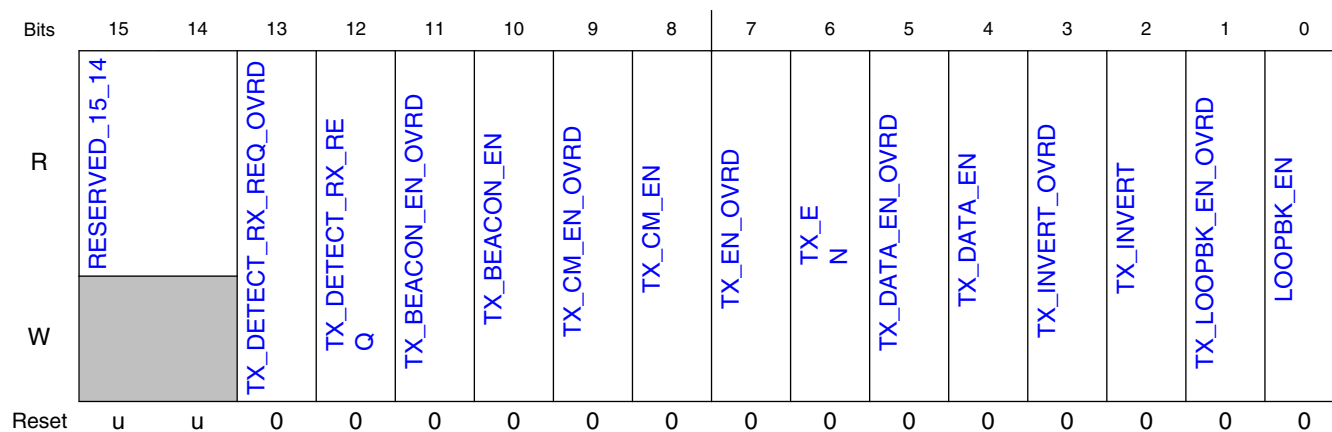
Field	Function
RESERVED_15_8	
7 NC	Reserved
6-0 SSC_CLK_DIV1 25	set ssc reference clock to be 20MHz

11.4.3.1.45 Override values for incoming TX controls from ASIC (LANE0_DIG_TX_OVRD_IN_LO)

11.4.3.1.45.1 Offset

Register	Offset
LANE0_DIG_TX_OVRD_IN_LO	2000h

11.4.3.1.45.2 Diagram



11.4.3.1.45.3 Fields

Field	Function
15-14 RESERVED_15_14	Reserved for Future use
13	Override enable for tx_detect_rx_req

Table continues on the next page...

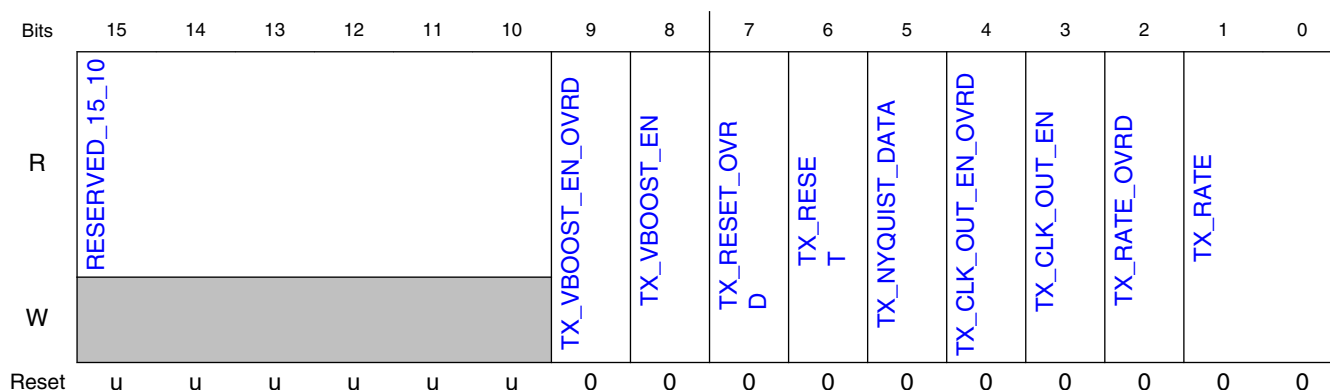
Field	Function
TX_DETECT_RX_REQ_OVRD	
12 TX_DETECT_RX_REQ	Override value for tx_detect_rx_req
11 TX_BEACON_EN_OVRD	Override enable for tx_beacon_en
10 TX_BEACON_EN	Override value for tx_beacon_en
9 TX_CM_EN_OVRD	Override enable for tx_cm_en
8 TX_CM_EN	Override value for tx_cm_en
7 TX_EN_OVRD	Override enable for tx_en
6 TX_EN	Override value for tx_en
5 TX_DATA_EN_OVRD	Override enable for tx_data_en
4 TX_DATA_EN	Override value for tx_data_en
3 TX_INVERT_OVRD	Override enable for tx_invert
2 TX_INVERT	Override value for tx_invert
1 TX_LOOPBK_EN_OVRD	Override enable for loopbk_en
0 LOOPBK_EN	Override value for loopbk_en

11.4.3.1.46 Override values for incoming TX controls from ASIC (LANE0_DIG_TX_OVRD_IN_HI)

11.4.3.1.46.1 Offset

Register	Offset
LANE0_DIG_TX_OVRD_IN_HI	2002h

11.4.3.1.46.2 Diagram



11.4.3.1.46.3 Fields

Field	Function
15-10 RESERVED_15_10	Reserved for Future use
9 TX_VBOOST_EN_OVRD	Override enable for tx_vboost_en
8 TX_VBOOST_EN	Override value for tx_vboost_en
7 TX_RESET_OVRD	Override enable for tx_reset
6 TX_RESET	Override value for tx_reset
5 TX_NYQUIST_DATA	Override incoming data to nyquist
4	Override enable for tx_clk_out_en.

Table continues on the next page...

PCI Express PHY (PCIe_PHY)

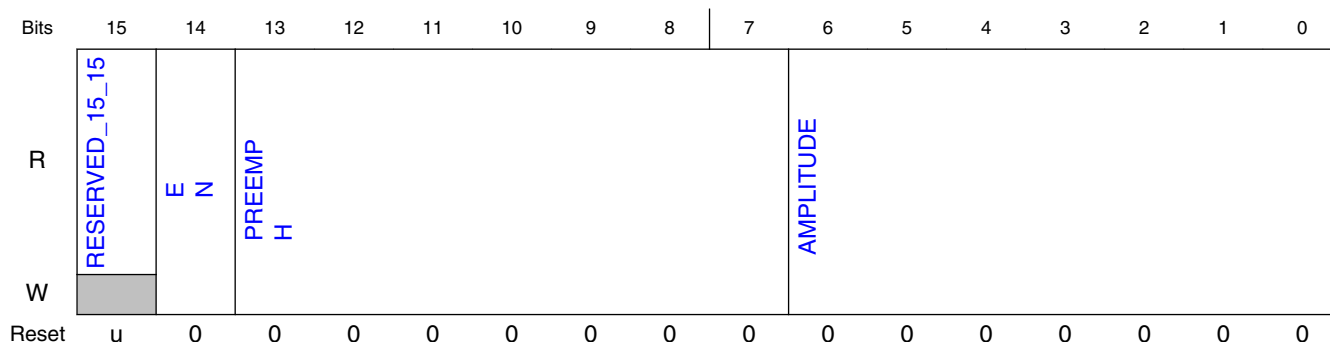
Field	Function
TX_CLK_OUT_EN_OVRD	
3 TX_CLK_OUT_EN	Override incoming tx_clk_out_en.
2 TX_RATE_OVRD	Override enable for tx_rate.
1-0 TX_RATE	Override incoming tx lane rate.

11.4.3.1.47 Override values for incoming TX drive controls from ASIC (LANE0_DIG_TX_OVRD_DRV_LO)

11.4.3.1.47.1 Offset

Register	Offset
LANE0_DIG_TX_OVRD_DRV_LO	2004h

11.4.3.1.47.2 Diagram



11.4.3.1.47.3 Fields

Field	Function
15 RESERVED_15_15	Reserved for Future use

Table continues on the next page...

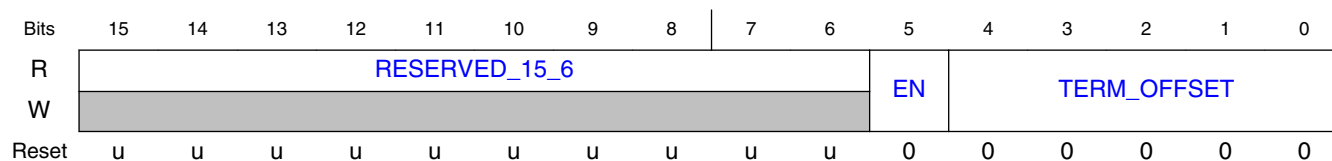
Field	Function
14 EN	Enable override values for all inputs controlled by this register
13-7 PREEMPH	Override value for transmit preemphasis.
6-0 AMPLITUDE	Override value for transmit amplitude.

11.4.3.1.48 Override values for incoming TX drive controls from ASIC (LANE0_DIG_TX_OVRD_DRV_HI)

11.4.3.1.48.1 Offset

Register	Offset
LANE0_DIG_TX_OVRD_DRV_HI	2006h

11.4.3.1.48.2 Diagram



11.4.3.1.48.3 Fields

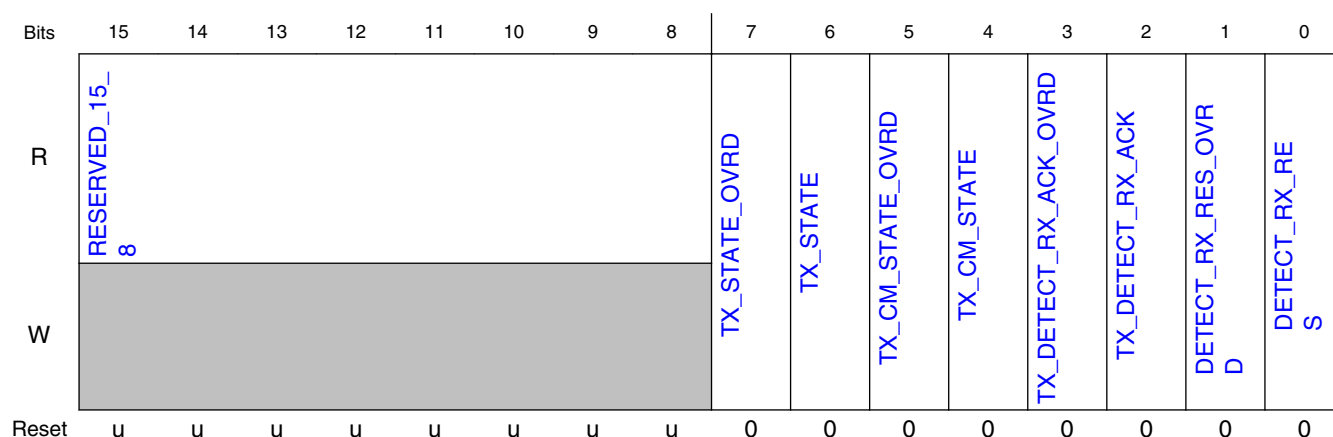
Field	Function
15-6 RESERVED_15_6	Reserved for Future use
5 EN	Enable override values for all inputs controlled by this register
4-0 TERM_OFFSET	Override value for tx_term_offset (tx_ana_vcmref_boost)

11.4.3.1.49 Override values for outgoing TX controls to ASIC (LANE0_DIG_TX_OVRD_OUT)

11.4.3.1.49.1 Offset

Register	Offset
LANE0_DIG_TX_OVRD_OUT	2008h

11.4.3.1.49.2 Diagram



11.4.3.1.49.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 TX_STATE_OVRD	Override enable for tx_state
6 TX_STATE	Override value for tx_state
5 TX_CM_STATE_OVRD	Override enable for tx_cm_state
4 TX_CM_STATE	Override value for tx_cm_state
3	Override enable for tx_detect_rx_ack

Table continues on the next page...

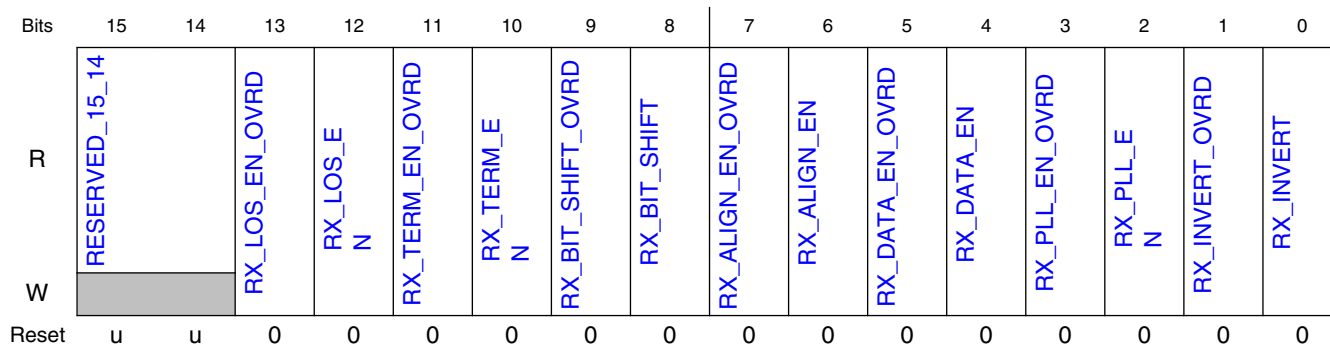
Field	Function
TX_DETECT_RX_ACK_OVRD	
2 TX_DETECT_RX_ACK	Override value for tx_detect_rx_ack
1 DETECT_RX_RES_OVRD	Override enable for tx_detect_rx_res
0 DETECT_RX_RES	Override value for tx_detect_rx_res

11.4.3.1.50 Override values for incoming RX controls from ASIC (LANE0_DIG_RX_OVRD_IN_LO)

11.4.3.1.50.1 Offset

Register	Offset
LANE0_DIG_RX_OVRD_IN_LO	200Ah

11.4.3.1.50.2 Diagram



11.4.3.1.50.3 Fields

Field	Function
15-14	Reserved for Future use

Table continues on the next page...

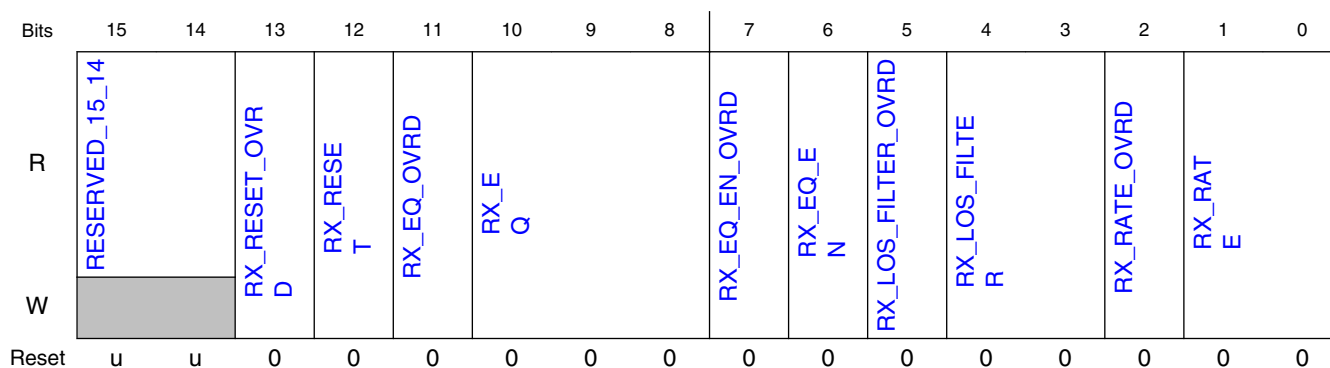
Field	Function
RESERVED_15_14	
13 RX_LOS_EN_OVRD	Override enable for rx_los_en
12 RX_LOS_EN	Override value for rx_los_en
11 RX_TERM_EN_OVRD	Override enable for rx_term_en
10 RX_TERM_EN	Override value for rx_term_en
9 RX_BIT_SHIFT_OVRD	Override enable for rx_bit_shift
8 RX_BIT_SHIFT	Override value for rx_bit_shift
7 RX_ALIGN_EN_OVRD	Override enable for rx_align_en
6 RX_ALIGN_EN	Override value for rx_align_en
5 RX_DATA_EN_OVRD	Override enable for rx_data_en
4 RX_DATA_EN	Override value for rx_data_en
3 RX_PLL_EN_OVRD	Override enable for rx_pll_en
2 RX_PLL_EN	Override value for rx_pll_en
1 RX_INVERT_OVRD	Override enable for rx_invert
0 RX_INVERT	Override value for rx_invert

11.4.3.1.51 Override values for incoming RX controls from ASIC (LANE0_DIG_RX_OVRD_IN_HI)

11.4.3.1.51.1 Offset

Register	Offset
LANE0_DIG_RX_OVRD_IN_HI	200Ch

11.4.3.1.51.2 Diagram



11.4.3.1.51.3 Fields

Field	Function
15-14 RESERVED_15_14	Reserved for Future use
13 RX_RESET_OVRD	Override enable for rx_reset
12 RX_RESET	Override value for rx_reset
11 RX_EQ_OVRD	Override enable for rx_eq
10-8 RX_EQ	Override value for rx_eq
7 RX_EQ_EN_OVRD	Override enable for rx_eq_en
6 RX_EQ_EN	Override value for rx_eq_en
5	Override enable for rx_los_filter

Table continues on the next page...

PCI Express PHY (PCle_PHY)

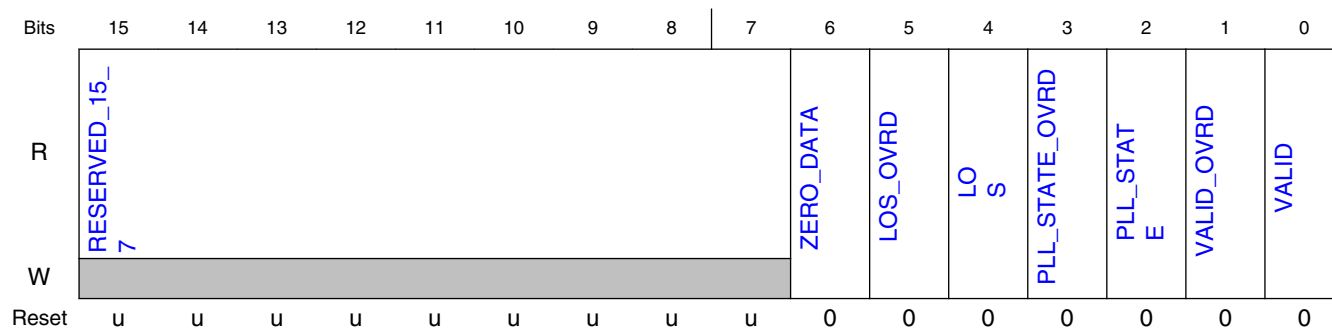
Field	Function
RX_LOS_FILTER_OVRD	
4-3 RX_LOS_FILTER	Override value for rx_los_filter
2 RX_RATE_OVRD	Override enable for rx_rate
1-0 RX_RATE	Override value for rx_rate

11.4.3.1.52 Override values for outgoing RX controls to ASIC (LANE0_DIG_RX_OVRD_OUT)

11.4.3.1.52.1 Offset

Register	Offset
LANE0_DIG_RX_OVRD_OUT	200Eh

11.4.3.1.52.2 Diagram



11.4.3.1.52.3 Fields

Field	Function
15-7 RESERVED_15_7	Reserved for Future use

Table continues on the next page...

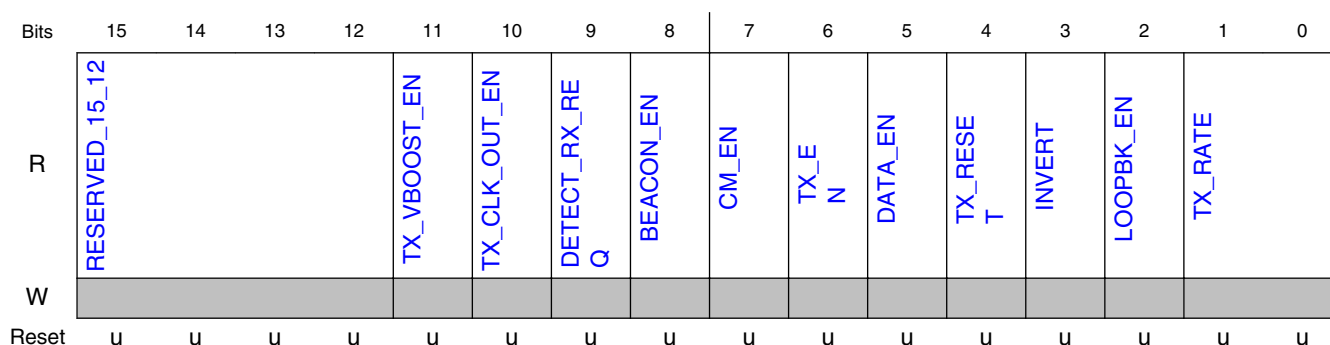
Field	Function
6 ZERO_DATA	Override data output to all zeros
5 LOS_OVRD	Override value for rx_los
4 LOS	Override value for rx_los
3 PLL_STATE_OVRD	Override enable for rx_pll_state
2 PLL_STATE	Override value for rx_pll_state
1 VALID_OVRD	Override enable for rx_valid
0 VALID	Override value for rx_valid

11.4.3.1.53 Current values for incoming TX controls from ASIC (LANE0_DIG_TX_ASIC_IN)

11.4.3.1.53.1 Offset

Register	Offset
LANE0_DIG_TX_ASIC_IN	2010h

11.4.3.1.53.2 Diagram



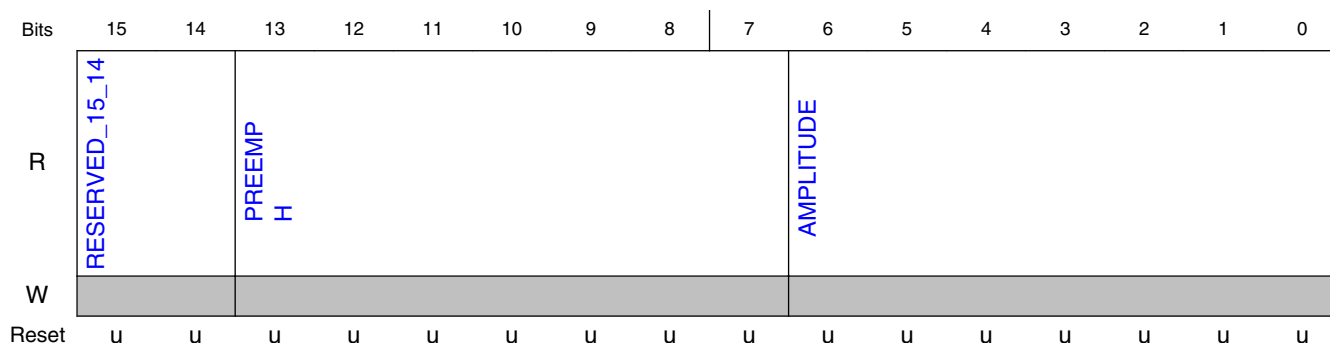
11.4.3.1.53.3 Fields

Field	Function
15-12 RESERVED_15_12	Reserved for Future use
11 TX_VBOOST_EN	Value from ASIC for tx_vboost_en (RU - read-only volatile)
10 TX_CLK_OUT_EN	Value from ASIC for tx_clk_out_en (RU - read-only volatile)
9 DETECT_RX_REQ	Value from ASIC for tx_detect_rx_req (RU - read-only volatile)
8 BEACON_EN	Value from ASIC for tx_beacon_en (RU - read-only volatile)
7 CM_EN	Value from ASIC for tx_cm_en (RU - read-only volatile)
6 TX_EN	Value from ASIC for tx_en (RU - read-only volatile)
5 DATA_EN	Value from ASIC for tx_data_en (RU - read-only volatile)
4 TX_RESET	Value from ASIC for tx_reset (RU - read-only volatile)
3 INVERT	Value from ASIC for tx_invert (RU - read-only volatile)
2 LOOPBK_EN	Value from ASIC for loopbk_en (RU - read-only volatile)
1-0 TX_RATE	Value from ASIC for tx_rate (RU - read-only volatile)

11.4.3.1.54 Current values for incoming TX level controls from ASIC (LANE0_DIG_TX_ASIC_DRV_LO)**11.4.3.1.54.1 Offset**

Register	Offset
LANE0_DIG_TX_ASIC_DRV_LO	2012h

11.4.3.1.54.2 Diagram



11.4.3.1.54.3 Fields

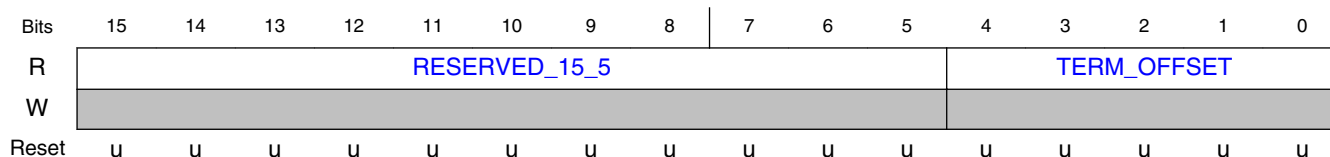
Field	Function
15-14 RESERVED_15_14	Reserved for Future use
13-7 PREEMPH	Value from ASIC for tx_preemph (RU - read-only volatile)
6-0 AMPLITUDE	Value from ASIC for tx_amplitude (RU - read-only volatile)

11.4.3.1.55 Current values for incoming TX level controls from ASIC (LANE0_DIG_TX_ASIC_DRV_HI)

11.4.3.1.55.1 Offset

Register	Offset
LANE0_DIG_TX_ASIC_DRV_HI	2014h

11.4.3.1.55.2 Diagram



11.4.3.1.55.3 Fields

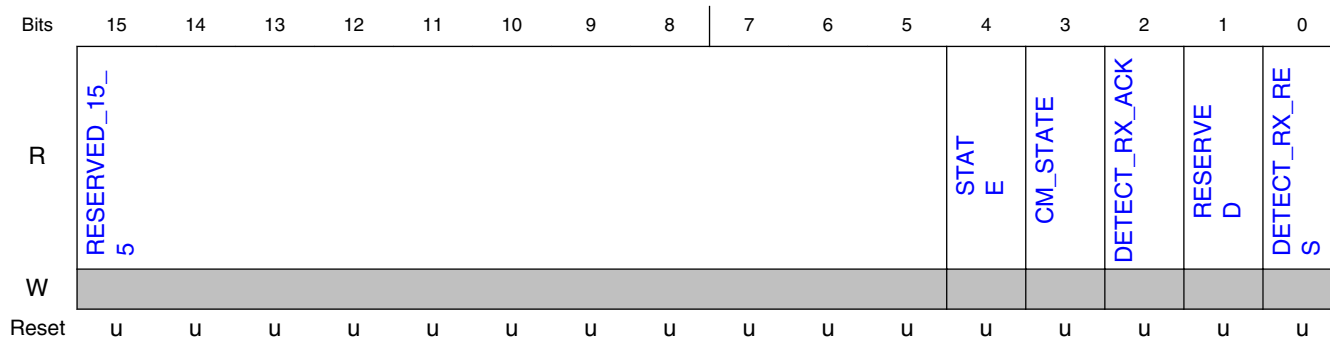
Field	Function
15-5 RESERVED_15_5	Reserved for Future use
4-0 TERM_OFFSET	Value from ASIC for tx_term_offset (RU - read-only volatile)

11.4.3.1.56 Current values for outgoing TX status controls from PHY (LANE0_DIG_TX_ASIC_OUT)

11.4.3.1.56.1 Offset

Register	Offset
LANE0_DIG_TX_ASIC_OUT	2016h

11.4.3.1.56.2 Diagram



11.4.3.1.56.3 Fields

Field	Function
15-5 RESERVED_15_5	Reserved for Future use
4 STATE	Value from PHY for tx_state (RU - read-only volatile)

Table continues on the next page...

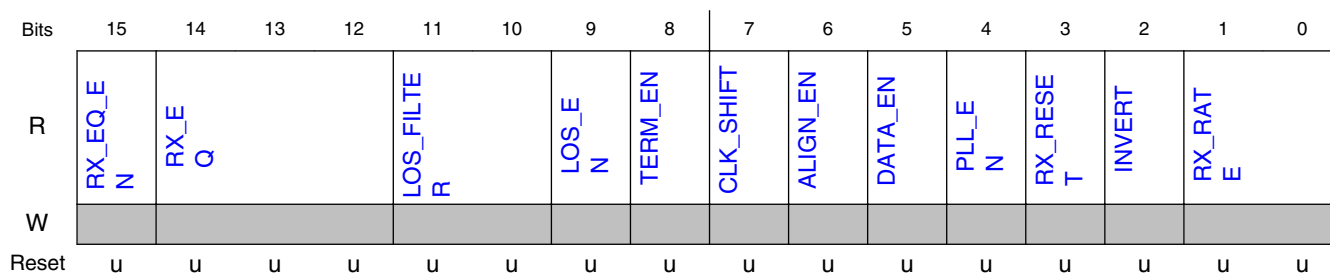
Field	Function
3 CM_STATE	Value from PHY for tx_cm_state (RU - read-only volatile)
2 DETECT_RX_ACK	Value from PHY for tx_detect_rx_ack (RU - read-only volatile)
1 RESERVED	Always returns 0 (RU - read-only volatile)
0 DETECT_RX_RES	Value from PHY for tx_detect_rx_res (RU - read-only volatile)

11.4.3.1.57 Current values for incoming RX controls from ASIC (LANE0_DIG_RX_ASIC_IN)

11.4.3.1.57.1 Offset

Register	Offset
LANE0_DIG_RX_ASIC_IN	2018h

11.4.3.1.57.2 Diagram



11.4.3.1.57.3 Fields

Field	Function
15 RX_EQ_EN	Value from ASIC for rx_eq_en (RU - read-only volatile)
14-12 RX_EQ	Value from ASIC for rx_eq (RU - read-only volatile)

Table continues on the next page...

PCI Express PHY (PCle_PHY)

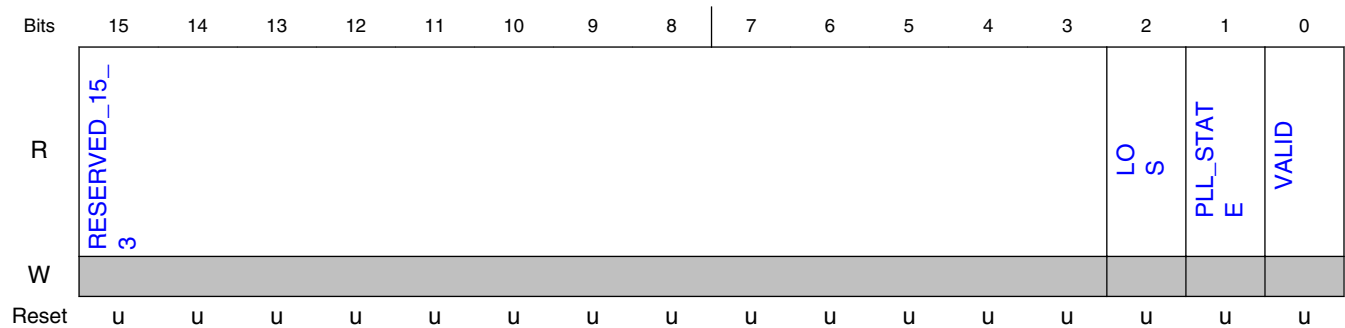
Field	Function
11-10 LOS_FILTER	Value from ASIC for rx_los_filter (RU - read-only volatile)
9 LOS_EN	Value from ASIC for rx_los_en (RU - read-only volatile)
8 TERM_EN	Value from ASIC for rx_term_en (RU - read-only volatile)
7 CLK_SHIFT	Value from ASIC for rx_bit_shift (RU - read-only volatile)
6 ALIGN_EN	Value from ASIC for rx_align_en (RU - read-only volatile)
5 DATA_EN	Value from ASIC for rx_data_en (RU - read-only volatile)
4 PLL_EN	Value from ASIC for rx_pll_en (RU - read-only volatile)
3 RX_RESET	Value from ASIC for rx_reset (RU - read-only volatile)
2 INVERT	Value from ASIC for rx_invert (RU - read-only volatile)
1-0 RX_RATE	Value from ASIC for rx_rate (RU - read-only volatile)

11.4.3.1.58 Current values for outgoing RX status controls from PHY (LANE0_DIG_RX_ASIC_OUT)

11.4.3.1.58.1 Offset

Register	Offset
LANE0_DIG_RX_ASIC_OUT	201Ah

11.4.3.1.58.2 Diagram



11.4.3.1.58.3 Fields

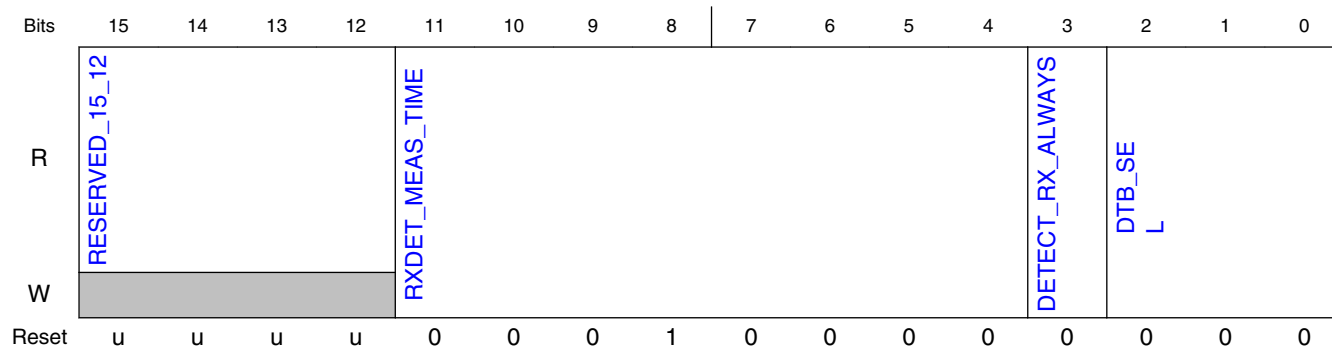
Field	Function
15-3 RESERVED_15_3	Reserved for Future use
2 LOS	Value from PHY for rx_los (RU - read-only volatile)
1 PLL_STATE	Value from PHY for rx_pll_state (RU - read-only volatile)
0 VALID	Value from PHY for rx_valid (RU - read-only volatile)

11.4.3.1.59 Transmit debug controls (LANE0_DIG_TX_DEBUG)

11.4.3.1.59.1 Offset

Register	Offset
LANE0_DIG_TX_DEBUG	2020h

11.4.3.1.59.2 Diagram



11.4.3.1.59.3 Fields

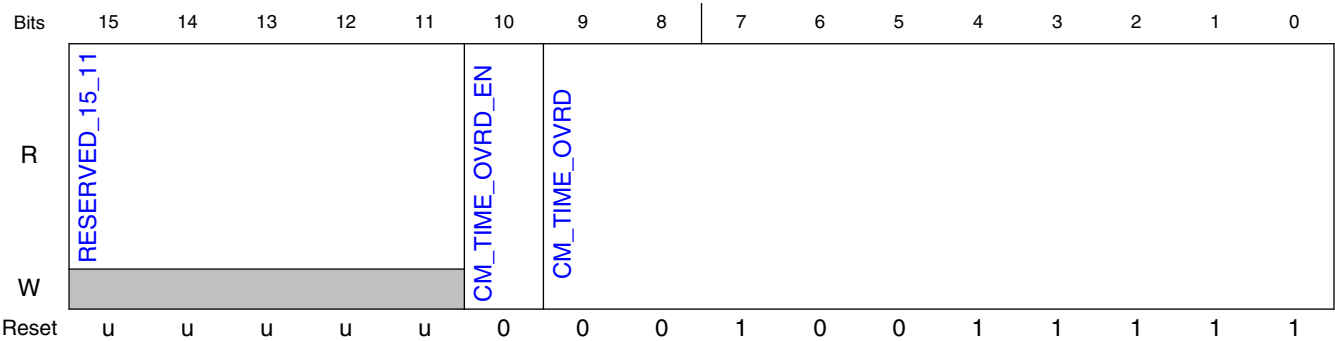
Field	Function
15-12 RESERVED_15_12	Reserved for Future use
11-4 RXDET_MEAS_TIME	Time to wait for rxdet measurement (* 4/2/1 refclk cycles based on multiplier)
3 DETECT_RX_ALWAYS	Always signal 1 for rx_detect ignoring analog
2-0 DTB_SEL	Selects data to drive on DTB 0 - disabled 1 - mag_update/sub signals 2 - mag_high/add signals 3 - clock aligner early/shift signals 4 - analog/asic clocks 5 - tx_state counter/lbert strobe 6 - noclck_ref_rst/dig_rst 7 - tx_ana_rxdetp_high/tx_ana_rxdetm_high

11.4.3.1.60 SSP_LANE0_DIG_TX_CM_WAIT_TIME_OVRD (LANE0_DIG_TX_CM_WAIT_TIME_OVRD)

11.4.3.1.60.1 Offset

Register	Offset
LANE0_DIG_TX_CM_WAIT_TIME_OVRD	2022h

11.4.3.1.60.2 Diagram



11.4.3.1.60.3 Fields

Field	Function
15-11 RESERVED_15_11	Reserved for Future use
10 CM_TIME_OVRD_EN	Override enable for common mode charging wait time.
9-0 CM_TIME_OVRD	Time to wait, in ref clock cycles, for LFPS common mode charging (This value, x32, is used for VCMHOLD charging method)

11.4.3.1.61 Calculated values for N_USE and N_TRISTATE.
(LANE0_DIG_TX_VMD_FSM_TX_VCM_0)

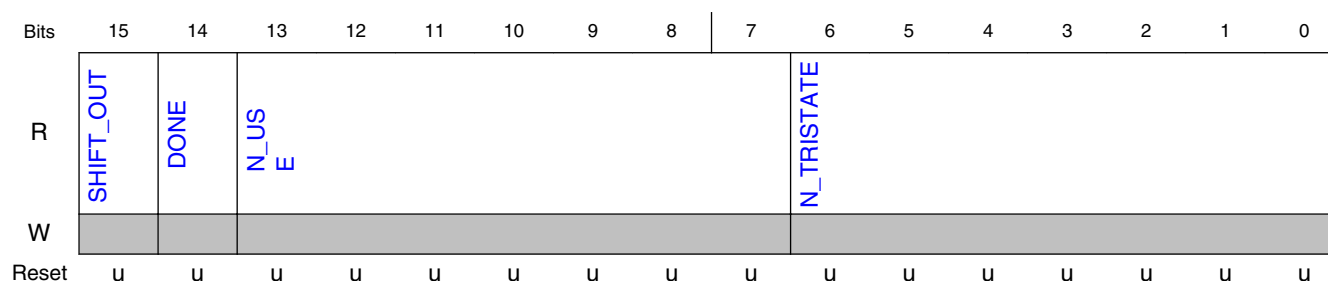
11.4.3.1.61.1 Offset

Register	Offset
LANE0_DIG_TX_VMD_FSM_TX_VCM_0	2024h

11.4.3.1.61.2 Function

Calculated values for N_USE and N_TRISTATE. Also TX ANA configuration shift register output.

11.4.3.1.61.3 Diagram



11.4.3.1.61.4 Fields

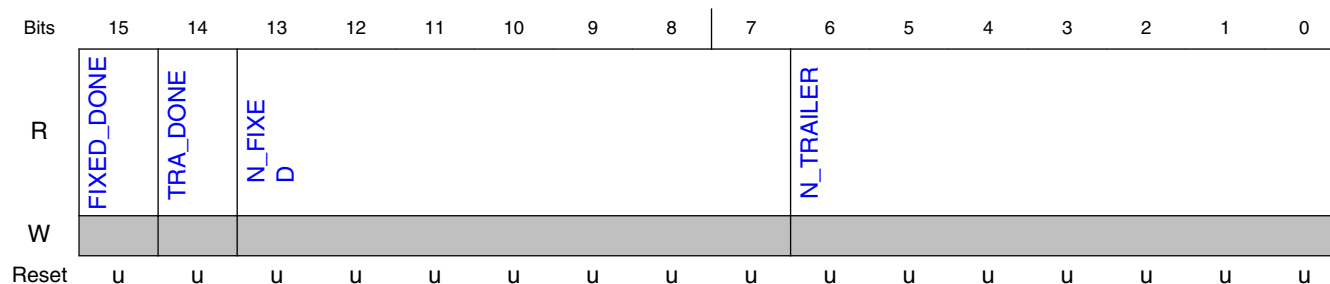
Field	Function
15 SHIFT_OUT	Current value from TX_ANAs configuration shift register. Current value from TX_ANAs configuration shift register. (RU - read-only volatile)
14 DONE	Configuration is done (RU - read-only volatile)
13-7 N_USE	Value from VMD for legs to use. Value from VMD for legs to use. (RU - read-only volatile)
6-0 N_TRISTATE	Value from VMD for number of tristate legs. Value from VMD for number of tristate legs. (RU - read-only volatile)

11.4.3.1.62 Calculated values for N_FIXED and N_TRAILER. (LANE0_DIG_TX_VMD_FSM_TX_VCM_1)

11.4.3.1.62.1 Offset

Register	Offset
LANE0_DIG_TX_VMD_FSM_TX_VCM_1	2026h

11.4.3.1.62.2 Diagram



11.4.3.1.62.3 Fields

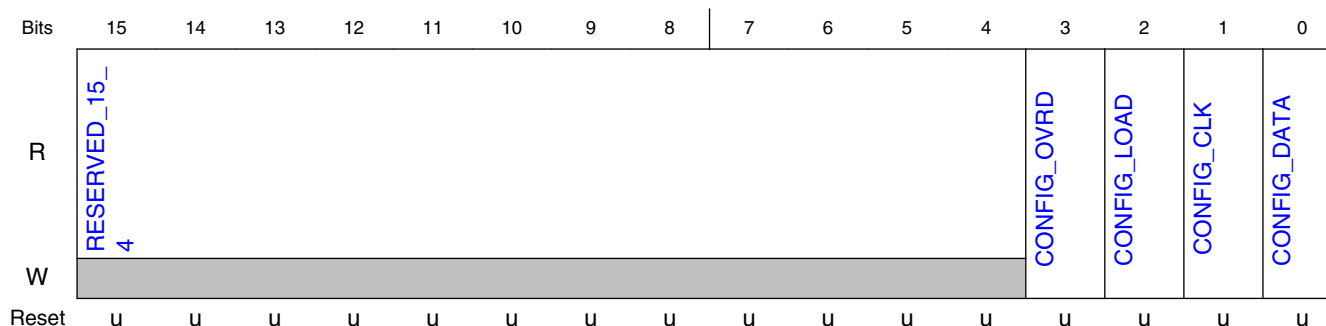
Field	Function
15 FIXED_DONE	N_FIXED Multiplication has completed. N_FIXED Multiplication has completed. (RU - read-only volatile)
14 TRA_DONE	N_TRAILER Multiplication has completed. N_TRAILER Multiplication has completed. (RU - read-only volatile)
13-7 N_FIXED	Value from VMD for number of fixed driver legs. Value from VMD for number of fixed driver legs. (RU - read-only volatile)
6-0 N_TRAILER	Value from VMD for number of trailer legs. Value from VMD for number of trailer legs. (RU - read-only volatile)

11.4.3.1.63 Access the TX_ANA VMD Control Shift Register (LANE0_DIG_TX_VMD_FSM_TX_VCM_DEBUG_IN)

11.4.3.1.63.1 Offset

Register	Offset
LANE0_DIG_TX_VMD_FSM_TX_VCM_DEBUG_IN	2028h

11.4.3.1.63.2 Diagram



11.4.3.1.63.3 Fields

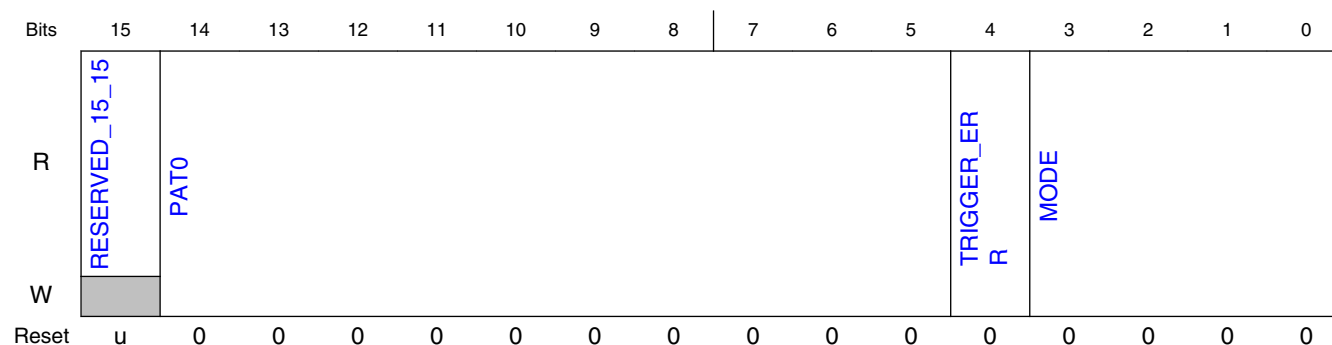
Field	Function
15-4 RESERVED_15_4	Reserved for Future use
3 CONFIG_OVRD	Override the Voltage Mode Driver Configuration FSM and access the shift chain directly. Override the Voltage Mode Driver Configuration FSM and access the shift chain directly. (volatile and 2 reads needed to read value)
2 CONFIG_LOAD	Override value for the Voltage Mode Driver Configuration FSM's config load. Override value for the Voltage Mode Driver Configuration FSM's config load. (volatile and 2 reads needed to read value)
1 CONFIG_CLK	Override value for the Voltage Mode Driver Configuration FSM's config clk. Override value for the Voltage Mode Driver Configuration FSM's config clk. (volatile and 2 reads needed to read value)
0 CONFIG_DATA	Override value for the Voltage Mode Driver Configuration FSM's config data. Override value for the Voltage Mode Driver Configuration FSM's config data. (volatile and 2 reads needed to read value)

11.4.3.1.64 Pattern Generator controls (LANE0_DIG_TX_LBERT_CTL)

11.4.3.1.64.1 Offset

Register	Offset
LANE0_DIG_TX_LBERT_CTL	202Ah

11.4.3.1.64.2 Diagram



11.4.3.1.64.3 Fields

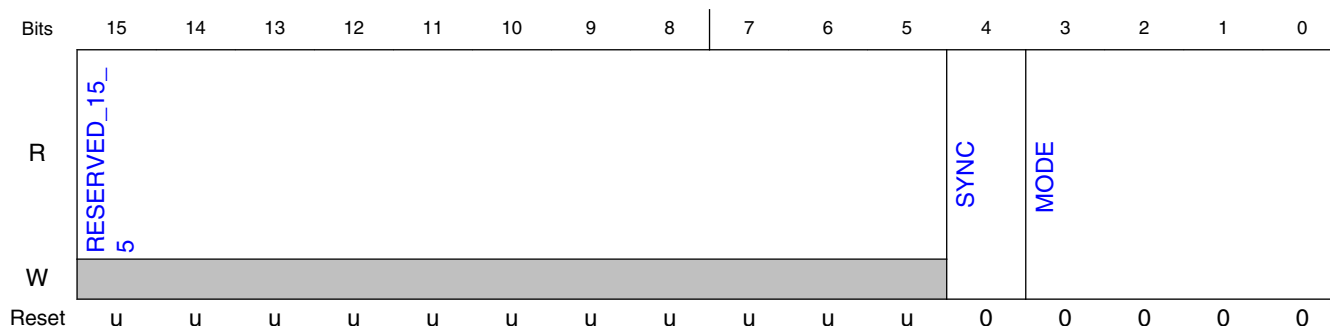
Field	Function
15 RESERVED_15_15	Reserved for Future use
14-5 PAT0	10 bit pattern for modes using PAT0 (5-7). 10 bit pattern for modes using PAT0 (5-7). Ignored for other MODEs.
4 TRIGGER_ERR	Single shot inversion the LSB of the current symbol. Single shot inversion the LSB of the current symbol. Any write of a 1 to this bit will insert an error. Maintain mode and pattern from setup.
3-0 MODE	Pattern to generate When changing modes, you must change to disabled first Pattern to generate When changing modes, you must change to disabled first 0 - disabled 1 - lfsr31. $X^{31} + X^{28} + 1$ 2 - lfsr23. $X^{23} + X^{18} + 1$ 3 - lfsr15. $X^{15} + X^{14} + 1$ 4 - lfsr7. $X^7 + X^6 + 1$ 5 - fixed word (PAT0) 6 - DC balanced word (PAT0, $\sim$ PAT0) 7 - Fixed pattern: (000, PAT0, 3ff, $\sim$ PAT0) 8 - USB3.0 TSEQ (rx equalizer training ordered sets) Transmits Even, then Odd. 9-15 - RESERVED

11.4.3.1.65 Pattern Matcher controls (LANE0_DIG_RX_LBERT_CTL)

11.4.3.1.65.1 Offset

Register	Offset
LANE0_DIG_RX_LBERT_CTL	202Ch

11.4.3.1.65.2 Diagram



11.4.3.1.65.3 Fields

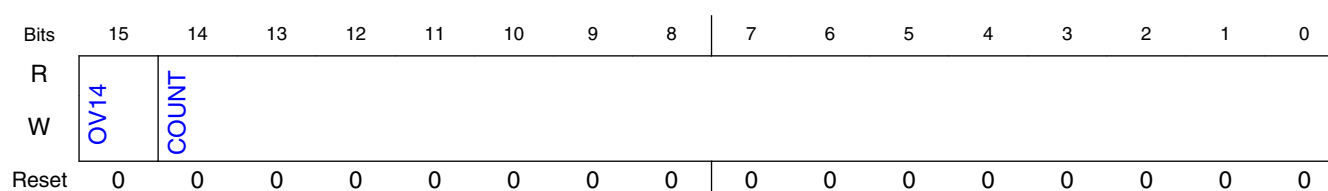
Field	Function
15-5 RESERVED_15_5	Reserved for Future use
4 SYNC	Synchronize pattern matcher with incoming data A write of a one to this bit will reset the error counter and start a synchronization of the PM. Synchronize pattern matcher with incoming data A write of a one to this bit will reset the error counter and start a synchronization of the PM. There is no need to write this back to zero to run normally.
3-0 MODE	MODE Pattern to match When changing modes, you must change to disabled first 0 - disabled 1 - lfsr31: $X^{31} + X^{28} + 1$ 2 - lfsr23: $X^{23} + X^{18} + 1$ 3 - lfsr15: $X^{15} + X^{14} + 1$ 4 - lfsr7 : $X^7 + X^6 + 1$ 5 - $d[n] = d[n-10]$ 6 - $d[n] = !d[n-10]$ 7 - $d[n] = !d[n-20]$ 8 - USB 3.0 TSEQ (NOT SUPPORTED IN PM. PG Only!)

11.4.3.1.66 Pattern match error counter (LANE0_DIG_RX_LBERT_ERR)

11.4.3.1.66.1 Offset

Register	Offset
LANE0_DIG_RX_LBERT_ERR	202Eh

11.4.3.1.66.2 Diagram



11.4.3.1.66.3 Fields

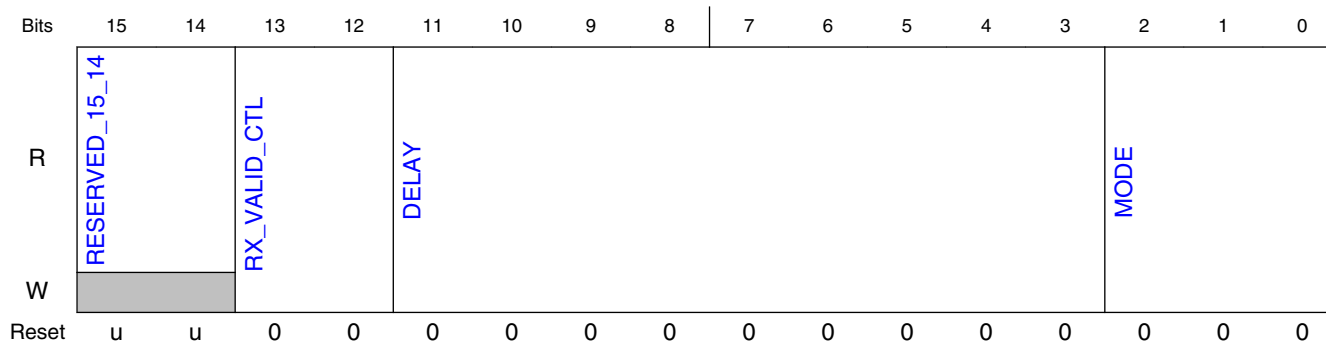
Field	Function
15 OV14	A read of this register, or a sync of the Pattern Matcher resets the error count. A read of this register, or a sync of the Pattern Matcher resets the error count. If all bits of this register are 1, the error counter has saturated. If the status is active, multiply COUNT by 128. If OV14=1 and COUNT=2 ¹⁵ -1, signals overflow of counter (volatile and 2 reads needed to read value)
14-0 COUNT	Current error count If OV14 field is active, then multiply count by 128 to get the actual count. Current error count If OV14 field is active, then multiply count by 128 to get the actual count. (volatile and 2 reads needed to read value)

11.4.3.1.67 Control bits for per-transceiver scope portion (LANE0_DIG_RX_SCOPE_CTL)

11.4.3.1.67.1 Offset

Register	Offset
LANE0_DIG_RX_SCOPE_CTL	2030h

11.4.3.1.67.2 Diagram



11.4.3.1.67.3 Fields

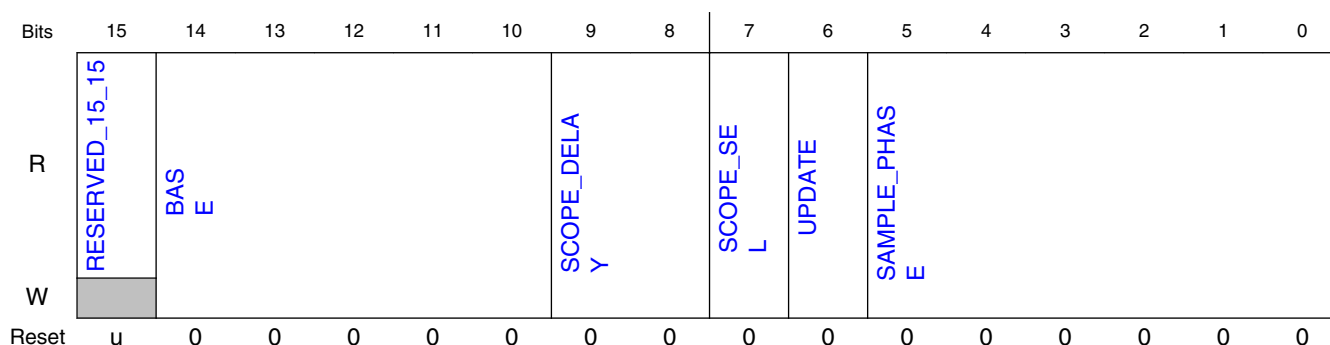
Field	Function
15-14 RESERVED_15_14	Reserved for Future use
13-12 RX_VALID_CTL	When to Start Sampling control based on rx_valid. When to Start Sampling control based on rx_valid. 0 = Ignore rx_valid. 1 = Heed rx_valid. 2 = Wait for rising_edge on rx_valid.
11-3 DELAY	number of cycles to skip between samples
2-0 MODE	sampling mode of counters NOTE: WORD is 20 bits. sampling mode of counters NOTE: WORD is 20 bits. 0 = off 1 = sample data every WORD*(1+DELAY) bits from the digital recovered path. 2 = sample data every WORD*(1+DELAY) + 1 bits from the digital recovered path. 3 = sample data every WORD*(1+DELAY) + 2 bits from the digital recovered path. 4 = sample data every WORD*(1+DELAY) bits from the analog probe. Asserts XOR and MASK increments accordingly.

11.4.3.1.68 Scope sampling phase. (LANE0_DIG_RX_SCOPE_PHASE)

11.4.3.1.68.1 Offset

Register	Offset
LANE0_DIG_RX_SCOPE_PHASE	2032h

11.4.3.1.68.2 Diagram



11.4.3.1.68.3 Fields

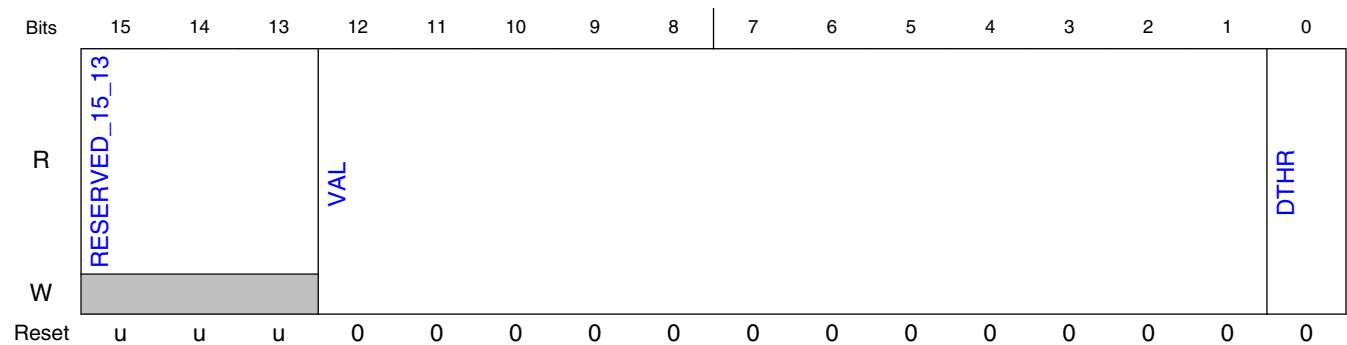
Field	Function
15 RESERVED_15_15	Reserved for Future use
14-10 BASE	which bit to sample when MODE = 1 or 4
9-8 SCOPE_DELAY	How many clocks to delay the analog scope_data.
7 SCOPE_SEL	Select Sampling Mode. Select Sampling Mode. 0 - Before AFE Sampling 1 - After AFE Sampling
6 UPDATE	Update Sampling phase. Update Sampling phase. Write a 1.
5-0 SAMPLE_PHASE	Sampling Phase

11.4.3.1.69 Current frequency integrator value. (LANE0_DIG_RX_DPLL_FREQ)

11.4.3.1.69.1 Offset

Register	Offset
LANE0_DIG_RX_DPLL_FREQ	2034h

11.4.3.1.69.2 Diagram



11.4.3.1.69.3 Fields

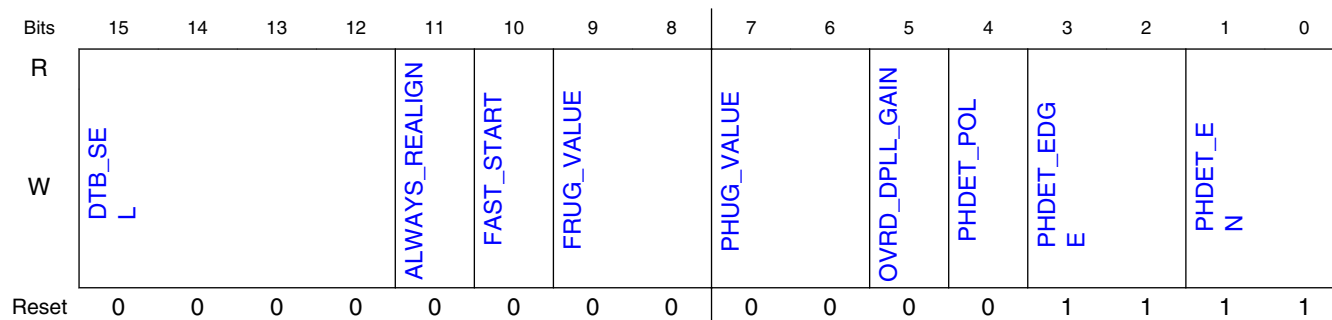
Field	Function
15-13 RESERVED_15_13	Reserved for Future use
12-1 VAL	Freq is 1. Freq is 1.526*VAL ppm from the reference When mppll_slow is set, the ppm is half the eqn above (volatile and 2 reads needed to read value)
0 DTHR	Bits below the useful resolution (volatile and 2 reads needed to read value)

11.4.3.1.70 Control bits for receiver in recovered domain (LANE0_DIG_RX_CDR_CTL)

11.4.3.1.70.1 Offset

Register	Offset
LANE0_DIG_RX_CDR_CTL	2036h

11.4.3.1.70.2 Diagram



11.4.3.1.70.3 Fields

Field	Function
15-12 DTB_SEL	select to drive various signals onto the dtb 0 - disabled 1 - pll_ana_rst,pll_count from rx_pwr_ctl 2 - com_good_high/low from aligner 3 - com_bad_high/low from aligner 4 - shift_in_prog,ana_odd_data from aligner 5 - low bits of XAUI align FSM state 6 - error_high/low from lbert_pm 7 - ana_los,los_filter from los block 8 - coast_dppll,eios_det from los block 9 - cdr_en and cdr_timeout from cdr_ctl A - cdr_en_adap and cdr_rx_valid from cdr_ctl B - cdr_en_eq and locked_vector_en from cdr_ctl C - dig_word_rst,rx_dig_en D - dig_dword_rst,dig_dword_clk
11 ALWAYS_REALIGN	realign on any misaligned comma
10 FAST_START	decrease startup steps by 50%
9-8 FRUG_VALUE	override value for FRUG
7-6 PHUG_VALUE	override value for PHUG
5 OVRD_DPLL_GAIN	Override PHUG and FRUG values

Table continues on the next page...

PCI Express PHY (PCIe_PHY)

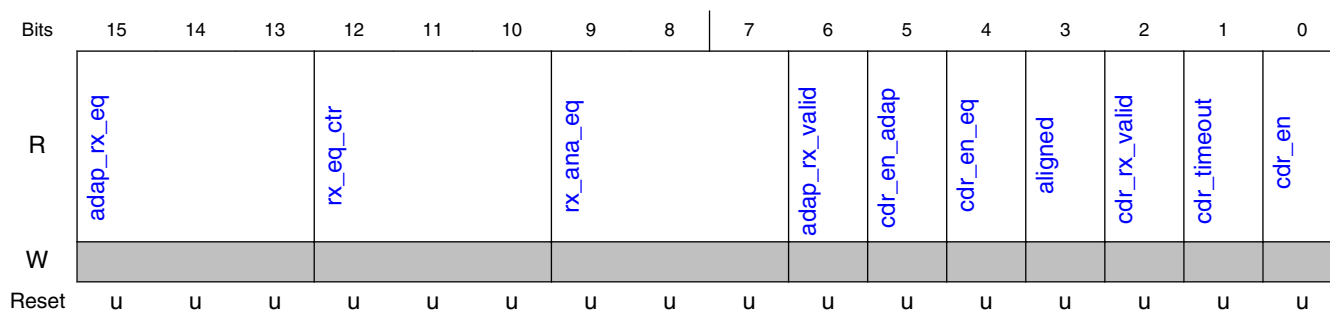
Field	Function
4 PHDET_POL	Reverse polarity of phase error
3-2 PHDET_EDGE	Edges to use for phase detection. Edges to use for phase detection. 11 - Use both edges 10 - Use rising edges only 01 - Use falling edges only 00 - Ignore all edges
1-0 PHDET_EN	Enable phase detector. Enable phase detector. top bit is odd slicers, bottom is even

11.4.3.1.71 Current values for CDR and Adaptation FSM. (LANE0_DIG_RX_CDR_CDR_FSM_DEBUG)

11.4.3.1.71.1 Offset

Register	Offset
LANE0_DIG_RX_CDR_CDR_FSM_DEBUG	2038h

11.4.3.1.71.2 Diagram



11.4.3.1.71.3 Fields

Field	Function
15-13 adap_rx_eq	Equalization setting from adaptation FSM. Equalization setting from adaptation FSM. (RU - read-only volatile)

Table continues on the next page...

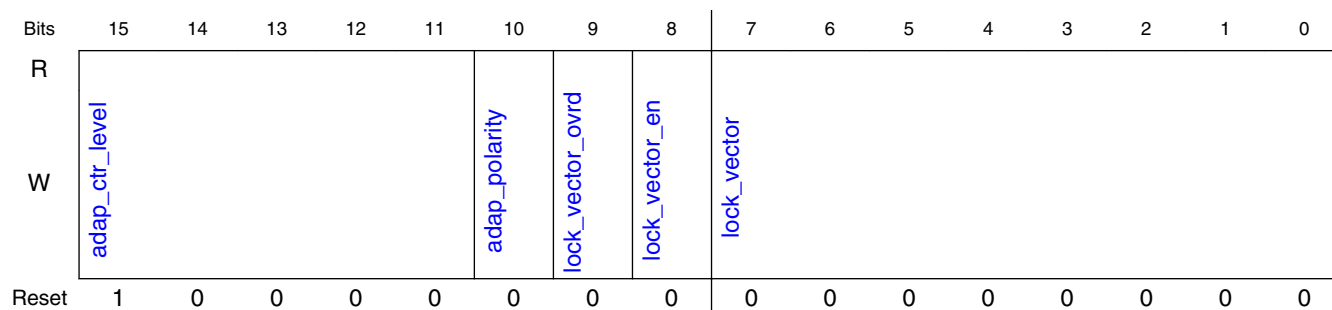
Field	Function
12-10 rx_eq_ctr	Initial centre point from equalization FSM. Initial centre point from equalization FSM. (RU - read-only volatile)
9-7 rx_ana_eq	Equalization setting to Analog. Equalization setting to Analog. (RU - read-only volatile)
6 adap_rx_valid	Adaptation has completed and locked. Adaptation has completed and locked. (RU - read-only volatile)
5 cdr_en_adap	Adapataion loop is enabling the CDR. Adapataion loop is enabling the CDR. (RU - read-only volatile)
4 cdr_en_eq	Equalization loop is enabling the CDR. Equalization loop is enabling the CDR. (RU - read-only volatile)
3 aligned	Datapath is bit-aligned. Datapath is bit-aligned. (RU - read-only volatile)
2 cdr_rx_valid	CDR has locked to incoming data stream. CDR has locked to incoming data stream. (RU - read-only volatile)
1 cdr_timeout	CDR has not locked to datastream and has timed-out. CDR has not locked to datastream and has timed-out. (RU - read-only volatile)
0 cdr_en	CDR has been enabled. CDR has been enabled. (RU - read-only volatile)

11.4.3.1.72 Control bits for receiver in recovered domain (LANE0_DIG_RX_CDR_LOCK_VEC_OVRD)

11.4.3.1.72.1 Offset

Register	Offset
LANE0_DIG_RX_CDR_LOCK_VEC_OVRD	203Ah

11.4.3.1.72.2 Diagram



11.4.3.1.72.3 Fields

Field	Function
15-11 adap_ctr_level	Amount of earlies that increment the adaptation counter (times 32, Default is 512). Amount of earlies that increment the adaptation counter (times 32, Default is 512). (volatile and 2 reads needed to read value)
10 adap_polarity	If asserted invert default adaptation adjustment for equalization. If asserted invert default adaptation adjustment for equalization. IF early decrease equalization. Normal mode is to decrease. (volatile and 2 reads needed to read value)
9 lock_vector_ovr d	Override enable for the rx_eq outputs. Override enable for the rx_eq outputs. (volatile and 2 reads needed to read value)
8 lock_vector_en	Override value for the locked_vector output completion. Override value for the locked_vector output completion. (volatile and 2 reads needed to read value)
7-0 lock_vector	Override value for the locked_vector. Override value for the locked_vector. (volatile and 2 reads needed to read value)

11.4.3.1.73 Current values for the locked vector. (LANE0_DIG_RX_CDR_LOCK_VEC)

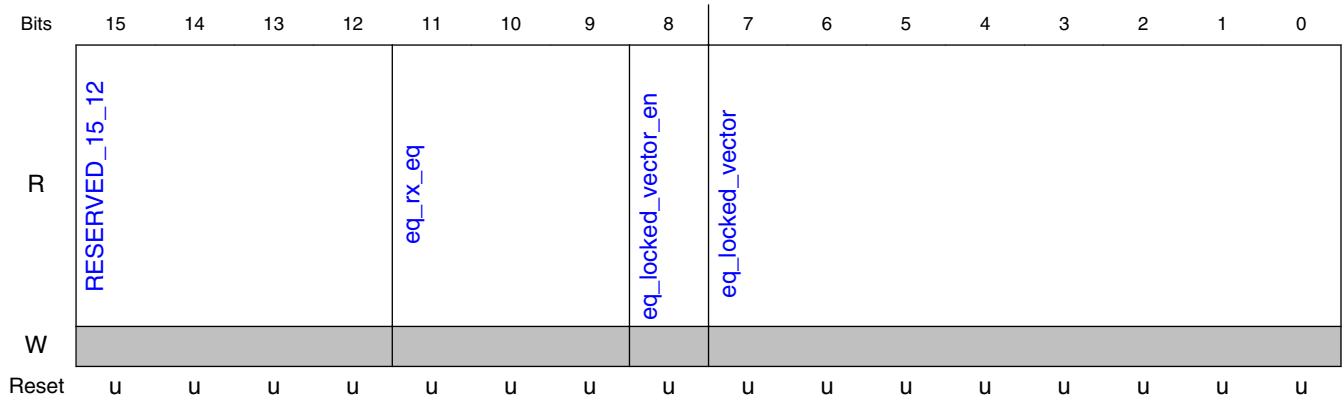
11.4.3.1.73.1 Offset

Register	Offset
LANE0_DIG_RX_CDR_L OCK_VEC	203Ch

11.4.3.1.73.2 Function

Current values for the locked vector..

11.4.3.1.73.3 Diagram



11.4.3.1.73.4 Fields

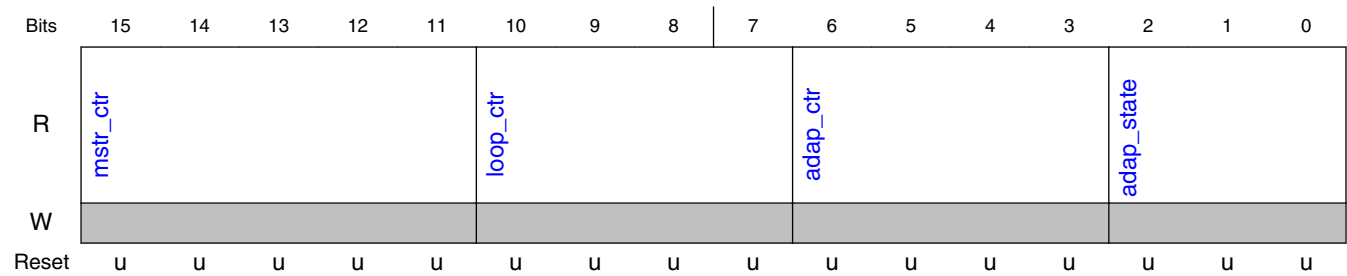
Field	Function
15-12 RESERVED_15_12	Reserved for Future use
11-9 eq_rx_eq	Equalization setting from the Equalization Loop. Equalization setting from the Equalization Loop. (RU - read-only volatile)
8 eq_locked_vect or_en	Equalization locked vector has been filled. Equalization locked vector has been filled. (RU - read-only volatile)
7-0 eq_locked_vect or	Results of equalization loop. Results of equalization loop. (RU - read-only volatile)

11.4.3.1.74 Observation of adaptation FSM register. (LANE0_DIG_RX_CDR_ADAP_FSM)

11.4.3.1.74.1 Offset

Register	Offset
LANE0_DIG_RX_CDR_A DAP_FSM	203Eh

11.4.3.1.74.2 Diagram



11.4.3.1.74.3 Fields

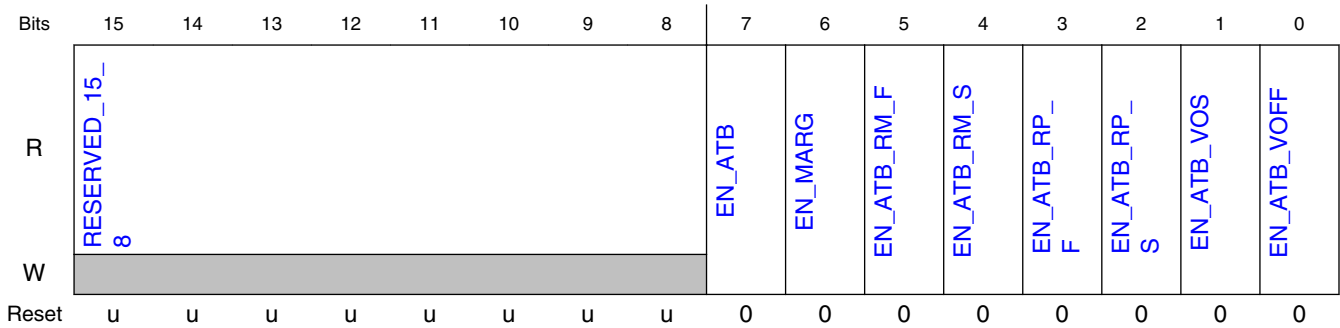
Field	Function
15-11 mstr_ctr	Master count register. Master count register. (RU - read-only volatile)
10-7 loop_ctr	Loop count register. Loop count register. (RU - read-only volatile)
6-3 adap_ctr	Adaptation count register. Adaptation count register. (RU - read-only volatile)
2-0 adap_state	Adaptation State. Adaptation State. NOTE: ADAP_RESET = 3'b000 ADAP_LOCK = 3'b001 ADAP_SUFF = 3'b010 ADAP_LOOP = 3'b011 ADAP_MSTR = 3'b100 ADAP_DONE = 3'b101 (RU - read-only volatile)

11.4.3.1.75 ATB Control Register (LANE0_ANA_RX_ATB0)

11.4.3.1.75.1 Offset

Register	Offset
LANE0_ANA_RX_ATB0	2040h

11.4.3.1.75.2 Diagram



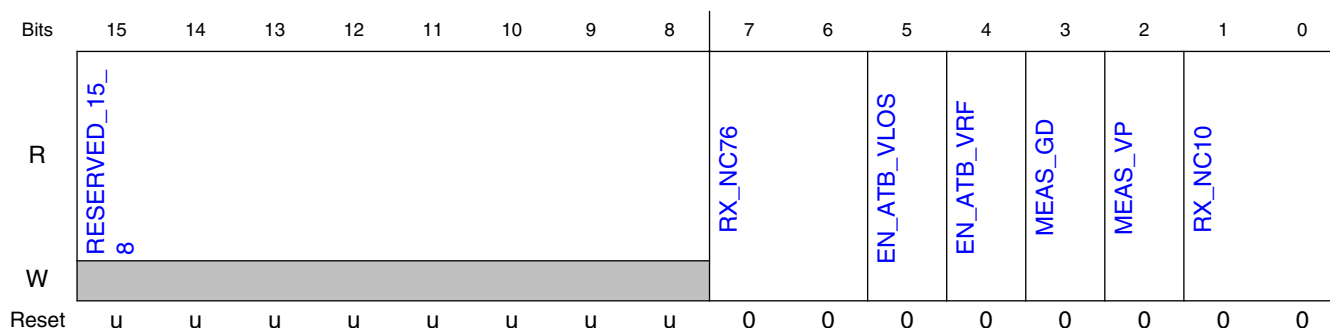
11.4.3.1.75.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 EN_ATB	Enable ATB sensing and forcing on internal RX nodes
6 EN_MARG	Enable margining mode in receiver; it puts EQ common mode output on atb_f_m
5 EN_ATB_RM_F	Enable atb_f_p on negative-side termination resistor
4 EN_ATB_RM_S	Enable atb_s_m on negative-side termination resistor
3 EN_ATB_RP_F	Enable atb_f_p on positive-side termination resistor
2 EN_ATB_RP_S	Enable atb_s_p on positive-side termination resistor
1 EN_ATB_VOS	Put rxafe filtered outputs vo_p on atb_s_p and vo_m on atb_s_m
0 EN_ATB_VOFF	Put rxafe voff_p on atb_s_p and voff_m on atb_s_m

11.4.3.1.76 ATB and Misc Control Register (LANE0_ANA_RX_ATB1)

11.4.3.1.76.1 Offset

Register	Offset
LANE0_ANA_RX_ATB1	2042h

11.4.3.1.76.2 Diagram**11.4.3.1.76.3 Fields**

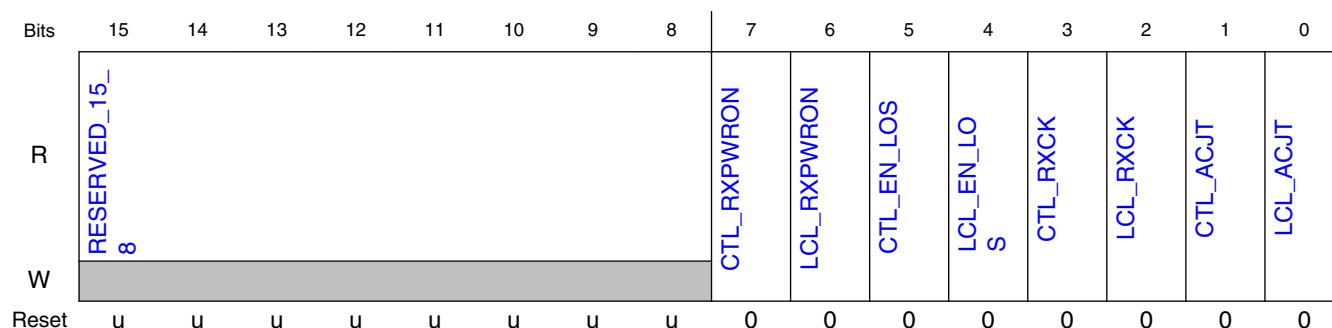
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7-6 RX_NC76	Reserved
5 EN_ATB_VLOS	Enable sensing of LOS reference voltage on atb_s_p
4 EN_ATB_VRF	Enable sensing of vref_rx (RX Bias B. Enable sensing of vref_rx (RX Bias B.G. Voltage) on atb_s_p
3 MEAS_GD	Enable sensing of local gd in RX; ties gd to atb_s_m
2 MEAS_VP	Enable sensing of local vp in RX; ties vp to atb_s_p
1-0 RX_NC10	Reserved

11.4.3.1.77 RX State Control Register (LANE0_ANA_RX_ENPWR0)

11.4.3.1.77.1 Offset

Register	Offset
LANE0_ANA_RX_ENPW R0	2044h

11.4.3.1.77.2 Diagram



11.4.3.1.77.3 Fields

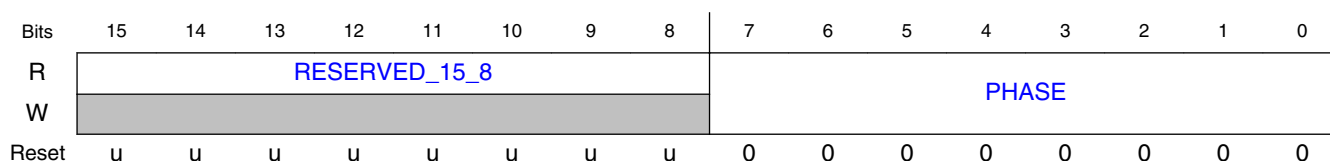
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 CTL_RXPWRO	MUX select for AFE en to select between CTL_RXPWRO=0 rx_afe_en (External signal) or CTL_RXPWRO=1 LCL_RXPWRO
6 LCL_RXPWRO	Enable/Disable RX AFE . Enable/Disable RX AFE .This bit is only selected when CTL_RXPWRO = 1
5 CTL_EN_LOS	MUX select for LOS en to select between CTL_EN_LOS=0 rx_los_en (External signal) or CTL_EN_LOS=1 LCL_EN_LOS
4 LCL_EN_LOS	Enable/Disable LOS. Enable/Disable LOS.This bit is only selected when CTL_EN_LOS = 1
3 CTL_RXCK	MUX select for RX Clock en to select between CTL_RXCK=0 rx_clock_en (External signal) or CTL_RXCK=1 LCL_RXCK.
2 LCL_RXCK	Enable/Disable en_rx_clock (RX clock enable). Enable/Disable en_rx_clock (RX clock enable).This bit is only selected when CTL_RXCK = 1
1 CTL_ACJT	MUX select for ACJTAG to select between CTL_ACJTG=0 rx_acjt_en (External signal) or CTL_ACJTG=1 LCL_ACJT.
0 LCL_ACJT	Enable/Disable acjtag block, This bit is only selected when CTL_ACJTG = 1

11.4.3.1.78 Phase select Value (LANE0_ANA_RX_PMIK_PHASE)

11.4.3.1.78.1 Offset

Register	Offset
LANE0_ANA_RX_PMIK_PHASE	2046h

11.4.3.1.78.2 Diagram



11.4.3.1.78.3 Fields

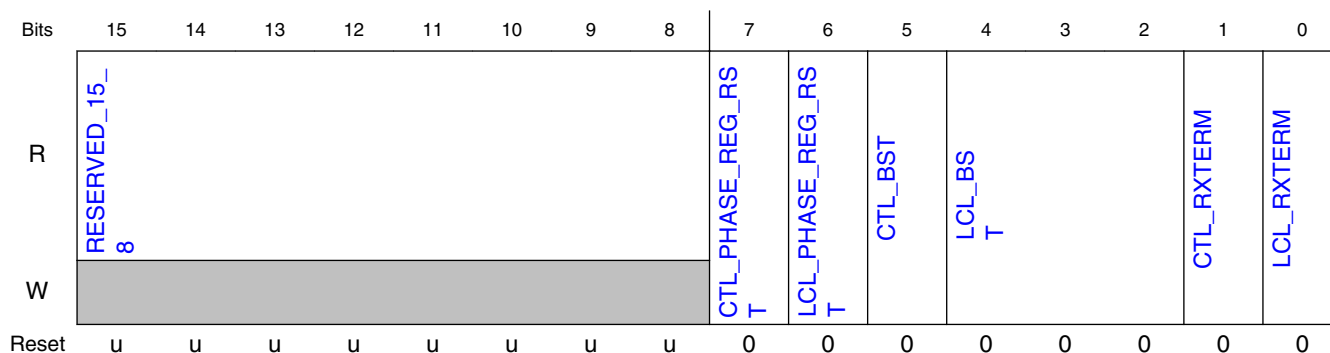
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7-0 PHASE	Write to bits 8 to 1 of the phase select register in the phase mixer through the CDR (USER2 volatile - read-write volatile and custom)

11.4.3.1.79 Additional RX State Control Register (LANE0_ANA_RX_ENPWR1)

11.4.3.1.79.1 Offset

Register	Offset
LANE0_ANA_RX_ENPWR1	2048h

11.4.3.1.79.2 Diagram



11.4.3.1.79.3 Fields

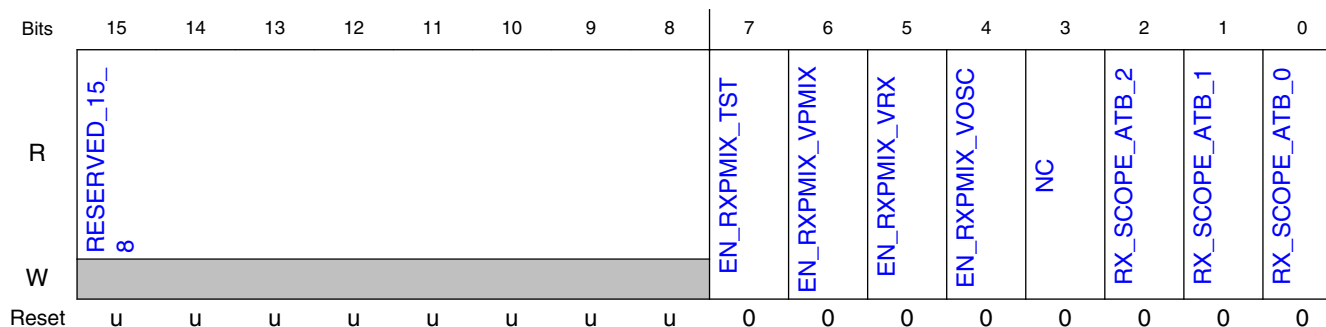
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 CTL_PHASE_REG_RST	MUX select for Phase Register Reset to select between CTL_PHASE_REG_RST=0 rx_phase_reg_rst (External signal) or CTL_PHASE_REG_RST=1 LCL_PHASE_REG_RST
6 LCL_PHASE_REG_RST	Reset phase register. Reset phase register. This bit is only selected when CTL_PHASE_REG_RST = 1
5 CTL_BST	MUX select for AFE Boost to select between CTL_BST=0 rx_aeq[2:0] (External signal) or CTL_BST=1 LCL_BST
4-2 LCL_BST	RX Boost (equalization) Value
1 CTL_RXTERM	MUX select for RX Term enable to select between CTL_RXTERM=0 rx_term_en (External signal) or CTL_RXTERM=1 LCL_RXTERM
0 LCL_RXTERM	Enable/disable RX termination when CTL_RXTERM = 1

11.4.3.1.80 Yet another RX State Control Register (LANE0_ANA_RX_ENPWR2)

11.4.3.1.80.1 Offset

Register	Offset
LANE0_ANA_RX_ENPWR2	204Ah

11.4.3.1.80.2 Diagram



11.4.3.1.80.3 Fields

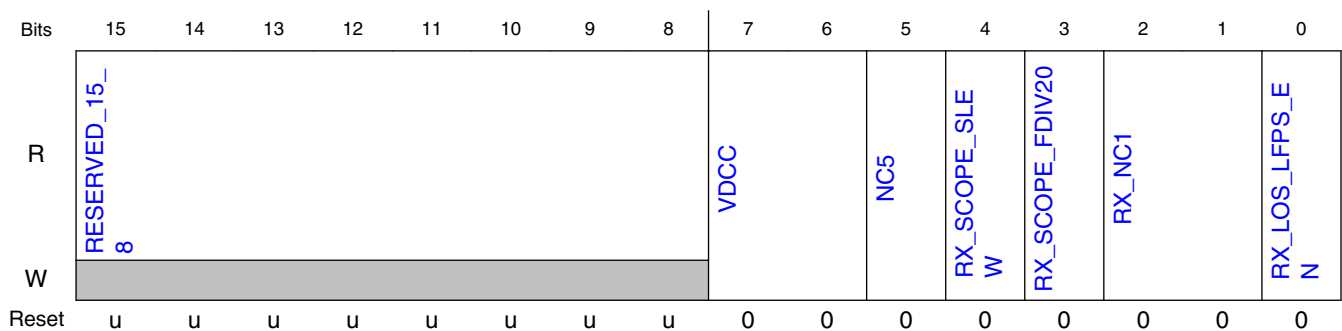
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 EN_RXPMIX_TST	Enable XOR gate to test linearity of RX phase mixer using atb_s_p and atb_s_m
6 EN_RXPMIX_VPMIX	Put vreg_pmix on atb_s_p
5 EN_RXPMIX_VRX	Put vreg_rx on atb_s_p
4 EN_RXPMIX_VOSC	Put vreg_vosc on atb_s_p
3 NC	Reserved
2 RX_SCOPE_ATB_2	Put XOR of RX SCOPE PMIX input and output on atb_s_p
1 RX_SCOPE_ATB_1	Put RX SCOPE regulated VP on atb_s_p
0 RX_SCOPE_ATB_0	Have RX SCOPE reference for regulator use atb_f_p as a reference instead of VP/VBG

11.4.3.1.81 RX Scope controls and clock DCC control voltage probe (LANE0_ANA_RX_SCOPE_VDCC)

11.4.3.1.81.1 Offset

Register	Offset
LANE0_ANA_RX_SCOPE_VDCC	204Ch

11.4.3.1.81.2 Diagram



11.4.3.1.81.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7-6 VDCC	Probe DCC control voltage
5 NC5	Reserved
4 RX_SCOPE_SLEW	Set high for low RX clock frequencies (625MHz) for RX Scope to work correctly
3 RX_SCOPE_FDIV20	Divide scope output clock by 20 instead of 10
2-1 RX_NC1	Reserved
0	Enables the RX LOS LFPS mode, use when you want to detect incoming LFPS signals, USB3.

PCI Express PHY (PCIe_PHY)

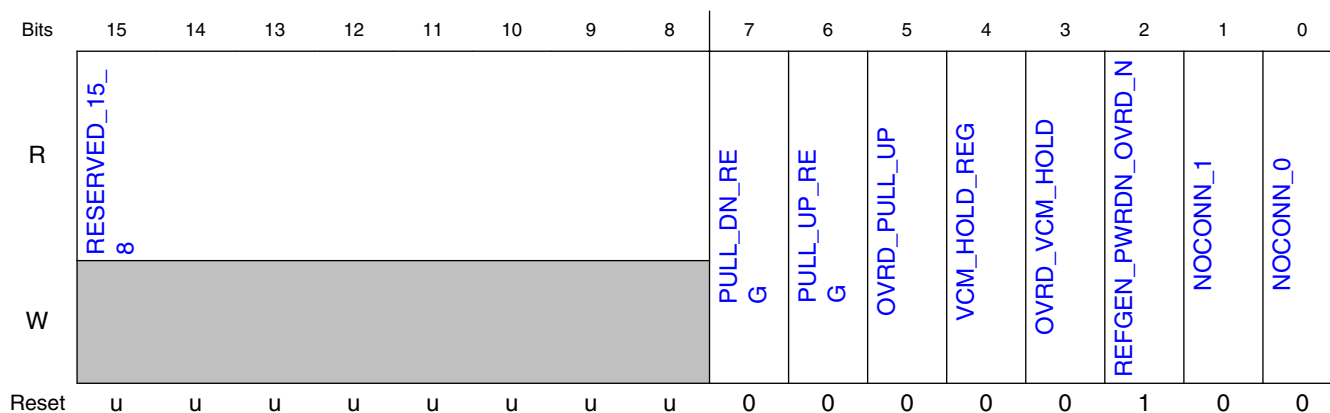
Field	Function
RX_LOS_LFPS_EN	Enables the RX LOS LFPS mode, use when you want to detect incoming LFPS signals, USB3.0

11.4.3.1.82 Register 3 (LANE0_ANA_TX_TXDRV_CNTRL)

11.4.3.1.82.1 Offset

Register	Offset
LANE0_ANA_TX_TXDRV_CNTRL	2056h

11.4.3.1.82.2 Diagram



11.4.3.1.82.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 PULL_DN_REG	register bit that causes the calibrated TX bits to pull down in a common mode fashion. register bit that causes the calibrated TX bits to pull down in a common mode fashion. If pull_dn_reg and tx_pull_up are both high, then pull_dn_reg wins
6 PULL_UP_REG	Register override for tx_pull_up. Register override for tx_pull_up. Used when ovr_d_pull_up is HIGH, causes calibrated TX bits to pull up in a common mode fashion, unless pull_dn_reg is HIGH
5	Selects local value of pull_up_reg instead of tx_pull_up

Table continues on the next page...

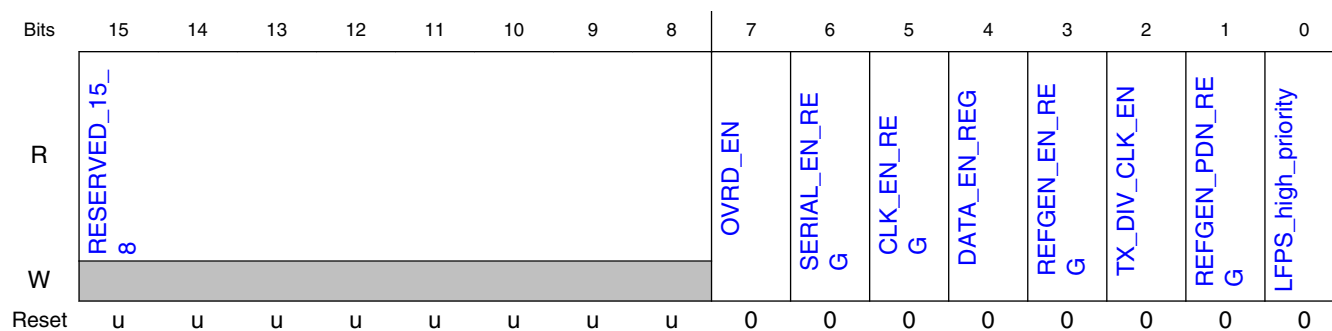
Field	Function
OVRD_PULL_UP	
4	Register override for tx_vcm_hold.
VCM_HOLD_REG	Register override for tx_vcm_hold. Used when ovr_d_vcm_hold is HIGH, controls the TX common mode hold circuitry
3	Selects local value of vcm_hold_reg instead of tx_vcm_hold to control state of TX common mode hold circuitry
OVRD_VCM_HOLD	
2	Set LOW for TX regulator not to turn on when RX clock is enabled.
REFGEN_PWRDN_OVRD_N	Set LOW for TX regulator not to turn on when RX clock is enabled. Default is for TX regulator to turn on when RX or TX clock is enabled
1	Reserved
NOCONN_1	
0	Reserved
NOCONN_0	

11.4.3.1.83 Register 4 (LANE0_ANA_TX_POWER_CTL)

11.4.3.1.83.1 Offset

Register	Offset
LANE0_ANA_TX_POWER_CTL	2058h

11.4.3.1.83.2 Diagram



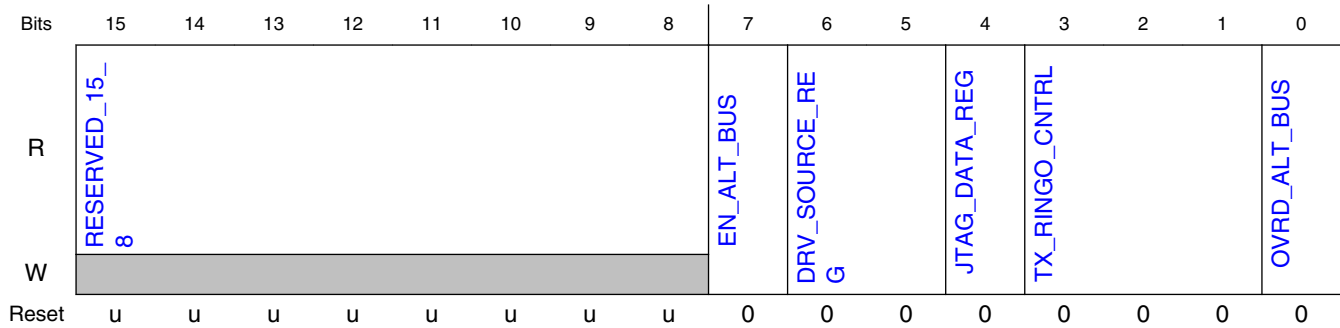
11.4.3.1.83.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 OVRD_EN	Enables local overrides for all signals in this register except for TX_DIV_CLK_EN which doesn't need this signal
6 SERIAL_EN_REG	Forces state of tx_serial_en when OVRD_EN is 1
5 CLK_EN_REG	Forces state of tx_clk_en when OVRD_EN is 1
4 DATA_EN_REG	Forces state of tx_data_en when OVRD_EN is HIGH
3 REFGEN_EN_REG	Register override value for tx_refgen_en. Register override value for tx_refgen_en. Controls the TX regulator, biasing circuitry and the rxdetect comparators when OVRD_EN is HIGH. This function may be overridden by the state of refgen_pwdn_reg
2 TX_DIV_CLK_EN	Enables the div clock that goes from the TX to the undersampler, more appropriately called tx_sampler_clk_en, this clock comes after the optional divide by 2 or 4, tx_clk_en must be HIGH to get a clock out
1 REFGEN_PDN_REG	Register override for refgen pwn. Register override for refgen pwn. Used when OVRD_EN is HIGH. Controls the TX regulator, biasing circuitry and the rxdetect comparators when OVRD_EN is HIGH
0 LFPS_high_priority	When this is set high LFPS priority becomes higher than tx_data

11.4.3.1.84 Register 5 (LANE0_ANA_TX_ALT_BLOCK)**11.4.3.1.84.1 Offset**

Register	Offset
LANE0_ANA_TX_ALT_BLOCK	205Ah

11.4.3.1.84.2 Diagram



11.4.3.1.84.3 Fields

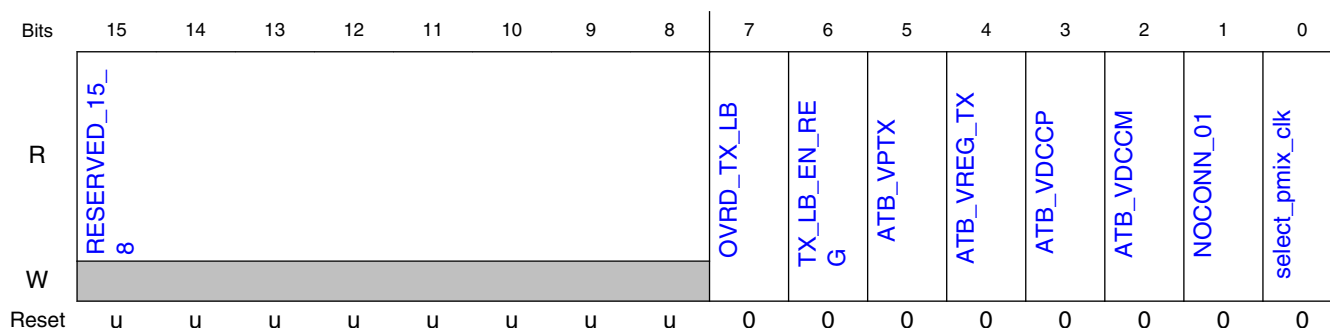
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 EN_ALT_BUS	Enables the TX for alt bus mode, powers up the TX regulator, biasing circuitry and the rxdetect comparators. Enables the TX for alt bus mode, powers up the TX regulator, biasing circuitry and the rxdetect comparators. Needed if manually running the alt bus features
6-5 DRV_SOURCE_REG	Forces state of tx_data_source when OVRD_ALT_BUS is HIGH. Forces state of tx_data_source when OVRD_ALT_BUS is HIGH. 00 = serializer data or alt oscillator vp/vph/vphreg/vptx if selected, 01 = jtag data differential, 10 = LFPS oscillator differential, 11 = jtag data common mode drive for test
4 JTAG_DATA_REG	Forces state of jtag_data when OVRD_ALT_BUS is HIGH
3-1 TX_RINGO_CNTRL	Enables and connects the various test oscillators to the transmit pins, must set drv_source_reg bus correctly. Enables and connects the various test oscillators to the transmit pins, must set drv_source_reg bus correctly. 0XX = All oscillators are disabled, 100 = HV oscillator enabled, 101 = SVT oscillator enabled, 110 = LVT oscillator enabled, 111 = ULVT oscillator enabled
0 OVRD_ALT_BUS	Enables local overrides for alt-bus control signals: DRV_SOURCE_REG and JTAG_DATA_REG

11.4.3.1.85 Register 6 (LANE0_ANA_TX_ALT_AND_LOOPBACK)

11.4.3.1.85.1 Offset

Register	Offset
LANE0_ANA_TX_ALT_A ND_LOOPBACK	205Ch

11.4.3.1.85.2 Diagram



11.4.3.1.85.3 Fields

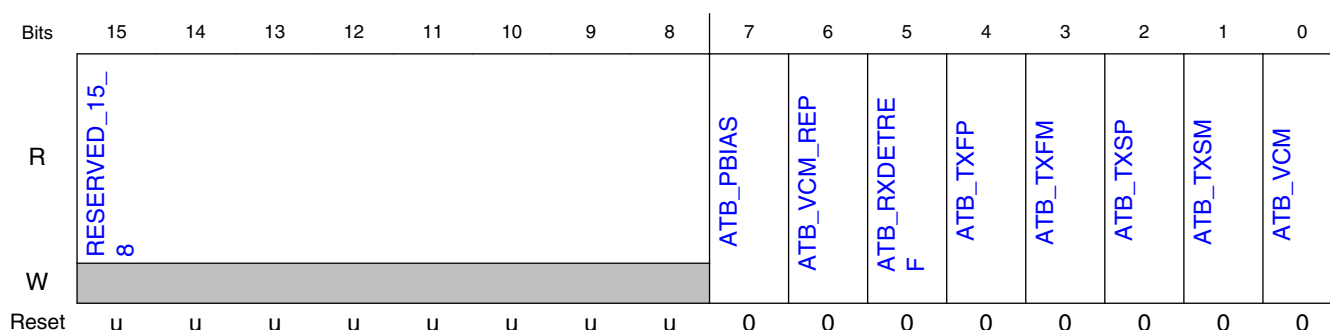
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 OVRD_TX_LB	Enables the override of the tx_lb_en
6 TX_LB_EN_REG	Forces state of tx_lb_en when OVRD_TX_LB is enabled
5 ATB_VPTX	Connects vptx to atb_s_p and gd to atb_s_m
4 ATB_VREG_TX	Connects TX regulator input reference to atb_s_p and gd to atb_s_m
3 ATB_VDCCP	Connects vdcc_p to atb_s_p
2 ATB_VDCCM	Connects vdcc_m to atb_s_m
1 NOCONN_01	Reserved
0 select_pmix_clk	select pmix_clk for TX clock for ATE test mode

11.4.3.1.86 Register 7 (LANE0_ANA_TX_TX_ATB_REG)

11.4.3.1.86.1 Offset

Register	Offset
LANE0_ANA_TX_TX_ATB_REG	205Eh

11.4.3.1.86.2 Diagram



11.4.3.1.86.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 ATB_PBIAS	Connect real pmos_bias voltage for TX PMOS driver pull up path to atb_s_p and local ground at the pmos_bias block to atb_s_m
6 ATB_VCM_REP	Connect common mode replica voltage in pmos_bias block to atb_s_p and local ground to atb_s_m
5 ATB_RXDETRE F	Connect RX detect block reference voltage to atb_s_p and local ground to atb_s_m
4 ATB_TXFP	Connect tx_p to atb_f_p
3 ATB_TXFM	Connect tx_m to atb_f_m
2 ATB_TXSP	Connect tx_p to atb_s_p

Table continues on the next page...

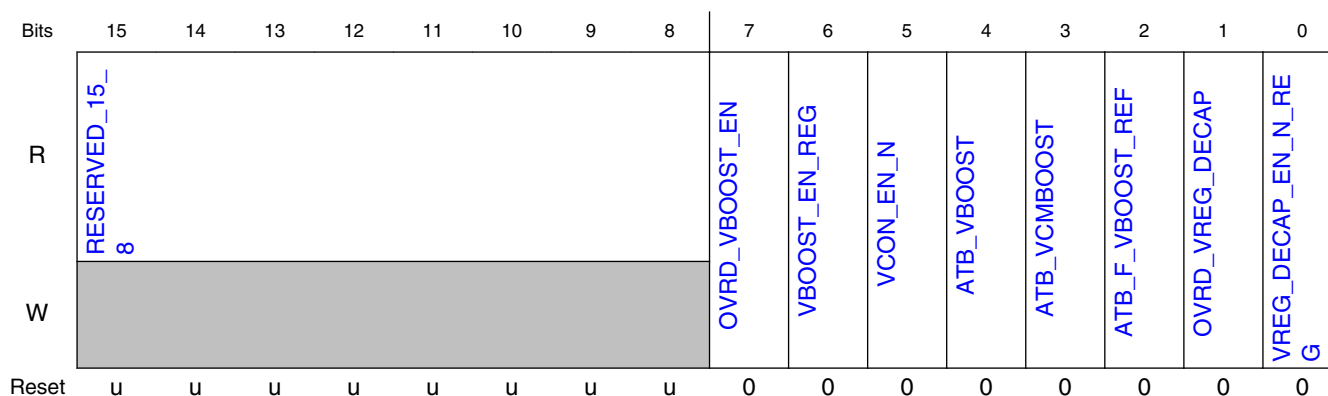
Field	Function
1 ATB_TXSM	Connect tx_m to atb_s_m
0 ATB_VCM	Connect tx_p/tx_m common mode voltage onto atb_s_p and local ground onto atb_s_m

11.4.3.1.87 Register 8 (LANE0_ANA_TX_TX_BOOST)

11.4.3.1.87.1 Offset

Register	Offset
LANE0_ANA_TX_TX_BOOST	2060h

11.4.3.1.87.2 Diagram



11.4.3.1.87.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 OVRD_VBOOST_EN	Enables the overrides on VBOOST_EN_REG
6	Forces state of tx_vboost_en, when OVRD_VBOOST_EN is HIGH

Table continues on the next page...

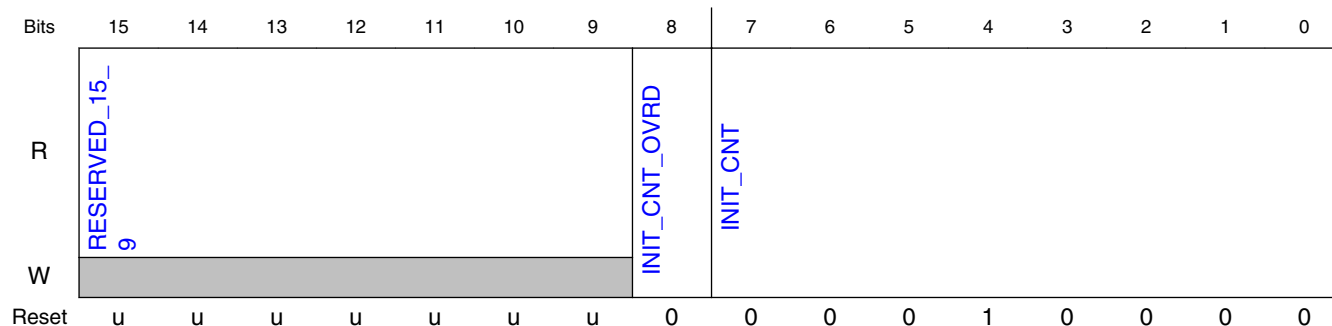
Field	Function
VBOOST_EN_REG	
5 VCON_EN_N	Forces state of vcon_en_n
4 ATB_VBOOST	Connects vboost to atb_s_p and local ground to Atb_s_m
3 ATB_VCMBOOST	Connects vcmboost to atb_s_p and local ground to atb_s_m.
2 ATB_F_VBOOST_REF	Enables forcing of tx_vboost_vref through atb_f_p from external
1 OVRD_VREG_DECAP	Enables override of vreg_decap_en_n signal
0 VREG_DECAP_EN_N_REG	Forces state of vreg_decap_en_n control signal. Forces state of vreg_decap_en_n control signal. Controls whether extra decoupling is switched in for TX regulator output.

11.4.3.1.88 Control bits for CDR init phase portion (LANE0_DIG_RX_CDR_FSM_INIT_CTL)

11.4.3.1.88.1 Offset

Register	Offset
LANE0_DIG_RX_CDR_FSM_INIT_CTL	2080h

11.4.3.1.88.2 Diagram



11.4.3.1.88.3 Fields

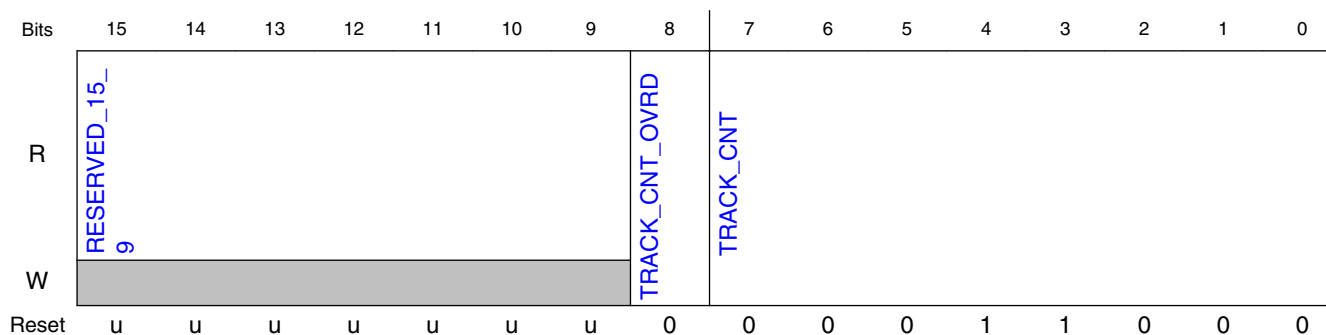
Field	Function
15-9 RESERVED_15_9	Reserved for Future use
8 INIT_CNT_OVRD	Active high override enable bit to select the override INIT_CNT register 0 = Disabled 1 = Enabled
7-0 INIT_CNT	Override value for CDR Init Phase

11.4.3.1.89 Control bits for CDR track phase portion (LANE0_DIG_RX_CDR_FSM_TRACK_CTL)

11.4.3.1.89.1 Offset

Register	Offset
LANE0_DIG_RX_CDR_FSM_TRACK_CTL	2082h

11.4.3.1.89.2 Diagram



11.4.3.1.89.3 Fields

Field	Function
15-9	Reserved for Future use

Table continues on the next page...

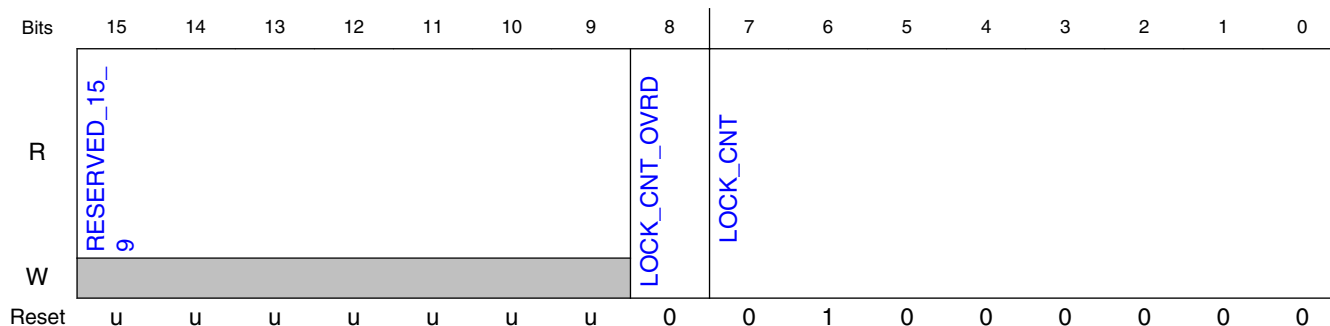
Field	Function
RESERVED_15_9	
8 TRACK_CNT_OVRD	Active high override enable bit to select the override TRACK_CNT register 0 = Disabled 1 = Enabled
7-0 TRACK_CNT	Override value for CDR Track Phase

11.4.3.1.90 Control bits for CDR lock phase portion (LANE0_DIG_RX_CDR_FSM_LOCK_CTL)

11.4.3.1.90.1 Offset

Register	Offset
LANE0_DIG_RX_CDR_FSM_LOCK_CTL	2084h

11.4.3.1.90.2 Diagram



11.4.3.1.90.3 Fields

Field	Function
15-9 RESERVED_15_9	Reserved for Future use
8 LOCK_CNT_OVRD	Active high override enable bit to select the override LOCK_CNT register 0 = Disabled 1 = Enabled

Table continues on the next page...

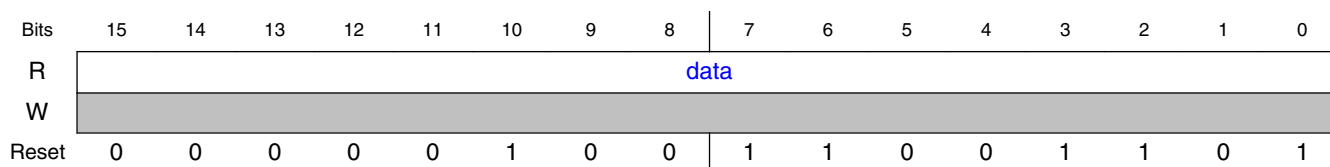
Field	Function
7-0 LOCK_CNT	Override value for CDR Lock Phase

11.4.3.1.91 Low 16 bits of IDCODE (SUPX_DIG_IDCODE_LO)

11.4.3.1.91.1 Offset

Register	Offset
SUPX_DIG_IDCODE_LO	1_0000h

11.4.3.1.91.2 Diagram



11.4.3.1.91.3 Fields

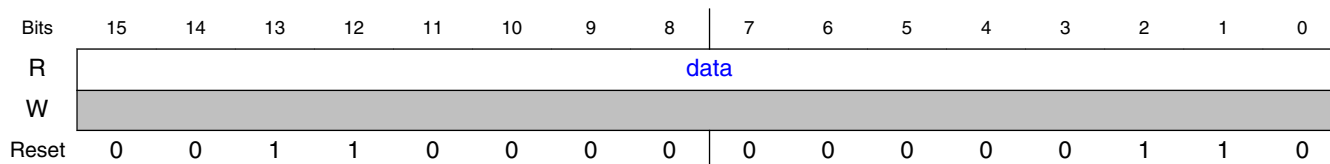
Field	Function
15-0 data	data

11.4.3.1.92 High 16 bits of IDCODE (SUPX_DIG_IDCODE_HI)

11.4.3.1.92.1 Offset

Register	Offset
SUPX_DIG_IDCODE_HI	1_0002h

11.4.3.1.92.2 Diagram



11.4.3.1.92.3 Fields

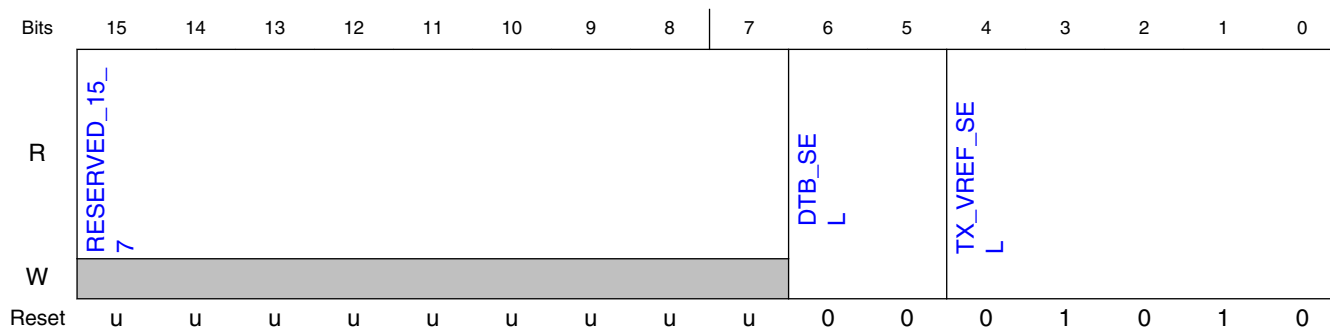
Field	Function
15-0	data
data	

11.4.3.1.93 Debug controls (SUPX_DIG_DEBUG)

11.4.3.1.93.1 Offset

Register	Offset
SUPX_DIG_DEBUG	1_0004h

11.4.3.1.93.2 Diagram



11.4.3.1.93.3 Fields

Field	Function
15-7	Reserved for Future use

Table continues on the next page...

PCI Express PHY (PCIe_PHY)

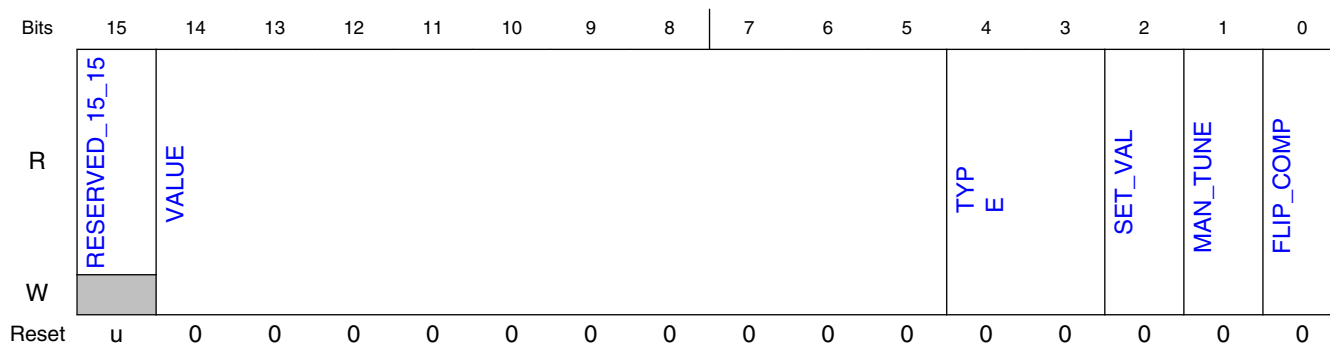
Field	Function
RESERVED_15_7	
6-5 DTB_SEL	The lane DTB's are OR'd together with the support DTB signals selected with the below encodings 0 - none 1 - reset_ctl DTB output 2 - scope DTB output 3 - rtune DTB output
4-0 TX_VREF_SEL	TX_VREF_SEL

11.4.3.1.94 Resistor tuning debug controls (SUPX_DIG_RTUNE_DEBUG)

11.4.3.1.94.1 Offset

Register	Offset
SUPX_DIG_RTUNE_DEBUG	1_0006h

11.4.3.1.94.2 Diagram



11.4.3.1.94.3 Fields

Field	Function
15 RESERVED_15_15	Reserved for Future use

Table continues on the next page...

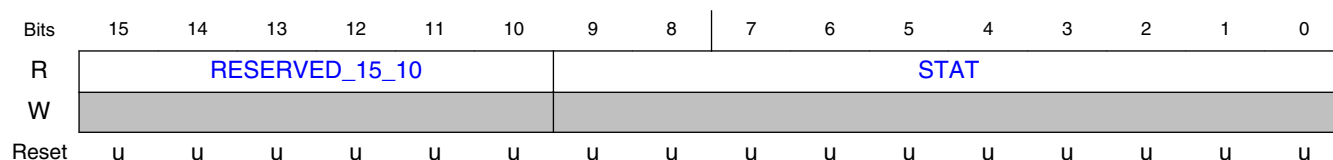
Field	Function
14-5 VALUE	Value to use when triggering SET_VAL field only the 6 LSB's are used when setting rx or tx cal values (USER2 - read-write and custom)
4-3 TYPE	Type of manual tuning or register read/write to execute 0 - ADC, or read/write rt_value 1 - Rx tune, or read/write rx_cal_val (only 6 bits) 2 - Tx tune, or read/write tx_cal_val (only 6 bits) 3 - Resref detect (no affect when triggering SET_VAL field) (USER2 - read-write and custom)
2 SET_VAL	Set value Write to a 1 to manually write the register specified by the TYPE field to the value in the VALUE field (USER2 - read-write and custom)
1 MAN_TUNE	Write to a 1 to do a manual tuning specified by TYPE field starting a manual tune while a tune is currently running can cause unpredictable results. Write to a 1 to do a manual tuning specified by TYPE field starting a manual tune while a tune is currently running can cause unpredictable results. For use only when you know the state of the rtune circuit. (USER2 - read-write and custom)
0 FLIP_COMP	Invert analog comparator output (USER2 - read-write and custom)

11.4.3.1.95 Resistor tuning register status (SUPX_DIG_RTUNE_STAT)

11.4.3.1.95.1 Offset

Register	Offset
SUPX_DIG_RTUNE_STAT	1_0008h

11.4.3.1.95.2 Diagram



11.4.3.1.95.3 Fields

Field	Function
15-10	Reserved for Future use

Table continues on the next page...

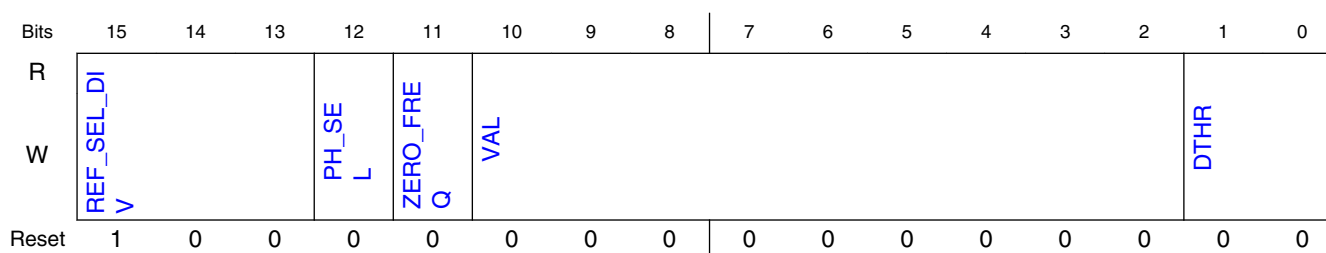
Field	Function
RESERVED_15_10	
9-0 STAT	Current value of the register specified by the TYPE field in the debug control register. Current value of the register specified by the TYPE field in the debug control register. Set the TYPE field in the DEBUG register, and then do a read of this register. (RU - read-only volatile)

11.4.3.1.96 Current MPLL phase selector value (SUPX_DIG_SS_PHASE)

11.4.3.1.96.1 Offset

Register	Offset
SUPX_DIG_SS_PHASE	1_000Ah

11.4.3.1.96.2 Diagram



11.4.3.1.96.3 Fields

Field	Function
15-13 REF_SEL_DIV	Reference Divisor to get better resolution and lower PPM errors when synthesizing from a non-integer reference clock. Reference Divisor to get better resolution and lower PPM errors when synthesizing from a non-integer reference clock. NOTE: Default value is 4. One is added in hardware to give a range of 1 to 8. (volatile and 2 reads needed to read value)
12 PH_SEL	Phase Mixer Clock select. Phase Mixer Clock select. 1: is pmix. 0 is VCO. (volatile and 2 reads needed to read value)
11 ZERO_FREQ	Zero frequency register. Zero frequency register. NOTES: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value)
10-2 VAL	Phase value from zero reference (volatile and 2 reads needed to read value)
1-0	Bits below the useful resolution (volatile and 2 reads needed to read value)

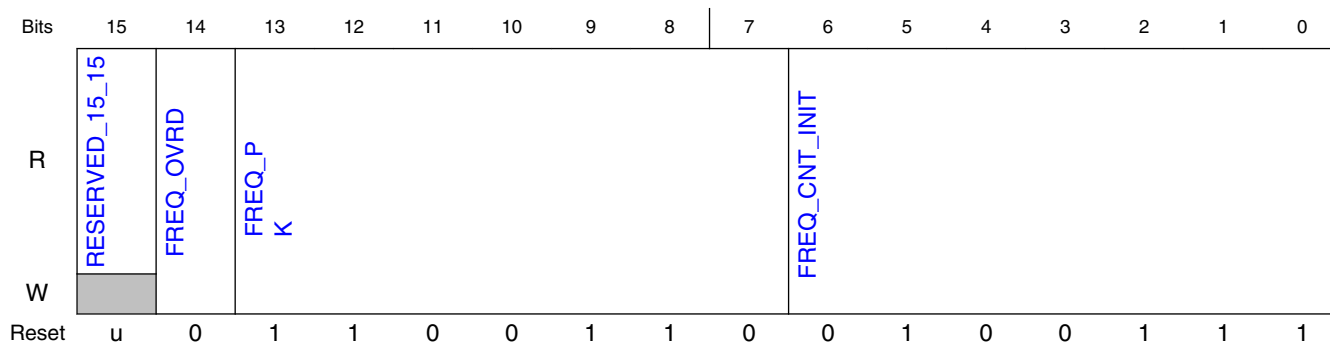
Field	Function
DTHR	

11.4.3.1.97 Frequency Control for Spread Spectrum (SUPX_DIG_SS_FREQ)

11.4.3.1.97.1 Offset

Register	Offset
SUPX_DIG_SS_FREQ	1_000Ch

11.4.3.1.97.2 Diagram



11.4.3.1.97.3 Fields

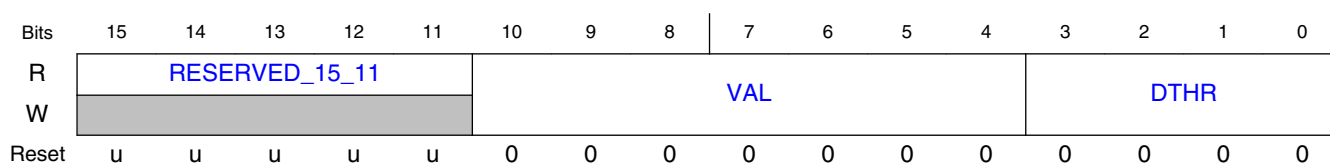
Field	Function
15 RESERVED_15_15	Reserved for Future use
14 FREQ_OVRD	Frequency Register Override. Frequency Register Override. NOTE: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value)
13-7 FREQ_PK	Peak Frequency Value (for changing direction) (volatile and 2 reads needed to read value)
6-0 FREQ_CNT_INIT	Initial Frequency Counter Value (volatile and 2 reads needed to read value)

11.4.3.1.98 Current HSP_PMI_X SSC phase sel value (SUPX_DIG_HSP_SS_PHASE)

11.4.3.1.98.1 Offset

Register	Offset
SUPX_DIG_HSP_SS_PHASE	1_000Eh

11.4.3.1.98.2 Diagram



11.4.3.1.98.3 Fields

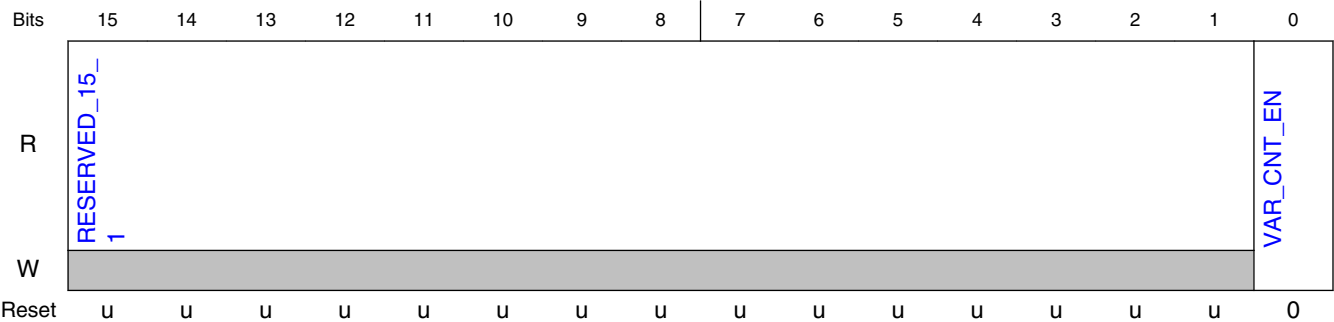
Field	Function
15-11 RESERVED_15_11	Reserved for Future use
10-4 VAL	SSC phase value from zero reference (volatile and 2 reads needed to read value)
3-0 DTHR	Bits below the useful resolution (volatile and 2 reads needed to read value)

11.4.3.1.99 Control for fixed or variable MPLL Count Selection (SUPX_DIG_MPLL_CNT_CTL)

11.4.3.1.99.1 Offset

Register	Offset
SUPX_DIG_MPLL_CNT_CTL	1_0010h

11.4.3.1.99.2 Diagram



11.4.3.1.99.3 Fields

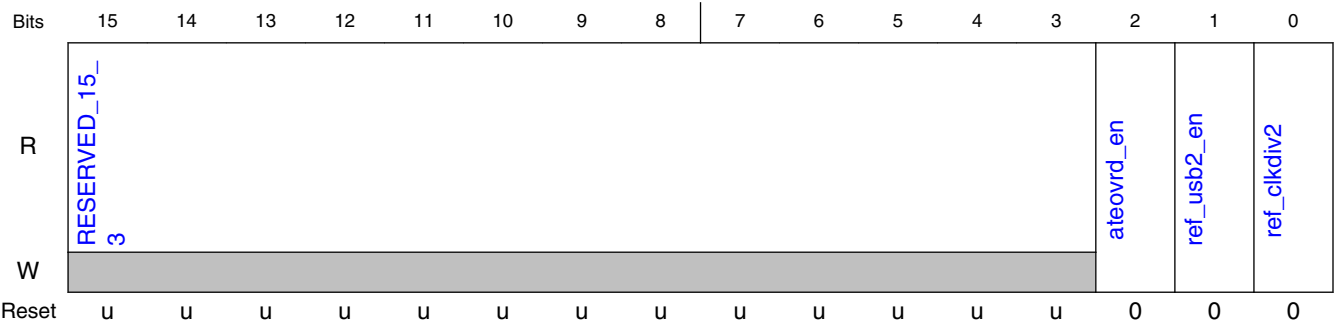
Field	Function
15-1 RESERVED_15_1	Reserved for Future use
0 VAR_CNT_EN	Selects variable count when enabled (Default = fixed count)

11.4.3.1.100 Override values for incoming controls from ASIC (SUPX_DIG_ATEOVRD)

11.4.3.1.100.1 Offset

Register	Offset
SUPX_DIG_ATEOVRD	1_0020h

11.4.3.1.100.2 Diagram



11.4.3.1.100.3 Fields

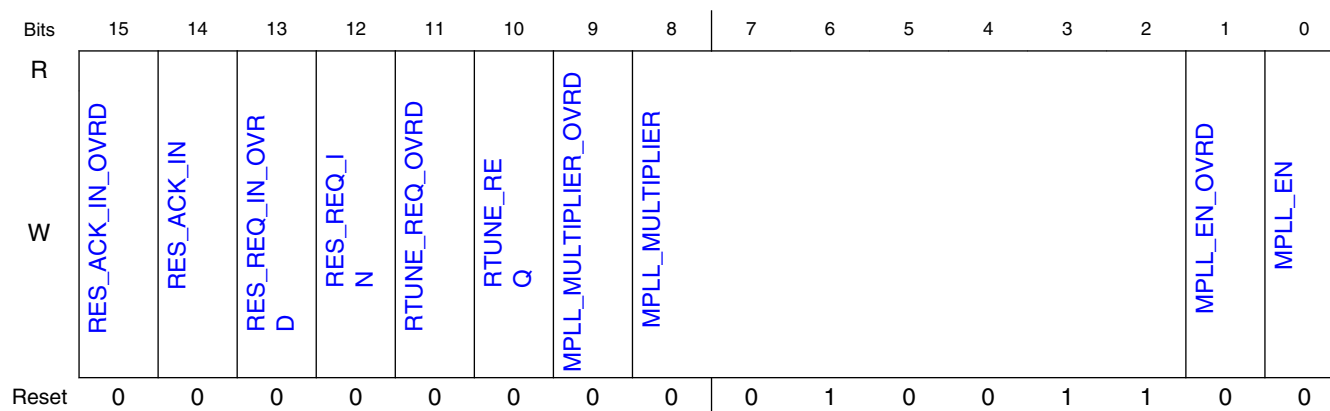
Field	Function
15-3 RESERVED_15_3	Reserved for Future use
2 ateovrd_en	Override enable for ATE signals. Override enable for ATE signals. NOTE: This register is only accessible through cregs. Does not retain state during reset or ATPG. (USER2 - read-write and custom)
1 ref_usb2_en	Override value for USB HSP PHY ref_clk enable. Override value for USB HSP PHY ref_clk enable. (USER2 - read-write and custom)
0 ref_clkdiv2	Override value for SSP ref_clk pre-scaler. Override value for SSP ref_clk pre-scaler. (USER2 - read-write and custom)

11.4.3.1.101 Override values for incoming RTUNE and MPLL controls from ASIC (SUPX_DIG_MPLL_OVRD_IN_LO)

11.4.3.1.101.1 Offset

Register	Offset
SUPX_DIG_MPLL_OVRD_IN_LO	1_0022h

11.4.3.1.101.2 Diagram



11.4.3.1.101.3 Fields

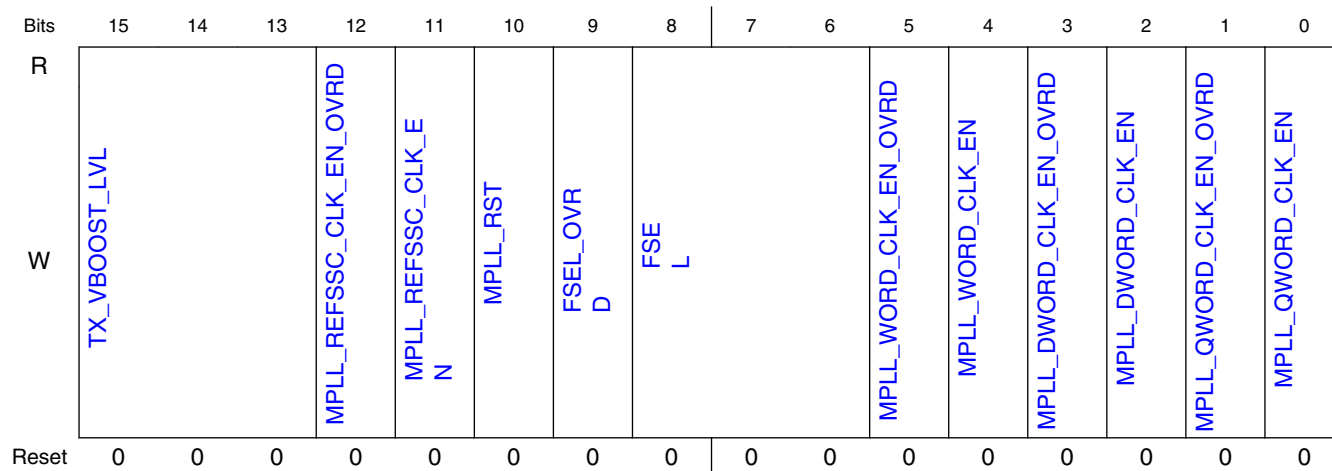
Field	Function
15 RES_ACK_IN_OVRD	Override enable for res_ack_in
14 RES_ACK_IN	Override value for res_ack_in
13 RES_REQ_IN_OVRD	Override enable for res_req_in
12 RES_REQ_IN	Override value for res_req_in
11 RTUNE_REQ_OVRD	Override enable for rtune_req
10 RTUNE_REQ	Override value for rtune_req
9 MPLL_MULTIPLIER_OVRD	Override enable for mpll_multipler
8-2 MPLL_MULTIPLIER	Override value for mpll_multipler
1 MPLL_EN_OVRD	Override enable for mpll_en
0 MPLL_EN	Override value for mpll_en

11.4.3.1.102 Override values for incoming VBOOST LVL and MPLL controls from ASIC (SUPX_DIG_MPLL_OVRD_IN_HI)

11.4.3.1.102.1 Offset

Register	Offset
SUPX_DIG_MPLL_OVRD_IN_HI	1_0024h

11.4.3.1.102.2 Diagram



11.4.3.1.102.3 Fields

Field	Function
15-13 TX_VBOOST_LVL	Override value for tx_vboost_lvl. Override value for tx_vboost_lvl. Override enabled using LEVEL_EN from LEVEL_OVRD_IN register.
12 MPLL_REFSSC_CLK_EN_OVRD	Override enable for mpll_refssc_clk_en
11 MPLL_REFSSC_CLK_EN	Override value for mpll_refssc_clk_en
10 MPLL_RST	Reset the MPLL state machine Writing the register with this bit set will reset the MPLL power up/down FSM, regardless of the current state of the register bit.
9 FSEL_OVRD	Override enable for fsel[2:0]
8-6 FSEL	Override value for fsel [2:0]
5 MPLL_WORD_CLK_EN_OVRD	Override enable for mpll_word_clk_en
4 MPLL_WORD_CLK_EN	Override value for mpll_word_clk_en
3	Override enable for mpll_dword_clk_en

Table continues on the next page...

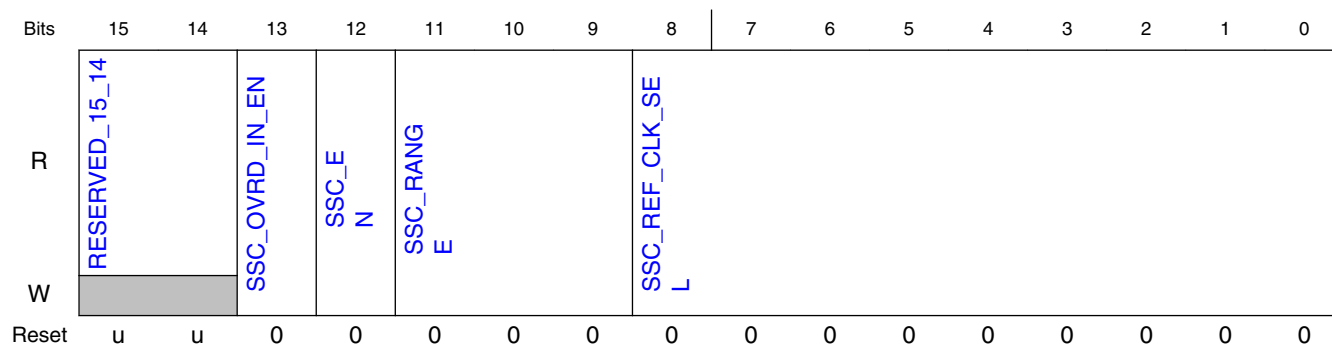
Field	Function
MPLL_DWORD_CLK_EN_OVRD	
2 MPLL_DWORD_CLK_EN	Override value for mpll_dword_clk_en
1 MPLL_QWORD_CLK_EN_OVRD	Override enable for mpll_qword_clk_en
0 MPLL_QWORD_CLK_EN	Override value for mpll_qword_clk_en

11.4.3.1.103 Override values for incoming SSC controls from ASIC (SUPX_DIG_SSC_OVRD_IN)

11.4.3.1.103.1 Offset

Register	Offset
SUPX_DIG_SSC_OVRD_IN	1_0026h

11.4.3.1.103.2 Diagram



11.4.3.1.103.3 Fields

Field	Function
15-14	Reserved for Future use

Table continues on the next page...

PCI Express PHY (PCIe_PHY)

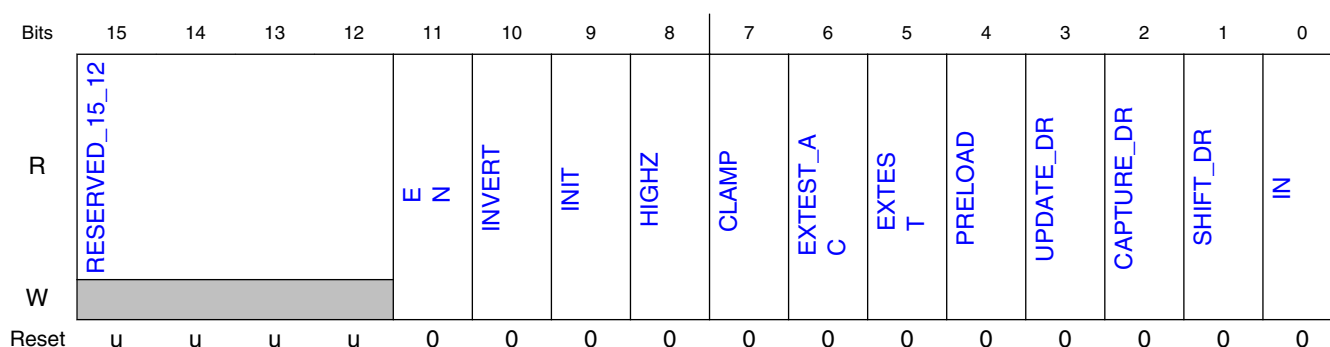
Field	Function
RESERVED_15_14	
13 SSC_OVRD_IN_EN	Override enable for Spread Spectrum generator.
12 SSC_EN	Override value for SSC enable.
11-9 SSC_RANGE	Override value for SSC modulation range.
8-0 SSC_REF_CLK_SEL	Override value for reference clock scaling

11.4.3.1.104 Override values for incoming BS controls from ASIC (SUPX_DIG_BS_OVRD_IN)

11.4.3.1.104.1 Offset

Register	Offset
SUPX_DIG_BS_OVRD_IN	1_0028h

11.4.3.1.104.2 Diagram



11.4.3.1.104.3 Fields

Field	Function
15-12	Reserved for Future use

Table continues on the next page...

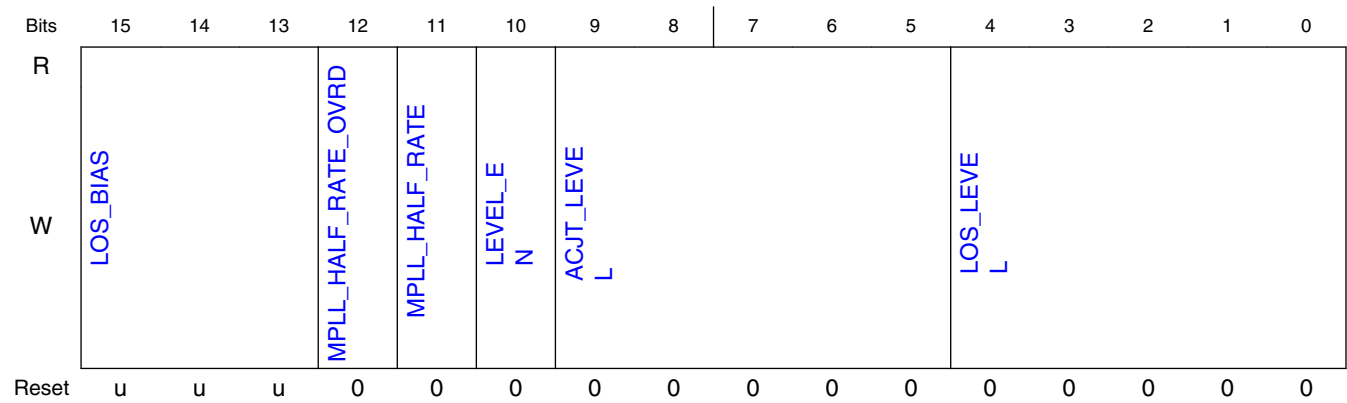
Field	Function
RESERVED_15_12	
11 EN	Enable override values for all inputs controlled by this register
10 INVERT	Override value for bs_invert
9 INIT	Override value for bs_init
8 HIGHZ	Override value for bs_highz
7 CLAMP	Override value for bs_clamp
6 EXTEST_AC	Override value for bs_extest_ac
5 EXTEST	Override value for bs_extest
4 PRELOAD	Override value for bs_preload
3 UPDATE_DR	Override value for bs_update_dr
2 CAPTURE_DR	Override value for bs_capture_dr
1 SHIFT_DR	Override value for bs_shift_dr
0 IN	Override value for bs_in

11.4.3.1.105 Override values for acjt_level and los_level (SUPX_DIG_LEVEL_OVRD_IN)

11.4.3.1.105.1 Offset

Register	Offset
SUPX_DIG_LEVEL_OVRD_IN	1_002Ah

11.4.3.1.105.2 Diagram



11.4.3.1.105.3 Fields

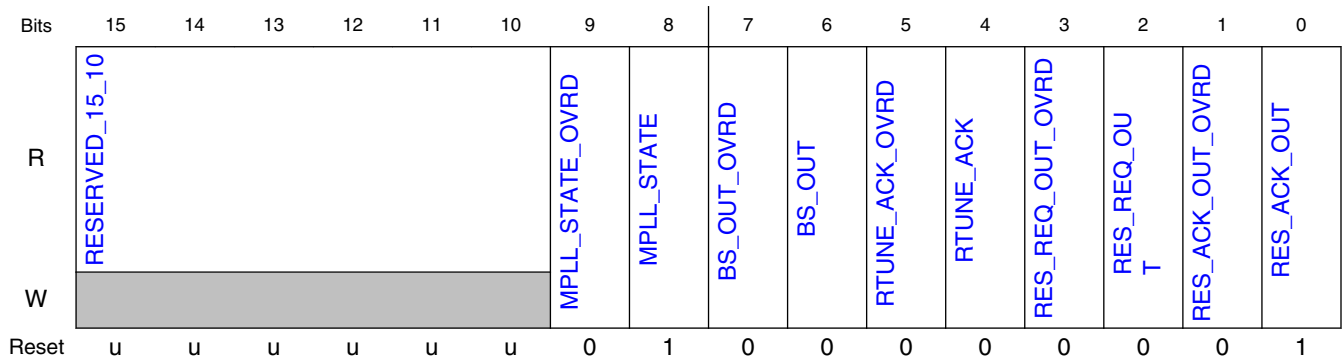
Field	Function
15-13 LOS_BIAS	Override value for los_bias
12 MPLL_HALF_RATE_OVRD	Enable override value for mpll_half_rate
11 MPLL_HALF_RATE	Override value for mpll_half_rate
10 LEVEL_EN	Enable override values for tx_vboost_lvl/los_bias/los_level/acjt_level inputs controlled by this and MPLL_OVRD_IN_HI registers
9-5 ACJT_LEVEL	Override value for acjt_level
4-0 LOS_LEVEL	Override value for los_level

11.4.3.1.106 Override values for support block (SUPX_DIG_SUP_OVRD_OUT)

11.4.3.1.106.1 Offset

Register	Offset
SUPX_DIG_SUP_OVRD_OUT	1_002Ch

11.4.3.1.106.2 Diagram



11.4.3.1.106.3 Fields

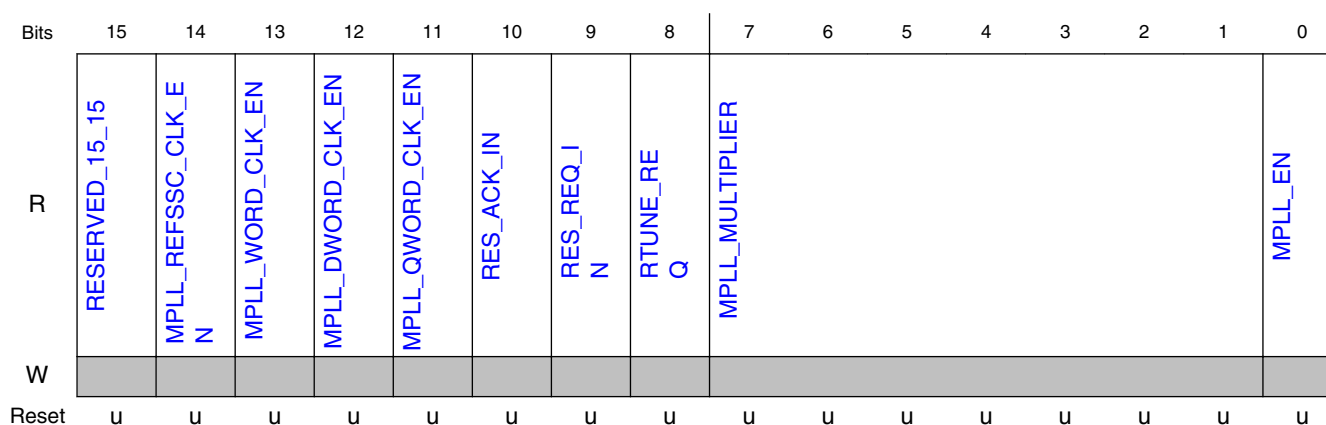
Field	Function
15-10 RESERVED_15_10	Reserved for Future use
9 MPLL_STATE_OVRD	Override enable for mpll_state output
8 MPLL_STATE	Override value for mpll_state output
7 BS_OUT_OVRD	Override enable for bs_out output
6 BS_OUT	Override value for bs_out output
5 RTUNE_ACK_OVRD	Override enable for rtune_ack output
4 RTUNE_ACK	Override value for rtune_ack output
3 RES_REQ_OUT_OVRD	Override enable for res_req_out output
2 RES_REQ_OUT	Override value for res_req_out output
1 RES_ACK_OUT_OVRD	Override enable for res_ack_out output
0 RES_ACK_OUT	Override value for res_ack_out output

11.4.3.1.107 Current values for incoming MPLL controls from ASIC (SUPX_DIG_MPLL_ASIC_IN)

11.4.3.1.107.1 Offset

Register	Offset
SUPX_DIG_MPLL_ASIC_IN	1_002Eh

11.4.3.1.107.2 Diagram



11.4.3.1.107.3 Fields

Field	Function
15 RESERVED_15_15	Reserved for Future use
14 MPLL_REFSSC_CLK_EN	Value from ASIC for mpll_refssc_clk_en (RU - read-only volatile)
13 MPLL_WORD_CLK_EN	Value from ASIC for mpll_word_clk_en (RU - read-only volatile)
12 MPLL_DWORD_CLK_EN	Value from ASIC for mpll_dword_clk_en (RU - read-only volatile)
11	Value from ASIC for mpll_qword_clk_en (RU - read-only volatile)

Table continues on the next page...

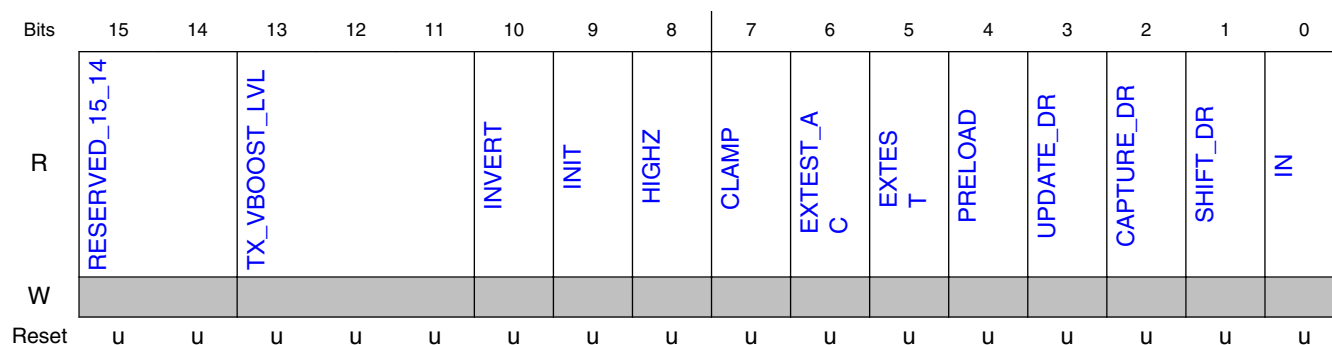
Field	Function
MPLL_QWORD_CLK_EN	
10 RES_ACK_IN	Value from ASIC for res_ack_in (RU - read-only volatile)
9 RES_REQ_IN	Value from ASIC for res_req_in (RU - read-only volatile)
8 RTUNE_REQ	Value from ASIC for rtune_req (RU - read-only volatile)
7-1 MPLL_MULTIPLIER	Value from ASIC for mpll_multiplier (RU - read-only volatile)
0 MPLL_EN	Value from ASIC for mpll_en (RU - read-only volatile)

11.4.3.1.108 Current values for incoming boundary scan controls from ASIC (SUPX_DIG_BS_ASIC_IN)

11.4.3.1.108.1 Offset

Register	Offset
SUPX_DIG_BS_ASIC_IN	1_0030h

11.4.3.1.108.2 Diagram



11.4.3.1.108.3 Fields

Field	Function
15-14	Reserved for Future use

Table continues on the next page...

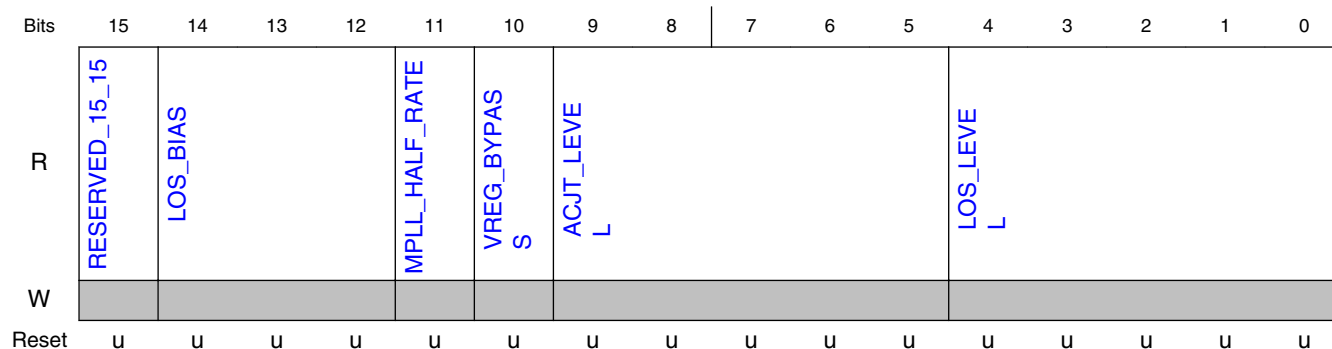
Field	Function
RESERVED_15_14	
13-11 TX_VBOOST_LVL	Value from ASIC for tx_vboost_lvl -- Not as bs_* signal, but we're out of bits. Value from ASIC for tx_vboost_lvl -- Not as bs_* signal, but we're out of bits. (RU - read-only volatile)
10 INVERT	Value from ASIC for bs_invert (RU - read-only volatile)
9 INIT	Value from ASIC for bs_init (RU - read-only volatile)
8 HIGHZ	Value from ASIC for bs_highz (RU - read-only volatile)
7 CLAMP	Value from ASIC for bs_clamp (RU - read-only volatile)
6 EXTEST_AC	Value from ASIC for bs_extest_ac (RU - read-only volatile)
5 EXTEST	Value from ASIC for bs_extest (RU - read-only volatile)
4 PRELOAD	Value from ASIC for bs_preload (RU - read-only volatile)
3 UPDATE_DR	Value from ASIC for bs_update_dr (RU - read-only volatile)
2 CAPTURE_DR	Value from ASIC for bs_capture_dr (RU - read-only volatile)
1 SHIFT_DR	Value from ASIC for bs_shift_dr (RU - read-only volatile)
0 IN	Value from ASIC for bs_in (RU - read-only volatile)

11.4.3.1.109 Current values for incoming los/acjt level, MPLL half rate mode and vreg bypass from ASIC (SUPX_DIG_LEVEL_ASIC_IN)

11.4.3.1.109.1 Offset

Register	Offset
SUPX_DIG_LEVEL_ASIC_IN	1_0032h

11.4.3.1.109.2 Diagram



11.4.3.1.109.3 Fields

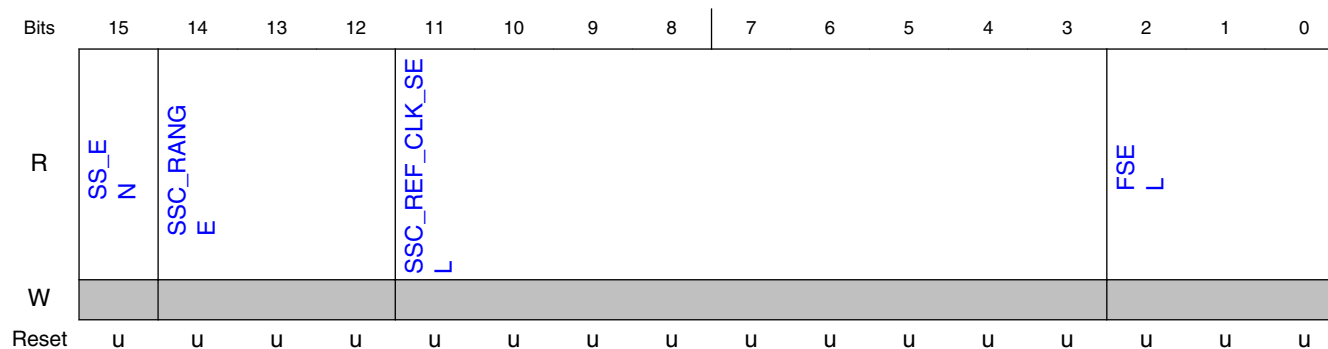
Field	Function
15 RESERVED_15_15	Reserved for Future use
14-12 LOS_BIAS	Value from ASIC for los_bias (RU - read-only volatile)
11 MPLL_HALF_RATE	Value from ASIC for mpll_half_rate (RU - read-only volatile)
10 VREG_BYPASS	Value from ASIC for vreg_bypass (RU - read-only volatile)
9-5 ACJT_LEVEL	Value from ASIC for acjt_level (RU - read-only volatile)
4-0 LOS_LEVEL	Value from ASIC for los_level (RU - read-only volatile)

11.4.3.1.110 Current values for incoming SSC controls from ASIC (SUPX_DIG_SSC_ASIC_IN)

11.4.3.1.110.1 Offset

Register	Offset
SUPX_DIG_SSC_ASIC_IN	1_0034h

11.4.3.1.110.2 Diagram



11.4.3.1.110.3 Fields

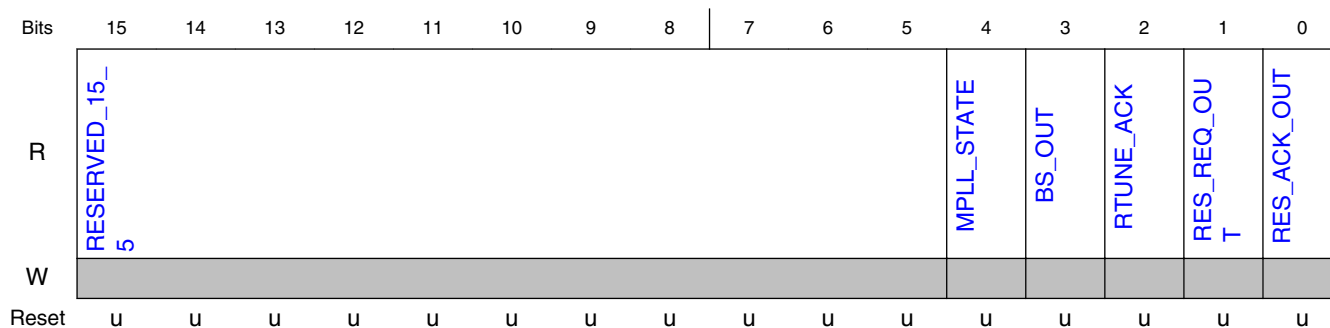
Field	Function
15 SS_EN	Value from ASIC for ssc_en (RU - read-only volatile)
14-12 SSC_RANGE	Value from ASIC for ssc_range (RU - read-only volatile)
11-3 SSC_REF_CLK_SEL	Value from ASIC for ssc_ref_clk_sel (RU - read-only volatile)
2-0 FSEL	Value from ASIC for fsel (RU - read-only volatile)

11.4.3.1.111 Current values for outgoing support block controls to ASIC (SUPX_DIG_SUP_ASIC_OUT)

11.4.3.1.111.1 Offset

Register	Offset
SUPX_DIG_SUP_ASIC_OUT	1_0036h

11.4.3.1.111.2 Diagram



11.4.3.1.111.3 Fields

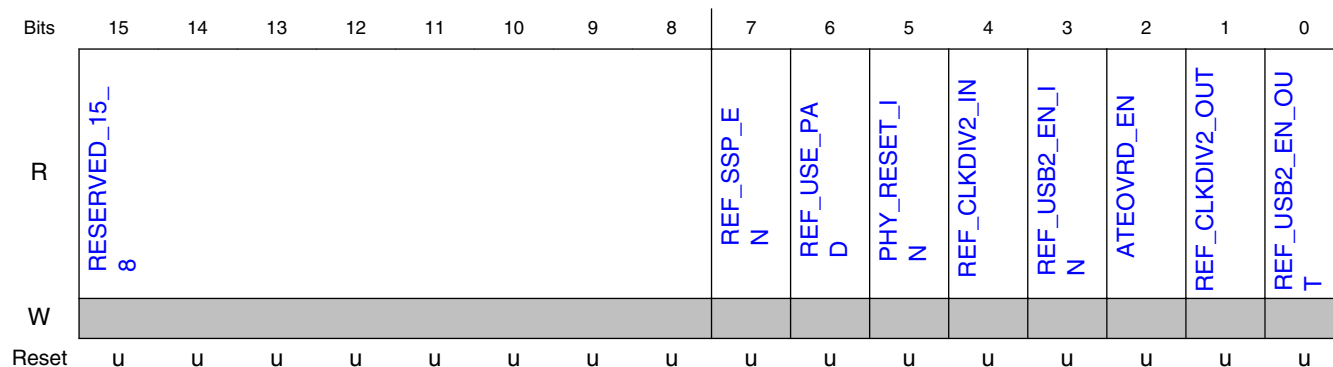
Field	Function
15-5 RESERVED_15_5	Reserved for Future use
4 MPLL_STATE	Value from PHY for mpll_state output (RU - read-only volatile)
3 BS_OUT	Value from PHY for bs_out output (RU - read-only volatile)
2 RTUNE_ACK	Value from PHY for rtune_ack output (RU - read-only volatile)
1 RES_REQ_OUT	Value from PHY for res_req_out output (RU - read-only volatile)
0 RES_ACK_OUT	Value from PHY for res_ack_out output (RU - read-only volatile)

11.4.3.1.112 SSP_SUPX_DIG_ATEOVRD_STATUS (SUPX_DIG_ATEOVRD_STATUS)

11.4.3.1.112.1 Offset

Register	Offset
SUPX_DIG_ATEOVRD_STATUS	1_0038h

11.4.3.1.112.2 Diagram



11.4.3.1.112.3 Fields

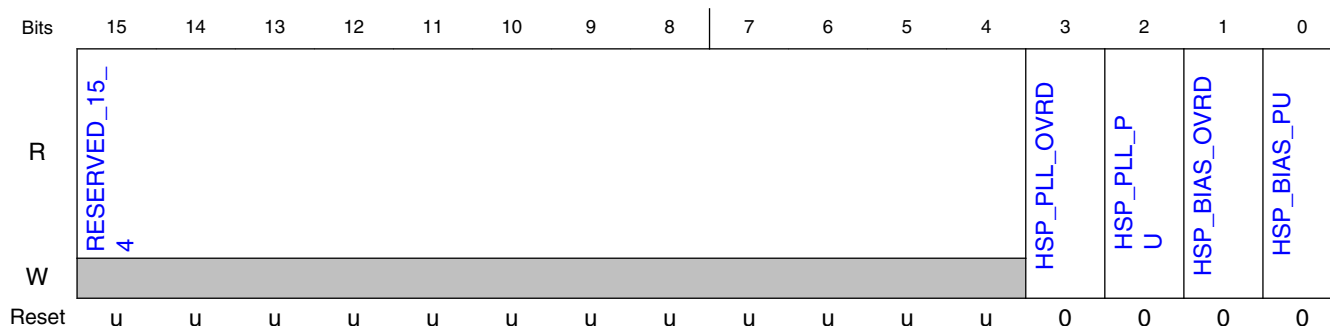
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 REF_SSP_EN	Value from ASIC for ref_ssp_en. Value from ASIC for ref_ssp_en. (RU - read-only volatile)
6 REF_USE_PAD	Value from ASIC for ref_use_pad. Value from ASIC for ref_use_pad. (RU - read-only volatile)
5 PHY_RESET_IN	Value from ASIC for phy_reset (RU - read-only volatile)
4 REF_CLKDIV2_IN	Value from ASIC for ref_clkdiv2 (RU - read-only volatile)
3 REF_USB2_EN_IN	Value from ASIC for ref_usb2_en (RU - read-only volatile)
2 ATEOVRD_EN	When set, values from ATEOVRD register are being sent to PHY (RU - read-only volatile)
1 REF_CLKDIV2_OUT	Value from ATEOVRD for ref_clkdiv2 (RU - read-only volatile)
0 REF_USB2_EN_OUT	Value from ATEOVRD for ref_usb2_en (RU - read-only volatile)

11.4.3.1.113 Override values for HSP (SUPX_DIG_HSP_OVRD_IN)

11.4.3.1.113.1 Offset

Register	Offset
SUPX_DIG_HSP_OVRD_IN	1_003Ah

11.4.3.1.113.2 Diagram



11.4.3.1.113.3 Fields

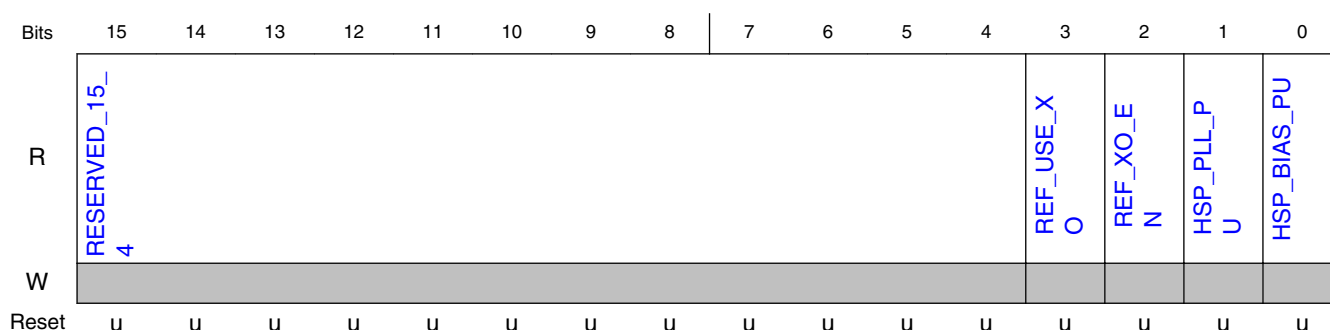
Field	Function
15-4 RESERVED_15_4	Reserved for Future use
3 HSP_PLL_OVRD	Override enable for HSP pll_pu signal
2 HSP_PLL_PU	Override value for HSP pll_pu signal
1 HSP_BIAS_OVRD	Override enable for HSP bias_pu signal
0 HSP_BIAS_PU	Override value for HSP bias_pu signal

11.4.3.1.114 Current values for incoming HSP & REF XO controls from ASIC (SUPX_DIG_HSP_ASIC_IN)

11.4.3.1.114.1 Offset

Register	Offset
SUPX_DIG_HSP_ASIC_IN	1_003Ch

11.4.3.1.114.2 Diagram



11.4.3.1.114.3 Fields

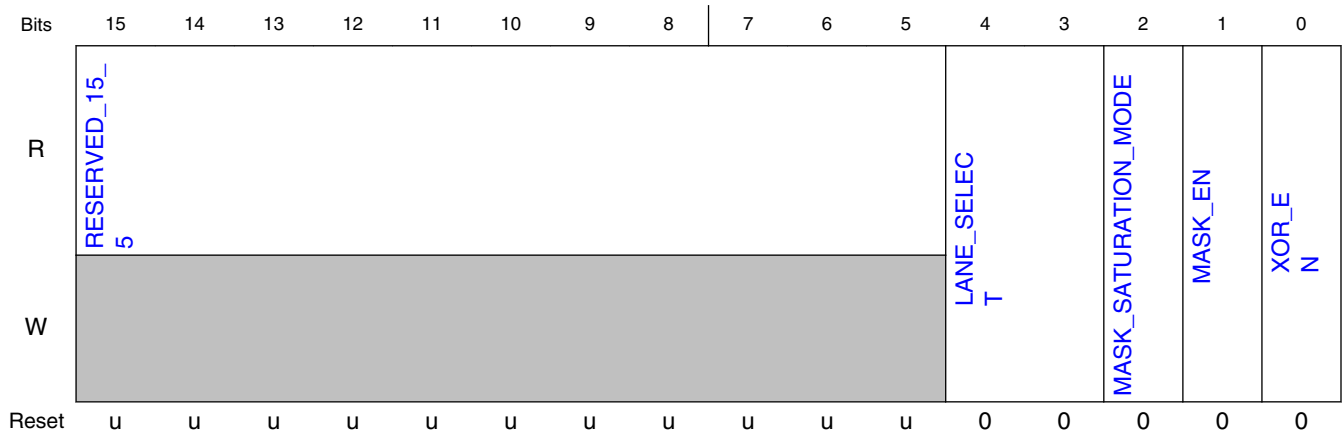
Field	Function
15-4 RESERVED_15_4	Reserved for Future use
3 REF_USE_XO	Value from ASIC for ref_use_xo (RU - read-only volatile)
2 REF_XO_EN	Value from ASIC for ref_xo_en (RU - read-only volatile)
1 HSP_PLL_PU	Value from ASIC for hsp_pll_pu (RU - read-only volatile)
0 HSP_BIAS_PU	Value from ASIC for hsp_bias_pu (RU - read-only volatile)

11.4.3.1.115 Scope Counter configuration (SUPX_DIG_SCOPE_ENABLES)

11.4.3.1.115.1 Offset

Register	Offset
SUPX_DIG_SCOPE_EN ABLES	1_0040h

11.4.3.1.115.2 Diagram



11.4.3.1.115.3 Fields

Field	Function
15-5 RESERVED_15_5	Reserved for Future use
4-3 LANE_SELECT	Scope Counter Input select
2 MASK_SATURATION_MODE	Method of mask saturation. Method of mask saturation. When 1 Saturate when the first mask_counter reaches sample_limit. When 0 Saturate when all mask_counters have reached sample_limit.
1 MASK_EN	Enable scope_mask input for tracking count values. Enable scope_mask input for tracking count values. Clears registers when deasserted.
0 XOR_EN	Use scope_xor input for count values.

11.4.3.1.116 Number of samples to count (SUPX_DIG_SCOPE_SAMPLES)

11.4.3.1.116.1 Offset

Register	Offset
SUPX_DIG_SCOPE_SAMPLES	1_0042h

11.4.3.1.116.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SAMPLES															
W																
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

11.4.3.1.116.3 Fields

Field	Function
15-0 SAMPLES	Number of Sample to count.

11.4.3.1.117 Results of scope counting (SUPX_DIG_SCOPE_COUNT)**11.4.3.1.117.1 Offset**

Register	Offset
SUPX_DIG_SCOPE_COUNT	1_0044h

11.4.3.1.117.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	COUNT															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.4.3.1.117.3 Fields

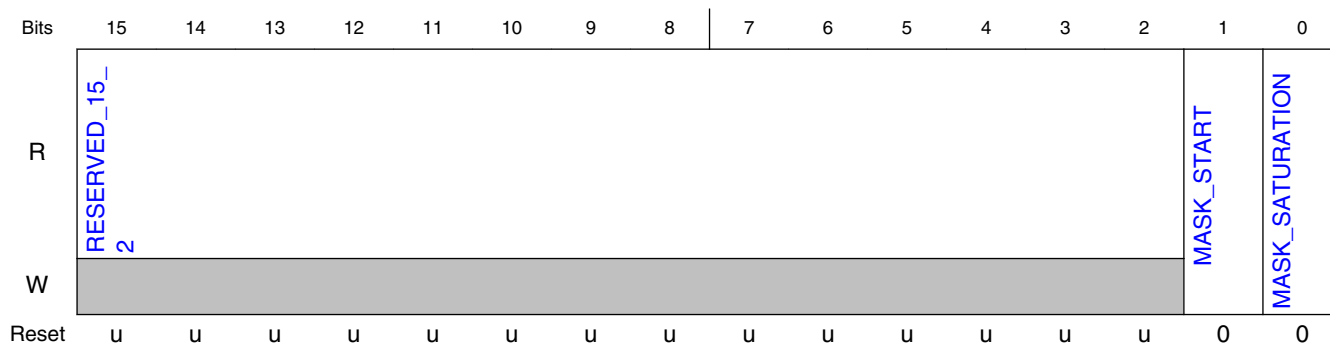
Field	Function
15-0 COUNT	A write to this register will start the counting process. The value of FFFF indicates counting still in progress. A write to this register will start the counting process The value of FFFF indicates counting still in progress. Asserting MASK_EN will also start the counting if in MASK mode. (RU - read-only volatile)

11.4.3.1.118 Masked scope counting Start and Status (SUPX_DIG_SCOPE_CTL)

11.4.3.1.118.1 Offset

Register	Offset
SUPX_DIG_SCOPE_CTL	1_0046h

11.4.3.1.118.2 Diagram



11.4.3.1.118.3 Fields

Field	Function
15-2 RESERVED_15_2	Reserved for Future use
1 MASK_START	>A write to this register will start the counting process. The value of FFFF indicates counting still in progress Write will trigger mask counting. A write to this register will start the counting process

Table continues on the next page...

Field	Function
	The value of FFFF indicates counting still in progress Write will trigger mask counting. (USER1 - read-write volatile)
0	When assert mask registers have saturated.
MASK_SATURATION	When assert mask registers have saturated. (USER1 - read-write volatile)

11.4.3.1.119 Current count value for mask_000 (SUPX_DIG_SCOPE_MASK_000)

11.4.3.1.119.1 Offset

Register	Offset
SUPX_DIG_SCOPE_MASK_000	1_0048h

11.4.3.1.119.2 Diagram



11.4.3.1.119.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_000	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.120 Current count value for mask_001 (SUPX_DIG_SCOPE_MASK_001)

11.4.3.1.120.1 Offset

Register	Offset
SUPX_DIG_SCOPE_MASK_001	1_004Ah

11.4.3.1.120.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MASK_VAL_001															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.4.3.1.120.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_001	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.121 Current count value for mask_010 (SUPX_DIG_SCOPE_MASK_010)

11.4.3.1.121.1 Offset

Register	Offset
SUPX_DIG_SCOPE_MASK_010	1_004Ch

11.4.3.1.121.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MASK_VAL_010															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.4.3.1.121.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_01 0	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.122 Current count value for mask_011 (SUPX_DIG_SCOPE_MASK_011)**11.4.3.1.122.1 Offset**

Register	Offset
SUPX_DIG_SCOPE_MASK_011	1_004Eh

11.4.3.1.122.2 Diagram**11.4.3.1.122.3 Fields**

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_01 1	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.123 Current count value for mask_100 (SUPX_DIG_SCOPE_MASK_100)

11.4.3.1.123.1 Offset

Register	Offset
SUPX_DIG_SCOPE_MASK_100	1_0050h

11.4.3.1.123.2 Diagram



11.4.3.1.123.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_100	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.124 Current count value for mask_101 (SUPX_DIG_SCOPE_MASK_101)

11.4.3.1.124.1 Offset

Register	Offset
SUPX_DIG_SCOPE_MASK_101	1_0052h

11.4.3.1.124.2 Diagram



11.4.3.1.124.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_10 1	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.125 Current count value for mask_110 (SUPX_DIG_SCOPE_MASK_110)**11.4.3.1.125.1 Offset**

Register	Offset
SUPX_DIG_SCOPE_MASK_110	1_0054h

11.4.3.1.125.2 Diagram**11.4.3.1.125.3 Fields**

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_11 0	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.126 Current count value for mask_111 (SUPX_DIG_SCOPE_MASK_111)

11.4.3.1.126.1 Offset

Register	Offset
SUPX_DIG_SCOPE_MASK_111	1_0056h

11.4.3.1.126.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MASK_VAL_111															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11.4.3.1.126.3 Fields

Field	Function
15-0	Starting count value of mask register.
MASK_VAL_111	Starting count value of mask register. (volatile and 2 reads needed to read value)

11.4.3.1.127 MPLL loop control bits (SUPX_ANA_MPLL_LOOP_CTL)

11.4.3.1.127.1 Offset

Register	Offset
SUPX_ANA_MPLL_LOOP_CTL	1_0060h

11.4.3.1.127.2 Diagram

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	RESERVED_15_8								PROP_CNTRL				INT_CNTRL		GD	VMB
W																
Reset	u	u	u	u	u	u	u	u	1	1	0	0	0	0	0	0

11.4.3.1.127.3 Fields

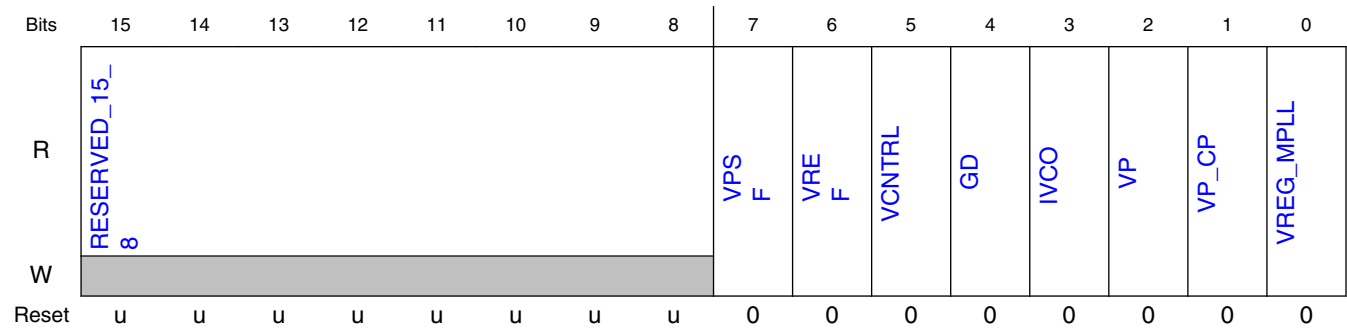
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7-4 PROP_CNTRL	Charge pump proportional current setting
3-2 INT_CNTRL	Charge pump integrating current setting
1 GD	Measure MPLL local GD
0 VMB	Measure MPLL Master Bias Voltage

11.4.3.1.128 ATB Select bits (SUPX_ANA_MPLL_ATB_MEAS1)

11.4.3.1.128.1 Offset

Register	Offset
SUPX_ANA_MPLL_ATB_MEAS1	1_0062h

11.4.3.1.128.2 Diagram



11.4.3.1.128.3 Fields

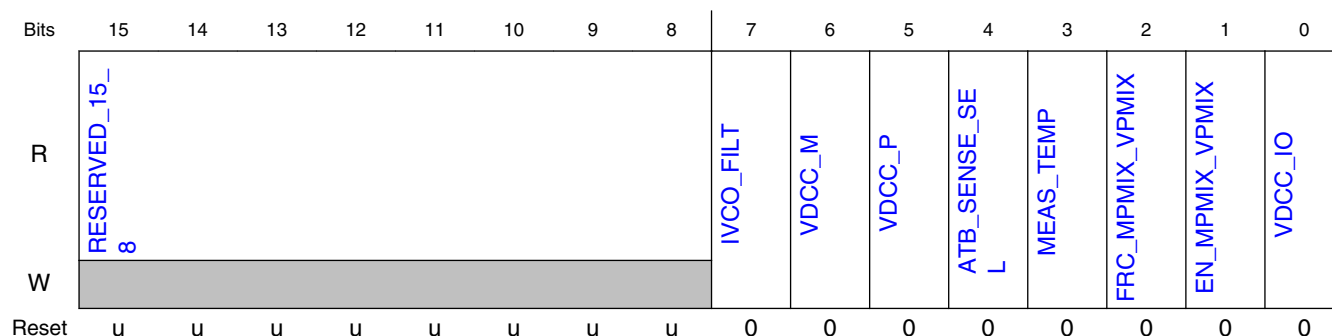
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 VPSF	puts vpsf on atb_s_p
6 VREF	puts vref from charge pump on atb_s_p
5 VCNTRL	puts vcntrl on atb_s_m
4 GD	puts gd on atb_s_m
3 IVCO	puts VCO current/93 on atb_f_m
2 VP	puts vpa on atb_s_m
1 VP_CP	puts vp_cp on atb_s_p
0 VREG_MPLL	puts vreg_mpll on atb_s_p

11.4.3.1.129 ATB Select bits (SUPX_ANA_MPLL_ATB_MEAS2)

11.4.3.1.129.1 Offset

Register	Offset
SUPX_ANA_MPLL_ATB_MEAS2	1_0064h

11.4.3.1.129.2 Diagram



11.4.3.1.129.3 Fields

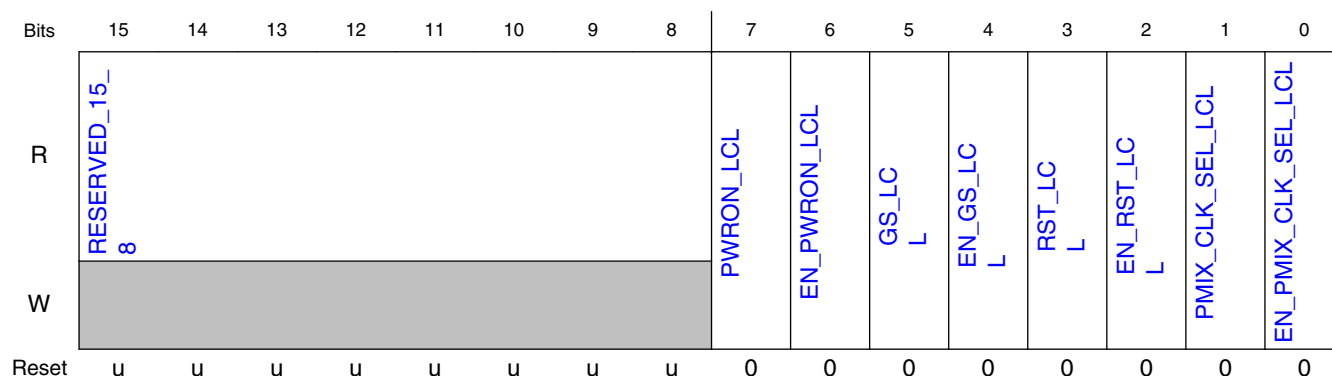
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 IVCO_FILT	puts filtered version of ivco on atb_s_p
6 VDCC_M	puts dcc output vdcc_p on atb_s_m
5 VDCC_P	puts dcc output vdcc_m on atb_s_p
4 ATB_SENSE_SE_L	connects internal atb sense bus to external bus
3 MEAS_TEMP	tells POR block to measure the temperature
2 FRC_MPMIX_VPMIX	Force mpll_pmix_vreg to use atb_s_m instead of vbg as its input
1 EN_MPMIX_VPMIX	Put vreg_pmix on atb_s_p
0 VDCC_IO	when asserted, bit 5, 6 of this register measures i phase clock DCC voltage, otherwise, q phase DCC voltage is measured

11.4.3.1.130 Local overrides (SUPX_ANA_MPLL_OVRD)

11.4.3.1.130.1 Offset

Register	Offset
SUPX_ANA_MPLL_OVRD	1_0066h

11.4.3.1.130.2 Diagram



11.4.3.1.130.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 PWRON_LCL	local power_on value
6 EN_PWRON_LCL	Enable local control of power_on
5 GS_LCL	local gear_shift value
4 EN_GS_LCL	Enable local control of gear_shift
3 RST_LCL	local reset value
2 EN_RST_LCL	enable local control of reset
1 PMIX_CLK_SEL_LCL	local pmix_clk_sel value

Table continues on the next page...

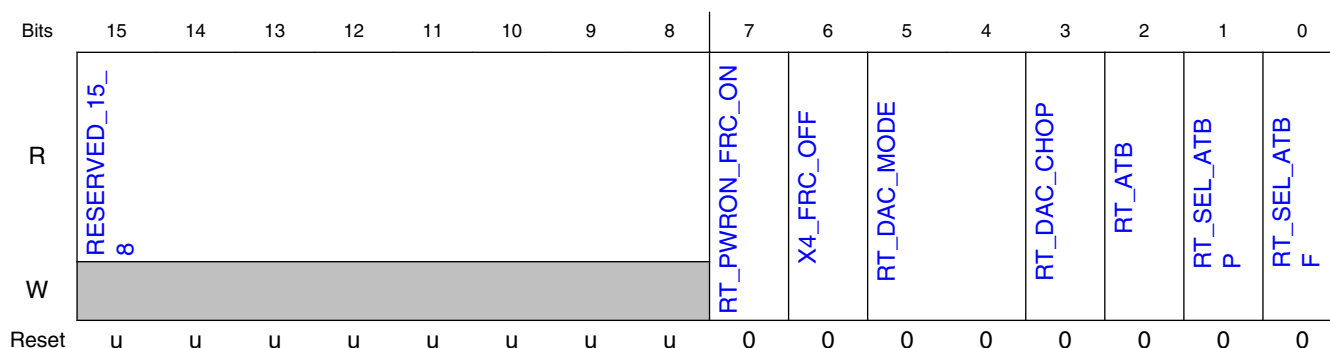
Field	Function
0 EN_PMIX_CLK_SEL_LCL	enable local control of pmix_clk_sel

11.4.3.1.131 RTUNE control bits (SUPX_ANA_RTUNE_RTUNE_CTRL)

11.4.3.1.131.1 Offset

Register	Offset
SUPX_ANA_RTUNE_RTUNE_CTRL	1_0068h

11.4.3.1.131.2 Diagram



11.4.3.1.131.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 RT_PWRON_FRC_ON	When set, forces RTUNE block to be on
6 X4_FRC_OFF	When set, do not multiply test current by 4
5-4 RT_DAC_MODE	Margin DAC mode control bits(00=powerdown, 01=DAC drives atb_s_p/m directly, 10=DAC drives atb_s_p/m to the RX in margining mode, 11=illegal state)

Table continues on the next page...

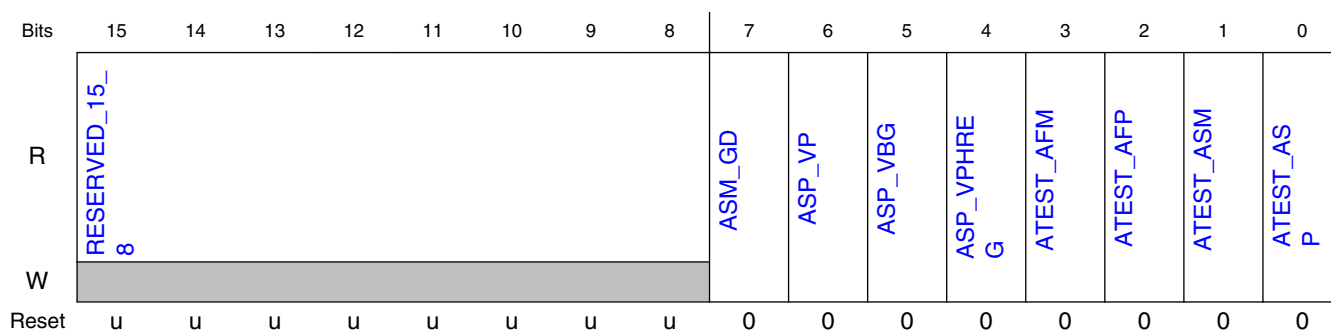
Field	Function
3 RT_DAC_CHOP	Margin DAC chop control bit. Margin DAC chop control bit. ref to ssp analog engineering doc 8.5
2 RT_ATB	RTUNE ATB mode control bit (1=RTUNE performs ADC on ATB input, 0=not accessing ATB. RTUNE ATB mode control bit (1=RTUNE performs ADC on ATB input, 0=not accessing ATB. refer to ssp analog engineering specification 8.1.4)
1 RT_SEL_ATBP	RTUNE ATB sense input select bit (1=atb_s_p, 0=atb_s_m)
0 RT_SEL_ATBF	RTUNE ATB input select bit (1=atb_fm, 0=atb_s_p/m)

11.4.3.1.132 ATB_SWITCHYARD control bits (SUPX_ANA_ATB_SWITCHYARD_CTRL)

11.4.3.1.132.1 Offset

Register	Offset
SUPX_ANA_ATB_SWITCHYARD_CTRL	1_006Ah

11.4.3.1.132.2 Diagram



11.4.3.1.132.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7	Connect gd to atb_s_m

Table continues on the next page...

PCI Express PHY (PCIe_PHY)

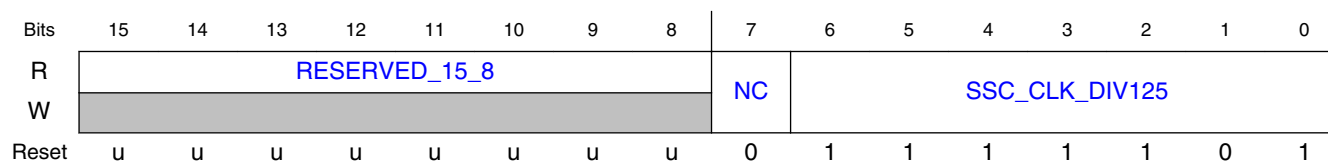
Field	Function
ASM_GD	
6 ASP_VP	Connect vp to atb_s_p
5 ASP_VBG	Connect vbg to atb_s_p, which is divided down value from a resistor stack
4 ASP_VPHREG	Connect vphreg to atb_s_p
3 ATEST_AFM	drive ATEST from atb_f_m
2 ATEST_AFP	drive ATEST from atb_f_p
1 ATEST_ASM	Connet ATEST to atb_s_m
0 ATEST_ASP	Connect ATEST to atb_s_p

11.4.3.1.133 controls sscf reference clock rate (SUPX_ANA_SSC_CLK_CNTRL)

11.4.3.1.133.1 Offset

Register	Offset
SUPX_ANA_SSC_CLK_CNTRL	1_006Ch

11.4.3.1.133.2 Diagram



11.4.3.1.133.3 Fields

Field	Function
15-8	Reserved for Future use

Table continues on the next page...

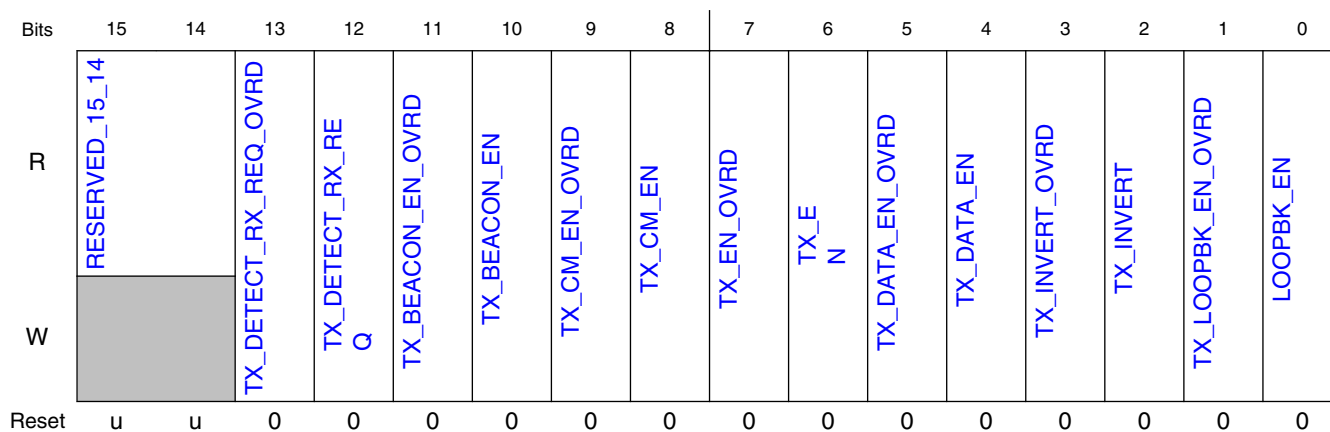
Field	Function
RESERVED_15_8	
7 NC	Reserved
6-0 SSC_CLK_DIV1 25	set ssc reference clock to be 20MHz

11.4.3.1.134 Override values for incoming TX controls from ASIC (LANEX_DIG_TX_OVRD_IN_LO)

11.4.3.1.134.1 Offset

Register	Offset
LANEX_DIG_TX_OVRD_IN_LO	1_2000h

11.4.3.1.134.2 Diagram



11.4.3.1.134.3 Fields

Field	Function
15-14 RESERVED_15_14	Reserved for Future use
13	Override enable for tx_detect_rx_req

Table continues on the next page...

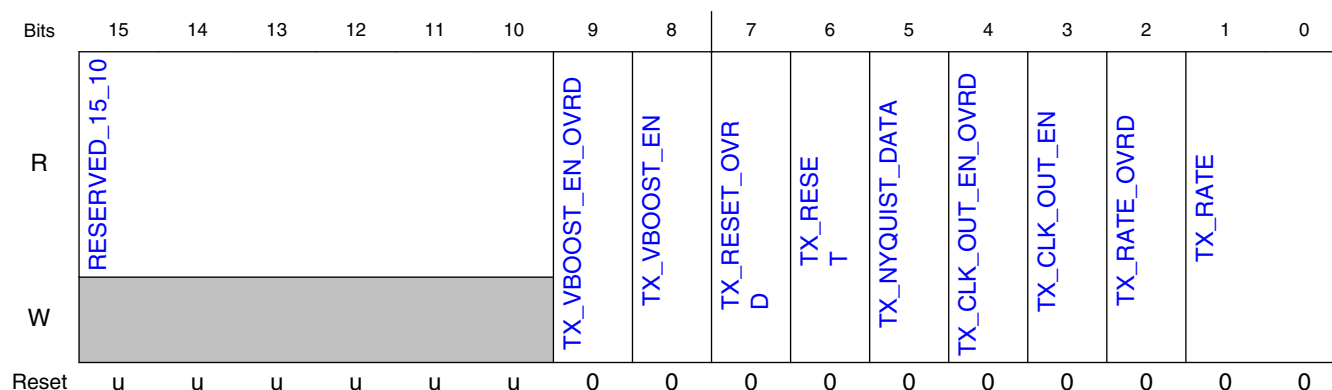
Field	Function
TX_DETECT_RX_REQ_OVRD	
12 TX_DETECT_RX_REQ	Override value for tx_detect_rx_req
11 TX_BEACON_EN_OVRD	Override enable for tx_beacon_en
10 TX_BEACON_EN	Override value for tx_beacon_en
9 TX_CM_EN_OVRD	Override enable for tx_cm_en
8 TX_CM_EN	Override value for tx_cm_en
7 TX_EN_OVRD	Override enable for tx_en
6 TX_EN	Override value for tx_en
5 TX_DATA_EN_OVRD	Override enable for tx_data_en
4 TX_DATA_EN	Override value for tx_data_en
3 TX_INVERT_OVRD	Override enable for tx_invert
2 TX_INVERT	Override value for tx_invert
1 TX_LOOPBK_EN_OVRD	Override enable for loopbk_en
0 LOOPBK_EN	Override value for loopbk_en

11.4.3.1.135 Override values for incoming TX controls from ASIC (LANEX_DIG_TX_OVRD_IN_HI)

11.4.3.1.135.1 Offset

Register	Offset
LANEX_DIG_TX_OVRD_IN_HI	1_2002h

11.4.3.1.135.2 Diagram



11.4.3.1.135.3 Fields

Field	Function
15-10 RESERVED_15_10	Reserved for Future use
9 TX_VBOOST_EN_OVRD	Override enable for tx_vboost_en
8 TX_VBOOST_EN	Override value for tx_vboost_en
7 TX_RESET_OVRD	Override enable for tx_reset
6 TX_RESET	Override value for tx_reset
5 TX_NYQUIST_DATA	Override incoming data to nyquist
4	Override enable for tx_clk_out_en.

Table continues on the next page...

PCI Express PHY (PCIe_PHY)

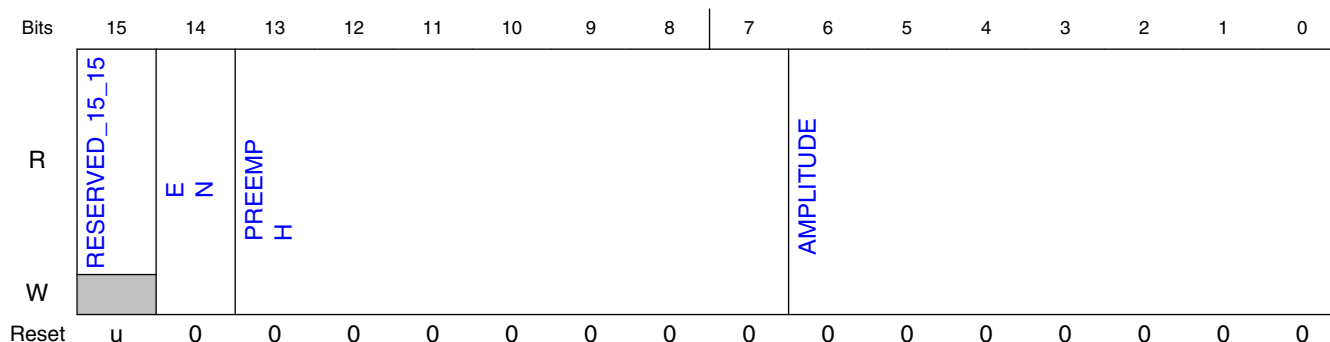
Field	Function
TX_CLK_OUT_EN_OVRD	
3 TX_CLK_OUT_EN	Override incoming tx_clk_out_en.
2 TX_RATE_OVRD	Override enable for tx_rate.
1-0 TX_RATE	Override incoming tx lane rate.

11.4.3.1.136 Override values for incoming TX drive controls from ASIC (LANEX_DIG_TX_OVRD_DRV_LO)

11.4.3.1.136.1 Offset

Register	Offset
LANEX_DIG_TX_OVRD_DRV_LO	1_2004h

11.4.3.1.136.2 Diagram



11.4.3.1.136.3 Fields

Field	Function
15 RESERVED_15_15	Reserved for Future use

Table continues on the next page...

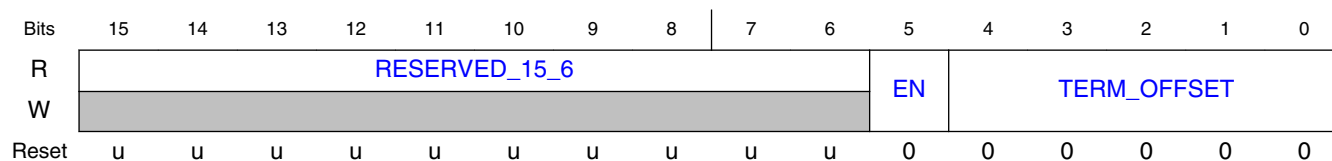
Field	Function
14 EN	Enable override values for all inputs controlled by this register
13-7 PREEMPH	Override value for transmit preemphasis.
6-0 AMPLITUDE	Override value for transmit amplitude.

11.4.3.1.137 Override values for incoming TX drive controls from ASIC (LANEX_DIG_TX_OVRD_DRV_HI)

11.4.3.1.137.1 Offset

Register	Offset
LANEX_DIG_TX_OVRD_DRV_HI	1_2006h

11.4.3.1.137.2 Diagram



11.4.3.1.137.3 Fields

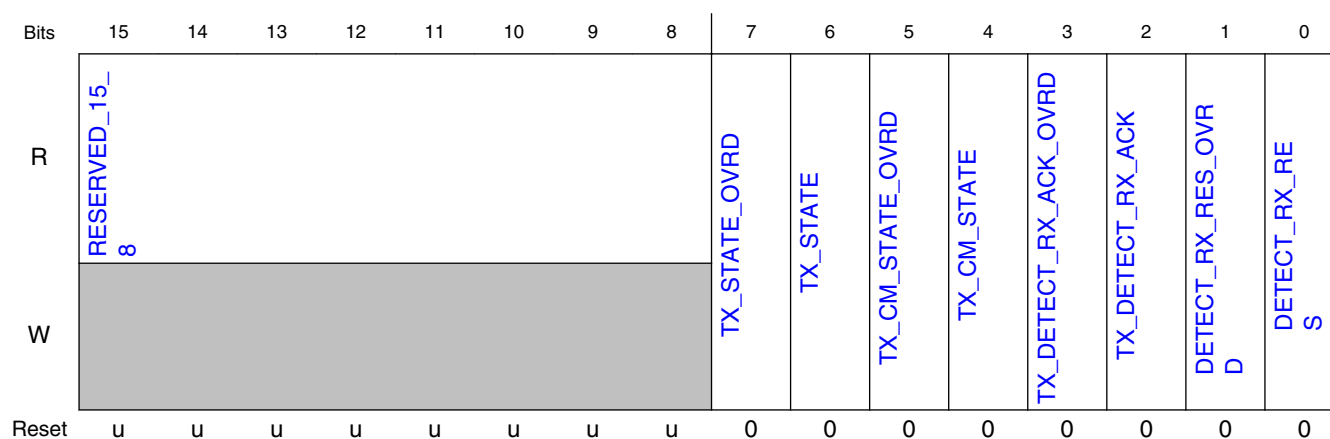
Field	Function
15-6 RESERVED_15_6	Reserved for Future use
5 EN	Enable override values for all inputs controlled by this register
4-0 TERM_OFFSET	Override value for tx_term_offset (tx_ana_vcmref_boost)

11.4.3.1.138 Override values for outgoing TX controls to ASIC (LANEX_DIG_TX_OVRD_OUT)

11.4.3.1.138.1 Offset

Register	Offset
LANEX_DIG_TX_OVRD_OUT	1_2008h

11.4.3.1.138.2 Diagram



11.4.3.1.138.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 TX_STATE_OVRD	Override enable for tx_state
6 TX_STATE	Override value for tx_state
5 TX_CM_STATE_OVRD	Override enable for tx_cm_state
4 TX_CM_STATE	Override value for tx_cm_state
3	Override enable for tx_detect_rx_ack

Table continues on the next page...

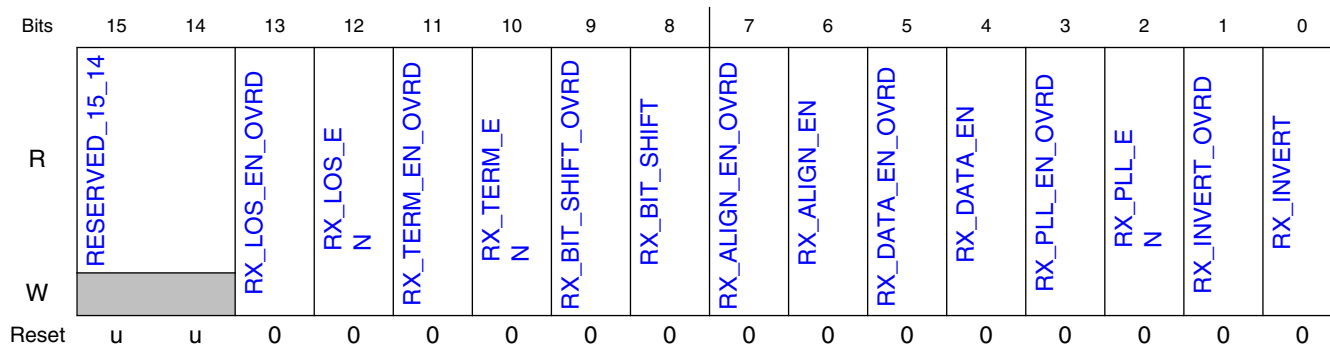
Field	Function
TX_DETECT_RX_ACK_OVRD	
2 TX_DETECT_RX_ACK	Override value for tx_detect_rx_ack
1 DETECT_RX_RES_OVRD	Override enable for tx_detect_rx_res
0 DETECT_RX_RES	Override value for tx_detect_rx_res

11.4.3.1.139 Override values for incoming RX controls from ASIC (LANEX_DIG_RX_OVRD_IN_LO)

11.4.3.1.139.1 Offset

Register	Offset
LANEX_DIG_RX_OVRD_IN_LO	1_200Ah

11.4.3.1.139.2 Diagram



11.4.3.1.139.3 Fields

Field	Function
15-14	Reserved for Future use

Table continues on the next page...

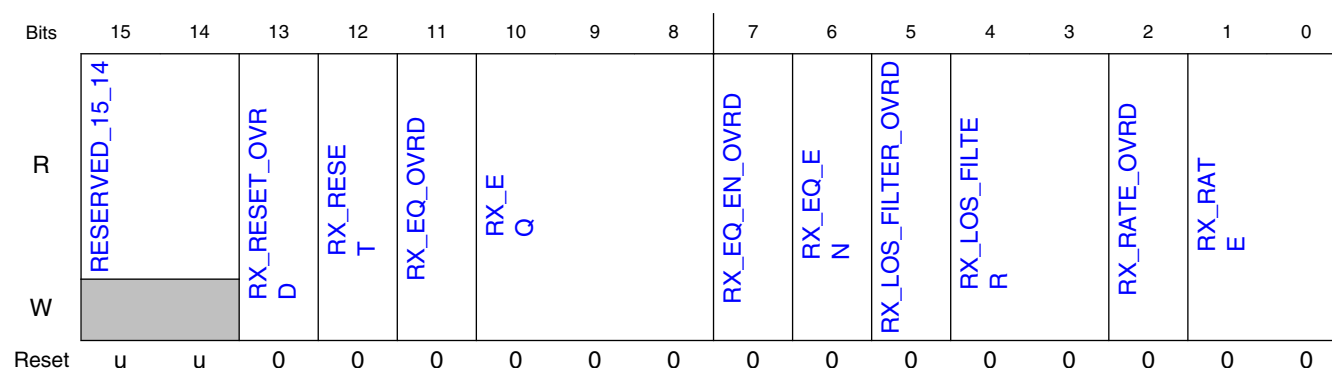
Field	Function
RESERVED_15_14	
13 RX_LOS_EN_OVRD	Override enable for rx_los_en
12 RX_LOS_EN	Override value for rx_los_en
11 RX_TERM_EN_OVRD	Override enable for rx_term_en
10 RX_TERM_EN	Override value for rx_term_en
9 RX_BIT_SHIFT_OVRD	Override enable for rx_bit_shift
8 RX_BIT_SHIFT	Override value for rx_bit_shift
7 RX_ALIGN_EN_OVRD	Override enable for rx_align_en
6 RX_ALIGN_EN	Override value for rx_align_en
5 RX_DATA_EN_OVRD	Override enable for rx_data_en
4 RX_DATA_EN	Override value for rx_data_en
3 RX_PLL_EN_OVRD	Override enable for rx_pll_en
2 RX_PLL_EN	Override value for rx_pll_en
1 RX_INVERT_OVRD	Override enable for rx_invert
0 RX_INVERT	Override value for rx_invert

11.4.3.1.140 Override values for incoming RX controls from ASIC (LANEX_DIG_RX_OVRD_IN_HI)

11.4.3.1.140.1 Offset

Register	Offset
LANEX_DIG_RX_OVRD_IN_HI	1_200Ch

11.4.3.1.140.2 Diagram



11.4.3.1.140.3 Fields

Field	Function
15-14 RESERVED_15_14	Reserved for Future use
13 RX_RESET_OVRD	Override enable for rx_reset
12 RX_RESET	Override value for rx_reset
11 RX_EQ_OVRD	Override enable for rx_eq
10-8 RX_EQ	Override value for rx_eq
7 RX_EQ_EN_OVRD	Override enable for rx_eq_en
6 RX_EQ_EN	Override value for rx_eq_en
5	Override enable for rx_los_filter

Table continues on the next page...

PCI Express PHY (PCIe_PHY)

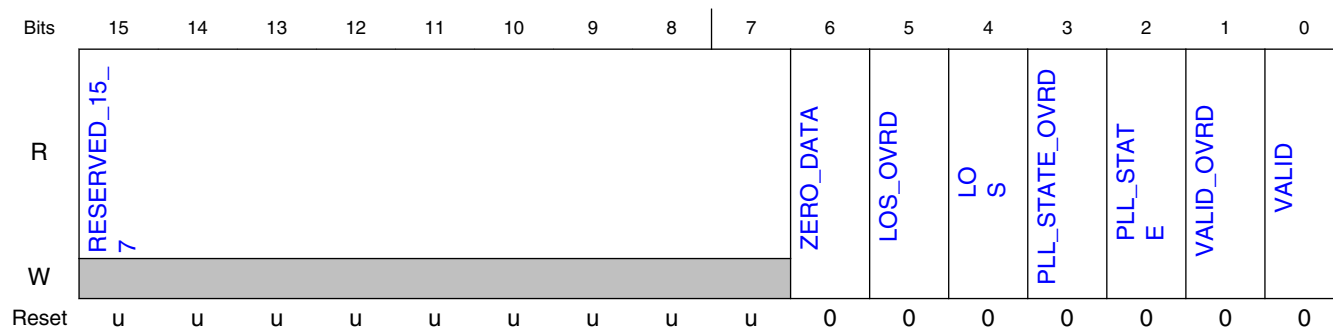
Field	Function
RX_LOS_FILTER_OVRD	
4-3 RX_LOS_FILTER	Override value for rx_los_filter
2 RX_RATE_OVRD	Override enable for rx_rate
1-0 RX_RATE	Override value for rx_rate

11.4.3.1.141 Override values for outgoing RX controls to ASIC (LANEX_DIG_RX_OVRD_OUT)

11.4.3.1.141.1 Offset

Register	Offset
LANEX_DIG_RX_OVRD_OUT	1_200Eh

11.4.3.1.141.2 Diagram



11.4.3.1.141.3 Fields

Field	Function
15-7 RESERVED_15_7	Reserved for Future use

Table continues on the next page...

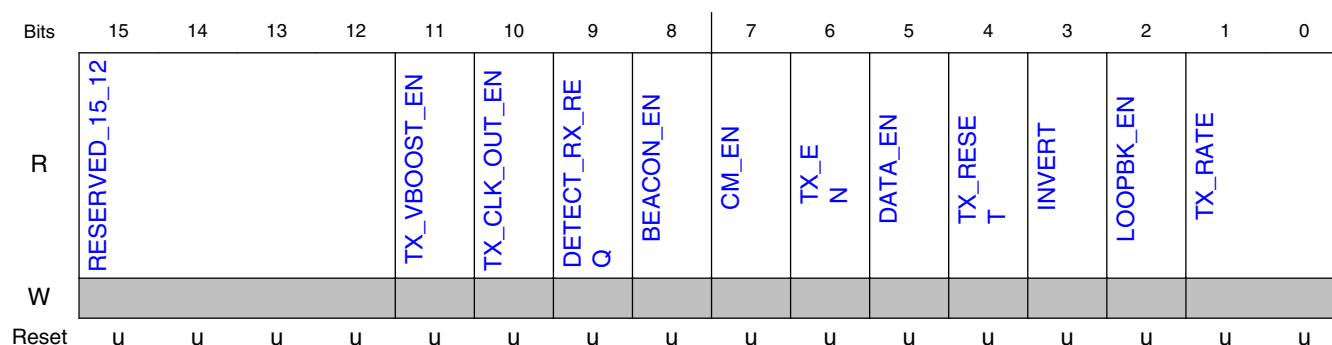
Field	Function
6 ZERO_DATA	Override data output to all zeros
5 LOS_OVRD	Override value for rx_los
4 LOS	Override value for rx_los
3 PLL_STATE_OVRD	Override enable for rx_pll_state
2 PLL_STATE	Override value for rx_pll_state
1 VALID_OVRD	Override enable for rx_valid
0 VALID	Override value for rx_valid

11.4.3.1.142 Current values for incoming TX controls from ASIC (LANEX_DIG_TX_ASIC_IN)

11.4.3.1.142.1 Offset

Register	Offset
LANEX_DIG_TX_ASIC_IN	1_2010h

11.4.3.1.142.2 Diagram



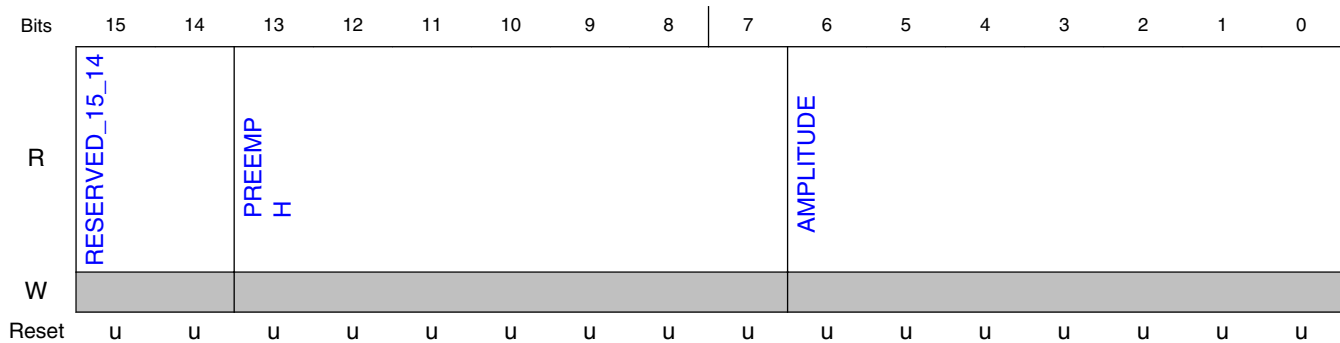
11.4.3.1.142.3 Fields

Field	Function
15-12 RESERVED_15_12	Reserved for Future use
11 TX_VBOOST_EN	Value from ASIC for tx_vboost_en (RU - read-only volatile)
10 TX_CLK_OUT_EN	Value from ASIC for tx_clk_out_en (RU - read-only volatile)
9 DETECT_RX_REQ	Value from ASIC for tx_detect_rx_req (RU - read-only volatile)
8 BEACON_EN	Value from ASIC for tx_beacon_en (RU - read-only volatile)
7 CM_EN	Value from ASIC for tx_cm_en (RU - read-only volatile)
6 TX_EN	Value from ASIC for tx_en (RU - read-only volatile)
5 DATA_EN	Value from ASIC for tx_data_en (RU - read-only volatile)
4 TX_RESET	Value from ASIC for tx_reset (RU - read-only volatile)
3 INVERT	Value from ASIC for tx_invert (RU - read-only volatile)
2 LOOPBK_EN	Value from ASIC for loopbk_en (RU - read-only volatile)
1-0 TX_RATE	Value from ASIC for tx_rate (RU - read-only volatile)

11.4.3.1.143 Current values for incoming TX level controls from ASIC (LANEX_DIG_TX_ASIC_DRV_LO)**11.4.3.1.143.1 Offset**

Register	Offset
LANEX_DIG_TX_ASIC_DRV_LO	1_2012h

11.4.3.1.143.2 Diagram



11.4.3.1.143.3 Fields

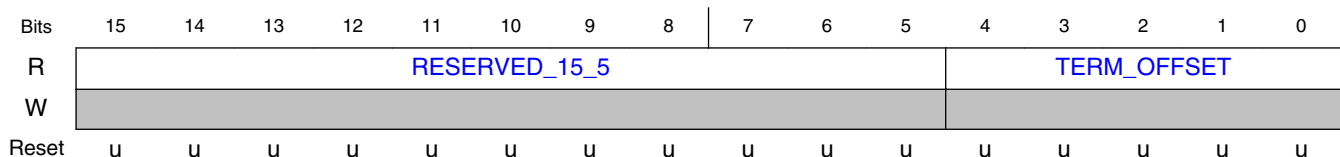
Field	Function
15-14 RESERVED_15_14	Reserved for Future use
13-7 PREMPH	Value from ASIC for tx_preemph (RU - read-only volatile)
6-0 AMPLITUDE	Value from ASIC for tx_amplitude (RU - read-only volatile)

11.4.3.1.144 Current values for incoming TX level controls from ASIC (LANEX_DIG_TX_ASIC_DRV_HI)

11.4.3.1.144.1 Offset

Register	Offset
LANEX_DIG_TX_ASIC_DRV_HI	1_2014h

11.4.3.1.144.2 Diagram



11.4.3.1.144.3 Fields

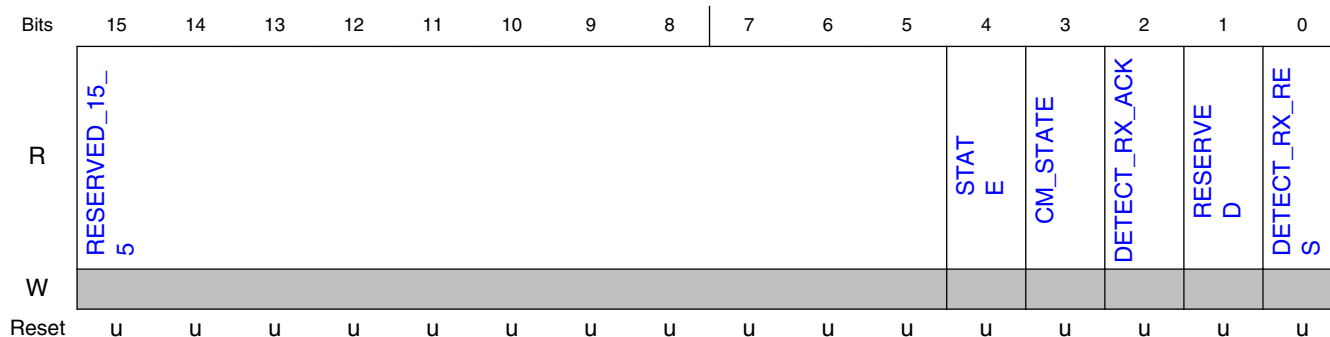
Field	Function
15-5 RESERVED_15_5	Reserved for Future use
4-0 TERM_OFFSET	Value from ASIC for tx_term_offset (RU - read-only volatile)

11.4.3.1.145 Current values for outgoing TX status controls from PHY (LANEX_DIG_TX_ASIC_OUT)

11.4.3.1.145.1 Offset

Register	Offset
LANEX_DIG_TX_ASIC_OUT	1_2016h

11.4.3.1.145.2 Diagram



11.4.3.1.145.3 Fields

Field	Function
15-5 RESERVED_15_5	Reserved for Future use
4 STATE	Value from PHY for tx_state (RU - read-only volatile)

Table continues on the next page...

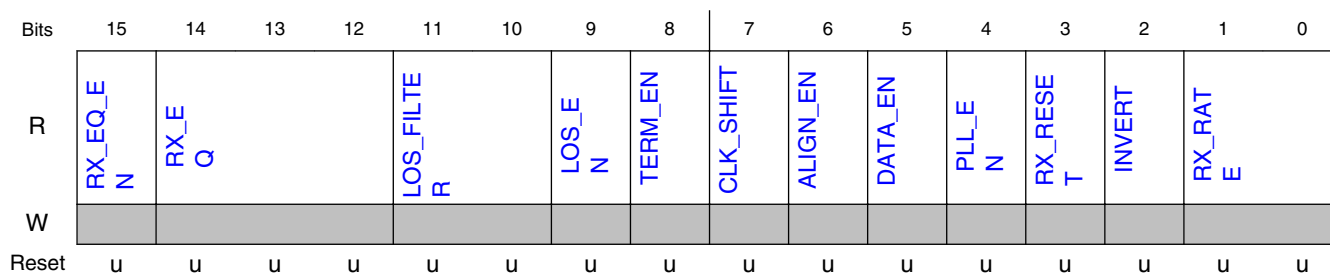
Field	Function
3 CM_STATE	Value from PHY for tx_cm_state (RU - read-only volatile)
2 DETECT_RX_ACK	Value from PHY for tx_detect_rx_ack (RU - read-only volatile)
1 RESERVED	Always returns 0 (RU - read-only volatile)
0 DETECT_RX_RES	Value from PHY for tx_detect_rx_res (RU - read-only volatile)

11.4.3.1.146 Current values for incoming RX controls from ASIC (LANEX_DIG_RX_ASIC_IN)

11.4.3.1.146.1 Offset

Register	Offset
LANEX_DIG_RX_ASIC_IN	1_2018h

11.4.3.1.146.2 Diagram



11.4.3.1.146.3 Fields

Field	Function
15 RX_EQ_EN	Value from ASIC for rx_eq_en (RU - read-only volatile)
14-12 RX_EQ	Value from ASIC for rx_eq (RU - read-only volatile)

Table continues on the next page...

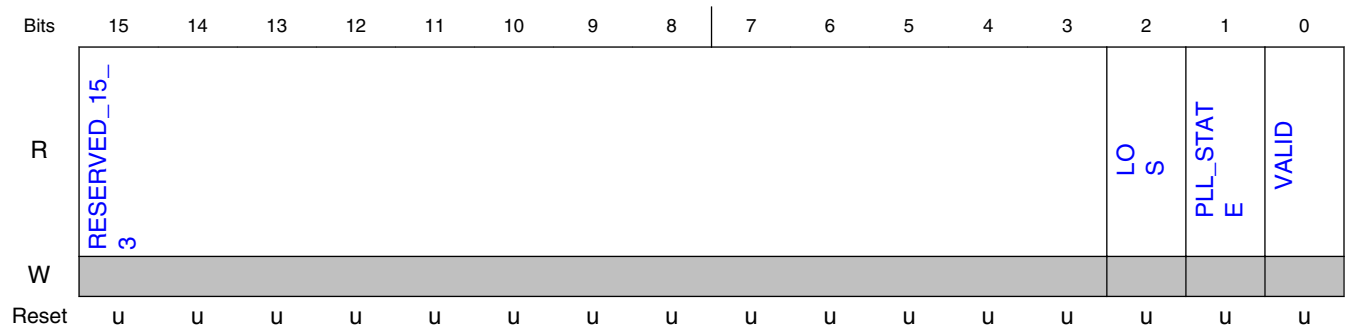
PCI Express PHY (PCle_PHY)

Field	Function
11-10 LOS_FILTER	Value from ASIC for rx_los_filter (RU - read-only volatile)
9 LOS_EN	Value from ASIC for rx_los_en (RU - read-only volatile)
8 TERM_EN	Value from ASIC for rx_term_en (RU - read-only volatile)
7 CLK_SHIFT	Value from ASIC for rx_bit_shift (RU - read-only volatile)
6 ALIGN_EN	Value from ASIC for rx_align_en (RU - read-only volatile)
5 DATA_EN	Value from ASIC for rx_data_en (RU - read-only volatile)
4 PLL_EN	Value from ASIC for rx_pll_en (RU - read-only volatile)
3 RX_RESET	Value from ASIC for rx_reset (RU - read-only volatile)
2 INVERT	Value from ASIC for rx_invert (RU - read-only volatile)
1-0 RX_RATE	Value from ASIC for rx_rate (RU - read-only volatile)

11.4.3.1.147 Current values for outgoing RX status controls from PHY (LANEX_DIG_RX_ASIC_OUT)**11.4.3.1.147.1 Offset**

Register	Offset
LANEX_DIG_RX_ASIC_OUT	1_201Ah

11.4.3.1.147.2 Diagram



11.4.3.1.147.3 Fields

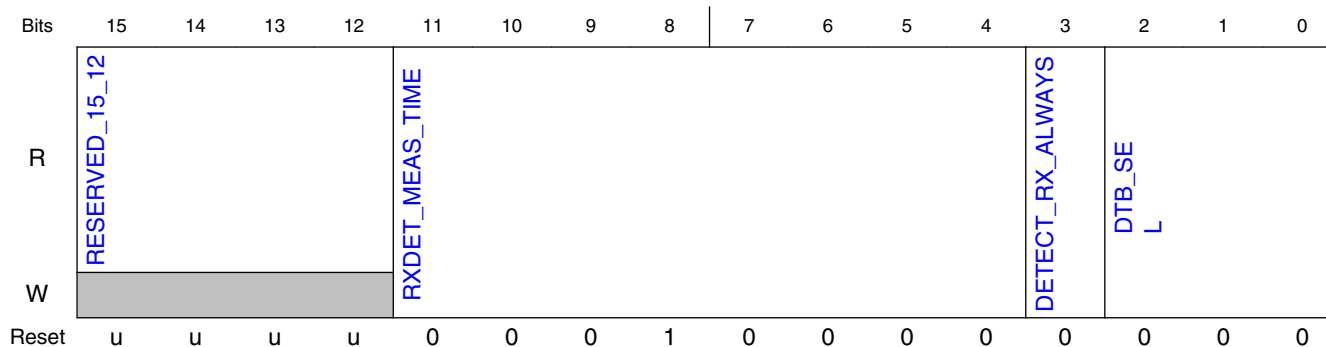
Field	Function
15-3 RESERVED_15_3	Reserved for Future use
2 LOS	Value from PHY for rx_los (RU - read-only volatile)
1 PLL_STATE	Value from PHY for rx_pll_state (RU - read-only volatile)
0 VALID	Value from PHY for rx_valid (RU - read-only volatile)

11.4.3.1.148 Transmit debug controls (LANEX_DIG_TX_DEBUG)

11.4.3.1.148.1 Offset

Register	Offset
LANEX_DIG_TX_DEBUG	1_2020h

11.4.3.1.148.2 Diagram



11.4.3.1.148.3 Fields

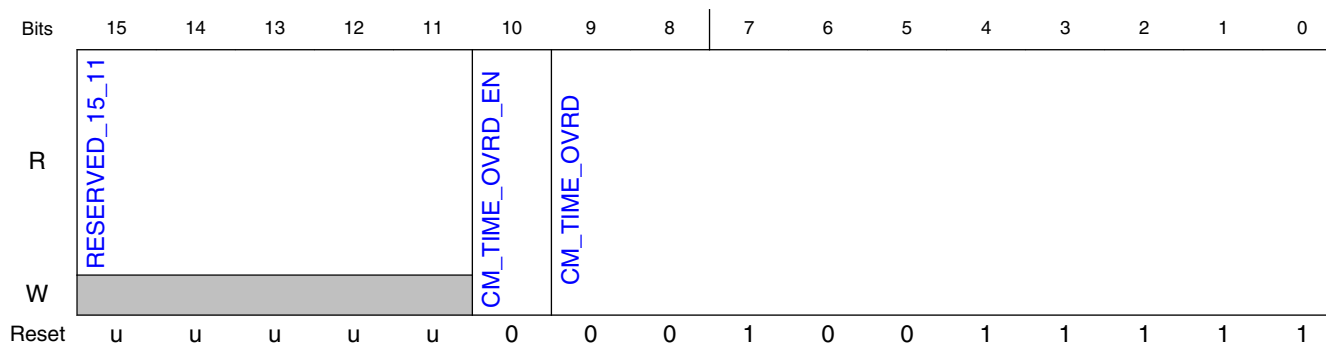
Field	Function
15-12 RESERVED_15_12	Reserved for Future use
11-4 RXDET_MEAS_TIME	Time to wait for rxdet measurement (* 4/2/1 refclk cycles based on multiplier)
3 DETECT_RX_ALWAYS	Always signal 1 for rx_detect ignoring analog
2-0 DTB_SEL	Selects data to drive on DTB 0 - disabled 1 - mag_update/sub signals 2 - mag_high/add signals 3 - clock aligner early/shift signals 4 - analog/asic clocks 5 - tx_state counter/lbert strobe 6 - noclck_ref_rst/dig_rst 7 - tx_ana_rxdetp_high/tx_ana_rxdetm_high

11.4.3.1.149 SSP_LANEX_DIG_TX_CM_WAIT_TIME_OVRD (LANEX_DIG_TX_CM_WAIT_TIME_OVRD)

11.4.3.1.149.1 Offset

Register	Offset
LANEX_DIG_TX_CM_WAIT_TIME_OVRD	1_2022h

11.4.3.1.149.2 Diagram



11.4.3.1.149.3 Fields

Field	Function
15-11 RESERVED_15_11	Reserved for Future use
10 CM_TIME_OVRD_EN	Override enable for common mode charging wait time.
9-0 CM_TIME_OVRD	Time to wait, in ref clock cycles, for LFPS common mode charging (This value, x32, is used for VCMHOLD charging method)

11.4.3.1.150 Calculated values for N_USE and N_TRISTATE. (LANEX_DIG_TX_VMD_FSM_TX_VCM_0)

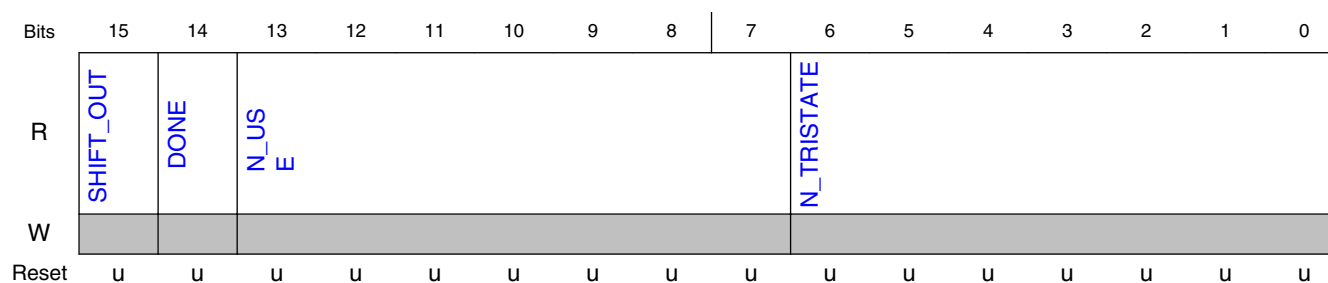
11.4.3.1.150.1 Offset

Register	Offset
LANEX_DIG_TX_VMD_FSM_TX_VCM_0	1_2024h

11.4.3.1.150.2 Function

Calculated values for N_USE and N_TRISTATE. Also TX ANA configuration shift register output.

11.4.3.1.150.3 Diagram



11.4.3.1.150.4 Fields

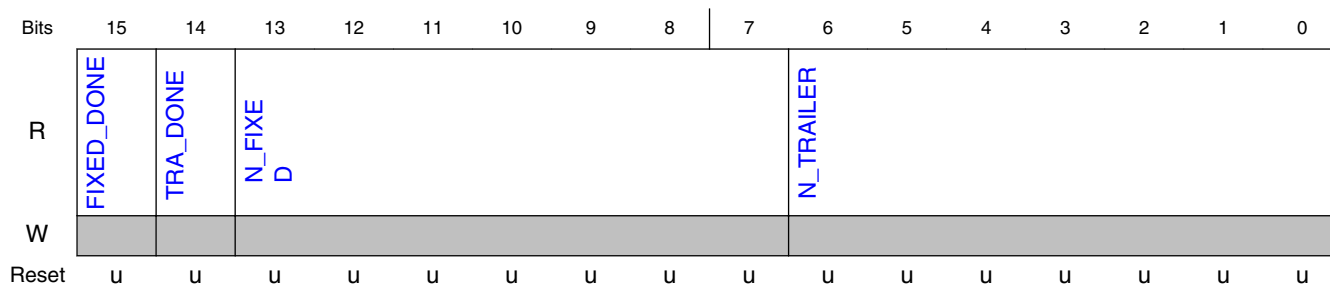
Field	Function
15 SHIFT_OUT	Current value from TX_ANAs configuration shift register. Current value from TX_ANAs configuration shift register. (RU - read-only volatile)
14 DONE	Configuration is done (RU - read-only volatile)
13-7 N_USE	Value from VMD for legs to use. Value from VMD for legs to use. (RU - read-only volatile)
6-0 N_TRISTATE	Value from VMD for number of tristate legs. Value from VMD for number of tristate legs. (RU - read-only volatile)

11.4.3.1.151 Calculated values for N_FIXED and N_TRAILER. (LANEX_DIG_TX_VMD_FSM_TX_VCM_1)

11.4.3.1.151.1 Offset

Register	Offset
LANEX_DIG_TX_VMD_FSM_TX_VCM_1	1_2026h

11.4.3.1.151.2 Diagram



11.4.3.1.151.3 Fields

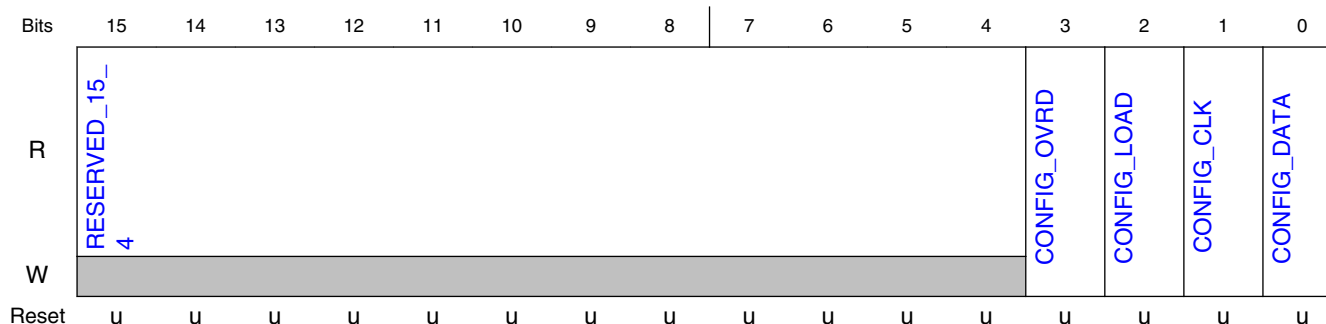
Field	Function
15 FIXED_DONE	N_FIXED Multiplication has completed. N_FIXED Multiplication has completed. (RU - read-only volatile)
14 TRA_DONE	N_TRAILER Multiplication has completed. N_TRAILER Multiplication has completed. (RU - read-only volatile)
13-7 N_FIXED	Value from VMD for number of fixed driver legs. Value from VMD for number of fixed driver legs. (RU - read-only volatile)
6-0 N_TRAILER	Value from VMD for number of trailer legs. Value from VMD for number of trailer legs. (RU - read-only volatile)

11.4.3.1.152 Access the TX_ANA VMD Control Shift Register (LANEX_DIG_TX_VMD_FSM_TX_VCM_DEBUG_IN)

11.4.3.1.152.1 Offset

Register	Offset
LANEX_DIG_TX_VMD_FSM_TX_VCM_DEBUG_IN	1_2028h

11.4.3.1.152.2 Diagram



11.4.3.1.152.3 Fields

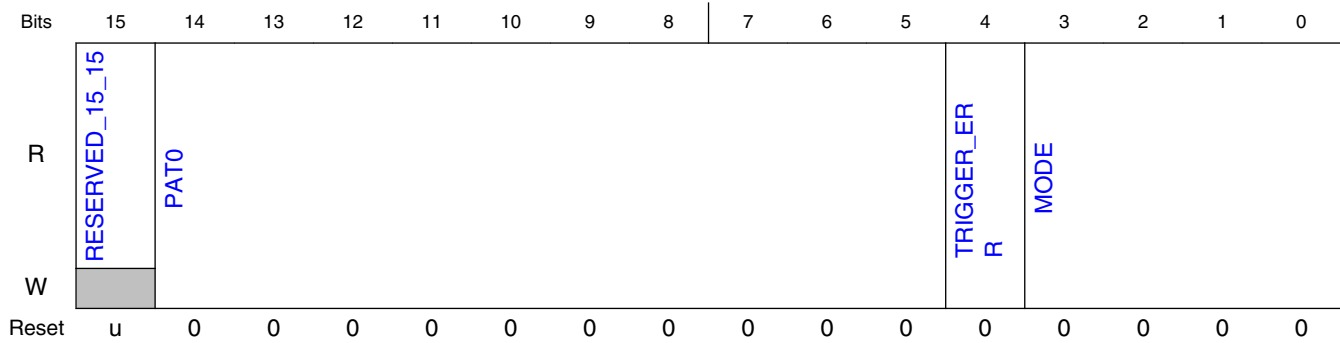
Field	Function
15-4 RESERVED_15_4	Reserved for Future use
3 CONFIG_OVRD	Override the Voltage Mode Driver Configuration FSM and access the shift chain directly. Override the Voltage Mode Driver Configuration FSM and access the shift chain directly. (volatile and 2 reads needed to read value)
2 CONFIG_LOAD	Override value for the Voltage Mode Driver Configuration FSM's config load. Override value for the Voltage Mode Driver Configuration FSM's config load. (volatile and 2 reads needed to read value)
1 CONFIG_CLK	Override value for the Voltage Mode Driver Configuration FSM's config clk. Override value for the Voltage Mode Driver Configuration FSM's config clk. (volatile and 2 reads needed to read value)
0 CONFIG_DATA	Override value for the Voltage Mode Driver Configuration FSM's config data. Override value for the Voltage Mode Driver Configuration FSM's config data. (volatile and 2 reads needed to read value)

11.4.3.1.153 Pattern Generator controls (LANEX_DIG_TX_LBERT_CTL)

11.4.3.1.153.1 Offset

Register	Offset
LANEX_DIG_TX_LBERT_CTL	1_202Ah

11.4.3.1.153.2 Diagram



11.4.3.1.153.3 Fields

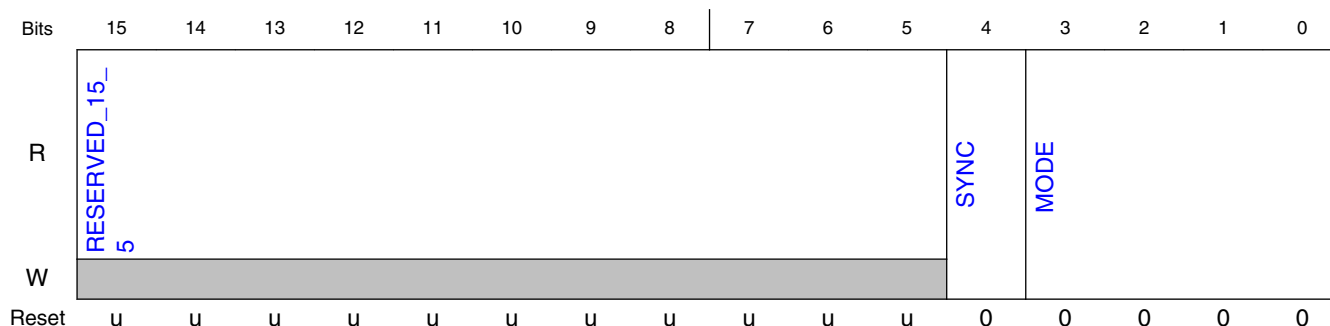
Field	Function
15 RESERVED_15_15	Reserved for Future use
14-5 PAT0	10 bit pattern for modes using PAT0 (5-7). 10 bit pattern for modes using PAT0 (5-7). Ignored for other MODEs.
4 TRIGGER_ERR	Single shot inversion the LSB of the current symbol. Single shot inversion the LSB of the current symbol. Any write of a 1 to this bit will insert an error. Maintain mode and pattern from setup.
3-0 MODE	Pattern to generate When changing modes, you must change to disabled first Pattern to generate When changing modes, you must change to disabled first 0 - disabled 1 - lfsr31. $X^{31} + X^{28} + 1$ 2 - lfsr23. $X^{23} + X^{18} + 1$ 3 - lfsr15. $X^{15} + X^{14} + 1$ 4 - lfsr7. $X^7 + X^6 + 1$ 5 - fixed word (PAT0) 6 - DC balanced word (PAT0, $\sim$ PAT0) 7 - Fixed pattern: (000, PAT0, 3ff, $\sim$ PAT0) 8 - USB3.0 TSEQ (rx equalizer training ordered sets) Transmits Even, then Odd. 9-15 - RESERVED

11.4.3.1.154 Pattern Matcher controls (LANEX_DIG_RX_LBERT_CTL)

11.4.3.1.154.1 Offset

Register	Offset
LANEX_DIG_RX_LBERT_CTL	1_202Ch

11.4.3.1.154.2 Diagram



11.4.3.1.154.3 Fields

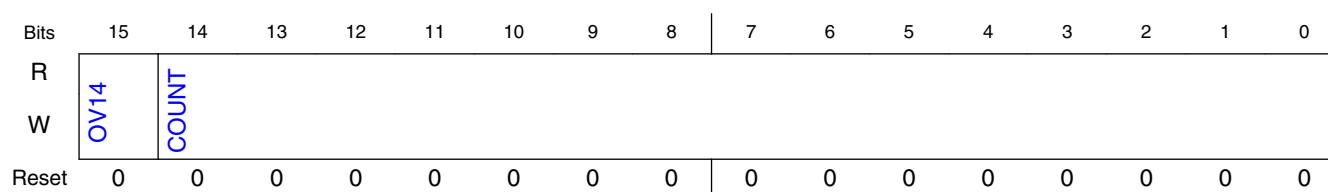
Field	Function
15-5 RESERVED_15_5	Reserved for Future use
4 SYNC	Synchronize pattern matcher with incoming data. A write of a one to this bit will reset the error counter and start a synchronization of the PM. Synchronize pattern matcher with incoming data A write of a one to this bit will reset the error counter and start a synchronization of the PM. There is no need to write this back to zero to run normally.
3-0 MODE	MODE Pattern to match When changing modes, you must change to disabled first 0 - disabled 1 - lfsr31: $X^{31} + X^{28} + 1$ 2 - lfsr23: $X^{23} + X^{18} + 1$ 3 - lfsr15: $X^{15} + X^{14} + 1$ 4 - lfsr7 : $X^7 + X^6 + 1$ 5 - $d[n] = d[n-10]$ 6 - $d[n] = !d[n-10]$ 7 - $d[n] = !d[n-20]$ 8 - USB 3.0 TSEQ (NOT SUPPORTED IN PM. PG Only!)

11.4.3.1.155 Pattern match error counter (LANEX_DIG_RX_LBERT_ERR)

11.4.3.1.155.1 Offset

Register	Offset
LANEX_DIG_RX_LBERT_ERR	1_202Eh

11.4.3.1.155.2 Diagram



11.4.3.1.155.3 Fields

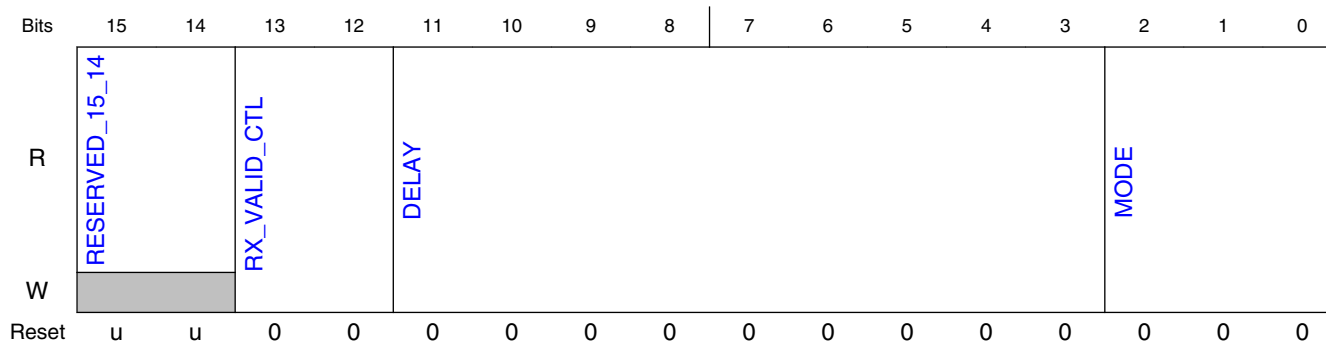
Field	Function
15 OV14	A read of this register, or a sync of the Pattern Matcher resets the error count. A read of this register, or a sync of the Pattern Matcher resets the error count. If all bits of this register are 1, the error counter has saturated. If the status is active, multiply COUNT by 128. If OV14=1 and COUNT=2 ¹⁵ -1, signals overflow of counter (volatile and 2 reads needed to read value)
14-0 COUNT	Current error count If OV14 field is active, then multiply count by 128 to get the actual count. Current error count If OV14 field is active, then multiply count by 128 to get the actual count. (volatile and 2 reads needed to read value)

11.4.3.1.156 Control bits for per-transceiver scope portion (LANEX_DIG_RX_SCOPE_CTL)

11.4.3.1.156.1 Offset

Register	Offset
LANEX_DIG_RX_SCOPE_CTL	1_2030h

11.4.3.1.156.2 Diagram



11.4.3.1.156.3 Fields

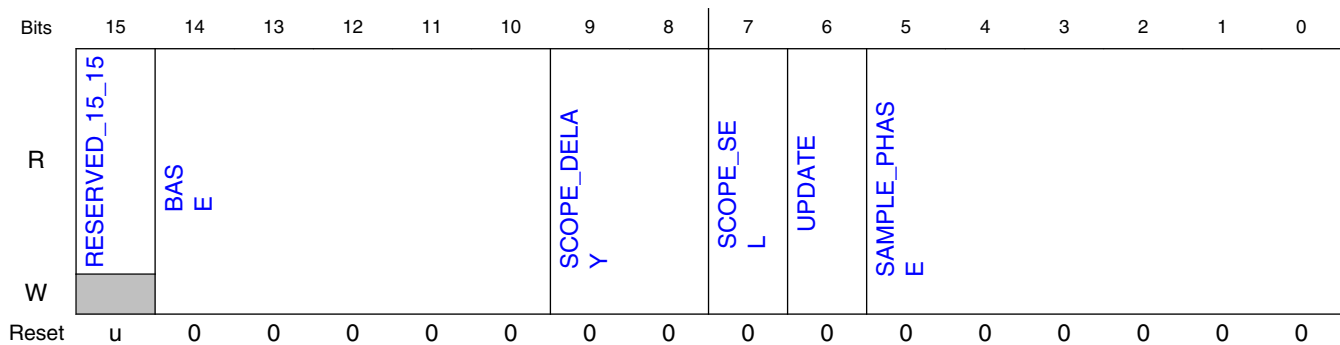
Field	Function
15-14 RESERVED_15_14	Reserved for Future use
13-12 RX_VALID_CTL	When to Start Sampling control based on rx_valid. When to Start Sampling control based on rx_valid. 0 = Ignore rx_valid. 1 = Heed rx_valid. 2 = Wait for rising_edge on rx_valid.
11-3 DELAY	number of cycles to skip between samples
2-0 MODE	sampling mode of counters NOTE: WORD is 20 bits. sampling mode of counters NOTE: WORD is 20 bits. 0 = off 1 = sample data every WORD*(1+DELAY) bits from the digital recovered path. 2 = sample data every WORD*(1+DELAY) + 1 bits from the digital recovered path. 3 = sample data every WORD*(1+DELAY) + 2 bits from the digital recovered path. 4 = sample data every WORD*(1+DELAY) bits from the analog probe. Asserts XOR and MASK increments accordingly.

11.4.3.1.157 Scope sampling phase. (LANEX_DIG_RX_SCOPE_PHASE)

11.4.3.1.157.1 Offset

Register	Offset
LANEX_DIG_RX_SCOPE_PHASE	1_2032h

11.4.3.1.157.2 Diagram



11.4.3.1.157.3 Fields

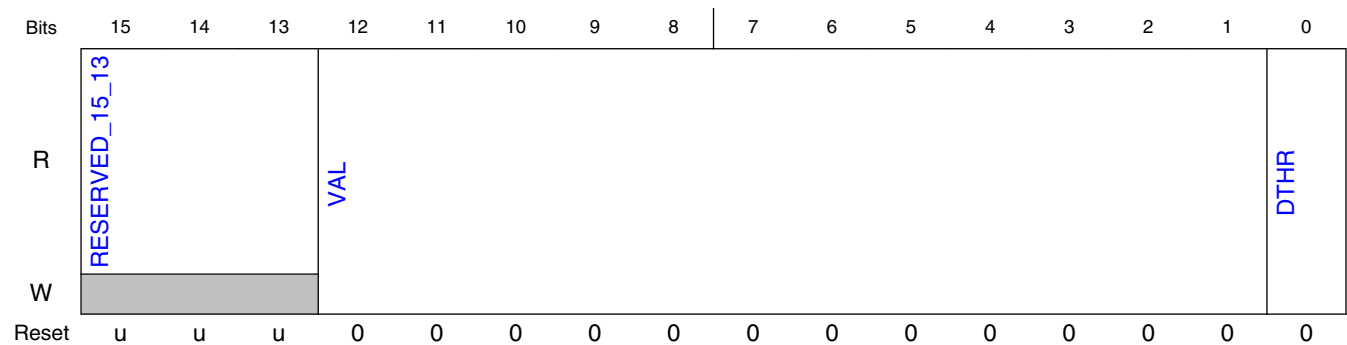
Field	Function
15 RESERVED_15_15	Reserved for Future use
14-10 BASE	which bit to sample when MODE = 1 or 4
9-8 SCOPE_DELAY	How many clocks to delay the analog scope_data.
7 SCOPE_SEL	Select Sampling Mode. Select Sampling Mode. 0 - Before AFE Sampling 1 - After AFE Sampling
6 UPDATE	Update Sampling phase. Update Sampling phase. Write a 1.
5-0 SAMPLE_PHASE	Sampling Phase

11.4.3.1.158 Current frequency integrator value. (LANEX_DIG_RX_DPLL_FREQ)

11.4.3.1.158.1 Offset

Register	Offset
LANEX_DIG_RX_DPLL_FREQ	1_2034h

11.4.3.1.158.2 Diagram



11.4.3.1.158.3 Fields

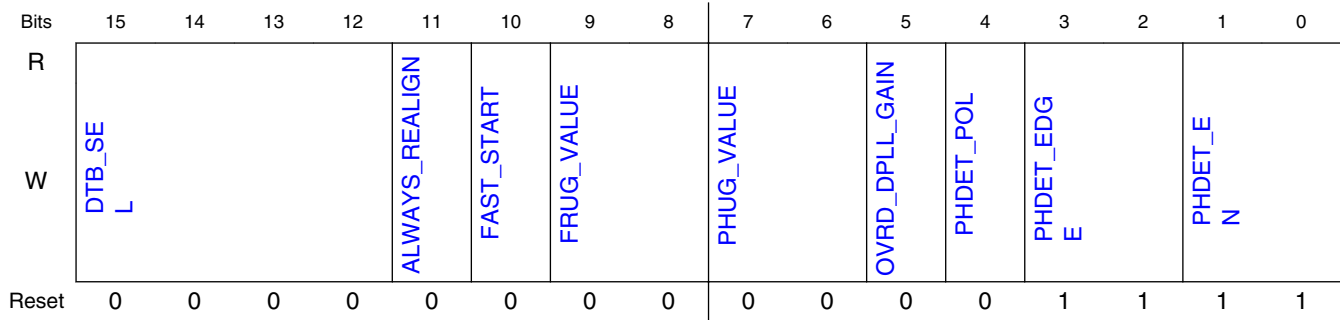
Field	Function
15-13 RESERVED_15_13	Reserved for Future use
12-1 VAL	Freq is 1. Freq is 1.526*VAL ppm from the reference When mppll_slow is set, the ppm is half the eqn above (volatile and 2 reads needed to read value)
0 DTHR	Bits below the useful resolution (volatile and 2 reads needed to read value)

11.4.3.1.159 Control bits for receiver in recovered domain (LANEX_DIG_RX_CDR_CTL)

11.4.3.1.159.1 Offset

Register	Offset
LANEX_DIG_RX_CDR_CTL	1_2036h

11.4.3.1.159.2 Diagram



11.4.3.1.159.3 Fields

Field	Function
15-12 DTB_SEL	select to drive various signals onto the dtb 0 - disabled 1 - pll_ana_rst,pll_count from rx_pwr_ctl 2 - com_good_high/low from aligner 3 - com_bad_high/low from aligner 4 - shift_in_prog,ana_odd_data from aligner 5 - low bits of XAUI align FSM state 6 - error_high/low from lbert_pm 7 - ana_los,los_filter from los block 8 - coast_dppll,eios_det from los block 9 - cdr_en and cdr_timeout from cdr_ctl A - cdr_en_adap and cdr_rx_valid from cdr_ctl B - cdr_en_eq and locked_vector_en from cdr_ctl C - dig_word_rst,rx_dig_en D - dig_dword_rst,dig_dword_clk
11 ALWAYS_REALIGN	realign on any misaligned comma
10 FAST_START	decrease startup steps by 50%
9-8 FRUG_VALUE	override value for FRUG
7-6 PHUG_VALUE	override value for PHUG
5 OVRD_DPLL_GAIN	Override PHUG and FRUG values

Table continues on the next page...

PCI Express PHY (PCIe_PHY)

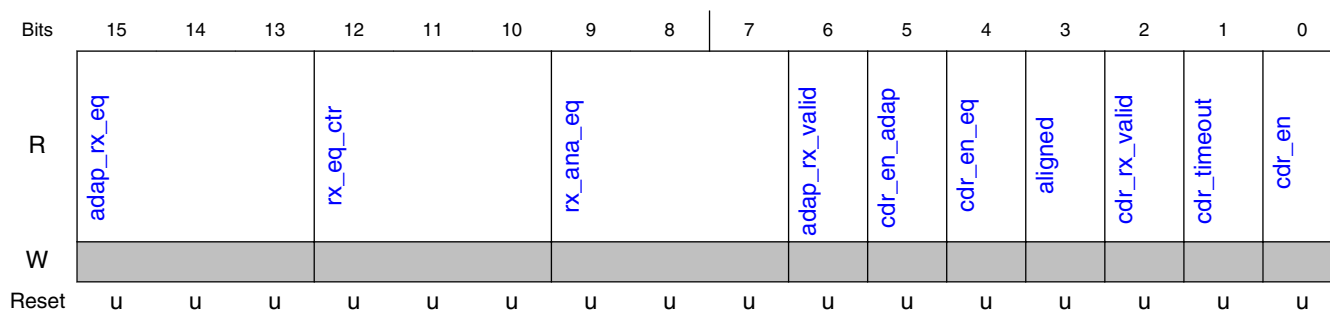
Field	Function
4 PHDET_POL	Reverse polarity of phase error
3-2 PHDET_EDGE	Edges to use for phase detection. Edges to use for phase detection. 11 - Use both edges 10 - Use rising edges only 01 - Use falling edges only 00 - Ignore all edges
1-0 PHDET_EN	Enable phase detector. Enable phase detector. top bit is odd slicers, bottom is even

11.4.3.1.160 Current values for CDR and Adaptation FSM. (LANEX_DIG_RX_CDR_CDR_FSM_DEBUG)

11.4.3.1.160.1 Offset

Register	Offset
LANEX_DIG_RX_CDR_CDR_FSM_DEBUG	1_2038h

11.4.3.1.160.2 Diagram



11.4.3.1.160.3 Fields

Field	Function
15-13 adap_rx_eq	Equalization setting from adaptation FSM. Equalization setting from adaptation FSM. (RU - read-only volatile)

Table continues on the next page...

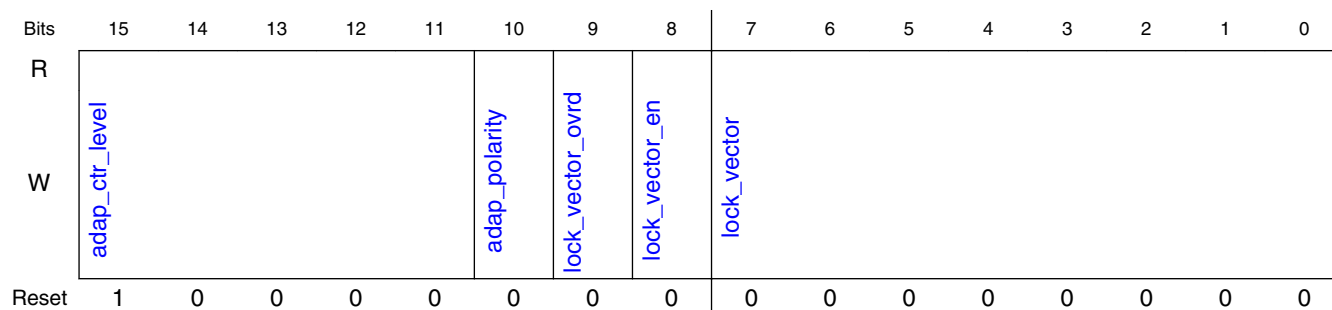
Field	Function
12-10 rx_eq_ctr	Initial centre point from equalization FSM. Initial centre point from equalization FSM. (RU - read-only volatile)
9-7 rx_ana_eq	Equalization setting to Analog. Equalization setting to Analog. (RU - read-only volatile)
6 adap_rx_valid	Adaptation has completed and locked. Adaptation has completed and locked. (RU - read-only volatile)
5 cdr_en_adap	Adapatation loop is enabling the CDR. Adapatation loop is enabling the CDR. (RU - read-only volatile)
4 cdr_en_eq	Equalization loop is enabling the CDR. Equalization loop is enabling the CDR. (RU - read-only volatile)
3 aligned	Datapath is bit-aligned. Datapath is bit-aligned. (RU - read-only volatile)
2 cdr_rx_valid	CDR has locked to incoming data stream. CDR has locked to incoming data stream. (RU - read-only volatile)
1 cdr_timeout	CDR has not locked to datastream and has timed-out. CDR has not locked to datastream and has timed-out. (RU - read-only volatile)
0 cdr_en	CDR has been enabled. CDR has been enabled. (RU - read-only volatile)

11.4.3.1.161 Control bits for receiver in recovered domain (LANEX_DIG_RX_CDR_LOCK_VEC_OVRD)

11.4.3.1.161.1 Offset

Register	Offset
LANEX_DIG_RX_CDR_LOCK_VEC_OVRD	1_203Ah

11.4.3.1.161.2 Diagram



11.4.3.1.161.3 Fields

Field	Function
15-11 adap_ctr_level	Ammount of earlies that increment the adaptation counter (times 32, Default is 512). Ammount of earlies that increment the adaptation counter (times 32, Default is 512). (volatile and 2 reads needed to read value)
10 adap_polarity	If asserted invert default adaptation adjustment for equalization. If asserted invert default adaptation adjustment for equalization. IF early decrease equalization. Normal mode is to decrease. (volatile and 2 reads needed to read value)
9 lock_vector_ovr d	Override enable for the rx_eq outputs. Override enable for the rx_eq outputs. (volatile and 2 reads needed to read value)
8 lock_vector_en	Override value for the locked_vector output completion. Override value for the locked_vector output completion. (volatile and 2 reads needed to read value)
7-0 lock_vector	Override value for the locked_vector. Override value for the locked_vector. (volatile and 2 reads needed to read value)

11.4.3.1.162 Current values for the locked vector. (LANEX_DIG_RX_CDR_LOCK_VEC)

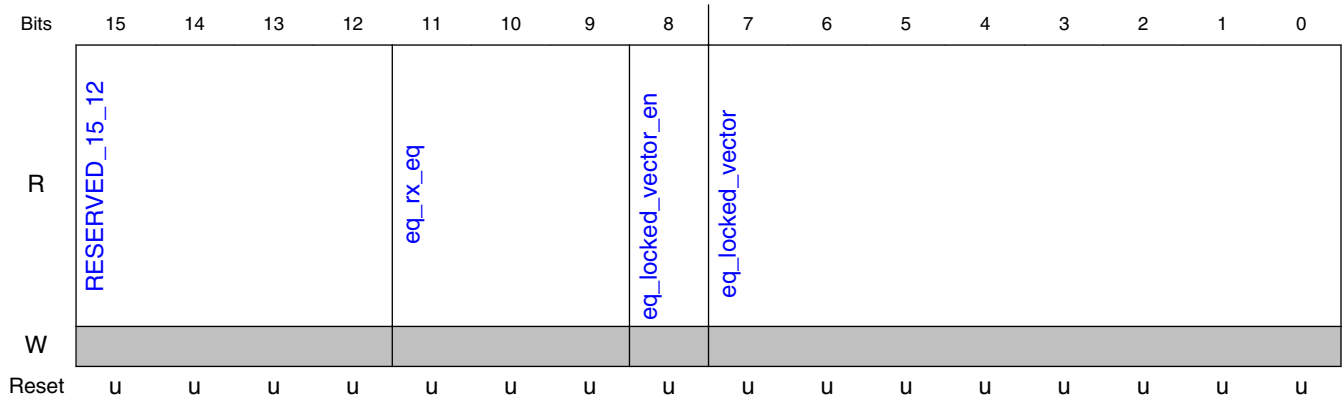
11.4.3.1.162.1 Offset

Register	Offset
LANEX_DIG_RX_CDR_L OCK_VEC	1_203Ch

11.4.3.1.162.2 Function

Current values for the locked vector..

11.4.3.1.162.3 Diagram



11.4.3.1.162.4 Fields

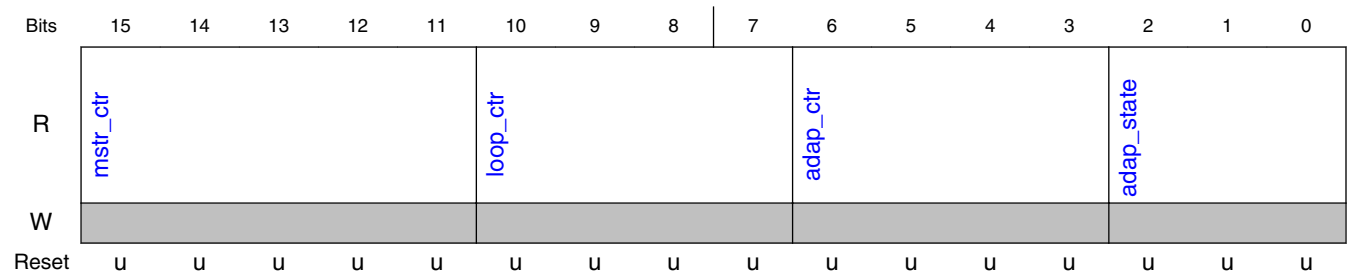
Field	Function
15-12 RESERVED_15_12	Reserved for Future use
11-9 eq_rx_eq	Equalization setting from the Equalization Loop. Equalization setting from the Equalization Loop. (RU - read-only volatile)
8 eq_locked_vect or_en	Equalization locked vector has been filled. Equalization locked vector has been filled. (RU - read-only volatile)
7-0 eq_locked_vect or	Results of equalization loop. Results of equalization loop. (RU - read-only volatile)

11.4.3.1.163 Observation of adaptation FSM register. (LANEX_DIG_RX_CDR_ADAP_FSM)

11.4.3.1.163.1 Offset

Register	Offset
LANEX_DIG_RX_CDR_ADAP_FSM	1_203Eh

11.4.3.1.163.2 Diagram



11.4.3.1.163.3 Fields

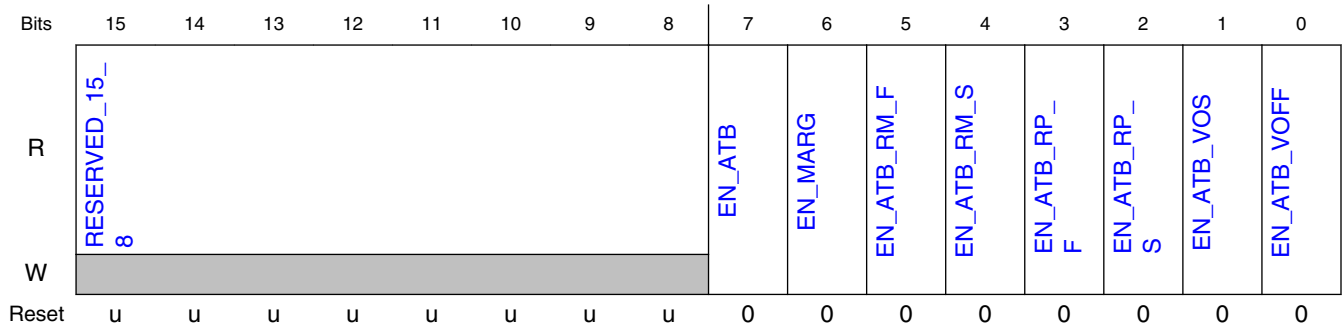
Field	Function
15-11 mstr_ctr	Master count register. Master count register. (RU - read-only volatile)
10-7 loop_ctr	Loop count register. Loop count register. (RU - read-only volatile)
6-3 adap_ctr	Adaptation count register. Adaptation count register. (RU - read-only volatile)
2-0 adap_state	Adaptation State. Adaptation State. NOTE: ADAP_RESET = 3'b000 ADAP_LOCK = 3'b001 ADAP_SUFF = 3'b010 ADAP_LOOP = 3'b011 ADAP_MSTR = 3'b100 ADAP_DONE = 3'b101 (RU - read-only volatile)

11.4.3.1.164 ATB Control Register (LANEX_ANA_RX_ATB0)

11.4.3.1.164.1 Offset

Register	Offset
LANEX_ANA_RX_ATB0	1_2040h

11.4.3.1.164.2 Diagram



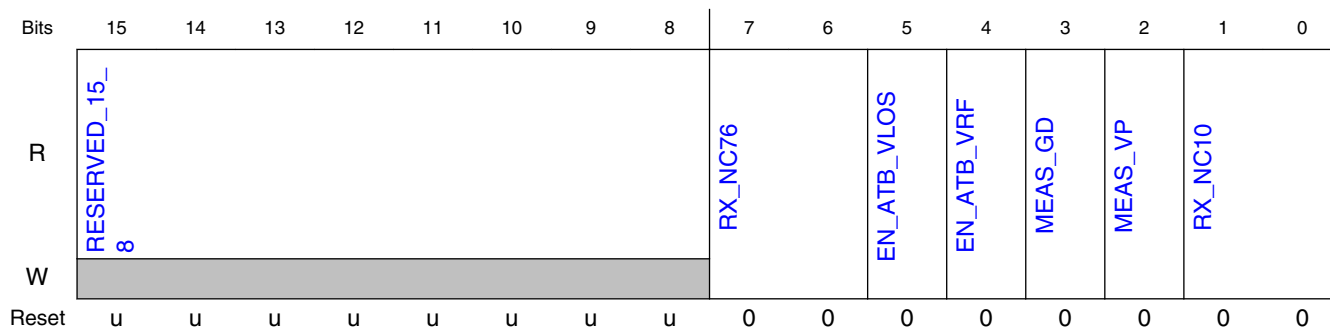
11.4.3.1.164.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 EN_ATB	Enable ATB sensing and forcing on internal RX nodes
6 EN_MARG	Enable margining mode in receiver; it puts EQ common mode output on atb_f_m
5 EN_ATB_RM_F	Enable atb_f_p on negative-side termination resistor
4 EN_ATB_RM_S	Enable atb_s_m on negative-side termination resistor
3 EN_ATB_RP_F	Enable atb_f_p on positive-side termination resistor
2 EN_ATB_RP_S	Enable atb_s_p on positive-side termination resistor
1 EN_ATB_VOS	Put rxafe filtered outputs vo_p on atb_s_p and vo_m on atb_s_m
0 EN_ATB_VOFF	Put rxafe voff_p on atb_s_p and voff_m on atb_s_m

11.4.3.1.165 ATB and Misc Control Register (LANEX_ANA_RX_ATB1)

11.4.3.1.165.1 Offset

Register	Offset
LANEX_ANA_RX_ATB1	1_2042h

11.4.3.1.165.2 Diagram**11.4.3.1.165.3 Fields**

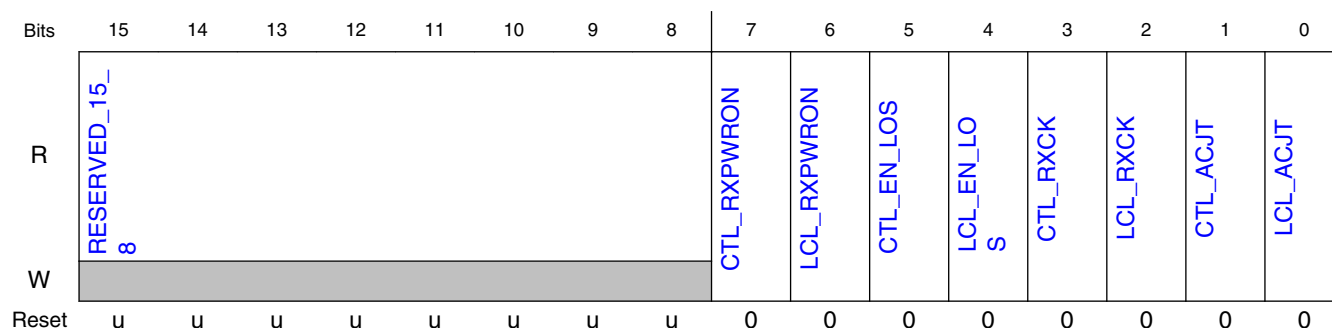
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7-6 RX_NC76	Reserved
5 EN_ATB_VLOS	Enable sensing of LOS reference voltage on atb_s_p
4 EN_ATB_VRF	Enable sensing of vref_rx (RX Bias B. Enable sensing of vref_rx (RX Bias B.G. Voltage) on atb_s_p
3 MEAS_GD	Enable sensing of local gd in RX; ties gd to atb_s_m
2 MEAS_VP	Enable sensing of local vp in RX; ties vp to atb_s_p
1-0 RX_NC10	Reserved

11.4.3.1.166 RX State Control Register (LANEX_ANA_RX_ENPWR0)

11.4.3.1.166.1 Offset

Register	Offset
LANEX_ANA_RX_ENPW R0	1_2044h

11.4.3.1.166.2 Diagram



11.4.3.1.166.3 Fields

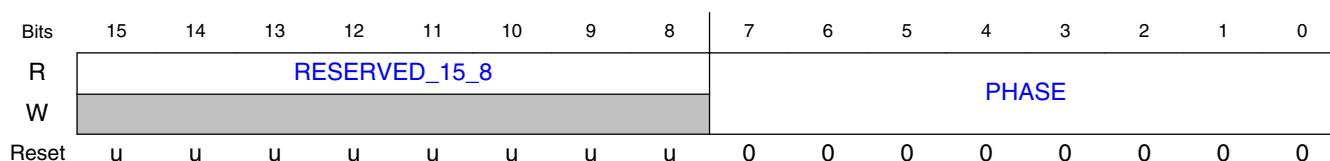
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 CTL_RXPWRO	MUX select for AFE en to select between CTL_RXPWRO=0 rx_afe_en (External signal) or CTL_RXPWRO=1 LCL_RXPWRO
6 LCL_RXPWRO	Enable/Disable RX AFE . Enable/Disable RX AFE .This bit is only selected when CTL_RXPWRO = 1
5 CTL_EN_LOS	MUX select for LOS en to select between CTL_EN_LOS=0 rx_los_en (External signal) or CTL_EN_LOS=1 LCL_EN_LOS
4 LCL_EN_LOS	Enable/Disable LOS. Enable/Disable LOS.This bit is only selected when CTL_EN_LOS = 1
3 CTL_RXCK	MUX select for RX Clock en to select between CTL_RXCK=0 rx_clock_en (External signal) or CTL_RXCK=1 LCL_RXCK.
2 LCL_RXCK	Enable/Disable en_rx_clock (RX clock enable). Enable/Disable en_rx_clock (RX clock enable).This bit is only selected when CTL_RXCK = 1
1 CTL_ACJT	MUX select for ACJTAG to select between CTL_ACJTG=0 rx_acjt_en (External signal) or CTL_ACJTG=1 LCL_ACJT.
0 LCL_ACJT	Enable/Disable acjtag block, This bit is only selected when CTL_ACJTG = 1

11.4.3.1.167 Phase select Value (LANEX_ANA_RX_PMI_X_PHASE)

11.4.3.1.167.1 Offset

Register	Offset
LANEX_ANA_RX_PMI_X_PHASE	1_2046h

11.4.3.1.167.2 Diagram



11.4.3.1.167.3 Fields

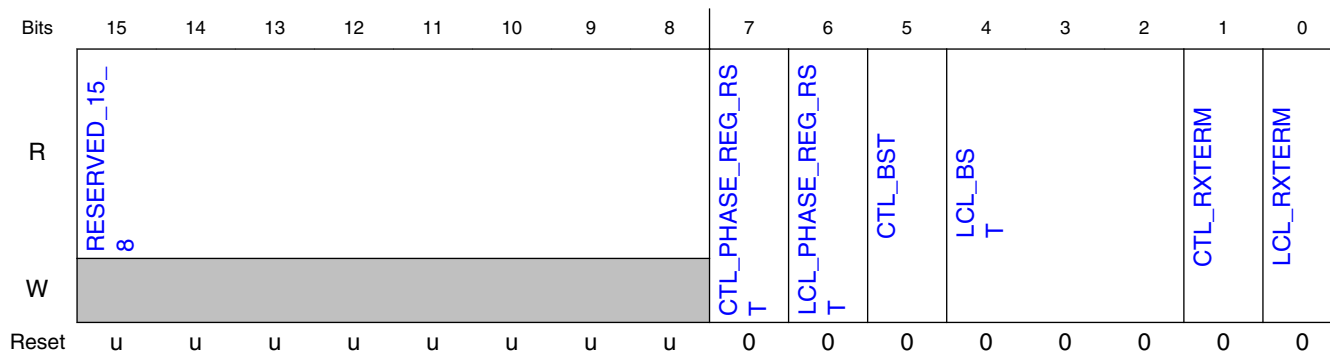
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7-0 PHASE	Write to bits 8 to 1 of the phase select register in the phase mixer through the CDR (USER2 volatile - read-write volatile and custom)

11.4.3.1.168 Additional RX State Control Register (LANEX_ANA_RX_ENPWR1)

11.4.3.1.168.1 Offset

Register	Offset
LANEX_ANA_RX_ENPWR1	1_2048h

11.4.3.1.168.2 Diagram



11.4.3.1.168.3 Fields

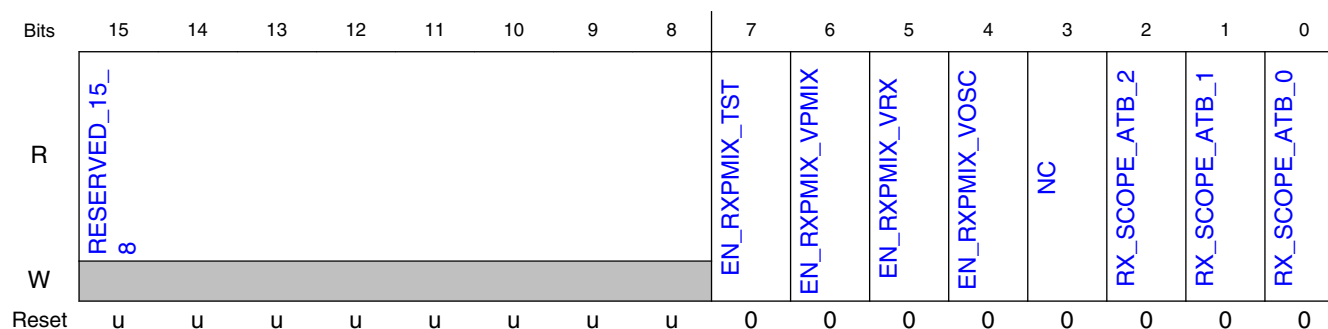
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 CTL_PHASE_REG_RST	MUX select for Phase Register Reset to select between CTL_PHASE_REG_RST=0 rx_phase_reg_rst (External signal) or CTL_PHASE_REG_RST=1 LCL_PHASE_REG_RST
6 LCL_PHASE_REG_RST	Reset phase register. Reset phase register. This bit is only selected when CTL_PHASE_REG_RST = 1
5 CTL_BST	MUX select for AFE Boost to select between CTL_BST=0 rx_aeq[2:0] (External signal) or CTL_BST=1 LCL_BST
4-2 LCL_BST	RX Boost (equalization) Value
1 CTL_RXTERM	MUX select for RX Term enable to select between CTL_RXTERM=0 rx_term_en (External signal) or CTL_RXTERM=1 LCL_RXTERM
0 LCL_RXTERM	Enable/disable RX termination when CTL_RXTERM = 1

11.4.3.1.169 Yet another RX State Control Register (LANEX_ANA_RX_ENPWR2)

11.4.3.1.169.1 Offset

Register	Offset
LANEX_ANA_RX_ENPWR2	1_204Ah

11.4.3.1.169.2 Diagram



11.4.3.1.169.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 EN_RXPMIX_TST	Enable XOR gate to test linearity of RX phase mixer using atb_s_p and atb_s_m
6 EN_RXPMIX_VPMIX	Put vreg_pmix on atb_s_p
5 EN_RXPMIX_VRX	Put vreg_rx on atb_s_p
4 EN_RXPMIX_VOOSC	Put vreg_vosc on atb_s_p
3 NC	Reserved
2 RX_SCOPE_ATB_2	Put XOR of RX SCOPE PMIX input and output on atb_s_p
1 RX_SCOPE_ATB_1	Put RX SCOPE regulated VP on atb_s_p
0 RX_SCOPE_ATB_0	Have RX SCOPE reference for regulator use atb_f_p as a reference instead of VP/VBG

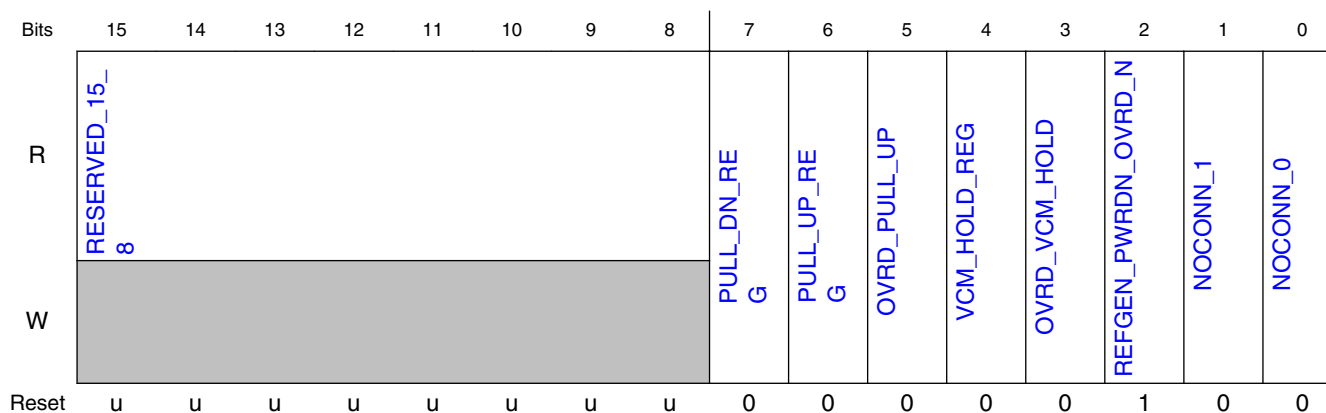
Field	Function
RX_LOS_LFPS_EN	Enables the RX LOS LFPS mode, use when you want to detect incoming LFPS signals, USB3.0

11.4.3.1.171 Register 3 (LANEX_ANA_TX_TXDRV_CNTRL)

11.4.3.1.171.1 Offset

Register	Offset
LANEX_ANA_TX_TXDRV_CNTRL	1_2056h

11.4.3.1.171.2 Diagram



11.4.3.1.171.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 PULL_DN_REG	register bit that causes the calibrated TX bits to pull down in a common mode fashion. register bit that causes the calibrated TX bits to pull down in a common mode fashion. If pull_dn_reg and tx_pull_up are both high, then pull_dn_reg wins
6 PULL_UP_REG	Register override for tx_pull_up. Register override for tx_pull_up. Used when ovr_d_pull_up is HIGH, causes calibrated TX bits to pull up in a common mode fashion, unless pull_dn_reg is HIGH
5	Selects local value of pull_up_reg instead of tx_pull_up

Table continues on the next page...

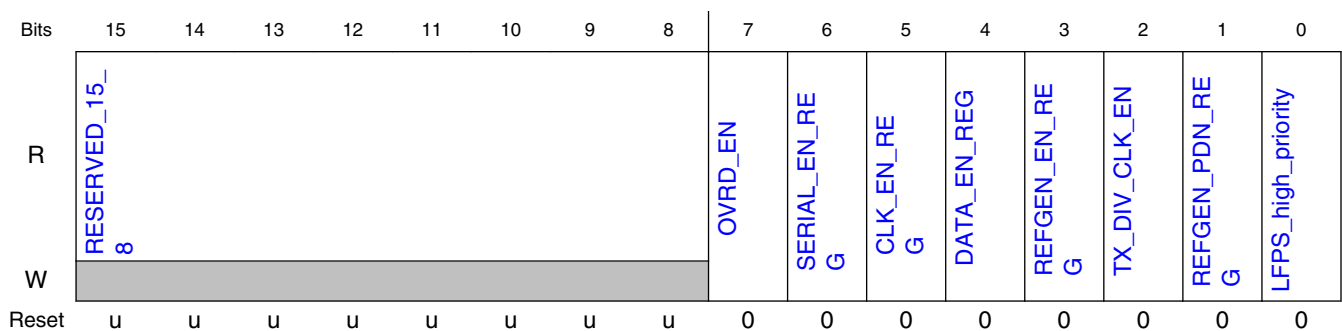
Field	Function
OVRD_PULL_UP	
4 VCM_HOLD_REG	Register override for tx_vcm_hold. Register override for tx_vcm_hold. Used when ovr_d_vcm_hold is HIGH, controls the TX common mode hold circuitry
3 OVRD_VCM_HOLD	Selects local value of vcm_hold_reg instead of tx_vcm_hold to control state of TX common mode hold circuitry
2 REFGEN_PWRDN_OVRD_N	Set LOW for TX regulator not to turn on when RX clock is enabled. Set LOW for TX regulator not to turn on when RX clock is enabled. Default is for TX regulator to turn on when RX or TX clock is enabled
1 NOCONN_1	Reserved
0 NOCONN_0	Reserved

11.4.3.1.172 Register 4 (LANEX_ANA_TX_POWER_CTL)

11.4.3.1.172.1 Offset

Register	Offset
LANEX_ANA_TX_POWER_CTL	1_2058h

11.4.3.1.172.2 Diagram



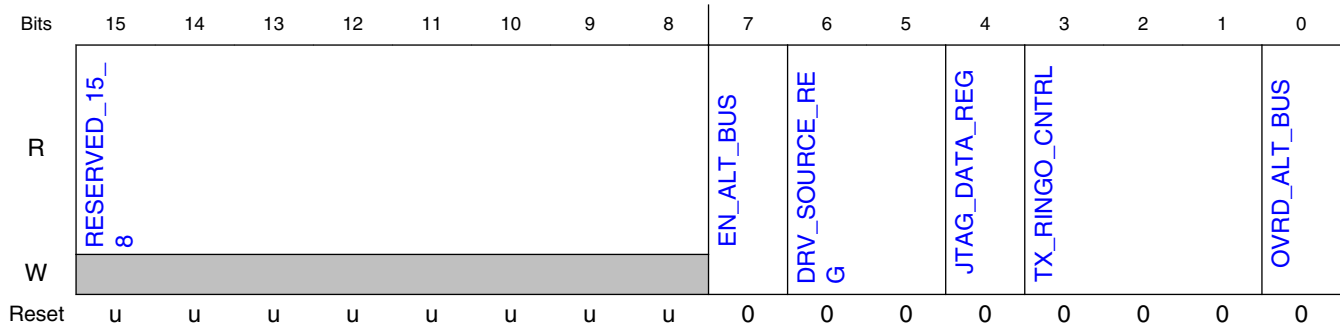
11.4.3.1.172.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 OVRD_EN	Enables local overrides for all signals in this register except for TX_DIV_CLK_EN which doesn't need this signal
6 SERIAL_EN_REG	Forces state of tx_serial_en when OVRD_EN is 1
5 CLK_EN_REG	Forces state of tx_clk_en when OVRD_EN is 1
4 DATA_EN_REG	Forces state of tx_data_en when OVRD_EN is HIGH
3 REFGEN_EN_REG	Register override value for tx_refgen_en. Register override value for tx_refgen_en. Controls the TX regulator, biasing circuitry and the rxdetect comparators when OVRD_EN is HIGH. This function may be overridden by the state of refgen_pwdn_reg
2 TX_DIV_CLK_EN	Enables the div clock that goes from the TX to the undersampler, more appropriately called tx_sampler_clk_en, this clock comes after the optional divide by 2 or 4, tx_clk_en must be HIGH to get a clock out
1 REFGEN_PDN_REG	Register override for refgen pwn. Register override for refgen pwn. Used when OVRD_EN is HIGH. Controls the TX regulator, biasing circuitry and the rxdetect comparators when OVRD_EN is HIGH
0 LFPS_high_priority	When this is set high LFPS priority becomes higher than tx_data

11.4.3.1.173 Register 5 (LANEX_ANA_TX_ALT_BLOCK)**11.4.3.1.173.1 Offset**

Register	Offset
LANEX_ANA_TX_ALT_BLOCK	1_205Ah

11.4.3.1.173.2 Diagram



11.4.3.1.173.3 Fields

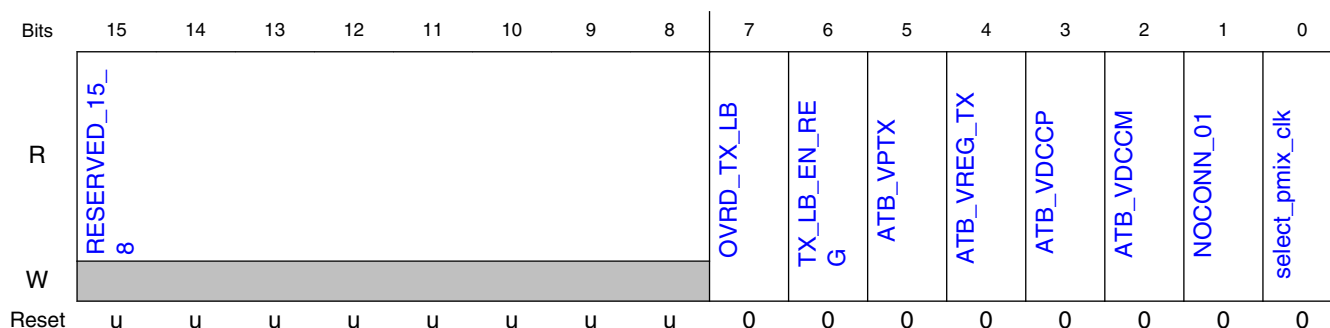
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 EN_ALT_BUS	Enables the TX for alt bus mode, powers up the TX regulator, biasing circuitry and the rxdetect comparators. Enables the TX for alt bus mode, powers up the TX regulator, biasing circuitry and the rxdetect comparators. Needed if manually running the alt bus features
6-5 DRV_SOURCE_REG	Forces state of tx_data_source when OVRD_ALT_BUS is HIGH. Forces state of tx_data_source when OVRD_ALT_BUS is HIGH. 00 = serializer data or alt oscillator vp/vph/vphreg/vptx if selected, 01 = jtag data differential, 10 = LFPS oscillator differential, 11 = jtag data common mode drive for test
4 JTAG_DATA_REG	Forces state of jtag_data when OVRD_ALT_BUS is HIGH
3-1 TX_RINGO_CNTRL	Enables and connects the various test oscillators to the transmit pins, must set drv_source_reg bus correctly. Enables and connects the various test oscillators to the transmit pins, must set drv_source_reg bus correctly. 0XX = All oscillators are disabled, 100 = HV oscillator enabled, 101 = SVT oscillator enabled, 110 = LVT oscillator enabled, 111 = ULVT oscillator enabled
0 OVRD_ALT_BUS	Enables local overrides for alt-bus control signals: DRV_SOURCE_REG and JTAG_DATA_REG

11.4.3.1.174 Register 6 (LANEX_ANA_TX_ALT_AND_LOOPBACK)

11.4.3.1.174.1 Offset

Register	Offset
LANEX_ANA_TX_ALT_A ND_LOOPBACK	1_205Ch

11.4.3.1.174.2 Diagram



11.4.3.1.174.3 Fields

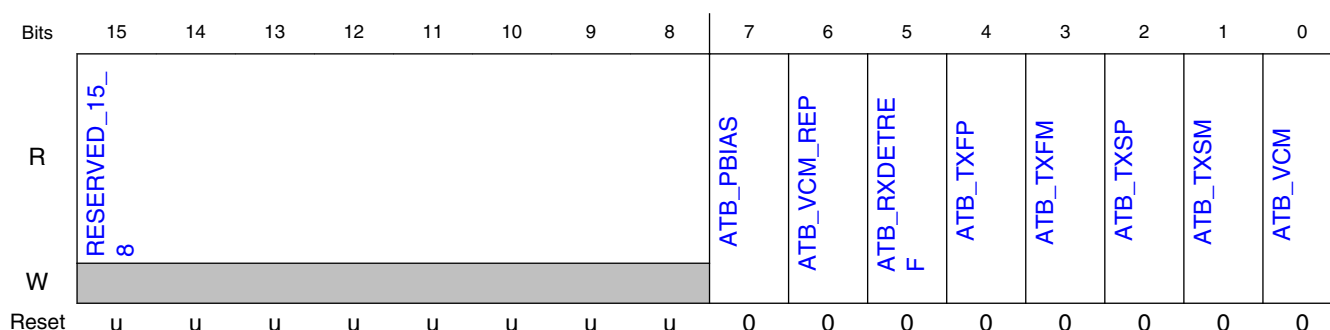
Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 OVRD_TX_LB	Enables the override of the tx_lb_en
6 TX_LB_EN_RE G	Forces state of tx_lb_en when OVRD_TX_LB is enabled
5 ATB_VPTX	Connects vptx to atb_s_p and gd to atb_s_m
4 ATB_VREG_TX	Connects TX regulator input reference to atb_s_p and gd to atb_s_m
3 ATB_VDCCP	Connects vdcc_p to atb_s_p
2 ATB_VDCCM	Connects vdcc_m to atb_s_m
1 NOCONN_01	Reserved
0 select_pmix_clk	select pmix_clk for TX clock for ATE test mode

11.4.3.1.175 Register 7 (LANEX_ANA_TX_TX_ATB_REG)

11.4.3.1.175.1 Offset

Register	Offset
LANEX_ANA_TX_TX_ATB_REG	1_205Eh

11.4.3.1.175.2 Diagram



11.4.3.1.175.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 ATB_PBIAS	Connect real pmos_bias voltage for TX PMOS driver pull up path to atb_s_p and local ground at the pmos_bias block to atb_s_m
6 ATB_VCM_REP	Connect common mode replica voltage in pmos_bias block to atb_s_p and local ground to atb_s_m
5 ATB_RXDETRE F	Connect RX detect block reference voltage to atb_s_p and local ground to atb_s_m
4 ATB_TXFP	Connect tx_p to atb_f_p
3 ATB_TXFM	Connect tx_m to atb_f_m
2 ATB_TXSP	Connect tx_p to atb_s_p

Table continues on the next page...

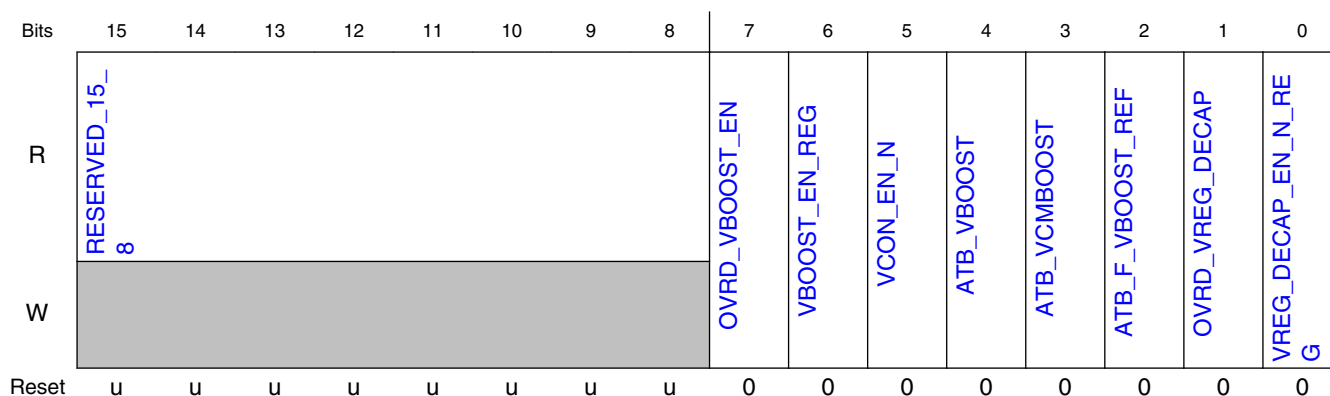
Field	Function
1 ATB_TXSM	Connect tx_m to atb_s_m
0 ATB_VCM	Connect tx_p/tx_m common mode voltage onto atb_s_p and local ground onto atb_s_m

11.4.3.1.176 Register 8 (LANEX_ANA_TX_TX_BOOST)

11.4.3.1.176.1 Offset

Register	Offset
LANEX_ANA_TX_TX_BOOST	1_2060h

11.4.3.1.176.2 Diagram



11.4.3.1.176.3 Fields

Field	Function
15-8 RESERVED_15_8	Reserved for Future use
7 OVRD_VBOOST_EN	Enables the overrides on VBOOST_EN_REG
6	Forces state of tx_vboost_en, when OVRD_VBOOST_EN is HIGH

Table continues on the next page...

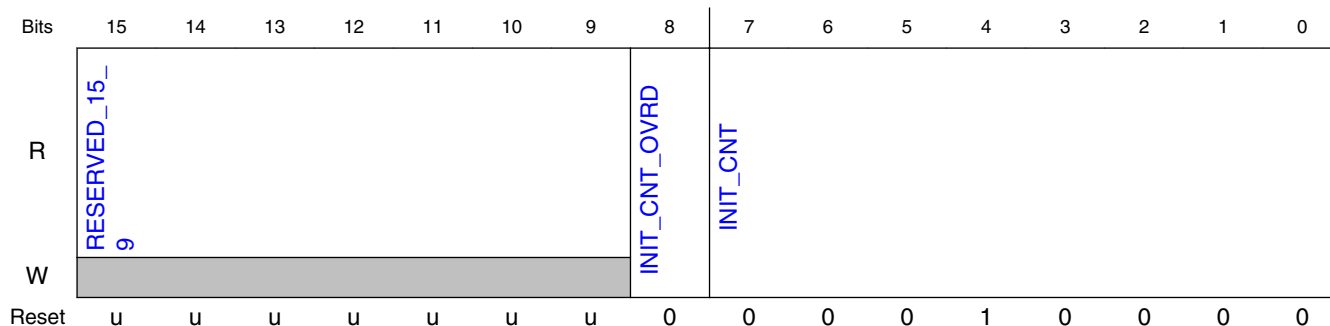
Field	Function
VBOOST_EN_REG	
5 VCON_EN_N	Forces state of vcon_en_n
4 ATB_VBOOST	Connects vboost to atb_s_p and local ground to Atb_s_m
3 ATB_VCMBOOST	Connects vcmboost to atb_s_p and local ground to atb_s_m.
2 ATB_F_VBOOST_REF	Enables forcing of tx_vboost_vref through atb_f_p from external
1 OVRD_VREG_DECAP	Enables override of vreg_decap_en_n signal
0 VREG_DECAP_EN_N_REG	Forces state of vreg_decap_en_n control signal. Forces state of vreg_decap_en_n control signal. Controls whether extra decoupling is switched in for TX regulator output.

11.4.3.1.177 Control bits for CDR init phase portion (LANEX_DIG_RX_CDR_FSM_INIT_CTL)

11.4.3.1.177.1 Offset

Register	Offset
LANEX_DIG_RX_CDR_FSM_INIT_CTL	1_2080h

11.4.3.1.177.2 Diagram

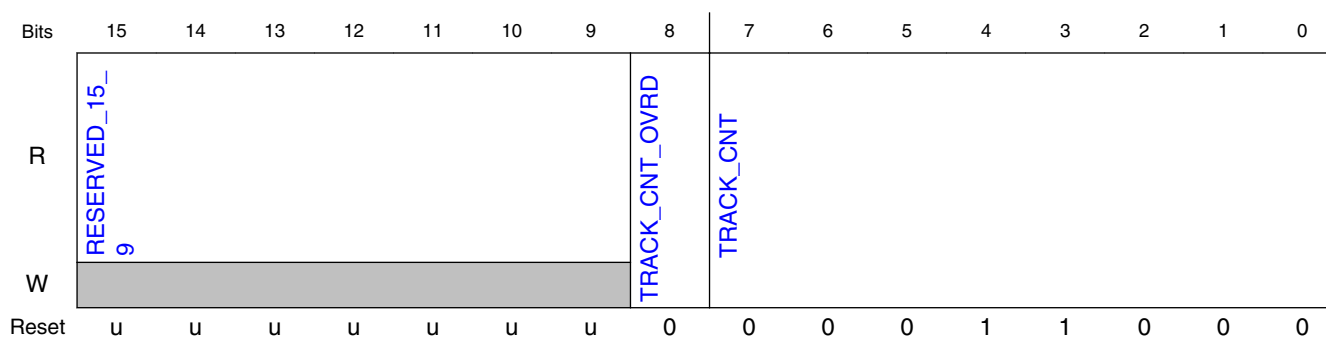


11.4.3.1.177.3 Fields

Field	Function
15-9 RESERVED_15_9	Reserved for Future use
8 INIT_CNT_OVRD	Active high override enable bit To select the override INIT_CNT register 0 = Disabled 1 = Enabled
7-0 INIT_CNT	Override value for CDR Init Phase

11.4.3.1.178 Control bits for CDR track phase portion (LANEX_DIG_RX_CDR_FSM_TRACK_CTL)**11.4.3.1.178.1 Offset**

Register	Offset
LANEX_DIG_RX_CDR_FSM_TRACK_CTL	1_2082h

11.4.3.1.178.2 Diagram**11.4.3.1.178.3 Fields**

Field	Function
15-9	Reserved for Future use

Table continues on the next page...

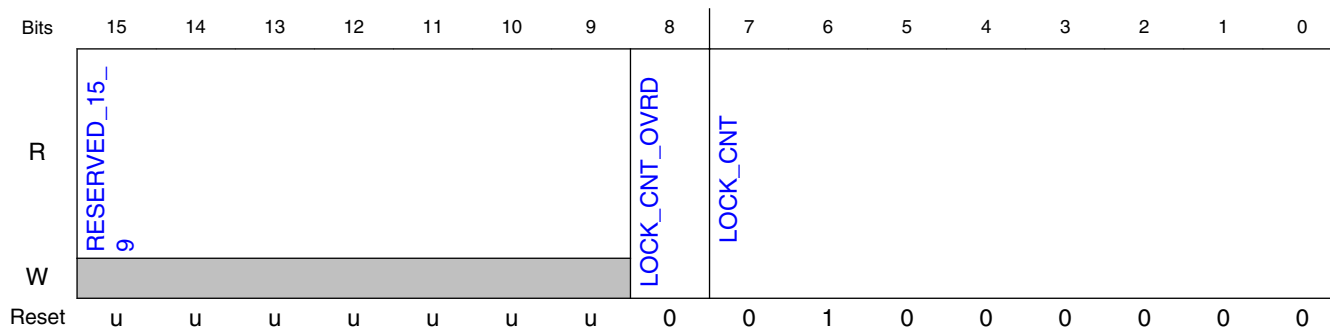
Field	Function
RESERVED_15_9	
8 TRACK_CNT_OVRD	Active high override enable bit To select the override TRACK_CNT register 0 = Disabled 1 = Enabled
7-0 TRACK_CNT	Override value for CDR Track Phase

11.4.3.1.179 Control bits for CDR lock phase portion (LANEX_DIG_RX_CDR_FSM_LOCK_CTL)

11.4.3.1.179.1 Offset

Register	Offset
LANEX_DIG_RX_CDR_FSM_LOCK_CTL	1_2084h

11.4.3.1.179.2 Diagram



11.4.3.1.179.3 Fields

Field	Function
15-9 RESERVED_15_9	Reserved for Future use
8	Active high override enable bit To select the override LOCK_CNT register

Table continues on the next page...

Field	Function
LOCK_CNT_OV RD	0 = Disabled 1 = Enabled
7-0 LOCK_CNT	Override value for CDR Lock Phase

11.5 Ethernet MAC (ENET)

11.5.1 Introduction

The MAC-NET core, in conjunction with a 10/100/1000-Mbit/s MAC, implements layer 3 network acceleration functions. These functions are designed to accelerate the processing of various common networking protocols, such as IP, TCP, UDP, and ICMP, providing wire speed services to client applications.

11.5.2 Overview

The core implements a triple-speed 10/100/1000-Mbit/s Ethernet MAC compliant with the IEEE802.3-2002 standard. The MAC layer provides compatibility with half- or full-duplex 10/100-Mbit/s and full-duplex gigabit Ethernet LANs.

The MAC operation is fully programmable and can be used in Network Interface Card (NIC), bridging, or switching applications. The core implements the remote network monitoring (RMON) counters according to IETF RFC 2819.

The core also implements a hardware acceleration block to optimize the performance of network controllers providing TCP/IP, UDP, and ICMP protocol services. The acceleration block performs critical functions in hardware, which are typically implemented with large software overhead.

The core implements programmable embedded FIFOs that can provide buffering on the receive path for lossless flow control.

Advanced power management features are available with magic packet detection and programmable power-down modes.

A unified DMA (uDMA), internal to the ENET module, optimizes data transfer between the ENET core and the SoC, and supports an enhanced buffer descriptor programming model to support IEEE 1588 functionality.

The programmable Ethernet MAC with IEEE 1588 integrates a standard IEEE 802.3 Ethernet MAC with a time-stamping module. The IEEE 1588 standard provides accurate clock synchronization for distributed control nodes for industrial automation applications.

11.5.2.1 Features

The MAC-NET core includes the following features.

11.5.2.1.1 Ethernet MAC features

- Implements the full 802.3 specification with preamble/SFD generation, frame padding generation, CRC generation and checking
- Supports zero-length preamble
- Dynamically configurable to support 10/100-Mbit/s and gigabit operation
- Supports 10/100 Mbit/s full-duplex and configurable half-duplex operation
- Supports gigabit full-duplex operation
- Compliant with the AMD magic packet detection with interrupt for node remote power management
- Seamless interface to commercial ethernet PHY devices via one of the following:
 - a 4-bit Media Independent Interface (MII) operating at 2.5/25 MHz.
 - a 4-bit non-standard MII-Lite (MII without the CRS and COL signals) operating at 2.5/25 MHz.
 - a 2-bit Reduced MII (RMII) operating at 50 MHz.
 - a (double data rate) 4-bit Reduced GMII (RGMII) operating at 125 MHz.
- Simple 64-Bit FIFO user-application interface
- CRC-32 checking at full speed with optional forwarding of the frame check sequence (FCS) field to the client
- CRC-32 generation and append on transmit or forwarding of user application provided FCS selectable on a per-frame basis
- In full-duplex mode:
 - Implements automated pause frame (802.3 x31A) generation and termination, providing flow control without user application intervention
 - Pause quanta used to form pause frames — dynamically programmable
 - Pause frame generation additionally controllable by user application offering flexible traffic flow control
 - Optional forwarding of received pause frames to the user application
 - Implements standard flow-control mechanism
- In half-duplex mode: provides full collision support, including jamming, backoff, and automatic retransmission
- Supports VLAN-tagged frames according to IEEE 802.1Q

- Programmable MAC address: Insertion on transmit; discards frames with mismatching destination address on receive (except broadcast and pause frames)
- Programmable promiscuous mode support to omit MAC destination address checking on receive
- Multicast and unicast address filtering on receive based on 64-entry hash table, reducing higher layer processing load
- Programmable frame maximum length providing support for any standard or proprietary frame length
- Statistics indicators for frame traffic and errors (alignment, CRC, length) and pause frames providing for IEEE 802.3 basic and mandatory management information database (MIB) package and remote network monitoring (RFC 2819)
- Simple handshake user application FIFO interface with fully programmable depth and threshold levels
- Provides separate status word for each received frame on the user interface providing information such as frame length, frame type, VLAN tag, and error information
- Multiple internal loopback options
- MDIO master interface for PHY device configuration and management supports two programmable MDIO base addresses, and standard (IEEE 802.3 Clause 22) and extended (Clause 45) MDIO frame formats
- Supports legacy FEC buffer descriptors
- Interrupt coalescing reduces the number of interrupts generated by the MAC, reducing CPU loading
- Traffic-shaping bandwidth distribution supports credit-based and round-robin-based policies. Either policy can be combined with time-based shaping.
- AVB (Audio Video Bridging, IEEE 802.1Qav) features:
 - Credit-based bandwidth distribution policy can be combined with time-based shaping
 - AVB endpoint talker and listener support
 - Support for arbitration between different priority traffic (for example, AVB class A, AVB class B, and non-AVB)

11.5.2.1.2 IP protocol performance optimization features

- Operates on TCP/IP and UDP/IP and ICMP/IP protocol data or IP header only
- Enables wire-speed processing
- Supports IPv4 and IPv6
- Transparent passing of frames of other types and protocols
- Supports VLAN tagged frames according to IEEE 802.1q with transparent forwarding of VLAN tag and control field

- Automatic IP-header and payload (protocol specific) checksum calculation and verification on receive
- Automatic IP-header and payload (protocol specific) checksum generation and automatic insertion on transmit configurable on a per-frame basis
- Supports IP and TCP, UDP, ICMP data for checksum generation and checking
- Supports full header options for IPv4 and TCP protocol headers
- Provides IPv6 support to datagrams with base header only — datagrams with extension headers are passed transparently unmodified/unchecked
- Provides statistics information for received IP and protocol errors
- Configurable automatic discard of erroneous frames
- Configurable automatic host-to-network (RX) and network-to-host (TX) byte order conversion for IP and TCP/UDP/ICMP headers within the frame
- Configurable padding remove for short IP datagrams on receive
- Configurable Ethernet payload alignment to allow for 32-bit word-aligned header and payload processing
- Programmable store-and-forward operation with clock and rate decoupling FIFOs

11.5.2.1.3 IEEE 1588 features

- Supports all IEEE 1588 frames.
- Allows reference clock to be chosen independently of network speed.
- Software-programmable precise time-stamping of ingress and egress frames
- Timer monitoring capabilities for system calibration and timing accuracy management
- Precise time-stamping of external events with programmable interrupt generation
- Programmable event and interrupt generation for external system control
- Supports hardware- and software-controllable timer synchronization.
- Provides a 4-channel IEEE 1588 timer. Each channel supports input capture and output compare using the 1588 counter.

11.5.2.2 Block diagram

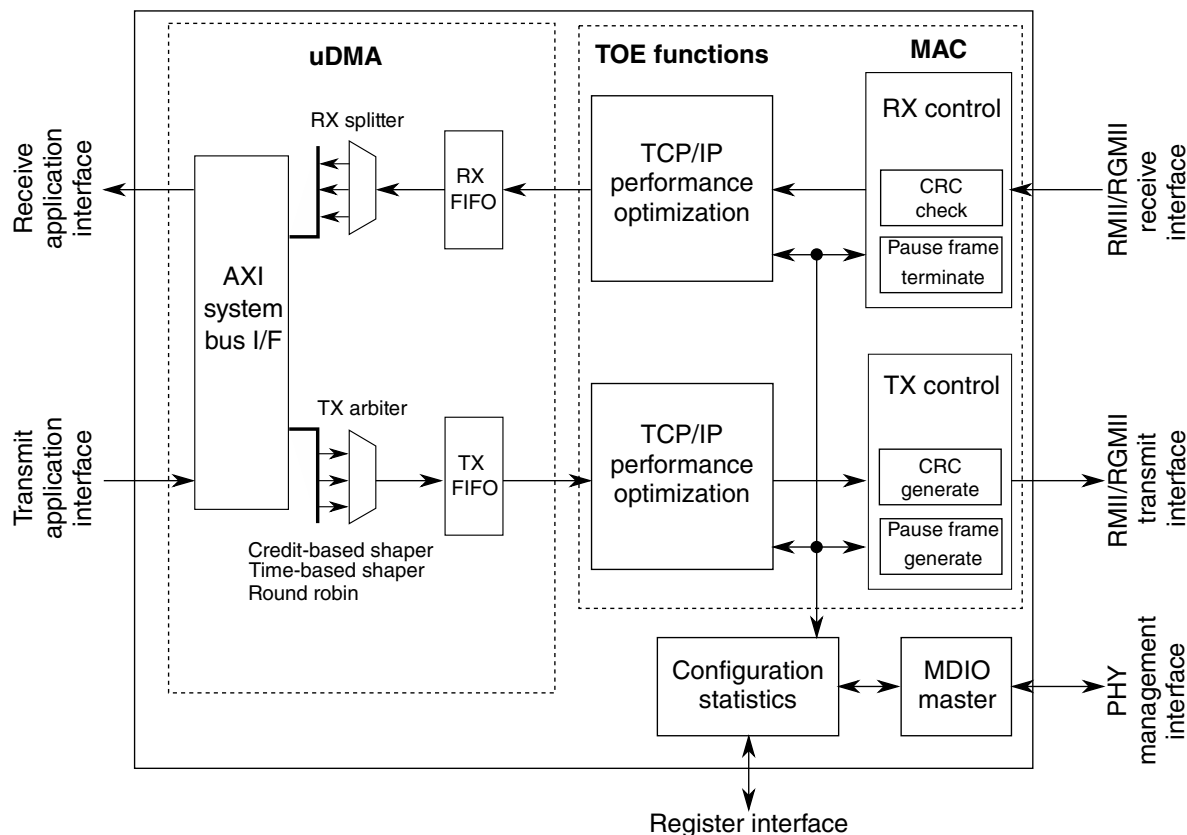


Figure 11-74. Ethernet MAC-NET core block diagram

11.5.3 External Signals and Clocks

The table found here describes the external signals of ENET.

Table 11-42. ENET External Signals

Signal	Description	Mode	Pad Name	Alt Mode	Direction
ENET1_MDC	Output clock provides a timing reference to the PHY for data transfers on the MDIO signal.	RMII/RGMII	ENET_MDC	ALT0	O
			GPIO1_IO06	ALT1	
			I2C1_SCL	ALT1	
ENET1_MDIO	Transfers control information between the external PHY and the media-access controller. Data is synchronous to MDC. This signal is an input after reset.	RMII/RGMII	ENET_MDIO	ALT0	I/O
			GPIO1_IO07	ALT1	
			I2C1_SDA	ALT1	
RGMII_TD3	Serial output Ethernet data. Only valid during TX_EN assertion. Only used for RGMII.	RGMII	ENET_TD3	ALT0	O
RGMII_TD2	Used as RGMII data.	RGMII	ENET_TD2	ALT0	O

Table continues on the next page...

Table 11-42. ENET External Signals (continued)

Signal	Description	Mode	Pad Name	Alt Mode	Direction
ENET1_TX_CLK	Used as RMII clock. There are two RMII clock schemes. 1. MAC generates output 50M reference clock for PHY, and MAC also uses this 50M clock. 2. MAC uses external 50M clock.	RMII	ENET_TD2	ALT1	I/O
RGMII_TD1	Serial output Ethernet data. Only valid during TX_EN assertion.	RMII/RGMII	ENET_TD1	ALT0	O
RGMII_TD0	Serial output Ethernet data. Only valid during TX_EN assertion.	RMII/RGMII	ENET_TD0	ALT0	O
RGMII_TX_CTL	In RGMII mode, contains TX_EN on the rising edge of RGMII_TXC, and a logical derivative of TX_EN and TX_ER (TX_EN XOR TX_ER) on the falling edge of RGMII_TXC. (RMII_TX_EN).	RMII/RGMII	ENET_TX_CTL	ALT0	O
RGMII_TXC	For RGMII, ENET_TXC works as RGMII_TXC in ALT0 mode.	RGMII	ENET_TXC	ALT0	O
ENET1_TX_ER	For RMII, ENET_TXC works as ENET1_TX_ER in ALT1 mode.	RMII	ENET_TXC	ALT1	O
RGMII_RX_CTL	In RGMII mode, contains RX_EN on the rising edge of RGMII_RXC, and a logical derivative of RX_EN and RX_ER (RX_EN XOR RX_ER) on the falling edge of RGMII_RXC. (RMII_RX_EN (CRS_DV)).	RMII/RGMII	ENET_RX_CTL	ALT0	I
RGMII_RXC	For RGMII, ENET_RXC works as RGMII_RXC in ALT0 mode.	RGMII	ENET_RXC	ALT0	I
ENET1_RX_ER	For RMII, ENET_RXC works as ENET1_RX_ER in ALT1 mode.	RMII	ENET_RXC	ALT1	I
RGMII_RD0	Contains the Ethernet input data transferred from the PHY to the media-access controller when RX_EN is asserted.	RMII/RGMII	ENET_RD0	ALT0	I
RGMII_RD1	Contains the Ethernet input data transferred from the PHY to the media-access controller when RX_EN is asserted.	RMII/RGMII	ENET_RD1	ALT0	I
RGMII_RD2	Contains the Ethernet input data transferred from the PHY to the media-access controller when RX_EN is asserted.	RGMII	ENET_RD2	ALT0	I
RGMII_RD3	Contains the Ethernet input data transferred from the PHY to the media-access controller when RX_EN is asserted.	RGMII	ENET_RD3	ALT0	I
ENET1_1588_EVENT0_IN	Capture/compare block input/output event bus signal. When configured for capture and a rising edge is detected, the current timer value is	RMII/RGMII	GPIO1_IO08	ALT1	I

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Table 11-42. ENET External Signals (continued)

Signal	Description	Mode	Pad Name	Alt Mode	Direction
	latched and transferred into the corresponding ENET_TCCRn register for inspection by software. When configured for compare, the corresponding signal 1588_EVENT is asserted for one cycle when the timer reaches the compare value programmed in register ENET_TCCRn. An interrupt or DMA request can be triggered if the corresponding bit in ENET_TCSRn[TIE] or ENET_TCSRn[TDRE] is set.				
ENET1_1588_EVENT0_OUT	Capture/compare block input/output event bus signal. When configured for capture and a rising edge is detected, the current timer value is latched and transferred into the corresponding ENET_TCCRn register for inspection by software. When configured for compare, the corresponding signal 1588_EVENT is asserted when the timer reaches the compare value programmed in register ENET_TCCRn. An interrupt or DMA request can be triggered if the corresponding bit in ENET_TCSRn[TIE] or ENET_TCSRn[TDRE] is set. Note: ENET_1588_EVENT0_OUT has a programmable output width, see IOMUXC_GPR0[CLK_STRETCH], delayed one clock cycle in relation to all other EVENTx_OUT signals.	RMII/RGMII	GPIO1_IO09	ALT1	O
ENET1_1588_EVENT1_IN	Capture/compare block input/output event bus signal. When configured for capture and a rising edge is detected, the current timer value is latched and transferred into the corresponding ENET_TCCRn register for inspection by software. When configured for compare, the corresponding signal 1588_EVENT is asserted for one cycle when the timer reaches the compare value programmed in register ENET_TCCRn. An interrupt or DMA request can be triggered if the corresponding bit in ENET_TCSRn[TIE] or ENET_TCSRn[TDRE] is set.	RMII/RGMII	I2C2_SCL	ALT1	I
ENET1_1588_EVENT1_OUT	Capture/compare block input/output event bus signal. When configured for capture and a rising edge is	RMII/RGMII	I2C2_SDA	ALT1	O

Table continues on the next page...

Table 11-42. ENET External Signals (continued)

Signal	Description	Mode	Pad Name	Alt Mode	Direction
	detected, the current timer value is latched and transferred into the corresponding ENET_TCCRn register for inspection by software. When configured for compare, the corresponding signal 1588_EVENT is asserted for one cycle when the timer reaches the compare value programmed in register ENET_TCCRn. An interrupt or DMA request can be triggered if the corresponding bit in ENET_TCSRn[TIE] or ENET_TCSRn[TDRE] is set.				
ENET_PHY_REF_CLK_ROOT	Reference clock for PHY.	RGMII	GPIO1_IO00	ALT1	O

The table below describes the clock sources for ENET. See Clock Controller Module (CCM) for clock setting, configuration, and gating information.

Table 11-43. ENET Clocks

Clock name	Clock Root	Description
ipp_ind_mac0_txclk	ENET_REF_CLK_ROOT	rmii/rgmii interface clock
ipg_clk_time	ENET_TIMER_CLK_ROOT	Time stamp clock

11.5.4 Functional description

This section provides a complete functional description of the MAC-NET core.

11.5.4.1 Ethernet MAC frame formats

The IEEE 802.3 standard defines the Ethernet frame format as follows:

- Minimum length of 64 bytes
- Maximum length of 1518 bytes excluding the preamble and the start frame delimiter (SFD) bytes

An Ethernet frame consists of the following fields:

- Seven bytes preamble

- Start frame delimiter (SFD)
- Two address fields
- Length or type field
- Data field
- Frame check sequence (CRC value)
- Extension field is defined only for Gigabit Ethernet half-duplex implementations and is not supported by the MAC core

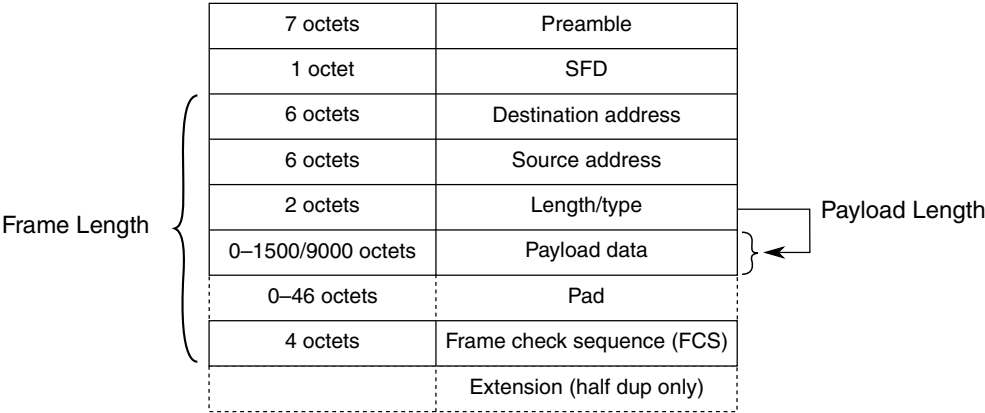


Figure 11-75. MAC frame format overview

Optionally, MAC frames can be VLAN-tagged with an additional four-byte field inserted between the MAC source address and the type/length field. VLAN tagging is defined by the IEEE P802.1q specification. VLAN-tagged frames have a maximum length of 1522 bytes, excluding the preamble and the SFD bytes.

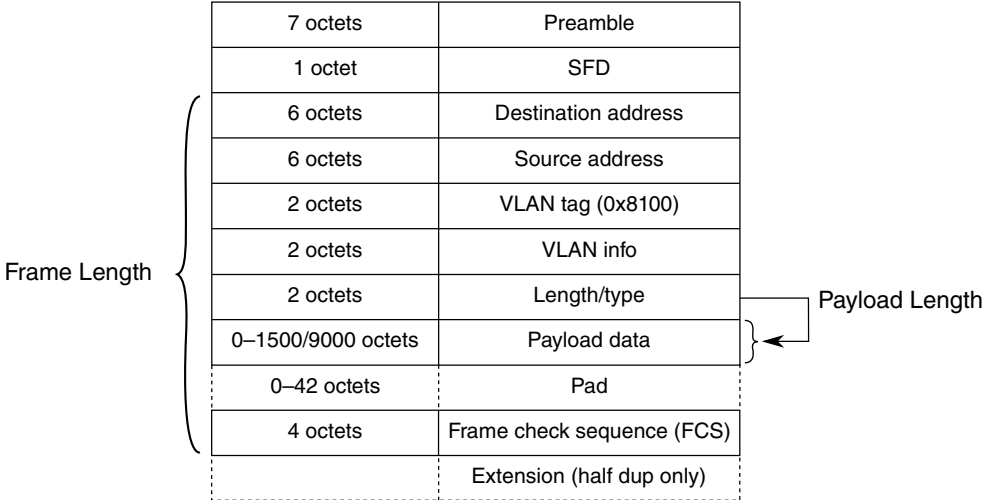


Figure 11-76. VLAN-tagged MAC frame format overview

Table 11-44. MAC frame definition

Term	Description
Frame length	Defines the length, in octets, of the complete frame without preamble and SFD. A frame has a valid length if it contains at least 64 octets and does not exceed the programmed maximum length.
Payload length	The length/type field indicates the length of the frame's payload section. The most significant byte is sent/received first. - If the length/type field is set to a value less than 46, the payload is padded so that the minimum frame length requirement (64 bytes) is met. For VLAN-tagged frames, a value less than 42 indicates a padded frame. - If the length/type field is set to a value larger than the programmed frame maximum length (e.g. 1518) it is interpreted as a type field.
Destination and source address	48-bit MAC addresses. The least significant byte is sent/received first and the first two least significant bits of the MAC address distinguish MAC frames, as detailed in MAC address check .

Note

Although the IEEE specification defines a maximum frame length, the MAC core provides the flexibility to program any value for the frame maximum length.

11.5.4.1.1 Pause Frames

The receiving device generates a pause frame to indicate a congestion to the emitting device, which should stop sending data.

Pause frames are indicated by the length/type set to 0x8808. The two first bytes of a pause frame following the type, defines a 16-bit opcode field set to 0x0001 always. A 16-bit pause quanta is defined in the frame payload bytes 2 (P1) and 3 (P2) as defined in the following table. The P1 pause quanta byte is the most significant.

Table 11-45. Pause Frame Format (Values in Hex)

1	2	3	4	5	6	7	8	9	10	11	12	13	14
55	55	55	55	55	55	55	D5	01	80	C2	00	00	01
Preamble							SFD	Multicast Destination Address					
15	16	17	18	19	20	21	22	23	24	25	26	27 –68	
00	00	00	00	00	00	88	08	00	01	hi	lo	00	
Source Address						Type		Opcode		P1	P2	pad (42)	
69	70	71	72										
26	6B	AE	0A										
CRC-32													

There is no payload length field found within a pause frame and a pause frame is always padded with 42 bytes (0x00).

If a pause frame with a pause value greater than zero (XOFF condition) is received, the MAC stops transmitting data as soon the current frame transfer is completed. The MAC stops transmitting data for the value defined in pause quanta. One pause quanta fraction refers to 512 bit times.

If a pause frame with a pause value of zero (XON condition) is received, the transmitter is allowed to send data immediately (see [Full-duplex flow control operation](#) for details).

11.5.4.1.2 Magic packets

A magic packet is a unicast, multicast, or broadcast packet, which carries a defined sequence in the payload section.

Magic packets are received and inspected only under specific conditions as described in [Magic packet detection](#).

The defined sequence to decode a magic packet is formed with a synchronization stream which consists of six consecutive 0xFF bytes, and is followed by sequence of sixteen consecutive unicast MAC addresses of the node to be awakened.

This sequence can be located anywhere in the magic packet payload. The magic packet is formed with a standard Ethernet header, optional padding, and CRC.

11.5.4.2 IP and higher layers frame format

The following sections use the term datagram to describe the protocol specific data unit that is found within the payload section of its container entity.

For example, an IP datagram specifies the payload section of an Ethernet frame. A TCP datagram specifies the payload section within an IP datagram.

11.5.4.2.1 Ethernet types

IP datagrams are carried in the payload section of an Ethernet frame. The Ethernet frame type/length field discriminates several datagram types.

The following table lists the types of interest:

Table 11-46. Ethernet type value examples

Type	Description
0x8100	VLAN-tagged frame. The actual type is found 4 octets later in the frame.
0x0800	IPv4
0x0806	ARP
0x86DD	IPv6

11.5.4.2.2 IPv4 datagram format

The following figure shows the IP Version 4 (IPv4) header, which is located at the beginning of an IP datagram. It is organized in 32-bit words. The first byte sent/received is the leftmost byte of the first word (in other words, version/IHL field).

The IP header can contain further options, which are always padded if necessary to guarantee the payload following the header is aligned to a 32-bit boundary.

The IP header is immediately followed by the payload, which can contain further protocol headers (for example, TCP or UDP, as indicated by the protocol field value). The complete IP datagram is transported in the payload section of an Ethernet frame.

Table 11-47. IPv4 header format

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Version				IHL				TOS								Length															
Fragment ID																Flags				Fragment offset											
TTL								Protocol								Header checksum															
Source address																															
Destination address																															
Options																															

Table 11-48. IPv4 header fields

Field name	Description
Version	4-bit IP version information. 0x4 for IPv4 frames.
IHL	4-bit Internet header length information. Determines number of 32-bit words found within the IP header. If no options are present, the default value is 0x5.
TOS	Type of service/DiffServ field.
Length	Total length of the datagram in bytes, including all octets of header and payload.
Fragment ID, flags, fragment offset	Fields used for IP fragmentation.
TTL	Time-to-live. In effect, is decremented at each router arrival. If zero, datagram must be discarded.
Protocol	Identifier of protocol that follows in the datagram.
Header checksum	Checksum of IP header. For computational purposes, this field's value is zero.
Source address	Source IP address.
Destination address	Destination IP address.

11.5.4.2.3 IPv6 datagram format

The following figure shows the IP version 6 (IPv6) header, which is located at the beginning of an IP datagram. It is organized in 32-bit words and has a fixed length of ten words (40 bytes). The next header field identifies the type of the header that follows the IPv6 header. It is defined similar to the protocol identifier within IPv4, with new definitions for identifying extension headers. These headers can be inserted between the IPv6 header and the protocol header, which will shift the protocol header accordingly. The accelerator currently only supports IPv6 without extension headers (in other words, the next header specifies TCP, UDP, or IMCP).

The first byte sent/received is the leftmost byte of the first word (in other words, version/traffic class fields).

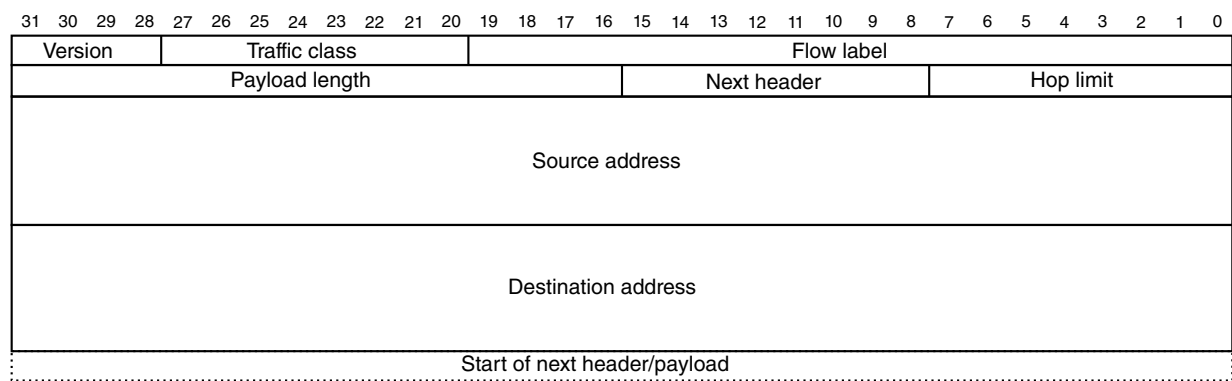


Figure 11-77. IPv6 header format

Table 11-49. IPv6 header fields

Field name	Description
Version	4-bit IP version information. 0x6 for all IPv6 frames.
Traffic class	8-bit field defining the traffic class.
Flow label	20-bit flow label identifying frames of the same flow.
Payload length	16-bit length of the datagram payload in bytes. It includes all octets following the IPv6 header.
Next header	Identifies the header that follows the IPv6 header. This can be the protocol header or any IPv6 defined extension header.
Hop limit	Hop counter, decremented by one by each station that forwards the frame. If hop limit is 0 the frame must be discarded.
Source address	128-bit IPv6 source address.
Destination address	128-bit IPv6 destination address.

11.5.4.2.4 Internet Control Message Protocol (ICMP) datagram format

An internet control message protocol (ICMP) is found following the IP header, if the protocol identifier is 1. The ICMP datagram has a four-octet header followed by additional message data.

Table 11-50. ICMP header format

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Type								Code								Checksum															
ICMP message data																															

Table 11-51. IP header fields

Field name	Description
Type	8-bit type information
Code	8-bit code that is related to the message type
Checksum	16-bit one's complement checksum over the complete ICMP datagram

11.5.4.2.5 User Datagram Protocol (UDP) datagram format

A user datagram protocol header is found after the IP header, when the protocol identifier is 17.

The payload of the datagram is after the UDP header. The header byte order follows the conventions given for the IP header above.

Table 11-52. UDP header format

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Source port																Destination port															
Length																Checksum															

Table 11-53. UDP header fields

Field name	Description
Source port	Source application port
Destination port	Destination application port
Length	Length of user data which immediately follows the header, including the UDP header (that is, minimum value is 8)
Checksum	Checksum over the complete datagram and some IP header information

11.5.4.2.6 TCP datagram format

A TCP header is found following the IP header, when the protocol identifier has a value of 6.

The TCP payload immediately follows the TCP header.

Table 11-54. TCP header format

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Source port																Destination port															
Sequence number																															
Acknowledgement number																															
Offset				Reserved				Flags				Window																			
Checksum																Urgent pointer															
Options																															

Table 11-55. TCP header fields

Field name	Description
Source port	Source application port
Destination port	Destination application port
Sequence number	Transmit sequence number
Ack. number	Receive sequence number
Offset	Data offset, which is number of 32-bit words within TCP header — if no options selected, defaults to value of 5
Flags	URG, ACK, PSH, RST, SYN, FIN flags
Window	TCP receive window size information
Checksum	Checksum over the complete datagram (TCP header and data) and IP header information
Options	Additional 32-bit words for protocol options

11.5.4.3 IEEE 1588 message formats

The following sections describe the IEEE 1588 message formats.

11.5.4.3.1 Transport encapsulation

The precision time protocol (PTP) datagrams are encapsulated in Ethernet frames using the UDP/IP transport mechanism, or optionally, with the newer 1588v2 directly in Ethernet frames (layer 2).

Typically, multicast addresses are used to allow efficient distribution of the synchronization messages.

11.5.4.3.1.1 UDP/IP

The 1588 messages (v1 and v2) can be transported using UDP/IP multicast messages.

Table 11-56 shows IP multicast groups defined for PTP. The table also shows their respective MAC layer multicast address mapping according to RFC 1112 (last three octets of IP follow the fixed value of 01-00-5E).

Table 11-56. UDP/IP multicast domains

Name	IP Address	MAC Address mapping
DefaultPTPdomain	224.0.1.129	01-00-5E-00-01-81
AlternatePTPdomain1	224.0.1.130	01-00-5E-00-01-82
AlternatePTPdomain2	224.0.1.131	01-00-5E-00-01-83
AlternatePTPdomain3	224.0.1.132	01-00-5E-00-01-84

Table 11-57. UDP port numbers

Message type	UDP port	Note
Event	319	Used for SYNC and DELAY_REQUEST messages
General	320	All other messages (for example, follow-up, delay-response)

11.5.4.3.1.2 Native Ethernet (PTPv2)

In addition to using UDP/IP frames, IEEE 1588v2 defines a native Ethernet frame format that uses ethertype = 0x88F7. The payload of the Ethernet frame immediately contains the PTP datagram, starting with the PTPv2 header.

Besides others, version 2 adds a peer delay mechanism to allow delay measurements between individual point-to-point links along a path over multiple nodes. The following multicast domains are also defined in PTPv2.

Table 11-58. PTPv2 multicast domains

Name	MAC address
Normal messages	01-1B-19-00-00-00
Peer delay messages	01-80-C2-00-00-0E

11.5.4.3.2 PTP header

All PTP frames contain a common header that determines the protocol version and the type of message, which defines the remaining content of the message.

All multi-octet fields are transmitted in big-endian order (the most significant byte is transmitted/received first).

The last four bits of versionPTP are at the same position (second byte) for PTPv1 and PTPv2 headers. This allows accurate identification by inspecting the first two bytes of the message.

11.5.4.3.2.1 PTPv1 header

Table 11-59. Common PTPv1 message header

Offset	Octets	Bits							
		7	6	5	4	3	2	1	0
0	2	versionPTP = 0x0001							
2	2	versionNetwork							
4	16	subdomain							
20	1	messageType							
21	1	sourceCommunicationTechnology							
22	6	sourceUuid							
28	2	sourcePortId							
30	2	sequenceId							
32	1	control							
33	1	0x00							
34	2	flags							
36	4	reserved							

The type of message is encoded in the messageType and control fields as shown in [Table 11-60](#) :

Table 11-60. PTPv1 message type identification

messageType	control	Message Name	Message
0x01	0x0	SYNC	Event message
0x01	0x1	DELAY_REQ	Event message
0x02	0x2	FOLLOW_UP	General message
0x02	0x3	DELAY_RESP	General message
0x02	0x4	MANAGEMENT	General message
other	other	—	Reserved

The field `sequenceId` is used to non-ambiguously identify a message.

11.5.4.3.2.2 PTPv2 header

Table 11-61. Common PTPv2 message header

Offset	Octets	Bits							
		7	6	5	4	3	2	1	0
0	1	transportSpecific				messageId			
1	1	reserved				versionPTP = 0x2			
2	2	messageLength							
4	1	domainNumber							
5	1	reserved							
6	2	flags							
8	8	correctionField							
16	4	reserved							
20	10	sourcePortIdentity							
30	2	sequenceId							
32	1	control							
33	1	logMeanMessageInterval							

The type of message is encoded in the field `messageId` as follows:

Table 11-62. PTPv2 message type identification

messageId	Message name	Message
0x0	SYNC	Event message
0x1	DELAY_REQ	Event message
0x2	PATH_DELAY_REQ	Event message
0x3	PATH_DELAY_RESP	Event message
0x4–0x7	—	Reserved
0x8	FOLLOW_UP	General message
0x9	DELAY_RESP	General message
0xa	PATH_DELAY_FOLLOW_UP	General message
0xb	ANNOUNCE	General message
0xc	SIGNALING	General message
0xd	MANAGEMENT	General message

The PTPv2 flags field contains further details on the type of message, especially if one-step or two-step implementations are used. The one- or two-step implementation is controlled by the `TWO_STEP` bit in the first octet of the flags field as shown below. Reserved bits are cleared.

Table 11-63. PTPv2 message flags field definitions

Bit	Name	Description
0	ALTERNATE_MASTER	See IEEE 1588 Clause 17.4
1	TWO_STEP	1 Two-step clock 0 One-step clock
2	UNICAST	1 Transport layer address uses a unicast destination address 0 Multicast is used
3	—	Reserved
4	—	Reserved
5	Profile specific	
6	Profile specific	
7	—	Reserved

11.5.4.4 MAC receive

The MAC receive engine performs the following tasks:

- Check frame framing
- Remove frame preamble and frame SFD field
- Discard frame based on frame destination address field
- Terminate pause frames
- Check frame length
- Remove payload padding if it exists
- Calculate and verify CRC-32
- Write received frames in the core receive FIFO

If the MAC is programmed to operate in half-duplex mode, it will also check if the frame is received with a collision.

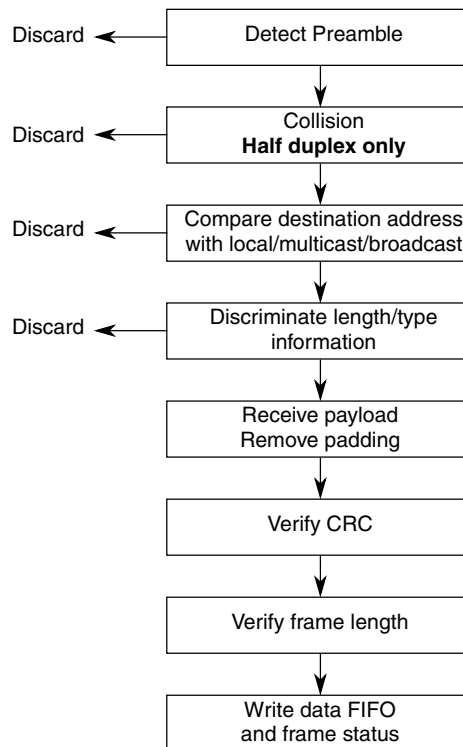


Figure 11-78. MAC receive flow

11.5.4.4.1 Collision detection in half-duplex mode

If the packet is received with a collision detected during reception of the first 64 bytes, the packet is discarded (if frame size was less than ~14 octets) or transmitted to the user application with an error and RxBD[CE] set.

11.5.4.4.2 Preamble processing

The IEEE 802.3 standard allows a maximum size of 56 bits (seven bytes) for the preamble, while the MAC core allows any preamble length, including zero length preamble.

The MAC core checks for the start frame delimiter (SFD) byte. If the next byte of the preamble, which is different from 0x55, is not 0xD5, the frame is discarded.

Although the IEEE specification dictates that the inner-packet gap should be at least 96 bits, the MAC core is designed to accept frames separated by only 64 10/100-Mbit/s operation (MII) bits.

The MAC core removes the preamble and SFD bytes.

11.5.4.4.3 MAC address check

The destination address bit 0 differentiates between multicast and unicast addresses.

- If bit 0 is 0, the MAC address is an individual (unicast) address.
- If bit 0 is 1, the MAC address defines a group (multicast) address.
- If all 48 bits of the MAC address are set, it indicates a broadcast address.

11.5.4.4.3.1 Unicast address check

If a unicast address is received, the destination MAC address is compared to the node MAC address programmed by the host in the PADDR1/2 registers.

If the destination address matches any of the programmed MAC addresses, the frame is accepted.

If Promiscuous mode is enabled (RCR[PROM] = 1) no address checking is performed and all unicast frames are accepted.

11.5.4.4.3.2 Multicast and unicast address resolution

The hash table algorithm used in the group and individual hash filtering operates as follows.

- The 48-bit destination address is mapped into one of 64 bits, represented by 64 bits in ENET n _GAUR/GALR (group address hash match) or ENET n _IAUR/IALR (individual address hash match).
- This mapping is performed by passing the 48-bit address through the on-chip 32-bit CRC generator and selecting the six most significant bits of the CRC-encoded result to generate a number between 0 and 63.
- The msb of the CRC result selects ENET n _GAUR (msb = 1) or ENET n _GALR (msb = 0).
- The five lsbs of the hash result select the bit within the selected register.
- If the CRC generator selects a bit set in the hash table, the frame is accepted; else, it is rejected.

For example, if eight group addresses are stored in the hash table and random group addresses are received, the hash table prevents roughly 56/64 (or 87.5%) of the group address frames from reaching memory. Those that do reach memory must be further filtered by the processor to determine if they truly contain one of the eight desired addresses.

The effectiveness of the hash table declines as the number of addresses increases.

The user must initialize the hash table registers. Use this CRC32 polynomial to compute the hash:

- $FCS(x) = x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x^1 + 1$

If Promiscuous mode is enabled ($ENETn_RCR[PROM] = 1$) all unicast and multicast frames are accepted regardless of $ENETn_GAUR/GALR$ and $ENETn_IAUR/IALR$ settings.

11.5.4.4.3.3 Broadcast address reject

All broadcast frames are accepted if BC_REJ is cleared or $ENETn_RCR[PROM]$ is set. If $PROM$ is cleared when $ENETn_RCR[BC_REJ]$ is set, all broadcast frames are rejected.

Table 11-64. Broadcast address reject programming

PROM	BC_REJ	Broadcast frames
0	0	Accepted
0	1	Rejected
1	0	Accepted
1	1	Accepted

11.5.4.4.3.4 Miss-bit implementation

For higher layer filtering purposes, $RxBD[M]$ indicates an address miss when the MAC operates in promiscuous mode and accepts a frame that would otherwise be rejected.

If a group/individual hash or exact match does not occur and Promiscuous mode is enabled ($RCR[PROM] = 1$), the frame is accepted and the M bit is set in the buffer descriptor; otherwise, the frame is rejected.

This means the status bit is set in any of the following conditions during Promiscuous mode:

- A broadcast frame is received when BC_REJ is set
- A unicast is received that does not match either:

- Node address (PALR[PADDR1] and PAUR[PADDR2])
- Hash table for unicast (IAUR[IADDR1] and IALR[IADDR2])
- A multicast is received that does not match the GAUR[GADDR1] and GALR[GADDR2] hash table entries

11.5.4.4.4 Frame length/type verification: payload length check

If the length/type is less than 0x600 and NLC is set, the MAC checks the payload length and reports any error in the frame status word and interrupt bit PLR.

If the length/type is greater than or equal to 0x600, the MAC interprets the field as a type and no payload length check is performed.

The length check is performed on VLAN and stacked VLAN frames. If a padded frame is received, no length check can be performed due to the extended frame payload because padded frames can never have a payload length error.

11.5.4.4.5 Frame length/type verification: frame length check

When the receive frame length exceeds MAX_FL bytes, the BABR interrupt is generated and the RxBD[LG] bit is set.

The frame is not truncated unless the frame length exceeds the value programmed in ENET_n_FTRL[TRUNC_FL]. If the frame is truncated, RxBD[TR] is set. In addition, a truncated frame always has the CRC error indication set (RxBD[CR]).

11.5.4.4.6 VLAN frames processing

VLAN frames have a length/type field set to 0x8100 immediately followed by a 16-Bit VLAN control information field.

VLAN-tagged frames are received as normal frames because the VLAN tag is not interpreted by the MAC function, and are pushed complete with the VLAN tag to the user application. If the length/type field of the VLAN-tagged frame, which is found four octets later in the frame, is less than 42, the padding is removed. In addition, the frame status word (RxBD[NO]) indicates that the current frame is VLAN tagged.

11.5.4.4.7 Pause frame termination

The receive engine terminates pause frames and does not transfer them to the receive FIFO. The quanta is extracted and sent to the MAC transmit path via a small internal clock rate decoupling asynchronous FIFO.

The quanta is written only if a correct CRC and frame length are detected by the control state machine. If not, the quanta is discarded and the MAC transmit path is not paused.

Good pause frames are ignored if $\text{ENET}_n\text{RCR}[\text{FCE}]$ is cleared and are forwarded to the client interface when $\text{ENET}_n\text{RCR}[\text{PAUFWD}]$ is set.

11.5.4.4.8 CRC check

The CRC-32 field is checked and forwarded to the core FIFO interface if $\text{ENET}_n\text{RCR}[\text{CRCFWD}]$ is cleared and $\text{ENET}_n\text{RCR}[\text{PADEN}]$ is cleared. When CRCFWD is set (regardless of PADEN), the CRC-32 field is checked and terminated (not transmitted to the FIFO).

The CRC polynomial, as specified in the 802.3 standard, is:

- $\text{FCS}(x) = x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x^1 + 1$

The 32 bits of the CRC value are placed in the frame check sequence (FCS) field with the x^{31} term as right-most bit of the first octet. The CRC bits are thus received in the following order: $x^{31}, x^{30}, \dots, x^1, x^0$.

If a CRC error is detected, the frame is marked invalid and $\text{RxBD}[\text{CR}]$ is set.

11.5.4.4.9 Frame padding removal

When a frame is received with a payload length field set to less than 46 (42 for VLAN-tagged frames and 38 for frames with stacked VLANs), the zero padding can be removed before the frame is written into the data FIFO depending on the setting of $\text{ENET}_n\text{RCR}[\text{PADEN}]$.

Note

If a frame is received with excess padding (in other words, the length field is set as mentioned above, but the frame has more than 64 octets) and padding removal is enabled, then the padding is removed as normal and no error is reported if the frame is otherwise correct (for example: good CRC, less than maximum length, and no other error).

11.5.4.4.10 Frame classification (AVB)

To support protocols such as Audio Video Bridging (AVB, IEEE 802.1Qav), normal traffic is separated from time-sensitive traffic immediately at the application interface to allow storing them in different queues for further processing. For every frame received, a

classification based on VLAN priority can be performed. When a frame is received, and it contains a VLAN tag, the priority is compared against the values set in the classification match registers (see [Receive Classification Match Register for Class n \(ENET_RCMRn\)](#)) and the following figure.

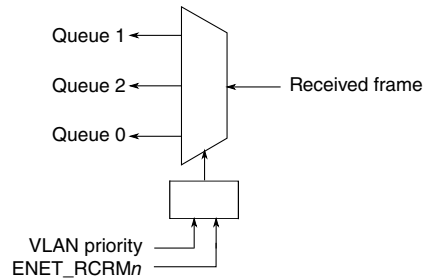


Figure 11-79. AVB frame classification

11.5.4.4.11 Receive flushing

RX flushing prevents frames in the RX FIFO from being blocked. Blocking can occur if the frame at the head of the RX FIFO cannot be forwarded because the ring it is associated with cannot accept it. This situation occurs when either the ring's `RxBD[EMPTY]` is not set or `ENET_RDARn` is not set. When RX flushing is enabled, via [QOS Scheme \(ENET_QOS\)](#), the blocking frame will be flushed (discarded).

11.5.4.5 MAC transmit

Frame transmission starts when the transmit FIFO holds enough data.

After a transfer starts, the MAC transmit function performs the following tasks:

- Generates preamble and SFD field before frame transmission
- Generates XOFF pause frames if the receive FIFO reports a congestion or if `ENETn_TCR[TFC_PAUSE]` is set with `ENETn_OPD[PAUSE_DUR]` set to a non-zero value
- Generates XON pause frames if the receive FIFO congestion condition is cleared or if `TFC_PAUSE` is set with `PAUSE_DUR` cleared
- Suspends Ethernet frame transfer (XOFF) if a non-zero pause quanta is received from the MAC receive path
- Adds padding to the frame if required

- Calculates and appends CRC-32 to the transmitted frame
- Sends the frame with correct inter-packet gap (IPG) (deferring)

When the MAC is configured to operate in half-duplex mode, the following additional tasks are performed:

- Collision detection
- Frame retransmit after back-off timer expires

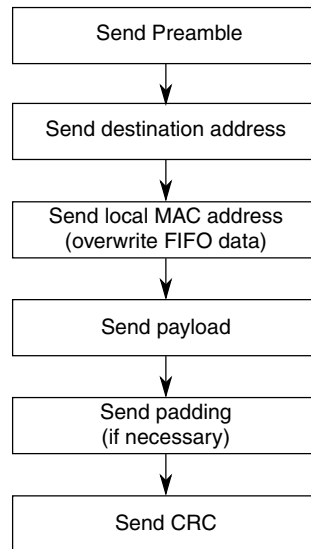


Figure 11-80. Frame transmit overview

11.5.4.5.1 Frame payload padding

The IEEE specification defines a minimum frame length of 64 bytes.

If the frame sent to the MAC from the user application has a size smaller than 60 bytes, the MAC automatically adds padding bytes (0x00) to comply with the Ethernet minimum frame length specification. Transmit padding is always performed and cannot be disabled.

If the MAC is not allowed to append a CRC (TxBD[TC] = 1), the user application is responsible for providing frames with a minimum length of 64 octets.

11.5.4.5.2 MAC address insertion

On each frame received from the core transmit FIFO interface, the source MAC address is either:

- Replaced by the address programmed in the PADDR1/2 fields (ENET n _TCR[ADDINS] = 1)
- Transparently forwarded to the Ethernet line (ENET n _TCR[ADDINS] = 0)

11.5.4.5.3 CRC-32 generation

The CRC-32 field is optionally generated and appended at the end of a frame.

The CRC polynomial, as specified in the 802.3 standard, is:

- $FCS(x) = x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x^1 + 1$

The 32 bits of the CRC value are placed in the FCS field so that the x^{31} term is the right-most bit of the first octet. The CRC bits are thus transmitted in the following order: x^{31} , x^{30} , ..., x^1 , x^0 .

11.5.4.5.4 Inter-packet gap (IPG)

In full-duplex mode, after frame transmission and before transmission of a new frame, an IPG (programmed in ENET n _TIPG) is maintained. The minimum IPG can be programmed between 8 and 26 byte-times (64 and 208 bit-times).

In half-duplex mode, the core constantly monitors the line. Actual transmission of the data onto the network occurs only if it has been idle for a 96-bit time period, and any back-off time requirements have been satisfied. In accordance with the standard, the core begins to measure the IPG from CRS/GMII_CRS de-assertion.

11.5.4.5.5 Collision detection and handling — half-duplex operation only

A collision occurs on a half-duplex network when concurrent transmissions from two or more nodes take place. During transmission, the core monitors the line condition and detects a collision when the PHY device asserts COL.

When the core detects a collision while transmitting, it stops transmission of the data and transmits a 32-bit jam pattern. If the collision is detected during the preamble or the SFD transmission, the jam pattern is transmitted after completing the SFD, which results in a minimum 96-bit fragment. The jam pattern is a fixed pattern that is not compared to the actual frame CRC, and has a very low probability (0.532) of having a jam pattern identical to the CRC.

If a collision occurs before transmission of 64 bytes (including preamble and SFD), the MAC core waits for the backoff period and retransmits the packet data (stored in a 64-byte re-transmit buffer) that has already been sent on the line. The backoff period is generated from a pseudo-random process (truncated binary exponential backoff).

If a collision occurs after transmission of 64 bytes (including preamble and SFD), the MAC discards the remainder of the frame, optionally sets the LC interrupt bit, and sets TxBD[LCE].

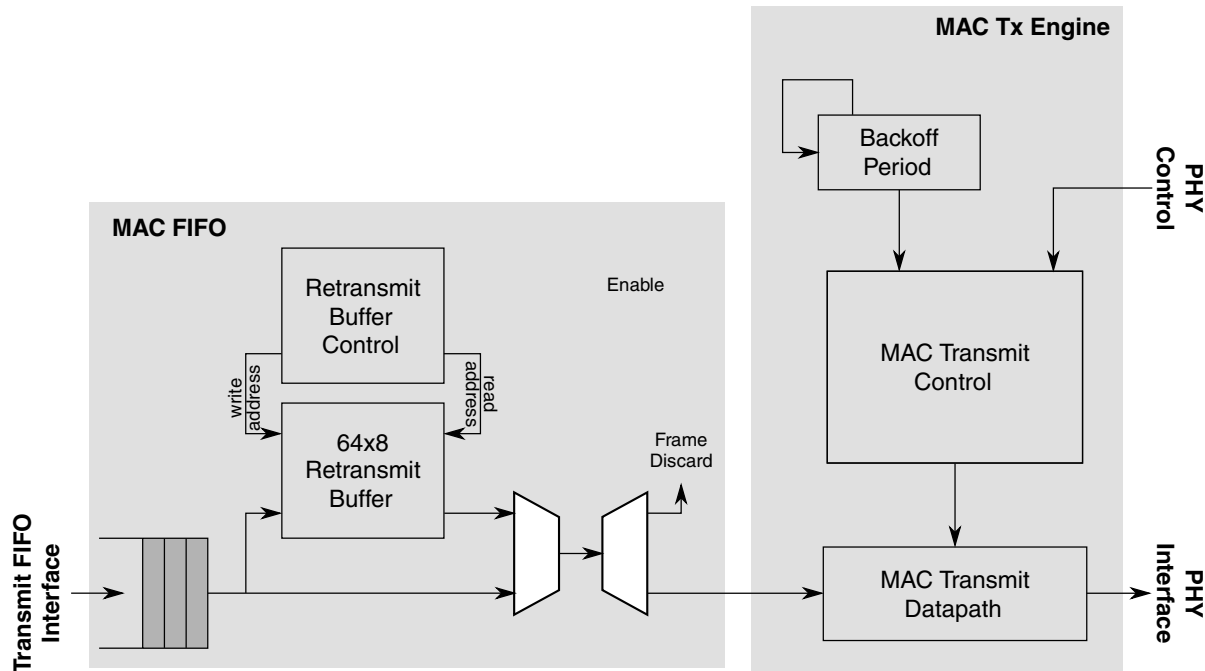


Figure 11-81. Packet re-transmit overview

The backoff time is represented by an integer multiple of slot times. One slot is equal to a 512-bit time period. The number of the delay slot times, before the n^{th} re-transmission attempt, is chosen as a uniformly-distributed random integer in the range:

- $0 < r < 2^k$
- $k = \min(n, N)$; where n is the number of retransmissions and $N = 10$

For example, after the first collision, the backoff period is 0 or 1 slot time. If a collision occurs on the first retransmission, the backoff period is 0, 1, 2, or 3, and so on.

The maximum backoff time (in 512-bit time slots) is limited by $N = 10$ as specified in the IEEE 802.3 standard.

If a collision occurs after 16 consecutive retransmissions, the core reports an excessive collision condition (ENET n _EIR[RL] interrupt field and TxBD[EE]) and discards the current packet from the FIFO.

In networks violating the standard requirements, a collision may occur after transmission of the first 64 bytes. In this case, the core stops the current packet transmission and discards the rest of the packet from the transmit FIFO. The core resumes transmission with the next packet available in the core transmit FIFO.

warning

Ethernet PHYs that support the SQE Test, or "heartbeat," feature must disable this feature. When this feature is enabled, the PHY asserts the collision signal after a frame is transmitted to indicate to the ENET that the PHY's collision logic is working. This may cause data corruption in the next frame from the ENET. This corrupted frame contains up to 21 zero bytes which start somewhere within the MAC destination address field. The ENET, however, will still generate a good FCS (CRC-32) but with corrupted data.

11.5.4.5.6 Rate limiting / traffic shaping support

The MAC-NET supports two methods to optimize frame traffic for either time-sensitive AVB frames (Class A and Class B) or best-effort non-AVB frames:

- Round-robin scheme
- Credit-based traffic shaping

To ensure that sufficient bandwidth is allocated for the AVB frames, the credit-based shaper must be used. Either method can be combined with a time-based shaper to ensure that a frame is always transmitted in its correct time slot.

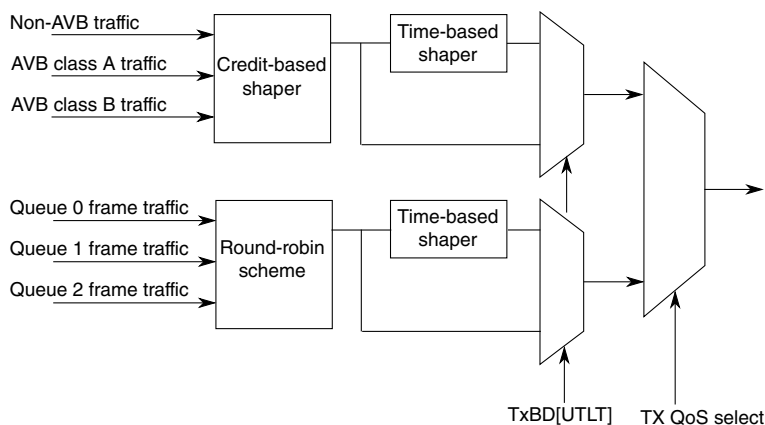


Figure 11-82. Transmit arbitration schemes

11.5.4.5.6.1 Round-robin policy

In the round-robin QoS scheme, each queue is given an equal opportunity to transmit one frame. For example, if queue n has a frame to transmit, the queue transmits its frame. After queue n has transmitted its frame, or if queue n does not have a frame to transmit, queue $n+1$ is then allowed to transmit its frame, and so on.

11.5.4.5.6.2 Credit-based shaper

The AVB credit based shaper acts independently, per class, to control the bandwidth distribution between normal traffic and time-sensitive traffic with respect to the total link bandwidth available. As per the IEEE 802.1Q, the maximum bandwidth distribution that can be allocated for the time-sensitive frames is 75%. The following example uses 70% for AVB frames and 30% for non-AVB frames:

Table 11-65. Bandwidth allocation example

AVB Class A	50%
AVB Class B	20%
non-AVB frames	30%
Total	100%

See [DMA Class Based Configuration \(ENET_DMA \$n\$ CFG\)](#) for information on how to allocate bandwidth. The following figure shows how frame traffic is controlled within the MAC-NET core.

11.5.4.5.6.3 Time-based shaper

The time-based shaper enables the user to specify when a frame can be transmitted. It is always used in combination with either the round-robin scheme or the credit-based shaper. Use of the time-based shaper can ensure that frames are always transmitted in the correct time slot.

11.5.4.5.6.3.1 Time-based shaper example

Time-based shaping involves the following procedure:

1. Read the current value of the timer. See [Timer Value Register \(ENET_ATVR\)](#) and [Adjustable Timer Control Register \(ENET_ATCR\)](#).
2. Calculate the frame launch time and write this value to the TLT field of the frame's TxBD. See [Enhanced transmit buffer descriptor](#).
3. Instruct the ENET to use the TLT by setting TxBD[UTLT].

The frame will be fetched and transmitted only if TxBD[TLT] is less than the current value of the timer. In other words, the timer must be past, that is, be greater than, the transmit launch time before the frame will be fetched and transmitted.

The transmit launch time must be not be greater than the current value of ENET_ATVR + (0.5 × ENET_ATPER). This means the application can not prepare frames with launch times beyond ENET_ATVR + (0.5 × ENET_ATPER). Because ENET_ATVR wraps, calculate TLT using the equation

$$(ENET_ATVR + (0.5 \times ENET_ATPER)) \bmod ENET_ATPER$$

For example, if ENET_ATPER is set to the recommended value of 1,000,000,000 ns (one second), then the user must not prepare future frames with launch times greater than (500,000,000 + ENET_ATVR).

As a specific example, if the current value of ENET_ATVR is 100,000, then you can prepare the following frames:

1. Frame 1 TxBD[TLT] = 100,000ns + 125,000ns
2. Frame 2 TxBD[TLT] = 100,000ns + 250,000ns
3. And so on

11.5.4.6 Full-duplex flow control operation

Three conditions are handled by the core's flow control engine:

- Remote device congestion — The remote device connected to the same Ethernet segment as the core reports congestion and requests that the core stop sending data.
- Core FIFO congestion — When the core's receive FIFO reaches a user-programmable threshold (RX section empty), the core sends a pause frame back to the remote device requesting the data transfer to stop.
- Local device congestion — Any device connected to the core can request (typically, via the host processor) the remote device to stop transmitting data.

11.5.4.6.1 Remote device congestion

When the MAC transmit control gets a valid pause quanta from the receive path and if ENET_n_RCR[FCE] is set, the MAC transmit logic:

- Completes the transfer of the current frame.

- Stops sending data for the amount of time specified by the pause quanta in 512 bit time increments.
- Sets ENET n _TCR[RFC_PAUSE].

Frame transfer resumes when the time specified by the quanta expires and if no new quanta value is received, or if a new pause frame with a quanta value set to 0x0000 is received. The MAC also resets RFC_PAUSE to zero.

If ENET n _RCR[FCE] cleared, the MAC ignores received pause frames.

Optionally and independent of ENET n _RCR[FCE], pause frames are forwarded to the client interface if PAUFWD is set.

11.5.4.6.2 Local device/FIFO congestion

The MAC transmit engine generates pause frames when the local receive FIFO is not able to receive more than a pre-defined number of words (FIFO programmable threshold) or when pause frame generation is requested by the local host processor.

- To generate a pause frame, the host processor sets ENET n _TCR[TFC_PAUSE]. A single pause frame is generated when the current frame transfer is completed and TFC_PAUSE is automatically cleared. Optionally, an interrupt is generated.
- An XOFF pause frame is generated when the receive FIFO asserts its section empty flag (internal). An XOFF pause frame is generated automatically, when the current frame transfer completes.
- An XON pause frame is generated when the receive FIFO deasserts its section empty flag (internal). An XON pause frame is generated automatically, when the current frame transfer completes.

When an XOFF pause frame is generated, the pause quanta (payload byte P1 and P2) is filled with the value programmed in ENET n _OPD[PAUSE_DUR].

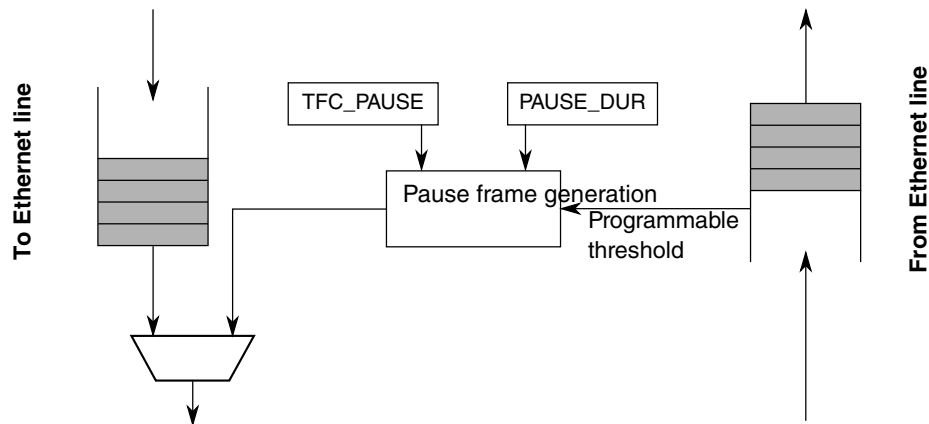


Figure 11-83. Pause frame generation overview

Note

Although the flow control mechanism should prevent any FIFO overflow on the MAC core receive path, the core receive FIFO is protected. When an overflow is detected on the receive FIFO, the current frame is truncated with an error indication set in the frame status word. The frame should subsequently be discarded by the user application.

11.5.4.7 Magic packet detection

Magic packet detection wakes a node that is put in power-down mode by the node management agent. Magic packet detection is supported only if the MAC is configured in sleep mode.

11.5.4.7.1 Sleep mode

To put the MAC in Sleep mode, set `ENETn_ECR[SLEEP]`. At the same time `ENETn_ECR[MAGICEN]` should also be set to enable magic packet detection.

In addition, if ENET is enabled, write 1 to `ENETn_ECR[SLEEP]` before entering into low power mode.

When the MAC is in Sleep mode:

- The transmit logic is disabled.
- The FIFO receive/transmit functions are disabled.
- The receive logic is kept in Normal mode, but it ignores all traffic from the line except magic packets. They are detected so that a remote agent can wake the node.

11.5.4.7.2 Magic packet detection

The core is designed to detect magic packets (see [Magic packets](#)) with the destination address set to:

- Any multicast address
- The broadcast address
- The unicast address programmed in PADDR1/2

When a magic packet is detected, EIR[WAKEUP] is set and none of the statistic registers are incremented.

11.5.4.7.3 Wakeup

When a magic packet is detected, indicated by ENET n _EIR[WAKEUP], ENET n _ECR[SLEEP] should be cleared to resume normal operation of the MAC. Clearing the SLEEP bit automatically masks ENET n _ECR[MAGICEN], disabling magic packet detection.

11.5.4.8 IP accelerator functions

The following sections describe the IP accelerator functions.

11.5.4.8.1 Checksum calculation

The IP and ICMP, TCP, UDP checksums are calculated with one's complement arithmetic summing up 16-bit values.

- For ICMP, the checksum is calculated over the complete ICMP datagram, in other words without IP header.
- For TCP and UDP, the checksums contain the header and data sections and values from the IP header, which can be seen as a pseudo-header that is not actually present in the data stream.

Table 11-66. IPv4 pseudo-header for checksum calculation

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Source address																															
Destination address																															
Zero								Protocol								TCP/UDP length															

Table 11-67. IPv6 pseudo-header for checksum calculation

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Source address																															
Destination address																															
TCP/UDP length																															
Zero																Next header															

The TCP/UDP length value is the length of the TCP or UDP datagram, which is equal to the payload of an IP datagram. It is derived by subtracting the IP header length from the complete IP datagram length that is given in the IP header (IPv4), or directly taken from the IP header (IPv6). The protocol field is the corresponding value from the IP header. The Zero fields are all zeroes.

For IPv6, the complete 128-bit addresses are considered. The next header value identifies the upper layer protocol as either TCP or UDP. It may differ from the next header value of the IPv6 header if extension headers are inserted before the protocol header.

The checksum calculation uses 16-bit words in network byte order: The first byte sent/received is the MSB, and the second byte sent/received is the LSB of the 16-bit value to add to the checksum. If the frame ends on an odd number of bytes, a zero byte is appended for checksum calculation only, and is not actually transmitted.

11.5.4.8.2 Additional padding processing

According to IEEE 802.3, any Ethernet frame must have a minimum length of 64 octets.

The MAC usually removes padding on receive when a frame with length information is received. Because IP frames have a type value instead of length, the MAC does not remove padding for short IP frames, as it is not aware of the frame contents.

The IP accelerator function can be configured to remove the Ethernet padding bytes that might follow the IP datagram.

On transmit, the MAC automatically adds padding as necessary to fill any frame to a 64-byte length.

11.5.4.8.3 32-bit Ethernet payload alignment

The data FIFOs allow inserting two additional arbitrary bytes in front of a frame. This extends the 14-byte Ethernet header to a 16-byte header, which leads to alignment of the Ethernet payload, following the Ethernet header, on a 32-bit boundary.

This function can be enabled for transmit and receive independently with the corresponding SHIFT16 bits in the ENET n _TACC and ENET n _RACC registers.

When enabled, the valid frame data is arranged as shown in [Table 11-68](#).

Table 11-68. 64-bit interface data structure with SHIFT16 enabled

63 56	55 48	47 40	39 32	31 24	23 16	15 8	7 0
Byte 5	Byte 4	Byte 3	Byte 2	Byte 1	Byte 0	Any value	Any value
Byte 13	Byte 12	Byte 11	Byte 10	Byte 9	Byte 8	Byte 7	Byte 6
...							

11.5.4.8.3.1 Receive processing

When ENET n _RACC[SHIFT16] is set, each frame is received with two additional bytes in front of the frame.

The user application must ignore these first two bytes and find the first byte of the frame in bits 23–16 of the first word from the RX FIFO.

Note

SHIFT16 must be set during initialization and kept set during the complete operation, because it influences the FIFO write behavior.

11.5.4.8.3.2 Transmit processing

When ENET n _TACC[SHIFT16] is set, the first two bytes of the first word written (bits 15–0) are discarded immediately by the FIFO write logic.

The SHIFT16 bit can be enabled/disabled for each frame individually if required, but can be changed only between frames.

11.5.4.8.4 Received frame discard

Because the receive FIFO must be operated in store and forward mode (ENET n _RSFL cleared), received frames can be discarded based on the following errors:

- The MAC function receives the frame with an error:
 - The frame has an invalid payload length
 - Frame length is greater than MAX_FL

- Frame received with a CRC-32 error
- Frame truncated due to receive FIFO overflow
- Frame is corrupted as PHY signaled an error (RX_ERR asserted during reception)
- An IP frame is detected and the IP header checksum is wrong
- An IP frame with a valid IP header and a valid IP header checksum is detected, the protocol is known but the protocol-specific checksum is wrong

If one of the errors occurs and the IP accelerator function is configured to discard frames (ENET n _RACC), the frame is automatically discarded. Statistics are maintained normally and are not affected by this discard function.

11.5.4.8.5 IPv4 fragments

When an IPv4 IP fragment frame is received, only the IP header is inspected and its checksum verified. 32-bit alignment operates the same way on fragments as it does on normal IP frames, as specified above.

The IP fragment frame payload is not inspected for any protocol headers. As such, a protocol header would only exist in the very first fragment. To assist in protocol-specific checksum verification, the one's-complement sum is calculated on the IP payload (all bytes following the IP header) and provided with the frame status word.

The frame fragment status field (RxB D [FRAG]) is set to indicate a fragment reception, and the one's-complement sum of the IP payload is available in RxB D [Payload checksum].

Note

After all fragments have been received and reassembled, the application software can take advantage of the payload checksum delivered with the frame's status word to calculate the protocol-specific checksum of the datagram.

For example, if a TCP payload is delivered by multiple IP fragments, the application software can calculate the pseudo-header checksum value from the first fragment, and add the payload checksums delivered with the status for all fragments to verify the TCP datagram checksum.

11.5.4.8.6 IPv6 support

The following sections describe the IPv6 support.

11.5.4.8.6.1 Receive processing

An Ethernet frame of type 0x86DD identifies an IP Version 6 frame (IPv6) frame. If an IPv6 frame is received, the first IP header is inspected (first ten words), which is available in every IPv6 frame.

If the receive SHIFT16 function is enabled, the IP header is aligned on a 32-bit boundary allowing more efficient processing (see [32-bit Ethernet payload alignment](#)).

For TCP and UDP datagrams, the pseudo-header checksum calculation is performed and verified.

To assist in protocol-specific checksum verification, the one's-complement sum is always calculated on the IP payload (all bytes following the IP header) and provided with the frame status word. For example, if extension headers were present, their sums can be subtracted in software from the checksum to isolate the TCP/UDP datagram checksum, if required.

11.5.4.8.6.2 Transmit processing

For IPv6 transmission, the SHIFT16 function is supported to process 32-bit aligned datagrams.

IPv6 has no IP header checksum; therefore, the IP checksum insertion configuration is ignored.

The protocol checksum is inserted only if the next header of the IP header is a known protocol (TCP, UDP, or ICMP). If a known protocol is detected, the checksum over all bytes following the IP header is calculated and inserted in the correct position.

The pseudo-header checksum calculation is performed for TCP and UDP datagrams accordingly.

11.5.4.9 Resets and stop controls

The following sections describe the resets and stop controls.

11.5.4.9.1 Hardware reset

To reset the Ethernet module, set ENET n _ECR[RESET].

11.5.4.9.2 Soft reset

When ENET n _ECR[ETHER_EN] is cleared during operation, the following occurs:

- uDMA, buffer descriptor, and FIFO control logic are reset, including the buffer descriptor and FIFO pointers.
- A currently ongoing transmit is terminated by asserting GMII/TXER to the PHY.
- A currently ongoing transmit FIFO write from the application is terminated by stopping the write to the FIFO, and all further data from the application is ignored. All subsequent writes are ignored until re-enabled.
- A currently ongoing receive FIFO read is terminated. The RxBD has arbitrary values in this case.

11.5.4.9.3 Hardware freeze

When the processor enters debug mode and ECR[DBGEN] is set, the MAC enters a freeze state where it stops all transmit and receive activities gracefully.

The following happens when the MAC enters hardware freeze:

- A currently ongoing receive transaction on the receive application interface is completed as normal. No further frames are read from the FIFO.
- A currently ongoing transmit transaction on the transmit application interface is completed as normal (in other words, until writing end-of-packet (EOP)).
- A currently ongoing frame receive is completed normally, after which no further frames are accepted from the MII/GMII.
- A currently ongoing frame transmit is completed normally, after which no further frames are transmitted.

11.5.4.9.4 Graceful stop

During a graceful stop, any currently ongoing transactions are completed normally and no further frames are accepted. The MAC can resume from a graceful stop without the need for a reset (for example, clearing ETHER_EN is not required).

The following conditions lead to a graceful stop of the MAC transmit or receive datapaths.

11.5.4.9.4.1 Graceful transmit stop (GTS)

When gracefully stopped, the MAC is no longer reading frame data from the transmit FIFO and has completed any ongoing transmission.

In any of the following conditions, the transmit datapath stops after an ongoing frame transmission has been completed normally.

- ENET n _TCR[GTS] is set by software.
- ENET n _TCR[TFC_PAUSE] is set by software requesting a pause frame transmission. The status (and register bit) is cleared after the pause frame has been sent.
- A pause frame was received stopping the transmitter. The stopped situation is terminated when the pause timer expires or a pause frame with zero quanta is received.
- MAC is placed in Sleep mode by software or the processor entering Stop mode (see [Sleep mode](#)).
- The MAC is in Hardware Freeze mode.

When the transmitter has reached its stopped state, the following events occur:

- The GRA interrupt is asserted, when transitioned into stopped.
- In Hardware Freeze mode, the GRA interrupt does not wait for the application write completion and asserts when the transmit state machine (in other words, line side of TX FIFO) reaches its stopped state.

11.5.4.9.4.2 Graceful receive stop (GRS)

When gracefully stopped, the MAC is no longer writing frames into the receive FIFO.

The receive datapath stops after any ongoing frame reception has been completed normally, if any of the following conditions occur:

- MAC is placed in Sleep mode either by the software or the processor is in Stop mode). The MAC continues to receive frames and search for magic packets if enabled (see [Magic packet detection](#)). However, no frames are written into the receive FIFO, and therefore are not forwarded to the application.
- The MAC is in Hardware Freeze mode. The MAC does not accept any frames from the MII/GMII.

When the receive datapath is stopped, the following events occur:

- If the RX is in the stopped state, RCR[GRS] is set
- The GRA interrupt is asserted when the transmitter and receiver are stopped
- Any ongoing receive transaction to the application (RX FIFO read) continues normally until the frame end of package (EOP) is reached. After this, the following occurs:
 - When Sleep mode is active, all further frames are discarded, flushing the RX FIFO
 - In Hardware Freeze mode, no further frames are delivered to the application and they stay in the receive FIFO.

Note

The assertion of GRS does not wait for an ongoing FIFO read transaction on the application side of the FIFO (FIFO read).

11.5.4.9.4.3 Graceful stop interrupt (GRA)

The graceful stopped interrupt (GRA) is asserted for the following conditions:

- In Sleep mode, the interrupt asserts only after both TX and RX datapaths are stopped.
- In Hardware Freeze mode, the interrupt asserts only after both TX and RX datapaths are stopped.
- The MAC transmit datapath is stopped for any other condition (GTS, TFC_PAUSE, pause received).

The GRA interrupt is triggered only once when the stopped state is entered. If the interrupt is cleared while the stop condition persists, no further interrupt is triggered.

11.5.4.10 IEEE 1588 functions

To allow for IEEE 1588 or similar time synchronization protocol implementations, the MAC is combined with a time-stamping module to support precise time-stamping of incoming and outgoing frames. Set `ENETn_ECR[EN1588]` to enable 1588 support.

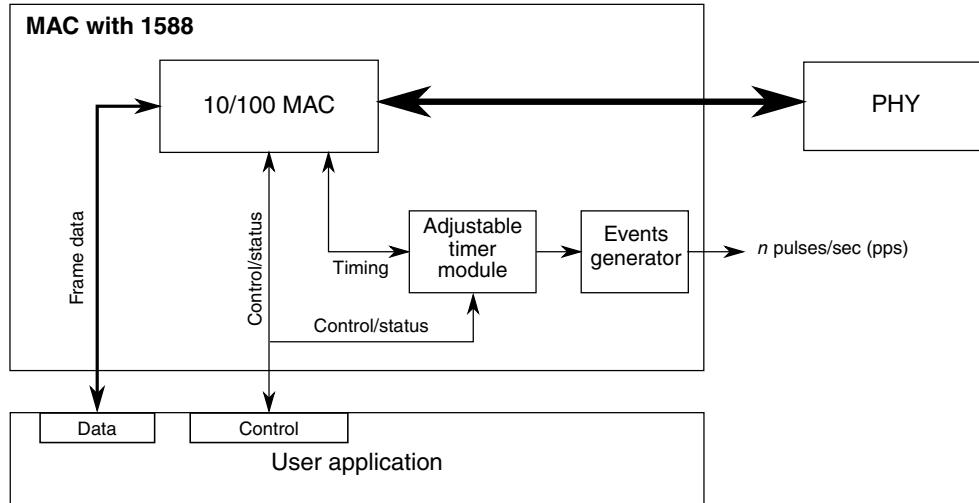


Figure 11-84. IEEE 1588 functions overview

11.5.4.10.1 Adjustable timer module

The adjustable timer module (TSM) implements the free-running counter (FRC), which generates the timestamps. The FRC operates with the time-stamping clock, which can be set to any value depending on your system requirements.

Through dedicated correction logic, the timer can be adjusted to allow synchronization to a remote master and provide a synchronized timing reference to the local system. The timer can be configured to cause an interrupt after a fixed time period, to allow synchronization of software timers or perform other synchronized system functions.

The timer is typically used to implement a period of one second; hence, its value ranges from 0 to $(1 \times 10^9) - 1$. The period event can trigger an interrupt, and software can maintain the seconds and hours time values as necessary.

11.5.4.10.1.1 Adjustable timer implementation

The adjustable timer consists of a programmable counter/accumulator and a correction counter. The periods of both counters and their increment rates are freely configurable, allowing very fine tuning of the timer.

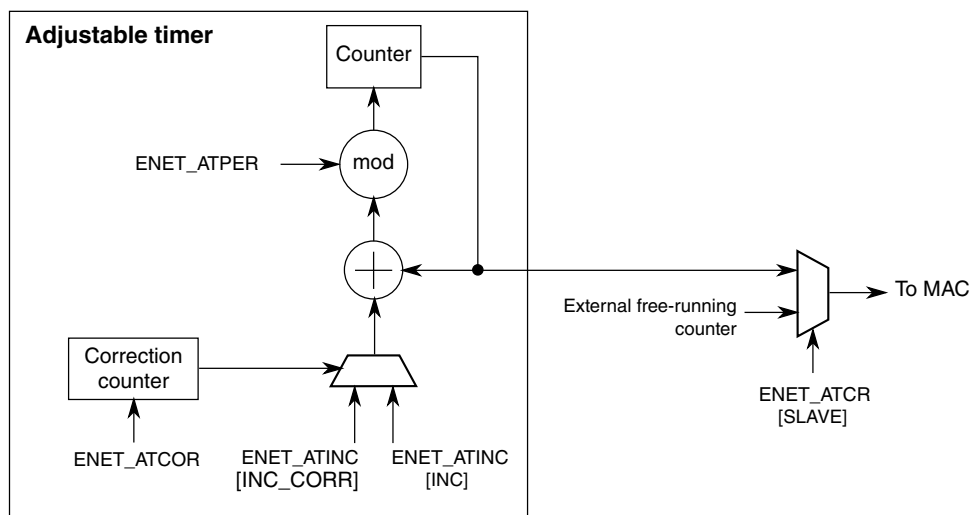


Figure 11-85. Adjustable timer implementation detail

The counter produces the current time. During each time-stamping clock cycle, a constant value is added to the current time as programmed in $\text{ENET}_n\text{_ATINC}$. The value depends on the chosen time-stamping clock frequency. For example, if it operates at 125 MHz, setting the increment to eight represents 8 ns.

The period, configured in $\text{ENET}_n\text{_ATPER}$, defines the modulo when the counter wraps. In a typical implementation, the period is set to 1×10^9 so that the counter wraps every second, and hence all timestamps represent the absolute nanoseconds within the one second period. When the period is reached, the counter wraps to start again respecting the period modulo. This means it does not necessarily start from zero, but instead the counter is loaded with the value $(\text{Current} + \text{Inc} - (1 \times 10^9))$, assuming the period is set to 1×10^9 .

The correction counter operates fully independently, and increments by one with each time-stamping clock cycle. When it reaches the value configured in $\text{ENET}_n\text{_ATCOR}$, it restarts and instructs the timer once to increment by the correction value, instead of the normal value.

The normal and correction increments are configured in $\text{ENET}_n\text{_ATINC}$. To speed up the timer, set the correction increment more than the normal increment value. To slow down the timer, set the correction increment less than the normal increment value.

The correction counter only defines the distance of the corrective actions, not the amount. This allows very fine corrections and low jitter (in the range of 1 ns) independent of the chosen clock frequency.

By enabling slave mode ($\text{ENET}_n\text{_ATCR}[\text{SLAVE}] = 1$), the timer is ignored and the current time is externally provided from one of the external modules. See the Chip Configuration details for which clock source is used. This is useful if multiple modules

within the system must operate from a single timer. When slave mode is enabled, you still must set `ENET $n$ _ATINC[INC]` to the value of the master, since it is used for internal comparisons.

11.5.4.10.2 Transmit timestamping

Only 1588 event frames need to be time-stamped on transmit. The client application (for example, the MAC driver) should detect 1588 event frames and set `TxBD[TS]` together with the frame.

If `TxBD[TS]` is set, the MAC records the timestamp for the frame in `ENET $n$ _ATSTMP`. `ENET $n$ _EIR[TS_AVAIL]` is set to indicate that a new timestamp is available.

Software implements a handshaking procedure by setting `TxBD[TS]` when it transmits the frame for which a timestamp is needed, and then waits for `ENET $n$ _EIR[TS_AVAIL]` to determine when the timestamp is available. The timestamp is then read from `ENET $n$ _ATSTMP`. This is done for all event frames. Other frames do not use `TxBD[TS]` and, therefore, do not interfere with the timestamp capture.

11.5.4.10.3 Receive timestamping

When a frame is received, the MAC latches the value of the timer when the frame's start of frame delimiter (SFD) field is detected, and provides the captured timestamp on `RxBD[1588 timestamp]`. This is done for all received frames.

11.5.4.10.4 Time synchronization

The adjustable timer module is available to synchronize the local clock of a node to a remote master. It implements a free running 32-bit counter, and also contains an additional correction counter.

The correction counter increases or decreases the rate of the free running counter, enabling very fine granular changes of the timer for synchronization, yet adding only very low jitter when performing corrections.

The application software implements, in a slave scenario, the required control algorithm, setting the correction to compensate for local oscillator drifts and locking the timer to the remote master clock on the network.

The timer and all timestamp-related information should be configured to show the true nanoseconds value of a second (in other words, the timer is configured to have a period of one second). Hence, the values range from 0 to $(1 \times 10^9) - 1$. In this application, the seconds counter is implemented in software using an interrupt function that is executed when the nanoseconds counter wraps at 1×10^9 .

11.5.4.10.5 Input Capture and Output Compare

The Input Capture Output Compare block can be used to provide precise hardware timing for input and output events.

11.5.4.10.5.1 Input capture

The $TCCR_n$ capture registers latch the time value when the corresponding external event occurs. An event can be a rising-, falling-, or either-edge of one of the 1588_TMR $_n$ signals. An event will cause the corresponding TCSR $_n$ [TF] timer flag to be set, indicating that an input capture has occurred. If the corresponding interrupt is enabled with the TCSR $_n$ [TIE] field, an interrupt can be generated.

11.5.4.10.5.2 Output compare

The $TCCR_n$ compare registers are loaded with the time at which the corresponding event should occur. When the ENET free-running counter value matches the output compare reference value in the $TCCR_n$ register, the corresponding flag, TCSR $_n$ [TF], is set, indicating that an output compare has occurred. The corresponding interrupt, if enabled by TCSR $_n$ [TIE], will be generated. The corresponding 1588_TMR $_n$ output signal will be asserted according to TCSR $_n$ [TMODE].

NOTE

If TCSR $_n$ [TMODE] is set to 10X1b or 1010b then the timer output pin toggle-on-overflow will occur only when PINPER, PEREN, and EN bits of the ATCR register are one.

11.5.4.10.5.3 DMA requests

A DMA request can be enabled by setting TCSR $_n$ [TDRE]. The corresponding DMA request is generated when the TCSR $_n$ [TF] timer flag is set. When the DMA has completed, the corresponding TCSR $_n$ [TF] flag is cleared.

11.5.4.11 FIFO thresholds

The core FIFO thresholds are fully programmable to dynamically change the FIFO operation.

For example, store and forward transfer can be enabled by a simple change in the FIFO threshold registers.

The thresholds are defined in 64-bit words.

The receive and transmit FIFOs both have a depth of 1024 words.

11.5.4.11.1 Receive FIFO

Four programmable thresholds are available, which can be set to any value to control the core operation as follows.

Table 11-69. Receive FIFO thresholds definition

Register	Description
ENET n _RSFL [RX_SECTION_F ULL]	When the FIFO level reaches the ENETn_RSFL value, the MAC status signal is asserted to indicate that data is available in the receive FIFO (cut-through operation). Once asserted, if the FIFO empties below the threshold set with ENETn_RAEM and if the end-of-frame is not yet stored in the FIFO, the status signal is deasserted again. If a frame has a size smaller than the threshold (in other words, an end-of-frame is available for the frame), the status is also asserted. To enable store and forward on the receive path, clear ENETn_RSFL. The MAC status signal is asserted only when a complete frame is stored in the receive FIFO. When programming a non-zero value to ENETn_RSFL (cut-through operation) it should be greater than ENETn_RAEM.
ENET n _RAEM [RX_ALMOST_E MPTY]	When the FIFO level reaches the ENETn_RAEM value, and the end-of-frame has not been received, the core receive read control stops the FIFO read (and subsequently stops transferring data to the MAC client application). It continues to deliver the frame, if again more data than the threshold or the end-of-frame is available in the FIFO. Set ENETn_RAEM to a minimum of six.
ENET n _RAFL [RX_ALMOST_F ULL]	When the FIFO level approaches the maximum and there is no more space remaining for at least ENETn_RAFL number of words, the MAC control logic stops writing data in the FIFO and truncates the receive frame to avoid FIFO overflow. The corresponding error status is set when the frame is delivered to the application. Set ENETn_RAFL to a minimum of 4.
ENET n _RSEM [RX_SECTION_E MPTY]	When the FIFO level reaches the ENETn_RSEM value, an indication is sent to the MAC transmit logic, which generates an XOFF pause frame. This indicates FIFO congestion to the remote Ethernet client. When the FIFO level goes below the value programmed in ENETn_RSEM, an indication is sent to the MAC transmit logic, which generates an XON pause frame. This indicates the FIFO congestion is cleared to the remote Ethernet client. Clearing ENETn_RSEM disables any pause frame generation.

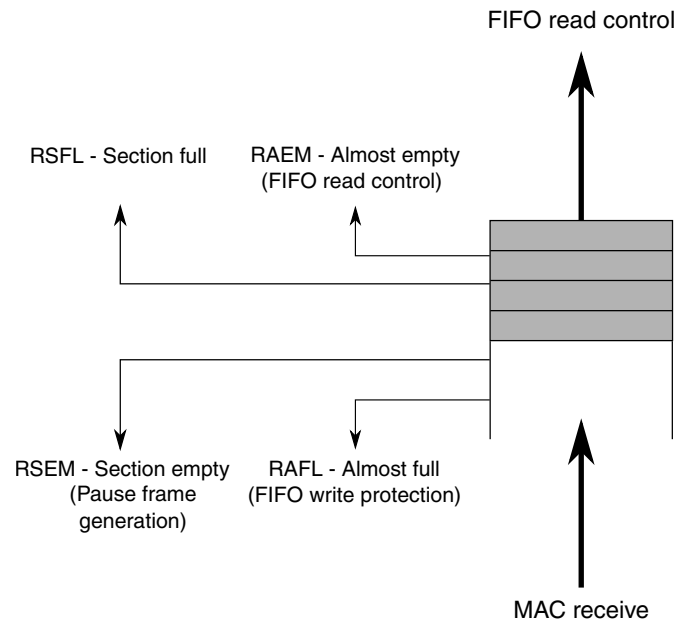


Figure 11-86. Receive FIFO overview

11.5.4.11.2 Transmit FIFO

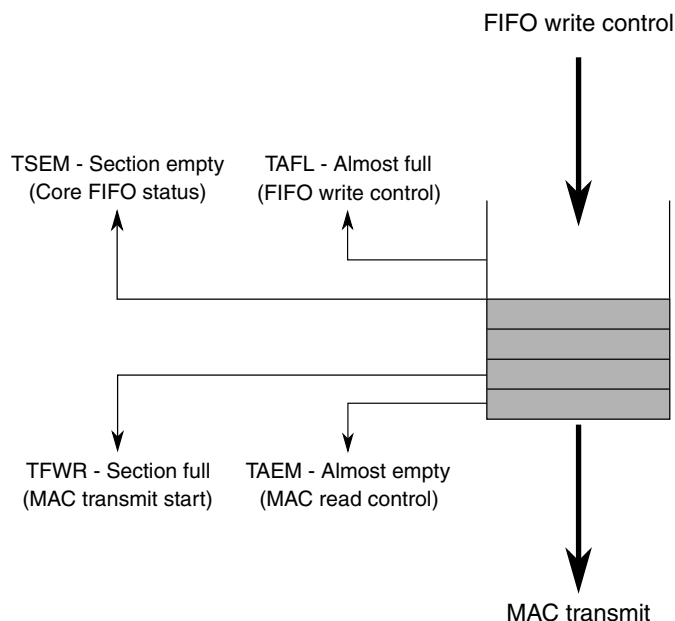
Four programmable thresholds are available which control the core operation as described below.

Table 11-70. Transmit FIFO thresholds definition

Register	Description
ENET n _TAEM [TX_ALMOST_EMPTY]	When the FIFO level reaches the ENET n _TAEM value and no end-of-frame is available for the frame, the MAC transmit logic avoids a FIFO underflow by stopping FIFO reads and transmitting the Ethernet frame with an MII error indication. Set ENET n _TAEM to a minimum of 4.
ENET n _TAFL [TX_ALMOST_FULL]	When the FIFO level approaches the maximum, so that there is no more space for at least ENET n _TAFL number of words, the MAC deasserts its control signal to the application. If the application does not react on this signal, the FIFO write control logic avoids FIFO overflow by truncating the current frame and setting the error status. As a result, the frame is transmitted with an GMII/MII error indication. Set ENET n _TAFL to a minimum of 4. Larger values allow more latency for the application to react on the MAC control signal deassertion, before the frame is truncated. A typical setting is 8, which offers 3–4 clock cycles of latency to the application to react on the MAC control signal deassertion.
ENET n _TSEM [TX_SECTION_EMPTY]	When the FIFO level reaches the ENET n _TSEM value, a MAC status signal is deasserted to indicate that the transmit FIFO is getting full. This gives the ENET module an indication to slow or stop its write transaction to avoid a buffer overflow. This is a pure indication function to the application. It has no effect within the MAC. When ENET n _TSEM is 0, the signal is never deasserted.
ENET n _TFWR	When the FIFO level reaches the ENET n _TFWR value and when STRFWD is cleared, the MAC transmit control logic starts frame transmission before the end-of-frame is available in the FIFO (cut-through operation).

Table 11-70. Transmit FIFO thresholds definition

Register	Description
	If a complete frame has a size smaller than the ENET_n_TFWR threshold, the MAC also transmits the frame to the line. To enable store and forward on the transmit path, set STRFWD. In this case, the MAC starts to transmit data only when a complete frame is stored in the transmit FIFO.

**Figure 11-87. Transmit FIFO overview**

11.5.4.12 Loopback options

The core implements external and internal loopback options, which are controlled by the ENET_n_RCR register fields found here.

The core implements external and internal loopback options, which are controlled by the following ENET_n_RCR register fields:

Table 11-71. Loopback options

Register field	Description
LOOP	Internal MII loopback. The MAC transmit is returned to the MAC receive. No data is transmitted to the external interfaces. In MII internal loopback, MII_TXCLK and MII_RXCLK must be provided with a clock signal (2.5 MHz for 10 Mbit/s, and 25 MHz for 100 Mbit/s))

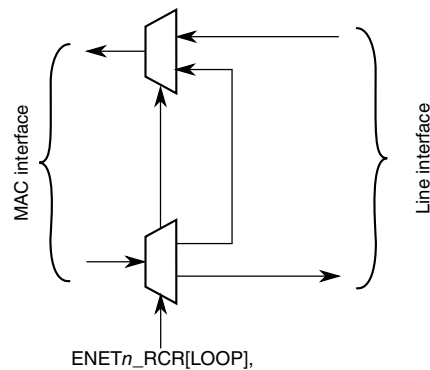


Figure 11-88. Loopback options

11.5.4.13 Legacy buffer descriptors

To support the Ethernet controller on previous chips, legacy FEC buffer descriptors are available. To enable legacy support, write 0 to ENETn_ECR[1588EN].

NOTE

- Legacy buffer descriptors are used in single-ring mode, that is, when DMA_nCFG[DMA_CLASS_EN] are zero. In multi-ring mode, legacy TxBDs are used only with the round-robin scheme.
- The legacy buffer descriptor tables show the byte order for big-endian chips. DBSWP must be set to 0 after reset to enable big-endian mode.

11.5.4.13.1 Legacy receive buffer descriptor

The following table shows the legacy FEC receive buffer descriptor. Table 11-75 contains the descriptions for each field.

Table 11-72. Legacy FEC receive buffer descriptor (RxBD)

	Byte 0								Byte 1							
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Offset + 0	E	RO1	W	RO2	L	—	—	M	BC	MC	LG	NO	—	CR	OV	TR
Offset + 2	Data length															
Offset + 4	Rx data buffer pointer — high halfword															
Offset + 6	Rx data buffer pointer — low halfword															

11.5.4.13.2 Legacy transmit buffer descriptor

The following table shows the legacy FEC transmit buffer descriptor. [Table 11-77](#) contains the descriptions for each field.

Table 11-73. Legacy FEC transmit buffer descriptor (TxBD)

	Byte 0								Byte 1							
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Offset + 0	R	TO1	W	TO2	L	TC	ABC ¹	—	—	—	—	—	—	—	—	—
Offset + 2	Data Length															
Offset + 4	Tx Data Buffer Pointer — high halfword															
Offset + 6	Tx Data Buffer Pointer — low halfword															

1. This field is not supported by the uDMA.

11.5.4.14 Enhanced buffer descriptors

This section provides a description of the enhanced operation of the driver/uDMA via the buffer descriptors.

It is followed by a detailed description of the receive and transmit descriptor fields. To enable the enhanced features, set ENET_n_ECR[1588EN].

NOTE

The enhanced buffer descriptor tables show the byte order for big-endian chips. [DBSWP](#) must be set to 0 after reset to enable big-endian mode.

11.5.4.14.1 Enhanced receive buffer descriptor

The following table shows the enhanced uDMA receive buffer descriptor. [Table 11-75](#) contains the descriptions for each field.

Table 11-74. Enhanced uDMA receive buffer descriptor (RxBd)

	Byte 0								Byte 1							
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Offset + 0	E	RO1	W	RO2	L	—	—	M	BC	MC	LG	NO	—	CR	OV	TR
Offset + 2	Data length															
Offset + 4	Rx data buffer pointer – high halfword															
Offset + 6	Rx data buffer pointer – low halfword															
Offset + 8	ME	—	—	—	—	PE	CE	UC	INT	—	—	—	—	—	—	—
Offset + A	VPCP			—	—	—	—	—	—	—	ICE	PCR	—	VLAN	IPV6	FRA G

Table continues on the next page...

Table 11-74. Enhanced uDMA receive buffer descriptor (RxBD) (continued)

Offset + C	Header length					—	—	—	Protocol type							
Offset + E	Payload checksum															
Offset + 10	BDU	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Offset + 12	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Offset + 14	1588 timestamp – high halfword															
Offset + 16	1588 timestamp – low halfword															
Offset + 18	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Offset + 1A	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Offset + 1C	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Offset + 1E	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table 11-75. Receive buffer descriptor field definitions

Word	Field	Description
Offset + 0	15	Empty. Written by the MAC (= 0) and user (= 1).
	E	0 The data buffer associated with this BD is filled with received data, or data reception has aborted due to an error condition. The status and length fields have been updated as required. 1 The data buffer associated with this BD is empty, or reception is currently in progress.
Offset + 0	14	Receive software ownership. This field is reserved for use by software. This read/write field is not modified by hardware, nor does its value affect hardware.
	RO1	
Offset + 0	13	Wrap. Written by user.
	W	0 The next buffer descriptor is found in the consecutive location. 1 The next buffer descriptor is found at the location defined in ENET _n _RDSR.
Offset + 0	12	Receive software ownership. This field is reserved for use by software. This read/write field is not modified by hardware, nor does its value affect hardware.
	RO2	
Offset + 0	11	Last in frame. Written by the uDMA.
	L	0 The buffer is not the last in a frame. 1 The buffer is the last in a frame.
Offset + 0	10–9	Reserved, must be cleared.
Offset + 0	8	Miss. Written by the MAC. This field is set by the MAC for frames accepted in promiscuous mode, but flagged as a miss by the internal address recognition. Therefore, while in promiscuous mode, you can use this field to quickly determine whether the frame was destined to this station. This field is valid only if the L and PROM fields are set to 1. 0 The frame was received because of an address recognition hit. 1 The frame was received because of promiscuous mode. The information needed for this field comes from the promiscuous_miss(ff_rx_err_stat[26]) sideband signal.
	M	
Offset + 0	7	Set if the DA is broadcast (FFFF_FFFF_FFFF).
	BC	
Offset + 0	6	Set if the DA is multicast and not BC.
	MC	

Table continues on the next page...

Table 11-75. Receive buffer descriptor field definitions (continued)

Word	Field	Description
Offset + 0	5 LG	Receive frame length violation. Written by the MAC. A frame length greater than RCR[MAX_FL] was recognized. This field is valid only if the L field is set. The receive data is not altered in any way unless the length exceeds TRUNC_FL bytes.
Offset + 0	4 NO	Receive non-octet aligned frame. Written by the MAC. A frame that contained a number of bits not divisible by 8 was received, and the CRC check that occurred at the preceding byte boundary generated an error or a PHY error occurred. This field is valid only if the L field is set. If this field is set, the CR field is not set.
Offset + 0	3	Reserved, must be cleared.
Offset + 0	2 CR	Receive CRC or frame error. Written by the MAC. This frame contains a PHY or CRC error and is an integral number of octets in length. This field is valid only if the L field is set.
Offset + 0	1 OV	Overflow. Written by the MAC. A receive FIFO overflow occurred during frame reception. If this field is set, the other status fields, M, LG, NO, and CR, lose their normal meaning and are zero. This field is valid only if the L field is set.
Offset + 0	0 TR	Set if the receive frame is truncated (frame length >TRUNC_FL). If the TR field is set, the frame must be discarded and the other error fields must be ignored because they may be incorrect.
Offset + 2	15–0 Data Length	Data length. Written by the MAC. Data length is the number of octets written by the MAC into this BD's data buffer if L is cleared (the value is equal to EMRBR), or the length of the frame including CRC if L is set. It is written by the MAC once as the BD is closed.
Offset + 4	15–0 Data buffer pointer high	Receive data buffer pointer, high halfword ¹
Offset + 6	15–0 Data buffer pointer low	Receive data buffer pointer, low halfword
Offset + 8	15 ME	MAC error. This field is written by the uDMA. This field means that the frame stored in the system memory was received with an error (typically, a receive FIFO overflow). This field is only valid when the L field is set.
Offset + 8	14–11	Reserved, must be cleared.
Offset + 8	10 PE	PHY Error. This field is written by the uDMA. Set to "1" when the frame was received with an Error character on the PHY interface. The frame is invalid. This field is valid only when the L field is set.
Offset + 8	9 CE	Collision. This field is written by the uDMA. Set when the frame was received with a collision detected during reception. The frame is invalid and sent to the user application. This field is valid only when the L field is set.
Offset + 8	8 UC	Unicast. This field is written by the uDMA, and means that the frame is unicast. This field is valid regardless of whether the L field is set.
Offset + 8	7 INT	Generate RXB/RXF interrupt. This field is set by the user to indicate that the uDMA is to generate an interrupt on the <i>dma_int_rxb</i> / <i>dma_int_rxfevent</i> .
Offset + 8	6–0	Reserved, must be cleared.
Offset + A	15–13 VPCP	VLAN priority code point. This field is written by the uDMA to indicate the frame priority level. Valid values are from 0 (best effort) to 7 (highest). This value can be used to prioritize different classes of traffic (e.g., voice, video, data). This field is only valid if the L field is set.
Offset + A	12–6	Reserved, must be cleared.

Table continues on the next page...

Table 11-75. Receive buffer descriptor field definitions (continued)

Word	Field	Description
Offset + A	5 ICE	IP header checksum error. This is an accelerator option. This field is written by the uDMA. Set when either a non-IP frame is received or the IP header checksum was invalid. An IP frame with less than 3 bytes of payload is considered to be an invalid IP frame. This field is only valid if the L field is set.
Offset + A	4 PCR	Protocol checksum error. This is an accelerator option. This field is written by the uDMA. Set when the checksum of the protocol is invalid or an unknown protocol is found and checksumming could not be performed. This field is only valid if the L field is set.
Offset + A	3	Reserved, must be cleared.
Offset + A	2 VLAN	VLAN. This is an accelerator option. This field is written by the uDMA. It means that the frame has a VLAN tag. This field is valid only if the L field is set.
Offset + A	1 IPV6	IPV6 Frame. This field is written by the uDMA. This field indicates that the frame has an IPv6 frame type. If this field is not set it means that an IPv4 or other protocol frame was received. This field is valid only if the L field is set.
Offset + A	0 FRAG	IPv4 Fragment. This is an accelerator option. This field is written by the uDMA. It indicates that the frame is an IPv4 fragment frame. This field is only valid when the L field is set.
Offset + C	15–11 Header length	Header length. This is an accelerator option. This field is written by the uDMA. This field is the sum of 32-bit words found within the IP and its following protocol headers. If an IP datagram with an unknown protocol is found, then the value is the length of the IP header. If no IP frame or an erroneous IP header is found, the value is 0. The following values are minimum values if no header options exist in the respective headers: • ICMP/IP: 6 (5 IP header, 1 ICMP header) • UDP/IP: 7 (5 IP header, 2 UDP header) • TCP/IP: 10 (5 IP header, 5 TCP header) This field is only valid if the L field is set.
Offset + C	10–8	Reserved, must be cleared.
Offset + C	7–0 Protocol type	Protocol type. This is an accelerator option. The 8-bit protocol field found within the IP header of the frame. It is valid only when ICE is cleared. This field is valid only when the L field is set.
Offset + E	15–0 Payload checksum	Internet payload checksum. This is an accelerator option. It is the one's complement sum of the payload section of the IP frame. The sum is calculated over all data following the IP header until the end of the IP payload. This field is valid only when the L field is set.
Offset + 10	15 BDU	Last buffer descriptor update done. Indicates that the last BD data has been updated by uDMA. This field is written by the user (=0) and uDMA (=1).
Offset + 10	14–0	Reserved, must be cleared.
Offset + 12	15–0	Reserved, must be cleared.
Offset + 14	15–0	This value is written by the uDMA. It is only valid if the L field is set.
Offset + 16	1588 timestamp	
Offset + 18	15–0	Reserved, must be cleared.
– Offset + 1E		

1. The receive buffer pointer, containing the address of the associated data buffer, must always be evenly divisible by 64. The buffer must reside in memory external to the MAC. The Ethernet controller never modifies this value.

11.5.4.14.2 Enhanced transmit buffer descriptor

Table 11-76. Enhanced transmit buffer descriptor (TxBD)

	Byte 0								Byte 1							
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Offset + 0	R	TO1	W	TO2	L	TC	—	—	—	—	—	—	—	—	—	—
Offset + 2	Data length															
Offset + 4	Tx Data Buffer Pointer – high halfword															
Offset + 6	Tx Data Buffer Pointer – low halfword															
Offset + 8	—	INT	TS	PINS	IINS	—	—	UTLT	FTYPE				—	—	—	—
Offset + A	TXE	—	UE	EE	FE	LCE	OE	TSE	—	—	—	—	—	—	—	—
Offset + C	TLT – high halfword															
Offset + E	TLT – low halfword															
Offset + 10	BDU	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Offset + 12	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Offset + 14	1588 timestamp – high halfword															
Offset + 16	1588 timestamp – low halfword															
Offset + 18	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Offset + 1A	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Offset + 1C	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Offset + 1E	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table 11-77. Enhanced transmit buffer descriptor field definitions

Word	Field	Description
Offset + 0	15	Ready. Written by the MAC and user.
	R	0 The data buffer associated with this BD is not ready for transmission. You are free to manipulate this BD or its associated data buffer. The MAC clears this field after the buffer has been transmitted or after an error condition is encountered. 1 The data buffer, prepared for transmission by you, has not been transmitted or currently transmits. You may write no fields of this BD after this field is set.
Offset + 0	14 TO1	Transmit software ownership. This field is reserved for software use. This read/write field is not modified by hardware and its value does not affect hardware.
Offset + 0	13	Wrap. Written by user.
	W	0 The next buffer descriptor is found in the consecutive location 1 The next buffer descriptor is found at the location defined in ETDSR.
Offset + 0	12 TO2	Transmit software ownership. This field is reserved for use by software. This read/write field is not modified by hardware and its value does not affect hardware.
Offset + 0	11	Last in frame. Written by user.
	L	0 The buffer is not the last in the transmit frame 1 The buffer is the last in the transmit frame

Table continues on the next page...

Table 11-77. Enhanced transmit buffer descriptor field definitions (continued)

Word	Field	Description
Offset + 0	10 TC	Transmit CRC. Written by user, and valid only when L is set. 0 End transmission immediately after the last data byte 1 Transmit the CRC sequence after the last data byte This field is valid only when the L field is set.
Offset + 0	9 ABC	Append bad CRC. Note: This field is not supported by the uDMA and is ignored.
Offset + 0	8–0	Reserved, must be cleared.
Offset + 2	15–0 Data Length	Data length, written by user. Data length is the number of octets the MAC should transmit from this BD's data buffer. It is never modified by the MAC.
Offset + 4	15–0 Data buffer pointer high	Tx data buffer pointer, high halfword. The buffer must reside in memory external to the MAC. This value is never modified by the Ethernet controller. NOTE: For optimal performance, make the transmit buffer pointer evenly divisible by 64.
Offset + 6	15–0 Data buffer pointer low	Tx data buffer pointer, low halfword
Offset + 8	15	Reserved, must be cleared.
Offset + 8	14 INT	Generate interrupt flags. This field is written by the user. This field is valid regardless of the L field and must be the same for all EBD for a given frame. The uDMA does not update this value.
Offset + 8	13 TS	Timestamp. This field is written by the user. This indicates that the uDMA is to generate a timestamp frame to the MAC. This field is valid regardless of the L field and must be the same for all EBD for the given frame. The uDMA does not update this value.
Offset + 8	12 PINS	Insert protocol specific checksum. This field is written by the user. If set, the MAC's IP accelerator calculates the protocol checksum and overwrites the corresponding checksum field with the calculated value. The checksum field must be cleared by the application generating the frame. The uDMA does not update this value. This field is valid regardless of the L field and must be the same for all EBD for a given frame.
Offset + 8	11 IINS	Insert IP header checksum. This field is written by the user. If set, the MAC's IP accelerator calculates the IP header checksum and overwrites the corresponding header field with the calculated value. The checksum field must be cleared by the application generating the frame. The uDMA does not update this value. This field is valid regardless of the L field and must be the same for all EBD for a given frame.
Offset + 8	10-9	Reserved, must be cleared.
Offset + 8	8 UTLT	Use transmit launch time. If set to 1, TLT high and low values are used to determine if the frame can be transmitted. This field must only be set in the <i>first</i> BD of a frame. TLT is always used in combination with either the round-robin scheme or the credit-based shaper. TLT can be used with single a BD queue. However, at least one DMA class, DMA _n CFG[DMA_CLASS_EN] must be enabled. For example, to use TLT with only queue 0, enable one of the DMA _n CFG[DMA_CLASS_EN] fields and clear both TDAR1 and TDAR2. Although this enables multi-queue, only the single queue 0 is used.

Table continues on the next page...

Table 11-77. Enhanced transmit buffer descriptor field definitions (continued)

Word	Field	Description
Offset + 8	7–4 FTYPE	Type of frame to be transmitted. If a credit-based scheme is used, this field must match the BD ring queue. 0x0 – Non-AVB. Corresponds to ENET_TDSR. 0x1 – AVB Class A. Corresponds to ENET_TDSR1. 0x2 – AVB Class B. Corresponds to ENET_TDSR2. All other values are reserved.
Offset + 8	3–0	Reserved, must be cleared.
Offset + A	15 TXE	Transmit error occurred. This field is written by the uDMA. This field indicates that there was a transmit error of some sort reported with the frame. Effectively this field is an OR of the other error fields including UE, EE, FE, LCE, OE, and TSE. This field is valid only when the L field is set.
Offset + A	14	Reserved, must be cleared.
Offset + A	13 UE	Underflow error. This field is written by the uDMA. This field indicates that the MAC reported an underflow error on transmit. This field is valid only when the L field is set.
Offset + A	12 EE	Excess Collision error. This field is written by the uDMA. This field indicates that the MAC reported an excess collision error on transmit. This field is valid only when the L field is set.
Offset + A	11 FE	Frame with error. This field is written by the uDMA. This field indicates that the MAC reported that the uDMA reported an error when providing the packet. This field is valid only when the L field is set.
Offset + A	10 LCE	Late collision error. This field is written by the uDMA. This field indicates that the MAC reported that there was a Late Collision on transmit. This field is valid only when the L field is set.
Offset + A	9 OE	Overflow error. This field is written by the uDMA. This field indicates that the MAC reported that there was a FIFO overflow condition on transmit. This field is only valid when the L field is set.
Offset + A	8 TSE	Timestamp error. This field is written by the uDMA. This field indicates that the MAC reported a different frame type then a timestamp frame. This field is valid only when the L field is set.
Offset + A	7–0	Reserved, must be cleared.
Offset + C	15–0 TLT high	Transmit launch time high. Specifies when frame can be transmitted.
Offset + E	15–0 TLT low	Transmit launch time low. Specifies when frame can be transmitted.
Offset + 10	15 BDU	Last buffer descriptor update done. Indicates that the last BD data has been updated by uDMA. This field is written by the user (=0) and uDMA (=1).
Offset + 10	14–0	Reserved, must be cleared.
Offset + 12	15–0	Reserved, must be cleared.
Offset + 14	1588 timestamp	This value is written by the uDMA . It is valid only when the L field is set.
Offset + 16		
Offset + 18–Offset + 1E	15–0	Reserved, must be cleared.

11.5.4.15 Client FIFO application interface

The FIFO interface is completely asynchronous from the Ethernet line, and the transmit and receive interface can operate at a different clock rate.

All transfers to/from the user application are handled independently of the core operation, and the core provides a simple interface to user applications based on a two-signal handshake.

11.5.4.15.1 Data structure description

The data structure defined in the following tables for the FIFO interface must be respected to ensure proper data transmission on the Ethernet line. Byte 0 is sent to and received from the line first.

Table 11-78. FIFO interface data structure

	63	56 55	48 47	40 39	32 31	24 23	16 15	8 7	0
Word 0	Byte 7	Byte 6	Byte 5	Byte 4	Byte 3	Byte 2	Byte 1	Byte 0	
Word 1	Byte 15	Byte 14	Byte 13	Byte 12	Byte 11	Byte 10	Byte 9	Byte 8	
...	...								

The size of a frame on the FIFO interface may not be a modulo of 64-bit.

The user application may not care about the Ethernet frame formats in full detail. It needs to provide and receive an Ethernet frame with the following structure:

- Ethernet MAC destination address
- Ethernet MAC source address
- Optional 802.1q VLAN tag (VLAN type and info field)
- Ethernet length/type field
- Payload

Frames on the FIFO interface do not contain preamble and SFD fields, which are inserted and discarded by the MAC on transmit and receive, respectively.

- On receive, CRC and frame padding can be stripped or passed through transparently.
- On transmit, padding and CRC can be provided by the user application, or appended automatically by the MAC independently for each frame. No size restrictions apply.

Note

On transmit, if ENET n _TCR[ADDINS] is set, bytes 6–11 of each frame can be set to any value, since the MAC overwrites the bytes with the MAC address programmed in the ENET n _PAUR and ENET n _PALR registers.

Table 11-79. FIFO interface frame format

Byte number	Field
0–5	Destination MAC address
6–11	Source MAC address
12–13	Length/type field
14–N	Payload data

VLAN-tagged frames are supported on both transmit and receive, and implement additional information (VLAN type and info).

Table 11-80. FIFO interface VLAN frame format

Byte number	Field
0–5	Destination MAC address
6–11	Source MAC address
12–15	VLAN tag and info
16–17	Length/type field
18–N	Payload data

Note

The standard defines that the LSB of the MAC address is sent/received first, while for all the other header fields — in other words, length/type, VLAN tag, VLAN info, and pause quanta — the MSB is sent/received first.

11.5.4.15.2 Data structure examples

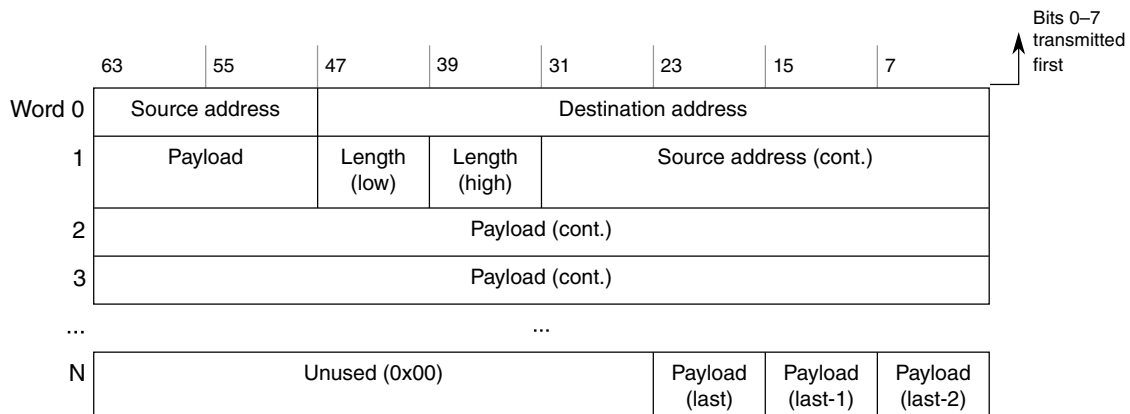


Figure 11-89. Normal Ethernet frame 64-bit mapping example

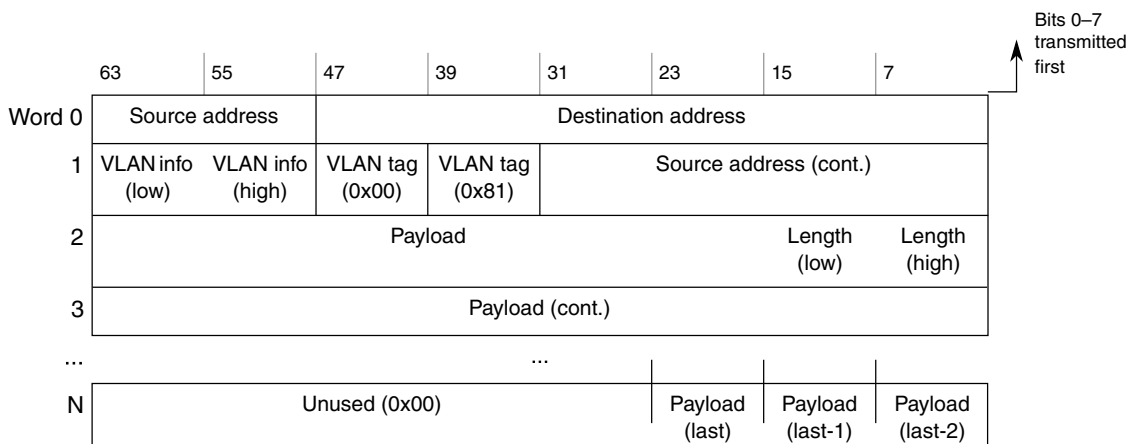


Figure 11-90. VLAN-tagged frame 64-bit mapping example

If CRC forwarding is enabled (CRCFWD = 0), the last four valid octets of the frame contain the FCS field. The non-significant bytes of the last word can have any value.

11.5.4.15.3 Frame status

A MAC layer status word and an accelerator status word is available in the receive buffer descriptor.

See [Enhanced buffer descriptors](#) for details.

The status is available with each frame with the last data of the frame.

If the frame status contains a MAC layer error (for example, CRC or length error), RxB[ME] is also set with the last data of the frame.

11.5.4.16 FIFO protection

The following sections describe the FIFO protection mechanisms.

11.5.4.16.1 Transmit FIFO underflow

During a frame transfer, when the transmit FIFO reaches the almost empty threshold with no end-of-frame indication stored in the FIFO, the MAC logic:

- Stops reading data from the FIFO
- Asserts the MII error signal (MII_TXER) (1 in Figure 11-91) to indicate that the fragment already transferred is not valid
- Deasserts the MII transmit enable signal (MII_TXEN) to terminate the frame transfer (2)

After an underflow, when the application completes the frame transfer (3), the MAC transmit logic discards any new data available in the FIFO until the end of packet is reached (4) and sets the enhanced TxBD[UE] field.

The MAC starts to transfer data on the MII interface when the application sends a new frame with a start of frame indication (5).

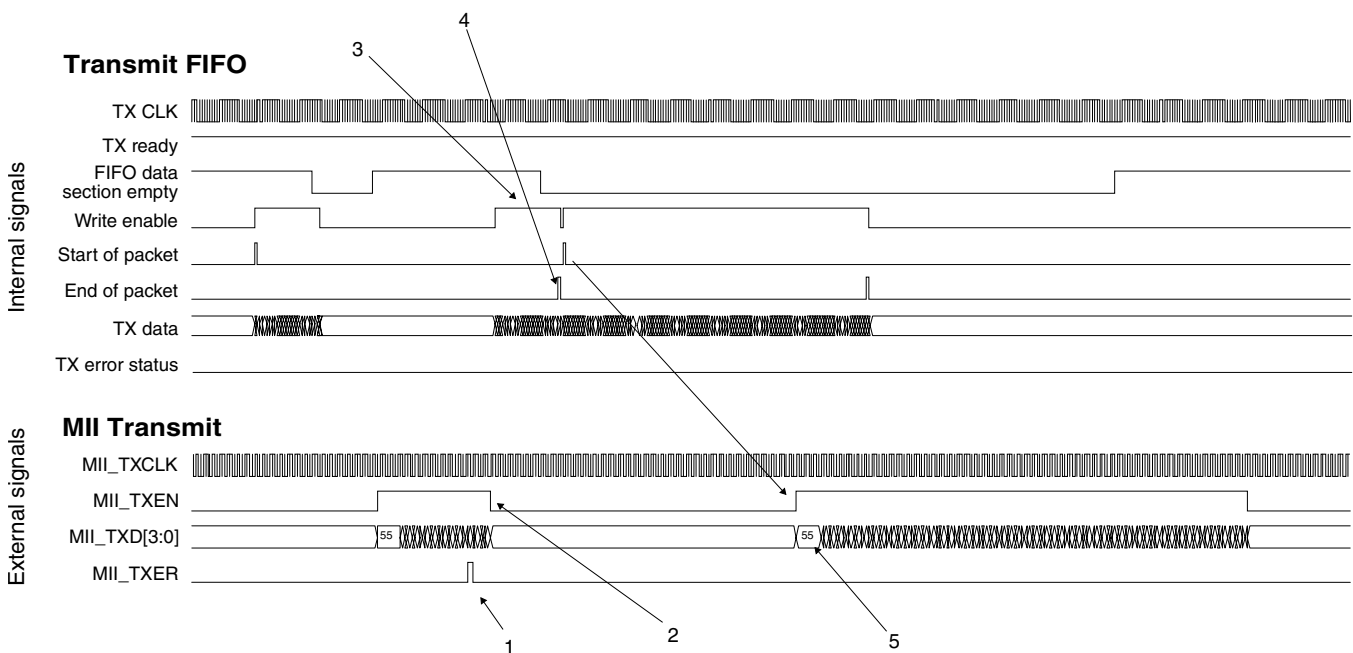


Figure 11-91. Transmit FIFO underflow protection

11.5.4.16.2 Transmit FIFO overflow

On the transmit path, when the FIFO reaches the programmable almost full threshold, the internal MAC ready signal is deasserted. The application should stop sending new data.

However, if the application keeps sending data, the transmit FIFO overflows, corrupting contents that were previously stored. The core logic sets the enhanced TxBD[OE] field for the next frame transmitted to indicate this overflow occurrence.

Note

Overflow is a fatal error and must be addressed by resetting the core or clearing ENET n _ECR[ETHER_EN], to clear the FIFOs and prepare for normal operation again.

11.5.4.16.3 Receive FIFO overflow

During a frame reception, if the client application is not able to receive data (1), the MAC receive control truncates the incoming frame when the FIFO reaches the programmable almost-full threshold to avoid an overflow.

The frame is subsequently received on the FIFO interface with an error indication (enhanced RxBD[ME] field set together with receive end-of-packet) (2) with the truncation error status field set (3).

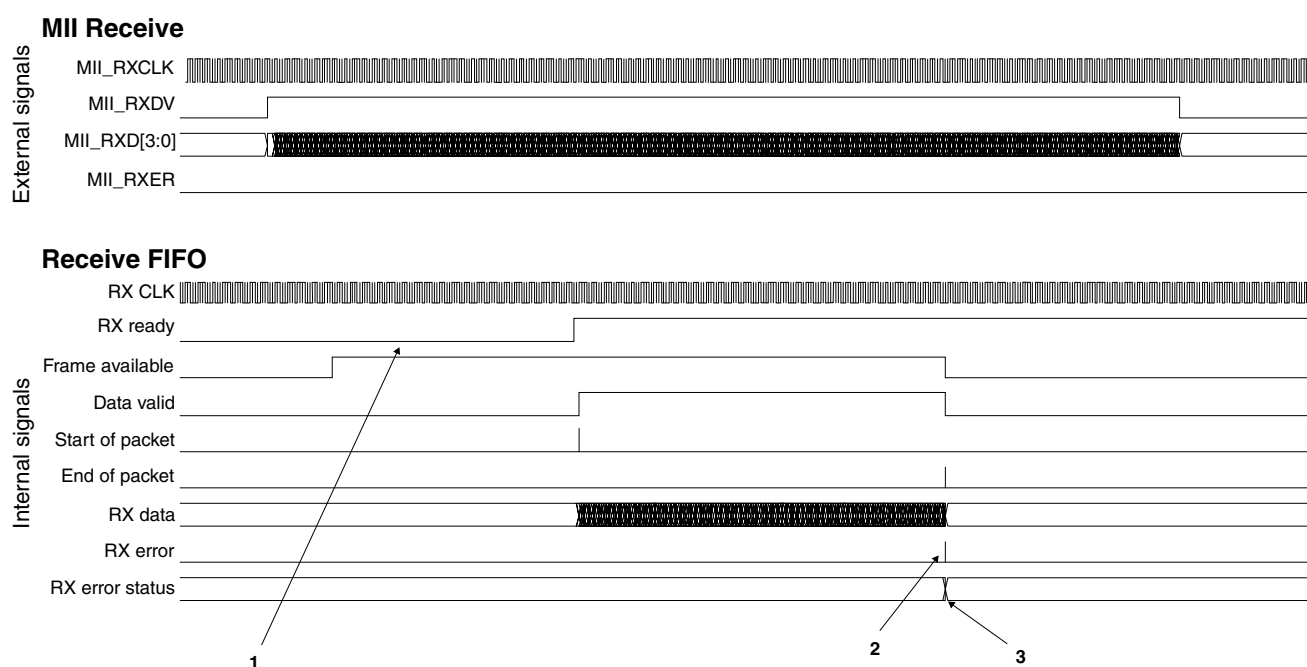


Figure 11-92. Receive FIFO overflow protection

11.5.4.17 PHY management interface

The MDIO interface is a two-wire management interface. The MDIO management interface implements a standardized method to access the PHY device management registers.

The core implements a master MDIO interface, which can be connected to up to 32 PHY devices.

11.5.4.17.1 MDIO clause 22 frame format

The core MDIO master controller communicates with the slave (PHY device) using frames that are defined in the following table.

A complete frame has a length of 64 bits made up of an optional 32-bit preamble, 14-bit command, 2-bit bus direction change, and 16-bit data. Each bit is transferred on the rising edge of the MDIO clock (MDC signal). The MDIO data signal is tri-stated between frames.

The core PHY management interface supports the standard MDIO specification (IEEE 802.3 Clause 22).

Table 11-81. MDIO clause 22 frame structure

ST	OP	PHYADR	REGADR	TA	DATA
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Table 11-82. MDIO frame field descriptions

Field	Description
ST (2 bits)	Start indication field, programmed with ENET n _MMFR[ST] and equal to 01 for Standard MDIO (Clause 22).
OP (2 bits)	Opcode defines type of operation. Programmed with ENET n _MMFR[OP]. 01 Write operation 10 Read operation
PHYADR (5 bits)	Five-bit PHY device address, programmed with ENET n _MMFR[PA]. Up to 32 devices can be addressed.
REGADR (5 bits)	Five-bit register address, programmed with ENET n _MMFR[RA]. Each PHY can implement up to 32 registers.
TA (2 bits)	Turnaround time, programmed with ENET n _MMFR[TA]. Two bit-times are reserved for read operations to switch the data bus from write to read. The PHY device presents its register contents in the data phase and drives the bus from the second bit of the turnaround phase.
Data (16 bits)	Data, set by ENET n _MMFR[DATA]. Written to or read from the PHY

11.5.4.17.2 MDIO clause 45 frame format

The extended MDIO frame structure defined in IEEE 802.3 Clause 45 introduces indirect addressing. First, a write transaction to an address register is done, followed by a write or read transaction which will put the 16-bit data in the register or retrieve the register contents respectively. A preamble of 32 bits of logical ones is sent prior to every transaction. The MDIO data signal is tri-stated between frames.

The extended MDIO defines four transactions, which are determined by the two-bit opcode field.

Table 11-83. MDIO clause 45 frame structure

ST	OP	PRTAD	DEVAD	TA	ADDR/DATA
----	----	-------	-------	----	-----------

All bits are transmitted from left to right (Preamble bits first) and all fields have their Most-Significant bit sent first (leftmost in above table). The complete frame has a length of 64 bits (32-bit preamble, 14-bit command, 2-bit bus direction change, 16-bit data). Each bit is transferred with the rising edge of the MDIO clock (MDC).

The fields and transactions are summarized in the following tables.

Table 11-84. MDIO clause 45 frame field descriptions

Field	Description
ST	Start indication. Indicates the end of the preamble and start of the frame. This value is 00 for extended MDIO (Clause 45) frames.
OP	Opcode defines if a read or write operation is performed and is programmed with ENET n _MMFR[OP]. See Table 11-85 for more information. 00 Address write 01 Write operation 10 Read inc. operation 11 Read operation
PRTAD	The port address specifies a MDIO port. Each Port can have up to 32 devices which each can have a separate set of registers.
DEVAD	Device address. Up to 32 devices can be addressed (within a port).
TA	Turnaround time, programmed with ENET n _MMFR[TA]. Two bit-times are reserved for read operations to switch the data bus from write to read. The PHY device presents its register contents in the data phase and drives the bus from the second bit of the turnaround phase.
ADDR/DATA	16-bit address (for address write) or data, set by ENET n _MMFR[DATA], written to or read from the PHY.

Table 11-85. MDIO Clause 45 Transactions

Transaction Type	Description
Address	A write transaction to the internal address register of the device/port. The data section of the frame contains the value to be stored in the device's internal address "pointer" register for further transactions.
Write	Data write to a register. The 16 bit data will be written to the register identified by the device-internal address.
Read	Data is read from the register identified by the device-internal address.
Read inc.	Read with address postincrement. The register identified by the device-internal address is read. After this, the device-internal address is incremented. If the address register is all '1' (0xFFFF) no increment is done (i.e. increment does not wrap around).

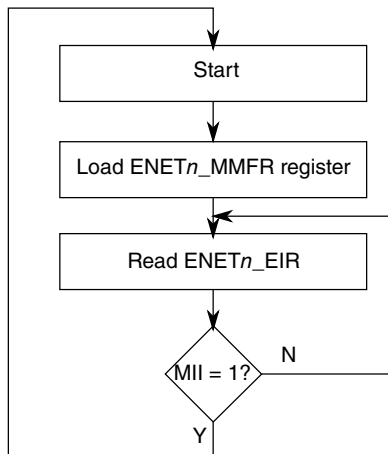
11.5.4.17.3 MDIO clock generation

The MDC clock is generated from the internal bus clock (i.e., IPS bus clock) divided by the value programmed in `ENETn_MSCR[MII_SPEED]`.

11.5.4.17.4 MDIO operation

To perform an MDIO access, set the MDIO command register (`ENETn_MMFR`) according to the description provided in MII Management Frame Register (`ENETn_MMFR`).

To check when the programmed access completes, read the `ENETn_EIR[MII]` field.

**Figure 11-93. MDIO access overview**

11.5.4.18 Ethernet interfaces

The following Ethernet interfaces are implemented:

- Fast Ethernet MII (Media Independent Interface)
- RMII 10/100 using interface converters/gaskets
- RGMII 10/100/1000 by way of interface converters/gaskets

The following table shows how to configure ENET registers to select each interface.

Mode	ECR[SPEED]	RCR[RMII_10T]	RCR[RMII_MODE]	RCR[RGMII_EN]
MII - 10 Mbit/s	0	—	0	0
MII - 100 Mbit/s	0	—	0	0
RMII - 10 Mbit/s	0	1	1	0
RMII - 100 Mbit/s	0	0	1	0
RGMII - 10 Mbit/s	0	1	0	1
RGMII - 100 Mbit/s	0	0	0	1
RGMII - 1000 Mbit/s	1	—	0	1

11.5.4.18.1 RMII interface

In RMII receive mode, for normal reception following assertion of CRS_DV, RXD[1:0] is 00 until the receiver determines that the receive event has a proper start-of-stream delimiter (SSD).

The preamble appears (RXD[1:0]=01) and the MACs begin capturing data following detection of SFD.

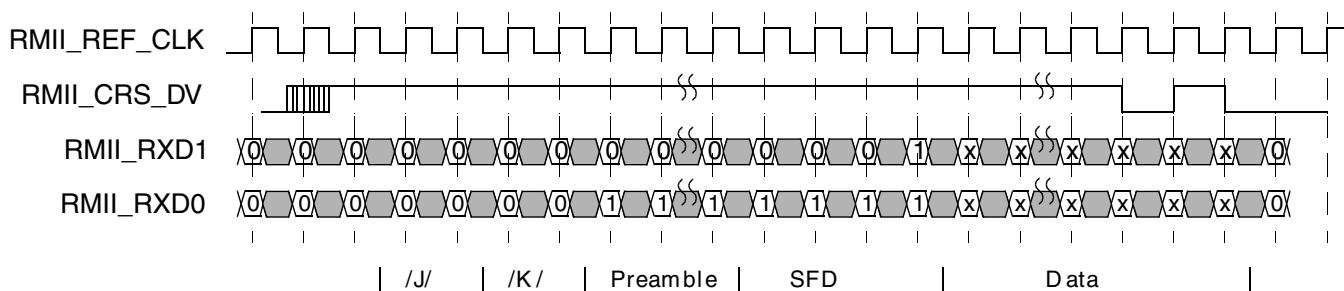


Figure 11-94. RMII receive operation

If a false carrier is detected (bad SSD), then RXD[1:0] is 10 until the end of the receive event. This is a unique pattern since a false carrier can only occur at the beginning of a packet where the preamble is decoded (RXD[1:0] = 01).

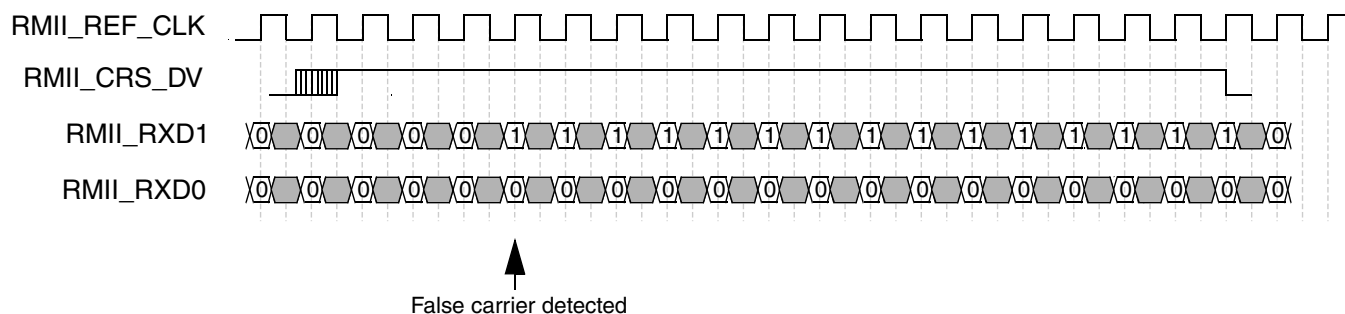


Figure 11-95. RMII receive operation with false carrier

In RMII transmit mode, TXD[1:0] provides valid data for each REF_CLK period while TXEN is asserted.

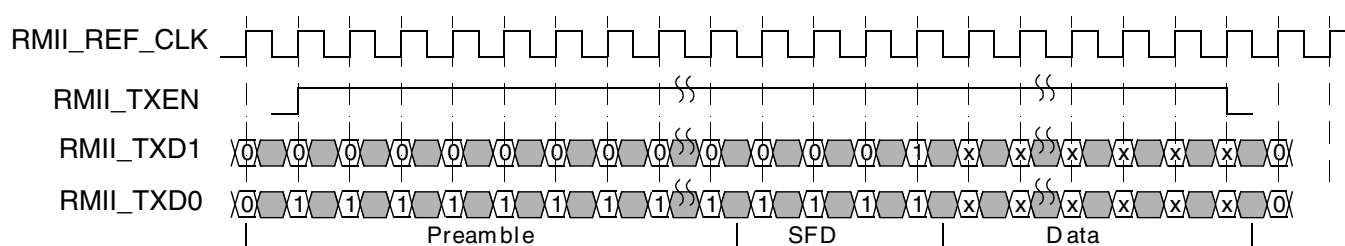


Figure 11-96. RMII transmit operation

11.5.4.18.2 RGMII interface

In RGMII modes, the data and control information is multiplexed by taking advantage of both edges of the reference clocks.

The data signals contain the lower four data bits on the rising edge and the upper four bits on the falling edge. The control signals are multiplexed into a single clock cycle using the same technique.

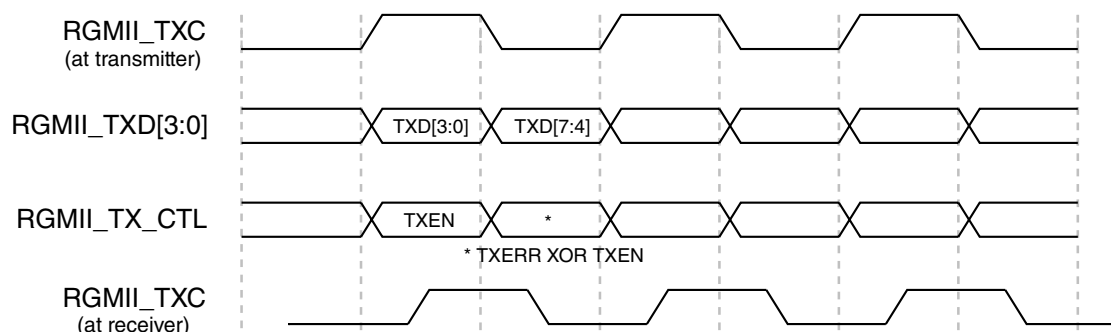


Figure 11-97. RGMII transmit operation

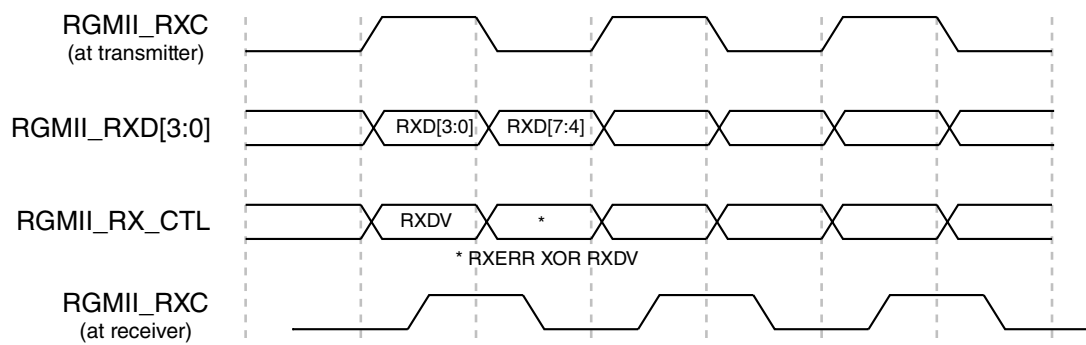


Figure 11-98. RGMII receive operation

11.5.4.18.3 MII Interface — transmit

On transmit, all data transfers are synchronous to MII_TXCLK rising edge. The MII data enable signal MII_TXEN is asserted to indicate the start of a new frame, and remains asserted until the last byte of the frame is present on the MII_TXD[3:0] bus.

Between frames, MII_TXEN remains deasserted.

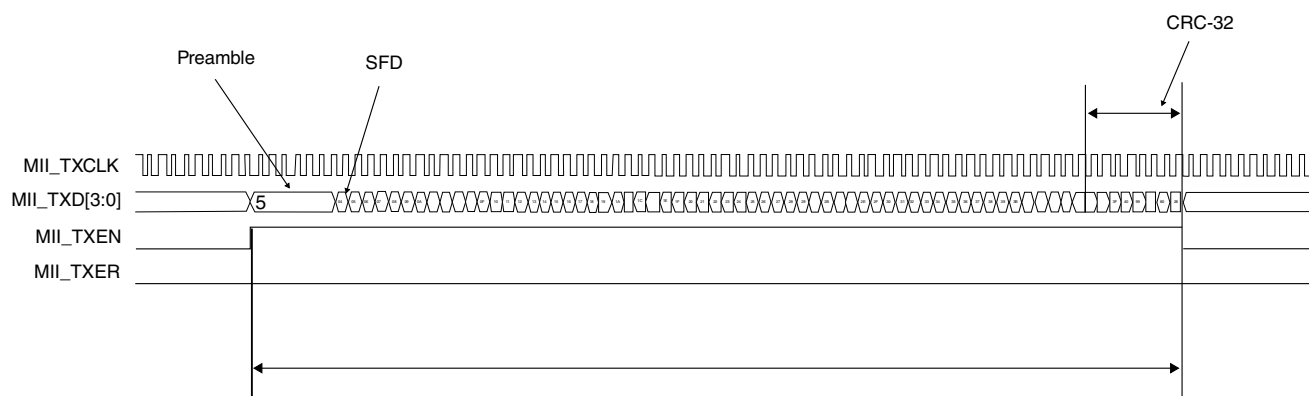


Figure 11-99. MII transmit operation

If a frame is received on the FIFO interface with an error (for example, RxBD[ME] set) the frame is subsequently transmitted with the MII_TXER error signal for one clock cycle at any time during the packet transfer.

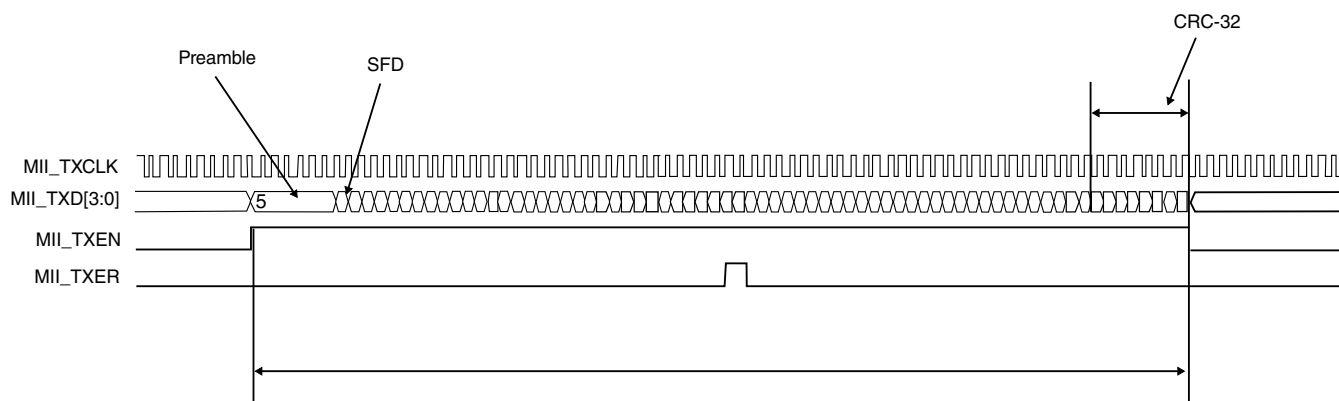


Figure 11-100. MII transmit operation — errored frame

11.5.4.18.3.1 Transmit with collision — half-duplex

When a collision is detected during a frame transmission (MII_COL asserted), the MAC stops the current transmission, sends a 32-bit jam pattern, and re-transmits the current frame.

(See [Collision detection in half-duplex mode](#) for details)

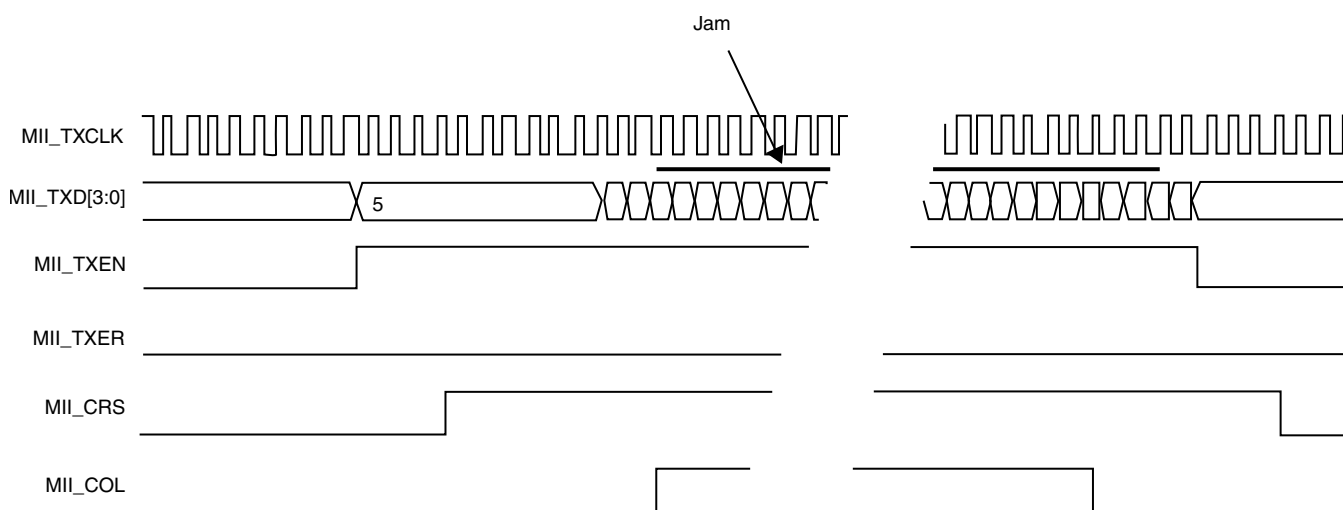


Figure 11-101. MII transmit operation — transmission with collision

11.5.4.18.4 MII interface — receive

On receive, all signals are sampled on the MII_RXCLK rising edge. The MII data enable signal, MII_RXDV, is asserted by the PHY to indicate the start of a new frame and remains asserted until the last byte of the frame is present on MII_RXD[3:0] bus.

Between frames, MII_RXDV remains deasserted.

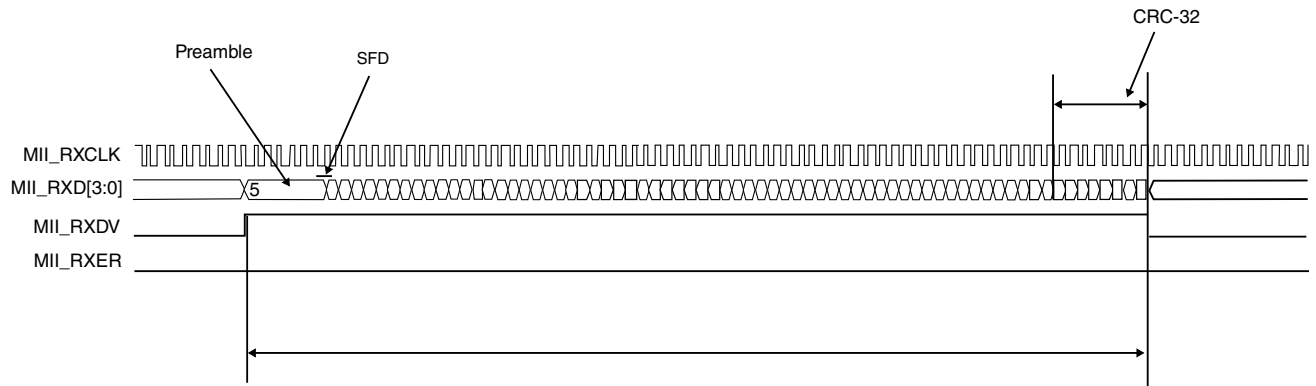


Figure 11-102. MII receive operation

If the PHY detects an error on the frame received from the line, the PHY asserts the MII error signal, MII_RXER, for at least one clock cycle at any time during the packet transfer.

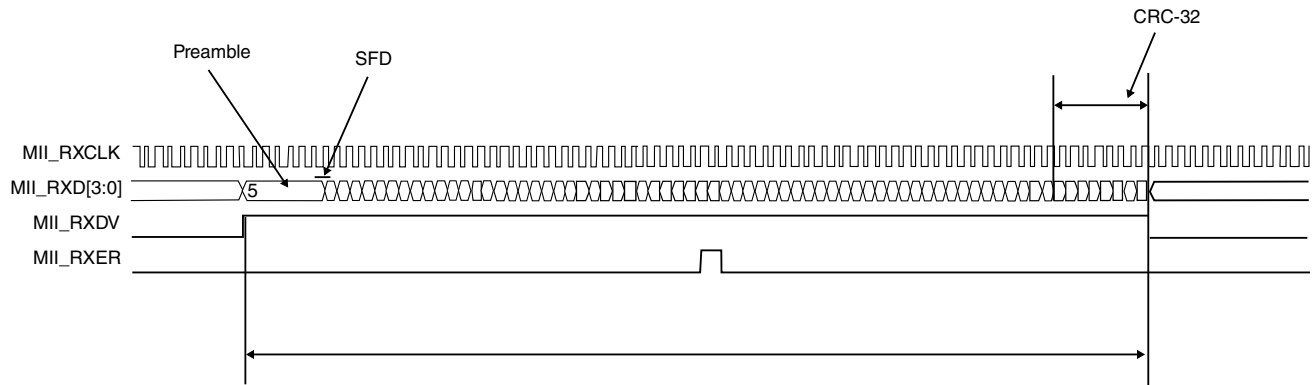


Figure 11-103. MII receive operation — errored frame

A frame received on the MII interface with a PHY error indication is subsequently transferred on the FIFO interface with RxBD[ME] set.

11.5.4.19 AVB configuration

The following steps give an example of how to initialize the ENET module for AVB.

1. Set up ENET_QOS:
 - Set TX_SCHEME to 000b, credit-based scheme.
 - Set RX_FLUSH0 to 1, enable RX flush for ring 0.
2. Set up TX BD ring and RX BD ring for each queue, 0-2.
 - Program ENET_MRBR, ENET_TDSR, and ENET_RDSR.
 - Program ENET_MRBR1, ENET_TDSR1, and ENET_RDSR1.

- Program ENET_MRBR2, ENET_TDSR2, and ENET_RDSR2.
- Program each TX and RX BD ring queue for all classes used in memory.

NOTE

If using credit-based scheme, ensure that enhanced transmit buffer descriptor FTYPE field matches BD ring queue, for example:

- FTYPE = 0h corresponds to ENET_TDSR
 - FTYPE = 1h corresponds to ENET_TDSR1
 - FTYPE = 2h corresponds to ENET_TDSR2
3. Program ENET_DMA1CFG and ENET_DMA2CFG for class 1 and class 2 corresponding to BD ring queue 1 and 2. DMA_CLASS_EN must be set to one for that class to be enabled. See [DMA Class Based Configuration \(ENET_DMA_nCFG\)](#) for information on how to program IDLE_SLOPE.
 4. Program ENET_RCMR1 and ENET_RCMR2 for class 1 and class 2 matching for receive.

NOTE

Even if a match occurs, if ENET_DMA_nCFG[DMA_CLASS_EN] is zero for the corresponding class, the RX frame will be automatically forwarded to BD ring queue 0.

5. Program the other ENET registers according to application requirements.
6. Program ENET_RDAR, ENET_RDAR1, ENET_RDAR2, ENET_TDAR, ENET_TDAR1, and ENET_TDAR2 to 0100_0000h according to the classes used.
7. Set ENET_ECR[ETHEREN] to one.

11.5.4.20 Interrupt coalescence

The purpose of the interrupt coalescing is to reduce the number of interrupts generated by the MAC so as to reduce the CPU loading.

To facilitate this interrupt coalescing for each queue, these registers are available with the same control and configuration fields.

- [Transmit Interrupt Coalescing Register \(ENET_TXIC_n\)](#) where n=0,1,2 for queue/class 0,1,2.
- [Receive Interrupt Coalescing Register \(ENET_RXIC_n\)](#) where n=0,1,2 for queue/class 0,1,2.

When coalescing is enabled by asserting the corresponding ICEN field and such interrupt is also enabled by the corresponding interrupt mask of the EIMR register, the MAC generates an interrupt when the threshold number of frames is reached (defined by ICFT) or when the threshold timer expires (defined by ICTT).

When coalescing is disabled by de-asserting ICEN, but interrupt is enabled by the corresponding interrupt mask of the EIMR register, the MAC generates an interrupt as they are received without using coalescing. Interrupt coalescing is done for each transmit and receive queue/class independently.

11.5.4.20.1 Interrupt coalescence setup

Interrupt coalescence supports both legacy and enhanced BDs. The following guidelines are recommended when setting up interrupt coalescence.

- When the MAC is configured for enhanced (IEEE 1588) mode, that is, enhanced BDs:
 - Set the INT bit in the enhanced received buffer descriptor to one.
 - Set the INT bit in the enhanced transmit buffer descriptor(s) to one.
- Clear the RXB, RXB1, and RXB2 fields in the EIMR register.
- Clear the TXB, TXB1, and TXB2 fields in the EIMR register.

11.5.4.20.2 Updating the frame count threshold on-the-fly

To update the ICFT field in the RXIC n and TXIC n registers:

1. Disable interrupt coalescence by clearing the appropriate ICEN field. This will allow the internal interrupt coalescence counter to reset to zero.

NOTE

When disabling interrupt coalescence, if an interrupt event is pending, that is, the interrupt counter is not zero, then an interrupt will occur.

2. Write the new threshold value to the ICFT field.
3. Set ICEN to one.

NOTE

The ICFT field can be updated on-the-fly without disabling the ICEN field. The hardware interrupt will continue and there is a possibility that an interrupt will occur depending on the state of the hardware counter and the previous ICFT value.

11.5.4.20.3 Updating the timer threshold on-the-fly

To update the ICTT field in the RXIC n and TXIC n registers:

1. Disable interrupt coalescence by clearing the appropriate ICEN field. This will allow the internal interrupt coalescence counter to reset to zero.

NOTE

When disabling interrupt coalescence, if an interrupt event is pending, that is, the interrupt counter is not zero, then an interrupt will occur.

2. Write the new timer value to the ICTT field.
3. Set ICEN to one.

11.5.5 Memory map/register definition

ENET registers must be read or written with 32-bit accesses. Non-32 bit accesses will terminate with an error.

Reserved bits should be written with 0 and ignored on read. Unused registers read zero and a write has no effect.

This table shows Ethernet registers organization.

Table 11-86. Register map summary

Offset Address	Section	Description
0x0000 – 0x01FF	Configuration	Core control and status registers
0x0200 – 0x03FF	Statistics counters	MIB and Remote Network Monitoring (RFC 2819) registers
0x0400 – 0x0430	1588 control	1588 adjustable timer (TSM) and 1588 frame control
0x0600 – 0x07FC	Capture/Compare block	Registers for the Capture/Compare block

ENET memory map

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
30BE_0004	Interrupt Event Register (ENET_EIR)	32	w1c	0000_0000h	11.5.5.1/3747
30BE_0008	Interrupt Mask Register (ENET_EIMR)	32	R/W	0000_0000h	11.5.5.2/3750
30BE_0010	Receive Descriptor Active Register - Ring 0 (ENET_RDAR)	32	R/W	0000_0000h	11.5.5.3/3754
30BE_0014	Transmit Descriptor Active Register - Ring 0 (ENET_TDAR)	32	R/W	0000_0000h	11.5.5.4/3755
30BE_0024	Ethernet Control Register (ENET_ECR)	32	R/W	See section	11.5.5.5/3756

Table continues on the next page...

ENET memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
30BE_0040	MII Management Frame Register (ENET_MMFR)	32	R/W	0000_0000h	11.5.5.6/3758
30BE_0044	MII Speed Control Register (ENET_MSCR)	32	R/W	0000_0000h	11.5.5.7/3759
30BE_0064	MIB Control Register (ENET_MIBC)	32	R/W	C000_0000h	11.5.5.8/3761
30BE_0084	Receive Control Register (ENET_RCR)	32	R/W	05EE_0001h	11.5.5.9/3762
30BE_00C4	Transmit Control Register (ENET_TCR)	32	R/W	0000_0000h	11.5.5.10/3765
30BE_00E4	Physical Address Lower Register (ENET_PALR)	32	R/W	0000_0000h	11.5.5.11/3767
30BE_00E8	Physical Address Upper Register (ENET_PAUR)	32	R/W	0000_8808h	11.5.5.12/3767
30BE_00EC	Opcode/Pause Duration Register (ENET_OPD)	32	R/W	0001_0000h	11.5.5.13/3768
30BE_00F0	Transmit Interrupt Coalescing Register (ENET_TXIC0)	32	R/W	0000_0000h	11.5.5.14/3768
30BE_00F4	Transmit Interrupt Coalescing Register (ENET_TXIC1)	32	R/W	0000_0000h	11.5.5.14/3768
30BE_00F8	Transmit Interrupt Coalescing Register (ENET_TXIC2)	32	R/W	0000_0000h	11.5.5.14/3768
30BE_0100	Receive Interrupt Coalescing Register (ENET_RXIC0)	32	R/W	0000_0000h	11.5.5.15/3769
30BE_0104	Receive Interrupt Coalescing Register (ENET_RXIC1)	32	R/W	0000_0000h	11.5.5.15/3769
30BE_0108	Receive Interrupt Coalescing Register (ENET_RXIC2)	32	R/W	0000_0000h	11.5.5.15/3769
30BE_0118	Descriptor Individual Upper Address Register (ENET_IAUR)	32	R/W	0000_0000h	11.5.5.16/3770
30BE_011C	Descriptor Individual Lower Address Register (ENET_IALR)	32	R/W	0000_0000h	11.5.5.17/3771
30BE_0120	Descriptor Group Upper Address Register (ENET_GAUR)	32	R/W	0000_0000h	11.5.5.18/3771
30BE_0124	Descriptor Group Lower Address Register (ENET_GALR)	32	R/W	0000_0000h	11.5.5.19/3772
30BE_0144	Transmit FIFO Watermark Register (ENET_TFWR)	32	R/W	0000_0000h	11.5.5.20/3772
30BE_0160	Receive Descriptor Ring 1 Start Register (ENET_RDSR1)	32	R/W	0000_0000h	11.5.5.21/3773
30BE_0164	Transmit Buffer Descriptor Ring 1 Start Register (ENET_TDSDR1)	32	R/W	0000_0000h	11.5.5.22/3774
30BE_0168	Maximum Receive Buffer Size Register - Ring 1 (ENET_MRBR1)	32	R/W	0000_0000h	11.5.5.23/3775

Table continues on the next page...

ENET memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
30BE_016C	Receive Descriptor Ring 2 Start Register (ENET_RDSCR2)	32	R/W	0000_0000h	11.5.5.24/3776
30BE_0170	Transmit Buffer Descriptor Ring 2 Start Register (ENET_TDSCR2)	32	R/W	0000_0000h	11.5.5.25/3776
30BE_0174	Maximum Receive Buffer Size Register - Ring 2 (ENET_MRBR2)	32	R/W	0000_0000h	11.5.5.26/3777
30BE_0180	Receive Descriptor Ring 0 Start Register (ENET_RDSCR)	32	R/W	0000_0000h	11.5.5.27/3778
30BE_0184	Transmit Buffer Descriptor Ring 0 Start Register (ENET_TDSCR)	32	R/W	0000_0000h	11.5.5.28/3779
30BE_0188	Maximum Receive Buffer Size Register - Ring 0 (ENET_MRBR)	32	R/W	0000_0000h	11.5.5.29/3779
30BE_0190	Receive FIFO Section Full Threshold (ENET_RSFL)	32	R/W	0000_0000h	11.5.5.30/3780
30BE_0194	Receive FIFO Section Empty Threshold (ENET_RSEM)	32	R/W	0000_0000h	11.5.5.31/3781
30BE_0198	Receive FIFO Almost Empty Threshold (ENET_RAEM)	32	R/W	0000_0004h	11.5.5.32/3781
30BE_019C	Receive FIFO Almost Full Threshold (ENET_RAFL)	32	R/W	0000_0004h	11.5.5.33/3782
30BE_01A0	Transmit FIFO Section Empty Threshold (ENET_TSEM)	32	R/W	0000_0000h	11.5.5.34/3782
30BE_01A4	Transmit FIFO Almost Empty Threshold (ENET_TAEM)	32	R/W	0000_0004h	11.5.5.35/3783
30BE_01A8	Transmit FIFO Almost Full Threshold (ENET_TAFL)	32	R/W	0000_0008h	11.5.5.36/3783
30BE_01AC	Transmit Inter-Packet Gap (ENET_TIPG)	32	R/W	0000_000Ch	11.5.5.37/3784
30BE_01B0	Frame Truncation Length (ENET_FTRL)	32	R/W	0000_07FFh	11.5.5.38/3784
30BE_01C0	Transmit Accelerator Function Configuration (ENET_TACC)	32	R/W	0000_0000h	11.5.5.39/3785
30BE_01C4	Receive Accelerator Function Configuration (ENET_RACC)	32	R/W	0000_0000h	11.5.5.40/3786
30BE_01C8	Receive Classification Match Register for Class n (ENET_RCMR1)	32	R/W	0000_0000h	11.5.5.41/3787
30BE_01CC	Receive Classification Match Register for Class n (ENET_RCMR2)	32	R/W	0000_0000h	11.5.5.41/3787
30BE_01D8	DMA Class Based Configuration (ENET_DMA1CFG)	32	R/W	0000_0000h	11.5.5.42/3788
30BE_01DC	DMA Class Based Configuration (ENET_DMA2CFG)	32	R/W	0000_0000h	11.5.5.42/3788
30BE_01E0	Receive Descriptor Active Register - Ring 1 (ENET_RDAR1)	32	R/W	0000_0000h	11.5.5.43/3790

Table continues on the next page...

ENET memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
30BE_01E4	Transmit Descriptor Active Register - Ring 1 (ENET_TDAR1)	32	R/W	0000_0000h	11.5.5.44/3791
30BE_01E8	Receive Descriptor Active Register - Ring 2 (ENET_RDAR2)	32	R/W	0000_0000h	11.5.5.45/3792
30BE_01EC	Transmit Descriptor Active Register - Ring 2 (ENET_TDAR2)	32	R/W	0000_0000h	11.5.5.46/3793
30BE_01F0	QOS Scheme (ENET_QOS)	32	R/W	0000_0000h	11.5.5.47/3793
30BE_0200	Reserved Statistic Register (ENET_RMON_T_DROP)	32	R	0000_0000h	11.5.5.48/3795
30BE_0204	Tx Packet Count Statistic Register (ENET_RMON_T_PACKETS)	32	R	0000_0000h	11.5.5.49/3795
30BE_0208	Tx Broadcast Packets Statistic Register (ENET_RMON_T_BC_PKT)	32	R	0000_0000h	11.5.5.50/3796
30BE_020C	Tx Multicast Packets Statistic Register (ENET_RMON_T_MC_PKT)	32	R	0000_0000h	11.5.5.51/3796
30BE_0210	Tx Packets with CRC/Align Error Statistic Register (ENET_RMON_T_CRC_ALIGN)	32	R	0000_0000h	11.5.5.52/3797
30BE_0214	Tx Packets Less Than Bytes and Good CRC Statistic Register (ENET_RMON_T_UNDERSIZE)	32	R	0000_0000h	11.5.5.53/3797
30BE_0218	Tx Packets GT MAX_FL bytes and Good CRC Statistic Register (ENET_RMON_T_OVERSIZE)	32	R	0000_0000h	11.5.5.54/3797
30BE_021C	Tx Packets Less Than 64 Bytes and Bad CRC Statistic Register (ENET_RMON_T_FRAG)	32	R	0000_0000h	11.5.5.55/3798
30BE_0220	Tx Packets Greater Than MAX_FL bytes and Bad CRC Statistic Register (ENET_RMON_T_JAB)	32	R	0000_0000h	11.5.5.56/3798
30BE_0224	Tx Collision Count Statistic Register (ENET_RMON_T_COL)	32	R	0000_0000h	11.5.5.57/3799
30BE_0228	Tx 64-Byte Packets Statistic Register (ENET_RMON_T_P64)	32	R	0000_0000h	11.5.5.58/3799
30BE_022C	Tx 65- to 127-byte Packets Statistic Register (ENET_RMON_T_P65TO127)	32	R	0000_0000h	11.5.5.59/3799
30BE_0230	Tx 128- to 255-byte Packets Statistic Register (ENET_RMON_T_P128TO255)	32	R	0000_0000h	11.5.5.60/3800
30BE_0234	Tx 256- to 511-byte Packets Statistic Register (ENET_RMON_T_P256TO511)	32	R	0000_0000h	11.5.5.61/3800
30BE_0238	Tx 512- to 1023-byte Packets Statistic Register (ENET_RMON_T_P512TO1023)	32	R	0000_0000h	11.5.5.62/3801
30BE_023C	Tx 1024- to 2047-byte Packets Statistic Register (ENET_RMON_T_P1024TO2047)	32	R	0000_0000h	11.5.5.63/3801
30BE_0240	Tx Packets Greater Than 2048 Bytes Statistic Register (ENET_RMON_T_P_GTE2048)	32	R	0000_0000h	11.5.5.64/3802
30BE_0244	Tx Octets Statistic Register (ENET_RMON_T_OCTETS)	32	R	0000_0000h	11.5.5.65/3802

Table continues on the next page...

ENET memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
30BE_0248	Reserved Statistic Register (ENET_IEEE_T_DROP)	32	R	0000_0000h	11.5.5.66/3802
30BE_024C	Frames Transmitted OK Statistic Register (ENET_IEEE_T_FRAME_OK)	32	R	0000_0000h	11.5.5.67/3803
30BE_0250	Frames Transmitted with Single Collision Statistic Register (ENET_IEEE_T_1COL)	32	R	0000_0000h	11.5.5.68/3803
30BE_0254	Frames Transmitted with Multiple Collisions Statistic Register (ENET_IEEE_T_MCOL)	32	R	0000_0000h	11.5.5.69/3804
30BE_0258	Frames Transmitted after Deferral Delay Statistic Register (ENET_IEEE_T_DEF)	32	R	0000_0000h	11.5.5.70/3804
30BE_025C	Frames Transmitted with Late Collision Statistic Register (ENET_IEEE_T_LCOL)	32	R	0000_0000h	11.5.5.71/3804
30BE_0260	Frames Transmitted with Excessive Collisions Statistic Register (ENET_IEEE_T_EXCOL)	32	R	0000_0000h	11.5.5.72/3805
30BE_0264	Frames Transmitted with Tx FIFO Underrun Statistic Register (ENET_IEEE_T_MACERR)	32	R	0000_0000h	11.5.5.73/3805
30BE_0268	Frames Transmitted with Carrier Sense Error Statistic Register (ENET_IEEE_T_CSERR)	32	R	0000_0000h	11.5.5.74/3806
30BE_026C	Reserved Statistic Register (ENET_IEEE_T_SQE)	32	R (reads 0)	0000_0000h	11.5.5.75/3806
30BE_0270	Flow Control Pause Frames Transmitted Statistic Register (ENET_IEEE_T_FDXFC)	32	R	0000_0000h	11.5.5.76/3806
30BE_0274	Octet Count for Frames Transmitted w/o Error Statistic Register (ENET_IEEE_T_OCTETS_OK)	32	R	0000_0000h	11.5.5.77/3807
30BE_0284	Rx Packet Count Statistic Register (ENET_RMON_R_PACKETS)	32	R	0000_0000h	11.5.5.78/3807
30BE_0288	Rx Broadcast Packets Statistic Register (ENET_RMON_R_BC_PKT)	32	R	0000_0000h	11.5.5.79/3808
30BE_028C	Rx Multicast Packets Statistic Register (ENET_RMON_R_MC_PKT)	32	R	0000_0000h	11.5.5.80/3808
30BE_0290	Rx Packets with CRC/Align Error Statistic Register (ENET_RMON_R_CRC_ALIGN)	32	R	0000_0000h	11.5.5.81/3808
30BE_0294	Rx Packets with Less Than 64 Bytes and Good CRC Statistic Register (ENET_RMON_R_UNDERSIZE)	32	R	0000_0000h	11.5.5.82/3809
30BE_0298	Rx Packets Greater Than MAX_FL and Good CRC Statistic Register (ENET_RMON_R_OVERSIZE)	32	R	0000_0000h	11.5.5.83/3809
30BE_029C	Rx Packets Less Than 64 Bytes and Bad CRC Statistic Register (ENET_RMON_R_FRAG)	32	R	0000_0000h	11.5.5.84/3810
30BE_02A0	Rx Packets Greater Than MAX_FL Bytes and Bad CRC Statistic Register (ENET_RMON_R_JAB)	32	R	0000_0000h	11.5.5.85/3810
30BE_02A4	Reserved Statistic Register (ENET_RMON_R_RESVD_0)	32	R (reads 0)	0000_0000h	11.5.5.86/3810
30BE_02A8	Rx 64-Byte Packets Statistic Register (ENET_RMON_R_P64)	32	R	0000_0000h	11.5.5.87/3811

Table continues on the next page...

ENET memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
30BE_02AC	Rx 65- to 127-Byte Packets Statistic Register (ENET_RMON_R_P65TO127)	32	R	0000_0000h	11.5.5.88/3811
30BE_02B0	Rx 128- to 255-Byte Packets Statistic Register (ENET_RMON_R_P128TO255)	32	R	0000_0000h	11.5.5.89/3812
30BE_02B4	Rx 256- to 511-Byte Packets Statistic Register (ENET_RMON_R_P256TO511)	32	R	0000_0000h	11.5.5.90/3812
30BE_02B8	Rx 512- to 1023-Byte Packets Statistic Register (ENET_RMON_R_P512TO1023)	32	R	0000_0000h	11.5.5.91/3812
30BE_02BC	Rx 1024- to 2047-Byte Packets Statistic Register (ENET_RMON_R_P1024TO2047)	32	R	0000_0000h	11.5.5.92/3813
30BE_02C0	Rx Packets Greater than 2048 Bytes Statistic Register (ENET_RMON_R_P_GTE2048)	32	R	0000_0000h	11.5.5.93/3813
30BE_02C4	Rx Octets Statistic Register (ENET_RMON_R_OCTETS)	32	R	0000_0000h	11.5.5.94/3814
30BE_02C8	Frames not Counted Correctly Statistic Register (ENET_IEEE_R_DROP)	32	R	0000_0000h	11.5.5.95/3814
30BE_02CC	Frames Received OK Statistic Register (ENET_IEEE_R_FRAME_OK)	32	R	0000_0000h	11.5.5.96/3814
30BE_02D0	Frames Received with CRC Error Statistic Register (ENET_IEEE_R_CRC)	32	R	0000_0000h	11.5.5.97/3815
30BE_02D4	Frames Received with Alignment Error Statistic Register (ENET_IEEE_R_ALIGN)	32	R	0000_0000h	11.5.5.98/3815
30BE_02D8	Receive FIFO Overflow Count Statistic Register (ENET_IEEE_R_MACERR)	32	R	0000_0000h	11.5.5.99/3816
30BE_02DC	Flow Control Pause Frames Received Statistic Register (ENET_IEEE_R_FDXFC)	32	R	0000_0000h	11.5.5.100/3816
30BE_02E0	Octet Count for Frames Received without Error Statistic Register (ENET_IEEE_R_OCTETS_OK)	32	R	0000_0000h	11.5.5.101/3816
30BE_0400	Adjustable Timer Control Register (ENET_ATCR)	32	R/W	0000_0000h	11.5.5.102/3817
30BE_0404	Timer Value Register (ENET_ATVR)	32	R/W	0000_0000h	11.5.5.103/3819
30BE_0408	Timer Offset Register (ENET_ATOFF)	32	R/W	0000_0000h	11.5.5.104/3819
30BE_040C	Timer Period Register (ENET_ATPER)	32	R/W	3B9A_CA00h	11.5.5.105/3820
30BE_0410	Timer Correction Register (ENET_ATCOR)	32	R/W	0000_0000h	11.5.5.106/3820
30BE_0414	Time-Stamping Clock Period Register (ENET_ATINC)	32	R/W	0000_0000h	11.5.5.107/3821
30BE_0418	Timestamp of Last Transmitted Frame (ENET_ATSTMP)	32	R	0000_0000h	11.5.5.108/3821
30BE_0604	Timer Global Status Register (ENET_TGSR)	32	R/W	0000_0000h	11.5.5.109/3822

Table continues on the next page...

ENET memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
30BE_0608	Timer Control Status Register (ENET_TCSR0)	32	R/W	0000_0000h	11.5.5.110/3823
30BE_060C	Timer Compare Capture Register (ENET_TCCR0)	32	R/W	0000_0000h	11.5.5.111/3824
30BE_0610	Timer Control Status Register (ENET_TCSR1)	32	R/W	0000_0000h	11.5.5.110/3823
30BE_0614	Timer Compare Capture Register (ENET_TCCR1)	32	R/W	0000_0000h	11.5.5.111/3824
30BE_0618	Timer Control Status Register (ENET_TCSR2)	32	R/W	0000_0000h	11.5.5.110/3823
30BE_061C	Timer Compare Capture Register (ENET_TCCR2)	32	R/W	0000_0000h	11.5.5.111/3824
30BE_0620	Timer Control Status Register (ENET_TCSR3)	32	R/W	0000_0000h	11.5.5.110/3823
30BE_0624	Timer Compare Capture Register (ENET_TCCR3)	32	R/W	0000_0000h	11.5.5.111/3824

11.5.5.1 Interrupt Event Register (ENET_EIR)

When an event occurs that sets a bit in EIR, an interrupt occurs if the corresponding bit in the interrupt mask register (EIMR) is also set. Writing a 1 to an EIR bit clears it; writing 0 has no effect. This register is cleared upon hardware reset.

NOTE

TxBD[INT] and RxBD[INT] must be set to 1 to allow setting the corresponding EIR register flags in enhanced mode, ENET_ECR[EN1588] = 1. Legacy mode does not require these flags to be enabled.

Address: 30BE_0000h base + 4h offset = 30BE_0004h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0	BABR	BABT	GRA	TXF	TXB	RXF	RXB	MII	EBERR	LC	RL	UN	PLR	WAKEUP	TS_AVAIL
W		w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Ethernet MAC (ENET)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	TS_TIMER	RXFLUSH_2	RXFLUSH_1	RXFLUSH_0					TXF2	TXB2	RXF2	RXB2	TXF1	TXB1	RXF1	RXB1
W	w1c	w1c	w1c	w1c		0		0	w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_EIR field descriptions

Field	Description
31 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
30 BABR	Babbling Receive Error Indicates a frame was received with length in excess of RCR[MAX_FL] bytes.
29 BABT	Babbling Transmit Error Indicates the transmitted frame length exceeds RCR[MAX_FL] bytes. Usually this condition is caused when a frame that is too long is placed into the transmit data buffer(s). Truncation does not occur.
28 GRA	Graceful Stop Complete This interrupt is asserted after the transmitter is put into a pause state after completion of the frame currently being transmitted. See Graceful Transmit Stop (GTS) for conditions that lead to graceful stop. NOTE: The GRA interrupt is asserted only when the TX transitions into the stopped state. If this bit is cleared by writing 1 and the TX is still stopped, the bit is not set again.
27 TXF	Transmit Frame Interrupt Indicates a frame has been transmitted and the last corresponding buffer descriptor has been updated.
26 TXB	Transmit Buffer Interrupt Indicates a transmit buffer descriptor has been updated.
25 RXF	Receive Frame Interrupt Indicates a frame has been received and the last corresponding buffer descriptor has been updated.
24 RXB	Receive Buffer Interrupt Indicates a receive buffer descriptor is not the last in the frame has been updated.
23 MII	MII Interrupt. Indicates that the MII has completed the data transfer requested.
22 EBERR	Ethernet Bus Error Indicates a system bus error occurred when a uDMA transaction is underway. When this bit is set, ECR[ETHEREN] is cleared, halting frame processing by the MAC. When this occurs, software must ensure proper actions, possibly resetting the system, to resume normal operation.

Table continues on the next page...

ENET_EIR field descriptions (continued)

Field	Description
21 LC	Late Collision Indicates a collision occurred beyond the collision window (slot time) in half-duplex mode. The frame truncates with a bad CRC and the remainder of the frame is discarded.
20 RL	Collision Retry Limit Indicates a collision occurred on each of 16 successive attempts to transmit the frame. The frame is discarded without being transmitted and transmission of the next frame commences. This error can only occur in half-duplex mode.
19 UN	Transmit FIFO Underrun Indicates the transmit FIFO became empty before the complete frame was transmitted. NOTE: In situations where the device has various masters generating high traffic, a FIFO underrun can occur on the transmit FIFO. To avoid transmit FIFO underrun, store and forward can be enabled in ENET_TFWR[STRFWD]. See STRFWD . Also, a higher priority can be set for ENET traffic using available means on the central bus fabric connecting the ENET module.
18 PLR	Payload Receive Error Indicates a frame was received with a payload length error. See Frame Length/Type Verification: Payload Length Check for more information.
17 WAKEUP	Node Wakeup Request Indication Read-only status bit to indicate that a magic packet has been detected. Will act only if ECR[MAGICEN] is set.
16 TS_AVAIL	Transmit Timestamp Available Indicates that the timestamp of the last transmitted timing frame is available in the ATSTMP register.
15 TS_TIMER	Timestamp Timer The adjustable timer reached the period event. A period event interrupt can be generated if ATCR[PEREN] is set and the timer wraps according to the periodic setting in the ATPER register. Set the timer period value before setting ATCR[PEREN].
14 RXFLUSH_2	RX DMA Ring 2 flush indication. This ring's RX frame has been flushed due to either RDAR2[RDAR] or RxBDE[E] being clear and only if QOS[RXFLUSH_2] is enabled.
13 RXFLUSH_1	RX DMA Ring 1 flush indication. This ring's RX frame has been flushed due to either RDAR1[RDAR] or RxBDE[E] being clear and only if QOS[RXFLUSH_1] is enabled.
12 RXFLUSH_0	RX DMA Ring 0 flush indication. . This ring's RX frame has been flushed due to either RDAR[RDAR] or RxBDE[E] being clear and only if QOS[RXFLUSH_0] is enabled.
11–9 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
8 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
7 TXF2	Transmit frame interrupt, class 2 This bit indicates that a frame has been transmitted and the last corresponding buffer descriptor has been updated (ring/class 2).
6 TXB2	Transmit buffer interrupt, class 2 This field indicates that a transmit buffer descriptor has been updated (ring/class 2).

Table continues on the next page...

ENET_EIR field descriptions (continued)

Field	Description
5 RXF2	Receive frame interrupt, class 2 This field indicates that a frame has been received and the last corresponding buffer descriptor has been updated (ring/class 2).
4 RXB2	Receive buffer interrupt, class 2 This field indicates that a receive buffer descriptor, that not the last in the frame, has been updated (ring/class 2).
3 TXF1	Transmit frame interrupt, class 1 This bit indicates that a frame has been transmitted and the last corresponding buffer descriptor has been updated (ring/class 1).
2 TXB1	Transmit buffer interrupt, class 1 This field indicates that a transmit buffer descriptor has been updated (ring/class 1).
1 RXF1	Receive frame interrupt, class 1 This field indicates that a frame has been received and the last corresponding buffer descriptor has been updated (ring/class 1).
0 RXB1	Receive buffer interrupt, class 1 This field indicates that a receive buffer descriptor, that not the last in the frame, has been updated (ring/class 1).

11.5.5.2 Interrupt Mask Register (ENET_EIMR)

EIMR controls which interrupt events are allowed to generate actual interrupts. A hardware reset clears this register. If the corresponding bits in the EIR and EIMR registers are set, an interrupt is generated. The interrupt signal remains asserted until a 1 is written to the EIR field (write 1 to clear) or a 0 is written to the EIMR field.

Address: 30BE_0000h base + 8h offset = 30BE_0008h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W	0	BABR	BABT	GRA	TXF	TXB	RXF	RXB	MII	EBERR	LC	RL	UN	PLR	WAKEUP	TS_AVAIL
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W	TS_TIMER	RXFLUSH_2	RXFLUSH_1	RXFLUSH_0					TXF2	TXB2	RXF2	RXB2	TXF1	TXB1	RXF1	RXB1
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_EIMR field descriptions

Field	Description
31 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
30 BABR	BABR Interrupt Mask Corresponds to interrupt source EIR[BABR] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR BABR field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared. 0 The corresponding interrupt source is masked. 1 The corresponding interrupt source is not masked.
29 BAPT	BAPT Interrupt Mask Corresponds to interrupt source EIR[BAPT] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR BAPT field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared. 0 The corresponding interrupt source is masked. 1 The corresponding interrupt source is not masked.
28 GRA	GRA Interrupt Mask Corresponds to interrupt source EIR[GRA] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR GRA field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared. 0 The corresponding interrupt source is masked. 1 The corresponding interrupt source is not masked.
27 TXF	TXF Interrupt Mask Corresponds to interrupt source EIR[TXF] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR TXF field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared. 0 The corresponding interrupt source is masked. 1 The corresponding interrupt source is not masked.
26 TXB	TXB Interrupt Mask Corresponds to interrupt source EIR[TXB] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR TXF field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared. 0 The corresponding interrupt source is masked. 1 The corresponding interrupt source is not masked.
25 RXF	RXF Interrupt Mask Corresponds to interrupt source EIR[RXF] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR RXF field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.

Table continues on the next page...

ENET_EIMR field descriptions (continued)

Field	Description
24 RXB	RXB Interrupt Mask Corresponds to interrupt source EIR[RXB] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR RXB field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
23 MII	MII Interrupt Mask Corresponds to interrupt source EIR[MII] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR MII field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
22 EBERR	EBERR Interrupt Mask Corresponds to interrupt source EIR[EBERR] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR EBERR field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
21 LC	LC Interrupt Mask Corresponds to interrupt source EIR[LC] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR LC field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
20 RL	RL Interrupt Mask Corresponds to interrupt source EIR[RL] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR RL field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
19 UN	UN Interrupt Mask Corresponds to interrupt source EIR[UN] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR UN field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
18 PLR	PLR Interrupt Mask Corresponds to interrupt source EIR[PLR] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR PLR field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
17 WAKEUP	WAKEUP Interrupt Mask Corresponds to interrupt source EIR[WAKEUP] register and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR WAKEUP field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
16 TS_AVAIL	TS_AVAIL Interrupt Mask Corresponds to interrupt source EIR[TS_AVAIL] register and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting

Table continues on the next page...

ENET_EIMR field descriptions (continued)

Field	Description
	source. The corresponding EIR TS_AVAIL field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
15 TS_TIMER	TS_TIMER Interrupt Mask Corresponds to interrupt source EIR[TS_TIMER] register and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. The corresponding EIR TS_TIMER field reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
14 RXFLUSH_2	Corresponds to interrupt source EIR[RXFLUSH_2] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. EIR[RXFLUSH_2] reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
13 RXFLUSH_1	Corresponds to interrupt source EIR[RXFLUSH_1] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. EIR[RXFLUSH_1] reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
12 RXFLUSH_0	Corresponds to interrupt source EIR[RXFLUSH_0] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. EIR[RXFLUSH_0] reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
11–9 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
8 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
7 TXF2	Transmit frame interrupt, class 2 Corresponds to interrupt source EIR[TXF2] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. EIR[TXF2] reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
6 TXB2	Transmit buffer interrupt, class 2 Corresponds to interrupt source EIR[TXB2] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. EIR[TXB2] reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
5 RXF2	Receive frame interrupt, class 2 Corresponds to interrupt source EIR[RXF2] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. EIR[RXF2] reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
4 RXB2	Receive buffer interrupt, class 2 Corresponds to interrupt source EIR[RXB2] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. EIR[RXB2] reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
3 TXF1	Transmit frame interrupt, class 1 Corresponds to interrupt source EIR[TXF1] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. EIR[TXF1] reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
2 TXB1	Transmit buffer interrupt, class 1

Table continues on the next page...

ENET_EIMR field descriptions (continued)

Field	Description
	Corresponds to interrupt source EIR[TXB1] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. EIR[TXB1] reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
1 RXF1	Receive frame interrupt, class 1 Corresponds to interrupt source EIR[RXF1] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. EIR[RXF1] reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.
0 RXB1	Receive buffer interrupt, class 1 Corresponds to interrupt source EIR[RXB1] and determines whether an interrupt condition can generate an interrupt. At every module clock, the EIR samples the signal generated by the interrupting source. EIR[RXB1] reflects the state of the interrupt signal even if the corresponding EIMR field is cleared.

11.5.5.3 Receive Descriptor Active Register - Ring 0 (ENET_RDAR)

RDAR is a command register, written by the user, to indicate that the receive descriptor ring 0 has been updated, that is, that the driver produced empty receive buffers with the empty bit set.

Address: 30BE_0000h base + 10h offset = 30BE_0010h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0							RDAR	0							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RDAR field descriptions

Field	Description
31–25 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
24 RDAR	Receive Descriptor Active

Table continues on the next page...

ENET_RDAR field descriptions (continued)

Field	Description
	Always set to 1 when this register is written, regardless of the value written. This field is cleared by the MAC device when no additional empty descriptors remain in the receive ring. It is also cleared when ECR[ETHEREN] transitions from set to cleared or when ECR[RESET] is set.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.4 Transmit Descriptor Active Register - Ring 0 (ENET_TDAR)

The TDAR is a command register that the user writes to indicate that the transmit descriptor ring 0 has been updated, that is, that transmit buffers have been produced by the driver with the ready bit set in the buffer descriptor.

The TDAR register is cleared at reset, when ECR[ETHEREN] transitions from set to cleared, or when ECR[RESET] is set.

Address: 30BE_0000h base + 14h offset = 30BE_0014h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
R	0								TDAR	0							
W																	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
R	0																
W																	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

ENET_TDAR field descriptions

Field	Description
31–25 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
24 TDAR	Transmit Descriptor Active Always set to 1 when this register is written, regardless of the value written. This bit is cleared by the MAC device when no additional ready descriptors remain in the transmit ring. Also cleared when ECR[ETHEREN] transitions from set to cleared or when ECR[RESET] is set.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.5 Ethernet Control Register (ENET_ECR)

ECR is a read/write user register, though hardware may also alter fields in this register. It controls many of the high level features of the Ethernet MAC, including legacy FEC support through the EN1588 field.

Address: 30BE_0000h base + 24h offset = 30BE_0024h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved														Reserved	
W																
Reset	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved				SVLANDBL	VLANUSE2ND	SVLANEN	DBSWP	Reserved	DBGEN	SPEED	EN1588	SLEEP	MAGICEN	ETHEREN	RESET
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_ECR field descriptions

Field	Description
31–18 Reserved	This field is reserved. Always write 01110000000000b to this field.
17–12 Reserved	This field is reserved. Always write 0 to this field.
11 SVLANDBL	S-VLAN double tag If enabled, S-VLAN detection requires a double-tagged frame to define a frame as being a VLAN frame. The following rules apply: • If the first tag is the S-VLAN type, it must be followed by a second tag with the C-VLAN type to declare the frame as VLAN frame. • If the first tag is the S-VLAN type but no 2nd tag follows, it is not considered a VLAN frame but instead treated as an untagged frame. • If the first tag is the C-VLAN type, it is considered a VLAN frame as normal. NOTE: VLANUSE2ND can be used to determine from which tag the data should be extracted. This applies only if SVLAN_EN = 1, ignored otherwise.
10 VLANUSE2ND	VLAN use second tag 0 Always extract data from the first VLAN tag if it exists. 1 When a double-tagged frame is detected, the data of the second tag is extracted for further processing. A double-tagged frame is defined as: • The first tag can be a C-VLAN or a S-VLAN (if SVLAN_ENA = 1) • The second tag must be a C-VLAN

Table continues on the next page...

ENET_ECR field descriptions (continued)

Field	Description
9 SVLANEN	S-VLAN enable Enable additional detection of S-VLAN tag according to IEEE802.1Q. 0 Only the EtherType 0x8100 will be considered for VLAN detection. 1 The EtherType 0x88a8 will be considered in addition to 0x8100 (C-VLAN) to identify a VLAN frame in receive. When a VLAN frame is identified, the two bytes following the VLAN type are extracted and used by the classification match comparators, RCMRn.
8 DBSWP	Descriptor Byte Swapping Enable Swaps the byte locations of the buffer descriptors. NOTE: This field resets to 0 and must not be changed. 0 The buffer descriptor bytes are not swapped to support big-endian devices. 1 The buffer descriptor bytes are swapped to support little-endian devices.
7 Reserved	This field is reserved. Always write 0 to this field.
6 DBGEN	Debug Enable Enables the MAC to enter hardware freeze mode when the device enters debug mode. 0 MAC continues operation in debug mode. 1 MAC enters hardware freeze mode when the processor is in debug mode.
5 SPEED	Selects between 10/100-Mbit/s and 1000-Mbit/s modes of operation. 0 10/100-Mbit/s mode 1 1000-Mbit/s mode
4 EN1588	EN1588 Enable Enables enhanced functionality of the MAC. 0 Legacy FEC buffer descriptors and functions enabled. 1 Enhanced frame time-stamping functions enabled.
3 SLEEP	Sleep Mode Enable 0 Normal operating mode. 1 Sleep mode.
2 MAGICEN	Magic Packet Detection Enable Enables/disables magic packet detection. NOTE: MAGICEN is relevant only if the SLEEP field is set. If MAGICEN is set, changing the SLEEP field enables/disables sleep mode and magic packet detection. NOTE: EIMR[WAKEUP] must be written to one if Magic packet wakeup is programed to wake up the chip from low power mode. 0 Magic detection logic disabled. 1 The MAC core detects magic packets and asserts EIR[WAKEUP] when a frame is detected.
1 ETHEREN	Ethernet Enable

Table continues on the next page...

ENET_ECR field descriptions (continued)

Field	Description
	Enables/disables the Ethernet MAC. When the MAC is disabled, the buffer descriptors for an aborted transmit frame are not updated. The uDMA, buffer descriptor, and FIFO control logic are reset, including the buffer descriptor and FIFO pointers. Hardware clears this field under the following conditions: • RESET is set by software • An error condition causes the EBERR field to set. NOTE: • ETHEREN must be set at the very last step during ENET configuration/setup/initialization, only <i>after</i> all other ENET-related registers have been configured. • If ETHEREN is cleared to 0 by software then next time ETHEREN is set, the EIR interrupts must cleared to 0 due to previous pending interrupts. 0 Reception immediately stops and transmission stops after a bad CRC is appended to any currently transmitted frame. 1 MAC is enabled, and reception and transmission are possible.
0 RESET	Ethernet MAC Reset When this field is set, it clears the ETHEREN field.

11.5.5.6 MII Management Frame Register (ENET_MMFR)

Writing to MMFR triggers a management frame transaction to the PHY device unless MSCR is programmed to zero.

If MSCR is changed from zero to non-zero during a write to MMFR, an MII frame is generated with the data previously written to the MMFR. This allows MMFR and MSCR to be programmed in either order if MSCR is currently zero.

If the MMFR register is written while frame generation is in progress, the frame contents are altered. Software must use the EIR[MII] interrupt indication to avoid writing to the MMFR register while frame generation is in progress.

Address: 30BE_0000h base + 40h offset = 30BE_0040h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ST		OP		PA				RA				TA		DATA																	
W	ST		OP		PA				RA				TA		DATA																	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

ENET_MMFR field descriptions

Field	Description
31–30 ST	Start Of Frame Delimiter See Table 11-81 (Clause 22) or Table 11-83 (Clause 45) for correct value.
29–28 OP	Operation Code

Table continues on the next page...

ENET_MMFR field descriptions (continued)

Field	Description
	See Table 11-81 (Clause 22) or Table 11-83 (Clause 45) for correct value.
27–23 PA	PHY Address See Table 11-81 (Clause 22) or Table 11-83 (Clause 45) for correct value.
22–18 RA	Register Address See Table 11-81 (Clause 22) or Table 11-83 (Clause 45) for correct value.
17–16 TA	Turn Around This field must be programmed to 10 to generate a valid MII management frame.
DATA	Management Frame Data This is the field for data to be written to or read from the PHY register.

11.5.5.7 MII Speed Control Register (ENET_MSCR)

MSCR provides control of the MII clock (MDC pin) frequency and allows a preamble drop on the MII management frame.

The MII_SPEED field must be programmed with a value to provide an MDC frequency of less than or equal to 2.5 MHz to be compliant with the IEEE 802.3 MII specification. The MII_SPEED must be set to a non-zero value to source a read or write management frame. After the management frame is complete, the MSCR register may optionally be cleared to turn off MDC. The MDC signal generated has a 50% duty cycle except when MII_SPEED changes during operation. This change takes effect following a rising or falling edge of MDC.

For example, if the internal module clock (that is, peripheral bus clock) is 25 MHz, programming MII_SPEED to 0x4 results in an MDC as given in the following equation:

$$\text{MII clock frequency} = 25 \text{ MHz} / ((4 + 1) \times 2) = 2.5 \text{ MHz}$$

The following table shows the optimum values for MII_SPEED as a function of IPS bus clock frequency.

Table 11-87. Programming Examples for MSCR

Internal module clock frequency	MSCR [MII_SPEED]	MDC frequency
25 MHz	0x4	2.50 MHz
33 MHz	0x6	2.36 MHz
40 MHz	0x7	2.50 MHz
50 MHz	0x9	2.50 MHz
66 MHz	0xD	2.36 MHz

Ethernet MAC (ENET)

Address: 30BE_0000h base + 44h offset = 30BE_0044h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0					HOLDTIME			DIS_PRE	MII_SPEED						0
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

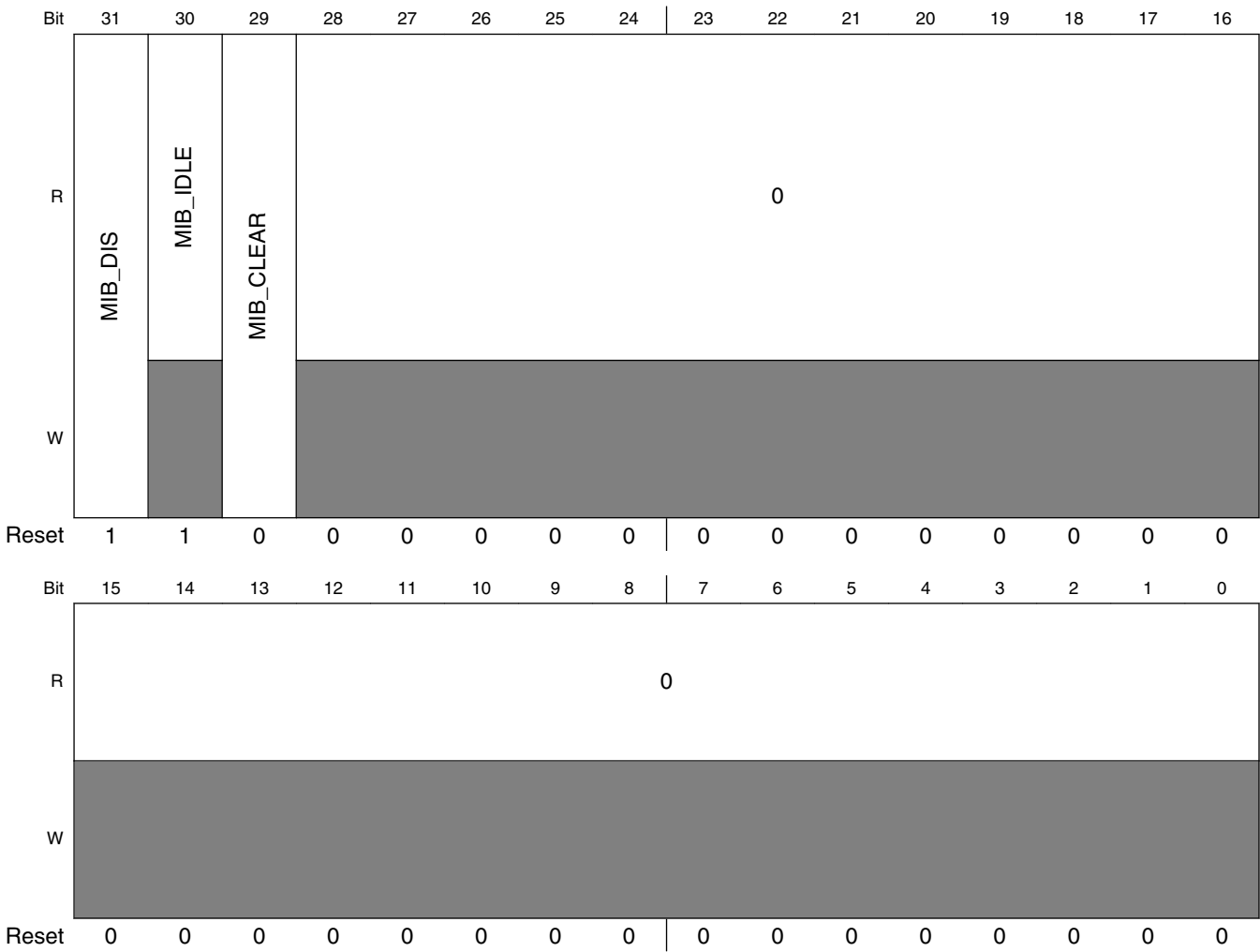
ENET_MSCR field descriptions

Field	Description
31–11 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
10–8 HOLDTIME	Hold time On MDIO Output IEEE802.3 clause 22 defines a minimum of 10 ns for the hold time on the MDIO output. Depending on the host bus frequency, the setting may need to be increased. 000 1 internal module clock cycle 001 2 internal module clock cycles 010 3 internal module clock cycles 111 8 internal module clock cycles
7 DIS_PRE	Disable Preamble Enables/disables prepending a preamble to the MII management frame. The MII standard allows the preamble to be dropped if the attached PHY devices do not require it. 0 Preamble enabled. 1 Preamble (32 ones) is not prepended to the MII management frame.
6–1 MII_SPEED	MII Speed Controls the frequency of the MII management interface clock (MDC) relative to the internal module clock. A value of 0 in this field turns off MDC and leaves it in low voltage state. Any non-zero value results in the MDC frequency of: $1/((\text{MII_SPEED} + 1) \times 2)$ of the internal module clock frequency
0 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.8 MIB Control Register (ENET_MIBC)

MIBC is a read/write register controlling and observing the state of the MIB block. Access this register to disable the MIB block operation or clear the MIB counters. The MIB_DIS field resets to 1.

Address: 30BE_0000h base + 64h offset = 30BE_0064h



ENET_MIBC field descriptions

Field	Description
31 MIB_DIS	Disable MIB Logic If this control field is set, 0 MIB logic is enabled. 1 MIB logic is disabled. The MIB logic halts and does not update any MIB counters.

Table continues on the next page...

ENET_MIBC field descriptions (continued)

Field	Description
30 MIB_IDLE	MIB Idle 0 The MIB block is updating MIB counters. 1 The MIB block is not currently updating any MIB counters.
29 MIB_CLEAR	MIB Clear NOTE: This field is not self-clearing. To clear the MIB counters set and then clear this field. 0 See note above. 1 All statistics counters are reset to 0.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.9 Receive Control Register (ENET_RCR)

Address: 30BE_0000h base + 84h offset = 30BE_0084h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
R	GRS	NLC	MAX_FL														
W																	
Reset	0	0	0	0	0	1	0	1	1	1	1	0	1	1	1	0	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
R	CFEN	CRCFWD	PAUFWD	PADEN				RMIL_10T	RMIL_MODE			FCE	BC_REJ	PROM	MIL_MODE	DRT	LOOP
W																	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	

ENET_RCR field descriptions

Field	Description
31 GRS	Graceful Receive Stopped Read-only status indicating that the MAC receive datapath is stopped.
30 NLC	Payload Length Check Disable Enables/disables a payload length check. 0 The payload length check is disabled. 1 The core checks the frame's payload length with the frame length/type field. Errors are indicated in the EIR[PLR] field.
29–16 MAX_FL	Maximum Frame Length Resets to decimal 1518. Length is measured starting at DA and includes the CRC at the end of the frame. Transmit frames longer than MAX_FL cause the BABT interrupt to occur. Receive frames longer than MAX_FL cause the BABR interrupt to occur and set the LG field in the end of frame receive buffer descriptor. The recommended default value to be programmed is 1518 or 1522 if VLAN tags are supported.
15 CFEN	MAC Control Frame Enable Enables/disables the MAC control frame. 0 MAC control frames with any opcode other than 0x0001 (pause frame) are accepted and forwarded to the client interface. 1 MAC control frames with any opcode other than 0x0001 (pause frame) are silently discarded.
14 CRCFWD	Terminate/Forward Received CRC Specifies whether the CRC field of received frames is transmitted or stripped. NOTE: If padding function is enabled (PADEN = 1), CRCFWD is ignored and the CRC field is checked and always terminated and removed. 0 The CRC field of received frames is transmitted to the user application. 1 The CRC field is stripped from the frame.
13 PAUFWD	Terminate/Forward Pause Frames Specifies whether pause frames are terminated or forwarded. 0 Pause frames are terminated and discarded in the MAC. 1 Pause frames are forwarded to the user application.
12 PADEN	Enable Frame Padding Remove On Receive Specifies whether the MAC removes padding from received frames. 0 No padding is removed on receive by the MAC. 1 Padding is removed from received frames.
11–10 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
9 RMII_10T	Enables 10-Mbit/s mode of the RMII or RGMII . 0 100-Mbit/s operation. 1 10-Mbit/s operation.

Table continues on the next page...

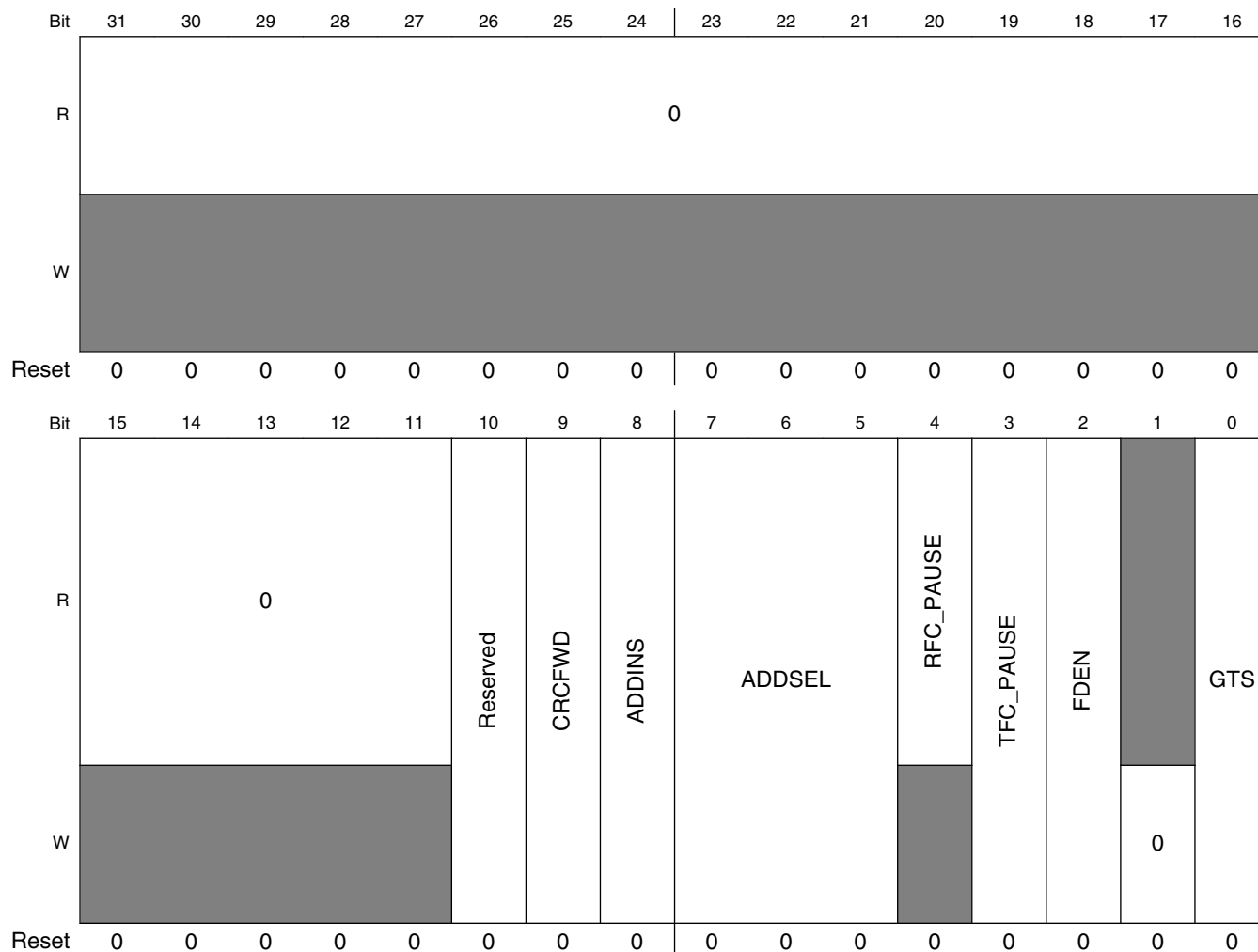
ENET_RCR field descriptions (continued)

Field	Description
8 RMII_MODE	RMII Mode Enable Specifies whether the MAC is configured for MII mode or RMII operation , when ECR[SPEED] is cleared . NOTE: Do not set both RCR[RGMIEN] and RCR[RMII_MODE]. 0 MAC configured for MII mode. 1 MAC configured for RMII operation.
7 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
6 RGMIEN	RGMI Mode Enable NOTE: Do not set both RCR[RGMIEN] and RCR[RMII_MODE]. 0 MAC configured for non-RGMI operation 1 MAC configured for RGMI operation. If ECR[SPEED] is set, the MAC is in RGMI 1000-Mbit/s mode. If ECR[SPEED] is cleared, the MAC is in RGMI 10/100-Mbit/s mode.
5 FCE	Flow Control Enable If set, the receiver detects PAUSE frames. Upon PAUSE frame detection, the transmitter stops transmitting data frames for a given duration.
4 BC_REJ	Broadcast Frame Reject If set, frames with destination address (DA) equal to 0xFFFF_FFFF_FFFF are rejected unless the PROM field is set. If BC_REJ and PROM are set, frames with broadcast DA are accepted and the MISS (M) is set in the receive buffer descriptor.
3 PROM	Promiscuous Mode All frames are accepted regardless of address matching. 0 Disabled. 1 Enabled.
2 MII_MODE	Media Independent Interface Mode This field must always be set. 0 Reserved. 1 MII or RMII mode, as indicated by the RMII_MODE field.
1 DRT	Disable Receive On Transmit 0 Receive path operates independently of transmit (i.e., full-duplex mode). Can also be used to monitor transmit activity in half-duplex mode. 1 Disable reception of frames while transmitting. (Normally used for half-duplex mode.)
0 LOOP	Internal Loopback This is an MII internal loopback, therefore MII_MODE must be written to 1 and RMII_MODE must be written to 0. 0 Loopback disabled. 1 Transmitted frames are looped back internal to the device and transmit MII output signals are not asserted. DRT must be cleared.

11.5.5.10 Transmit Control Register (ENET_TCR)

TCR is read/write and configures the transmit block. This register is cleared at system reset. FDEN can only be modified when ECR[ETHEREN] is cleared.

Address: 30BE_0000h base + C4h offset = 30BE_00C4h



ENET_TCR field descriptions

Field	Description
31–11 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
10 Reserved	This field is reserved. This field is read/write and must be set to 0.
9 CRCFWD	Forward Frame From Application With CRC

Table continues on the next page...

ENET_TCR field descriptions (continued)

Field	Description
	0 TxBD[TC] controls whether the frame has a CRC from the application. 1 The transmitter does not append any CRC to transmitted frames, as it is expecting a frame with CRC from the application.
8 ADDINS	Set MAC Address On Transmit 0 The source MAC address is not modified by the MAC. 1 The MAC overwrites the source MAC address with the programmed MAC address according to ADDSEL.
7–5 ADDSEL	Source MAC Address Select On Transmit If ADDINS is set, indicates the MAC address that overwrites the source MAC address. 000 Node MAC address programmed on PADDR1/2 registers. 100 Reserved. 101 Reserved. 110 Reserved.
4 RFC_PAUSE	Receive Frame Control Pause This status field is set when a full-duplex flow control pause frame is received and the transmitter pauses for the duration defined in this pause frame. This field automatically clears when the pause duration is complete.
3 TFC_PAUSE	Transmit Frame Control Pause Pauses frame transmission. When this field is set, EIR[GRA] is set. With transmission of data frames stopped, the MAC transmits a MAC control PAUSE frame. Next, the MAC clears TFC_PAUSE and resumes transmitting data frames. If the transmitter pauses due to user assertion of GTS or reception of a PAUSE frame, the MAC may continue transmitting a MAC control PAUSE frame. 0 No PAUSE frame transmitted. 1 The MAC stops transmission of data frames after the current transmission is complete.
2 FDEN	Full-Duplex Enable If this field is set, frames transmit independent of carrier sense and collision inputs. Only modify this bit when ECR[ETHEREN] is cleared.
1 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
0 GTS	Graceful Transmit Stop When this field is set, MAC stops transmission after any frame currently transmitted is complete and EIR[GRA] is set. If frame transmission is not currently underway, the GRA interrupt is asserted immediately. After transmission finishes, clear GTS to restart. The next frame in the transmit FIFO is then transmitted. If an early collision occurs during transmission when GTS is set, transmission stops after the collision. The frame is transmitted again after GTS is cleared. There may be old frames in the transmit FIFO that transmit when GTS is reasserted. To avoid this, clear ECR[ETHEREN] following the GRA interrupt.

11.5.5.11 Physical Address Lower Register (ENET_PALR)

PALR contains the lower 32 bits (bytes 0, 1, 2, 3) of the 48-bit address used in the address recognition process to compare with the destination address (DA) field of receive frames with an individual DA. In addition, this register is used in bytes 0 through 3 of the six-byte source address field when transmitting PAUSE frames.

Address: 30BE_0000h base + E4h offset = 30BE_00E4h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PADDR1																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_PALR field descriptions

Field	Description
PADDR1	Pause Address Bytes 0 (bits 31:24), 1 (bits 23:16), 2 (bits 15:8), and 3 (bits 7:0) of the 6-byte individual address are used for exact match and the source address field in PAUSE frames.

11.5.5.12 Physical Address Upper Register (ENET_PAUR)

PAUR contains the upper 16 bits (bytes 4 and 5) of the 48-bit address used in the address recognition process to compare with the destination address (DA) field of receive frames with an individual DA. In addition, this register is used in bytes 4 and 5 of the six-byte source address field when transmitting PAUSE frames. Bits 15:0 of PAUR contain a constant type field (0x8808) for transmission of PAUSE frames.

Address: 30BE_0000h base + E8h offset = 30BE_00E8h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PADDR2																TYPE															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0

ENET_PAUR field descriptions

Field	Description
31–16 PADDR2	Bytes 4 (bits 31:24) and 5 (bits 23:16) of the 6-byte individual address used for exact match, and the source address field in PAUSE frames.
TYPE	Type Field In PAUSE Frames These fields have a constant value of 0x8808.

11.5.5.13 Opcode/Pause Duration Register (ENET_OPD)

OPD is read/write accessible. This register contains the 16-bit opcode and 16-bit pause duration fields used in transmission of a PAUSE frame. The opcode field is a constant value, 0x0001. When another node detects a PAUSE frame, that node pauses transmission for the duration specified in the pause duration field. The lower 16 bits of this register are not reset and you must initialize it.

Address: 30BE_0000h base + ECh offset = 30BE_00ECh

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	OPCODE																PAUSE_DUR															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_OPD field descriptions

Field	Description
31–16 OPCODE	Opcode Field In PAUSE Frames These fields have a constant value of 0x0001.
PAUSE_DUR	Pause Duration Pause duration field used in PAUSE frames.

11.5.5.14 Transmit Interrupt Coalescing Register (ENET_TXICn)

See [Interrupt coalescence](#) for more information.

Address: 30BE_0000h base + F0h offset + (4d × i), where i=0d to 2d

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	ICEN	ICCS	Reserved		ICFT								Reserved			
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ICTT															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_TXICn field descriptions

Field	Description
31 ICEN	Interrupt Coalescing Enable

Table continues on the next page...

ENET_TXICn field descriptions (continued)

Field	Description
	0 Disable Interrupt coalescing. 1 Enable Interrupt coalescing.
30 ICCS	Interrupt Coalescing Timer Clock Source Select 0 Use MII/GMII TX clocks. 1 Use ENET system clock.
29–28 Reserved	This field must be set to 0. This field is reserved.
27–20 ICFT	Interrupt coalescing frame count threshold This value determines the number of frames needed to be transmitted for raising an interrupt. Frame counter restarts after reaching this threshold value or after the expiring of the coalescing timer. Must be greater than zero to avoid unpredictable behavior.
19–16 Reserved	This field must be set to 0. This field is reserved.
ICTT	Interrupt coalescing timer threshold Interrupt coalescing timer threshold in units of 64 clock periods. This value determines the maximum amount of time after transmitting a frame before raising an interrupt. The threshold timer is disabled after expiring or number of frame transmission defined by ICFT and starts again upon transmission of the next first frame. Must be greater than zero to avoid unpredictable behavior.

11.5.5.15 Receive Interrupt Coalescing Register (ENET_RXICn)

See [Interrupt coalescence](#) for more information.

Address: 30BE_0000h base + 100h offset + (4d × i), where i=0d to 2d

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W																
	ICEN	ICCS	Reserved											Reserved		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RXICn field descriptions

Field	Description
31 ICEN	Interrupt Coalescing Enable 0 Disable Interrupt coalescing. 1 Enable Interrupt coalescing.

Table continues on the next page...

ENET_RXICn field descriptions (continued)

Field	Description
30 ICCS	Interrupt Coalescing Timer Clock Source Select 0 Use MII/GMII TX clocks. 1 Use ENET system clock.
29–28 Reserved	This field must be set to 0. This field is reserved.
27–20 ICFT	Interrupt coalescing frame count threshold This value determines the number of frames needed to be received for raising an interrupt. Frame counter restarts after reaching this threshold value or after the expiring of the coalescing timer. Must be greater than zero to avoid unpredictable behavior.
19–16 Reserved	This field must be set to 0. This field is reserved.
ICTT	Interrupt coalescing timer threshold Interrupt coalescing timer threshold in units of 64 clock periods. This value determines the maximum amount of time after receiving a frame before raising an interrupt. The threshold timer is disabled after expiring or number of frame reception defined by ICFT and starts again upon reception of the next first frame. Must be greater than zero to avoid unpredictable behavior.

11.5.5.16 Descriptor Individual Upper Address Register (ENET_IAUR)

IAUR contains the upper 32 bits of the 64-bit individual address hash table. The address recognition process uses this table to check for a possible match with the destination address (DA) field of receive frames with an individual DA. This register is not reset and you must initialize it.

Address: 30BE_0000h base + 118h offset = 30BE_0118h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IAUR field descriptions

Field	Description
IADDR1	Contains the upper 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a unicast address. Bit 31 of IADDR1 contains hash index bit 63. Bit 0 of IADDR1 contains hash index bit 32.

11.5.5.17 Descriptor Individual Lower Address Register (ENET_IALR)

IALR contains the lower 32 bits of the 64-bit individual address hash table. The address recognition process uses this table to check for a possible match with the DA field of receive frames with an individual DA. This register is not reset and you must initialize it.

Address: 30BE_0000h base + 11Ch offset = 30BE_011Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	IADDR2																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IALR field descriptions

Field	Description
IADDR2	Contains the lower 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a unicast address. Bit 31 of IADDR2 contains hash index bit 31. Bit 0 of IADDR2 contains hash index bit 0.

11.5.5.18 Descriptor Group Upper Address Register (ENET_GAUR)

GAUR contains the upper 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a multicast address. You must initialize this register.

Address: 30BE_0000h base + 120h offset = 30BE_0120h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	GADDR1																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

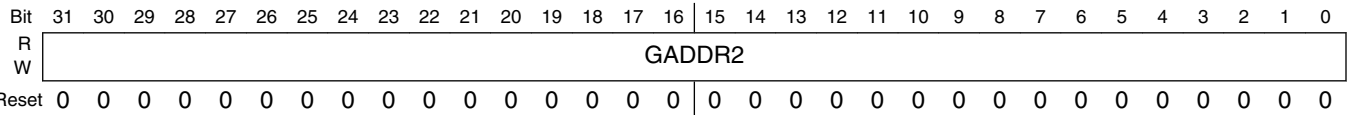
ENET_GAUR field descriptions

Field	Description
GADDR1	Contains the upper 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a multicast address. Bit 31 of GADDR1 contains hash index bit 63. Bit 0 of GADDR1 contains hash index bit 32.

11.5.5.19 Descriptor Group Lower Address Register (ENET_GALR)

GALR contains the lower 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a multicast address. You must initialize this register.

Address: 30BE_0000h base + 124h offset = 30BE_0124h



ENET_GALR field descriptions

Field	Description
GADDR2	Contains the lower 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a multicast address. Bit 31 of GADDR2 contains hash index bit 31. Bit 0 of GADDR2 contains hash index bit 0.

11.5.5.20 Transmit FIFO Watermark Register (ENET_TFWR)

If TFWR[STRFWD] is cleared, TFWR[TFWR] controls the amount of data required in the transmit FIFO before transmission of a frame can begin. This allows you to minimize transmit latency (TFWR = 00 or 01) or allow for larger bus access latency (TFWR = 11) due to contention for the system bus. Setting the watermark to a high value minimizes the risk of transmit FIFO underrun due to contention for the system bus. The byte counts associated with the TFWR field may need to be modified to match a given system requirement, for example, worst-case bus access latency by the transmit data uDMA channel.

When the FIFO level reaches the value the TFWR field and when the STR_FWD is set to '0', the MAC transmit control logic starts frame transmission even before the end-of-frame is available in the FIFO (cut-through operation).

If a complete frame has a size smaller than the threshold programmed with TFWR, the MAC also transmits the Frame to the line.

To enable store and forward on the Transmit path, set STR_FWD to '1'. In this case, the MAC starts to transmit data only when a complete frame is stored in the Transmit FIFO.

Address: 30BE_0000h base + 144h offset = 30BE_0144h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
R	0																
W																	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
R	0								STRFWD	0		TFWR					
W																	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

ENET_TFWR field descriptions

Field	Description
31–9 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
8 STRFWD	Store And Forward Enable 0 Reset. The transmission start threshold is programmed in TFWR[TFWR]. 1 Enabled.
7–6 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TFWR	Transmit FIFO Write If TFWR[STRFWD] is cleared, this field indicates the number of bytes, in steps of 64 bytes, written to the transmit FIFO before transmission of a frame begins. NOTE: If a frame with less than the threshold is written, it is still sent independently of this threshold setting. The threshold is relevant only if the frame is larger than the threshold given. 000000 64 bytes written. 000001 64 bytes written. 000010 128 bytes written. 000011 192 bytes written. 111111 4032 bytes written.

11.5.5.21 Receive Descriptor Ring 1 Start Register (ENET_RDSR1)

RDSR1 points to the beginning of circular receive buffer descriptor queue 1 in external memory. This pointer must be 64-bit aligned (bits 2–0 must be zero); however, for optimal performance the pointer should be 512-bit aligned, that is, evenly divisible by 64.

NOTE

This register must be initialized prior to operation.

Ethernet MAC (ENET)

Address: 30BE_0000h base + 160h offset = 30BE_0160h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	R_DES_START															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	R_DES_START														0	
W															0	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RDSR1 field descriptions

Field	Description
31–3 R_DES_START	Pointer to the beginning of the receive buffer descriptor queue 1.
2 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.22 Transmit Buffer Descriptor Ring 1 Start Register (ENET_TDSR1)

TDSR1 provides a pointer to the beginning of the circular transmit buffer descriptor queue 1 in external memory. This pointer must be 64-bit aligned (bits 2–0 must be zero); however, for optimal performance the pointer should be 512-bit aligned, that is, evenly divisible by 64.

NOTE

This register must be initialized prior to operation.

Address: 30BE_0000h base + 164h offset = 30BE_0164h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	X_DES_START															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	X_DES_START														0	
W															0	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_TDSR1 field descriptions

Field	Description
31–3 X_DES_START	Pointer to the beginning of transmit buffer descriptor queue 1.

Table continues on the next page...

ENET_TDSR1 field descriptions (continued)

Field	Description
2 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.23 Maximum Receive Buffer Size Register - Ring 1 (ENET_MRBR1)

MRBR1 is a user-programmable register that dictates the maximum size of all ring-1 receive buffers. This value should take into consideration that the receive CRC is always written into the last receive buffer.

- To allow one maximum size frame per buffer, MRBR1 must be set to RCR[MAX_FL] or larger.
- R_BUF_SIZE is concatenated with the four least-significant bits of this register and are used as the maximum receive buffer size.
- To properly align the buffer, MRBR1 must be evenly divisible by 64. To ensure this, set the lower two bits of R_BUF_SIZE to zero. The lower four bits are already set to zero by the device.
- To minimize bus usage (descriptor fetches), set MRBR1 greater than or equal to 256 bytes.

NOTE

This register must be initialized before operation.

Address: 30BE_0000h base + 168h offset = 30BE_0168h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																R_BUF_SIZE				0											
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_MRBR1 field descriptions

Field	Description
31–11 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
10–4 R_BUF_SIZE	Receive buffer size in bytes. This value, concatenated with the four least-significant bits of this register (which are always zero), is the effective maximum receive buffer size.
Reserved	This field, which is always zero, is the four least-significant bits of the maximum receive buffer size. This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.24 Receive Descriptor Ring 2 Start Register (ENET_RDSR2)

RDSR points to the beginning of circular receive buffer descriptor queue 2 in external memory. This pointer must be 64-bit aligned (bits 2–0 must be zero); however, for optimal performance the pointer should be 512-bit aligned, that is, evenly divisible by 64.

NOTE

This register must be initialized prior to operation

Address: 30BE_0000h base + 16Ch offset = 30BE_016Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	R_DES_START															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	R_DES_START												0		0	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RDSR2 field descriptions

Field	Description
31–3 R_DES_START	Pointer to the beginning of receive buffer descriptor queue 2.
2 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.25 Transmit Buffer Descriptor Ring 2 Start Register (ENET_TDSR2)

TDSR2 provides a pointer to the beginning of circular transmit buffer descriptor queue 2 in external memory. This pointer must be 64-bit aligned (bits 2–0 must be zero); however, for optimal performance the pointer should be 512-bit aligned, that is, evenly divisible by 64.

NOTE

This register must be initialized prior to operation

Address: 30BE_0000h base + 170h offset = 30BE_0170h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	X_DES_START															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	X_DES_START														0	
W														0		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_TDSR2 field descriptions

Field	Description
31–3 X_DES_START	Pointer to the beginning of transmit buffer descriptor queue 2.
2 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.26 Maximum Receive Buffer Size Register - Ring 2 (ENET_MRBR2)

MRBR2 is a user-programmable register that dictates the maximum size of all ring-2 receive buffers. This value should take into consideration that the receive CRC is always written into the last receive buffer.

- To allow one maximum size frame per buffer, MRBR2 must be set to RCR[MAX_FL] or larger.
- R_BUF_SIZE is concatenated with the four least-significant bits of this register and are used as the maximum receive buffer size.
- To properly align the buffer, MRBR2 must be evenly divisible by 64. To ensure this, set the lower two bits of R_BUF_SIZE to zero. The lower four bits are already set to zero by the device.
- To minimize bus usage (descriptor fetches), set MRBR2 greater than or equal to 256 bytes.

NOTE

This register must be initialized prior to operation

Address: 30BE_0000h base + 174h offset = 30BE_0174h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																R_BUF_SIZE								0							
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

ENET_MRBR2 field descriptions

Field	Description
31–11 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
10–4 R_BUF_SIZE	Receive buffer size in bytes. This value, concatenated with the four least-significant bits of this register (which are always zero), is the effective maximum receive buffer size.
Reserved	This field, which is always zero, is the four least-significant bits of the maximum receive buffer size. This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.27 Receive Descriptor Ring 0 Start Register (ENET_RDSR)

RDSR points to the beginning of the circular receive buffer descriptor queue in external memory. This pointer must be 64-bit aligned (bits 2–0 must be zero); however, for optimal performance the pointer should be 512-bit aligned, that is, evenly divisible by 64.

NOTE

This register must be initialized prior to operation

Address: 30BE_0000h base + 180h offset = 30BE_0180h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	R_DES_START															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	R_DES_START												0		0	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RDSR field descriptions

Field	Description
31–3 R_DES_START	Pointer to the beginning of the receive buffer descriptor queue. 0
2 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.28 Transmit Buffer Descriptor Ring 0 Start Register (ENET_TDSR)

TDSR provides a pointer to the beginning of the circular transmit buffer descriptor queue 0 in external memory. This pointer must be 64-bit aligned (bits 2–0 must be zero); however, for optimal performance the pointer should be 512-bit aligned, that is, evenly divisible by 64.

NOTE

This register must be initialized prior to operation.

Address: 30BE_0000h base + 184h offset = 30BE_0184h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	X_DES_START															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	X_DES_START												0		0	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_TDSR field descriptions

Field	Description
31–3 X_DES_START	Pointer to the beginning of the transmit buffer descriptor queue.
2 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.29 Maximum Receive Buffer Size Register - Ring 0 (ENET_MRBR)

The MRBR is a user-programmable register that dictates the maximum size of all ring-0 receive buffers. This value should take into consideration that the receive CRC is always written into the last receive buffer.

- R_BUF_SIZE is concatenated with the four least-significant bits of this register and are used as the maximum receive buffer size.
- To allow one maximum size frame per buffer, MRBR must be set to RCR[MAX_FL] or larger.

Ethernet MAC (ENET)

- To properly align the buffer, MRBR must be evenly divisible by 64. To ensure this, set the lower two bits of R_BUF_SIZE to zero. The lower four bits of this register are already set to zero by the device.
- To minimize bus usage (descriptor fetches), set MRBR greater than or equal to 256 bytes.

NOTE

This register must be initialized before operation.

Address: 30BE_0000h base + 188h offset = 30BE_0188h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																R_BUF_SIZE								0							
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_MRBR field descriptions

Field	Description
31–11 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
10–4 R_BUF_SIZE	Receive buffer size in bytes. This value, concatenated with the four least-significant bits of this register (which are always zero), is the effective maximum receive buffer size.
Reserved	This field, which is always zero, is the four least-significant bits of the maximum receive buffer size. This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.30 Receive FIFO Section Full Threshold (ENET_RSFL)

Address: 30BE_0000h base + 190h offset = 30BE_0190h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																RX_SECTION_FULL															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RSFL field descriptions

Field	Description
31–10 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
RX_SECTION_FULL	Value Of Receive FIFO Section Full Threshold Value, in 64-bit words, of the receive FIFO section full threshold. Clear this field to enable store and forward on the RX FIFO. When programming a value greater than 0 (cut-through operation), it must be greater than RAEM[RX_ALMOST_EMPTY]. When the FIFO level reaches the value in this field, data is available in the Receive FIFO (cut-through operation).

11.5.5.31 Receive FIFO Section Empty Threshold (ENET_RSEM)

Address: 30BE_0000h base + 194h offset = 30BE_0194h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0											STAT_SECTION_EMPTY				0																
W												STAT_SECTION_EMPTY								RX_SECTION_EMPTY												
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RSEM field descriptions

Field	Description
31–21 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
20–16 STAT_SECTION_EMPTY	RX Status FIFO Section Empty Threshold Defines number of frames in the receive FIFO, independent of its size, that can be accepted. If the limit is reached, reception will continue normally, however a pause frame will be triggered to indicate a possible congestion to the remote device to avoid FIFO overflow. A value of 0 disables automatic pause frame generation
15–10 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
RX_SECTION_EMPTY	Value Of The Receive FIFO Section Empty Threshold Value, in 64-bit words, of the receive FIFO section empty threshold. When the FIFO has reached this level, a pause frame will be issued. A value of 0 disables automatic pause frame generation. When the FIFO level goes below the value programmed in this field, an XON pause frame is issued to indicate the FIFO congestion is cleared to the remote Ethernet client. NOTE: The section-empty threshold indications from both FIFOs are OR'ed to cause XOFF pause frame generation.

11.5.5.32 Receive FIFO Almost Empty Threshold (ENET_RAEM)

Address: 30BE_0000h base + 198h offset = 30BE_0198h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																RX_ALMOST_EMPTY															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

ENET_RAEM field descriptions

Field	Description
31–10 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

Table continues on the next page...

ENET_RAEM field descriptions (continued)

Field	Description
RX_ALMOST_EMPTY	Value Of The Receive FIFO Almost Empty Threshold Value, in 64-bit words, of the receive FIFO almost empty threshold. When the FIFO level reaches the value programmed in this field and the end-of-frame has not been received for the frame yet, the core receive read control stops FIFO read (and subsequently stops transferring data to the MAC client application). It continues to deliver the frame, if again more data than the threshold or the end-of-frame is available in the FIFO. A minimum value of 4 should be set.

11.5.5.33 Receive FIFO Almost Full Threshold (ENET_RAFL)

Address: 30BE_0000h base + 19Ch offset = 30BE_019Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																RX_ALMOST_FULL															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

ENET_RAFL field descriptions

Field	Description
31–10 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
RX_ALMOST_FULL	Value Of The Receive FIFO Almost Full Threshold Value, in 64-bit words, of the receive FIFO almost full threshold. When the FIFO level comes close to the maximum, so that there is no more space for at least RX_ALMOST_FULL number of words, the MAC stops writing data in the FIFO and truncates the received frame to avoid FIFO overflow. The corresponding error status will be set when the frame is delivered to the application. A minimum value of 4 should be set.

11.5.5.34 Transmit FIFO Section Empty Threshold (ENET_TSEM)

Address: 30BE_0000h base + 1A0h offset = 30BE_01A0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TX_SECTION_EMPTY															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_TSEM field descriptions

Field	Description
31–10 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TX_SECTION_EMPTY	Value Of The Transmit FIFO Section Empty Threshold

Table continues on the next page...

ENET_TSEM field descriptions (continued)

Field	Description
	Value, in 64-bit words, of the transmit FIFO section empty threshold. See Transmit FIFO for more information.

11.5.5.35 Transmit FIFO Almost Empty Threshold (ENET_TAEM)

Address: 30BE_0000h base + 1A4h offset = 30BE_01A4h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TX_ALMOST_EMPTY															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

ENET_TAEM field descriptions

Field	Description
31–10 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TX_ALMOST_EMPTY	Value of Transmit FIFO Almost Empty Threshold Value, in 64-bit words, of the transmit FIFO almost empty threshold. When the FIFO level reaches the value programmed in this field, and no end-of-frame is available for the frame, the MAC transmit logic, to avoid FIFO underflow, stops reading the FIFO and transmits a frame with an MII error indication. See Transmit FIFO for more information. A minimum value of 4 should be set.

11.5.5.36 Transmit FIFO Almost Full Threshold (ENET_TAFL)

Address: 30BE_0000h base + 1A8h offset = 30BE_01A8h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TX_ALMOST_FULL															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

ENET_TAFL field descriptions

Field	Description
31–10 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TX_ALMOST_FULL	Value Of The Transmit FIFO Almost Full Threshold Value, in 64-bit words, of the transmit FIFO almost full threshold. A minimum value of six is required . A recommended value of at least 8 should be set allowing a latency of two clock cycles to the application. If more latency is required the value can be increased as necessary (latency = TAFL - 5).

Table continues on the next page...

ENET_TAFL field descriptions (continued)

Field	Description
	When the FIFO level comes close to the maximum, so that there is no more space for at least TX_ALMOST_FULL number of words, the pin ff_tx_rdy is deasserted. If the application does not react on this signal, the FIFO write control logic, to avoid FIFO overflow, truncates the current frame and sets the error status. As a result, the frame will be transmitted with an GMII/MII error indication. See Transmit FIFO for more information.
	NOTE: A FIFO overflow is a fatal error and requires a global reset on the transmit datapath or at least deassertion of ETHEREN.

11.5.5.37 Transmit Inter-Packet Gap (ENET_TIPG)

Address: 30BE_0000h base + 1ACh offset = 30BE_01ACh

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																IPG															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0

ENET_TIPG field descriptions

Field	Description
31–5 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
IPG	Transmit Inter-Packet Gap Indicates the IPG, in bytes, between transmitted frames. Valid values range from 8 to 26. If the written value is less than 8 or greater than 26, the internal (effective) IPG is 12. NOTE: The IPG value read will be the value that was written, even if it is out of range.

11.5.5.38 Frame Truncation Length (ENET_FTRL)

Address: 30BE_0000h base + 1B0h offset = 30BE_01B0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TRUNC_FL															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	

ENET_FTRL field descriptions

Field	Description
31–14 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TRUNC_FL	Frame Truncation Length

Table continues on the next page...

ENET_FTRL field descriptions (continued)

Field	Description
	Indicates the value a receive frame is truncated, if it is greater than this value. Must be greater than or equal to RCR[MAX_FL].
	NOTE: Truncation happens at TRUNC_FL. However, when truncation occurs, the application (FIFO) may receive less data, guaranteeing that it never receives more than the set limit.

11.5.5.39 Transmit Accelerator Function Configuration (ENET_TACC)

TACC controls accelerator actions when sending frames. The register can be changed before or after each frame, but it must remain unmodified during frame writes into the transmit FIFO.

The TFWR[STRFWD] field must be set to use the checksum feature.

Address: 30BE_0000h base + 1C0h offset = 30BE_01C0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W	0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R													PROCHK	IPCHK		SHIFT16
W	0														0	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_TACC field descriptions

Field	Description
31–5 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
4 PROCHK	Enables insertion of protocol checksum. 0 Checksum not inserted. 1 If an IP frame with a known protocol is transmitted, the checksum is inserted automatically into the frame. The checksum field must be cleared. The other frames are not modified.
3 IPCHK	Enables insertion of IP header checksum. 0 Checksum is not inserted. 1 If an IP frame is transmitted, the checksum is inserted automatically. The IP header checksum field must be cleared. If a non-IP frame is transmitted the frame is not modified.

Table continues on the next page...

ENET_TACC field descriptions (continued)

Field	Description
2–1 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
0 SHIFT16	TX FIFO Shift-16 0 Disabled. 1 Indicates to the transmit data FIFO that the written frames contain two additional octets before the frame data. This means the actual frame begins at bit 16 of the first word written into the FIFO. This function allows putting the frame payload on a 32-bit boundary in memory, as the 14-byte Ethernet header is extended to a 16-byte header.

11.5.5.40 Receive Accelerator Function Configuration (ENET_RACC)

Address: 30BE_0000h base + 1C4h offset = 30BE_01C4h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W	0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R									SHIFT16	LINEDIS				PRODIS	IPDIS	PADREM
W	0										0					
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RACC field descriptions

Field	Description
31–8 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
7 SHIFT16	RX FIFO Shift-16 When this field is set, the actual frame data starts at bit 16 of the first word read from the RX FIFO aligning the Ethernet payload on a 32-bit boundary. NOTE: This function only affects the FIFO storage and has no influence on the statistics, which use the actual length of the frame received. 0 Disabled. 1 Instructs the MAC to write two additional bytes in front of each frame received into the RX FIFO.
6 LINEDIS	Enable Discard Of Frames With MAC Layer Errors 0 Frames with errors are not discarded. 1 Any frame received with a CRC, length, or PHY error is automatically discarded and not forwarded to the user application interface.

Table continues on the next page...

ENET_RACC field descriptions (continued)

Field	Description
5–3 Reserved	This field is reserved. This write-only field is reserved. It must always be written with the value 0.
2 PRODIS	Enable Discard Of Frames With Wrong Protocol Checksum 0 Frames with wrong checksum are not discarded. 1 If a TCP/IP, UDP/IP, or ICMP/IP frame is received that has a wrong TCP, UDP, or ICMP checksum, the frame is discarded. Discarding is only available when the RX FIFO operates in store and forward mode (RSFL cleared).
1 IPDIS	Enable Discard Of Frames With Wrong IPv4 Header Checksum 0 Frames with wrong IPv4 header checksum are not discarded. 1 If an IPv4 frame is received with a mismatching header checksum, the frame is discarded. IPv6 has no header checksum and is not affected by this setting. Discarding is only available when the RX FIFO operates in store and forward mode (RSFL cleared).
0 PADREM	Enable Padding Removal For Short IP Frames 0 Padding not removed. 1 Any bytes following the IP payload section of the frame are removed from the frame.

11.5.5.41 Receive Classification Match Register for Class n (ENET_RCMRn)

This match register allows specifying up to four priorities, which are tested (OR'ed) simultaneously. The match detection uses the extracted VLAN field according to the rules for VLAN detection configured through the ECR register. If both match registers, RCMR1 and RCMR2, report a match at the same time, only the class 1 match is indicated as the final result.

Address: 30BE_0000h base + 1C8h offset + (4d × i), where i=0d to 1d

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															MATCHEN
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0	CMP3			0	CMP2			0	CMP1			0	CMP0		
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RCMR_n field descriptions

Field	Description
31–17 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
16 MATCHEN	Match Enable NOTE: A comparison is done only on incoming VLAN frames. If no VLAN frame is received no match will occur. If both match registers have overlapping compare values and hence can match both on the same frame, only class 1 will be indicated and the class 2 match is ignored. 0 Disabled (default): no compares will occur and the classification indicator for this class will never assert. 1 The register contents are valid and a comparison with all compare values is done when a VLAN frame is received.
15 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
14–12 CMP3	Compare 3 Fourth value to compare against. If unused it must be set to the same value as CMP0.
11 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
10–8 CMP2	Compare 2 Third value to compare against. If unused it must be set to the same value as CMP0.
7 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
6–4 CMP1	Compare 1 Second value to compare against. If unused it must be set to the same value as CMP0.
3 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
CMP0	Compare 0 A three-bit value that will be compared with the frame's VLAN priority field (if a VLAN frame is received). All four compare values, CMP0..3, will be used in parallel. If any of the values match, a match for the class is reported (if MATCHEN is 1). NOTE: To implement a single priority match, all four compare values must be set to the same value.

11.5.5.42 DMA Class Based Configuration (ENET_DMAnCFG)

The DMA class based configuration registers are used to configure the DMA controller interface to support the additional class 1 (buffer descriptor ring 1) and class 2 (buffer descriptor ring 2) traffic and define configuration options such as bandwidth allocation as needed.

NOTE

The registers are cleared when ECR[ETHEREN] becomes 0.

Address: 30BE_0000h base + 1D8h offset + (4d × i), where i=0d to 1d

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0								Reserved							
W															CALC_NOIPG	DMA_CLASS_EN
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	IDLE_SLOPE															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_DMA_nCFG field descriptions

Field	Description
31–24 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
23–18 Reserved	This field is reserved. NOTE: Write only zeroes to this field.
17 CALC_NOIPG	Calculate no IPG Disable inclusion of IPG bytes for bandwidth calculations. 0 The traffic shaper function should consider 12 octets of IPG in addition to the frame data transferred for a frame when doing bandwidth calculations. This is the default. 1 Addition of 12 bytes for the IPG should be omitted when calculating the bandwidth (for traffic shaping, when writing a frame into the transmit FIFO, the shaper will usually consider 12 bytes of IPG for every frame as part of the bandwidth allocated by the frame. This addition can be suppressed, meaning short frames will become more bandwidth than large frames due to the relation of data to IPG overhead).
16 DMA_CLASS_EN	DMA class enable 0 The DMA controller's channel for the class is not used. NOTE: Disabling the DMA controller of a class also requires disabling the class match comparator for the class (see registers RCMR _n). When class 1 and class 2 queues are disabled then their frames will be placed in queue 0. 1 Enable the DMA controller to support the corresponding descriptor ring for this class of traffic.
IDLE_SLOPE	Idle slope 16-bit value to define the per class idle slope setting used by the credit based shaper defining allocated bandwidth for the class. This value is used to calculate the BW (bandwidth) fraction, given by the equation, BW fraction = 1/(1+512/IDLE_SLOPE). Idle slope is restricted to certain values. For values less than 128, idle slope = 2 ⁿ , where n = 0, 1, 2, ...6. For values equal to or greater than 128, idle slope = 128×m, where m = 1, 2, 3, ...12.

Table continues on the next page...

ENET_DMAnCFG field descriptions (continued)

Field	Description
	Example 1. BW fraction = $0.20 = 1/(1+(512/128))$; therefore idleslope = 128. Example 2. BW fraction = $0.33 = 1/(1+(512/256))$; therefore idleslope = 256. Example 3. BW fraction = $0.75 = 1/(1+(512/1536))$; therefore idleslope = 1536. NOTE: For AVB applications, the BW fraction of class 1 and class 2 combined must not exceed .75.

11.5.5.43 Receive Descriptor Active Register - Ring 1 (ENET_RDAR1)

RDAR1 is a command register, written by the user, to indicate that the receive descriptor ring 1 has been updated, that is, that the driver produced empty receive buffers with the empty bit set.

Address: 30BE_0000h base + 1E0h offset = 30BE_01E0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0							RDAR	0							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RDAR1 field descriptions

Field	Description
31–25 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
24 RDAR	Receive Descriptor Active Always set to 1 when this register is written, regardless of the value written. This field is cleared by the MAC device when no additional empty descriptors remain in the receive ring. It is also cleared when ECR[ETHEREN] transitions from set to cleared or when ECR[RESET] is set.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.44 Transmit Descriptor Active Register - Ring 1 (ENET_TDAR1)

TDAR1 is a command register that the user writes to indicate that transmit descriptor ring 1 has been updated, that is, that transmit buffers have been produced by the driver with the ready bit set in the buffer descriptor.

The TDAR register is cleared at reset, when ECR[ETHEREN] transitions from set to cleared, or when ECR[RESET] is set.

Address: 30BE_0000h base + 1E4h offset = 30BE_01E4h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
R	0								TDAR	0							
W																	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
R	0																
W																	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

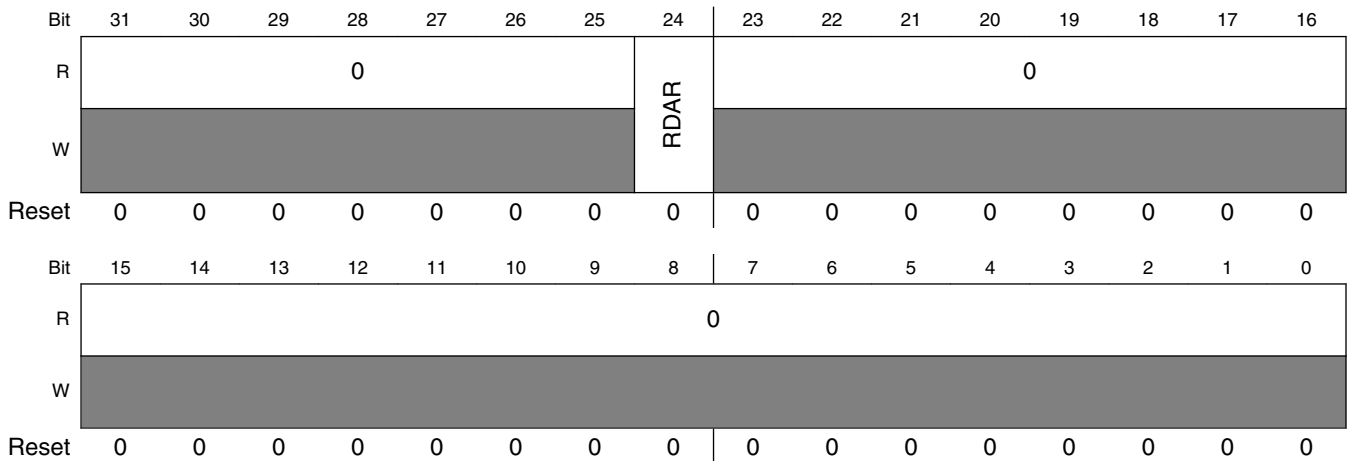
ENET_TDAR1 field descriptions

Field	Description
31–25 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
24 TDAR	Transmit Descriptor Active Always set to 1 when this register is written, regardless of the value written. This bit is cleared by the MAC device when no additional ready descriptors remain in the transmit ring. Also cleared when ECR[ETHEREN] transitions from set to cleared or when ECR[RESET] is set.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.45 Receive Descriptor Active Register - Ring 2 (ENET_RDAR2)

RDAR2 is a command register, written by the user, to indicate that the receive descriptor ring 2 has been updated, that is, that the driver produced empty receive buffers with the empty bit set.

Address: 30BE_0000h base + 1E8h offset = 30BE_01E8h



ENET_RDAR2 field descriptions

Field	Description
31–25 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
24 RDAR	Receive Descriptor Active Always set to 1 when this register is written, regardless of the value written. This field is cleared by the MAC device when no additional empty descriptors remain in the receive ring. It is also cleared when ECR[ETHEREN] transitions from set to cleared or when ECR[RESET] is set.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.46 Transmit Descriptor Active Register - Ring 2 (ENET_TDAR2)

TDAR2 is a command register that the user writes to indicate that transmit descriptor ring 2 has been updated, that is, that transmit buffers have been produced by the driver with the ready bit set in the buffer descriptor.

Address: 30BE_0000h base + 1ECh offset = 30BE_01ECh

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0							TDAR	0							
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_TDAR2 field descriptions

Field	Description
31–25 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
24 TDAR	Transmit Descriptor Active Always set to 1 when this register is written, regardless of the value written. This bit is cleared by the MAC device when no additional ready descriptors remain in the transmit ring. Also cleared when ECR[ETHEREN] transitions from set to cleared or when ECR[RESET] is set.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.47 QOS Scheme (ENET_QOS)

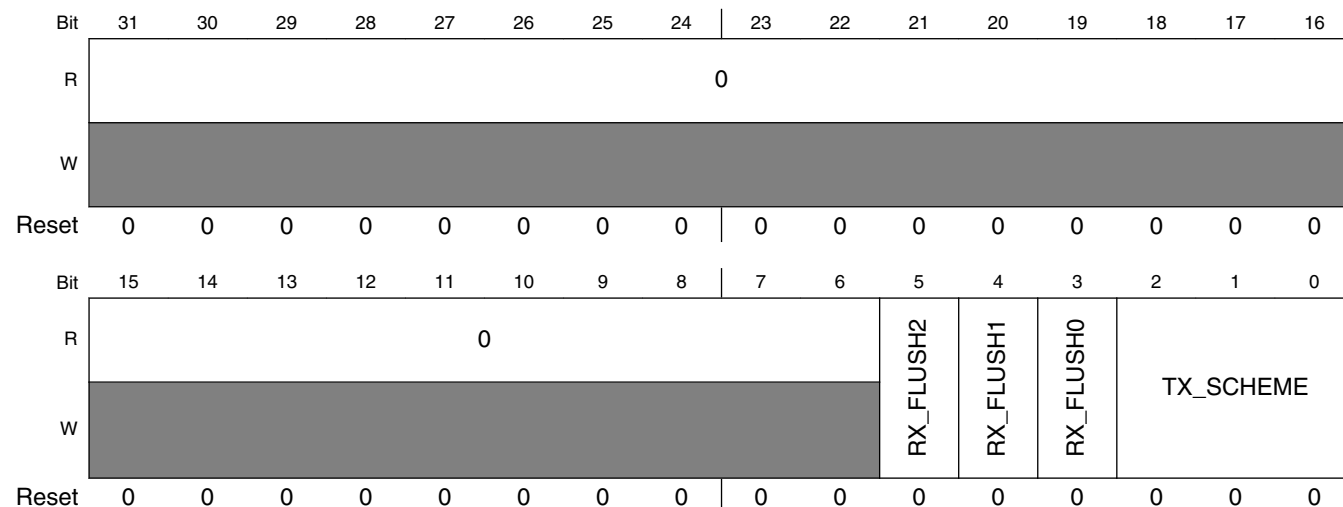
This register sets the QOS scheme.

NOTE

When both class 1 and class 2 are disabled, RX flushing for these rings must also be disabled.

Ethernet MAC (ENET)

Address: 30BE_0000h base + 1F0h offset = 30BE_01F0h



ENET_QOS field descriptions

Field	Description
31–6 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
5 RX_FLUSH2	RX Flush Ring 2 Enable or disable RX Flush for ring 2. See Receive flushing . 0 Disable 1 Enable
4 RX_FLUSH1	RX Flush Ring 1 Enable or disable RX Flush for ring 1. See Receive flushing . 0 Disable 1 Enable
3 RX_FLUSH0	RX Flush Ring 0 Enable or disable RX Flush for ring 0. See Receive flushing . 0 Disable 1 Enable
TX_SCHEME	TX scheme configuration Configuration information for DMA to select transmitter queue selection/arbitration scheme. 000 Credit-based scheme 001 Round-robin scheme 010-111 Reserved

11.5.5.48 Reserved Statistic Register (ENET_RMON_T_DROP)

Address: 30BE_0000h base + 200h offset = 30BE_0200h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_DROP field descriptions

Field	Description
Reserved	This read-only field always has the value 0. This field is reserved.

11.5.5.49 Tx Packet Count Statistic Register (ENET_RMON_T_PACKETS)

Address: 30BE_0000h base + 204h offset = 30BE_0204h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_PACKETS field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Packet count Transmit packet count

11.5.5.50 Tx Broadcast Packets Statistic Register (ENET_RMON_T_BC_PKT)

RMON Tx Broadcast Packets

Address: 30BE_0000h base + 208h offset = 30BE_0208h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_BC_PKT field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Broadcast packets

11.5.5.51 Tx Multicast Packets Statistic Register (ENET_RMON_T_MC_PKT)

Address: 30BE_0000h base + 20Ch offset = 30BE_020Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_MC_PKT field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Multicast packets

11.5.5.52 Tx Packets with CRC/Align Error Statistic Register (ENET_RMON_T_CRC_ALIGN)

Address: 30BE_0000h base + 210h offset = 30BE_0210h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_CRC_ALIGN field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Packets with CRC/align error

11.5.5.53 Tx Packets Less Than Bytes and Good CRC Statistic Register (ENET_RMON_T_UNDERSIZE)

Address: 30BE_0000h base + 214h offset = 30BE_0214h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_UNDERSIZE field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of transmit packets less than 64 bytes with good CRC

11.5.5.54 Tx Packets GT MAX_FL bytes and Good CRC Statistic Register (ENET_RMON_T_OVERSIZE)

Address: 30BE_0000h base + 218h offset = 30BE_0218h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_OVERSIZE field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of transmit packets greater than MAX_FL bytes with good CRC

11.5.55 Tx Packets Less Than 64 Bytes and Bad CRC Statistic Register (ENET_RMON_T_FRAG)

Address: 30BE_0000h base + 21Ch offset = 30BE_021Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_FRAG field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of packets less than 64 bytes with bad CRC

11.5.56 Tx Packets Greater Than MAX_FL bytes and Bad CRC Statistic Register (ENET_RMON_T_JAB)

Address: 30BE_0000h base + 220h offset = 30BE_0220h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_JAB field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of transmit packets greater than MAX_FL bytes and bad CRC

11.5.5.57 Tx Collision Count Statistic Register (ENET_RMON_T_COL)

Address: 30BE_0000h base + 224h offset = 30BE_0224h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_COL field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of transmit collisions

11.5.5.58 Tx 64-Byte Packets Statistic Register (ENET_RMON_T_P64)

Address: 30BE_0000h base + 228h offset = 30BE_0228h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_P64 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of 64-byte transmit packets

11.5.5.59 Tx 65- to 127-byte Packets Statistic Register (ENET_RMON_T_P65TO127)

Address: 30BE_0000h base + 22Ch offset = 30BE_022Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_P65TO127 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of 65- to 127-byte transmit packets

11.5.5.60 Tx 128- to 255-byte Packets Statistic Register (ENET_RMON_T_P128TO255)

Address: 30BE_0000h base + 230h offset = 30BE_0230h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_P128TO255 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of 128- to 255-byte transmit packets

11.5.5.61 Tx 256- to 511-byte Packets Statistic Register (ENET_RMON_T_P256TO511)

Address: 30BE_0000h base + 234h offset = 30BE_0234h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_P256TO511 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of 256- to 511-byte transmit packets

11.5.5.62 Tx 512- to 1023-byte Packets Statistic Register (ENET_RMON_T_P512TO1023)

Address: 30BE_0000h base + 238h offset = 30BE_0238h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_P512TO1023 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of 512- to 1023-byte transmit packets

11.5.5.63 Tx 1024- to 2047-byte Packets Statistic Register (ENET_RMON_T_P1024TO2047)

Address: 30BE_0000h base + 23Ch offset = 30BE_023Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_P1024TO2047 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of 1024- to 2047-byte transmit packets

11.5.5.64 Tx Packets Greater Than 2048 Bytes Statistic Register (ENET_RMON_T_P_GTE2048)

Address: 30BE_0000h base + 240h offset = 30BE_0240h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																TXPKTS															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_P_GTE2048 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
TXPKTS	Number of transmit packets greater than 2048 bytes

11.5.5.65 Tx Octets Statistic Register (ENET_RMON_T_OCTETS)

Address: 30BE_0000h base + 244h offset = 30BE_0244h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	TXOCTS																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_T_OCTETS field descriptions

Field	Description
TXOCTS	Number of transmit octets

11.5.5.66 Reserved Statistic Register (ENET_IEEE_T_DROP)

Address: 30BE_0000h base + 248h offset = 30BE_0248h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_DROP field descriptions

Field	Description
Reserved	This read-only field always has the value 0. This field is reserved.

11.5.5.67 Frames Transmitted OK Statistic Register (ENET_IEEE_T_FRAME_OK)

Address: 30BE_0000h base + 24Ch offset = 30BE_024Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_FRAME_OK field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of frames transmitted OK NOTE: Does not increment for the broadcast frames when broadcast reject is enabled and promiscuous mode is disabled within the receive control register (RCR).

11.5.5.68 Frames Transmitted with Single Collision Statistic Register (ENET_IEEE_T_1COL)

Address: 30BE_0000h base + 250h offset = 30BE_0250h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_1COL field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of frames transmitted with one collision

11.5.5.69 Frames Transmitted with Multiple Collisions Statistic Register (ENET_IEEE_T_MCOL)

Address: 30BE_0000h base + 254h offset = 30BE_0254h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_MCOL field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of frames transmitted with multiple collisions

11.5.5.70 Frames Transmitted after Deferral Delay Statistic Register (ENET_IEEE_T_DEF)

Address: 30BE_0000h base + 258h offset = 30BE_0258h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_DEF field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of frames transmitted with deferral delay

11.5.5.71 Frames Transmitted with Late Collision Statistic Register (ENET_IEEE_T_LCOL)

Address: 30BE_0000h base + 25Ch offset = 30BE_025Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_LCOL field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of frames transmitted with late collision

11.5.5.72 Frames Transmitted with Excessive Collisions Statistic Register (ENET_IEEE_T_EXCOL)

Address: 30BE_0000h base + 260h offset = 30BE_0260h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_EXCOL field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of frames transmitted with excessive collisions

11.5.5.73 Frames Transmitted with Tx FIFO Underrun Statistic Register (ENET_IEEE_T_MACERR)

Address: 30BE_0000h base + 264h offset = 30BE_0264h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_MACERR field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of frames transmitted with transmit FIFO underrun

11.5.5.74 Frames Transmitted with Carrier Sense Error Statistic Register (ENET_IEEE_T_CSERR)

Address: 30BE_0000h base + 268h offset = 30BE_0268h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_CSERR field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of frames transmitted with carrier sense error

11.5.5.75 Reserved Statistic Register (ENET_IEEE_T_SQE)

Address: 30BE_0000h base + 26Ch offset = 30BE_026Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_SQE field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	This read-only field is reserved and always has the value 0. NOTE: Counter not implemented as no SQE information is available.

11.5.5.76 Flow Control Pause Frames Transmitted Statistic Register (ENET_IEEE_T_FDXFC)

Address: 30BE_0000h base + 270h offset = 30BE_0270h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_FDXFC field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of flow-control pause frames transmitted

11.5.5.77 Octet Count for Frames Transmitted w/o Error Statistic Register (ENET_IEEE_T_OCTETS_OK)

Address: 30BE_0000h base + 274h offset = 30BE_0274h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	COUNT																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_T_OCTETS_OK field descriptions

Field	Description
COUNT	Octet count for frames transmitted without error NOTE Counts total octets (includes header and FCS fields).

11.5.5.78 Rx Packet Count Statistic Register (ENET_RMON_R_PACKETS)

Address: 30BE_0000h base + 284h offset = 30BE_0284h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

ENET_RMON_R_PACKETS field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of packets received

11.5.5.79 Rx Broadcast Packets Statistic Register (ENET_RMON_R_BC_PKT)

Address: 30BE_0000h base + 288h offset = 30BE_0288h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_BC_PKT field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of receive broadcast packets

11.5.5.80 Rx Multicast Packets Statistic Register (ENET_RMON_R_MC_PKT)

Address: 30BE_0000h base + 28Ch offset = 30BE_028Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_MC_PKT field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of receive multicast packets

11.5.5.81 Rx Packets with CRC/Align Error Statistic Register (ENET_RMON_R_CRC_ALIGN)

Address: 30BE_0000h base + 290h offset = 30BE_0290h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_CRC_ALIGN field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of receive packets with CRC or align error

11.5.5.82 Rx Packets with Less Than 64 Bytes and Good CRC Statistic Register (ENET_RMON_R_UNDERSIZE)

Address: 30BE_0000h base + 294h offset = 30BE_0294h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_UNDERSIZE field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of receive packets with less than 64 bytes and good CRC

11.5.5.83 Rx Packets Greater Than MAX_FL and Good CRC Statistic Register (ENET_RMON_R_OVERSIZE)

Address: 30BE_0000h base + 298h offset = 30BE_0298h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_OVERSIZE field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of receive packets greater than MAX_FL and good CRC

11.5.5.84 Rx Packets Less Than 64 Bytes and Bad CRC Statistic Register (ENET_RMON_R_FRAG)

Address: 30BE_0000h base + 29Ch offset = 30BE_029Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_FRAG field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of receive packets with less than 64 bytes and bad CRC

11.5.5.85 Rx Packets Greater Than MAX_FL Bytes and Bad CRC Statistic Register (ENET_RMON_R_JAB)

Address: 30BE_0000h base + 2A0h offset = 30BE_02A0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_JAB field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of receive packets greater than MAX_FL and bad CRC

11.5.5.86 Reserved Statistic Register (ENET_RMON_R_RESVD_0)

Address: 30BE_0000h base + 2A4h offset = 30BE_02A4h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																0															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_RESVD_0 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

11.5.5.87 Rx 64-Byte Packets Statistic Register (ENET_RMON_R_P64)

Address: 30BE_0000h base + 2A8h offset = 30BE_02A8h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_P64 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of 64-byte receive packets

11.5.5.88 Rx 65- to 127-Byte Packets Statistic Register (ENET_RMON_R_P65TO127)

Address: 30BE_0000h base + 2ACh offset = 30BE_02ACh

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_P65TO127 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of 65- to 127-byte receive packets

11.5.5.89 Rx 128- to 255-Byte Packets Statistic Register (ENET_RMON_R_P128TO255)

Address: 30BE_0000h base + 2B0h offset = 30BE_02B0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_P128TO255 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of 128- to 255-byte receive packets

11.5.5.90 Rx 256- to 511-Byte Packets Statistic Register (ENET_RMON_R_P256TO511)

Address: 30BE_0000h base + 2B4h offset = 30BE_02B4h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_P256TO511 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of 256- to 511-byte receive packets

11.5.5.91 Rx 512- to 1023-Byte Packets Statistic Register (ENET_RMON_R_P512TO1023)

Address: 30BE_0000h base + 2B8h offset = 30BE_02B8h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_P512TO1023 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of 512- to 1023-byte receive packets

11.5.5.92 Rx 1024- to 2047-Byte Packets Statistic Register (ENET_RMON_R_P1024TO2047)

Address: 30BE_0000h base + 2BCh offset = 30BE_02BCh

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_P1024TO2047 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of 1024- to 2047-byte receive packets

11.5.5.93 Rx Packets Greater than 2048 Bytes Statistic Register (ENET_RMON_R_P_GTE2048)

Address: 30BE_0000h base + 2C0h offset = 30BE_02C0h

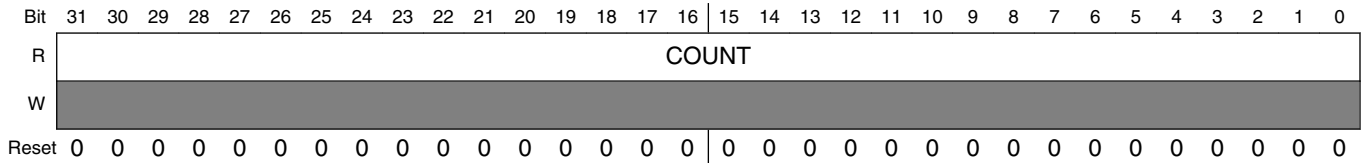
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_RMON_R_P_GTE2048 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of greater-than-2048-byte receive packets

11.5.5.94 Rx Octets Statistic Register (ENET_RMON_R_OCTETS)

Address: 30BE_0000h base + 2C4h offset = 30BE_02C4h



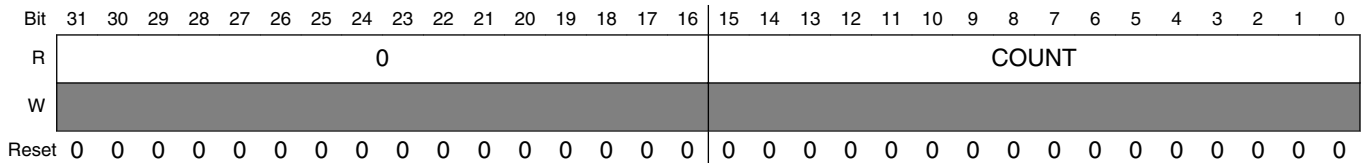
ENET_RMON_R_OCTETS field descriptions

Field	Description
COUNT	Number of receive octets

11.5.5.95 Frames not Counted Correctly Statistic Register (ENET_IEEE_R_DROP)

Counter increments if a frame with invalid or missing SFD character is detected and has been dropped. None of the other counters increments if this counter increments.

Address: 30BE_0000h base + 2C8h offset = 30BE_02C8h

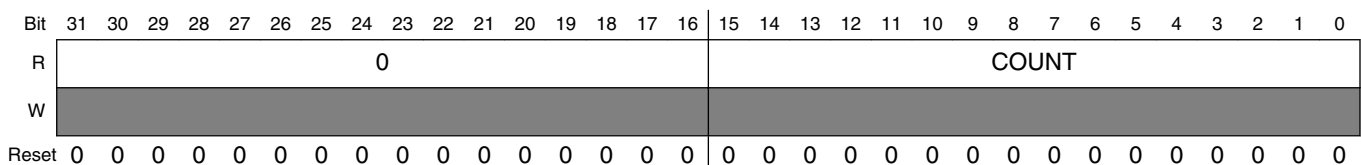


ENET_IEEE_R_DROP field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Frame count

11.5.5.96 Frames Received OK Statistic Register (ENET_IEEE_R_FRAME_OK)

Address: 30BE_0000h base + 2CCh offset = 30BE_02CCh



ENET_IEEE_R_FRAME_OK field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of frames received OK

11.5.5.97 Frames Received with CRC Error Statistic Register (ENET_IEEE_R_CRC)

Address: 30BE_0000h base + 2D0h offset = 30BE_02D0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_R_CRC field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of frames received with CRC error

11.5.5.98 Frames Received with Alignment Error Statistic Register (ENET_IEEE_R_ALIGN)

Address: 30BE_0000h base + 2D4h offset = 30BE_02D4h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_R_ALIGN field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of frames received with alignment error

11.5.5.99 Receive FIFO Overflow Count Statistic Register (ENET_IEEE_R_MACERR)

Address: 30BE_0000h base + 2D8h offset = 30BE_02D8h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_R_MACERR field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Receive FIFO overflow count

11.5.5.100 Flow Control Pause Frames Received Statistic Register (ENET_IEEE_R_FDXFC)

Address: 30BE_0000h base + 2DCh offset = 30BE_02DCh

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_R_FDXFC field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Number of flow-control pause frames received

11.5.5.101 Octet Count for Frames Received without Error Statistic Register (ENET_IEEE_R_OCTETS_OK)

Address: 30BE_0000h base + 2E0h offset = 30BE_02E0h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	COUNT																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_IEEE_R_OCTETS_OK field descriptions

Field	Description
COUNT	Number of octets for frames received without error NOTE: Counts total octets (includes header and FCS fields). Does not increment for the broadcast frames when broadcast reject is enabled and promiscuous mode is disabled within the receive control register (RCR).

11.5.5.102 Adjustable Timer Control Register (ENET_ATCR)

ATCR command fields can trigger the corresponding events directly. It is not necessary to preserve any of the configuration fields when a command field is set in the register, that is, no read-modify-write is required.

NOTE

The CAPTURE and RESTART fields and bits 12 and 10 must be 0 in order to write to the other fields in this register.

Address: 30BE_0000h base + 400h offset = 30BE_0400h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0		SLAVE	Reserved	CAPTURE	Reserved	RESTART		PINPER			PEREN	OFFRST	OFFEN		EN
W							0			0	1				0	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_ATCR field descriptions

Field	Description
31–14 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
13 SLAVE	Enable Timer Slave Mode 0 The timer is active and all configuration fields in this register are relevant. 1 The internal timer is disabled and the externally provided timer value is used. All other fields, except CAPTURE, in this register have no effect. CAPTURE can still be used to capture the current timer value.

Table continues on the next page...

ENET_ATCR field descriptions (continued)

Field	Description
12 Reserved	This field is reserved. Always write 0 to this field.
11 CAPTURE	Capture Timer Value When this field is set, all other fields are ignored during a write. This field automatically clears to 0 after the command completes. NOTE: To ensure that the correct time value is read from the ATVR register, a minimum amount of time must elapse from issuing this command to reading the ATVR register. This minimum time is defined by the greater of either six register clock cycles or six 1588/timestamp clock cycles. 0 No effect. 1 The current time is captured and can be read from the ATVR register.
10 Reserved	This field is reserved. Always write 0 to this field.
9 RESTART	Reset Timer Resets the timer to zero. This has no effect on the counter enable. If the counter is enabled when this field is set, the timer is reset to zero and starts counting from there. When set, all other fields are ignored during a write. This field automatically clears to 0 after the command completes. RESTART should be used when the timer is enabled. NOTE: The Reset Timer command requires at least 6 clock cycles of either the register clock or the 1588/timestamp clock, whichever is greater, to complete.
8 Reserved	This field is reserved.
7 PINPER	Enables event signal output assertion on period event. NOTE: Not all devices contain the event signal output. See the chip configuration details. 0 Disable. 1 Enable.
6 Reserved	This field is reserved.
5 Reserved	This field is reserved. NOTE: This field must be written always with one.
4 PEREN	Enable Periodical Event 0 Disable. 1 A period event interrupt can be generated (EIR[TS_TIMER]) and the event signal output is asserted when the timer wraps around according to the periodic setting ATPER. The timer period value must be set before setting this bit. NOTE: Not all devices contain the event signal output. See the chip configuration details.
3 OFFRST	Reset Timer On Offset Event 0 The timer is not affected and no action occurs, besides clearing OFFEN, when the offset is reached. 1 If OFFEN is set, the timer resets to zero when the offset setting is reached. The offset event does not cause a timer interrupt.
2 OFFEN	Enable One-Shot Offset Event

Table continues on the next page...

ENET_ATCR field descriptions (continued)

Field	Description
0	Disable.
1	The timer can be reset to zero when the given offset time is reached (offset event). The field is cleared when the offset event is reached, so no further event occurs until the field is set again. The timer offset value must be set before setting this field.
1 Reserved	This field is reserved.
0 EN	Enable Timer
0	The timer stops at the current value.
1	The timer starts incrementing.

11.5.5.103 Timer Value Register (ENET_ATVR)

Address: 30BE_0000h base + 404h offset = 30BE_0404h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_ATVR field descriptions

Field	Description
ATIME	A write sets the timer. A read returns the last captured value. To read the current value, issue a capture command (i.e., set ATCR[CAPTURE]) prior to reading this register.

11.5.5.104 Timer Offset Register (ENET_ATOFF)

Address: 30BE_0000h base + 408h offset = 30BE_0408h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_ATOFF field descriptions

Field	Description
OFFSET	Offset value for one-shot event generation. When the timer reaches the value, an event can be generated to reset the counter. If the increment value in ATINC is given in true nanoseconds, this value is also given in true nanoseconds.

11.5.5.105 Timer Period Register (ENET_ATPER)

Address: 30BE_0000h base + 40Ch offset = 30BE_040Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PERIOD																															
W	PERIOD																															
Reset	0	0	1	1	1	0	1	1	1	0	0	1	1	0	1	0	1	1	0	0	1	0	1	0	0	0	0	0	0	0	0	0

ENET_ATPER field descriptions

Field	Description
PERIOD	Value for generating periodic events. Each instance the timer reaches this value, the period event occurs and the timer restarts. If the increment value in ATINC is given in true nanoseconds, this value is also given in true nanoseconds. The value should be initialized to 1,000,000,000 (1×10^9) to represent a timer wrap around of one second. The increment value set in ATINC should be set to the true nanoseconds of the period of clock ts_clk, hence implementing a true 1 second counter. NOTE: The value of PERIOD has the following constraint: $2^{32} - \text{ENET_ATINC}[\text{INC_COR}] - 3 \times \text{ENET_ATINC}[\text{INC}] \geq \text{PERIOD} > 0$.

11.5.5.106 Timer Correction Register (ENET_ATCOR)

Address: 30BE_0000h base + 410h offset = 30BE_0410h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_ATCOR field descriptions

Field	Description
31 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COR	Correction Counter Wrap-Around Value Defines after how many timer clock cycles (ts_clk) the correction counter should be reset and trigger a correction increment on the timer. The amount of correction is defined in ATINC[INC_CORR]. A value of 0 disables the correction counter and no corrections occur. NOTE: This value is given in clock cycles, not in nanoseconds as all other values.

11.5.5.107 Time-Stamping Clock Period Register (ENET_ATINC)

Address: 30BE_0000h base + 414h offset = 30BE_0414h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0	INC_CORR							0	INC						
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_ATINC field descriptions

Field	Description
31–15 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
14–8 INC_CORR	Correction Increment Value This value is added every time the correction timer expires (every clock cycle given in ATCOR). A value less than INC slows down the timer. A value greater than INC speeds up the timer.
7 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
INC	Clock Period Of The Timestamping Clock (ts_clk) In Nanoseconds The timer increments by this amount each clock cycle. For example, set to 10 for 100 MHz, 8 for 125 MHz, 5 for 200 MHz. NOTE: For highest precision, use a value that is an integer fraction of the period set in ATPER.

11.5.5.108 Timestamp of Last Transmitted Frame (ENET_ATSTMP)

Address: 30BE_0000h base + 418h offset = 30BE_0418h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	TIMESTAMP																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_ATSTMP field descriptions

Field	Description
TIMESTAMP	Timestamp of the last frame transmitted by the core that had TxB[TS] set . This register is only valid when EIR[TS_AVAIL] is set.

11.5.5.109 Timer Global Status Register (ENET_TGSR)

Address: 30BE_0000h base + 604h offset = 30BE_0604h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0												TF3	TF2	TF1	TF0
W													w1c	w1c	w1c	w1c
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_TGSR field descriptions

Field	Description
31–4 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
3 TF3	Copy Of Timer Flag For Channel 3 0 Timer Flag for Channel 3 is clear 1 Timer Flag for Channel 3 is set
2 TF2	Copy Of Timer Flag For Channel 2 0 Timer Flag for Channel 2 is clear 1 Timer Flag for Channel 2 is set
1 TF1	Copy Of Timer Flag For Channel 1 0 Timer Flag for Channel 1 is clear 1 Timer Flag for Channel 1 is set
0 TF0	Copy Of Timer Flag For Channel 0 0 Timer Flag for Channel 0 is clear 1 Timer Flag for Channel 0 is set

11.5.5.110 Timer Control Status Register (ENET_TCSRn)

Address: 30BE_0000h base + 608h offset + (8d × i), where i=0d to 3d

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved					0			TF	TIE	TMODE				0	TDRE
W									w1c							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_TCSRn field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
15–11 Reserved	This field is reserved. This field must be written with 0.
10–8 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
7 TF	Timer Flag Sets when input capture or output compare occurs. This flag is double buffered between the module clock and 1588 clock domains. When this field is 1, it can be cleared to 0 by writing 1 to it. 0 Input Capture or Output Compare has not occurred. 1 Input Capture or Output Compare has occurred.
6 TIE	Timer Interrupt Enable 0 Interrupt is disabled 1 Interrupt is enabled
5–2 TMODE	Timer Mode Updating the Timer Mode field takes a few cycles to register because it is synchronized to the 1588 clock. The version of Timer Mode returned on a read is from the 1588 clock domain. When changing Timer Mode, always disable the channel and read this register to verify the channel is disabled first. 0000 Timer Channel is disabled. 0001 Timer Channel is configured for Input Capture on rising edge. 0010 Timer Channel is configured for Input Capture on falling edge. 0011 Timer Channel is configured for Input Capture on both edges. 0100 Timer Channel is configured for Output Compare - software only. 0101 Timer Channel is configured for Output Compare - toggle output on compare. 0110 Timer Channel is configured for Output Compare - clear output on compare. 0111 Timer Channel is configured for Output Compare - set output on compare. 1000 Reserved

Table continues on the next page...

ENET_TCSR_n field descriptions (continued)

Field	Description
	1010 Timer Channel is configured for Output Compare - clear output on compare, set output on overflow. 10X1 Timer Channel is configured for Output Compare - set output on compare, clear output on overflow. 110X Reserved 1110 Timer Channel is configured for Output Compare - pulse output low on compare for one 1588-clock cycle. 1111 Timer Channel is configured for Output Compare - pulse output high on compare for one 1588-clock cycle.
1 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
0 TDRE	Timer DMA Request Enable 0 DMA request is disabled 1 DMA request is enabled

11.5.5.111 Timer Compare Capture Register (ENET_TCCR_n)

Address: 30BE_0000h base + 60Ch offset + (8d × i), where i=0d to 3d

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	TCC																															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENET_TCCR_n field descriptions

Field	Description
TCC	Timer Capture Compare This register is double buffered between the module clock and 1588 clock domains. When configured for compare, the 1588 clock domain updates with the value in the module clock domain whenever the Timer Channel is first enabled and on each subsequent compare. Write to this register with the first compare value before enabling the Timer Channel. When the Timer Channel is enabled, write the second compare value either immediately, or at least before the first compare occurs. After each compare, write the next compare value before the previous compare occurs and before clearing the Timer Flag. The compare occurs one 1588 clock cycle after the IEEE 1588 Counter increments past the compare value in the 1588 clock domain. If the compare value is less than the value of the 1588 Counter when the Timer Channel is first enabled, then the compare does not occur until following the next overflow of the 1588 Counter. If the compare value is greater than the IEEE 1588 Counter when the 1588 Counter overflows, or the compare value is less than the value of the IEEE 1588 Counter after the overflow, then the compare occurs one 1588 clock cycle following the overflow. When configured for capture, the value of the IEEE 1588 Counter is captured into the 1588 clock domain and then updated into the module clock domain, provided the Timer Flag is clear. Always read the capture value before clearing the Timer Flag.

Chapter 12

Timers

12.1 General Purpose Timer (GPT)

12.1.1 Overview

This chapter describes the General Purpose Timer (GPT) module interface. It is also a reference for software driver programming. The GPT has a 32-bit up-counter. The timer counter value can be captured in a register using an event on an external pin. The capture trigger can be programmed to be a rising or/and falling edge. The GPT can also generate an event on the output compare pins and an interrupt when the timer reaches a programmed value. The GPT has a 12-bit prescaler, which provides a programmable clock frequency derived from multiple clock sources.

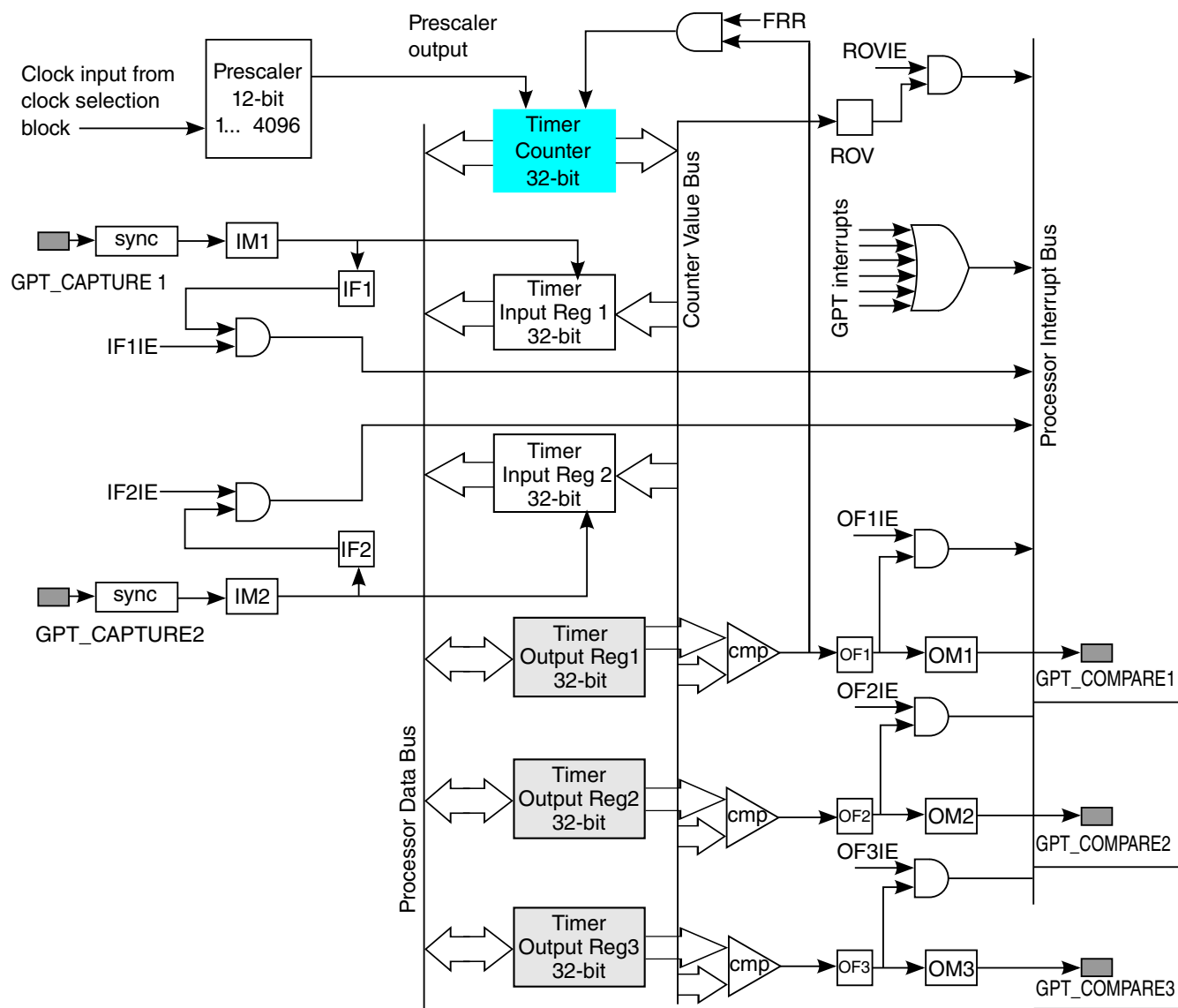


Figure 12-1. GPT Block Diagram

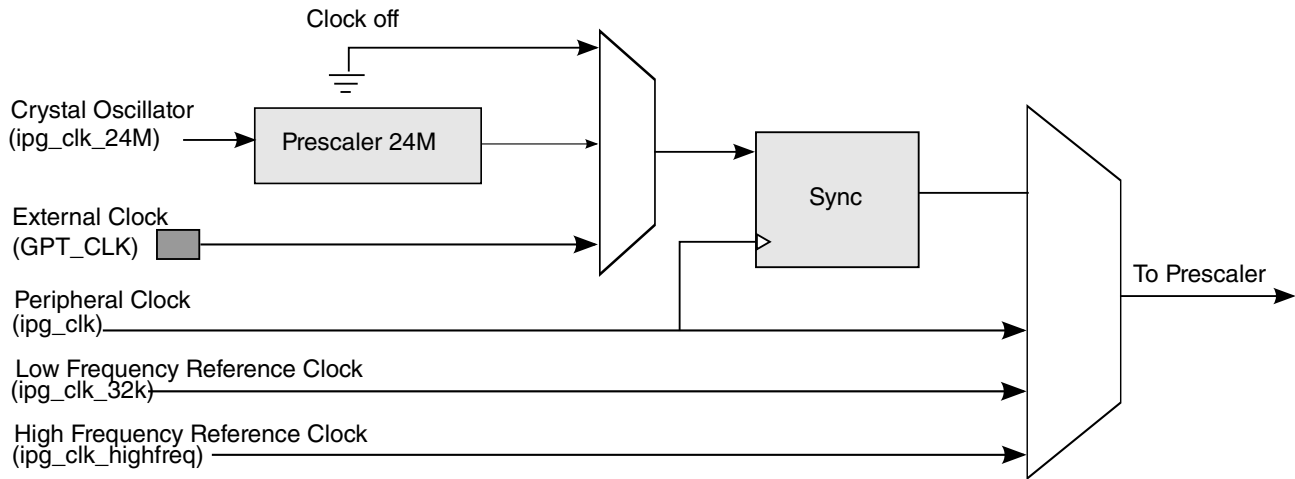


Figure 12-2. GPT Counter Clocks Diagram

12.1.1.1 Features

- One 32-bit up-counter with clock source selection, including external clock.
- Two input capture channels with a programmable trigger edge.
- Three output compare channels with a programmable output mode. A "forced compare" feature is also available.
- Can be programmed to be *active* in low power and debug modes.
- Interrupt generation at capture, compare, and rollover events.
- Restart or free-run modes for counter operations.

12.1.1.2 Modes and Operation

The GPT supports the modes described in the indicated sections:

- [Operating Modes](#)
 - [Restart Mode](#)
 - [Free-Run Mode](#)

12.1.2 External Signals

The GPT follows the IP Bus protocol for interfacing with the processor core. The GPT does not have *any interface signals with any other module inside the chip*, except for the clock and reset inputs (from the clock and reset controller module) and for the interrupt signals *to* the processor interrupt handler. There are functional and clock inputs, and functional output signals going outside the chip boundary.

There are six signals (three input, three output) in the GPT module that *can be* connected to the chip pads.

12.1.2.1 External Clock Input

The GPT counter can be operated using an external clock from outside the device, and this is the input pin used for that purpose. The external clock input (GPT_CLK) is treated as asynchronous to the peripheral clock (ipg_clk). To ensure proper operations of GPT, the external clock input frequency should be less than 1/4 of frequency of the peripheral clock (ipg_clk). Hysteresis characteristics on this pad will be required because this is a clock input.

12.1.2.2 Input Capture Trigger Signals

The GPT counter value can be stored in a register, triggered by an event from *outside the device*. A positive or/and negative edge on these signals GPT_CAPTURE1 , GPT_CAPTURE2 can trigger this capture event. These signals are treated as asynchronous to the peripheral clock (ipg_clk). Only those transitions which occur *at least a single clock cycle* (the clock selected to run the counter) *after the previous recorded transition* are guaranteed to trigger a capture event.

12.1.2.3 Output Compare Signals

The output compare signals: GPT_COMPARE1, GPT_COMPARE2, GPT_COMPARE3, indicate that output compare events have gone through a specified transition.

12.1.3 Clocks

The clock that is input to the prescaler can be selected from 4 clock sources. The following table describes the clock sources for GPT. Please see Clock Controller Module (CCM) for clock setting, configuration and gating information.

Table 12-1. GPT Clocks

Clock name	Clock Root	Description
ipg_clk		Peripheral clock
ipg_clk_32k		Low-frequency reference clock (32 kHz)

Table continues on the next page...

Table 12-1. GPT Clocks (continued)

Clock name	Clock Root	Description
ipg_clk_highfreq		High-frequency reference clock
ipg_clk_s		Peripheral access clock

- High-Frequency Clock (ipg_clk_highfreq)

Provided by the Clock Controller Module (CCM), the High Frequency Clock is intended to be ON in Normal Power mode when the Peripheral Clock (ipg_clk) is turned OFF, thereby enabling the GPT to be operated using the High Frequency Clock *in Normal Power mode*. The CCM is expected to provide this clock *after* synchronizing it to the System Bus Clock (ahb_clk) in Normal functional mode; the CCM is also expected to switch to the *unsynchronized* version of the High Frequency Clock in a Low Power mode.

- Low-Reference Clock (ipg_clk_32k)

This 32 kHz Low Reference Clock (provided by the CCM) is intended to be ON in Low Power mode when the Peripheral Clock (ipg_clk) is turned OFF, thereby enabling the GPT to be operated using the Low Reference Clock in Low Power mode. The CCM is expected to provide the Low Reference Clock *after* synchronizing it to the System Bus Clock (ahb_clk) in Normal functional mode; the CCM is also expected to switch to the *unsynchronized* version of the Low Reference Clock in a Low Power mode.

- Peripheral Clock (ipg_clk)

If the Peripheral Clock (ipg_clk) or the External Clock (GPT_CLK) is selected (CLKSRC=001 or 011) as Clock Source, then the Peripheral Clock will be ON in normal GPT operations. In Low Power modes, if the GPT is programmed to be disabled (STOPEN or WAITEN or DOZEN=0), then the Peripheral Clock (ipg_clk) can be switched OFF.

- External Clock (GPT_CLK)

The External Clock comes from *outside the device* and can be selected to run the GPT counter. The External Clock is treated as *asynchronous to the Peripheral Clock*, (ipg_clk) and is synchronized to the Peripheral Clock (ipg_clk), *inside* the module. Therefore, the External Clock frequency is limited to $< 1/4$ frequency of the Peripheral Clock (ipg_clk), for proper GPT operations. Note that in Low Power modes, *if* the Peripheral Clock (ipg_clk) is not available, then the External Clock *cannot be used* to run the counter.

- Crystal Oscillator Clock (ipg_clk_24M)

This 24 MHz Crystal Oscillator Clock (provided by the CCM) is intended to be used against frequency change of Peripheral Clock (ipg_clk) changes to provide a more accurate timer clock for operation system. The CCM is expected to provide the 24 MHz Crystal Oscillator Clock *without* synchronizing it to the System Bus Clock (ahb_clk) in Normal functional mode. Synchronization is done in GPT module. Before synchronization, the 24 MHz Crystal Oscillator Clock is divided by a 24 MHz clock prescaler, to make sure the clock frequency less than half of System Bus Clock (ahb_clk).

The clock input source is configured using the clock source field (CLKSRC, in the GPT_CR control register). The clock input to the prescaler can be disabled by programming the CLKSRC bits (of the GPT_CR control register) to 000. **The CLKSRC field value should be changed only after disabling the GPT** (by setting the EN bit in the GPT_CR to 0).

The PRESCALER field selects the divide ratio of the input clock that drives the main counter. The prescaler can divide the input clock by a value (from 1 to 4096) and can be changed *at any time*. A change in the value of the PRESCALER field *immediately affects* the output clock frequency.

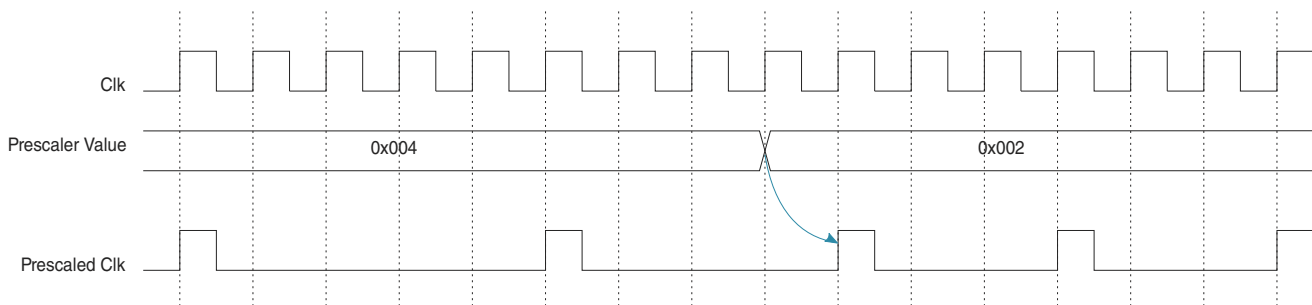


Figure 12-3. Prescaler Value Change Timing Diagram

12.1.4 Functional Description

This section provides a complete functional description of the GPT.

12.1.4.1 Operating Modes

The GPT counter can be programmed to work in either of two modes: Restart mode or Free-Run mode.

12.1.4.1.1 Restart Mode

In Restart mode (selectable through the GPT Control Register GPT_CR), when the counter reaches the compared value, the counter resets and starts again from 0x00000000. The Restart feature is associated only with Compare Channel 1.

Any write access to the Compare register of Channel 1 will reset the GPT counter. This is done to avoid possibly missing a compare event when compare value is changed from a higher value to lower value while counting is proceeding.

For the other two compare channels, when the compare event occurs the counter is *not reset*.

12.1.4.1.2 Free-Run Mode

In Free-Run mode, when compare events occur for all 3 channels, the counter is *not reset*; instead the counter continues to count until 0xffffffff, and then rolls over (to 0x00000000).

12.1.4.2 Operation

The General Purpose Timer (GPT) has a single counter (GPT_CNT) that is a 32-bit free-running *up-counter*, which starts counting *after it is enabled by software* (EN=1). The counter's clock source is the output of the prescaler labelled "Prescaler output" in [Figure 12-1](#).

- If the GPT timer is disabled (EN=0), then the Main Counter *and* Prescaler Counter freeze their current count values. The ENMOD bit determines the value of the GPT counter when the EN bit is set and the Counter is enabled again.
 - If the ENMOD bit is set (=1), then the Main Counter and Prescaler Counter values are reset to 0, when GPT is enabled (EN=1).
 - If ENMOD bit is programmed to 0, then the Main Counter and Prescaler Counter restart counting from their frozen values, when GPT is enabled again (EN=1).
- If GPT is programmed to be disabled in a low power mode (STOP/WAIT), then the Main Counter and Prescaler Counter freeze at their current count values *when* GPT enters low power mode. When GPT exits a low power mode, the Main Counter and Prescaler Counter start counting from their frozen values *regardless* of the ENMOD bit value. Note that the GPT_CNT can be read *at any time* by the processor, and that *both* Input Capture Channels use the *same* counter (GPT_CNT).

- A hardware reset resets all the GPT registers to their respective reset values. All registers except the Output Compare Registers (OCR1, OCR2, OCR3) obtain a value of 0x0. The Compare registers are reset to 0xFFFF_FFFF.
- The software reset (SWR bit in the GPT_CR control register) resets *all* of the register bits *except* the EN, ENMOD, STOPEN, WAITEN, and DBGEN bits. The state of these bits is not affected by a software reset. Note that a software reset can be given *while the GPT is disabled*.

12.1.4.2.1 Input Capture

There are two Input Capture Channels, and each Input Capture Channel has a dedicated capture pin, capture register and input edge detection/selection logic. Each input capture function has an associated status flag, and can cause the processor to make an interrupt service request.

When a selected edge transition occurs on an Input Capture pin, the contents of the GPT_CNT is captured on the corresponding capture register and the appropriate interrupt status flag is set. An interrupt request can be generated when the transition is detected *if* its corresponding enable bit is set (in the Interrupt Register). The capture can be programmed to occur on the input pin's rising edge, falling edge, on both rising and falling edges, or the capture can be disabled. The events are synchronized with the clock that was selected to run the counter. Only those transitions that occur at least one clock cycle (clock selected to run the counter) *after* the previous recorded transition will be guaranteed to trigger a capture event. There can be up to one clock cycle of uncertainty in the latching of the input transition. The Input Capture registers can be read *at any time* without affecting their values.

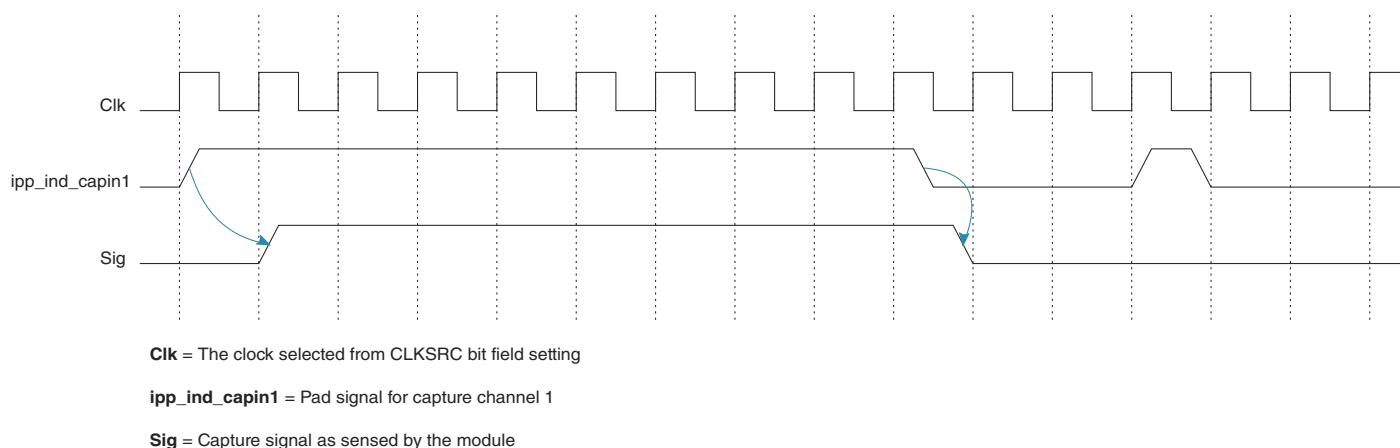


Figure 12-4. Input Capture Event Timing

12.1.4.2.2 Output Compare

The three Output Compare Channels *use the same counter* (GPT_CNT) as the Input Capture Channels. When the programmed content of an Output Compare register matches the value in GPT_CNT, an output compare status flag is set and an interrupt is generated (if the corresponding bit is set in the interrupt register). Consequently, the Output Compare timer pin will be set, cleared, toggled, not affected at all or provide an active-low pulse for one input clock period (subject to the restriction on the maximum frequency allowed on the pad) according to the mode bits (that were programmed).

There is also a "forced-compare" feature that allows the software to generate a compare event when required, *without the condition of the counter value that is equal to the compare value*. The action taken as a result of a forced compare is the same as when an output compare match occurs, *except that the status flags are not set and no interrupt can be generated*. Forced channels take programmed action immediately after the write to the force-compare bits. These bits are self-negating and always read as zeros.

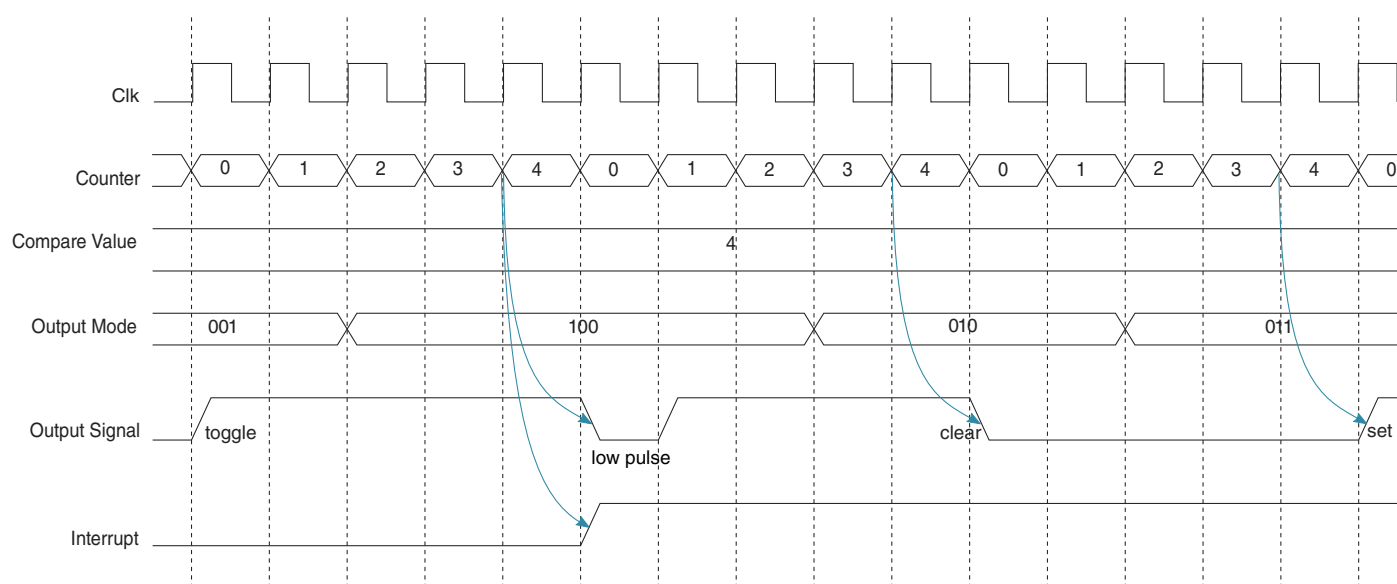


Figure 12-5. Output Compare and Interrupt Timing

12.1.4.2.3 Interrupts

There are 6 different interrupts that are generated by the GPT. If the selected clock for running the counter is available, then *all interrupts can be generated in Low Power and Debug modes*.

- Rollover Interrupt

The Rollover Interrupt is generated when the GPT counter reaches 0xffffffff, then resets to 0x00000000 and continues counting. The Rollover Interrupt is enabled by the ROVIE bit in the GPT_IR register; the associated status bit is the ROV bit in the GPT_SR register.

- Input Capture Interrupt 1, 2

After a capture event occurs, the associated Input Capture Channel generates an interrupt. The "capture event" interrupts are enabled by the IF2IE and IF1IE bits (in the GPT_IR register); the associated status bits are IF2 and IF1 (in the GPT_SR register). The capture of the counter value because of a capture event is *not affected by a pending capture interrupt*. The Capture register is updated with a new counter value when a capture event occurs, regardless of whether that Capture Channels' interrupt has been serviced or not.

- Output Compare Interrupt 1, 2, 3

After a compare event occurs, the associated Output Compare Channel generates an interrupt. The "compare event" interrupts are enabled by the OF3IE, OF2IE, and OF1IE bits (in the GPT_IR register); the associated status bits are OF3, OF2, and OF1 (in the GPT_SR register). A "forced compare" does not generate an interrupt.

A *cumulative* interrupt line is also present, which is asserted whenever any of the above interrupts are posted. The cumulative interrupt line has *no* associated enables or status bits.

12.1.4.2.4 Low Power Mode Behavior

In Low Power modes, if the clock from the selected clock source is available (except for the External Clock (GPT_CLK), which can be used *only if* the Peripheral Clock (ipg_clk) is available), the counter will continue to run depending on whether the control bit for that mode is set. If the clock is not present or if the corresponding low power bit in the GPT_CR control register is 0, the Main Counter and the Prescaler Counter freeze at their current values and resume counting (from their frozen values) when the Low Power mode is exited.

12.1.4.2.5 Debug Mode Behavior

In Debug mode, the modules in the device have the option of continuing to run or be halted.

- If the DBGEN bit is set, then the GPT timer will continue to run in Debug mode.
- If the DBGEN bit is not set (in the GPT_CR control register), then the GPT timer is halted.

12.1.5 Initialization/ Application Information

12.1.5.1 Selecting the Clock Source

The CLKSRC field in the GPT_CR register selects the clock source. The CLKSRC field value should be changed only after disabling the GPT (EN=0).

The software sequence to be followed while changing clock source is:

1. Disable GPT by setting EN=0 in GPT_CR register.
2. Disable GPT interrupt register (GPT_IR).
3. Configure Output Mode to unconnected/ disconnected—Write zeros in OM3, OM2, and OM1 in GPT_CR
4. Disable Input Capture Modes—Write zeros in IM1 and IM2 in GPT_CR
5. Change clock source CLKSRC to the desired value in GPT_CR register.
6. Assert the SWR bit in GPT_CR register.
7. Clear GPT status register (GPT_SR) (i.e., w1c).
8. Set ENMOD=1 in GPT_CR register, to bring GPT counter to 0x00000000.
9. Enable GPT (EN=1) in GPT_CR register.
10. Enable GPT interrupt register (GPT_IR).

12.1.6 GPT Memory Map/Register Definition

The GPT has 10 user-accessible 32-bit registers, which are used to configure, operate, and monitor the state of the GPT.

An IP bus write access to the GPT Control Register (GPT_CR) and the GPT Output Compare Register1 (GPT_OCR1) results in *one cycle of wait state*, while other valid IP bus accesses incur 0 wait states.

Irrespective of the Response Select signal value, a Write access to the GPT Status Registers (Read-only registers GPT_ICR1, GPT_ICR2, GPT_CNT) will generate a bus exception.

General Purpose Timer (GPT)

- If the Response Select signal is driven Low, then the Read/Write access to the *unimplemented* address space of GPT (*ips_addr* is greater than or equal to \$BASE + \$028) will generate a bus exception.
- If the Response Select is driven High, then the Read/Write access to the unimplemented address space of GPT will *not* generate any error response (like a bus exception).

GPT memory map

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
302D_0000	GPT Control Register (GPT1_CR)	32	R/W	0000_0000h	12.1.6.1/3839
302D_0004	GPT Prescaler Register (GPT1_PR)	32	R/W	0000_0000h	12.1.6.2/3843
302D_0008	GPT Status Register (GPT1_SR)	32	R/W	0000_0000h	12.1.6.3/3844
302D_000C	GPT Interrupt Register (GPT1_IR)	32	R/W	0000_0000h	12.1.6.4/3845
302D_0010	GPT Output Compare Register 1 (GPT1_OCR1)	32	R/W	FFFF_FFFFh	12.1.6.5/3846
302D_0014	GPT Output Compare Register 2 (GPT1_OCR2)	32	R/W	FFFF_FFFFh	12.1.6.6/3847
302D_0018	GPT Output Compare Register 3 (GPT1_OCR3)	32	R/W	FFFF_FFFFh	12.1.6.7/3847
302D_001C	GPT Input Capture Register 1 (GPT1_ICR1)	32	R	0000_0000h	12.1.6.8/3848
302D_0020	GPT Input Capture Register 2 (GPT1_ICR2)	32	R	0000_0000h	12.1.6.9/3848
302D_0024	GPT Counter Register (GPT1_CNT)	32	R	0000_0000h	12.1.6.10/3849
302E_0000	GPT Control Register (GPT2_CR)	32	R/W	0000_0000h	12.1.6.1/3839
302E_0004	GPT Prescaler Register (GPT2_PR)	32	R/W	0000_0000h	12.1.6.2/3843
302E_0008	GPT Status Register (GPT2_SR)	32	R/W	0000_0000h	12.1.6.3/3844
302E_000C	GPT Interrupt Register (GPT2_IR)	32	R/W	0000_0000h	12.1.6.4/3845
302E_0010	GPT Output Compare Register 1 (GPT2_OCR1)	32	R/W	FFFF_FFFFh	12.1.6.5/3846
302E_0014	GPT Output Compare Register 2 (GPT2_OCR2)	32	R/W	FFFF_FFFFh	12.1.6.6/3847
302E_0018	GPT Output Compare Register 3 (GPT2_OCR3)	32	R/W	FFFF_FFFFh	12.1.6.7/3847
302E_001C	GPT Input Capture Register 1 (GPT2_ICR1)	32	R	0000_0000h	12.1.6.8/3848

Table continues on the next page...

GPT memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
302E_0020	GPT Input Capture Register 2 (GPT2_ICR2)	32	R	0000_0000h	12.1.6.9/3848
302E_0024	GPT Counter Register (GPT2_CNT)	32	R	0000_0000h	12.1.6.10/3849
302F_0000	GPT Control Register (GPT3_CR)	32	R/W	0000_0000h	12.1.6.1/3839
302F_0004	GPT Prescaler Register (GPT3_PR)	32	R/W	0000_0000h	12.1.6.2/3843
302F_0008	GPT Status Register (GPT3_SR)	32	R/W	0000_0000h	12.1.6.3/3844
302F_000C	GPT Interrupt Register (GPT3_IR)	32	R/W	0000_0000h	12.1.6.4/3845
302F_0010	GPT Output Compare Register 1 (GPT3_OCR1)	32	R/W	FFFF_FFFFh	12.1.6.5/3846
302F_0014	GPT Output Compare Register 2 (GPT3_OCR2)	32	R/W	FFFF_FFFFh	12.1.6.6/3847
302F_0018	GPT Output Compare Register 3 (GPT3_OCR3)	32	R/W	FFFF_FFFFh	12.1.6.7/3847
302F_001C	GPT Input Capture Register 1 (GPT3_ICR1)	32	R	0000_0000h	12.1.6.8/3848
302F_0020	GPT Input Capture Register 2 (GPT3_ICR2)	32	R	0000_0000h	12.1.6.9/3848
302F_0024	GPT Counter Register (GPT3_CNT)	32	R	0000_0000h	12.1.6.10/3849
306E_0000	GPT Control Register (GPT6_CR)	32	R/W	0000_0000h	12.1.6.1/3839
306E_0004	GPT Prescaler Register (GPT6_PR)	32	R/W	0000_0000h	12.1.6.2/3843
306E_0008	GPT Status Register (GPT6_SR)	32	R/W	0000_0000h	12.1.6.3/3844
306E_000C	GPT Interrupt Register (GPT6_IR)	32	R/W	0000_0000h	12.1.6.4/3845
306E_0010	GPT Output Compare Register 1 (GPT6_OCR1)	32	R/W	FFFF_FFFFh	12.1.6.5/3846
306E_0014	GPT Output Compare Register 2 (GPT6_OCR2)	32	R/W	FFFF_FFFFh	12.1.6.6/3847
306E_0018	GPT Output Compare Register 3 (GPT6_OCR3)	32	R/W	FFFF_FFFFh	12.1.6.7/3847
306E_001C	GPT Input Capture Register 1 (GPT6_ICR1)	32	R	0000_0000h	12.1.6.8/3848
306E_0020	GPT Input Capture Register 2 (GPT6_ICR2)	32	R	0000_0000h	12.1.6.9/3848
306E_0024	GPT Counter Register (GPT6_CNT)	32	R	0000_0000h	12.1.6.10/3849

GPT memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
306F_0000	GPT Control Register (GPT5_CR)	32	R/W	0000_0000h	12.1.6.1/3839
306F_0004	GPT Prescaler Register (GPT5_PR)	32	R/W	0000_0000h	12.1.6.2/3843
306F_0008	GPT Status Register (GPT5_SR)	32	R/W	0000_0000h	12.1.6.3/3844
306F_000C	GPT Interrupt Register (GPT5_IR)	32	R/W	0000_0000h	12.1.6.4/3845
306F_0010	GPT Output Compare Register 1 (GPT5_OCR1)	32	R/W	FFFF_FFFFh	12.1.6.5/3846
306F_0014	GPT Output Compare Register 2 (GPT5_OCR2)	32	R/W	FFFF_FFFFh	12.1.6.6/3847
306F_0018	GPT Output Compare Register 3 (GPT5_OCR3)	32	R/W	FFFF_FFFFh	12.1.6.7/3847
306F_001C	GPT Input Capture Register 1 (GPT5_ICR1)	32	R	0000_0000h	12.1.6.8/3848
306F_0020	GPT Input Capture Register 2 (GPT5_ICR2)	32	R	0000_0000h	12.1.6.9/3848
306F_0024	GPT Counter Register (GPT5_CNT)	32	R	0000_0000h	12.1.6.10/3849
3070_0000	GPT Control Register (GPT4_CR)	32	R/W	0000_0000h	12.1.6.1/3839
3070_0004	GPT Prescaler Register (GPT4_PR)	32	R/W	0000_0000h	12.1.6.2/3843
3070_0008	GPT Status Register (GPT4_SR)	32	R/W	0000_0000h	12.1.6.3/3844
3070_000C	GPT Interrupt Register (GPT4_IR)	32	R/W	0000_0000h	12.1.6.4/3845
3070_0010	GPT Output Compare Register 1 (GPT4_OCR1)	32	R/W	FFFF_FFFFh	12.1.6.5/3846
3070_0014	GPT Output Compare Register 2 (GPT4_OCR2)	32	R/W	FFFF_FFFFh	12.1.6.6/3847
3070_0018	GPT Output Compare Register 3 (GPT4_OCR3)	32	R/W	FFFF_FFFFh	12.1.6.7/3847
3070_001C	GPT Input Capture Register 1 (GPT4_ICR1)	32	R	0000_0000h	12.1.6.8/3848
3070_0020	GPT Input Capture Register 2 (GPT4_ICR2)	32	R	0000_0000h	12.1.6.9/3848
3070_0024	GPT Counter Register (GPT4_CNT)	32	R	0000_0000h	12.1.6.10/3849

12.1.6.1 GPT Control Register (GPTx_CR)

The GPT Control Register (GPT_CR) is used to program and configure GPT operations. An IP Bus Write to the GPT Control Register occurs after one cycle of wait state, while an IP Bus Read occurs after 0 wait states.

Address: Base address + 0h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0	0	0	OM3			OM2			OM1			IM2		IM1	
W	FO3	FO2	FO1													
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SWR	0				EN_24M	FRR	CLKSRC			STOPEN	DOZEEN	WAITEN	DBGEN	ENMOD	EN
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GPTx_CR field descriptions

Field	Description
31 FO3	FO3 Force Output Compare Channel 3 FO2 Force Output Compare Channel 2 FO1 Force Output Compare Channel 1 The FOn bit causes the pin action <i>programmed</i> for the timer Output Compare <i>n</i> pin (according to the OMn bits in this register). • The OFn flag (OF3, OF2, OF1) in the status register is not affected. • This bit is self-negating and always read as zero. 0 Writing a 0 has no effect. 1 Causes the programmed pin action on the timer Output Compare <i>n</i> pin; the OFn flag is not set.
30 FO2	See FO3

Table continues on the next page...

GPTx_CR field descriptions (continued)

Field	Description
29 FO1	See F03
28–26 OM3	OM3 (bits 28-26) controls the Output Compare Channel 3 operating mode. OM2 (bits 25-23) controls the Output Compare Channel 2 operating mode. OM1 (bits 22-20) controls the Output Compare Channel 1 operating mode. The OMn bits specify the response that a compare event will generate on the output pin of Output Compare Channel n. - The toggle, clear, and set options cause a change on the output pin <i>only</i> if a compare event occurs. - When OMn is programmed as 1xx (active low pulse), the output pin is set to one immediately on the next input clock; a low pulse (that is an input clock in width) occurs when there is a compare event. Note that here, "input clock" refers to the clock selected by the CLKSRC bits of the GPT Control Register. 000 Output disconnected. No response on pin. 001 Toggle output pin 010 Clear output pin 011 Set output pin 1xx Generate an active low pulse (that is one input clock wide) on the output pin.
25–23 OM2	See OM3
22–20 OM1	See OM3
19–18 IM2	IM2 (bits 19-18, Input Capture Channel 2 operating mode) IM1 (bits 17-16, Input Capture Channel 1 operating mode) The IMn bit field determines the transition on the input pin (for Input capture channel n), which will trigger a capture event. 00 capture disabled 01 capture on rising edge only 10 capture on falling edge only 11 capture on both edges
17–16 IM1	See IM2
15 SWR	Software reset. This is the software reset of the GPT module. It is a self-clearing bit. - The SWR bit is set when the module is in reset state. - The SWR bit is cleared when the reset procedure finishes. - Setting the SWR bit resets all of the registers to their default reset values, except for the EN, ENMOD, STOPEN, DOZEEN, WAITEN, and DBGEN bits in the GPT Control Register (this control register). 0 GPT is not in reset state 1 GPT is in reset state
14–11 Reserved	This read-only field is reserved and always has the value 0.
10 EN_24M	Enable 24 MHz clock input from crystal.

Table continues on the next page...

GPTx_CR field descriptions (continued)

Field	Description
	- A hardware reset resets the EN_24M bit. - A software reset <i>does not affect</i> the EN_24M bit. 0 24M clock disabled 1 24M clock enabled
9 FRR	Free-Run or Restart mode. The FRR bit determines the behavior of the GPT when a compare event in channel 1 occurs. - In Restart mode, after a compare event, the counter resets to 0x00000000 and resumes counting (after the occurrence of a compare event). - In Free-Run mode, after a compare event, the counter continues counting until 0xFFFFFFFF and then rolls over to 0. 0 Restart mode 1 Free-Run mode
8–6 CLKSRC	Clock Source select. The CLKSRC bits select which clock will go to the prescaler (and subsequently be used to run the GPT counter). - The CLKSRC bit field value should only be changed after disabling the GPT by clearing the EN bit in this register (GPT_CR). - A software reset does not affect the CLKSRC bit. 000 No clock 001 Peripheral Clock (ipg_clk) 010 High Frequency Reference Clock (ipg_clk_highfreq) 011 External Clock 100 Low Frequency Reference Clock (ipg_clk_32k) 101 Crystal oscillator as Reference Clock (ipg_clk_24M) others Reserved
5 STOPEN	GPT Stop Mode enable. The STOPEN read/write control bit enables GPT operation <i>during Stop mode</i>. - A hardware reset resets the STOPEN bit. - A software reset <i>does not affect</i> the STOPEN bit. 0 GPT is disabled in Stop mode. 1 GPT is enabled in Stop mode.
4 DOZEEN	GPT Doze Mode Enable. - A hardware reset resets the DOZEEN bit. - A software reset <i>does not affect</i> the DOZEEN bit. 0 GPT is disabled in doze mode. 1 GPT is enabled in doze mode.
3 WAITEN	GPT Wait Mode enable. The WAITEN read/write control bit enables GPT operation <i>during Wait mode</i>. - A hardware reset resets the WAITEN bit. - A software reset <i>does not affect</i> the WAITEN bit. 0 GPT is disabled in wait mode. 1 GPT is enabled in wait mode.

Table continues on the next page...

GPTx_CR field descriptions (continued)

Field	Description
2 DBGEN	GPT debug mode enable. The DBGEN read/write control bit enables GPT operation <i>during Debug mode</i>. - A hardware reset resets the DBGEN bit. - A software reset <i>does not affect</i> the DBGEN bit. 0 GPT is disabled in debug mode. 1 GPT is enabled in debug mode.
1 ENMOD	GPT Enable mode. When the GPT is disabled (EN=0), then both the Main Counter and Prescaler Counter <i>freeze their current count values</i>. The ENMOD bit determines the value of the GPT counter when Counter is enabled again (if the EN bit is set). - If the ENMOD bit is 1, then the Main Counter and Prescaler Counter values are reset to 0 after GPT is enabled (EN=1). - If the ENMOD bit is 0, then the Main Counter and Prescaler Counter restart counting <i>from their frozen values</i> after GPT is enabled (EN=1). - If GPT is programmed to be disabled in a low power mode (STOP/WAIT), then the Main Counter and Prescaler Counter <i>freeze at their current count values</i> when the GPT enters low power mode. - When GPT exits low power mode, the Main Counter and Prescaler Counter start counting from their frozen values, regardless of the ENMOD bit value. - Setting the SWR bit will clear the Main Counter and Prescaler Counter values, regardless of the value of EN or ENMOD bits. - A hardware reset resets the ENMOD bit. - A software reset <i>does not affect</i> the ENMOD bit. 0 GPT counter will retain its value when it is disabled. 1 GPT counter value is reset to 0 when it is disabled.
0 EN	GPT Enable. The EN bit is the GPT module enable bit. Before setting the EN bit, we recommend that <i>all registers be properly programmed</i>. - A hardware reset resets the EN bit. - A software reset <i>does not affect</i> the EN bit. 0 GPT is disabled. 1 GPT is enabled.

12.1.6.2 GPT Prescaler Register (GPTx_PR)

The GPT Prescaler Register (GPT_PR) contains bits that determine the *divide value* of the clock that runs the counter.

Address: Base address + 4h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PRESCALER24M				PRESCALER											
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GPTx_PR field descriptions

Field	Description
31–16 Reserved	This read-only field is reserved and always has the value 0.
15–12 PRESCALER24M	Prescaler bits. 24M crystal clock is divided by [PRESCALER24M + 1] before selected by the CLKSRC field. If 24M crystal clock is not selected, this feild takes no effect. 0x0 Divide by 1 0x1 Divide by 2 ... 0xF Divide by 16
PRESCALER	Prescaler bits. The clock selected by the CLKSRC field is divided by [PRESCALER + 1], and then used to run the counter. - A change in the value of the PRESCALER bits cause the Prescaler counter to reset and a new count period to start immediately. - See Figure 12-3 for the timing diagram. 0x000 Divide by 1 0x001 Divide by 2 ... 0xFFFF Divide by 4096

12.1.6.3 GPT Status Register (GPTx_SR)

The GPT Status Register (GPT_SR) contains bits that indicate that a counter has rolled over, and if any event has occurred on the Input Capture and Output Compare channels. The bits are cleared by writing a 1 to them.

Address: Base address + 8h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0								ROV		IF2	IF1	OF3	OF2	OF1	
W									w1c		w1c	w1c	w1c	w1c	w1c	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GPTx_SR field descriptions

Field	Description
31–6 Reserved	This read-only field is reserved and always has the value 0.
5 ROV	Rollover Flag. The ROV bit indicates that the counter has reached its <i>maximum possible value</i> and <i>rolled over</i> to 0 (from which the counter continues counting). The ROV bit is only set if the counter has reached 0xFFFFFFFF in both Restart and Free-Run modes. 0 Rollover has not occurred. 1 Rollover has occurred.
4 IF2	IF2 Input capture 2 Flag IF1 Input capture 1 Flag The IFn bit indicates that a capture event has occurred on Input Capture channel n. 0 Capture event has not occurred. 1 Capture event has occurred.
3 IF1	See IF2
2 OF3	OF3 Output Compare 3 Flag OF2 Output Compare 2 Flag OF1 Output Compare 1 Flag The OFn bit indicates that a compare event has occurred on Output Compare channel n. 0 Compare event has not occurred. 1 Compare event has occurred.

Table continues on the next page...

GPTx_SR field descriptions (continued)

Field	Description
1 OF2	See OF3
0 OF1	See OF3

12.1.6.4 GPT Interrupt Register (GPTx_IR)

The GPT Interrupt Register (GPT_IR) contains bits that control whether interrupts are generated after rollover, input capture and output compare events.

Address: Base address + Ch offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0										ROVIE	IF2IE	IF1IE	OF3IE	OF2IE	OF1IE
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GPTx_IR field descriptions

Field	Description
31–6 Reserved	This read-only field is reserved and always has the value 0.
5 ROVIE	Rollover Interrupt Enable. The ROVIE bit controls the Rollover interrupt. 0 Rollover interrupt is disabled. 1 Rollover interrupt enabled.
4 IF2IE	IF2IE Input capture 2 Interrupt Enable IF1IE Input capture 1 Interrupt Enable The IF n IE bit controls the IF n IE Input Capture n Interrupt Enable. 0 IF2IE Input Capture n Interrupt Enable is disabled. 1 IF2IE Input Capture n Interrupt Enable is enabled.

Table continues on the next page...

GPTx_IR field descriptions (continued)

Field	Description
3 IF1IE	See IF2IE
2 OF3IE	OF3IE Output Compare 3 Interrupt Enable OF2IE Output Compare 2 Interrupt Enable OF1IE Output Compare 1 Interrupt Enable The OF n IE bit controls the Output Compare Channel n interrupt. 0 Output Compare Channel n interrupt is disabled. 1 Output Compare Channel n interrupt is enabled.
1 OF2IE	See OF3IE
0 OF1IE	See OF3IE

12.1.6.5 GPT Output Compare Register 1 (GPTx_OCR1)

The GPT Compare Register 1 (GPT_OCR1) holds the value that determines when a compare event will be generated on Output Compare Channel 1. Any write access to the Compare register of Channel 1 while in Restart mode (FRR=0) will reset the GPT counter.

An IP Bus Write access to the GPT Output Compare Register1 (GPT_OCR1) occurs *after* one cycle of wait state; an IP Bus Read access occurs *immediately* (0 wait states).

Address: Base address + 10h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	COMP																															
W																																
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	

GPTx_OCR1 field descriptions

Field	Description
COMP	Compare Value. When the counter value equals the COMP bit field value, a compare event is generated on Output Compare Channel 1.

12.1.6.6 GPT Output Compare Register 2 (GPTx_OCR2)

The GPT Compare Register 2 (GPT_OCR2) holds the value that determines when a compare event will be generated on Output Compare Channel 2.

Address: Base address + 14h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
	COMP																															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	

GPTx_OCR2 field descriptions

Field	Description
COMP	Compare Value. When the counter value equals the COMP bit field value, a compare event is generated on Output Compare Channel 2.

12.1.6.7 GPT Output Compare Register 3 (GPTx_OCR3)

The GPT Compare Register 3 (GPT_OCR3) holds the value that determines when a compare event will be generated on Output Compare Channel 3.

Address: Base address + 18h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R																																
W																																
	COMP																															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	

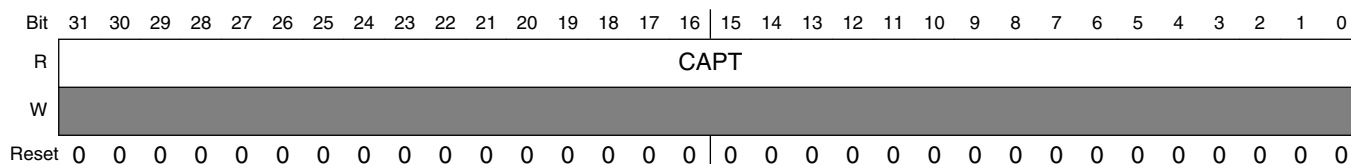
GPTx_OCR3 field descriptions

Field	Description
COMP	Compare Value. When the counter value equals the COMP bit field value, a compare event is generated on Output Compare Channel 3.

12.1.6.8 GPT Input Capture Register 1 (GPTx_ICR1)

The GPT Input Capture Register 1 (GPT_ICR1) is a read-only register that holds the value *that was in the counter during the last capture event* on Input Capture Channel 1.

Address: Base address + 1Ch offset



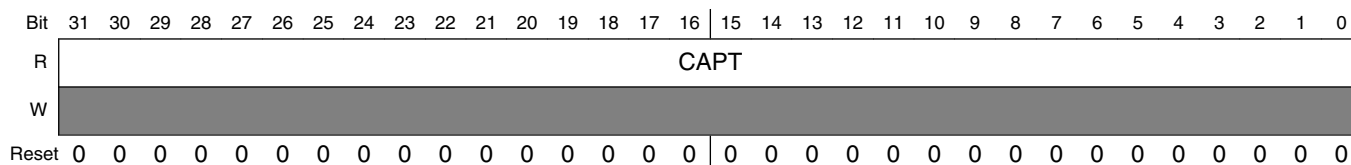
GPTx_ICR1 field descriptions

Field	Description
CAPT	Capture Value. After a capture event on Input Capture Channel 1 occurs, the current value of the counter is loaded into GPT Input Capture Register 1.

12.1.6.9 GPT Input Capture Register 2 (GPTx_ICR2)

The GPT Input capture Register 2 (GPT_ICR2) is a read-only register which holds the value that was in the counter during the last capture event on input capture channel 2.

Address: Base address + 20h offset



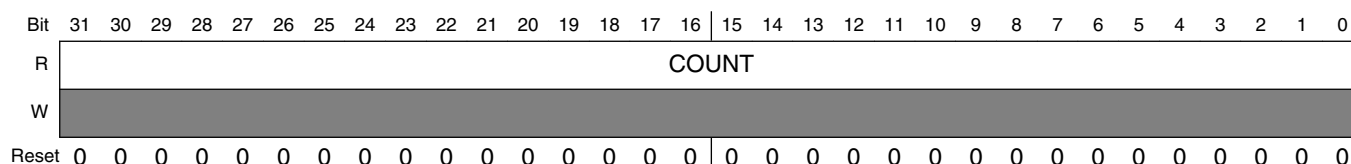
GPTx_ICR2 field descriptions

Field	Description
CAPT	Capture Value. After a capture event on Input Capture Channel 2 occurs, the current value of the counter is loaded into GPT Input Capture Register 2.

12.1.6.10 GPT Counter Register (GPTx_CNT)

The GPT Counter Register (GPT_CNT) is the main counter's register. GPT_CNT is a read-only register and can be read *without affecting the counting process* of the GPT.

Address: Base address + 24h offset



GPTx_CNT field descriptions

Field	Description
COUNT	Counter Value. The COUNT bits show the current count value of the GPT counter.

12.2 Pulse Width Modulation (PWM)

12.2.1 Overview

The Pulse Width Modulation (PWM) has a 16-bit counter, and is optimized to generate sound from stored sample audio images and it can also generate tones. It uses 16-bit resolution and a 4 x 16 data FIFO.

This section presents an overview of the PWM. A block diagram of the PWM module is shown in the figure below.

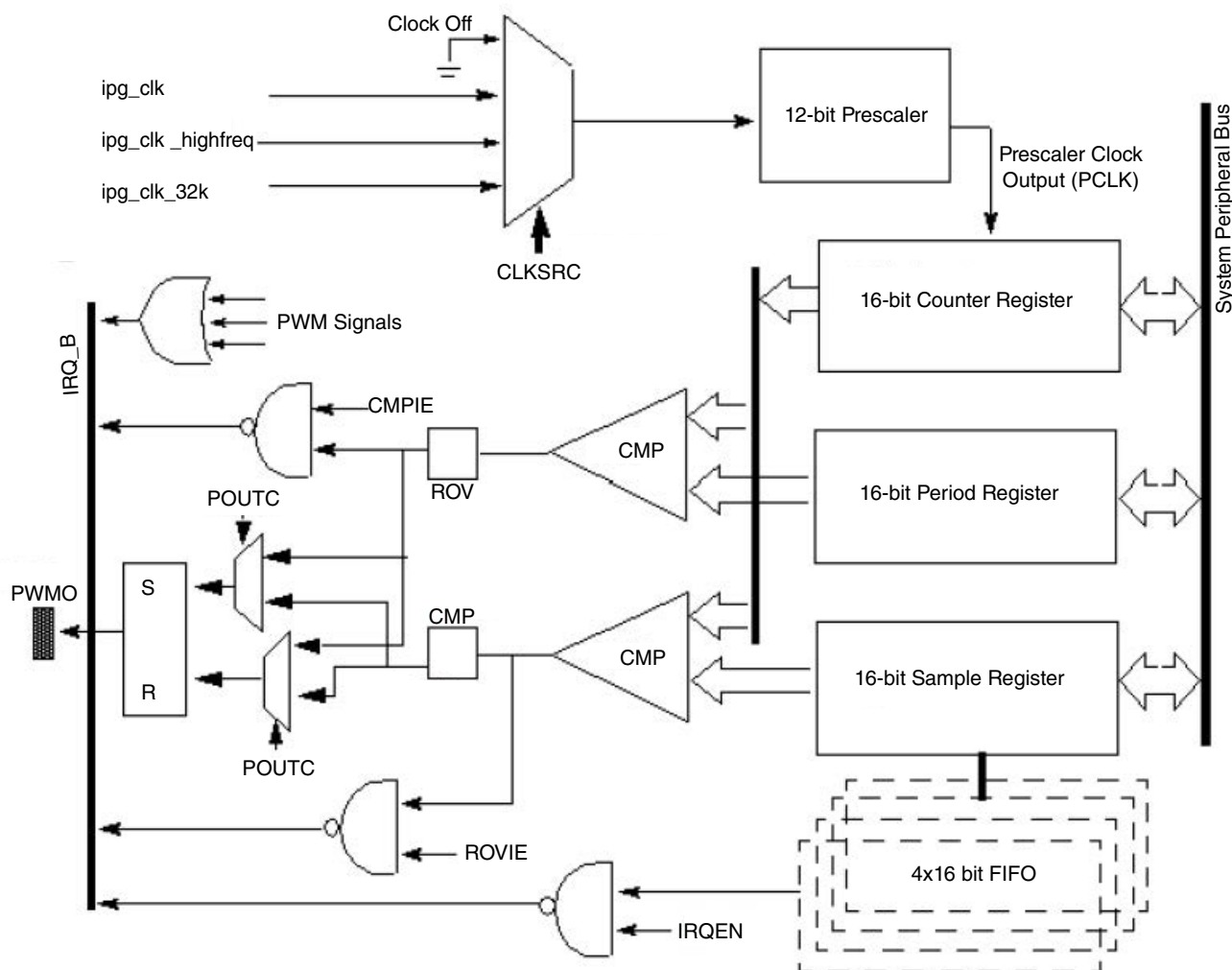


Figure 12-6. Pulse-Width Modulator Block Diagram

The following features characterize the PWM:

- 16-bit up-counter with clock source selection
- 4 x 16 FIFO to minimize interrupt overhead
- 12-bit prescaler for division of clock
- Sound and melody generation
- Active high or active low configured output
- Can be programmed to be active in low-power mode
- Can be programmed to be active in debug mode
- Interrupts at compare and rollover

12.2.2 External Signals

The PWM follows IP Bus protocol when interfacing with the processor core. PWM does not have any interface signals with any other block inside the chip except for clock and reset inputs from the Clock Control Module (CCM), System Reset Controller (SRC), and interrupt signals to the processor interrupt handler. There is a single output signal.

12.2.3 Clocks

The table found here describes the clock sources for PWM. Please see [Clock Controller Module \(CCM\)](#) for clock setting, configuration and gating information.

Table 12-2. PWM Clocks

Clock name	Clock Root	Description
ipg_clk	ipg_clk_root	Peripheral clock
ipg_clk_32k	ckil_sync_clk_root	low-frequency reference clock (32 kHz)
ipg_clk_highfreq	perclk_clk_root	high-frequency reference clock
ipg_clk_s	ipg_clk_root	Peripheral access clock

The clock that feeds the prescaler can be selected from:

- High-frequency reference clock (ipg_clk_highfreq) pat_ref or CKIH

This is a high frequency clock, provided by the Clock Control Module (CCM). This clock should be on in the low power mode when the ipg_clk is turned off. Thus, the PWM can be run on this clock in the low power mode. The CCM is expected to provide this clock after synchronizing it to ahb_clk in the normal functional mode and then switch to the unsynchronized version in the low power mode.

- Low-frequency reference clock (ipg_clk_32k, CKIL)

This is the 32 KHz low reference clock which is provided by the CCM. This clock should be on in the low power mode when ipg_clk is turned off. Thus, PWM can be run on this clock in the low power mode. The CCM is expected to provide this clock after synchronizing it to ahb_clk in the normal functional mode and then switch to the unsynchronized version in the low power mode.

- Peripheral clock (ipg_clk)

This clock should be on in normal operations. In low power mode, it can be switched off.

- Peripheral access clock (ipg_clk_s)

This clock is used for register read/write.

The clock input source is determined by the PWM control register field `PWM_CR[CLKSRC]`. The `CLKSRC` value should only be changed when the PWM is disabled.

A change in the value of the `PRESCALER` field of the control register is immediately reflected on its output clock frequency.

12.2.4 Functional Description

The following sections detail the PWM operation and function.

12.2.4.1 Operation

The output of the PWM is a toggling signal whose frequency and duty cycle can be modulated by programming the appropriate registers. It has a 16-bit up counter which counts from `0x0000` until the counter value equals the `PWM_PR + 1`. After this match occurs the counter is reset to `0x0000`.

At the beginning of a count period cycle, the `PWMO` pin is set to one (default) and the counter begins counting up from `0x0000`. The sample value in the sample FIFO is compared on each count of prescaler clock. When the sample and count values match, the `PWMO` signal is cleared to zero (default). The counter continues counting until the period match occurs and subsequently another period cycle begins.

When the PWM is enabled, the counter starts running and generates an output with the reset values in the period and sample registers. It is recommended that the programming of these registers be done before PWM is enabled.

A hardware reset results in all the PWM count and sample registers being cleared and the FIFO being flushed. The control register shows that FIFO is empty and it can be written into, and the PWM is disabled. A software reset has the same results, however the state of the `DBGEN`, `STOPEN`, `DOZEN`, and `WAITEN` bits in the control register are not affected. Software reset can be asserted even when the PWM is in disabled state.

12.2.4.1.1 FIFO

Digital sample values can be loaded into the pulse-width modulator as 16-bit words. The endianness can be changed using the `BCTR` and `HCTR` bits of the control register. A 4-word (16-bit) FIFO minimizes interrupt overhead. A maskable interrupt is generated when the number of data words fall below the water level set by the `FWM` field in the control register.

A write to the PWM_SAR sample register results in the value being stored into the FIFO if it is not full. A write when the FIFO is full sets FWE (FIFO write error) bit in the status register and the FIFO contents remain unchanged. The FIFO can be written at any time, but can be read only when the PWM is enabled. The PWM_SR[FIFOAV] field shows how many data words are currently contained in the FIFO and whether or not it can be written into.

A read on the sample register yields the current FIFO value that is being used, or will be used, by the PWM for generation on the output signal. Therefore, a write and a subsequent read on the sample register may result in different values being obtained.

12.2.4.1.2 Rollover and Compare Event

The counter is reset to 0x0000 after its value equals the PWM_PR[PERIOD] + 1 and resumes counting thereafter. This event is referred to as a rollover. For example, if PWM_PR[PERIOD] = 0x0000, the counter is reset when it equals 0x0001. When PWM_PR[PERIOD] = 0xFFFF or 0xFFFE, the counter is reset when it equals 0xFFFF. For more information, see the PWM Period Register (PWM_PR) description.

During a rollover event the output is either set (default), reset or has no effect according to the programming of the POUTC field in the control register. This event can also generate an interrupt if the respective interrupt enable bit is set in the control register.

When the counter value reaches the sample value, the output of the PWM is reset (default), set or has no effect according to the programming of the POUTC field of control register. This event is referred to as a compare event. This event can also generate an interrupt if the respective interrupt enable bit is set in the control register.

If the rollover event sets the PWM output signal, the compare event will reset it and vice versa for a particular programming configuration of POUTC field.

12.2.4.1.3 Low Power Mode Behavior

In low power mode, if the clock from the selected clock source is available, the PWM counter continues to run and an output is produced, depending on whether the control bit for that mode is set or not. In the absence of the clock itself, or if the corresponding low power bit in the control register is 0, the counter is reset and resumes counting when it exits the low power mode.

12.2.4.1.4 Debug Mode Behavior

In debug mode, PWM has the option of continuing to run or be halted. If the DBGEN bit is not set in the PWM_PWMCR, the PWM is halted. If the DBGEN bit is set, then the PWM will continue to run in the debug mode.

12.2.5 Enable Sequence for the PWM

The sequence found here should be used to enable the PWM.

1. Configure the desired settings for the PWM Control Register (PWMx_PWMCR) while keeping the PWM disabled (PWMx_PWMCR[0]=0).
2. Enable the desired interrupts in the PWM Interrupt Register (PWMx_PWMIR).
3. One to three initial samples may be written to the PWM Sample Register (PWMx_PWMSAR). The initial sample values will be loaded into the PWM FIFO even if the PWM is not yet enabled. Do not write a 4th sample because the FIFO will become full and trigger a FIFO Write Error (FWE). This error will prevent the PWM from starting once it is enabled.
4. Check the FIFO Write Error status bit (FWE), the Compare status bit (CMP) and the Roll-over status bit (ROV) in the PWM Status Register (PWMx_PWMSR) to make sure they are all zero. Any non-zero status bits should be cleared by writing a 1 to them.
5. Write the desired period to the PWM Period Register (PWMx_PWMPR).
6. Enable the PWM by writing a 1 to the PWM Enable bit, PWMx_PWMCR[0], while maintaining the other register bits in their previously configured state.

12.2.6 Disable Sequence for the PWM

The PWM can be disabled at any time by clearing the PWM enable bit, PWMx_PWMCR[0] to 0.

Any data remaining in the FIFO will not be produced at the PWM output after the PWM has been disabled and will remain in the FIFO until the PWM is enabled again. A software reset (setting PWMx_PWMCR[3] to 1) or a hardware reset will clear the FIFO and any remaining data will be lost.

12.2.7 PWM Memory Map/Register Definition

The PWM includes six user-accessible 32-bit registers.

PWM memory map

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
3066_0000	PWM Control Register (PWM1_PWMCR)	32	R/W	0000_0000h	12.2.7.1/3856
3066_0004	PWM Status Register (PWM1_PWMSR)	32	w1c	0000_0008h	12.2.7.2/3858
3066_0008	PWM Interrupt Register (PWM1_PWMIR)	32	R/W	0000_0000h	12.2.7.3/3860
3066_000C	PWM Sample Register (PWM1_PWMSAR)	32	R/W	0000_0000h	12.2.7.4/3860
3066_0010	PWM Period Register (PWM1_PWMPR)	32	R/W	0000_FFFEh	12.2.7.5/3861
3066_0014	PWM Counter Register (PWM1_PWMCNR)	32	R	0000_0000h	12.2.7.6/3862
3067_0000	PWM Control Register (PWM2_PWMCR)	32	R/W	0000_0000h	12.2.7.1/3856
3067_0004	PWM Status Register (PWM2_PWMSR)	32	w1c	0000_0008h	12.2.7.2/3858
3067_0008	PWM Interrupt Register (PWM2_PWMIR)	32	R/W	0000_0000h	12.2.7.3/3860
3067_000C	PWM Sample Register (PWM2_PWMSAR)	32	R/W	0000_0000h	12.2.7.4/3860
3067_0010	PWM Period Register (PWM2_PWMPR)	32	R/W	0000_FFFEh	12.2.7.5/3861
3067_0014	PWM Counter Register (PWM2_PWMCNR)	32	R	0000_0000h	12.2.7.6/3862
3068_0000	PWM Control Register (PWM3_PWMCR)	32	R/W	0000_0000h	12.2.7.1/3856
3068_0004	PWM Status Register (PWM3_PWMSR)	32	w1c	0000_0008h	12.2.7.2/3858
3068_0008	PWM Interrupt Register (PWM3_PWMIR)	32	R/W	0000_0000h	12.2.7.3/3860
3068_000C	PWM Sample Register (PWM3_PWMSAR)	32	R/W	0000_0000h	12.2.7.4/3860
3068_0010	PWM Period Register (PWM3_PWMPR)	32	R/W	0000_FFFEh	12.2.7.5/3861
3068_0014	PWM Counter Register (PWM3_PWMCNR)	32	R	0000_0000h	12.2.7.6/3862
3069_0000	PWM Control Register (PWM4_PWMCR)	32	R/W	0000_0000h	12.2.7.1/3856
3069_0004	PWM Status Register (PWM4_PWMSR)	32	w1c	0000_0008h	12.2.7.2/3858
3069_0008	PWM Interrupt Register (PWM4_PWMIR)	32	R/W	0000_0000h	12.2.7.3/3860
3069_000C	PWM Sample Register (PWM4_PWMSAR)	32	R/W	0000_0000h	12.2.7.4/3860

Table continues on the next page...

PWM memory map (continued)

Absolute address (hex)	Register name	Width (in bits)	Access	Reset value	Section/page
3069_0010	PWM Period Register (PWM4_PWMPR)	32	R/W	0000_FFFEh	12.2.7.5/3861
3069_0014	PWM Counter Register (PWM4_PWMCNR)	32	R	0000_0000h	12.2.7.6/3862

12.2.7.1 PWM Control Register (PWMx_PWMCR)

The PWM control register (PWM_PWMCR) is used to configure the operating settings of the PWM. It contains the prescaler for the clock division.

Address: Base address + 0h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0				FWM		STOPEN	DOZEN	WAITEN	DBGEN	BCTR	HCTR	POUTC		CLKSRC	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	PRESCALER												SWR	REPEAT		EN
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PWMx_PWMCR field descriptions

Field	Description
31–28 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
27–26 FWM	FIFO Water Mark. These bits are used to set the data level at which the FIFO empty flag will be set and the corresponding interrupt generated 00 FIFO empty flag is set when there are more than or equal to 1 empty slots in FIFO 01 FIFO empty flag is set when there are more than or equal to 2 empty slots in FIFO 10 FIFO empty flag is set when there are more than or equal to 3 empty slots in FIFO 11 FIFO empty flag is set when there are more than or equal to 4 empty slots in FIFO
25 STOPEN	Stop Mode Enable. This bit keeps the PWM functional while in stop mode. When this bit is cleared, the input clock is gated off in stop mode. This bit is not affected by software reset. It is cleared by hardware reset.

Table continues on the next page...

PWMx_PWMCR field descriptions (continued)

Field	Description
	0 Inactive in stop mode 1 Active in stop mode
24 DOZEN	Doze Mode Enable. This bit keeps the PWM functional in doze mode. When this bit is cleared, the input clock is gated off in doze mode. This bit is not affected by software reset. It is cleared by hardware reset. 0 Inactive in doze mode 1 Active in doze mode
23 WAITEN	Wait Mode Enable. This bit keeps the PWM functional in wait mode. When this bit is cleared, the input clock is gated off in wait mode. This bit is not affected by software reset. It is cleared by hardware reset. 0 Inactive in wait mode 1 Active in wait mode
22 DBGEN	Debug Mode Enable. This bit keeps the PWM functional in debug mode. When this bit is cleared, the input clock is gated off in debug mode. This bit is not affected by software reset. It is cleared by hardware reset. 0 Inactive in debug mode 1 Active in debug mode
21 BCTR	Byte Data Swap Control. This bit determines the byte ordering of the 16-bit data when it goes into the FIFO from the sample register. 0 byte ordering remains the same 1 byte ordering is reversed
20 HCTR	Half-word Data Swap Control. This bit determines which half word data from the 32-bit IP Bus interface is written into the lower 16 bits of the sample register. 0 Half word swapping does not take place 1 Half words from write data bus are swapped
19–18 POUTC	PWM Output Configuration. This bit field determines the mode of PWM output on the output pin. 00 Output pin is set at rollover and cleared at comparison 01 Output pin is cleared at rollover and set at comparison 10 PWM output is disconnected 11 PWM output is disconnected
17–16 CLKSRC	Select Clock Source. These bits determine which clock input will be selected for running the counter. After reset the system functional clock is selected. The input clock can also be turned off if these bits are set to 00. This field value should only be changed when the PWM is disabled 00 Clock is off 01 ipg_clk 10 ipg_clk_highfreq 11 ipg_clk_32k
15–4 PRESCALER	Counter Clock Prescaler Value. This bit field determines the value by which the clock will be divided before it goes to the counter. 0x000 Divide by 1 0x001 Divide by 2 0xff Divide by 4096
3 SWR	Software Reset. PWM is reset when this bit is set to 1. It is a self clearing bit. A write 1 to this bit is a single wait state write cycle. When the block is in reset state this bit is set and is cleared when the reset

Table continues on the next page...

PWMx_PWMCR field descriptions (continued)

Field	Description
	procedure is over. Setting this bit resets all the registers to their reset values except for the DBGEN, STOPEN, DOZEN, and WAITEN bits in this control register. 0 PWM is out of reset 1 PWM is undergoing reset
2–1 REPEAT	Sample Repeat. This bit field determines the number of times each sample from the FIFO is to be used. 00 Use each sample once 01 Use each sample twice 10 Use each sample four times 11 Use each sample eight times
0 EN	PWM Enable. This bit enables the PWM. If this bit is not enabled, the clock prescaler and the counter is reset. When the PWM is enabled, it begins a new period, the output pin is set to start a new period while the prescaler and counter are released and counting begins. To make the PWM work with softreset and disable/enable, users can do software reset by setting the SWR bit, wait software reset done, configure the registers, and then enable the PWM by setting this bit to "1" Users can also disable/enable the PWM if PWM would like to be stopped and resumed with same registers configurations . 0 PWM disabled 1 PWM enabled

12.2.7.2 PWM Status Register (PWMx_PWMSR)

The PWM status register (PWM_PWMSR) contains seven bits which display the state of the FIFO and the occurrence of rollover and compare events. The FIFOAV bit is read-only but the other four bits can be cleared by writing 1 to them. The FE, ROV, and CMP bits are associated with FIFO-Empty, Roll-over, and Compare interrupts, respectively.

Address: Base address + 4h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0								FWE		CMP	ROV	FE	FIFOAV		
W									w1c		w1c	w1c	w1c			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

PWMx_PWMSR field descriptions

Field	Description
31–7 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
6 FWE	FIFO Write Error Status. This bit shows that an attempt has been made to write FIFO when it is full. 0 FIFO write error not occurred 1 FIFO write error occurred
5 CMP	Compare Status. This bit shows that a compare event has occurred. 0 Compare event not occurred 1 Compare event occurred
4 ROV	Roll-over Status. This bit shows that a roll-over event has occurred. 0 Roll-over event not occurred 1 Roll-over event occurred
3 FE	FIFO Empty Status Bit. This bit indicates the FIFO data level in comparison to the water level set by FWM field in the control register. 0 Data level is above water mark 1 When the data level falls below the mark set by FWM field
FIFOAV	FIFO Available. These read-only bits indicate the data level remaining in the FIFO. An attempted write to these bits will not affect their value and no transfer error is generated. 000 No data available 001 1 word of data in FIFO 010 2 words of data in FIFO 011 3 words of data in FIFO 100 4 words of data in FIFO 101 unused 110 unused 111 unused

12.2.7.3 PWM Interrupt Register (PWMx_PWMIR)

The PWM Interrupt register (PWM_PWMIR) contains three bits which control the generation of the compare, rollover and FIFO empty interrupts.

Address: Base address + 8h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0													CIE	RIE	FIE
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PWMx_PWMIR field descriptions

Field	Description
31–3 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
2 CIE	Compare Interrupt Enable. This bit controls the generation of the Compare interrupt. 0 Compare Interrupt not enabled 1 Compare Interrupt enabled
1 RIE	Roll-over Interrupt Enable. This bit controls the generation of the Rollover interrupt. 0 Roll-over interrupt not enabled 1 Roll-over Interrupt enabled
0 FIE	FIFO Empty Interrupt Enable. This bit controls the generation of the FIFO Empty interrupt. 0 FIFO Empty interrupt disabled 1 FIFO Empty interrupt enabled

12.2.7.4 PWM Sample Register (PWMx_PWMSAR)

The PWM sample register (PWM_PWMSAR) is the input to the FIFO. 16-bit words are loaded into the FIFO. The FIFO can be written at any time, but can be read only when the PWM is enabled. The PWM will run at the last set duty-cycle setting if all the values of the FIFO has been utilized, until the FIFO is reloaded or the PWM is disabled. When a new value is written, the duty cycle changes after the current period is over.

A value of zero in the sample register will result in the PWM0 output signal always being low/high (POUTC = 00 it will be low and POUTC = 01 it will be high), and no output waveform will be produced. If the value in this register is higher than the PERIOD + 1, the output will never be set/reset depending on POUTC value.

Address: Base address + Ch offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																SAMPLE															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PWMx_PWMSAR field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
SAMPLE	Sample Value. This is the input to the 4x16 FIFO. The value in this register denotes the value of the sample being currently used.

12.2.7.5 PWM Period Register (PWMx_PWMPR)

The PWM period register (PWM_PWMPR) determines the period of the PWM output signal. After the counter value matches PERIOD + 1, the counter is reset to start another period.

$$\text{PWM0 (Hz)} = \text{PCLK(Hz)} / (\text{period} + 2)$$

A value of zero in the PWM_PWMPR will result in a period of two clock cycles for the output signal. Writing 0xFFFF to this register will achieve the same result as writing 0xFFFE.

A change in the period value due to a write in PWM_PWMPR results in the counter being reset to zero and the start of a new count period.

NOTE

Settings PWM_PWMPR to 0xFFFF when PWMx_PWMCR REPEAT bits are set to non-zero values is not allowed.

Address: Base address + 10h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																PERIOD															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0

PWMx_PWMPR field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
PERIOD	Period Value. These bits determine the Period of the count cycle. The counter counts up to [Period Value] +1 and is then reset to 0x0000.

12.2.7.6 PWM Counter Register (PWMx_PWMCNR)

The read-only pulse-width modulator counter register (PWM_PWMCNR) contains the current count value and can be read at any time without disturbing the counter.

Address: Base address + 14h offset

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0																COUNT															
W																																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PWMx_PWMCNR field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
COUNT	Counter Value. These bits are the counter register value and denotes the current count state the counter register is in.